BRILLIANT PUBLIC SCHOOL, SITAMARHI

(Affiliated up to +2 level to C.B.S.E., New Delhi)

Class-XI

IIT-JEE Advanced Mathematics

Study Package

Session: 2014-15

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STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS-XI

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STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 1 XI M 1. Trigonometric Ratio and Identity

Index:

1. Key Concepts

2. Exercise I to V

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Trigonometric Ratios & Identities

1. Basic Trigonometric Identities:

(a) $\sin^2\theta + \cos^2\theta = 1;\ -1 \le \sin\theta \le 1;\ -1 \le \cos\theta \le 1\ \ \forall\ \theta \in R$

(b) $\sec^2\theta - \tan^2\theta = 1\ ;\ |\sec\theta| \ge 1\ \ \forall\ \ \theta \in R - \left\{(2n+1)\frac{\pi}{2}, n \in I\right\}$

(c) $\text{cosec}^2\theta - \cot^2\theta = 1\ ;\ |\text{cosec}\,\theta| \ge 1\ \ \forall\ \theta \in R - \{n\pi,\ n \in I\}$

Solved Example # 1

Prove that

(i) $\cos^4A - \sin^4A + 1 = 2\cos^2A$

(ii) $\dfrac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \dfrac{1+\sin A}{\cos A}$

Solution

(i) $\cos^4A - \sin^4A + 1$

$= (\cos^2A - \sin^2A)(\cos^2A + \sin^2A) + 1$

$= \cos^2A - \sin^2A + 1 \qquad [\because\ \cos^2A + \sin^2A = 1]$

$= 2\cos^2A$

(ii) $\dfrac{\tan A + \sec A - 1}{\tan A - \sec A + 1}$

$= \dfrac{\tan A + \sec A - (\sec^2 A - \tan^2 A)}{\tan A - \sec A + 1}$

$= \dfrac{(\tan A + \sec A)(1 - \sec A + \tan A)}{\tan A - \sec A + 1}$

$= \tan A + \sec A = \dfrac{1+\sin A}{\cos A}$

Solved Example # 2

If $\sin x + \sin^2x = 1$, then find the value of

$$\cos^{12}x + 3\cos^{10}x + 3\cos^8x + \cos^6x - 1$$

Solution

$\cos^{12}x + 3\cos^{10}x + 3\cos^8x + \cos^6x - 1$

$= (\cos^4x + \cos^2x)^3 - 1$

$= (\sin^2x + \sin x)^3 - 1 \qquad [\because\ \cos^2x = \sin x]$

$= 1 - 1 = 0$

Solved Example # 3

If $\tan\theta = m - \dfrac{1}{4m}$, then show that $\sec\theta - \tan\theta = -2m$ or $\dfrac{1}{2m}$

Solution

Depending on quadrant in which θ falls, $\sec\theta$ can be $\pm\ \dfrac{4m^2+1}{4m}$

So, if $\sec\theta = \dfrac{4m^2+1}{4m} = m + \dfrac{1}{4m}$

$\Rightarrow$ $\sec\theta - \tan\theta = \frac{1}{2m}$ and if $\sec\theta = -\left(m + \frac{1}{4m}\right)$

$\Rightarrow$ $\sec\theta - \tan\theta = -2m$

Self Practice Problem

1. Prove the followings :

(i) $\cos^6 A + \sin^6 A + 3\sin^2 A \cos^2 A = 1$

(ii) $\sec^2 A + \text{cosec}^2 A = (\tan A + \cot A)^2$

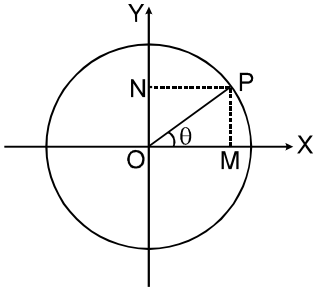
(iii) $\sec^2 A \, \text{cosec}^2 A = \tan^2 A + \cot^2 A + 2$

(iv) $(\tan\alpha + \text{cosec}\,\beta)^2 - (\cot\beta - \sec\alpha)^2 = 2\tan\alpha\cot\beta(\text{cosec}\,\alpha + \sec\beta)$

(v) $\left(\frac{1}{\sec^2\alpha - \cos^2\alpha} + \frac{1}{\text{cosec}^2\alpha - \sin^2\alpha}\right)\cos^2\alpha \sin^2\alpha = \frac{1 - \sin^2\alpha\cos^2\alpha}{2 + \sin^2\alpha\cos^2\alpha}$

2. If $\sin\theta = \frac{m^2 + 2mn}{m^2 + 2mn + 2n^2}$, then prove that $\tan\theta = \frac{m^2 + 2mn}{2mn + 2n^2}$

2. Circular Definition Of Trigonometric Functions:



$\sin\theta = \frac{PM}{OP}$ $\quad \cos\theta = \frac{OM}{OP}$

$\tan\theta = \frac{\sin\theta}{\cos\theta}$, $\cos\theta \neq 0$

$\cot\theta = \frac{\cos\theta}{\sin\theta}$, $\sin\theta \neq 0$

$\sec\theta = \frac{1}{\cos\theta}$, $\cos\theta \neq 0$ $\quad$ $\text{cosec}\,\theta = \frac{1}{\sin\theta}$, $\sin\theta \neq 0$

3. Trigonometric Functions Of Allied Angles:

If θ is any angle, then $-\theta$, $90 \pm \theta$, $180 \pm \theta$, $270 \pm \theta$, $360 \pm \theta$ etc. are called **Allied Angles**.

(a) $\sin(-\theta) = -\sin\theta$; $\cos(-\theta) = \cos\theta$

(b) $\sin(90° - \theta) = \cos\theta$; $\cos(90° - \theta) = \sin\theta$

(c) $\sin(90° + \theta) = \cos\theta$; $\cos(90° + \theta) = -\sin\theta$

(d) $\sin(180° - \theta) = \sin\theta$; $\cos(180° - \theta) = -\cos\theta$

(e) $\sin(180° + \theta) = -\sin\theta$; $\cos(180° + \theta) = -\cos\theta$

(f) $\sin(270° - \theta) = -\cos\theta$; $\cos(270° - \theta) = -\sin\theta$

(g) $\sin(270° + \theta) = -\cos\theta$; $\cos(270° + \theta) = \sin\theta$

(h) $\tan(90° - \theta) = \cot\theta$; $\cot(90° - \theta) = \tan\theta$

Solved Example # 4

Prove that

(i) $\cot A + \tan(180º + A) + \tan(90º + A) + \tan(360º - A) = 0$

(ii) $\sec(270º - A)\sec(90º - A) - \tan(270º - A)\tan(90º + A) + 1 = 0$

Solution

(i) $\cot A + \tan(180º + A) + \tan(90º + A) + \tan(360º - A)$
$= \cot A + \tan A - \cot A - \tan A = 0$

(ii) $\sec(270º - A)\sec(90º - A) - \tan(270º - A)\tan(90º + A) + 1$
$= -\text{cosec}^2 A + \cot^2 A + 1 = 0$

Self Practice Problem

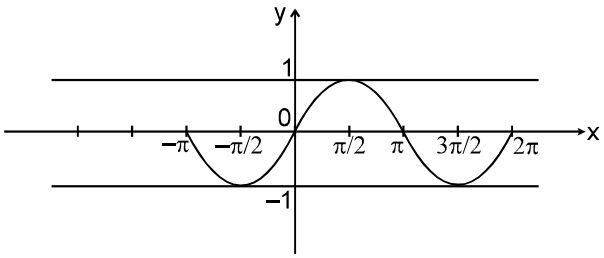
3. Prove that

(i) $\sin 420º \cos 390º + \cos(-300º)\sin(-330º) = 1$

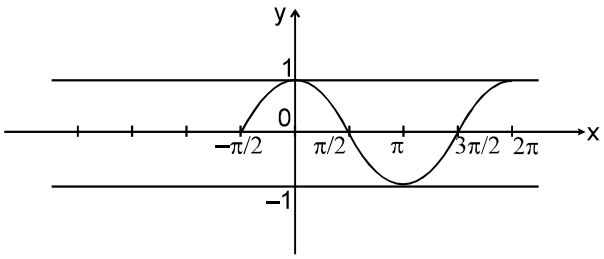
(ii) $\tan 225º \cot 405º + \tan 765º \cot 675º = 0$

4. Graphs of Trigonometric functions:

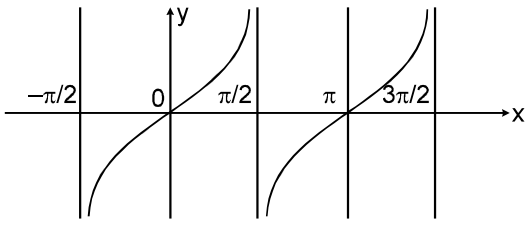
(a) $y = \sin x \quad x \in R;\ y \in [-1, 1]$



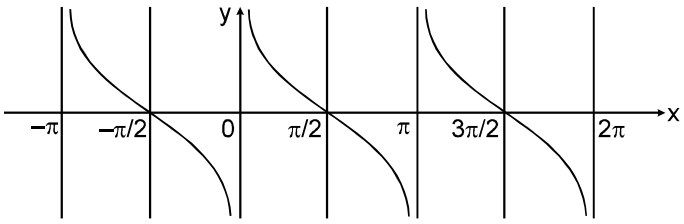
(b) $y = \cos x \quad x \in R;\ y \in [-1, 1]$



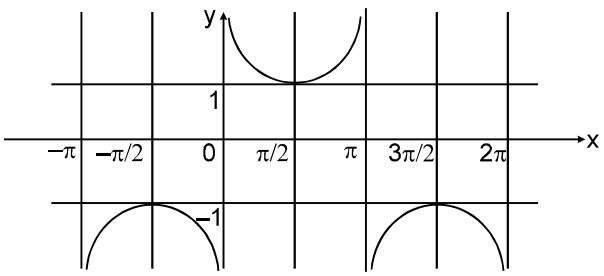
(c) $y = \tan x \quad x \in R - (2n + 1)\,\pi/2,\ n \in I\ ;\ y \in R$



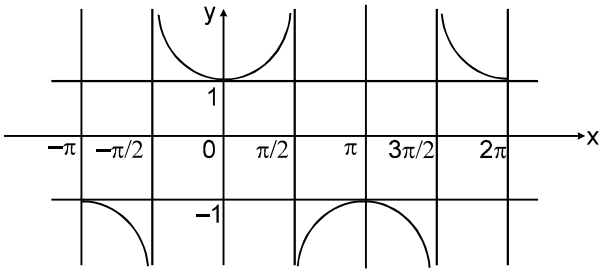
(d) $y = \cot x \quad x \in R - n\pi\ ,\ n \in I;\ y \in R$



(e) $y = \text{cosec } x \quad x \in R\ - n\pi\ ,\ n \in I\ ;\ y \in (-\infty, -1] \cup [1, \infty)$



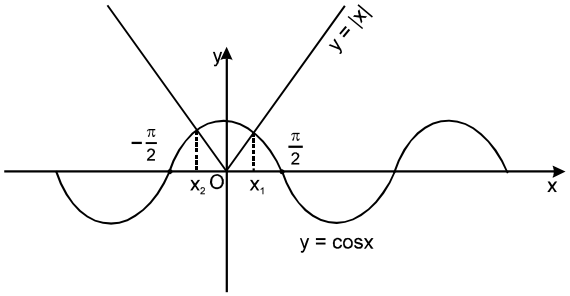
(f) $y = \sec x \quad x \in R - (2n + 1)\,\pi/2,\ n \in I\ ;\ y \in (-\infty, -1] \cup [1, \infty)$



Solved Example # 5

Find number of solutions of the equation cos x = |x|

Solution



Clearly graph of cos x & |x| intersect at two points. Hence no. of solutions is 2

Solved Example # 6

Find range of $y = \sin^2 x + 2 \sin x + 3 \ \forall \ x \in R$

Solution

We know $-1 \le \sin x \le 1$

$\Rightarrow \quad 0 \le \sin x + 1 \le 2$

$\Rightarrow \quad 2 \le (\sin x + 1)^2 + 2 \le 6$

Hence range is $y \in [2, 6]$

Self Practice Problem

4. Show that the equation $\sec^2\theta = \dfrac{4xy}{(x+y)^2}$ is only possible when $x = y \neq 0$

5. Find range of the followings.

(i) $y = 2 \sin^2 x + 5 \sin x + 1 \ \forall \ x \in R$ **Answer** $[-2, 8]$

(ii) $y = \cos^2 x - \cos x + 1 \quad \forall \ x \in R$ **Answer** $\left[\dfrac{3}{4}, 3\right]$

6. Find range of $y = \sin x, \ x \in \left[\dfrac{2\pi}{3}\ 2\pi\right]$ **Answer** $\left[-1, \dfrac{\sqrt{3}}{2}\right]$

5. Trigonometric Functions of Sum or Difference of Two Angles:

(a) $\sin (A \pm B) = \sin A \cos B \pm \cos A \sin B$

(b) $\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$

(c) $\sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A = \sin (A+B).\ \sin (A - B)$

(d) $\cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A = \cos (A+B).\ \cos (A - B)$

(e) $\tan (A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

(f) $\cot (A \pm B) = \dfrac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$

(g) $\tan(A + B + C) = \dfrac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$.

Solved Example # 7

Prove that

(i) $\sin(45^\circ + A)\cos(45^\circ - B) + \cos(45^\circ + A)\sin(45^\circ - B) = \cos(A - B)$

(ii) $\tan\left(\dfrac{\pi}{4} + \theta\right)\tan\left(\dfrac{3\pi}{4} + \theta\right) = -1$

Solution

(i) Clearly $\sin(45^\circ + A)\cos(45^\circ - B) + \cos(45^\circ + A)\sin(45^\circ - B)$
$= \sin(45^\circ + A + 45^\circ - B)$
$= \sin(90^\circ + A - B)$
$= \cos(A - B)$

(ii) $\tan\left(\dfrac{\pi}{4} + \theta\right) \times \tan\left(\dfrac{3\pi}{4} + \theta\right)$

$= \dfrac{1 + \tan\theta}{1 - \tan\theta} \times \dfrac{-1 + \tan\theta}{1 + \tan\theta} = -1$

Self Practice Problem

7. If $\sin\alpha = \dfrac{3}{5}$, $\cos\beta = \dfrac{5}{13}$, then find $\sin(\alpha + \beta)$ **Answer** $-\dfrac{33}{65}, \dfrac{63}{65}$

8. Find the value of $\sin 105^\circ$ **Answer** $\dfrac{\sqrt{3}+1}{2\sqrt{2}}$

9. Prove that $1 + \tan A \tan\dfrac{A}{2} = \tan A \cot\dfrac{A}{2} - 1 = \sec A$

6. Factorisation of the Sum or Difference of Two Sines or Cosines:

(a) $\sin C + \sin D = 2\sin\dfrac{C+D}{2}\cos\dfrac{C-D}{2}$ (b) $\sin C - \sin D = 2\cos\dfrac{C+D}{2}\sin\dfrac{C-D}{2}$

(c) $\cos C + \cos D = 2\cos\dfrac{C+D}{2}\cos\dfrac{C-D}{2}$ (d) $\cos C - \cos D = -2\sin\dfrac{C+D}{2}\sin\dfrac{C-D}{2}$

Solved Example # 8

Prove that $\sin 5A + \sin 3A = 2\sin 4A \cos A$

Solution

L.H.S. $\sin 5A + \sin 3A = 2\sin 4A \cos A$ = R.H.S.

$[\because \sin C + \sin D = 2\sin\dfrac{C+D}{2}\cos\dfrac{C-D}{2}]$

Solved Example # 9

Find the value of $2\sin 3\theta \cos\theta - \sin 4\theta - \sin 2\theta$

Solution

$2\sin 3\theta \cos\theta - \sin 4\theta - \sin 2\theta = 2\sin 3\theta \cos\theta - [2\sin 3\theta \cos\theta] = 0$

Self Practice Problem

10. Proved that

(i) $\cos 8x - \cos 5x = -2 \sin \frac{13x}{2} \sin \frac{3x}{2}$ (ii) $\frac{\sin A + \sin 2A}{\cos A - \cos 2A} = \cot \frac{A}{2}$

(iii) $\frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = \tan 4A$

(iv) $\frac{\sin A + 2\sin 3A + \sin 5A}{\sin 3A + 2\sin 5A + \sin 7A} = \frac{\sin 3A}{\sin 5A}$

(v) $\frac{\sin A - \sin 5A + \sin 9A - \sin 13A}{\cos A - \cos 5A - \cos 9A + \cos 13A} = \cot 4A$

7. Transformation of Products into Sum or Difference of Sines & Cosines:

(a) $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$ (b) $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$

(c) $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$ (d) $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$

Solved Example # 10

Prove that

(i) $\frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta} = \tan 2\theta$

(ii) $\frac{\tan 5\theta + \tan 3\theta}{\tan 5\theta - \tan 3\theta} = 4 \cos 2\theta \cos 4\theta$

Solution

(i) $\frac{2\sin 8\theta \cos \theta - 2\sin 6\theta \cos 3\theta}{2\cos 2\theta \cos \theta - 2\sin 3\theta \sin 4\theta}$

$= \frac{\sin 9\theta + \sin 7\theta - \sin 9\theta - \sin 3\theta}{\cos 3\theta + \cos \theta - \cos \theta + \cos 7\theta} = \frac{2\sin 2\theta \cos 5\theta}{2\cos 5\theta \cos 2\theta} = \tan 2\theta$

(ii) $\frac{\tan 5\theta + \tan 3\theta}{\tan 5\theta - \tan 3\theta} = \frac{\sin 5\theta \cos 3\theta + \sin 3\theta \cos 5\theta}{\sin 5\theta \cos 3\theta - \sin 3\theta \cos 5\theta} = \frac{\sin 8\theta}{\sin 2\theta} = 4 \cos 2\theta \cos 4\theta$

Self Practice Problem

11. Prove that $\sin \frac{\theta}{2} \sin \frac{7\theta}{2} + \sin \frac{3\theta}{2} \sin \frac{11\theta}{2} = \sin 2\theta \sin 5\theta$

12. Prove that $\cos A \sin (B - C) + \cos B \sin (C - A) + \cos C \sin (A - B) = 0$

13. Prove that $2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} = 0$

8. Multiple and Sub-multiple Angles :

(a) $\sin 2A = 2 \sin A \cos A$; $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$

(b) $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2 \sin^2 A$; $2 \cos^2 \frac{\theta}{2} = 1 + \cos \theta$, $2 \sin^2 \frac{\theta}{2} = 1 - \cos \theta$.

(c) $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$; $\tan \theta = \frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}$

(d) $\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$, $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$

(e) $\sin 3A = 3\sin A - 4\sin^3 A$ (f) $\cos 3A = 4\cos^3 A - 3\cos A$

(g) $\tan 3A = \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

Solved Example # 11

Prove that

(i) $\dfrac{\sin 2A}{1+\cos 2A} = \tan A$

(ii) $\tan A + \cot A = 2 \operatorname{cosec} 2A$

(iii) $\dfrac{1-\cos A+\cos B-\cos(A+B)}{1+\cos A-\cos B-\cos(A+B)} = \tan\dfrac{A}{2}\cot\dfrac{B}{2}$

Solution

(i) L.H.S. $\dfrac{\sin 2A}{1+\cos 2A} = \dfrac{2\sin A\cos A}{2\cos^2 A} = \tan A$

(ii) L.H.S. $\tan A + \cot A = \dfrac{1+\tan^2 A}{\tan A} = 2\left(\dfrac{1+\tan^2 A}{2\tan A}\right) = \dfrac{2}{\sin 2A} = 2\operatorname{cosec} 2A$

(iii) L.H.S. $\dfrac{1-\cos A+\cos B-\cos(A+B)}{1+\cos A-\cos B-\cos(A+B)}$

$$= \frac{2\sin^2\frac{A}{2} + 2\sin\frac{A}{2}\sin\left(\frac{A}{2}+B\right)}{2\cos^2\frac{A}{2} - 2\cos\frac{A}{2}\cos\left(\frac{A}{2}+B\right)}$$

$$= \tan\frac{A}{2}\left[\frac{\sin\frac{A}{2}+\sin\left(\frac{A}{2}+B\right)}{\cos\frac{A}{2}-\cos\left(\frac{A}{2}+B\right)}\right] = \tan\frac{A}{2}\left[\frac{2\sin\frac{A+B}{2}\cos\left(\frac{B}{2}\right)}{2\sin\frac{A+B}{2}\sin\left(\frac{B}{2}\right)}\right]$$

$$= \tan\frac{A}{2}\cot\frac{B}{2}$$

Self Practice Problem

14. Prove that $\dfrac{\sin\theta+\sin 2\theta}{1+\cos\theta+\cos 2\theta} = \tan\theta$

15. Prove that $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ = \dfrac{3}{16}$

16. Prove that $\tan 3A \tan 2A \tan A = \tan 3A - \tan 2A - \tan A$

17. Prove that $\tan\left(45^\circ + \dfrac{A}{2}\right) = \sec A + \tan A$

9. Important Trigonometric Ratios:

(a) $\sin n\pi = 0$; $\cos n\pi = (-1)^n$; $\tan n\pi = 0$, where $n \in I$

(b) $\sin 15^\circ$ or $\sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75^\circ$ or $\cos\frac{5\pi}{12}$;

$\cos 15^\circ$ or $\cos\frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75^\circ$ or $\sin\frac{5\pi}{12}$;

$\tan 15^\circ = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2-\sqrt{3} = \cot 75^\circ$; $\tan 75^\circ = \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2+\sqrt{3} = \cot 15^\circ$

(c) $\sin\frac{\pi}{10}$ or $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$ & $\cos 36^\circ$ or $\cos\frac{\pi}{5} = \frac{\sqrt{5}+1}{4}$

10. Conditional Identities:

If $A + B + C = \pi$ then :

(i) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

(ii) $\sin A + \sin B + \sin C = 4 \cos\frac{A}{2} \cos\frac{B}{2} \cos\frac{C}{2}$

(iii) $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$

(iv) $\cos A + \cos B + \cos C = 1 + 4 \sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2}$

(v) $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

(vi) $\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = 1$

(vii) $\cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2} = \cot\frac{A}{2} \cdot \cot\frac{B}{2} \cdot \cot\frac{C}{2}$

(viii) $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$

(ix) $A + B + C = \frac{\pi}{2}$ then $\tan A \tan B + \tan B \tan C + \tan C \tan A = 1$

Solved Example # 12

If $A + B + C = 180^\circ$, Prove that, $\sin^2 A + \sin^2 B + \sin^2 C = 2 + 2\cos A \cos B \cos C$.

Solution.

Let $S = \sin^2 A + \sin^2 B + \sin^2 C$

so that $2S = 2\sin^2 A + 1 - \cos 2B + 1 - \cos 2C$

$= 2\sin^2 A + 2 - 2\cos(B + C)\cos(B - C)$

$= 2 - 2\cos^2 A + 2 - 2\cos(B + C)\cos(B - C)$

$\therefore$ $S = 2 + \cos A [\cos(B - C) + \cos(B + C)]$

since $\cos A = -\cos(B+C)$

$\therefore$ $S = 2 + 2 \cos A \cos B \cos C$

Solved Example # 13

If $x + y + z = xyz$, Prove that $\frac{2x}{1-x^2} + \frac{2y}{1-y^2} + \frac{2z}{1-z^2} = \frac{2x}{1-x^2} \cdot \frac{2y}{1-y^2} \cdot \frac{2z}{1-z^2}$.

Solution.

Put $x = \tan A$, $y = \tan B$ and $z = \tan C$,

so that we have

$\tan A + \tan B + \tan C = \tan A \tan B \tan C \Rightarrow A + B + C = n\pi$, where $n \in I$

Hence

L.H.S.

$\therefore \quad \frac{2x}{1-x^2} + \frac{2y}{1-y^2} + \frac{2z}{1-z^2} = \frac{2\tan A}{1-\tan^2 A} + \frac{2\tan B}{1-\tan^2 B} + \frac{2\tan C}{1-\tan^2 C}$.

$= \tan 2A + \tan 2B + \tan 2C \qquad [\because A + B + C = n\pi]$

$= \tan 2A \tan 2B \tan 2C$

$= \frac{2x}{1-x^2} \cdot \frac{2y}{1-y^2} \cdot \frac{2z}{1-z^2}$

Self Practice Problem

18. If $A + B + C = 180°$, prove that

(i) $\sin(B + 2C) + \sin(C + 2A) + \sin(A + 2B) = 4\sin\frac{B-C}{2}\sin\frac{C-A}{2}\sin\frac{A-B}{2}$

(ii) $\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = 8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$.

19. If $A + B + C = 2S$, prove that

(i) $\sin(S - A)\sin(S - B) + \sin S \sin(S - C) = \sin A \sin B$.

(ii) $\sin(S - A) + \sin(S - B) + \sin(S - C) - \sin S = 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$.

11. Range of Trigonometric Expression:

$E = a\sin\theta + b\cos\theta$

$E = \sqrt{a^2+b^2}\sin(\theta + \alpha)$, where $\tan\alpha = \frac{b}{a}$

$= \sqrt{a^2+b^2}\cos(\theta - \beta)$, where $\tan\beta = \frac{a}{b}$

Hence for any real value of θ, $-\sqrt{a^2+b^2} \le E \le \sqrt{a^2+b^2}$

Solved Example # 14

Find maximum and minimum values of following :

(i) $3\sin x + 4\cos x$

(ii) $1 + 2\sin x + 3\cos^2 x$

Solution.

(i) We know

$-\sqrt{3^2+4^2} \le 3\sin x + 4\cos x \le \sqrt{3^2+4^2}$

$-5 \le 3\sin x + 4\cos x \le 5$

(ii) $1 + 2\sin x + 3\cos^2 x$

$= -3\sin^2 x + 2\sin x + 4$

$= -3\left(\sin^2 x - \frac{2\sin x}{3}\right) + 4$

$= -3\left(\sin x - \frac{1}{3}\right)^2 + \frac{13}{3}$

Now $\quad 0 \le \left(\sin x - \frac{1}{3}\right)^2 \le \frac{16}{9}$

$\Rightarrow \quad -\frac{16}{3} \le -3\left(\sin x - \frac{1}{3}\right)^2 \le 0$

$$-1 \le -3\left(\sin x - \frac{1}{3}\right)^2 + \frac{13}{3} \le \frac{13}{3}$$

Self Practice Problem

20. Find maximum and minimum values of following

(i) $3 + (\sin x - 2)^2$ **Answer** max = 12, min = 4.

(ii) $10\cos^2 x - 6\sin x \cos x + 2\sin^2 x$ **Answer** max = 11, min = 1.

(iii) $\cos\theta + 3\sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right) + 6$ **Answer** max = 11, min = 1

12. Sine and Cosine Series:

$$\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots\ldots + \sin\left(\alpha + \overline{n-1}\beta\right) = \frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}\sin\left(\alpha + \frac{n-1}{2}\beta\right)$$

$$\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots\ldots + \cos\left(\alpha + \overline{n-1}\beta\right) = \frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}\cos\left(\alpha + \frac{n-1}{2}\beta\right)$$

Solved Example # 15

Find the summation of the following

(i) $\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7}$

(ii) $\cos\frac{\pi}{7} + \cos\frac{2\pi}{7} + \cos\frac{3\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{5\pi}{7} + \cos\frac{6\pi}{7}$

(iii) $\cos\frac{\pi}{11} + \cos\frac{3\pi}{11} + \cos\frac{5\pi}{11} + \cos\frac{7\pi}{11} + \cos\frac{9\pi}{11}$

Solution.

(i) $$\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} = \frac{\cos\frac{\left(\frac{2\pi}{7}+\frac{6\pi}{7}\right)}{2}\sin\frac{3\pi}{7}}{\sin\frac{\pi}{7}}$$

$$= \frac{\cos\frac{4\pi}{7}\sin\frac{3\pi}{7}}{\sin\frac{\pi}{7}}$$

$$= \frac{-\cos\frac{3\pi}{7}\sin\frac{3\pi}{7}}{\sin\frac{\pi}{7}}$$

$$= -\frac{\sin\frac{6\pi}{7}}{2\sin\frac{\pi}{7}} = -\frac{1}{2}$$

(ii) $\cos\frac{\pi}{7} + \cos\frac{2\pi}{7} + \cos\frac{3\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{5\pi}{7} + \cos\frac{6\pi}{7}$

$$= \frac{\cos\left(\frac{\frac{\pi}{7}+\frac{6\pi}{7}}{2}\right)\sin\frac{6\pi}{14}}{\sin\frac{\pi}{14}} = \frac{\cos\frac{\pi}{2}\sin\frac{6\pi}{14}}{\sin\frac{\pi}{14}} = 0$$

(iii) $\cos\frac{\pi}{11} + \cos\frac{3\pi}{11} + \cos\frac{5\pi}{11} + \cos\frac{7\pi}{11} + \cos\frac{9\pi}{11}$

$$= \frac{\cos\frac{10\pi}{22}\sin\frac{5\pi}{11}}{\sin\frac{\pi}{11}} = \frac{\sin\frac{10\pi}{11}}{2\sin\frac{\pi}{11}} = \frac{1}{2}$$

Self Practice Problem

Find sum of the following series :

21. $\cos\frac{\pi}{2n+1} + \cos\frac{3\pi}{2n+1} + \cos\frac{5\pi}{2n+1}$ + ... + to n terms. **Answer** $\frac{1}{2}$

22. $\sin2\alpha + \sin3\alpha + \sin4\alpha$ + ... + $\sin n\alpha$, where $(n + 2)\alpha = 2\pi$ **Answer** 0.

SHORT REVISION

Trigonometric Ratios & Identities

1. BASIC TRIGONOMETRIC IDENTITIES :

(a) $\sin^2\theta + \cos^2\theta = 1$; $-1 \le \sin\theta \le 1$;

$-1 \le \cos\theta \le 1 \quad \forall \quad \theta \in R$

(b) $\sec^2\theta - \tan^2\theta = 1$; $|\sec\theta| \ge 1 \quad \forall \quad \theta \in R$

(c) $\text{cosec}^2\theta - \cot^2\theta = 1$; $|\text{cosec}\,\theta| \ge 1 \quad \forall \quad \theta \in R$

2. IMPORTANT T′ RATIOS:

(a) $\sin n\pi = 0$; $\cos n\pi = (-1)^n$; $\tan n\pi = 0$ where $n \in I$

(b) $\sin\dfrac{(2n+1)\pi}{2} = (-1)^n$ & $\cos\dfrac{(2n+1)\pi}{2} = 0$ where $n \in I$

(c) $\sin 15^\circ$ or $\sin\dfrac{\pi}{12} = \dfrac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75^\circ$ or $\cos\dfrac{5\pi}{12}$;

$\cos 15^\circ$ or $\cos\dfrac{\pi}{12} = \dfrac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75^\circ$ or $\sin\dfrac{5\pi}{12}$;

$\tan 15^\circ = \dfrac{\sqrt{3}-1}{\sqrt{3}+1} = 2-\sqrt{3} = \cot 75^\circ$; $\tan 75^\circ = \dfrac{\sqrt{3}+1}{\sqrt{3}-1} = 2+\sqrt{3} = \cot 15^\circ$

(d) $\sin\dfrac{\pi}{8} = \dfrac{\sqrt{2-\sqrt{2}}}{2}$; $\cos\dfrac{\pi}{8} = \dfrac{\sqrt{2+\sqrt{2}}}{2}$; $\tan\dfrac{\pi}{8} = \sqrt{2}-1$; $\tan\dfrac{3\pi}{8} = \sqrt{2}+1$

(e) $\sin\dfrac{\pi}{10}$ or $\sin 18^\circ = \dfrac{\sqrt{5}-1}{4}$ & $\cos 36^\circ$ or $\cos\dfrac{\pi}{5} = \dfrac{\sqrt{5}+1}{4}$

3. TRIGONOMETRIC FUNCTIONS OF ALLIED ANGLES :

If θ is any angle, then $-\theta$, $90 \pm \theta$, $180 \pm \theta$, $270 \pm \theta$, $360 \pm \theta$ etc. are called **ALLIED ANGLES.**

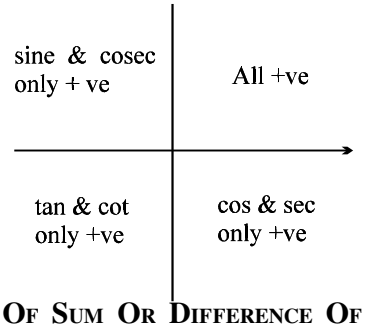
(a) $\sin(-\theta) = -\sin\theta$; $\cos(-\theta) = \cos\theta$

(b) $\sin(90^\circ - \theta) = \cos\theta$; $\cos(90^\circ - \theta) = \sin\theta$

(c) $\sin(90^\circ + \theta) = \cos\theta$; $\cos(90^\circ + \theta) = -\sin\theta$ **(d)** $\sin(180^\circ - \theta) = \sin\theta$; $\cos(180^\circ - \theta) = -\cos\theta$

(e) $\sin(180^\circ + \theta) = -\sin\theta$; $\cos(180^\circ + \theta) = -\cos\theta$

(f) $\sin(270^\circ - \theta) = -\cos\theta$; $\cos(270^\circ - \theta) = -\sin\theta$ **(g)** $\sin(270^\circ + \theta) = -\cos\theta$; $\cos(270^\circ + \theta) = \sin\theta$



4. TRIGONOMETRIC FUNCTIONS OF SUM OR DIFFERENCE OF TWO ANGLES :

(a) $\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$ **(b)** $\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$

(c) $\sin^2A - \sin^2B = \cos^2B - \cos^2A = \sin(A+B)\,.\,\sin(A-B)$

(d) $\cos^2A - \sin^2B = \cos^2B - \sin^2A = \cos(A+B)\,.\,\cos(A-B)$

(e) $\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$ **(f)** $\cot(A \pm B) = \dfrac{\cot A\cot B \mp 1}{\cot B \pm \cot A}$

5. FACTORISATION OF THE SUM OR DIFFERENCE OF TWO SINES OR COSINES :

(a) $\sin C + \sin D = 2\sin\dfrac{C+D}{2}\cos\dfrac{C-D}{2}$ **(b)** $\sin C - \sin D = 2\cos\dfrac{C+D}{2}\sin\dfrac{C-D}{2}$

(c) $\cos C + \cos D = 2\cos\dfrac{C+D}{2}\cos\dfrac{C-D}{2}$ **(d)** $\cos C - \cos D = -2\sin\dfrac{C+D}{2}\sin\dfrac{C-D}{2}$

6. TRANSFORMATION OF PRODUCTS INTO SUM OR DIFFERENCE OF SINES & COSINES :

(a) $2\sin A\cos B = \sin(A+B) + \sin(A-B)$ **(b)** $2\cos A\sin B = \sin(A+B) - \sin(A-B)$

(c) $2\cos A\cos B = \cos(A+B) + \cos(A-B)$ **(d)** $2\sin A\sin B = \cos(A-B) - \cos(A+B)$

7. MULTIPLE ANGLES AND HALF ANGLES :

(a) $\sin 2A = 2\sin A\cos A$; $\sin\theta = 2\sin\dfrac{\theta}{2}\cos\dfrac{\theta}{2}$

(b) $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$;

$\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = 2\cos^2\frac{\theta}{2} - 1 = 1 - 2\sin^2\frac{\theta}{2}$.

$2\cos^2 A = 1 + \cos 2A$, $2\sin^2 A = 1 - \cos 2A$; $\tan^2 A = \frac{1-\cos 2A}{1+\cos 2A}$

$2\cos^2\frac{\theta}{2} = 1 + \cos\theta$, $2\sin^2\frac{\theta}{2} = 1 - \cos\theta$.

(c) $\tan 2A = \frac{2\tan A}{1-\tan^2 A}$; $\tan\theta = \frac{2\tan(\theta/2)}{1-\tan^2(\theta/2)}$

(d) $\sin 2A = \frac{2\tan A}{1+\tan^2 A}$, $\cos 2A = \frac{1-\tan^2 A}{1+\tan^2 A}$ **(e)** $\sin 3A = 3\sin A - 4\sin^3 A$

(f) $\cos 3A = 4\cos^3 A - 3\cos A$ **(g)** $\tan 3A = \frac{3\tan A - \tan^3 A}{1-3\tan^2 A}$

8. THREE ANGLES :

(a) $\tan(A+B+C) = \frac{\tan A+\tan B+\tan C-\tan A\tan B\tan C}{1-\tan A\tan B-\tan B\tan C-\tan C\tan A}$

NOTE IF : **(i)** $A+B+C = \pi$ then $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

(ii) $A+B+C = \frac{\pi}{2}$ then $\tan A\tan B + \tan B\tan C + \tan C\tan A = 1$

(b) If $A + B + C = \pi$ then : **(i)** $\sin 2A + \sin 2B + \sin 2C = 4\sin A \sin B \sin C$

(ii) $\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

9. MAXIMUM & MINIMUM VALUES OF TRIGONOMETRIC FUNCTIONS:

(a) Min. value of $a^2\tan^2\theta + b^2\cot^2\theta = 2ab$ where $\theta \in R$

(b) Max. and Min. value of $a\cos\theta + b\sin\theta$ are $\sqrt{a^2+b^2}$ and $-\sqrt{a^2+b^2}$

(c) If $f(\theta) = a\cos(\alpha+\theta) + b\cos(\beta+\theta)$ where a, b, α and β are known quantities then

$-\sqrt{a^2+b^2+2ab\cos(\alpha-\beta)} \le f(\theta) \le \sqrt{a^2+b^2+2ab\cos(\alpha-\beta)}$

(d) If $\alpha,\beta \in \left(0,\frac{\pi}{2}\right)$ and $\alpha+\beta = \sigma$ (constant) then the maximum values of the expression

$\cos\alpha\cos\beta$, $\cos\alpha + \cos\beta$, $\sin\alpha + \sin\beta$ and $\sin\alpha\sin\beta$

occurs when $\alpha = \beta = \sigma/2$.

(e) If $\alpha,\beta \in \left(0,\frac{\pi}{2}\right)$ and $\alpha+\beta = \sigma$(constant) then the minimum values of the expression

$\sec\alpha + \sec\beta$, $\tan\alpha + \tan\beta$, $\operatorname{cosec}\alpha + \operatorname{cosec}\beta$ occurs when $\alpha = \beta = \sigma/2$.

(f) If A, B, C are the angles of a triangle then maximum value of

$\sin A + \sin B + \sin C$ and $\sin A \sin B \sin C$ occurs when $A = B = C = 60^0$

(g) In case a quadratic in $\sin\theta$ or $\cos\theta$ is given then the maximum or minimum values can be interpreted by making a perfect square.

10. Sum of sines or cosines of n angles,

$$\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots\ldots + \sin\left(\alpha + \overline{n-1}\,\beta\right) = \frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}\sin\left(\alpha+\frac{n-1}{2}\beta\right)$$

$$\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots\ldots + \cos\left(\alpha + \overline{n-1}\,\beta\right) = \frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}\cos\left(\alpha+\frac{n-1}{2}\beta\right)$$

EXERCISE–I

Q.1 Prove that $\cos^2\alpha + \cos^2(\alpha+\beta) - 2\cos\alpha\cos\beta\cos(\alpha+\beta) = \sin^2\beta$

Q.2 Prove that $\cos 2\alpha = 2\sin^2\beta + 4\cos(\alpha+\beta)\sin\alpha\sin\beta + \cos 2(\alpha+\beta)$

Q.3 Prove that , $\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha = \cot\alpha$.

Q.4 Prove that : (a) $\tan 20^\circ . \tan 40^\circ . \tan 60^\circ . \tan 80^\circ = 3$

(b) $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ = 4$. (c) $\sin^4\frac{\pi}{16} + \sin^4\frac{3\pi}{16} + \sin^4\frac{5\pi}{16} + \sin^4\frac{7\pi}{16} = \frac{3}{2}$

Q.5 Calculate without using trigonometric tables :

(a) $\operatorname{cosec} 10^\circ - \sqrt{3}\sec 10^\circ$ (b) $4\cos 20^\circ - \sqrt{3}\cot 20^\circ$ (c) $\frac{2\cos 40^\circ - \cos 20^\circ}{\sin 20^\circ}$

(d) $2\sqrt{2}\sin 10^\circ\left[\frac{\sec 5^\circ}{2} + \frac{\cos 40^\circ}{\sin 5^\circ} - 2\sin 35^\circ\right]$ (e) $\cos^6\frac{\pi}{16} + \cos^6\frac{3\pi}{16} + \cos^6\frac{5\pi}{16} + \cos^6\frac{7\pi}{16}$

(f) $\tan 10^\circ - \tan 50^\circ + \tan 70^\circ$

Q.6(a) If $X = \sin\left(\theta+\frac{7\pi}{12}\right) + \sin\left(\theta-\frac{\pi}{12}\right) + \sin\left(\theta+\frac{3\pi}{12}\right)$, $\underset{14}{Y} = \cos\left(\theta+\frac{7\pi}{12}\right) + \cos\left(\theta-\frac{\pi}{12}\right) + \cos\left(\theta+\frac{3\pi}{12}\right)$

then prove that $\frac{X}{Y}-\frac{Y}{X}=2\tan 2\theta$.

(b) Prove that $\sin^2 12^\circ+\sin^2 21^\circ+\sin^2 39^\circ+\sin^2 48^\circ = 1+\sin^2 9^\circ+\sin^2 18^\circ$.

Q.7 Show that : (a) $\cot 7\frac{1^\circ}{2}$ or $\tan 82\frac{1^\circ}{2}=\left(\sqrt{3}+\sqrt{2}\right)\left(\sqrt{2}+1\right)$ or $\sqrt{2}+\sqrt{3}+\sqrt{4}+\sqrt{6}$

(b) $\tan 142\frac{1^\circ}{2}=2+\sqrt{2}-\sqrt{3}-\sqrt{6}$.

Q.8 If $m\tan(\theta-30^\circ)=n\tan(\theta+120^\circ)$, show that $\cos 2\theta=\frac{m+n}{2(m-n)}$.

Q.9 If $\tan\left(\frac{\pi}{4}+\frac{y}{2}\right)=\tan^3\left(\frac{\pi}{4}+\frac{x}{2}\right)$, prove that $\frac{\sin y}{\sin x}=\frac{3+\sin^2 x}{1+3\sin^2 x}$.

Q.10 If $\cos(\alpha+\beta)=\frac{4}{5}$; $\sin(\alpha-\beta)=\frac{5}{13}$ & α,β lie between 0 & $\frac{\pi}{4}$, then find the value of $\tan 2\alpha$.

Q.11 Prove that if the angles α & β satisfy the relation $\frac{\sin\beta}{\sin(2\alpha+\beta)}=\frac{n}{m}\left(|m|>|n|\right)$ then $\frac{1+\frac{\tan\beta}{\tan\alpha}}{m+n}=\frac{1-\tan\alpha\tan\beta}{m-n}$.

Q.12 (a) If $y=10\cos^2 x-6\sin x\cos x+2\sin^2 x$, then find the greatest & least value of y.
(b) If $y=1+2\sin x+3\cos^2 x$, find the maximum & minimum values of y $\forall$ $x\in R$.
(c) If $y=9\sec^2 x+16\operatorname{cosec}^2 x$, find the minimum value of y $\forall\, x\in R$.

(d) Prove that $3\cos\left(\theta+\frac{\pi}{3}\right)+5\cos\theta+3$ lies from - 4 & 10 .

(e) Prove that $\left(2\sqrt{3}+4\right)\sin\theta+4\cos\theta$ lies between $-2\left(2+\sqrt{5}\right)$ & $2\left(2+\sqrt{5}\right)$.

Q.13 If $A+B+C=\pi$, prove that $\sum\left(\frac{\tan A}{\tan B.\tan C}\right)=\sum(\tan A)-2\sum(\cot A)$.

Q.14 If $\alpha+\beta=c$ where $\alpha,\beta>0$ each lying between 0 and $\pi/2$ and c is a constant, find the maximum or minimum value of

(a) $\sin\alpha+\sin\beta$ (b) $\sin\alpha\sin\beta$
(c) $\tan\alpha+\tan\beta$ (d) $\operatorname{cosec}\alpha+\operatorname{cosec}\beta$

Q.15 Let $A_1, A_2, \ldots\ldots, A_n$ be the vertices of an n-sided regular polygon such that ;
$\frac{1}{A_1A_2}=\frac{1}{A_1A_3}+\frac{1}{A_1A_4}$. Find the value of n.

Q.16 Prove that : $\operatorname{cosec}\theta+\operatorname{cosec}2\theta+\operatorname{cosec}2^2\theta+\ldots\ldots+\operatorname{cosec}2^{n-1}\theta=\cot(\theta/2)-\cot 2^{n-1}\theta$

Q.17 For all values of α,β,γ prove that;

$\cos\alpha+\cos\beta+\cos\gamma+\cos(\alpha+\beta+\gamma)=4\cos\frac{\alpha+\beta}{2}.\cos\frac{\beta+\gamma}{2}.\cos\frac{\gamma+\alpha}{2}$.

Q.18 Show that $\frac{1+\sin A}{\cos A}+\frac{\cos B}{1-\sin B}=\frac{2\sin A-2\sin B}{\sin(A-B)+\cos A-\cos B}$.

Q.19 If $\tan\beta=\frac{\tan\alpha+\tan\gamma}{1+\tan\alpha.\tan\gamma}$, prove that $\sin 2\beta=\frac{\sin 2\alpha+\sin 2\gamma}{1+\sin 2\alpha.\sin 2\gamma}$.

Q.20 If $\alpha+\beta=\gamma$, prove that $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1+2\cos\alpha\cos\beta\cos\gamma$.

Q.21 If $\alpha+\beta+\gamma=\frac{\pi}{2}$, show that $\frac{\left(1-\tan\frac{\alpha}{2}\right)\left(1-\tan\frac{\beta}{2}\right)\left(1-\tan\frac{\gamma}{2}\right)}{\left(1+\tan\frac{\alpha}{2}\right)\left(1+\tan\frac{\beta}{2}\right)\left(1+\tan\frac{\gamma}{2}\right)}=\frac{\sin\alpha+\sin\beta+\sin\gamma-1}{\cos\alpha+\cos\beta+\cos\gamma}$.

Q.22 If $A+B+C=\pi$ and $\cot\theta=\cot A+\cot B+\cot C$, show that ,
$\sin(A-\theta).\sin(B-\theta).\sin(C-\theta)=\sin^3\theta$.

Q.23 If $P=\cos\frac{\pi}{19}+\cos\frac{3\pi}{19}+\cos\frac{5\pi}{19}+\ldots\ldots+\cos\frac{17\pi}{19}$ and

$Q=\cos\frac{2\pi}{21}+\cos\frac{4\pi}{21}+\cos\frac{6\pi}{21}+\ldots\ldots+\cos\frac{20\pi}{21}$, then find P – Q.

Q.24 If A, B, C denote the angles of a triangle ABC then prove that the triangle is right angled if and only if $\sin 4A+\sin 4B+\sin 4C=0$.

Q.25 Given that $(1+\tan 1^\circ)(1+\tan 2^\circ)\ldots\ldots(1+\tan 45^\circ)=2^n$, find *n*.

EXERCISE–II

Q.1 If $\tan\alpha=p/q$ where $\alpha=6\beta$, α being an acute angle, prove that;
$\frac{1}{2}(p\operatorname{cosec}2\beta-q\sec 2\beta)=\sqrt{p^2+q^2}$.

Q.2 Let $A_1, A_2, A_3 \ldots\ldots\ldots A_n$ are the vertices of a regular n sided polygon inscribed in a circle of radius R. If $(A_1A_2)^2+(A_1A_3)^2+\ldots\ldots\ldots+(A_1A_n)^2=14R^2$, find the number of sides in the polygon.

Q.3 Prove that: $\frac{\cos 3\theta+\cos 3\phi}{2\cos(\theta-\phi)-1}=(\cos\theta+\cos\phi)\cos(\theta+\phi)-(\sin\theta+\sin\phi)\sin(\theta+\phi)$

Q.4 Without using the surd value for $\sin 18^0$ or $\cos 36^0$, prove that $4\sin 36^0\cos 18^0=\sqrt{5}$

Q.5 Show that, $\frac{\sin x}{\cos 3x}+\frac{\sin 3x}{\cos 9x}+\frac{\sin 9x}{\cos 27x}=\frac{1}{2}(\tan 27x-\tan x)$

Q.6 Let $x_1=\prod_{r=1}^{5}\cos\frac{r\pi}{11}$ and $x_2=\sum_{r=1}^{5}\cos\frac{r\pi}{11}$, then show that

$x_1\cdot x_2=\frac{1}{64}\left(\operatorname{cosec}\frac{\pi}{22}-1\right)$, where $\boldsymbol{\Pi}$ denotes the continued product.

Q.7 If $\theta=\frac{2\pi}{7}$, prove that $\tan\theta\,.\tan 2\theta+\tan 2\theta\,.\tan 4\theta+\tan 4\theta\,.\tan\theta=-7$.

Q.8 For $0<x<\frac{\pi}{4}$ prove that, $\frac{\cos x}{\sin^2 x(\cos x-\sin x)}>8$.

Q.9 (a) If $\alpha=\frac{2\pi}{7}$ prove that, $\sin\alpha+\sin 2\alpha+\sin 4\alpha=\frac{\sqrt{7}}{2}$ (b) $\sin\frac{\pi}{7}\,.\sin\frac{2\pi}{7}\,.\sin\frac{3\pi}{7}=\frac{\sqrt{7}}{8}$

Q.10 Let $k=1^\circ$, then prove that $\sum_{n=0}^{88}\frac{1}{\cos nk\cdot\cos(n+1)k}=\frac{\cos k}{\sin^2 k}$

Q.11 Prove that the value of $\cos A+\cos B+\cos C$ lies between 1 & $\frac{3}{2}$ where $A+B+C=\pi$.

Q.12 If $\cos A=\tan B$, $\cos B=\tan C$ and $\cos C=\tan A$, then prove that $\sin A=\sin B=\sin C=2\sin 18^\circ$.

Q.13 Show that $\frac{3+\cos x}{\sin x}$ $\forall\, x\in R$ can not have any value between $-2\sqrt{2}$ and $2\sqrt{2}$. What inference can you draw about the values of $\frac{\sin x}{3+\cos x}$?

Q.14 If $(1+\sin t)(1+\cos t)=\frac{5}{4}$. Find the value of $(1-\sin t)(1-\cos t)$.

Q.15 Prove that from the equality $\frac{\sin^4\alpha}{a}+\frac{\cos^4\alpha}{b}=\frac{1}{a+b}$ follows the relation; $\frac{\sin^8\alpha}{a^3}+\frac{\cos^8\alpha}{b^3}=\frac{1}{(a+b)^3}$.

Q.16 Prove that the triangle ABC is equilateral iff, $\cot A+\cot B+\cot C=\sqrt{3}$.

Q.17 Prove that the average of the numbers $n\sin n^\circ$, $n=2, 4, 6, \ldots\ldots, 180$, is $\cot 1^\circ$.

Q.18 Prove that : $4\sin 27^\circ=\left(5+\sqrt{5}\right)^{1/2}-\left(3-\sqrt{5}\right)^{1/2}$.

Q.19 If $A+B+C=\pi$; prove that $\tan^2\frac{A}{2}+\tan^2\frac{B}{2}+\tan^2\frac{C}{2}\geq 1$.

Q.20 If $A+B+C=\pi$ $(A, B, C>0)$, prove that $\sin\frac{A}{2}\,.\sin\frac{B}{2}\,.\sin\frac{C}{2}\leq\frac{1}{8}$.

Q.21 Show that elliminating x & y from the equations, $\sin x+\sin y=a$;

$\cos x+\cos y=b$ & $\tan x+\tan y=c$ gives $\frac{8ab}{\left(a^2+b^2\right)^2-4a^2}=c$.

Q.22 Determine the smallest positive value of x (in degrees) for which

$\tan(x+100^\circ)=\tan(x+50^\circ)\tan x\ \tan(x-50^\circ)$.

Q.23 Evaluate : $\sum_{n=1}^{n}\frac{\tan\frac{x}{2^n}}{2^{n-1}\cos\frac{x}{2^{n-1}}}$

Q.24 If $\alpha+\beta+\gamma=\pi$ & $\tan\left(\frac{\beta+\gamma-\alpha}{4}\right)\cdot\tan\left(\frac{\gamma+\alpha-\beta}{4}\right)\cdot\tan\left(\frac{\alpha+\beta-\gamma}{4}\right)=1$, then prove that;

$1+\cos\alpha+\cos\beta+\cos\gamma=0$.

Q.25 $\forall\, x\in R$, find the range of the function, $f(x)=\cos x\,(\sin x+\sqrt{\sin^2 x+\sin^2\alpha}\,)$; $\alpha\in[0,\pi]$

EXERCISE–III

Q.1 $\sec^2\theta=\frac{4xy}{(x+y)^2}$ is true if and only if : [JEE '96, 1]

(A) $x+y\neq 0$ (B) $x=y$, $x\neq 0$ (C) $x=y$ (D) $x\neq 0$, $y\neq 0$

Q.2 (a) Let n be an odd integer. If $\sin n\theta=\sum_{r=0}^{n}b_r\sin^r\theta$, for every value of θ, then :

(A) $b_0=1, b_1=3$ (B) $b_0=0, b_1=n$

(C) $b_0 = -1,\ b_1 = n$ (D) $b_0 = 0,\ b_1 = n^2 - 3n + 3$

(b) Let $A_0A_1A_2A_3A_4A_5$ be a regular hexagon inscribed in a circle of unit radius. Then the product of the lengths of the line segments A_0A_1, A_0A_2 & A_0A_4 is :

(A) $\frac{3}{4}$ (B) $3\sqrt{3}$ (C) 3 (D) $\frac{3\sqrt{3}}{2}$

(c) Which of the following number(s) is/are rational ? [JEE '98, 2 + 2 + 2 = 6 out of 200]

(A) $\sin 15^\circ$ (B) $\cos 15^\circ$ (C) $\sin 15^\circ \cos 15^\circ$ (D) $\sin 15^\circ \cos 75^\circ$

Q.3 For a positive integer n, let $f_n(\theta) = \left(\tan\frac{\theta}{2}\right)(1+\sec\theta)(1+\sec 2\theta)(1+\sec 4\theta)\ldots(1+\sec 2^n\theta)$ Then

(A) $f_2\left(\frac{\pi}{16}\right) = 1$ (B) $f_3\left(\frac{\pi}{32}\right) = 1$ (C) $f_4\left(\frac{\pi}{64}\right) = 1$ (D) $f_5\left(\frac{\pi}{128}\right) = 1$ [JEE '99,3]

Q.4(a) Let $f(\theta) = \sin\theta\ (\sin\theta + \sin 3\theta)$. Then $f(\theta)$: [JEE 2000 Screening. 1 out of 35]

(A) ≥ 0 only when $\theta \geq 0$ (B) ≤ 0 for all real θ

(C) ≥ 0 for all real θ (D) ≤ 0 only when $\theta \leq 0$.

(b) In any triangle ABC, prove that, $\cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2} = \cot\frac{A}{2}\cot\frac{B}{2}\cot\frac{C}{2}$. [JEE 2000]

Q.5(a) Find the maximum and minimum values of $27^{\cos 2x} \cdot 81^{\sin 2x}$.

(b) Find the smallest positive values of x & y satisfying, $x - y = \frac{\pi}{4}$, $\cot x + \cot y = 2$. [REE 2000, 3]

Q.6 If $\alpha + \beta = \frac{\pi}{2}$ and $\beta + \gamma = \alpha$ then $\tan\alpha$ equals [JEE 2001 (Screening), 1 out of 35]

(A) $2(\tan\beta + \tan\gamma)$ (B) $\tan\beta + \tan\gamma$ (C) $\tan\beta + 2\tan\gamma$ (D) $2\tan\beta + \tan\gamma$

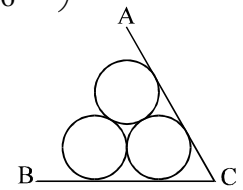
Q.7 If θ and ϕ are acute angles satisfying $\sin\theta = \frac{1}{2}$, $\cos\phi = \frac{1}{3}$, then $\theta + \phi \in$ [JEE 2004 (Screening)]

(A) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$ (B) $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ (C) $\left(\frac{2\pi}{3}, \frac{5\pi}{6}\right)$ (D) $\left(\frac{5\pi}{6}, \pi\right)$

Q.8 In an equilateral triangle, 3 coins of radii 1 unit each are kept so that they touch each other and also the sides of the triangle. Area of the triangle is

(A) $4 + 2\sqrt{3}$ (B) $6 + 4\sqrt{3}$

(C) $12 + \frac{7\sqrt{3}}{4}$ (D) $3 + \frac{7\sqrt{3}}{4}$



[JEE 2005 (Screening)]

Q.9 Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$, then

(A) $t_1 > t_2 > t_3 > t_4$ (B) $t_4 > t_3 > t_1 > t_2$ (C) $t_3 > t_1 > t_2 > t_4$ (D) $t_2 > t_3 > t_1 > t_4$

[JEE 2006, 3]

ANSWER SHEET (EXERCISE–I)

Q 5. (a) 4 **(b)** -1 **(c)** $\sqrt{3}$ **(d)** 4 **(e)** $\frac{5}{4}$ **(f)** $\sqrt{3}$ **Q 10.** $\frac{56}{33}$

Q 12. (a) $y_{max} = 11$; $y_{min} = 1$ **(b)** $y_{max} = \frac{13}{3}$; $y_{min} = -1$, (c) 49

Q14. (a) max = 2 sin (c/2), (b) max. = $\sin^2$ (c/2), (c) min = 2 tan (c/2), (d) min = 2 cosec (c/2)

Q 15. n = 7 **Q23.** 1 Q.25 $n = 23$

EXERCISE –II

Q.2 n = 7 **Q.13** $\left[-\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}\right]$ **Q.14** $\frac{13}{4} - \sqrt{10}$ **Q.22** $x = 30^\circ$

Q 23. $\frac{2}{\sin 2x} - \frac{1}{2^{n-1}\sin\frac{x}{2^{n-1}}}$ **Q.25** $-\sqrt{1+\sin^2\alpha} \leq y \leq \sqrt{1+\sin^2\alpha}$

EXERCISE–III

Q.1 B **Q.2 (a)** B, **(b)** C, **(c)** C **Q.3** A, B, C, D **Q.4 (a)** C

Q.5 (a) max. = 3^5 & min. = 3^{-5} ; (b) $x = \frac{5\pi}{12}$; $y = \frac{\pi}{6}$ **Q.6** C **Q.7** B

Q.8 B **Q.9** B

EXERCISE–IV (Objective)

Part : (A) Only one correct option

1. $\frac{\tan\left(x-\frac{\pi}{2}\right)\cos\left(\frac{3\pi}{2}+x\right)-\sin^3\left(\frac{7\pi}{2}-x\right)}{\cos\left(x-\frac{\pi}{2}\right).\tan\left(\frac{3\pi}{2}+x\right)}$ when simplified reduces to:

(A) $\sin x \cos x$ (B) $-\sin^2 x$ (C) $-\sin x \cos x$ (D) $\sin^2 x$

2. The expression $3\left[\sin^4\left(\frac{3\pi}{2}-\alpha\right)+\sin^4(3\pi+\alpha)\right]-2\left[\sin^6\left(\frac{\pi}{2}+\alpha\right)+\sin^6(5\pi+\alpha)\right]$ is equal to

(A) 0 (B) 1 (C) 3 (D) $\sin 4\alpha + \sin 6\alpha$

3. If tan A & tan B are the roots of the quadratic equation $x^2 - ax + b = 0$, then the value of $\sin^2(A + B)$.

(A) $\frac{a^2}{a^2+(1-b)^2}$ (B) $\frac{a^2}{a^2+b^2}$ (C) $\frac{a^2}{(b+c)^2}$ (D) $\frac{a^2}{b^2(1-a)^2}$

4. The value of $\log_2[\cos^2(\alpha+\beta)+\cos^2(\alpha-\beta)-\cos 2\alpha.\cos 2\beta]$:

(A) depends on α & β both (B) depends on α but not on β

(C) depends on β but not on α (D) independent of both α & β.

5. $\frac{\cos 20°+8\sin 70°\sin 50°\sin 10°}{\sin^2 80°}$ is equal to:

(A) 1 (B) 2 (C) 3/4 (D) none

6. If $\cos A = 3/4$, then the value of $16\cos^2(A/2) - 32\sin(A/2)\sin(5A/2)$ is

(A) – 4 (B) – 3 (C) 3 (D) 4

7. If $y = \cos^2(45° + x) + (\sin x - \cos x)^2$ then the maximum & minimum values of y are:

(A) 2 & 0 (B) 3 & 0 (C) 3 & 1 (D) none

8. The value of $\cos\frac{\pi}{19}+\cos\frac{3\pi}{19}+\cos\frac{5\pi}{19}+\ldots\ldots+\cos\frac{17\pi}{19}$ is equal to:

(A) 1/2 (B) 0 (C) 1 (D) none

9. The greatest and least value of $\log_{\sqrt{2}}\left(\sin x - \cos x + 3\sqrt{2}\right)$ are respectively:

(A) 2 & 1 (B) 5 & 3 (C) 7 & 5 (D) 9 & 7

10. In a right angled triangle the hypotenuse is $2\sqrt{2}$ times the perpendicular drawn from the opposite vertex. Then the other acute angles of the triangle are

(A) $\frac{\pi}{3}$ & $\frac{\pi}{6}$ (B) $\frac{\pi}{8}$ & $\frac{3\pi}{8}$ (C) $\frac{\pi}{4}$ & $\frac{\pi}{4}$ (D) $\frac{\pi}{5}$ & $\frac{3\pi}{10}$

11. $\frac{1}{\cos 290°}+\frac{1}{\sqrt{3}\sin 250°}=$

(A) $\frac{2\sqrt{3}}{3}$ (B) $\frac{4\sqrt{3}}{3}$ (C) $\sqrt{3}$ (D) none

12. If $\frac{3\pi}{4} < \alpha < \pi$, then $\sqrt{2\cot\alpha+\frac{1}{\sin^2\alpha}}$ is equal to

(A) $1 + \cot\alpha$ (B) $-1 - \cot\alpha$ (C) $1 - \cot\alpha$ (D) $-1 + \cot\alpha$

13. If $x \in \left(\pi, \frac{3\pi}{2}\right)$ then $4\cos^2\left(\frac{\pi}{4}-\frac{x}{2}\right)+\sqrt{4\sin^4 x+\sin^2 2x}$ is always equal to

(A) 1 (B) 2 (C) – 2 (D) none of these

14. If $2\cos x + \sin x = 1$, then value of $7\cos x + 6\sin x$ is equal to

(A) 2 or 6 (B) 1 or 3 (C) 2 or 3 (D) none of these

15. If $\text{cosec } A + \cot A = \frac{11}{2}$, then tan A is

(A) $\frac{21}{22}$ (B) $\frac{15}{16}$ (C) $\frac{44}{117}$ (D) $\frac{117}{43}$

16. If $\cot\alpha + \tan\alpha = m$ and $\frac{1}{\cos\alpha} - \cos\alpha = n$, then

(A) $m(mn^2)^{1/3} - n(nm^2)^{1/3} = 1$ (B) $m(m^2n)^{1/3} - n(nm^2)^{1/3} = 1$

(C) $n(mn^2)^{1/3} - m(nm^2)^{1/3} = 1$ (D) $n(m^2n)^{1/3}\ m(mn^2)^{1/3} = 1$

17. The expression $\frac{\cos 6x+6\cos 4x+15\cos 2x+10}{\cos 5x+5\cos 3x+10\cos x}$ is equal to

(A) cos 2x (B) 2 cos x (C) $\cos^2 x$ (D) 1 + cos x

18. If $\frac{\sin A}{\sin B}=\frac{\sqrt{3}}{2}$ and $\frac{\cos A}{\cos B}=\frac{\sqrt{5}}{2}$, $0 < A, B < \pi/2$, then tan A + tan B is equal to

(A) $\sqrt{3}/\sqrt{5}$ (B) $\sqrt{5}/\sqrt{3}$ (C) 1 (D) $(\sqrt{5}+\sqrt{3})/\sqrt{5}$

19. If $\sin 2\theta = k$, then the value of $\frac{\tan^3\theta}{1+\tan^2\theta}+\frac{\cot^3\theta}{1+\cot^2\theta}$ is equal to

(A) $\frac{1-k^2}{k}$ (B) $\frac{2-k^2}{k}$ (C) $k^2 + 1$ (D) $2 - k^2$

Part : (B) May have more than one options correct

20. Which of the following is correct ?
(A) $\sin 1° > \sin 1$ (B) $\sin 1° < \sin 1$ (C) $\cos 1° > \cos 1$ (D) $\cos 1° < \cos 1$

21. If $3\sin\beta = \sin(2\alpha + \beta)$, then $\tan(\alpha + \beta) - 2\tan\alpha$ is
(A) independent of α (B) independent of β
(C) dependent of both α and β (D) independent of α but dependent of β

22. It is known that $\sin\beta = \frac{4}{5}$ & $0 < \beta < \pi$ then the value of $\frac{\sqrt{3}\sin(\alpha+\beta) - \frac{2}{\cos\frac{\pi}{6}}\cos(\alpha+\beta)}{\sin\alpha}$ is:

(A) independent of α for all β in $(0, \pi)$ (B) $\frac{5}{\sqrt{3}}$ for $\tan\beta > 0$

(C) $\frac{\sqrt{3}(7+24\cot\alpha)}{15}$ for $\tan\beta < 0$ (D) none

23. If the sides of a right angled triangle are $\{\cos2\alpha + \cos2\beta + 2\cos(\alpha+\beta)\}$ and $\{\sin2\alpha + \sin2\beta + 2\sin(\alpha+\beta)\}$, then the length of the hypotenuse is:

(A) $2[1+\cos(\alpha-\beta)]$ (B) $2[1-\cos(\alpha+\beta)]$ (C) $4\cos^2\frac{\alpha-\beta}{2}$ (D) $4\sin^2\frac{\alpha+\beta}{2}$

24. If $x = \sec\phi - \tan\phi$ & $y = \operatorname{cosec}\phi + \cot\phi$ then:

(A) $x = \frac{y+1}{y-1}$ (B) $y = \frac{1+x}{1-x}$ (C) $x = \frac{y-1}{y+1}$ (D) $xy + x - y + 1 = 0$

25. $(a+2)\sin\alpha + (2a-1)\cos\alpha = (2a+1)$ if $\tan\alpha =$

(A) $\frac{3}{4}$ (B) $\frac{4}{3}$ (C) $\frac{2a}{a^2+1}$ (D) $\frac{2a}{a^2-1}$

26. If $\tan x = \frac{2b}{a-c}$, $(a \neq c)$
$y = a\cos^2x + 2b\sin x\cos x + c\sin^2x$
$z = a\sin^2x - 2b\sin x\cos x + c\cos^2x$, then
(A) $y = z$ (B) $y + z = a + c$ (C) $y - z = a - c$ (D) $y - z = (a-c)^2 + 4b^2$

27. $\left(\frac{\cos A+\cos B}{\sin A-\sin B}\right)^n + \left(\frac{\sin A+\sin B}{\cos A-\cos B}\right)^n$

(A) $2\tan^n\frac{A-B}{2}$ (B) $2\cot^n\frac{A-B}{2}$: n is even (C) 0 : n is odd (D) none

28. The equation $\sin^6x + \cos^6x = a^2$ has real solution if

(A) $a \in (-1, 1)$ (B) $a \in \left(-1, -\frac{1}{2}\right)$ (C) $a \in \left(-\frac{1}{2}\frac{1}{2}\right)$ (D) $a \in \left(\frac{1}{2}, 1\right)$

EXERCISE–IV (Subjective)

1. The minute hand of a watch is 1.5 cm long. How far does its tip move in 50 minutes? (Use $\pi = 3.14$).

2. If the arcs of the same length in two circles subtend angles 75° and 120° at the centre, find the ratio of their radii.

3. Sketch the following graphs :

(i) $y = 3\sin 2x$ (ii) $y = 2\tan x$ (iii) $y = \sin\frac{x}{2}$

4. Prove that $\cos\left(\frac{3\pi}{2}+\theta\right)\cos(2\pi+\theta)\left[\cot\left(\frac{3\pi}{2}-\theta\right)+\cot(2\pi+\theta)\right] = 1$.

5. Prove that $\cos 2\theta\cos\frac{\theta}{2} - \cos 3\theta\cos\frac{9\theta}{2} = \sin 5\theta\sin\frac{5\theta}{2}$.

6. If $\tan x = \frac{3}{4}$, $\pi < x < \frac{3\pi}{2}$, find the value of $\sin\frac{x}{2}$ and $\cos\frac{x}{2}$.

7. prove that $\left\{\frac{1-\cot^2\left(\frac{\alpha-\pi}{4}\right)}{1+\cot^2\left(\frac{\alpha-\pi}{4}\right)} + \cos\frac{\alpha}{2}\cot 4\alpha\right\}\sec\frac{9\alpha}{2} = \operatorname{cosec} 4\alpha$.

8. Prove that, $\sin 3x.\sin^3 x + \cos 3x.\cos^3 x = \cos^3 2x$.

9. If $\tan\alpha = \frac{p}{q}$ where $\alpha = 6\beta$, α being an acute angle, prove that; $\frac{1}{2}(p\operatorname{cosec} 2\beta - q\sec 2\beta) = \sqrt{p^2+q^2}$.

10. If $\tan\beta = \frac{\tan\alpha+\tan\gamma}{1+\tan\alpha.\tan\gamma}$, prove that $\sin 2\beta = \frac{\sin 2\alpha+\sin 2\gamma}{1+\sin 2\alpha.\sin 2\gamma}$.

11. Show that: (i) $\cot 7\frac{1°}{2}$ or $\tan 82\frac{1°}{2} = \left(\sqrt{3}+\sqrt{2}\right)\left(\sqrt{2}+1\right)$ or $\sqrt{2}+\sqrt{3}+\sqrt{4}+\sqrt{6}$

(ii) $\tan 142\frac{1°}{2} = 2+\sqrt{2}-\sqrt{3}-\sqrt{6}$. (iii) $4\sin 27° = \left(5+\sqrt{5}\right)^{1/2} - \left(3-\sqrt{5}\right)^{1/2}$

12. Prove that, $\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha = \cot\alpha$.

13. If $\cos(\beta-\gamma) + \cos(\gamma-\alpha) + \cos(\alpha-\beta) = \frac{-3}{2}$, prove that
$\cos\alpha + \cos\beta + \cos\gamma = 0$, $\sin\alpha + \sin\beta + \sin\gamma = 0$.

14. Prove that from the equality $\frac{\sin^4\alpha}{a} + \frac{\cos^4\alpha}{b} = \frac{1}{a+b}$ follows the relation $\frac{\sin^8\alpha}{a^3} + \frac{\cos^8\alpha}{b^3} = \frac{1}{(a+b)^3}$

15. Prove that: $\operatorname{cosec}\theta + \operatorname{cosec} 2\theta + \operatorname{cosec} 2^2\theta + \ldots + \operatorname{cosec} 2^{n-1}\theta = \cot(\theta/2) - \cot 2^{n-1}\theta$. Hence or otherwise prove that $\operatorname{cosec}\frac{4\pi}{15} + \operatorname{cosec}\frac{8\pi}{15} + \operatorname{cosec}\frac{16\pi}{15} + \operatorname{cosec}\frac{32\pi}{15} = 0$

16. Let $A_1, A_2, \ldots\ldots, A_n$ be the vertices of an n–sided regular polygon such that; $\frac{1}{A_1A_2} = \frac{1}{A_1A_3} + \frac{1}{A_1A_4}$.
Find the value of n.

17. If $A + B + C = \pi$, then prove that

(i) $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} \geq 1$ (ii) $\sin\frac{A}{2}.\sin\frac{B}{2}.\sin\frac{C}{2} \leq \frac{1}{8}$.

(iii) $\cos A + \cos B + \cos C \leq \frac{3}{2}$

18. If $\frac{ax}{\cos\theta} + \frac{by}{\sin\theta} = a^2 - b^2$, $\frac{ax\sin\theta}{\cos^2\theta} - \frac{by\cos\theta}{\sin^2\theta} = 0$. Show that $(ax)^{2/3} + (by)^{2/3} = (a^2 - b^2)^{2/3}$

19. If $P_n = \cos^n\theta + \sin^n\theta$ and $Q_n = \cos^n\theta - \sin^n\theta$, then show that
$P_n - P_{n-2} = -\sin^2\theta\cos^2\theta P_{n-4}$
$Q_n - Q_{n-2} = -\sin^2\theta\cos^2\theta Q_{n-4}$ and hence show that
$P_4 = 1 - 2\sin^2\theta\cos^2\theta$, $Q_4 = \cos^2\theta - \sin^2\theta$

20. If $\sin(\theta+\alpha) = a$ & $\sin(\theta+\beta) = b$ $(0 < \alpha, \beta, \theta < \pi/2)$ then find the value of
$\cos 2(\alpha-\beta) - 4ab\cos(\alpha-\beta)$

21. If $A + B + C = \pi$, prove that
$\tan B\tan C + \tan C\tan A + \tan A\tan B = 1 + \sec A.\sec B.\sec C$.

22. If $\tan^2\alpha + 2\tan\alpha.\tan 2\beta = \tan^2\beta + 2\tan\beta.\tan 2\alpha$, then prove that each side is equal to 1 or $\tan\alpha = \pm\tan\beta$.

Answers

EXERCISE–IV

1. D **2.** B **3.** A **4.** D **5.** B **6.** C **7.** B

8. A **9.** B **10.** B **11.** B **12.** B **13.** B **14.** A

15. C **16.** A **17.** B **18.** D **19.** B **20.** BC

21. AB **22.** BC **23.** AC **24.** BCD **25.** BD **26.** BC

27. BC **28.** BD

EXERCISE–V

1. 7.85 cm **2.** $r_1 : r_2 = 8 : 5$

6. $\sin\frac{x}{2} = \frac{3}{\sqrt{10}}$ and $\cos\frac{x}{2} = -\frac{1}{\sqrt{10}}$

16. n = 7 **20.** $1 - 2a^2 - 2b^2$

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 2 XI M 2. Trigonometric Equations

Index:

1. Key Concepts

2. Exercise I to III

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Trigonometric Equation

1. Trigonometric Equation :

An equation involving one or more trigonometric ratios of an unknown angle is called a trigonometric equation.

2. Solution of Trigonometric Equation :

A solution of trigonometric equation is the value of the unknown angle that satisfies the equation.

e.g. if $\sin\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{4}, ...$

Thus, the trigonometric equation may have infinite number of solutions (because of their periodic nature) and can be classified as :

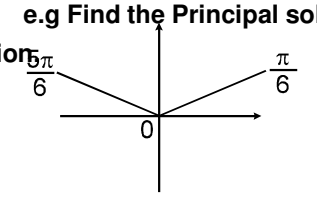
(i) Principal solution (ii) General solution.

2.1 Principal solutions:

The solutions of a trigonometric equation which lie in the interval $[0, 2\pi)$ are called **Principal solutions.**

e.g Find the Principal solutions of the equation $\sin x = \frac{1}{2}$.

Solution.



$\because$ $\sin x = \frac{1}{2}$

$\because$ there exists two values

i.e. $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ which lie in $[0, 2\pi)$ and whose sine is $\frac{1}{2}$

$\therefore$ Principal solutions of the equation $\sin x = \frac{1}{2}$ are $\frac{\pi}{6}, \frac{5\pi}{6}$ **Ans.**

2.2 General Solution :

The expression involving an integer 'n' which gives all solutions of a trigonometric equation is called **General solution**.

General solution of some standard trigonometric equations are given below.

3. General Solution of Some Standard Trigonometric Equations :

(i)	If $\sin\theta = \sin\alpha$	$\Rightarrow \theta = n\pi + (-1)^n \alpha$	where $\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $n \in I$.
(ii)	If $\cos\theta = \cos\alpha$	$\Rightarrow \theta = 2n\pi \pm \alpha$	where $\alpha \in [0, \pi]$, $n \in I$.
(iii)	If $\tan\theta = \tan\alpha$	$\Rightarrow \theta = n\pi + \alpha$	where $\alpha \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $n \in I$.
(iv)	If $\sin^2\theta = \sin^2\alpha$	$\Rightarrow \theta = n\pi \pm \alpha$, $n \in I$.	
(v)	If $\cos^2\theta = \cos^2\alpha$	$\Rightarrow \theta = n\pi \pm \alpha$, $n \in I$.	
(vi)	If $\tan^2\theta = \tan^2\alpha$	$\Rightarrow \theta = n\pi \pm \alpha$, $n \in I$.	[**Note:** α is called the principal angle]

Some Important deductions :

(i)	$\sin\theta = 0$	$\Rightarrow$	$\theta = n\pi$, $n \in I$
(ii)	$\sin\theta = 1$	$\Rightarrow$	$\theta = (4n + 1)\frac{\pi}{2}$, $n \in I$
(iii)	$\sin\theta = -1$	$\Rightarrow$	$\theta = (4n - 1)\frac{\pi}{2}$, $n \in I$
(iv)	$\cos\theta = 0$	$\Rightarrow$	$\theta = (2n + 1)\frac{\pi}{2}$, $n \in I$
(v)	$\cos\theta = 1$	$\Rightarrow$	$\theta = 2n\pi$, $n \in I$
(vi)	$\cos\theta = -1$	$\Rightarrow$	$\theta = (2n + 1)\pi$, $n \in I$
(vii)	$\tan\theta = 0$	$\Rightarrow$	$\theta = n\pi$, $n \in I$

Solved Example # 1

Solve $\sin\theta = \frac{\sqrt{3}}{2}$.

Solution.

$\because \quad \sin\theta = \frac{\sqrt{3}}{2}$

$\Rightarrow \quad \sin\theta = \sin\frac{\pi}{3}$

$\therefore \quad \boldsymbol{\theta = n\pi + (-1)^n \frac{\pi}{3}, n \in I}$ **Ans.**

Solved Example # 2

Solve $\boldsymbol{\sec 2\theta = -\frac{2}{\sqrt{3}}}$

Solution.

$\because \quad \sec 2\theta = -\frac{2}{\sqrt{3}}$

$\Rightarrow \quad \cos 2\theta = -\frac{\sqrt{3}}{2} \quad \Rightarrow \quad \cos 2\theta = \cos\frac{5\pi}{6}$

$\Rightarrow \quad 2\theta = 2n\pi \pm \frac{5\pi}{6}, n \in I$

$\Rightarrow \quad \boldsymbol{\theta = n\pi \pm \frac{5\pi}{12}, n \in I}$ **Ans.**

Solved Example # 3

Solve $\boldsymbol{\tan\theta = 2}$

Solution.

$\because \quad \tan\theta = 2$ .....(i)

Let $2 = \tan\alpha$

$\Rightarrow \quad \tan\theta = \tan\alpha$

$\Rightarrow \quad \boldsymbol{\theta = n\pi + \alpha}$**, where** $\boldsymbol{\alpha = \tan^{-1}(2), n \in I}$

Self Practice Problems:

1. Solve $\cot\theta = -1$

2. Solve $\cos 3\theta = -\frac{1}{2}$

Ans. (1) $\theta = n\pi - \frac{\pi}{4}, n \in I$ (2) $\frac{2n\pi}{3} \pm \frac{2\pi}{9}, n \in I$

Solved Example # 4

Solve $\boldsymbol{\cos^2\theta = \frac{1}{2}}$

Solution.

$\because \quad \cos^2\theta = \frac{1}{2}$

$\Rightarrow \quad \cos^2\theta = \left(\frac{1}{\sqrt{2}}\right)^2$

$\Rightarrow \quad \cos^2\theta = \cos^2\frac{\pi}{4}$

$\Rightarrow \quad \boldsymbol{\theta = n\pi \pm \frac{\pi}{4}, n \in I}$ **Ans.**

Solved Example # 5

Solve $\boldsymbol{4\tan^2\theta = 3\sec^2\theta}$

Solution.

$\because \quad 4\tan^2\theta = 3\sec^2\theta$ .....(i)

For equation (i) to be defined $\theta \neq (2n+1)\frac{\pi}{2}, n \in I$

$\because$ equation (i) can be written as:

$\frac{4\sin^2\theta}{\cos^2\theta} = \frac{3}{\cos^2\theta}$ $\quad \because \quad \theta \neq (2n+1)\frac{\pi}{2}, n \in I$

$\therefore \quad \cos^2\theta \neq 0$

$\Rightarrow \quad 4\sin^2\theta = 3$

$\Rightarrow \quad \sin^2\theta = \left(\frac{\sqrt{3}}{2}\right)^2$

$\Rightarrow \quad \sin^2\theta = \sin^2 \frac{\pi}{3}$

$\Rightarrow \quad \boldsymbol{\theta = n\pi \pm \frac{\pi}{3}, n \in I}$ **Ans.**

Self Practice Problems :

1. Solve $7\cos^2\theta + 3\sin^2\theta = 4$.

2. Solve $2\sin^2 x + \sin^2 2x = 2$

Ans. (1) $n\pi \pm \frac{\pi}{3}, n \in I$ (2) $(2n + 1)\frac{\pi}{2}, n \in I$ or $n\pi \pm \frac{\pi}{4}, n \in I$

Types of Trigonometric Equations :

Type -1

Trigonometric equations which can be solved by use of factorization.

Solved Example # 6

Solve $(2\sin x - \cos x)(1 + \cos x) = \sin^2 x$.

Solution.

$\because \quad (2\sin x - \cos x)(1 + \cos x) = \sin^2 x$

$\Rightarrow \quad (2\sin x - \cos x)(1 + \cos x) - \sin^2 x = 0$

$\Rightarrow \quad (2\sin x - \cos x)(1 + \cos x) - (1 - \cos x)(1 + \cos x) = 0$

$\Rightarrow \quad (1 + \cos x)(2\sin x - 1) = 0$

$\Rightarrow \quad 1 + \cos x = 0$ or $2\sin x - 1 = 0$

$\Rightarrow \quad \cos x = -1$ or $\sin x = \frac{1}{2}$

$\Rightarrow \quad x = (2n + 1)\pi, n \in I$ or $\sin x = \sin \frac{\pi}{6}$ $\Rightarrow x = n\pi + (-1)^n \frac{\pi}{6}, n \in I$

$\therefore$ Solution of given equation is

$\boldsymbol{(2n + 1)\pi, n \in I}$ or $\boldsymbol{n\pi + (-1)^n \frac{\pi}{6}, n \in I}$ **Ans.**

Self Practice Problems :

1. Solve $\cos^3 x + \cos^2 x - 4\cos^2 \frac{x}{2} = 0$

2. Solve $\cot^2\theta + 3\operatorname{cosec}\theta + 3 = 0$

Ans. (1) $(2n + 1)\pi, n \in I$

(2) $2n\pi - \frac{\pi}{2}, n \in I$ or $n\pi + (-1)^{n+1}\frac{\pi}{6}, n \in I$

Type - 2

Trigonometric equations which can be solved by reducing them in quadratic equations.

Solved Example # 7

Solve $2\cos^2 x + 4\cos x = 3\sin^2 x$

Solution.

$\because \quad 2\cos^2 x + 4\cos x - 3\sin^2 x = 0$

$\Rightarrow \quad 2\cos^2 x + 4\cos x - 3(1 - \cos^2 x) = 0$

$\Rightarrow \quad 5\cos^2 x + 4\cos x - 3 = 0$

$\Rightarrow \quad \left\{\cos x - \left(\frac{-2+\sqrt{19}}{5}\right)\right\}\left\{\cos x - \left(\frac{-2-\sqrt{19}}{5}\right)\right\} = 0$ .....(ii)

$\because \quad \cos x \in [-1, 1] \; \forall \; x \in R$

$\therefore \quad \cos x \neq \frac{-2-\sqrt{19}}{5}$

$\therefore$ equation (ii) will be true if

$\cos x = \frac{-2+\sqrt{19}}{5}$

$\Rightarrow \quad \cos x = \cos\alpha, \quad$ where $\cos\alpha = \frac{-2+\sqrt{19}}{5}$

$\Rightarrow \quad \boldsymbol{x = 2n\pi \pm \alpha}$ **where** $\boldsymbol{\alpha = \cos^{-1}\left(\frac{-2+\sqrt{19}}{5}\right), n \in I}$ **Ans.**

Self Practice Problems : **1.** Solve $\cos2\theta - (\sqrt{2}+1)\left(\cos\theta - \frac{1}{\sqrt{2}}\right) = 0$

2. Solve $4\cos\theta - 3\sec\theta = \tan\theta$

Ans. (1) $2n\pi \pm \frac{\pi}{3}, n \in I$ or $2n\pi \pm \frac{\pi}{4}, n \in I$

(2) $n\pi + (-1)^n \alpha$ where $\alpha = \sin^{-1}\left(\frac{-1-\sqrt{17}}{8}\right), n \in I$

or $n\pi + (-1)^n \beta$ where $\beta = \sin^{-1}\left(\frac{-1+\sqrt{17}}{8}\right), n \in I$

Type - 3

Trigonometric equations which can be solved by transforming a sum or difference of trigonometric ratios into their product.

Solved Example # 8

Solve $\cos3x + \sin2x - \sin4x = 0$

Solution.

$\cos3x + \sin2x - \sin4x = 0 \Rightarrow \cos3x + 2\cos3x.\sin(-x) = 0$

$\Rightarrow \cos3x - 2\cos3x.\sin x = 0 \Rightarrow \cos3x(1 - 2\sin x) = 0$

$\Rightarrow \cos3x = 0$ or $1 - 2\sin x = 0$

$\Rightarrow 3x = (2n+1)\frac{\pi}{2}, n \in I$ or $\sin x = \frac{1}{2}$

$\Rightarrow x = (2n+1)\frac{\pi}{6}, n \in I$ or $x = n\pi + (-1)^n \frac{\pi}{6}, n \in I$

$\therefore$ solution of given equation is

$(2n+1)\frac{\pi}{6}, n \in I$ or $n\pi + (-1)^n \frac{\pi}{6}, n \in I$ Ans.

Self Practice Problems :

1. Solve $\sin7\theta = \sin3\theta + \sin\theta$

2. Solve $5\sin x + 6\sin2x + 5\sin3x + \sin4x = 0$

3. Solve $\cos\theta - \sin3\theta = \cos2\theta$

Ans. (1) $\frac{n\pi}{3}, n \in I$ or $\frac{n\pi}{2} \pm \frac{\pi}{12}, n \in I$

(2) $\frac{n\pi}{2}, n \in I$ or $2n\pi \pm \frac{2\pi}{3}, n \in I$

(3) $\frac{2n\pi}{3}, n \in I$ or $2n\pi - \frac{\pi}{2}, n \in I$ or $n\pi + \frac{\pi}{4}, n \in I$

Type - 4

Trigonometric equations which can be solved by transforming a product of trigonometric ratios into their sum or difference.

Solved Example # 9

Solve $\sin5x.\cos3x = \sin6x.\cos2x$

Solution.

$\because \sin5x.\cos3x = \sin6x.\cos2x \Rightarrow 2\sin5x.\cos3x = 2\sin6x.\cos2x$

$\Rightarrow \sin8x + \sin2x = \sin8x + \sin4x \Rightarrow \sin4x - \sin2x = 0$

$\Rightarrow 2\sin2x.\cos2x - \sin2x = 0 \Rightarrow \sin2x(2\cos2x - 1) = 0$

$\Rightarrow \sin2x = 0$ or $2\cos2x - 1 = 0$

$\Rightarrow 2x = n\pi, n \in I$ or $\cos2x = \frac{1}{2}$

$\Rightarrow x = \frac{n\pi}{2}, n \in I$ or $2x = 2n\pi \pm \frac{\pi}{3}, n \in I$

$\Rightarrow x = n\pi \pm \frac{\pi}{6}, n \in I$

$\therefore$ Solution of given equation is

$\frac{n\pi}{2}, n \in I$ or $n\pi \pm \frac{\pi}{6}, n \in I$ Ans.

Type - 5

Trigonometric Equations of the form $a\sin x + b\cos x = c$, where $a, b, c \in R$, can be solved by dividing both sides of the equation by $\sqrt{a^2+b^2}$.

Solved Example # 10

Solve $\sin x + \cos x = \sqrt{2}$

Solution.

$\because$ $\sin x + \cos x = \sqrt{2}$ .....(i)

Here $a = 1, b = 1$.

$\therefore$ divide both sides of equation (i) by $\sqrt{2}$, we get

$$\sin x \cdot \frac{1}{\sqrt{2}} + \cos x \cdot \frac{1}{\sqrt{2}} = 1$$

$$\Rightarrow \quad \sin x \cdot \sin\frac{\pi}{4} + \cos x \cdot \cos\frac{\pi}{4} = 1$$

$$\Rightarrow \quad \cos\left(x - \frac{\pi}{4}\right) = 1$$

$$\Rightarrow \quad x - \frac{\pi}{4} = 2n\pi,\ n \in I$$

$$\Rightarrow \quad x = 2n\pi + \frac{\pi}{4},\ n \in I$$

$\therefore$ Solution of given equation is

$2n\pi + \frac{\pi}{4}$, $n \in I$ **Ans.**

Note : Trigonometric equation of the form $a\sin x + b\cos x = c$ can also be solved by changing **sinx** and **cosx** into their **corresponding tangent of half the angle.**

Solved Example # 11

Solve $3\cos x + 4\sin x = 5$

Solution.

$\because$ $3\cos x + 4\sin x = 5$ .....(i)

$\because$ $\cos x = \dfrac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}$ & $\sin x = \dfrac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}$

$\therefore$ equation (i) becomes

$$\Rightarrow \quad 3\left(\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right) + 4\left(\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right) = 5 \quad \text{.....(ii)}$$

Let $\tan\frac{x}{2} = t$

$\therefore$ equation (ii) becomes

$$3\left(\frac{1-t^2}{1+t^2}\right) + 4\left(\frac{2t}{1+t^2}\right) = 5$$

$\Rightarrow$ $4t^2 - 4t + 1 = 0$

$\Rightarrow$ $(2t-1)^2 = 0$

$$\Rightarrow \quad t = \frac{1}{2} \qquad \because \quad t = \tan\frac{x}{2}$$

$$\Rightarrow \quad \tan\frac{x}{2} = \frac{1}{2}$$

$$\Rightarrow \quad \tan\frac{x}{2} = \tan\alpha, \text{ where } \tan\alpha = \frac{1}{2}$$

$$\Rightarrow \quad \frac{x}{2} = n\pi + \alpha$$

$\Rightarrow$ **$x = 2n\pi + 2\alpha$ where $\alpha = \tan^{-1}\left(\frac{1}{2}\right)$, $n \in I$** **Ans.**

Self Practice Problems :

1. Solve $\sqrt{3}\cos x + \sin x = 2$

2. Solve $\sin x + \tan\frac{x}{2} = 0$

Ans. (1) $2n\pi + \frac{\pi}{6}, n \in I$ (2) $x = 2n\pi, n \in I$

Type - 6

Trigonometric equations of the form $P(\sin x \pm \cos x, \sin x \cos x) = 0$, where $p(y, z)$ is a polynomial, can be solved by using the substitution $\sin x \pm \cos x = t$.

Solved Example # 12

Solve $\sin x + \cos x = 1 + \sin x.\cos x$

Solution.

$\because$ $\sin x + \cos x = 1 + \sin x.\cos x$ .....(i)

Let $\sin x + \cos x = t$

$\Rightarrow$ $\sin^2 x + \cos^2 x + 2\sin x.\cos x = t^2$

$\Rightarrow$ $\sin x.\cos x = \frac{t^2 - 1}{2}$

Now put $\sin x + \cos x = t$ and $\sin x.\cos x = \frac{t^2 - 1}{2}$ in (i), we get

$t = 1 + \frac{t^2 - 1}{2}$

$\Rightarrow$ $t^2 - 2t + 1 = 0$

$\Rightarrow$ $t = 1$ $\because$ $t = \sin x + \cos x$

$\Rightarrow$ $\sin x + \cos x = 1$ .....(ii)

divide both sides of equation (ii) by $\sqrt{2}$, we get

$\Rightarrow$ $\sin x.\frac{1}{\sqrt{2}} + \cos x.\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$

$\Rightarrow$ $\cos\left(x - \frac{\pi}{4}\right) = \cos\frac{\pi}{4}$

$\Rightarrow$ $x - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}$

(i) if we take positive sign, we get

$x = 2n\pi + \frac{\pi}{2}, n \in I$ Ans.

(ii) if we take negative sign, we get

$x = 2n\pi, n \in I$ Ans.

Self Practice Problems:

1. Solve $\sin 2x + 5\sin x + 1 + 5\cos x = 0$

2. Solve $3\cos x + 3\sin x + \sin 3x - \cos 3x = 0$

3. Solve $(1 - \sin 2x)(\cos x - \sin x) = 1 - 2\sin^2 x$.

Ans. (1) $n\pi - \frac{\pi}{4}, n \in I$ (2) $n\pi - \frac{\pi}{4}, n \in I$

(3) $2n\pi + \frac{\pi}{2}, n \in I$ or $2n\pi, n \in I$ or $n\pi + \frac{\pi}{4}, n \in I$

Type - 7

Trigonometric equations which can be solved by the use of boundness of the trigonometric ratios sinx and cosx.

Solved Example # 13

Solve $\sin x\left(\cos\frac{x}{4} - 2\sin x\right) + \left(1 + \sin\frac{x}{4} - 2\cos x\right)\cos x = 0$

Solution.

$\because$ $\sin x\left(\cos\frac{x}{4} - 2\sin x\right) + \left(1 + \sin\frac{x}{4} - 2\cos x\right)\cos x = 0$ .....(i)

$\Rightarrow$ $\sin x.\cos\frac{x}{4} - 2\sin^2 x + \cos x + \sin\frac{x}{4}.\cos x - 2\cos^2 x = 0$

$\Rightarrow$ $\left(\sin x.\cos\frac{x}{4} + \sin\frac{x}{4}.\cos x\right) - 2(\sin^2 x + \cos^2 x) + \cos x = 0$

$\Rightarrow$ $\sin\frac{5x}{4} + \cos x = 2$ .....(ii)

Now equation (ii) will be true if

$\sin\frac{5x}{4} = 1$ **and** $\cos x = 1$

$\Rightarrow \quad \frac{5x}{4} = 2n\pi + \frac{\pi}{2}, n \in I$ **and** $x = 2m\pi, m \in I$

$\Rightarrow \quad x = \frac{(8n+2)\pi}{5}, n \in I$ .....(iii) **and** $x = 2m\pi, m \in I$ .....(iv)

Now to find general solution of equation (i)

$\frac{(8n+2)\pi}{5} = 2m\pi$

$\Rightarrow \quad 8n + 2 = 10m$

$\Rightarrow \quad n = \frac{5m-1}{4}$

if	$m = 1$	then	$n = 1$
if	$m = 5$	then	$n = 6$
	.....	.....	.....
	.....	.....	.....
if	$m = 4p - 3, p \in I$	then	$n = 5p - 4, p \in I$

$\therefore$ general solution of given equation can be obtained by substituting either $m = 4p - 3$ in equation (iv) or $n = 5p - 4$ in equation (iii)

$\therefore$ general solution of equation (i) is
$(8p - 6)\pi, p \in I$ **Ans.**

Self Practice Problems :

1. Solve $\sin 3x + \cos 2x = -2$

2. Solve $\sqrt{3\sin 5x - \cos^2 x - 3} = 1 - \sin x$

Ans. (1) $(4p-3)\frac{\pi}{2}, p \in I$ (2) $2m\pi + \frac{\pi}{2}, m \in I$

SHORT REVISION

TRIGONOMETRIC EQUATIONS & *INEQUATIONS*

THINGS TO REMEMBER :

1. If $\sin\theta = \sin\alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$ where $\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], n \in I$.

2. If $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha$ where $\alpha \in [0, \pi], n \in I$.

3. If $\tan\theta = \tan\alpha \Rightarrow \theta = n\pi + \alpha$ where $\alpha \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), n \in I$.

4. If $\sin^2\theta = \sin^2\alpha \Rightarrow \theta = n\pi \pm \alpha$.

5. $\cos^2\theta = \cos^2\alpha \Rightarrow \theta = n\pi \pm \alpha$.

6. $\tan^2\theta = \tan^2\alpha \Rightarrow \theta = n\pi \pm \alpha$. [Note : α is called the principal angle]

7. TYPES OF TRIGONOMETRIC EQUATIONS :

(a) Solutions of equations by factorising . Consider the equation ;
$(2\sin x - \cos x)(1 + \cos x) = \sin^2 x$; $\cot x - \cos x = 1 - \cot x \cos x$

(b) Solutions of equations reducible to quadratic equations. Consider the equation :
$3\cos^2 x - 10\cos x + 3 = 0$ and $2\sin^2 x + \sqrt{3}\sin x + 1 = 0$

(c) Solving equations by introducing an Auxilliary argument . Consider the equation :
$\sin x + \cos x = \sqrt{2}$; $\sqrt{3}\cos x + \sin x = 2$; $\sec x - 1 = (\sqrt{2} - 1)\tan x$

(d) Solving equations by Transforming a sum of Trigonometric functions into a product.
Consider the example : $\cos 3x + \sin 2x - \sin 4x = 0$;
$\sin^2 x + \sin^2 2x + \sin^2 3x + \sin^2 4x = 2$; $\sin x + \sin 5x = \sin 2x + \sin 4x$

(e) Solving equations by transforming a product of trigonometric functions into a sum.

Consider the equation :

$\sin 5x \cdot \cos 3x = \sin 6x \cdot \cos 2x$; $8 \cos x \cos 2x \cos 4x = \dfrac{\sin 6x}{\sin x}$; $\sin 3\theta = 4 \sin\theta \sin 2\theta \sin 4\theta$

(f) Solving equations by a change of variable :

(i) Equations of the form of $a \cdot \sin x + b \cdot \cos x + d = 0$, where a, b & d are real numbers & $a, b \neq 0$ can be solved by changing $\sin x$ & $\cos x$ into their corresponding tangent of half the angle. Consider the equation $3 \cos x + 4 \sin x = 5$.

(ii) Many equations can be solved by introducing a new variable . eg. the equation $\sin^4 2x + \cos^4 2x = \sin 2x \cdot \cos 2x$ changes to

$2(y+1)\left(y - \dfrac{1}{2}\right) = 0$ by substituting, $\sin 2x \cdot \cos 2x = y$.

(g) Solving equations with the use of the Boundness of the functions $\sin x$ & $\cos x$ or by making two perfect squares. Consider the equations :

$\sin x \left(\cos\dfrac{x}{4} - 2\sin x\right) + \left(1 + \sin\dfrac{x}{4} - 2\cos x\right) \cdot \cos x = 0$;

$\sin^2 x + 2\tan^2 x + \dfrac{4}{\sqrt{3}} \tan x - \sin x + \dfrac{11}{12} = 0$

8. TRIGONOMETRIC INEQUALITIES : There is no general rule to solve a Trigonometric inequations and the same rules of algebra are valid except the domain and range of trigonometric functions should be kept in mind.

Consider the examples : $\log_2\left(\sin\dfrac{x}{2}\right) < -1$; $\sin x\left(\cos x + \dfrac{1}{2}\right) \leq 0$; $\sqrt{5 - 2\sin 2x} \geq 6\sin x - 1$

EXERCISE–I

Q.1 Solve the equation for x, $5^{\frac{1}{2}} + 5^{\frac{1}{2} + \log_5(\sin x)} = 15^{\frac{1}{2} + \log_{15}\cos x}$

Q.2 Find all the values of θ satisfying the equation; $\sin\theta + \sin 5\theta = \sin 3\theta$ such that $0 \leq \theta \leq \pi$.

Q.3 Find all value of θ, between 0 & π, which satisfy the equation; $\cos\theta \cdot \cos 2\theta \cdot \cos 3\theta = 1/4$.

Q.4 Solve for x, the equation $\sqrt{13 - 18\tan x} = 6\tan x - 3$, where $-2\pi < x < 2\pi$.

Q.5 Determine the smallest positive value of x which satisfy the equation, $\sqrt{1 + \sin 2x} - \sqrt{2}\cos 3x = 0$.

Q.6 $2\sin\left(3x + \dfrac{\pi}{4}\right) = \sqrt{1 + 8\sin 2x \cdot \cos^2 2x}$

Q.7 Find the general solution of the trigonometric equation $3^{\left(\frac{1}{2} + \log_3(\cos x + \sin x)\right)} - 2^{\log_2(\cos x - \sin x)} = \sqrt{2}$.

Q.8 Find all values of θ between $0°$ & $180°$ satisfying the equation;
$\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0$.

Q.9 Find the solution set of the equation, $\log_{\frac{-x^2-6x}{10}}(\sin 3x + \sin x) = \log_{\frac{-x^2-6x}{10}}(\sin 2x)$.

Q.10 Find the value of θ, which satisfy $3 - 2\cos\theta - 4\sin\theta - \cos 2\theta + \sin 2\theta = 0$.

Q.11 Find the general solution of the equation, $\sin \pi x + \cos \pi x = 0$. Also find the sum of all solutions in [0, 100].

Q.12 Find the least positive angle measured in degrees satisfying the equation
$\sin^3 x + \sin^3 2x + \sin^3 3x = (\sin x + \sin 2x + \sin 3x)^3$.

Q.13 Find the general values of θ for which the quadratic function

$(\sin\theta)\, x^2 + (2\cos\theta)x + \dfrac{\cos\theta + \sin\theta}{2}$ is the square of a linear function.

Q.14 Prove that the equations (a) $\sin x \cdot \sin 2x \cdot \sin 3x = 1$ (b) $\sin x \cdot \cos 4x \cdot \sin 5x = -1/2$ have no solution.

Q.15 Let $f(x) = \sin^6 x + \cos^6 x + k(\sin^4 x + \cos^4 x)$ for some real number k. Determine

(a) all real numbers k for which $f(x)$ is constant for all values of x.

(b) all real numbers k for which there exists a real number 'c' such that $f(c) = 0$.

(c) If $k = -0.7$, determine all solutions to the equation $f(x) = 0$.

Q.16 If α and β are the roots of the equation, $a\cos\theta + b\sin\theta = c$ then match the entries of **column-I** with the entries of **column-II**.

Column-I		**Column-II**	
(A)	$\sin\alpha + \sin\beta$	(P)	$\dfrac{2b}{a+c}$
(B)	$\sin\alpha \,.\, \sin\beta$	(Q)	$\dfrac{c-a}{c+a}$
(C)	$\tan\dfrac{\alpha}{2} + \tan\dfrac{\beta}{2}$	(R)	$\dfrac{2bc}{a^2+b^2}$
(D)	$\tan\dfrac{\alpha}{2} \,.\, \tan\dfrac{\beta}{2} =$	(S)	$\dfrac{c^2-a^2}{a^2+b^2}$

Q.17 Find all the solutions of, $4\cos^2 x \sin x - 2\sin^2 x = 3\sin x$.

Q.18 Solve for x, $(-\pi \le x \le \pi)$ the equation; $2(\cos x + \cos 2x) + \sin 2x\,(1 + 2\cos x) = 2\sin x$.

Q.19 Solve the inequality $\sin 2x > \sqrt{2}\,\sin^2 x + (2 - \sqrt{2}\,)\cos^2 x$.

Q.20 Find the set of values of 'a' for which the equation, $\sin^4 x + \cos^4 x + \sin 2x + a = 0$ possesses solutions. Also find the general solution for these values of 'a'.

Q.21 Solve: $\tan^2 2x + \cot^2 2x + 2\tan 2x + 2\cot 2x = 6$.

Q.22 Solve: $\tan^2 x \,.\, \tan^2 3x \,.\, \tan 4x = \tan^2 x - \tan^2 3x + \tan 4x$.

Q.23 Find the set of values of x satisfying the equality

$\sin\left(x - \dfrac{\pi}{4}\right) - \cos\left(x + \dfrac{3\pi}{4}\right) = 1$ and the inequality $\dfrac{2\cos 7x}{\cos 3 + \sin 3} > 2^{\cos 2x}$.

Q.24 Let S be the set of all those solutions of the equation, $(1 + k)\cos x \cos(2x - \alpha) = (1 + k\cos 2x)\cos(x - \alpha)$ which are independent of k & α. Let H be the set of all such solutions which are dependent on k & α. Find the condition on k & α such that H is a non-empty set, state S. If a subset of H is $(0, \pi)$ in which $k = 0$, then find all the permissible values of α.

Q.25 Solve for x & y, $\begin{aligned} x\cos^3 y + 3x\cos y \sin^2 y &= 14 \\ x\sin^3 y + 3x\cos^2 y \sin y &= 13 \end{aligned}$

Q.26 Find the value of α for which the three elements set $S = \{\sin\alpha, \sin 2\alpha, \sin 3\alpha\}$ is equal to the three element set $T = \{\cos\alpha, \cos 2\alpha, \cos 3\alpha\}$.

Q.27 Find all values of 'a' for which every root of the equation, $a\cos 2x + |a|\cos 4x + \cos 6x = 1$ is also a root of the equation, $\sin x \cos 2x = \sin 2x \cos 3x - \dfrac{1}{2}\sin 5x$, and conversely, every root

of the second equation is also a root of the first equation.

Q.28 Solve the equations for 'x' given in **column-I** and match with the entries of **column-II**.

	Column-I		**Column-II**
(A)	$\cos 3x \cdot \cos^3 x + \sin 3x \cdot \sin^3 x = 0$	(P)	$n\pi \pm \frac{\pi}{3}$
(B)	$\sin 3\alpha = 4 \sin\alpha \sin(x+\alpha) \sin(x-\alpha)$ where α is a constant $\neq n\pi$.	(Q)	$n\pi + \frac{\pi}{4}$, $n \in I$
(C)	$\lvert 2\tan x - 1 \rvert + \lvert 2\cot x - 1 \rvert = 2$.	(R)	$\frac{n\pi}{4} + \frac{\pi}{8}$, $n \in I$
(D)	$\sin^{10}x + \cos^{10}x = \frac{29}{16}\cos^4 2x$.	(S)	$\frac{n\pi}{2} \pm \frac{\pi}{4}$

EXERCISE–II

Q.1 Solve the following system of equations for x and y [REE '2001(mains), 3]

$5^{(\cos ec^2 x - 3\sec^2 y)} = 1$ and $2^{(2\cos ecx + \sqrt{3}\lvert \sec y \rvert)} = 64$.

Q.2 The number of integral values of k for which the equation $7\cos x + 5\sin x = 2k + 1$ has a solution is

(A) 4 (B) 8 (C) 10 (D) 12 [JEE 2002 (Screening), 3]

Q.3 $\cos(\alpha - \beta) = 1$ and $\cos(\alpha + \beta) = 1/e$, where $\alpha, \beta \in [-\pi, \pi]$, numbers of pairs of α, β which satisfy both the equations is

(A) 0 (B) 1 (C) 2 (D) 4 [JEE 2005 (Screening)]

Q.4 If $0 < \theta < 2\pi$, then the intervals of values of θ for which $2\sin^2\theta - 5\sin\theta + 2 > 0$, is

(A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$ (B) $\left(\frac{\pi}{8}, \frac{5\pi}{6}\right)$ (C) $\left(0, \frac{\pi}{8}\right) \cup \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$ (D) $\left(\frac{41\pi}{48}, \pi\right)$ [JEE 2006, 3]

Q.5 The number of solutions of the pair of equations

$2\sin^2\theta - \cos 2\theta = 0$

$2\cos^2\theta - 3\sin\theta = 0$

in the interval $[0, 2\pi]$ is

(A) zero (B) one (C) two (D) four [JEE 2007, 3]

ANSWER EXERCISE–I

Q.1 $x = 2n\pi + \frac{\pi}{6}$, $n \in I$ **Q.2** $0, \frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6}$ & π **Q.3** $\frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$

Q.4 $\alpha - 2\pi$; $\alpha - \pi$, α , $\alpha + \pi$, where $\tan\alpha = \frac{2}{3}$ **Q.5** $x = \pi/16$

Q.6 $x = 2n\pi + \frac{\pi}{12}$ or $2n\pi + \frac{17\pi}{12}$; $n \in I$ **Q.7** $x = 2n\pi + \frac{\pi}{12}$ **Q.8** $30°, 45°, 90°, 135°, 150°$

Q.9 $x = -\frac{5\pi}{3}$ **Q.10** $\theta = 2n\pi$ **or** $2n\pi + \frac{\pi}{2}$; $n \in I$ **Q.11** $x = n - \frac{1}{4}$, $n \in I$; sum = 5025 **Q.12** $72°$

Q.13 $2n\pi + \frac{\pi}{4}$ or $(2n+1)\pi - \tan^{-1}2$, $n \in I$ **Q.15** (a) $-\frac{3}{2}$; (b) $k \in \left[-1, -\frac{1}{2}\right]$; (c) $x = \frac{n\pi}{2} \pm \frac{\pi}{6}$

Q.16 (A) R; (B) S; (C) P; (D) Q **Q.17** $n\pi$; $n\pi + (-1)^n \frac{\pi}{10}$ **or** $n\pi + (-1)^n\left(-\frac{3\pi}{10}\right)$

Q.18 $\frac{\pm\pi}{3}, \frac{-\pi}{2}, \pm\pi$

Q.19 $n\pi + \frac{\pi}{8} < x < n\pi + \frac{\pi}{4}$

Q.20 $\frac{1}{2}\left[n\pi + (-1)^n \sin^{-1}\left(1-\sqrt{2a+3}\right)\right]$ where $n \in I$ and $a \in \left[-\frac{3}{2}, \frac{1}{2}\right]$

Q.21 $x = \frac{n\pi}{4} + (-1)^n \frac{\pi}{8}$ **or** $\frac{n\pi}{4} + (-1)^{n+1} \frac{\pi}{24}$

Q.22 $\frac{(2n+1)\pi}{4}$, $k\pi$, where $n, k \in I$

Q.23 $x = 2n\pi + \frac{3\pi}{4}$, $n \in I$

Q.24 (i) $|k \sin\alpha| \le 1$ (ii) $S = n\pi$, $n \in I$ (iii) $\alpha \in (-m\pi, 2\pi - m\pi)$ $m \in I$

Q.25 $x = \pm 5\sqrt{5}$ & $y = n\pi + \tan^{-1}\frac{1}{2}$ **Q.26** $\frac{n\pi}{2} + \frac{\pi}{8}$

Q.27 $a = 0$ or $a < -1$ **Q.28** (A) S; (B) P; (C) Q; (D) R

EXERCISE–II

Q.1 $x = n\pi + (-1)^n \frac{\pi}{6}$ and $y = m\pi \pm \frac{\pi}{6}$ where m & n are integers.

Q.2 B **Q.3** D **Q.4** A **Q.5** C

Exercise - 1 (Objective Questions)

Part : (A) Only one correct option

1. The solution set of the equation $4\sin\theta.\cos\theta - 2\cos\theta - 2\sqrt{3}\sin\theta + \sqrt{3} = 0$ in the interval $(0, 2\pi)$ is

(A) $\left\{\frac{3\pi}{4}, \frac{7\pi}{4}\right\}$ (B) $\left\{\frac{\pi}{3}, \frac{5\pi}{3}\right\}$ (C) $\left\{\frac{3\pi}{4}, \pi, \frac{\pi}{3}, \frac{5\pi}{3}\right\}$ (D) $\left\{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{11\pi}{6}\right\}$

2. All solutions of the equation, $2\sin\theta + \tan\theta = 0$ are obtained by taking all integral values of m and n in:

(A) $2n\pi + \frac{2\pi}{3}$, $n \in I$ (B) $n\pi$ or $2m\pi \pm \frac{2\pi}{3}$ where $n, m \in I$

(C) $n\pi$ or $m\pi \pm \frac{\pi}{3}$ where $n, m \in I$ (D) $n\pi$ or $2m\pi \pm \frac{\pi}{3}$ where $n, m \in I$

3. If $20\sin^2\theta + 21\cos\theta - 24 = 0$ & $\frac{7\pi}{4} < \theta < 2\pi$ then the values of $\cot\frac{\theta}{2}$ is:

(A) 3 (B) $\frac{\sqrt{15}}{3}$ (C) $-\frac{\sqrt{15}}{3}$ (D) -3

4. The general solution of $\sin x + \sin 5x = \sin 2x + \sin 4x$ is:

(A) $2n\pi$; $n \in I$ (B) $n\pi$; $n \in I$ (C) $n\pi/3$; $n \in I$ (D) $2n\pi/3$; $n \in I$

5. A triangle ABC is such that $\sin(2A + B) = \frac{1}{2}$. If A, B, C are in A.P. then the angle A, B, C are respectively.

(A) $\frac{5\pi}{12}, \frac{\pi}{4}, \frac{\pi}{3}$ (B) $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{12}$ (C) $\frac{\pi}{3}, \frac{\pi}{4}, \frac{5\pi}{12}$ (D) $\frac{\pi}{3}, \frac{5\pi}{12}, \frac{\pi}{4}$

6. The maximum value of $3\sin x + 4\cos x$ is

(A) 3 (B) 4 (C) 5 (D) 7

7. If $\sin\theta + 7\cos\theta = 5$, then $\tan(\theta/2)$ is a root of the equation

(A) $x^2 - 6x + 1 = 0$ (B) $6x^2 - x - 1 = 0$ (C) $6x^2 + x + 1 = 0$ (D) $x^2 - x + 6 = 0$

8. $\frac{\sin^3\theta - \cos^3\theta}{\sin\theta - \cos\theta} - \frac{\cos\theta}{\sqrt{1+\cot^2\theta}} - 2\tan\theta\cot\theta = -1$ if:

(A) $\theta \in \left(0, \frac{\pi}{2}\right)$ (B) $\theta \in \left(\frac{\pi}{2}, \pi\right)$ (C) $\theta \in \left(\pi, \frac{3\pi}{2}\right)$ (D) $\theta \in \left(\frac{3\pi}{2}, 2\pi\right)$

9. The number of integral values of a for which the equation $\cos 2x + a \sin x = 2a - 7$ possesses a solution is
(A) 2 (B) 3 (C) 4 (D) 5

10. The principal solution set of the equation, $2\cos x = \sqrt{2 + 2\sin 2x}$ is

(A) $\left\{\frac{\pi}{8}, \frac{13\pi}{8}\right\}$ (B) $\left\{\frac{\pi}{4}, \frac{13\pi}{8}\right\}$ (C) $\left\{\frac{\pi}{4}, \frac{13\pi}{10}\right\}$ (D) $\left\{\frac{\pi}{8}, \frac{13\pi}{10}\right\}$

11. The number of all possible triplets (a_1, a_2, a_3) such that : $a_1 + a_2 \cos 2x + a_3 \sin^2 x = 0$ for all x is
(A) 0 (B) 1 (C) 2 (D) infinite

12. If $2\tan^2 x - 5\sec x - 1 = 0$ has 7 different roots in $\left[0, \frac{n\pi}{2}\right]$, $n \in N$, then greatest value of n is
(A) 8 (B) 10 (C) 13 (D) 15

13. The solution of $|\cos x| = \cos x - 2\sin x$ is

(A) $x = n\pi, n \in I$ (B) $x = n\pi + \frac{\pi}{4}, n \in I$

(C) $x = n\pi + (-1)^n \frac{\pi}{4}, n \in I$ (D) $(2n+1)\pi + \frac{\pi}{4}, n \in I$

14. The arithmetic mean of the roots of the equation $4\cos^3 x - 4\cos^2 x - \cos(\pi + x) - 1 = 0$ in the interval [0, 315] is equal to
(A) 49π (B) 50π (C) 51π (D) 100π

15. Number of solutions of the equation $\cos 6x + \tan^2 x + \cos 6x \cdot \tan^2 x = 1$ in the interval $[0, 2\pi]$ is :
(A) 4 (B) 5 (C) 6 (D) 7

Part : (B) May have more than one options correct

16. $\sin x - \cos^2 x - 1$ assumes the least value for the set of values of x given by:
(A) $x = n\pi + (-1)^{n+1}(\pi/6)$, $n \in I$ (B) $x = n\pi + (-1)^n(\pi/6)$, $n \in I$
(C) $x = n\pi + (-1)^n(\pi/3)$, $n \in I$ (D) $x = n\pi - (-1)^n(\pi/6)$, $n \in I$

17. $\cos 4x \cos 8x - \cos 5x \cos 9x = 0$ if
(A) $\cos 12x = \cos 14x$ (B) $\sin 13x = 0$
(C) $\sin x = 0$ (D) $\cos x = 0$

18. The equation $2\sin\frac{x}{2} \cdot \cos^2 x + \sin^2 x = 2\sin\frac{x}{2} \cdot \sin^2 x + \cos^2 x$ has a root for which
(A) $\sin 2x = 1$ (B) $\sin 2x = -1$ (C) $\cos x = \frac{1}{2}$ (D) $\cos 2x = -\frac{1}{2}$

19. $\sin^2 x + 2\sin x \cos x - 3\cos^2 x = 0$ if
(A) $\tan x = 3$ (B) $\tan x = -1$
(C) $x = n\pi + \pi/4$, $n \in I$ (D) $x = n\pi + \tan^{-1}(-3)$, $n \in I$

20. $\sin^2 x - \cos 2x = 2 - \sin 2x$ if
(A) $x = n\pi/2$, $n \in I$ (B) $x = n\pi - \pi/2$, $n \in I$
(C) $x = (2n+1)\pi/2$, $n \in I$ (D) $x = n\pi + (-1)^n \sin^{-1}(2/3)$, $n \in I$

Exercise - 2 (Subjective Questions)

1. Solve $\cot\theta = \tan 8\theta$

2. Solve $\cot\left(\frac{x}{2}\right) - \operatorname{cosec}\left(\frac{x}{2}\right) = \cot x$

3. Solve $\cot^2\theta + \left(\sqrt{3} + \frac{1}{\sqrt{3}}\right)\cot\theta + 1 = 0$.

4. Solve $\cos 2\theta + 3\cos\theta = 0$.

5. Solve the equation: $\sin 6x = \sin 4x - \sin 2x$.

6. Solve: $\cos\theta + \sin\theta = \cos 2\theta + \sin 2\theta$.

7. Solve $4 \sin x \,.\, \sin 2x \,.\, \sin 4x = \sin 3x$.

8. Solve $\sin^2 n\theta - \sin^2(n-1)\theta = \sin^2\theta$, where n is constant and $n \neq 0, 1$

9. Solve $\tan\theta + \tan 2\theta + \sqrt{3}\tan\theta\tan 2\theta = \sqrt{3}$.

10. Solve: $\sin^3 x \cos 3x + \cos^3 x \sin 3x + 0.375 = 0$

11. Solve the equation, $\frac{\sin^3\frac{x}{2} - \cos^3\frac{x}{2}}{2 + \sin x} = \frac{\cos x}{3}$.

12. Solve the equation: $\sin 5x = 16\sin^5 x$.

13. If $\tan\theta + \sin\phi = \frac{3}{2}$ & $\tan^2\theta + \cos^2\phi = \frac{7}{4}$ then find the general value of θ & ϕ .

14. Solve for x, the equation $\sqrt{13 - 18\tan x} = 6\tan x - 3$, where $-2\pi < x < 2\pi$.

15. Find the general solution of $\sec 4\theta - \sec 2\theta = 2$.

16. Solve the equation $\frac{\sqrt{3}}{2}\sin x - \cos x = \cos^2 x$.

17. Solve for x: $2\sin\left(3x + \frac{\pi}{4}\right) = \sqrt{1 + 8\sin 2x \,.\, \cos^2 2x}$.

18. Solve the equation for $0 \le \theta \le 2\pi$; $\left(\sin 2\theta + \sqrt{3}\cos 2\theta\right)^2 - 5 = \cos\left(\frac{\pi}{6} - 2\theta\right)$.

19. Solve: $\tan^2 x \,.\, \tan^2 3x \,.\, \tan 4x = \tan^2 x - \tan^2 3x + \tan 4x$.

20. Find the values of x, between 0 & 2π, satisfying the equation; $\cos 3x + \cos 2x = \sin\frac{3x}{2} + \sin\frac{x}{2}$.

21. Solve: $\cos\frac{2x}{3}\ \cos 6x = -1$.

22. Solve the equation, $\sin^2 4x + \cos^2 x = 2 \sin 4x \cos^4 x$.

Answers

EXERCISE # 1

1. D **2.** B **3.** D **4.** C **5.** B **6.** C **7.** B

8. B **9.** D **10.** A **11.** D **12.** D **13.** D **14.** C

15. D **16.** AD **17.** ABC **18.** ABCD **19.** CD

20. BC

EXERCISE # 2

1. $\left(n+\frac{1}{2}\right)\frac{\pi}{9}, n \in I$

2. $x = 4n\pi \pm \frac{2\pi}{3}, n \in I$

3. $\theta = n\pi - \frac{\pi}{3}, n \in I$ or $n\pi - \frac{\pi}{6}, n \in I$

4. $2n\pi \pm \alpha$ where $\alpha = \cos^{-1}\left(\frac{\sqrt{17}-3}{4}\right), n \in I$

5. $\frac{n\pi}{4}, n \in I$ or $n\pi \pm \frac{\pi}{6}, n \in I$

6. $2n\pi, n \in I$ or $\frac{2n\pi}{3} + \frac{\pi}{6}, n \in I$

7. $x = n\pi, n \in I$ **or** $\frac{n\pi}{3} \pm \frac{\pi}{9}, n \in I$

8. $m\pi, m \in I$ or $\frac{m\pi}{n-1}, m \in I$ or $\left(m+\frac{1}{2}\right)\frac{\pi}{n}, m \in I$

9. $\left(n+\frac{1}{3}\right)\frac{\pi}{3}, n \in I$

10. $x = \frac{n\pi}{4} + (-1)^{n+1}\frac{\pi}{24}$, $n \in I$

11. $x = (4n+1)\frac{\pi}{2}, n \in I$

12. $x = n\pi$; $x = n\pi \pm \frac{\pi}{6}, n \in I$

13. $\theta = n\pi + \frac{\pi}{4}, \phi = n\pi + (-1)^n\frac{\pi}{6}, n \in I$

14. $\alpha - 2\pi;\ \alpha - \pi, \alpha, \alpha + \pi$, where $\tan\alpha = \frac{2}{3}$

15. $\frac{2n\pi}{5} \pm \frac{\pi}{10}$ or $2n\pi \pm \frac{\pi}{2}, n \in I$

16. $x = (2n+1)\pi, , n \in I$ or $2n\pi \pm \frac{\pi}{3}, n \in I$

17. $(24\ell + 1)\frac{\pi}{12}, \ell \in I$ or $x = (24k-7)\frac{\pi}{12}, k \in I$

18. $\theta = \frac{7\pi}{12}, \frac{19\pi}{12}$

19. $\frac{(2n+1)\pi}{4}, k\pi$, where $n, k \in I$

20. $\frac{\pi}{7}, \frac{5\pi}{7}, \pi, \frac{9\pi}{7}, \frac{13\pi}{7}$

21. ϕ

22. $x = (2n+1)\frac{\pi}{2}, n \in I$

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 3 XI M 3. Properties of Triangle

Index:

1. Key Concepts

2. Exercise I to V

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Properties & Solution of Triangle

1. Sine Rule:

In any triangle ABC, the sines of the angles are proportional to the opposite sides i.e. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$.

Example : In any ΔABC, prove that $\frac{a+b}{c} = \frac{\cos\left(\frac{A-B}{2}\right)}{\sin\frac{C}{2}}$.

Solution. $\because$ We have to prove $\frac{a+b}{c} = \frac{\cos\left(\frac{A-B}{2}\right)}{\sin\frac{C}{2}}$.

$\because$ From **sine rule**, we know that

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$ (let)

$\Rightarrow$ $a = k \sin A$, $b = k \sin B$ and $c = k \sin C$

$\because$ L.H.S. $= \frac{a+b}{c}$

$= \frac{k(\sin A + \sin B)}{k \sin C}$ $= \frac{\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{\sin\frac{C}{2}\cos\frac{C}{2}}$

$= \frac{\cos\frac{C}{2}\cos\left(\frac{A-B}{2}\right)}{\sin\frac{C}{2}\cos\frac{C}{2}}$ $= \frac{\cos\left(\frac{A-B}{2}\right)}{\sin\frac{C}{2}}$

= R.H.S.

Hence L.H.S. = R.H.S. **Proved**

Example : In any ΔABC, prove that

$(b^2 - c^2)\cot A + (c^2 - a^2)\cot B + (a^2 - b^2)\cot C = 0$

Solution. $\because$ We have to prove that

$(b^2 - c^2)\cot A + (c^2 - a^2)\cot B + (a^2 - b^2)\cot C = 0$

$\because$ from **sine rule**, we know that

$a = k \sin A$, $b = k \sin B$ and $c = k \sin C$

$\therefore$ $(b^2 - c^2)\cot A = k^2(\sin^2 B - \sin^2 C)\cot A$

$\because$ $\sin^2 B - \sin^2 C = \sin(B + C)\sin(B - C)$

$\therefore$ $(b^2 - c^2)\cot A = k^2 \sin(B + C)\sin(B - C)\cot A$ $\quad\because$ $B + C = \pi - A$

$\therefore$ $(b^2 - c^2)\cot A = k^2 \sin A \sin(B - C)\frac{\cos A}{\sin A}$ $\quad\because$ $\cos A = -\cos(B + C)$

$= -k^2 \sin(B - C)\cos(B + C)$

$= -\frac{k^2}{2}[2\sin(B - C)\cos(B + C)]$

$\Rightarrow$ $(b^2 - c^2)\cot A = -\frac{k^2}{2}[\sin 2B - \sin 2C]$ .....(i)

Similarly $(c^2 - a^2)\cot B = -\frac{k^2}{2}[\sin 2C - \sin 2A]$ .....(ii)

and $(a^2 - b^2)\cot C = -\frac{k^2}{2}[\sin 2A - \sin 2B]$ .....(iii)

adding equations (i), (ii) and (iii), we get

$(b^2 - c^2)\cot A + (c^2 - a^2)\cot B + (a^2 - b^2)\cot C = 0$ **Hence Proved**

Self Practice Problems

In any ΔABC, prove that

1. $a \sin\left(\frac{A}{2} + B\right) = (b + c)\sin\left(\frac{A}{2}\right)$.

2. $\frac{a^2 \sin(B-C)}{\sin B + \sin C} + \frac{b^2 \sin(C-A)}{\sin C + \sin A} + \frac{c^2 \sin(A-B)}{\sin A + \sin B} = 0$

3. $\frac{c}{a-b} = \frac{\tan\frac{A}{2} + \tan\frac{B}{2}}{\tan\frac{A}{2} - \tan\frac{B}{2}}$.

2. Cosine Formula:

(i) $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ or $a^2 = b^2 + c^2 - 2bc\cos A = b^2 + c^2 + 2bc\cos(B + C)$

(ii) $\cos B = \frac{c^2 + a^2 - b^2}{2ca}$ **(iii)** $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

Example : In a triangle ABC if a = 13, b = 8 and c = 7, then find sin A.

Solution. $\because \quad \cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{64+49-169}{2.8.7}$

$\Rightarrow \quad \cos A = -\frac{1}{2} \quad \Rightarrow \quad A = \frac{2\pi}{3}$

$\therefore \quad \sin A = \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$ **Ans.**

***Example :** In a ΔABC, prove that $a(b\cos C - c\cos B) = b^2 - c^2$

Solution. $\because$ We have to prove $a(b\cos C - c\cos B) = b^2 - c^2$.

$\because$ from **cosine rule** we know that

$\cos C = \frac{a^2+b^2-c^2}{2ab}$ & $\cos B = \frac{a^2+c^2-b^2}{2ac}$

$\therefore \quad$ L.H.S. $= a\left\{b\left(\frac{a^2+b^2-c^2}{2ab}\right) - c\left(\frac{a^2+c^2-b^2}{2ac}\right)\right\}$

$= \frac{a^2+b^2-c^2}{2} - \frac{(a^2+c^2-b^2)}{2}$

$= (b^2 - c^2)$ $= $ R.H.S.

Hence L.H.S. = R.H.S. **Proved**

Example : If in a ΔABC, $\angle A = 60°$ then find the value of $\left(1+\frac{a}{c}+\frac{b}{c}\right)\left(1+\frac{c}{b}-\frac{a}{b}\right)$.

Solution. $\because \quad \angle A = 60°$

$\because \quad \left(1+\frac{a}{c}+\frac{b}{c}\right)\left(1+\frac{c}{b}-\frac{a}{b}\right) = \left(\frac{c+a+b}{c}\right)\left(\frac{b+c-a}{b}\right)$

$= \frac{(b+c)^2 - a^2}{bc}$

$= \frac{(b^2+c^2-a^2)+2bc}{bc}$

$= \frac{b^2+c^2-a^2}{bc} + 2$

$= 2\left(\frac{b^2+c^2-a^2}{2bc}\right) + 2$

$= 2\cos A + 2 \qquad \because \angle A = 60° \quad \Rightarrow \quad \cos A = \frac{1}{2}$

$\therefore \quad \left(1+\frac{a}{c}+\frac{b}{c}\right)\left(1+\frac{c}{b}-\frac{a}{b}\right) = 3$ **Ans.**

Self Practice Problems :

1. The sides of a triangle ABC are a, b, $\sqrt{a^2+ab+b^2}$, then prove that the greatest angle is 120°.

2. In a triangle ABC prove that $a(\cos B + \cos C) = 2(b + c)\sin^2\frac{A}{2}$.

3. Projection Formula:

(i) $a = b\cos C + c\cos B$ **(ii)** $b = c\cos A + a\cos C$ **(iii)** $c = a\cos B + b\cos A$

Example : In a triangle ABC prove that $a(b\cos C - c\cos B) = b^2 - c^2$

Solution. $\because$ L.H.S. $= a(b\cos C - c\cos B)$

$= b(a\cos C) - c(a\cos B)$ .....(i)

$\because$ From **projection rule**, we know that

$b = a\cos C + c\cos A \quad \Rightarrow \quad a\cos C = b - c\cos A$

& $c = a\cos B + b\cos A \quad \Rightarrow \quad a\cos B = c - b\cos A$

Put values of $a\cos C$ and $a\cos B$ in equation (i), we get

L.H.S. $= b(b - c\cos A) - c(c - b\cos A)$

$= b^2 - bc\cos A - c^2 + bc\cos A$

$= b^2 - c^2$

$=$ R.H.S.

Hence L.H.S. = R.H.S. **Proved**

Note: We have also proved $a(b\cos C - c\cos B) = b^2 - c^2$ by using **cosine – rule** in solved ***Example.**

Example : In a ΔABC prove that $(b + c)\cos A + (c + a)\cos B + (a + b)\cos C = a + b + c$.

Solution. $\because$ L.H.S. $= (b + c)\cos A (c + a)\cos B + (a + B)\cos C$

$= b\cos A + c\cos A + c\cos B + a\cos B + a\cos C + b\cos C$
$= (b\cos A + a\cos B) + (c\cos A + a\cos C) + (c\cos B + b\cos C)$
$= a + b + c$
$= R.H.S.$

Hence L.H.S. = R.H.S. **Proved**

Self Practice Problems

In a ΔABC, prove that

1. $2\left(b\cos^2\frac{C}{2} + c\cos^2\frac{B}{2}\right) = a + b + c.$

2. $\frac{\cos B}{\cos C} = \frac{c - b\cos A}{b - c\cos A}$.

3. $\frac{\cos A}{c\cos B + b\cos C} + \frac{\cos B}{a\cos C + c\cos A} + \frac{\cos C}{a\cos B + b\cos A} = \frac{a^2 + b^2 + c^2}{2abc}$.

4. Napier's Analogy - tangent rule:

(i) $\tan\frac{B - C}{2} = \frac{b - c}{b + c}\cot\frac{A}{2}$ **(ii)** $\tan\frac{C - A}{2} = \frac{c - a}{c + a}\cot\frac{B}{2}$

(iii) $\tan\frac{A - B}{2} = \frac{a - b}{a + b}\cot\frac{C}{2}$

Example : Find the unknown elements of the ΔABC in which $a = \sqrt{3} + 1$, $b = \sqrt{3} - 1$, $C = 60°$.

Solution. $\because$ $a = \sqrt{3} + 1$, $b = \sqrt{3} - 1$, $C = 60°$

$\because$ $A + B + C = 180°$

$\therefore$ $A + B = 120°$ .....(i)

$\because$ From law of tangent, we know that

$$\tan\left(\frac{A - B}{2}\right) = \frac{a - b}{a + b}\cot\frac{C}{2}$$

$$= \frac{(\sqrt{3} + 1) - (\sqrt{3} - 1)}{(\sqrt{3} + 1) + (\sqrt{3} - 1)}\cot 30°$$

$$= \frac{2}{2\sqrt{3}}\cot 30°$$

$\Rightarrow$ $\tan\left(\frac{A - B}{2}\right) = 1$

$\therefore$ $\frac{A - B}{2} = \frac{\pi}{4} = 45°$

$\Rightarrow$ $A - B = 90°$ .....(ii)

From equation (i) and (ii), we get

$A = 105°$ and $B = 15°$

Now,

$\because$ From **sine-rule,** we know that $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

$\therefore$ $c = \frac{a\sin C}{\sin A} = \frac{(\sqrt{3} + 1)\sin 60°}{\sin 105°}$

$= \frac{(\sqrt{3} + 1)\frac{\sqrt{3}}{2}}{\frac{\sqrt{3} + 1}{2\sqrt{2}}}$ $\because$ $\sin 105° = \frac{\sqrt{3} + 1}{2\sqrt{2}}$

$\Rightarrow$ $c = \sqrt{6}$

$\therefore$ $c = \sqrt{6}$, $A = 105°$, $B = 15°$ **Ans.**

Self Practice Problem

1. In a ΔABC if $b = 3$, $c = 5$ and $\cos(B - C) = \frac{7}{25}$, then find the value of $\tan\frac{A}{2}$.

Ans. $\frac{1}{3}$

2. If in a ΔABC, we define $x = \tan\left(\frac{B-C}{2}\right)\tan\frac{A}{2}$, $y = \tan\left(\frac{C-A}{2}\right)\tan\frac{B}{2}$ and $z = \tan\left(\frac{A-B}{2}\right)\tan\frac{C}{2}$ then show that $x + y + z = -xyz$.

5. Trigonometric Functions of Half Angles:

(i) $\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$; $\sin\frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}$; $\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

(ii) $\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$; $\cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}$; $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$

(iii) $\tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \frac{\Delta}{s(s-a)}$ where $s = \frac{a+b+c}{2}$ is semi perimetre of triangle.

(iv) $\sin A = \frac{2}{bc}\sqrt{s(s-a)(s-b)(s-c)} = \frac{2\Delta}{bc}$

6. Area of Triangle (Δ)

$$\Delta = \frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B = \sqrt{s(s-a)(s-b)(s-c)}$$

Example : In a ΔABC if a, b, c are in A.P. then find the value of $\tan\frac{A}{2}.\tan\frac{C}{2}$.

Solution. $\because$ $\tan\frac{A}{2} = \frac{\Delta}{s(s-a)}$ and $\tan\frac{C}{2} = \frac{\Delta}{s(s-c)}$

$\therefore$ $\tan\frac{A}{2}.\tan\frac{C}{2} = \frac{\Delta^2}{s^2(s-a)(s-c)}$ $\quad\because\quad \Delta^2 = s(s-a)(s-b)(s-c)$

$\therefore$ $\tan\frac{A}{2}.\tan\frac{C}{2} = \frac{s-b}{s} = 1 - \frac{b}{s}$ .....(i)

$\because$ it is given that a, b, c are in A.P.

$\Rightarrow$ $2b = a + c$

$\because$ $s = \frac{a+b+c}{2} = \frac{3b}{2}$

$\therefore$ $\frac{b}{s} = \frac{2}{3}$ put in equation (i)

$\therefore$ $\tan\frac{A}{2}.\tan\frac{C}{2} = 1 - \frac{2}{3}$

$\Rightarrow$ $\tan\frac{A}{2}.\tan\frac{C}{2} = \frac{1}{3}$ **Ans.**

Example : In a ΔABC if $b\sin C(b\cos C + c\cos B) = 42$, then find the area of the ΔABC.

Solution. $\because$ $b\sin C(b\cos C + c\cos B) = 42$ .....(i) given

$\because$ From **projection rule**, we know that
$a = b\cos C + c\cos B$ put in (i), we get
$ab\sin C = 42$ .....(ii)

$\because$ $\Delta = \frac{1}{2}ab\sin C$

$\therefore$ $\Delta = 21$ sq. unit **Ans.**

Example : In any ΔABC prove that $(a + b + c)\left(\tan\frac{A}{2} + \tan\frac{B}{2}\right) = 2c\cot\frac{C}{2}$.

Solution. $\because$ L.H.S. $= (a + b + c)\left(\tan\frac{A}{2} + \tan\frac{B}{2}\right)$

$\because$ $\tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$ and $\tan\frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$

$\therefore$ L.H.S. $= (a + b + c)\left[\sqrt{\frac{(s-b)(s-c)}{s(s-a)}} + \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}\right]$

$= 2s\sqrt{\frac{s-c}{s}}\left[\sqrt{\frac{s-b}{s-a}} + \sqrt{\frac{s-a}{s-b}}\right]$

$$= 2\sqrt{s(s-c)}\left[\frac{s-b+s-a}{\sqrt{(s-a)(s-b)}}\right] \qquad \because \quad 2s = a+b+c$$

$$\therefore \quad 2s-b-a = c$$

$$= 2\sqrt{s(s-c)}\left[\frac{c}{\sqrt{(s-a)(s-b)}}\right]$$

$$= 2c\sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \qquad \because \quad \cot\frac{C}{2} = \sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$$

$$= 2c\cot\frac{C}{2}$$

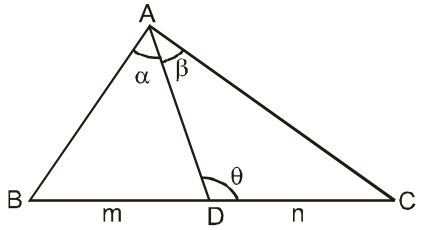
$$= \text{R.H.S.}$$

Hence L.H.S. = R.H.S. **Proved**

7. m - n Rule:

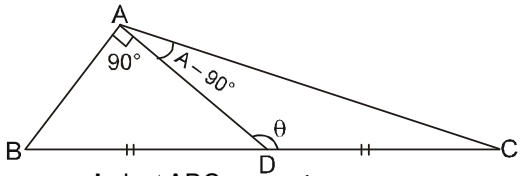
$$(m+n)\cot\theta = m\cot\alpha - n\cot\beta$$

$$= n\cot B - m\cot C$$



Example : If the median AD of a triangle ABC is perpendicular to AB, prove that tan A + 2tan B = 0.

Solution. From the figure, we see that $\theta = 90° + B$ (as θ is external angle of ΔABD)



Now if we apply **m-n rule** in ΔABC, we get

$(1 + 1)\cot(90 + B) = 1.\cot 90° - 1.\cot(A - 90°)$

$\Rightarrow \quad -2\tan B = \cot(90° - A)$

$\Rightarrow \quad -2\tan B = \tan A$

$\Rightarrow \quad \tan A + 2\tan B = 0$ **Hence proved.**

Example : The base of a triangle is divided into three equal parts. If t_1, t_2, t_3 be the tangents of the angles subtended by these parts at the opposite vertex, prove that

$$4\left(1+\frac{1}{t_2^2}\right) = \left(\frac{1}{t_1}+\frac{1}{t_2}\right)\left(\frac{1}{t_2}+\frac{1}{t_3}\right).$$

Solution. Let point D and E divides the base BC into three equal parts i.e. BD = DE = DC = d (Let) and let α, β and γ be the angles subtended by BD, DE and EC respectively at their opposite vertex.

$\Rightarrow \quad t_1 = \tan\alpha$, $t_2 = \tan\beta$ and $t_3 = \tan\gamma$

Now in ΔABC

$\because \quad BE : EC = 2d : d = 2 : 1$

$\therefore \quad$ from **m-n rule**, we get

$(2 + 1)\cot\theta = 2\cot(\alpha + \beta) - \cot\gamma$

$\Rightarrow \quad 3\cot\theta = 2\cot(\alpha + \beta) - \cot\gamma$ .....(i)

again

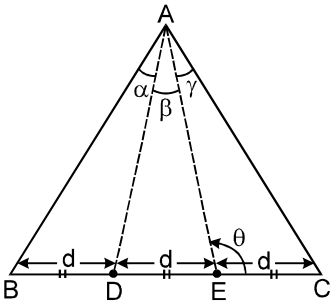
$\because \quad$ in ΔADC

$\because \quad DE : EC = x : x = 1 : 1$

$\therefore \quad$ if we apply **m-n rule** in ΔADC, we get

$(1 + 1)\cot\theta = 1.\cot\beta - 1\cot\gamma$

$2\cot\theta = \cot\beta - \cot\gamma$ .....(ii)



from (i) and (ii), we get

$$\frac{3\cot\theta}{2\cot\theta} = \frac{2\cot(\alpha+\beta)-\cot\gamma}{\cot\beta-\cot\gamma}$$

$\Rightarrow \quad 3\cot\beta - 3\cot\gamma = 4\cot(\alpha + \beta) - 2\cot\gamma$

$\Rightarrow \quad 3\cot\beta - \cot\gamma = 4\cot(\alpha + \beta)$

$$\Rightarrow \quad 3\cot\beta - \cot\gamma = 4\left\{\frac{\cot\alpha.\cot\beta - 1}{\cot\beta + \cot\alpha}\right\}$$

$\Rightarrow \quad 3\cot^2\beta + 3\cot\alpha\cot\beta - \cot\beta\cot\gamma - \cot\alpha\cot\gamma = 4\cot\alpha\cot\beta - 4$

$\Rightarrow \quad 4 + 3\cot^2\beta = \cot\alpha\cot\beta + \cot\beta\cot\gamma + \cot\alpha\cot\gamma$

$\Rightarrow \quad 4 + 4\cot^2\beta = \cot\alpha\cot\beta + \cot\alpha\cot\gamma + \cot\beta\cot\gamma + \cot^2\beta$

$\Rightarrow \quad 4(1 + \cot^2\beta) = (\cot\alpha + \cot\beta)(\cot\beta + \cot\gamma)$

$$\Rightarrow \quad 4\left(1+\frac{1}{\tan^2\beta}\right) = \left(\frac{1}{\tan\alpha}+\frac{1}{\tan\beta}\right)\left(\frac{1}{\tan\beta}+\frac{1}{\tan\gamma}\right)$$

$\Rightarrow \quad 4\left(1+\frac{1}{t_2^2}\right) = \left(\frac{1}{t_1}+\frac{1}{t_2}\right)\left(\frac{1}{t_2}+\frac{1}{t_3}\right)$ **Hence proved**

Self Practice Problems :

1. In a ΔABC, the median to the side BC is of length $\frac{1}{\sqrt{11-6\sqrt{3}}}$ and it divides angle A into the angles of $30°$ and $45°$. Prove that the side BC is of length 2 units.

8. Radius of Circumcirlce :

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C} = \frac{abc}{4\Delta}$$

Example : In a ΔABC prove that $\sin A + \sin B + \sin C = \frac{s}{R}$

Solution. In a ΔABC, we know that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$\therefore \quad \sin A = \frac{a}{2R}$, $\sin B = \frac{b}{2R}$ and $\sin C = \frac{c}{2R}$.

$\therefore \quad \sin A + \sin B + \sin C = \frac{a+b+c}{2R}$ $\quad \because \quad a + b + c = 2s$

$= \frac{2s}{2R}$ $\quad \Rightarrow \quad \sin A + \sin B + \sin C = \frac{s}{R}$.

Example : In a ΔABC if a = 13 cm, b = 14 cm and c = 15 cm, then find its circumradius.

Solution. $\because \quad R = \frac{abc}{4\Delta}$ .....(i)

$\because \quad \Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$\because \quad s = \frac{a+b+c}{2} = 21$ cm

$\therefore \quad \Delta = \sqrt{21.8.7.6} = \sqrt{7^2.4^2.3^2}$

$\Rightarrow \quad \Delta = 84\ cm^2$

$\therefore \quad R = \frac{13.14.15}{4.84} = \frac{65}{8}$ cm

$\therefore \quad R = \frac{65}{8}$ cm.

Example : In a ΔABC prove that $s = 4R\cos\frac{A}{2}.\cos\frac{B}{2}.\cos\frac{C}{2}$.

Solution. In a ΔABC,

$\because \quad \cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$, $\cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}$ and $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$ and $R = \frac{abc}{4\Delta}$

$\because \quad$ R.H.S. $= 4R\cos\frac{A}{2}.\cos\frac{B}{2}.\cos\frac{C}{2}$.

$= \frac{abc}{\Delta}.s\sqrt{\frac{s(s-a)(s-b)(s-c)}{(abc)^2}}$ $\quad \because \quad \Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$= s$

$=$ L.H.S.

Hence R.H.L $=$ L.H.S. proved

Example : In a ΔABC, prove that $\frac{1}{s-a}+\frac{1}{s-b}+\frac{1}{s-c}-\frac{1}{s} = \frac{4R}{\Delta}$.

Solution. $\frac{1}{s-a}+\frac{1}{s-b}+\frac{1}{s-c}-\frac{1}{s} = \frac{4R}{\Delta}$

$\because \quad$ L.H.S. $= \left(\frac{1}{s-a}+\frac{1}{s-b}\right)+\left(\frac{1}{s-c}-\frac{1}{s}\right)$

$= \frac{2s-a-b}{(s-a)(s-b)} + \frac{(s-s+c)}{s(s-c)}$ $\quad \because \quad 2s = a + b + c$

$= \frac{c}{(s-a)(s-b)} + \frac{c}{s(s-c)}$

$= c\left[\frac{s(s-c)+(s-a)(s-b)}{s(s-a)(s-b)(s-c)}\right] = c\left[\frac{2s^2-s(a+b+c)+ab}{\Delta^2}\right]$

$\therefore$ L.H.S. $= c\left[\frac{2s^2 - s(2s) + ab}{\Delta^2}\right] = \frac{abc}{\Delta^2} = \frac{4R\Delta}{\Delta^2} = \frac{4R}{\Delta}$ $\quad\because\quad R = \frac{abc}{4\Delta}$

$\Rightarrow \quad abc = 4R\Delta$

$\therefore$ L.H.S. $= \frac{4R}{\Delta}$

Self Practice Problems :

In a ΔABC, prove the followings :

1. $a \cot A + b \cot B + \cos C = 2(R + r)$.

2. $4\left(\frac{s}{a} - 1\right)\left(\frac{s}{b} - 1\right)\left(\frac{s}{c} - 1\right) = \frac{r}{R}$.

3. If α, β, γ are the distances of the vertices of a triangle from the corresponding points of contact with the incircle, then prove that $\frac{\alpha\beta y}{\alpha + \beta + y} = r^2$

9. Radius of The Incircle :

(i) $r = \frac{\Delta}{s}$ **(ii)** $r = (s-a)\tan\frac{A}{2} = (s-b)\tan\frac{B}{2} = (s-c)\tan\frac{C}{2}$

(iii) $r = \frac{a\sin\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{A}{2}}$ & so on **(iv)** $r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

10. Radius of The Ex- Circles :

(i) $r_1 = \frac{\Delta}{s-a}$; $r_2 = \frac{\Delta}{s-b}$; $r_3 = \frac{\Delta}{s-c}$ **(ii)** $r_1 = s\tan\frac{A}{2}$; $r_2 = s\tan\frac{B}{2}$; $r_3 = s\tan\frac{C}{2}$

(iii) $r_1 = \frac{a\cos\frac{B}{2}\cos\frac{C}{2}}{\cos\frac{A}{2}}$ & so on **(iv)** $r_1 = 4R\sin\frac{A}{2}.\cos\frac{B}{2}.\cos\frac{C}{2}$

Example : In a ΔABC, prove that $r_1 + r_2 + r_3 - r = 4R = 2a\,\text{cosec}A$

Solution. $\because$ L.H.S $= r_1 + r_2 + r_3 - r$

$= \frac{\Delta}{s-a} + \frac{\Delta}{s-b} + \frac{\Delta}{s-c} - \frac{\Delta}{s}$

$= \Delta\left(\frac{1}{s-a} + \frac{1}{s-b}\right) + \Delta\left(\frac{1}{s-c} - \frac{1}{s}\right)$

$= \Delta\left[\left(\frac{s-b+s-a}{(s-a)(s-b)}\right) + \left(\frac{s-s+c}{s(s-c)}\right)\right]$

$= \Delta\left[\frac{c}{(s-a)(s-b)} + \frac{c}{s(s-c)}\right]$

$= c\Delta\left[\frac{s(s-c)+(s-a)(s-b)}{s(s-a)(s-b)(s-c)}\right]$

$= c\Delta\left[\frac{2s^2 - s(a+b+c) + ab}{\Delta^2}\right]$ $\quad\because\quad a + b + c = 2s$

$= \frac{abc}{\Delta}$ $\quad\because\quad R = \frac{abc}{4\Delta}$

$= 4R = 2a\,\text{cosec}A$ $\quad\because\quad \frac{a}{\sin A} = 2R = a\,\text{cosec}A$

$= \text{R.H.S.}$

Hence L.H.S. $=$ R.H.S. **proved**

Example : If the area of a ΔABC is 96 sq. unit and the radius of the escribed circles are respectively 8, 12 and 24. Find the perimeter of ΔABC.

Solution. $\because$ $\Delta = 96$ sq. unit

$r_1 = 8$, $r_2 = 12$ and $r_3 = 24$

$\because \quad r_1 = \frac{\Delta}{s-a} \quad \Rightarrow \quad s - a = 12 \quad$ .....(i)

$\because \quad r_2 = \frac{\Delta}{s-b} \quad \Rightarrow \quad s - b = 8 \quad$ .....(ii)

$\because \quad r_3 = \frac{\Delta}{s-c} \quad \Rightarrow \quad s - c = 4 \quad$ .....(iii)

$\therefore$ adding equations (i), (ii) & (iii), we get

$3s - (a + b + c) = 24$

$s = 24$

$\therefore$ perimeter of $\Delta ABC = 2s = 48$ unit.

Self Practice Problems

In a ΔABC prove that

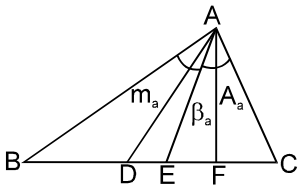
1. $r_1r_2 + r_2r_3 + r_3r_1 = s^2$

2. $rr_1 + rr_2 + rr_3 = ab + bc + ca - s^2$

3. If A, A_1, A_2 and A_3 are the areas of the inscribed and escribed circles respectively of a ΔABC, then prove that $\frac{1}{\sqrt{A}} = \frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}}$.

4. $\frac{r_1 - r}{a} + \frac{r_2 - r}{b} = \frac{c}{r_3}$.

11. Length of Angle Bisectors, Medians & Altitudes :



(i) Length of an angle bisector from the angle A = $\beta_a = \frac{2bc\cos\frac{A}{2}}{b+c}$;

(ii) Length of median from the angle A = $m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

& (iii) Length of altitude from the angle A = $A_a = \frac{2\Delta}{a}$

NOTE : $m_a^2 + m_b^2 + m_c^2 = \frac{3}{4}(a^2 + b^2 + c^2)$

Example : AD is a median of the ΔABC. If AE and AF are medians of the triangles ABD and ADC respectively, and AD = m_1, AE = m_2, AF = m_3, then prove that $m_2^2 + m_3^2 - 2m_1^2 = \frac{a^2}{8}$.

Solution. $\because$ In ΔABC

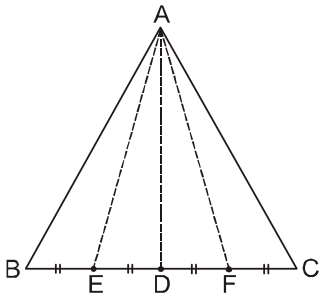
$AD^2 = \frac{1}{4}(2b^2 + 2c^2 - a^2) = m_1^2$ .....(i)

$\because$ In ΔABD, $AE^2 = m_2^2 = \frac{1}{4}(2c^2 + 2AD^2 - \frac{a^2}{4})$ .....(ii)

Similarly in ΔADC, $AF^2 = m_3^2 = \frac{1}{4}\left(2AD^2 + 2b^2 - \frac{a^2}{4}\right)$ .....(iii)

by adding equations (ii) and (iii), we get

$\because$ $m_2^2 + m_3^2 = \frac{1}{4}\left(4AD^2 + 2b^2 + 2c^2 - \frac{a^2}{2}\right)$



$= AD^2 + \frac{1}{4}\left(2b^2 + 2c^2 - \frac{a^2}{2}\right)$

$= AD^2 + \frac{1}{4}\left(2b^2 + 2c^2 - a^2 + \frac{a^2}{2}\right)$

$= AD^2 + \frac{1}{4}(2b^2 + 2c^2 - a^2) + \frac{a^2}{8}$

$= AD^2 + AD^2 + \frac{a^2}{8}$

$= 2AD^2 + \frac{a^2}{8}$ $\quad\because\quad AD^2 = m_1^2$

$= 2m_1^2 + \frac{a^2}{8}$

$\therefore$ $m_2^2 + m_3^2 - 2m_1^2 = \frac{a^2}{8}$ **Hence Proved**

Self Practice Problem :

3. In a ΔABC a = 5, b = 4, c = 3. 'G' is the centroid of triangle, then find circumradius of ΔGAB.

Ans. $\frac{5}{12}\sqrt{13}$

12. The Distances of The Special Points from Vertices and Sides of Triangle:

(i) Circumcentre (O) : $OA = R$ & $O_a = R\cos A$

(ii) Incentre (I) : $IA = r\,\text{cosec}\frac{A}{2}$ & $I_a = r$

(iii) Excentre (I_1) : $I_1A = r_1\,\text{cosec}\frac{A}{2}$ & $I_{1a} = r_1$

(iv) Orthocentre (H) : $HA = 2R\cos A$ & $H_a = 2R\cos B\cos C$

(v) Centroid (G) : $GA = \frac{1}{3}\sqrt{2b^2+2c^2-a^2}$ & $G_a = \frac{2\Delta}{3a}$

Example : If x, y and z are respectively the distances of the vertices of the ΔABC from its orthocentre, then prove that

(i) $\frac{a}{x}+\frac{b}{y}+\frac{c}{z} = \frac{abc}{xyz}$ (ii) $x\ y + z = 2(R + r)$

Solution. $\because$ $x = 2R\cos A,\ y = 2R\cos B,\ z = 2R\cos C$ and
and $a = 2R\sin A,\ b = 2R\sin B,\ c = 2R\sin C$

$\therefore$ $\frac{a}{x}+\frac{b}{y}+\frac{c}{z} = \tan A + \tan B + \tan C$ .....(i)

& $\frac{abc}{xyz} = \tan A.\tan B.\tan C$ .....(ii)

$\because$ We know that in a ΔABC $\Sigma\tan A = \Pi\tan A$

$\therefore$ From equations (i) and (ii), we get

$$\frac{a}{x}+\frac{b}{y}+\frac{c}{z} = \frac{abc}{xyz}$$

$\because$ $x + y + z = 2R(\cos A + \cos B + \cos C)$

$\because$ in a ΔABC $\cos A + \cos B + \cos C = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$\therefore$ $x + y + z = 2R\left(1+4\sin\frac{A}{2}.\sin\frac{B}{2}.\sin\frac{C}{2}\right)$

$= 2\left(R+4R\sin\frac{A}{2}.\sin\frac{B}{2}.\sin\frac{C}{2}\right)$ $\because$ $r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$\therefore$ $x + y + z = 2(R + r)$

Self Practice Problems

1. If I be the incentre of ΔABC, then prove that $IA\,.\,IB\,.\,IC = abc\tan\frac{A}{2}\tan\frac{B}{2}\tan\frac{C}{2}$.

2. If x, y, z are respectively be the perpendiculars from the circumcentre to the sides of ΔABC, then prove that $\frac{a}{x}+\frac{b}{y}+\frac{c}{z} = \frac{abc}{4xyz}$.

13. Orthocentre and Pedal Triangle:

The triangle KLM which is formed by joining the feet of the altitudes is called the Pedal Triangle.
(i) Its angles are $\pi - 2A$, $\pi - 2B$ and $\pi - 2C$.

(ii) Its sides are $a\cos A = R\sin 2A$,
$b\cos B = R\sin 2B$ and
$c\cos C = R\sin 2C$

(iii) Circumradii of the triangles PBC, PCA, PAB and ABC are equal.

14. Excentral Triangle:

The triangle formed by joining the three excentres I_1, I_2 and I_3 of ΔABC is called the excentral or excentric triangle.
(i) ΔABC is the pedal triangle of the $\Delta I_1I_2I_3$. (ii) Its angles are

$\frac{\pi}{2}-\frac{A}{2}, \frac{\pi}{2}-\frac{B}{2}$ & $\frac{\pi}{2}-\frac{C}{2}$.

(iii) Its sides are $4R\cos\frac{A}{2}$,

$4R\cos\frac{B}{2}$ & $4R\cos\frac{C}{2}$.

(iv) $II_1 = 4R\sin\frac{A}{2}$;

$II_2 = 4R\sin\frac{B}{2}$; $II_3 = 4R\sin\frac{C}{2}$.

(v) Incentre I of ΔABC is the orthocentre of the excentral $\Delta I_1I_2I_3$.

15. Distance Between Special Points :

(i) Distance between circumcentre and orthocentre

$OH^2 = R^2(1 - 8\cos A\cos B\cos C)$

(ii) Distance between circumcentre and incentre

$$OI^2 = R^2\left(1 - 8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right) = R^2 - 2Rr$$

(iii) Distance between circumcentre and centroid

$$OG^2 = R^2 - \frac{1}{9}(a^2 + b^2 + c^2)$$

Example : In I is the incentre and I_1, I_2, I_3 are the centres of escribed circles of the ΔABC, prove that
(i) $II_1 \cdot II_2 \cdot II_3 = 16R^2r$ (ii) $II_1^2 + I_2I_3^2 = II_2^2 + I_3I_1^2 = II_3^2 + I_1I_2^2$

Solution. (i) $\because$ We know that

$$II_1 = a\sec\frac{A}{2},\ II_2 = b\sec\frac{B}{2} \quad \text{and} \quad II_3 = c\sec\frac{C}{2}$$

$\because$ $I_1I_2 = c.\ \text{cosec}\frac{C}{2}$, $I_2I_3 = a\ \text{cosec}\frac{A}{2}$ and $I_3I_1 = b\ \text{cosec}\frac{B}{2}$

$\because$ $II_1 \cdot II_2 \cdot II_3 = abc\sec\frac{A}{2}\sec\frac{B}{2}.\sec\frac{C}{2}$ .....(i)

$\because$ $a = 2R\sin A$, $b = 2R\sin B$ and $c = 2R\sin C$

$\therefore$ equation (i) becomes

$\because$ $II_1 . II_2 . II_3 = (2R\sin A)(2R\sin B)(2R\sin C)\sec\frac{A}{2}\sec\frac{B}{2}\sec\frac{C}{2}$

$$= 8R^3 \cdot \frac{\left(2\sin\frac{A}{2}\cos\frac{A}{2}\right)\left(2\sin\frac{B}{2}\cos\frac{B}{2}\right)\left(2\sin\frac{C}{2}\cos\frac{C}{2}\right)}{\cos\frac{A}{2}.\cos\frac{B}{2}.\cos\frac{C}{2}}$$

$$= 64R^3\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \qquad \because \quad r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

$\therefore$ $II_1 . II_2 . II_3 = 16R^2r$ **Hence Proved**

(ii) $II_1^2 + I_2I_3^2 = II_2^2 + I_3I_1^2 = II_3^2 + I_1I_2^2$

$\because$ $II_1^2 + I_2I_3^2 = a^2\sec^2\frac{A}{2} + a^2\ \text{cosec}^2\frac{A}{2} = \frac{a^2}{\sin^2\frac{A}{2}\cos^2\frac{A}{2}}$

$\because$ $a = 2R\sin A = 4R\sin\frac{A}{2}\cos\frac{A}{2}$ $\therefore$ $II_1^2 + I_2I_3^2 = \frac{16R^2\sin^2\frac{A}{2}.\cos^2\frac{A}{2}}{\sin^2\frac{A}{2}.\cos^2\frac{A}{2}} = 16R^2$

Similarly we can prove $II_2^2 + I_3I_1^2 = II_3^2 + I_1I_2^2 = 16R^2$

Hence $II_1^2 + I_2I_3^2 = II_2^2 + I_3I_1^2 = II_3^2 + I_1I_2^2$

Self Practice Problem :

1. In a ΔABC, if b = 2 cm, c = $\sqrt{3}$ cm and $\angle A = \frac{\pi}{6}$, then find distance between its circumcentre and incentre.

Ans. $\sqrt{2-\sqrt{3}}$ cm

SHORT REVISION

SOLUTIONS OF TRIANGLE

I. SINE FORMULA : In any triangle ABC, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$.

II. COSINE FORMULA : **(i)** $\cos A = \frac{b^2+c^2-a^2}{2bc}$ or $a^2 = b^2 + c^2 - 2bc.\cos A$

(ii) $\cos B = \frac{c^2+a^2-b^2}{2ca}$ **(iii)** $\cos C = \frac{a^2+b^2-c^2}{2ab}$

III. PROJECTION FORMULA : **(i)** $a = b\cos C + c\cos B$ **(ii)** $b = c\cos A + a\cos C$

(iii) $c = a\cos B + b\cos A$

IV. NAPIER'S ANALOGY – TANGENT RULE : **(i)** $\tan\frac{B-C}{2} = \frac{b-c}{b+c}\cot\frac{A}{2}$

(ii) $\tan\frac{C-A}{2} = \frac{c-a}{c+a}\cot\frac{B}{2}$ **(iii)** $\tan\frac{A-B}{2} = \frac{a-b}{a+b}\cot\frac{C}{2}$

V. TRIGONOMETRIC FUNCTIONS OF HALF ANGLES :

(i) $\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$; $\sin\frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}$; $\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

(ii) $\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$; $\cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}$; $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$

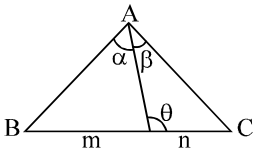
(iii) $\tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \frac{\Delta}{s(s-a)}$ where $s = \frac{a+b+c}{2}$ & Δ = area of triangle.

(iv) Area of triangle $= \sqrt{s(s-a)(s-b)(s-c)}$.

VI. m – n RULE : In any triangle,

$(m+n)\cot\theta = m\cot\alpha - n\cot\beta$

$= n\cot B - m\cot C$



VII. $\frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B$ = area of triangle ABC.

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

Note that $R = \frac{abc}{4\Delta}$; Where R is the radius of circumcircle & Δ is area of triangle

VIII. Radius of the incircle 'r' is given by:

(a) $r = \frac{\Delta}{s}$ where $s = \frac{a+b+c}{2}$ **(b)** $r = (s-a)\tan\frac{A}{2} = (s-b)\tan\frac{B}{2} = (s-c)\tan\frac{C}{2}$

(c) $r = \frac{a\sin\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{A}{2}}$ & so on **(d)** $r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

IX. Radius of the Ex–circles r_1, r_2 & r_3 are given by :

(a) $r_1 = \frac{\Delta}{s-a}$; $r_2 = \frac{\Delta}{s-b}$; $r_3 = \frac{\Delta}{s-c}$ **(b)** $r_1 = s\tan\frac{A}{2}$; $r_2 = s\tan\frac{B}{2}$; $r_3 = s\tan\frac{C}{2}$

(c) $r_1 = \frac{a\cos\frac{B}{2}\cos\frac{C}{2}}{\cos\frac{A}{2}}$ & so on **(d)** $r_1 = 4R\sin\frac{A}{2}.\cos\frac{B}{2}.\cos\frac{C}{2}$;

$r_2 = 4R\sin\frac{B}{2}.\cos\frac{A}{2}.\cos\frac{C}{2}$; $r_3 = 4R\sin\frac{C}{2}.\cos\frac{A}{2}.\cos\frac{B}{2}$

X. LENGTH OF ANGLE BISECTOR & MEDIANS :

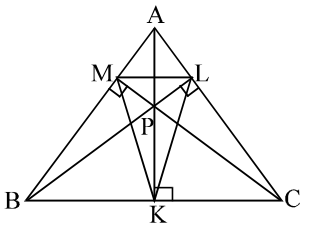
If m_a and β_a are the lengths of a median and an angle bisector from the angle A then,

$$m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2} \quad \text{and} \quad \beta_a = \frac{2bc\cos\frac{A}{2}}{b+c}$$

Note that $m_a^2 + m_b^2 + m_c^2 = \frac{3}{4}(a^2 + b^2 + c^2)$

XI. ORTHOCENTRE AND PEDAL TRIANGLE :

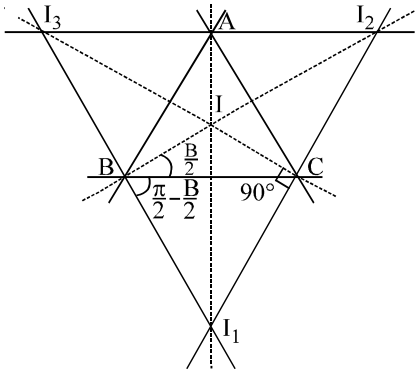
The triangle KLM which is formed by joining the feet of the altitudes is called the pedal triangle.



- the distances of the orthocentre from the angular points of the ΔABC are $2R\cos A$, $2R\cos B$ and $2R\cos C$
- the distances of P from sides are $2R\cos B\cos C$, $2R\cos C\cos A$ and $2R\cos A\cos B$
- the sides of the pedal triangle are $a\cos A\ (= R\sin 2A)$, $b\cos B\ (= R\sin 2B)$ and $c\cos C\ (= R\sin 2C)$ and its angles are $\pi - 2A$, $\pi - 2B$ and $\pi - 2C$.
- circumradii of the triangles PBC, PCA, PAB and ABC are equal .

XII EXCENTRAL TRIANGLE :

The triangle formed by joining the three excentres I_1, I_2 and I_3 of ΔABC is called the excentral or excentric triangle.



Note that :

- Incentre I of ΔABC is the orthocentre of the excentral $\Delta I_1I_2I_3$.
- ΔABC is the pedal triangle of the $\Delta I_1I_2I_3$.
- the sides of the excentral triangle are

  $4R\cos\frac{A}{2}$, $4R\cos\frac{B}{2}$ and $4R\cos\frac{C}{2}$

  and its angles are $\frac{\pi}{2} - \frac{A}{2}$, $\frac{\pi}{2} - \frac{B}{2}$ and $\frac{\pi}{2} - \frac{C}{2}$.
- $II_1 = 4R\sin\frac{A}{2}$; $II_2 = 4R\sin\frac{B}{2}$; $II_3 = 4R\sin\frac{C}{2}$.

XIII. THE DISTANCES BETWEEN THE SPECIAL POINTS :

(a) The distance between circumcentre and orthocentre is $= R\sqrt{1 - 8\cos A\cos B\cos C}$

(b) The distance between circumcentre and incentre is $= \sqrt{R^2 - 2Rr}$

(c) The distance between incentre and orthocentre is $\sqrt{2r^2 - 4R^2\cos A\cos B\cos C}$

XIV. Perimeter (P) and area (A) of a regular polygon of n sides inscribed in a circle of radius r are given by

$$P = 2nr\sin\frac{\pi}{n} \qquad \text{and} \qquad A = \frac{1}{2}nr^2\sin\frac{2\pi}{n}$$

Perimeter and area of a regular polygon of n sides circumscribed about a given circle of radius r is given by

$$P = 2nr\tan\frac{\pi}{n} \qquad \text{and} \qquad A = nr^2\tan\frac{\pi}{n}$$

EXERCISE–I

With usual notations, prove that in a triangle ABC:

Q.1 $\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3} = 0$

Q.2 $a\cot A + b\cot B + c\cot C = 2(R + r)$

Q.3 $\frac{r_1}{(s-b)(s-c)} + \frac{r_2}{(s-c)(s-a)} + \frac{r_3}{(s-a)(s-b)} = \frac{3}{r}$

Q.4 $\frac{r_1 - r}{a} + \frac{r_2 - r}{b} = \frac{c}{r_3}$

Q.5 $\frac{abc}{s}\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2} = \Delta$

Q.6 $(r_1 + r_2)\tan\frac{C}{2} = (r_3 - r)\cot\frac{C}{2} = c$

Q.7 $(r_1 - r)(r_2 - r)(r_3 - r) = 4Rr^2$

Q.8 $(r + r_1)\tan\frac{B-C}{2} + (r + r_2)\tan\frac{C-A}{2} + (r + r_3)\tan\frac{A-B}{2} = 0$

Q.9 $\frac{1}{r^2}+\frac{1}{r_1^2}+\frac{1}{r_2^2}+\frac{1}{r_3^2}=\frac{a^2+b^2+c^2}{\Delta^2}$

Q.10 $(r_3+r_1)(r_3+r_2)\sin C = 2r_3\sqrt{r_2r_3+r_3r_1+r_1r_2}$

Q.11 $\frac{1}{bc}+\frac{1}{ca}+\frac{1}{ab}=\frac{1}{2Rr}$

Q.12 $\left(\frac{1}{r}-\frac{1}{r_1}\right)\left(\frac{1}{r}-\frac{1}{r_2}\right)\left(\frac{1}{r}-\frac{1}{r_3}\right)=\frac{4R}{r^2s^2}$

Q.13 $\frac{bc-r_2r_3}{r_1}=\frac{ca-r_3r_1}{r_2}=\frac{ab-r_1r_2}{r_3}=r$

Q.14 $\left(\frac{1}{r}+\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}\right)^2=\frac{4}{r}\left(\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}\right)$

Q.15 $Rr(\sin A+\sin B+\sin C)=\Delta$

Q.16 $2R\cos A = 2R+r-r_1$

Q.17 $\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}=\frac{s^2}{\Delta}$

Q.18 $\cot A+\cot B+\cot C=\frac{a^2+b^2+c^2}{4\Delta}$

Q.19 Given a triangle ABC with sides a = 7, b = 8 and c = 5. If the value of the expression $\left(\sum\sin A\right)\left(\sum\cot\frac{A}{2}\right)$ can be expressed in the form $\frac{p}{q}$ where $p, q \in N$ and $\frac{p}{q}$ is in its lowest form find the value of $(p+q)$.

Q.20 If $r_1 = r+r_2+r_3$ then prove that the triangle is a right angled triangle.

Q.21 If two times the square of the diameter of the circumcircle of a triangle is equal to the sum of the squares of its sides then prove that the triangle is right angled.

Q.22 In acute angled triangle ABC, a semicircle with radius r_a is constructed with its base on BC and tangent to the other two sides. r_b and r_c are defined similarly. If r is the radius of the incircle of triangle ABC then prove that, $\frac{2}{r}=\frac{1}{r_a}+\frac{1}{r_b}+\frac{1}{r_c}$.

Q.23 Given a right triangle with $\angle A = 90°$. Let M be the mid-point of BC. If the inradii of the triangle ABM and ACM are r_1 and r_2 then find the range of r_1/r_2.

Q.24 If the length of the perpendiculars from the vertices of a triangle A, B, C on the opposite sides are p_1, p_2, p_3 then prove that $\frac{1}{p_1}+\frac{1}{p_2}+\frac{1}{p_3}=\frac{1}{r}=\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}$.

Q.25 Prove that in a triangle $\frac{bc}{r_1}+\frac{ca}{r_2}+\frac{ab}{r_3}=2R\left[\left(\frac{a}{b}+\frac{b}{a}\right)+\left(\frac{b}{c}+\frac{c}{b}\right)+\left(\frac{c}{a}+\frac{a}{c}\right)-3\right]$.

EXERCISE–II

Q.1 With usual notation, if in a ΔABC, $\frac{b+c}{11}=\frac{c+a}{12}=\frac{a+b}{13}$; then prove that, $\frac{\cos A}{7}=\frac{\cos B}{19}=\frac{\cos C}{25}$.

Q.2 For any triangle ABC, if B = 3C, show that $\cos C=\sqrt{\frac{b+c}{4c}}$ & $\sin\frac{A}{2}=\frac{b-c}{2c}$.

Q.3 In a triangle ABC, BD is a median. If $l(BD)=\frac{\sqrt{3}}{4}\cdot l(AB)$ and $\angle DBC=\frac{\pi}{2}$. Determine the $\angle ABC$.

Q.4 ABCD is a trapezium such that AB, DC are parallel & BC is perpendicular to them. If angle $ADB=\theta$, $BC=p$ & $CD=q$, show that $AB=\frac{(p^2+q^2)\sin\theta}{p\cos\theta+q\sin\theta}$.

Q.5 If sides a, b, c of the triangle ABC are in A.P., then prove that $\sin^2\frac{A}{2}\operatorname{cosec}2A$; $\sin^2\frac{B}{2}\operatorname{cosec}2B$; $\sin^2\frac{C}{2}\operatorname{cosec}2C$ are in H.P.

Q.6 Find the angles of a triangle in which the altitude and a median drawn from the same vertex divide the angle at that vertex into 3 equal parts.

Q.7 In a triangle ABC, if $\tan\frac{A}{2}$, $\tan\frac{B}{2}$, $\tan\frac{C}{2}$ are in AP. Show that $\cos A$, $\cos B$, $\cos C$ are in AP.

Q.8 ABCD is a rhombus. The circumradii of ΔABD and ΔACD are 12.5 and 25 respectively. Find the area of rhombus.

Q.9 In a triangle ABC if $a^2 + b^2 = 101c^2$ then find the value of $\frac{\cot C}{\cot A + \cot B}$.

Q.10 The two adjacent sides of a cyclic quadrilateral are 2 & 5 and the angle between them is $60°$. If the area of the quadrilateral is $4\sqrt{3}$, find the remaining two sides.

Q.11 If I be the in–centre of the triangle ABC and x, y, z be the circum radii of the triangles IBC, ICA & IAB, show that $4R^3 - R(x^2 + y^2 + z^2) - xyz = 0$.

Q.12 Sides a, b, c of the triangle ABC are in H.P., then prove that
$\operatorname{cosec} A(\operatorname{cosec} A + \cot A)$; $\operatorname{cosec} B(\operatorname{cosec} B + \cot B)$ & $\operatorname{cosec} C(\operatorname{cosec} C + \cot C)$ are in A.P.

Q.13 In a Δ ABC, (i) $\frac{a}{\cos A} = \frac{b}{\cos B}$ (ii) $2\sin A\cos B = \sin C$

(iii) $\tan^2\frac{A}{2} + 2\tan\frac{A}{2}\tan\frac{C}{2} - 1 = 0$, prove that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

Q.14 The sequence $a_1, a_2, a_3, \ldots\ldots$ is a geometric sequence.
The sequence $b_1, b_2, b_3, \ldots\ldots$ is a geometric sequence.

$b_1 = 1$; $b_2 = \sqrt[4]{7} - \sqrt[4]{28} + 1$; $a_1 = \sqrt[4]{28}$ and $\sum_{n=1}^{\infty}\frac{1}{a_n} = \sum_{n=1}^{\infty} b_n$

If the area of the triangle with sides lengths a_1, a_2 and a_3 can be expressed in the form of p/q where p and q are relatively prime, find $(p + q)$.

Q.15 If p_1, p_2, p_3 are the altitudes of a triangle from the vertices A, B, C & Δ denotes the area of the triangle, prove that $\frac{1}{p_1} + \frac{1}{p_2} - \frac{1}{p_3} = \frac{2ab}{(a+b+c)\Delta}\cos^2\frac{C}{2}$.

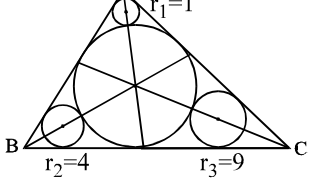
Q.16 The triangle ABC (with side lengths a, b, c as usual) satisfies
$\log a^2 = \log b^2 + \log c^2 - \log(2bc\cos A)$. What can you say about this triangle?

Q.17 With reference to a given circle, A_1 and B_1 are the areas of the inscribed and circumscribed regular polygons of n sides, A_2 and B_2 are corresponding quantities for regular polygons of 2n sides. Prove that
(1) A_2 is a geometric mean between A_1 and B_1.
(2) B_2 is a harmonic mean between A_2 and B_1.

Q.18 The sides of a triangle are consecutive integers n, $n + 1$ and $n + 2$ and the largest angle is twice the smallest angle. Find n.

Q.19 The triangle ABC is a right angled triangle, right angle at A. The ratio of the radius of the circle circumscribed to the radius of the circle escribed to the hypotenuse is, $\sqrt{2} : \left(\sqrt{3} + \sqrt{2}\right)$. Find the acute angles B & C. Also find the ratio of the two sides of the triangle other than the hypotenuse.

Q.20 ABC is a triangle. Circles with radii as shown are drawn inside the triangle each touching two sides and the incircle. Find the radius of the incircle of the ΔABC.



Q.21 Line *l* is a tangent to a unit circle S at a point P. Point A and the circle S are on the same side of *l*, and the distance from A to *l* is 3. Two tangents from point A intersect line *l* at the point B and C respectively. Find the value of (PB)(PC).

Q.22 Let ABC be an acute triangle with orthocenter H. D, E, F are the feet of the perpendiculars from A, B, and C on the opposite sides. Also R is the circumradius of the triangle ABC.
Given $(AH)(BH)(CH) = 3$ and $(AH)^2 + (BH)^2 + (CH)^2 = 7$. Find

(a) the ratio $\dfrac{\prod \cos A}{\sum \cos^2 A}$, (b) the product (HD)(HE)(HF) (c) the value of R.

EXERCISE–III

Q.1 The radii r_1, r_2, r_3 of escribed circles of a triangle ABC are in harmonic progression. If its area is 24 sq. cm and its perimeter is 24 cm, find the lengths of its sides. [REE '99, 6]

Q.2(a) In a triangle ABC, Let $\angle C = \dfrac{\pi}{2}$. If 'r' is the inradius and 'R' is the circumradius of the triangle, then $2(r + R)$ is equal to:

(A) $a + b$ (B) $b + c$ (C) $c + a$ (D) $a + b + c$

(b) In a triangle ABC, $2ac \sin\dfrac{1}{2}(A - B + C) =$

(A) $a^2 + b^2 - c^2$ (B) $c^2 + a^2 - b^2$ (C) $b^2 - c^2 - a^2$ (D) $c^2 - a^2 - b^2$

[JEE '2000 (Screening) 1 + 1]

Q.3 Let ABC be a triangle with incentre 'I' and inradius 'r'. Let D, E, F be the feet of the perpendiculars from I to the sides BC, CA & AB respectively. If r_1, r_2 & r_3 are the radii of circles inscribed in the quadrilaterals AFIE, BDIF & CEID respectively, prove that

$$\frac{r_1}{r - r_1} + \frac{r_2}{r - r_2} + \frac{r_3}{r - r_3} = \frac{r_1 r_2 r_3}{(r - r_1)(r - r_2)(r - r_3)}.$$ [JEE '2000, 7]

Q.4 If Δ is the area of a triangle with side lengths a, b, c, then show that: $\Delta \le \dfrac{1}{4}\sqrt{(a + b + c)abc}$

Also show that equality occurs in the above inequality if and only if $a = b = c$. [JEE ' 2001]

Q.5 Which of the following pieces of data does NOT uniquely determine an acute–angled triangle ABC (R being the radius of the circumcircle)?

(A) a, sinA, sinB (B) a, b, c (C) a, sinB, R (D) a, sinA, R

[JEE ' 2002 (Scr), 3]

Q.6 If I_n is the area of n sided regular polygon inscribed in a circle of unit radius and O_n be the area of the polygon circumscribing the given circle, prove that

$$I_n = \frac{O_n}{2}\left(1 + \sqrt{1 - \left(\frac{2I_n}{n}\right)^2}\right)$$ [JEE 2003, Mains, 4 out of 60]

Q.7 The ratio of the sides of a triangle ABC is $1 : \sqrt{3} : 2$. The ratio A : B : C is

(A) 3 : 5 : 2 (B) $1 : \sqrt{3} : 2$ (C) 3 : 2 : 1 (D) 1 : 2 : 3

[JEE 2004 (Screening)]

Q.8(a) In ΔABC, a, b, c are the lengths of its sides and A, B, C are the angles of triangle ABC. The correct relation is

(A) $(b - c)\sin\left(\dfrac{B - C}{2}\right) = a\cos\left(\dfrac{A}{2}\right)$ (B) $(b - c)\cos\left(\dfrac{A}{2}\right) = a\sin\left(\dfrac{B - C}{2}\right)$

(C) $(b+c)\sin\left(\frac{B+C}{2}\right)=a\cos\left(\frac{A}{2}\right)$ (D) $(b-c)\cos\left(\frac{A}{2}\right)=2a\sin\left(\frac{B+C}{2}\right)$

[JEE 2005 (Screening)]

(b) Circles with radii 3, 4 and 5 touch each other externally if P is the point of intersection of tangents to these circles at their points of contact. Find the distance of P from the points of contact.

[JEE 2005 (Mains), 2]

Q.9(a) Given an isosceles triangle, whose one angle is 120° and radius of its incircle is $\sqrt{3}$. Then the area of triangle in sq. units is

(A) $7 + 12\sqrt{3}$ (B) $12 - 7\sqrt{3}$ (C) $12 + 7\sqrt{3}$ (D) 4π

[JEE 2006, 3]

(b) Internal bisector of $\angle A$ of a triangle ABC meets side BC at D. A line drawn through D perpendicular to AD intersects the side AC at E and the side AB at F. If a, b, c represent sides of ΔABC then

(A) AE is HM of b and c (B) AD = $\frac{2bc}{b+c}\cos\frac{A}{2}$

(C) EF = $\frac{4bc}{b+c}\sin\frac{A}{2}$ (D) the triangle AEF is isosceles [JEE 2006, 5]

Q.10 Let ABC and ABC′ be two non-congruent triangles with sides AB = 4, AC = AC′ = $2\sqrt{2}$ and angle B = 30°. The absolute value of the difference between the areas of these triangles is

[JEE 2009, 5]

EXERCISE–I

Q.19 107 **Q.23** $\left(\frac{1}{2}, 2\right)$

EXERCISE–II

Q.3 120° **Q.6** $\pi/6,\ \pi/3,\ \pi/2$ **Q.8** 400 **Q.9** 50 **Q.10** 3 cms & 2 cms

Q.14 9 **Q.16** triangle is isosceles **Q.18** 4 **Q.19** B = $\frac{5\pi}{12}$; C = $\frac{\pi}{12}$; $\frac{b}{c}=2+\sqrt{3}$

Q.20 r = 11 **Q.21** 3 **Q.22** (a) $\frac{3}{14R}$, (b) $\frac{9}{8R^3}$, (c) $\frac{3}{2}$

EXERCISE–III

Q.1 6, 8, 10 cms **Q.2** **(a)** A, **(b)** B **Q.5** D **Q.7** D **Q.8** **(a)** B; **(b)** $\sqrt{5}$

Q.9 **(a)** C, **(b)** A, B, C, D **Q.10** 4

P. T. O.

Exercise - 1 (Objective Questions)

Part : (A) Only one correct option

1. In a triangle ABC, $(a + b + c)(b + c - a) = k.\ bc$, if :
(A) $k < 0$ (B) $k > 6$ (C) $0 < k < 4$ (D) $k > 4$

2. In a ΔABC, $A = \frac{2\pi}{3}$, $b - c = 3\sqrt{3}$ cm and ar $(\Delta ABC) = \frac{9\sqrt{3}}{2}$ cm². Then a is
(A) $6\sqrt{3}$ cm (B) 9 cm (C) 18 cm (D) none of these

3. If R denotes circumradius, then in ΔABC, $\frac{b^2 - c^2}{2aR}$ is equal to
(A) cos (B – C) (B) sin (B – C) (C) cos B – cos C (D) none of these

4. If the radius of the circumcircle of an isosceles triangle PQR is equal to PQ (= PR), then the angle P is
(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{2\pi}{3}$

5. In a Δ ABC, the value of $\frac{a\cos A + b\cos B + c\cos C}{a+b+c}$ is equal to:
(A) $\frac{r}{R}$ (B) $\frac{R}{2r}$ (C) $\frac{R}{r}$ (D) $\frac{2r}{R}$

6. In a right angled triangle R is equal to
(A) $\frac{s+r}{2}$ (B) $\frac{s-r}{2}$ (C) s – r (D) $\frac{s+r}{a}$

7. In a ΔABC, the inradius and three exradii are r, r_1, r_2 and r_3 respectively. In usual notations the value of $r.\ r_1.\ r_2.\ r_3$ is equal to
(A) 2Δ (B) Δ^2 (C) $\frac{abc}{4R}$ (D) none of these

8. In a triangle if $r_1 > r_2 > r_3$, then
(A) a > b > c (B) a < b < c (C) a > b and b < c (D) a < b and b > c

9. With usual notation in a Δ ABC $\left(\frac{1}{r_1}+\frac{1}{r_2}\right)\left(\frac{1}{r_2}+\frac{1}{r_3}\right)\left(\frac{1}{r_3}+\frac{1}{r_1}\right) = \frac{KR^3}{a^2b^2c^2}$,
where 'K' has the value equal to:
(A) 1 (B) 16 (C) 64 (D) 128

10. The product of the arithmetic mean of the lengths of the sides of a triangle and harmonic mean of the lengths of the altitudes of the triangle is equal to:
(A) Δ (B) 2Δ (C) 3Δ (D) 4Δ

11. In a triangle ABC, right angled at B, the inradius is:
(A) $\frac{AB+BC-AC}{2}$ (B) $\frac{AB+AC-BC}{2}$ (C) $\frac{AB+BC+AC}{2}$ (D) None

12. The distance between the middle point of BC and the foot of the perpendicular from A is :
(A) $\frac{-a^2+b^2+c^2}{2a}$ (B) $\frac{b^2-c^2}{2a}$ (C) $\frac{b^2+c^2}{\sqrt{bc}}$ (D) none of these

13. In a triangle ABC, B = 60° and C = 45°. Let D divides BC internally in the ratio 1 : 3, then, $\frac{\sin\angle BAD}{\sin\angle CAD} =$
(A) $\sqrt{\frac{2}{3}}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{\sqrt{6}}$ (D) $\frac{1}{3}$

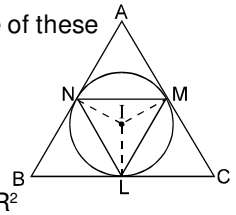
14. Let f, g, h be the lengths of the perpendiculars from the circumcentre of the Δ ABC on the sides a, b and c respectively. If $\frac{a}{f}+\frac{b}{g}+\frac{c}{h} = \lambda\ \frac{abc}{fgh}$ then the value of λ is:
(A) 1/4 (B) 1/2 (C) 1 (D) 2

15. A triangle is inscribed in a circle. The vertices of the triangle divide the circle into three arcs of length 3, 4 and 5 units. Then area of the triangle is equal to:

(A) $\frac{9\sqrt{3}(1+\sqrt{3})}{\pi^2}$ (B) $\frac{9\sqrt{3}(\sqrt{3}-1)}{\pi^2}$ (C) $\frac{9\sqrt{3}(1+\sqrt{3})}{2\pi^2}$ (D) $\frac{9\sqrt{3}(\sqrt{3}-1)}{2\pi^2}$

16. If in a triangle ABC, the line joining the circumcentre and incentre is parallel to BC, then cos B + cos C is equal to:
(A) 0 (B) 1 (C) 2 (D) none of these

17. If the incircle of the Δ ABC touches its sides respectively at L, M and N and if x, y, z be the circumradii of the triangles MIN, NIL and LIM where I is the incentre then the product xyz is equal to:



(A) $R r^2$ (B) $r R^2$ (C) $\frac{1}{2} R r^2$ (D) $\frac{1}{2} r R^2$

18. If in a ΔABC, $\frac{r}{r_1} = \frac{1}{2}$, then the value of $\tan\frac{A}{2}\left(\tan\frac{B}{2}+\tan\frac{C}{2}\right)$ is equal to :

(A) 2 (B) $\frac{1}{2}$ (C) 1 (D) None of these

19. In any ΔABC, then minimum value of $\frac{r_1 r_2 r_3}{r^3}$ is equal to
(A) 3 (B) 9 (C) 27 (D) None of these

20. In a acute angled triangle ABC, AP is the altitude. Circle drawn with AP as its diameter cuts the sides AB and AC at D and E respectively, then length DE is equal to

(A) $\frac{\Delta}{2R}$ (B) $\frac{\Delta}{3R}$ (C) $\frac{\Delta}{4R}$ (D) $\frac{\Delta}{R}$

21. AA_1, BB_1 and CC_1 are the medians of triangle ABC whose centroid is G. If the concyclic, then points A, C_1, G and B_1 are
(A) $2b^2 = a^2 + c^2$ (B) $2c^2 = a^2 + b^2$ (C) $2a^2 = b^2 + c^2$ (D) None of these

22. In a ΔABC, a, b, A are given and c_1, c_2 are two values of the third side c. The sum of the areas of two triangles with sides a, b, c_1 and a, b, c_2 is

(A) $\frac{1}{2}$ $b^2 \sin 2A$ (B) $\frac{1}{2}$ $a^2 \sin 2A$ (C) $b^2 \sin 2A$ (D) none of these

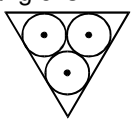
23. In a triangle ABC, let $\angle C = \frac{\pi}{2}$. If r is the inradius and R is the circumradius of the triangle, then 2(r + R) is equal to **[IIT - 2000]**
(A) a + b – c (B) b + c (C) c + a (D) a + b + c

24. Which of the following pieces of data does NOT uniquely determine an acute - angled triangle ABC (R being the radius of the circumcircle)? **[IIT - 2002]**
(A) a , sin A, sin B (B) a, b, c (C) a, sin B, R (D) a, sin A, R

25. If the angles of a triangle are in the ratio 4 : 1 : 1, then the ratio of the longest side to the perimeter is
(A) $\sqrt{3} : (2 + \sqrt{3})$ (B) 1 : 6 (C) $1 : 2 + \sqrt{3}$ (D) 2 : 3 **[IIT - 2003]**

26. The sides of a triangle are in the ratio $1 : \sqrt{3} : 2$, then the angle of the triangle are in the ratio **[IIT - 2004]**
(A) 1 : 3 : 5 (B) 2 : 3 : 4 (C) 3 : 2 : 1 (D) 1 : 2 : 3

27. In an equilateral triangle, 3 coincs of radii 1 unit each are kept so that they touche each other and also the sides of the triangle. Area of the triangle is **[IIT - 2005]**



(A) $4 + 2\sqrt{3}$ (B) $6 + 4\sqrt{3}$ (C) $12 + \frac{7\sqrt{3}}{4}$ (D) $3 + \frac{7\sqrt{3}}{4}$

28. If P is a point on C_1 and Q is a point on C_2, then $\frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2}$ equals
(A) 1/2 (B) 3/4 (C) 5/6 (D) 7/8

29. A circle C touches a line L and circle C_1 externally. If C and C_1 are on the same side of the line L, then locus of the centre of circle C is
(A) an ellipse (B) a circle (C) a parabola (D) a hyperbola

30. Let ℓ be a line through A and parallel to BD. A point S moves such that its distance from the line BD and the vertex A are equal. If the locus of S meets AC in A_1, and ℓ in A_2 and A_3, then area of $\Delta A_1 A_2 A_3$ is
(A) 0.5 (unit)^2 (B) 0.75 (unit)^2 (C) 1 (unit)^2 (D) (2/3) (unit)^2

Part : (B) May have more than one options correct

31. In a ΔABC, following relations hold good. In which case(s) the triangle is a right angled triangle?
(A) $r_2 + r_3 = r_1 - r$ (B) $a^2 + b^2 + c^2 = 8R^2$ (C) $r_1 = s$ (D) $2R = r_1 - r$

32. In a triangle ABC, with usual notations the length of the bisector of angle A is :

(A) $\dfrac{2bc\cos\frac{A}{2}}{b+c}$ (B) $\dfrac{2bc\sin\frac{A}{2}}{b+c}$ (C) $\dfrac{abc\,\mathrm{cosec}\,\frac{A}{2}}{2R(b+c)}$ (D) $\dfrac{2\Delta}{b+c}\cdot\mathrm{cosec}\,\dfrac{A}{2}$

33. AD, BE and CF are the perpendiculars from the angular points of a Δ ABC upon the opposite sides, then :

(A) $\dfrac{\text{Perimeter of } \Delta DEF}{\text{Perimeter of } \Delta ABC} = \dfrac{r}{R}$ (B) Area of $\Delta DEF = 2\Delta \cos A \cos B \cos C$

(C) Area of $\Delta AEF = \Delta \cos^2 A$ (D) Circum radius of $\Delta DEF = \dfrac{R}{2}$

34. The product of the distances of the incentre from the angular points of a Δ ABC is:

(A) $4R^2r$ (B) $4Rr^2$ (C) $\dfrac{(abc)R}{s}$ (D) $\dfrac{(abc)r}{s}$

35. In a triangle ABC, points D and E are taken on side BC such that BD = DE = EC. If angle ADE = angle AED = θ, then:
(A) $\tan\theta = 3\tan B$ (B) $3\tan\theta = \tan C$

(C) $\dfrac{6\tan\theta}{\tan^2\theta - 9} = \tan A$ (D) angle B = angle C

36. With usual notation, in a Δ ABC the value of $\Pi(r_1 - r)$ can be simplified as:

(A) $abc\,\Pi\tan\dfrac{A}{2}$ (B) $4rR^2$ (C) $\dfrac{(abc)^2}{R(a+b+c)^2}$ (D) $4Rr^2$

Exercise - 2 (Subjective Questions)

1. If in a triangle ABC, $\dfrac{\cos A + 2\cos C}{\cos A + 2\cos B} = \dfrac{\sin B}{\sin C}$, prove that the triangle ABC is either isosceles or right angled.

2. In a triangle ABC, if $a\tan A + b\tan B = (a+b)\tan\left(\dfrac{A+B}{2}\right)$, prove that triangle is isosceles.

3. If $\left(1-\dfrac{r_1}{r_2}\right)\left(1-\dfrac{r_1}{r_3}\right) = 2$ then prove that the triangle is the right triangle.

4. In a Δ ABC, $\angle C = 60°$ & $\angle A = 75°$. If D is a point on AC such that the area of the Δ BAD is $\sqrt{3}$ times the area of the Δ BCD, find the $\angle$ ABD.

5. The radii r_1, r_2, r_3 of escribed circles of a triangle ABC are in harmonic progression. If its area is 24 sq. cm and its perimeter is 24 cm, find the lengths of its sides.

6. ABC is a triangle. D is the middle point of BC. If AD is perpendicular to AC, then prove that $\cos A.\cos C = \dfrac{2(c^2 - a^2)}{3ac}$.

7. Two circles, of radii a and b, cut each other at an angle θ. Prove that the length of the common chord is $\dfrac{2ab\sin\theta}{\sqrt{a^2 + b^2 + 2ab\cos\theta}}$.

8. In the triangle ABC, lines OA, OB and OC are drawn so that the angles OAB, OBC and OCA are each equal to ω, prove that
(i) $\cot\omega = \cot A + \cot B + \cot C$
(ii) $\mathrm{cosec}^2\omega = \mathrm{cosec}^2 A + \mathrm{cosec}^2 B + \mathrm{cosec}^2 C$

9. In a plane of the given triangle ABC with sides a, b, c the points A′, B′, C′ are taken so that the Δ A′BC, Δ AB′C and Δ ABC′ are equilateral triangles with their circum radii R_a, R_b, R_c; in–radii r_a, r_b, r_c & ex–radii r_a', r_b' & r_c' respectively. Prove that;

(i) $\Pi r_a : \Pi R_a : \Pi r_a' = 1 : 8 : 27$ (ii) $r_1 r_2 r_3 = \dfrac{\left[\sum(3R_a + 6r_a + 2r_a')\right]^3}{648\sqrt{3}}\Pi\tan\dfrac{A}{2}$

10. The triangle ABC is a right angled triangle, right angle at A. The ratio of the radius of the circle

circumscribed to the radius of the circle escribed to the hypotenuse is, $\sqrt{2} : \left(\sqrt{3} + \sqrt{2}\right)$. Find the acute angles B & C. Also find the ratio of the two sides of the triangle other than the hypotenuse.

11. The triangle ABC is a right angled triangle, right angle at A. The ratio of the radius of the circle circumscribed to the radius of the circle escribed to the hypotenuse is, $\sqrt{2} : \left(\sqrt{3} + \sqrt{2}\right)$. Find the acute angles B & C. Also find the ratio of the two sides of the triangle other than the hypotenuse.

12. If the circumcentre of the Δ ABC lies on its incircle then prove that,

$$\cos A + \cos B + \cos C = \sqrt{2}$$

13. Three circles, whose radii area a, b and c, touch one another externally and the tangents at their points of contact meet in a point; prove that the distance of this point from either of their points of contacts is $\left(\frac{abc}{a+b+c}\right)^{\frac{1}{2}}$.

Answers

EXERCISE # 1

1. C **2.** B **3.** B **4.** D **5.** A **6.** B **7.** B

8. A **9.** C **10.** B **11.** A **12.** B **13.** C **14.** A

15. A **16.** B **17.** C **18.** B **19.** C **20.** D **21.** C

22. A **23.** A **24.** D **25.** A **26.** A **27.** B **28.** B

29. C **30.** C **31.** ABCD **32.** ACD **33.** ABCD

34. BD **35.** ACD **36.** ACD

EXERCISE # 2

4. $\angle ABD = 30°$ **5.** 6, 8, 10 cms

10. $B = \frac{5\pi}{12}$, $C = \frac{\pi}{12}$, $\frac{b}{c} = 2+\sqrt{3}$

11. $B = \frac{5\pi}{12}$, $C = \frac{\pi}{12}$, $\frac{b}{c} = 2+\sqrt{3}$

Some questions (Assertion–Reason type) are given below. Each question contains **Statement – 1 (Assertion)** and **Statement – 2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. So select the correct choice :

(A) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is** a correct explanation for **Statement – 1**.

(B) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is NOT** a correct explanation for **Statement – 1**.

(C) **Statement – 1** is True, **Statement – 2** is False.(D) **Statement – 1** is False, **Statement – 2** is True.

576. **Statement-1:** The value of $\tan^{-1}2+\tan^{-1}3=\frac{3\pi}{4}$

Statement-2: If $x>0, y>0, xy>1$, then $\tan^{-1}x+\tan^{-1}y=\pi+\tan^{-1}\left(\frac{x+y}{1-xy}\right)$.

577. **Statement-1:** $\frac{7\pi}{6}$ is the principal value of $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$ **Statement-2:** $\cos^{-1}(\cos x)=x$ if $x\in[0,\pi]$

578. **Statement-1:** The value of $\cot^{-1}(-1)$ is $\frac{3\pi}{4}$ **Statement-2:** $\cot^{-1}(-x)=\pi-\cot^{-1}x,\ x\in R$

579. **Statement-1:** If $x+\frac{1}{x}=2$ then the principal value of $\sin^{-1}x$ is $\frac{\pi}{2}$

Statement-2: $\sin^{-1}(\sin x)=x \quad \forall\ x\in R$.

580. **Statement-1:** If A, B, C are the angles of a triangle such that angle A is obtuse then tan B tan C > 1.

Statement-2: In any triangle $\tan A=\frac{\tan B+\tan C}{\tan B\tan C-1}$.

581. Let $f(\theta)=\sin\theta.\sin(\pi/3+\theta).\sin(\pi/3-\theta)$

Statement-1: $f(\theta)\le 1/4$

Statement-2: $f(\theta)=1/4\sin 2$

582. **Statement–1 :** Number of ordered pairs (θ, x) satisfying $2\sin\theta=e^{x}+e^{-x}$, $\theta\in[0,3\pi]$ is 2.

Statement–2 : Number of values of x for which $\sin 2x+\cos 4x=2$ is zero.

583. **Statement–1 :** The number of values of $x\in[0,4\pi]$ satisfying $|\sqrt{3}\cos x-\sin x|\ge 2$ is 2.

Statement–2 : $|\cos(x+\pi/6)|=1\Rightarrow$ number of solutions of $|\sqrt{3}\cos x-\sin x|\ge 2$ is 4

584. **Statement–1 :** Number of solutions of $\sin^{-1}(\sin x)=2\pi-x$; $x\in[3\pi/2, 5\pi/2]$ is 1

Statement–2 : $\sin^{-1}(\sin x)=x$, $x\in[-\pi/2,\pi/2]$

585. **Statement–1 :** Number of ordered pairs (x, y) satisfying $\sin^{-1}x=\pi-\sin^{-1}y$ and $\cos^{-1}x+\cos^{-1}y=0$ simultaneously is 1

Statement–2 : Ordered pairs (x, y) satisfying $\sin^{-1}x=\pi-\sin^{-1}y$ and $\cos^{-1}x+\cos^{-1}y=0$ will lie on $x^2+y^2=2$.

586. **Statement–1 :** The equation $k\cos x-3\sin x=k+1$ is solvable only if k belongs to the interval $(-\infty, 4]$

Statement–2 : $-\sqrt{a^2+b^2}\le a\sin x\pm b\cos x\le\sqrt{a^2+b^2}$.

587. **Statement–1 :** The equation $2\sec^2x-3\sec x+1=0$ has no solution in the interval $(0, 2\pi)$

Statement–2 : $\sec x\le -1$ as $\sec x\ge 1$.

588. **Statement–1 :** The number of solution of the equation $|\sin x|=|x|$ is only one.

Statement–2 : The number of point of intersection of the two curves $y=|\sin x|$ and $y=|x|$ is three.

589. **Statement–1 :** The equation $\sin x=1$ has infinite number of solution.

Statement–2 : The domain of $f(x)=\sin x$ is $(-\infty,\infty)$.

590. **Statement–1 :** There is no solution of the equation $|\sin x|+|\cos x|=\tan^2x+\cot^2x$.

Statement–2 : $0\le|\sin x|+|\cos x|\le\sqrt{2}$ and $\tan^2x+\cot^2x\ge 2$.

591. **Statement–1 :** The equation $\sin^2x+\cos^2y=2\sec^2z$ is only solvable when $\sin x=1$, $\cos y=1$ an $\sec z=1$ where $x, y, z\in R$.

Statement–2 : Maximum value of sin x and cos y is 1 and minimum value of sec z is 1.

592. **Statement–1 :** If $\cot^{-1}x<n$, $n\in R$ then $x<\cot(n)$

Statement–2 : $\cot^{-1}(x)$ is an decreasing function.

593. **Statement–1 :**If $\sin\theta=a$ for exactly one value of $\theta\in\left[0,\frac{7\pi}{3}\right]$, then a can take infinite value in the interval $[-1, 1]$.

Statement–2 : $-1\le\sin\theta\le 1$.

594. **Statement–1 :** $\tan 5°$ is an irrational number. **Statement–2 :** $\tan 15°$ is an irrational number.

595. Let θ be an acute angle

Statement–1 : $\sin^6\theta+\cos^6\theta\le 1$. **Statement–2 :** $\sin\theta+\cos\theta\le 1$

596. **S–1 :** $\sin\frac{\pi}{18}$ is a root of $8x^3-6x+1=0$. **S–2 :** For any $\theta\in R$, $\sin 3\theta=3\sin\theta-4\sin^3\theta$.

597. Let f be any one of the six trigonometric functions. Let A, B ∈ R satisfying f(2A) = f(2B).
Statement–1 : A = nπ + B, for some n ∈ I. **Statement–2** : 2π is one of the period of f.

598. Let x ∈ [-1, 1]

Statement–1 : $2\sin^{-1} x = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$. **Statement–2** : $-1 \le 2x\sqrt{1-x^2} \le 1$.

599. Let $f(x) = \cos^{-1} x$
Statement–1 : f is a decreasing function.
Statement–2 : f(– x) = π – f(x).

600. **Statement–1** : The total number of 2 real roots of the equation $x^2 \tan x = 1$ lies between the interval (0, 2π).
Statement–2 : The total number of solution of equation $|\cos x - \sin x| = 2\cos x$ in [0, 2π] is 3.

601. **Statement–1** : The number of real solutions of equation $\sin e^x \cos e^x = 2^{x-2} + 2^{-x-2}$ is 0.
Statement–2 : The number of solutions of the eqution $1 + \sin x \sin^2 \frac{x}{2} = 0$ n [- π, π] is 0.

602. **Statement–1** : Equation $\tan^{-1}\left(x + \frac{2}{x}\right) - \tan^{-1}\left(\frac{4}{x}\right) - \tan^{-1}\left(x - \frac{2}{x}\right) = 0$ has 3 real roots.
Statement–2 : the number of real solution of $\sqrt{1+\cos 2x} = \sqrt{2}\sin^{-1}(\sin x)$; $x \in [-\pi, \pi]$ is 2.

603. **Statement–1** : If
$$\tan^{-1}\frac{1}{1+2} + \tan^{-1}\frac{1}{1+2.3} + \tan^{-1}\frac{1}{1+3.4} \ldots + \tan^{-1}\frac{1}{1+n(n+1)} = \tan^{-1}\theta, \text{ then } \theta = \frac{n}{n+1}.$$
Statement–2 : The sum of series $\cos^{-1} 2 + \cot^{-1} 8 + \cot^{-1} 18 + \ldots$ is $\frac{\pi}{4}$.

604. **Statement-1:** If $\tan\theta + \sec\theta = \sqrt{3}$, $0 < \theta < \pi$, then $\theta = \pi/6$
Statement-2: General solution of $\cos\theta = \cos\alpha$ is $\theta = \alpha$, if $0 < \alpha < \pi/2$

605. **Statement-1:** If $x < 0$, $\tan^{-1}x + \tan^{-1}\frac{1}{x} = \pi/2$
Statement-2: $\tan^{-1}x + \cos^{-1}x = \pi/2, \forall x \in R$

606. **Statement-1:** $\sin^{-1}(\sin 10) = 10$
Statement-2: For principal value $\sin^{-1}(\sin x) = x$

607. **Statement-1:** $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7} = -\frac{1}{8}$
Statement-2: $\cos\theta \cos 2\theta \cos 2^3\theta \ldots \cos 2^{n-1}\theta = -\frac{1}{2^n}$ if $\theta = \frac{\pi}{2^n - 1}, n \in N,\ n \ge 2$.

TRI

608. **Statement-1:** sin3 < sin1 < sin2 is true
Statement-2: sinx is positive in first and second quadrants.

609. **Statement-1:** The equation $2\sin^2 x - (P + 3)\sin x + (2P - 2) = 0$ possesses a real solution if $P \in [-1, 3]$
Statement-2 : $-1 \le \sin x \le 1$

610. **Statement-1:** The maximum value of $3\sin\theta + 4\cos\left(\theta + \frac{\pi}{4}\right)$ is 5 here $\theta \in R$.
Statement-2:: $-\sqrt{a^2+b^2} \le a\sin\theta + b\cos\theta \le \sqrt{a^2+b^2}$

611. **Statement-1:** If $A + B + C = \pi$, $\cos A + \cos B + \cos C \le 3/2$

Statement-2:: If $A + B + C = \pi$, $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \le \frac{1}{8}$

612. **Statement-1:** The maximum & minimum values of the function $f(x) = \frac{1}{6\sin x - 8\cos x + 5}$ does not exists.

Statement-2: The given function is an unbounded function.

613. **Statement-1:** If $x < 0$ $\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right) = \pi/2$

Statement-2: $\tan^{-1}x + \cot^{-1}x = \pi/2 \ \forall\ x \in R$.

614. **Statement-1:** In any triangle square of the length of the bisector AD is $bc\left(1 - \frac{a^2}{(b+c)^2}\right)$

Statement-2: In any triangle length of bisector $AD = \frac{bc}{b+c}\cos\frac{A}{2}$

615. **Statement-1:** If in a triangle ABC, $C = 2a\cos B$, then the triangle is isosceles.
Statement-2: Triangle ABC, the two sides are equal i.e. $a = b$.

616. **Statement-1:** If the radius of the circumcircle of an isosceles triangle pqR is equal to pq = PR then the angle $p = 2\pi/3$.
Statement-2: OPQ and oPR will be equilateral i.e., $\angle OPq = 60°$, $\angle OPR = 60°$

617. **Statement-1:** The minimum value of the expression $\sin\alpha + \sin\beta + \sin\gamma$ is negative, where α, β, γ are real numbers such that $\alpha + \beta + \gamma = \pi$.

Statement-2: If α, β, γ are angle of a triangle then $\sin\alpha + \sin\beta + \sin\gamma = 4\cos\frac{\alpha}{2}\cos\frac{\beta}{2}\cos\frac{\gamma}{2}$.

618. **Statement-1:** If in a triangle $\sin^2A + \sin^2B + \sin^2C = 2$ then one of the angles must be 90°.
Statement-2: In any triangle $\sin^2A + \sin^2B + \sin^2C = 2 + 2\cos A \cos B \cos C$.

619. **Statement-1:** If in a ΔABC $a \to 2c$ and $b \to 3c$ then $\cos B$ must tend to –1.

Statement-1: In a ΔABC $\cos B = \frac{c^2 + a^2 - b^2}{2ac}$.

620. **Statement-1:** $\cos(45 - A)\cos(45 - B) - \sin(45 - A) \times \sin(45 - B) = \sin(A + B)$.
Statement-2: $\cos(90 - \theta) = -\sin\theta$.

621. **Statement-1:** The maximum and minimum values of $7\cos\theta + 24\sin\theta$ are 25 and – 25 respectively.

Statement-2: $\sqrt{a^2 + b^2} \le a\cos\theta \pm b\sin\theta \le \sqrt{a^2 + b^2}$ for all θ.

622. **Statement-1:** If $\sin^{-1}x + \sin^{-1}(1 - x) = \sin^{-1}\sqrt{1 - x^2}$ then $x = 0, \frac{1}{2}$

Statement-2: $\sin^{-1}\sin x = x \ \ \forall x \in R$

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623. **Statement-1:** The numbers $\sin 18°$ and $-\sin 54°$ are roots of same quadratic equation with integer coefficients.
Statement-2: If $x = 18°$, then $5x = 90°$, if $y = -54°$, then $5y = -270°$.

Inverse Trigonometry

624. **Statement-1:** The number of solution of the equation $\cos\left(\pi\sqrt{x - 4}\right)\cos\left(\pi\sqrt{x}\right) = 1$ is one.

Statement-2: $\cos x = \cos\alpha \Rightarrow x = 2n\pi \pm \alpha\ n \in I$

Inverse Trigonometric Function

627. **Statement-1:** The range of $\sin^{-1}x + \cos^{-1}x + \tan^{-1}x$ is $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

Statement-2: $\sin^{-1}x + \cos^{-1}x = \pi/2$ for every $x \in R$.

628. **Statement-1:** $\sin^{-1}(\sin 10) = 10$

Statement-2: $\sin^{-1}(\sin x) = x$ for $-\pi/2 \le x \le \pi/2$

629. **Statement-1:** If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$, the value of $\cos^{-1}x + \cos^{-1}y$ is $\pi/3$.

Statement-2: $\sin^{-1}x + \cos^{-1}x = \pi/2 \ \forall x \in [-1, 1]$.

630. **Statement-1:** $7\pi/6$ is the principal value of $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$

Statement-2: $\cos^{-1}(\cos x) = x$, if $x \in [0, \pi]$.

631. **Statement-1:** $\sqrt{3}\cos\theta + \sin\theta = 5$ has no solution.

Statement-2: $a\cos\theta + b\sin\theta = c$ has solution if $|c| \le \sqrt{a^2 + b^2}$

632. **Statement-1:** The equation $\sin^4x + \cos^4x + \sin 2x + a = 0$ is valid if $a \in \left[-\frac{3}{2}, \frac{1}{2}\right]$

Statement-2: If discriminant of a quadratic equation is –ve. Then its roots are real.

633. **Statement-1:** In a ΔABC $\cos A\cos B + \sin A\sin B\sin C = 1$ then ΔABC must be isosceles as well as right angled triangle.

Statement-2: In a ΔABC if $A = \frac{\pi}{4}$ $\tan A \tan B = k$. then k must satisfy $k^2 - 6k + 1 \ge 0$

634. **Statement-1:** If r_1, r_2, r_3 in a ΔABC are in H.P. then sides a, b, c are in A.P.

Statement-2:: $r_1 = \frac{\Delta}{s-a}$, $r_2 = \frac{\Delta}{s-b}$, $r_3 = \frac{\Delta}{s-c}$.

Answer Key

576. A	577. D	578. A	579. C	580. D	581.C	582.B
583. D	584. B	585. B	586. A	587. B	588. C	589.A
590. C	591. A	592. D	593. D	594. A	595. C	596. A
597. A	598. D	599. B	600. C	601. B	602. D	603. D
604. A	605. D	606. D	607. D	608. B	609. A	610. D
611. B	612. A	613. D	614. C	615. A	616. A	617. B
618. A	619. A	620. C	621. A	622. C	623. A	624. B
627. C	628. D	629. A	630. D	631. A		

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 4 XI M 4. Functions

Index:

1. Key Concepts

2. Exercise I to V

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Functions

A. Definition :

Function is a special case of relation, from a non empty set A to a non empty set B, that associates each member of A to a unique member of B. Symbolically, we write f: A $\to$ B. We read it as "f is a function from A to B".

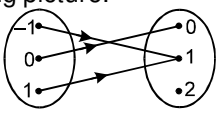
Set 'A' is called **domain** of f and set 'B' is called **co-domain** of f.

For example, let A $\equiv$ {–1, 0, 1} and B $\equiv$ {0, 1, 2}. Then A $\times$ B $\equiv$ {(–1, 0), (–1, 1), (–1, 2), (0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2)}

Now, " f : A $\to$ B defined by $f(x) = x^2$ " is the function such that

f $\equiv$ {(–1, 1), (0, 0), (1, 1)}

f can also be show diagramatically by following picture.



Every function say f : A $\to$ B satisfies the following conditions:

(a) $f \subseteq A \times B$, (b) $\forall\ a \in A \Rightarrow (a, f(a)) \in f$ and (c) $(a, b) \in f\ \&\ (a, c) \in f \Rightarrow b = c$

Illustration # 1: (i) Which of the following correspondences can be called a function ?

(A) $f(x) = x^3$; $\{-1, 0, 1\} \to \{0, 1, 2, 3\}$

(B) $f(x) = \pm\sqrt{x}$; $\{0, 1, 4\} \to \{-2, -1, 0, 1, 2\}$

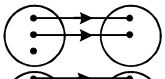
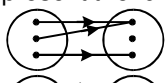
(C) $f(x) = \sqrt{x}$; $\{0, 1, 4\} \to \{-2, -1, 0, 1, 2\}$

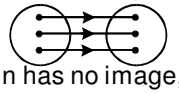
(D) $f(x) = -\sqrt{x}$; $\{0, 1, 4\} \to \{-2, -1, 0, 1, 2\}$

Solution: f(x) in (C) & (D) are functions as definition of function is satisfied. while in case of (A) the given relation is not a function, as f(–1) $\notin$ codomain. Hence definition of function is not satisfied.

While in case of (B), the given relation is not a function, as f(1) = $\pm$ 1 and f(4) = $\pm$ 2 i.e. element 1 as well as 4 in domain is related with two elements of codomain. Hence definition of function is not satisfied.

(ii) Which of the following pictorial diagrams represent the function

(A)  (B) 

(C)  (D) 

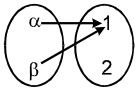
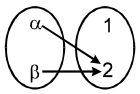
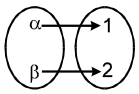
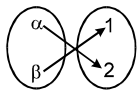
Solution: B & D. In (A) one element of domain has no image, while in (C) one element of domain has two images in codomain

Assignment: 1. Let g(x) be a function defined on [–1, 1]. If the area of the equilateral triangle with two of its vertices at (0,0) & (x,g(x)) is $\sqrt{3}/4$ sq. units, then the function g(x) may be.

(A) $g(x) = \pm\sqrt{(1-x^2)}$ (B*) $g(x) = \sqrt{(1-x^2)}$ (C*) $g(x) = -\sqrt{(1-x^2)}$ (D) $g(x) = \sqrt{(1+x^2)}$

2. Represent all possible functions defined from {α, β} to {1, 2}

Answer (1) B

(2) (i)  (ii)  (iii)  (iv) 

B. Domain, Co-domain & Range of a Function :

Let f: A $\to$ B, then the set A is known as the domain of f & the set B is known as co–domain of f. If a member 'a' of A is associated to the member 'b' of B, then 'b' is called **the f-image** of 'a' and we write b = f (a). Further 'a' is called **a pre-image** of 'b'. The set {f(a): $\forall$ a $\in$ A} is called **the range** of f and is denoted by f(A). Clearly f(A) $\subseteq$ B.

Sometimes if only definition of f (x) is given (domain and codomain are not mentioned), then domain is set of those values of 'x' for which f (x) is defined, while codomain is considered to be ($-\infty$, ∞)

A function whose domain and range both are sets of real numbers is called **a real function**. Conventionally the word **"FUNCTION"** is used only as the meaning of real function.

Illustration # 2 : Find the domain of following functions :

(i) $f(x) = \sqrt{x^2 - 5}$ (ii) $\sin^{-1}(2x - 1)$

Solution :(i) $f(x) = \sqrt{x^2 - 5}$ is real iff $x^2 - 5 \ge 0$ $\Rightarrow$ $|x| \ge \sqrt{5}$ $\Rightarrow$ $x \le -\sqrt{5}$ or $x \ge \sqrt{5}$

$\therefore$ the domain of f is $(-\infty, -\sqrt{5}\,] \cup [\sqrt{5}, \infty)$

(ii) $-1 \le 2x - 1 \le +1$ $\therefore$ domain is $x \in [0, 1]$

Algebraic Operations on Functions :

If f & g are real valued functions of x with domain set A and B respectively, then both f & g are defined in A $\cap$ B. Now we define f + g, f – g, (f · g) & (f /g) as follows:

(i) $(f \pm g)(x) = f(x) \pm g(x)$ ┐
(ii) $(f.g)(x) = f(x).\ g(x)$ ┘ domain in each case is A $\cap$ B

(iii) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ domain is $\{x \mid x \in A \cap B \text{ such that } g(x) \ne 0\}$.

Note : ☞ For domain of $\phi(x) = \{f(x)\}^{g(x)}$, conventionally, the conditions are f(x) > 0 and g(x) must be defined.

☞ For domain of $\phi(x) = {}^{f(x)}C_{g(x)}$ or $\phi(x) = {}^{f(x)}P_{g(x)}$ conditions of domain are $f(x) \ge g(x)$ and $f(x) \in N$ and $g(x) \in W$

Illustration # 3: Find the domain of following functions :

(i) $f(x) = \sqrt{\sin x} - \sqrt{16 - x^2}$ (ii) $f(x) = \frac{3}{\sqrt{4 - x^2}} \log(x^3 - x)$ (iii) $f(x) = x^{\cos^{-1} x}$

Solution: (i) $\sqrt{\sin x}$ is real iff $\sin x \geq 0 \Leftrightarrow x \in [2n\pi, 2n\pi + \pi]$, $n \in I$.

$\sqrt{16 - x^2}$ is real iff $16 - x^2 \geq 0 \Leftrightarrow -4 \leq x \leq 4$.

Thus the domain of the given function is $\{x : x \in [2n\pi, 2n\pi + \pi], n \in I\} \cap [-4, 4] = [-4, -\pi] \cup [0, \pi]$.

(ii) Domain of $\sqrt{4 - x^2}$ is $[-2, 2]$ but $\sqrt{4 - x^2} = 0$ for $x = \pm 2$ $\Rightarrow$ $x \in (-2, 2)$

$\log(x^3 - x)$ is defined for $x^3 - x > 0$ i.e. $x(x - 1)(x + 1) > 0$.

$\therefore$ domain of $\log(x^3 - x)$ is $(-1, 0) \cup (1, \infty)$.

Hence the domain of the given function is $\{(-1, 0) \cup (1, \infty)\} \cap (-2, 2) = (-1, 0) \cup (1, 2)$.

(iii) $x > 0$ and $-1 \leq x \leq 1$ $\therefore$ domain is $(0, 1]$

Assignment :

3. Find the domain of following functions.

(i) $f(x) = \frac{1}{\log(2 - x)} + \sqrt{x + 1}$ (ii) $f(x) = \sqrt{1 - x} - \sin^{-1} \frac{2x - 1}{3}$

Ans. (i) $[-1, 1) \cup (1, 2)$ (ii) $[-1, 1]$

Methods of determining range :

(i) **Representing x in terms of y**

Definition of the function is usually represented as y (i.e. f(x) which is dependent variable) in terms of an expression of x (which is independent variable). To find range rewrite given definition so as to represent x in terms of an expression of y and thus obtain range (possible values of y).

If $y = f(x) \Leftrightarrow x = g(y)$, then domain of g(y) represents possible values of y, i.e. range of f(x).

Illustration # 4: Find the range of $f(x) = \frac{x^2 + x + 1}{x^2 + x - 1}$

Solution $f(x) = \frac{x^2 + x + 1}{x^2 + x - 1}$ $\{x^2 + x + 1$ and $x^2 + x - 1$ have no common factor$\}$

$y = \frac{x^2 + x + 1}{x^2 + x - 1} \Rightarrow yx^2 + yx - y = x^2 + x + 1$

$\Rightarrow (y - 1)x^2 + (y - 1)x - y - 1 = 0$

If $y = 1$, then the above equation reduces to $-2 = 0$. Which is not true.

Further if $y \neq 1$, then $(y - 1)x^2 + (y - 1)x - y - 1 = 0$ is a quadratic and has real roots if $(y - 1)^2 - 4(y - 1)(-y - 1) \geq 0$ i.e. if $y \leq -3/5$ or $y \geq 1$ but $y \neq 1$

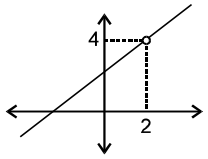
Thus the range is $(-\infty, -3/5] \cup (1, \infty)$

(ii) **Graphical Method :** Values covered on y-axis by the graph of function is range

Illustration # 5: Find the range of $f(x) = \frac{x^2 - 4}{x - 2}$

Solution $f(x) = \frac{x^2 - 4}{x - 2} = x + 2; x \neq 2$

$\therefore$ graph of f(x) would be



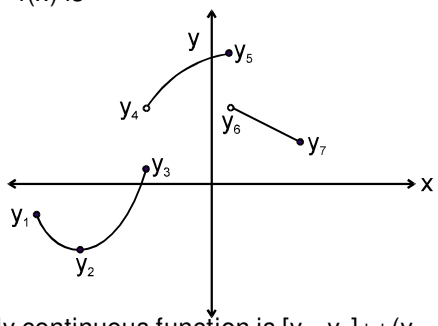
Thus the range of $f(x)$ is $R - \{4\}$

(iii) **Using Monotonocity/Maxima-Minima**

(a) **Continuous function:** If $y = f(x)$ is continuous in its domain then range of f(x) is $y \in [\min f(x), \max. f(x)]$

(b) **Sectionally continuous function:** In case of sectionally continuous functions, range will be union of $[\min f(x), \max. f(x)]$ over all those intervals where f(x) is continuous, as shown by following example.

Let graph of function $y = f(x)$ is



Then range of above sectionally continuous function is $[y_2, y_3] \cup (y_4, y_5] \cup (y_6, y_7]$

Note : ☞ In case of monotonic functions minimum and maximum values lie at end points of interval.

Illustration # 6: Find the range of following functions :

(i) $y = \ell n(2x - x^2)$ (ii) $y = \sec^{-1}(x^2 + 3x + 1)$

Solution : (i) **Step – 1**

Using maxima-minima, we have $2x - x^2 \in (-\infty, 1]$

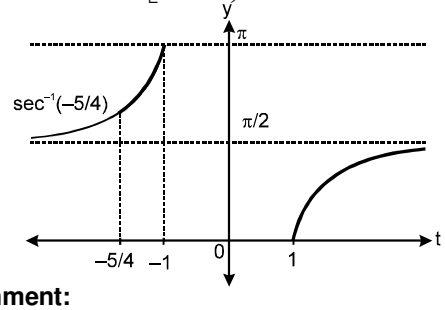
Step – 2 For log to be defined accepted values are $2x - x^2 \in (0, 1]$ $\{$i.e. domain $(0, 1]\}$

Now, using monotonocity $\ell n(2x - x^2) \in (-\infty, 0]$

$\therefore$ range is $(-\infty, 0]$ **Ans.**

(ii) $y = \sec^{-1}(x^2 + 3x + 1)$ Let $t = x^2 + 3x + 1$ for $x \in R$

then $t \in \left[-\frac{5}{4}, \infty\right)$ but $y = \sec^{-1}(t)$ $\Rightarrow$ $t \in \left[-\frac{5}{4}, -1\right] \cup [1, \infty)$



from graph range is $y \in \left[0, \frac{\pi}{2}\right) \cup \left[\sec^{-1}\left(-\frac{5}{4}\right), \pi\right]$

Assignment:

4. Find domain and range of following functions.

(i) $y = x^3$ (ii) $y = \frac{x^2 - 2x + 5}{x^2 + 2x + 5}$

Answer (i) domain R; range R (ii) domain R ; range $\left[\frac{3-\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2}\right]$

(iii) $y = \frac{1}{\sqrt{x^2 - x}}$ **Answer** domain R – [0, 1] ; range $(0, \infty)$

(iv) $y = \cot^{-1}(2x - x^2)$ **Answer** domain R ; range $\left[\frac{\pi}{4}, \pi\right)$

(v) $y = \ell n \sin^{-1}\left(x^2 + x + \frac{3}{4}\right)$ **Answer** domain $x \in \left[\frac{-2-\sqrt{5}}{4}, \frac{-2+\sqrt{5}}{4}\right]$; range $\left[\ell n\frac{\pi}{6}, \ell n\frac{\pi}{2}\right]$

C. Classification of Functions :

Functions can be classified as :

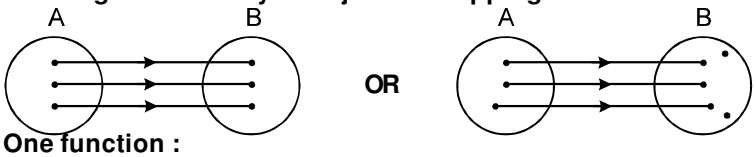
(i) One – One Function (Injective Mapping) and Many – One Function:

One – One Function :

A function $f : A \to B$ is said to be a one-one function or injective mapping if different elements of A have different f images in B.

Thus for $x_1, x_2 \in A$ & $f(x_1), f(x_2) \in B$, $f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$ or $x_1 \neq x_2 \Leftrightarrow f(x_1) \neq f(x_2)$.

Diagrammatically an injective mapping can be shown as

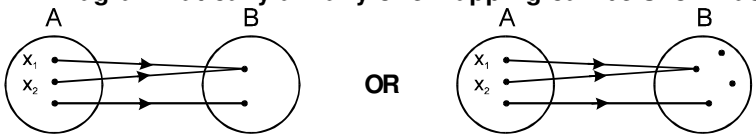


Many – One function :

A function $f : A \to B$ is said to be a many one function if two or more elements of A have the same f image in B.

Thus $f : A \to B$ is many one iff there exists atleast two elements $x_1, x_2 \in A$, such that $f(x_1) = f(x_2)$ but $x_1 \neq x_2$.

Diagrammatically a many one mapping can be shown as



Note : ☞ If a function is one–one, it cannot be many–one and vice versa.

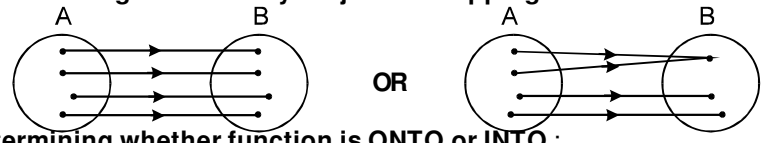
Methods of determining whether function is ONE-ONE or MANY-ONE :

(a) If $x_1, x_2 \in A$ & $f(x_1), f(x_2) \in B$, $f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$ or $x_1 \neq x_2 \Leftrightarrow f(x_1) \neq f(x_2)$, then function is ONE-ONE otherwise MANY-ONE. (b) If there exists a straight line parallel to x-axis, which cuts the graph of the function atleast at two points, then the function is MANY-ONE, otherwise ONE-ONE. (c) If either $f'(x) \geq 0$, $\forall$ x $\in$ complete domain or $f'(x) \leq 0$ $\forall$ x $\in$ complete domain, where equality can hold at discrete point(s) only, then function is ONE-ONE, otherwise MANY-ONE.

(ii) Onto function (Surjective mapping) and Into function : Onto function :

If the function $f : A \to B$ is such that each element in B (co–domain) must have atleast one pre–image in A, then we say that f is a function of A 'onto' B. Thus $f : A \to B$ is surjective iff $\forall$ b $\in$ B, there exists some a $\in$ A such that f (a) = b.

Diagrammatically surjective mapping can be shown as



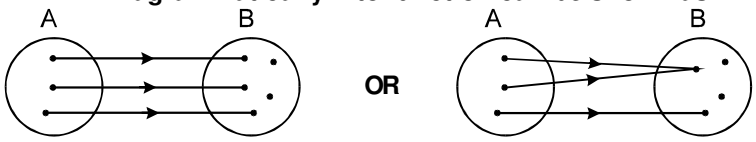
Method of determining whether function is ONTO or INTO :

Find the range of given function. If range ≡ co–domain, then f(x) is onto, otherwise into

Into function :

If $f : A \to B$ is such that there exists atleast one element in co–domain which is not the image of any element in domain, then f(x) is into.

Diagrammatically into function can be shown as



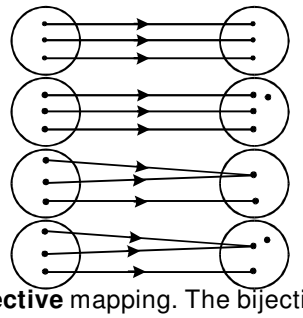
Note : ☞ If a function is onto, it cannot be into and vice versa.
Thus a function can be one of these four types:

(a) one–one onto (injective & surjective)

(b) one–one into (injective but not surjective)

(c) many–one onto (surjective but not injective)

(d) many–one into (neither surjective nor injective)



Note : ☞ If f is both injective & surjective, then it is called a **bijective** mapping. The bijective functions are also named as invertible, non singular or biuniform functions.

☞ If a set A contains 'n' distinct elements then the number of different functions defined from $A \to A$ is n^n and out of which n! are one one.

Illustration # 7 (i) Find whether $f(x) = x + \cos x$ is one-one.

Solution The domain of f(x) is R. $f'(x) = 1 - \sin x$.

$\therefore$ $f'(x) \geq 0 \ \forall \ x \in$ complete domain and equality holds at discrete points only

$\therefore$ f(x) is strictly increasing on R. Hence f(x) is one-one.

(ii) Identify whether the function $f(x) = -x^3 + 3x^2 - 2x + 4$; $R \to R$ is ONTO or INTO

Solution As codomain ≡ range, therefore given function is ONTO

(iii) $f(x) = x^2 - 2x + 3$; $[0, 3] \to A$. Find whether f(x) is injective or not. Also find the set A, if f(x) is surjective.

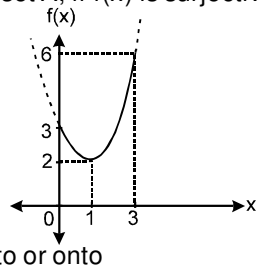
Solution $f'(x) = 2(x - 1)$; $0 \leq x \leq 3$

$\therefore$ $f'(x) = \begin{cases} -ve & ; \ 0 \leq x < 1 \\ +ve & ; \ 1 < x < 3 \end{cases}$

$\therefore$ f(x) is not monotonic. Hence it is not injective.

For f(x) to be surjective, A should be equal to its range. By graph range is [2, 6]

$\therefore$ A ≡ [2, 6]



Assignment:

5. For each of the following functions find whether it is one-one or many-one and also into or onto

(i) $f(x) = 2 \tan x$; $(\pi/2, 3\pi/2) \to R$

Answer one-one onto

(ii) $f(x) = \dfrac{1}{1+x^2}$; $(-\infty, 0) \to R$

Answer one-one into

(iii) $f(x) = x^2 + \ell n \, x$

Answer one-one onto

D. Various Types of Functions :

(i) Polynomial Function : If a function f is defined by $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + ... + a_{n-1} x + a_n$ where n is a **non negative integer** and $a_0, a_1, a_2, ..., a_n$ are real numbers and $a_0 \neq 0$, then f is called a polynomial function of degree n.

Note : ☞ There are two polynomial functions, satisfying the relation; $f(x).f(1/x) = f(x) + f(1/x)$, which are $f(x) = 1 \pm x^n$

(ii) Algebraic Function : y is an algebraic function of x, if it is a function that satisfies an algebraic equation of the form, $P_0(x) y^n + P_1(x) y^{n-1} + ... + P_{n-1}(x) y + P_n(x) = 0$ where n is a positive integer and $P_0(x)$, $P_1(x)$..... are polynomials in x. e.g. $y = |x|$ is an algebraic function, since it satisfies the equation $y^2 - x^2 = 0$.

Note : ☞ All polynomial functions are algebraic but not the converse.

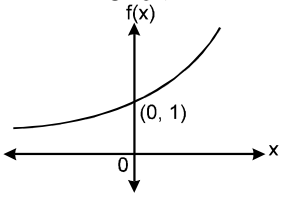
☞ A function that is not algebraic is called **Transcendental Function**.

(iii) Fractional / Rational Function : A rational function is a function of the form, $y = f(x) = \dfrac{g(x)}{h(x)}$, where g (x) & h (x) are polynomials and $h(x) \neq 0$.

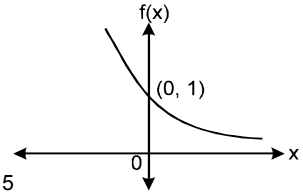
(iv) Exponential Function :

A function $f(x) = a^x = e^{x \ln a}$ $(a > 0, a \neq 1, x \in R)$ is called an exponential function. Graph of exponential function can be as follows :

Case - I
For a > 1

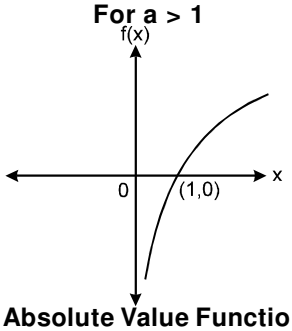


Case - II
For 0 < a < 1

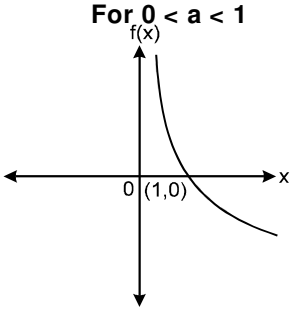


(v) Logarithmic Function : $f(x) = \log_a x$ is called logarithmic function where $a > 0$ and $a \neq 1$ and $x > 0$. Its graph can be as follows

Case- I
For a > 1

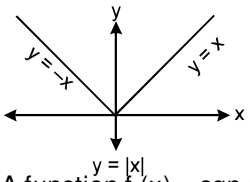


Case- II
For 0 < a < 1



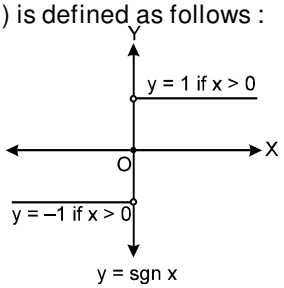
(vi) Absolute Value Function / Modulus Function :

The symbol of modulus function is $f(x) = |x|$ and is defined as: $y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$.



(vi) Signum Function : A function $f(x) = \text{sgn}(x)$ is defined as follows :

$$f(x) = \text{sgn}(x) = \begin{cases} 1 & \text{for } x > 0 \\ 0 & \text{for } x = 0 \\ -1 & \text{for } x < 0 \end{cases}$$

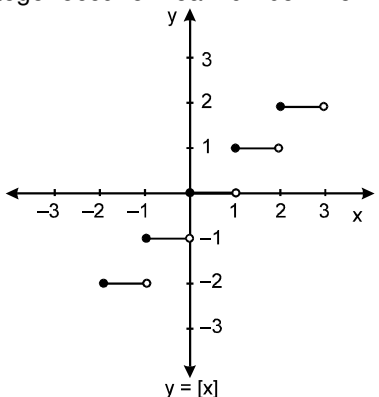


It is also written as $\text{sgn } x = \begin{cases} \frac{|x|}{x}; & x \neq 0 \\ 0; & x = 0 \end{cases}$

Note : $\text{sgn } f(x) = \begin{cases} \frac{|f(x)|}{f(x)}; & f(x) \neq 0 \\ 0; & f(x) = 0 \end{cases}$

(vii) Greatest Integer Function or Step Up Function :
The function $y = f(x) = [x]$ is called the greatest integer function where [x] equals to the greatest integer less than or equal to x. For example :
for $-1 \leq x < 0$; $[x] = -1$; for $0 \leq x < 1$; $[x] = 0$
for $1 \leq x < 2$; $[x] = 1$; for $2 \leq x < 3$; $[x] = 2$ and so on.

Alternate Definition :
The greatest integer occur on real number line while moving L.H.S. of x (starting from x) is [x]

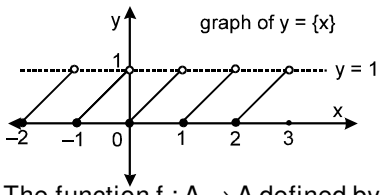


Properties of greatest integer function :

(a) $x - 1 < [x] \leq x$ (b) $[x \pm m] = [x] \pm m$ iff m is an integer.

(c) $[x] + [y] \leq [x + y] \leq [x] + [y] + 1$ (d) $[x] + [-x] = \begin{cases} 0; & \text{if } x \text{ is an integer} \\ -1 & \text{otherwise} \end{cases}$

(viii) Fractional Part Function: It is defined as, $y = \{x\} = x - [x]$.
e.g. the fractional part of the number 2.1 is $2.1 - 2 = 0.1$ and the fractional part of -3.7 is 0.3. The period of this function is 1 and graph of this function is as shown.



(ix) Identity function : The function $f : A \to A$ defined by, $f(x) = x \ \forall \ x \in A$ is called the identity function on A and is denoted by I_A. It is easy to observe that identity function is a bijection.

(x) Constant function : A function $f : A \to B$ is said to be a constant function, if every element of A has the same f image in B. Thus $f : A \to B$; $f(x) = c$, $\forall \ x \in A$, $c \in B$ is a constant function.

Illustration # 8 (i) Let {x} & [x] denote the fractional and integral part of a real number x respectively. Solve $4\{x\} = x + [x]$

Solution As $x = [x] + \{x\}$

$\therefore$ Given equation $\Rightarrow$ $4\{x\} = [x] + \{x\} + [x]$ $\Rightarrow$ $\{x\} = \dfrac{2[x]}{3}$

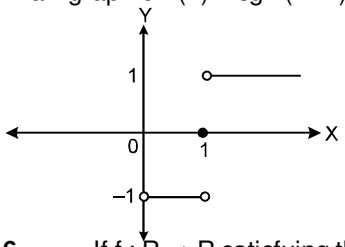
As [x] is always an integer and $\{x\} \in [0, 1)$, possible values are

[x]	{x}	x = [x] + {x}
0	0	0
1	$\frac{2}{3}$	$\frac{5}{3}$

$\therefore$ There are two solution of given equation $x = 0$ and $x = \dfrac{5}{3}$

(ii) Draw graph of $f(x) = \text{sgn}(\ell n\, x)$

Solution



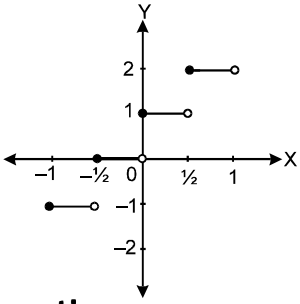
Assignment: 6. If $f : R \to R$ satisfying the conditions $f(0) = 1$, $f(1) = 2$ and $f(x + 2) = 2f(x) + f(x + 1)$, then find f (6).

Answer 64

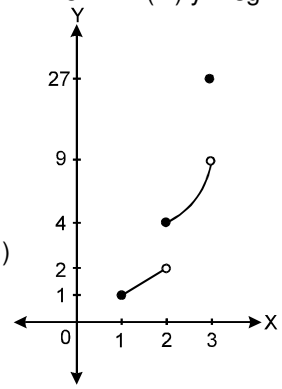
7. Draw the graph of following functions where [.] denotes greatest integer function

(i) $y = [2x] + 1$ (ii) $y = x^{[x]}$, $1 \le x \le 3$ (iii) $y = \text{sgn}(x^2 - x)$

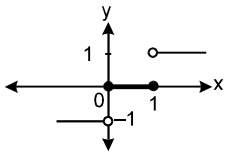
Answer (i)



(ii)



(iii)



E. Odd & Even Functions :

(i) If $f(-x) = f(x)$ for all x in the domain of 'f' then f is said to be an even function. If $f(x) - f(-x) = 0 \Rightarrow f(x)$ is even. e.g. $f(x) = \cos x$; $g(x) = x^2 + 3$.

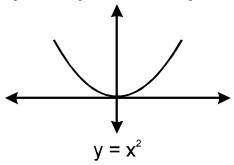
(ii) If $f(-x) = -f(x)$ for all x in the domain of 'f' then f is said to be an odd function.
If $f(x) + f(-x) = 0 \Rightarrow f(x)$ is odd. e.g. $f(x) = \sin x$; $g(x) = x^3 + x$.

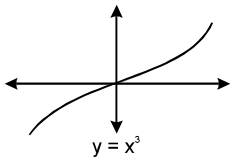
Note : ☞ A function may neither be odd nor even. (e.g. $f(x) = e^x$, $\cos^{-1}x$)

☞ If an odd function is defined at x = 0, then f(0) = 0

Properties of Even/Odd Function

(a) Every even function is symmetric about the y–axis & every odd function is symmetric about the origin. For example graph of $y = x^2$ is symmetric about y-axis, while graph of $y = x^3$ is symmetric about origin





(b) All functions (whose domain is symmetrical about origin) can be expressed as the sum of an even & an odd function, as follows

$$f(x) = \underbrace{\frac{f(x)+f(-x)}{2}}_{\text{Even}} + \underbrace{\frac{f(x)-f(-x)}{2}}_{\text{Odd}}$$

(c) The only function which is defined on the entire number line and is even & odd at the same time is f(x) = 0.

(d) If f and g both are even or both are odd then the function f.g will be even but if any one of them is odd then f.g will be odd. (e) If f(x) is even then f′(x) is odd but converse need not be true.

Illustration # 9: Show that $\log\left(x+\sqrt{x^2+1}\right)$ is an odd function.

Solution Let $f(x) = \log\left(x+\sqrt{x^2+1}\right)$. Then $f(-x) = \log\left(-x+\sqrt{(-x)^2+1}\right)$

$$= \log \frac{\left(\sqrt{x^2+1}-x\right)\left(\sqrt{x^2+1}+x\right)}{\sqrt{x^2+1}+x} = \log \frac{1}{\sqrt{x^2+1}+x} - \log\left(x+\sqrt{x^2+1}\right) = -f(x)$$

Hence f(x) is an odd function.

Illustration # 10 Show that $a^x + a^{-x}$ is an even function.

Solution Let $f(x) = a^x + a^{-x}$ Then $f(-x) = a^{-x} + a^{-(-x)} = a^{-x} + a^x = f(x)$.

Hence f(x) is an even function

Illustration # 11 Show that $\cos^{-1} x$ is neither odd nor even.

Solution Let $f(x) = \cos^{-1}x$. Then $f(-x) = \cos^{-1}(-x) = \pi - \cos^{-1} x$ which is neither equal to f(x) nor equal to f(–x).

Hence $\cos^{-1} x$ is neither odd nor even

Assignment: 8. Determine whether following functions are even or odd?

(i) $\dfrac{e^x+e^{-x}}{e^x-e^{-x}}$ **Answer** Odd

(ii) $\log\left(\sqrt{x^2+1}-x\right)$ **Answer** Odd

(iii) $x \log\left(x+\sqrt{x^2+1}\right)$ **Answer** Even

(iv) $\sin^{-1} 2x\sqrt{1-x^2}$ **Answer** Odd

Even extension / Odd extension :

Let the defincition of the function f(x) is given only for $x \geq 0$. Even extension of this function implies to define the function for $x < 0$ assuming it to be even. In order to get even extension replace x by –x in the given defincition Similarly, odd extension implies to define the function for $x < 0$ assuming it to be odd. In order to get odd extension, multiply the definition of even extension by –1

Illustration # 12 What is even and odd extension of $f(x) = x^3 - 6x^2 + 5x - 11$; $x > 0$

Solution Even extension

$f(x) = -x^3 - 6x^2 + 5x - 11$; $x < 0$

Odd extension

$f(x) = x^3 + 6x^2 + 5x + 11$; $x < 0$

F. Periodic Function :

A function f(x) is called periodic with a period T if there exists a real number $T > 0$ such that for each x in the domain of f the numbers $x - T$ and $x + T$ are also in the domain of f and $f(x) = f(x + T)$ for all x in the domain of 'f'. Domain of a periodic function is always unbounded. Graph of a periodic function with period T is repeated after every interval of 'T'.

e.g. The function sin x & cos x both are periodic over 2π & tan x is periodic over π.

The **least positive period is called the principal or fundamental period** of f or simply the period of f.

Note : ☞ $f(T) = f(0) = f(-T)$, where 'T' is the period.

☞ Inverse of a periodic function does not exist. ☞ Every constant function is always periodic, with no fundamental period.

Properties of Periodic Function

(a) If f(x) has a period T, then $\dfrac{1}{f(x)}$ and $\sqrt{f(x)}$ also have a period T.

(b) If f(x) has a period T then f (ax + b) has a period $\dfrac{T}{|a|}$.

(c) If f (x) has a period T_1 & g (x) also has a period T_2 then period of $f(x) \pm g(x)$ or f(x) . g(x) or $\dfrac{f(x)}{g(x)}$ is L.C.M. of T_1 & T_2 provided their L.C.M. exists. However that L.C.M. (if exists) need not to be fundamental period.

If L.C.M. does not exists $f(x) \pm g(x)$ or f(x) . g(x) or $\dfrac{f(x)}{g(x)}$ is aperiodic.

e.g. |sinx| has the period π, | cosx | also has the period π

$\therefore$ |sinx| + |cosx| also has a period π. But the fundamental period of |sinx| + |cosx| is $\dfrac{\pi}{2}$.

Illustration # 13 Find period of following functions

(i) $f(x) = \sin\dfrac{x}{2} + \cos\dfrac{x}{3}$ (ii) $f(x) = \{x\} + \sin x$

(iii) $f(x) = \cos x \cdot \cos 3x$ (iv) $f(x) = \sin\dfrac{3x}{2} - \cos\dfrac{x}{3} - \tan\dfrac{2x}{3}$

Solution (i) Period of $\sin\dfrac{x}{2}$ is 4π while period of $\cos\dfrac{x}{3}$ is 6π . Hence period of $\sin\dfrac{x}{2} + \cos\dfrac{x}{3}$ is 12π {L.C.M. of 4 & 6 is 12}

(ii) Period of $\sin x = 2p$

Period of {x} = 1 but L.C.M. of 2π & 1 is not possible

$\therefore$ it is aperiodic

(iii) $f(x) = \cos x \cdot \cos 3x$

period of f(x) is L.C.M. of $\left(2\pi, \frac{2\pi}{3}\right) = 2\pi$

but 2π may or may not be fundamental periodic, but fundamental period = $\frac{2\pi}{n}$, where $n \in N$. Hence cross-checking for n = 1, 2, 3, ...we find π to be fundamental period $f(\pi + x) = (-\cos x)(-\cos 3x) = f(x)$

(iv) Period of f(x) is L.C.M. of $\frac{2\pi}{3/2}, \frac{2\pi}{1/3}, \frac{\pi}{3/2}$

= L.C.M. of $\frac{4\pi}{3}, 6\pi, \frac{2\pi}{3}$ $= 12\pi$

NOTE : $\textbf{L.C.M. of } \left(\frac{a}{b}, \frac{p}{q}, \frac{\ell}{m}\right) = \frac{\textbf{L.C.M. (a, p, } \ell\textbf{)}}{\textbf{H.C.F. (b, q, m)}}$

Assignment: 9. Find the period of following function.

(i) $f(x) = \sin x + |\sin x|$ **Answer** 2π

(ii) $f(x) = \sqrt{3}\cos x - \sin\frac{x}{3}$ **Answer** 6π

(iii) $\sin\frac{2x}{5} - \cos\frac{3x}{7}$ **Answer** 70π

(iv) $f(x) = \sin^2 x + \cos^4 x$ **Answer** π

G. Composite Function :

Let $f: X \to Y_1$ and $g: Y_2 \to Z$ be two functions and the set $D = \{x \in X: f(x) \in Y_2\}$. If $D \neq \phi$, then the function h defined on D by $h(x) = g\{f(x)\}$ is called composite function of g and f and is denoted by gof. It is also called function of a function.

Note : ☞ Domain of gof is D which is a subset of X (the domain of f). Range of gof is a subset of the range of g. If D = X, then $f(x) \subseteq Y_2$.

Properties of Composite Functions :

(a) In general $gof \neq fog$ (i.e. not commutative)

(b) The composite of functions are associative i.e. if three functions f, g, h are such that fo (goh) & (fog) oh are defined, then fo (goh) = (fog) oh.

(c) If f and g both are one-one, then gof and fog would also be one-one.

(d) If f and g both are onto, then gof or fog may or may not be onto.

(e) The composite of two bijections is a bijection iff f & g are two bijections such that gof is defined, then gof is also a bijection only **when co-domain of f is equal to the domain of g**.

(f) If g is a function such that gof is defined on the domain of f and f is periodic with T, then gof is also periodic with T as one of its periods. Further if

g is one-one, then T is the period of gof

g is also periodic with T′ as the period and the range of f is a sub-set of [0, T′], then T is the period of gof

Illustration # 14 Describe fog and gof wherever is possible for the following functions

(i) $f(x) = \sqrt{x+3}$, $g(x) = 1 + x^2$ (ii) $f(x) = \sqrt{x}$, $g(x) = x^2 - 1$.

Solution (i) Domain of f is $[-3, \infty)$, range of f is $[0, \infty)$.

Domain of g is R, range of g is $[1, \infty)$.

Since range of f is a subset of domain of g,

$\therefore$ domain of gof is $[-3, \infty)$ {equal to the domain of f }

$gof(x) = g\{f(x)\} = g(\sqrt{x+3}) = 1 + (x+3) = x + 4$. Range of gof is $[1, \infty)$.

Further since range of g is a subset of domain of f,

$\therefore$ domain of fog is R {equal to the domain of g}

$fog(x) = f\{g(x)\} = f(1 + x^2) = \sqrt{x^2+4}$ Range of fog is $[2, \infty)$.

(ii) $f(x) = \sqrt{x}$, $g(x) = x^2 - 1$.

Domain of f is $[0, \infty)$, range of f is $[0, \infty)$.

Domain of g is R, range of g is $[-1, \infty)$.

Since range of f is a subset of the domain of g,

$\therefore$ domain of gof is $[0, \infty)$ and $g\{f(x)\} = g(\sqrt{x}) = x - 1$. Range of gof is $[-1, \infty)$

Further since range of g is not a subset of the domain of f

i.e. $[-1, \infty) \not\subset [0, \infty)$

$\therefore$ fog is not defined on whole of the domain of g.

Domain of fog is {$x \in R$, the domain of g : $g(x) \in [0, \infty)$, the domain of f}.

Thus the domain of fog is $D = \{x \in R: 0 \le g(x) < \infty\}$

i.e. $D = \{x \in R: 0 \le x^2 - 1\} = \{x \in R: x \le -1 \text{ or } x \ge 1\} = (-\infty, -1] \cup [1, \infty)$

$fog(x) = f\{g(x)\} = f(x^2-1) = \sqrt{x^2-1}$ Its range is $[0, \infty)$.

(iii) Let $f(x) = e^x$; $R^+ \to R$ and $g(x) = \sin^{-1} x$; $[-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Find domain and range of fog (x)

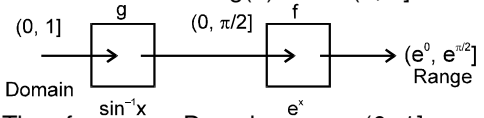
Solution

Domain of f(x) : $(0, \infty)$ Range of g(x) : $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

values in range of g(x) which are accepted by f(x) are $\left(0, \frac{\pi}{2}\right]$

$\Rightarrow$ $0 < g(x) \le \frac{\pi}{2}$ $0 < \sin^{-1} x \le \frac{\pi}{2}$ $0 < x \le 1$

Hence domain of fog(x) is $x \in (0, 1]$



Therefore Domain : $(0, 1]$
Range : $(1, e^{\pi/2}]$

Example of composite function of non-uniformly defined functions :

Illustration # 15 If $f(x) = | |x - 3| - 2 |$ $0 \le x \le 4$
$g(x) = 4 - |2 - x|$ $-1 \le x \le 3$
then find fog(x) and draw rough sketch of fog(x).

Solution $f(x) = | | x - 3| - 2|$ $0 \le x \le 4$

$$= \begin{cases} |x-1| & 0 \le x < 3 \\ |x-5| & 3 \le x \le 4 \end{cases}$$

$$= \begin{cases} 1-x & 0 \le x < 1 \\ x-1 & 1 \le x < 3 \\ 5-x & 3 \le x \le 4 \end{cases}$$

$g(x) = 4 - |2 - x|$ $-1 \le x \le 3$

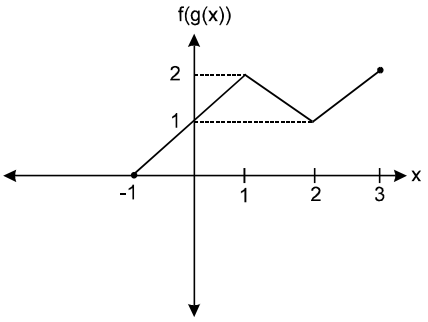
$$= \begin{cases} 4-(2-x) & -1 \le x < 2 \\ 4-(x-2) & 2 \le x \le 3 \end{cases}$$

$$= \begin{cases} 2+x & -1 \le x < 2 \\ 6-x & 2 \le x \le 3 \end{cases}$$

$$\therefore \quad fog(x) = \begin{cases} 1-g(x) & 0 \le g(x) < 1 \\ g(x)-1 & 1 \le g(x) < 3 \\ 5-g(x) & 3 \le g(x) \le 4 \end{cases}$$

$$= \begin{cases} 1-(2+x) & 0 \le 2+x < 1 \text{ and } -1 \le x < 2 \\ 2+x-1 & 1 \le 2+x < 3 \text{ and } -1 \le x < 2 \\ 5-(2+x) & 3 \le 2+x \le 4 \text{ and } -1 \le x < 2 \\ 1-6+x & 0 \le 6-x < 1 \text{ and } 2 \le x \le 3 \\ 6-x-1 & 1 \le 6-x \le 3 \text{ and } 2 \le x \le 3 \\ 5-6+x & 3 \le 6-x \le 4 \text{ and } 2 \le x \le 3 \end{cases}$$

$$= \begin{cases} -1-x & -2 \le x < 1 \text{ and } -1 \le x < 2 \\ 1+x & -1 \le x < 1 \text{ and } -1 \le x < x \\ 3-x & 1 \le x \le 2 \text{ and } -1 \le x < 2 \\ x-5 & -6 \le -x < -5 \text{ and } 2 \le x \le 3 \\ 5-x & -5 \le -x < -3 \text{ and } 2 \le x \le 3 \\ x-1 & -3 \le -x \le -2 \text{ and } 2 \le x \le 3 \end{cases}$$



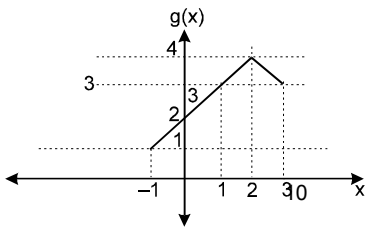
$$= \begin{cases} -1-x & -2 \le x < -1 \text{ and } -1 \le x < 2 \\ 1+x & -1 \le x < 1 \text{ and } -1 \le x < 2 \\ 3-x & 1 \le x \le 2 \text{ and } -1 \le x < 2 \\ x-5 & 5 < x \le 6 \text{ and } 2 \le x \le 3 \\ 5-x & 3 < x \le 5 \text{ and } 2 \le x \le 3 \\ x-1 & 2 \le x \le 3 \text{ and } 2 \le x \le 3 \end{cases}$$

$$= \begin{cases} 1+x & -1 \le x < 1 \\ 3-x & 1 \le x < 2 \\ x-1 & 2 \le x \le 3 \end{cases}$$

Alternate method for finding fog

$$g(x) = \begin{cases} 2+x & -1 \le x < 2 \\ 6-x & 2 \le x \le 3 \end{cases}$$

graph of g(x) is



$$\therefore \quad fog(x) = \begin{cases} 1-g(x) & 0 \le g(x) < 1 \\ g(x)-1 & 1 \le g(x) < 3 \\ 5-g(x) & 3 \le g(x) \le 4 \end{cases}$$

$$= \begin{cases} 1-g(x) & \text{for no value} \\ g(x)-1 & -1 \le x < 1 \\ 5-g(x) & 1 \le x \le 3 \end{cases} = \begin{cases} 2+x-1 & -1 \le x < 1 \\ 5-(2+x) & 1 \le x < 2 \\ 5-(6-x) & 2 \le x \le 3 \end{cases} = \begin{cases} x+1 & -1 \le x < 1 \\ 3-x & 1 \le x < 2 \\ x-1 & 2 \le x \le 3 \end{cases}$$

Assignment: **10.** Define fog(x) and gof(x). Also their Domain & Range.

(i) $f(x) = [x], g(x) = \sin x$ (ii) $f(x) = \tan x, x \in (-\pi/2, \pi/2); g(x) = \sqrt{1-x^2}$

Answer (i) gof = sin [x]
domain : R range { sin a : a ∈ I}
fog = [sin x] domain : R range : {–1, 0, 1}

Answer (ii) gof = $\sqrt{1-\tan^2 x}$

domain : $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ range : [0, 1]

fog = $\tan\sqrt{1-x^2}$ domain : [–1, 1] range [0, tan 1]

11. Let $f(x) = e^x : R^+ \to R$ and $g(x) = x^2 - x : R \to R$. Find domain and range of fog (x) & gof (x)

Answer

fog (x)	gof f(x)
Domain : $(-\infty, 0) \cup (1, \infty)$	Domain : $(0, \infty)$
Range : $[1, \infty)$	Range : $\left[-\frac{1}{4}, \infty\right)$

H. Inverse of a Function :

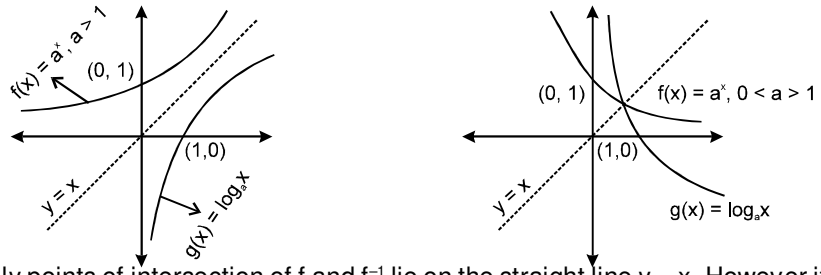
Let $f : A \to B$ be a function. Then f is invertible iff there is a function $g : B \to A$ such that go f is an identity function on A and fog is an identity function on B. Then g is called inverse of f and is denoted by f^{-1}.

For a function to be invertible it must be bijective

Note : ☞ The inverse of a bijection is unique. ☞ Inverse of an even function is not defined.

Properties of Inverse Function :

(a) The graphs of f & g are the mirror images of each other in the line y = x. For example $f(x) = a^x$ and $g(x) = \log_a x$ are inverse of each other, and their graphs are mirror images of each other on the line y = x as shown below.



(b) Normally points of intersection of f and f^{-1} lie on the straight line y = x. However it must be noted that f(x) and $f^{-1}(x)$ may intersect otherwise also.

(c) In general fog(x) and gof(x) are not equal but if they are equal then in majority of cases either f and g are inverse of each other or atleast one of f and g is an identity function.

(d) If f & g are two bijections $f : A \to B$, $g : B \to C$ then the inverse of gof exists and $(gof)^{-1} = f^{-1} o g^{-1}$.

(e) If f(x) and g are inverse function of each other then $f'(g(x)) = \frac{1}{g'(x)}$

Illustration # 16 (i) Determine whether $f(x) = \frac{2x+3}{4}$; $R \to R$, is invertible or not? If so find it.

Solution: As given function is one-one and onto, therefore it is invertible. $y = \frac{2x+3}{4}$

$\Rightarrow \quad x = \frac{4y-3}{2} \qquad \therefore \quad f^{-1}(x) = \frac{4x-3}{2}$

(ii) Is the function $f(x) = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$ invertible?

Solution: Domain of f is [–1, 1] and f is continuous

$$f'(x) = \frac{2\left(1-2x^2\right)}{\left|1-2x^2\right|\sqrt{1-x^2}} = \begin{cases} \frac{2}{\sqrt{1-x^2}} & \text{if } \frac{-1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}} \\ \frac{-2}{\sqrt{1-x^2}} & \text{if } x < \frac{-1}{\sqrt{2}} \text{ or } x > \frac{1}{\sqrt{2}} \end{cases}$$

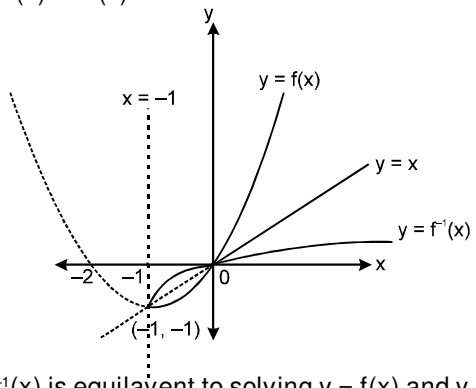
$\therefore$ f(x) is increasing in $\left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and is decreasing in each of the intervals

$\left(-1, \frac{-1}{\sqrt{2}}\right)$ and $\left(\frac{-1}{\sqrt{2}}, 1\right)$

∴ f(x) is not one-one, so is not invertible.

(iii) Let $f(x) = x^2 + 2x$; $x \geq -1$. Draw graph of $f^{-1}(x)$ also find the number of solutions of the equation, $f(x) = f^{-1}(x)$

Solution



$f(x) = f^{-1}(x)$ is equilavent to solving $y = f(x)$ and $y = x$

$\Rightarrow \quad x^2 + 2x = x \quad \Rightarrow \quad x(x + 1) = 0 \quad \Rightarrow \quad x = 0, -1$

Hence two solution for $f(x) = f^{-1}(x)$

(iv) If $y = f(x) = x^2 - 3x + 1$, $x \geq 2$. Find the value of $g'(1)$ where g is inverse of f

Solution $y = 1 \quad \Rightarrow \quad x^2 - 3x + 1 = 1 \quad \Rightarrow \quad x(x - 3) = 0 \quad \Rightarrow \quad x = 0, 3$

But $x \geq 2 \quad \therefore \quad x = 3$

Now $g(f(x)) = x$

Differentiating both sides w.r.t. x

$\Rightarrow \quad g'(f(x)).\, f'(x) = 1 \quad \Rightarrow \quad g'(f(x)) = \dfrac{1}{f'(x)}$

$\Rightarrow \quad g'(f(3)) = \dfrac{1}{f'(3)} \quad \Rightarrow \quad g'(1) = = \dfrac{1}{6-3}$ (As $f'(x) = 2x - 3$) $= \dfrac{1}{3}$

Alternate Method

$y = x^2 - 3x + 1$

$x^2 - 3x + 1 - y = 0$

$x = \dfrac{3 \pm \sqrt{9 - 4(1-y)}}{2}$

$= \dfrac{3 \pm \sqrt{5 + 4y}}{2}$

$x \geq 2$

$x = \dfrac{3 + \sqrt{5 + 4y}}{2}$

$g(x) = \dfrac{3 + \sqrt{5 + 4y}}{2}$

$g'(x) = 0 + \dfrac{1}{x\sqrt{5 + 4x}}\, x \qquad g'(1) = \dfrac{1}{\sqrt{5+4}} = \dfrac{1}{\sqrt{9}} = \dfrac{1}{3}$

Assignment: **12.** Determine $f^{-1}(x)$, if given function is invertible

(i) $f : (-\infty, -1) \to (-\infty, -2)$ defined $f(x) = -(x + 1)^2 - 2$

(ii) $f : \left[\dfrac{\pi}{6}, \dfrac{7\pi}{6}\right] \to [-1, 1]$ defined by $f(x) = \sin\left(x + \dfrac{\pi}{3}\right)$

Answer (i) $-1 + \sqrt{-x-2}$ (ii) $\dfrac{2\pi}{3} - \sin^{-1}x$

I. Equal or Identical Function :

Two functions f & g are said to be identical (or equal) iff :

(i) The domain of f ≡ the domain of g. (ii) The range of f ≡ the range of g and

(iii) $f(x) = g(x)$, for every x belonging to their common domain. e.g. $f(x) = \dfrac{1}{x}$ & $g(x) = \dfrac{x}{x^2}$ are identical functions.

But $f(x) = x$ and $g(x) = \dfrac{x^2}{x}$ are not identical functions.

Illustration # 17 Examine whether following pair of functions are identical or not

(i) $f(x) = \dfrac{x^2}{x}$ & $g(x) = x$ **Answer** No, as domain of f(x) is R – {0} while domain of g(x) is R

(ii) $f(x) = \sin^2x + \cos^2x$ & $g(x) = \sec^2x - \tan^2x$

Answer No, as domain are not same. Domain of f(x) is R while that of g(x) is $R - \left\{(2n+1)\dfrac{\pi}{2}; n \in I\right\}$

Assignment: **13.** Examine whether following pair of functions are identical or not

(i) $f(x) = \text{sgn}\,(x)$ & $g(x) = \begin{cases} \dfrac{x}{|x|} & x \neq 0 \\ 0 & x = 0 \end{cases}$

(ii) $f(x) = \sin^{-1}x + \cos^{-1}x$ & $g(x) = \frac{\pi}{2}$ **Answer** (i) Yes (ii) No

J. General : If x, y are independent variables, then:

(i) $f(xy) = f(x) + f(y) \Rightarrow f(x) = k \ln x$ or $f(x) = 0$. (ii) $f(xy) = f(x).f(y) \Rightarrow f(x) = x^n$, $n \in R$

(iii) $f(x + y) = f(x).f(y) \Rightarrow f(x) = a^{kx}$. (iv) $f(x + y) = f(x) + f(y) \Rightarrow f(x) = kx$, where k is a constant.

(v) $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) \Rightarrow f(x) = 1 \pm x^n$ where $n \in N$

Illustration # 18 If f(x) is a polynomial function satisfying $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) \forall x \in R - \{0\}$ and $f(2) = 9$, then find f (3)

Solution $f(x) = 1 \pm x^n$ As $f(2) = 9$ $\therefore$ $f(x) = 1 + x^3$

Hence $f(3) = 1 + 3^3 = 28$

Assignment: 14. If f(x) is a polynomial function satisfying $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) \forall x \in R - \{0\}$ and $f(3) = -8$, then find f(4) **Answer** – 15

15. If $f(x + y) = f(x) \cdot f(y)$ for all real x, y and $f(0) \neq 0$ then prove that the function, $g(x) = \frac{f(x)}{1+f^2(x)}$ is an even function

SHORT REVISION (FUNCTIONS)

THINGS TO REMEMBER :

1. GENERAL DEFINITION :

If to every value (Considered as real unless other–wise stated) of a variable x, which belongs to some collection (Set) E, there corresponds one and only one finite value of the quantity y, then y is said to be a function (Single valued) of x or a dependent variable defined on the set E ; x is the argument or independent variable .

If to every value of x belonging to some set E there corresponds one or several values of the variable y, then y is called a multiple valued function of x defined on E.Conventionally the word **"FUNCTION"** is used only as the meaning of a single valued function, if not otherwise stated.

Pictorially : $\xrightarrow[\text{input}]{x}$ $\boxed{f}$ $\xrightarrow[\text{output}]{f(x)=y}$, y is called the image of x & x is the pre-image of y under f.

Every function from $A \to B$ satisfies the following conditions .

(i) $f \subset A \times B$ **(ii)** $\forall\ a \in A \Rightarrow (a, f(a)) \in f$ and

(iii) $(a, b) \in f\ \&\ (a, c) \in f \Rightarrow b = c$

2. DOMAIN, CO–DOMAIN & RANGE OF A FUNCTION :

Let $f: A \to B$, then the set A is known as the domain of f & the set B is known as co-domain of f . The set of all f images of elements of A is known as the range of f . Thus :

Domain of $f = \{a \mid a \in A, (a, f(a)) \in f\}$

Range of $f = \{f(a) \mid a \in A, f(a) \in B\}$

It should be noted that range is a subset of co–domain . If only the rule of function is given then the domain of the function is the set of those real numbers, where function is defined. For a continuous function, the interval from minimum to maximum value of a function gives the range.

3. IMPORTANT TYPES OF FUNCTIONS :

(i) POLYNOMIAL FUNCTION :

If a function f is defined by $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + ... + a_{n-1}x + a_n$ where n is a non negative integer and $a_0, a_1, a_2, ..., a_n$ are real numbers and $a_0 \neq 0$, then f is called a polynomial function of degree n .

NOTE : (a) A polynomial of degree one with no constant term is called an odd linear function . i.e. $f(x) = ax,\ a \neq 0$

(b) There are two polynomial functions, satisfying the relation ; $f(x).f(1/x) = f(x) + f(1/x)$. They are :

(i) $f(x) = x^n + 1$ & **(ii)** $f(x) = 1 - x^n$, where n is a positive integer .

(ii) ALGEBRAIC FUNCTION :

y is an algebraic function of x, if it is a function that satisfies an algebraic equation of the form $P_0(x)\ y^n + P_1(x)\ y^{n-1} + ... + P_{n-1}(x)\ y + P_n(x) = 0$ Where n is a positive integer and $P_0(x), P_1(x)$ ..... are Polynomials in x.

e.g. $y = |x|$ is an algebraic function, since it satisfies the equation $y^2 - x^2 = 0$.

Note that all polynomial functions are Algebraic but not the converse. A function that is not algebraic is called **TRANSCEDENTAL FUNCTION** .

(iii) FRACTIONAL RATIONAL FUNCTION :

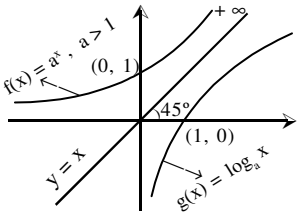
A rational function is a function of the form. $y = f(x) = \frac{g(x)}{h(x)}$, where

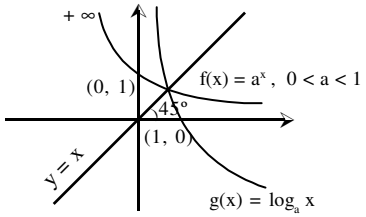
$g(x)$ & $h(x)$ are polynomials & $h(x) \neq 0$.

(IV) EXPONENTIAL FUNCTION :

A function $f(x) = a^x = e^{x \ln a}\ (a > 0,\ a \neq 1, x \in R)$ is called an exponential function. The inverse of the exponential function is called the logarithmic function . i.e. $g(x) = \log_a x$.

Note that f(x) & g(x) are inverse of each other & their graphs are as shown .





(v) ABSOLUTE VALUE FUNCTION :

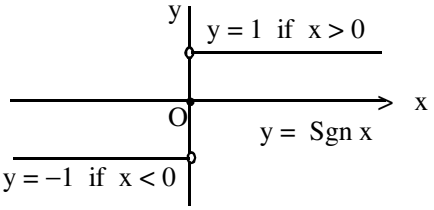
A function $y = f(x) = |x|$ is called the absolute value function or Modulus function. It is defined as

: $y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

(vi) SIGNUM FUNCTION :

A function $y = f(x) = \text{Sgn}(x)$ is defined as follows :

$$y = f(x) = \begin{cases} 1 & \text{for } x > 0 \\ 0 & \text{for } x = 0 \\ -1 & \text{for } x < 0 \end{cases}$$



It is also written as $\text{Sgn } x = |x|/x$;
$x \neq 0$; $f(0) = 0$

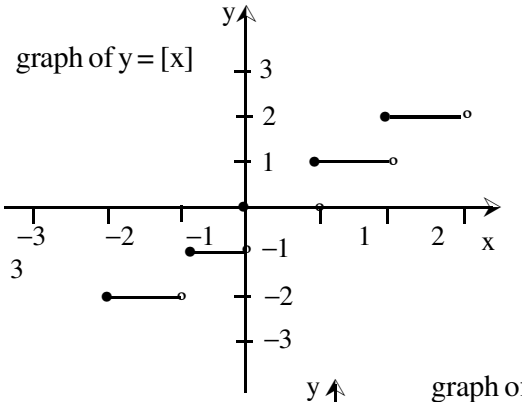
(vii) GREATEST INTEGER OR STEP UP FUNCTION :

The function $y = f(x) = [x]$ is called the greatest integer function where $[x]$ denotes the greatest integer less than or equal to x . Note that for :

$-1 \leq x < 0$; $[x] = -1$ $\quad$ $0 \leq x < 1$; $[x] = 0$
$1 \leq x < 2$; $[x] = 1$ $\quad$ $2 \leq x < 3$; $[x] = 2$
and so on .

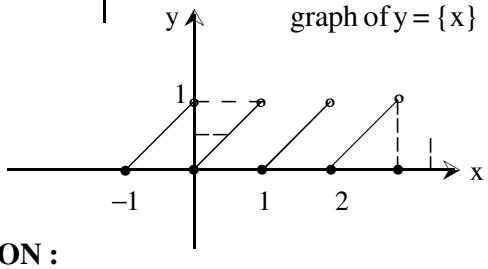
Properties of greatest integer function :

(a) $[x] \leq x < [x]+1$ and
$x - 1 < [x] \leq x$, $0 \leq x - [x] < 1$
(b) $[x+m] = [x]+m$ if m is an integer .
(c) $[x]+[y] \leq [x+y] \leq [x]+[y]+1$
(d) $[x]+[-x] = 0$ if x is an integer
$= -1$ otherwise .



(viii) FRACTIONAL PART FUNCTION :

It is defined as :
$g(x) = \{x\} = x - [x]$.
e.g. the fractional part of the no. 2.1 is
$2.1 - 2 = 0.1$ and the fractional part of -3.7 is 0.3.
The period of this function is 1 and graph of this function is as shown .



4. DOMAINS AND RANGES OF COMMON FUNCTION :

Function ($y = f(x)$)	Domain (i.e. values taken by x)	Range (i.e. values taken by $f(x)$)
A. Algebraic Functions		
(i) x^n , $(n \in N)$	R = (set of real numbers)	R , if n is odd $R^+ \cup \{0\}$, if n is even
(ii) $\frac{1}{x^n}$, $(n \in N)$	$R - \{0\}$	$R - \{0\}$, if n is odd

			R^+ , if n is even
(iii)	$x^{1/n}$, $(n \in N)$	R , if n is odd	R , if n is odd
		$R^+ \cup \{0\}$, if n is even	$R^+ \cup \{0\}$, if n is even
(iv)	$\frac{1}{x^{1/n}}$, $(n \in N)$	$R - \{0\}$, if n is odd	$R - \{0\}$, if n is odd
		R^+ , if n is even	R^+ , if n is even

B. Trigonometric Functions

(i)	sin x	R	[–1, + 1]
(ii)	cos x	R	[–1, + 1]
(iii)	tan x	$R - (2k + 1)\frac{\pi}{2}, k \in I$	R
(iv)	sec x	$R - (2k + 1)\frac{\pi}{2}, k \in I$	$(-\infty, -1] \cup [1, \infty)$
(v)	cosec x	$R - k\pi, k \in I$	$(-\infty, -1] \cup [1, \infty)$
(vi)	cot x	$R - k\pi, k \in I$	R

C. Inverse Circular Functions (Refer after Inverse is taught)

(i)	$\sin^{-1} x$	[–1, + 1]	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(ii)	$\cos^{-1} x$	[–1, + 1]	[0, π]
(iii)	$\tan^{-1} x$	R	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(iv)	$\text{cosec}^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
(v)	$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
(vi)	$\cot^{-1} x$	R	(0, π)

Function	**Domain**	**Range**
(y = f (x))	(i.e. values taken by x)	(i.e. values taken by f (x))

D. Exponential Functions

(i)	e^x	R	R^+
(ii)	$e^{1/x}$	$R - \{0\}$	$R^+ - \{1\}$
(iii)	a^x , $a > 0$	R	R^+
(iv)	$a^{1/x}$, $a > 0$	$R - \{0\}$	$R^+ - \{1\}$

E. Logarithmic Functions

(i)	$\log_a x$, $(a > 0)$ $(a \neq 1)$	R^+	R
(ii)	$\log_x a = \frac{1}{\log_a x}$ $(a > 0)$ $(a \neq 1)$	$R^+ - \{1\}$	$R - \{0\}$

F. Integral Part Functions Functions

(i)	$[x]$	R	I
(ii)	$\frac{1}{[x]}$	R – [0, 1)	$\left\{\frac{1}{n}, n \in I - \{0\}\right\}$

G. Fractional Part Functions

(i)	$\{x\}$	R	[0, 1)
(ii)	$\frac{1}{\{x\}}$	R – I	$(1, \infty)$

H. Modulus Functions

(i)	$\lvert x \rvert$	R	$R^+ \cup \{0\}$
(ii)	$\frac{1}{\lvert x \rvert}$	R – { 0 }	R^+

I. Signum Function

$\text{sgn}(x) = \frac{\lvert x \rvert}{x}, x \neq 0$ $= 0, x = 0$	R	{–1, 0 , 1}

J. Constant Function

say $f(x) = c$	R	{ c }

5. EQUAL OR IDENTICAL FUNCTION :

Two functions f & g are said to be equal if :

(i) The domain of f = the domain of g.

(ii) The range of f = the range of g and

(iii) $f(x) = g(x)$, for every x belonging to their common domain. eg.

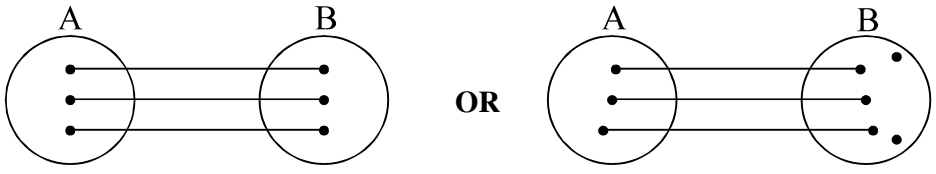
$f(x) = \frac{1}{x}$ & $g(x) = \frac{x}{x^2}$ are identical functions .

6. CLASSIFICATION OF FUNCTIONS :

One–One Function (Injective mapping) :

A function $f: A \to B$ is said to be a one–one function or injective mapping if different elements of A have different f images in B . Thus for $x_1, x_2 \in A$ & $f(x_1)$, $f(x_2) \in B$, $f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$ or $x_1 \neq x_2 \Leftrightarrow f(x_1) \neq f(x_2)$.

Diagramatically an injective mapping can be shown as



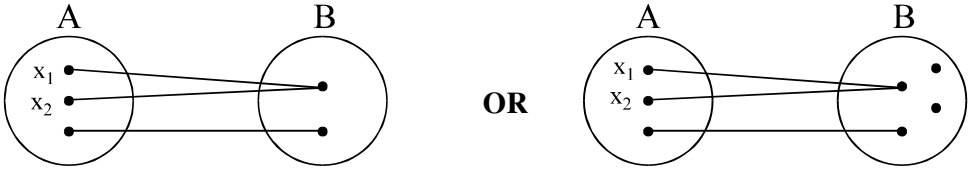
Note : (i) Any function which is entirely increasing or decreasing in whole domain, then f(x) is one–one .

(ii) If any line parallel to x–axis cuts the graph of the function atmost at one point, then the function is one–one .

Many–one function :

A function $f: A \to B$ is said to be a many one function if two or more elements of A have the same f image in B . Thus $f: A \to B$ is many one if for some $x_1, x_2 \in A$, $f(x_1) = f(x_2)$ but $x_1 \neq x_2$.

Diagramatically a many one mapping can be shown as



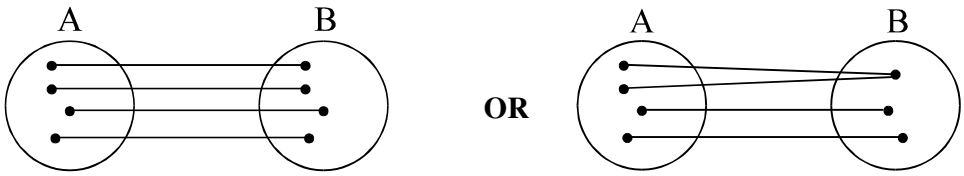
Note : **(i)** Any continuous function which has atleast one local maximum or local minimum, then f(x) is many–one . In other words, if a line parallel to x–axis cuts the graph of the function atleast at two points, then f is many–one .

(ii) If a function is one–one, it cannot be many–one and vice versa .

Onto function (Surjective mapping) :

If the function $f: A \to B$ is such that each element in B (co–domain) is the f image of atleast one element in A, then we say that f is a function of A 'onto' B . Thus $f: A \to B$ is surjective iff $\forall\ b \in B$, $\exists$ some $a \in A$ such that $f(a) = b$.

Diagramatically surjective mapping can be shown as

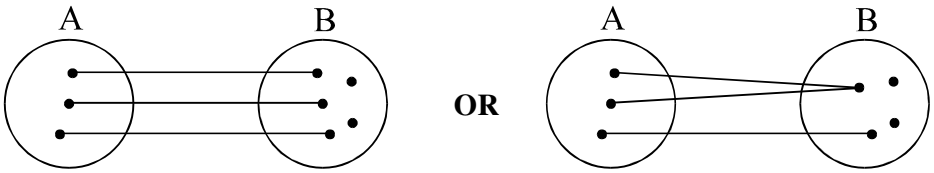


Note that : if range = co–domain, then f(x) is onto.

Into function :

If $f: A \to B$ is such that there exists atleast one element in co–domain which is not the image of any element in domain, then f(x) is into .

Diagramatically into function can be shown as



Note that : If a function is onto, it cannot be into and vice versa . A polynomial of degree even will always be into.

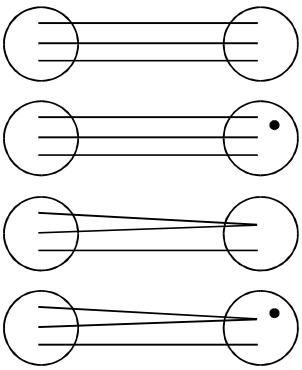
Thus a function can be one of these four types :

(a) one–one onto (injective & surjective)

(b) one–one into (injective but not surjective)

(c) many–one onto (surjective but not injective)

(d) many–one into (neither surjective nor injective)



Note : **(i)** If f is both injective & surjective, then it is called a **Bijective** mapping.
The bijective functions are also named as invertible, non singular or biuniform functions.

(ii) If a set A contains n distinct elements then the number of different functions defined from $A \to A$ is n^n & out of it n ! are one one.

Identity function :

The function $f: A \to A$ defined by $f(x) = x\ \forall\ x \in A$ is called the identity of A and is denoted by I_A.
It is easy to observe that identity function is a bijection .

Constant function :

A function $f: A \to B$ is said to be a constant function if every element of A has the same f image in B . Thus $f: A \to B$; $f(x) = c$, $\forall\ x \in A$, $c \in B$ is a constant function. Note that the range of a constant function is a singleton and a constant function may be one-one or many-one, onto or into .

7. ALGEBRAIC OPERATIONS ON FUNCTIONS :

If f & g are real valued functions of x with domain set A, B respectively, then both f & g are defined in $A \cap B$. Now we define $f+g$, $f-g$, $(f.g)$ & (f/g) as follows :

(i) $(f \pm g)(x) = f(x) \pm g(x)$

(ii) $(f.g)(x) = f(x) \,.\, g(x)$

domain in each case is $A \cap B$

(iii) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ domain is $\{x \mid x \in A \cap B \ \text{s.t}\ g(x) \neq 0\}$.

8. COMPOSITE OF UNIFORMLY & NON-UNIFORMLY DEFINED FUNCTIONS :

Let $f: A \to B$ & $g: B \to C$ be two functions . Then the function $gof: A \to C$ defined by $(gof)(x) = g(f(x))$ $\forall\ x \in A$ is called the composite of the two functions f & g .

Diagramatically $\xrightarrow{x}$ [f] $\xrightarrow{f(x)}$ [g] $\longrightarrow g(f(x))$.

Thus the image of every $x \in A$ under the function gof is the g–image of the f–image of x .

Note that gof is defined only if $\forall\ x \in A$, $f(x)$ is an element of the domain of g so that we can take its g-image. Hence for the product gof of two functions f & g, the range of f must be a subset of the domain of g.

PROPERTIES OF COMPOSITE FUNCTIONS :

(i) The composite of functions is not commutative i.e. $gof \neq fog$.

(ii) The composite of functions is associative i.e. if f, g, h are three functions such that $fo(goh)$ & $(fog)oh$ are defined, then $fo(goh) = (fog)oh$.

(iii) The composite of two bijections is a bijection i.e. if f & g are two bijections such that gof is defined, then gof is also a bijection.

9. HOMOGENEOUS FUNCTIONS :

A function is said to be homogeneous with respect to any set of variables when each of its terms is of the same degree with respect to those variables .

For example $5x^2 + 3y^2 - xy$ is homogeneous in x & y . Symbolically if,

$f(tx, ty) = t^n \,.\, f(x, y)$ then $f(x, y)$ is homogeneous function of degree n .

10. BOUNDED FUNCTION :

A function is said to be bounded if $|f(x)| \leq M$, where M is a finite quantity .

11. IMPLICIT & EXPLICIT FUNCTION :

A function defined by an equation not solved for the dependent variable is called an **IMPLICIT FUNCTION** . For eg. the equation $x^3 + y^3 = 1$ defines y as an implicit function. If y has been expressed in terms of x alone then it is called an **EXPLICIT FUNCTION**.

12. INVERSE OF A FUNCTION :

Let $f: A \to B$ be a one–one & onto function, then their exists a unique function $g: B \to A$ such that $f(x) = y \Leftrightarrow g(y) = x$, $\forall\ x \in A$ & $y \in B$. Then g is said to be inverse of f . Thus $g = f^{-1}: B \to A = \{(f(x), x) \mid (x, f(x)) \in f\}$.

PROPERTIES OF INVERSE FUNCTION :

(i) The inverse of a bijection is unique .

(ii) If $f: A \to B$ is a bijection & $g: B \to A$ is the inverse of f, then $fog = I_B$ and $gof = I_A$, where I_A & I_B are identity functions on the sets A & B respectively.

Note that the graphs of f & g are the mirror images of each other in the line $y = x$. As shown in the figure given below a point (x', y') corresponding to $y = x^2$ $(x \geq 0)$ changes to (y', x') corresponding to $y = +\sqrt{x}$, the changed form of $x = \sqrt{y}$.

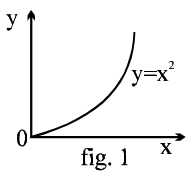


fig. 1

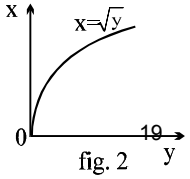


fig. 2

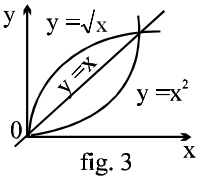


fig. 3

(iii) The inverse of a bijection is also a bijection .

(iv) If f & g are two bijections $f: A \to B$, $g: B \to C$ then the inverse of gof exists and $(gof)^{-1} = f^{-1} o g^{-1}$.

13. ODD & EVEN FUNCTIONS :

If $f(-x) = f(x)$ for all x in the domain of 'f' then f is said to be an even function.

e.g. $f(x) = \cos x$; $g(x) = x^2 + 3$.

If $f(-x) = -f(x)$ for all x in the domain of 'f' then f is said to be an odd function.

e.g. $f(x) = \sin x$; $g(x) = x^3 + x$.

NOTE : (a) $f(x) - f(-x) = 0 \Rightarrow f(x)$ is even & $f(x) + f(-x) = 0 \Rightarrow f(x)$ is odd .

(b) A function may neither be odd nor even .

(c) Inverse of an even function is not defined .

(d) Every even function is symmetric about the y–axis & every odd function is symmetric about the origin .

(e) Every function can be expressed as the sum of an even & an odd function.

e.g. $f(x) = \underbrace{\frac{f(x)+f(-x)}{2}}_{\text{EVEN}} + \underbrace{\frac{f(x)-f(-x)}{2}}_{\text{ODD}}$

(f) The only function which is defined on the entire number line & is even and odd at the same time is $f(x) = 0$.

(g) If f and g both are even or both are odd then the function f.g will be even but if any one of them is odd then f.g will be odd .

14. PERIODIC FUNCTION :

A function f(x) is called periodic if there exists a positive number T ($T > 0$) called the period of the function such that $f(x+T) = f(x)$, for all values of x within the domain of x.

e.g. The function sin x & cos x both are periodic over 2π & tan x is periodic over π .

NOTE : (a) $f(T) = f(0) = f(-T)$, where 'T' is the period .

(b) Inverse of a periodic function does not exist .

(c) Every constant function is always periodic, with no fundamental period .

(d) If f(x) has a period T & g(x) also has a period T then it does not mean that $f(x)+g(x)$ must have a period T . e.g. $f(x) = |\sin x| + |\cos x|$.

(e) If f(x) has a period p, then $\frac{1}{f(x)}$ and $\sqrt{f(x)}$ also has a period p .

(f) if f(x) has a period T then f(ax + b) has a period T/a ($a > 0$) .

15. GENERAL :

If x, y are independent variables, then :

(i) $f(xy) = f(x) + f(y) \Rightarrow f(x) = k \ln x$ or $f(x) = 0$.

(ii) $f(xy) = f(x) \cdot f(y) \Rightarrow f(x) = x^n$, $n \in R$

(iii) $f(x+y) = f(x) \cdot f(y) \Rightarrow f(x) = a^{kx}$.

(iv) $f(x+y) = f(x) + f(y) \Rightarrow f(x) = kx$, where k is a constant .

EXERCISE–1

Q.1 Find the domains of definitions of the following functions :

(Read the symbols [*] and {*} as greatest integers and fractional part functions respectively.)

(i) $f(x) = \sqrt{\cos 2x} + \sqrt{16 - x^2}$

(ii) $f(x) = \log_7 \log_5 \log_3 \log_2 (2x^3 + 5x^2 - 14x)$

(iii) $f(x) = \ln\left(\sqrt{x^2 - 5x - 24} - x - 2\right)$

(iv) $f(x) = \sqrt{\frac{1-5^x}{7^{-x}-7}}$

(v) $y = \log_{10} \sin(x-3) + \sqrt{16 - x^2}$

(vi) $f(x) = \log_{100x}\left(\frac{2\log_{10} x + 1}{-x}\right)$

(vii) $f(x)=\frac{1}{\sqrt{4x^2-1}}+\ln x(x^2-1)$ (viii) $f(x)=\sqrt{\log_{\frac{1}{2}}\frac{x}{x^2-1}}$

(ix) $f(x)=\sqrt{x^2-|x|}+\frac{1}{\sqrt{9-x^2}}$ (x) $f(x)=\sqrt{(x^2-3x-10).ln^2(x-3)}$

(xi) $f(x)=\sqrt{\log_x(\cos 2\pi x)}$ (xii) $f(x)=\frac{\sqrt{\cos x-\frac{1}{2}}}{\sqrt{6+35x-6x^2}}$

(xiii) $f(x)=\sqrt{\log_{1/3}\left(\log_4\left([x]^2-5\right)\right)}$ (xiv) $f(x)=\frac{1}{[x]}+\log_{(2\{x\}-5)}(x^2-3x+10)+\frac{1}{\sqrt{1-|x|}}$,

(xv) $f(x)=\log_x \sin x$

(xvi) $f(x)=\log_2\left(-\log_{1/2}\left(1+\frac{1}{\sin\left(\frac{x^\circ}{100}\right)}\right)\right)+\sqrt{\log_{10}(\log_{10}x)-\log_{10}(4-\log_{10}x)-\log_{10}3}$

(xvii) $f(x)=\frac{1}{[x]}+\log_{1-\{x\}}(x^2-3x+10)+\frac{1}{\sqrt{2-|x|}}+\frac{1}{\sqrt{\sec(\sin x)}}$

(xviii) $f(x)=\sqrt{(5x-6-x^2)\left[\{ln\{x\}\}\right]}+\sqrt{(7x-5-2x^2)}+\left(ln\left(\frac{7}{2}-x\right)\right)^{-1}$

(xix) If $f(x)=\sqrt{x^2-5x+4}$ & $g(x)=x+3$, then find the domain of $\frac{f}{g}(x)$.

Q.2 Find the domain & range of the following functions .
(Read the symbols [*] and {*} as greatest integers and fractional part functions respectively.)

(i) $y=\log_{\sqrt{5}}\left(\sqrt{2}(\sin x-\cos x)+3\right)$ (ii) $y=\frac{2x}{1+x^2}$ (iii) $f(x)=\frac{x^2-3x+2}{x^2+x-6}$

(iv) $f(x)=\frac{x}{1+|x|}$ (v) $y=\sqrt{2-x}+\sqrt{1+x}$

(vi) $f(x)=\log_{(\text{cosec}\,x-1)}(2-[\sin x]-[\sin x]^2)$ (vii) $f(x)=\frac{\sqrt{x+4}-3}{x-5}$

Q.3 Draw graphs of the following function , where [ ] denotes the greatest integer function.
(i) $f(x)=x+[x]$
(ii) $y=(x)^{[x]}$ where $x=[x]+(x)$ & $x>0$ & $x\le 3$
(iii) $y=\text{sgn}\,[x]$ (iv) $\text{sgn}\,(x-|x|)$

Q.4 Classify the following functions f(x) definzed in $R\to R$ as injective, surjective, both or none .

(a) $f(x)=\frac{x^2+4x+30}{x^2-8x+18}$ (b) $f(x)=x^3-6x^2+11x-6$ (c) $f(x)=(x^2+x+5)(x^2+x-3)$

Q.5 Let $f(x)=\frac{1}{1-x}$. Let $f_2(x)$ denote $f[f(x)]$ and $f_3(x)$ denote $f[f\{f(x)\}]$. Find $f_{3n}(x)$ where n is a natural number. Also state the domain of this composite function.

Q.6 If $f(x)=\sin^2x+\sin^2\left(x+\frac{\pi}{3}\right)+\cos x\cos\left(x+\frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right)=1$, then find $(gof)(x)$.

Q.7 The function f(x) is defined on the interval [0,1]. Find the domain of definition of the functions.
(a) $f(\sin x)$ (b) $f(2x+3)$

Q.8**(i)** Find whether the following functions are even or odd or none

(a) $f(x)=\log\left(x+\sqrt{1+x^2}\right)$ (b) $f(x)=\frac{x\left(a^x+1\right)}{a^x-1}$ (c) $f(x)=\sin x+\cos x$

(d) $f(x)=x\sin^2x-x^3$ (e) $f(x)=\sin x-\cos 2x$ (f) $f(x)=\frac{\left(1+2^x\right)^2}{2^x}$

(g) $f(x)=\frac{x}{e^x-1}+\frac{x}{2}+1$ (h) $f(x)=[(x+1)^2]^{1/3}+[(x-1)^2]^{1/3}$

(ii) If f is an even function defined on the interval (–5, 5), then find the 4 real values of x satisfying the equation $f(x)=f\left(\frac{x+1}{x+2}\right)$..

Q.9 Write explicitly, functions of y defined by the following equations and also find the domains of definition of the given implicit functions :
(a) $10^x+10^y=10$ (b) $x+|y|=2y$

Q.10 Show if $f(x)=\sqrt[n]{a-x^n}$, $x>0$ $n\geq 2$, $n\in N$, then $(fof)(x)=x$. Find also the inverse of f(x).

Q.11 (a) Represent the function $f(x)=3^x$ as the sum of an even & an odd function.

(b) For what values of $p\in z$, the function $f(x)=\sqrt[n]{x^p}$, $n\in N$ is even.

Q.12 A function f defined for all real numbers is defined as follows for $x\geq 0$: $f(x)=\begin{cases}x, & 0\leq x\leq 1\\ 1, & x>1\end{cases}$
How is f defined for $x\leq 0$ if : (a) f is even (b) f is odd?

Q.13 If $f(x)=\max\left(x,\frac{1}{x}\right)$ for $x>0$ where max (a, b) denotes the greater of the two real numbers a and b.

Define the function $g(x)=f(x).f\left(\frac{1}{x}\right)$ and plot its graph.

Q.14 The function $f(x)$ has the property that for each real number x in its domain, 1/x is also in its domain and $f(x)+f\left(\frac{1}{x}\right)=x$. Find the largest set of real numbers that can be in the domain of $f(x)$?

Q.15 Compute the inverse of the functions:

(a) $f(x)=ln\left(x+\sqrt{x^2+1}\right)$ (b) $f(x)=2^{\frac{x}{x-1}}$ (c) $y=\frac{10^x-10^{-x}}{10^x+10^{-x}}$

Q.16 A function $f:\left[\frac{1}{2},\infty\right)\to\left[\frac{3}{4},\infty\right)$ defined as, $f(x)=x^2-x+1$. Then solve the equation $f(x)=f^{-1}(x)$.

Q.17 Function f & g are defined by $f(x)=\sin x$, $x\in R$; $g(x)=\tan x$, $x\in R-\left(K+\frac{1}{2}\right)\pi$
where $K\in I$. Find **(i)** periods of fog & gof. **(ii)** range of the function fog & gof.

Q.18 Find the period for each of the following functions :
(a) $f(x)=\sin^4x+\cos^4x$ (b) $f(x)=|\cos x|$ (c) $f(x)=|\sin x|+|\cos x|$

(d) $f(x)=\cos\frac{3}{5}x-\sin\frac{2}{7}x$.

Q.19 Prove that the functions ; (a) $f(x)=\cos\sqrt{x}$ (b) $f(x)=\sin\sqrt{x}$
(c) $f(x)=x+\sin x$ (d) $f(x)=\cos x^2$ are not periodic .

Q.20 Find out for what integral values of n the number 3π is a period of the function :
$f(x)=\cos nx\,.\sin(5/n)x$.

EXERCISE–2

Q.1 Let f be a one–one function with domain {x,y,z} and range {1,2,3}. It is given that exactly one of the following statements is true and the remaining two are false .
$f(x)=1$; $f(y)\neq 1$; $f(z)\neq 2$. Determine $f^{-1}(1)$

Q.2 Solve the following problems from (a) to (e) on functional equation.

(a) The function $f(x)$ defined on the real numbers has the property that $f(f(x))\cdot(1+f(x))=-f(x)$ for all x in the domain of f. If the number 3 is in the domain and range of f, compute the value of $f(3)$.

(b) Suppose f is a real function satisfying $f(x+f(x)) = 4f(x)$ and $f(1) = 4$. Find the value of $f(21)$.

(c) Let 'f' be a function defined from $R^+ \to R^+$. If $[f(xy)]^2 = x\left(f(y)\right)^2$ for all positive numbers x and y and $f(2) = 6$, find the value of $f(50)$.

(d) Let $f(x)$ be a function with two properties
(i) for any two real number x and y, $f(x+y) = x + f(y)$ and
(ii) $f(0) = 2$.
Find the value of $f(100)$.

(e) Let f be a function such that $f(3) = 1$ and $f(3x) = x + f(3x-3)$ for all x. Then find the value of $f(300)$.

Q.3(a) A function f is defined for all positive integers and satisfies $f(1) = 2005$ and $f(1)+f(2)+ \ldots + f(n) = n^2 f(n)$ for all $n > 1$. Find the value of $f(2004)$.

(b) If a, b are positive real numbers such that $a - b = 2$, then find the smallest value of the constant L for which $\sqrt{x^2+ax} - \sqrt{x^2+bx} < L$ for all $x > 0$.

(c) Let $f(x) = x^2 + kx$; k is a real number. The set of values of k for which the equation $f(x) = 0$ and $f\left(f(x)\right) = 0$ have same real solution set.

(d) If $f(2x+1) = 4x^2 + 14x$, then find the sum of the roots of the equation $f(x) = 0$.

Q.4 Let $f(x) = \dfrac{ax+b}{4x+c}$ for real a, b and c with $a \neq 0$. If the vertical asymptote of $y = f(x)$ is $x = -\dfrac{5}{4}$ and the vertical asymptote of $y = f^{-1}(x)$ is $x = \dfrac{3}{4}$, find the value(s) that b can take on.

Q.5 A function $f : R \to R$ satisfies the condition, $x^2 f(x) + f(1-x) = 2x - x^4$. Find $f(x)$ and its domain and range.

Q.6 Suppose p(x) is a polynomial with integer coefficients. The remainder when p(x) is divided by $x - 1$ is 1 and the remainder when p(x) is divided by $x - 4$ is 10. If r (x) is the remainder when p(x) is divided by $(x-1)(x-4)$, find the value of r (2006).

Q.7 Prove that the function defined as , $f(x) = \begin{cases} e^{-\sqrt{|\ln\{x\}|}} - \{x\}^{\sqrt{\frac{1}{|\ln\{x\}|}}} & \text{where ever it exists} \\ \{x\} & \text{otherwise, then} \end{cases}$

$f(x)$ is odd as well as even. (where $\{x\}$ denotes the fractional part function)

Q.8 In a function $2f(x) + xf\left(\dfrac{1}{x}\right) - 2f\left(\left|\sqrt{2}\sin\left(\pi\left(x+\dfrac{1}{4}\right)\right)\right|\right) = 4\cos^2\dfrac{\pi x}{2} + x\cos\dfrac{\pi}{x}$
Prove that (i) $f(2) + f(1/2) = 1$ and (ii) $f(2) + f(1) = 0$

Q.9 A function f, defined for all $x, y \in R$ is such that $f(1) = 2$; $f(2) = 8$ & $f(x+y) - kxy = f(x) + 2y^2$, where k is some constant. Find $f(x)$ & show that :

$$f(x+y)\, f\left(\frac{1}{x+y}\right) = k \text{ for } x + y \neq 0.$$

Q.10 Let 'f' be a real valued function defined for all real numbers x such that for some positive constant 'a' the equation $f(x+a) = \dfrac{1}{2} + \sqrt{f(x) - \left(f(x)\right)^2}$ holds for all x. Prove that the function f is periodic.

Q.11 If $f(x) = -1 + |x-2|$, $0 \le x \le 4$
$g(x) = 2 - |x|$, $-1 \le x \le 3$
Then find $fog(x)$ & $gof(x)$. Draw rough sketch of the graphs of $fog(x)$ & $gof(x)$.

Q.12 Find the domain of definition of the implicit function defined by the implicit equation,

$$3^y + 2^{x^4} = 2^{4x^2 - 1}.$$

Q.13 Let {x} & [x] denote the fractional and integral part of a real number x respectively. Solve $4\{x\} = x + [x]$

Q.14 Let $f(x) = \dfrac{9^x}{9^x + 3}$ then find the value of the sum $f\left(\dfrac{1}{2006}\right) + f\left(\dfrac{2}{2006}\right) + f\left(\dfrac{3}{2006}\right) + \ldots + f\left(\dfrac{2005}{2006}\right)$

Q.15 Let $f(x) = (x + 1)(x + 2)(x + 3)(x + 4) + 5$ where $x \in [-6, 6]$. If the range of the function is [a, b] where $a, b \in N$ then find the value of $(a + b)$.

Q.16 Find a formula for a function g (x) satisfying the following conditions
(a) domain of g is $(-\infty, \infty)$
(b) range of g is $[-2, 8]$
(c) g has a period π and
(d) $g(2) = 3$

Q.17 The set of real values of 'x' satisfying the equality $\left[\dfrac{3}{x}\right] + \left[\dfrac{4}{x}\right] = 5$ (where [ ] denotes the greatest integer function) belongs to the interval $\left(a, \dfrac{b}{c}\right]$ where $a, b, c \in N$ and $\dfrac{b}{c}$ is in its lowest form. Find the value of $a + b + c + abc$.

Q.18 Find the set of real x for which the function $f(x) = \dfrac{1}{\left[|x - 1|\right] + \left[|12 - x|\right] - 11}$ is not defined, where [x] denotes the greatest integer function.

Q.19 A is a point on the circumference of a circle. Chords AB and AC divide the area of the circle into three equal parts . If the angle BAC is the root of the equation, $f(x) = 0$ then find $f(x)$.

Q.20 If for all real values of u & v, $2f(u)\cos v = f(u + v) + f(u - v)$, prove that, for all real values of x
(i) $f(x) + f(-x) = 2a\cos x$ (ii) $f(\pi - x) + f(-x) = 0$
(iii) $f(\pi - x) + f(x) = -2b\sin x$. Deduce that $f(x) = a\cos x - b\sin x$, a, b are arbitrary constants.

EXERCISE–3

Q.1 If the functions f, g, h are defined from the set of real numbers R to R such that ;

$f(x) = x^2 - 1$, $g(x) = \sqrt{x^2 + 1}$, $h(x) = \begin{cases} 0, & \text{if } x \le 0 \\ x, & \text{if } x \ge 0 \end{cases}$; then find the composite function ho(fog) & determine whether the function (fog) is invertible & the function h is the identity function. [REE '97, 6]

Q.2(a) If $g(f(x)) = |\sin x|$ & $f(g(x)) = \left(\sin\sqrt{x}\right)^2$, then :

(A) $f(x) = \sin^2 x$, $g(x) = \sqrt{x}$ (B) $f(x) = \sin x$, $g(x) = |x|$

(C) $f(x) = x^2$, $g(x) = \sin\sqrt{x}$ (D) f & g cannot be determined

(b) If $f(x) = 3x - 5$, then $f^{-1}(x)$

(A) is given by $\dfrac{1}{3x - 5}$ (B) is given by $\dfrac{x + 5}{3}$

(C) does not exist because f is not one–one (D) does not exist because f is not onto

[JEE'98, 2 + 2]

Q.3 If the functions f & g are defined from the set of real numbers R to R such that $f(x) = e^x$, $g(x) = 3x - 2$, then find functions fog & gof. Also find the domains of functions $(fog)^{-1}$ & $(gof)^{-1}$.
[REE '98, 6]

Q.4 If the function $f : [1, \infty) \to [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$, then $f^{-1}(x)$ is : [JEE '99, 2]

(A) $\left(\dfrac{1}{2}\right)^{x(x-1)}$ (B) $\dfrac{1}{2}\left(1 + \sqrt{1 + 4\log_2 x}\right)$ (C) $\dfrac{1}{2}\left(1 - \sqrt{1 + 4\log_2 x}\right)$ (D) not defined

Q.5 The domain of definition of the function, $y(x)$ given by the equation, $2^x + 2^y = 2$ is :

(A) $0 < x \le 1$ (B) $0 \le x \le 1$ (C) $-\infty < x \le 0$ (D) $-\infty < x < 1$

Q.6 Given $x = \{1, 2, 3, 4\}$, find all one–one, onto mappings, $f : X \to X$ such that, $f(1) = 1$, $f(2) \ne 2$ and $f(4) \ne 4$. [REE 2000, 3 out of 100]

Q.7(a) Let $g(x) = 1 + x - [x]$ & $f(x) = \begin{cases} -1 & , \ x < 0 \\ 0 & , \ x = 0 \\ 1 & , \ x > 0 \end{cases}$. Then for all x, $f(g(x))$ is equal to

(A) x (B) 1 (C) $f(x)$ (D) $g(x)$

(b) If $f : [1, \infty) \to [2, \infty)$ is given by, $f(x) = x + \frac{1}{x}$, then $f^{-1}(x)$ equals

(A) $\frac{x + \sqrt{x^2 - 4}}{2}$ (B) $\frac{x}{1 + x^2}$ (C) $\frac{x - \sqrt{x^2 - 4}}{2}$ (D) $1 - \sqrt{x^2 - 4}$

(c) The domain of definition of $f(x) = \frac{\log_2 (x + 3)}{x^2 + 3x + 2}$ is :

(A) $R \setminus \{-1, -2\}$ (B) $(-2, \infty)$ (C) $R \setminus \{-1, -2, -3\}$ (D) $(-3, \infty) \setminus \{-1, -2\}$

(d) Let $E = \{1, 2, 3, 4\}$ & $F = \{1, 2\}$. Then the number of onto functions from E to F is

(A) 14 (B) 16 (C) 12 (D) 8

(e) Let $f(x) = \frac{\alpha x}{x + 1}$, $x \ne -1$. Then for what value of α is $f(f(x)) = x$?

(A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) 1 (D) -1.

Q.8(a) Suppose $f(x) = (x + 1)^2$ for $x \ge -1$. If $g(x)$ is the function whose graph is the reflection of the graph of $f(x)$ with respect to the line $y = x$, then $g(x)$ equals

(A) $-\sqrt{x} - 1, x \ge 0$ (B) $\frac{1}{(x+1)^2}, x \ge -1$ (C) $\sqrt{x+1}, x \ge -1$ (D) $\sqrt{x} - 1, x \ge 0$

(b) Let function $f : R \to R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$. Then f is

(A) one to one and onto (B) one to one but NOT onto

(C) onto but NOT one to one (D) neither one to one nor onto

Q.9(a) Range of the function $f(x) = \frac{x^2 + x + 2}{x^2 + x + 1}$ is

(A) $[1, 2]$ (B) $[1, \infty)$ (C) $\left[2, \frac{7}{3}\right]$ (D) $\left(1, \frac{7}{3}\right]$

(b) Let $f(x) = \frac{x}{1 + x}$ defined from $(0, \infty) \to [0, \infty)$ then by $f(x)$ is

(A) one- one but not onto (B) one- one and onto

(C) Many one but not onto (D) Many one and onto [JEE 2003 (Scr),3+3]

Q.10 Let $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$. Thus $g(f(x))$ is invertible for $x \in$

(A) $\left[-\frac{\pi}{2}, 0\right]$ (B) $\left[-\frac{\pi}{2}, \pi\right]$ (C) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ (D) $\left[0, \frac{\pi}{2}\right]$ [JEE 2004 (Screening)]

Q.11(a) If the functions $f(x)$ and $g(x)$ are defined on $R \to R$ such that

$f(x) = \begin{cases} 0, & x \in \text{rational} \\ x, & x \in \text{irrational} \end{cases}$, $g(x) = \begin{cases} 0, & x \in \text{irrational} \\ x, & x \in \text{rational} \end{cases}$

then $(f - g)(x)$ is

(A) one-one and onto (B) neither one-one nor onto (C) one-one but not onto (D) onto but not one-one

(b) X and Y are two sets and $f : X \to Y$. If $\{f(c) = y; c \subset X, y \subset Y\}$ and $\{f^{-1}(d) = x; d \subset Y, x \subset X\}$, then the true statement is

(A) $f(f^{-1}(b)) = b$ (B) $f^{-1}(f(a)) = a$

(C) $f(f^{-1}(b)) = b, b \subset y$ (D) $f^{-1}(f(a)) = a, a \subset x$ [JEE 2005 (Scr.)]

ANSWER KEY
FUNCTIONS
EXERCISE–1

Q 1. **(i)** $\left[-\frac{5\pi}{4},\frac{-3\pi}{4}\right]\cup\left[-\frac{\pi}{4},\frac{\pi}{4}\right]\cup\left[\frac{3\pi}{4},\frac{5\pi}{4}\right]$ **(ii)** $\left(-4,-\frac{1}{2}\right)\cup(2,\infty)$ **(iii)** $(-\infty,-3]$

(iv) $(-\infty,-1)\cup[0,\infty)$ **(v)** $(3-2\pi<x<3-\pi)\ U\ (3<x\le 4)$ **(vi)** $\left(0,\frac{1}{100}\right)\cup\left(\frac{1}{100},\frac{1}{\sqrt{10}}\right)$

(vii) $(-1<x<-1/2)\ U\ (x>1)$ **(viii)** $\left[\frac{1-\sqrt{5}}{2},0\right)\cup\left[\frac{1+\sqrt{5}}{2},\infty\right)$ **(ix)** $(-3,-1]\ U\ \{0\}\ U\ [1,3)$

(x) $\{4\}\cup[5,\infty)$ **(xi)** $(0,1/4)\ U\ (3/4,1)\ U\ \{x : x\in N, x\ge 2\}$ **(xii)** $\left(-\frac{1}{6},\frac{\pi}{3}\right]\cup\left[\frac{5\pi}{3},6\right)$

(xiii) $[-3,-2)\cup[3,4)$ **(xiv)** ϕ

(xv) $2K\pi<x<(2K+1)\pi$ but $x\ne 1$ where K is non–negative integer

(xvi) $\{x\mid 1000\le x<10000\}$ **(xvii)** $(-2,-1)\cup(-1,0)\cup(1,2)$ **(xviii)** $(1,2)\cup\left(2,\frac{5}{2}\right)$

(xix) $(-\infty,-3)\cup(-3,1]\cup[4,\infty)$

Q 2.

(i) $D : x\in R$ $R : [0,2]$ **(ii)** $D = R$; range $[-1,1]$

(iii) $D : \{x\mid x\in R;\ x\ne -3;\ x\ne 2\}$ $R : \{f(x)\mid f(x)\in R, f(x)\ne 1/5;\ f(x)\ne 1\}$

(iv) $D : R$; $R : (-1,1)$ **(v)** $D : -1\le x\le 2$ $R : \left[\sqrt{3},\sqrt{6}\right]$

(vi) $D : x\in(2n\pi,(2n+1)\pi)-\left\{2n\pi+\frac{\pi}{6},2n\pi+\frac{\pi}{2},2n\pi+\frac{5\pi}{6}, n\in I\right\}$ and

$R : \log_a 2$; $a\in(0,\infty)-\{1\}\Rightarrow$ Range is $(-\infty,\infty)-\{0\}$

(vii) $D : [-4,\infty)-\{5\}$; $R : \left(0,\frac{1}{6}\right)\cup\left(\frac{1}{6},\frac{1}{3}\right]$

Q.4 **(a)** neither surjective nor injective **(b)** surjective but not injective

(c) neither injective nor surjective

Q.5 $f_{3n}(x) = x$; Domain $= R-\{0,1\}$

Q.6 1 **Q.7** **(a)** $2K\pi\le x\le 2K\pi+\pi$ where $K\in I$ **(b)** $[-3/2,-1]$

Q.8 **(i)** (a) odd, (b) even, (c) neither odd nor even, (d) odd, (e) neither odd nor even, (f) even, (g) even, (h) even; **(ii)** $\frac{-1+\sqrt{5}}{2},\frac{-1-\sqrt{5}}{2},\frac{-3+\sqrt{5}}{2},\frac{-3-\sqrt{5}}{2}$

Q.9 **(a)** $y=\log(10-10^x), -\infty<x<1$

(b) $y = x/3$ when $-\infty<x<0$ & $y = x$ when $0\le x<+\infty$

Q.10 $f^{-1}(x)=(a-x^n)^{1/n}$

Q.12 **(a)** $f(x) = 1$ for $x<-1$ & $-x$ for $-1\le x\le 0$; **(b)** $f(x) = -1$ for $x<-1$ and x for $-1\le x\le 0$

Q.13 $g(x)=\begin{cases}\frac{1}{x^2} & \text{if } 0<x\le 1\\ x^2 & \text{if } x>1\end{cases}$ **Q.14** $\{-1,1\}$

Q.15 **(a)** $\frac{e^x-e^{-x}}{2}$; **(b)** $\frac{\log_2 x}{\log_2 x-1}$; **(c)** $\frac{1}{2}\log\frac{1+x}{1-x}$ **Q.16** $x = 1$

Q.17 **(i)** period of fog is π, period of gof is 2π; **(ii)** range of fog is $[-1,1]$, range of gof is $[-\tan1,\tan1]$

Q.18 **(a)** $\pi/2$ **(b)** π **(c)** $\pi/2$ **(d)** 70π
Q.20 $\pm 1, \pm 3, \pm 5, \pm 15$

EXERCISE–2

Q 1. $f^{-1}(1) = y$
Q.2 (a) – 3/4, (b) 64, (c) 30, (d) 102, (e) 5050

Q.3 (a) $\frac{1}{1002}$, (b) 1, (c) [0, 4), (d) – 5

Q 4. b can be any real number except $\frac{15}{4}$ **Q5.** $f(x) = 1 - x^2$, D = $x \in R$; range $= (-\infty, 1]$

Q.6 6016 **Q 9.** $f(x) = 2x^2$

Q 11. $fog(x) = \begin{matrix} -(1+x) & , & -1 \le x \le 0 \\ x-1 & , & 0 < x \le 2 \end{matrix}$; $gof(x) = \begin{matrix} x+1 & , & 0 \le x < 1 \\ 3-x & , & 1 \le x \le 2 \\ x-1 & , & 2 < x \le 3 \\ 5-x & , & 3 < x \le 4 \end{matrix}$;

$fof(x) = \begin{matrix} x & , & 0 \le x \le 1 \\ 4-x & , & 3 \le x \le 4 \end{matrix}$; $gog(x) = \begin{matrix} -x & , & -1 \le x \le 0 \\ x & , & 0 < x \le 2 \\ 4-x & , & 2 < x \le 3 \end{matrix}$

Q 12. $\left(-\frac{\sqrt{3}+1}{\sqrt{2}}, \frac{1-\sqrt{3}}{\sqrt{2}}\right) \cup \left(\frac{\sqrt{3}-1}{\sqrt{2}}, \frac{\sqrt{3}+1}{\sqrt{2}}\right)$ **Q.13** x = 0 or 5/3

Q.14 1002.5 **Q.15** 5049 **Q.16** $g(x) = 3 + 5\sin(n\pi + 2x - 4), n \in I$

Q.17 20 **Q 18.** $(0, 1) \cup \{1, 2, ..., 12\} \cup (12, 13)$ **Q 19.** $f(x) = \sin x + x - \frac{\pi}{3}$

EXERCISE–3

Q.1 $(hofog)(x) = h(x^2) = x^2$ for $x \in R$, Hence h is not an identity function , fog is not invertible
Q.2 **(a)** A, **(b)** B
Q.3 $(fog)(x) = e^{3x-2}$; $(gof)(x) = 3e^x - 2$;
Domain of $(fog)^{-1}$ = range of fog = $(0, \infty)$; Domain of $(gof)^{-1}$ = range of gof = $(-2, \infty)$
Q.4 B **Q.5** D
Q.6 {(1, 1), (2, 3), (3, 4), (4, 2)} ; {(1, 1), (2, 4), (3, 2), (4, 3)} and {(1, 1), (2, 4), (3, 3), (4, 2)}
Q.7 **(a)** B, **(b)** A, **(c)** D, **(d)** A, **(e)** D **Q.8** **(a)** D ; **(b)** A
Q.9 **(a)** D , **(b)** A **Q.10** C **Q.11** **(a)** A ; **(b)** D

Exercise-4

Part : (A) Only one correct option

1. The domain of the function $f(x) = \frac{\sqrt{-\log_{0.3}(x-1)}}{\sqrt{x^2+2x+8}}$ is
(A) (1, 4) (B) (– 2, 4) (C) (2, 4) (D) [2, ∞)

2. The function $f(x) = \cot^{-1}\sqrt{(x+3)x} + \cos^{-1}\sqrt{x^2+3x+1}$ is defined on the set S, where S is equal to:
(A) {0, 3} (B) (0, 3) (C) {0, – 3} (D) [– 3, 0]

3. The range of the function $f(x) = \sin^{-1}\left[x^2+\frac{1}{2}\right] + \cos^{-1}\left[x^2 - \frac{1}{2}\right]$, where [ ] is the greatest integer function, is:
(A) $\left\{\frac{\pi}{2}, \pi\right\}$ (B) $\left\{0, \frac{\pi}{2}\right\}$ (C) $\{\pi\}$ (D) $\left(0, \frac{\pi}{2}\right)$

4. Range of $f(x) = \log_{\sqrt{5}}\{\sqrt{2}(\sin x - \cos x) + 3\}$ is
(A) [0, 1] (B) [0, 2] (C) $\left[0, \frac{3}{2}\right]$ (D) none of these

5. Range of $f(x) = 4^x + 2^x + 1$ is
(A) $(0, \infty)$ (B) $(1, \infty)$ (C) $(2, \infty)$ (D) $(3, \infty)$

6. If x and y satisfy the equation $y = 2[x] + 3$ and $y = 3[x - 2]$ simultaneously, the $[x + y]$ is
(A) 21 (B) 9 (C) 30 (D) 12

7. The function $f : [2, \infty) \to Y$ defined by $f(x) = x^2 - 4x + 5$ is both one–one & onto if
(A) $Y = R$ (B) $Y = [1, \infty)$ (C) $Y = [4, \infty)$ (D) $Y = [5, \infty)$

8. Let S be the set of all triangles and R^+ be the set of positive real numbers. Then the function, $f : S \to R^+$, $f(\Delta)$ = area of the Δ, where $\Delta \in S$ is :
(A) injective but not surjective (B) surjective but not injective
(C) injective as well as surjective (D) neither injective nor surjective

9. Let f(x) be a function whose domain is $[-5, 7]$. Let $g(x) = |2x + 5|$. Then domain of (fog) (x) is
(A) $[-4, 1]$ (B) $[-5, 1]$ (C) $[-6, 1]$ (D) none of these

10. The inverse of the function $y = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$ is

(A) $\dfrac{1}{2} \log \dfrac{1+x}{1-x}$ (B) $\dfrac{1}{2} \log \dfrac{2+x}{2-x}$ (C) $\dfrac{1}{2} \log \dfrac{1-x}{1+x}$ (D) $2 \log (1 + x)$

11. The fundamental period of the function,
$f(x) = x + a - [x + b] + \sin \pi x + \cos 2\pi x + \sin 3\pi x + \cos 4\pi x + \ldots\ldots + \sin (2n - 1)\pi x$
$+ \cos 2 n\pi x$ for every $a, b \in R$ is: (where [ ] denotes the greatest integer function)
(A) 2 (B) 4 (C) 1 (D) 0

12. The period of $e^{\cos^4 \pi x + x - [x] + \cos \pi x}$ is _____(where [ ] denotes the greatest integer function)
(A) 1 (B) 2 (C) 3 (D) 4

13. If $y = f(x)$ satisfies the condition $f\left(x + \frac{1}{x}\right) = x^2 + \dfrac{1}{x^2}$ $(x \neq 0)$ then f(x) =
(A) $-x^2 + 2$ (B) $-x^2 - 2$ (C) $x^2 + 2$ (D) $x^2 - 2$

14. Given the function $f(x) = \dfrac{a^x + a^{-x}}{2}$ $(a > 0)$. If $f(x + y) + f(x - y) = k\, f(x).\, f(y)$ then k has the value equal to:
(A) 1 (B) 2 (C) 4 (D) 1/2

15. A function $f : R \to R$ satisfies the condition, $x^2 f(x) + f(1 - x) = 2x - x^4$. Then f(x) is:
(A) $-x^2 - 1$ (B) $-x^2 + 1$ (C) $x^2 - 1$ (D) $-x^4 + 1$

16. The domain of the function, $f(x) = \sqrt{\dfrac{1}{(|x| - 1)\cos^{-1}(2x + 1)\ \tan 3x}}$ is:

(A) $(-1, 0)$ (B) $(-1, 0) - \left\{-\dfrac{\pi}{6}\right\}$ (C) $(-1, 0] - \left\{-\dfrac{\pi}{6}, -\dfrac{\pi}{2}\right\}$ (D) $\left(-\dfrac{\pi}{6}, 0\right)$

17. If $f(x) = 2[x] + \cos x$, then $f: R \to R$ is: (where [ ] denotes greatest integer function)
(A) one–one and onto (B) one–one and into (C) many–one and into (D) many–one and onto

18. If $q^2 - 4pr = 0$, $p > 0$, then the domain of the function, $f(x) = \log (px^3 + (p + q)x^2 + (q + r)x + r)$ is:

(A) $R - \left\{-\dfrac{q}{2p}\right\}$ (B) $R - \left[(-\infty, -1] \cup \left\{-\dfrac{q}{2p}\right\}\right]$ (C) $R - \left[(-\infty, -1) \cap \left\{-\dfrac{q}{2p}\right\}\right]$ (D) none of these

19. If $[2 \cos x] + [\sin x] = -3$, then the range of the function, $f(x) = \sin x + \sqrt{3} \cos x$ in $[0, 2\pi]$ is:
(where [ ] denotes greatest integer function)
(A) $[-2, -1)$ (B) $(-2, -1]$ (C) $(-2, -1)$ (D) $[-2, -\sqrt{3})$

20. The domain of the function $f(x) = \log_{1/2}\left(-\log_2\left(1 + \dfrac{1}{\sqrt[4]{x}}\right) - 1\right)$ is:

(A) $0 < x < 1$ (B) $0 < x \leq 1$ (C) $x \geq 1$ (D) null set

21. The range of the functions $f(x) = \log_{\sqrt{2}}\left(2 - \log_2\left(16\sin^2 x + 1\right)\right)$ is
(A) $(-\infty, 1)$ (B) $(-\infty, 2)$ (C) $(-\infty, 1]$ (D) $(-\infty, 2]$

22. The domain of the function, $f(x) = \sin^{-1}\left(\dfrac{1 + x^3}{2x^{3/2}}\right) + \sqrt{\sin(\sin x)} + \log_{(3\{x\} + 1)}(x^2 + 1)$,
where $\{x\}$ represents fractional part function is:
(A) $x \in \{1\}$ (B) $x \in R - \{1, -1\}$ (C) $x > 3$, $x \neq 1$ (D) none of these

23. The minimum value of $f(x) = a \tan^2 x + b \cot^2 x$ equals the maximum value of $g(x) = a \sin^2 x + b \cos^2 x$ where $a > b > 0$, when
(A) $4a = b$ (B) $3a = b$ (C) $a = 3b$ (D) $a = 4b$

24. Let $f(2, 4) \to (1, 3)$ be a function defined by $f(x) = x - \left[\dfrac{x}{2}\right]$ (where [.] denotes the greatest integer function), then $f^{-1}(x)$ is equal to :

(A) $2x$ (B) $x + \left[\dfrac{x}{2}\right]$ (C) $x + 1$ (D) $x - 1$

25. The image of the interval R when the mapping $f: R \to R$ given by $f(x) = \cot^{-1}(x^2 - 4x + 3)$ is

(A) $\left[\dfrac{\pi}{4}, \dfrac{3\pi}{4}\right]$ (B) $\left[\dfrac{\pi}{4}, \pi\right)$ (C) $(0, \pi)$ (D) $\left(0, \dfrac{3\pi}{4}\right]$

26. If the graph of the function $f(x) = \frac{a^x - 1}{x^n (a^x + 1)}$ is symmetric about y-axis, then n is equal to:

(A) 2 (B) 2 / 3 (C) 1 / 4 (D) – 1 / 3

27. If $f(x) = \cot^{-1}x : R^+ \to \left(0, \frac{\pi}{2}\right)$

and $g(x) = 2x - x^2 : R \to R$. Then the range of the function f(g(x)) wherever define is

(A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left(0, \frac{\pi}{4}\right)$ (C) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right)$ (D) $\left(\frac{\pi}{4}\right)$

28. Let $f: (e^2, \infty) \to R$ be defined by $f(x) = \ell n\,(\ell n(\ell n\, x))$, then

(A) f is one one but not onto (B) f is on to but not one - one (C) f is one-one and onto (D) f is neither one-one nor onto

29. Let $f: (e, \infty) \to R$ be defined by $f(x) = \ell n\,(\ell n(\ell n\, x))$, then

(A) f is one one but not onto (B) f is on to but not one - one

(C) f is one-one and onto (D) f is neither one-one nor onto

30. Let $f(x) = \sin x$ and $g(x) = |\ell n\, x|$ if composite functions fog(x) and gof (x) are defined and have ranges R_1 & R_2 respectively then.

(A) $R_1 = \{u: -1 < u < 1\}$ $R_2 = \{v: 0 < v < \infty\}$

(B) $R_1 = \{u: -\infty < u \le 0\}$ $R_2 = \{v: -1 \le v \le 1\}$

(C) $R_1 = \{u: 0 \le u < \infty\}$ $R_2 = \{v: -1 < v < 1;\ v \ne 0\}$

(D) $R_1 = \{u: -1 \le u \le 1\}$ $R_2 = \{v: 0 \le v < \infty\}$

31. Function $f: (-\infty, 1) \to (0, e^5]$ defined by $f(x) = e^{-(x^2 - 3x + 2)}$ is

(A) many one and onto (B) many one and into (C) one one and onto (D) one one and into

32. The number of solutions of the equation $[\sin^{-1} x] = x - [x]$, where [.] denotes the greatest integer function is

(A) 0 (B) 1 (C) 2 (D) infinitely many

33. The function $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$ is

(A) an odd function (B) an even function

(C) neither an odd nor an even function (D) a periodic function

Part : (B) May have more than one options correct

34. For the function $f(x) = \ell n \quad (\sin^{-1} \ell og_2 x)$,

(A) Domain is $\left[\frac{1}{2}, 2\right]$ (B) Range is $\left(-\infty, \ell n \frac{\pi}{2}\right]$ (C) Domain is (1, 2] (D) Range is R

35. A function 'f' from the set of natural numbers to integers defined by,

$$f(n) = \begin{cases} \frac{n-1}{2} & , \text{ when n is odd} \\ -\frac{n}{2} & , \text{ when n is even} \end{cases} \text{ is:}$$

(A) one-one (B) many-one (C) onto (D) into

36. Domain of $f(x) = \sin^{-1}[2 - 4x^2]$ where [x] denotes greatest integer function is:

(A) $\left(-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right) - \{0\}$ (B) $\left[-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right] - \{0\}$ (C) $\left[-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right]$ (D) $\left(-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$

37. If $F(x) = \frac{\sin \pi [x]}{\{x\}}$, then F (x) is:

(A) periodic with fundamental period 1 (B) even

(C) range is singleton

(D) identical to $\text{sgn}\left(\text{sgn}\frac{\{x\}}{\sqrt{\{x\}}}\right) - 1$, where {x} denotes fractional part function and [ ] denotes greatest integer function and sgn (x) is a signum function.

38. $D \equiv [-1, 1]$ is the domain of the following functions, state which of them are injective.

(A) $f(x) = x^2$ (B) $g(x) = x^3$ (C) $h(x) = \sin 2x$ (D) $k(x) = \sin(\pi x/2)$

Exercise-5

1. Find the domain of the function $f(x) = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$

2. Find the domain of the function $f(x) = \sqrt{1-2x} + 3\sin^{-1}\left(\frac{3x-1}{2}\right)$

3. Find the inverse of the following functions. $f(x) = \ell n\,(x + \sqrt{1+x^2})$

4. Let $f: \left[-\frac{\pi}{3}, \frac{\pi}{6}\right] \to B$ defined by $f(x) = 2\cos^2 x + \sqrt{3}\sin 2x + 1$. Find the B such that f^{-1} exists. Also find $f^{-1}(x)$.

5. Find for what values of x, the following functions would be identical.

$f(x) = \log(x-1) - \log(x-2)$ and $g(x) = \log\left(\frac{x-1}{x-2}\right)$.

6. If $f(x) = \frac{4^x}{4^x+2}$, then show that $f(x) + f(1-x) = 1$

7. Let f(x) be a polynomial function satisfying the relation $f(x).\ f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)\ \forall\ x \in R - \{0\}$ and $f(3) = -26$. Determine $f'(1)$.

8. Find the domain of definitions of the following functions.

(i) $f(x) = \sqrt{3 - 2^x - 2^{1-x}}$ (ii) $f(x) = \sqrt{\frac{x-2}{x+2}} + \sqrt{\frac{1-x}{1+x}}$

(iii) $f(x) = \ell og_{10}(1 - \ell og_{10}(x^2 - 5x + 16))$

9. Find the range of the following functions.

(i) $f(x) = \frac{x^2-2x+4}{x^2+2x+4}$ (ii) $f(x) = \sin \ell og\left(\frac{\sqrt{4-x^2}}{1-x}\right)$

(iii) $f(x) = x^4 - 2x^2 + 5$ (iv) $f(x) = \sin^2 x + \cos^4 x$

10. Solve the following equation for x (where [x] & {x} denotes integral and fractional part of x)
$2x + 3[x] - 4\{-x\} = 4$

11. Draw the graph of following functions where [.] denotes greatest integer function and { .} denotes fractional part function.

(i) $y = \{\sin x\}$ (ii) $y = [x] + \sqrt{\{x\}}$

12. Draw the graph of the function $f(x) = \left|x^2 - 4|x| + 3\right|$ and also find the set of values of 'a' for which the equation f(x) = a has exactly four distinct real roots.

13. Examine whether the following functions are even or odd or none.

(i) $f(x) = \frac{(1+2^x)^7}{2^x}$ (ii) $f(x) = \begin{cases} x|x|, & x \le -1 \\ [1+x]+[1-x], & -1 < x < 1 \\ -x|x|, & x \ge 1 \end{cases}$

(iii) $f(x) = \frac{2x(\sin x + \tan x)}{2\left[\frac{x+2\pi}{\pi}\right] - 3}$, where [ ] denotes greatest integer function.

14. Find the period of the following functions.

(i) $f(x) = 1 - \frac{\sin^2 x}{1+\cot x} - \frac{\cos^2 x}{1+\tan x}$

(ii) $f(x) = \tan\frac{\pi}{2}[x]$, where [.] denotes greatest integer function.

(iii) $f(x) = \frac{1}{2}\left(\frac{|\sin x|}{\cos x} + \frac{\sin x}{|\cos x|}\right)$ (iv) $f(x) = \frac{\sin x + \sin 3x}{\cos x + \cos 3x}$

15. If $f(x) = \begin{cases} 1+x^2 & x \le 1 \\ x+1 & 1 < x \le 2 \end{cases}$ and $g(x) = 1 - x\ ; -2 \le x \le 1$ then define the function fog(x).

16. Find the set of real x for which the function, $f(x) = \frac{1}{[|x-1|] + [|12-x|] - 11}$ is not defined, where [x] denotes the greatest integer not greater than x.

17. Given the functions $f(x) = e^{\cos^{-1}\left(\sin\left(x+\frac{\pi}{3}\right)\right)}$, $g(x) = \text{cosec}^{-1}\left(\frac{4-2\cos x}{3}\right)$ & the function $h(x) = f(x)$ defined only for those values of x, which are common to the domains of the functions f(x) and g(x). Calculate the range of the function h(x).

18. Let 'f' be a real valued function defined for all real numbers x such that for some positive constant 'a' the equation $f(x+a) = \frac{1}{2} + \sqrt{f(x) - (f(x))^2}$ holds for all x. Prove that the function f is periodic.

19. If $f(x) = -1 + |x-2|,\ 0 \le x \le 4$
$g(x) = 2 - |x|,\ -1 \le x \le 3$

Then find fog (x), gof (x), fof(x) & gog(x). Draw rough sketch of the graphs of fog (x) & gof (x).

20. Find the integral solutions to the equation [x] [y] = x + y. Show that all the non-integral solutions lie on exactly two lines. Determine these lines. Here [.] denotes greatest integer function.

Exercise-4

1.	D	**2.**	C	**3.**	C	**4.**	B	**5.**	B	**6.**	C	**7.**	B
8.	B	**9.**	C	**10.**	A	**11.**	A	**12.**	B	**13.**	D	**14.**	B
15.	B	**16.**	D	**17.**	C	**18.**	B	**19.**	D	**20.**	D	**21.**	D
22.	D	**23.**	D	**24.**	C	**25.**	D	**26.**	D	**27.**	C	**28.**	A
29.	C	**30.**	D	**31.**	D	**32.**	B	**33.**	B	**34.**	BC		
35.	AC	**36.**	B	**37.**	ABCD			**38.**	BD				

Exercise-5

1. $[-2, 0) \cup (0, 1)$ **2.** $\left[-\frac{1}{3}, \frac{1}{2}\right]$

3. $f^{-1} = \frac{e^x - e^{-x}}{2}$

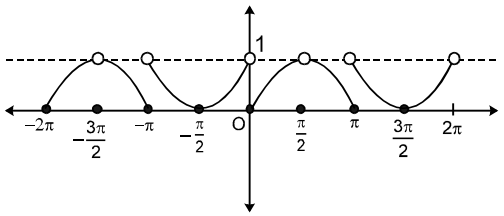
4. $B = [0, 4]$; $f^{-1}(x) = \frac{1}{2}\left(\sin^{-1}\left(\frac{x-2}{2}\right) - \frac{\pi}{6}\right)$

5. $(2, \infty)$ **7.** -3 **8.** (i) $[0, 1]$ (ii) ϕ (iii) $(2, 3)$

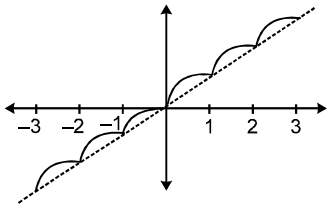
9. (i) $\left[\frac{1}{3}, 3\right]$ (ii) $[-1, 1]$ (iii) $[4, \infty)$ (iv) $\left[\frac{3}{4}, 1\right]$

10. $\left\{\frac{3}{2}\right\}$

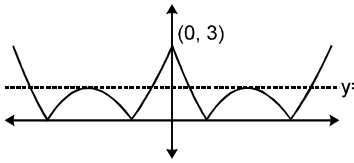
11. (i)



(ii)



12.



$a \in (1, 3) \cup \{0\}$

13. (i) neither even nor odd (ii) even (iii) odd

14. (i) π (ii) 2 (iii) 2π (iv) π

15. $f(g(x)) = \begin{cases} 2 - 2x + x^2 & 0 \le x \le 1 \\ 2 - x & -1 \le x < 0 \end{cases}$

16. (0, 1) U {1, 2,....., 12} U (12, 13) **17.** $\left[e^{\frac{\pi}{6}}, e^{\pi}\right]$

18. Period 2 a

19. $\text{fog}(x) = \begin{cases} -(1+x) & , \ -1 \le x \le 0 \\ x - 1 & , \ 0 < x \le 2 \end{cases}$;

$$\text{gof}(x) = \begin{cases} x+1 & , \ 0 \le x < 1 \\ 3 - x & , \ 1 \le x \le 2 \\ x - 1 & , \ 2 < x \le 3 \\ 5 - x & , \ 3 < x \le 4 \end{cases}$$

$$\text{fof}(x) = \begin{cases} x & , \ 0 \le x \le 2 \\ 4 - x & , \ 2 < x \le 2 \end{cases};$$

$$\text{gog}(x) = \begin{cases} -x & , \ -1 \le x \le 0 \\ x & , \ 0 < x \le 2 \\ 4 - x & , \ 2 < x \le 3 \end{cases}$$

20. Integral solution (0, 0); (2, 2). x + y = 6, x + y = 0

FUNCTIONS (ASSERTION AND REASON)

Some questions (Assertion–Reason type) are given below. Each question contains **Statement – 1 (Assertion)** and **Statement – 2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. So select the correct choice :

Choices are :

(A) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is** a correct explanation for **Statement – 1**.

(B) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is NOT** a correct explanation for **Statement – 1**.

(C) **Statement – 1** is True, **Statement – 2** is False.

(D) **Statement – 1** is False, **Statement – 2** is True.

1. Let $f(x) = \cos 3\pi x + \sin \sqrt{3}\pi x$.

Statement – 1 : f(x) is not a periodic function.

Statement – 2 : L.C.M. of rational and irrational does not exist

2. **Statement – 1:** If $f(x) = ax + b$ and the equation $f(x) = f^{-1}(x)$ is satisfied by every real value of x, then $a \in R$ and $b = -1$.

Statement – 2: If $f(x) = ax + b$ and the equation $f(x) = f^{-1}(x)$ is satisfied by every real value of x, then $a = -1$ and $b \in R$.

3. **Statements-1:** If $f(x) = x$ and $F(x) = \frac{x^2}{x}$, then F(x) = f(x) always

Statements-2: At x = 0, F(x) is not defined.

4. **Statement–1 :** If $f(x) = \frac{1}{1-x}$, $x \neq 0, 1$, then the graph of the function $y = f(f(f(x)), x > 1$ is a straight line **Statement–2 :** f(f(x))) = x

5. Let f(1 + x) = f(1 – x) and f(4 + x) = f(4– x)

Statement–1 : f(x) is periodic with period 6

Statement–2 : 6 is not necessarily fundamental period of f(x)

6. **Statement–1 :** Period of the function $f(x) = \sqrt{1+\sin 2x} + e^{\{x\}}$ does not exist

Statement–2 : LCM of rational and irrational does not exist

7. **Statement–1 :** Domain of $f(x) = \frac{1}{\sqrt{|x|-x}}$ is $(-\infty, 0)$ **Statement–2:** $|x| - x > 0$ for $x \in R^-$

8. **Statement–1 :** Range of $f(x) = \sqrt{4-x^2}$ is [0, 2]

Statement–2 : f(x) is increasing for $0 \le x \le 2$ and decreasing for $-2 \le x \le 0$.

9. Let $a, b \in R$, $a \neq b$ and let $f(x) = \frac{a+x}{b+x}$.

Statement–1 : f is a one–one function. **Statement–2** : Range of f is R – {1}

10. **Statement–1** : sin x + cos (πx) is a non–periodic function.

Statement–2 : Least common multiple of the periods of sin x and cos (πx) is an irrational number.

11. **Statement–1:** The graph of f(x) is symmetrical about the line x = 1, then, f(1 + x) = f(1 – x).

Statement–2 : even functions are symmetric about the y-axis.

12. **Statement–1 :** Period of $f(x) = \sin\frac{\pi x}{(n-1)!} + \cos\frac{\pi x}{n!}$ is 2(n)!

Statement–2 : Period of $|\cos x| + |\sin x| + 3$ is π.

13. **Statement–1 :** Number of solutions of $\tan(|\tan^{-1}x|) = \cos|x|$ equals 2 **Statement–2** : ?

14. **Statement–1 :** Graph of an even function is symmetrical about y–axis

Statement–2 : If f(x) = cosx has x (+)ve solution then total number of solution of the above equation is 2n. (when f(x) is continuous even function).

15. If f is a polynomial function satisfying $2 + f(x).f(y) = f(x) + f(y) + f(xy)\ \forall\ x, y \in R$

Statement-1: f(2) = 5 which implies f(5) = 26

Statement-2: If f(x) is a polynomial of degree 'n' satisfying f(x) + f(1/x) = f(x). f(1/x), then

$f(x) = 1\ x^n + 1$

16. **Statement-1:** The range of the function $\sin^{-1} + \cos^{-1}x + \tan^{-1}x$ is $[\pi/4, 3\pi/4]$

Statement-2: $\sin^{-1}x$, $\cos^{-1}x$ are defined for $|x| \le 1$ and $\tan^{-1}x$ is defined for all 'x'.

17. A function f(x) is defined as $f(x) = \begin{cases} 0 & \text{where x is rational} \\ 1 & \text{where x is irrational} \end{cases}$

Statement-1 : f(x) is discontinuous at xll $x \in R$

Statement-2 : In the neighbourhood of any rational number there are irrational numbers and in the vincity of any irrational number there are rational numbers.

18. Let $f(x) = \sin\left(2\sqrt{3}\,\pi x\right) + \cos\left(3\sqrt{3}\,\pi x\right)$

Statement-1 : f(x) is a periodic function

Statement-2: LCM of two irrational numbers of two similar kind exists.

19. **Statements-1:** The domain of the function $f(x) = \cos^{-1}x + \tan^{-1}x + \sin^{-1}x$ is [-1, 1]

Statements-2: $\sin^{-1}x$, $\cos^{-1}x$ are defined for $|x| \le 1$and $\tan^{-1}x$ is defined for all x.

20. **Statement-1 :** The period of f(x) = = sin2x cos [2x] – cos2x sin [2x] is 1/2

Statements-2: The period of x – [x] is 1, where [·] denotes greatest integer function.

21. **Statements-1:** If the function $f : R \to R$ be such that f(x) = x – [x], where [·] denotes the greatest integer less than or equal to x, then $f^{-1}(x)$ is equals to [x] + x

Statements-2: Function 'f ' is invertible iff is one-one and onto.

22. **Statements-1 :** Period of $f(x) = \sin 4\pi\{x\} + \tan\pi[x]$ were, [·] & {·} denote we G.I.F. & fractional part respectively is 1.

Statements-2: A function f(x) is said to be periodic if there exist a positive number T independent of x such that f(T + x) = f(x). The smallest such positive value of T is called the period or fundamental period.

23. **Statements-1:** $f(x) = \frac{x+1}{x-1}$ is one-one function

Statements-2: $\frac{x+1}{x-1}$ is monotonically decreasing function and every decreasing function is one-one.

24. **Statements-1:** f(x) = sin2x (|sinx| - |cosx|) is periodic with fundamental period $\pi/2$

Statements-2: When two or more than two functions are given in subtraction or multiplication form we take the L.C.M. of fundamental periods of all the functions to find the period.

25. **Statements-1:** $e^x = \ln x$ has one solution.

Statements-2: If $f(x) = x \Rightarrow f(x) = f^{-1}(x)$ have a solution on $y = x$.

26. **Statements-1:** $F(x) = x + \sin x$. $G(x) = -x$

$H(x) = F(X) + G(x)$, is a periodic function.

Statements-2: If $F(x)$ is a non-periodic function & $g(x)$ is a non-periodic function then $h(x) = f(x) \pm g(x)$ will be a periodic function.

27. **Statements-1:** $f(x) = \begin{cases} x+1, & x \geq 0 \\ x-1, & x < 0 \end{cases}$ is an odd function.

Statements-2: If $y = f(x)$ is an odd function and $x = 0$ lies in the domain of $f(x)$ then $f(0) = 0$

28. **Statements-1:** $f(x) = \begin{cases} x; & x \in Q \\ -x; & x \in Q^C \end{cases}$ is one to one and non-monotonic function.

Statements-2: Every one to one function is monotonic.

29. **Statement–1 :** Let $f : [1, 2] \cup [5, 6] \to [1, 2] \cup [5, 6]$ defined as $f(x) = \begin{cases} x+4, & x \in [1, 2] \\ -x+7, & x \in [5, 6] \end{cases}$ then the equation $f(x) = f^{-1}(x)$ has two solutions.

Statements-2: $f(x) = f^{-1}(x)$ has solutions only on $y = x$ line.

30. **Statements-1:** The function $\frac{px+q}{rx+s}$ $(ps - qr \neq 0)$ cannot attain the value p/r.

Statements-2: The domain of the function $g(y) = \frac{q-sy}{ry-p}$ is all real except a/c.

31. **Statements-1:** The period of $f(x) = \sin [2] x\cos [2x] - \cos 2x \sin [2x]$ is 1/2

Statements-2: The period of $x - [x]$ is 1.

32. **Statements-1:** If f is even function, g is odd function then $\frac{b}{g}$ $(g \neq 0)$ is an odd function.

Statements-2: If $f(-x) = -f(x)$ for every x of its domain, then $f(x)$ is called an odd function and if $f(-x) = f(x)$ for every x of its domain, then $f(x)$ is called an even function.

33. **Statements-1:** $f : A \to B$ and $g : B \to C$ are two function then $(gof)^{-1} = f^{-1} og^{-1}$.

Statements-2: $f : A \to B$ and $g : B \to C$ are bijections then f^{-1} & g^{-1} are also bijections.

34. **Statements-1:** The domain of the function $f(x) = \sqrt{\log_2 \sin x}$ is $(4n + 1)\frac{\pi}{2}, n \in N$.

Statements-2: Expression under even root should be ≥ 0

35. **Statements-1:** The function $f : R \to R$ given $f(x) = \log_a(x + \sqrt{x^2+1})$ $a > 0, a \neq 1$ is invertible.

Statements-2: f is many one into.

36. **Statements-1:** $\phi(x) = \sin (\cos x)$ $x \in \left[0, \frac{\pi}{2}\right]$ is a one-one function.

Statements-2: $\phi'(x) \le \forall x \in \left[0, \frac{\pi}{2}\right]$

37. **Statements-1:** For the equation $kx^2 + (2 - k)x + 1 = 0$ $k \in R - \{0\}$ exactly one root lie in (0, 1).

Statements-2: If $f(k_1)\, f(k_2) < 0$ (f(x) is a polynomial) then exactly one root of f(x) = 0 lie in (k_1, k_2).

38. **Statements-1:** Domain of $f(x) = \sin^{-1}\left(\frac{1+x^2}{2x}\right)$ is $\{-1, 1\}$

Statements-2: $x + \frac{1}{x} \ge 2$ when $x > 0$ and $x + \frac{1}{x} \le -2$ when $x < 0$.

39. **Statements-1:** Range of $f(x) = |x|(|x| + 2) + 3$ is $[3, \infty)$

Statements-2: If a function f(x) is defined $\forall\ x \in R$ and for $x \ge 0$ if $a \le f(x) \le b$ and f(x) is even function than range of f(x) f(x) is [a, b].

40. **Statements-1:** Period of $\{x\} = 1$. **Statements-2:** Period of $[x] = 1$

41. **Statements-1:** Domain of $f = \phi$. If $f(x) = \frac{1}{\sqrt{[x]-x}}$

Statements-2: $[x] \le x\ \forall\ x \in R$

42. **Statements-1:** The domain of the function $\sin^{-1}x + \cos^{-1}x + \tan^{-1}x$ is $[-1, 1]$

Statements-2: $\sin^{-1}x$, $\cos^{-1}x$ are defined for $|x| \le 1$ and $\tan^{-1}x$ is defined for all 'x'

ANSWER KEY

1. A	**2. D**	**3. A**	**4. C**	**5. A**	**6. A**	**7. A**
8. C	**9. B**	**10. C**	**11. A**	**12. C**	**13. B**	**14. A**
15. A	**16. A**	**17. A**	**18. A**	**19. A**	**20. A**	**21. D**
22. A	**23. A**	**24. A**	**25. D**	**26. C**	**27. D**	**28. C**
29. C	**30. A**	**31. A**	**32. A**	**33. D**	**34. A**	**35. C**
36. A	**37. C**	**38. A**	**39. A**	**40. A**	**41. A**	**42. A**

SOLUTIONS

4. $f(f(x)) = \frac{1}{1-f(x)} = \frac{1}{1-\frac{1}{1-x}} = \frac{x-1}{x}$

$\therefore\ f(f(f(x))) = \frac{1}{1-f(f(x))} = \frac{1}{1-\frac{x-1}{x}} = x$ Ans. C

5. $f(1 + x) = f(1 - x)$ ... (1)

$f(4 + x) = f(4 - x)$ ... (2)

$x \to 1 - x$ in (1) $\Rightarrow$ $f(1 - x) = f(x)$ ... (3)

$x \to 4 - x$ in (2) $\Rightarrow$ $f(2 - x)\, f(8 - x) = f(x)$ ... (4)

(1) and (4) $\Rightarrow$ f(2 – x) = f(8 – x) .... (5)

Use x $\rightarrow$ x – x in (5), we get

f(x) = f(6 + x)

$\Rightarrow$ f(x) is periodic with period 6

Obviously 6 is not necessary the fundamental period. Ans. A

6. L.C.M. of $\{\pi, 1\}$ does not exist $\therefore$ (A) is the correct option.

7. (a) Clearly both are true and statement – II is correct explantion of Statement – I .

8. (c) $f'(x) = \frac{-x}{\sqrt{4-x^2}}$ $\therefore$ f(x) is increasing for $-2 \le x \le 0$ and decreasing for $0 \le x \le 2$.

9. Suppose a > b. Statement – II is true as $f'(x) = \frac{b-a}{(b+x)^2}$, which is always negative and hence monotonic in its continuous part. Also $\lim_{x \to -b^+} f(x) = \infty$ and $\lim_{x \to -b^-} f(x) = -\infty$. Moreover $\lim_{x \to \infty} f(x) = 1+$ and $\lim_{x \to -\infty} f(x) = -1-$. Hence range of f is R – {1}.

F is obviously one–one as $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.

However statement – II is not a correct reasoning for statement – I

Hence (b) is the correct answer.

10. Statement – I is true, as period of sin x and cos πx are 2π and 2 respectively whose L.C.M does not exist.

Obviously statement – II is false Hence (c) is the correct answer.

11. Graph of f(x) is symmetric about the line x = 0 if f(- x) = f(x) i.e. if f(0 – x) = f(0 + x)

$\therefore$ Graph of y = f(x) is symmetric about x = 1, if f(1 + x) = f(1 – x).

Hence (a) is the correct answer.

12. Period of $\sin \frac{\pi x}{(n-1)!} = 2(n-1)!$ Period of $\cos \frac{\pi x}{n!} = 2(n)!$

$\Rightarrow$ Period of f(x) = L.C.M of 2(n – 1)! And 2(n)! = 2(n!)

Now, $f(x) = |\cos x| + |\sin x| + 3 = \sqrt{1+|\sin 2x|} + 3$

$\therefore$ f(x) is periodic function with period = $\frac{\pi}{2}$. Hence (c) is the correct answer.

13. $\tan(|\tan^{-1}x|) = |x|$, since $|\tan^{-1}x| = \tan^{-1}|x|$

Obviously cos|x| and |x| meets at exactly two points

$\therefore$ (B) is the correct option.

14. (A)Since cos n is also even function. Therefore solution of cosx = f(x) is always sym. also out y–axis.

19. (a) Both A and R are obviously correct. **20.** (a) f(x) = x [x]

f(x + 1) = x + 1 – ([x] + 1) = x – [x]

So, period of x – [x] is 1.

Let f(x) = sin (2x – [2x])

$f\left(x+\frac{1}{2}\right)=\sin\left(2\left(x+\frac{1}{2}\right)-\left[2\left(x+\frac{1}{2}\right)\right]\right)$

$= \sin(2x + 1 - [2x] - 1)$ $= \sin(2x - [2x])$ So, period is 1/2

21. $f(1) = 1 - 1 = 0$ $f(0) = 0$

$\therefore$ f is not one-one $\therefore f^{-1}(x)$ is not defined Ans. (D)

22. Clearly $\tan \pi[x] = 0 \ \forall \ x \in R$ and period of $\sin 4\pi\{x\} = 1$. Ans. (A)

23. $f(x) = \frac{x+1}{x-1}$ $f'(x) = \frac{(x-1)-(x+1)}{(x-1)^2} = \frac{-2}{(x-1)^2} < 0$

So f(x) is monotonically decreasing & every monotonic function is one-one.

So 'a' is correct.

24. $f(x) = \sin 2x\,(|\sin x| - |\cos x|)$ is periodic with period $\pi/2$ because $f(\pi/2 + x) = \sin 2(\pi/2 + x)\,(|\sin(\pi/2 + x)| - |\cos(\pi/2 + x)|)$

$= \sin(\pi + 2x)\,(|\cos x| - |\sin x|)$ $= -\sin 2x\,(|\cos x| - |\sin x|)$

$= \sin 2x\,(|\sin x| - |\cos x|)$

Sometimes $f(x + r) = f(x)$ where r is less than the L.C.M. of periods of all the function, but according to definition of periodicity, period must be least and positive, so 'r' is the fundamental period.

So 'f' is correct.

27. (D) If f(x) is an odd function, then $f(x) + f(-x) = 0 \ \forall \ x \in D_f$

28. (C) For one to one function if $x_1 \neq x_2$

$\Rightarrow f(x_1) \neq f(x_2)$ for all $x_1, x_2 \in D_f$ $\sqrt{3} > 1$

but $f(\sqrt{3}) < f(1)$ and $3 > 1$

$f(5) > f(1)$ f(x) is one-to-one but non-monotonic

29. (C) $\left(\frac{3}{2}, \frac{11}{2}\right)$ and $\left(\frac{11}{2}, \frac{3}{2}\right)$ both lie on y = f(x) then they will also lie on $y = f^{-1}(x) \Rightarrow$ there are two solutions and they do not lie on y = x.

30. If we take $y = \frac{px+q}{rx+s}$ then $x = \frac{q-sx}{rx-p} \Rightarrow$ x does not exist if y = p/r

Thus statement-1 is correct and follows from statement-2 (A)

31. $f(x) = \sin(2x - [2x]$ $f(x + 1/2) = \sin\left(2x+1-\left[2\left(x+\frac{1}{2}\right)\right]\right)$

$= \sin(2x + 1 - [2x] - 1]$ $= \sin(2x - [2x].)$ i.e., period is 1/2.

$f(x) = x - [x]$

$f(x + 1) = x + 1 - ([x] + 1) = x - [x]$ i.e., period is 1. (A)

32. (A) Let $h(x) = \frac{f(x)}{g(x)}$

$h(-x) = \frac{f(-x)}{g(-x)} = \frac{f(x)}{g(-x)} = \frac{f(x)}{-g(x)} = -h(x)$

$\therefore$ h(x) = $\frac{f}{g}$ is an odd function.

33. (D) Assertion : f : A $\to$ B, g : B $\to$ C are two functions then $(gof)^{-1} \neq f^{-1} og^{-1}$ (since functions need not posses inverses. Reason : Bijective functions are invertibles.

34. (A) for f(x) to be real $\log_2(\sin x) \geq 0$

$\Rightarrow \sin x \geq 2^o$ $\Rightarrow \sin x = 1 \Rightarrow x = (4n + 1)\frac{\pi}{2}, n \in N.$

35. (C) f is injective since $x \neq y$ $(x, y \in R)$

$\Rightarrow \log_a\left\{x+\sqrt{x^2+1}\right\} \neq \log_a\left\{y+\sqrt{y^2+1}\right\}$

$\Rightarrow f(x) \neq f(y)$

f is onto because $\log_a\left(x+\sqrt{x^2+1}\right) = y \Rightarrow x = \frac{a^y - a^{-y}}{2}$.

40. Since $\{x\} = x - [x]$

$\therefore \{x + 1\} = x + 1 - [x + 1]$

$= x + 1 - [x] - 1$ $= x - [x] = [x]$

Period of $[x] = 1$ Ans (A)

41. $f(x) = \frac{1}{\sqrt{[x]-x}}$ $[x] - x \neq 0$

$[x] \neq x \to [x] > x$ It is imposible or $[x] \leq x$

So the domain of f is ϕ because reason $[x] \leq x$ Ans. (A)

Imp. Que. From Competitive exams

1. If $f(x) = \frac{\cos^2 x + \sin^4 x}{\sin^2 x + \cos^4 x}$ for $x \in R$, then $f(2002) =$ **[EAMCET 2002]**

(a) 1 (b) 2 (c) 3 (d) 4

2. If $f : R \to R$ satisfies $f(x + y) = f(x) + f(y)$, for all $x, y \in R$ and $f(1) = 7$, then $\sum_{r=1}^{n} f(r)$ is **[AIEEE 2003]**

(a) $\frac{7n}{2}$ (b) $\frac{7(n+1)}{2}$ (c) $7n(n+1)$ (d) $\frac{7n(n+1)}{2}$

3. Suppose $f : [2, 2] \to R$ is defined by $f(x) = \begin{cases} -1 & \text{for } -2 \leq x \leq 0 \\ x - 1 & \text{for } 0 \leq x \leq 2 \end{cases}$, then $\{x \in (-2, 2) : x \leq 0$ and $f(|x|) = x\} =$

(a) $\{-1\}$ (b) $\{0\}$ (c) $\{-1/2\}$ (d) ϕ **[EAMCET 2003]**

4. If $f(x) = \text{sgn}(x^3)$, then **[DCE 2001]**

(a) f is continuous but not derivable at $x = 0$ (b) $f'(0^+) = 2$

(c) $f'(0^-) = 1$ (d) f is not derivable at $x = 0$

5. If $f : R \to R$ and $g : R \to R$ are given by $f(x) = |x|$ and $g(x) = |x|$ for each $x \in R$, then $\{x \in R : g(f(x)) \leq f(g(x))\} =$

(a) $Z \cup (-\infty, 0)$ (b) $(-\infty, 0)$ (c) Z (d) R **[EAMCET 2003]**

6. For a real number x, $[x]$ denotes the integral part of x. The value of

$\left[\frac{1}{2}\right]+\left[\frac{1}{2}+\frac{1}{100}\right]+\left[\frac{1}{2}+\frac{2}{100}\right]+....+\left[\frac{1}{2}+\frac{99}{100}\right]$ is **[IIT Screening 1994]**

(a) 49 (b) 50 (c) 48 (d) 51

7. If function $f(x)=\frac{1}{2}-\tan\left(\frac{\pi x}{2}\right)$; $(-1<x<1)$ and $g(x)=\sqrt{3+4x-4x^2}$, then the domain of *gof* is **[IIT 1990]**

(a) $(-1, 1)$ (b) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (c) $\left[-1, \frac{1}{2}\right]$ (d) $\left[-\frac{1}{2}, -1\right]$

8. The domain of the function $f(x)=\frac{1}{\log_{10}(1-x)}+\sqrt{x+2}$ is **[DCE 2000]**

(a) $]-3, -2.5[\cup]-2.5, -2[$ (b) $[-2, 0[\cup]0, 1[$ (c) $]0, 1[$ (d) None of these

9. The domain of definition of the function $y(x)$ given by $2^x+2^y=2$ is **[IIT Screening 2000; DCE 2001]**

(a) $(0, 1]$ (b) $[0, 1]$ (c) $(-\infty, 0]$ (d) $(-\infty, 1)$

10. Let $f(x)=(1+b^2)x^2+2bx+1$ and $m(b)$ the minimum value of $f(x)$ for a given b. As b varies, the range of $m(b)$ is **[IIT Screening 2001]**

(a) $[0, 1]$ (b) $\left(0, \frac{1}{2}\right]$ (c) $\left[\frac{1}{2}, 1\right]$ (d) $(0, 1]$

11. The range of the function $f(x)={}^{7-x}P_{x-3}$ is **[AIEEE 2004]**

(a) $\{1, 2, 3, 4, 5\}$ (b) $(1, 2, 3, 4, 5, 6)$ (c) $\{1, 2, 3, 4\}$ (d) $\{1, 2, 3\}$

12. Let $2\sin^2 x+3\sin x-2>0$ and $x^2-x-2<0$ (x is measured in radians). Then x lies in the interval **[IIT 1994]**

(a) $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$ (b) $\left(-1, \frac{5\pi}{6}\right)$ (c) $(-1, 2)$ (d) $\left(\frac{\pi}{6}, 2\right)$

13. Let $f(x)=(x+1)^2-1$, $(x\geq -1)$. Then the set $S=\{x : f(x)=f^{-1}(x)\}$ is **[IIT 1995]**

(a) Empty (b) $\{0, -1\}$ (c) $\{0, 1, -1\}$ (d) $\left\{0, -1, \frac{-3+i\sqrt{3}}{2}, \frac{-3-i\sqrt{3}}{2}\right\}$

14. If f is an even function defined on the interval $(-5, 5)$, then four real values of x satisfying the equation $f(x)=f\left(\frac{x+1}{x+2}\right)$ are

(a) $\frac{-3-\sqrt{5}}{2}, \frac{-3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2}$ (b) $\frac{-5+\sqrt{3}}{2}, \frac{-3+\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}$

(c) $\frac{3-\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2}, \frac{-3-\sqrt{5}}{2}, \frac{5+\sqrt{3}}{2}$ (d) $-3-\sqrt{5}, -3+\sqrt{5}, 3-\sqrt{5}, 3+\sqrt{5}$ **[IIT 1996]**

15. If $f(x)=\sin^2 x+\sin^2\left(x+\frac{\pi}{3}\right)+\cos x\cos\left(x+\frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right)=1$, then $(gof)(x)=$ **[IIT 1996]**

(a) –2 (b) –1 (c) 2 (d) 1

16. If $g(f(x))=|\sin x|$ and $f(g(x))=(\sin\sqrt{x})^2$, then **[IIT 1998]**

(a) $f(x)=\sin^2 x$, $g(x)=\sqrt{x}$ (b) $f(x)=\sin x$, $g(x)=|x|$ (c) $f(x)=x^2$, $g(x)=\sin\sqrt{x}$ (d) *f* and *g* cannot be determined

17. If $f(x)=3x+10$, $g(x)=x^2-1$, then $(fog)^{-1}$ is equal to **[UPSEAT 2001]**

(a) $\left(\frac{x-7}{3}\right)^{1/2}$ (b) $\left(\frac{x+7}{3}\right)^{1/2}$ (c) $\left(\frac{x-3}{7}\right)^{1/2}$ (d) $\left(\frac{x+3}{7}\right)^{1/2}$

18. If $f : R\to R$ and $g : R\to R$ are defined by $f(x)=2x+3$ and $g(x)=x^2+7$, then the values of x such that $g(f(x))=8$ are

(a) 1, 2 (b) –1, 2 (c) –1, –2 (d) 1, –2 **[EAMCET 2000, 03]**

19. $\lim_{x\to 1}(1-x)\tan\left(\frac{\pi x}{2}\right)=$ **[IIT 1978, 84; RPET 1997, 2001; UPSEAT 2003; Pb. CET 2003]**

(a) $\frac{\pi}{2}$ (b) π (c) $\frac{2}{\pi}$ (d) 0

20. True statement for $\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{2+3x}-\sqrt{2-3x}}$ is **[BIT Ranchi 1982]**

(a) Does not exist (b) Lies between 0 and $\frac{1}{2}$ (c) Lies between $\frac{1}{2}$ and 1 (d) Greater then 1

21. $\lim_{x\to\infty}\frac{x^n}{e^x}=0$ for **[IIT 1992]**

(a) No value of n (b) n is any whole number (c) $n=0$ only (d) $n=2$ only

22. $\lim_{n\to\infty}\sin[\pi\sqrt{n^2+1}]=$

(a) ∞ (b) 0 (c) Does not exist (d) None of these

23. If [.] denotes the greatest integer less than or equal to x, then the value of $\lim_{x\to 1}(1-x+[x-1]+[1-x])$ is

(a) 0 (b) 1 (c) –1 (d) None of these

24. The values of a and b such that $\lim_{x\to 0}\frac{x(1+a\cos x)-b\sin x}{x^3}=1$, are **[Roorkee 1996]**

(a) $\frac{5}{2},\frac{3}{2}$ (b) $\frac{5}{2},-\frac{3}{2}$ (c) $-\frac{5}{2},-\frac{3}{2}$ (d) None of these

25. If $\lim_{x\to a}\frac{a^x-x^a}{x^x-a^a}=-1$, then **[EAMCET 2003]**

(a) $a=1$ (b) $a=0$ (c) $a=e$ (d) None of these

26. If $x_1=3$ and $x_{n+1}=\sqrt{2+x_n}$, $n\geq 1$, then $\lim_{n\to\infty}x_n$ is equal to

(a) –1 (b) 2 (c) $\sqrt{5}$ (d) 3

27. The value of $\lim_{x\to\frac{\pi}{2}}\frac{\int_{\pi/2}^{x}t\,dt}{\sin(2x-\pi)}$ is **[MP PET 1998]**

(a) ∞ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{8}$

The $\lim_{x\to 0}(\cos x)^{\cot x}$ is **[RPET 1999]**(a)–1(b)0c)1(d) None of these

28. The integer n for which $\lim_{x\to 0}\frac{(\cos x-1)(\cos x-e^x)}{x^n}$ is a finite non-zero number is **[IIT Screening 2002]**

(a) 1 (b) 2 (c) 3 (d) 4

29. If f is strictly increasing function, then $\lim_{x\to 0}\frac{f(x^2)-f(x)}{f(x)-f(0)}$ is equal to **[IIT Screening 2004]**

(a) 0 (b) 1 (c) –1 (d) 2

30. If $f(x)=\begin{cases}x^2-3, & 2<x<3\\ 2x+5, & 3<x<4\end{cases}$, the equation whose roots are $\lim_{x\to 3^-}f(x)$ and $\lim_{x\to 3^+}f(x)$ is **[Orissa JEE 2004]**

(a) $x^2-7x+3=0$ (b) $x^2-20x+66=0$ (c) $x^2-17x+66=0$ (d) $x^2-18x+60=0$

31. The function $f(x)=[x]\cos\left[\frac{2x-1}{2}\right]\pi$, where [.] denotes the greatest integer function, is discontinuous at **[IIT 1995]**

(a) All x (b) No x (c) All integer points (d) x which is not an integer

32. Let $f(x)$ be defined for all $x>0$ and be continuous. Let $f(x)$ satisfy $f\left(\frac{x}{y}\right)=f(x)-f(y)$ for all x, y and $f(e)=1$, then **[IIT 1995]**

(a) $f(x) = \ln x$ (b) $f(x)$ is bounded (c) $f\left(\frac{1}{x}\right) \to 0$ as $x \to 0$ (d) $x\, f(x) \to 1$ as $x \to 0$

33. The value of p for which the function $f(x) = \begin{cases} \dfrac{(4^x - 1)^3}{\sin\frac{x}{p}\log\left[1 + \frac{x^2}{3}\right]}, x \neq 0 \\ 12(\log 4)^3, \; x = 0 \end{cases}$ may be continuous at $x = 0$, is **[Orissa JEE 2004]**

(a) 1 (b) 2 (c) 3 (d) None of these

34. The function $f(x) = [x]^2 - [x^2]$, (where [y] is the greatest integer less than or equal to y),is discontinuous at **[IIT 1999]**

(a) All integers (b) All integers except 0 and 1 (c) All integers except 0 (d) All integers except 1

35. If $f(x) = \begin{cases} xe^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then $f(x)$ is **[AIEEE 2003]**

(a) Continuous as well as differentiable for all x (b) Continuous for all x but not differentiable at $x = 0$

(c) Neither differentiable nor continuous at $x = 0$ (d) Discontinuous every where

36. Let $f(x) = \frac{1 - \tan x}{4x - \pi}$, $x \neq \frac{\pi}{4}$, $x \in \left[0, \frac{\pi}{2}\right]$, If $f(x)$ is continuous in $\left[0, \frac{\pi}{2}\right]$, then $f\left(\frac{\pi}{4}\right)$ is **[AIEEE 2004]**

(a) -1 (b) $\frac{1}{2}$ (c) $-\frac{1}{2}$ (d) 1

37. Let $g(x) = x.f(x)$, where $f(x) = \begin{cases} x\sin\frac{1}{x}, x \neq 0 \\ 0, x = 0 \end{cases}$ at $x = 0$ **[IIT Screening 1994; UPSEAT 2004]**

(a) g is differentiable but g' is not continuous (b) g is differentiable while f is not

(c) Both f and g are differentiable (d) g is differentiable and g' is continuous

38. The function $f(x) = \max[(1 - x), (1 + x), 2]$, $x \in (-\infty, \infty)$, is **[IIT 1995]**

(a) Continuous at all points (b)Differentiable at all points(c) Differentiable at all points except at $x = 1$ and $x = -1$

(d) Continuous at all points except at $x = 1$ and $x = -1$ where it is discontinuous

39. The function $f(x) = |x| + |x - 1|$ is **[RPET 1996; Kurukshetra CEE 2002]**

(a) Continuous at $x = 1$, but not differentiable at $x = 1$ (b) Both continuous and differentiable at $x = 1$

(c) Not continuous at $x = 1$ (d) Not differentiable at $x = 1$

ANSWER: Imp. Que. From Competitive exams

1	a	2	d	3	c	4	d	5	d
6	b	7	a	8	b	9	d	10	d
11	d	12	d	13	d	14	a	15	d
16	a	17	a	18	c	19	c	20	b
21	b	22	b	23	c	24	c	25	a
26	b	27	c	28	c	29	c	30	c
31	c	32	c	33	a	34	d	35	d
36	b	37	c	38	a,b	39	a,c	40	a

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 5 XI M 5. Complex Numbers

Index:

1. Key Concepts

2. Exercise I to V

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Complex Numbers

1. The complex number system

There is no real number x which satisfies the polynomial equation $x^2 + 1 = 0$. To permit solutions of this and similar equations, the set of complex numbers is introduced.

We can consider a complex number as having the form a + bi where a and b are real number and i, which is called the imaginary unit, has the property that $i^2 = -1$.

It is denoted by z i.e. z = a + ib. 'a' is called as real part of z which is denoted by (Re z) and 'b' is called as imaginary part of z which is denoted by (Im z).

Any complex number is :

(i) Purely real, if b = 0 ; (ii) Purely imaginary, if a = 0

(iii) Imaginary, if $b \neq 0$.

NOTE : (a) The set R of real numbers is a proper subset of the Complex Numbers. Hence the complete number system is $N \subset W \subset I \subset Q \subset R \subset C$.

(b) Zero is purely real as well as purely imaginary but not imaginary.

(c) $i = \sqrt{-1}$ is called the imaginary unit.

Also $i^2 = -1$; $i^3 = -i$; $i^4 = 1$ etc.

(d) $\sqrt{a}\ \sqrt{b} = \sqrt{ab}$ only if atleast one of a or b is non - negative.

(e) is z = a + ib, then a – ib is called complex conjugate of z and written as $\bar{z} = a - ib$

Self Practice Problems

1. Write the following as complex number

(i) $\sqrt{-16}$ (ii) $\sqrt{x}$, (x > 0)

(iii) $-b + \sqrt{-4ac}$, (a, c> 0)

Ans. (i) $0 + i\sqrt{16}$ (ii) $\sqrt{x} + 0i$ (iii) $-b + i\sqrt{4ac}$

2. Write the following as complex number

(i) $\sqrt{x}$ (x < 0) (ii) roots of $x^2 - (2\cos\theta)x + 1 = 0$

2. Algebraic Operations:

Fundamental operations with complex numbers

In performing operations with complex numbers we can proceed as in the algebra of real numbers, replacing i^2 by – 1 when it occurs.

1. Addition (a + bi) + (c + di) = a + bi + c + di = (a + c) + (b + d) i
2. Subtraction (a + bi) – c + di) = a + bi – c – di = (a – c) + (b – d) i
3. Multiplication (a + bi) (c + di) = ac + adi + bci + bdi^2 = (ac – bd) + (ad+ bc)i
4. Division

$$\frac{a+bi}{c+di} = \frac{a+bi}{c+di} \cdot \frac{c-bi}{c-di} = \frac{ac-adi+bci-bdi^2}{c^2-d^2i^2}$$

$$= \frac{ac+bd+(bc-ad)i}{c^2-d^2} = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}\,i$$

Inequalities in complex numbers are not defined. There is no validity if we say that complex number is positive or negative.

e.g. z > 0, 4 + 2i < 2 + 4 i are meaningless.

In real numbers if $a^2 + b^2 = 0$ then a = 0 = b however in complex numbers,

$z_1^2 + z_2^2 = 0$ does not imply $z_1 = z_2 = 0$.

Example : Find multiplicative inverse of 3 + 2i.

Solution Let z be the multiplicative inverse of 3 + 2i. then

$\Rightarrow$ z . (3 + 2i) = 1

$$\Rightarrow \quad z = \frac{1}{3+2i} = \frac{3-2i}{(3+2i)(3-2i)}$$

$$\Rightarrow \quad z = \frac{3}{13} - \frac{2}{13}\,i$$

$$\left(\frac{3}{13} - \frac{2}{13}i\right)$$ **Ans.**

Self Practice Problem

1. Simplify $i^{n+100} + i^{n+50} + i^{n+48} + i^{n+46}$, $n \in I$.

Ans. 0

3. Equality In Complex Number:

Two complex numbers $z_1 = a_1 + ib_1$ & $z_2 = a_2 + ib_2$ are equal if and only if their real and imaginary parts are equal respectively

i.e. $z_1 = z_2 \Rightarrow Re(z_1) = Re(z_2)$ and $I_m(z_1) = I_m(z_2)$.

Example: Find the value of x and y for which $(2 + 3i)\,x^2 - (3 - 2i)\,y = 2x - 3y + 5i$ where $x, y \in R$.
Solution $(z + 3i)x^2 - (3 - 2i)y = 2x - 3y + 5i$

$\Rightarrow \quad 2x^2 - 3y = 2x - 3y$

$\Rightarrow \quad x^2 - x = 0$

$\Rightarrow \quad x = 0, 1 \quad$ and $\quad 3x^2 + 2y = 5$

$\Rightarrow \quad$ if $x = 0, y = \frac{5}{2} \quad$ and $\quad$ if $x = 1, y = 1$

$\therefore \quad x = 0, y = \frac{5}{2} \quad$ and $\quad x = 1, y = 1$

are two solutions of the given equation which can also be represented as $\left(0, \frac{5}{2}\right)$ & $(1, 1)$

$\left(0, \frac{5}{2}\right), (1, 1)$ **Ans.**

Example: Find the value of expression $x^4 - 4x^3 + 3x^2 - 2x + 1$ when $x = 1 + i$ is a factor of expression.
Solution. $x = 1 + i$

$\Rightarrow \quad x - 1 = i$

$\Rightarrow \quad (x - 1)^2 = -1$

$\Rightarrow \quad x^2 - 2x + 2 = 0$

Now $\quad x^4 - 4x^3 + 3x^2 - 2x + 1$

$= (x^2 - 2x + 2)(x^2 - 3x - 3) - 4x + 7$

$\therefore \quad$ when $x = 1 + i \quad$ i.e. $\quad x^2 - 2x + 2 = 0$

$x^4 - 4x^3 + 3x^2 - 2x + 1 = 0 - 4(1 + i) + 7$

$= -4 + 7 - 4i$

$= 3 - 4i$ **Ans.**

Example: Solve for z if $z^2 + |z| = 0$
Solution. Let $z = x + iy$

$\Rightarrow \quad (x + iy)^2 + \sqrt{x^2 + y^2} = 0$

$\Rightarrow \quad x^2 - y^2 + \sqrt{x^2 + y^2} = 0 \quad$ and $\quad 2xy = 0$

$\Rightarrow \quad x = 0$ or $y = 0$

when $x = 0 \quad -y^2 + |y| = 0$

$\Rightarrow \quad y = 0, 1, -1$

$\Rightarrow \quad z = 0, i, -i$

when $y = 0 \quad x^2 + |x| = 0$

$\Rightarrow \quad x = 0 \quad \Rightarrow \quad z = 0$ **Ans.** $z = 0, z = i, z = -i$

Example: Find square root of $9 + 40i$
Solution. Let $(x + iy)^2 = 9 + 40i$

$\therefore \quad x^2 - y^2 = 9 \quad$ .....(i)

and $\quad xy = 20 \quad$ .....(ii)

squing (i) and adding with 4 times the square of (ii)

we get $x^4 + y^4 - 2x^2y^2 + 4x^2y^2 = 81 + 1600$

$\Rightarrow \quad (x^2 + y^2)^2 = 168$

$\Rightarrow \quad x^2 + y^2 = 4 \quad$ .....(iii)

from (i) + (iii) we get $\quad x^2 = 25 \quad \Rightarrow \quad x = \pm 5$

and $y = 16 \quad \Rightarrow \quad y = \pm 4$

from equation (ii) we can see that

x & y are of same sign

$\therefore \quad x + iy = +(5 + 4i)$ or $= (5 + 4i)$

$\therefore \quad$ sq. roots of $a + 40i = \pm (5 + 4i)$ **Ans.** $\pm (5 + 4i)$

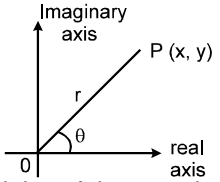
Self Practice Problem

1. Solve for z : $\bar{z} = i z^2$ **Ans.** $\pm \frac{\sqrt{3}}{2} - \frac{1}{2}i, 0, i$

4. Representation Of A Complex Number:

(a) Cartesian Form (Geometric Representation) :

Every complex number $z = x + iy$ can be represented by a point on the Cartesian plane known as complex plane (Argand diagram) by the ordered pair (x, y).



Length OP is called modulus of the complex number which is denoted by $|z|$ & θ is called the argument or amplitude.

$|z| = \sqrt{x^2 + y^2}$ & $\theta = \tan^{-1}\frac{y}{x}$ (angle made by OP with positive x–axis)

NOTE : (i) Argument of a complex number is a many valued function. If θ is the argument of a complex number then $2n\pi + \theta$; $n \in I$ will also be the argument of that complex number. Any two arguments of a complex number differ by $2n\pi$.

(ii) The unique value of θ such that $-\pi < \theta \le \pi$ is called the principal value of the argument.
Unless otherwise stated, amp z implies principal value of the argument.

(iii) By specifying the modulus & argument a complex number is defined completely. For the complex number 0 + 0 i the argument is not defined and this is the only complex number which is only given by its modulus.

(b) Trignometric/Polar Representation :
$z = r(\cos\theta + i\sin\theta)$ where $|z| = r$; $\arg z = \theta$; $\bar{z} = r(\cos\theta - i\sin\theta)$

NOTE : $\cos\theta + i\sin\theta$ is also written as CiS θ or $e^{i\theta}$.

Also $\cos x = \frac{e^{ix}+e^{-ix}}{2}$ & $\sin x = \frac{e^{ix}-e^{-ix}}{2}$ are known as Euler's identities.

(c) **Euler's Representation :**
$z = re^{i\theta}$; $|z| = r$; $\arg z = \theta$; $\bar{z} = re^{-i\theta}$

(d) **Vectorial Representation :**
Every complex number can be considered as if it is the position vector of a point. If the point P represents the complex number z then, $\vec{OP} = z$ & $|\vec{OP}| = |z|$.

Example: Express the complex number $z = -1 + \sqrt{2}\,i$ in polar form.

Solution. $z = -1 + i\sqrt{2}$

$|z| = \sqrt{(-1)^2 + (\sqrt{2})^2} = \sqrt{1+2} = \sqrt{3}$

$\arg z = \pi - \tan^{-1}\left(\frac{\sqrt{2}}{1}\right) = \pi - \tan^{-1}\sqrt{2} = \theta$ (say)

$\therefore \quad z = \sqrt{3}\,(\cos\theta + i\sin\theta)$ where $\theta = \pi - \tan^{-1}\sqrt{2}$

Self Practice Problems

1. Find the principal argument and |z|

$$z = \frac{-1(9+i)}{2-i}$$

Ans. $-\tan^{-1}\frac{17}{11}, \sqrt{\frac{8^2}{5}}$

2. Find the |z| and principal argument of the complex number $z = 6(\cos 310^\circ - i\sin 310^\circ)$
Ans. 6, 50°

5. Modulus of a Complex Number :

If $z = a + ib$, then it's modulus is denoted and defined by $|z| = \sqrt{a^2+b^2}$. Infact |z| is the distance of z from origin. Hence $|z_1 - z_2|$ is the distance between the points represented by z_1 and z_2.

Properties of modulus

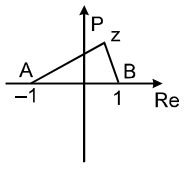
(i) $|z_1 z_2| = |z_1| \cdot |z_2|$ (ii) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$ (provided $z_2 \neq 0$)

(iii) $|z_1 + z_2| \le |z_1| + |z_2|$ (iv) $|z_1 - z_2| \ge ||z_1| - |z_2||$

(Equality in (iii) and (iv) holds if and only if origin, z_1 and z_2 are collinear with z_1 and z_2 on the same side of origin).

Example: If $|z - 5 - 7i| = 9$, then find the greatest and least values of $|z - 2 - 3i|$.
Solution. We have $9 = |z - (5 + 7i)|$ = distance between z and 5 + 7i.
Thus locus of z is the circle of radius 9 and centre at 5 + 7i. For such a z (on the circle), we have to find its greatest and least distance as from 2 + 3i, which obviously 14 and 4.

Example: Find the minimum value of $|1 + z| + |1 - z|$.
Solution $|1 + z| + |1 - z| \ge |1 + z + 1 - z|$ (triangle inequality)
$\Rightarrow \quad |1 + z| + |1 - z| \ge 2$
$\therefore$ minimum value of $(|1 + z| + |1 - z|) = 2$
Geometrically $|z + 1| + |1 - 2| = |z + 1| + |z - 1|$ which represents sum of distances of z from 1 and -1
it can be seen easily that minimu (PA + PB) = AB = 2



Ans. $2^{1/4}e^{1\left(\frac{\pi}{8}+n\pi\right)}$

Example: $\left|z-\frac{2}{z}\right| = 1$ then find the maximum and minimum value of $|z|$

Solution. $\left|z-\frac{2}{z}\right| = 1 \quad \left||z|-\left|\frac{2}{z}\right|\right| \le \left|z-\frac{2}{2}\right| \le |z| + \left|-\frac{2}{z}\right|$

Let $|z| = r$

$\Rightarrow \quad \left|r-\frac{2}{r}\right| \le 1 \le r + \frac{2}{r}$

$r + \frac{2}{r} \ge 1 \quad \Rightarrow \quad r \in R^+$ .....(i)

and $\left|r-\frac{2}{r}\right| \le 1 \Rightarrow \quad -1 \le r - \frac{2}{r} \le 1$

$\Rightarrow \quad r \in (1, 2)$ .....(ii)

$\therefore$ from (i) and (ii) $r \in (1, 2)$

Ans. $r \in (1, 2)$

Self Practice Problem

1. $|z - 3| < 1$ and $|z - 4i| > M$ then find the positive real value of M for which these exist at least one complex number z satisfy both the equation.
Ans. $M \in (0, 6)$

6. Agrument of a Complex Number :

Argument of a non-zero complex number P(z) is denoted and defined by arg(z) = angle which OP makes with the positive direction of real axis.
If OP = |z| = r and arg(z) = θ, then obviously $z = r(\cos\theta + i\sin\theta)$, called the polar form of z. In what follows, 'argument of z' would mean principal argument of z(i.e. argument lying in $(-\pi, \pi]$ unless the context requires otherwise. Thus argument of a complex number $z = a + ib = r(\cos\theta + i\sin\theta)$ is the value of θ satisfying $r\cos\theta = a$ and $r\sin\theta = b$.

Thus the argument of $z = \theta, \pi - \theta, -\pi + \theta, -\theta, \theta = \tan^{-1}\left|\frac{b}{a}\right|$, according as $z = a + ib$ lies in I, II, III or IVth quadrant.

Properties of arguments

(i) $\arg(z_1z_2) = \arg(z_1) + \arg(z_2) + 2m\pi$ for some integer m.

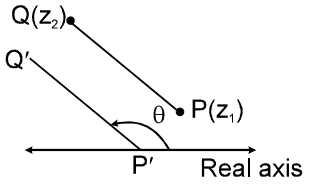
(ii) $\arg(z_1/z_2) = \arg(z_1) - \arg(z_2) + 2m\pi$ for some integer m.

(iii) $\arg(z^2) = 2\arg(z) + 2m\pi$ for some integer m.

(iv) $\arg(z) = 0 \quad \Leftrightarrow \quad$ z is real, for any complex number $z \neq 0$

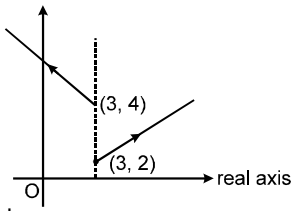
(v) $\arg(z) = \pm\pi/2 \quad \Leftrightarrow \quad$ z is purely imaginary, for any complex number $z \neq 0$

(vi) $\arg(z_2 - z_1)$ = angle of the line segment
P′Q′ || PQ, where P′ lies on real axis, with the real axis.



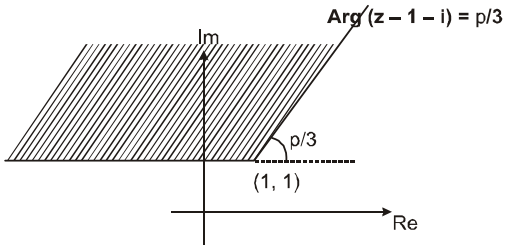
Example: Solve for z, which satisfy $\text{Arg}(z - 3 - 2i) = \frac{\pi}{6}$ and $\text{Arg}(z - 3 - 4i) = \frac{2\pi}{3}$.

Solution From the figure, it is clear that there is no z, which satisfy both ray

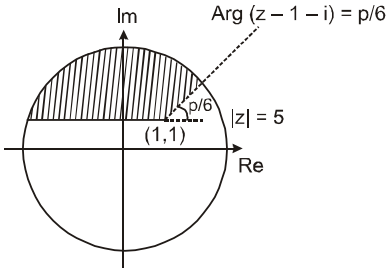


Example: Sketch the region given by
(i) $\text{Arg}(z - 1 - i) \ge \pi/3$
(ii) $|z| = \le 5$ & $\text{Arg}(z - i - 1) > \pi/3$

Solution (i)



(ii)



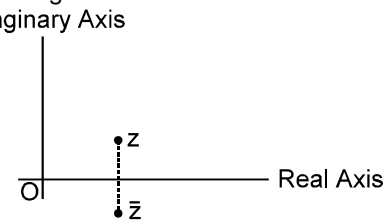
Self Practice Problems

1. Sketch the region given by
(i) $|\text{Arg}(z - i - 2)| < \pi/4$ (ii) $\text{Arg}(z + 1 - i) \le \pi/6$

2. Consider the region $|z - 15i| \le 10$. Find the point in the region which has
(i) max $|z|$ (ii) min $|z|$
(iii) max arg (z) (iv) min arg (z)

7. Conjugate of a complex Number

Conjugate of a complex number $z = a + b$ is denoted and defined by $\bar{z} = a - ib$.
In a complex number if we replace i by – i, we get conjugate of the complex number. $\bar{z}$ is the mirror image of z about real axis on Argand's Plane.



Properties of conjugate

(i) $|z| = |\bar{z}|$ (ii) $z\bar{z} = |z|^2$
(iii) $\overline{(z_1 + z_2)} = (\bar{z}_1) + (\bar{z}_2)$ (iv) $\overline{(z_1 - z_2)} = (\bar{z}_1) - (\bar{z}_2)$
(v) $\overline{(z_1 z_2)} = \bar{z}_1 \bar{z}_2$ (vi) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{(\bar{z}_1)}{(\bar{z}_2)}$ $(z_2 \neq 0)$
(vii) $|z_1 + z_2|^2 = (z_1 + z_2)\ \overline{(z_1 + z_2)} = |z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + \bar{z}_1 z_2$
(viii) $\overline{(\bar{z}_1)} = z$ (ix) If $w = f(z)$, then $\bar{w} = f(\bar{z})$
(x) $\arg(z) + \arg(\bar{z}) = 0$

Example: If $\frac{z-1}{z+1}$ is purely imaginary, then prove that $|z| = 1$

Solution. $\text{Re}\left(\frac{z-1}{z+1}\right) = 0$

$\Rightarrow \quad \frac{z-1}{z+1} + \left(\overline{\frac{z-1}{z+1}}\right) = 0 \quad \Rightarrow \quad \frac{z-1}{z+1} + \frac{\bar{z}-1}{\bar{z}+1} = 0$

$\Rightarrow \quad z\bar{z} - \bar{z} + z - 1 + z\bar{z} - z + \bar{z} - 1 = 0$

$\Rightarrow \quad z\bar{z} = 1 \quad \Rightarrow \quad |z|^2 = 1$

$\Rightarrow \quad |z| = 1$ **Hence proved**

Self Practice Problem

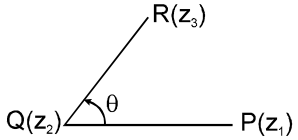
1. If $\frac{z_1 - 2z_2}{2 - z_1\bar{z}_2}$ is unmodulus and z_2 is not unimodulus then find $|z_1|$.
Ans. $|z_1| = 2$

8. Rotation theorem

(i) $P(z_1)$ and $Q(z_2)$ are two complex numbers such that $|z_1| = |z_2|$, then $z_2 = z_1 e^{i\theta}$ where $\theta = \angle POQ$

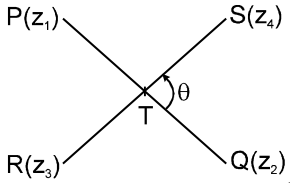
(ii) If $P(z_1)$, $Q(z_2)$ and $R(z_3)$ are three complex numbers and $\angle PQR = \theta$, then

$$\left(\frac{z_3 - z_2}{z_1 - z_2}\right) = \left|\frac{z_3 - z_2}{z_1 - z_2}\right| e^{i\theta}$$

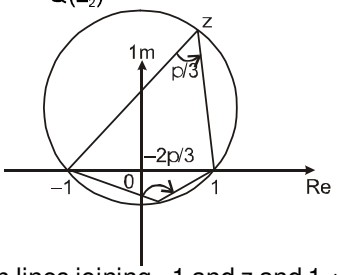


(iii) If $P(z_1)$, $Q(z_2)$, $R(z_3)$ and $S(z_4)$ are four complex numbers and $\angle STQ = \theta$, then

$$\frac{z_3 - z_2}{z_1 - z_2} = \left|\frac{z_3 - z_4}{z_1 - z_2}\right| e^{i\theta}$$



Example: If $\arg\left(\frac{z-1}{z+i}\right) = \frac{\pi}{3}$ then interrupter the locus.



Solution $\arg\left(\frac{z-1}{z+i}\right) = \frac{\pi}{3}$

$\Rightarrow \quad \arg\left(\frac{1-z}{-1-z}\right) = \frac{\pi}{3}$

Here $\arg\left(\frac{1-z}{-1-z}\right)$ represents the angle between lines joining –1 and z and 1 + z. As this angle is constant, the locus of z will be a of a circle segment. (angle in a segment is count). It can be seen that locus is not the complete side as in the major are $\arg\left(\frac{1-z}{-1-z}\right)$ will be equal to $-\frac{2\pi}{3}$.

Now try to geometrically find out radius and centre of this circle.

centre $\equiv \left(0, \frac{1}{\sqrt{3}}\right)$ Radius $\equiv \frac{2}{\sqrt{3}}$ **Ans.**

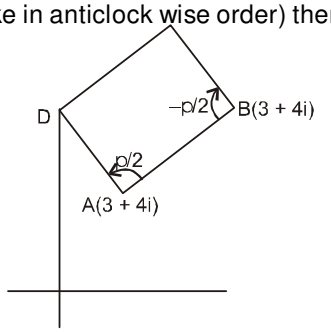
Example: If A(z + 3i) and B(3 + 4i) are two vertices of a square ABCD (take in anticlock wise order) then find C and D.

Solution. Let affix of C and D are $z_3 + z_4$ respectively
Considering $\angle DAB = 90^\circ$ + AD = AB

we get $\frac{z_4 - (2+3i)}{AD} = \frac{(3+4i)-(2+3i)}{AB} \quad e^{\frac{i\pi}{2}}$

$\Rightarrow \quad z_4 - (2 + 3i) = (1 + i)\, i$

$\Rightarrow \quad Z_4 = 2 + 3i + i - 1 = 1 + zi$



and $\frac{z_3 - (3+4i)}{CB} = \frac{(z+3i)-(3-4i)}{AB} \; e^{-\frac{i\pi}{2}}$

$\Rightarrow \quad z_3 = 3 + 4i - (1 + i)(-i)$

$z_3 = 3 + 4i + i - 1 = z + 5i$

Self Practice Problems

1. z_1, z_2, z_3, z_4 are the vertices of a square taken in anticlockwise order then prove that
$2z_2 = (1 + i)\, z_1 + (1 - i)\, z_3$
Ans. $(1 + i)\, z_1 + (1 - i) z_3$

2. Check that z_1z_2 and z_3z_4 are parallel or, not
where, $z_1 = 1 + i$ $z_3 = 4 + 2i$
$z_2 = 2 - i$ $z_4 = 1 - i$
Ans. Hence, z_1z_2 and z_3z_4 are not parallel.

3. P is a point on the argand diagram on the circle with OP as diameter "two point Q and R are taken such that $\angle POQ = \angle QOR$
If O is the origin and P, Q, R are represented by complex z_1, z_2, z_3 respectively then show that
$z_2^2 \cos 2\theta = z_1z_3\cos^2\theta$
Ans. $z_1z_3 \cos^2\theta$

9. Demoivre's Theorem:

Case I

Statement :
If n is any integer then
(i) $(\cos\theta + i \sin\theta)^n = \cos n\theta + i \sin n\theta$
(ii) $(\cos\theta_1 + i \sin\theta_1)(\cos\theta_2) + i \sin\theta_2)(\cos\theta_3 + i \sin\theta_2)(\cos\theta_3 + i \sin\theta_3)$ .....$(\cos\theta_n + i \sin\theta_n)$
$= \cos(\theta_1 + \theta_2 + \theta_3 + \ldots\ldots\ldots \theta_n) + i \sin(\theta_1 + \theta_2 + \theta_3 + \ldots\ldots + \theta_n)$

Case II
Statement : If $p, q \in Z$ and $q \neq 0$ then

$$(\cos\theta + i \sin\theta)^{p/q} = \cos\left(\frac{2k\pi + p\theta}{q}\right) + i \sin\left(\frac{2k\pi + p\theta}{q}\right)$$

where k = 0, 1, 2, 3, ..., q – 1

NOTE : Continued product of the roots of a complex quantity should be determined using theory of equations.

10. Cube Root Of Unity :

(i) The cube roots of unity are 1, $\frac{-1+i\sqrt{3}}{2}$, $\frac{-1-i\sqrt{3}}{2}$.

(ii) ,, ω is one of the imaginary cube roots of unity then $1 + \omega + \omega^2 = 0$. In general $1 + \omega^r + \omega^{2r} = 0$; where $r \in I$ but is not the multiple of 3.

(iii) In polar form the cube roots of unity are :

$\cos 0 + i \sin 0$; $\cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3}$, $\cos\frac{4\pi}{3} + i \sin\frac{4\pi}{3}$

(iv) The three cube roots of unity when plotted on the argand plane constitute the verties of an equilateral triangle.

(v) The following factorisation should be remembered :
($a, b, c \in R$ & ω is the cube root of unity)

$a^3 - b^3 = (a - b)(a - \omega b)(a - \omega^2 b)$; $x^2 + x + 1 = (x - \omega)(x - \omega^2)$;

$a^3 + b^3 = (a + b)(a + \omega b)(a + \omega^2 b)$; $a^2 + ab + b^2 = (a - bw)(a - bw^2)$

$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)$

Example: Find the value of $\omega^{192} + \omega^{194}$

Solution. $\omega^{192} + \omega^{194}$

$= 1 + \omega^2 = -\omega$

Ans. $-\omega$

Example: If 1, ω, ω^2 are cube roots of unity prove

(i) $(1 - \omega + \omega^2)(1 + \omega - \omega^2) = 4$

(ii) $(1 - \omega + \omega^2)^5 + (1 + \omega - \omega^2)^5 = 32$

(iii) $(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^8) = 9$

(iv) $(1 - \omega + \omega^2)(1 - \omega^2 + \omega^4)(1 - \omega^4 + \omega^8)$ ..... to 2n factors $= 2^{2n}$

Solution. (i) $(1 - \omega + \omega^2)(1 + \omega - \omega^2)$

$= (-2\omega)(-2\omega^2)$

$= 4$

Self Practice Problem

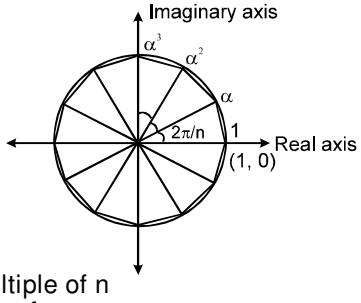
1. Find $\sum_{r=0}^{10}(1+\omega^r+\omega^{2r})$

Ans. 12

11. n^{th} Roots of Unity :

If 1, α_1, α_2, α_3.... α_{n-1} are the n, n^{th} root of unity then :

(i) They are in G.P. with common ratio $e^{i(2\pi/n)}$ &



(ii) $1^p + \alpha_1^p + \alpha_2^p + ... + \alpha_{n-1}^p = 0$ if p is not an integral multiple of n
$= n$ if p is an integral multiple of n

(iii) $(1 - \alpha_1)(1 - \alpha_2)..... (1 - \alpha_{n-1}) = n$ &
$(1 + \alpha_1)(1 + \alpha_2)..... (1 + \alpha_{n-1}) = 0$ if n is even and 1 if n is odd.

(iv) $1.\ \alpha_1\ \alpha_2\ \alpha_3 ..... \alpha_{n-1} = 1$ or -1 according as n is odd or even.

Example: Find the roots of the equation $z^6 + 64 = 0$ where real part is positive.

Solution. $z^6 = -64$

$z^6 = z^6 . e^{+i(2n+1)\pi}$ $x \in z$

$\Rightarrow$ $z = z\ e^{i(2n+1)\frac{\pi}{6}}$

$\therefore$ $z = 2\ e^{i\frac{\pi}{6}}, 2e^{i\frac{\pi}{2}}, ze^{i\frac{\pi}{2}}, ze^{i\frac{5\pi}{6}} = e^{i\frac{7\pi}{6}}, ze^{i\frac{3\pi}{2}}, ze^{i\frac{11\pi}{2}}$

$\therefore$ roots with +ve real part are $= e^{\frac{i\pi}{6}} + e^{i\frac{11\pi}{6}}$

$2e^{i\left(-\frac{\pi}{6}\right)}$ **Ans.**

Example: Find the value $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7}-\cos\frac{2\pi k}{7}\right)$

Solution. $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7}\right)-\sum_{k=1}^{6}\left(\cos\frac{2\pi k}{7}\right)$

$= \sum_{k=0}^{6}\sin\frac{2\pi k}{7}-\sum_{k=0}^{6}\cos\frac{2\pi k}{7}+1$

$= \sum_{k=0}^{6}$ (Sum of imaginary part of seven seventh roots of unity)

$-\sum_{k=0}^{6}$ (Sum of real part of seven seventh roots of unity) + 1

$= 0 - 0 + 1 = 1$

i **Ans.**

Self Practice Problems

1. Resolve $z^7 - 1$ into linear and quadratic factor with real coefficient.

Ans. $(z-1)\left(z^2-2\cos\frac{2\pi}{7}z+1\right).\left(z^2-2\cos\frac{4\pi}{7}z+1\right).\left(z^2-2\cos\frac{6\pi}{7}z+1\right)$

2. Find the value of $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}$.

Ans. $-\frac{1}{2}$

12. The Sum Of The Following Series Should Be Remembered :

(i) $\cos\theta+\cos 2\theta+\cos 3\theta+\ldots..+\cos n\theta=\frac{\sin(n\theta/2)}{\sin(\theta/2)}\cos\left(\frac{n+1}{2}\right)\theta.$

(ii) $\sin\theta+\sin 2\theta+\sin 3\theta+\ldots..+\sin n\theta=\frac{\sin(n\theta/2)}{\sin(\theta/2)}\sin\left(\frac{n+1}{2}\right)\theta.$

NOTE : If $\theta = (2\pi/n)$ then the sum of the above series vanishes.

13. Logarithm Of A Complex Quantity :

(i) $\text{Log}_e(\alpha+i\beta)=\frac{1}{2}\text{Log}_e(\alpha^2+\beta^2)+i\left(2n\pi+\tan^{-1}\frac{\beta}{\alpha}\right)$ where $n\in I$.

(ii) i^i represents a set of positive real numbers given by $e^{-\left(2n\pi+\frac{\pi}{2}\right)}$, $n\in I$.

Example: Find the value of

(i) $\log(1+\sqrt{3}\,i)$ **Ans.** $\log 2+i\left(2n\pi+\frac{\pi}{3}\right)$

(ii) $\log(-1)$ **Ans.** $i\pi$

(iii) z^i **Ans.** $\cos(\ln 2)+i\sin(\ln 2)=e^{i(\ln 2)}$

(iv) i^i **Ans.** $e^{-(4n+1).\frac{\pi}{2}}$

(v) $|(1+i)^i|$ **Ans.** $e^{-(8n+1).\frac{\pi}{4}}$

(vi) $\arg((1+i)^i)$ **Ans.** $\frac{1}{2}\ell n(2).$

Solution. (i) $\log(1+\sqrt{3}\,i) = \log\left(2e^{i\left(\frac{\pi}{3}+2n\pi\right)}\right)$

$= \log 2 + i\left(\frac{\pi}{3}+2n\pi\right)$

(iii) $2^i = e^{i\ell n\,2} = \cos(\ell n\,2)\cos(\ell n\,2)+i\sin(\ell n\,2)]$

Self Practice Problem

1. Find the real part of cos (1 + i)

Ans. $\frac{1-e^2}{2ei}$

14. Geometrical Properties :

Distance formula :

If z_1 and z_2 are affixies of the two points ↓ P and Q respectively then distance between P + Q is given by $|z_1 - z_2|$.

Section formula

If z_1 and z_2 are affixes of the two points P and Q respectively and point C devides the line joining P and Q internally in the ratio m : n then affix z of C is given by

$$z = \frac{mz_2 + nz_1}{m+n}$$

If C devides PQ in the ratio m : n externally then

$$z = \frac{mz_2 - nz_1}{m-n}$$

(b) If a, b, c are three real numbers such that $az_1 + bz_2 + cz_3 = 0$; where a + b + c = 0 and a,b,c are not all simultaneously zero, then the complex numbers z_1, z_2 & z_3 are collinear.

(1) If the vertices A, B, C of a Δ represent the complex nos. z_1, z_2, z_3 respectively and a, b, c are the length of sides then,

(i) Centroid of the Δ ABC = $\frac{z_1 + z_2 + z_3}{3}$:

(ii) Orthocentre of the Δ ABC =

$$\frac{(a\sec A)z_1 + (b\sec B)z_2 + (c\operatorname{sec}C)z_3}{a\sec A + b\sec B + c\operatorname{sec}C} \text{ or } \frac{z_1\tan A + z_2\tan B + z_3\tan C}{\tan A + \tan B + \tan C}$$

(iii) Incentre of the Δ ABC = $(az_1 + bz_2 + cz_3) \div (a + b + c)$.

(iv) Circumcentre of the Δ ABC = :
$(Z_1 \sin 2A + Z_2 \sin 2B + Z_3 \sin 2C) \div (\sin 2A + \sin 2B + \sin 2C)$.

(2) amp(z) = θ is a ray emanating from the origin inclined at an angle θ to the x–axis.

(3) $|z - a| = |z - b|$ is the perpendicular bisector of the line joining a to b.

(4) The equation of a line joining z_1 & z_2 is given by, $z = z_1 + t(z_1 - z_2)$ where t is a real parameter.

(5) $z = z_1(1 + it)$ where t is a real parameter is a line through the point z_1 & perpendicular to the line joining z_1 to the origin.

(6) The equation of a line passing through z_1 & z_2 can be expressed in the determinant form as

$\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$. This is also the condition for three complex numbers to be collinear. The above equation on manipulating, takes the form $\bar{\alpha}z + \alpha\bar{z} + r = 0$ where r is real and α is a non zero complex constant.

NOTE : If we replace z by $ze^{i\theta}$ and $\bar{z}$ by $\bar{z}e^{-i\theta}$ then we get equation of a straight line which. Passes through the foot of the perpendicular from origin to given straight line and makes an angle θ with the given straightl line.

(7) The equation of circle having centre z_0 & radius ρ is :
$|z - z_0| = \rho$ or $z\bar{z} - z_0\bar{z} - \bar{z}_0 z + \bar{z}_0 z_0 - \rho^2 = 0$ which is of the form
$z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + k = 0$, k is real. Centre is $-\alpha$ & radius = $\sqrt{\alpha\bar{\alpha} - k}$.
Circle will be real if $\alpha\bar{\alpha} - k \geq 0$..

(8) The equation of the circle described on the line segment joining z_1 & z_2 as diameter is
$\arg\frac{z - z_2}{z - z_1} = \pm\frac{\pi}{2}$ or $(z - z_1)(\bar{z} - \bar{z}_2) + (z - z_2)(\bar{z} - \bar{z}_1) = 0$.

(9) Condition for four given points z_1, z_2, z_3 & z_4 to be concyclic is the number
$\frac{z_3 - z_1}{z_3 - z_2} \cdot \frac{z_4 - z_2}{z_4 - z_1}$ should be real. Hence the equation of a circle through 3 non collinear

Successful People Replace the words like; "wish", "try" & "should" with "I Will". Ineffective People don't.

points z_1, z_2 & z_3 can be taken as $\frac{(z-z_2)(z_3-z_1)}{(z-z_1)(z_3-z_2)}$ is real

$\Rightarrow \quad \frac{(z-z_2)(z_3-z_1)}{(z-z_1)(z_3-z_2)} = \frac{(\bar{z}-\bar{z}_2)(\bar{z}_3-\bar{z}_1)}{(\bar{z}-\bar{z}_1)(\bar{z}_3-\bar{z}_2)}$.

(10) $\text{Arg}\left(\frac{z-z_1}{z-z_2}\right) = \theta$ represent (i) a line segment if $\theta = \pi$

(ii) Pair of ray if $\theta = 0$ (iii) a part of circle, if $0 < \theta < \pi$.

(11) Area of triangle formed by the points z_1, z_2 & z_3 is $\left|\frac{1}{4i}\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix}\right|$

(12) Perpendicular distance of a point z_0 from the line $\bar{\alpha}z + \alpha\bar{z} + r = 0$ is $\frac{|\bar{\alpha}z_0 + \alpha\bar{z}_0 + r|}{2|\alpha|}$

(13) (i) Complex slope of a line $\bar{\alpha}z + \alpha\bar{z} + r = 0$ is $\omega = -\frac{\alpha}{\bar{\alpha}}$.

(ii) Complex slope of a line joining by the points z_1 & z_2 is $\omega = \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}$

(iii) Complex slope of a line making θ angle with real axis = $e^{2i\theta}$

(14) ω_1 & ω_2 are the compelx slopes of two lines.
(i) If lines are parallel then $\omega_1 = \omega_2$
(ii) If lines are perpendicular then $\omega_1 + \omega_2 = 0$

(15) If $|z - z_1| + |z - z_2| = K > |z_1 - z_2|$ then locus of z is an ellipse whose focii are z_1 & z_2

(16) If $|z - z_0| = \left|\frac{\bar{\alpha}z + \alpha\bar{z} + r}{2|\alpha|}\right|$ then locus of z is parabola whose focus is z_0 and directrix is the line $\bar{\alpha}z_0 + \bar{\alpha}z_0 + r = 0$

(17) If $\left|\frac{z-z_1}{z-z_2}\right| = k \neq 1, 0$, then locus of z is circle.

(18) If $\big||z-z_1| - |z-z_2|\big| = K < |z_1 - z_2|$ then locus of z is a hyperbola, whose focii are z_1 & z_2.

Match the following columns :

	Column - I		**Column - II**
(i)	If $\|z - 3+2i\| - \|z + i\| = 0$, then locus of z represents	(i)	circle
(ii)	If $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$, then locus of z represents...	(ii)	Straight line
(iii)	if $\|z - 8 - 2i\| + \|z - 5 - 6i\| = 5$ then locus of z represents	(iii)	Ellipse
(iv)	If $\arg\left(\frac{z-3+4i}{z+2-5i}\right) = \frac{5\pi}{6}$, then locus of z represents	(iv)	Hyperbola
(v)	If $\|z - 1\| + \|z + i\| = 10$ then locus of z represents	(v)	Major Arc
(vi)	$\|z - 3 + i\| - \|z + 2 - i\| = 1$ then locus of z represents	(vi)	Minor arc
(vii)	$\|z - 3i\| = 25$	(vii)	Perpendicular bisector of a line segment
(viii)	$\arg\left(\frac{z-3+5i}{z+i}\right) = \pi$	(viii)	Line segment

Ans.									
I	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
II	(vii)	(v)	(viii)	(vi)	(iii)	(iv)	(i)	(ii)

15. (a) Reflection points for a straight line :

Two given points P & Q are the reflection points for a given straight line if the given line is the right bisector of the segment PQ. Note that the two points denoted by the complex numbers z_1 & z_2 will be the reflection points for the straight line $\bar{\alpha} z + \alpha \bar{z} + r = 0$ if and only if, $\bar{\alpha} z_1 + \alpha \bar{z}_2 + r = 0$, where r is real and α is non zero complex constant.

(b) **Inverse points w.r.t. a circle :**

Two points P & Q are said to be inverse w.r.t. a circle with centre 'O' and radius ρ, if:
(i) the point O, P, Q are collinear and P, Q are on the same side of O.
(ii) OP. OQ = ρ^2.

Note : that the two points z_1 & z_2 will be the inverse points w.r.t. the circle $z\bar{z} + \bar{\alpha} z + \alpha \bar{z} + r = 0$ if and only if $z_1 \bar{z}_2 + \bar{\alpha} z_1 + \alpha \bar{z}_2 + r = 0$.

16. Ptolemy's Theorem:

It states that the product of the lengths of the diagonals of a convex quadrilateral inscribed in a circle is equal to the sum of the products of lengths of the two pairs of its opposite sides.
i.e. $|z_1 - z_3|\,|z_2 - z_4| = |z_1 - z_2|\,|z_3 - z_4| + |z_1 - z_4|\,|z_2 - z_3|$.

Example: If $\cos\alpha + \cos\beta + \cos\gamma = 0$ and also $\sin\alpha + \sin\beta + \sin\gamma = 0$, then prove that

(i) $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = \sin 2\alpha + \sin 2\beta + \sin 2\gamma = 0$

(ii) $\sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3 \sin(\alpha + \beta + \gamma)$

(iii) $\cos 3\alpha + \cos 3\beta + \cos 3\gamma = 3 \cos(\alpha + \beta + \gamma)$

Solution. Let $z_1 = \cos\alpha + i \sin\alpha$, $z_2 = \cos\beta + i \sin\beta$,
$z_3 = \cos\gamma + i \sin\gamma$.

$\therefore$ $z_1 + z_2 + z_3 = (\cos\alpha + \cos\beta + \cos\gamma) + i(\sin\alpha + \sin\beta + \sin\gamma)$
$= 0 + i \cdot 0 = 0$ (1)

(i) Also $\frac{1}{z_1} = (\cos\alpha + i \sin\alpha)^{-1} = \cos\alpha - i \sin\alpha$

$\frac{1}{z_2} = \cos\beta - i \sin\beta$, $\frac{1}{z_3} - \cos\gamma - \sin\gamma$

$\therefore$ $\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} = (\cos\alpha + \cos\beta + \cos\gamma) - i(\sin\alpha + \sin\beta + \sin\gamma)$ (2)

$= 0 - i \cdot 0 = 0$

Now $z_1^2 + z_2^2 + z_3^3 = (z_1 + z_2 + z_3)^2 - 2(z_1z_2 + z_2z_3 + z_3z_1)$

$= 0 - 2z_1z_2z_3\left(\frac{1}{z_3} + \frac{1}{z_1} + \frac{1}{z_2}\right)$

$= 0 - 2z_1 z_2 z_3 \cdot 0 = 0$, using (1) and (2)

or $(\cos\alpha + i \sin\alpha)^2 + (\cos\beta + i \sin\beta)^2 + (\cos\gamma + i \sin\gamma)^2 = 0$

or $\cos 2\alpha + i \sin 2\alpha)^2 + \cos 2\beta + i \sin 2\beta + \cos 2\gamma + i \sin 2\gamma = 0 + i.0$

Equation real and imaginary parts on both sides, $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = 0$ and $\sin 2\alpha + \sin 2\beta + \sin 2\gamma = 0$

(ii) $z_1^3 + z_2^3 + z_3^3 = (z_1 + z_2)^3 - 3z_1z_2(z_1 + z_2) + z_3^3$
$= (-z_3)^3 - 3z_1z_2(-z_3) + z_3^3$, using (1)
$= 3z_1z_2z_3$

$\therefore$ $(\cos\alpha + i \sin\alpha)^3 + (\cos\beta + i \sin\beta)^3 + (\cos\gamma + i \sin\gamma)^3$
$= 3(\cos\alpha + i \sin\alpha)(\cos\beta + i \sin\beta)(\cos\gamma + i \sin\gamma)$

or $\cos 3\alpha + i \sin 3\alpha + \cos 3\beta + i \sin 3\beta + \cos 3\gamma + i \sin 3\gamma$
$= 3\{\cos(\alpha + \beta + \gamma) + i \sin(\alpha + \beta + \gamma)$

Equation imaginary parts on both sides, $\sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3 \sin(\alpha + \beta + \gamma)$

Alternative method

Let $C \equiv \cos\alpha + \cos\beta + \cos\gamma = 0$
$S \equiv \sin\alpha + \sin\beta + \sin\gamma = 0$
$C + iS = e^{i\alpha} + e^{i\beta} + e^{i\gamma} = 0$ (1)
$C - iS = e^{-i\alpha} + e^{-i\beta} + e^{-i\gamma} = 0$ (2)

From (1) $\Rightarrow$ $(e^{-i\alpha})^2 + (e^{-i\beta})^2 + (e^{-i\gamma})^2 = (e^{i\alpha})(e^{i\beta}) + (e^{i\beta})(e^{i\gamma}) + (e^{i\gamma})(e^{i\alpha})$
$\Rightarrow$ $e^{i2\alpha} + e^{i2\beta} + e^{i2\gamma} = e^{i\alpha} e^{i\beta} e^{i\gamma}(e^{-2\gamma} + e^{-i\alpha} + e^{i\beta})$
$\Rightarrow$ $e^{i(2\alpha)} + e^{i2\beta} + e^{i2\gamma} = 0$ (from 2)

Comparing the real and imaginary parts we
$\cos 2\alpha + \cos 2\beta + \cos 2\gamma - \sin 2\alpha + \sin 2\beta + \sin 2\gamma = 0$
Also from (1) $(e^{i\alpha})^3 + (e^{i\beta})^3 + (e^{i\gamma})^3 = 3e^{i\alpha}\ e^{i\beta}\ e^{i\gamma}$
$\Rightarrow$ $e^{i3\alpha} + e^{i3\beta} + e^{i3\gamma} = 3e^{i(\alpha+\beta+\gamma)}$
Comparing the real and imaginary parts we obtain the results.

Example: If z_1 and z_2 are two complex numbers and $c > 0$, then prove that

$$|z_1 + z_2|^2 \le (I + C)\,|z_1|^2 + (I + C^{-1})\,|z_2|^2$$

Solution. We have to prove :

$$|z_1 + z_2|^2 \le (1 + c)\,|z_1|^2 + (1 + c^{-1})\,|z_2|^2$$

i.e. $|z_1|^3 + |z_2|^2 + z_1\bar{z}_2 + \bar{z}_2 z_2 \le (1 + c)\,|z_1|^2 + (1 + c^{-1})\,|z_2|^3$

or $z_1\bar{z}_2 + \bar{z}_2 z_2 \le c|z_1|^2 + c^{-1}|z_2|^2$ or $c|z_1|^2 + \dfrac{1}{c}\,|z_2|^2 - z_1\bar{z}_2 - \bar{z}_2 z_2 \ge 0$

(using $\text{Re}\,(z_1\bar{z}_2) \le |z_1\bar{z}_2|$)

or $\left(\sqrt{c}\,|z_1| - \dfrac{1}{\sqrt{c}}\,|z_2|\right)^2 \ge 0$ which is always true.

Example: If $\theta, \in [\pi/6, \pi/3]$, i = 1, 2, 3, 4, 5, and $z^4\cos\theta_1 + z^3\cos\theta_2 + z^3\cos\theta_3 . + z\cos\theta_4 + \cos\theta_5 = 2\sqrt{3}$, then show that $|z| > \dfrac{3}{4}$

Solution. Given that

$$\cos\theta_1 . z^4 + \cos\theta_2 . z^3 + \cos\theta_3 . z^2 + \cos\theta_4 . z + \cos\theta_5 = 2\sqrt{3}$$

or $|\cos\theta_1 . z^4 + \cos\theta_2 . z^3 + \cos\theta_3 . z^2 + \cos\theta_4 . z + \cos\theta_5| = 2\sqrt{3}$

$2\sqrt{3} \le |\cos\theta_1 . z^4| + |\cos\theta_2 . z^3| + |\cos\theta_3 . z^2| + \cos\theta_4 . z| + |\cos\theta_5|$

$\because$ $\theta i \in [\pi/6, \pi/3]$

$\therefore$ $\dfrac{1}{2} \le \cos\theta_i \le \dfrac{\sqrt{3}}{2}$

$$2\sqrt{3} \le \frac{\sqrt{3}}{2}\,|z|^4 + \frac{\sqrt{3}}{2}|z|^3 + \frac{\sqrt{3}}{2}\,|z|^2 + \frac{\sqrt{3}}{2}\,|z| + \frac{\sqrt{3}}{2}$$

$3 \le |z|^4 + |z|^3 + |z|^2 + |z|$

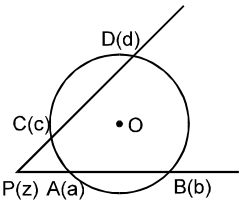
$3 < |z| + |z|^2 + |z|^3 + |z|^4 + |z|^5 + \ldots\ldots\infty$

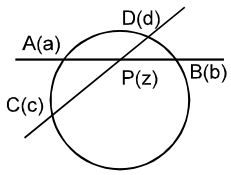
$3 < \dfrac{|z|}{1 - |z|}$ $\quad 3 - e\,|z| < |z|$

$4|z| > 3$ $\quad \therefore$ $\quad |z| > \dfrac{3}{4}$

Example: Two different non parallel lines cut the circle $|z| = r$ in point a, b, c, d respectively. Prove that these lines meet in the point z given by $z = \dfrac{a^{-1} + b^{-1} - c^{-1} - d^{-1}}{a^{-1}b^{-1} - c^{-1}d^{-1}}$

Solution. Since point P, A, B are collinear





$$\therefore \quad \begin{vmatrix} z & \bar{z} & 1 \\ a & \bar{a} & 1 \\ b & \bar{b} & 1 \end{vmatrix} = 0 \Rightarrow \quad z\left(\bar{a} - \bar{b}\right) - \bar{z}\,(a - b) + \left(a\bar{b} - a\bar{b}\right) = 0 \qquad \text{(i)}$$

Similarlym, since points P, C, D are collinear

$\therefore$ $z\left(\bar{a} - \bar{b}\right)(c - d) - z\left(\bar{c} - \bar{d}\right)(a - b) = \left(c\bar{d} - \bar{c}d\right)(a - b) - \left(a\bar{b} - \bar{a}b\right)(c - d)$ (iii)

$\because$ $z\bar{z} = r^2 = k$ (say) $\quad \therefore$ $\quad \bar{a} = \dfrac{k}{a}, \ \bar{b} = \dfrac{k}{b}, \ \bar{c} = \dfrac{k}{c}$ etc.

From equation (iii) we get

$$z\left(\frac{k}{a} - \frac{k}{b}\right)(c - d) - z\left(\frac{k}{c} - \frac{k}{d}\right)(a - b) = \left(\frac{ck}{d} - \frac{kd}{c}\right)(a - b) - \left(\frac{ak}{b} - \frac{bk}{a}\right)(c - d)$$

$\therefore$ $z = \dfrac{a^{-1} + b^{-1} - c^{-1} - d^{-1}}{a^{-1}b^{-1} - c^{-1}d^{-1}}$

Short Revision

1. DEFINITION :

Complex numbers are definited as expressions of the form a + ib where a, b ∈ R & i = $\sqrt{-1}$. It is denoted by z i.e. z = a + ib. 'a' is called as real part of z (Re z) and 'b' is called as imaginary part of z (Im z).

EVERY COMPLEX NUMBER CAN BE REGARDED AS

Purely real	Purely imaginary	Imaginary
if b = 0	if a = 0	if b ≠ 0

Note :

(a) The set R of real numbers is a proper subset of the Complex Numbers. Hence the Complete Number system is $N \subset W \subset I \subset Q \subset R \subset C$.

(b) Zero is both purely real as well as purely imaginary but not imaginary.

(c) $i = \sqrt{-1}$ is called the imaginary unit. Also $i^2 = -1$; $i^3 = -i$; $i^4 = 1$ etc.

(d) $\sqrt{a}\,\sqrt{b} = \sqrt{ab}$ only if atleast one of either a or b is non-negative.

2. CONJUGATE COMPLEX :

If z = a + ib then its conjugate complex is obtained by changing the sign of its imaginary part & is denoted by $\bar{z}$. i.e. $\bar{z} = a - ib$.

Note that :

(i) $z + \bar{z} = 2\,\text{Re}(z)$ **(ii)** $z - \bar{z} = 2i\,\text{Im}(z)$ **(iii)** $z\bar{z} = a^2 + b^2$ which is real

(iv) If z lies in the 1st quadrant then $\bar{z}$ lies in the 4th quadrant and $-\bar{z}$ lies in the 2nd quadrant.

3. ALGEBRAIC OPERATIONS :

The algebraic operations on complex numbers are similiar to those on real numbers treating *i* as a polynomial. Inequalities in complex numbers are not defined. There is no validity if we say that complex number is positive or negative.

e.g. $z > 0$, $4 + 2i < 2 + 4i$ are meaningless .

However in real numbers if $a^2 + b^2 = 0$ then a = 0 = b but in complex numbers,

$z_1^2 + z_2^2 = 0$ does not imply $z_1 = z_2 = 0$.

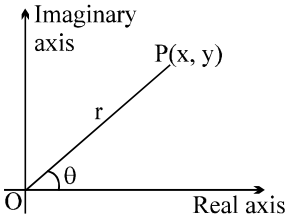
4. EQUALITY IN COMPLEX NUMBER :

Two complex numbers $z_1 = a_1 + ib_1$ & $z_2 = a_2 + ib_2$ are equal if and only if their real & imaginary parts coincide.

5. REPRESENTATION OF A COMPLEX NUMBER IN VARIOUS FORMS :

(a) Cartesian Form (Geometric Representation) :

Every complex number z = x + iy can be represented by a point on the cartesian plane known as complex plane (Argand diagram) by the ordered pair (x, y).



length OP is called modulus of the complex number denoted by $|z|$ & θ is called the argument or amplitude .

eg. $|z| = \sqrt{x^2 + y^2}$ &

$\theta = \tan^{-1}\frac{y}{x}$ (angle made by OP with positive x–axis)

NOTE :(i) $|z|$ is always non negative . Unlike real numbers $|z| = \begin{cases} z & \text{if } z > 0 \\ -z & \text{if } z < 0 \end{cases}$ is **not correct**

(ii) Argument of a complex number is a many valued function . If θ is the argument of a complex number then $2n\pi + \theta$; $n \in I$ will also be the argument of that complex number. Any two arguments of a complex number differ by $2n\pi$.

(iii) The unique value of θ such that $-\pi < \theta \le \pi$ is called the principal value of the argument.

(iv) Unless otherwise stated, amp z implies principal value of the argument.

(v) By specifying the modulus & argument a complex number is defined completely. For the complex number 0 + 0i the argument is not defined and this is the only complex number which is given by its modulus.

(vi) There exists a one-one correspondence between the points of the plane and the members of the set of complex numbers.

(b) Trignometric / Polar Representation :

$z = r(\cos\theta + i\sin\theta)$ where $|z| = r$; $\arg z = \theta$; $\bar{z} = r(\cos\theta - i\sin\theta)$

Note: $\cos\theta + i\sin\theta$ is also written as $CiS\,\theta$.

Also $\cos x = \dfrac{e^{ix} + e^{-ix}}{2}$ & $\sin x = \dfrac{e^{ix} - e^{-ix}}{2}$ are known as Euler's identities.

(c) Exponential Representation :

$z = re^{i\theta}$; $|z| = r$; $\arg z = \theta$; $\bar{z} = re^{-i\theta}$

6. IMPORTANT PROPERTIES OF CONJUGATE / MODULI / AMPLITUDE :

If $z, z_1, z_2 \in C$ then ;

(a) $z + \bar{z} = 2\,Re(z)$; $z - \bar{z} = 2i\,Im(z)$; $\overline{(\bar{z})} = z$; $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$;

$\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$; $\overline{z_1 z_2} = \bar{z}_1 \,.\, \bar{z}_2$ $\quad \overline{\left(\dfrac{z_1}{z_2}\right)} = \dfrac{\bar{z}_1}{\bar{z}_2}$; $z_2 \neq 0$

(b) $|z| \geq 0$; $|z| \geq Re(z)$; $|z| \geq Im(z)$; $|z| = |\bar{z}| = |-z|$; $z\bar{z} = |z|^2$;

$|z_1 z_2| = |z_1| \,.\, |z_2|$; $\left|\dfrac{z_1}{z_2}\right| = \dfrac{|z_1|}{|z_2|}$, $z_2 \neq 0$, $|z^n| = |z|^n$;

$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\left[|z_1|^2 + |z_2|^2\right]$

$\big||z_1| - |z_2|\big| \leq |z_1 + z_2| \leq |z_1| + |z_2|$ **[TRIANGLE INEQUALITY]**

(c) **(i)** $\text{amp}(z_1 \,.\, z_2) = \text{amp}\ z_1 + \text{amp}\ z_2 + 2k\pi$. $k \in I$

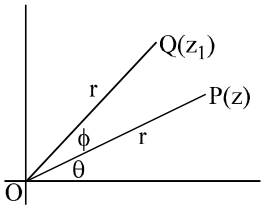
(ii) $\text{amp}\left(\dfrac{z_1}{z_2}\right) = \text{amp}\ z_1 - \text{amp}\ z_2 + 2k\pi$; $k \in I$

(iii) $\text{amp}(z^n) = n\,\text{amp}(z) + 2k\pi$.
where proper value of k must be chosen so that RHS lies in $(-\pi, \pi]$.

(7) VECTORIAL REPRESENTATION OF A COMPLEX :

Every complex number can be considered as if it is the position vector of that point. If the point P represents the complex number z then, $\vec{OP} = z$ & $|\vec{OP}| = |z|$.

NOTE :



(i) If $\vec{OP} = z = re^{i\theta}$ then $\vec{OQ} = z_1 = re^{i(\theta+\phi)} = z \,.\, e^{i\phi}$. If $\vec{OP}$ and $\vec{OQ}$ are of unequal magnitude then $\overset{\wedge}{OQ} = \overset{\wedge}{OP}\, e^{i\phi}$

(ii) If A, B, C & D are four points representing the complex numbers z_1, z_2, z_3 & z_4 then

AB || CD if $\dfrac{z_4 - z_3}{z_2 - z_1}$ is purely real ; AB $\perp$ CD if $\dfrac{z_4 - z_3}{z_2 - z_1}$ is purely imaginary]

(iii) If z_1, z_2, z_3 are the vertices of an equilateral triangle where z_0 is its circumcentre then

(a) $z_1^2 + z_2^2 + z_3^2 - z_1 z_2 - z_2 z_3 - z_3 z_1 = 0$ **(b)** $z_1^2 + z_2^2 + z_3^2 = 3\,z_0^2$

8. DEMOIVRE'S THEOREM : Statement : $\cos n\theta + i\sin n\theta$ is the value or one of the values of $(\cos\theta + i\sin\theta)^n$ ¥ $n \in Q$. The theorem is very useful in determining the roots of any complex quantity **Note :** Continued product of the roots of a complex quantity should be determined using theory of equations.

9. CUBE ROOT OF UNITY : (i) The cube roots of unity are 1 , $\dfrac{-1+i\sqrt{3}}{2}$, $\dfrac{-1-i\sqrt{3}}{2}$.

(ii) If w is one of the imaginary cube roots of unity then $1 + w + w^2 = 0$. In general $1 + w^r + w^{2r} = 0$; where $r \in I$ but is not the multiple of 3.

(iii) In polar form the cube roots of unity are :

$\cos 0 + i\sin 0$; $\cos\dfrac{2\pi}{3} + i\sin\dfrac{2\pi}{3}$, $\cos\dfrac{4\pi}{3} + i\sin\dfrac{4\pi}{3}$

(iv) The three cube roots of unity when plotted on the argand plane constitute the verties of an equilateral triangle.

(v) The following factorisation should be remembered :
(a, b, c ∈ R & ω is the cube root of unity)

$a^3 - b^3 = (a - b)(a - \omega b)(a - \omega^2 b)$; $x^2 + x + 1 = (x - \omega)(x - \omega^2)$;
$a^3 + b^3 = (a + b)(a + \omega b)(a + \omega^2 b)$;
$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)$

10. n^{th} ROOTS OF UNITY :

If $1, \alpha_1, \alpha_2, \alpha_3 ..... \alpha_{n-1}$ are the n, n^{th} root of unity then :

(i) They are in G.P. with common ratio $e^{i(2\pi/n)}$ **&**

(ii) $1^p + \alpha_1^p + \alpha_2^p + ... + \alpha_{n-1}^p = 0$ if p is not an integral multiple of n
$= n$ if p is an integral multiple of n

(iii) $(1 - \alpha_1)(1 - \alpha_2) ..... (1 - \alpha_{n-1}) = n$ **&**
$(1 + \alpha_1)(1 + \alpha_2) ..... (1 + \alpha_{n-1}) = 0$ if n is even and 1 if n is odd.

(iv) $1 . \alpha_1 . \alpha_2 . \alpha_3 ..... \alpha_{n-1} = 1$ or -1 according as n is odd or even.

11. THE SUM OF THE FOLLOWING SERIES SHOULD BE REMEMBERED :

(i) $\cos\theta + \cos 2\theta + \cos 3\theta + ... + \cos n\theta = \dfrac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\left(\dfrac{n+1}{2}\right)\theta.$

(ii) $\sin\theta + \sin 2\theta + \sin 3\theta + ... + \sin n\theta = \dfrac{\sin(n\theta/2)}{\sin(\theta/2)} \sin\left(\dfrac{n+1}{2}\right)\theta.$

Note : If $\theta = (2\pi/n)$ then the sum of the above series vanishes.

12. STRAIGHT LINES & CIRCLES IN TERMS OF COMPLEX NUMBERS :

(A) If z_1 & z_2 are two complex numbers then the complex number $z = \dfrac{nz_1 + mz_2}{m+n}$ divides the joins of z_1 & z_2 in the ratio m : n.

Note:(i) If a, b, c are three real numbers such that $az_1 + bz_2 + cz_3 = 0$;
where $a + b + c = 0$ and a,b,c are not all simultaneously zero, then the complex numbers z_1, z_2 & z_3 are collinear.

(ii) If the vertices A, B, C of a Δ represent the complex nos. z_1, z_2, z_3 respectively, then :

(a) Centroid of the $\Delta ABC = \dfrac{z_1 + z_2 + z_3}{3}$:

(b) Orthocentre of the $\Delta ABC =$
$\dfrac{(a\sec A)z_1 + (b\sec B)z_2 + (c\sec C)z_3}{a\sec A + b\sec B + c\sec C}$ **OR** $\dfrac{z_1\tan A + z_2\tan B + z_3\tan C}{\tan A + \tan B + \tan C}$

(c) Incentre of the $\Delta ABC = (az_1 + bz_2 + cz_3) \div (a + b + c)$.

(d) Circumcentre of the $\Delta ABC =$:
$(Z_1 \sin 2A + Z_2 \sin 2B + Z_3 \sin 2C) \div (\sin 2A + \sin 2B + \sin 2C)$.

(B) $\text{amp}(z) = \theta$ is a ray emanating from the origin inclined at an angle θ to the x–axis.

(C) $|z - a| = |z - b|$ is the perpendicular bisector of the line joining a to b.

(D) The equation of a line joining z_1 & z_2 is given by ;
$z = z_1 + t(z_1 - z_2)$ where t is a perameter.

(E) $z = z_1(1 + it)$ where t is a real parameter is a line through the point z_1 & perpendicular to oz_1.

(F) The equation of a line passing through z_1 & z_2 can be expressed in the determinant form as

$\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$. This is also the condition for three complex numbers to be collinear.

(G) Complex equation of a straight line through two given points z_1 & z_2 can be written as $z(\bar{z}_1 - \bar{z}_2) - \bar{z}(z_1 - z_2) + (z_1\bar{z}_2 - \bar{z}_1 z_2) = 0$, which on manipulating takes the form as $\bar{\alpha}z + \alpha\bar{z} + r = 0$ where r is real and α is a non zero complex constant.

(H) The equation of circle having centre z_0 & radius ρ is :
$|z - z_0| = \rho$ or $z\bar{z} - z_0\bar{z} - \bar{z}_0 z + \bar{z}_0 z_0 - \rho^2 = 0$ which is of the form

$z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0$, r is real centre $-\alpha$ & radius $\sqrt{\alpha\bar{\alpha} - r}$.

Circle will be real if $\alpha\bar{\alpha} - r \geq 0$.

(I) The equation of the circle described on the line segment joining z_1 & z_2 as diameter is :

(i) $\arg\dfrac{z - z_2}{z - z_1} = \pm\dfrac{\pi}{2}$ or $(z - z_1)(\bar{z} - \bar{z}_2) + (z - z_2)(\bar{z} - \bar{z}_1) = 0$

(J) Condition for four given points z_1, z_2, z_3 & z_4 to be concyclic is, the number

$\frac{z_3 - z_1}{z_3 - z_2} \cdot \frac{z_4 - z_2}{z_4 - z_1}$ is real. Hence the equation of a circle through 3 non collinear points z_1, z_2 & z_3 can be taken as $\frac{(z - z_2)(z_3 - z_1)}{(z - z_1)(z_3 - z_2)}$ is real $\Rightarrow \frac{(z - z_2)(z_3 - z_1)}{(z - z_1)(z_3 - z_2)} = \frac{(\bar{z} - \bar{z}_2)(\bar{z}_3 - \bar{z}_1)}{(\bar{z} - \bar{z}_1)(\bar{z}_3 - \bar{z}_2)}$

13.(a) Reflection points for a straight line :
Two given points P & Q are the reflection points for a given straight line if the given line is the right bisector of the segment PQ. Note that the two points denoted by the complex numbers z_1 & z_2 will be the reflection points for the straight line $\bar{\alpha} z + \alpha \bar{z} + r = 0$ if and only if ; $\bar{\alpha} z_1 + \alpha \bar{z}_2 + r = 0$, where r is real and α is non zero complex constant.

(b) Inverse points w.r.t. a circle :
Two points P & Q are said to be inverse w.r.t. a circle with centre 'O' and radius ρ, if :
(i) the point O, P, Q are collinear and on the same side of O. **(ii)** $OP \cdot OQ = \rho^2$.
Note that the two points z_1 & z_2 will be the inverse points w.r.t. the circle $z\bar{z} + \bar{\alpha} z + \alpha \bar{z} + r = 0$ if and only if $z_1 \bar{z}_2 + \bar{\alpha} z_1 + \alpha \bar{z}_2 + r = 0$.

14. PTOLEMY'S THEOREM : It states that the product of the lengths of the diagonals of a convex quadrilateral inscribed in a circle is equal to the sum of the lengths of the two pairs of its opposite sides. i.e. $|z_1 - z_3|\,|z_2 - z_4| = |z_1 - z_2|\,|z_3 - z_4| + |z_1 - z_4|\,|z_2 - z_3|$.

15. LOGARITHM OF A COMPLEX QUANTITY :

(i) $\text{Log}_e(\alpha + i\beta) = \frac{1}{2}\text{Log}_e(\alpha^2 + \beta^2) + i\left(2n\pi + \tan^{-1}\frac{\beta}{\alpha}\right)$ where $n \in I$.

(ii) i^i represents a set of positive real numbers given by $e^{-\left(2n\pi + \frac{\pi}{2}\right)}$, $n \in I$.

VERY ELEMENTARY EXERCISE

Q.1 Simplify and express the result in the form of $a + bi$

(a) $\left(\frac{1+2i}{2+i}\right)^2$ (b) $-i(9 + 6i)(2 - i)^{-1}$ (c) $\left(\frac{4i^3 - i}{2i + 1}\right)^2$ (d) $\frac{3+2i}{2-5i} + \frac{3-2i}{2+5i}$ (e) $\frac{(2+i)^2}{2-i} - \frac{(2-i)^2}{2+i}$

Q.2 Given that $x, y \in R$, solve : (a) $(x + 2y) + i(2x - 3y) = 5 - 4i$ (b) $(x + iy) + (7 - 5i) = 9 + 4i$
(c) $x^2 - y^2 - i(2x + y) = 2i$ (d) $(2 + 3i)x^2 - (3 - 2i)y = 2x - 3y + 5i$
(e) $4x^2 + 3xy + (2xy - 3x^2)i = 4y^2 - (x^2/2) + (3xy - 2y^2)i$

Q.3 Find the square root of : (a) $9 + 40i$ (b) $-11 - 60i$ (c) $50i$

Q.4 (a) If $f(x) = x^4 + 9x^3 + 35x^2 - x + 4$, find $f(-5 + 4i)$
(b) If $g(x) = x^4 - x^3 + x^2 + 3x - 5$, find $g(2 + 3i)$

Q.5 Among the complex numbers z satisfying the condition $|z + 3 - \sqrt{3}i| = \sqrt{3}$, find the number having the least positive argument.

Q.6 Solve the following equations over C and express the result in the form $a + ib$, $a, b \in R$.
(a) $ix^2 - 3x - 2i = 0$ (b) $2(1 + i)x^2 - 4(2 - i)x - 5 - 3i = 0$

Q.7 Locate the points representing the complex number z on the Argand plane:
(a) $|z + 1 - 2i| = \sqrt{7}$; **(b)** $|z - 1|^2 + |z + 1|^2 = 4$; **(c)** $\left|\frac{z-3}{z+3}\right| = 3$; **(d)** $|z - 3| = |z - 6|$

Q.8 If a & b are real numbers between 0 & 1 such that the points $z_1 = a + i$, $z_2 = 1 + bi$ & $z_3 = 0$ form an equilateral triangle, then find the values of 'a' and 'b'.

Q.9 For what real values of x & y are the numbers $-3 + ix^2y$ & $x^2 + y + 4i$ conjugate complex?

Q.10 Find the modulus, argument and the principal argument of the complex numbers.
(i) $6(\cos 310° - i \sin 310°)$ (ii) $-2(\cos 30° + i \sin 30°)$ (iii) $\frac{2+i}{4i + (1+i)^2}$

Q.11 If $(x + iy)^{1/3} = a + bi$; prove that $4(a^2 - b^2) = \frac{x}{a} + \frac{y}{b}$.

Q.12(a) If $\frac{a + ib}{c + id} = p + qi$, prove that $p^2 + q^2 = \frac{a^2 + b^2}{c^2 + d^2}$.

(b) Let z_1, z_2, z_3 be the complex numbers such that
$z_1 + z_2 + z_3 = z_1z_2 + z_2z_3 + z_3z_1 = 0$. Prove that $|z_1| = |z_2| = |z_3|$.

Q.13 Let z be a complex number such that $z \in c\backslash R$ and $\frac{1 + z + z^2}{1 - z + z^2} \in R$, then prove that $|z| = 1$.

Q.14 Prove the identity, $|1 - z_1\bar{z}_2|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2)$

Q.15 For any two complex numbers, prove that $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\left[|z_1|^2 + |z_2|^2\right]$. Also give the geometrical interpretation of this identity.

Q.16 (a) Find all non–zero complex numbers Z satisfying $\bar{Z} = iZ^2$.

(b) If the complex numbers $z_1, z_2, \ldots\ldots\ldots\ldots z_n$ lie on the unit circle $|z| = 1$ then show that $|z_1 + z_2 + \ldots\ldots\ldots\ldots + z_n| = |z_1^{-1} + z_2^{-1} + \ldots\ldots\ldots\ldots + z_n^{-1}|$.

Q.17 Find the Cartesian equation of the locus of 'z' in the complex plane satisfying, $|z - 4| + |z + 4| = 16$.

Q.18 If ω is an imaginary cube root of unity then prove that :

(a) $(1 + \omega - \omega^2)^3 - (1 - \omega + \omega^2)^3 = 0$ (b) $(1 - \omega + \omega^2)^5 + (1 + \omega - \omega^2)^5 = 32$

(c) If ω is the cube root of unity, Find the value of, $(1 + 5\omega^2 + \omega^4)(1 + 5\omega^4 + \omega^2)(5\omega^3 + \omega + \omega^2)$.

Q.19 If ω is a cube root of unity, prove that ; (i) $(1 + \omega - \omega^2)^3 - (1 - \omega + \omega^2)^3$

(ii) $\dfrac{a + b\omega + c\omega^2}{c + a\omega + b\omega^2} = \omega^2$ (iii) $(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^8) = 9$

Q.20 If $x = a + b$; $y = a\omega + b\omega^2$; $z = a\omega^2 + b\omega$, show that

(i) $xyz = a^3 + b^3$ (ii) $x^2 + y^2 + z^2 = 6ab$ (iii) $x^3 + y^3 + z^3 = 3(a^3 + b^3)$

Q.21 If $(w \neq 1)$ is a cube root of unity then $\begin{vmatrix} 1 & 1+i+w^2 & w^2 \\ 1-i & -1 & w^2-1 \\ -i & -i+w-1 & -1 \end{vmatrix} =$

(A) 0 (B) 1 (C) i (D) w

Q.22(a) $(1 + w)^7 = A + Bw$ where w is the imaginary cube root of a unity and $A, B \in R$, find the ordered pair (A, B).

(b) The value of the expression ;

$1.(2 - w)(2 - w^2) + 2.(3 - w)(3 - w^2) + \ldots\ldots\ldots\ldots + (n - 1).(n - w)(n - w^2)$, where w is an imaginary cube root of unity is _____.

Q.23 If $n \in N$, prove that $(1 + i)^n + (1 - i)^n = 2^{\frac{n}{2}+1} \cdot \cos\dfrac{n\pi}{4}$.

Q.24 Show that the sum $\displaystyle\sum_{k=1}^{2n}\left(\sin\frac{2\pi k}{2n+1} - i\cos\frac{2\pi k}{2n+1}\right)$ simplifies to a pure imaginary number.

Q.25 If $x = \cos\theta + i\sin\theta$ & $1 + \sqrt{1 - a^2} = na$, prove that $1 + a\cos\theta = \dfrac{a}{2n}(1 + nx)\left(1 + \dfrac{n}{x}\right)$.

Q.26 The number t is real and not an integral multiple of $\pi/2$. The complex number x_1 and x_2 are the roots of the equation, $\tan^2(t) \cdot x^2 + \tan(t) \cdot x + 1 = 0$

Show that $(x_1)^n + (x_2)^n = 2\left(\cos\dfrac{2n\pi}{3}\right)\cot^n(t)$.

EXERCISE-1

Q.1 Simplify and express the result in the form of $a + bi$:

(a) $-i(9 + 6i)(2 - i)^{-1}$ (b) $\left(\dfrac{4i^3 - i}{2i + 1}\right)^2$ (c) $\dfrac{3 + 2i}{2 - 5i} + \dfrac{3 - 2i}{2 + 5i}$

(d) $\dfrac{(2 + i)^2}{2 - i} - \dfrac{(2 - i)^2}{2 + i}$ (e) $\sqrt{i} + \sqrt{-i}$

Q.2 Find the modulus, argument and the principal argument of the complex numbers.

(i) $z = 1 + \cos\left(\dfrac{10\pi}{9}\right) + i\sin\left(\dfrac{10\pi}{9}\right)$ (ii) $(\tan 1 - i)^2$

(iii) $z = \dfrac{\sqrt{5 + 12i} + \sqrt{5 - 12i}}{\sqrt{5 + 12i} - \sqrt{5 - 12i}}$ (iv) $\dfrac{i - 1}{i\left(1 - \cos\dfrac{2\pi}{5}\right) + \sin\dfrac{2\pi}{5}}$

Q.3 Given that $x, y \in R$, solve :

(a) $(x + 2y) + i(2x - 3y) = 5 - 4i$ (b) $\dfrac{x}{1 + 2i} + \dfrac{y}{3 + 2i} = \dfrac{5 + 6i}{8i - 1}$

(c) $x^2 - y^2 - i(2x + y) = 2i$ (d) $(2 + 3i)x^2 - (3 - 2i)y = 2x - 3y + 5i$

(e) $4x^2 + 3xy + (2xy - 3x^2)i = 4y^2 - (x^2/2) + (3xy - 2y^2)i$

Q.4(a) Let Z is complex satisfying the equation, $z^2 - (3 + i)z + m + 2i = 0$, where $m \in R$.

Suppose the equation has a real root, then find the value of *m*.

(b) a, b, c are real numbers in the polynomial, $P(Z) = 2Z^4 + aZ^3 + bZ^2 + cZ + 3$
If two roots of the equation $P(Z) = 0$ are 2 and i, then find the value of 'a'.

Q.5(a) Find the real values of x & y for which $z_1 = 9y^2 - 4 - 10\,ix$ and $z_2 = 8y^2 - 20\,i$ are conjugate complex of each other.

(b) Find the value of $x^4 - x^3 + x^2 + 3x - 5$ if $x = 2 + 3i$

Q.6 Solve the following for z : (a) $z^2 - (3 - 2\,i)z = (5i - 5)$ (b) $|z| + z = 2 + i$

Q.7(a) If $iZ^3 + Z^2 - Z + i = 0$, then show that $|Z| = 1$.

(b) Let z_1 and z_2 be two complex numbers such that $\left|\frac{z_1 - 2z_2}{2 - z_1\bar{z}_2}\right| = 1$ and $|z_2| \neq 1$, find $|z_1|$.

(c) Let $z_1 = 10 + 6i$ & $z_2 = 4 + 6i$. If z is any complex number such that the argument of, $\frac{z - z_1}{z - z_2}$ is $\frac{\pi}{4}$, then prove that $|z - 7 - 9i| = 3\sqrt{2}$.

Q.8 Show that the product,

$$\left[1+\left(\frac{1+i}{2}\right)\right]\left[1+\left(\frac{1+i}{2}\right)^2\right]\left[1+\left(\frac{1+i}{2}\right)^{2^2}\right].....\left[1+\left(\frac{1+i}{2}\right)^{2^n}\right]$$ is equal to $\left(1 - \frac{1}{2^{2^n}}\right)(1+i)$ where $n \geq 2$.

Q.9 Let a & b be complex numbers (which may be real) and let,
$Z = z^3 + (a + b + 3i)\,z^2 + (ab + 3\,ia + 2\,ib - 2)\,z + 2\,abi - 2a$.
(i) Show that Z is divisible by, $z + b + i$. (ii) Find all complex numbers z for which $Z = 0$.
(iii) Find all purely imaginary numbers a & b when $z = 1 + i$ and Z is a real number.

Q.10 Interpret the following locii in $z \in C$.

(a) $1 < |z - 2i| < 3$ (b) $\text{Re}\left(\frac{z + 2i}{iz + 2}\right) \leq 4 \quad (z \neq 2i)$

(c) $\text{Arg}(z + i) - \text{Arg}(z - i) = \pi/2$ (d) $\text{Arg}(z - a) = \pi/3$ where $a = 3 + 4i$.

Q.11 Prove that the complex numbers z_1 and z_2 and the origin form an isosceles triangle with vertical angle $2\pi/3$ if $z_1^2 + z_2^2 + z_1 z_2 = 0$.

Q.12 P is a point on the Aragand diagram. On the circle with OP as diameter two points Q & R are taken such that $\angle POQ = \angle QOR = \theta$. If 'O' is the origin & P, Q & R are represented by the complex numbers z_1, Z_2 & Z_3 respectively, show that : $Z_2^2 \cdot \cos 2\theta = Z_1 \cdot Z_3 \cos^2\theta$.

Q.13 Let z_1, z_2, z_3 are three pair wise distinct complex numbers and t_1, t_2, t_3 are non-negative real numbers such that $t_1 + t_2 + t_3 = 1$. Prove that the complex number $z = t_1z_1 + t_2z_2 + t_3z_3$ lies inside a triangle with vertices z_1, z_2, z_3 or on its boundry.

Q.14 If $a\,\text{CiS}\,\alpha$, $b\,\text{CiS}\,\beta$, $c\,\text{CiS}\,\gamma$ represent three distinct collinear points in an Argand's plane, then prove the following :

(i) $\Sigma\, ab \sin(\alpha - \beta) = 0$.

(ii) $(a\,\text{CiS}\,\alpha)\sqrt{b^2 + c^2 - 2bc\cos(\beta - \gamma)} \pm (b\,\text{CiS}\,\beta)\sqrt{a^2 + c^2 - 2ac\cos(\alpha - \gamma)}$
$\mp (c\,\text{CiS}\,\gamma)\sqrt{a^2 + b^2 - 2ab\cos(\alpha - \beta)} = 0$.

Q.15 Find all real values of the parameter a for which the equation $(a - 1)z^4 - 4z^2 + a + 2 = 0$ has only pure imaginary roots.

Q.16 Let $A \equiv z_1$; $B \equiv z_2$; $C \equiv z_3$ are three complex numbers denoting the vertices of an acute angled triangle. If the origin 'O' is the orthocentre of the triangle, then prove that

$$z_1\bar{z}_2 + \bar{z}_1 z_2 = z_2\bar{z}_3 + \bar{z}_2 z_3 = z_3\bar{z}_1 + \bar{z}_3 z_1$$

hence show that the ΔABC is a right angled triangle $\Leftrightarrow z_1\bar{z}_2 + \bar{z}_1 z_2 = z_2\bar{z}_3 + \bar{z}_2 z_3 = z_3\bar{z}_1 + \bar{z}_3 z_1 = 0$

Q.17 If the complex number P(w) lies on the standard unit circle in an Argand's plane and $z = (aw + b)(w - c)^{-1}$ then, find the locus of z and interpret it. Given a, b, c are real.

Q.18(a) Without expanding the determinant at any stage , find $K \in R$ such that

$$\begin{vmatrix} 4i & 8+i & 4+3i \\ -8+i & 16i & i \\ -4+Ki & i & 8i \end{vmatrix}$$ has purely imaginary value.

(b) If A, B and C are the angles of a triangle

$D = \begin{vmatrix} e^{-2iA} & e^{iC} & e^{iB} \\ e^{iC} & e^{-2iB} & e^{iA} \\ e^{iB} & e^{iA} & e^{-2iC} \end{vmatrix}$ where $i = \sqrt{-1}$ then find the value of D.

Q.19 If w is an imaginary cube root of unity then prove that :
(a) $(1 - w + w^2)(1 - w^2 + w^4)(1 - w^4 + w^8)$ ..... to 2n factors $= 2^{2n}$.
(b) If w is a complex cube root of unity, find the value of
$(1 + w)(1 + w^2)(1 + w^4)(1 + w^8)$ ..... to n factors .

Q.20 Prove that $\left(\frac{1+\sin\theta+i\cos\theta}{1+\sin\theta-i\cos\theta}\right)^n = \cos\left(\frac{n\pi}{2}-n\theta\right)+i\sin\left(\frac{n\pi}{2}-n\theta\right)$. Hence deduce that

$\left(1+\sin\frac{\pi}{5}+i\cos\frac{\pi}{5}\right)^5 + i\left(1+\sin\frac{\pi}{5}-i\cos\frac{\pi}{5}\right)^5 = 0$

Q.21 If $\cos(\alpha-\beta)+\cos(\beta-\gamma)+\cos(\gamma-\alpha) = -3/2$ then prove that :
(a) $\Sigma\cos 2\alpha = 0 = \Sigma\sin 2\alpha$ (b) $\Sigma\sin(\alpha+\beta) = 0 = \Sigma\cos(\alpha+\beta)$ (c) $\Sigma\sin^2\alpha = \Sigma\cos^2\alpha = 3/2$
(d) $\Sigma\sin 3\alpha = 3\sin(\alpha+\beta+\gamma)$ (e) $\Sigma\cos 3\alpha = 3\cos(\alpha+\beta+\gamma)$
(f) $\cos^3(\theta+\alpha)+\cos^3(\theta+\beta)+\cos^3(\theta+\gamma) = 3\cos(\theta+\alpha).\cos(\theta+\beta).\cos(\theta+\gamma)$ where $\theta \in R$.

Q.22 Resolve $Z^5 + 1$ into linear & quadratic factors with real coefficients. Deduce that : $4\cdot\sin\frac{\pi}{10}\cdot\cos\frac{\pi}{5} = 1$.

Q.23 If $x = 1+i\sqrt{3}$; $y = 1-i\sqrt{3}$ & $z = 2$, then prove that $x^p + y^p = z^p$ for every prime $p > 3$.

Q.24 If the expression $z^5 - 32$ can be factorised into linear and quadratic factors over real coefficients as $(z^5 - 32) = (z-2)(z^2 - pz + 4)(z^2 - qz + 4)$ then find the value of $(p^2 + 2p)$.

Q.25(a) Let $z = x + iy$ be a complex number, where x and y are real numbers. Let A and B be the sets defined by $A = \{z : |z| \le 2\}$ and $B = \{z : (1-i)z + (1+i)\bar{z} \ge 4\}$. Find the area of the region $A \cap B$.

(b) For all real numbers x, let the mapping $f(x) = \frac{1}{x-i}$, where $i = \sqrt{-1}$. If there exist real number a, b, c and d for which $f(a), f(b), f(c)$ and $f(d)$ form a square on the complex plane. Find the area of the square.

EXERCISE-2

Q.1 If $\begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix} = 0$; where p , q , r are the moduli of non–zero complex numbers u, v, w respectively, prove that, $\arg\frac{w}{v} = \arg\left(\frac{w-u}{v-u}\right)^2$.

Q.2 The equation $x^3 = 9 + 46i$ where $i = \sqrt{-1}$ has a solution of the form $a + bi$ where a and b are integers. Find the value of $(a^3 + b^3)$.

Q.3 Show that the locus formed by z in the equation $z^3 + iz = 1$ never crosses the co-ordinate axes in the Argand's plane. Further show that $|z| = \sqrt{\frac{-\mathrm{Im}(z)}{2\,\mathrm{Re}(z)\,\mathrm{Im}(z)+1}}$

Q.4 If ω is the fifth root of 2 and $x = \omega + \omega^2$, prove that $x^5 = 10x^2 + 10x + 6$.

Q.5 Prove that , with regard to the quadratic equation $z^2 + (p + ip')z + q + iq' = 0$ where p , p′, q , q′ are all real.
(i) if the equation has one real root then $q'^2 - pp'q' + qp'^2 = 0$.
(ii) if the equation has two equal roots then $p^2 - p'^2 = 4q$ & $pp' = 2q'$.
State whether these equal roots are real or complex.

Q.6 If the equation $(z+1)^7 + z^7 = 0$ has roots $z_1, z_2, \ldots z_7$, find the value of
(a) $\sum_{r=1}^{7}\mathrm{Re}(Z_r)$ and (b) $\sum_{r=1}^{7}\mathrm{Im}(Z_r)$

Q.7 Find the roots of the equation $Z^n = (Z+1)^n$ and show that the points which represent them are collinear on the complex plane. Hence show that these roots are also the roots of the equation

$\left(2\sin\frac{m\pi}{n}\right)^2\bar{Z}^2 + \left(2\sin\frac{m\pi}{n}\right)^2\bar{Z} + 1 = 0.$

Q.8 Dividing f(z) by $z - i$, we get the remainder i and dividing it by $z + i$, we get the remainder

$1 + i$. Find the remainder upon the division of $f(z)$ by $z^2 + 1$.

Q.9 Let z_1 & z_2 be any two arbitrary complex numbers then prove that :

$$|z_1 + z_2| \geq \frac{1}{2}\left(|z_1| + |z_2|\right)\left|\frac{z_1}{|z_1|} + \frac{z_2}{|z_2|}\right|.$$

Q.10 If Z_r, $r = 1, 2, 3, \ldots\ldots 2m$, $m \in N$ are the roots of the equation

$Z^{2m} + Z^{2m-1} + Z^{2m-2} + \ldots\ldots\ldots\ldots + Z + 1 = 0$ then prove that $\sum_{r=1}^{2m} \frac{1}{Z_r - 1} = -m$

Q.11 If $(1 + x)^n = C_0 + C_1 x + C_2 x^2 + \ldots. + C_n x^n$ $(n \in N)$, prove that :

(a) $C_0 + C_4 + C_8 + \ldots. = \frac{1}{2}\left[2^{n-1} + 2^{n/2} \cos\frac{n\pi}{4}\right]$ (b) $C_1 + C_5 + C_9 + \ldots. = \frac{1}{2}\left[2^{n-1} + 2^{n/2} \sin\frac{n\pi}{4}\right]$

(c) $C_2 + C_6 + C_{10} + \ldots\ldots = \frac{1}{2}\left[2^{n-1} - 2^{n/2} \cos\frac{n\pi}{4}\right]$ (d) $C_3 + C_7 + C_{11} + \ldots. = \frac{1}{2}\left[2^{n-1} - 2^{n/2} \sin\frac{n\pi}{4}\right]$

(e) $C_0 + C_3 + C_6 + C_9 + \ldots\ldots\ldots = \frac{1}{3}\left[2^n + 2\cos\frac{n\pi}{3}\right]$

Q.12 Let z_1, z_2, z_3, z_4 be the vertices A, B, C, D respectively of a square on the Argand diagram taken in anticlockwise direction then prove that :

(i) $2z_2 = (1 + i) z_1 + (1 - i) z_3$ & (ii) $2z_4 = (1 - i) z_1 + (1 + i) z_3$

Q.13 Show that all the roots of the equation $\left(\frac{1 + ix}{1 - ix}\right)^n = \frac{1 + ia}{1 - ia}$ $a \in R$ are real and distinct.

Q.14 Prove that:

(a) $\cos x + {}^nC_1 \cos 2x + {}^nC_2 \cos 3x + \ldots\ldots + {}^nC_n \cos(n + 1)x = 2^n \cdot \cos^n \frac{x}{2} \cdot \cos\left(\frac{n + 2}{2}\right)x$

(b) $\sin x + {}^nC_1 \sin 2x + {}^nC_2 \sin 3x + \ldots\ldots + {}^nC_n \sin(n + 1)x = 2^n \cdot \cos^n \frac{x}{2} \cdot \sin\left(\frac{n + 2}{2}\right)x$

(c) $\cos\left(\frac{2\pi}{2n + 1}\right) + \cos\left(\frac{4\pi}{2n + 1}\right) + \cos\left(\frac{6\pi}{2n + 1}\right) + \ldots\ldots + \cos\left(\frac{2n\pi}{2n + 1}\right) = -\frac{1}{2}$ When $n \in N$.

Q.15 Show that all roots of the equation $a_0 z^n + a_1 z^{n-1} + \ldots\ldots + a_{n-1} z + a_n = n$,

where $|a_i| \leq 1$, $i = 0, 1, 2, \ldots.$, n lie outside the circle with centre at the origin and radius $\frac{n - 1}{n}$.

Q.16 The points A, B, C depict the complex numbers z_1, z_2, z_3 respectively on a complex plane & the angle B & C of the triangle ABC are each equal to $\frac{1}{2}(\pi - \alpha)$. Show that

$$(z_2 - z_3)^2 = 4 (z_3 - z_1)(z_1 - z_2) \sin^2 \frac{\alpha}{2}.$$

Q.17 Show that the equation $\frac{A_1^2}{x - a_1} + \frac{A_2^2}{x - a_2} + \ldots\ldots + \frac{A_n^2}{x - a_n} = k$ has no imaginary root, given that:

$a_1, a_2, a_3 \ldots. a_n$ & $A_1, A_2, A_3 \ldots\ldots A_n$, k are all real numbers.

Q.18 Let a, b, c be distinct complex numbers such that $\frac{a}{1 - b} = \frac{b}{1 - c} = \frac{c}{1 - a} = k$. Find the value of k.

Q.19 Let α, β be fixed complex numbers and z is a variable complex number such that,

$$|z - \alpha|^2 + |z - \beta|^2 = k.$$

Find out the limits for 'k' such that the locus of z is a circle. Find also the centre and radius of the circle.

Q.20 C is the complex number. $f : C \to R$ is defined by $f(z) = |z^3 - z + 2|$. What is the maximum value of f on the unit circle $|z| = 1$?

Q.21 Let $f(x) = \log_{\cos 3x}(\cos 2ix)$ if $x \neq 0$ and $f(0) = K$ (where $i = \sqrt{-1}$) is continuous at $x = 0$ then find the value of K. Use of L Hospital's rule or series expansion not allowed.

Q.22 If z_1, z_2 are the roots of the equation $az^2 + bz + c = 0$, with a, b, $c > 0$; $2b^2 > 4ac > b^2$; $z_1 \in$ third quadrant ; $z_2 \in$ second quadrant in the argand's plane then, show that

$$\arg\left(\frac{z_1}{z_2}\right) = 2\cos^{-1}\left(\frac{b^2}{4ac}\right)^{1/2}$$

Q.23 Find the set of points on the argand plane for which the real part of the complex number $(1 + i)\, z^2$ is positive where $z = x + iy$, $x, y \in R$ and $i = \sqrt{-1}$.

Q.24 If a and b are positive integer such that $N = (a + ib)^3 - 107i$ is a positive integer. Find N.

Q.25 If the biquadratic $x^4 + ax^3 + bx^2 + cx + d = 0$ $(a, b, c, d \in R)$ has 4 non real roots, two with sum $3 + 4i$ and the other two with product $13 + i$. Find the value of 'b'.

EXERCISE-3

Q.1 Evaluate: $\sum_{p=1}^{32} (3p + 2)\left(\sum_{q=1}^{10}\left(\sin\frac{2q\pi}{11} - i\cos\frac{2q\pi}{11}\right)\right)^p$. [REE '97, 6]

Q.2(a) Let z_1 and z_2 be roots of the equation $z^2 + pz + q = 0$, where the co–efficients p and q may be complex numbers. Let A and B represent z_1 and z_2 in the complex plane. If $\angle AOB = \alpha \neq 0$ and OA = OB, where O is the origin. Prove that $p^2 = 4q\cos^2\left(\frac{\alpha}{2}\right)$. [JEE '97 , 5]

(b) Prove that $\sum_{k=1}^{n-1} (n-k)\cos\frac{2k\pi}{n} = -\frac{n}{2}$ where $n \geq 3$ is an integer. [JEE '97, 5]

Q.3(a) If ω is an imaginary cube root of unity, then $(1 + \omega - \omega^2)^7$ equals
(A) 128ω (B) -128ω (C) $128\omega^2$ (D) $-128\omega^2$

(b) The value of the sum $\sum_{n=1}^{13}\left(i^n + i^{n+1}\right)$, where $i = \sqrt{-1}$, equals
(A) i (B) $i - 1$ (C) $-i$ (D) 0 [JEE' 98, 2 + 2]

Q.4 Find all the roots of the equation $(3z - 1)^4 + (z - 2)^4 = 0$ in the simplified form of $a + ib$. [REE '98, 6]

Q.5(a) If $i = \sqrt{-1}$, then $4 + 5\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{334} + 3\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{365}$ is equal to :
(A) $1 - i\sqrt{3}$ (B) $-1 + i\sqrt{3}$ (C) $i\sqrt{3}$ (D) $-i\sqrt{3}$

(b) For complex numbers z & ω, prove that, $|z|^2\,\omega - |\omega|^2 z = z - \omega$ if and only if, $z = \omega$ or $z\bar{\omega} = 1$ [JEE '99, 2 + 10 (out of 200)]

Q.6 If $\alpha = e^{\frac{2\pi i}{7}}$ and $f(x) = A_0 + \sum_{k=1}^{20} A_k x^k$, then find the value of, $f(x) + f(\alpha x) + ... + f(\alpha^6 x)$ independent of α. [REE '99, 6]

Q.7(a) If z_1, z_2, z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = 1$, then $|z_1 + z_2 + z_3|$ is :
(A) equal to 1 (B) less than 1 (C) greater than 3 (D) equal to 3

(b) If $\arg(z) < 0$, then $\arg(-z) - \arg(z) =$
(A) π (B) $-\pi$ (C) $-\frac{\pi}{2}$ (D) $\frac{\pi}{2}$
[JEE 2000 (Screening) 1 + 1 out of 35]

Q.8 Given, $z = \cos\frac{2\pi}{2n+1} + i\sin\frac{2\pi}{2n+1}$, 'n' a positive integer, find the equation whose roots are, $\alpha = z + z^3 + ... + z^{2n-1}$ & $\beta = z^2 + z^4 + ... + z^{2n}$.
[REE 2000 (Mains) 3 out of 100]

Q.9(a) The complex numbers z_1, z_2 and z_3 satisfying $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$ are the vertices of a triangle which is
(A) of area zero (B) right-angled isosceles
(C) equilateral (D) obtuse – angled isosceles

(b) Let z_1 and z_2 be nth roots of unity which subtend a right angle at the origin. Then n must be of the form
(A) $4k + 1$ (B) $4k + 2$ (C) $4k + 3$ (D) $4k$
[JEE 2001 (Scr) 1 + 1 out of 35]

Q.10 Find all those roots of the equation $z^{12} - 56z^6 - 512 = 0$ whose imaginary part is positive.
[REE 2000, 3 out of 100]

Q.11(a) Let $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$. Then the value of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1-\omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{vmatrix}$ is
(A) 3ω (B) $3\omega(\omega - 1)$ (C) $3\omega^2$ (D) $3\omega(1 - \omega)$

(b) For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$, the minimum value of $|z_1 - z_2|$ is
(A) 0 (B) 2 (C) 7 (D) 17
[JEE 2002 (Scr) 3+3]

(c) Let a complex number α, $\alpha \neq 1$, be a root of the equation
$z^{p+q} - z^p - z^q + 1 = 0$ where p, q are distinct primes.
Show that either $1 + \alpha + \alpha^2 + ... + \alpha^{p-1} = 0$ or $1 + \alpha + \alpha^2 + ... + \alpha^{q-1} = 0$, but not both together.
[JEE 2002, (5)]

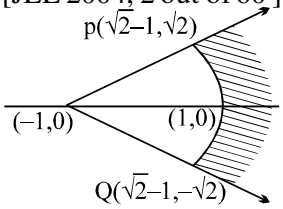
Q.12(a) If z_1 and z_2 are two complex numbers such that $|z_1| < 1 < |z_2|$ then prove that $\left|\frac{1 - z_1\bar{z}_2}{z_1 - z_2}\right| < 1$.

(b) Prove that there exists no complex number z such that $|z| < \frac{1}{3}$ and $\sum_{r=1}^{n} a_r z^r = 1$ where $|a_r| < 2$.
[JEE-03, 2 + 2 out of 60]

Q.13(a) ω is an imaginary cube root of unity. If $(1 + \omega^2)^m = (1 + \omega^4)^m$, then least positive integral value of m is
(A) 6 (B) 5 (C) 4 (D) 3
[JEE 2004 (Scr)]

(b) Find centre and radius of the circle determined by all complex numbers $z = x + iy$ satisfying $\left|\frac{(z-\alpha)}{(z-\beta)}\right| = k$,
where $\alpha = \alpha_1 + i\alpha_2$, $\beta = \beta_1 + i\beta_2$ are fixed complex and $k \neq 1$. [JEE 2004, 2 out of 60]

Q.14(a) The locus of z which lies in shaded region is best represented by
(A) $z : |z + 1| > 2, |\arg(z + 1)| < \pi/4$
(B) $z : |z - 1| > 2, |\arg(z - 1)| < \pi/4$
(C) $z : |z + 1| < 2, |\arg(z + 1)| < \pi/2$
(D) $z : |z - 1| < 2, |\arg(z - 1)| < \pi/2$



(b) If a, b, c are integers not all equal and w is a cube root of unity ($w \neq 1$), then the minimum value of $|a + bw + cw^2|$ is
(A) 0 (B) 1 (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$
[JEE 2005 (Scr), 3 + 3]

(c) If one of the vertices of the square circumscribing the circle $|z - 1| = \sqrt{2}$ is $2 + \sqrt{3}\,i$. Find the other vertices of square. [JEE 2005 (Mains), 4]

Q.15 If $w = \alpha + i\beta$ where $\beta \neq 0$ and $z \neq 1$, satisfies the condition that $\frac{w - \bar{w}z}{1 - z}$ is purely real, then the set of values of z is
(A) $\{z : |z| = 1\}$ (B) $\{z : z = \bar{z}\}$ (C) $\{z : z \neq 1\}$ (D) $\{z : |z| = 1, z \neq 1\}$
[JEE 2006, 3]

ANSWER KEY

VERY ELEMENTARY EXERCISE

Q.1 **(a)** $\frac{7}{25} + \frac{24}{25}i$; **(b)** $\frac{21}{5} - \frac{12}{5}i$; **(c)** $3 + 4i$; **(d)** $-\frac{8}{29} + 0i$; **(e)** $\frac{22}{5}i$

Q.2 **(a)** $x = 1, y = 2$; **(b)** (2, 9); **(c)** (–2 , 2) or $\left(-\frac{2}{3}, -\frac{2}{3}\right)$; **(d)** (1 ,1) $\left(0, \frac{5}{2}\right)$ **(e)** $x = K$, $y = \frac{3K}{2}$, $K \in R$

Q.3 **(a)** $\pm(5+4i)$; **(b)** $\pm(5-6i)$ **(c)** $\pm 5(1+i)$ **Q.4** **(a)** -160 ; **(b)** $-(77+108\,i)$

Q.5 $-\frac{3}{2}+\frac{3\sqrt{3}}{2}i$ **Q.6** **(a)** $-i, -2i$ **(b)** $\frac{3-5i}{2}$ or $-\frac{1+i}{2}$

Q.7 **(a)** on a circle of radius $\sqrt{7}$ with centre $(-1, 2)$ **;** **(b)** on a unit circle with centre at origin
(c) on a circle with centre $(-15/4, 0)$ & radius $9/4$ **; (d)** a straight line

Q.8 $a = b = 2-\sqrt{3}$; **Q.9** $x = 1,\ y = -4$ or $x = -1,\ y = -4$

Q.10 **(i)** Modulus $= 6$, Arg $= 2k\pi + \frac{5\pi}{18}$ $(K \in I)$, Principal Arg $= \frac{5\pi}{18}$ $(K \in I)$

(ii) Modulus $= 2$, Arg $= 2k\pi + \frac{7\pi}{6}$, Principal Arg $= -\frac{5\pi}{6}$

(iii) Modulus $= \frac{\sqrt{5}}{6}$, Arg $= 2k\pi - \tan^{-1}2$ $(K \in I)$, Principal Arg $= -\tan^{-1}2$

Q.16 **(a)** $\frac{\sqrt{3}}{2}-\frac{i}{2}, -\frac{\sqrt{3}}{2}-\frac{i}{2}, i$; **Q.17** $\frac{x^2}{64}+\frac{y^2}{48}=1$; **Q.18** **(c)** 64 ; **Q.21** A

Q.22 **(a)** $(1, 1)$; **(b)** $\left[\frac{n(n+1)}{2}\right]^2 - n$

EXERCISE-1

Q.1 (a) $\frac{21}{5}-\frac{12}{5}i$ **(b)** $3+4i$ **(c)** $-\frac{8}{29}+0\,i$ **(d)** $\frac{22}{5}i$ **(e)** $\pm\sqrt{2}+0i$ or $0\pm\sqrt{2}\,i$

Q.2 (i) Principal Arg $z = -\frac{4\pi}{9}$; $|z| = 2\cos\frac{4\pi}{9}$; Arg $z = 2k\pi - \frac{4\pi}{9}$ $k \in I$

(ii) Modulus $= \sec^2 1$, Arg $= 2n\pi + (2-\pi)$, Principal Arg $= (2-\pi)$

(iii) Principal value of Agr $z = -\frac{\pi}{2}$ & $|z| = \frac{3}{2}$; Principal value of Arg $z = \frac{\pi}{2}$ & $|z| = \frac{2}{3}$

(iv) Modulus $= \frac{1}{\sqrt{2}}\operatorname{cosec}\frac{\pi}{5}$, Arg $z = 2n\pi + \frac{11\pi}{20}$, Principal Arg $= \frac{11\pi}{20}$

Q.3(a) $x = 1, y = 2$; **(b)** $x = 1$ & $y = 2$; **(c)** $(-2, 2)$ or $\left(-\frac{2}{3}, -\frac{2}{3}\right)$; **(d)** $(1, 1)$ $\left(0, \frac{5}{2}\right)$; **(e)** $x = K$, $y = \frac{3K}{2}$ $K \in R$

Q.4 **(a)** 2, **(b)** $-11/2$ **Q.5** **(a)** $[(-2, 2)$; $(-2, -2)]$ **(b)** $-(77+108\,i)$

Q.6 **(a)** $z = (2+i)$ or $(1-3i)$; **(b)** $z = \frac{3+4i}{4}$

Q.7 **(b)** 2

Q.9 **(ii)** $z = -(b+i)$; $-2\,i$, $-a$ **(iii)** $\left(-\frac{2ti}{3t+5}, ti\right)$ where $t \in R - \left\{-\frac{5}{3}\right\}$

Q.10 **(a)** The region between the co encentric circles with centre at $(0, 2)$ & radii 1 & 3 units

(b) region outside or on the circle with centre $\frac{1}{2}+2i$ and radius $\frac{1}{2}$.

(c) semi circle (in the 1st & 4th quadrant) $x^2 + y^2 = 1$ **(d)** a ray emanating from the point $(3+4i)$ directed away from the origin & having equation $\sqrt{3}x - y + 4 - 3\sqrt{3} = 0$

Q.15 $[-3, -2]$ **Q.17** $(1-c^2)|z|^2 - 2(a+bc)(\text{Re } z) + a^2 - b^2 = 0$

Q.18 (a) $K = 3$ **, (b)** -4 **Q.19** **(b)** one if n is even ; $-w^2$ if n is odd

Q.22 $(Z+1)(Z^2 - 2Z\cos 36° + 1)(Z^2 - 2Z\cos 108° + 1)$ **Q.24** 4

Q.25 **(a)** $\pi - 2$; **(b)** $1/2$

EXERCISE-2

Q.2 35 **Q.6** (a) $-\frac{7}{2}$, (b) zero **Q.8** $\frac{iz}{2}+\frac{1}{2}+i$ **Q.18** $-\omega$ or $-\omega^2$

Q.19 $k > \frac{1}{2}|\alpha-\beta|^2$ **Q.20** $|f(z)|$ is maximum when $z = \omega$, where ω is the cube root unity and $|f(z)| = \sqrt{13}$

Q.21 $K = -\frac{4}{9}$

Q.23 required set is constituted by the angles without their boundaries, whose sides are the straight lines $y = (\sqrt{2}-1)\ x$ and $y + (\sqrt{2}+1)\ x = 0$ containing the $x-$axis

Q.24 198 **Q.25** 51

EXERCISE-3

Q.1 $48(1-i)$ **Q.3 (a)** D **(b)** B

Q.4 $\mathbf{Z} = \dfrac{(29+20\sqrt{2})+i(\pm 15+25\sqrt{2})}{82}$, $\dfrac{(29-20\sqrt{2})+i(\pm 15-25\sqrt{2})}{82}$ **Q.5 (a)** C

Q.6 $7A_0 + 7A_7 x^7 + 7A_{14} x^{14}$ **Q.7 (a)** A **(b)** A **Q.8** $z^2 + z + \dfrac{\sin^2 n\theta}{\sin^2\theta} = 0$, where $\theta = \dfrac{2\pi}{2n+1}$

Q.9 (a) C, **(b)** D **Q.10** $\pm 1 + i\sqrt{3}, \dfrac{(\pm\sqrt{3}+i)}{\sqrt{2}}, \sqrt{2}\,i$ **Q.11** **(a)** B **; (b)** B

Q.13 **(a)** D ; **(b)** Centre $\equiv \dfrac{k^2\beta-\alpha}{k^2-1}$, Radius $= \dfrac{1}{(k^2-1)}\sqrt{|\alpha-k^2\beta|^2-\left(k^2.|\beta|^2-|\alpha|^2\right)\left(k^2-1\right)}$

Q.14 **(a)** A, **(b)** B, **(c)** $z_2 = -\sqrt{3}\,i$; $z_3 = \left(1-\sqrt{3}\right)+i$; $z_4 = \left(1+\sqrt{3}\right)-i$ **Q.15** D

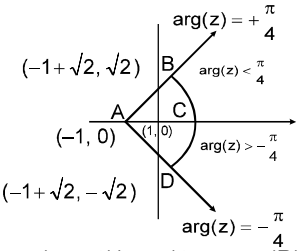
EXERCISE-4

Part : (A) Only one correct option

1. If $|z| = 1$ and $\omega = \dfrac{z-1}{z+1}$ (where $z \neq -1$), the Re(ω) is **[IIT – 2003, 3]**

(A) 0 (B) $-\dfrac{1}{|z+1|^2}$ (C) $\left|\dfrac{z}{z+1}\right| \cdot \dfrac{1}{|z+1|^2}$ (D) $\dfrac{\sqrt{2}}{|z+1|^2}$

2. The locus of z which lies in shaded region (excluding the boundaries) is best represented by



[IIT – 2005, 3]

(A) $z : |z+1| > 2$ and $|\arg(z+1)| < \pi/4$ (B) $z : |z-1| > 2$ and $|\arg(z-1)| < \pi/4$
(C) $z : |z+1| < 2$ and $|\arg(z+1)| < \pi/2$ (D) $z : |z-1| < 2$ and $|\arg(z+1)| < \pi/2$

3. If $w = \alpha, + i\beta$, where $\beta \neq 0$ and $z \neq 1$, satisfies the condition that $\left(\dfrac{w-\overline{w}z}{1-z}\right)$ is purely real, then the set of values of z is **[IIT – 2006, (3, –1)]**

(A) $\{z : |z| = 1\}$ (B) $\{z : z = \overline{z}\}$ (C) $\{z : z \neq 1\}$ (D) $\{z : |z| = 1, z \neq 1\}$

4. If $(\sqrt{3}+i)^{100} = 2^{99}(a+ib)$, then b is equal to

(A) $\sqrt{3}$ (B) $\sqrt{2}$ (C) 1 (D) none of these

5. If $\text{Re}\left(\dfrac{z-8i}{z+6}\right) = 0$, then z lies on the curve

(A) $x^2 + y^2 + 6x - 8y = 0$ (B) $4x - 3y + 24 = 0$ (C) 4ab (D) none of these

6. If n_1, n_2 are positive integers then : $(1+i)^{n_1} + (1+i^3)^{n_1} + (1-i^5)^{n_2} + (1-i^7)^{n_2}$ is a real number if and only if

(A) $n_1 = n_2 + 1$ (B) $n_1 + 1 = n_2$
(C) $n_1 = n_2$ (D) n_1, n_2 are any two positive integers

7. The three vertices of a triangle are represented by the complex numbers, 0, z_1 and z_2. If the triangle is equilateral, then

(A) $z_1^2 - z_2^2 = z_1 z_2$ (B) $z_2^2 - z_1^2 = z_1 z_2$ (C) $z_1^2 + z_2^2 = z_1 z_2$ (D) $z_1^2 + z_2^2 + z_1 z_2 = 0$

8. If $x^2 - x + 1 = 0$ then the value of $\sum_{n=1}^{5}\left(x^n + \dfrac{1}{x^n}\right)^2$ is

(A) 8 (B) 10 (C) 12 (D) none of these

Successful People Replace the words like; "wish", "try" & "should" with "I Will". Ineffective People don't.

9. If α is nonreal and $\alpha = \sqrt[5]{1}$ then the value of $2^{|1+\alpha+\alpha^2+\alpha^{-2}-\alpha^{-1}|}$ is equal to
(A) 4 (B) 2 (C) 1 (D) none of these

10. If $z = x + iy$ and $z^{1/3} = a - ib$ then $\frac{x}{a} - \frac{y}{b} = k\left(a^2 - b^2\right)$ where k =
(A) 1 (B) 2 (C) 3 (D) 4

11. $\left[\frac{-1+i\sqrt{3}}{2}\right]^6 + \left[\frac{-1-i\sqrt{3}}{2}\right]^6 + \left[\frac{-1+i\sqrt{3}}{2}\right]^5 + \left[\frac{-1-i\sqrt{3}}{2}\right]^5$ is equal to :
(A) 1 (B) – 1 (C) 2 (D) none

12. Expressed in the form $r(\cos\theta + i\sin\theta)$, $-2 + 2i$ becomes :
(A) $2\sqrt{2}\left[\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right]$ (B) $2\sqrt{2}\left[\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right]$
(C) $2\sqrt{2}\left[\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right]$ (D) $\sqrt{2}\left[\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right]$

13. The number of solutions of the equation in z, $z\bar{z} - (3 + i)z - (3 - i)\bar{z} - 6 = 0$ is :
(A) 0 (B) 1 (C) 2 (D) infinite

14. If $|z| = \max\{|z - 1|, |z + 1|\}$ then
(A) $|z + \bar{z}| = \frac{1}{2}$ (B) $z + \bar{z} = 1$ (C) $|z + \bar{z}| = 1$ (D) none of these

15. If P, P′ represent the complex number z_1 and its additive inverse respectively then the complex equation of the circle with PP′ as a diameter is
(A) $\frac{z}{z_1} = \left(\frac{z_1}{z}\right)$ (B) $z\bar{z} + z_1\bar{z}_1 = 0$ (C) $z\bar{z}_1 + \bar{z}z_1 = 0$ (D) none of these

16. The points $z_1 = 3 + \sqrt{3}\,i$ and $z_2 = 2\sqrt{3} + 6i$ are given on a complex plane. The complex number lying on the bisector of the angle formed by the vectors z_1 and z_2 is :
(A) $z = \frac{(3+2\sqrt{3})}{2} + \frac{\sqrt{3}+2}{2}\,i$ (B) $z = 5 + 5i$
(C) $z = -1 - i$ (D) none

17. The expression $\left[\frac{1+i\tan\alpha}{1-i\tan\alpha}\right]^n - \frac{1+i\tan n\alpha}{1-i\tan n\alpha}$ when simplified reduces to :
(A) zero (B) $2\sin n\alpha$ (C) $2\cos n\alpha$ (D) none

18. All roots of the equation, $(1 + z)^6 + z^6 = 0$:
(A) lie on a unit circle with centre at the origin (B) lie on a unit circle with centre at (– 1, 0)
(C) lie on the vertices of a regular polygon with centre at the origin (D) are collinear

19. Points z_1 & z_2 are adjacent vertices of a regular octagon. The vertex z_3 adjacent to z_2 ($z_3 \neq z_1$) is represented by :
(A) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 + z_2)$ (B) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 - z_2)$
(C) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_2 - z_1)$ (D) none of these

20. If $z = x + iy$ then the equation of a straight line $Ax + By + C = 0$ where A, B, C $\in$ R, can be written on the complex plane in the form $\bar{a}z + a\bar{z} + 2C = 0$ where 'a' is equal to :
(A) $\frac{(A + iB)}{2}$ (B) $\frac{A - iB}{2}$ (C) $A + iB$ (D) none

21. The points of intersection of the two curves $|z - 3| = 2$ and $|z| = 2$ in an argand plane are:
(A) $\frac{1}{2}\left(7 \pm i\sqrt{3}\right)$ (B) $\frac{1}{2}\left(3 \pm i\sqrt{7}\right)$ (C) $\frac{3}{2} \pm i\sqrt{\frac{7}{2}}$ (D) $\frac{7}{2} \pm i\sqrt{\frac{3}{2}}$

22. The equation of the radical axis of the two circles represented by the equations, $|z - 2| = 3$ and $|z - 2 - 3i| = 4$ on the complex plane is :
(A) $3iz - 3i\bar{z} - 2 = 0$ (B) $3iz - 3i\bar{z} + 2 = 0$ (C) $iz - i\bar{z} + 1 = 0$ (D) $2iz - 2i\bar{z} + 3 = 0$

23. If $\prod_{p=1}^{r} e^{ip\theta} = 1$ where Π denotes the continued product, then the most general value of θ is :
(A) $\frac{2n\pi}{r(r-1)}$ (B) $\frac{2n\pi}{r(r+1)}$ (C) $\frac{4n\pi}{r(r-1)}$ (D) $\frac{4n\pi}{r(r+1)}$

24. The set of values of $a \in R$ for which $x^2 + i(a - 1)x + 5 = 0$ will have a pair of conjugate imaginary roots is
(A) R (B) {1} (C) $|a|\ a^2 - 2a + 21 > 0\}$ (D) none of these

Successful People Replace the words like; "wish", "try" & "should" with "I Will". Ineffective People don't.

25. If $|z_1 - 1| < 1$, $|z_2 - 2| < 2$, $|z_3 - 3| < 3$ then $|z_1 + z_2 + z_3|$
(A) is less than 6 (B) is more than 3
(C) is less than 12 (D) lies between 6 and 12

26. If $z_1, z_2, z_3, \ldots\ldots\ldots, z_n$ lie on the circle $|z| = 2$, then the value of
$E = |z_1 + z_2 + \ldots\ldots + z_n| - 4\left|\frac{1}{z_1} + \frac{1}{z_2} + \ldots\ldots + \frac{1}{z_n}\right|$ is
(A) 0 (B) n (C) –n (D) none of these

Part : (B) May have more than one options correct

27. If z_1 lies on $|z| = 1$ and z_2 lies on $|z| = 2$, then
(A) $3 \le |z_1 - 2z_2| \le 5$ (B) $1 \le |z_1 + z_2| \le 3$
(C) $|z_1 - 3z_2| \ge 5$ (D) $|z_1 - z_2| \ge 1$

28. If z_1, z_2, z_3, z_4 are root of the equation $a_0z^4 + z_1z^3 + z_2z^2 + z_3z + z_4 = 0$, where a_0, a_1, a_2, a_3 and a_4 are real, then
(A) $\bar{z}_1, \bar{z}_2, \bar{z}_3, \bar{z}_4$ are also roots of the equation (B) z_1 is equal to at least one of $\bar{z}_1, \bar{z}_2, \bar{z}_3, \bar{z}_4$
(C) $-\bar{z}_1, -\bar{z}_2, -\bar{z}_3, -\bar{z}_4$ are also roots of the equation (D) none of these

29. If $a^3 + b^3 + 6abc = 8c^3$ & ω is a cube root of unity then :
(A) a, c, b are in A.P. (B) a, c, b are in H.P.
(C) $a + b\omega - 2c\omega^2 = 0$ (D) $a + b\omega^2 - 2c\omega = 0$

30. The points z_1, z_2, z_3 on the complex plane are the vertices of an equilateral triangle if and only if :
(A) $\Sigma (z_1 - z_2)(z_2 - z_3) = 0$ (B) $z_1^2 + z_2^2 + z_3^2 = 2(z_1z_2 + z_2z_3 + z_3z_1)$
(C) $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$ (D) $2(z_1^2 + z_2^2 + z_3^2) = z_1z_2 + z_2z_3 + z_3z_1$

31. If $|z_1 + z_2| = |z_1 - z_2|$ then
(A) $|\text{amp } z_1 - \text{amp } z_2| = \frac{\pi}{2}$ (B) $|\text{amp } z_1 - \text{amp}_2| = \pi$
(C) $\frac{z_1}{z_2}$ is purely real (D) $\frac{z_1}{z_2}$ is purely imaginary

EXERCISE-5

1. Given that $x, y \in R$, solve : $4x^2 + 3xy + (2xy - 3x^2)i = 4y^2 - (x^2/2) + (3xy - 2y^2)i$

2. If α & β are any two complex numbers, prove that :
$$\left|\alpha - \sqrt{\alpha^2 - \beta^2}\right| + \left|\alpha + \sqrt{\alpha^2 - \beta^2}\right| = |\alpha + \beta| + |\alpha - \beta|.$$

3. If α, β are the numbers between 0 and 1, such that the points $z_1 = \alpha + i$, $z_2 = 1 + \beta i$ and $z_3 = 0$ form an equilateral triangle, then find α and β.

4. ABCD is a rhombus. Its diagonals AC and BD intersect at the point M and satisfy BD = 2AC. If the points D and M represent the complex numbers 1 + i and 2 - i respectively, then find the complex number corresponding to A.

5. Show that the sum of the pth powers of nth roots of unity :
(a) is zero, when p is not a multiple of n. (b) is equal to n, when p is a multiple of n.

6. If $(1 + x)^n = p_0 + p_1x + p_2x^2 + p_3x^3 + \ldots\ldots$, then prove that :
(a) $p_0 - p_2 + p_4 - \ldots\ldots = 2^{n/2}\cos\frac{n\pi}{4}$ (b) $p_1 - p_3 + p_5 - \ldots\ldots = 2^{n/2}\sin\frac{n\pi}{4}$

7. Prove that, $\log_e\left(\frac{1}{1 - e^{i\theta}}\right) = \log_e\left(\frac{1}{2}\operatorname{cosec}\frac{\theta}{2}\right) + i\left(\frac{\pi}{2} - \frac{\theta}{2}\right)$

8. If $i^{i^{i \ldots\ldots \infty}} = A + iB$, principal values only being considered, prove that
(a) $\tan\frac{1}{2}\pi A = \frac{B}{A}$ (b) $A^2 + B^2 = e^{-\pi B}$

9. Prove that the roots of the equation, $(x - 1)^n = x^n$ are $\frac{1}{2}\left(1 + i\cot\frac{r\pi}{n}\right)$, where
$r = 0, 1, 2, \ldots\ldots (n - 1)$ & $n \in N$.

10. If $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -3/2$ then prove that :
(a) $\Sigma \cos 2\alpha = 0 = \Sigma \sin 2\alpha$ (b) $\Sigma \sin(\alpha + \beta) = 0 = \Sigma \cos(\alpha + \beta)$
(c) $\Sigma \sin 3\alpha = 3\sin(\alpha + \beta + \gamma)$ (d) $\Sigma \cos 3\alpha = 3\cos(\alpha + \beta + \gamma)$
(e) $\Sigma \sin^2\alpha = \Sigma \cos^2\alpha = 3/2$
(f) $\cos^3(\theta + \alpha) + \cos^3(\theta + \beta) + \cos^3(\theta + \gamma) = 3\cos(\theta + \alpha).\cos(\theta + \beta).\cos(\theta + \gamma)$
where $\theta \in R$.

11. If α, β, γ are roots of $x^3 - 3x^2 + 3x + 7 = 0$ (and ω is imaginary cube root of unity), then find the value of $\frac{\alpha-1}{\beta-1} + \frac{\beta-1}{\gamma-1} + \frac{\gamma-1}{\alpha-1}$.

12. Given that, $|z-1| = 1$, where 'z' is a point on the argand plane. Show that $\frac{z-2}{z} = i \tan(\arg z)$.

13. P is a point on the Argand diagram. On the circle with OP as diameter two points Q & R are taken such that $\angle POQ = \angle QOR = \theta$. If 'O' is the origin & P, Q & R are represented by the complex numbers Z_1, Z_2 & Z_3 respectively, show that : $Z_2^2 \cdot \cos 2\theta = Z_1 Z_3 \cos^2\theta$.

14. Find an expression for $\tan 7\theta$ in terms of $\tan\theta$, using complex numbers. By considering $\tan 7\theta = 0$, show that $x = \tan^2(3\pi/7)$ satisfies the cubic equation $x^3 - 21x^2 + 35x - 7 = 0$.

15. If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$ $(n \in N)$, prove that : $C_2 + C_6 + C_{10} + ... = \frac{1}{2}\left[2^{n-1} - 2^{n/2}\cos\frac{n\pi}{4}\right]$

16. Prove that : $\cos\left(\frac{2\pi}{2n+1}\right) + \cos\left(\frac{4\pi}{2n+1}\right) + \cos\left(\frac{6\pi}{2n+1}\right) + ... + \cos\left(\frac{2n\pi}{2n+1}\right) = -\frac{1}{2}$ When $n \in N$.

17. Show that all the roots of the equation $a_1z^3 + a_2z^2 + a_3z + a_4 = 3$, where $|a_i| \le 1$, $i = 1, 2, 3, 4$ lie outside the circle with centre origin and radius 2/3.

18. Prove that $\sum_{k=1}^{n-1}(n-k)\cos\frac{2k\pi}{n} = -\frac{n}{2}$, where $n \ge 3$ is an integer

19. Show that the equation $\frac{A_1^2}{x-a_1} + \frac{A_2^2}{x-a_2} + ... + \frac{A_n^2}{x-a_n} = k$ has no imaginary root, given that :
$a_1, a_2, a_3 ... a_n$ & $A_1, A_2, A_3 .... A_n$, k are all real numbers.

20. Let z_1, z_2, z_3 be three distinct complex numbers satisfying, $|z_1-1| = |z_2-1| = |z_3-1|$. Let A, B & C be the points represented in the Argand plane corresponding to z_1, z_2 and z_3 resp. Prove that $z_1 + z_2 + z_3 = 3$ if and only if D ABC is an equilateral triangle.

21. Let α, β be fixed complex numbers and z is a variable complex number such that,

$|z-\alpha|^2 + |z-\beta|^2 = k$.

Find out the limits for 'k' such that the locus of z is a circle. Find also the centre and radius of the circle.

22. If $1, \alpha_1, \alpha_2, \alpha_3, ..., \alpha_{n-1}$ are the n, n^{th} roots of unity, then prove that
$(1-\alpha_1)(1-\alpha_2)(1-\alpha_3)..... (1-\alpha_{n-1}) = n$.

Hence prove that $\sin\frac{\pi}{n} \cdot \sin\frac{2\pi}{n} \cdot \sin\frac{3\pi}{n} ..... \sin\frac{(n-1)\pi}{n} = \frac{n}{2^{n-1}}$.

23. Find the real values of the parameter 'a' for which at least one complex number $z = x + iy$ satisfies both the equality $|z - ai| = a + 4$ and the inequality $|z-2| < 1$.

24. Prove that, with regard to the quadratic equation $z^2 + (p + ip')z + q + iq' = 0$; where p, p', q, q' are all real.
(a) if the equation has one real root then $q'^2 - pp'q' + qp'^2 = 0$.
(b) if the equation has two equal roots then $p^2 - p'^2 = 4q$ & $pp' = 2q'$.
State whether these equal roots are real or complex.

25. The points A, B, C depict the complex numbers z_1, z_2, z_3 respectively on a complex plane & the angle B & C of the triangle ABC are each equal to $\frac{1}{2}(\pi - \alpha)$. Show that

$(z_2 - z_3)^2 = 4(z_3 - z_1)(z_1 - z_2)\sin^2\frac{\alpha}{2}$.

26. If z_1, z_2 & z_3 are the affixes of three points A, B & C respectively and satisfy the condition $|z_1 - z_2| = |z_1| + |z_2|$ and $|(2 - i)z_1 + iz_3| = |z_1| + |(1 - i)z_1 + iz_3|$ then prove that Δ ABC in a right angled.

27. If $1, \alpha_1, \alpha_2, \alpha_3, \alpha_4$ be the roots of $x^5 - 1 = 0$, then prove that

$$\frac{\omega-\alpha_1}{\omega^2-\alpha_1} \cdot \frac{\omega-\alpha_2}{\omega^2-\alpha_2} \cdot \frac{\omega-\alpha_3}{\omega^2-\alpha_3} \cdot \frac{\omega-\alpha_4}{\omega^2-\alpha_4} = \omega.$$

28. If one the vertices of the square circumscribing the circle $|z-1| = \sqrt{2}$ is $2 + \sqrt{3}\,i$. Find the other vertices of the square. **[IIT – 2005, 4]**

EXERCISE-4

1.	A	**2.**	C	**3.**	D	**4.**	A
5.	A	**6.**	D	**7.**	C	**8.**	A
9.	A	**11.**	D	**12.**	A	**13.**	B
14.	D	**15.**	D	**16.**	A	**17.**	B
18.	A	**19.**	D	**20.**	C	**21.**	C
22.	B	**23.**	B	**24.**	D	**25.**	B
26.	C	**27.**	A	**28.**	ABCD	**29.**	AB
30.	ACD	**31.**	AC	**10.**	AD		

EXERCISE-5

1. $x = K,\ y = \frac{3K}{2}\ K \in R$

3. $2-\sqrt{3}, 2-\sqrt{3}$

4. $3 - \frac{i}{2}$ or $1 - \frac{3}{2}i$

11. $3\ \omega^2$

21. $k > \frac{1}{2}|\alpha - \beta|^2$

23. $\left(-\frac{21}{10}, -\frac{5}{6}\right)$

28. $-i\sqrt{3}\ ,\ 1-\sqrt{3}\ +i,\ 1+\sqrt{3}\ -i$

Some questions (Assertion–Reason type) are given below. Each question contains **Statement – 1 (Assertion)** and **Statement – 2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. So select the correct choice :

Choices are :

(A) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is** a correct explanation for **Statement – 1**.
(B)**Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is NOT** a correct explanation for **Statement – 1**.
(C) **Statement – 1** is True, **Statement – 2** is False.
(D) **Statement – 1** is False, **Statement – 2** is True.

344. Let $z = e^{i\theta} = \cos\theta + i\sin\theta$
Statement 1: Value of $e^{iA} . e^{iB} . e^{iC} = -1$ if $A + B + C = \pi$. **Statement 2:** $\arg(z) = \theta$ and $|z| = 1$.

345 Let $a_1, a_2, \ldots, a_n \in R^+$
Statement–1 : Minimum value of $\frac{a_1}{a_2}+\frac{a_2}{a_3}+\ldots.+\frac{a_{n-1}}{a_n}+\frac{a_n}{a_1}$
Statement–2 : For positive real numbers, A.M ≥ G.M.

346. Let $\log\left(\frac{5c}{a}\right), \log\left(\frac{3b}{5c}\right)$ and $\log\left(\frac{a}{3b}\right)$ then A.P., where a, b, c are in G.P. If a, b, c represents the sides of a triangle. Then : **Statement–1 :** Triangle represented by the sides a, b, c will be an isosceles triangle
Statement–2 : $b + c < a$

347. Let Z_1, Z_2 be two complex numbers represented by points on the curves $|z| = \sqrt{2}$ and $|z - 3 - 3i| = 2\sqrt{2}$. Then
Statement–1 : min $|z_1 - z_2| = 0$ and max $|z_1 - z_2| = 6\sqrt{2}$
Statement–2 : Two curves $|z| = \sqrt{2}$ and $|z - 3 - 3i| = 2\sqrt{2}$ touch each other externally

348. **Statement–1 :** If $|z - i| \le 2$ and $z_0 = 5 + 3i$, then the maximum value of $|iz + z_0|$ is 7
Statement–2 : For the complex numbers z_1 and z_2 $|z_1 + z_2| \le |z_1| + |z_2|$

349. Let z_1 and z_2 be complex number such that $|z_1 + z_2| = |z_1| + |z_2|$
Statement–1 : $\arg\left(\frac{z_1}{z_2}\right) = 0$
Statement–2 : z_1, z_2 and origin are collinear and z_1, z_2 are on the same side of origin.

350. Let fourth roots of unity be z_1, z_2, z_3 and z_4 respectively
Statement–1 : $z_1^2 + z_2^2 + z_3^2 + z_4^2 = 0$ **Statement–2 :** $z_1 + z_2 + z_3 + z_4 = 0$.

351. Let $z_1, z_2, \ldots, z_n$ be the roots of $z^n = 1$, $n \in N$.
Statement–1 : $z_1 . z_2 \ldots z_n = (-1)^n$ **Statement–2 :** Product of the roots of the equation $a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \ldots + a_1 x + a_0 = 0$, $a_n \ne 0$, is $(-1)^n . \frac{a_0}{a_n}$.

352. Let z_1, z_2, z_3 and z_4 be the complex numbers satisfying $z_1 - z_2 = z_4 - z_3$.
Statement–1 : z_1, z_2, z_3, z_4 are the vertices of a parallelogram
Statement–2 : $\frac{z_1 + z_3}{2} = \frac{z_2 + z_4}{2}$.

353. **Statement–1 :** The minimum value of $|z| + |z - i|$ is 0.
Statement–2 : For any two complex number z_1 and z_2, $|z_1 + z_2| \le |z_1| + |z_2|$.

354. **Statement–1 :** Let z_1 and z_2 are two complex numbers such that $|z_1 - z_2| = |z_1 + z_2|$ then the orthocenter of ΔAOB is $\frac{z_1 + z_2}{2}$. (where O is the origin)
Statement–2 : In case of right angled triangle, orthocenter is that point at which triangle is right angled.

355. **Statement–1 :** If ω is complex cube root of unity then $(x - y)(x\omega - y)(x\omega^2 - y)$ is equal to $x^3 + y^2$
Statement–2 : If ω is complex cube root of unity then $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$

356. **Statement-1 :** If $|z| \le 4$, then greatest value of $|z + 3 - 4i|$ is 9.
Statement-2 : $\forall z_1, z_2 \in C$, $|z_1 + z_2| \le |z_1| + |z_2|$

357. **Statement-1:** The slope of line $(2 - 3i)\,z + (2 + 3i)\,\bar{z} - 1 = 0$ is $\frac{2}{3}$

Statement-2:: The slope of line $\bar{a}z + a\bar{z} + b = 0$ $b \in R$ & a be any non-zero complex. Constant is $-\frac{Re(a)}{Im(a)}$

358. **Statement-1:** The value of $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7} - i\cos\frac{2\pi k}{7}\right)$ is i

Statement-2: The roots of the equation $z^n = 1$ are called the nth roots of unity where $z = \left(\frac{\cos 2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right)$ where k = 0, 1, 2, ... (n – 1)

359. **Statement-1:** $|z_1 - a| < a$, $|z_2 - b| < b$ $|z_3 - c| < c$, where a, b, c are +ve real nos, then $|z_1 + z_2 + z_3|$ is greater than $2|a + b + c|$ **Statement-2:** $|z_1 \pm z_2| \le |z_1| + |z_2|$

360. **Statement-1:** $(\cos 2 + i\sin 2)^{\pi} = 1$

Statement-2: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ it is not true when n is irrational number.

361. **Statement-1 :** If $\alpha_1, \alpha_2, \alpha_3 \ldots \alpha_8$ be the 8th root of unity, then $\alpha_1^{16} + \alpha_2^{16} + \alpha_3^{16} + \ldots + \alpha_8^{16} = 8$

Statement-2 : In case of sum of pth power of nth roots of unity sum = 0 if $p \neq kn$ where p, k, n are integers sum = n if p = kn.

362. **Statement-1:** Locus of z, satisfying the equation $|z - 1| + |z - 8| = 16$ is an ellipse of eccentricity 7/16

Statement-2:: Sum of focal distances of any point is constant for an ellipse

363. **Statement-1:** $\arg\left(\frac{z_2}{z_1}\right) = \arg z_2 - \arg z_1$ & $\arg z^n = n(\arg z)$ **Statement-2:** If $|z| = 1$, then $\arg(z^2 + \bar{z}) = \frac{1}{2}\arg z$.

364. **Statement-1:** If $|z - z + i| \le 2$ then $\sqrt{5} - 2 \le |z| \le \sqrt{5} + 2$

Statement-2: If $|z - 2 + i| \le 2$ the z lies inside or on the circle having centre (2, –1) & radius 2.

365. **Statement-1:** The area of the triangle on argand plane formed by the complex numbers z, iz and z + iz is $\frac{1}{2}|z|^2$

Statement-2: The angle between the two complex numbers z and iz is $\frac{\pi}{2}$.

366. **Statement-1:** If $\left|\frac{zz_1 - z_2}{zz_1 + z_2}\right| = k$, $(z_1, z_2 \neq 0)$, then locus of z is circle.

Statement-2 : As, $\left|\frac{z - z_1}{z - z_2}\right| = \lambda$ represents a circle if, $\lambda \notin \{0, 1\}$

367. **Statement-1:** If z_1 and z_2 are two complex numbers such that $|z_1| = |z_2| + |z_1 - z_2|$, then $\text{Im}\left(\frac{z_1}{z_2}\right) = 0$.

Statement-2: $\arg(z) = 0 \Rightarrow$ z is purely real.

368. **Statement-1:** If $\alpha = \cos\left(\frac{2\pi}{7}\right) + i\sin\left(\frac{2\pi}{7}\right)$, $p = \alpha + \alpha^2 + \alpha^4$, $q = \alpha^3 + \alpha^5 + \alpha^6$, then the equation whose roots are p and q is $x^2 + x + 2 = 0$

Statement-2: If α is a root of $z^7 = 1$, then $1 + \alpha + \alpha^2 + \ldots. + \alpha^6 = 0$.

369. **Statement-1:** If $|z| < \sqrt{2} - 1$ then $|z^2 + 2z\cos\alpha|$ is less than one.

Statement-2: $|z_1 + z_2| < |z_1| + |z_2|$. Also $|\cos\alpha| \le 1$.

370. **Statement-1:** The number of complex number satisfying the equation $|z|^2 + P|z| + q = 0$ $(p, q, \in R)$ is atmost 2.

Statement-2 : A quadratic equation in which all the co-efficients are non-zero real can have exactly two roots.

371. **Statement-1:** If $\left|\beta + \frac{1}{\beta}\right| = 1 (\beta \neq 0)$ is a complex number, then the maximum value of $|\beta|$ is $\frac{\sqrt{5} + 1}{2}$.

Statement-2 :: On the locus $\left|\beta + \frac{1}{\beta}\right| = 1$ the farthest distance from origin is $\frac{\sqrt{5} + 1}{2}$.

372. **Statement-1:** The locus of z moving in the Argand plane such that arg $\left(\frac{z-2}{z+2}\right)=\frac{\pi}{2}$ is a circle.

Statement-2: This is represent a circle, whose centre is origin and radius is 2.

ANSWER

344. B	**345. A**	**346. D**	**347. A**	**348. A**	**349. A**	**350. B**
351. D	**352. A**	**353. D**	**354. D**	**355. D**	**356. A**	**357.. A**
358. A	**359. D**	**360. D**	**361. A**	**362. A**	**363. B**	**364. A**
365. A	**366. D**	**367. A**	**368. A**	**369. A**	**370. D**	**371. A**
372. A						

SOLUTION

345. Using AM ≥ GM $\frac{a_1}{a_2}+\frac{a_2}{a_3}+...+\frac{a_{n-1}}{a_n}+\frac{a_n}{a_1} \geq n\left(\frac{a_1}{a_2}.\frac{a_2}{a_3}....\frac{a_n}{a_1}\right)^{1/n} \Rightarrow \frac{a_1}{a_2}+\frac{a_2}{a_3}+...+\frac{a_n}{a_1} \geq n$

Hence (A) is correct option.

346. $2\log\frac{3b}{5c}=\log\frac{5c}{a}+\log\frac{a}{3b} \Rightarrow \left(\frac{3b}{5c}\right)^2=\frac{5c}{a}.\frac{a}{3b} \Rightarrow 3b = 5c$

Also, $b^2 = ac \Rightarrow 9ac = 25c^2$ or $9a = 25c \therefore \frac{9a}{5}=5c=3b \Rightarrow \frac{a}{5}=\frac{b}{3}=\frac{c}{9/5} \Rightarrow b + c < a$

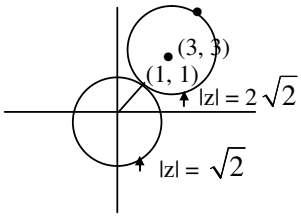
∴ (D) is the correct answer

347. From the diagram it is clear that both circles touch each other externally

∴ min $|z_1 - z_2| = 0$

max $|z_1 - z_2| = \sqrt{36+36}=6\sqrt{2}$]

Hence (A) is correct option.



348. $|iz + z_0| = |i(z - i) - 1 + 5 + 3i| = |i (z–i) + 4 + 3i|$

$\leq |i|\,|z - i| + |4 + 3i| \leq 7$ Hence (A) is the correct option.

349. (A) arg (z_1) = arg (z_2)

$\therefore \arg\left(\frac{z_1}{z_2}\right)=\arg(z_1)-\arg(z_2)=0.$

350. (B) Fourth roots of unity are – 1, 1, – i and i

$\therefore z_1^2+z_2^2+z_3^2+z_4^2=0$ and $z_1+z_2+z_3+z_4=0.$

351. Statement – II is true (a known fact).

Hence if $z_1, z_2, \ldots, z_n$ are roots of $z^n - 1 = 0$, then $z_1 . z_2 \ldots z_n = (-1)^n . \frac{(-1)}{1}=(-1)^{n+1}$,

which is never equal to $(-1)^n$ Hence (d) is the correct answer.

352. Both statements – I and II are true and statement – II is the correct reasoning of statement – I, because $\frac{z_1+z_3}{2}=\frac{z_2+z_4}{2} \Rightarrow$ mid point of join of z_1, z_3 and z_2, z_4 are same, which is the necessary and sufficient condition for a quadrilateral ABCD, when A ≡ A(z_1), B ≡ B(z_2), C ≡ C(z_3), D ≡ D(z_4) to be a parallelogram Hence (A) is the correct answer.

353. $|z+i-z| \leq |z|+|i-z| \Rightarrow |z|+|z-i| \geq |i|=1$

∴ Hence (d) is the correct answer.

354. $|z_1-z_2|^2=|z_1+z_2|^2$

$\Rightarrow z_1\bar{z}_2+\bar{z}_1z_2=0 \Rightarrow |z_1-z_2|^2=|z_1|^2+|z_2|^2$

$\Rightarrow \Delta AOB$ is right angled at O.

∵ orthocenter is the origin. ∴ Hence (d) is the correct answer.

355. (D) $(x - y)(x\omega - y)(x\omega^2 - y)$

$= x^3\omega^2 - x^2y\omega - x^2y\omega^2 + xy^2 - x^2y\omega + xy^2\omega + xy^2\omega^2 - y^3 = x^3 - y^3$

356. Option (A) is correct
Since
$|z + 3 - 4i| \le |z| + |3-4i| = 9$ $(\because |z| \le 4)$.

357. Option (A) is correct.

358. $\sum_{k=1}^{6}(-i)\left(\cos\frac{2\pi k}{7} - i\sin\frac{2\pi k}{7}\right)$

$= (-i))\sum_{k<1}^{6} z^k = (-i)\left(\frac{z - z^7}{1 - z}\right)$ $[\because z^7 = 1]$

$= (-i)(-1) = i$ Ans. (A)

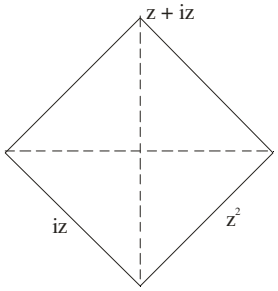
359. $|z_1 + z_2 + z_3| = |z_1 - a + z_2 - b + z_3 - c + (a + b + c)|$
$\le |z_1 - a| + |z_2 - b| + |z_3 - c| + |a + b + c| \le 2|a + b + c|$ Ans. (D)

360. $(\cos 2 + i \sin 2)^{\pi}$ can not be evaluated because demoviers theorem does not hold for irrational index.
'd' is correct.

361. $1, \alpha, \alpha^2, \ldots \alpha^7$ are 8, 8th root of unity then after raising 16th power, we get $1, \alpha^{16}, \alpha^{32}, \alpha^{48} \ldots \alpha^{112}$
$1 + \alpha^{16} + \alpha^{32} + \alpha^{48} + \ldots + \alpha^{112}$
Now $\alpha^8 = 1$
So $\alpha^{16} = 1$
$1 + 1 + 1 + \ldots + 1 = 8$
'A' is correct.

365. (A)

$$\frac{1}{2}|z||iz|$$
$$= \frac{|z|^2}{2}$$



366. (D)

$$\left|\frac{zz_1 - z_2}{z_1 z + z_2}\right| = k \Rightarrow \left|\frac{z - \frac{z_2}{z_1}}{z + \frac{z_2}{z_1}}\right| = k$$

Clearly, if $k \ne 0, 1$; then z would lie on a circle. If $k = 1$, z would lie on the perpendicular bisector of line segment joining $\frac{z_2}{z_1}$ and $\frac{-z_2}{z_1}$ and represents a point, if $k = 0$.

367. We have, $\arg(z) = 0 \Rightarrow z$ is purely real. R is true
Also, $|z_1| = |z_2| + |z_1 - z_2|$
$\Rightarrow (|z_1|^2 + |z_2|^2 - 2|z_1||z_2| \cos(\theta_1 - \theta_2)$
$= |z_1|^2 + |z_2|^2 - 2|z_1||z_2|$
$\Rightarrow \cos(\theta_1 - \theta_2) = 1 \Rightarrow \theta_1 - \theta_2 = 0$
$\Rightarrow \arg\left(\frac{z_1}{z_2}\right) = 0 \Rightarrow \frac{z_1}{z_2}$ is purely real.

$I_m\left(\frac{z_1}{z_2}\right) = 0$ (A)

368. (A)
α is seventh root of unity $\Rightarrow 1 + \alpha + \alpha^2 + \ldots + \alpha^6 = 0$
$\Rightarrow p + q = -1.$
$pq = \alpha^4 + \alpha^6 + \alpha^7 + \alpha^5 + \alpha^7 + \alpha^8 + \alpha^7 + \alpha^9 + \alpha^{10} = 3 - 1 = 2.$

$\therefore x^2 + x + 2 = 0$ is the req. equation.

Both A and R are true and R is correct explanation of A.

369. (A)

$|z^2 + 2z\cos\alpha| < |z^2| + |2z\cos\alpha| < |z^2| + 2|z|\,|\cos\alpha|$

$< (\sqrt{2}-1)^2 + 2(\sqrt{2}-1) < 1.$

($\because |\cos\alpha| \le 1$).

372. $\dfrac{z-2}{z+2} = \left|\dfrac{z-2}{z+2}\right| e^{i\pi/2} = \left|\dfrac{z-2z+2}{}\,i\right|$ ... (i)

therefore $\dfrac{\overline{z}-2}{\overline{z}+2} = \left|\dfrac{\overline{z}-2}{\overline{z}+2}\right|(-1) = -\left|\dfrac{z-2}{z+2}\right| i$ ... (ii)

Then adding (i) & (ii)

$\dfrac{z-2}{z+2} + \dfrac{\overline{z}-2}{\overline{z}+2} = 0$

i.e., $(z-2)\ \overline{z}+2) + (z+2)(\overline{z} - 2) = 0,\ 2z\overline{z} - 8 = 0$

$|z|^2 = 4 \ \therefore x^2 + y^2 = 4.$

Ans. (a)

Imp. Que. From Competitive Exams

1. The number of real values of a satisfying the equation $a^2 - 2a\sin x + 1 = 0$ is

(a) Zero (b) One

(c) Two (d) Infinite

2. For positive integers n_1, n_2 the value of the expression $(1+i)^{n_1} + (1+i^3)^{n_1} + (1+i^5)^{n_2} + (1+i^7)^{n_2}$ where $i = \sqrt{-1}$ is a real number if and only if **[IIT 1996]**

(a) $n_1 = n_2 + 1$ (b) $n_1 = n_2 - 1$

(c) $n_1 = n_2$ (d) $n_1 > 0, n_2 > 0$

3. Given that the equation $z^2 + (p+iq)z + r + is = 0,$ where p, q, r, s are real and non-zero has a real root, then

(a) $pqr = r^2 + p^2 s$ (b) $prs = q^2 + r^2 p$

(c) $qrs = p^2 + s^2 q$ (d) $pqs = s^2 + q^2 r$

4. If $x = -5 + 2\sqrt{-4},$ then the value of the expression $x^4 + 9x^3 + 35x^2 - x + 4$ is **[IIT 1972]**

(a) 160 (b) –160

(c) 60 (d) –60

5. If $\sqrt{3} + i = (a+ib)(c+id)$, then $\tan^{-1}\left(\dfrac{b}{a}\right) + \tan^{-1}\left(\dfrac{d}{c}\right)$ has the value

(a) $\dfrac{\pi}{3} + 2n\pi, n \in I$ (b) $n\pi + \dfrac{\pi}{6}, n \in I$

(c) $n\pi - \dfrac{\pi}{3}, n \in I$ (d) $2n\pi - \dfrac{\pi}{3}, n \in I$

6. If $a = \cos\alpha + i\sin\alpha,\ b = \cos\beta + i\sin\beta,$

$c = \cos\gamma + i\sin\gamma$ and $\dfrac{b}{c} + \dfrac{c}{a} + \dfrac{a}{b} = 1,$ then $\cos(\beta-\gamma) + \cos(\gamma-\alpha) + \cos(\alpha-\beta)$ is equal to **[RPET 2001]**

(a) 3/2 (b) – 3/2

(c) 0 (d) 1

7. If $(1+i)(1+2i)(1+3i).....(1+ni) = a+ib$, then $2.5.10....(1+n^2)$ is equal to

[Karnataka CET 2002; Kerala (Engg.) 2002]

(a) a^2-b^2 (b) a^2+b^2

(c) $\sqrt{a^2+b^2}$ (d) $\sqrt{a^2-b^2}$

8. If z is a complex number, then the minimum value of $|z|+|z-1|$ is **[Roorkee 1992]**

(a) 1 (b) 0

(c) 1/2 (d) None of these

9. For any two complex numbers z_1 and z_2 and any real numbers a and b; $|(az_1-bz_2)|^2+|(bz_1+az_2)|^2=$ **[IIT 1988]**

(a) $(a^2+b^2)(|z_1|+|z_2|)$ (b) $(a^2+b^2)(|z_1|^2+|z_2|^2)$

(c) $(a^2+b^2)(|z_1|^2-|z_2|^2)$ (d) None of these

10. The locus of z satisfying the inequality $\log_{1/3}|z+1|>\log_{1/3}|z-1|$ is

(a) $R(z)<0$ (b) $R(z)>0$

(c) $I(z)<0$ (d) None of these

11. If $z_1=a+ib$ and $z_2=c+id$ are complex numbers such that $|z_1|=|z_2|=1$ and $R(z_1\overline{z_2})=0$, then the pair of complex numbers $w_1=a+ic$ and $w_2=b+id$ satisfies **[IIT 1985]**

(a) $|w_1|=1$ (b) $|w_2|=1$

(c) $R(w_1\overline{w_2})=0$, (d) All the above

12. Let z and w be two complex numbers such that $|z|\le 1$, $|w|\le 1$ and $|z+iw|=|z-i\overline{w}|=2$. Then z is equal to **[IIT 1995]**

(a) 1 or i (b) i or $-i$

(c) 1 or -1 (d) i or -1

13. The maximum distance from the origin of coordinates to the point z satisfying the equation $\left|z+\frac{1}{z}\right|=a$ is

(a) $\frac{1}{2}(\sqrt{a^2+1}+a)$ (b) $\frac{1}{2}(\sqrt{a^2+2}+a)$

(c) $\frac{1}{2}(\sqrt{a^2+4}+a)$ (d) None of these

14. Find the complex number z satisfying the equations $\left|\frac{z-12}{z-8i}\right|=\frac{5}{3}, \left|\frac{z-4}{z-8}\right|=1$ **[Roorkee 1993]**

(a) 6 (b) $6\pm 8i$

(c) $6+8i, 6+17i$ (d) None of these

15. If z_1, z_2, z_3 are complex numbers such that $|z_1|=|z_2|=|z_3|=\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right|=1$, then $|z_1+z_2+z_3|$ is **[MP PET 2004; IIT Screening 2000]**

(a) Equal to 1 (b) Less than 1

(c) Greater than 3 (d) Equal to 3

16. If $z_1=10+6i, z_2=4+6i$ and z is a complex number such that $amp\left(\frac{z-z_1}{z-z_2}\right)=\frac{\pi}{4}$, then the value of $|z-7-9i|$ is equal to **[IIT 1990]**

(a) $\sqrt{2}$ (b) $2\sqrt{2}$

(c) $3\sqrt{2}$ (d) $2\sqrt{3}$

17. If z_1, z_2, z_3 be three non-zero complex number, such that $z_2\ne z_1, a=|z_1|, b=|z_2|$ and $c=|z_3|$ suppose that $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}=0$, then $arg\left(\frac{z_3}{z_2}\right)$ is equal to

(a) $arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right)^2$ (b) $arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right)$

(c) $arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)^2$ (d) $arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$

18. Let z and w be the two non-zero complex numbers such that $|z| = |w|$ and $arg\, z + arg\, w = \pi$. Then z is equal to

[IIT 1995; AIEEE 2002]

(a) w (b) $-w$

(c) $\bar{w}$ (d) $-\bar{w}$

19. If $|z - 25i| \le 15$, then $|\max .amp(z) - \min .amp(z)| =$

(a) $\cos^{-1}\left(\frac{3}{5}\right)$ (b) $\pi - 2\cos^{-1}\left(\frac{3}{5}\right)$

(c) $\frac{\pi}{2} + \cos^{-1}\left(\frac{3}{5}\right)$ (d) $\sin^{-1}\left(\frac{3}{5}\right) - \cos^{-1}\left(\frac{3}{5}\right)$

20. If z_1, z_2 and z_3, z_4 are two pairs of conjugate complex numbers, then $arg\left(\frac{z_1}{z_4}\right) + arg\left(\frac{z_2}{z_3}\right)$ equals

(a) o (b) $\frac{\pi}{2}$

(c) $\frac{3\pi}{2}$ (d) π

21. Let z, w be complex numbers such that $\bar{z} + i\bar{w} = 0$ and $arg\ zw = \pi$. Then $arg\ z$ equals **[AIEEE 2004]**

(a) $5\pi/4$ (b) $\pi/2$

(c) $3\pi/4$ (d) $\pi/4$

22. If $(1+x)^n = C_0 + C_1 x + C_2 x^2 + ... + C_n x^n$, then the value of $C_0 - C_2 + C_4 - C_6 + ...$ is

(a) 2^n (b) $2^n \cos\frac{n\pi}{2}$

(c) $2^n \sin\frac{n\pi}{2}$ (d) $2^{n/2}\cos\frac{n\pi}{4}$

23. If $x = \cos\theta + i\sin\theta$ and $y = \cos\phi + i\sin\phi$, then $x^m y^n + x^{-m} y^{-n}$ is equal to

(a) $\cos(m\theta + n\phi)$

(b) $\cos(m\theta - n\phi)$

(c) $2\cos(m\theta + n\phi)$

(d) $2\cos(m\theta - n\phi)$

24. The value of $\sum_{r=1}^{8}\left(\sin\frac{2r\pi}{9} + i\cos\frac{2r\pi}{9}\right)$ is

(a) -1 (b) 1

(c) i (d) $-i$

25. If a, b, c and u, v, w are complex numbers representing the vertices of two triangles such that $c = (1-r)a + rb$ and $w = (1-r)u + rv$, where r is a complex number, then the two triangles

(a) Have the same area (b) Are similar

(c) Are congruent (d) None of these

26. Suppose z_1, z_2, z_3 are the vertices of an equilateral triangle inscribed in the circle $|z| = 2$. If $z_1 = 1 + i\sqrt{3}$, then values of z_3 and z_2 are respectively **[IIT 1994]**

(a) $-2, 1 - i\sqrt{3}$ (b) $2, 1 + i\sqrt{3}$

(c) $1 + i\sqrt{3}, -2$ (d) None of these

27. If the complex number z_1, z_2 the origin form an equilateral triangle then $z_1^2 + z_2^2 =$ **[IIT 1983]**

(a) $z_1 z_2$ (b) $z_1 \overline{z_2}$

(c) $\overline{z_2} z_1$ (d) $|z_1|^2 = |z_2|^2$

28. If at least one value of the complex number $z = x + iy$ satisfy the condition $|z + \sqrt{2}| = a^2 - 3a + 2$ and the inequality $|z + i\sqrt{2}| < a^2$, then

(a) $a > 2$ (b) $a = 2$

(c) $a < 2$ (d) None of these

29. If z, iz and $z + iz$ are the vertices of a triangle whose area is 2 units, then the value of $|z|$ is **[RPET 2000]**

(a) – 2 (b) 2

(c) 4 (d) 8

30. If $z^2 + z|z| + |z|^2 = 0$, then the locus of z is

(a) A circle (b) A straight line

(c) A pair of straight lines (d) None of these

31. If $\cos\alpha + \cos\beta + \cos\gamma = \sin\alpha + \sin\beta + \sin\gamma = 0$ then $\cos 3\alpha + \cos 3\beta + \cos 3\gamma$ equals to **[Karnataka CET 2000]**

(a) 0 (b) $\cos(\alpha + \beta + \gamma)$

(c) $3\cos(\alpha + \beta + \gamma)$ (d) $3\sin(\alpha + \beta + \gamma)$

32. If $z_r = \cos\frac{r\alpha}{n^2} + i\sin\frac{r\alpha}{n^2}$, where r = 1, 2, 3,....,n, then $\lim_{n\to\infty} z_1 z_2 z_3 ... z_n$ is equal to **[UPSEAT 2001]**

(a) $\cos\alpha + i\sin\alpha$ (b) $\cos(\alpha/2) - i\sin(\alpha/2)$

(c) $e^{i\alpha/2}$ (d) $\sqrt[3]{e^{i\alpha}}$

33. If the cube roots of unity be $1, \omega, \omega^2$, then the roots of the equation $(x-1)^3 + 8 = 0$ are **[IIT 1979; MNR 1986; DCE 2000; AIEEE 2005]**

(a) $-1, 1 + 2\omega, 1 + 2\omega^2$

(b) $-1, 1 - 2\omega, 1 - 2\omega^2$

(c) $-1, -1, -1$

(d) None of these

34. If $1, \omega, \omega^2, \omega^3 ....., \omega^{n-1}$ are the n, n^{th} roots of unity, then $(1-\omega)(1-\omega^2).....(1-\omega^{n-1})$ equals **[MNR 1992; IIT 1984; DCE 2001; MP PET 2004]**

(a) 0 (b) 1

(c) n (d) n^2

35. The value of the expression $1.(2-\omega)(2-\omega^2) + 2.(3-\omega)(3-\omega^2) + ...$

$.... + (n-1).(n-\omega)(n-\omega^2)$,

where ω is an imaginary cube root of unity, is **[IIT 1996]**

(a) $\frac{1}{2}(n-1)n(n^2+3n+4)$

(b) $\frac{1}{4}(n-1)n(n^2+3n+4)$

(c) $\frac{1}{2}(n+1)n(n^2+3n+4)$

(d) $\frac{1}{4}(n+1)n(n^2+3n+4)$

36. If $i=\sqrt{-1}$, then $4+5\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^{334}+3\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^{365}$ is equal to **[IIT 1999]**

(a) $1-i\sqrt{3}$ (b) $-1+i\sqrt{3}$

(c) $i\sqrt{3}$ (d) $-i\sqrt{3}$

37. If $a=\cos(2\pi/7)+i\sin(2\pi/7)$, then the quadratic equation whose roots are $\alpha=a+a^2+a^4$ and $\beta=a^3+a^5+a^6$ is

[RPET 2000]

(a) $x^2-x+2=0$ (b) $x^2+x-2=0$

(c) $x^2-x-2=0$ (d) $x^2+x+2=0$

38. Let z_1 and z_2 be n^{th} roots of unity which are ends of a line segment that subtend a right angle at the origin. Then n must be of the form **[IIT Screening 2001; Karnataka 2002]**

(a) $4k+1$ (b) $4k+2$

(c) $4k+3$ (d) $4k$

39. Let ω is an imaginary cube roots of unity then the value of

$2(\omega+1)(\omega^2+1)+3(2\omega+1)(2\omega^2+1)+.....+(n+1)(n\omega+1)(n\omega^2+1)$ is **[Orissa JEE 2002]**

(a) $\left[\frac{n(n+1)}{2}\right]^2+n$ (b) $\left[\frac{n(n+1)}{2}\right]^2$

(c) $\left[\frac{n(n+1)}{2}\right]^2-n$ (d) None of these

40. ω is an imaginary cube root of unity. If $(1+\omega^2)^m=(1+\omega^4)^m$, then least positive integral value of m is

[IIT Screening 2004]

(a) 6 (b) 5

(c) 4 (d) 3

ANSWER

1	c	2	d	3	d	4	b	5	b
6	d	7	b	8	a	9	b	10	a
11	d	12	c	13	c	14	c	15	a
16	c	17	c	18	d	19	b	20	a
21	c	22	d	23	c	24	d	25	b
26	a	27	a	28	a	29	b	30	c
31	c	32	c	33	b	34	c	35	b
36	c	37	D	38	d	39	a	40	d

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 6 XI M 6. Quadratic Equations

Index:

1. Key Concepts

2. Exercise I to VI

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Quadratic Equation

1. Equation v/s Identity:

A quadratic equation is satisfied by exactly two values of 'x' which may be real or imaginary. The equation, $ax^2 + bx + c = 0$ is:

★	a quadratic equation if	$a \neq 0$	Two Roots
★	a linear equation if	$a = 0, b \neq 0$	One Root
★	a contradiction if	$a = b = 0, c \neq 0$	No Root
★	an identity if	$a = b = c = 0$	Infinite Roots

If a quadratic equation is satisfied by three distinct values of 'x', then it is an identity.

Solved Example # 1: (i) $3x^2 + 2x - 1 = 0$ is a quadratic equation here $a = 3$.
(ii) $(x + 1)^2 = x^2 + 2x + 1$ is an identity in x.

Solution.: Here highest power of x in the given relation is 2 and this relation is satisfied by three different values $x = 0$, $x = 1$ and $x = -1$ and hence it is an identity because a polynomial equation of nth degree cannot have more than n distinct roots.

2. Relation Between Roots & Co-efficients:

(i) The solutions of quadratic equation, $ax^2 + bx + c = 0$, $(a \neq 0)$ is given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The expression, $b^2 - 4ac \equiv D$ is called discriminant of quadratic equation.

(ii) If α, β are the roots of quadratic equation, $ax^2 + bx + c = 0, a \neq 0$. Then:

(a) $\alpha + \beta = -\frac{b}{a}$ (b) $\alpha\beta = \frac{c}{a}$ (c) $|\alpha - \beta| = \frac{\sqrt{D}}{|a|}$

(iii) A quadratic equation whose roots are α & β, is $(x - \alpha)(x - \beta) = 0$ i.e.
$x^2 -$ (sum of roots) $x +$ (product of roots) $= 0$

Solved Example # 2: If α and β are the roots of $ax^2 + bx + c = 0$, find the equation whose roots are $\alpha+2$ and $\beta+2$.

Solution. Replacing x by $x - 2$ in the given equation, the required equation is
$a(x - 2)^2 + b(x - 2) + c = 0$ i.e., $ax^2 - (4a - b)x + (4a - 2b + c) = 0$.

Solved Example # 3 The coefficient of x in the quadratic equation $x^2 + px + q = 0$ was taken as 17 in place of 13, its roots were found to be -2 and -15. Find the roots of the original equation.

Solution. Here $q = (-2) \times (-15) = 30$, correct value of $p = 13$. Hence original equation is
$x^2 + 13x + 30 = 0$ as $(x + 10)(x + 3) = 0$ $\therefore$ roots are $-10, -3$

Self Practice Problems : 1. If α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$ then find the quadratic equation whose roots are

(i) $2\alpha, 2\beta$ (ii) α^2, β^2 (iii) $\alpha + 1, \beta + 1$ (iv) $\frac{1+\alpha}{1-\alpha}, \frac{1+\beta}{1-\beta}$ (v) $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$

2. If r be the ratio of the roots of the equation $ax^2 + bx + c = 0$, show that $\frac{(r+1)^2}{r} = \frac{b^2}{ac}$.

Ans.(1) (i) $ax^2 + 2bx + 4c = 0$ (ii) $a^2x^2 + (2ac - b^2)x + c^2 = 0$
(iii) $ax^2 - (2a - b)x + a + c - b = 0$ (iv) $(a + b + c)x^2 - 2(a - c)x + a - b + c = 0$
(v) $acx^2 - (b^2 - 2ac)x + ac = 0$

3. Nature of Roots:

Consider the quadratic equation, $ax^2 + bx + c = 0$ having α, β as its roots; $D \equiv b^2 - 4ac$

- $D = 0$ → Roots are equal $\alpha = \beta = -b/2a$
- $D \neq 0$ → Roots are unequal
  - $a, b, c \in R$ & $D > 0$ → Roots are real
    - $a, b, c \in Q$ & D is a perfect square ⇒ Roots are rational
      - $a = 1, b, c \in I$ & D is a perfect square ⇒ Roots are integral.
    - $a, b, c \in Q$ & D is not a perfect square ⇒ Roots are irrational i.e. $\alpha = p + \sqrt{q}, \beta = p - \sqrt{q}$
  - $a, b, c \in R$ & $D < 0$ → Roots are imaginary $\alpha = p + iq, \beta = p - iq$

Solved Example # 4: For what values of m the equation $(1 + m)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has equal roots.

Solution.
Given equation is $(1 + m)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ .....(i)
Let D be the discriminant of equation (i).
Roots of equation (i) will be equal if $D = 0$.
or, $4(1 + 3m)^2 - 4(1 + m)(1 + 8m) = 0$

or, $4(1 + 9m^2 + 6m - 1 - 9m - 8m^2) = 0$

or, $m^2 - 3m = 0$ or, $m(m - 3) = 0$ $\therefore$ $m = 0, 3$.

Solved Example # 5: Find all the integral values of a for which the quadratic equation $(x - a)(x - 10) + 1 = 0$ has integral roots.

Solution.: Here the equation is $x^2 - (a + 10)x + 10a + 1 = 0$. Since integral roots will always be rational it means D should be a perfect square.

From (i) $D = a^2 - 20a + 96$.

$\Rightarrow$ $D = (a - 10)^2 - 4$ $\Rightarrow$ $4 = (a - 10)^2 - D$

If D is a perfect square it means we want difference of two perfect square as 4 which is possible only when $(a - 10)^2 = 4$ and $D = 0$.

$\Rightarrow$ $(a - 10) = \pm 2$ $\Rightarrow$ $a = 12, 8$

Solved Example # 6: If the roots of the equation $(x - a)(x - b) - k = 0$ be c and d, then prove that the roots of the equation $(x - c)(x - d) + k = 0$, are a and b.

Solution. By given condition

$(x - a)(x - b) - k \equiv (x - c)(x - d)$ or $(x - c)(x - d) + k \equiv (x - a)(x - b)$

Above shows that the roots of $(x - c)(x - d) + k = 0$ are a and b.

Self Practice Problems :

3. Let $4x^2 - 4(\alpha - 2)x + \alpha - 2 = 0$ ($\alpha \in R$) be a quadratic equation. Find the value of α for which

(i) Both roots are real and distinct. (ii) Both roots are equal.

(iii) Both roots are imaginary (iv) Both roots are opposite in sign.

(v) Both roots are equal in magnitude but opposite in sign.

4. Find the values of a, if $ax^2 - 4x + 9 = 0$ has integral roots.

5. If $P(x) = ax^2 + bx + c$, and $Q(x) = -ax^2 + dx + c$, $ac \neq 0$ then prove that $P(x) \cdot Q(x) = 0$ has atleast two real roots.

Ans. (1) (i) $(-\infty, 2) \cup (3, \infty)$ (ii) $\alpha \in \{2, 3\}$

(iii) $(2, 3)$ (iv) $(-\infty, 2)$ (v) ϕ

(2) $a = \frac{1}{3}, -\frac{1}{4}$

4. Common Roots:

Consider two quadratic equations, $a_1x^2 + b_1x + c_1 = 0$ & $a_2x^2 + b_2x + c_2 = 0$.

(i) If two quadratic equations have both roots common, then the equation are identical and their co-efficient are in proportion. i.e. $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

(ii) If only one root is common, then the common root 'α' will be: $\alpha = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} = \frac{b_1c_2 - b_2c_1}{c_1a_2 - c_2a_1}$

Hence the condition for one common root is:

$$a_1\left[\frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}\right]^2 + b_1\left[\frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}\right] + c_1 = 0$$

$$\equiv (c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$$

Note : If $f(x) = 0$ & $g(x) = 0$ are two polynomial equation having some common root(s) then those common root(s) is/are also the root(s) of $h(x) = a\,f(x) + bg(x) = 0$.

Solved Example # 7: If $x^2 - ax + b = 0$ and $x^2 - px + q = 0$ have a root in common and the second equation has equal roots, show that $b + q = \frac{ap}{2}$.

Solution. Given equations are : $x^2 - ax + b = 0$ and $x^2 - px + q = 0$.

Let α be the common root. Then roots of equation (2) will be α and α. Let β be the other root of equation (1). Thus roots of equation (1) are α, β and those of equation (2) are α, α.

Now $\alpha + \beta = a$ ..... (iii)

$\alpha\beta = b$ ..... (iv)

$2\alpha = p$ ..... (v)

$\alpha^2 = q$ ..... (vi)

L.H.S. $= b + q = \alpha\beta + \alpha^2 = \alpha(\alpha + \beta)$ ..... (vii)

and R.H.S. $= \frac{ap}{2} = \frac{(\alpha + \beta)\,2\alpha}{2} = \alpha(\alpha + \beta)$ ..... (viii)

from (7) and (8), L.H.S. = R.H.S.

Solved Example # 8: If $a, b, c \in R$ and equations $ax^2 + bx + c = 0$ and $x^2 + 2x + 9 = 0$ have a common root, show that $a : b : c = 1 : 2 : 9$.

Solution. Given equations are : $x^2 + 2x + 9 = 0$ .....(i)

and $ax^2 + bx + c = 0$ .....(ii)

Clearly roots of equation (i) are imaginary since equation (i) and (ii) have a common root, therefore common root must be imaginary and hence both roots will be common.

Therefore equations (i) and (ii) are identical

$\therefore$ $\frac{a}{1} = \frac{b}{2} = \frac{c}{9}$ $\therefore$ $a : b : c = 1 : 2 : 9$

Self Practice Problems : **6.** If the equation $x^2 + bx + ac = 0$ and $x^2 + cx + ab = 0$ have a common root then prove that the equation containing other roots will be given by $x^2 + ax + bc = 0$.

7. If the equations $ax^2 + bx + c = 0$ and $x^3 + 3x^2 + 3x + 2 = 0$ have two common roots then show that $a = b = c$.

8. If $ax^2 + 2bx + c = 0$ and $a_1x^2 + 2b_1x + c_1 = 0$ have a common root and $\frac{a}{a_1}, \frac{b}{b_1}, \frac{c}{c_1}$ are in A.P. show that a_1, b_1, c_1 are in G.P.

5. Factorisation of Quadratic Expressions:

★ The condition that a quadratic expression $f(x) = ax^2 + bx + c$ a perfect square of a linear expression, is $D \equiv b^2 - 4ac = 0$.

★ The condition that a quadratic expression $f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ may be resolved into two linear factors is that;

$$\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \text{ OR } \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0.$$

Solved Example # 9: Determine a such that $x^2 - 11x + a$ and $x^2 - 14x + 2a$ may have a common factor.

Solution. Let $x - \alpha$ be a common factor of $x^2 - 11x + a$ and $x^2 - 14x + 2a$.

Then $x = \alpha$ will satisfy the equations $x^2 - 11x + a = 0$ and $x^2 - 14x + 2a = 0$.

$\therefore$ $\alpha^2 - 11\alpha + a = 0$ and $\alpha^2 - 14\alpha + 2a = 0$

Solving (i) and (ii) by cross multiplication method, we get $a = 24$.

Sol. Ex. 10: Show that the expression $x^2 + 2(a + b + c)x + 3(bc + ca + ab)$ will be a perfect square if $a = b = c$.

Solution. Given quadratic expression will be a perfect square if the discriminant of its corresponding equation is zero.

i.e. $4(a + b + c)^2 - 4.3(bc + ca + ab) = 0$

or $(a + b + c)^2 - 3(bc + ca + ab) = 0$

or $\frac{1}{2}((a - b)^2 + (b - c)^2 + (c - a)^2) = 0$

which is possible only when $a = b = c$.

Self Practice Problems :

9. For what values of k the expression $(4 - k)x^2 + 2(k + 2)x + 8k + 1$ will be a perfect square ?

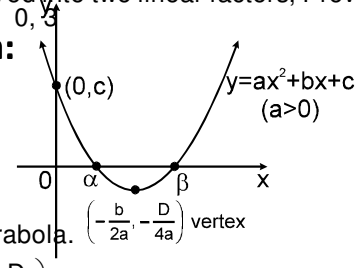
10. If $x - \alpha$ be a factor common to $a_1x^2 + b_1x + c$ and $a_2x^2 + b_2x + c$ prove that $\alpha(a_1 - a_2) = b_2 - b_1$.

11. If $3x^2 + 2\alpha xy + 2y^2 + 2ax - 4y + 1$ can be resolved into two linear factors, Prove that α is a root of the equation $x^2 + 4ax + 2a^2 + 6 = 0$. **Ans.** (1) 0, 3

6. Graph of Quadratic Expression:

$y = f(x) = ax^2 + bx + c$

or $\left(y + \frac{D}{4a}\right) = a\left(x + \frac{b}{2a}\right)^2$



★ the graph between x, y is always a parabola.

★ the co–ordinate of vertex are $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$

★ If $a > 0$ then the shape of the parabola is concave upwards & if $a < 0$ then the shape of the parabola is concave downwards.

★ the parabola intersect the y–axis at point (0, c).

★ the x–co–ordinate of point of intersection of parabola with x–axis are the real roots of the quadratic equation $f(x) = 0$. Hence the parabola may or may not intersect the x–axis at real points.

7. Range of Quadratic Expression $f(x) = ax^2 + bx + c$.

(i) Absolute Range:

If $a > 0 \Rightarrow f(x) \in \left[-\frac{D}{4a}, \infty\right)$

$a < 0 \Rightarrow f(x) \in \left(-\infty, -\frac{D}{4a}\right]$

Hence maximum and minimum values of the expression $f(x)$ is $-\frac{D}{4a}$ in respective cases and it occurs at $x = -\frac{b}{2a}$ (at vertex).

(ii) Range in restricted domain: Given $x \in [x_1, x_2]$

(a) If $-\frac{b}{2a} \notin [x_1, x_2]$ then,

$$f(x) \in \left[\min\left\{f(x_1), f(x_2)\right\}, \quad \max\left\{f(x_1), f(x_2)\right\}\right]$$

(b) If $-\frac{b}{2a} \in [x_1, x_2]$ then,

$$f(x) \in \left[\min\left\{f(x_1), f(x_2), -\frac{D}{4a}\right\}, \quad \max\left\{f(x_1), f(x_2), -\frac{D}{4a}\right\}\right]$$

Solved Example # 11 If $c < 0$ and $ax^2 + bx + c = 0$ does not have any real roots then prove that

(i) $a - b + c < 0$ (ii) $9a + 3b + c < 0$.

Solution.

$c < 0$ and $D < 0 \Rightarrow f(x) = ax^2 + bx + c < 0$ for all $x \in R$

$\Rightarrow f(-1) = a - b + c < 0$

and $f(3) = 9a + 3b + c < 0$

Solved Example # 12 Find the maximum and minimum values of $f(x) = x^2 - 5x + 6$.

Solution.

minimum of $f(x) = -\frac{D}{4a}$ at $x = -\frac{b}{2a}$

$= -\left(\frac{25-24}{4}\right)$ at $x = \frac{5}{2}$ $= -\frac{1}{4}$

maximum of $f(x) = \infty$ Hence range is $\left[-\frac{1}{4}, \infty\right)$.

Solved Example # 13 : Find the range of rational expression $y = \frac{x^2-x+1}{x^2+x+1}$ if x is real.

Solution. $y = \frac{x^2-x+1}{x^2+x+1}$

$\Rightarrow$ $(y-1)x^2 + (y+1)x + y - 1 = 0$

$\therefore$ x is real $\therefore$ $D \geq 0$

$\Rightarrow$ $(y+1)^2 - 4(y-1)^2 \geq 0$ $\Rightarrow$ $(y-3)(3y-1) \leq 0$ $\Rightarrow$ $y \in \left[\frac{1}{3}, 3\right]$.

Solved Example # 14: Find the range of $y = \frac{x+2}{2x^2+3x+6}$, if x is real.

Solution.: $y = \frac{x+2}{2x^2+3x+6}$

$\Rightarrow$ $2yx^2 + 3yx + 6y = x + 2$ $\Rightarrow$ $2yx^2 + (3y-1)x + 6y - 2 = 0$

$\therefore$ x is real

$D \geq 0$

$\Rightarrow$ $(3y-1)^2 - 8y(6y-2) \geq 0$ $\Rightarrow$ $(3y-1)(13y+1) \leq 0$

$y \in \left[-\frac{1}{13}, \frac{1}{3}\right]$.

Self Practice Problems :

12. If $c > 0$ and $ax^2 + 2bx + 3c = 0$ does not have any real roots then prove that
(i) $a - 2b + 3c > 0$ (ii) $a + 4b + 12c > 0$

13. If $f(x) = (x-a)(x-b)$, then show that $f(x) \geq -\frac{(a-b)^2}{4}$.

14. For what least integral value of k the quadratic polynomial $(k-2)x^2 + 8x + k + 4 > 0 \ \forall\ x \in R$.

15. Find the range in which the value of function $\frac{x^2+34x-71}{x^2+2x-7}$ lies $\forall\ x \in R$.

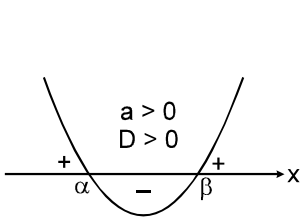
16. Find the interval in which 'm' lies so that the function $y = \frac{mx^2+3x-4}{-4x^2+3x+m}$ can take all real values $\forall\ x \in R$.

Ans. (14) k = 5. (15) $(-\infty, 5] \cup [9, \infty)$ (16) $m \in [1, 7]$

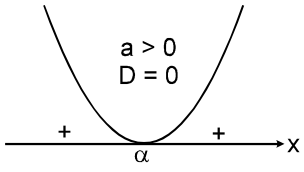
8. Sign of Quadratic Expressions:

The value of expression, $f(x) = ax^2 + bx + c$ at $x = x_0$ is equal to y–co–ordinate of a point on parabola $y = ax^2 + bx + c$ whose x–co–ordinate is x_0. Hence if the point lies above the x–axis for some $x = x_0$, then $f(x_0) > 0$ and vice–versa.

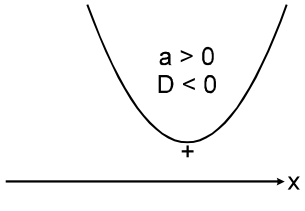
We get six different positions of the graph with respect to x–axis as shown.



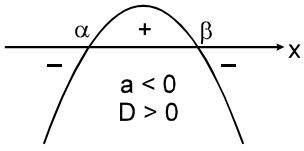
Roots are real & distinct

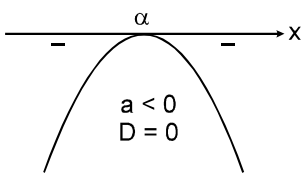


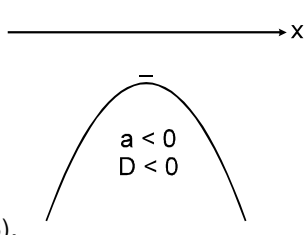
Roots are equal



Roots are imaginary







NOTE:
(i) $\forall\ x \in R$, $y > 0$ only if $a > 0$ & $D \equiv b^2 - 4ac < 0$ (figure 3).
(ii) $\forall\ x \in R$, $y < 0$ only if $a < 0$ & $D \equiv b^2 - 4ac < 0$ (figure 6).

9. Solution of Quadratic Inequalities:

The values of 'x' satisfying the inequality, $ax^2 + bx + c > 0$ $(a \neq 0)$ are:

(i) If $D > 0$, i.e. the equation $ax^2 + bx + c = 0$ has two different roots $\alpha < \beta$.
Then $a > 0 \Rightarrow x \in (-\infty, \alpha) \cup (\beta, \infty)$

$a < 0 \Rightarrow x \in (\alpha, \beta)$

(ii) If D = 0, i.e. roots are equal, i.e. $\alpha = \beta$.
Then $a > 0 \Rightarrow x \in (-\infty, \alpha) \cup (\alpha, \infty)$
$a < 0 \Rightarrow x \in \phi$

(iii) If D < 0, i.e. the equation $ax^2 + bx + c = 0$ has no real root.
Then $a > 0 \Rightarrow x \in R$
$a < 0 \Rightarrow x \in \phi$

(iv) Inequalities of the form $\frac{P(x)\ Q(x)\ R(x).....}{A(x)\ B(x)\ C(x).....} \begin{matrix} \le \\ = \\ > \end{matrix} 0$ can be quickly solved using the method of intervals, where A, B, C....., P, Q, R..... are linear functions of 'x'.

Solved Example # 15 Solve $\frac{x^2+6x-7}{x^2+1} \le 2$

Solution. $\Rightarrow \quad x^2 + 6x - 7 \le 2x^2 + 2$
$\Rightarrow \quad x^2 - 6x + 9 \ge 0 \quad \Rightarrow \quad (x-3)^2 \ge 0 \quad \Rightarrow \quad x \in R$

Solved Example # 16: Solve $\frac{x^2+x+1}{|x+1|} > 0$.

Solution. $\therefore \quad |x + 1| > 0$
$\forall\ x \in R - \{-1\}$
$\therefore \quad x^2 + x + 1 > 0 \quad \therefore \quad D = 1 - 4 = -3 < 0$
$\therefore \quad x^2 + x + 1 > 0\ \forall\ x \in R \quad \therefore \quad x \in (-\infty, -1) \cup (-1, \infty)$

Solved Example # 17 $\left|\frac{x^2-3x-1}{x^2+x+1}\right| < 3.$

Solution. $\frac{|x^2-3x-1|}{x^2+x+1} < 3.$

$\because$ in $x^2 + x + 1$
$D = 1 - 4 = -3 < 0$
$\therefore \quad x^2 + x + 1 > 0\ \forall\ x \in R \quad \therefore \quad |x^2 - 3x - 1| < 3(x^2 + x + 1)$
$\Rightarrow \quad (x^2 - 3x - 1)^2 - \{3(x^2 + x + 1)\}^2 < 0$
$\Rightarrow \quad (4x^2 + 2)(-2x^2 - 6x - 4) < 0$
$\Rightarrow \quad (2x^2 + 1)(x + 2)(x + 1) > 0 \quad \Rightarrow \quad x \in (-\infty, -2) \cup (-1, \infty)$

Self Practice Problems :

17. (i) $|x^2 + x| - 5 < 0$ (ii) $x^2 - 7x + 12 < |x - 4|$

18. Solve $\frac{2x}{x^2-9} \le \frac{1}{x+2}$

19. Solve the inequation $(x^2 + 3x + 1)(x^2 + 3x - 3) \ge 5$

20. Find the value of parameter 'α' for which the inequality $\left|\frac{x^2+\alpha x+1}{x^2+x+1}\right| < 3$ is satisfied $\forall\ x \in R$

21. Solve $\left|\frac{x^2-5x+4}{x^2-4}\right| \le 1$

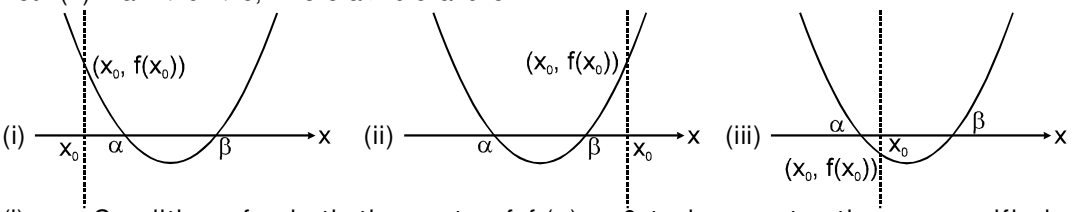
Ans. (17) (i) $\left(-\left(\frac{1+\sqrt{21}}{2}\right), \left(\frac{\sqrt{21}-1}{2}\right)\right)$ (ii) $(2, 4)$

(18) $(-\infty, -3) \cup (-2, 3)$ (19) $(-\infty, -4] \cup [-2, -1] \cup [1, \infty)$

(20) $(-1, 5)$ (21) $\left[0, \frac{8}{5}\right] \cup \left[\frac{5}{2}, \infty\right)$

10. Location Of Roots:

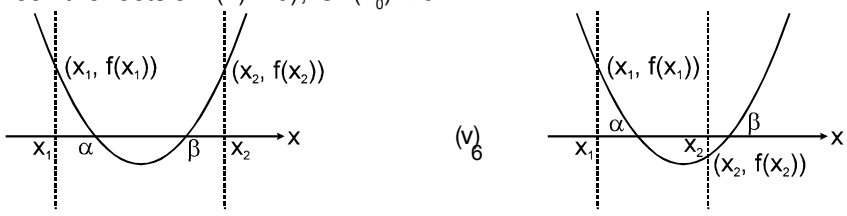
Let $f(x) = ax^2 + bx + c$, where $a > 0$ & $a, b, c \in R$.



(i) Conditions for both the roots of $f(x) = 0$ to be greater than a specified number 'x_0' are $b^2 - 4ac \ge 0$; $f(x_0) > 0$ & $(-b/2a) > x_0$.

(ii) Conditions for both the roots of $f(x) = 0$ to be smaller than a specified number 'x_0' are $b^2 - 4ac \ge 0$; $f(x_0) > 0$ & $(-b/2a) < x_0$.

(iii) Conditions for both roots of $f(x) = 0$ to lie on either side of the number 'x_0' (in other words the number 'x_0' lies between the roots of $f(x) = 0$), is $f(x_0) < 0$.

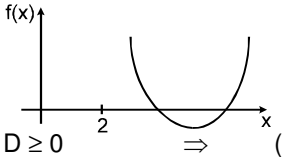


(iv) Conditions that both roots of $f(x) = 0$ to be confined between the numbers x_1 and x_2, $(x_1 < x_2)$ are $b^2 - 4ac \geq 0$; $f(x_1) > 0$; $f(x_2) > 0$ & $x_1 < (-b/2a) < x_2$.

(v) Conditions for exactly one root of $f(x) = 0$ to lie in the interval (x_1, x_2) i.e. $x_1 < x < x_2$ is $f(x_1) . f(x_2) < 0$.

Ex.10.1 $x^2 - (m - 3)x + m = 0$

(a) Find values of m so that both the roots are greater than 2.



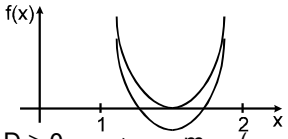
Condition - I $D \geq 0 \Rightarrow (m-3)^2 - 4m \geq 0 \Rightarrow m^2 - 10m + 9 \geq 0$
$\Rightarrow (m-1)(m-9) \geq 0 \Rightarrow m \in (-\infty, 1] \cup [9, \infty)$ .....(i)

Condition - II $f(2) > 0 \Rightarrow 4 - (m-3)2 + m > 0 \Rightarrow m < 10$...(ii),

Condition - III $-\frac{b}{2a} > 2 \Rightarrow \frac{m-3}{2} > 2 \Rightarrow m > 7$.....(iii)

Intersection of (i), (ii) and (iii) gives $m \in [9, 10)$ **Ans.**

(b) Find the values of m so that both roots lie in the interval (1, 2)

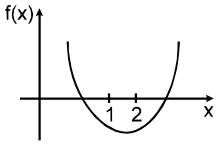


Condition - I $D \geq 0 \Rightarrow m \in (-\infty, 1] \cup [9, \infty)$
Condition - II $f(1) > 0 \Rightarrow 1 - (m-3) + m > 0 \Rightarrow 4 > 0 \Rightarrow m \in R$
Condition - III $f(2) > 0 \Rightarrow m < 10$

Condition - IV $1 < -\frac{b}{2a} < 2 \Rightarrow 1 < \frac{m-3}{2} < 2 \Rightarrow 5 < m < 7$

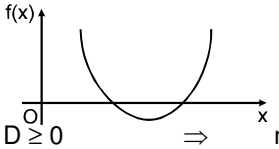
intersection gives $m \in \phi$ **Ans.**

(c) One root is greater than 2 and other smaller than 1



Condition - I $f(1) < 0 \Rightarrow 4 < 0 \Rightarrow m \in \phi$
Condition - II $f(2) < 0 \Rightarrow m > 10$
Intersection gives $m \in \phi$ **Ans.**

(d) Find the value of m for which both roots are positive.

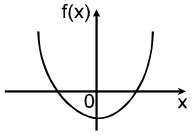


Condition - I $D \geq 0 \Rightarrow m \in (-\infty, 1] \cup [9, \infty)$
Condition - II $f(0) > 0 \Rightarrow m > 0$

Condition - III $-\frac{b}{2a} > 0 \Rightarrow \frac{m-3}{2} > 0 \Rightarrow m > 3$

intersection gives $m \in [9, \infty)$ **Ans.**

(e) Find the values of m for which one root is (positive) and other is (negative).

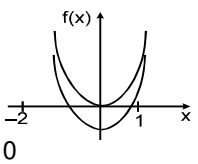


Condition - I $f(0) < 0 \Rightarrow m < 0$ **Ans.**

(f) Roots are equal in magnitude and opposite in sign.
sum of roots = 0 $\Rightarrow m = 3$
and $f(0) < 0 \Rightarrow m < 0$
$\therefore$ $m \in \phi$ **Ans.**

Ex.10.2 Find all the values of 'a' for which both the roots of the equation $(a-2)x^2 + 2ax + (a+3) = 0$ lies in the interval $(-2, 1)$.

Sol. **Case - I**



When $a - 2 > 0$
$\Rightarrow a > 2$

Condition - I $f(-2) > 0 \Rightarrow (a-2)4 - 4a + a + 3 > 0 \Rightarrow a - 5 > 0 \Rightarrow a > 5$

Condition - II $f(1)>0 \Rightarrow 4a+1>0 \Rightarrow a>-\frac{1}{4}$

Condition - III $D \geq 0 \Rightarrow 4a^2-4(a+3)(a-2) \geq 0 \Rightarrow a \leq 6$

Condition - IV $-\frac{b}{2a}<1 \Rightarrow \frac{2(a-1)}{a-2}>0 \Rightarrow a \in (-\infty, 1) \cup (4, \infty)$

Condition - V $-2<-\frac{b}{2a} \Rightarrow \frac{-2a}{2(a-2)}>-2 \Rightarrow \frac{a-4}{a-2}>0$

Intersection gives $a \in (5, 6]$. **Ans.**

Case-II when $a-2<0$

$a<2$

Condition - I $f(-2)<0 \Rightarrow a<5$

Condition - II $f(1)<0, \Rightarrow a<-\frac{1}{4}$

Condition - III $-2<-\frac{b}{2a}<1 \Rightarrow a \in (-\infty, 1) \cup (4, \infty)$

Condition - IV $D \geq 0 \Rightarrow a \leq 6$

intersection gives $a \in \left(-\infty, -\frac{1}{4}\right)$

complete solution is $a \in \left(-\infty, -\frac{1}{4}\right) \cup (5, 6]$ **Ans.**

Self Practice Problems :

22. Let $4x^2-4(\alpha-2)x+\alpha-2=0$ ($\alpha \in R$) be a quadratic equation find the value of α for which
(a) Both the roots are positive (b) Both the roots are negative
(c) Both the roots are opposite in sign. (d) Both the roots are greater than 1/2.
(e) Both the roots are smaller than 1/2.
(f) One root is small than 1/2 and the other root is greater than 1/2.
Ans. (a) $[3, \infty)$ (b) ϕ (c) $(-\infty, 2)$ (d) ϕ (e) $(-\infty, 2]$ (f) $(3, \infty)$

23. Find the values of the parameter a for which the roots of the quadratic equation $x^2+2(a-1)x+a+5=0$ are
(i) positive (ii) negative (iii) opposite in sign.
Ans. (i) $(-5, -1]$ (ii) $[4, \infty)$ (iii) $(-\infty, -5)$

24. Find the values of P for which both the roots of the equation $4x^2-20px+(25p^2+15p-66)=0$ are less than 2.
Ans. $(-\infty, -1)$

25. Find the values of α for which 6 lies between the roots of the equation $x^2+2(\alpha-3)x+9=0$.
Ans. $\left(-\infty, -\frac{3}{4}\right)$.

26. Let $4x^2-4(\alpha-2)x+\alpha-2=0$ ($\alpha \in R$) be a quadratic equation find the value of α for which
(i) Exactly one root lies in $\left(0, \frac{1}{2}\right)$. (ii) Both roots lies in $\left(0, \frac{1}{2}\right)$.
(iii) At least one root lies in $\left(0, \frac{1}{2}\right)$. (iv) One root is greater than 1/2 and other root is smaller than 0.
Ans. (i) $(-\infty, 2) \cup (3, \infty)$ (ii) ϕ (iii) $(-\infty, 2) \cup (3, \infty)$ (iv) ϕ

27. In what interval must the number 'a' vary so that both roots of the equation $x^2-2ax+a^2-1=0$ lies between -2 and 4. **Ans.** $(-1, 3)$

28. Find the values of k, for which the quadratic expression $ax^2+(a-2)x-2$ is negative for exactly two integral values of x. **Ans.** [1, 2)

11. Theory Of Equations:

If $\alpha_1, \alpha_2, \alpha_3, \ldots \alpha_n$ are the roots of the equation;

$f(x)=a_0x^n+a_1x^{n-1}+a_2x^{n-2}+\ldots+a_{n-1}x+a_n=0$ where $a_0, a_1, \ldots a_n$ are all real & $a_0 \neq 0$ then,

$\sum \alpha_1=-\frac{a_1}{a_0}$, $\sum \alpha_1\alpha_2=+\frac{a_2}{a_0}$, $\sum \alpha_1\alpha_2\alpha_3=-\frac{a_3}{a_0}$, ..., $\alpha_1\alpha_2\alpha_3 \ldots \alpha_n=(-1)^n\frac{a_n}{a_0}$

NOTE :
(i) If α is a root of the equation $f(x)=0$, then the polynomial $f(x)$ is exactly divisible by $(x-\alpha)$ or $(x-\alpha)$ is a factor of $f(x)$ and conversely.
(ii) Every equation of nth degree ($n \geq 1$) has exactly n roots & if the equation has more than n roots, it is an identity.
(iii) If the coefficients of the equation $f(x)=0$ are all real and $\alpha+i\beta$ is its root, then $\alpha-i\beta$ is also a root. i.e. imaginary roots occur in conjugate pairs.
(iv) An equation of odd degree will have odd number of real roots and an equation of even degree will have even numbers of real roots.
(v) If the coefficients in the equation are all rational & $\alpha+\sqrt{\beta}$ is one of its roots, then $\alpha-\sqrt{\beta}$ is also a root where $\alpha, \beta \in Q$ & β is not a perfect square.
(vi) If there be any two real numbers 'a' & 'b' such that $f(a)$ & $f(b)$ are of opposite signs, then $f(x)=0$ must have odd number of real roots (also atleast one real root) between 'a' and 'b'.
(vii) Every equation $f(x)=0$ of degree odd has atleast one real root of a sign opposite to that of its last term. (If coefficient of highest degree term is positive).

Ex.11.1 $2x^3 + 3x^2 + 5x + 6 = 0$ has roots α, β, γ then find $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$.

$\therefore \quad \alpha + \beta + \gamma = = -\frac{3}{2} \qquad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{5}{2}, \qquad \alpha\beta\gamma = -\frac{6}{2} = -3.$

Ex.11.2 Find the roots of $4x^3 + 20x^2 - 23x + 6 = 0$. If two roots are equal.

Let roots be α, α and β

$\therefore \quad \alpha + \alpha + \beta = -\frac{20}{4}$

$\Rightarrow \quad 2\alpha + \beta = -5 \qquad$ .....(i)

$\therefore \quad \alpha \cdot \alpha + \alpha\beta + \alpha\beta = -\frac{23}{4}$

$\Rightarrow \quad \alpha^2 + 2\alpha\beta = -\frac{23}{4} \qquad \& \qquad \alpha^2\beta = -\frac{6}{4}$

from equation (i)

$$\alpha^2 + 2\alpha(-5 - 2\alpha) = -\frac{23}{4}$$

$\Rightarrow \quad \alpha^2 - 10\alpha - 4\alpha^2 = -\frac{23}{4} \quad \Rightarrow \quad 12\alpha^2 + 40\alpha - 23 = 0$

$\therefore \quad \alpha = 1/2, -\frac{23}{6} \qquad$ when $\alpha = \frac{1}{2}$

from equation (i) $\qquad \alpha^2\beta = \frac{1}{4}(-5 - 1) = -\frac{3}{2}$

when $\alpha = -\frac{23}{6}$

$$\alpha^2\beta = \frac{23 \times 23}{36}\left(-5 - 2x\left(-\frac{23}{6}\right)\right) \neq -\frac{3}{2}$$

$\Rightarrow \quad \alpha = \frac{1}{2}, \qquad \beta = -6$

Hence roots of equation $= \frac{1}{2}, \frac{1}{2}, -6$ **Ans.**

Self Practice Problems :

29. Find the relation between p, q and r if the roots of the cubic equation $x^3 - px^2 + qx - r = 0$ are such that they are in A.P. **Ans.** $2p^3 - 9pq + 27r = 0$

30. If α, β, γ are the roots of the cubic $x^3 + qx + r = 0$ then find the equation whose roots are

(a) $\alpha + \beta, \beta + \gamma, \gamma + \alpha$ **Ans.** $x^3 + qx - r = 0$

(b) $\alpha\beta, \beta\gamma, \gamma\alpha$ **Ans.** $x^3 - qx^2 - r^2 = 0$

(c) $\alpha^2, \beta^2, \gamma^2$ **Ans.** $x^3 + 2qx^2 + q^2x - r^2 = 0$

(d) $\alpha^3, \beta^3, \gamma^3$ **Ans.** $x^3 + 3x^2r + (3r^2 + q^3)x + r^3 = 0$

SHORT REVISION

The general form of a quadratic equation in x is, $ax^2 + bx + c = 0$, where $a, b, c \in R$ & $a \neq 0$.

RESULTS :

1. The solution of the quadratic equation, $ax^2 + bx + c = 0$ is given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

The expression $b^2 - 4ac = D$ is called the discriminant of the quadratic equation.

2. If α & β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then;

(i) $\alpha + \beta = -b/a$ (ii) $\alpha\beta = c/a$ (iii) $\alpha - \beta = \sqrt{D}/a$.

3. **NATURE OF ROOTS:**

(A) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in R$ & $a \neq 0$ then ;

(i) $D > 0 \Leftrightarrow$ roots are real & distinct (unequal).

(ii) $D = 0 \Leftrightarrow$ roots are real & coincident (equal).

(iii) $D < 0 \Leftrightarrow$ roots are imaginary .

(iv) If $p + iq$ is one root of a quadratic equation, then the other must be the conjugate $p - iq$ & vice versa. $(p, q \in R \ \& \ i = \sqrt{-1})$.

(B) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in Q$ & $a \neq 0$ then;

(i) If $D > 0$ & is a perfect square, then roots are rational & unequal.

(ii) If $\alpha = p + \sqrt{q}$ is one root in this case, (where p is rational & $\sqrt{q}$ is a surd)

then the other root must be the conjugate of it i.e. $\beta = p - \sqrt{q}$ & vice versa.

4. A quadratic equation whose roots are α & β is $(x - \alpha)(x - \beta) = 0$ i.e. $x^2 - (\alpha + \beta)x + \alpha\beta = 0$ i.e. $x^2 - (\text{sum of roots})x + \text{product of roots} = 0$.

5. Remember that a quadratic equation cannot have three different roots & if it has, it becomes an identity.

6. **Consider the quadratic expression,** $y = ax^2 + bx + c$, $a \neq 0$ & $a, b, c \in R$ then ;

(i) The graph between x, y is always a parabola. If $a > 0$ then the shape of the parabola is concave upwards & if $a < 0$ then the shape of the parabola is concave downwards.

(ii) $\forall\ x \in R,\ y > 0$ only if $a > 0\ \&\ b^2 - 4ac < 0$ (figure 3).
(iii) $\forall\ x \in R,\ y < 0$ only if $a < 0\ \&\ b^2 - 4ac < 0$ (figure 6).
Carefully go through the 6 different shapes of the parabola given below.

7. SOLUTION OF QUADRATIC INEQUALITIES:
$ax^2 + bx + c > 0\ (a \neq 0)$.
(i) If $D > 0$, then the equation $ax^2 + bx + c = 0$ has two different roots $x_1 < x_2$.
Then $a > 0 \Rightarrow x \in (-\infty, x_1) \cup (x_2, \infty)$
$a < 0 \Rightarrow x \in (x_1, x_2)$
(ii) If $D = 0$, then roots are equal, i.e. $x_1 = x_2$.
In that case $a > 0 \Rightarrow x \in (-\infty, x_1) \cup (x_1, \infty)$
$a < 0 \Rightarrow x \in \phi$
(iii) Inequalities of the form $\frac{P(x)}{Q(x)}$ 0 can be quickly solved using the method of intervals.

8. MAXIMUM & MINIMUM VALUE of $y = ax^2 + bx + c$ occurs at $x = -(b/2a)$ according as;
$a < 0$ or $a > 0$. $y \in \left[\frac{4ac - b^2}{4a}, \infty\right)$ if $a > 0$ & $y \in \left(-\infty, \frac{4ac - b^2}{4a}\right]$ if $a < 0$.

9. COMMON ROOTS OF 2 QUADRATIC EQUATIONS [ONLY ONE COMMON ROOT] :
Let α be the common root of $ax^2 + bx + c = 0$ & $a'x^2 + b'x + c' = 0$. Therefore
$a\alpha^2 + b\alpha + c = 0$; $a'\alpha^2 + b'\alpha + c' = 0$. By Cramer's Rule $\frac{\alpha^2}{bc' - b'c} = \frac{\alpha}{a'c - ac'} = \frac{1}{ab' - a'b}$
Therefore, $\alpha = \frac{ca' - c'a}{ab' - a'b} = \frac{bc' - b'c}{a'c - ac'}$.
So the condition for a common root is $(ca' - c'a)^2 = (ab' - a'b)(bc' - b'c)$.

10. The condition that a quadratic function $f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ may be resolved into two linear factors is that ;
$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ **OR** $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0.$

11. THEORY OF EQUATIONS :
If $\alpha_1, \alpha_2, \alpha_3, ...\alpha_n$ are the roots of the equation;
$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + ... + a_{n-1}x + a_n = 0$ where $a_0, a_1, ... a_n$ are all real & $a_0 \neq 0$ then,
$\sum \alpha_1 = -\frac{a_1}{a_0}$, $\sum \alpha_1\alpha_2 = +\frac{a_2}{a_0}$, $\sum \alpha_1\alpha_2\alpha_3 = -\frac{a_3}{a_0}$, ..., $\alpha_1\alpha_2\alpha_3.....\alpha_n = (-1)^n\frac{a_n}{a_0}$

Note : **(i)** If α is a root of the equation $f(x) = 0$, then the polynomial $f(x)$ is exactly divisible by $(x - \alpha)$ or $(x - \alpha)$ is a factor of $f(x)$ and conversely .
(ii) Every equation of nth degree $(n \geq 1)$ has exactly n roots & if the equation has more than n roots, it is an identity.
(iii) If the coefficients of the equation $f(x) = 0$ are all real and $\alpha + i\beta$ is its root, then $\alpha - i\beta$ is also a root. i.e. **imaginary roots occur in conjugate pairs**.
(iv) If the coefficients in the equation are all rational & $\alpha + \sqrt{\beta}$ is one of its roots, then $\alpha - \sqrt{\beta}$ is also a root where $\alpha, \beta \in Q$ & β is not a perfect square.
(v) If there be any two real numbers 'a' & 'b' such that $f(a)$ & $f(b)$ are of opposite signs, then $f(x) = 0$ must have atleast one real root between 'a' and 'b'.
(vi) Every equation $f(x) = 0$ of degree odd has atleast one real root of a sign opposite to that of its last term.

12. LOCATION OF ROOTS : Let $f(x) = ax^2 + bx + c$, where $a > 0$ & $a, b, c \in R$.
(i) Conditions for both the roots of $f(x) = 0$ to be greater than a specified number 'd' are $b^2 - 4ac \geq 0$; $f(d) > 0$ & $(-b/2a) > d$.
(ii) Conditions for both roots of $f(x) = 0$ to lie on either side of the number 'd' (in other words the number 'd' lies between the roots of $f(x) = 0$) is $f(d) < 0$.
(iii) Conditions for exactly one root of $f(x) = 0$ to lie in the interval (d, e) i.e. $d < x < e$ are $b^2 - 4ac > 0$ & $f(d).f(e) < 0$.
(iv) Conditions that both roots of $f(x) = 0$ to be confined between the numbers p & q are $(p < q)$. $b^2 - 4ac \geq 0$; $f(p) > 0$; $f(q) > 0$ & $p < (-b/2a) < q$.

13. LOGARITHMIC INEQUALITIES
(i) For $a > 1$ the inequality $0 < x < y$ & $\log_a x < \log_a y$ are equivalent.
(ii) For $0 < a < 1$ the inequality $0 < x < y$ & $\log_a x > \log_a y$ are equivalent.
(iii) If $a > 1$ then $\log_a x < p \Rightarrow 0 < x < a^p$
(iv) If $a > 1$ then $\log_a x > p \Rightarrow x > a^p$
(v) If $0 < a < 1$ then $\log_a x < p \Rightarrow x > a^p$
(vi) If $0 < a < 1$ then $\log_a x > p \Rightarrow 0 < x < a^p$

EXERCISE–1

Q.1 If the roots of the equation $[1/(x+p)] + [1/(x+q)] = 1/r$ are equal in magnitude but opposite in sign, show that $p + q = 2r$ & that the product of the roots is equal to $(-1/2)(p^2 + q^2)$.

Q.2 If $x^2 - x\cos(A + B) + 1$ is a factor of the expression,
$2x^4 + 4x^3\sin A\sin B - x^2(\cos 2A + \cos 2B) + 4x\cos A\cos B - 2$. Then find the other factor.

Q.3 α, β are the roots of the equation $K(x^2 - x) + x + 5 = 0$. If K_1 & K_2 are the two values of K for which the roots α, β are connected by the relation $(\alpha/\beta) + (\beta/\alpha) = 4/5$. Find the value of $(K_1/K_2) + (K_2/K_1)$.

Q.4 If the quadratic equations, $x^2 + bx + c = 0$ and $bx^2 + cx + 1 = 0$ have a common root then prove that either $b + c + 1 = 0$ or $b^2 + c^2 + 1 = bc + b + c$.

Q.5 If the roots of the equation $\left(1 - q + \frac{p^2}{2}\right)x^2 + p(1+q)x + q(q-1) + \frac{p^2}{2} = 0$ are equal then show that

$p^2 = 4q$.

Q.6 If one root of the equation $ax^2 + bx + c = 0$ be the square of the other, prove that $b^3 + a^2c + ac^2 = 3abc$.

Q.7 Find the range of values of a, such that $f(x) = \dfrac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32}$ is always negative.

Q.8 Find a quadratic equation whose sum and product of the roots are the values of the expressions (cosec $10° - \sqrt{3}$ sec$10°$) and (0.5 cosec$10° - 2$ sin$70°$) respectively. Also express the roots of this quadratic in terms of tangent of an angle lying in $\left(0, \dfrac{\pi}{2}\right)$.

Q.9 Find the least value of $\dfrac{6x^2 - 22x + 21}{5x^2 - 18x + 17}$ for all real values of x, using the theory of quadratic equations.

Q.10 Find the least value of $(2p^2 + 1)x^2 + 2(4p^2 - 1)x + 4(2p^2 + 1)$ for real values of p and x.

Q.11 If α be a root of the equation $4x^2 + 2x - 1 = 0$ then prove that $4\alpha^3 - 3\alpha$ is the other root.

Q.12(a) If α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$ then which of the following expressions in α, β will denote the symmetric functions of roots. Give proper reasoning. (i) $f(\alpha, \beta) = \alpha^2 - \beta$ (ii) $f(\alpha, \beta) = \alpha^2\beta + \alpha\beta^2$ (iii) $f(\alpha, \beta) = ln\ \dfrac{\alpha}{\beta}$

(iv) $f(\alpha, \beta) = \cos(\alpha - \beta)$

(b) If α, β are the roots of the equation $x^2 - px + q = 0$, then find the quadratic equation the roots of which are $(\alpha^2 - \beta^2)(\alpha^3 - \beta^3)$ & $\alpha^3\beta^2 + \alpha^2\beta^3$.

Q.13 If α, β are the roots of $ax^2 + bx + c = 0$ & $\alpha', -\beta$ are the roots of $a'x^2 + b'x + c' = 0$, show that α, α' are the roots of $\left[\dfrac{b}{a} + \dfrac{b'}{a'}\right]^{-1} x^2 + x + \left[\dfrac{b}{c} + \dfrac{b'}{c'}\right]^{-1} = 0$.

Q.14 If α, β are the roots of $x^2 - px + 1 = 0$ & γ, δ are the roots of $x^2 + qx + 1 = 0$, show that $(\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta) = q^2 - p^2$.

Q.15 Show that if p, q, r & s are real numbers & $pr = 2(q + s)$, then at least one of the equations $x^2 + px + q = 0$, $x^2 + rx + s = 0$ has real roots.

Q.16 If a & b are positive numbers, prove that the equation $\dfrac{1}{x} + \dfrac{1}{x-a} + \dfrac{1}{x+b} = 0$ has two real roots, one between a/3 & 2a/3 and the other between $-2b/3$ & $-b/3$.

Q.17 If the roots of $x^2 - ax + b = 0$ are real & differ by a quantity which is less than c ($c > 0$), prove that b lies between $(1/4)(a^2 - c^2)$ & $(1/4)a^2$.

Q.18 At what values of 'a' do all the zeroes of the function , $f(x) = (a-2)x^2 + 2ax + a + 3$ lie on the interval $(-2, 1)$?

Q.19 If one root of the quadratic equation $ax^2 + bx + c = 0$ is equal to the nth power of the other, then show that $(ac^n)^{1/(n+1)} + (a^nc)^{1/(n+1)} + b = 0$.

Q.20 If p, q, r and s are distinct and different from 2, show that if the points with co-ordinates $\left(\dfrac{p^4}{p-2}, \dfrac{p^3-5}{p-2}\right), \left(\dfrac{q^4}{q-2}, \dfrac{q^3-5}{q-2}\right), \left(\dfrac{r^4}{r-2}, \dfrac{r^3-5}{r-2}\right)$ and $\left(\dfrac{s^4}{s-2}, \dfrac{s^3-5}{s-2}\right)$ are collinear then $pqrs = 5(p + q + r + s) + 2(pqr + qrs + rsp + spq)$.

Q.21 The quadratic equation $x^2 + px + q = 0$ where p and q are integers has rational roots. Prove that the roots are all integral.

Q.22 If the quadratic equations $x^2 + bx + ca = 0$ & $x^2 + cx + ab = 0$ have a common root, prove that the equation containing their other root is $x^2 + ax + bc = 0$.

Q.23 If α, β are the roots of $x^2 + px + q = 0$ & $x^{2n} + p^nx^n + q^n = 0$ where n is an even integer, show that $\alpha/\beta, \beta/\alpha$ are the roots of $x^n + 1 + (x+1)^n = 0$.

Q.24 If α, β are the roots of the equation $x^2 - 2x + 3 = 0$ obtain the equation whose roots are $\alpha^3 - 3\alpha^2 + 5\alpha - 2$, $\beta^3 - \beta^2 + \beta + 5$.

Q.25 If each pair of the following three equations $x^2 + p_1x + q_1 = 0$, $x^2 + p_2x + q_2 = 0$ & $x^2 + p_3x + q_3 = 0$ has exactly one root common , prove that; $(p_1 + p_2 + p_3)^2 = 4[p_1p_2 + p_2p_3 + p_3p_1 - q_1 - q_2 - q_3]$.

Q.26 Show that the function $z = 2x^2 + 2xy + y^2 - 2x + 2y + 2$ is not smaller than -3.

Q.27 Find all real numbers x such that, $\left(x - \dfrac{1}{x}\right)^{\frac{1}{2}} + \left(1 - \dfrac{1}{x}\right)^{\frac{1}{2}} = x$.

Q.28 Find the values of 'a' for which $-3 < [(x^2 + ax - 2)/(x^2 + x + 1)] < 2$ is valid for all real x.

Q.29 Find the minimum value of $\dfrac{\left(x + \dfrac{1}{x}\right)^6 - \left(x^6 + \dfrac{1}{x^6}\right) - 2}{\left(x + \dfrac{1}{x}\right)^3 + x^3 + \dfrac{1}{x^3}}$ for $x > 0$.

Q.30 Find the product of the real roots of the equation, $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$

EXERCISE–2

Q.1 Solve the following where $x \in R$.

(a) $(x-1)|x^2 - 4x + 3| + 2x^2 + 3x - 5 = 0$ (b) $3|x^2 - 4x + 2| = 5x - 4$

(c) $|x^3 + 1| + x^2 - x - 2 = 0$ (d) $2^{|x+2|} - |2^{x+1} - 1| = 2^{x+1} + 1$

(e) For $a \le 0$, determine all real roots of the equation $x^2 - 2a|x - a| - 3a^2 = 0$.

Q.2 Let a, b, c, d be distinct real numbers and a and b are the roots of quadratic equation $x^2 - 2cx - 5d = 0$. If c and d are the roots of the quadratic equation $x^2 - 2ax - 5b = 0$ then find the numerical value of $a + b + c + d$.

Q.3 Let $f(x) = ax^2 + bx + c = 0$ has an irrational root r. If $u = \frac{p}{q}$ be any rational number, where a, b, c, p and q are integer. Prove that $\frac{1}{q^2} \le |f(u)|$.

Q.4 Let a, b, c be real. If $ax^2 + bx + c = 0$ has two real roots α & β, where $\alpha < -1$ & $\beta > 1$ then show that $1 + c/a + |b/a| < 0$.

Q.5 If α, β are the roots of the equation, $x^2 - 2x - a^2 + 1 = 0$ and γ, δ are the roots of the equation, $x^2 - 2(a+1)x + a(a-1) = 0$ such that $\alpha, \beta \in (\gamma, \delta)$ then find the values of 'a'.

Q.6 Two roots of a biquadratic $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$ have their product equal to (-32). Find the value of k.

Q.7 If by eleminating x between the equation $x^2 + ax + b = 0$ & $xy + l(x+y) + m = 0$, a quadratic in y is formed whose roots are the same as those of the original quadratic in x. Then prove either $a = 2l$ & $b = m$ or $b + m = al$.

Q.8 If x be real, prove that $\frac{x^2 - 2x\cos\alpha + 1}{x^2 - 2x\cos\beta + 1}$ lies between $\frac{\sin^2\frac{\alpha}{2}}{\sin^2\frac{\beta}{2}}$ and $\frac{\cos^2\frac{\alpha}{2}}{\cos^2\frac{\beta}{2}}$.

Q.9 Solve the equations, $ax^2 + bxy + cy^2 = bx^2 + cxy + ay^2 = d$.

Q.10 Find the values of K so that the quadratic equation $x^2 + 2(K-1)x + K + 5 = 0$ has atleast one positive root.

Q.11 Find the values of 'b' for which the equation $2\log_{\frac{1}{25}}(bx + 28) = -\log_5(12 - 4x - x^2)$ has only one solution.

Q.12 Find all the values of the parameter 'a' for which both roots of the quadratic equation $x^2 - ax + 2 = 0$ belong to the interval $(0, 3)$.

Q.13 Find all the values of the parameters c for which the inequality has at least one solution.

$$1 + \log_2\left(2x^2 + 2x + \frac{7}{2}\right) \ge \log_2\left(cx^2 + c\right).$$

Q.14 Find the values of K for which the equation $x^4 + (1-2K)x^2 + K^2 - 1 = 0$;
(a) has no real solution (b) has one real solution

Q.15 Find all the values of the parameter 'a' for which the inequality
$a.9^x + 4(a-1)3^x + a - 1 > 0$ is satisfied for all real values of x.

Q.16 Find the complete set of real values of 'a' for which both roots of the quadratic equation
$(a^2 - 6a + 5)x^2 - \sqrt{a^2 + 2a}\,x + (6a - a^2 - 8) = 0$ lie on either side of the origin.

Q.17 If $g(x) = x^3 + px^2 + qx + r$ where p, q and r are integers. If g (0) and g (–1) are both odd, then prove that the equation $g(x) = 0$ cannot have three integral roots.

Q.18 Find all numbers p for each of which the least value of the quadratic trinomial $4x^2 - 4px + p^2 - 2p + 2$ on the interval $0 \le x \le 2$ is equal to 3.

Q.19 Let $P(x) = x^2 + bx + c$, where b and c are integer. If P (x) is a factor of both $x^4 + 6x^2 + 25$ and $3x^4 + 4x^2 + 28x + 5$, find the value of P(1).

Q.20 Let x be a positive real. Find the maximum possible value of the expression

$$y = \frac{x^2 + 2 - \sqrt{x^4 + 4}}{x}.$$

EXERCISE–3

Solve the inequality. Where ever base is not given take it as 10.

Q.1 $(\log_2 x)^4 - \left(\log_{\frac{1}{2}}\frac{x^5}{4}\right)^2 - 20\log_2 x + 148 < 0$.

Q.2 $x^{1/\log x} \cdot \log x < 1$

Q.3 $(\log 100x)^2 + (\log 10x)^2 + \log x \le 14$

Q.4 $\log_{1/2}(x+1) > \log_2(2-x)$.

Q.5 $\log_x 2 \cdot \log_{2x} 2 \cdot \log_2 4x > 1$.

Q.6 $\log_{1/5}(2x^2 + 5x + 1) < 0$.

Q.7 $\log_{1/2} x + \log_3 x > 1$.

Q.8 $\log_{x^2}(2+x) < 1$

Q.9 $\log_x \frac{4x+5}{6-5x} < -1$

Q.10 $(\log_{|x+6|} 2) \cdot \log_2(x^2 - x - 2) \ge 1$

Q.11 $\log_3 \frac{|x^2 - 4x| + 3}{x^2 + |x-5|} \ge 0$

Q.12 $\log_{[(x+6)/3]}[\log_2\{(x-1)/(2+x)\}] > 0$

Q.13 Find out the values of 'a' for which any solution of the inequality, $\frac{\log_3(x^2 - 3x + 7)}{\log_3(3x+2)} < 1$ is also a solution of the inequality, $x^2 + (5-2a)x \le 10a$.

Q.14 Solve the inequality $\log_{\log_2\left(\frac{x}{2}\right)}(x^2 - 10x + 22) > 0$.

Q.15 Find the set of values of 'y' for which the inequality, $2\log_{0.5} y^2 - 3 + 2x\log_{0.5} y^2 - x^2 > 0$ is valid for atleast one real value of 'x'.

EXERCISE–4

Q.1 Prove that the values of the function $\frac{\sin x \cos 3x}{\sin 3x \cos x}$ do not lie from $\frac{1}{3}$ & 3 for any real x.[JEE '97 , 5]

Q.2 The sum of all the real roots of the equation $|x-2|^2 + |x-2| - 2 = 0$ is _____. [JEE '97, 2]

Q.3 Let S be a square of unit area. Consider any quadrilateral which has one vertex on each side of S. If a, b, c & d denote the lengths of the sides of the quadrilateral, prove that: $2 \le a^2 + b^2 + c^2 + d^2 \le 4$.

Q.4 In a college of 300 students, every student reads 5 news papers & every news paper is read by 60 students. The number of news papers is:
(A) atleast 30 (B) atmost 20 (C) exactly 25 (D) none of the above

Q.5 If α, β are the roots of the equation $x^2 - bx + c = 0$, then find the equation whose roots are, $(\alpha^2 + \beta^2)$ $(\alpha^3 + \beta^3)$ & $\alpha^5\beta^3 + \alpha^3\beta^5 - 2\alpha^4\beta^4$.

Q.6(i) Let $\alpha + i\beta$; $\alpha, \beta \in R$, be a root of the equation $x^3 + qx + r = 0$; $q, r \in R$. Find a real cubic equation, independent of α & β, whose one root is 2α.

(ii) Find the values of α & β, $0 < \alpha, \beta < \pi/2$, satisfying the following equation,
$\cos\alpha \cos\beta \cos(\alpha + \beta) = -1/8$. [REE '99, 3 + 6]

Q.7(i) In a triangle PQR, $\angle R = \frac{\pi}{2}$. If $\tan\left(\frac{P}{2}\right)$ & $\tan\left(\frac{Q}{2}\right)$ are the roots of the equation
$ax^2 + bx + c = 0$ $(a \ne 0)$ then :
(A) $a + b = c$ (B) $b + c = a$ (C) $a + c = b$ (D) $b = c$

(ii) If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real & less than 3 then
(A) $a < 2$ (B) $2 \le a \le 3$ (C) $3 < a \le 4$ (D) $a > 4$ [JEE '99, 2 + 2]

Q.8 If α, β are the roots of the equation, $(x - a)(x - b) + c = 0$, find the roots of the equation, $(x-\alpha)(x-\beta) = c$. [REE 2000 (Mains), 3]

Q.9(a) For the equation, $3x^2 + px + 3 = 0$, $p > 0$ if one of the roots is square of the other, then p is equal to:
(A) 1/3 (B) 1 (C) 3 (D) 2/3

(b) If α & β $(\alpha < \beta)$, are the roots of the equation, $x^2 + bx + c = 0$, where $c < 0 < b$, then
(A) $0 < \alpha < \beta$ (B) $\alpha < 0 < \beta < |\alpha|$
(C) $\alpha < \beta < 0$ (D) $\alpha < 0 < |\alpha| < \beta$

(c) If $b > a$, then the equation, $(x - a)(x - b) - 1 = 0$, has :
(A) both roots in [a, b] (B) both roots in $(-\infty, a)$
(C) both roots in $[b, \infty)$ (D) one root in $(-\infty, a)$ & other in $(b, +\infty)$
[JEE 2000 Screening, 1 + 1 + 1 out of 35]

(d) If α, β are the roots of $ax^2 + bx + c = 0$, $(a \ne 0)$ and $\alpha + \delta$, $\beta + \delta$, are the roots of, $Ax^2 + Bx + C = 0$, $(A \ne 0)$ for some constant δ, then prove that,
$$\frac{b^2 - 4ac}{a^2} = \frac{B^2 - 4AC}{A^2}.$$ [JEE 2000, Mains, 4 out of 100]

Q.10 The number of integer values of m, for which the x co-ordinate of the point of intersection of the lines $3x + 4y = 9$ and $y = mx + 1$ is also an integer, is [JEE 2001, Screening, 1 out of 35]
(A) 2 (B) 0 (C) 4 (D) 1

Q.11 Let a, b, c be real numbers with $a \ne 0$ and let α, β be the roots of the equation $ax^2 + bx + c = 0$. Express the roots of $a^3x^2 + abcx + c^3 = 0$ in terms of α, β.
[JEE 2001, Mains, 5 out of 100]

Q.12 The set of all real numbers x for which $x^2 - |x + 2| + x > 0$, is
(A) $(-\infty, -2) \cup (2, \infty)$ (B) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
(C) $(-\infty, -1) \cup (1, \infty)$ (D) $(\sqrt{2}, \infty)$ [JEE 2002 (screening), 3]

Q.13 If $x^2 + (a - b)x + (1 - a - b) = 0$ where $a, b \in R$ then find the values of 'a' for which equation has unequal real roots for all values of 'b'. [JEE 2003, Mains-4 out of 60]
[Based on M. R. test]

Q.14(a) If one root of the equation $x^2 + px + q = 0$ is the square of the other, then
(A) $p^3 + q^2 - q(3p + 1) = 0$ (B) $p^3 + q^2 + q(1 + 3p) = 0$
(C) $p^3 + q^2 + q(3p - 1) = 0$ (D) $p^3 + q^2 + q(1 - 3p) = 0$

(b) If $x^2 + 2ax + 10 - 3a > 0$ for all $x \in R$, then
(A) $-5 < a < 2$ (B) $a < -5$ (C) $a > 5$ (D) $2 < a < 5$
[JEE 2004 (Screening)]

Q.15 Find the range of values of *t* for which $2\sin t = \frac{1 - 2x + 5x^2}{3x^2 - 2x - 1}$, $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
[JEE 2005(Mains), 2]

Q.16(a) Let a, b, c be the sides of a triangle. No two of them are equal and $\lambda \in R$. If the roots of the equation $x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are real, then
(A) $\lambda < \frac{4}{3}$ (B) $\lambda > \frac{5}{3}$ (C) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$ (D) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$
[JEE 2006, 3]

(b) If roots of the equation $x^2 - 10cx - 11d = 0$ are a, b and those of $x^2 - 10ax - 11b = 0$ are c, d, then find the value of $a + b + c + d$. (a, b, c and d are distinct numbers) [JEE 2006, 6]

EXERCISE–5

Part : (A) Only one correct option

1. The roots of the quadratic equation $(a + b - 2c)\,x^2 - (2a - b - c)\,x + (a - 2b + c) = 0$ are

(A) $a + b + c$ and $a - b + c$ (B) $\frac{1}{2}$ and $a - 2b + c$

(C) $a - 2b + c$ and $\frac{1}{a+b-c}$ (D) none of these

2. The roots of the equation $2^{x+2}.\ 3^{\frac{3x}{x-1}} = 9$ are given by

(A) $1 - \log_2 3,\ 2$ (B) $\log_2 (2/3),\ 1$ (C) $-2,\ 2$ (D) $-2,\ 1 - \frac{\log 3}{\log 2}$

3. Two real numbers α & β are such that $\alpha + \beta = 3$ & $|\alpha - \beta| = 4$, then α & β are the roots of the quadratic equation:

(A) $4x^2 - 12x - 7 = 0$ (B) $4x^2 - 12x + 7 = 0$ (C) $4x^2 - 12x + 25 = 0$ (D) none of these

4. Let a, b and c be real numbers such that $4a + 2b + c = 0$ and $ab > 0$. Then the equation $ax^2 + bx + c = 0$ has

(A) real roots (B) imaginary roots (C) exactly one root (D) none of these

5. If $e^{\cos x} - e^{-\cos x} = 4$, then the value of cos x is

(A) $\log\left(2+\sqrt{5}\right)$ (B) $-\log\left(2+\sqrt{5}\right)$ (C) $\log\left(-2+\sqrt{5}\right)$ (D) none of these

6. The number of the integer solutions of $x^2 + 9 < (x + 3)^2 < 8x + 25$ is :

(A) 1 (B) 2 (C) 3 (D) none

7. If $(x + 1)^2$ is greater than $5x - 1$ & less than $7x - 3$ then the integral value of x is equal to

(A) 1 (B) 2 (C) 3 (D) 4

8. The set of real 'x' satisfying, $\left|\,|x-1|-1\,\right| \le 1$ is:

(A) [0, 2] (B) [– 1, 3] (C) [– 1, 1] (D) [1, 3]

9. Let $f(x) = x^2 + 4x + 1$. Then

(A) $f(x) > 0$ for all x (B) $f(x) > 1$ when $x \ge 0$ (C) $f(x) \ge 1$ when $x \le -4$ (D) $f(x) = f(-x)$ for all x

10. If x is real and $k = \frac{x^2 - x + 1}{x^2 + x + 1}$ then:

(A) $\frac{1}{3} \le k \le 3$ (B) $k \ge 5$ (C) $k \le 0$ (D) none

11. If x is real, then $\frac{x^2 - x + c}{x^2 + x + 2c}$ can take all real values if :

(A) $c \in [0, 6]$ (B) $c \in [-6, 0]$ (C) $c \in (-\infty, -6) \cup (0, \infty)$ (D) $c \in (-6, 0)$

12. The solution set of the inequality $\frac{x^4 - 3x^3 + 2x^2}{x^2 - x - 30} \ge 0$ is:

(A) $(-\infty, -5) \cup (1, 2) \cup (6, \infty) \cup \{0\}$ (B) $(-\infty, -5) \cup [1, 2] \cup (6, \infty) \cup \{0\}$

(C) $(-\infty, -5] \cup [1, 2] \cup [6, \infty) \cup \{0\}$ (D) none of these

13. If $x - y$ and $y - 2x$ are two factors of the expression $x^3 - 3x^2y + \lambda xy^2 + \mu y^3$, then

(A) $\lambda = 11,\ \mu = -3$ (B) $\lambda = 3,\ \mu = -11$ (C) $\lambda = \frac{11}{4},\ \mu = -\frac{3}{4}$ (D) none of these

14. If α, β are the roots of the equation, $x^2 - 2mx + m^2 - 1 = 0$ then the range of values of m for which $\alpha, \beta \in (-2, 4)$ is:

(A) $(-1, 3)$ (B) $(1, 3)$ (C) $(\infty, -1) \cup ((3, \infty)$ (D) none

15. If the inequality $(m - 2)x^2 + 8x + m + 4 > 0$ is satisfied for all $x \in R$ then the least integral m is:

(A) 4 (B) 5 (C) 6 (D) none

16. For all $x \in R$, if $mx^2 - 9mx + 5m + 1 > 0$, then m lies in the interval

(A) $-(4/61, 0)$ (B) $[0, 4/61)$ (C) $(4/61, 61/4)$ (D) $(-61/4, 0]$

17. Let $a > 0$, $b > 0$ & $c > 0$. Then both the roots of the equation $ax^2 + bx + c = 0$

(A) are real & negative (B) have negative real parts (C) are rational numbers (D) none

18. The value of 'a' for which the sum of the squares of the roots of the equation, $x^2 - (a - 2)x - a - 1 = 0$ assume the

least value is:
(A) 0 (B) 1 (C) 2 (D) 3

19. Consider $y = \frac{2x}{1+x^2}$, then the range of expression, $y^2 + y - 2$ is:

(A) $[-1, 1]$ (B) $[0, 1]$ (C) $[-9/4, 0]$ (D) $[-9/4, 1]$

20. If both roots of the quadratic equation $x^2 + x + p = 0$ exceed p where $p \in R$ then p must lie in the interval:
(A) $(-\infty, 1)$ (B) $(-\infty, -2)$ (C) $(-\infty, -2) \cup (0, 1/4)$ (D) $(-2, 1)$

21. If a, b, p, q are non–zero real numbers, the two equations, $2a^2x^2 - 2abx + b^2 = 0$ and $p^2x^2 + 2pqx + q^2 = 0$ have:
(A) no common root (B) one common root if $2a^2 + b^2 = p^2 + q^2$
(C) two common roots if $3pq = 2ab$ (D) two common roots if $3qb = 2ap$

22. If α, β & γ are the roots of the equation, $x^3 - x - 1 = 0$ then, $\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma}$ has the value equal to:

(A) zero (B) -1 (C) -7 (D) 1

23. The equations $x^3 + 5x^2 + px + q = 0$ and $x^3 + 7x^2 + px + r = 0$ have two roots in common. If the third root of each equation is represented by x_1 and x_2 respectively, then the ordered pair (x_1, x_2) is:
(A) $(-5, -7)$ (B) $(1, -1)$ (C) $(-1, 1)$ (D) $(5, 7)$

24. If α, β are roots of the equation $ax^2 + bx + c = 0$ then the equation whose roots are $2\alpha + 3\beta$ and $3\alpha + 2\beta$ is
(A) $abx^2 - (a + b)cx + (a + b)^2 = 0$ (B) $acx^2 - (a + c)bx + (a + c)^2 = 0$
(C) $acx^2 + (a + c)bx - (a + c)^2 = 0$ (D) none of these

25. If coefficients of the equation $ax^2 + bx + c = 0$, $a \neq 0$ are real and roots of the equation are non-real complex and $a + c < b$, then
(A) $4a + c > 2b$ (B) $4a + c < 2b$ (C) $4a + c = 2b$ (D) none of these

26. The set of possible values of λ for which $x^2 - (\lambda^2 - 5\lambda + 5)x + (2\lambda^2 - 3\lambda - 4) = 0$ has roots, whose sum and product are both less than 1, is

(A) $\left(-1, \frac{5}{2}\right)$ (B) $(1, 4)$ (C) $\left[1, \frac{5}{2}\right]$ (D) $\left(1, \frac{5}{2}\right)$

27. Let conditions C_1 and C_2 be defined as follows : $C_1 : b^2 - 4ac \geq 0$, C_2 : a, –b, c are of same sign. The roots of $ax^2 + bx + c = 0$ are real and positive, if
(A) both C_1 and C_1 are satisfied (B) only C_2 is satisfied
(C) only C_1 is satisfied (D) none of these

Part : (B) May have more than one options correct

28. If a, b are non-zero real numbers, and α, β the roots of $x^2 + ax + b = 0$, then

(A) α^2, β^2 are the roots of $x^2 - (2b - a^2)x + a^2 = 0$ (B) $\frac{1}{\alpha}, \frac{1}{\beta}$ are the roots of $bx^2 + ax + 1 = 0$

(C) $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of $bx^2 + (2b - a^2)x + b = 0$ (D) $-\alpha, -\beta$ are the roots of $x^2 + ax - b = 0$

29. $x^2 + x + 1$ is a factor of $ax^3 + bx^2 + cx + d = 0$, then the real root of above equation is
(a, b, c, d $\in$ R)
(A) $-d/a$ (B) d/a (C) $(b - a)/a$ (D) $(a - b)/a$

30. If $(x^2 + x + 1) + (x^2 + 2x + 3) + (x^2 + 3x + 5) + \ldots\ldots + (x^2 + 20x + 39) = 4500$, then x is equal to:
(A) 10 (B) -10 (C) 20.5 (D) -20.5

31. $\cos\alpha$ is a root of the equation $25x^2 + 5x - 12 = 0$, $-1 < x < 0$, then the value of $\sin 2\alpha$ is:
(A) 24/25 (B) $-12/25$ (C) $-24/25$ (D) 20/25

32. If the quadratic equations, $x^2 + abx + c = 0$ and $x^2 + acx + b = 0$ have a common root then the equation containing their other roots is/are:
(A) $x^2 + a(b + c)x - a^2bc = 0$ (B) $x^2 - a(b + c)x + a^2bc = 0$
(C) $a(b + c)x^2 - (b + c)x + abc = 0$ (D) $a(b + c)x^2 + (b + c)x - abc = 0$

EXERCISE–6

1. Solve the equation, $x(x + 1)(x + 2)(x + 3) = 120$.

2. Solve the following where $x \in R$.
(a) $(x - 1)|x^2 - 4x + 3| + 2x^2 + 3x - 5 = 0$ (b) $(x + 3) \cdot |x + 2| + |2x + 3| + 1 = 0$
(c) $|(x + 3)| \cdot (x + 1) + |2x + 5| = 0$ (d) $2^{|x+2|} - |2^{x+1} - 1| = 2^{x+1} + 1$

3. If ' x ' is real, show that, $\frac{(x - 1)(x + 1)(x + 4)(x + 6) + 25}{7x^2 + 8x + 4} \geq 0$.

4. Find the value of x which satisfy inequality $\frac{x-2}{x+2} > \frac{2x-3}{4x-1}$.

5. Find the range of the expression $f(x) = \sin^2 x - \sin x + 1 \ \forall \ x \in R$.

6. Find the range of the quadratic expression $f(x) = x^2 - 2x + 3 \ \forall \ x \in [0, 2]$.

7. Prove that the function $y = (x^2 + x + 1)/(x^2 + 1)$ cannot have values greater than 3/2 and values smaller than 1/2 for $\forall \ x \in R$.

8. If x be real, show that $\frac{x^2-2x+9}{x^2+2x+9}$ lies in $\left[\frac{1}{2}, 2\right]$.

9. For what values of k the expression $3x^2 + 2xy + y^2 + 4x + y + k$ can be resolved into two linear factors.

10. Show that one of the roots of the equation, $a x^2 + b x + c = 0$ may be reciprocal of one of the roots of $a_1 x^2 + b_1 x + c_1 = 0$ if $(a a_1 - c c_1)^2 = (b c_1 - a b_1)(b_1 c - a_1 b)$.

11. Let $\alpha + i\beta$; $\alpha, \beta \in R$, be a root of the equation $x^3 + qx + r = 0$; $q, r \in R$. Find a real cubic equation, independent of α and β, whose one root is 2α.

12. If a, b are the roots of $x^2 + px + 1 = 0$ and c, d are the roots of $x^2 + qx + 1 = 0$. Show that $q^2 - p^2 = (a - c)(b - c)(a + d)(b + d)$.

13. If α, β are the roots of the equation $x^2 - px + q = 0$, then find the quadratic equation the roots of which are $(\alpha^2 - \beta^2)$ $(\alpha^3 - \beta^3)$ & $\alpha^3 \beta^2 + \alpha^2 \beta^3$.

14. If 'x' is real, find values of 'k' for which, $\left|\frac{x^2+kx+1}{x^2+x+1}\right| < 2$ is valid.

15. Solve the inequality, $\frac{1}{x-1} - \frac{4}{x-2} + \frac{4}{x-3} - \frac{1}{x-4} < \frac{1}{30}$.

16. The equations $x^2 - ax + b = 0$ & $x^3 - px^2 + qx = 0$, where $b \neq 0$, $q \neq 0$ have one common root & the second equation has two equal roots. Prove that $2(q + b) = ap$.

17. Find the real values of 'm' for which the equation, $\left(\frac{x}{1+x^2}\right)^2 - (m-3)\left(\frac{x}{1+x^2}\right) + m = 0$ has atleast one real root ?

18. Let a and b be two roots of the equation $x^3 + px^2 + qx + r = 0$ satisfying the relation $ab + 1 = 0$. Prove that $r^2 + pr + q + 1 = 0$.

ANSWER KEY

EXERCISE–1

Q.2 $2x^2 + 2x\cos(A - B) - 2$ **Q.3** 254 **Q.7** $a \in \left(-\infty, -\frac{1}{2}\right)$

Q.8 $x^2 - 4x + 1 = 0$; $\alpha = \tan\left(\frac{\pi}{12}\right)$; $\beta = \tan\left(\frac{5\pi}{12}\right)$ **Q.9** 1 **Q.10** minimum value 3 when $x = 1$ and $p = 0$

Q.12 **(a)** (ii) and (iv) ; **(b)** $x^2 - p(p^4 - 5p^2q + 5q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$

Q.18 $\left(-\infty, -\frac{1}{4}\right) \cup \{2\} \cup (5, 6]$ **Q.24** $x^2 - 3x + 2 = 0$ **Q.27** $x = \frac{\sqrt{5}+1}{2}$ **Q.28** $-2 < a < 1$

Q.29 $y_{min} = 6$ **Q.30** 20

EXERCISE–2

Q.1 **(a)** $x = 1$; **(b)** $x = 2$ or 5; **(c)** $x = -1$ or 1; **(d)** $x \geq -1$ or $x = -3$; **(e)** $x = (1 - \sqrt{2})a$ or $(\sqrt{6} - 1)a$

Q.2 30 **Q.5** $a \in \left(-\frac{1}{4}, 1\right)$ **Q.6** k = 86

Q.9 $x^2 = y^2 = d/(a+b+c)$; $x/(c-a) = y/(a-b) = K$ where $K^2 a(a^2 + b^2 + c^2 - ab - bc - ca) = d$

Q 10. $K \leq -1$ **Q 11.** $(-\infty, -14) \cup \{4\} \cup \left[\frac{14}{3}, \infty\right)$ **Q 12.** $2\sqrt{2} \leq a < \frac{11}{3}$

Q.13 $(0, 8]$ **Q 14. (a)** $K < -1$ or $K > 5/4$ **(b)** $K = -1$ **Q 15.** $[1, \infty)$

Q 16. $(-\infty, -2] \cup [0, 1) \cup (2, 4) \cup (5, \infty)$ **Q18.** $a = 1 - \sqrt{2}$ or $5 + \sqrt{10}$

Q.19 $P(1)=4$ **Q 20.** $2\left(\sqrt{2}-1\right)$ where $x=\sqrt{2}$

EXERCISE–3

Q 1. $x\in\left(\frac{1}{16},\frac{1}{8}\right)\cup(8,16)$ **Q 2.** $(0,1)\cup(1,10^{1/10})$ **Q 3.** $\frac{1}{\sqrt{10}^{9}}\le x\le 10$

Q 4. $-1<x<\frac{1-\sqrt{5}}{2}$ or $\frac{1+\sqrt{5}}{2}<x<2$ **Q 5.** $2^{-\sqrt{2}}<x<2^{-1}$; $1<x<2^{\sqrt{2}}$ **Q 6.** $(-\infty,-2.5)\cup(0,\infty)$

Q 7. $0<x<3^{1/1-\log 3}$ (where base of log is 2) **Q 8.** $-2<x<-1, -1<x<0, 0<x<1, x>2$ **Q 9.** $\frac{1}{2}<x<1$

Q 10. $x<-7$, $-5<x\le -2$, $x\ge 4$ **Q 11.** $x\le -\frac{2}{3}$; $\frac{1}{2}\le x\le 2$ **Q12.** $(-6,-5)\cup(-3,-2)$ **Q 13.** $a\ge\frac{5}{2}$

Q 14. $x\in\left(3,5-\sqrt{3}\right)\cup(7,\infty)$ **Q 15.** $\left(-\infty,-2\sqrt{2}\right)\cup\left(-\frac{1}{\sqrt{2}},0\right)\cup\left(0,\frac{1}{\sqrt{2}}\right)\cup\left(2\sqrt{2},\infty\right)$

EXERCISE–4

Q.2 4 **Q.4** C

Q.5 $x^2-(x_1+x_2)x+x_1x_2=0$ where $x_1=(b^2-2c)(b^3-3cb)$; $x_2=c^3(b^2-4c)$

Q.6 (i) $x^3+qx-r=0$, **(ii)** $\alpha=\beta=\pi/3$, **Q.7 (i)** A, **(ii)** A, **Q.8** (a, b) **Q.9 (a)** C, **(b)** B, **(c)** D

Q.10 A **Q.11** $\gamma=\alpha^2\beta$ and $\delta=\alpha\beta^2$ or $\gamma=\alpha\beta^2$ and $\delta=\alpha^2\beta$ **Q.12** B **Q.13** $a>1$

Q.14 (a) D ; (b) A **Q.15** $\left[-\frac{\pi}{2},-\frac{\pi}{10}\right]\cup\left[\frac{3\pi}{10},\frac{\pi}{2}\right]$ **Q.16** **(a)** A, **(b)** 1210

EXERCISE–5

1.	D	**2.**	D	**3.**	A	**4.**	A	**5.**	D	**6.**	D	**7.**	C	**8.**	B
9.	C	**10.**	A	**11.**	B	**12.**	B	**13.**	C	**14.**	A	**15.**	B	**16.**	B
17.	B	**18.**	B	**19.**	C	**20.**	B	**21.**	A	**22.**	C	**23.**	A	**24.**	D
25.	B	**26.**	D	**27.**	A	**28.**	BC	**29.**	AD	**30.**	AD	**31.**	AC	**32.**	BD

EXERCISE–6

1. $\{2,-5\}$ **2.** (a) $x=1$ (b) $x=(-7-\sqrt{17})/2$

(c) $x=-2,-4,-(1+\sqrt{3})$ (d) $x\ge -1$, $x=-3$

4. $x\in(-\infty,-2)\cup\left(\frac{1}{4},1\right)\cup(4,\infty)$ **5.** $\left[\frac{3}{4},3\right]$ **6.** [2, 3] **9.** $k=\frac{11}{8}$ **11.** $x^3+qx-r=0$

13. $x^2-p(p^4-5p^2q+5q^2)x+p^2q^2(p^2-4q)(p^2-q)=0$ **14.** $k\in(0,4)$

15. $(-\infty,-2)\cup(-1,1)\cup(2,3)\cup(4,6)\cup(7,\infty)$ **17.** $\left[\frac{-7}{2},\frac{5}{6}\right]$

QUADRATIC EQUATIONS

Some questions (Assertion–Reason type) are given below. Each question contains **Statement – 1 (Assertion)** and **Statement – 2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. So select the correct choice :

Choices are :

(A) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is** a correct explanation for **Statement – 1**.

(B) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is NOT** a correct explanation for **Statement – 1**.

(C) **Statement – 1** is True, **Statement – 2** is False.

(D) **Statement – 1** is False, **Statement – 2** is True.

1. **Statement-1:** If $x \in R$, $2x^2 + 3x + 5$ is positive.
Statement-2: If $\Delta < 0$, $ax^2 + bx + c$, 'a' have same sign $\forall x \in R$.

2. **Statement-1:** If $1+\sqrt{2}$ is a root of $x^2 - 2x - 1 = 0$, then $1-\sqrt{2}$ will be the other root.
Statement-2: Irrational roots of a quadratic equation with rational coefficients always occur in conjugate pair.

3. **Statement-1:** The roots of the equation $2x^2 + 3i\,x + 2 = 0$ are always conjugate pair.
Statement-2: Imaginary roots of a quadratic equation with real coefficients always occur in conjugate pair.

4. Consider the equation $(a^2 - 3a + 2)\,x^2 + (a^2 - 5a + 6)x + a^2 - 1 = 0$
Statement – 1: If a = 1, then above equation is true for all real x.
Statement – 2: If a = 1, then above equation will have two real and distinct roots.

5. Consider the equation $(a + 2)x^2 + (a - 3)\,x = 2a - 1$
Statement–1 : Roots of above equation are rational if 'a' is rational and not equal to –2.
Statement–2 : Roots of above equation are rational for all rational values of 'a'.

6. Let $f(x) = x^2 = -x^2 + (a + 1)\,x + 5$
Statement–1 : f(x) is positive for same $\alpha < x < \beta$ and for all $a \in R$
Statement–2 : f(x) is always positive for all $x \in R$ and for same real 'a'.

7. Consider $f(x) = (x^2 + x + 1)\,a^2 - (x^2 + 2)\,a$
$-3\,(2x^2 + 3x + 1) = 0$
Statement–1 : Number of values of 'a' for which f(x) = 0 will be an identity in x is 1.
Statement–2 : a = 3 the only value for which f(x) = 0 will represent an identity.

8. Let a, b, c be real such that $ax^2 + bx + c = 0$ and $x^2 + x + 1 = 0$ have a common root
Statement–1 : a = b = c
Statement–2 : Two quadratic equations with real coefficients can not have only one imaginary root common.

9. **Statement–1** : The number of values of a for which $(a^2 - 3a + 2)\,x^2 + (a^2 - 5a + b)\,x + a^2 - 4 = 0$ is an identity in x is 1.
Statement–2 : If $ax^2 + bx + c = 0$ is an identity in x then a = b = c = 0.

10. Let $a \in (-\infty, 0)$.
Statement–1 : $ax^2 - x + 4 < 0$ for all $x \in R$
Statement–2 : If roots of $ax^2 + bx + c = 0$, $b, c \in R$ are imaginary then signs of $ax^2 + bx + c$ and a are same for all $x \in R$.

11. Let $a, b, c \in R$, $a \neq 0$.
Statement–1 : Difference of the roots of the equation $ax^2 + bx + c = 0$
= Difference of the roots of the equation $-ax^2 + bx - c = 0$
Statement–2 : The two quadratic equations over reals have the same difference of roots if product of the coefficient of the two equations are the same.

12. **Statement–1** : If the roots of $x^5 - 40x^4 + Px^3 + Qx^2 + Rx + S = 0$ are in G.P. and sum of their reciprocal is 10, then $|S| = 32$.
Statement–2 : $x_1 . x_2 . x_3 . x_4 . x_5 = S$, where x_1, x_2, x_3, x_4, x_5 are the roots of given equation.

13. **Statement–1** : If $0 < \alpha < \frac{\pi}{4}$, then the equation $(x - \sin\alpha)\,(x - \cos\alpha) - 2 = 0$ has both roots in $(\sin\alpha, \cos\alpha)$

Statement–2 : If f(a) and f(b) possess opposite signs then there exist at least one solution of the equation f(x) = 0 in open interval (a, b).

14. **Statement–1 :** If $a \geq 1/2$ then $\alpha < 1 < p$ where α , β are roots of equation $-x^2 + ax + a = 0$
Statement–2 : Roots of quadratic equation are rational if discriminant is perfect square.

15. **Statement-1 :** The number of real roots of $|x|^2 + |x| + 2 = 0$ is zero. **Statement-2 :** $\forall x \in R$, $|x| \geq 0$.

16. **Statement-1:** If all real values of x obtained from the equation $4^x - (a - 3)\,2^x + (a - 4) = 0$ are non-positive, then $a \in (4, 5]$
Statement-2: If $ax^2 + bx + c$ is non-positive for all real values of x, then $b^2 - 4ac$ must be –ve or zero and 'a' must be –ve.

17. **Statement-1:** If a , b , c , $d \in R$ such that $a < b < c < d$, then the equation $(x - a)(x - c) + 2(x - b)(x - d) = 0$ are real and distinct.
Statement-2: If f(x) = 0 is a polynomial equation and a, b are two real numbers such that f(a) f(b) < 0 has at least one real root.

18. **Statement-1:** $f(x) = \dfrac{x^2 + x + 1}{x^2 + 2x + 5} > 0 \ \forall x \in R$
Statement-2: $ax^2 + bx + c > 0 \ \forall x \in R$ if $a > 0$ and $b^2 - 4ac < 0$.

19. **Statement-1:** If $a + b + c = 0$ then $ax^2 + bx + c = 0$ must have '1' as a root of the equation
Statement-2: If $a + b + c = 0$ then $ax^2 + bx + c = 0$ has roots of opposite sign.

20. **Statement-1:** $ax^2 + bx + c = 0$ is a quadratic equation with real coefficients, if $2 + \sqrt{3}$ is one root then other root can be any other real number.

Statement-2: If $P + \sqrt{q}$ is a real root of a quadratic equation, then $P - \sqrt{q}$ is other root only when the coefficients of equation are rational

21. **Statement-1:** If $px^2 + qx + r = 0$ is a quadratic equation (p, q, $r \in R$) such that its roots are α, β & $p + q + r < 0$, $p - q + r < 0$ & $r > 0$, then $3[\alpha] + 3[\beta] = -3$, where $[\cdot]$ denotes G.I.F.
Statement-2: If for any two real numbers a & b, function f(x) is such that $f(a).f(b) < 0 \Rightarrow$ f(x) has at least one real root lying between (a, b)

22. **Statement-1:** If $x = 2 + \sqrt{3}$ is a root of a quadratic equation then another root of this equation must be $x = 2 + \sqrt{3}$
Statement-2: If $ax^2 + bx + c = 0$, a, b, $c \in Q$, having irrational roots then they are in conjugate pairs.

23. **Statement-1:** If roots of the quadratic equation $ax^2 + bx + c = 0$ are distinct natural number then both roots of the equation $cx^2 + bx + a = 0$ cannot be natural numbers.

Statement-2: If α, β be the roots of $ax^2 + bx + c = 0$ then $\dfrac{1}{\alpha}, \dfrac{1}{\beta}$ are the roots of $cx^2 + bx + a = 0$.

24. **Statement-1:** The $(x - p)(x - r) + \lambda (x - q)(x - s) = 0$ where $p < q < r < s$ has non real roots if $\lambda > 0$.
Statement-2: The equation (p, q, $r \in R$) $\beta x^2 + qx + r = 0$ has non-real roots if $q^2 - 4pr < 0$.

25. **Statement-1:** One is always one root of the equation $(l - m)x^2 + (m - n)x + (n - l) = 0$, where l, m, $n \in R$.
Statement-2: If $a + b + c = 0$ in the equation $ax^2 + bx + c = 0$, then 1 is the one root.

26. **Statement-1:** If $(a^2 - 4)x^2 + (a^2 - 3a + 2)x + (a^2 - 7a + 0) = 0$ is an identity, then the value of a is 2.
Statement-2: If a = b = 0 then $ax^2 + bx + c = 0$ is an identity.

27. **Statement-1:** $x^2 + 2x + 3 > 0 \ \forall \ x \in R$
Statement-2: $ax^2 + bx + c > 0 \ \forall \ x \in R$ if $b^2 - 4ac < 0$ and $a > 0$.

28. **Statement-1:** Maximum value of $\dfrac{1}{2^{x^2 - x + 1}}$ is $\dfrac{1}{2^{3/4}}$

Statement-2: Minimum value of $ax^2 + bx + c$ $(a > 0)$ occurs at $x = -\dfrac{b}{2a}$.

29. **Statement-1:** If quadratic equation $ax^2 + bx - 2 = 0$ have non-real roots then $a < 0$
Statement-2: For the quadratic expression $f(x) = ax^2 + bx + c$ if $b^2 - 4ac < 0$ then f(x) = 0 have non real roots.

30. **Statement-1:** Roots of equation $x^5 - 40x^4 + Px^3 + Qx^2 + Rx + S = 0$ are in G.P. and sum of their reciprocal is equal to 10 then $|s| = 32$.
Statement-2: If x_1, x_2, x, x_4 are roots of equation

$ax^4 + bx^3 + cx^2 + dx + e = 0$ $(a \neq 0)$

$x_1 + x_2 + x_3 + x_4 = -b/a$

$$\sum x_1 x_2 = \frac{c}{a}$$

$$\sum x_1 x_2 x_3 = -\frac{d}{a} \qquad x_1 x_2 x_3 x_4 = \frac{e}{a}$$

31. **Statement-1:** The real values of a form which the quadratic equation $2x^2 - (a^3 + 8a - 1) + a^2 - 4a = 0$. Possesses roots of opposite signs are given by $0 < a < 4$.
Statement-2: Disc ≥ 0 and product of root is < 2

ANSWER KEY

1. A	**2. A**	**3. D**	**4. C**	**5. C**	**6. C**	**7. D**	**8. A**	**9. A**	**10. D**	**11. C**	**12. C**
13. D	**14. B**	**15. A**	**16. B**	**17. A**	**18. A**	**19. C**	**20. A**	**21. A**	**22. A**	**23. A**	**24. D**
25. A	**26. C**	**27. A**	**28. A**	**29. A**	**30. A**	**31. A**					

Solution

5. Obviously x = 1 is one of the root

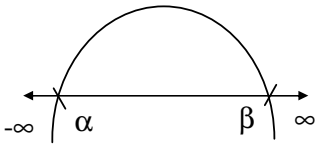
$\therefore$ Other root = $-\ -\dfrac{2a-1}{a+2}$ = rational for all rational $a \neq -2$.

(C) is correct option.

6. Here f(x) is a downward parabola

$D = (a + 1)^2 + 20 > 0$

From the graph clearly st (1) is true but st (2) is false



7. $f(x) = 0$ represents an identity if $a^2 - a - 6 = 0 \Rightarrow a = 3, -2$

$a^2 - a - 6 = 0 \Rightarrow a = 3, -2$

$a^2 - a = 0 \Rightarrow a = 3, -3$

$a^2 - 2a - 3 = 0 \Rightarrow a = 3, -1 \Rightarrow a = 3$ is the only values. Ans.: D

8. (A)

$x^2 + x + 1 = 0$

$D = -3 < 0$ $\therefore x^2 + x + 1 = 0$ and $ax^2 + bx + c = 0$ have both the roots common

$\Rightarrow a = b = c$.

9. (A)

$(a^2 - 3a + 2)x^2 + (a^2 - 5a + 6)x + a^2 - 4 = 0$

Clearly only for a = 2, it is an identify.

10. Statement – II is true as if $ax^2 + bx + c = 0$ has imaginary roots, then for no real x, $ax^2 + bx + c$ is zero, meaning thereby $ax^2 + bx + c$ is always of one sign. Further $\lim\limits_{x\to\infty}\left(ax^2+bx+c\right)$ = signum (a). ∞

statement – I is false, because roots of $ax^2 - x + 4 = 0$ are real for any $a \in (-\infty, 0)$ and hence $ax^2 - x + 4$ takes zero, positive and negative values.

Hence (d) is the correct answer.

11. Statement–I is true, as Difference of the roots of a quadratic equation is always $\sqrt{D}$, D being the discriminant of the quadratic equation and the two given equations have the same discriminant.

Statement – II is false as if two quadratic equations over reals have the same product of the coefficients, their discriminents need not be same.

Hence (c) is the correct answer.

12. Roots of the equation $x^5 - 40x^4 + px^3 + qx^2 + rx + s = 0$ are in G.P., let roots be a, ar, ar^2, ar^3, ar^4

$\therefore a + ar + ar^2 + ar^3 + ar^4 = 40$ . . . (i)

and $\dfrac{1}{a}+\dfrac{1}{ar}+\dfrac{1}{ar^2}+\dfrac{1}{ar^3}+\dfrac{1}{ar^4}=10$ . . . (ii)

from (i) and (ii); $ar^2 = \pm 2$ . . . (iii)

Now, - S = product of roots = $a^5r^{10} = (ar^2)^5 = \pm 32$.

$\therefore |s| = 32$. $\therefore$ Hence (c) is the correct answer.

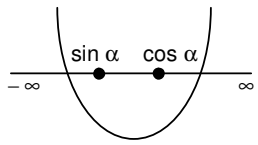
13. Let, $f(x) = (x - \sin\alpha)(x - \cos\alpha) - 2$

then, $f(\sin\alpha) = -2 < 0$; $f(\cos\alpha) = -2 < 0$

Also as $0 < \alpha < \dfrac{\pi}{4}$; $\therefore \sin\alpha < \cos\alpha$

There-fore equation $f(x) = 0$ has one root in $(-\infty, \sin\alpha)$ and other in $(\cos\alpha, \infty)$

Hence (c) is the correct answer.



Hence (d) is the correct answer.

14. (B) $x^2 - ax - a = 0$ $g(1) < 0 \Rightarrow a > 1/2$

15. equation can be written as $(2^x)^2 - (a-4)\,2^x - (a-4) = 0$
$\Rightarrow 2^x = 1$ & $2^x = a - 4$
Since $x \le 0$ and $2^x = a - 4$ [$\because$ x is non positive] $\therefore$
$0 < a - 4 \le 1 \Rightarrow 4 < a \le 5$
i.e., $a \in (4, 5]$
Hence ans. (B).

16. (A) Let $f(x) = (x-a)(x-c) + 2(x-b)(x-d)$
Then $f(a) = 2(a-b)(a-d) > 0$
$f(b) = (b-a)(b-c) < 0$
$f(d) = (d-a)(d-b) > 0$
Hence a root of $f(x) = 0$ lies between a & b and another root lies between (b & d).
Hence the roots of the given equation are real and distinct.

17. $x^2 + x + 1 > 0\ \forall x \in R$
$a = 1 > 0$
$b^2 - 4ac = 1 - 4 = -3 < 0$
$x^2 + 2x + 5 > 0\ \forall x \in R$
$a = 1 > 0$
$b^2 - 4ac = 4 - 20 = -16 < 0$
So $\dfrac{x^2+x+1}{x^2+2x+5} > 0\ \forall x \in R$ 'a' is correct

18. $ax^2 + bx + c = 0$
Put $x = 1$
$a + b + c = 0$ which is given
So clearly '1' is the root of the equation
Nothing can be said about the sign of the roots.
'c' is correct.

19. (A) If the coefficients of quadratic equation are not rational then root may be $2+\sqrt{3}$ and $2+\sqrt{3}$.

20. (D) R is obviously true. So test the statement let $f(x) = (x-p)(x-r) + \lambda(x-q)(x-s) = 0$
Then $f(p) = \lambda(p-q)(p-s)$
$f(r) = \lambda(r-q)(r-s)$
If $\lambda > 0$ then $f(p) > 0$, $f(r) < 0$
$\Rightarrow$ There is a root between p & r
Thus statement-1 is false.

21. (A) Both Statement-1 and Statement-2 are true and Statement-2 is the correct explanation of Statement-1.

22. (C) Clearly Statement-1 is true but Statement-2 is false.
$\because$ $ax^2 + bx + c = 0$ is an identity when $a = b = c = 0$.

23. (A) for $x^2 + 2x + 3$
$a > 0$ and $D < 0$

24. (A) $x^2 - x + 1$

$$= \left(x - \frac{1}{2}\right)^2 + \frac{3}{4}$$

25. The roots of the given equation will be of opposite signs. If they are real and their product is negative
$D \ge 0$ and product of root is < 0

$\Rightarrow (a^3 - 8a - 1)^2 - 8(a^2 - 4a) \ge 0$ and $\dfrac{a^2-4a}{2} < 0$

$\Rightarrow a^2 - 4a < 0$
$\Rightarrow 0 < a < 4$.
Ans. (a)

Que. from Compt. Exams

1. If $x = \sqrt{1+\sqrt{1+\sqrt{1+\text{.....to infinity}}}}$, then $x =$

(a) $\dfrac{1+\sqrt{5}}{2}$ (b) $\dfrac{1-\sqrt{5}}{2}$

(c) $\dfrac{1\pm\sqrt{5}}{2}$ (d) None of these

2. For the equation $|x^2| + |x| - 6 = 0$, the roots are **[EAMCET 1988, 93]**

(a) One and only one real number (b) Real with sum one
(c) Real with sum zero (d) Real with product zero

3. If $ax^2+bx+c=0$, then $x=$ **[MP PET 1995]**

(a) $\frac{b\pm\sqrt{b^2-4ac}}{2a}$ (b) $\frac{-b\pm\sqrt{b^2-ac}}{2a}$

(c) $\frac{2c}{-b\pm\sqrt{b^2-4ac}}$ (d) None of these

4. If the equations $2x^2+3x+5\lambda=0$ and $x^2+2x+3\lambda=0$ have a common root, then $\lambda=$ **[RPET 1989]**

(a) 0 (b) –1 (c) 0,–1 (d) 2,–1

5. If the equation $x^2+\lambda x+\mu=0$ has equal roots and one root of the equation $x^2+\lambda x-12=0$ is 2, then $(\lambda,\mu)=$

(a) (4, 4) (b) (–4,4) (c) (4,–4) (d) (–4,–4)

6. If x is real and $k=\frac{x^2-x+1}{x^2+x+1}$, then **[MNR 1992; RPET 1997]**

(a) $\frac{1}{3}\le k\le 3$ (b) $k\ge 5$ (c) $k\le 0$ (d) None of these

7. If $a<b<c<d$, then the roots of the equation $(x-a)(x-c)+2(x-b)(x-d)=0$ are **[IIT 1984]**

(a) Real and distinct (b) Real and equal (c) Imaginary (d) None of these

8. If the roots of the equation $qx^2+px+q=0$ where p, q are real, be complex, then the roots of the equation $x^2-4qx+p^2=0$ are

(a) Real and unequal (b) Real and equal (c) Imaginary (d) None of these

9. The values of 'a' for which $(a^2-1)x^2+2(a-1)x+2$ is positive for any x are **[UPSEAT 2001]**

(a) $a\ge 1$ (b) $a\le 1$ (c) $a>-3$ (d) $a<-3$ or $a>1$

10. If the roots of equation $\frac{x^2-bx}{ax-c}=\frac{m-1}{m+1}$ are equal but opposite in sign, then the value of m will be **[RPET 1988, 2001; MP PET 1996, 2002; Pb. CET 2000]**

(a) $\frac{a-b}{a+b}$ (b) $\frac{b-a}{a+b}$ (c) $\frac{a+b}{a-b}$ (d) $\frac{b+a}{b-a}$

11. The coefficient of x in the equation $x^2+px+q=0$ was taken as 17 in place of 13, its roots were found to be –2 and –15, The roots of the original equation are **[IIT 1977, 79]**

(a) 3, 10 (b) $-3,-10$ (c) $-5,-18$ (d) None of these

12. If one root of the equation $ax^2+bx+c=0$ be n times the other root, then

(a) $na^2=bc(n+1)^2$ (b) $nb^2=ac(n+1)^2$ (c) $nc^2=ab(n+1)^2$ (d) None of these

13. If one root of the quadratic equation $ax^2+bx+c=0$ is equal to the nth power of the other root, then the value of $(ac^n)^{\frac{1}{n+1}}+(a^nc)^{\frac{1}{n+1}}=$ **[IIT 1983]**

(a) b (b) $-b$ (c) $b^{\frac{1}{n+1}}$ (d) $-b^{\frac{1}{n+1}}$

14. If $\sin\alpha,\cos\alpha$ are the roots of the equation $ax^2+bx+c=0$, then **[MP PET 1993]**

(a) $a^2-b^2+2ac=0$ (b) $(a-c)^2=b^2+c^2$ (c) $a^2+b^2-2ac=0$ (d) $a^2+b^2+2ac=0$

15. If both the roots of the quadratic equation

$x^2-2kx+k^2+k-5=0$

are less than 5, then k lies in the interval **[AIEEE 2005]**

(a) $(-\infty,4)$ (b) $[4,5]$ (c) $(5,6]$ (d) $(6,\infty)$

16. If the roots of the equations $x^2-bx+c=0$ and $x^2-cx+b=0$ differ by the same quantity, then $b+c$ is equal to **[BIT Ranchi 1969; MP PET 1993]**

(a) 4 (b) 1 (c) 0 (d) –4

17. If the product of roots of the equation

$x^2-3kx+2e^{2\log k}-1=0$

is 7, then its roots will real when **[IIT 1984]**

(a) $k=1$ (b) $k=2$ (c) $k=3$ (d) None of these

18. If a root of the given equation $a(b-c)x^2+b(c-a)x+c(a-b)=0$ is 1, then the other will be **[RPET 1986]**

(a) $\frac{a(b-c)}{b(c-a)}$ (b) $\frac{b(c-a)}{a(b-c)}$ (c) $\frac{c(a-b)}{a(b-c)}$ (d) None of these

19. In a triangle ABC the value of $\angle A$ is given by $5\cos A+3=0$, then the equation whose roots are $\sin A$ and $\tan A$ will be **[Roorkee 1972]**

(a) $15x^2-8x+16=0$ (b) $15x^2+8x-16=0$ (c) $15x^2-8\sqrt{2}x+16=0$ (d) $15x^2-8x-16=0$

20. If one root of the equation $ax^2+bx+c=0$ the square of the other, then $a(c-b)^3=cX$, where X is

(a) a^3+b^3 (b) $(a-b)^3$ (c) a^3-b^3 (d) None of these

21. If 8, 2 are the roots of $x^2+ax+\beta=0$ and 3, 3 are the roots of $x^2+\alpha x+b=0$, then the roots of $x^2+ax+b=0$ are

(a) $8,-1$ (b) $-9,2$ (c) $-8,-2$ (d) $9,1$ **[EAMCET 1987]**

22. The set of values of x which satisfy $5x+2<3x+8$ and $\frac{x+2}{x-1}<4$, is **[EAMCET 1989]**

(a) $(2,3)$ (b) $(-\infty,1)\cup(2,3)$ (c) $(-\infty,1)$ (d) $(1,3)$

23. If α,β are the roots of $x^2-ax+b=0$ and if $\alpha^n+\beta^n=V_n$, then **[RPET 1995; Karnataka CET 2000; Pb. CET 2002]**

(a) $V_{n+1}=aV_n+bV_{n-1}$ (b) $V_{n+1}=aV_n+aV_{n-1}$ (c) $V_{n+1}=aV_n-bV_{n-1}$ (d) $V_{n+1}=aV_{n-1}-bV_n$

24. The value of 'c' for which $|\alpha^2-\beta^2|=\frac{7}{4}$, where α and β are the roots of $2x^2+7x+c=0$, is

(a) 4 (b) 0 (c) 6 (d) 2

25. For what value of λ the sum of the squares of the roots of $x^2+(2+\lambda)x-\frac{1}{2}(1+\lambda)=0$ is minimum **[AMU 1999]**

(a) 3/2 (b) 1 (c) 1/2 (d) 11/4

26. The product of all real roots of the equation $x^2-|x|-6=0$ is **[Roorkee 2000]**

(a) -9 (b) 6 (c) 9 (d) 36

27. For the equation $3x^2+px+3=0, p>0$ if one of the root is square of the other, then p is equal to **[IIT Screening 2000]**

(a) $\frac{1}{3}$ (b) 1 (c) 3 (d) $\frac{2}{3}$

28. If α,β be the roots of $x^2+px+q=0$ and $\alpha+h,\beta+h$ are the roots of $x^2+rx+s=0$, then **[AMU 2001]**

(a) $\frac{p}{r}=\frac{q}{s}$ (b) $2h=\left[\frac{p}{q}+\frac{r}{s}\right]$ (c) $p^2-4q=r^2-4s$ (d) $pr^2=qs^2$

29. If $x^2+px+q=0$ is the quadratic equation whose roots are $a-2$ and $b-2$ where a and b are the roots of $x^2-3x+1=0$, then **[Kerala (Engg.) 2002]**

(a) $p=1,q=5$ (b) $p=1,q=-5$ (c) $p=-1,\ q=1$ (d) None of these

30. The value of 'a' for which one root of the quadratic equation $(a^2-5a+3)x^2+(3a-1)x+2=0$ is twice as large as the other, is **[AIEEE 2003]**

(a) $\frac{2}{3}$ (b) $-\frac{2}{3}$ (c) $\frac{1}{3}$ (d) $-\frac{1}{3}$

31. If a,b,c are in G.P., then the equations $ax^2+2bx+c=0$ and $dx^2+2ex+f=0$ have a common root if $\frac{d}{a},\frac{e}{b},\frac{f}{c}$ are in **[IIT 1985; Pb. CET 2000; DCE 2000]**

(a) A.P. (b) G.P. (c) H.P. (d) None of these

32. The value of 'a' for which the equations $x^2-3x+a=0$ and $x^2+ax-3=0$ have a common root is **[Pb. CET 1999]**

(a) 3 (b) 1 (c) -2 (d) 2

33. If $(x+1)$ is a factor of

$x^4-(p-3)x^3-(3p-5)x^2+(2p-7)x+6$, then $p=$ **[IIT 1975]**

(a) 4 (b) 2 (c) 1 (d) None of these

34. The roots of the equation

$4x^4-24x^3+57x^2+18x-45=0$,

If one of them is $3+i\sqrt{6}$, are

(a) $3-i\sqrt{6},\pm\sqrt{\frac{3}{2}}$ (b) $3-i\sqrt{6},\pm\frac{3}{\sqrt{2}}$ (c) $3-i\sqrt{6},\pm\frac{\sqrt{3}}{2}$ (d) None of these

35. The values of a for which $2x^2-2(2a+1)x+a(a+1)=0$ may have one root less than a and other root greater than a are given by **[UPSEAT 2001]**

(a) $1>a>0$ (b) $-1<a<0$ (c) $a\geq 0$ (d) $a>0$ or $a<-1$

ANSWER KEY(Que. from Compt. Exams)

1	a	2	c	3	c	4	c	5	a
6	a	7	a	8	a	9	d	10	a
11	b	12	b	13	b	14	a	15	a
16	d	17	b	18	c	19	b	20	b
21	d	22	b	23	c	24	c	25	c
26	a	27	c	28	c	29	d	30	a
31	a	32	d	33	a	34	c	35	d

36	d	37	b	38	c	39	a	40	a

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 7 XI M 7. Permutations and Combinations

Index:

1. Key Concepts

2. Exercise I to V

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Permutation and Combination

Permutations are arrangements and combinations are selections. In this chapter we discuss the methods of counting of arrangements and selections. The basic results and formulas are as follows:

1. Fundamental Principle of Counting :

(i) Principle of Multiplication: If an event can occur in 'm' different ways, following which another event can occur in 'n' different ways, then total number of different ways of simultaneous occurrence of both the events in a definite order is $m \times n$.

(ii) Principle of Addition: If an event can occur in 'm' different ways, and another event can occur in 'n' different ways, then exactly one of the events can happen in m + n ways.

Example # 1 There are 8 buses running from Kota to Jaipur and 10 buses running from Jaipur to Delhi. In how many ways a person can travel from Kota to Delhi via Jaipur by bus.

Solution. Let E_1 be the event of travelling from Kota to Jaipur & E_2 be the event of travelling from Jaipur to Delhi by the person.

E_1 can happen in 8 ways and E_2 can happen in 10 ways.

Since both the events E_1 and E_2 are to be happened in order, simultaneously, the number of ways = $8 \times 10 = 80$.

Example # 2 How many numbers between 10 and 10,000 can be formed by using the digits 1, 2, 3, 4, 5 if
(i) No digit is repeated in any number. (ii) Digits can be repeated.

Solution.

(i) Number of two digit numbers = $5 \times 4 = 20$
Number of three digit numbers = $5 \times 4 \times 3 = 60$
Number of four digit numbers = $5 \times 4 \times 3 \times 2 = 120$ Total = 200

(ii) Number of two digit numbers = $5 \times 5 = 25$
Number of three digit numbers = $5 \times 5 \times 5 = 125$
Number of four digit numbers = $5 \times 5 \times 5 \times 5 = 625$ Total = 775

Self Practice Problems :

1. How many 4 digit numbers are there, without repetition of digits, if each number is divisible by 5. **Ans.** 952

2. Using 6 different flags, how many different signals can be made by using atleast three flags, arranging one above the other. **Ans.** 1920

2. Arrangement : If ${}^nP_r$ denotes the number of permutations of n different things, taking r at a time, then

$${}^nP_r = n\,(n-1)\,(n-2)\ldots\ldots(n-r+1) = \frac{n!}{(n-r)!}$$

NOTE : (i) factorials of negative integers are not defined. (ii) $0! = 1! = 1$;
(iii) ${}^nP_n = n! = n.(n-1)!$ (iv) $(2n)! = 2^n.\,n!\,[1.3.5.7\ldots(2n-1)]$

Example # 3: How many numbers of three digits can be formed using the digits 1, 2, 3, 4, 5, without repetition of digits. How many of these are even.

Solution.: Three places are to be filled with 5 different objects.

$\therefore$ Number of ways = ${}^5P_3 = 5 \times 4 \times 3 = 60$

For the 2nd part, unit digit can be filled in two ways & the remaining two digits can be filled in ${}^4P_2$ ways.

$\therefore$ Number of even numbers = $2 \times {}^4P_2 = 24$.

Example # 4: If all the letters of the word 'QUEST' are arranged in all possible ways and put in dictionary order, then find the rank of the given word.

Solution.: Number of words beginning with E = ${}^4P_4 = 24$
Number of wards beginning with QE = ${}^3P_3 = 6$
Number of words beginning with QS = 6
Number of words beginning withQT = 6.
Next word is 'QUEST' $\therefore$ its rank is $24 + 6 + 6 + 6 + 1 = 43$.

Self Practice Problems :

3. Find the sum of all four digit numbers (without repetition of digits) formed using the digits 1, 2, 3, 4, 5. **Ans.** 399960

4. Find 'n', if ${}^{n-1}P_3 : {}^nP_4 = 1 : 9$. **Ans.** 9

5. Six horses take part in a race. In how many ways can these horses come in the first, second and third place, if a particular horse is among the three winners (Assume No Ties). **Ans.** 60

3. Circular Permutation : The number of circular permutations of n different things taken all at a time is; $(n-1)!$. If clockwise & anti–clockwise circular permutations are considered to be same, then it is $\frac{(n-1)!}{2}$.

Note: Number of circular permutations of n things when p alike and the rest different taken all at a time distinguishing clockwise and anticlockwise arrangement is $\frac{(n-1)!}{p!}$.

Example # 5: In how many ways can we arrange 6 different flowers in a circle. In how many ways we can form a garland using these flowers.

Solution.: The number of circular arrangements of 6 different flowers = $(6-1)! = 120$

When we form a garland, clockwise and anticlockwise arrangements are similar. Therefore, the number of ways of forming garland = $\frac{1}{2}(6-1)! = 60$.

Example # 6: In how many ways 6 persons can sit at a round table, if two of them prefer to sit together.

Solution.: Let $P_1, P_2, P_3, P_4, P_5, P_6$ be the persons, where P_1, P_2 want to sit together.

Regard these person as 5 objects. They can be arranged in a circle in $(5-1)! = 24$. Now P_1P_2 can be arranged in 2! ways. Thus the total number of ways = $24 \times 2 = 48$.

Self Practice Problems : 6. In how many ways the letters of the word 'MONDAY' can be written around a circle if the vowels are to be separated in any arrangement. **Ans.** 72

7. In how many ways we can form a garland using 3 different red flowers, 5 different yellow flowers and 4 different blue flowers, if flowers of same colour must be together. **Ans.** 17280.

4. Selection : If ${}^nC_r$ denotes the number of combinations of n different things taken r at a time, then

$${}^nC_r = \frac{n!}{r!\,(n-r)!} = \frac{{}^nP_r}{r!} \quad \text{where } r \le n\ ;\ n \in N \text{ and } r \in W.$$

NOTE : (i) ${}^nC_r = {}^nC_{n-r}$ (ii) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$ (iii) ${}^nC_r = 0$ if $r \notin \{0, 1, 2, 3\ldots\ldots, n\}$

Example # 7 Fifteen players are selected for a cricket match.

(i) In how many ways the playing 11 can be selected
(ii) In how many ways the playing 11 can be selected including a particular player.
(iii) In how many ways the playing 11 can be selected excluding two particular players.

Solution. (i) 11 players are to be selected from 15
Number of ways = $^{15}C_{11} = 1365$.

(ii) Since one player is already included, we have to select 10 from the remaining 14
Number of ways = $^{14}C_{10} = 1001$.

(iii) Since two players are to be excluded, we have to select 11 from the remaining 13.
Number of ways = $^{13}C_{11} = 78$.

Example # 8 If $^{49}C_{3r-2} = {}^{49}C_{2r+1}$, find 'r'.

Solution. $^nC_r = {}^nC_s$ if either r = s or r + s = n.

Thus $3r - 2 = 2r + 1 \Rightarrow r = 3$

or $3r - 2 + 2r + 1 = 49 \Rightarrow 5r - 1 = 49 \Rightarrow r = 10$

$\therefore$ r = 3, 10

Example # 9 A regular polygon has 20 sides. How many triangles can be drawn by using the vertices, but not using the sides.

Solution. The first vertex can be selected in 20 ways. The remaining two are to be selected from 17 vertices so that they are not consecutive. This can be done in $^{17}C_2 - 16$ ways.

$\therefore$ The total number of ways = $20 \times (^{17}C_2 - 16)$

But in this method, each selection is repeated thrice.

$\therefore$ Number of triangles = $\frac{20 \times (^{17}C_2 - 16)}{3} = 800$.

Example # 10 10 persons are sitting in a row. In how many ways we can select three of them if adjacent persons are not selected.

Solution. Let $P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}$ be the persons sitting in this order.
If three are selected (non consecutive) then 7 are left out.
Let PPPPPPP be the left out & q, q, q be the selected. The number of ways in which these 3 q's can be placed into the 8 positions between the P's (including extremes) is the number ways of required selection. Thus number of ways = $^8C_3 = 56$.

Example # 11 In how many ways we can select 4 letters from the letters of the word MISSISSIPPI.

Solution.
M
IIII
SSSS
PP
Number of ways of selecting 4 alike letters = $^2C_1 = 2$.
Number of ways of selecting 3 alike and 1 different letters = $^2C_1 \times {}^3C_1 = 6$
Number of ways of selecting 2 alike and 2 alike letters = $^3C_2 = 3$
Number of ways of selecting 2 alike & 2 different = $^3C_1 \times {}^3C_2 = 9$
Number of ways of selecting 4 different = $^4C_4 = 1$ Total = 21

Self Practice Problems :8. In how many ways 7 persons can be selected from among 5 Indian, 4 British & 2 Chinese, if atleast two are to be selected from each country. **Ans.** 100

9. 10 points lie in a plane, of which 4 points are collinear. Barring these 4 points no three of the 10 points are collinear. How many quadrilaterals can be drawn. **Ans.** 185.

10. In how many ways 5 boys & 5 girls can sit at a round table so that girls & boys sit alternate. **Ans.** 2880

11. In how many ways 4 persons can occupy 10 chairs in a row, if no two sit on adjacent chairs. **Ans.** 840.

12. In how many ways we can select 3 letters of the word PROPORTION. **Ans.** 36

5. The number of permutations of 'n' things, taken all at a time, when 'p' of them are similar & of one type, q of them are similar & of another type, 'r' of them are similar & of a third type & the remaining n – (p + q + r) are all different is $\frac{n!}{p!\,q!\,r!}$.

Example # 12 In how many ways we can arrange 3 red flowers, 4 yellow flowers and 5 white flowers in a row. In how many ways this is possible if the white flowers are to be separated in any arrangement (Flowers of same colour are identical).

Solution. Total we have 12 flowers 3 red, 4 yellow and 5 white.

Number of arrangements = $\frac{12!}{3!\,4!\,5!} = 27720$.

For the second part, first arrange 3 red & 4 yellow

This can be done in $\frac{7!}{3!\,4!} = 35$ ways

Now select 5 places from among 8 places (including extremes) & put the white flowers there.
This can be done in $^8C_5 = 56$.

$\therefore$ The number of ways for the 2nd part = $35 \times 56 = 1960$.

Example # 13 In how many ways the letters of the word "ARRANGE" can be arranged without altering the relative positions of vowels & consonants.

Solution. The consonants in their positions can be arranged in $\frac{4!}{2!} = 12$ ways.

The vowels in their positions can be arranged in $\frac{3!}{2!} = 3$ ways

$\therefore$ Total number of arrangements = $12 \times 3 = 26$

Self Practice Problems : 13. How many words can be formed using the letters of the word ASSESSMENT if each word begin with A and end with T. **Ans.** 840

14. If all the letters of the word ARRANGE are arranged in all possible ways, in how many of words we will have the A's not together and also the R's not together. **Ans.** 660

15. How many arrangements can be made by taking four letters of the word MISSISSIPPI. **Ans.** 176.

6. **Formation of Groups :** Number of ways in which (m + n + p) different things can be divided into three different groups containing m, n & p things respectively is $\frac{(m+n+p)!}{m!\,n!\,p!}$,

If m = n = p and the groups have identical qualitative characteristic then the number of groups = $\frac{(3n)!}{n!\, n!\, n!\, 3!}$.

However, if 3n things are to be divided equally among three people then the number of ways = $\frac{(3n)!}{(n!)^3}$.

Ex.14. 12 different toys are to be distributed to three children equally. In how many ways this can be done.
Solution. The problem is to divide 12 different things into three different groups.

Number of ways = $\frac{12!}{4!\,4!\,4!}$ = 34650.

Example # 15 In how many ways 10 persons can be divided into 5 pairs.
Solution. We have each group having 2 persons and the qualitative characteristic are same (Since there is no purpose mentioned or names for each pair).

Thus the number of ways = $\frac{10!}{(2!)^5\, 5!}$ = 945.

Self Practice Problems : 16. 9 persons enter a lift from ground floor of a building which stops in 10 floors (excluding ground floor). If is known that persons will leave the lift in groups of 2, 3, & 4 in different floors. In how many ways this can happen. **Ans.** 907200

17. In how many ways one can make four equal heaps using a pack of 52 playing cards. **Ans.** $\frac{52!}{(13!)^4\, 4!}$

18. In how many ways 11 different books can be parcelled into four packets so that three of the packets contain 3 books each and one of 2 books, if all packets have the same destination. **Ans.** $\frac{11!}{(3!)^4\, 2}$

7. Selection of one or more objects

(a) Number of ways in which atleast one object be selected out of 'n' distinct objects is
$${}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots\ldots\ldots\ldots + {}^nC_n = 2^n - 1$$

(b) Number of ways in which atleast one object may be selected out of 'p' alike objects of one type 'q' alike objects of second type and 'r' alike of third type is
$$(p + 1)(q + 1)(r + 1) - 1$$

(c) Number of ways in which atleast one object may be selected from 'n' objects where 'p' alike of one type 'q' alike of second type and 'r' alike of third type and rest n – (p + q + r) are different, is
$$(p + 1)(q + 1)(r + 1)\, 2^{n-(p+q+r)} - 1$$

Example # 16 There are 12 different books on a shelf. In how many ways we can select atleast one of them.
Solution. We may select 1 book, 2 books,....., 12 books.
∴ The number of ways = ${}^{12}C_1 + {}^{12}C_2 + \ldots\ldots + {}^{12}C_{12} = 2^{12} - 1. = 4095$

Example # 17 There are 12 fruits in a basket of which 5 are apples, 4 mangoes and 3 bananas (fruits of same species are identical). How many ways are there to select atleast one fruit.
Solution. Let x be the number of apples being selected
y be the number of mangoes being selected and
z be the number of bananas being selected.
Then x = 0, 1, 2, 3, 4, 5
y = 0, 1, 2, 3, 4
z = 0, 1, 2, 3
Total number of triplets (x, y, z) is 6 × 5 × 4 = 120
Exclude (0, 0, 0) ∴ Number of combinations = 120 – 1 = 119.

Self Practice Problems

19. In a shelf there are 5 physics, 4 chemistry and 3 mathematics books. How many combinations are there if (i) books of same subject are different (ii) books of same subject are identical.
Ans. (i) 4095 (ii) 119

20. From 5 apples, 4 mangoes & 3 bananas in how many ways we can select atleast two fruits of each variety if (i) fruits of same species are identical (ii) fruits of same species are different.
Ans. (i) 24 (ii) 1144

8. Multinomial Theorem: Coefficient of x^r in expansion of $(1 - x)^{-n} = {}^{n+r-1}C_r$ $(n \in N)$
Number of ways in which it is possible to make a selection from m + n + p = N things, where p are alike of one kind, m alike of second kind & n alike of third kind taken r at a time is given by coefficient of x^r in the expansion of
$$(1 + x + x^2 + \ldots\ldots + x^p)(1 + x + x^2 + \ldots\ldots + x^m)(1 + x + x^2 + \ldots\ldots + x^n).$$

(i) For example the number of ways in which a selection of four letters can be made from the letters of the word **PROPORTION** is given by coefficient of x^4 in
$$(1 + x + x^2 + x^3)(1 + x + x^2)(1 + x + x^2)(1 + x)(1 + x)(1 + x).$$

(ii) **Method of fictious partition :**
Number of ways in which n identical things may be distributed among p persons if each person may receive none, one or more things is; ${}^{n+p-1}C_n$.

Example # 18: Find the number of solutions of the equation x + y + z = 6, where x, y, z ∈ W.
Solution. Number of solutions = coefficient of x^6 in $(1 + x + x^2 + \ldots\ldots x^6)^3$
= coefficient of x^6 in $(1 - x^7)^3 (1 - x)^{-3}$
= coefficient of x^6 in $(1 - x)^{-3}$
$$= \binom{3+6-1}{6} = {}^8C_6 = 28.$$

Example # 19: In a bakery four types of biscuits are available. In how many ways a person can buy 10 biscuits if he decide to take atleast one biscuit of each variety.
Solution. Let x be the number of biscuits the person select from first variety, y from the second, z from the third and w from the fourth variety. Then the number of ways = number of solutions of the equation x + y + z + w = 10.
where x = 1, 2, ...,7
y = 1, 2, ...,7
z = 1, 2, ...,7
w = 1, 2, ...,7

This is equal to = coefficient of x^{10} in $(x + x^2 + ... + x^7)^4$
= coefficient of x^6 in $(1 + x + ... + x^6)^4$
= coefficient of x^6 in $(1 - x^7)^4 (1 - x)^{-4}$
= coefficient x^6 in $(1 - x)^{-4}$ $= \begin{pmatrix} 4+6-1 \\ 6 \end{pmatrix} = 84.$

Self Practice Problems:

21. Three distinguishable dice are rolled. In how many ways we can get a total 15. **Ans.** 10.

22. In how many ways we can give 5 apples, 4 mangoes and 3 oranges (fruits of same species are similar) to three persons if each may receive none, one or more. **Ans.** 3150

9. Let $N = p^a \cdot q^b \cdot r^c .....$ where p, q, r..... are distinct primes & a, b, c..... are natural numbers then :

(a) The total numbers of divisors of N including 1 & N is $= (a + 1) (b + 1) (c + 1).....$

(b) The sum of these divisors is =
$(p^0 + p^1 + p^2 + ... + p^a) (q^0 + q^1 + q^2 + ... + q^b) (r^0 + r^1 + r^2 + ... + r^c).....$

(c) Number of ways in which N can be resolved as a product of two factors is

$$= \begin{cases} \frac{1}{2}(a+1)(b+1)(c+1).... & \text{if N is not a perfect square} \\ \frac{1}{2}[(a+1)(b+1)(c+1)....+1] & \text{if N is a perfect square} \end{cases}$$

(d) Number of ways in which a composite number N can be resolved into two factors which are relatively prime (or coprime) to each other is equal to 2^{n-1} where n is the number of different prime factors in N.

Example # 20 Find the number of divisors of 1350. Also find the sum of all divisors.

Solution. $1350 = 2 \times 3^3 \times 5^2$

$\therefore$ Number of divisors $= (1+ 1) (3 + 1) (2 + 1) = 24$

sum of divisors $= (1 + 2) (1 + 3 + 3^2 + 3^3) (1 + 5 + 5^2) = 3720.$

Example # 21 In how many ways 8100 can be resolved into product of two factors.

Solution. $8100 = 2^2 \times 3^4 \times 5^2$

Number of ways $= \frac{1}{2} ((2 + 1) (4 + 1) (2 + 1) + 1) = 23$

Self Practice Problems :

23. How many divisors of 9000 are even but not divisible by 4. Also find the sum of all such divisors.
Ans. 12, 4056.

24. In how many ways the number 8100 can be written as product of two coprime factors. **Ans.** 4

10. Let there be 'n' types of objects, with each type containing atleast r objects. Then the number of ways of arranging r objects in a row is n^r.

Example # 22 How many 3 digit numbers can be formed by using the digits 0, 1, 2, 3, 4, 5. In how many of these we have atleast one digit repeated.

Solution. We have to fill three places using 6 objects (repeatation allowed), 0 cannot be at 100th place.

The number of numbers = 180. $\overset{5}{\square}\overset{6}{\square}\overset{6}{\square}$

Number of numbers in which no digit is repeated = 100 $\overset{5}{\square}\overset{5}{\square}\overset{4}{\square}$

$\therefore$ Number of numbers in which atleast one digit is repeated = 180 – 100 = 80

Example # 23 How many functions can be defined from a set A containing 5 elements to a set B having 3 elements. How many these are surjective functions.

Solution. Image of each element of A can be taken in 3 ways.

$\therefore$ Number of functions from A to B $= 3^5 = 243.$

Number of into functions from A to B $= 2^5 + 2^5 + 2^5 - 3 = 93.$

$\therefore$ Number of onto functions = 150.

Self Practice Problems : 25. Find the sum of all three digit numbers those can be formed by using the digits. 0, 1, 2, 3, 4.
Ans. 27200.

26. How many functions can be defined from a set A containing 4 elements to a set B containing 5 elements. How many of these are injective functions. **Ans.** 625, 120

27. In how many ways 5 persons can enter into a auditorium having 4 entries. **Ans.** 1024.

11. Dearrangement :

Number of ways in which 'n' letters can be put in 'n' corresponding envelopes such that no letter goes to correct envelope is

$$n!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}.....+(-1)^n\frac{1}{n!}\right)$$

Example # 24 In how many ways we can put 5 writings into 5 corresponding envelopes so that no writing go to the corresponding envelope.

Solution. The problem is the number of dearragements of 5 digits.

This is equal to $5!\left(\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}\right) = 44.$

Example # 25 Four slip of papers with the numbers 1, 2, 3, 4 written on them are put in a box. They are drawn one by one (without replacement) at random. In how many ways it can happen that the ordinal number of atleast one slip coincide with its own number.

Solution. Total number of ways = 4 ! = 24.

The number of ways in which ordinal number of any slip does not coincide with its own number is the

number of dearrangements of 4 objects $= 4!\left(\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\right) = 9$

Thus the required number of ways. = 24 – 9 = 15

Self Practice Problems:

28. In a match column question, Column I contain 10 questions and Column II contain 10 answers written in some arbitrary order. In how many ways a student can answer this question so that exactly 6 of his matchings are correct. **Ans.** 1890

29. In how many ways we can put 5 letters into 5 corresponding envelopes so that atleast one letter go to wrong envelope. **Ans.** 119

SHORT REVISION

DEFINITIONS :

1. **PERMUTATION :** Each of the arrangements in a definite order which can be made by taking some or all of a number of things is called a **PERMUTATION**.
2. **COMBINATION :** Each of the groups or selections which can be made by taking some or all of a number of things without reference to the order of the things in each group is called a **COMBINATION**.

FUNDAMENTAL PRINCIPLE OF COUNTING :

If an event can occur in 'm' different ways, following which another event can occur in '*n*' different ways, then the total number of different ways of simultaneous occurrence of both events in a definite order is $m \times n$. This can be extended to any number of events.

RESULTS : **(i)** A Useful Notation : $n! = n(n-1)(n-2)\ldots\ldots 3.2.1$; $n! = n.(n-1)!$

$0! = 1! = 1$; $(2n)! = 2^n . n! [1.3.5.7\ldots(2n-1)]$ Note that factorials of negative integers are not defined.

(ii) If ${}^nP_r$ denotes the number of permutations of n different things, taking r at a time, then

${}^nP_r = n(n-1)(n-2)\ldots\ldots(n-r+1) = \dfrac{n!}{(n-r)!}$ Note that , ${}^nP_n = n!$.

(iii) If ${}^nC_r$ denotes the number of combinations of n different things taken r at a time, then

${}^nC_r = \dfrac{n!}{r!(n-r)!} = \dfrac{{}^nP_r}{r!}$ where $r \le n$; $n \in N$ and $r \in W$.

(iv) The number of ways in which $(m+n)$ different things can be divided into two groups containing m & n things respectively is : $\dfrac{(m+n)!}{m!n!}$ If $m = n$, the groups are equal & in this case the number of subdivision is $\dfrac{(2n)!}{n!n!2!}$; for in any one way it is possible to interchange the two groups without obtaining a new distribution. However, if 2n things are to be divided equally between two persons then the number of ways $= \dfrac{(2n)!}{n!n!}$.

(v) Number of ways in which $(m+n+p)$ different things can be divided into three groups containing m , n & p things respectively is $\dfrac{(m+n+p)!}{m!n!p!}$, $m \ne n \ne p$. If $m = n = p$ then the number of groups $= \dfrac{(3n)!}{n!n!n!3!}$.

However, if 3n things are to be divided equally among three people then the number of ways $= \dfrac{(3n)!}{(n!)^3}$.

(vi) The number of permutations of *n* things taken all at a time when *p* of them are similar & of one type, q of them are similar & of another type, *r* of them are similar & of a third type & the remaining $n-(p+q+r)$ are all different is : $\dfrac{n!}{p!q!r!}$.

(vii) The number of circular permutations of n different things taken all at a time is ; $(n-1)!$. If clockwise & anti–clockwise circular permutations are considered to be same, then it is $\dfrac{(n-1)!}{2}$.

Note : Number of circular permutations of n things when p alike and the rest different taken all at a time distinguishing clockwise and anticlockwise arrangement is $\dfrac{(n-1)!}{p!}$.

(viii) Given n different objects, the number of ways of selecting atleast one of them is , ${}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots\ldots + {}^nC_n = 2^n - 1$. This can also be stated as the total number of combinations of n distinct things.

(ix) Total number of ways in which it is possible to make a selection by taking some or all out of $p+q+r+\ldots\ldots$ things , where p are alike of one kind, q alike of a second kind , r alike of third kind & so on is given by : $(p+1)(q+1)(r+1)\ldots\ldots\ldots -1$.

(x) Number of ways in which it is possible to make a selection of $m+n+p = N$ things , where p are alike of one kind, m alike of second kind & n alike of third kind taken r at a time is given by coefficient of x^r in the expansion of

$(1+x+x^2+\ldots\ldots+x^p)(1+x+x^2+\ldots\ldots+x^m)(1+x+x^2+\ldots\ldots+x^n)$.

Note : Remember that coefficient of x^r in $(1-x)^{-n} = {}^{n+r-1}C_r$ $(n \in N)$. For example the number of ways in which a selection of four letters can be made from the letters of the word **PROPORTION** is given by coefficient of x^4 in $(1+x+x^2+x^3)(1+x+x^2)(1+x+x^2)(1+x)(1+x)(1+x)$.

(xi) Number of ways in which n distinct things can be distributed to p persons if there is no restriction to the number of things received by men $= p^n$.

(xii) Number of ways in which n identical things may be distributed among p persons if each person may receive none, one or more things is ; ${}^{n+p-1}C_p$.

(xiii) **a.** ${}^nC_r = {}^nC_{n-r}$; ${}^nC_0 = {}^nC_n = 1$; **b.** ${}^nC_x = {}^nC_y \Rightarrow x = y$ or $x + y = n$

c. ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

(xiv) ${}^nC_r$ is maximum if : (a) $r = \dfrac{n}{2}$ if n is even. (b) $r = \dfrac{n-1}{2}$ or $\dfrac{n+1}{2}$ if n is odd.

(xv) Let $N = p^a . q^b . r^c \ldots\ldots$ where p , q , r..... are distinct primes & a , b , c..... are natural numbers then:

(a) The total numbers of divisors of N including 1 & N is $= (a+1)(b+1)(c+1)\ldots\ldots$

(b) The sum of these divisors is

$= (p^0 + p^1 + p^2 + \ldots + p^a)(q^0 + q^1 + q^2 + \ldots + q^b)(r^0 + r^1 + r^2 + \ldots + r^c)\ldots$

(c) Number of ways in which N can be resolved as a product of two factors is $= \begin{cases} \frac{1}{2}(a+1)(b+1)(c+1)\ldots & \text{if N is not a perfect square} \\ \frac{1}{2}\left[(a+1)(b+1)(c+1)\ldots+1\right] & \text{if N is a perfect square} \end{cases}$

(d) Number of ways in which a composite number N can be resolved into two factors which are relatively prime (or coprime) to each other is equal to 2^{n-1} where n is the number of different prime factors in N.
[Refer Q.No.28 of Ex–I]

(xvi) Grid Problems and tree diagrams.

DEARRANGEMENT : Number of ways in which n letters can be placed in n directed letters so that no letter goes into its own envelope is $= n!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}+.....+(-1)^n\frac{1}{n!}\right]$.

(xvii) Some times students find it difficult to decide whether a problem is on permutation or combination or both. Based on certain words / phrases occuring in the problem we can fairly decide its nature as per the following table :

PROBLEMS OF COMBINATIONS	**PROBLEMS OF PERMUTATIONS**
■ Selections , choose	■ Arrangements
■ Distributed group is formed	■ Standing in a line seated in a row
■ Committee	■ problems on digits
■ Geometrical problems	■ Problems on letters from a word

EXERCISE–1

Q.1 The straight lines l_1 , l_2 & l_3 are parallel & lie in the same plane. A total of m points are taken on the line l_1 , n points on l_2 & k points on l_3. How many maximum number of triangles are there whose vertices are at these points ?

Q.2 How many five digits numbers divisible by 3 can be formed using the digits 0, 1, 2, 3, 4, 7 and 8 if each digit is to be used atmost once.

Q.3 There are 2 women participating in a chess tournament. Every participant played 2 games with the other participants. The number of games that the men played between themselves exceeded by 66 as compared to the number of games that the men played with the women. Find the number of participants & the total numbers of games played in the tournament.

Q.4 All the 7 digit numbers containing each of the digits 1, 2, 3, 4, 5, 6, 7 exactly once, and not divisible by 5 are arranged in the increasing order. Find the $(2004)^{th}$ number in this list.

Q.5 5 boys & 4 girls sit in a straight line. Find the number of ways in which they can be seated if 2 girls are together & the other 2 are also together but separate from the first 2.

Q.6 A crew of an eight oar boat has to be chosen out of 11 men five of whom can row on stroke side only, four on the bow side only, and the remaining two on either side. How many different selections can be made?

Q.7 An examination paper consists of 12 questions divided into parts A & B.
Part-A contains 7 questions & Part–B contains 5 questions. A candidate is required to attempt 8 questions selecting atleast 3 from each part. In how many maximum ways can the candidate select the questions ?

Q.8 In how many ways can a team of 6 horses be selected out of a stud of 16 , so that there shall always be 3 out of A B C A′ B′ C′ , but never A A′ , B B′ or C C′ together.

Q.9 During a draw of lottery, tickets bearing numbers 1, 2, 3,....., 40, 6 tickets are drawn out & then arranged in the descending order of their numbers. In how many ways, it is possible to have 4th ticket bearing number 25.

Q.10 Find the number of distinct natural numbers upto a maximum of 4 digits and divisible by 5, which can be formed with the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 each digit not occuring more than once in each number.

Q.11 The Indian cricket team with eleven players, the team manager, the physiotherapist and two umpires are to travel from the hotel where they are staying to the stadium where the test match is to be played. Four of them residing in the same town own cars, each a four seater which they will drive themselves. The bus which was to pick them up failed to arrive in time after leaving the opposite team at the stadium. In how many ways can they be seated in the cars ? In how many ways can they travel by these cars so as to reach in time, if the seating arrangement in each car is immaterial and all the cars reach the stadium by the same route.

Q.12 There are n straight lines in a plane, no 2 of which parallel , & no 3 pass through the same point. Their point of intersection are joined. Show that the number of fresh lines thus introduced is $\frac{n(n-1)(n-2)(n-3)}{8}$.

Q.13 In how many ways can you divide a pack of 52 cards equally among 4 players. In how many ways the cards can be divided in 4 sets, 3 of them having 17 cards each & the 4th with 1 card.

Q.14 A firm of Chartered Accountants in Bombay has to send 10 clerks to 5 different companies, two clerks in each. Two of the companies are in Bombay and the others are outside. Two of the clerks prefer to work in Bombay while three others prefer to work outside. In how many ways can the assignment be made if the preferences are to be satisfied.

Q.15 A train going from Cambridge to London stops at nine intermediate stations. 6 persons enter the train during the journey with 6 different tickets of the same class. How many different sets of ticket may they have had?

Q.16 Prove that if each of m points in one straight line be joined to each of n in another by straight lines terminated by the points, then excluding the given points, the lines will intersect $\frac{1}{4}mn(m-1)(n-1)$ times.

Q.17 How many arrangements each consisting of 2 vowels & 2 consonants can be made out of the letters of the word **'DEVASTATION'**?

Q.18 Find the number of words each consisting of 3 consonants & 3 vowels that can be formed from the letters of the word **"Circumference".** In how many of these c's will be together.

Q.19 There are 5 white , 4 yellow , 3 green , 2 blue & 1 red ball. The balls are all identical except for colour. These are to be arranged in a line in 5 places. Find the number of distinct arrangements.

Q.20 How many 4 digit numbers are there which contains not more than 2 different digits?

Q.21 In how many ways 8 persons can be seated on a round table
(a) If two of them (say A and B) must not sit in adjacent seats?
(b) If 4 of the persons are men and 4 ladies and if no two men are to be in adjacent seats.

(c) If 8 persons constitute 4 married couples and if no husband and wife, as well as no two men, are to be in adjacent seats?

Q.22 (i) If 'n' things are arranged in circular order , then show that the number of ways of selecting four of the things no two of which are consecutive is $\frac{n(n-5)(n-6)(n-7)}{4!}$

(ii) If the 'n' things are arranged in a row, then show that the number of such sets of four is $\frac{(n-3)(n-4)(n-5)(n-6)}{4!}$

Q.23(a)How many divisors are there of the number x = 21600. Find also the sum of these divisors.

(b)In how many ways the number 7056 can be resolved as a product of 2 factors.

(c)Find the number of ways in which the number 300300 can be split into 2 factors which are relatively prime.

Q.24 How many ten digits whole number satisfy the following property they have 2 and 5 as digits, and there are no consecutive 2's in the number (i.e. any two 2's are separated by at least one 5).

Q.25 How many different ways can 15 Candy bars be distributed between Ram, Shyam, Ghanshyam and Balram, if Ram can not have more than 5 candy bars and Shyam must have at least two. Assume all Candy bars to be alike.

Q.26 Find the number of distinct throws which can be thrown with 'n' six faced normal dice which are indistinguishable among themselves.

Q.27 How many integers between 1000 and 9999 have exactly one pair of equal digit such as 4049 or 9902 but not 4449 or 4040?

Q.28 In a certain town the streets are arranged like the lines of a chess board. There are 6 streets running north & south and 10 running east & west. Find the number of ways in which a man can go from the north-west corner to the south-east corner covering the shortest possible distance in each case.

Q.29 (i) Prove that : ${}^{n}P_r = {}^{n-1}P_r + r.\ {}^{n-1}P_{r-1}$ (ii) If ${}^{20}C_{r+2} = {}^{20}C_{2r-3}$ find ${}^{12}C_r$

(iii) Find the ratio ${}^{20}C_r$ to ${}^{25}C_r$ when each of them has the greatest value possible.

(iv) Prove that ${}^{n-1}C_3 + {}^{n-1}C_4 > {}^{n}C_3$ if $n > 7$. (v) Find r if ${}^{15}C_{3r} = {}^{15}C_{r+3}$

Q.30 There are 20 books on Algebra & Calculus in our library. Prove that the greatest number of selections each of which consists of 5 books on each topic is possible only when there are 10 books on each topic in the library.

EXERCISE–2

Q.1 Find the number of ways in which 3 distinct numbers can be selected from the set $\{3^1, 3^2, 3^3, \ldots\ldots 3^{100}, 3^{101}\}$ so that they form a G.P.

Q.2 Let n & k be positive integers such that $n \geq \frac{k(k+1)}{2}$. Find the number of solutions $(x_1, x_2, \ldots, x_k)$, $x_1 \geq 1$, $x_2 \geq 2$, $\ldots$, $x_k \geq k$, all integers, satisfying $x_1 + x_2 + \ldots + x_k = n$.

Q.3 There are counters available in 7 different colours. Counters are all alike except for the colour and they are atleast ten of each colour. Find the number of ways in which an arrangement of 10 counters can be made. How many of these will have counters of each colour.

Q.4 For each positive integer k, let S_k denote the increasing arithmetic sequence of integers whose first term is 1 and whose common difference is k. For example, S_3 is the sequence 1, 4, 7, 10..... Find the number of values of k for which S_k contain the term 361.

Q.5 Find the number of 7 lettered words each consisting of 3 vowels and 4 consonants which can be formed using the letters of the word "DIFFERENTIATION".

Q.6 A shop sells 6 different flavours of ice-cream. In how many ways can a customer choose 4 ice-cream cones if

(i) they are all of different flavours (ii) they are non necessarily of different flavours

(iii) they contain only 3 different flavours (iv) they contain only 2 or 3 different flavours?

Q.7 6 white & 6 black balls of the same size are distributed among 10 different urns. Balls are alike except for the colour & each urn can hold any number of balls. Find the number of different distribution of the balls so that there is atleast 1 ball in each urn.

Q.8 There are 2n guests at a dinner party. Supposing that the master and mistress of the house have fixed seats opposite one another, and that there are two specified guests who must not be placed next to one another. Show that the number of ways in which the company can be placed is $(2n-2)!.(4n^2 - 6n + 4)$.

Q.9 Each of 3 committees has 1 vacancy which is to be filled from a group of 6 people. Find the number of ways the 3 vacancies can be filled if ;

(i) Each person can serve on atmost 1 committee.

(ii) There is no restriction on the number of committees on which a person can serve.

(iii) Each person can serve on atmost 2 committees.

Q.10 How many 15 letter arrangements of 5 A's, 5 B's and 5 C's have no A's in the first 5 letters, no B's in the next 5 letters, and no C's in the last 5 letters.

Q.11 5 balls are to be placed in 3 boxes. Each box can hold all 5 balls. In how many different ways can we place the balls so that no box remains empty if,

(i) balls & boxes are different (ii) balls are identical but boxes are different

(iii) balls are different but boxes are identical (iv) balls as well as boxes are identical

(v) balls as well as boxes are identical but boxes are kept in a row.

Q.12 In how many other ways can the letters of the word **MULTIPLE** be arranged;

(i) without changing the order of the vowels (ii) keeping the position of each vowel fixed &

(iii) without changing the relative order/position of vowels & consonants.

Q.13 Find the number of ways in which the number 30 can be partitioned into three unequal parts, each part being a natural number. What this number would be if equal parts are also included.

Q.14 In an election for the managing committee of a reputed club , the number of candidates contesting elections exceeds the number of members to be elected by r (r > 0). If a voter can vote in 967 different ways to elect the managing committee by voting atleast 1 of them & can vote in 55 different ways to elect (r – 1) candidates by

voting in the same manner. Find the number of candidates contesting the elections & the number of candidates losing the elections.

Q.15 Find the number of three digits numbers from 100 to 999 inclusive which have any one digit that is the average of the other two.

Q.16 Prove by combinatorial argument that :

(a) ${}^{n+1}C_r = {}^nC_r + {}^nC_{r-1}$ (b) ${}^{n+m}c_r = {}^nc_0 \cdot {}^mc_r + {}^nc_1 \cdot {}^mc_{r-1} + {}^nc_2 \cdot {}^mc_{r-2} + \ldots\ldots + {}^nc_r \cdot {}^mc_0$.

Q.17 A man has 3 friends. In how many ways he can invite one friend everyday for dinner on 6 successive nights so that no friend is invited more than 3 times.

Q.18 12 persons are to be seated at a square table, three on each side. 2 persons wish to sit on the north side and two wish to sit on the east side. One other person insists on occupying the middle seat (which may be on any side). Find the number of ways they can be seated.

Q.19 There are 15 rowing clubs; two of the clubs have each 3 boats on the river; five others have each 2 and the remaining eight have each 1; find the number of ways in which a list can be formed of the order of the 24 boats, observing that the second boat of a club cannot be above the first and the third above the second. How many ways are there in which a boat of the club having single boat on the river is at the third place in the list formed above?

Q.20 25 passengers arrive at a railway station & proceed to the neighbouring village. At the station there are 2 coaches accommodating 4 each & 3 carts accommodating 3 each. Find the number of ways in which they can proceed to the village assuming that the conveyances are always fully occupied & that the conveyances are all distinguishable from each other.

Q.21 An 8 oared boat is to be manned by a crew chosen from 14 men of which 4 can only steer but can not row & the rest can row but cannot steer. Of those who can row, 2 can row on the bow side. In how many ways can the crew be arranged.

Q.22 How many 6 digits odd numbers greater than 60,0000 can be formed from the digits 5, 6, 7, 8, 9, 0 if (i) repetitions are not allowed (ii) repetitions are allowed.

Q.23 Find the sum of all numbers greater than 10000 formed by using the digits 0 , 1 , 2 , 4 , 5 no digit being repeated in any number.

Q.24 The members of a chess club took part in a round robin competition in which each plays every one else once. All members scored the same number of points, except four juniors whose total score were 17.5. How many members were there in the club? Assume that for each win a player scores 1 point , for draw 1/2 point and zero for losing.

Q.25 There are 3 cars of different make available to transport 3 girsls and 5 boys on a field trip. Each car can hold up to 3 children. Find (a) the number of ways in which they can be accomodated.

(b) the numbers of ways in which they can be accomodated if 2 or 3 girls are assigned to one of the cars.
In both the cars internal arrangement of childrent inside the car is to be considered as immaterial.

Q.26 Six faces of an ordinary cubical die marked with alphabets A, B, C, D, E and F is thrown n times and the list of n alphabets showing up are noted. Find the total number of ways in which among the alphabets A, B, C, D, E and F only three of them appear in the list.

Q.27 Find the number of integer betwen 1 and 10000 with at least one 8 and at least one 9 as digits.

Q.28 The number of combinations n together of 3n letters of which n are'a' and n are'b' and the rest unlike is $(n+2).2^{n-1}$.

Q.29 In Indo–Pak one day International cricket match at Sharjah , India needs 14 runs to win just before the start of the final over. Find the number of ways in which India just manages to win the match (i.e. scores exactly 14 runs) , assuming that all the runs are made off the bat & the batsman can not score more than 4 runs off any ball.

Q.30 A man goes in for an examination in which there are 4 papers with a maximum of m marks for each paper; show that the number of ways of getting 2m marks on the whole is $\frac{1}{3}(m+1)(2m^2+4m+3)$.

EXERCISE–3

Q.1 Find the total number of ways of selecting five letters from the letters of the word INDEPENDENT.[REE '97, 6]

Q.2 Select the correct alternative(s). [JEE '98, 2 + 2]

(i) Number of divisors of the form $4n+2$ ($n \geq 0$) of the integer 240 is
(A) 4 (B) 8 (C) 10 (D) 3

(ii) An n-digit number is a positive number with exactly 'n' digits. Nine hundred distinct n-digit numbers are to be formed using only the three digits 2, 5 & 7. The smallest value of n for which this is possible is :
(A) 6 (B) 7 (C) 8 (D) 9

Q.3 How many different nine digit numbers can be formed from the number 223355888 by rearranging its digits so that the odd digits occupy even positions ? [JEE '2000, (Scr)]
(A) 16 (B) 36 (C) 60 (D) 180

Q.4 Let T_n denote the number of triangles which can be formed using the vertices of a regular polygon of ' n ' sides. If $T_{n+1} - T_n = 21$, then ' n ' equals: [JEE '2001, (Scr)]
(A) 5 (B) 7 (C) 6 (D) 4

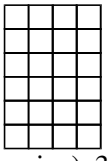
Q.5 The number of arrangements of the letters of the word BANANA in which the two N's do not appear adjacently is [JEE 2002 (Screening), 3]
(A) 40 (B) 60 (C) 80 (D) 100

Q.6 Number of points with integral co-ordinates that lie inside a triangle whose co-ordinates are (0, 0), (0, 21) and (21,0) [JEE 2003 (Screening), 3]
(A) 210 (B) 190 (C) 220 (D) None

Q.7 Using permutation or otherwise, prove that $\frac{(n^2)!}{(n!)^n}$ is an integer, where n is a positive integer. [JEE 2004, 2 out of 60]

Q.8 A rectangle with sides $2m-1$ and $2n-1$ is divided into squares of unit length by drawing

parallel lines as shown in the diagram, then the number of rectangles possible with odd side lengths is



(A) $(m+n+1)^2$ (B) 4^{m+n-1}
(C) m^2n^2 (D) $mn(m+1)(n+1)$

[JEE 2005 (Screening), 3]

Q.9 If *r*, *s*, *t* are prime numbers and p, q are the positive integers such that their LCM of p, q is is $r^2t^4s^2$, then the numbers of ordered pair of (p, q) is
(A) 252 (B) 254 (C) 225 (D) 224 [JEE 2006, 3]

EXERCISE–4

Part : (A) Only one correct option

1. There are 2 identical white balls, 3 identical red balls and 4 green balls of different shades. The number of ways in which they can be arranged in a row so that atleast one ball is separated from the balls of the same colour, is:
(A) 6 (7 ! – 4 !) (B) 7 (6 ! – 4 !) (C) 8 ! – 5 ! (D) none

2. The number of permutations that can be formed by arranging all the letters of the word 'NINETEEN' in which no two E's occur together is

(A) $\frac{8!}{3!\,3!}$ (B) $\frac{5!}{3!\times {}^6C_2}$ (C) $\frac{5!}{3!}\times {}^6C_3$ (D) $\frac{8!}{5!}\times {}^6C_3$.

3. The number of ways in which n different things can be given to r persons when there is no restriction as to the number of things each may receive is:
(A) ${}^nC_r$ (B) ${}^nP_r$ (C) n^r (D) r^n

4. The number of divisors of $a^pb^qc^rd^s$ where a, b, c, d are primes & p, q, r, s $\in$ N, excluding 1 and the number itself is:
(A) pqrs (B) (p + 1) (q + 1) (r + 1) (s + 1) – 4
(C) pqrs – 2 (D) (p + 1) (q + 1) (r + 1) (s + 1) – 2

5. The number of ordered triplets of positive integers which are solutions of the equation x + y + z = 100 is:
(A) 3125 (B) 5081 (C) 6005 (D) 4851

6. Number of ways in which 7 people can occupy six seats, 3 seats on each side in a first class railway compartment if two specified persons are to be always included and occupy adjacent seats on the same side, is (k). 5 ! then k has the value equal to:
(A) 2 (B) 4 (C) 8 (D) none

7. Number of different words that can be formed using all the letters of the word "DEEPMALA" if two vowels are together and the other two are also together but separated from the first two is:
(A) 960 (B) 1200 (C) 2160 (D) 1440

8. Six persons A, B, C, D, E and F are to be seated at a circular table. The number of ways this can be done if A must have either B or C on his right and B must have either C or D on his right is:
(A) 36 (B) 12 (C) 24 (D) 18

9. The number of ways in which 15 apples & 10 oranges can be distributed among three persons, each receiving none, one or more is:
(A) 5670 (B) 7200 (C) 8976 (D) none of these

10. The number of permutations which can be formed out of the letters of the word "SERIES" taking three letters together is:
(A) 120 (B) 60 (C) 42 (D) none

11. Seven different coins are to be divided amongst three persons. If no two of the persons receive the same number of coins but each receives atleast one coin & none is left over, then the number of ways in which the division may be made is:
(A) 420 (B) 630 (C) 710 (D) none

12. The streets of a city are arranged like the lines of a chess board. There are m streets running North to South & 'n' streets running East to West. The number of ways in which a man can travel from NW to SE corner going the shortest possible distance is:

(A) $\sqrt{m^2+n^2}$ (B) $\sqrt{(m-1)^2\,.\,(n-1)^2}$ (C) $\frac{(m+n)!}{m!\,.\,n!}$ (D) $\frac{(m+n-2)!}{(m-1)!\,.\,(n-1)!}$

13. In a conference 10 speakers are present. If S_1 wants to speak before S_2 & S_2 wants to speak after S_3, then the number of ways all the 10 speakers can give their speeches with the above restriction if the remaining seven speakers have no objection to speak at any number is:

(A) ${}^{10}C_3$ (B) ${}^{10}P_8$ (C) ${}^{10}P_3$ (D) $\frac{10!}{3}$

14. Two variants of a test paper are distributed among 12 students. Number of ways of seating of the students in two rows so that the students sitting side by side do not have identical papers & those sitting in the same column have the same paper is:

(A) $\frac{12!}{6!\,6!}$ (B) $\frac{(12)!}{2^5\,.\,6!}$ (C) $(6!)^2.\,2$ (D) $12!\times 2$

15. Sum of all the numbers that can be formed using all the digits 2, 3, 3, 4, 4, 4 is:
(A) 22222200 (B) 11111100 (C) 55555500 (D) 20333280

16. There are m apples and n oranges to be placed in a line such that the two extreme fruits being both oranges. Let P denotes the number of arrangements if the fruits of the same species are different and Q the corresponding figure when the fruits of the same species are alike, then the ratio P/Q has the value equal to:
(A) ${}^nP_2.\,{}^mP_m.\,(n-2)!$ (B) ${}^mP_2.\,{}^nP_n.\,(n-2)!$ (C) ${}^nP_2.\,{}^nP_n.\,(m-2)!$ (D) none

17. The number of integers which lie between 1 and 10^6 and which have the sum of the digits equal to 12 is:
(A) 8550 (B) 5382 (C) 6062 (D) 8055

18. Number of ways in which a pack of 52 playing cards be distributed equally among four players so that each may have the Ace, King, Queen and Jack of the same suit is:

(A) $\frac{36!}{(9!)^4}$ (B) $\frac{36!\,.\,4!}{(9!)^4}$ (C) $\frac{36!}{(9!)^4 10\,4!}$ (D) none

19. A five letter word is to be formed such that the letters appearing in the odd numbered positions are taken from the

letters which appear without repetition in the word "MATHEMATICS". Further the letters appearing in the even numbered positions are taken from the letters which appear with repetition in the same word "MATHEMATICS". The number of ways in which the five letter word can be formed is:
(A) 720 (B) 540 (C) 360 (D) none

20. Number of ways of selecting 5 coins from coins three each of Rs. 1, Rs. 2 and Rs. 5 if coins of the same denomination are alike, is:
(A) 9 (B) 12 (C) 21 (D) none

21. Number of ways in which all the letters of the word " ALASKA " can be arranged in a circle distinguishing between the clockwise and anticlockwise arrangement is:
(A) 60 (B) 40 (C) 20 (D) none of these

22. If r, s, t are prime numbers and p, q are the positive integers such that the LCM of p, q is $r^2t^4s^2$, then the number of ordered pair (p, q) is **[IIT – 2006]**
(A) 252 (B) 254 (C) 225 (D) 224

Part : (B) May have more than one options correct

23. ${}^{n+1}C_6 + {}^nC_4 > {}^{n+2}C_5 - {}^nC_5$ for all 'n' greater than:
(A) 8 (B) 9 (C) 10 (D) 11

24. In an examination, a candidate is required to pass in all the four subjects he is studying. The number of ways in which he can fail is
(A) ${}^4P_1 + {}^4P_2 + {}^4P_3 + {}^4P_4$ (B) $4^4 - 1$ (C) $2^4 - 1$ (D) ${}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4$

25. The kindergarten teacher has 25 kids in her class. She takes 5 of them at a time, to zoological garden as often as she can, without taking the same 5 kids more than once. Then the number of visits, the teacher makes to the garden exceeds that of a kid by:
(A) ${}^{25}C_5 - {}^{24}C_4$ (B) ${}^{24}C_5$ (C) ${}^{25}C_5 - {}^{24}C_5$ (D) ${}^{24}C_4$

26. The number of ways of arranging the letters AAAAA, BBB, CCC, D, EE & F in a row if the letter C are separated from one another is:
(A) ${}^{13}C_3 \cdot \dfrac{12!}{5!\,3!\,2!}$ (B) $\dfrac{13!}{5!\,3!\,3!\,2!}$ (C) $\dfrac{14!}{3!\,3!\,2!}$ (D) 11. $\dfrac{13!}{6!}$

27. There are 10 points $P_1, P_2, \ldots, P_{10}$ in a plane, no three of which are collinear. Number of straight lines which can be determined by these points which do not pass through the points P_1 or P_2 is:
(A) ${}^{10}C_2 - 2.{}^9C_1$ (B) 27 (C) ${}^8C_2$ (D) ${}^{10}C_2 - 2.{}^9C_1 + 1$

28. Number of quadrilaterals which can be constructed by joining the vertices of a convex polygon of 20 sides if none of the side of the polygon is also the side of the quadrilateral is:
(A) ${}^{17}C_4 - {}^{15}C_2$ (B) $\dfrac{{}^{15}C_3 \cdot 20}{4}$ (C) 2275 (D) 2125

29. You are given 8 balls of different colour (black, white,...). The number of ways in which these balls can be arranged in a row so that the two balls of particular colour (say red & white) may never come together is:
(A) $8! - 2.7!$ (B) $6.7!$ (C) $2.6!.{}^7C_2$ (D) none

30. A man is dealt a poker hand (consisting of 5 cards) from an ordinary pack of 52 playing cards. The number of ways in which he can be dealt a "straight" (a straight is five consecutive values not of the same suit, eg. {Ace, 2, 3, 4, 5}, {2, 3, 4, 5, 6}..... & {10, J, Q, K, Ace}) is
(A) $10(4^5 - 4)$ (B) $4!.2^{10}$ (C) 10.2^{10} (D) 10200

31. Number of ways in which 3 numbers in A.P. can be selected from 1, 2, 3,..... n is:
(A) $\left(\dfrac{n-1}{2}\right)^2$ if n is even (B) $\dfrac{n(n-2)}{4}$ if n is odd (C) $\dfrac{(n-1)^2}{4}$ if n is odd (D) $\dfrac{n(n-2)}{4}$ if n is even

32. Consider the expansion $(a_1 + a_2 + a_3 + \ldots\ldots + a_p)^n$ where $n \in N$ and $n \le p$. The correct statement(s) is/are:
(A) number of different terms in the expansion is ${}^{n+p-1}C_n$
(B) co-efficient of any term in which none of the variables $a_1, a_2 \ldots, a_p$ occur more than once is 'n'
(C) co-efficient of any term in which none of the variables $a_1, a_2, \ldots, a_p$ occur more than once is n ! if n = p
(D) Number of terms in which none of the variables $a_1, a_2, \ldots\ldots, a_p$ occur more than once is $\binom{p}{n}$.

EXERCISE–5

1. In a telegraph communication how many words can be communicated by using atmost 5 symbols. (only dot and dash are used as symbols)

2. If all the letters of the word 'AGAIN' are arranged in all possible ways & put in dictionary order, what is the 50th word.

3. A committee of 6 is to be chosen from 10 persons with the condition that if a particular person 'A' is chosen, then another particular person B must be chosen.

4. A family consists of a grandfather, m sons and daughters and 2n grand children. They are to be seated in a row for dinner. The grand children wish to occupy the n seats at each end and the grandfather refuses to have a grand children on either side of him. In how many ways can the family be made to sit?

5. The sides AB, BC & CA of a triangle ABC have 3, 4 & 5 interior points respectively on them. Find the number of triangles that can be constructed using these interior points as vertices.

6. How many five digits numbers divisible by 3 can be formed using the digits 0, 1, 2, 3, 4, 7 and 8 if, each digit is to be used atmost one.

7. In how many other ways can the letters of the word MULTIPLE be arranged ; (i) without changing the order of the vowels (ii) keeping the position of each vowel fixed (iii) without changing the relative order/position of vowels & consonants.

8. There are p intermediate stations on a railway line from one terminus to another. In how many ways can a train stop at 3 of these intermediate stations if no 2 of these stopping stations are to be consecutive?

9. Find the number of positive integral solutions of $x + y + z + w = 20$ under the following conditions:
(i) Zero values of x, y, z, w are include (ii) Zero values are excluded
(iii) No variable may exceed 10; Zero values excluded (iv) Each variable is an odd number
(v) x, y, z, w have different values (zero excluded).

10. Find the number of words each consisting of 3 consonants & 3 vowels that can be formed from the letters of the word **"CIRCUMFERENCE".** In how many of these C's will be together.

11. If 'n' distinct things are arranged in a circle, show that the number of ways of selecting three of these things so that no two of them are next to each other is, $\dfrac{1}{6}\, n\,(n-4)^{11}(n-5)$.

12. In maths paper there is a question on "Match the column" in which column A contains 6 entries & each entry of column A corresponds to exactly one of the 6 entries given in column B written randomly. 2 marks are awarded for each correct matching & 1 mark is deducted from each incorrect matching. A student having no subjective knowledge decides to match all the 6 entries randomly. Find the number of ways in which he can answer, to get atleast 25 % marks in this question.

13. Show that the number of combinations of n letters together out of 3n letters of which n are a and n are b and the rest unlike is, $(n + 2).\ 2^{n-1}$.

14. Find the number of positive integral solutions of, (i) $x^2 - y^2 = 352706$ (ii) $xyz = 21600$

15. There are 'n' straight line in a plane, no two of which are parallel and no three pass through the same point. Their points of intersection are joined. Show that the number of fresh lines thus introduced is, $\frac{1}{8}\ n\ (n-1)\ (n-2)\ (n-3)$.

16. A forecast is to be made of the results of five cricket matches, each of which can be a win or a draw or a loss for Indian team. Find
(i) number of forecasts with exactly 1 error (ii) number of forecasts with exactly 3 errors
(iii) number of forecasts with all five errors

17. Prove by permutation or otherwise $\frac{(n^2)!}{(n!)^n}$ is an integer $(n \in I^+)$. **[IIT – 2004]**

18. If total number of runs scored in n matches is $\left(\frac{n+1}{4}\right)(2^{n+1} - n - 2)$ where $n > 1$, and the rund scored in the kth match are given by $k.\ 2^{n+1-k}$, where $1 \le k \le n$. Find n **[IIT – 2005]**

EXERCISE–1

Q.1 ${}^{m+n+k}C_3 - ({}^mC_3 + {}^nC_3 + {}^kC_3)$ **Q.2** 744 **Q.3** 13 , 156

Q.4 4316527 **Q.5** 43200 **Q.6** 145 **Q.7** 420

Q.8 960 **Q.9** ${}^{24}C_2\ .\ {}^{15}C_3$ **Q.10** 1106 **Q.11** $12!\ ;\ \frac{11!\ .\ 4!}{(3!)^4\ 2!}$

Q.13 $\frac{52!}{(13!)^4}\ ;\ \frac{52!}{3!(17!)^3}$ **Q.14** 5400 **Q.15** ${}^{45}C_6$ **Q.17** 1638

Q.18 22100 , 52 **Q.19** 2111 **Q.20** 576 **Q.21** (a) $5 \cdot (6!)$, (b) $3! \cdot 4!$, (c) 12

Q.23 **(a)** 72 ; 78120 ; **(b)** 23 ; **(c)** 32 **Q.24** 143 **Q.25** 440 **Q.26** ${}^{n+5}C_5$

Q.27 3888 **Q.28** $\frac{(14)!}{5!9!}$ **Q.29** **(ii)** 792 ; **(iii)** $\frac{143}{4025}$; **(v)** r = 3

EXERCISE–2

Q.1 2500 **Q.2** ${}^mC_{k-1}$ where $m = (1/2)\ (2n - k^2 + k - 2)$

Q.3 $7^{10}\ ;\ \left(\frac{49}{6}\right)\ 10\ !$ **Q.4** 24 **Q.5** 532770 **Q.6** (i) 15, (ii) 126, (iii) 60, (iv) 105

Q.7 26250 **Q.9** 120, 216, 210 **Q.10** 2252 **Q.11** **(i)** 150 ; **(ii)** 6 ; **(iii)** 25 ; **(iv)** 2 ; **(v)** 6

Q.12 **(i)** 3359 ; **(ii)** 59 ; **(iii)** 359 **Q.13** 61, 75 **Q.14** 10, 3

Q.15 121 **Q.17** 510 **Q.18** 2 ! 3 ! 8 !

Q.19 $\frac{24!}{(3!)^2(2!)^5}\ ;\ {}^8C_1 \cdot \frac{23!}{(3!)^2(2!)^5}$ **Q.20** $\frac{(25)\ !}{(3!)^3\ (4!)^4\ .\ 4}$ **Q.21** $4\ .\ (4!)^2\ .\ {}^8C_4\ .\ {}^6C_2$

Q.22 240 , 15552 **Q.23** 3119976 **Q.24** 27 **Q.25** (a) 1680; (b) 1140

Q.26 ${}^6C_3[3^n - {}^3C_1(2^n - 2) - {}^3C_2]$ **Q.27** 974 **Q.29** 1506

EXERCISE–3

Q.1 72 **Q.2** **(i)** A ; **(ii)** B **Q.3** C **Q.4** B

Q.5 A **Q.6** B **Q.8** C **Q.9** C

EXERCISE–4

1. A **2.** C **3.** D **4.** D **5.** D **6.** C **7.** D **8.** D **9.** C **10.** C **11.** B

12. D **13.** D **14.** D **15.** A **16.** A **17.** C **18.** B **19.** B **20.** B **21.** C **22.** C

23. BCD **24.** CD **25.** AB **26.** AD **27.** CD **28.** AB **29.** ABC **30.** AD **31.** CD

32. ACD

EXERCISE–5

1. 62 **2.** NAAIG **3.** 154 **4.** (2n)! m! (m – 1) **5.** 205

6. 744 **7.** (i) 3359 (ii) 59 (iii) 359 **8.** ${}^{p-2}C_3$

9. (i) ${}^{23}C_3$ (ii) ${}^{19}C_3$ (iii) ${}^{19}C_3 - 4.{}^9C_3$ (iv) ${}^{11}C_8$ (v) 552

10. 22100, 52 **12.** 56 ways **14.** (i) Zero (ii) 1260 **16.** (i) 10 (ii) 80 (iii) 32 **18.** 7

EXERCISE–4

Part : (A) Only one correct option

1. There are 2 identical white balls, 3 identical red balls and 4 green balls of different shades. The number of ways in which they can be arranged in a row so that atleast one ball is separated from the balls of the same colour, is:
(A) 6 (7 ! – 4 !) (B) 7 (6 ! – 4 !) (C) 8 ! – 5 ! (D) none

2. The number of permutations that can be formed by arranging all the letters of the word 'NINETEEN' in which no two E's occur together is
(A) $\frac{8!}{3!\,3!}$ (B) $\frac{5!}{3! \times {}^6C_2}$ (C) $\frac{5!}{3!} \times {}^6C_3$ (D) $\frac{8!}{5!} \times {}^6C_3$.

3. The number of ways in which n different things can be given to r persons when there is no restriction as to the number of things each may receive is:
(A) ${}^nC_r$ (B) ${}^nP_r$ (C) n^r (D) r^n

4. The number of divisors of $a^pb^qc^rd^s$ where a, b, c, d are primes & p, q, r, s $\in$ N, excluding 1 and the number itself is:
(A) pqrs (B) (p + 1) (q + 1) (r + 1) (s + 1) – 4
(C) pqrs – 2 (D) (p + 1) (q + 1) (r + 1) (s + 1) – 2

5. The number of ordered triplets of positive integers which are solutions of the equation x + y + z = 100 is:
(A) 3125 (B) 5081 (C) 6005 (D) 4851

6. Number of ways in which 7 people can occupy six seats, 3 seats on each side in a first class railway compartment if two specified persons are to be always included and occupy adjacent seats on the same side, is (k). 5 ! then k has the value equal to:
(A) 2 (B) 4 (C) 8 (D) none

7. Number of different words that can be formed using all the letters of the word "DEEPMALA" if two vowels are together and the other two are also together but separated from the first two is:
(A) 960 (B) 1200 (C) 2160 (D) 1440

8. Six persons A, B, C, D, E and F are to be seated at a circular table. The number of ways this can be done if A must have either B or C on his right and B must have either C or D on his right is:
(A) 36 (B) 12 (C) 24 (D) 18

9. The number of ways in which 15 apples & 10 oranges can be distributed among three persons, each receiving none, one or more is:
(A) 5670 (B) 7200 (C) 8976 (D) none of these

10. The number of permutations which can be formed out of the letters of the word "SERIES" taking three letters together is:
(A) 120 (B) 60 (C) 42 (D) none

11. Seven different coins are to be divided amongst three persons. If no two of the persons receive the same number of coins but each receives atleast one coin & none is left over, then the number of ways in which the division may be made is:
(A) 420 (B) 630 (C) 710 (D) none

12. The streets of a city are arranged like the lines of a chess board. There are m streets running North to South & 'n' streets running East to West. The number of ways in which a man can travel from NW to SE corner going the shortest possible distance is:
(A) $\sqrt{m^2+n^2}$ (B) $\sqrt{(m-1)^2\,.\,(n-1)^2}$ (C) $\frac{(m+n)!}{m!\,.\,n!}$ (D) $\frac{(m+n-2)!}{(m-1)!\,.\,(n-1)!}$

13. In a conference 10 speakers are present. If S_1 wants to speak before S_2 & S_2 wants to speak after S_3, then the number of ways all the 10 speakers can give their speeches with the above restriction if the remaining seven speakers have no objection to speak at any number is:
(A) ${}^{10}C_3$ (B) ${}^{10}P_8$ (C) ${}^{10}P_3$ (D) $\frac{10!}{3}$

14. Two variants of a test paper are distributed among 12 students. Number of ways of seating of the students in two rows so that the students sitting side by side do not have identical papers & those sitting in the same column have the same paper is:

(A) $\frac{12!}{6!\,6!}$ (B) $\frac{(12)!}{2^5 . 6!}$ (C) $(6!)^2 . 2$ (D) $12! \times 2$

15. Sum of all the numbers that can be formed using all the digits 2, 3, 3, 4, 4, 4 is:
(A) 22222200 (B) 11111100 (C) 55555500 (D) 20333280

16. There are m apples and n oranges to be placed in a line such that the two extreme fruits being both oranges. Let P denotes the number of arrangements if the fruits of the same species are different and Q the corresponding figure when the fruits of the same species are alike, then the ratio P/Q has the value equal to:
(A) ${}^nP_2 . {}^mP_m . (n-2)!$ (B) ${}^mP_2 . {}^nP_n . (n-2)!$ (C) ${}^nP_2 . {}^nP_n . (m-2)!$ (D) none

17. The number of integers which lie between 1 and 10^6 and which have the sum of the digits equal to 12 is:
(A) 8550 (B) 5382 (C) 6062 (D) 8055

18. Number of ways in which a pack of 52 playing cards be distributed equally among four players so that each may have the Ace, King, Queen and Jack of the same suit is:

(A) $\frac{36!}{(9!)^4}$ (B) $\frac{36! . 4!}{(9!)^4}$ (C) $\frac{36!}{(9!)^4 . 4!}$ (D) none

19. A five letter word is to be formed such that the letters appearing in the odd numbered positions are taken from the letters which appear without repetition in the word "MATHEMATICS". Further the letters appearing in the even numbered positions are taken from the letters which appear with repetition in the same word "MATHEMATICS". The number of ways in which the five letter word can be formed is:
(A) 720 (B) 540 (C) 360 (D) none

20. Number of ways of selecting 5 coins from coins three each of Rs. 1, Rs. 2 and Rs. 5 if coins of the same denomination are alike, is:
(A) 9 (B) 12 (C) 21 (D) none

21. Number of ways in which all the letters of the word "ALASKA" can be arranged in a circle distinguishing between the clockwise and anticlockwise arrangement is:
(A) 60 (B) 40 (C) 20 (D) none of these

22. If r, s, t are prime numbers and p, q are the positive integers such that the LCM of p, q is $r^2t^4s^2$, then the number of ordered pair (p, q) is **[IIT – 2006]**
(A) 252 (B) 254 (C) 225 (D) 224

Part : (B) May have more than one options correct

23. ${}^{n+1}C_6 + {}^nC_4 > {}^{n+2}C_5 - {}^nC_5$ for all 'n' greater than:
(A) 8 (B) 9 (C) 10 (D) 11

24. In an examination, a candidate is required to pass in all the four subjects he is studying. The number of ways in which he can fail is
(A) ${}^4P_1 + {}^4P_2 + {}^4P_3 + {}^4P_4$ (B) $4^4 - 1$
(C) $2^4 - 1$ (D) ${}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4$

25. The kindergarten teacher has 25 kids in her class. She takes 5 of them at a time, to zoological garden as often as she can, without taking the same 5 kids more than once. Then the number of visits, the teacher makes to the garden exceeds that of a kid by:
(A) ${}^{25}C_5 - {}^{24}C_4$ (B) ${}^{24}C_5$ (C) ${}^{25}C_5 - {}^{24}C_5$ (D) ${}^{24}C_4$

26. The number of ways of arranging the letters AAAAA, BBB, CCC, D, EE & F in a row if the letter C are separated from one another is:

(A) ${}^{13}C_3 . \frac{12!}{5!\,3!\,2!}$ (B) $\frac{13!}{5!\,3!\,3!\,2!}$ (C) $\frac{14!}{3!\,3!\,2!}$ (D) $11 . \frac{13!}{6!}$

27. There are 10 points $P_1, P_2, ..., P_{10}$ in a plane, no three of which are collinear. Number of straight lines which can be determined by these points which do not pass through the points P_1 or P_2 is:
(A) ${}^{10}C_2 - 2 . {}^9C_1$ (B) 27 (C) ${}^8C_2$ (D) ${}^{10}C_2 - 2 . {}^9C_1 + 1$

28. Number of quadrilaterals which can be constructed by joining the vertices of a convex polygon of 20 sides if none of the side of the polygon is also the side of the quadrilateral is:

(A) ${}^{17}C_4 - {}^{15}C_2$ (B) $\frac{{}^{15}C_3 . 20}{4}$ (C) 2275 (D) 2125

29. You are given 8 balls of different colour (black, white, ...). The number of ways in which these balls can

be arranged in a row so that the two balls of particular colour (say red & white) may never come together is:
(A) $8! - 2.7!$ (B) $6.7!$ (C) $2.6!.{}^{7}C_{2}$ (D) none

30. A man is dealt a poker hand (consisting of 5 cards) from an ordinary pack of 52 playing cards. The number of ways in which he can be dealt a "straight" (a straight is five consecutive values not of the same suit, eg. {Ace, 2, 3, 4, 5}, {2, 3, 4, 5, 6}..... & {10, J, Q, K, Ace}) is
(A) $10(4^5 - 4)$ (B) $4!.2^{10}$ (C) 10.2^{10} (D) 10200

31. Number of ways in which 3 numbers in A.P. can be selected from 1, 2, 3,..... n is:
(A) $\left(\frac{n-1}{2}\right)^2$ if n is even (B) $\frac{n(n-2)}{4}$ if n is odd
(C) $\frac{(n-1)^2}{4}$ if n is odd (D) $\frac{n(n-2)}{4}$ if n is even

32. Consider the expansion $(a_1 + a_2 + a_3 + ... + a_p)^n$ where $n \in N$ and $n \le p$. The correct statement(s) is/ are:
(A) number of different terms in the expansion is ${}^{n+p-1}C_n$
(B) co-efficient of any term in which none of the variables $a_1, a_2 ..., a_p$ occur more than once is 'n'
(C) co-efficient of any term in which none of the variables $a_1, a_2, ..., a_p$ occur more than once is n ! if n = p
(D) Number of terms in which none of the variables $a_1, a_2, ..., a_p$ occur more than once is $\binom{p}{n}$.

EXERCISE–5

1. In a telegraph communication how many words can be communicated by using atmost 5 symbols. (only dot and dash are used as symbols)

2. If all the letters of the word 'AGAIN' are arranged in all possible ways & put in dictionary order, what is the 50th word.

3. A committee of 6 is to be chosen from 10 persons with the condition that if a particular person 'A' is chosen, then another particular person B must be chosen.

4. A family consists of a grandfather, m sons and daughters and 2n grand children. They are to be seated in a row for dinner. The grand children wish to occupy the n seats at each end and the grandfather refuses to have a grand children on either side of him. In how many ways can the family be made to sit?

5. The sides AB, BC & CA of a triangle ABC have 3, 4 & 5 interior points respectively on them. Find the number of triangles that can be constructed using these interior points as vertices.

6. How many five digits numbers divisible by 3 can be formed using the digits 0, 1, 2, 3, 4, 7 and 8 if, each digit is to be used atmost one.

7. In how many other ways can the letters of the word MULTIPLE be arranged ; (i) without changing the order of the vowels (ii) keeping the position of each vowel fixed (iii) without changing the relative order/ position of vowels & consonants.

8. There are p intermediate stations on a railway line from one terminus to another. In how many ways can a train stop at 3 of these intermediate stations if no 2 of these stopping stations are to be consecutive?

9. Find the number of positive integral solutions of $x + y + z + w = 20$ under the following conditions:
(i) Zero values of x, y, z, w are include
(ii) Zero values are excluded
(iii) No variable may exceed 10; Zero values excluded
(iv) Each variable is an odd number
(v) x, y, z, w have different values (zero excluded).

10. Find the number of words each consisting of 3 consonants & 3 vowels that can be formed from the letters of the word **"CIRCUMFERENCE".** In how many of these C's will be together.

11. If ' n ' distinct things are arranged in a circle, show that the number of ways of selecting three of these

things so that no two of them are next to each other is, $\frac{1}{6}$ n (n – 4) (n – 5).

12. In maths paper there is a question on "Match the column" in which column A contains 6 entries & each entry of column A corresponds to exactly one of the 6 entries given in column B written randomly. 2 marks are awarded for each correct matching & 1 mark is deducted from each incorrect matching. A student having no subjective knowledge decides to match all the 6 entries randomly. Find the number of ways in which he can answer, to get atleast 25 % marks in this question.

13. Show that the number of combinations of n letters together out of 3n letters of which n are a and n are b and the rest unlike is, $(n + 2).\ 2^{n-1}$.

14. Find the number of positive integral solutions of, (i) $x^2 - y^2 = 352706$ (ii) $xyz = 21600$

15. There are 'n' straight line in a plane, no two of which are parallel and no three pass through the same point. Their points of intersection are joined. Show that the number of fresh lines thus introduced is,

$\frac{1}{8}$ n (n – 1) (n – 2) (n – 3).

16. A forecast is to be made of the results of five cricket matches, each of which can be a win or a draw or a loss for Indian team. Find

(i) number of forecasts with exactly 1 error
(ii) number of forecasts with exactly 3 errors
(iii) number of forecasts with all five errors

17. Prove by permutation or otherwise $\frac{(n^2)!}{(n!)^n}$ is an integer ($n \in I^+$). **[IIT – 2004]**

18. If total number of runs scored in n matches is $\left(\frac{n+1}{4}\right)(2^{n+1} - n - 2)$ where $n > 1$, and the rund scored in the kth match are given by $k.\ 2^{n+1-k}$, where $1 \le k \le n$. Find n **[IIT – 2005]**

ANSWER KEY

EXERCISE–4

1. A **2.** C **3.** D **4.** D **5.** D **6.** C **7.** D

8. D **9.** C **10.** C **11.** B **12.** D **13.** D **14.** D

15. A **16.** A **17.** C **18.** B **19.** B **20.** B **21.** C

22. C **23.** BCD **24.** CD **25.** AB **26.** AD

27. CD **28.** AB **29.** ABC **30.** AD **31.** CD **32.** ACD

EXERCISE–5

1. 62 **2.** NAAIG **3.** 154

4. (2n)! m! (m – 1) **5.** 205 **6.** 744

7. (i) 3359 (ii) 59 (iii) 359

8. ${}^{p-2}C_3$

9. (i) ${}^{23}C_3$ (ii) ${}^{19}C_3$ (iii) ${}^{19}C_3 - 4.{}^{9}C_3$ (iv) ${}^{11}C_8$ (v) 552

10. 22100, 52 **12.** 56 ways

14. (i) Zero (ii) 1260 **16.** (i) 10 (ii) 80 (iii) 32

18. 7

Some questions (Assertion–Reason type) are given below. Each question contains **Statement – 1 (Assertion)** and **Statement – 2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. So select the correct choice :

(A) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is** a correct explanation for **Statement – 1**.
(B) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is NOT** a correct explanation for **Statement – 1**.
(C) **Statement – 1** is True, **Statement – 2** is False.
(D) **Statement – 1** is False, **Statement – 2** is True.

399. **Statement-1:** $51 \times 52 \times 53 \times 54 \times 55 \times 56 \times 57 \times 58$ is divisible by 40320
Statement-2: The product of r consecutive natural numbers is always divisible by r!

400. **Statement-1:** Domain is $\{d_1, d_2, d_3, d_4\}$, range is $\{r_1, r_2, r_3\}$. Number of into functions which can be made is 45.
Statement-2: Numbers of into function = number of all functions – number of onto functions.
$= \mathbf{3^4 -} 3({}^4C_2 \cdot {}^2C_1) = 81 - 36 = 45$ of d_1, d_2, d_3, d_4 any two correspond to r_1, remaining two to r_2, r_3 one with each
$\therefore {}^4C_2 \times {}^2C_1 = 12$, total $= 12 \times 3 = 36$ = number of onto functions.

401. **Statement-1:** The smallest number which has 24 divisors is 420.
Statement-2: $24 = 3 \times 2 \times 2 = (2+1)(1+1)(1+1)(1+1)$, therefore, prime factors of the number are 2, 2, 3, 5, 7 & their product is 420.

402. Consider the word 'SMALL'
Statement–1 : Total number of 3 letter words from the letters of the given word is 13.
Statement–2 : Number of words having all the letters distinct = 4 and number of words having two are alike and third different = 9

403. **Statement–1 :** Number of non integral solution of the equation $x_1 + x_2 + x_3 = 10$ is equal to 34.
S–2 : Number of non integral solution of the equation $x_1 + x_2 + x_3 + \ldots x_n = r$ is equal to ${}^{n+r-1}C_r$

404. **Statement–1 :** ${}^{10}C_r = {}^{10}C_4 \Rightarrow r = 4$ or 6 **Statement–2 :** ${}^nC_r = {}^nC_{n-r}$

405. **Statement–1 :** The number of ways of arranging n boys and n girls in a circle such that no two boys are consecutive, is $(\lfloor n-1)^2$.
Statement–2 : The number of ways of arranging n distinct objects in a circle is $\lfloor n-1$

406. **Statement–1 :** The number of ways of selecting 5 students from 12 students (of which six are boys and six are girls), such that in the selection there are at least three girls is ${}^6C_3 \times {}^9C_2$.
Statement–2 : If a work has two independent parts, of which first part can be done in m way and for each choice of first part, the second part can be done in n ways, then the work can be completed in $m \times n$ ways.

407. **Statement–1 :** The number of ways of writing 1400 as a product of two positive integers is 12.
Statement–2 : 1400 is divisible by exactly three prime numbers.

408. **Statement–1 :** The number of selections of four letters taken from the word 'PARALLEL' must be 15.
Statement–2 : Coefficient of x^4 in the expansion of $(1-x)^{-3}$ is 15.

409. **Statement–1 :** Total number of permutation of n things of which p are alike of one kind, q are alike of 2nd kind, r are alike of 3rd kind and rest are all difference is $\frac{n!}{p!q!r!}$.
Statement–2 : Total number of selection from n identical object is n.

410. **Statement–1 :** A polygon has 44 diagonals and number of sides are 11.
Statement–2 : From n distinct object r object can be selected in ${}^nC_r$ ways.

411. Let $y = x + 3$, $y = 2x + 3$, $y = 3x + 2$ and $y + x = 3$ are four straight lines
Statement-1 : The number of triangles formed is ${}^4C_3$
Statement-2 : Number of distinct point of intersection between various lines will determine the number of possible triangle.

412. **Statement-1 :** The total number of positive integral solutions (zero included) of $x + y + z + \omega = 20$ without restriction is ${}^{23}C_{20}$
Statement-2 : Number of ways of distributing n identical items among m persons when each person gets zero or more items = ${}^{m+n-1}C_n$

413. **Statement-1 :** The total ways of selection of 5 objects out of $n(n \geq 5)$ identical objects is one.
Statement-2: If objects are identical then total ways of selection of any number of objects from given objects is one.

414. **Statement-1:** The total number of different 3-digits number of type N = abc, where $a < b < c$ is 84.
Statement-2: O cannot appear at any position, so total numbers are ${}^{9}C_3$.

415. **Statement-1:** The number of positive integral solutions of the equation $x_1x_2x_3x_4x_5 = 1050$ is 1875.
Statement-2: The total number of divisor of 1050 is 25.

416. **Statement-1:** $\left(\sum_{r=0}^{100} {}^{500-r}C_3\right) + {}^{400}C_4 = {}^{501}C_4$ **Statement-2 :** ${}^{n}C_r + {}^{n}C_{r-1} = {}^{n+1}C_r$

417. **Statement-1 :** $\frac{(n^2)!}{(n!)^n}$ is a natural number for all $n \in N$

S-2 : The number of ways of distributing mn things in m groups each containing n things is $\frac{(mn)!}{(n!)^m}$.

418. **Statement-1:** The number of divisors of 10, 800 is 60.
Statement-2: The number of odd divisors of 10, 800 is 12.

419. **Statement-1:** Number of onto functions from $A \to B$ where A contains n elements 2B contains m elements (where $n \geq m$) $= m^n - {}^{m}C_1 (m-1)^n + {}^{m}C_2 (m-2)^n + ...$
Statement-2: Number of ways of putting 5 identical balls in 3 different boxes when empty boxes are not allowed are 6.

420. **Statement-1 :** 4 persons can be seated in a row containing 12 chairs, such that no two of them are consecutive in ${}^{9}C_4 \times 4!$ ways
S-2: Number of non-negative integral solutions of equation $x_1+x_2+...+ x_r = n$ is = ${}^{n+r-1}C_{r-1}$.

421. **Statement-1:** The number of selections of four letters taken from the word PARALLEL must be 22.
Statement-2: Coefficient of x^4 in the expansion of $(1 - x)^3$ is 10.

422. **Statement-1:** Number of permutations of n dissimilar things taken 'n' at a time is ${}^{n}P_n$.
Statement-2: $n(A) = n(B) = n$ then the total number of functions from A to B are n!

423. **Statement-1:** Number of permutations of n dissimilar things taken n at a time in ${}^{n}P_n$.
Statement-2: $n(A) = n(B) = n$ then the total number of functions from A to B are n!

424. **Statement-1:** ${}^{n}C_r = {}^{n}C_p \Rightarrow r = p$ or $r + p = n$ **Statement-2:** ${}^{n}C_r = {}^{n}C_{n-r}$

425. **S-1:** The total number of words with letters of the word civilization (all taken at a time) is 19958393.
Statement-2: The number of permutations of n distinct objects (r taken at a time) is ${}^{n}p_{r+1}$.

426. **S-1:** The number of ways in which 81 different beads can be arranged to form a necklace is $\frac{80}{2!}$
Statement-2: Number of circular arrangements of n different objects is $(n - 1)!$.

427. **Statement-1:** There are 9^n, n digit numbers in which no two consecutive digits are same.
Statement-2: The n digits number in which no two consecutive digits are equal cannot contain zero.

428. **Statement-1:** $\frac{(n+2)!}{(n-1)!}$ is divisible by 6.**S-2:** : Product of three consecutive integer is divisible by 6.

Answer

399. A	**400. A**	**401. C**	**402. A**	**403. D**	**404. A**	**405. D**
406. D	**407. B**	**408. D**	**409. C**	**410. A**	**411. A**	**412. A**
413. A	**414. A**	**415. C**	**416. A**	**417. A**	**418. B**	**419. B**
420. A	**421. C**	**422. C**	**423. C**	**424. A**	**425. C**	**426. A**
427. C	**428. A**					

Details Solution

Number of words having all the letters distinct = ${}^4P_1 = 4$

Number of words having two are alike and third different = ${}^1C_1 \cdot {}^3C_1 \cdot \frac{3!}{2!} = 9$

$\therefore$ (A) is the correct option.

403. (D) Number of solution = ${}^{12}C_{10} = 66$.

404. (A) $r = 4$

or $r = 10 - 4 = 6$.

405. Statement – II is true as on fixing one object anywhere in the circle, the remaining n – 1 objects can be arranged in $\underline{|n-1}$ ways

Statement – II is false, as after arranging boys on the circle in $\underline{|n-1}$ ways, girls can be arranged in between the boys in $\underline{|n}$ ways (for any arrangement of boys).

Hence number of arrangements is $\underline{|n}\ \underline{|n-1}$.

Hence (D) is the correct answer.

406. Statement – II is true, known as the rule of product.

Statement – I is not true, as the two parts of the work are not independent . Three girls can be chosen out of six girls in ${}^6C_3$ ways, but after this choosing 3 students out of remaining nine students depends on the first part.

Hence (D) is the correct answer.

407. Since, $1400 = 2^3.5^2.7^1$

$\Rightarrow$ Total no. of factors = (3 + 1) (2 + 1) (1 + 1) = 24

$\Rightarrow$ No. of ways of expressing 1400 as a product of two numbers = $\frac{1}{2} \times 24 = 12$.

But this does not follow from statement – II which is obviously true.

Hence (b) is the correct answer.

408. Statement – I is false since the number of selection of four letters from 'PARALLEL' is 22.

1. 3 alike, 1 diff. = ${}^1c_1 \times {}^4c_1 = 4$
2. 2 alike, 2 alike = ${}^2c_2 = 1$
3. 2 alike, 2 diff. = ${}^2c_1 \times {}^4c_2 = 12$
4. All diff. = ${}^5c_4 = 5$

Total selection = 22

Statement – II is true, since

$(1 - x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + 15x^4 + \ldots$ Hence (D) is the correct answer.

410. (A) Let no of sides are n.

${}^nC_2 - n = 44$ $\Rightarrow n = -8$ or 11 $\Rightarrow n = 11$.

415. $x_1x_2x_3x_4 = 1050 = 2 \times 3 \times 5^2 \times 7$

Thus 5^2 can as sign in ${}^5C_1 + {}^5C_2 = 15$ ways

We can assign 2, 3, or 7 to any. of 5 variables.

Hence req. number of solutions.

$= 5 \times 5 \times 5 \times 15 = 1875$ Ans. (C)

416. $({}^{400}C_4 + {}^{400}C_3) + {}^{401}C_3 + \ldots + {}^{500}C_3$

$= ({}^{401}C_4 + {}^{401}C_3) + {}^{402}C_3 + \ldots + {}^{500}C_3$

$\ldots = ({}^{500}C_4 + {}^{500}C_3) = {}^{501}C_4$ Ans. (A)

417. The number of ways of distributing mn things in m groups each containing n things is $\frac{(mn)!}{(n!)^m}$

here if m = n, then $\frac{(n^2)!}{(n!)^n}$ which must be a natural number.

418. If n = 10, 800 $= 2^4 \times 3^3 \times 5^2$

Number of divisors depends upon all possible selection of prime factors. So clearly (4 + 1) (3 + 1) (2+1) = 5 ×4 ×3 = 60 for odd divisors, only selection of odd prime factors, (3 + 1) (2 + 1) = 12

b is correct.

421. (C) A is true since number of selection of four letters from PARALLEL is 22. (3 alike 1 different 4 cases; 2 alike and 2 alike one case;2 alike 2 different $2 \times {}^4C_2 = 12$ and all different ${}^5C_4 = 5$ total selections = 4 + 1 + 12 + 5 = 22). R is false since $(1 - x)^{-3} = 1 + 3x + 6x^2 + 10\,x^3 + 15x^4 + ...$

422. ${}^nP_n = n!$ but number of function from A to B is n^n. (C)

423. (C) ${}^nP_n = n!$, but the number of functions from A to B is n^n.

424. (A) Statement-1 is true,

Statement-2 is true, Also Statement-2 is the correct explanation of Statement-1.

425. (C)

In the given word 4 are there so required number of permutations is $\frac{12!}{4!} = 19958392$

426. (A) Since clockwise and anticlockwise arrangements are not different so required number of arrangements is $\frac{80}{2!}$.

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 8 XI M 8. Binomial Theorem

Index:

1. Key Concepts

2. Exercise I to VIII

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Binomial Theorem

1. Binomial Expression :

Any algebraic expression which contains two dissimilar terms is called binomial expression. For example : $x + y$, $x^2y + \frac{1}{xy^2}$, $3 - x$, $\sqrt{x^2+1} + \frac{1}{(x^3+1)^{1/3}}$ etc.

2. Statement of Binomial theorem :

If $a, b \in R$ and $n \in N$, then ;

$(a + b)^n = {}^nC_0\, a^n b^0 + {}^nC_1\, a^{n-1} b^1 + {}^nC_2\, a^{n-2} b^2 + \ldots + {}^nC_r\, a^{n-r} b^r + \ldots + {}^nC_n\, a^0 b^n$

or $(a + b)^n = \sum_{r=0}^{n} {}^nC_r\, a^{n-r} b^r$

Now, putting $a = 1$ and $b = x$ in the binomial theorem -

or $(1 + x)^n = {}^nC_0 + {}^nC_1\, x + {}^nC_2\, x^2 + \ldots + {}^nC_r\, x^r + \ldots + {}^nC_n\, x^n$

$(1 + x)^n = \sum_{r=0}^{n} {}^nC_r\, x^r$

Solved Example # 1: Expand the following binomials :

(i) $(x - 3)^5$ (ii) $\left(1 - \frac{3x^2}{2}\right)^4$

Solution.

(i) $(x-3)^5 = {}^5C_0 x^5 + {}^5C_1 x^4 (-3)^1 + {}^5C_2 x^3 (-3)^2 + {}^5C_3 x^2 (-3)^3 + {}^5C_4 x (-3)^4 + {}^5C_5 (-3)^5$

$= x^5 - 15x^4 + 90x^3 - 270x^2 + 405x - 243$

(ii) $\left(1 - \frac{3x^2}{2}\right)^4 = {}^4C_0 + {}^4C_1 \left(-\frac{3x^2}{2}\right) + {}^4C_2 \left(-\frac{3x^2}{2}\right)^2 + {}^4C_3 \left(-\frac{3x^2}{2}\right)^3 + {}^4C_4 \left(-\frac{3x^2}{2}\right)^4$

$= 1 - 6x^2 + \frac{27}{2} x^4 - \frac{27}{2} x^6 + \frac{81}{16} x^8$

Solved Example # 2: Expand the binomial $\left(\frac{2x}{3} + \frac{3y}{2}\right)^{20}$ up to four terms

Solution.

$\left(\frac{2x}{3} + \frac{3y}{2}\right)^{20} = {}^{20}C_0 \left(\frac{2x}{3}\right)^{20} + {}^{20}C_1 \left(\frac{2x}{3}\right)^{19} \left(\frac{3y}{2}\right) + {}^{20}C_2 \left(\frac{2x}{3}\right)^{18} \left(\frac{3y}{2}\right)^2 + {}^{20}C_3 \left(\frac{2x}{3}\right)^{17} \left(\frac{3y}{2}\right)^3 + \ldots.$

$= \left(\frac{2x}{3}\right)^{20} + 20 . \left(\frac{2}{3}\right)^{18} x^{19} y + 190 . \left(\frac{2}{3}\right)^{16} x^{18} y^2 + 1140 \left(\frac{2}{3}\right)^{14} x^{17} y^3 + \ldots..$

Self practice problems

1. Write the first three terms in the expansion of $\left(2 - \frac{y}{3}\right)^6$.

2. Expand the binomial $\left(\frac{x^2}{3} + \frac{3}{x}\right)^5$.

Ans. (1) $64 - 64y + \frac{80}{3} y^2$ (2) $\frac{x^{10}}{243} + \frac{5}{27} x^7 + \frac{10}{3} x^4 + 30x + \frac{135}{x^2} + \frac{243}{x^5}$.

3. Properties of Binomial Theorem :

(i) The number of terms in the expansion is $n + 1$.

(ii) The sum of the indices of x and y in each term is n.

(iii) The binomial coefficients $({}^nC_0, {}^nC_1 \ldots\ldots\ldots {}^nC_n)$ of the terms equidistant from the beginning and the end are equal, i.e. ${}^nC_0 = {}^nC_n$, ${}^nC_1 = {}^nC_{n-1}$ etc. $\{\because {}^nC_r = {}^nC_{n-r}\}$

Solved Example # 3: The number of dissimilar terms in the expansion of $(1 - 3x + 3x^2 - x^3)^{20}$ is

(A) 21 (B) 31 (C) 41 (D) 61

Solution. $(1 - 3x + 3x^2 - x^3)^{20} = [(1 - x)^3]^{20} = (1 - x)^{60}$

Therefore number of dissimilar terms in the expansion of $(1 - 3x + 3x^2 - x^3)^{20}$ is **61.**

4. Some important terms in the expansion of $(x + y)^n$:

(i) General term :

$(x + y)^n = {}^nC_0\, x^n y^0 + {}^nC_1\, x^{n-1} y^1 + \ldots\ldots\ldots\ldots + {}^nC_r\, x^{n-r} y^r + \ldots\ldots\ldots + {}^nC_n\, x^0 y^n$

$(r + 1)$th term is called general term.

$T_{r+1} = {}^nC_r\, x^{n-r} y^r$

Solved Example # 4

Find (i) 28th term of $(5x + 8y)^{30}$ (ii) 7th term of $\left(\frac{4x}{5} - \frac{5}{2x}\right)^9$

Solution. (i) $T_{27+1} = {}^{30}C_{27}\,(5x)^{30-27}\,(8y)^{27}$

$= \dfrac{30!}{3!\,27!}\,(5x)^3 \,.\, (8y)^{27}$ **Ans.**

(ii) 7th term of $\left(\dfrac{4x}{5} - \dfrac{5}{2x}\right)^9$

$T_{6+1} = {}^9C_6 \left(\dfrac{4x}{5}\right)^{9-6} \left(-\dfrac{5}{2x}\right)^6$

$= \dfrac{9!}{3!\,6!} \left(\dfrac{4x}{5}\right)^3 \left(\dfrac{5}{2x}\right)^6 = \dfrac{10500}{x^3}$ **Ans.**

Solved Example # 5 : Find the number of rational terms in the expansion of $(9^{1/4} + 8^{1/6})^{1000}$.

Solution. The general term in the expansion of $\left(9^{1/4} + 8^{1/6}\right)^{1000}$ is

$T_{r+1} = {}^{1000}C_r \left(9^{\frac{1}{4}}\right)^{1000-r} \left(8^{\frac{1}{6}}\right)^r = {}^{1000}C_r \; 3^{\frac{1000-r}{2}} \; 2^{\frac{r}{2}}$

The above term will be rational if exponent of 3 and 2 are integres

It means $\dfrac{1000-r}{2}$ and $\dfrac{r}{2}$ must be integers

The possible set of values of r is {0, 2, 4, ..., 1000}

Hence, number of rational terms is **501 Ans.**

(ii) Middle term (s) :

(a) If n is even, there is only one middle term, which is $\left(\dfrac{n+2}{2}\right)$th term.

(b) If n is odd, there are two middle terms, which are $\left(\dfrac{n+1}{2}\right)$th and $\left(\dfrac{n+1}{2}+1\right)$th terms.

Solved Example # 6 : Find the middle term(s) in the expansion of

(i) $\left(1 - \dfrac{x^2}{2}\right)^{14}$ **(ii)** $\left(3a - \dfrac{a^3}{6}\right)^9$

Solution. (i) $\left(1 - \dfrac{x^2}{2}\right)^{14}$

Here, n is even, therefore middle term is $\left(\dfrac{14+2}{2}\right)$th term.

It means T_8 is middle term

$T_8 = {}^{14}C_7 \left(-\dfrac{x^2}{2}\right)^7 = -\,\mathbf{\dfrac{429}{16}}\,x^{14}$. **Ans.**

(ii) $\left(3a - \dfrac{a^3}{6}\right)^9$

Here, n is odd therefore, middle terms are $\left(\dfrac{9+1}{2}\right)$th & $\left(\dfrac{9+1}{2}+1\right)$th.

It means T_5 & T_6 are middle terms

$T_5 = {}^9C_4\,(3a)^{9-4} \left(-\dfrac{a^3}{6}\right)^4 = \mathbf{\dfrac{189}{8}}\,a^{17}$ **Ans.**

$T_6 = {}^9C_5\,(3a)^{9-5} \left(-\dfrac{a^3}{6}\right)^5 = -\,\mathbf{\dfrac{21}{16}}\,a^{19}$. **Ans.**

(iii) Term containing specified powers of x in $\left(ax^{\alpha} \pm \dfrac{b}{x^{\beta}}\right)^n$

Solved Example # 7: Find the coefficient of x^{32} and x^{-17} in $\left(x^4 - \dfrac{1}{x^3}\right)^{15}$.

Solution.: Let (r + 1)th term contains x^m

$T_{r+1} = {}^{15}C_r\,(x^4)^{15-r} \left(-\dfrac{1}{x^3}\right)^r$

$= {}^{15}C_r\; x^{60-7r}\,(-1)^r$

(i) for x^{32}, $60 - 7r = 32$

$\Rightarrow \quad 7r = 28 \quad \Rightarrow \quad r = 4. \quad (T_5)$

$T_5 = {}^{15}C_4\; x^{32}\,(-1)^4$

Hence, coefficient of x^{32} is **1365 Ans.**

(ii) for x^{-17}, $60 - 7r = -17$

$\Rightarrow$ $r = 11$ (T_{12})

$T_{12} = {}^{15}C_{11}\, x^{-17}(-1)^{11}$

Hence, coefficient of x^{-17} is **– 1365** **Ans.**

(iv) **Numerically greatest term in the expansion of $(x + y)^n$, $n \in N$**

Let T_r and T_{r+1} be the rth and (r + 1)th terms respectively

$T_r = {}^nC_{r-1}\, x^{n-(r-1)}\, y^{r-1}$

$T_{r+1} = {}^nC_r\, x^{n-r}\, y^r$

Now, $$\left|\frac{T_{r+1}}{T_r}\right| = \left|\frac{{}^nC_r}{{}^nC_{r-1}}\frac{x^{n-r}\,y^r}{x^{n-r+1}y^{r-1}}\right| = \frac{n-r+1}{r}\cdot\left|\frac{y}{x}\right|$$

Consider $$\left|\frac{T_{r+1}}{T_r}\right| \geq 1$$

$$\left(\frac{n-r+1}{r}\right)\left|\frac{y}{x}\right| \geq 1$$

$$\frac{n+1}{r} - 1 \geq \left|\frac{x}{y}\right|$$

$$r \leq \frac{n+1}{1+\left|\frac{x}{y}\right|}$$

Case - I When $\dfrac{n+1}{1+\left|\frac{x}{y}\right|}$ is an integer (say m), then

(i) $T_{r+1} > T_r$ when $r < m$ $(r = 1, 2, 3 \ldots., m - 1)$
i.e. $T_2 > T_1, T_3 > T_2, \ldots\ldots, T_m > T_{m-1}$

(ii) $T_{r+1} = T_r$ when $r = m$
i.e. $T_{m+1} = T_m$

(iii) $T_{r+1} < T_r$ when $r > m$ $(r = m + 1, m + 2, \ldots\ldots\ldots n)$
i.e. $T_{m+2} < T_{m+1}, T_{m+3} < T_{m+2}, \ldots\ldots\ldots T_{n+1} < T_n$

Conclusion : When $\dfrac{n+1}{1+\left|\frac{x}{y}\right|}$ is an integer, equal to m, then T_m and T_{m+1} will be numerically greatest terms (both terms are equal in magnitude)

Case - II When $\dfrac{n+1}{1+\left|\frac{x}{y}\right|}$ is not an integer (Let its integral part be m), then

(i) $T_{r+1} > T_r$ when $r < \dfrac{n+1}{1+\left|\frac{x}{y}\right|}$ $(r = 1, 2, 3, \ldots\ldots\ldots, m-1, m)$

i.e. $T_2 > T_1, T_3 > T_2, \ldots\ldots\ldots\ldots, T_{m+1} > T_m$

(ii) $T_{r+1} < T_r$ when $r > \dfrac{n+1}{1+\left|\frac{x}{y}\right|}$ $(r = m + 1, m + 2, \ldots\ldots\ldots\ldots n)$

i.e. $T_{m+2} < T_{m+1}, T_{m+3} < T_{m+2}, \ldots\ldots\ldots\ldots, T_{n+1} < T_n$

Conclusion : When $\dfrac{n+1}{1+\left|\frac{x}{y}\right|}$ is not an integer and its integral part is m, then T_{m+1} will be the numerically greatest term.

Solved Example # 8 Find the numerically greatest term in the expansion of $(3 - 5x)^{15}$ when $x = \dfrac{1}{5}$.

Solution. Let r^{th} and $(r + 1)^{th}$ be two consecutive terms in the expansion of $(3 - 5x)^{15}$

$T_r \geq T_{r-1}$

${}^{15}C_r\, 3^{15-r}\,(|-5x|)^r \geq {}^{15}C_{r-1}\, 3^{15-(r-1)}\,(|-5x|)^{r-1}$

$$\frac{(15)!}{(15-r)!\,r!}\,|-5x| \geq \frac{3.(15)!}{(16-r)!\,(r-1)!}$$

$$5 \cdot \frac{1}{5}(16 - r) \geq 3r$$

$16 - r \geq 3r$

$4r \leq 16$

$r \leq 4$

Explanation: For $r \leq 4$, $T_{r+1} \geq T_r$ $\Rightarrow$ $T_2 > T_1$

$T_3 > T_2$

$T_4 > T_3$

$T_5 = T_4$ For $r > 5$, $T_{r+1} < T_r$

$T_6 < T_5$ $T_7 < T_6$

and so on Hence, T_4 and T_5 are numerically greatest terms and both are equal.

Self practice problems :

3. Find the term independent of x in $\left(x^2 - \frac{3}{x}\right)^9$

4. The sum of all rational terms in the expansion of $(3^{1/5} + 2^{1/3})^{15}$ is
(A) 60 (B) 59 (C) 95 (D) 105

5. Find the coefficient of x^{-1} in $(1 + 3x^2 + x^4)\left(1+\frac{1}{x}\right)^8$

6. Find the middle term(s) in the expansion of $(1 + 3x + 3x^2 + x^3)^{2n}$

7. Find the numerically greatest term in the expansion of $(7 - 5x)^{11}$ where $x = \frac{2}{3}$.

Ans. (3) 28.3^7 (4) B (5) 232

(6) ${}^{6n}C_{3n} \cdot x^{3n}$ (7) $T_4 = \frac{440}{9} \times 7^8 \times 5^3$.

5. Multinomial Theorem:

As we know the Binomial Theorem –

$$(x + y)^n = \sum_{r=0}^{n} {}^nC_r \; x^{n-r} y^r$$

$$= \sum_{r=0}^{n} \frac{n!}{(n-r)!\,r!} \; x^{n-r} y^r$$

putting $n - r = r_1$, $r = r_2$

therefore, $(x + y)^n = \sum_{r_1+r_2=n} \frac{n!}{r_1!\; r_2!} \; x^{r_1} \cdot y^{r_2}$

Total number of terms in the expansion of $(x + y)^n$ is equal to number of non-negative integral solution of $r_1 + r_2 = n$ i.e. ${}^{n+2-1}C_{2-1} = {}^{n+1}C_1 = n + 1$

In the same fashion we can write the multinomial theorem

$$(x_1 + x_2 + x_3 + \ldots\ldots\ldots x_k)^n = \sum_{r_1+r_2+\ldots+r_k=n} \frac{n!}{r_1!\, r_2! \ldots r_k!} \; x_1^{r_1} \cdot x_2^{r_2} \ldots x_k^{r_k}$$

Here total number of terms in the expansion of $(x_1 + x_2 + \ldots\ldots\ldots + x_k)^n$ is equal to number of non-negative integral solution of $r_1 + r_2 + \ldots\ldots\ldots + r_k = n$ i.e. ${}^{n+k-1}C_{k-1}$

Solved Example # 9 Find the coeff. of $a^2 b^3 c^4 d$ in the expansion of $(a - b - c + d)^{10}$

Solution. $(a - b - c + d)^{10} = \sum_{r_1+r_2+r_3+r_4=10} \frac{(10)!}{r_1!\, r_2! r_3!\, r_4!} \; (a)^{r_1} (-b)^{r_2} (-c)^{r_3} (d)^{r_4}$

we want to get $a^2 b^3 c^4 d$ this implies that $r_1 = 2, r_2 = 3, r_3 = 4, r_4 = 1$

$\therefore$ coeff. of $a^2 b^3 c^4 d$ is

$$\frac{(10)!}{2!\,3!\,4!\,1!} \; (-1)^3 (-1)^4 = \mathbf{-12600}$$ **Ans.**

Solved Example # 10 In the expansion of $\left(1+x+\frac{7}{x}\right)^{11}$ find the term independent of x.

Solution.

$$\left(1+x+\frac{7}{x}\right)^{11} = \sum_{r_1+r_2+r_3=11} \frac{(11)!}{r_1!\, r_2!\, r_3!} \; (1)^{r_1} (x)^{r_2} \left(\frac{7}{x}\right)^{r_3}$$

The exponent 11 is to be divided among the base variables 1, x and $\frac{7}{x}$ in such a way so that we get x^0.

Therefore, possible set of values of (r_1, r_2, r_3) are (11, 0, 0), (9, 1, 1), (7, 2, 2), (5, 3, 3), (3, 4, 4), (1, 5, 5)

Hence the required term is

$$\frac{(11)!}{(11)!}(7^0) + \frac{(11)!}{9!\,1!\,1!}\,7^1 + \frac{(11)!}{7!\,2!\,2!}\,7^2 + \frac{(11)!}{5!\,3!\,3!}\,7^3 + \frac{(11)!}{3!\,4!\,4!}\,7^4 + \frac{(11)!}{1!\,5!\,5!}\,7^5$$

$$= 1 + \frac{(11)!}{9!\,2!} \cdot \frac{2!}{1!\,1!}\,7^1 + \frac{(11)!}{7!\,4!} \cdot \frac{4!}{2!\,2!}\,7^2 + \frac{(11)!}{5!\,6!} \cdot \frac{6!}{3!\,3!}\,7^3$$

$$+ \frac{(11)!}{3!\,8!} \cdot \frac{8!}{4!\,4!}\,7^4 + \frac{(11)!}{1!\,10!} \cdot \frac{(10)!}{5!\,5!}\,7^5$$

$$= 1 + {}^{11}C_2 \cdot {}^2C_1 \cdot 7^1 + {}^{11}C_4 \cdot {}^4C_2 \cdot 7^2 + {}^{11}C_6 \cdot {}^6C_3 \cdot 7^3 + {}^{11}C_8 \cdot {}^8C_4 \cdot 7^4 + {}^{11}C_{10} \cdot {}^{10}C_5 \cdot 7^5$$

$$= \mathbf{1 + \sum_{r=1}^{5} {}^{11}C_{2r} \cdot {}^{2r}C_r \cdot 7^r}$$ **Ans.**

Self practice problems :

8. The number of terms in the expansion of $(a + b + c + d + e + f)^n$ is
(A) ${}^{n+4}C_4$ (B) ${}^{n+3}C_n$ (C) ${}^{n+5}C_n$ (D) $n + 1$

9. Find the coefficient of $x^3 y^4 z^2$ in the expansion of $(2x - 3y + 4z)^9$

10. Find the coefficient of x^4 in $(1 + x - 2x^2)^7$

Ans. (8) C (9) $\frac{9!}{3!\,4!\,2!}$ (10) $2^3\,3^4\,4^2 - 91$

6. Application of Binomial Theorem :

(i) If $(\sqrt{A}+B)^n = I + f$ where I and n are positive integers, n being odd and $0 < f < 1$ then $(I + f)\,f = k^n$ where $A - B^2 = k > 0$ and $\sqrt{A} - B < 1$.
If n is an even integer, then $(I + f)(1 - f) = k^n$

Solved Example # 11

If n is positive integer, then prove that the integral part of $(7 + 4\sqrt{3})^n$ is an odd number.

Solution. Let $(7 + 4\sqrt{3})^n = I + f$ .....(i)
where I & f are its integral and fractional parts respectively.
It means $0 < f < 1$

Now, $0 < 7 - 4\sqrt{3} < 1$

$0 < (7 - 4\sqrt{3})^n < 1$

Let $(7 - 4\sqrt{3})^n = f'$ .....(ii)
$\Rightarrow$ $0 < f' < 1$
Adding (i) and (ii)

$I + f + f' = (7 + 4\sqrt{3})^n + (7 - 4\sqrt{3})^n$

$= 2\,[{}^nC_0\,7^n + {}^nC_2\,7^{n-2}\,(4\sqrt{3})^2 + \ldots\ldots]$

$I + f + f'$ = even integer ($f + f'$ must be an integer)
$0 < f + f' < 2 \Rightarrow f + f' = 1$
$I + 1$ = even integer
therefore I is an odd integer.

Solved Example # 12

Show that the integer just above $(\sqrt{3} + 1)^{2n}$ is divisible by 2^{n+1} for all $n \in N$.

Solution. Let $(\sqrt{3} + 1)^{2n} = (4 + 2\sqrt{3})^n = 2^n\,(2 + \sqrt{3})^n = I + f$ .....(i)
where I and f are its integral & fractional parts respectively
$0 < f < 1$.

Now $0 < \sqrt{3} - 1 < 1$

$0 < (\sqrt{3} - 1)^{2n} < 1$

Let $(\sqrt{3} - 1)^{2n} = (4 - 2\sqrt{3})^n = 2^n\,(2 - \sqrt{3})^n = f'$. .....(ii)
$0 < f' < 1$
adding (i) and (ii)

$I + f + f' = (\sqrt{3} + 1)^{2n} + (\sqrt{3} - 1)^{2n}$

$= 2^n\,[(2 + \sqrt{3})^n + (2 - \sqrt{3})^n]$

$= 2.2^n\,[{}^nC_0\,2^n + {}^nC_2\,2^{n-2}\,(\sqrt{3})^2 + \ldots\ldots]$

$I + f + f' = 2^{n+1}\,k$ (where k is a positive integer)
$0 < f + f' < 2 \Rightarrow f + f' = 1$
$I + 1 = 2^{n+1}\,k$.

$I + 1$ is the integer just above $(\sqrt{3} + 1)^{2n}$ and which is divisible by 2^{n+1}.

(ii) Cheking divisibility

Solved Example # 13: Show that $9^n + 7$ is divisible by 8, where n is a positive integer.

Solution. $9^n + 7 = (1 + 8)^n + 7$
$= {}^nC_0 + {}^nC_1 . 8 + {}^nC_2 . 8^2 + \ldots\ldots + {}^nC_n\,8^n + 7.$
$= 8.\,C_1 + 8^2.\,C_2 + \ldots\ldots + C_n . 8^n + 8.$
$= 8\lambda$ where, λ is a positive integer, Hence, $9^n + 7$ is divisible by 8.

(iii) Finding remainder

Solved Example # 14

What is the remainder when 5^{99} is divided by 13.

Solution.: $5^{99} = 5.5^{98} = 5.\,(25)^{49}$
$= 5\,(26 - 1)^{49}$
$= 5\,[{}^{49}C_0\,(26)^{49} - {}^{49}C_1\,(26)^{48} + \ldots\ldots\ldots + {}^{49}C_{48}\,(26)^1 - {}^{49}C_{49}\,(26)^0]$
$= 5\,[{}^{49}C_0\,(26)^{49} - {}^{49}C_1\,(26)^{48} + \ldots\ldots\ldots + {}^{49}C_{48}\,(26)^1 - 1]$
$= 5\,[{}^{49}C_0\,(26)^{49} - {}^{49}C_1\,(26)^{48} + \ldots\ldots\ldots + {}^{49}C_{48}\,(26)^1 - 13] + 60$
$= 13\,(k) + 52 + 8$ (where k is a positive integer)
$= 13\,(k + 4) + 8$ Hence, remainder is 8. **Ans.**

(iv) Finding last digit, last two digits and last there digits of the given number.

Solved Example # 15 Find the last two digits of the number $(17)^{10}$.

Solution. $(17)^{10} = (289)^5$
$= (290 - 1)^5$
$= {}^5C_0\,(290)^5 - {}^5C_1\,(290)^4 + \ldots\ldots\ldots + {}^5C_4\,(290)^1 - {}^5C_5\,(290)^0$
$= {}^5C_0\,(290)^5 - {}^5C_1 . (290)^4 + \ldots\ldots\ldots {}^5C_3\,(290)^2 + 5 \times 290 - 1$
= A multiple of 1000 + 1449 Hence, last two digits are 49 **Ans.**

Note : We can also conclude that last three digits are 449.

(v) Comparison between two numbers

Solved Example # 16 : Which number is larger $(1.01)^{1000000}$ or 10,000 ?

Solution.: By Binomial Theorem
$(1.01)^{1000000} = (1 + 0.01)^{1000000}$
$= 1 + {}^{1000000}C_1\,(0.01)$ + other positive terms
$= 1 + 1000000 \times 0.01$ + other positive terms
$= 1 + 10000$ + other positive terms, Hence $(1.01)^{1000000} > 10{,}000$

Self practice problems :

11. If n is positive integer, prove that the integral part of $(5\sqrt{5} + 11)^{2n+1}$ is an even number.

12. If $(7 + 4\sqrt{3})^n = \alpha + \beta$, where α is a positive integer and β is a proper fraction then prove that $(1 - \beta)(\alpha + \beta) = 1$.

13. If n is a positive integer then show that $3^{2n+1} + 2^{n+2}$ is divisible by 7.

14. What is the remainder when 7^{103} is divided by 25 .

15. Find the last digit, last two digits and last three digits of the number $(81)^{25}$.

16. Which number is larger $(1.2)^{4000}$ or 800

Ans. (14) 18 (15) 1, 01, 001 (16) $(1.2)^{4000}$.

7. Properties of Binomial Coefficients :

$(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots\ldots\ldots + C_r x^r + \ldots\ldots\ldots + C_n x^n$ .....(1)

(1) The sum of the binomial coefficients in the expansion of $(1 + x)^n$ is 2^n

Putting x = 1 in (1)

$${}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots\ldots\ldots + {}^nC_n = 2^n \quad \ldots\ldots(2)$$

or $$\sum_{r=0}^{n} {}^nC_r = 2^n$$

(2) Again putting x = –1 in (1), we get

$${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \ldots\ldots\ldots\ldots + (-1)^n\,{}^nC_n = 0 \quad \ldots\ldots(3)$$

or $$\sum_{r=0}^{n} (-1)^r\,{}^nC_r = 0$$

(3) The sum of the binomial coefficients at odd position is equal to the sum of the binomial coefficients at even position and each is equal to 2^{n-1}.

from (2) and (3)

$${}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots\ldots\ldots\ldots = 2^{n-1}$$

$${}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots\ldots\ldots\ldots = 2^{n-1}$$

(4) Sum of two consecutive binomial coefficients

$${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

$$\text{L.H.S.} = {}^nC_r + {}^nC_{r-1} = \frac{n!}{(n-r)!\,r!} + \frac{n!}{(n-r+1)!\,(r-1)!}$$

$$= \frac{n!}{(n-r)!\,(r-1)!}\left[\frac{1}{r} + \frac{1}{n-r+1}\right] = \frac{n!}{(n-r)!\,(r-1)!}\,\frac{(n+1)}{r(n-r+1)}$$

$$= \frac{(n+1)!}{(n-r+1)!\,r!} = {}^{n+1}C_r = \text{R.H.S.}$$

(5) Ratio of two consecutive binomial coefficients

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$$

(6) $${}^nC_r = \frac{n}{r}\,{}^{n-1}C_{r-1} = \frac{n(n-1)}{r(r-1)}\,{}^{n-2}C_{r-2} = \ldots\ldots\ldots\ldots = \frac{n(n-1)(n-2)\ldots\ldots(n-(r-1))}{r\,(r-1)(r-2)\ldots\ldots 2\,.1}$$

Solved Example # 17

If $(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots\ldots\ldots\ldots + c_nx^n$, then show that

(i) $C_0 + 3C_1 + 3^2C_2 + \ldots\ldots\ldots + 3^n\,C_n = 4^n$.

(ii) $C_0 + 2C_1 + 3.\,C_2 + \ldots\ldots\ldots + (n + 1)\,C_n = 2^{n-1}\,(n + 2)$.

(iii) $C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \frac{C_3}{4} + \ldots\ldots\ldots + (-1)^n\,\frac{C_n}{n+1} = \frac{1}{n+1}$.

Solution. (i) $(1 + x)^n = C_0 + C_1\,x + C_2x^2 + \ldots\ldots\ldots\ldots + C_nx^n$

put x = 3

$C_0 + 3\,.\,C_1 + 3^2\,.\,C_2 + \ldots\ldots\ldots + 3^n\,.\,C_n = 4^n$

(ii) **I Method : By Summation**

$$\text{L.H.S.} = {}^nC_0 + 2.\,{}^nC_1 + 3\,.\,{}^nC_2 + \ldots\ldots\ldots + (n + 1).\,{}^nC_n.$$

$$= \sum_{r=0}^{n} (r+1)\,.\,{}^nC_r$$

$$= \sum_{r=0}^{n} r.\,{}^nC_r + \sum_{r=0}^{n} {}^nC_r \qquad = n\sum_{r=0}^{n} {}^{n-1}C_{r-1} + \sum_{r=0}^{n} {}^nC_r$$

$= n\,.\,2^{n-1} + 2^n = 2^{n-1}\,(n + 2)$. **RHS**

II Method : By Differentiation

$(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots\ldots\ldots\ldots + C_nx^n$

Multiplying both sides by x,

$x(1 + x)^n = C_0x + C_1x^2 + C_2x^3 + \ldots\ldots\ldots + C_n\,x^{n+1}$.

Differentiating both sides

$(1 + x)^n + x\,n\,(1 + x)^{n-1} = C_0 + 2.\,C_1 + 3\,.\,C_2x^2 + \ldots\ldots\ldots + (n + 1)C_nx^n$.

putting x = 1, we get

$C_0 + 2.C_1 + 3\,.\,C_2 + \ldots\ldots\ldots + (n + 1)\,C_n = 2^n + n\,.\,2^{n-1}$

$C_0 + 2.C_1 + 3\,.\,C_2 + \ldots\ldots\ldots + (n + 1)\,C_n = 2^{n-1}\,(n + 2)$ **Proved**

(iii) **I Method : By Summation**

$$\text{L.H.S.} = C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \frac{C_3}{4} + \ldots\ldots\ldots + (-1)^n.\,\frac{C_n}{n+1}$$

$= \sum_{r=0}^{n}(-1)^r \cdot \frac{{}^nC_r}{r+1}$

$= \frac{1}{n+1}\sum_{r=0}^{n}(-1)^r \cdot {}^{n+1}C_{r+1} \qquad \left\{\frac{n+1}{r+1}\cdot {}^nC_r = {}^{n+1}C_{r+1}\right\}$

$= \frac{1}{n+1}[{}^{n+1}C_1 - {}^{n+1}C_2 + {}^{n+1}C_3 - \ldots\ldots\ldots\ldots + (-1)^n \cdot {}^{n+1}C_{n+1}]$

$= \frac{1}{n+1}[-{}^{n+1}C_0 + {}^{n+1}C_1 - {}^{n+1}C_2 + \ldots\ldots\ldots + (-1)^n \cdot {}^{n+1}C_{n+1} + {}^{n+1}C_0]$

$= \frac{1}{n+1}$ = R.H.S. $\left\{-{}^{n+1}C_0 + {}^{n+1}C_1 - {}^{n+1}C_2 + \ldots + (-1)^n \ {}^{n+1}C_{n+1} = 0\right\}$

II Method : By Integration

$(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots\ldots + C_nx^n$.

Integrating both sides, with in the limits – 1 to 0.

$$\left[\frac{(1+x)^{n+1}}{n+1}\right]_{-1}^{0} = \left[C_0x + C_1\frac{x^2}{2} + C_2\frac{x^3}{3} + \ldots.. + C_n\frac{x^{n+1}}{n+1}\right]_{-1}^{0}$$

$$\frac{1}{n+1} - 0 = 0 - \left[-C_0 + \frac{C_1}{2} - \frac{C_2}{3} + \ldots.. + (-1)^{n+1}\frac{C_n}{n+1}\right]$$

$C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \ldots\ldots\ldots + (-1)^n \frac{C_n}{n+1} = \frac{1}{n+1}$ **Proved**

Solved Example # 18 If $(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots\ldots\ldots + C_nx^n$, then prove that

(i) $C_0^2 + C_1^2 + C_2^2 + \ldots\ldots + C_n^2 = {}^{2n}C_n$

(ii) $C_0C_2 + C_1C_3 + C_2C_4 + \ldots\ldots\ldots + C_{n-2}C_n = {}^{2n}C_{n-2}$ or ${}^{2n}C_{n+2}$

(iii) $1.C_0^2 + 3.C_1^2 + 5.C_2^2 + \ldots\ldots\ldots + (2n+1).C_n^2 = 2n.\ {}^{2n-1}C_n + {}^{2n}C_n$.

Solution. (i) $(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots\ldots\ldots + C_nx^n$. .....(i)

$(x + 1)^n = C_0x^n + C_1x^{n-1} + C_2x^{n-2} + \ldots\ldots + C_nx^0$ .....(ii)

Multiplying (i) and (ii)

$(C_0 + C_1x + C_2x^2 + \ldots\ldots\ldots + C_nx^n)(C_0x^n + C_1x^{n-1} + \ldots\ldots\ldots + C_nx^0) = (1 + x)^{2n}$

Comparing coefficient of x^n,

$C_0^2 + C_1^2 + C_2^2 + \ldots\ldots\ldots + C_n^2 = {}^{2n}C_n$

(ii) From the product of (i) and (ii) comparing coefficients of x^{n-2} or x^{n+2} both sides,

$C_0C_2 + C_1C_3 + C_2C_4 + \ldots\ldots\ldots + C_{n-2}C_n = {}^{2n}C_{n-2}$ or ${}^{2n}C_{n+2}$.

(iii) **I Method : By Summation**

L.H.S. $= 1.C_0^2 + 3.C_1^2 + 5.C_2^2 + \ldots\ldots\ldots + (2n+1)C_n^2$.

$$= \sum_{r=0}^{n}(2r+1)\,{}^nC_r^2$$

$$= \sum_{r=0}^{n}2r \cdot ({}^nC_r)^2 + \sum_{r=0}^{n}({}^nC_r)^2 \qquad = 2\sum_{r=1}^{n} n \cdot {}^{n-1}C_{r-1}\,{}^nC_r + {}^{2n}C_n$$

$(1 + x)^n = {}^nC_1 + {}^nC_4x + {}^nC_2x^2 + \ldots\ldots\ldots\ldots {}^nC_nx^n$ .....(i)

$(x + 1)^{n-1} = {}^{n-1}C_0x^{n-1} + {}^{n-1}C_1x^{n-2} + \ldots\ldots\ldots + {}^{n-1}C_{n-1}x^0$ .....(ii)

Multiplying (i) and (ii) and comparing coeffcients of x^n.

${}^{n-1}C_0 \cdot {}^nC_1 + {}^{n-1}C_1 \cdot {}^nC_2 + \ldots\ldots\ldots + {}^{n-1}C_{n-1} \cdot {}^nC_n = {}^{2n-1}C_n$

$$\sum_{r=0}^{n} {}^{n-1}C_{r-1} \cdot {}^nC_r = {}^{2n-1}C_n$$

Hence, required summation is

$2n.\ {}^{2n-1}C_n + {}^{2n}C_n$ = R.H.S.

II Method : By Differentiation

$(1 + x^2)^n = C_0 + C_1x^2 + C_2x^4 + C_3x^6 + \ldots\ldots\ldots\ldots + C_nx^{2n}$

Multiplying both sides by x

$x(1 + x^2)^n = C_0x + C_1x^3 + C_2x^5 + \ldots\ldots\ldots\ldots + C_nx^{2n+1}$.

Differentiating both sides

$x \cdot n(1 + x^2)^{n-1} \cdot 2x + (1 + x^2)^n = C_0 + 3.C_1x^2 + 5.C_2x^4 + \ldots\ldots + (2n + 1)C_nx^{2n}$ .....(i)

$(x^2 + 1)^n = C_0x^{2n} + C_1x^{2n-2} + C_2x^{2n-4} + \ldots\ldots\ldots + C_n$ .....(ii)

Multiplying (i) & (ii)

$(C_0 + 3C_1x^2 + 5C_2x^4 + \ldots\ldots\ldots + (2n + 1)C_nx^{2n})(C_0x^{2n} + C_1x^{2n-2} + \ldots\ldots\ldots + C_n)$

$= 2nx^2(1 + x^2)^{2n-1} + (1 + x^2)^{2n}$

comparing coefficient of x^{2n},

$C_0^2 + 3C_1^2 + 5C_2^2 + \ldots\ldots\ldots + (2n + 1)C_n^2 = 2n \cdot {}^{2n-1}C_{n-1} + {}^{2n}C_n$.

$C_0^2 + 3C_1^2 + 5C_2^2 + \ldots\ldots\ldots + (2n + 1)C_n^2 = 2n \cdot {}^{2n-1}C_n + {}^{2n}C_n$. **Proved**

Solved Example # 19

Find the summation of the following series –

(i) ${}^mC_m + {}^{m+1}C_m + {}^{m+2}C_m + \ldots\ldots\ldots\ldots + {}^nC_m$ (ii) ${}^nC_3 + 2 \cdot {}^{n+1}C_3 + 3 \cdot {}^{n+2}C_3 + \ldots\ldots\ldots + n \cdot {}^{2n-1}C_3$

Solution. **(i) I Method : Using property, ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$**

${}^mC_m + {}^{m+1}C_m + {}^{m+2}C_m + \ldots\ldots\ldots\ldots + {}^nC_m$

$= \underbrace{{}^{m+1}C_{m+1} + {}^{m+1}C_m} + {}^{m+2}C_m + \ldots\ldots\ldots\ldots + {}^nC_m \quad \{\because {}^mC_m = {}^{m+1}C_{m+1}\}$

$= \underbrace{{}^{m+2}C_{m+1} + {}^{m+2}C_m} + \ldots\ldots\ldots\ldots + {}^{n}C_m$

$= {}^{m+3}C_{m+1} + \ldots\ldots\ldots\ldots + {}^{n}C_m \qquad = {}^{n}C_{m+1} + {}^{n}C_m = {}^{n+1}C_{m+1}$ **Ans.**

II Method

${}^{m}C_m + {}^{m+1}C_m + {}^{m+2}C_m + \ldots\ldots\ldots + {}^{n}C_m$

The above series can be obtained by writing the coefficient of x^m in

$(1 + x)^m + (1 + x)^{m+1} + \ldots\ldots\ldots + (1 + x)^n$

Let $S = (1 + x)^m + (1 + x)^{m+1} + \ldots\ldots\ldots\ldots + (1 + x)^n$

$= \dfrac{(1+x)^m\left[(1+x)^{n-m+1} - 1\right]}{x} \qquad = \dfrac{(1+x)^{n+1} - (1+x)^m}{x}$

x^m : S (coefficient of x^m in S)

$x^m : \dfrac{(1+x)^{n+1} - (1+x)^m}{x}$

Hence, required summation of the series is ${}^{n+1}C_{m+1}$ **Ans.**

(ii)

${}^{n}C_3 + 2 \cdot {}^{n+1}C_3 + 3 \cdot {}^{n+2}C_3 + \ldots\ldots\ldots + n \cdot {}^{2n-1}C_3$

The above series can be obatined by writing the coefficient of x^3 in

$(1 + x)^n + 2 \cdot (1 + x)^{n+1} + 3 \cdot (1 + x)^{n+2} + \ldots\ldots\ldots + n \cdot (1 + x)^{2n-1}$

Let $S = (1 + x)^n + 2 \cdot (1 + x)^{n+1} + 3 \cdot (1 + x)^{n+2} + \ldots\ldots\ldots + n\,(1 + x)^{2n-1}$ .....(i)

$(1 + x)S = (1 + x)^{n+1} + 2\,(1 + x)^{n+2} + \ldots\ldots\ldots\ldots + (n - 1)\,(1 + x)^{2n-1} + n(1 + x)^{2n}$ .....(ii)

Subtracting (ii) from (i)

$-xS = (1 + x)^n + (1 + x)^{n+1} + (1 + x)^{n+2} + \ldots\ldots\ldots\ldots + (1 + x)^{2n-1} - n(1 + x)^{2n}$

$= \dfrac{(1+x)^n\left[(1+x)^n - 1\right]}{x} - n\,(1 + x)^{2n}$

$S = \dfrac{-(1+x)^{2n} + (1+x)^n}{x^2} + \dfrac{n(1+x)^{2n}}{x}$

x^3 : S (coefficient of x^3 in S)

$x^3 : \dfrac{-(1+x)^{2n} + (1+x)^n}{x^2} + \dfrac{n(1+x)^{2n}}{x}$

Hence, required summation of the series is $-{}^{2n}C_5 + {}^{n}C_5 + n \cdot {}^{2n}C_4$ **Ans.**

Self practice problems :

17. Prove the following

(i) $C_0 + 3C_1 + 5C_2 + \ldots\ldots\ldots\ldots + (2n + 1)\,C_n = 2^n\,(n + 1)$

(ii) $4C_0 + \dfrac{4^2}{2} \cdot C_1 + \dfrac{4^3}{3} C_2 + \ldots\ldots\ldots\ldots + \dfrac{4^{n+1}}{n+1} C_n = \dfrac{5^{n+1} - 1}{n+1}$

(iii) ${}^{n}C_0 \cdot {}^{n+1}C_n + {}^{n}C_1 \cdot {}^{n}C_{n-1} + {}^{n}C_2 \cdot {}^{n-1}C_{n-2} + \ldots\ldots\ldots + {}^{n}C_n \cdot {}^{1}C_0 = 2^{n-1}\,(n + 2)$

(iv) ${}^{2}C_2 + {}^{3}C_2 + \ldots\ldots\ldots + {}^{n}C_2 = {}^{n+1}C_3$

8. Binomial Theorem For Negative Integer Or Fractional Indices

If $n \in R$ then,

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \ldots\ldots\ldots\ldots$$

$$\ldots\ldots\ldots\ldots + \frac{n(n-1)(n-2)\ldots\ldots(n-r+1)}{r!} x^r + \ldots\ldots\ldots\ldots \infty.$$

Remarks: (i) The above expansion is valid for any rational number other than a whole number if $|x| < 1$.

(ii) When the index is a negative integer or a fraction then number of terms in the expansion of $(1 + x)^n$ is infinite, and the symbol ${}^{n}C_r$ cannot be used to denote the coefficient of the general term.

(iii) The first terms must be unity in the expansion, when index 'n' is a negative integer or fraction

$$(x + y)^n = \begin{cases} x^n\left(1+\dfrac{y}{x}\right)^n = x^n\left\{1 + n \cdot \dfrac{y}{x} + \dfrac{n(n-1)}{2!}\left(\dfrac{y}{x}\right)^2 + \ldots\ldots\right\} \text{ if } \left|\dfrac{y}{x}\right| < 1 \\ y^n\left(1+\dfrac{x}{y}\right)^n = y^n\left\{1 + n \cdot \dfrac{x}{y} + \dfrac{n(n-1)}{2!}\left(\dfrac{x}{y}\right)^2 + \ldots\ldots\right\} \text{if } \left|\dfrac{x}{y}\right| < 1 \end{cases}$$

(iv) The general term in the expansion of $(1 + x)^n$ is $T_{r+1} = \dfrac{n(n-1)(n-2)\ldots\ldots(n-r+1)}{r!} x^r$

(v) When 'n' is any rational number other than whole number then approximate value of $(1 + x)^n$ is $1 + nx$ (x^2 and higher powers of x can be neglected)

(vi) Expansions to be remembered ($|x| < 1$)

(a) $(1 + x)^{-1} = 1 - x + x^2 - x^3 + \ldots\ldots\ldots + (-1)^r x^r + \ldots\ldots\ldots\infty$

(b) $(1 - x)^{-1} = 1 + x + x^2 + x^5 + \ldots\ldots\ldots + x^r + \ldots\ldots\ldots\infty$

(c) $(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \ldots\ldots\ldots + (-1)^r\,(r + 1)\,x^r + \ldots\ldots\ldots\infty$

(d) $(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \ldots\ldots\ldots\ldots + (r + 1)x^r + \ldots\ldots\ldots \infty$

Solved Example # 20: Prove that the coefficient of x^r in $(1 - x)^{-n}$ is ${}^{n+r-1}C_r$

Soltion.: $(r + 1)$th term in the expansion of $(1 - x)^{-n}$ can be written as

$$T_{r+1} = \frac{-n(-n-1)(-n-2)\ldots\ldots(-n-r+1)}{r!} (-x)^r$$

$$= (-1)^r \frac{n(n+1)(n+2).....(n+r-1)}{r!} (-x)^r$$

$$= \frac{n(n+1)(n+2).....(n+r-1)}{r!} x^r$$

$$= \frac{(n-1)!\, n(n+1).....(n+r-1)}{(n-1)!\, r!} x^r$$

Hence, coefficient of x^r is $\frac{(n+r-1)!}{(n-1)!\, r!} = {}^{n+r-1}C_r$ **Proved**

Solved Example # 21: If x is so small such that its square and higher powers may be neglected then find the value of $\frac{(1-3x)^{1/2} + (1-x)^{5/3}}{(4+x)^{1/2}}$

Solution. $\frac{(1-3x)^{1/2} + (1-x)^{5/3}}{(4+x)^{1/2}}$

$$= \frac{1-\frac{3}{2}x+1-\frac{5x}{3}}{2\left(1+\frac{x}{4}\right)^{1/2}} \qquad = \frac{1}{2}\left(2-\frac{19}{6}x\right)\left(1+\frac{x}{4}\right)^{-1/2}$$

$$= \frac{1}{2}\left(2-\frac{19}{6}x\right)\left(1-\frac{x}{8}\right) \qquad = \frac{1}{2}\left(2-\frac{x}{4}-\frac{19}{6}x\right)$$

$$= 1 - \frac{x}{8} - \frac{19}{12}x \qquad = \mathbf{1 - \frac{41}{24}\, x} \quad \textbf{Ans.}$$

Self practice problems :

18. Find the possible set of values of x for which expansion of $(3-2x)^{1/2}$ is valid in ascending powers of x.

19. If $y = \frac{3}{5} + \frac{1.3}{2!}\left(\frac{2}{5}\right)^2 + \frac{1.3.5}{3!}\left(\frac{2}{5}\right)^3 + ...$, then find the value of $y^2 + 2y$

20. The coefficient of x^{100} in $\frac{3-5x}{(1-x)^2}$ is

(A) 100 (B) –57 (C) –197 (D) 53

Ans. (18) $x \in \left(-\frac{3}{2}, \frac{3}{2}\right)$ (19) 4 (20) C

Short Revision

BINOMIAL EXPONENTIAL & LOGARITHMIC SERIES

1. BINOMIAL THEOREM : The formula by which any positive integral power of a binomial expression can be expanded in the form of a series is known as **BINOMIAL THEOREM** .
If $x, y \in R$ and $n \in N$, then ;

$$(x+y)^n = {}^nC_0\, x^n + {}^nC_1\, x^{n-1}\, y + {}^nC_2\, x^{n-2}y^2 + \ldots\ldots + {}^nC_r\, x^{n-r}y^r + \ldots\ldots + {}^nC_n y^n = \sum_{r=0}^{n} {}^nC_r\, x^{n-r}\, y^r.$$

This theorem can be proved by Induction .

OBSERVATIONS :

(i) The number of terms in the expansion is $(n+1)$ i.e. one or more than the index .

(ii) The sum of the indices of x & y in each term is n .

(iii) The binomial coefficients of the terms ${}^nC_0$, ${}^nC_1$ .... **equidistant** from the beginning and the end are equal.

2. IMPORTANT TERMS IN THE BINOMIAL EXPANSION ARE :

(i) General term **(ii)** Middle term

(iii) Term independent of x & **(iv)** Numerically greatest term

(i) The general term or the $(r+1)^{th}$ term in the expansion of $(x+y)^n$ is given by ;
$T_{r+1} = {}^nC_r\, x^{n-r}\, .\, y^r$

(ii) The middle term(s) is the expansion of $(x+y)^n$ is (are) :

(a) If n is even , there is only one middle term which is given by ;
$T_{(n+2)/2} = {}^nC_{n/2}\, .\, x^{n/2}\, .\, y^{n/2}$

(b) If n is odd , there are two middle terms which are :
$T_{(n+1)/2}$ & $T_{[(n+1)/2]+1}$

(iii) Term independent of x contains no x ; Hence find the value of r for which the exponent of x is zero.

(iv) To find the Numerically greatest term is the expansion of $(1+x)^n$, $n \in N$ find $\frac{T_{r+1}}{T_r} = \frac{{}^nC_r x^r}{{}^nC_{r-1}x^{r-1}} = \frac{n-r+1}{r}x$. Put the absolute value of x & find the value of r Consistent with the inequality $\frac{T_{r+1}}{T_r} > 1$.

Note that the Numerically greatest term in the expansion of $(1-x)^n$, $x>0$, $n \in N$ is the same as the greatest term in $(1+x)^n$.

3. If $\left(\sqrt{A}+B\right)^n = I + f$, where I & n are positive integers, n being odd and $0 < f < 1$, then $(I+f)\,.\,f = K^n$ where $A - B^2 = K > 0$ & $\sqrt{A} - B < 1$.
If n is an even integer, then $(I+f)(1-f) = K^n$.

4. BINOMIAL COEFFICIENTS : **(i)** $C_0 + C_1 + C_2 + \ldots\ldots + C_n = 2^n$

(ii) $C_0 + C_2 + C_4 + \ldots\ldots = C_1 + C_3 + C_5 + \ldots\ldots = 2^{n-1}$

(iii) $C_0^2 + C_1^2 + C_2^2 + \ldots + C_n^2 = {}^{2n}C_n = \frac{(2n)!}{n!\,n!}$

(iv) $C_0.C_r + C_1.C_{r+1} + C_2.C_{r+2} + \ldots + C_{n-r}.C_n = \frac{(2n)!}{(n+r)(n-r)!}$

REMEMBER : **(i)** $(2n)! = 2^n\,.\,n!\,[1.\,3.\,5 \ldots\ldots (2n-1)]$

5. BINOMIAL THEOREM FOR NEGATIVE OR FRACTIONAL INDICES :

If $n \in Q$, then $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots\ldots\infty$ Provided $|x| < 1$.

Note : (i) When the index n is a positive integer the number of terms in the expansion of $(1+x)^n$ is finite i.e. $(n+1)$ & the coefficient of successive terms are :
${}^nC_0$, ${}^nC_1$, ${}^nC_2$, ${}^nC_3$ ..... ${}^nC_n$.

(ii) When the index is other than a positive integer such as negative integer or fraction, the number of terms in the expansion of $(1+x)^n$ is infinite and the symbol ${}^nC_r$ cannot be used to denote the Coefficient of the general term .

(iii) Following expansion should be remembered ($|x| < 1$).

(a) $(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \ldots\ldots\infty$ **(b)** $(1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \ldots\ldots\infty$

(c) $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \ldots\ldots\infty$ **(d)** $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \ldots\ldots\infty$

(iv) The expansions in ascending powers of x are only valid if x is 'small'. If x is large i.e. $|x| > 1$ then we may find it convinient to expand in powers of $\frac{1}{x}$, which then will be small.

6. APPROXIMATIONS : $(1+x)^n = 1 + nx + \frac{n(n-1)}{1.2}x^2 + \frac{n(n-1)(n-2)}{1.2.3}x^3 \ldots\ldots$

If $x < 1$, the terms of the above expansion go on decreasing and if x be very small, a stage may be reached when we may neglect the terms containing higher powers of x in the expansion. Thus, if x be so small that its squares and higher powers may be neglected then $(1+x)^n = 1 + nx$, approximately.

This is an approximate value of $(1+x)^n$.

7. EXPONENTIAL SERIES :

(i) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + ...\infty$; where x may be any real or complex & $e = \underset{n\to\infty}{\text{Limit}} \left(1+\frac{1}{n}\right)^n$

(ii) $a^x = 1 + \frac{x}{1!} ln\, a + \frac{x^2}{2!} ln^2 a + \frac{x^3}{3!} ln^3 a + ...\infty$ where $a > 0$

Note : **(a)** $e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + ...\infty$

(b) e is an irrational number lying between 2.7 & 2.8. Its value correct upto 10 places of decimal is 2.7182818284.

(c) $e + e^{-1} = 2\left(1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + ...\infty\right)$ **(d)** $e - e^{-1} = 2\left(1 + \frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + ...\infty\right)$

(e) Logarithms to the base 'e' are known as the Napierian system, so named after Napier, their inventor. They are also called **Natural Logarithm**.

8. LOGARITHMIC SERIES :

(i) $ln\,(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...\infty$ where $-1 < x \le 1$

(ii) $ln\,(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + ...\infty$ where $-1 \le x < 1$

(iii) $ln\,\frac{(1+x)}{(1-x)} = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + ...\infty\right)$ $|x| < 1$

REMEMBER : **(a)** $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + ... \infty = ln\,2$ **(b)** $e^{ln\,x} = x$

(c) $ln 2 = 0.693$ **(d)** $ln 10 = 2.303$

EXERCISE - 1

Q.1 Find the coefficients : (i) x^7 in $\left(ax^2 + \frac{1}{bx}\right)^{11}$ (ii) x^{-7} in $\left(ax - \frac{1}{bx^2}\right)^{11}$

(iii) Find the relation between a & b, so that these coefficients are equal.

Q.2 If the coefficients of $(2r+4)^{th}$, $(r-2)^{th}$ terms in the expansion of $(1+x)^{18}$ are equal, find r.

Q.3 If the coefficients of the rth, $(r+1)^{th}$ & $(r+2)^{th}$ terms in the expansion of $(1+x)^{14}$ are in AP, find r.

Q.4 Find the term independent of x in the expansion of (a) $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$ (b) $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$

Q.5 Find the sum of the series $\sum_{r=0}^{n} (-1)^r .{}^nC_r \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \frac{15^r}{2^{4r}} + ...\text{up to m terms}\right]$

Q.6 If the coefficients of 2nd, 3rd & 4th terms in the expansion of $(1+x)^{2n}$ are in AP, show that $2n^2 - 9n + 7 = 0$.

Q.7 Given that $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + ... + a_{2n} x^{2n}$, find the values of :
(i) $a_0 + a_1 + a_2 + ... + a_{2n}$; (ii) $a_0 - a_1 + a_2 - a_3 ..... + a_{2n}$; (iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + a_{2n}^2$

Q.8 If a, b, c & d are the coefficients of any four consecutive terms in the expansion of $(1+x)^n$, $n \in N$, prove that $\frac{a}{a+b} + \frac{c}{c+d} = \frac{2b}{b+c}$.

Q.9 Find the value of x for which the fourth term in the expansion, $\left(5^{\frac{2}{5}\log_5 \sqrt{4^x+44}} + \frac{1}{5^{\log_5 \sqrt[3]{2^{x-1}+7}}}\right)^8$ is 336.

Q.10 Prove that : ${}^{n-1}C_r + {}^{n-2}C_r + {}^{n-3}C_r + ... + {}^rC_r = {}^nC_{r+1}$.

Q.11 (a) Which is larger : $(99^{50} + 100^{50})$ or $(101)^{50}$.

(b) Show that ${}^{2n-2}C_{n-2} + 2.{}^{2n-2}C_{n-1} + {}^{2n-2}C_n > \frac{4n}{n+1}$, $n \in N$, $n > 2$

Q.12 In the expansion of $\left(1 + x + \frac{7}{x}\right)^{11}$ find the term not containing x.

Q.13 Show that coefficient of x^5 in the expansion of $(1+x^2)^5 . (1+x)^4$ is 60.

Q.14 Find the coefficient of x^4 in the expansion of :
(i) $(1 + x + x^2 + x^3)^{11}$ (ii) $(2 - x + 3x^2)^6$

Q.15 Find numerically the greatest term in the expansion of :
(i) $(2+3x)^9$ when $x=\frac{3}{2}$ (ii) $(3-5x)^{15}$ when $x=\frac{1}{5}$

Q.16 Given $s_n = 1+q+q^2+.....+q^n$ & $S_n = 1+\frac{q+1}{2}+\left(\frac{q+1}{2}\right)^2+....+\left(\frac{q+1}{2}\right)^n$, $q \neq 1$, prove that ${}^{n+1}C_1 + {}^{n+1}C_2.s_1 + {}^{n+1}C_3.s_2 +....+ {}^{n+1}C_{n+1}.s_n = 2^n . S_n$.

Q.17 Prove that the ratio of the coefficient of x^{10} in $(1-x^2)^{10}$ & the term independent of x in $\left(x-\frac{2}{x}\right)^{10}$ is 1 : 32 .

Q.18 Find the term independent of x in the expansion of $(1+x+2x^3)\left(\frac{3x^2}{2}-\frac{1}{3x}\right)^9$.

Q.19 In the expansion of the expression $(x+a)^{15}$, if the eleventh term is the geometric mean of the eighth and twelfth terms, which term in the expansion is the greatest ?

Q.20 Let $(1+x^2)^2 . (1+x)^n = \sum_{K=0}^{n+4} a_K . x^K$. If a_1, a_2 & a_3 are in AP, find n.

Q.21 If the coefficient of a^{r-1}, a^r, a^{r+1} in the expansion of $(1+a)^n$ are in arithmetic progression, prove that $n^2 - n(4r+1) + 4r^2 - 2 = 0$.

Q.22 If ${}^nJ_r = \frac{(1-x^n)(1-x^{n-1})(1-x^{n-2})\ldots\ldots\ldots\ldots(1-x^{n-r+1})}{(1-x)(1-x^2)(1-x^3)\ldots\ldots\ldots\ldots(1-x^r)}$, prove that ${}^nJ_{n-r} = {}^nJ_r$.

Q.23 Prove that $\sum_{K=0}^{n} {}^nC_K \sin Kx . \cos(n-K)x = 2^{n-1} \sin nx$.

Q.24 The expressions $1+x$, $1+x+x^2$, $1+x+x^2+x^3$,..... $1+x+x^2+$..... $+x^n$ are multiplied together and the terms of the product thus obtained are arranged in increasing powers of x in the form of $a_0 + a_1x + a_2x^2 +$....., then,
(a) how many terms are there in the product.
(b) show that the coefficients of the terms in the product, equidistant from the beginning and end are equal.
(c) show that the sum of the odd coefficients = the sum of the even coefficients = $\frac{(n+1)!}{2}$

Q.25 Find the coeff. of (a) x^6 in the expansion of $(ax^2+bx+c)^9$.
(b) $x^2y^3z^4$ in the expansion of $(ax-by+cz)^9$.
(c) $a^2b^3c^4d$ in the expansion of $(a-b-c+d)^{10}$.

Q.26 If $\sum_{r=0}^{2n} a_r(x-2)^r = \sum_{r=0}^{2n} b_r(x-3)^r$ & $a_k = 1$ for all $k \geq n$, then show that $b_n = {}^{2n+1}C_{n+1}$.

Q.27 If $P_k(x) = \sum_{i=0}^{i=k-1} x^i$ then prove that, $\sum_{k=1}^{n} {}^nC_k P_k(x) = 2^{n-1} \cdot P_n\left(\frac{1+x}{2}\right)$

Q.28 Find the coefficient of x^r in the expression of :
$(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + ... + (x+2)^{n-1}$

Q.29(a) Find the index n of the binomial $\left(\frac{x}{5}+\frac{2}{5}\right)^n$ if the 9th term of the expansion has numerically the greatest coefficient $(n \in N)$.
(b) For which positive values of x is the fourth term in the expansion of $(5+3x)^{10}$ is the greatest.

Q.30 Prove that $\frac{(72)!}{(36!)^2} - 1$ is divisible by 73.

Q.31 If the 3rd, 4th, 5th & 6th terms in the expansion of $(x+y)^n$ be respectively a, b, c & d then prove that $\frac{b^2-ac}{c^2-bd} = \frac{5a}{3c}$.

Q.32 Find x for which the $(k+1)^{th}$ term of the expansion of $(x+y)^n$ is the greatest if $x+y=1$ and $x>0$, $y>0$.

Q.33 If x is so small that its square and higher powers may be neglected, prove that :
(i) $\frac{(1-3x)^{1/2}+(1-x)^{5/3}}{(4+x)^{1/2}} = 1-\left(\frac{41}{24}\right)x$ (ii) $\frac{\left(1-\frac{3x}{7}\right)^{1/3}+\left(1-\frac{3x}{5}\right)^{-5}}{\left(1+\frac{x}{2}\right)^{1/3}+\left(1-\frac{7x}{3}\right)^{1/7}} = 1+\left(\frac{10}{7}\right)x+\left(\frac{1}{12}\right)x$ or $1+\left(\frac{127}{84}\right)x$

Q.34 (a) If $x = \frac{1}{3}+\frac{1.3}{3.6}+\frac{1.3.5}{3.6.9}+\frac{1.3.5.7}{3.6.9.12}+..... \infty$ then prove that $x^2+2x-2=0$.

(b) If $y = \frac{2}{5}+\frac{1.3}{2!}\left(\frac{2}{5}\right)^2+\frac{1.3.5}{3!}\left(\frac{2}{5}\right)^3+.....$ then find the value of y^2+2y.

Q.35 If $p=q$ nearly and $n>1$, show that $\frac{(n+1)p+(n-1)q}{(n-1)p+(n+1)q} = \left(\frac{p}{q}\right)^{1/n}$.

EXERCISE - 2

Q.1 Show that the integral part in each of the following is odd. $n \in N$

(A) $\left(5 + 2\sqrt{6}\right)^n$ (B) $\left(8 + 3\sqrt{7}\right)^n$ (C) $\left(6 + \sqrt{35}\right)^n$

Q.2 Show that the integral part in each of the following is even. $n \in N$

(A) $\left(3\sqrt{3} + 5\right)^{2n+1}$ (B) $\left(5\sqrt{5} + 11\right)^{2n+1}$

Q.3 If $\left(7 + 4\sqrt{3}\right)^n = p+\beta$ where n & p are positive integers and β is a proper fraction show that $(1 - \beta)(p + \beta) = 1$.

Q.4 If x denotes $\left(2 + \sqrt{3}\right)^n$, $n \in N$ & [x] the integral part of x then find the value of : $x - x^2 + x[x]$.

Q.5 If $P = \left(8 + 3\sqrt{7}\right)^n$ and $f = P - [P]$, where [ ] denotes greatest integer function.
Prove that : $P(1 - f) = 1$ $(n \in N)$

Q.6 If $\left(6\sqrt{6} + 14\right)^{2n+1} = N$ & F be the fractional part of N, prove that $NF = 20^{2n+1}$ $(n \in N)$

Q.7 Prove that if p is a prime number greater than 2, then the difference $\left[\left(2 + \sqrt{5}\right)^p\right] - 2^{p+1}$ is divisible by p, where [ ] denotes greatest integer.

Q.8 Prove that the integer next above $\left(\sqrt{3} + 1\right)^{2n}$ contains 2^{n+1} as factor $(n \in N)$

Q.9 Let I denotes the integral part & F the proper fractional part of $\left(3 + \sqrt{5}\right)^n$ where $n \in N$ and if ρ denotes the rational part and σ the irrational part of the same, show that

$$\rho = \frac{1}{2}(I + 1) \text{ and } \sigma = \frac{1}{2}(I + 2F - 1).$$

Q.10 Prove that $\frac{{}^{2n}C_n}{n+1}$ is an integer, $\forall\, n \in N$.

EXERCISE - 3

(NOT IN THE SYLLABUS OF IIT-JEE)

PROBLEMS ON EXPONENTIAL & LOGARITHMIC SERIES

For Q.1 TO Q.15, Prove That :

Q.1 $\left(1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + ...\right)^2 - \left(1 + \frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + ...\right)^2 = 1$

Q.2 $\frac{e - 1}{e + 1} = \left(\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + ...\right) \div \left(\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + ...\right)$

Q.3 $\frac{e^2 - 1}{e^2 + 1} = \left(\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + ...\right) \div \left(1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + ...\right)$

Q.4 $1 + \frac{1+2}{2!} + \frac{1+2+3}{3!} + \frac{1+2+3+4}{4!} + ... \infty = \left(\frac{3}{2}\right) e$

Q.5 $\frac{1}{1.3} + \frac{1}{1.2.3.5} + \frac{1}{1.2.3.4.5.7} + ... \infty = \frac{1}{e}$

Q.6 $1 + \frac{1+2}{2!} + \frac{1+2+2^2}{3!} + \frac{1+2+2^2+2^3}{4!} + ... = e^2 - e$

Q.7 $1 + \frac{2^3}{2!} + \frac{3^3}{3!} + \frac{4^3}{4!} + ... = 5e$ Q 8. $\frac{2}{1!} + \frac{3}{2!} + \frac{6}{3!} + \frac{11}{4!} + \frac{18}{5!} + ... = 3(e - 1)$

Q.9 $\frac{1}{2.3} + \frac{1}{4.5} + \frac{1}{6.7} + ... \infty = 1 - \log_e 2$ Q 10. $1 + \frac{1}{3.2^2} + \frac{1}{5.2^4} + \frac{1}{7.2^6} + ... \infty = \log_e 3$

Q.11 $\frac{1}{1.2} + \frac{1}{3.4} + \frac{1}{5.6} + ... = \frac{1}{2} + \frac{1}{1.2.3} + \frac{1}{3.4.5} + \frac{1}{5.6.7} + ... = \ln 2$

Q.12 $\frac{1}{2} - \frac{1}{2.2^2} + \frac{1}{3.2^3} - \frac{1}{4.2^4} + ... = \ln 3 - \ln 2$ Q 13. $\frac{1}{3} + \frac{1}{3.3^3} + \frac{1}{5.3^5} + \frac{1}{7.3^7} + ... = \left(\frac{1}{2}\right) l\text{n } 2$

Q.14 $\frac{1}{2}\left(\frac{1}{2} + \frac{1}{3}\right) - \frac{1}{4}\left(\frac{1}{2^2} + \frac{1}{3^2}\right) + \frac{1}{6}\left(\frac{1}{2^3} + \frac{1}{3^3}\right) - ..... = \ln\sqrt{2}$

Q.15 If $y = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...$ where $|x| < 1$, then prove that $x = y + \frac{y^2}{2!} + \frac{y^3}{3!} + \frac{y^4}{4!} + ...$

EXERCISE - 4

If $C_0, C_1, C_2, ..., C_n$ are the combinatorial coefficients in the expansion of $(1 + x)^n$, $n \in N$, then prove the following :

Q.1 $C_0^2 + C_1^2 + C_2^2 + ... + C_n^2 = \frac{(2n)!}{n!\,n!}$ Q.2 $C_0\,C_1 + C_1\,C_2 + C_2\,C_3 + ... + C_{n-1}\,C_n = \frac{(2n)!}{(n+1)!\,(n-1)!}$

Q.3 $C_1 + 2C_2 + 3C_3 + ... + n\,.\,C_n = n\,.\,2^{n-1}$

Q.4 $C_0 + 2C_1 + 3C_2 + ... + (n+1)C_n = (n+2)2^{n-1}$

Q.5 $C_0 + 3C_1 + 5C_2 + ... + (2n+1)C_n = (n+1)\,2^n$

Q.6 $(C_0+C_1)(C_1+C_2)(C_2+C_3) ..... (C_{n-1}+C_n) = \frac{C_0\,.\,C_1\,.\,C_2\,....\,C_{n-1}\,(n+1)^n}{n!}$

Q.7 $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + ... + \frac{n\,.\,C_n}{C_{n-1}} = \frac{n(n+1)}{2}$ Q 8. $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + ... + \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$

Q.9 $2\,.\,C_o + \frac{2^2\,.\,C_1}{2} + \frac{2^3\,.\,C_2}{3} + \frac{2^4\,.\,C_3}{4} + ... \frac{2^{n+1}\,.\,C_n}{n+1} = \frac{3^{n+1}-1}{n+1}$

Q.10 $C_oC_r + C_1C_{r+1} + C_2C_{r+2} + ... + C_{n-r}\,C_n = \frac{2n!}{(n-r)!\,(n+r)!}$

Q.11 $C_o - \frac{C_1}{2} + \frac{C_2}{3} - ..... + (-1)^n\,\frac{C_n}{n+1} = \frac{1}{n+1}$

Q.12 $C_o - C_1 + C_2 - C_3 + ... + (-1)^r\,.\,C_r = \frac{(-1)^r\,(n-1)!}{r!\,.\,(n-r-1)!}$

Q.13 $C_o - 2C_1 + 3C_2 - 4C_3 + ... + (-1)^n\,(n+1)\,C_n = 0$

Q.14 $C_o^2 - C_1^2 + C_2^2 - C_3^2 + ... + (-1)^n\,C_n^2 = 0$ or $(-1)^{n/2}\,C_{n/2}$ according as n is odd or even.

Q.15 If n is an integer greater than 1, show that ;
$a - {}^nC_1(a-1) + {}^nC_2(a-2) - ..... + (-1)^n\,(a-n) = 0$

Q.16 $(n-1)^2\,.\,C_1 + (n-3)^2\,.\,C_3 + (n-5)^2\,.\,C_5 + ... = n\,(n+1)2^{n-3}$

Q.17 $1\,.\,C_o^2 + 3\,.\,C_1^2 + 5\,.\,C_2^2 + ... + (2n+1)\,C_n^2 = \frac{(n+1)\,(2n)!}{n!\,n!}$

Q.18 If $a_0, a_1, a_2, ...$ be the coefficients in the expansion of $(1 + x + x^2)^n$ in ascending powers of x, then prove that : (i) $a_0\,a_1 - a_1\,a_2 + a_2\,a_3 - .... = 0$.
(ii) $a_0a_2 - a_1a_3 + a_2a_4 - ..... + a_{2n-2}\,a_{2n} = a_{n+1}$ or a_{n-1}.
(iii) $E_1 = E_2 = E_3 = 3^{n-1}$; where $E_1 = a_0 + a_3 + a_6 + ...$; $E_2 = a_1 + a_4 + a_7 + ...$ & $E_3 = a_2 + a_5 + a_8 + ...$

Q.19 Prove that : $\sum_{r=0}^{n-2}\left({}^nC_r\,.\,{}^nC_{r+2}\right) = \frac{(2n)!}{(n-2)!\,(n+2)!}$

Q.20 If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_n\,x^n$, then show that the sum of the products of the C_i's taken two at a time, represented by $\underset{0 \le i < j \le n}{\Sigma\,\Sigma}\,C_i\,C_j$ is equal to $2^{2n-1} - \frac{2n!}{2(n!)^2}$.

Q.21 $\sqrt{C_1} + \sqrt{C_2} + \sqrt{C_3} + ... + \sqrt{C_n} \le 2^{n-1} + \frac{n-1}{2}$

Q.22 $\sqrt{C_1} + \sqrt{C_2} + \sqrt{C_3} + ... + \sqrt{C_n} \le \left[n\left(2^n-1\right)\right]^{1/2}$ for $n \ge 2$.

EXERCISE - 5

Q.1 If $(1+x)^{15} = C_0 + C_1.\,x + C_2.\,x^2 + ... + C_{15}.\,x^{15}$, then find the value of :
$C_2 + 2C_3 + 3C_4 + ... + 14C_{15}$

Q.2 If $(1 + x + x^2 + ... + x^p)^n = a_0 + a_1x + a_2x^2 + ... + a_{np}.\,x^{np}$, then find the value of :
$a_1 + 2a_2 + 3a_3 + ... + np\,.\,a_{np}$

Q.3 $1^2.\,C_0 + 2^2.\,C_1 + 3^2.\,C_2 + 4^2.\,C_3 + ... + (n+1)^2\,C_n = 2^{n-2}\,(n+1)\,(n+4)$.

Q.4 $\sum_{r=0}^{n} r^2\,.\,C_r = n(n+1)\,2^{n-2}$

Q.5 Given $p+q = 1$, show that $\sum_{r=0}^{n} r^2\,.\,{}^nC_r\,.\,p^r\,.\,q^{n-r} = np\left[(n-1)\,p+1\right]$

Q.6 Show that $\sum_{r=0}^{n} C_r\left(2r-n\right)^2 = n\,.\,2^n$ where C_r denotes the combinatorial coeff. in the expansion of $(1+x)^n$.

Q.7 $C_0 + \frac{C_1}{2}x + \frac{C_2}{3}x^2 + \frac{C_3}{4}x^3 + ... + \frac{C_n}{n+1}\,.\,x^n = \frac{(1+x)^{n+1}-1}{(n+1)x}$

Q.8 Prove that , $2\,.\,C_0 + \frac{2^2}{2}\,.C_1 + \frac{2^3}{3}\,.C_2 + ... + \frac{2^{11}}{11}\,.C_{10} = \frac{3^{11}-1}{11}$

Q.9 If $(1+x)^n = \sum_{r=0}^{n} C_r\,.\,x^r$ then prove that ;

$$\frac{2^2.C_0}{1.2}+\frac{2^3.C_1}{2.3}+\frac{2^4.C_2}{3.4}+.....+\frac{2^{n+2}.C_n}{(n+1)(n+2)}=\frac{3^{n+2}-2n-5}{(n+1)(n+2)}$$

Q.10 $\frac{C_0}{1}+\frac{C_2}{3}+\frac{C_4}{5}+.....=\frac{2^n}{n+1}$

Q.11 $\frac{C_0}{1}-\frac{C_1}{5}+\frac{C_2}{9}-\frac{C_3}{13}+.....(-1)^n\frac{C_n}{4n+1}=\frac{4^n.n!}{1.5.9.13.....(4n-3)(4n+1)}$

Q.12 $\frac{C_0}{2}+\frac{C_1}{3}+\frac{C_2}{4}+\frac{C_3}{5}+.....+\frac{C_n}{n+2}=\frac{1+n.2^{n+1}}{(n+1)(n+2)}$

Q.13 $\frac{C_0}{2}-\frac{C_1}{3}+\frac{C_2}{4}-\frac{C_3}{5}+.....+(-1)^n.\frac{C_n}{n+2}=\frac{1}{(n+1)(n+2)}$

Q.14 $\frac{C_1}{1}-\frac{C_2}{2}+\frac{C_3}{3}-\frac{C_4}{4}+.....+(-1)^{n-1}.\frac{C_n}{n}=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+.....+\frac{1}{n}$

Q.15 If $(1+x)^n=C_0+C_1x+C_2x^2+.....+C_nx^n$, then show that :

$$C_1(1-x)-\frac{C_2}{2}(1-x)^2+\frac{C_3}{3}(1-x)^3-....+(-1)^{n-1}\frac{1}{n}(1-x)^n=(1-x)+\frac{1}{2}(1-x^2)+\frac{1}{3}(1-x^3)+.....+\frac{1}{n}(1-x^n)$$

Q.16 Prove that , $\frac{1}{2}{}^nC_1-\frac{2}{3}{}^nC_2+\frac{3}{4}{}^nC_3-\frac{4}{5}{}^nC_4+.....+\frac{(-1)^{n+1}\,n}{n+1}.{}^nC_n=\frac{1}{n+1}$

Q.17 If $n\in N$; show that $\frac{{}^nC_0}{x}-\frac{{}^nC_1}{x+1}+\frac{{}^nC_2}{x+2}-.....+(-1)^n\frac{{}^nC_n}{x+n}=\frac{n!}{x(x+1)(x+2)....(x+n)}$

Q.18 Prove that , $({}^{2n}C_1)^2+2.({}^{2n}C_2)^2+3.({}^{2n}C_3)^2+...+2n.({}^{2n}C_{2n})^2=\frac{(4n-1)!}{[(2n-1)!]^2}$

Q.19 If $(1+x+x^2)^n=\sum_{r=0}^{2n}a_r\,x^r$, $n\in N$, then prove that

$$(r+1)\,a_{r+1}=(n-r)\,a_r+(2n-r+1)\,a_{r-1}.\qquad(0<r<2n)$$

Q.20 Prove that the sum to $(n+1)$ terms of $\frac{C_0}{n(n+1)}-\frac{C_1}{(n+1)(n+2)}+\frac{C_2}{(n+2)(n+3)}-.....$ equals

$\int_0^1 x^{n-1}.(1-x)^{n+1}\,.\,dx$ & evaluate the integral.

EXERCISE - 6

Q.1 The sum of the rational terms in the expansion of $\left(\sqrt{2}+3^{1/5}\right)^{10}$ is ___. [JEE '97, 2]

Q.2 If $a_n=\sum_{r=0}^{n}\frac{1}{{}^nC_r}$, then $\sum_{r=0}^{n}\frac{r}{{}^nC_r}$ equals [JEE'98, 2]

(A) $(n-1)a_n$ (B) $n\,a_n$ (C) $n\,a_n/2$ (D) None of these

Q.3 Find the sum of the series $\frac{3}{1!}+\frac{5}{2!}+\frac{9}{3!}+\frac{15}{4!}+\frac{23}{5!}+.....\infty$ [REE '98, 6]

Q.4 If in the expansion of $(1+x)^m(1-x)^n$, the co-efficients of x and x^2 are 3 and -6 respectively, then m is : [JEE '99, 2 (Out of 200)]

(A) 6 (B) 9 (C) 12 (D) 24

Q.5(i) For $2\le r\le n$, $\binom{n}{r}+2\binom{n}{r-1}+\binom{n}{r-2}=$

(A) $\binom{n+1}{r-1}$ (B) $2\binom{n+1}{r+1}$ (C) $2\binom{n+2}{r}$ (D) $\binom{n+2}{r}$

(ii) In the binomial expansion of $(a-b)^n$, $n\ge5$, the sum of the 5th and 6th terms is zero . Then $\frac{a}{b}$ equals:

[JEE '2000 (Screening) , 1 + 1]

(A) $\frac{n-5}{6}$ (B) $\frac{n-4}{5}$ (C) $\frac{5}{n-4}$ (D) $\frac{6}{n-5}$

Q.6 For any positive integers m, n (with $n\ge m$), let $\binom{n}{m}={}^nC_m$. Prove that

$$\binom{n}{m}+\binom{n-1}{m}+\binom{n-2}{m}+.....+\binom{m}{m}=\binom{n+1}{m+1}$$

Hence or otherwise prove that ,

$$\binom{n}{m}+2\binom{n-1}{m}+3\binom{n-2}{m}+.....+(n-m+1)\binom{m}{m}=\binom{n+2}{m+2}\,.$$

[JEE '2000 (Mains), 6]

Q.7 Find the largest co-efficient in the expansion of $(1 + x)^n$, given that the sum of co-efficients of the terms in its expansion is 4096. [REE '2000 (Mains)]

Q.8 In the binomial expansion of $(a - b)^n$, $n \geq 5$, the sum of the 5th and 6th terms is zero. Then $\frac{a}{b}$ equals

(A) $\frac{n-5}{6}$ (B) $\frac{n-4}{5}$ (C) $\frac{5}{n-4}$ (D) $\frac{6}{n-5}$

[JEE '2001 (Screening), 3]

Q.9 Find the coeffcient of x^{49} in the polynomial [REE '2001 (Mains), 3]

$\left(x - \frac{C_1}{C_0}\right)\left(x - 2^2 \cdot \frac{C_2}{C_1}\right)\left(x - 3^2 \cdot \frac{C_3}{C_2}\right) \ldots\ldots\ldots\ldots \left(x - 50^2 \cdot \frac{C_{50}}{C_{49}}\right)$ where $C_r = {}^{50}C_r$.

Q.10 The sum $\sum_{i=0}^{m} \binom{10}{i}\binom{20}{m-i}$, (where $\binom{p}{q} = 0$ if $P < q$) is maximum when m is

(A) 5 (B) 10 (C) 15 (D) 20

Q.11(a) Coefficient of t^{24} in the expansion of $(1+t^2)^{12}(1+t^{12})(1+t^{24})$ is

(A) ${}^{12}C_6 + 2$ (B) ${}^{12}C_6 + 1$ (C) ${}^{12}C_6$ (D) none

[JEE 2003, Screening 3 out of 60]

(b) Prove that : $2^K \cdot \binom{n}{0}\binom{n}{K} - 2^{K-1}\binom{n}{1}\binom{n-1}{K-1} + 2^{K-2}\binom{n}{2}\binom{n-2}{K-2} \ldots\ldots (-1)^K\binom{n}{K}\binom{n-K}{0} = \binom{n}{K}$.

[JEE 2003, Mains-2 out of 60]

Q.12 ${}^{n-1}C_r = (K^2 - 3).{}^nC_{r+1}$, if $K \in$

(A) $[-\sqrt{3}, \sqrt{3}]$ (B) $(-\infty, -2)$ (C) $(2, \infty)$ (D) $(\sqrt{3}, 2]$

[JEE 2004 (Screening)]

Q.13 The value of $\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} \ldots\ldots + \binom{30}{20}\binom{30}{30}$ *is, where* $\binom{n}{r} = {}^nC_r$.

(A) $\binom{30}{10}$ *(B)* $\binom{30}{15}$ *(C)* $\binom{60}{30}$ *(D)* $\binom{31}{10}$

[JEE 2005 (Screening)]

EXERCISE - 7

Part : (A) Only one correct option

1. In the expansion of $\left(3 - \sqrt{\frac{17}{4} + 3\sqrt{2}}\right)^{15}$, the 11th term is a:

(A) positive integer (B) positive irrational number (C) negative integer (D) negative irrational number.

2. If the second term of the expansion $\left[a^{1/13} + \frac{a}{\sqrt{a^{-1}}}\right]^n$ is $14a^{5/2}$ then the value of $\frac{{}^nC_3}{{}^nC_2}$ is:

(A) 4 (B) 3 (C) 12 (D) 6

3. The value of, $\frac{18^3 + 7^3 + 3.18.7.25}{3^6 + 6.243.2 + 15.81.4 + 20.27.8 + 15.9.16 + 6.3.32 + 64}$ is :

(A) 1 (B) 2 (C) 3 (D) none

4. Let the co-efficients of x^n in $(1 + x)^{2n}$ & $(1 + x)^{2n-1}$ be P & Q respectively, then $\left(\frac{P+Q}{Q}\right)^5 =$

(A) 9 (B) 27 (C) 81 (D) none of these

5. If the sum of the co-efficients in the expansion of $(1 + 2x)^n$ is 6561, then the greatest term in the expansion for $x = 1/2$ is :

(A) 4th (B) 5th (C) 6th (D) none of these

6. Find numerically the greatest term in the expansion of $(2 + 3x)^9$, when $x = 3/2$.

(A) ${}^9C_6 . 2^9 . (3/2)^{12}$ (B) ${}^9C_3 . 2^9 . (3/2)^6$ (C) ${}^9C_5 . 2^9 . (3/2)^{10}$ (D) ${}^9C_4 . 2^9 . (3/2)^8$

7. The numbers of terms in the expansion of $\left(a^3 + \frac{1}{a^3} + 1\right)^{100}$ is

(A) 201 (B) 300 (C) 200 (D) ${}^{100}C_3$

8. The coefficient of x^{10} in the expansion of $(1 + x^2 - x^3)^8$ is

(A) 476 (B) 496 (C) 506 (D) 528

9. $(1 + x)(1 + x + x^2)(1 + x + x^2 + x^3) \ldots\ldots (1 + x + x^2 + \ldots\ldots + x^{100})$ when written in the ascending power of x then the highest exponent of x is

(A) 505 (B) 5050 (C) 100 (D) 50

10. If $x = \left(7 + 4\sqrt{3}\right)^{2n} = [x] + f$, then $x(1 - f) =$

(A) 2 (B) 0 (C) 1 (D) 2520

11. The remainder when 2^{2003} is divided by 17 is

(A) 1 (B) 2 (C) 8 (D) none of these

12. The last two digits of the number 3^{400} are:

(A) 81 (B) 43 (C) 29 (D) 01

13. The value of $\binom{50}{0}\binom{50}{1} + \binom{50}{1}\binom{50}{2} + \ldots\ldots\ldots + \binom{50}{49}\binom{50}{50}$ is, where ${}^{n}C_{r} = \binom{n}{r}$

(A) $\binom{100}{50}$ (B) $\binom{100}{51}$ (C) $\binom{50}{25}$ (D) $\binom{50}{25}^2$

14. The value of the expression $\left(\sum_{r=0}^{10} {}^{10}C_r\right)\left(\sum_{K=0}^{10}(-1)^K \frac{{}^{10}C_K}{2^K}\right)$ is –

(A) 2^{10} (B) 2^{20} (C) 1 (D) 2^5

15. If $|x| < 1$, then the co-efficient of x^n in the expansion of $(1 + x + x^2 + x^3 + \ldots\ldots)^2$ is

(A) n (B) n – 1 (C) n + 2 (D) n + 1

16. The number of values of 'r' satisfying the equation, ${}^{39}C_{3r-1} - {}^{39}C_{r^2} = {}^{39}C_{r^2-1} - {}^{39}C_{3r}$ is :

(A) 1 (B) 2 (C) 3 (D) 4

17. Number of elements in set of value of r for which, ${}^{18}C_{r-2} + 2.\,{}^{18}C_{r-1} + {}^{18}C_r \geq {}^{20}C_{13}$ is satisfied

(A) 4 elements (B) 5 elements (C) 7 elements (D) 10 elements

18. The co-efficient of x^5 in the expansion of, $(1 + x)^{21} + (1 + x)^{22} + \ldots\ldots + (1 + x)^{30}$ is :

(A) ${}^{51}C_5$ (B) ${}^{9}C_5$ (C) ${}^{31}C_6 - {}^{21}C_6$ (D) ${}^{30}C_5 + {}^{20}C_5$

19. If $(1 + x)^{10} = a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{10}x^{10}$, then $(a_0 - a_2 + a_4 + a_6 + a_8 - a_{10})^2 + (a_1 - a_3 + a_5 - a_7 + a_9)^2$ is equal to

(A) 3^{10} (B) 2^{10} (C) 2^9 (D) none of these

20. The value of $\sum_{r=1}^{10} r \cdot \frac{{}^{n}C_r}{{}^{n}C_{r-1}}$ is equal to

(A) 5 (2n – 9) (B) 10 n (C) 9 (n – 4) (D) none of these

21. If $C_0, C_1, C_2, \ldots\ldots\ldots C_n$ are the Binomial coefficients in the expansion of $(1 + x)^n$. n being even, then $C_0 + (C_0 + C_1) + (C_0 + C_1 + C_2) + \ldots\ldots\ldots + (C_0 + C_1 + C_2 + \ldots\ldots\ldots + C_{n-1})$ is equal to

(A) n. 2^n (B) n. 2^{n-1} (C) n. 2^{n-2} (D) n. 2^{n-3}

22. If $(1 + x + 2x^2)^{20} = a_0 + a_1x + a_2x^2 + \ldots\ldots\ldots + a_{40}x^{40}$, then $a_0 + a_2 + a_4 \ldots\ldots + a_{38}$ equals

(A) $2^{19}(2^{30} + 1)$ (B) $2^{19}(2^{20} - 1)$ (C) $2^{20}(2^{19} - 1)$ (D) none of these

23. Co-efficient of x^{15} in $(1 + x + x^3 + x^4)^n$ is :

(A) $\sum_{r=0}^{5} {}^{n}C_{5-r} \cdot {}^{n}C_{3r}$ (B) $\sum_{r=0}^{5} {}^{n}C_{5r}$ (C) $\sum_{r=0}^{5} {}^{n}C_{3r}$ (D) $\sum_{r=0}^{3} {}^{n}C_{3-r} \cdot {}^{n}C_{5r}$

24. The sum of the coefficients of all the integral powers of x in the expansion of $\left(1 + 2\sqrt{x}\right)^{40}$ is

(A) $3^{40} + 1$ (B) $3^{40} - 1$ (C) $\frac{1}{2}(3^{40} - 1)$ (D) $\frac{1}{2}(3^{40} + 1)$

25. If $\{x\}$ denotes the fractional part of 'x', then $\left\{\frac{3^{1001}}{82}\right\} =$

(A) 9/82 (B) 81/82 (C) 3/82 (D) 1/82

26. The coefficient of the term independent of x in the expansion of $\left(\frac{x+1}{x^{\frac{2}{3}} - x^{\frac{1}{3}} + 1} - \frac{x-1}{x - x^{\frac{1}{2}}}\right)^{10}$ is

(A) 70 (B) 112 (C) 105 (D) 210

27. The coefficient of x^n in polynomial $(x + {}^{2n+1}C_0)(x + {}^{2n+1}C_1)(x + {}^{2n+1}C_2)\ldots\ldots(x + {}^{2n+1}C_n)$ is

(A) 2^{n+1} (B) $2^{2n+1} - 1$ (C) 2^{2n} (D) none of these

28. In the expansion of $(1 + x)^n (1 + y)^n (1 + z)^n$, the sum of the co-efficients of the terms of degree 'r' is :

(A) ${}^{n^3}C_r$ (B) ${}^{n}C_{r^3}$ (C) ${}^{3n}C_r$ (D) $3 \cdot {}^{2n}C_r$

29. $\sum_{r=1}^{n}\left(\sum_{p=0}^{r-1} {}^{n}C_r \, {}^{r}C_p \, 2^p\right)$ is equal to

(A) $4^n - 3^n + 1$ (B) $4^n - 3^n - 1$ (C) $4^n - 3^n + 2$ (D) $4^n - 3^n$

30. If $(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$, then show that the sum of the products of the C_i's taken two at a time, represented by $\sum\sum_{0 \leq i < j \leq n} C_i \, C_j$ is equal to

(A) $2^{2n+1} - \frac{2n!}{2(n!)}$ (B) $2^{2n-1} - \frac{2n!}{2(n!)^2}$ (C) $2^{2n-1} - \frac{2n!}{2(n!)}$ (D) $2^{2n+1} - \frac{2n!}{2(n!)^2}$

Part : (B) May have more than one options correct

31. In the expansion of $(x + y + z)^{25}$

(A) every term is of the form ${}^{25}C_r \cdot {}^{r}C_k \cdot x^{25-r} \cdot y^{r-k} \cdot z^k$

(B) the coefficient of $x^8 y^9 z^9$ is 0 (C) the number of terms is 325
(D) none of these

32. $7^9 + 9^7$ is divisible by
(A) 16 (B) 24 (C) 64 (D) 72

EXERCISE - 8

1. Find the value of 'x' for which the fourth term in the expansion, $\left(5^{\frac{2}{5}\log_5\sqrt{4^x+44}} + \frac{1}{5^{\log_5\sqrt[3]{2^{x-1}+7}}}\right)^8$ is 336.

2. In the binomial expansion of $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}}\right)^n$, the ratio of the 7th term from the begining to the 7th term from the end is 1 : 6 ; find n.

3. Find the terms independent of 'x' in the expansion of the expression, $(1 + x + 2x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$.

4. If in the expansion of $(1 - x)^{2n-1}$, the co-efficient of x^r is denoted by a_r, then prove that $a_{r-1} + a_{2n-r} = 0$.

5. Show that the term independent of x in the expansion of $\left(1 + x + \frac{6}{x}\right)^{10}$ is, $1 + \sum_{r=1}^{5} {}^{10}C_{2r}\, {}^{2r}C_r\, 6^r$.

6. Find the coefficient of $a^5 b^4 c^7$ in the expansion of $(bc + ca + ab)^8$.

7. If $(1 + 2x + 3x^2)^{10} = a_0 + a_1x + a_2x^2 + ... + a_{20}x^{20}$, then calculate a_1, a_2, a_4.

8. If $\left(3\sqrt{3} + 5\right)^n = p + f$, where p is an integer and f is a proper fraction then find the value of $\left(3\sqrt{3} - 5\right)^n$, $n \in N$.

9. Write down the binomial expansion of $(1 + x)^{n+1}$, when x = 8. Deduce that $9^{n+1} - 8n - 9$ is divisible by 64, whenever n is a positive integer.

10. Prove that $53^{53} - 33^{33}$ is divisible by 10.

11. Which is larger : $(99^{50} + 100^{50})$ or $(101)^{50}$.

If $C_0, C_1, C_2, ..., C_n$ are the combinatorial co-efficients in the expansion of $(1 + x)^n$, $n \in N$, then prove the followings: (Q. No. 12 - 14)

12. $2.C_o + \frac{2^2.C_1}{2} + \frac{2^3.C_2}{3} + \frac{2^4.C_3}{4} + ...\frac{2^{n+1}.C_n}{n+1} = \frac{3^{n+1} - 1}{n+1}$

13. $\frac{C_1}{C_0} + 2.\frac{C_2}{C_1} + 3\frac{C_3}{C_2} + ... + n\frac{C_n}{C_{n-1}} = \frac{n(n+1)}{2}$

14. $1^2.C_0 + 2^2.C_1 + 3^2.C_2 + 4^2.C_3 + ... + (n+1)^2 C_n = 2^{n-2}(n+1)(n+4)$.

15. Assuming 'x' to be so small that x^2 and higher powers of 'x' can be neglected, show that, $\frac{\left(1 + \frac{3}{4}x\right)^{-4}(16 - 3x)^{1/2}}{(8 + x)^{2/3}}$ is approximately equal to, $1 - \frac{305}{96}x$.

16. If $\sum_{r=0}^{n} (-1)^r.{}^nC_r\left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + ... \text{ to m terms}\right] = k\left(1 - \frac{1}{2^{mn}}\right)$, then find the value of k.

17. Find the coefficient of x^{50} in the expression:
$(1 + x)^{1000} + 2x.(1 + x)^{999} + 3x^2(1 + x)^{998} + ... + 1001\, x^{1000}$

18. Given $s_n = 1 + q + q^2 + ... + q^n$ & $S_n = 1 + \frac{q+1}{2} + \left(\frac{q+1}{2}\right)^2 + ... + \left(\frac{q+1}{2}\right)^n$, $q \neq 1$,
prove that ${}^{n+1}C_1 + {}^{n+1}C_2.s_1 + {}^{n+1}C_3.s_2 + ... + {}^{n+1}C_{n+1}.s_n = 2^n.S_n$.

19. Show that if the greatest term in the expansion of $(1 + x)^{2n}$ has also the greatest co-efficient, then 'x' lies between, $\frac{n}{n+1}$ & $\frac{n+1}{n}$.

20. Find the remainder when $32^{32^{32}}$ is divided by 7.

21. If $(1 + x + x^2 + ... + x^p)^n = a_0 + a_1x + a_2x^2 + ... + a_{np}.x^{np}$, then find the value of :
$a_1 + 2a_2 + 3a_3 + ... + np.a_{np}$.

22. Prove that, $({}^{2n}C_1)^2 + 2.({}^{2n}C_2)^2 + 3.({}^{2n}C_3)^2 + ... + 2n.({}^{2n}C_{2n})^2 = \frac{(4n-1)!}{\{(2n-1)!\}^2}$

23. If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, then show that:

$$C_1(1-x) - \frac{C_2}{2}(1-x)^2 + \frac{C_3}{3}(1-x)^3 - ..... + (-1)^{n-1}\frac{1}{n}(1-x)^n$$
$$= (1-x) + \frac{1}{2}(1-x^2) + \frac{1}{3}(1-x^3) + ... + \frac{1}{n}(1-x^n)$$

24. Prove that $\sum_{r=0}^{n} r^2\, {}^nC_r\, p^r q^{n-r} = npq + n^2p^2$ if $p + q = 1$.

25. If $\sum_{r=0}^{2n} a_r (x-2)^r = \sum_{r=0}^{2n} b_r (x-3)^r$ & $a_k = 1$ for all $k \geq n$, then show that $b_n = {}^{2n+1}C_{n+1}$.

26. If $a_0, a_1, a_2, \ldots$ be the coefficients in the expansion of $(1 + x + x^2)^n$ in ascending powers of x, then prove that :
(i) $a_0 a_1 - a_1 a_2 + a_2 a_3 - \ldots = 0$ (ii) $a_0 a_2 - a_1 a_3 + a_2 a_4 - \ldots + a_{2n-2} a_{2n} = a_{n+1}$
(iii) $E_1 = E_2 = E_3 = 3^{n-1}$; where $E_1 = a_0 + a_3 + a_6 + \ldots$; $E_2 = a_1 + a_4 + a_7 + \ldots$ & $E_3 = a_2 + a_5 + a_8 + \ldots$

27. If $(1 + x)^n = p_0 + p_1 x + p_2 x^2 + p_3 x^3 + \ldots\ldots$, then prove that :

(a) $p_0 - p_2 + p_4 - \ldots\ldots = 2^{n/2} \cos \frac{n\pi}{4}$ (b) $p_1 - p_3 + p_5 - \ldots\ldots = 2^{n/2} \sin \frac{n\pi}{4}$

28. If $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \ldots + C_n x^n$, then show that the sum of the products of the C_i's taken two at a time, represented by $\sum\sum_{0 \leq i < j \leq n} C_i \, C_j$ is equal to $2^{2n-1} - \frac{2n!}{2\,(n!)^2}$.

ANSWER KEY *EXERCISE - 1*

Q 1. **(i)** ${}^{11}C_5 \frac{a^6}{b^5}$ **(ii)** ${}^{11}C_6 \frac{a^5}{b^6}$ **(iii)** $ab = 1$ **Q 2.** $r = 6$ **Q 3.** $r = 5$ or 9 **Q 4.** (a) $\frac{5}{12}$ (b) T_6 **=7**

Q 5. $\frac{\left(2^{mn}-1\right)}{\left(2^n-1\right)\left(2^{mn}\right)}$ **Q 7. (i)** 3^n **(ii)** 1, **(iii)** a_n **Q 9.** $x = 0$ or 1 **Q 10.** $x = 0$ or 2

Q 11. **(a)** 101^{50} (Prove that $101^{50} - 99^{50} = 100^{50} +$ some +ive qty) **Q 12.** $1 + \sum_{k=1}^{5} {}^{11}C_{2k} \cdot {}^{2k}C_k \, 7^k$

Q 14. **(i)** 990 **(ii)** 3660 **Q 15. (i)** $T_7 = \frac{7.3^{13}}{2}$ **(ii)** 455×3^{12} **Q 18.** $\frac{17}{54}$

Q.19 T_8 **Q.20** $n = 2$ or 3 or 4 **Q.24** **(a)** $\frac{n^2+n+2}{2}$

Q 25. **(a)** $84b^6c^3 + 630ab^4c^4 + 756a^2b^2c^5 + 84a^3c^6$ **; (b)** $-1260 \cdot a^2b^3c^4$ **; (c)** -12600

Q 28. ${}^nC_r (3^{n-r} - 2^{n-r})$ **Q 29.** (a) $n = 12$ (b) $\frac{5}{8} < x < \frac{20}{21}$ **Q.32** $\frac{n-k}{n}$

Q 34. **(a)** Hint : Add 1 to both sides & compare the RHS series with the expansion $(1+y)^n$ to get n & y **(b)** 4

EXERCISE - 2

Q.4 1

EXERCISE - 5

Q 1. divide expansion of $(1+x)^{15}$ both sides by x & diff. w.r.t.x , put $x = 1$ to get 212993

Q 2. Differentiate the given expn. & put $x = 1$ to get the result $\frac{np}{2}(p+1)^n$

Q 9. Integrate the expn. of $(1 + x)^n$. Determine the value of constant of integration by putting $x = 0$. Integrate the result again between 0 & 2 to get the result.

Q 10. Consider $\frac{1}{2}[(1+x)^n + (1-x)^n] = C_0 + C_2x^2 + C_4x^4 + \ldots\ldots$ Integrate between 0 & 1.

Q 12. Multiply both sides by x the expn. $(1+x)^n$. Integrate both sides between 0 & 1.

Q 14. Note that $\frac{(1-x)^n - 1}{x} = -C_1 + C_2x - C_3x^2 + \ldots + C_n \cdot x^{n-1}$. Integrate between 1 & 0

Q 20. $\frac{(n-1)!\,(n+1)!}{(2n+1)!}$

EXERCISE - 6

Q.1 41 **Q.2** C **Q.3** $4e - 3$ **Q.4** C **Q.5** **(i)** D **(ii)** B
Q.7 ${}^{12}C_6$ **Q.8** B **Q.9** -22100 **Q.10** C
Q.11 **(a)** A **Q.12** D **Q.13** A

EXERCISE - 7

1.	B	**2.**	A	**3.**	A	**4.**	D	**5.**	B	**6.**	A	**7.**	A	**8.**	A
9.	B	**10.**	C	**11.**	C	**12.**	D	**13.**	B	**14.**	C	**15.**	D	**16.**	B
17.	C	**18.**	C	**19.**	B	**20.**	A	**21.**	B	**22.**	B	**23.**	A	**24.**	D
25.	C	**26.**	D	**27.**	C	**28.**	C	**29.**	D	**30.**	B	**31.**	AB	**32.**	AC

EXERCISE - 8

1. $x = 0$ or 1 **2.** $n = 9$ **3.** $\frac{17}{54}$ **6.** 280 **7.** $a_1 = 20$, $a_2 = 210$, $a_4 = 8085$

8. 1 – f, if n is even and f, if n is odd **11.** 101^{50} **16.** $\frac{1}{2^n - 1}$ **17.** ${}^{1002}C_{50}$

20. 4 **21.** $\frac{np}{2}(p+1)^n$

Some questions (Assertion–Reason type) are given below. Each question contains **Statement – 1 (Assertion)** and **Statement – 2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. So select the correct choice :***Choices are :***

(A)**Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is** a correct explanation for **Statement – 1**.

(B)**Statement – 1** is True, **Statmnt – 2** is True; **Statement – 2 is NOT** a correct explanation for **Statement – 1**.

(C) **Statement – 1** is True, **Statement – 2** is False.

(D) Statement – 1 is False, Statement – 2 is True.

BINOMIAL THEOREM

373. **Statement-1:** The binomial theorem provides an expansion for the expression $(a + b)^n$. where a, b, $n \in R$.
Statement-2: All coefficients in a binomial expansion may be obtained by Pascal's triangle.

374. **Statement-1:** If n is an odd prime then integral part of $(\sqrt{5}+2)^n - 2^{n+1}([x]$ is divisible by 20 n.
Statement-2: If n is prime then ${}^nC_1, {}^nC_2, {}^nC_3, \ldots .. {}^nC_{n-1}$ must be divisible by n.

375. **Statement–1 :** 2^{60} when divided by 7 leaves the reminder 1.
Statement–2 : $(1 + x)^n = 1 + n_1x$, where $n, n_1 \in N$.

376. **Statement–1 :** ${}^{21}C_0 + {}^{21}C_1 + \ldots + {}^{21}C_{10} = 2^{20}$
Statement–2 : ${}^{2n+1}C_0 + {}^{2n+1}C_1 + \ldots {}^{2n+1}C_{2n+1} = 2^{2n+1}$ and ${}^nC_r = {}^nC_{n-r}$

377. Let n be a positive integers and k be a whole number, $k \le 2n$.
Statement–1 : The maximum value of ${}^{2n}C_k$ is ${}^{2n}C_n$.
Statement–2 : $\frac{{}^{2n}C_{k+1}}{{}^{2n}C_k} > 1$, for k = 0, 1, 2, . . . , n – 1.

378. Let n be a positive integer. **Statement–1 :** $3^{2n+2} - 8n - 9$ is divisible by 64.
Statement–2 : $3^{2n+2} - 8n - 9 = (1+8)^{n+1} - 8n - 9$ and in the binomial expansion of $(1+8)^{n+1}$, sum of first two terms is 8n + 9 and after that each term is a multiple of 8^2.

379. **Statement–1 :** If n is an odd prime, then integral part of $\left(\sqrt{5}+2\right)^n$ is divisible by 20n.
Statement–2 : If n is prime, then ${}^nc_1, {}^nc_2, {}^nc_3 \ldots {}^nc_{n-1}$ must be divisible by n.

380. **Statement–1 :** The coefficient of x^{203} in the expression $(x - 1)(x^2 - 2)(x^2 - 3) \ldots (x^{20} - 20)$ must be 13.
Statement–2 : The coefficient of x^8 in the expression $(2 + x)^2 (3 + x)^3 (4 + x)^4$ is equal to 30.

381. **Statement–1 :** ${C_0}^2 + {C_1}^2 + {C_2}^2 + {C_3}^2 + \ldots + {C_n}^2 = \frac{2n!}{(n!)^2}$ **Statement–2 :** ${}^nC_0 - {}^nC_1 + {}^nC_2 \ldots + (-1)^n {}^nC_n = 0$

382. **Statement–1 :** Some of coefficient $(x - 2y + 4z)^n$ is 3^n
Statement–2 : Some of coefficient of $(c_0x_0 + c_1x_1 + c_2x_2 + \ldots + c_nx^n)^n$ is 2^n

383. **Statement-1:** The greatest coefficient in the expansion of $(a_1 + a_2 + a_3 + a_4)^{17}$ is $\frac{17!}{(3!)^3\, 4!}$
Statement-2: The number of distinct terms in $(1 + x + x^2 + x^3 + x^4 + x^5)^{100}$ is 501.

384. **Statement-1:** The co-efficient of x^5 in the expansion of $(1 + x^2)^5 (1 + x)^4$ is 120
Statement-2: The sum of the coefficients in the expansion of $(1 + 2x - 3y + 5z)^3$ is 125.

385. **Statement-1:** The number of distinct terms in $(1 + x + x^2 + x^3 + x^4)^{1000}$ is 4001
Statement-2: The number of distinct terms in the expansion $(a_1 + a_2 + \ldots + a_m)^n$ is ${}^{n+m-1}C_{m-1}$

386. **Statement-1:** In the expansion of $(1 + x)^{30}$, greatest binomial coefficient is ${}^{30}C_{15}$
Statement-2: In the expansion of $(1 + x)^{30}$, the binomial coefficients of equidistant terms from end & beginning are equal.

387. **Statement-1:** Integral part of $\left(\sqrt{3}+1\right)^{2n+1}$ is even where $n \in I$.
Statement-2: Integral part of any integral power of the expression of the form of $p + \sqrt{q}$ is even.

388. **Statement-1 :** $\sum_{r=4}^{20} {}^{r}C_4 = {}^{21}C_4$ **Statement-2:** $1 + x + x^2 + x^3 + ... + x^{n-1} = \frac{1-x^n}{1-x}$ = sum of n terms of GP.

389. **Statement-1:** Last two digits of the number $(13)^{41}$ are 31.
Statement-2: When a number in divided by 1000, the remainder gives the last three digits.

390. **Statement-1:** ${}^{n}C_0 + {}^{n}C_1 + {}^{n}C_2 + ... + {}^{n}C_n = 2^n$ where $n \in N$.
Statement-2: The all possible selections of n distinct objects are 2^n.

391. **Statement-1 :** The integral part of $(5+2\sqrt{6})^n$ is odd, where $n \in N$.
Statement-2 : $(x + a)^n - (x - a)^n = 2[{}^{n}C_0x^n + {}^{n}C_2x^{x-2} a^2 + {}^{n}C_4 + x^{n-4} a^4 + ...]$

392. **Statement-1:** If n is even than ${}^{2n}C_1 + {}^{2n}C_3 + {}^{2n}C_5 + ... + {}^{2n}C_{n-1} = 2^{2n-1}$
Statement-2: ${}^{2n}C_1 + {}^{2n}C_3 + {}^{2n}C_5 + ... + {}^{2n}C_{2n-1} = 2^{2n-1}$

393. **Statement-1 :** Any positive integral power of $\left(\sqrt{2}-1\right)$ can be expressed as $\sqrt{N}-\sqrt{N-1}$ for some natural number N > 1.
Statement-2 : Any positive integral power of $\sqrt{2}-1$ can be expressed as A + B $\sqrt{2}$ where A and B are integers.

394. **Statement-1 :** The term independent of x in the expansion of $\left(x+\frac{1}{x}+3\right)^m$ is $\frac{4m!}{(2m!)^2}$.
Statement-2: The Coefficient of x^b in the expansion of $(1 + x)^n$ is ${}^{n}C_b$.

395. **Statement-1:** The coefficient of x^8 in the expansion of $(1 + 3x + 3x^2 + x^3)^{17}$ is ${}^{51}C_2$.
Statement-2 : Coefficient of x^r in the expansion of $(1 + x)^n$ is ${}^{n}C_r$.

396. **Statement-1:** If $(1 + x)^n = c_0+c_1x + c_2x^2 + \ldots + c_nx^n$ then
$c_0 - 2.c_1 + 3.c_2 - ..... + (-1)^n (n + 1)c_n = 0$
Statement-2: Coefficients of equidistant terms in the expansion of $(x + a)^n$ where $n \in N$ are equal.

397. **Statement-1:** $\sum_{k=1}^{n} k\left({}^{n}C_n\right)^2 = n\ {}^{2n-1}C_{n-1}$
Statement-2: If 2^{2003} is divided by 15 then remainder is 8.

398. **Statement-1:** The co-efficient of $(1 + x^2)^5 (1 + x)^4$ is 120.
Statement-2: The integral part of $(\sqrt{5}+2)^{10}$ is odd.

ANSWER

373. D	374. A	375. A	376. A	377. A	378. A	379. A	380. C	381. B	382. C	383. D		
384. D	385. B	386. B	387. C	388. D	389. D	390. A	391. B.	392. D	393. A	394. D	395. D	
396. B	397. B	398. D	399. A									

QUE. FROM COMPT. EXAMS.

1. The value of $(\sqrt{2}+1)^6 + (\sqrt{2}-1)^6$ will be **[RPET 1997]**
(a) – 198 (b) 198 (c) 99 (d) – 99

2. If $(1+ax)^n = 1 + 8x + 24x^2 + ...$, then the value of a and n is **[IIT 1983; Pb. CET 1994, 99]**
(a) 2, 4 (b) 2, 3 (c) 3, 6 (d) 1, 2

3. The coefficient of x^5 in the expansion of $(1 + x^2)^5 (1 + x)^4$ is **[EAMCET 1996; UPSEAT 2001; Pb. CET 2002]**
(a) 30 (b) 60 (c) 40 (d) None of these

4. If $\frac{(1-3x)^{1/2} + (1-x)^{5/3}}{\sqrt{4-x}}$ is approximately equal to $a + bx$ for small values of x, then $(a,b) =$
(a) $\left(1,\frac{35}{24}\right)$ (b) $\left(1,-\frac{35}{24}\right)$ (c) $\left(2,\frac{35}{12}\right)$ (d) $\left(2,-\frac{35}{12}\right)$

5. The value of x in the expression $[x + x^{\log_{10}(x)}]^5$, if the third term in the expansion is 10,00,000 **[Roorkee 1992]**
(a) 10 (b) 11 (c) 12 (d) None of these

6. If the coefficient of the middle term in the expansion of $(1+x)^{2n+2}$ is p and the coefficients of middle terms in the expansion of $(1+x)^{2n+1}$ are q and r, then

(a) $p+q=r$ (b) $p+r=q$ (c) $p=q+r$ (d) $p+q+r=0$

7. In the polynomial $(x-1)(x-2)(x-3)\ldots\ldots\ldots\ldots(x-100)$, the coefficient of x^{99} is **[AMU 2002]**

(a) 5050 (b) -5050 (c) 100 (d) 99

8. The coefficient of x^{100} in the expansion of $\sum_{j=0}^{200}(1+x)^j$ is **[UPSEAT 2004]**

(a) $\binom{200}{100}$ (b) $\binom{201}{102}$ (c) $\binom{200}{101}$ (d) $\binom{201}{100}$

9. If the coefficient of x^7 in $\left(ax^2+\frac{1}{bx}\right)^{11}$ is equal to the coefficient of x^{-7} in $\left(ax-\frac{1}{bx^2}\right)^{11}$, then ab = **[MP PET 1999; AMU 2001; Pb. CET 2002; AIEEE 2005]**

(a) 1 (b) 1/2 (c) 2 (d) 3

10. If the coefficient of x in the expansion of $\left(x^2+\frac{k}{x}\right)^5$ is 270, then k = **[EAMCET 2002]**

(a) 1 (b) 2 (c) 3 (d) 4

11. The coefficients of three successive terms in the expansion of $(1+x)^n$ are 165, 330 and 462 respectively, then the value of n will be **[UPSEAT 1999]**

(a) 11 (b) 10 (c) 12 (d) 8

12. If the coefficient of $(2r+4)^{th}$ and $(r-2)^{th}$ terms in the expansion of $(1+x)^{18}$ are equal, then r= **[MP PET 1997; Pb. CET 2001]**

(a) 12 (b) 10 (c) 8 (d) 6

13. The middle term in the expansion of $(1+x)^{2n}$ is **[Pb. CET 1998]**

(a) $\frac{1.3.5....(5n-1)}{n!}x^n$ (b) $\frac{2.4.6....2n}{n!}x^{2n+1}$ (c) $\frac{1.3.5....(2n-1)}{n!}x^n$ (d) $\frac{1.3.5....(2n-1)}{n!}2^n x^n$

14. The value of $\binom{30}{0}\binom{30}{10}-\binom{30}{1}\binom{30}{11}+\binom{30}{2}\binom{30}{12}+\ldots\ldots+\binom{30}{20}\binom{30}{30}$ **[IIT Screening 2005]**

(a) ${}^{60}C_{20}$ (b) ${}^{30}C_{10}$ (c) ${}^{60}C_{30}$ (d) ${}^{40}C_{30}$

15. Middle term in the expansion of $(1+3x+3x^2+x^3)^6$ is **[MP PET 1997]**

(a) 4th (b) 3rd (c) 10th (d) None of these

16. Two middle terms in the expansion of $\left(x-\frac{1}{x}\right)^{11}$ are

(a) $231x$ and $\frac{231}{x}$ (b) $462x$ and $\frac{462}{x}$ (c) $-462x$ and $\frac{462}{x}$ (d) None of these

17. The term independent of y in the expansion of $(y^{-1/6}-y^{1/3})^9$ is **[BIT Ranchi 1980]**

(a) 84 (b) 8.4 (c) 0.84 (d) -84

18. The coefficient of the term independent of x in the expansion of $(1+x+2x^3)\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$ is **[DCE 1994]**

(a) $\frac{1}{3}$ (b) $\frac{19}{54}$ (c) $\frac{17}{54}$ (d) $\frac{1}{4}$

19. The term independent of x in $\left[\frac{\sqrt{x}}{3}+\frac{\sqrt{3}}{x^2}\right]^{10}$ is **[EAMCET 1984; RPET 2000]**

(a) $\frac{2}{3}$ (b) $\frac{5}{3}$ (c) $\frac{4}{3}$ (d) None of these

20. The term independent of x in $\left(\sqrt{x}-\frac{2}{x}\right)^{18}$ is **[EAMCET 1990]**

(a) ${}^{18}C_6\,2^6$ (b) ${}^{18}C_6\,2^{12}$ (c) ${}^{18}C_{18}\,2^{18}$ (d) None of these

21. The largest term in the expansion of $(3+2x)^{50}$ where $x=\frac{1}{5}$ is **[IIT Screening 1993]**

(a) 5th (b) 51st (c) 7th (d) 6th

22. $\frac{C_1}{C_0}+2\frac{C_2}{C_1}+3\frac{C_3}{C_2}+....+15\frac{C_{15}}{C_{14}}=$ **[IIT 1962]**

(a) 100 (b) 120 (c) −120 (d) None of these

23. $\binom{n}{0}+2\binom{n}{1}+2^2\binom{n}{2}+.....+2^n\binom{n}{n}$ is equal to **[AMU 2000]**

(a) 2^n (b) 0 (c) 3^n (d) None of these

24. If C_r stands for ${}^nC_r$, the sum of the given series

$\frac{2(n/2)!(n/2)!}{n!}[C_0^2-2C_1^2+3C_2^2-.....+(-1)^n(n+1)C_n^2]$, Where n is an even positive integer, is **[IIT 1986]**

(a) 0 (b) $(-1)^{n/2}(n+1)$ (c) $(-1)^n(n+2)$ (d) $(-1)^{n/2}(n+2)$

25. Sum of odd terms is A and sum of even terms is B in the expansion $(x+a)^n$, then **[RPET 1987; UPSEAT 2004]**

(a) $AB=\frac{1}{4}(x-a)^{2n}-(x+a)^{2n}$ (b) $2AB=(x+a)^{2n}-(x-a)^{2n}$

(c) $4AB=(x+a)^{2n}-(x-a)^{2n}$ (d) None of these

26. In the expansion of $(x+a)^n$, the sum of odd terms is P and sum of even terms is Q, then the value of (P^2-Q^2) will be **[RPET 1997; Pb. CET 1998]**

(a) $(x^2+a^2)^n$ (b) $(x^2-a^2)^n$ (c) $(x-a)^{2n}$ (d) $(x+a)^{2n}$

27. The sum of the coefficients in the expansion of $(1+x-3x^2)^{2163}$ will be **[IIT 1982]**

(a) 0 (b) 1 (c) −1 (d) 2^{2163}

28. If the sum of the coefficients in the expansion of $(1-3x+10x^2)^n$ is a and if the sum of the coefficients in the expansion of $(1+x^2)^n$ is b, then **[UPSEAT 2001]**

(a) $a=3b$ (b) $a=b^3$ (c) $b=a^3$ (d) None of these

29. The sum of the coefficients in the expansion of $(x+y)^n$ is 4096. The greatest coefficient in the expansion is **[Kurukshetra CEE 1998; AIEEE 2002]**

(a) 1024 (b) 924 (c) 824 (d) 724

30. If the sum of the coefficients in the expansion of $(\alpha x^2-2x+1)^{35}$ is equal to the sum of the coefficients in the expansion of $(x-\alpha y)^{35}$, then $\alpha=$

(a) 0 (b) 1

(c) May be any real number (d) No such value exist

31. For every natural number n, $3^{2n+2}-8n-9$ is divisible by **[IIT 1977]**

(a) 16 (b) 128 (c) 256 (d) None of these

32. The least remainder when 17^{30} is divided by 5 is

[Karnataka CET 2003]

(a) 1 (b) 2 (c) 3 (d) 4

33. The value of the natural numbers n such that the inequality $2^n > 2n+1$ is valid is **[MNR 1994]**

(a) For $n \geq 3$ (b) For $n < 3$ (c) For mn (d) For any n

34. Let $P(n)$ be a statement and let $P(n) \Rightarrow p(n+1)$ for all natural numbers n, then $P(n)$ is true

(a) For all n (b) For all $n > 1$

(c) For all $n > m$, m being a fixed positive integer

(d) Nothing can be said

35. $(1+x)^n - nx - 1$ is divisible by (where $n \in N$)

(a) $2x$ (b) x^2 (c) $2x^3$ (d) All of these

ANSWER KEY

1	b	2	a	3	b	4	b	5	a
6	c	7	b	8	a	9	a	10	c
11	a	12	d	13	d	14	b	15	c
16	c	17	d	18	c	19	b	20	a
21	c	22	b	23	c	24	d	25	c
26	b	27	c	28	b	29	b	30	b
31	a	32	d	33	a	34	d	35	b

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 9 XI M 9. Progressions

Index:

1. Key Concepts

2. Exercise I to V

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Sequence & Progression

Sequence : A sequence is a function whose domain is the set N of natural numbers. Since the domain for every sequence is the set N of natural numbers, therefore a sequence is represented by its range. If $f : N \to R$, then $f(n) = t_n$ $n \in N$ is called a sequence and is denoted by
$\{f(1), f(2), f(3),.....\} =$
$\{t_1, t_2, t_3, ...\} = \{t_n\}$

Real Sequence : A sequence whose range is a subset of R is called a real sequence.

Examples : (i) 2, 5, 8, 11, ... (ii) 4, 1, – 2, – 5, ...
(iii) 3, –9, 27, – 81, ...

Types of Sequence : On the basis of the number of terms there are two types of sequence.

(i) Finite sequences : A sequence is said to be finite if it has finite number of terms.

(ii) Infinite sequences : A sequenceis said to be infinite if it has infinite number of terms.

Solved Example # 1 Write down the sequence whose n^{th} term is

(i) $\frac{2^n}{n}$ (ii) $\frac{3+(-1)^n}{3^n}$

Solution. (i) Let $t_n = \frac{2^n}{n}$

put n = 1, 2, 3, 4, ... we get

$t_1 = 2, t_2 = 2, t_3 = \frac{8}{3}, t_4 = 4$

so the sequence is $2, 2, \frac{8}{3}, 4, ...$

(ii) Let $t_n = \frac{3+(-1)^n}{3^n}$

put n = 1, 2, 3, 4, ...

so the sequence is $\frac{2}{3}, \frac{4}{9}, \frac{2}{27}, \frac{4}{81}, ...$

Series By adding or substracting the terms of a sequence, we get an expression which is called a series. If $a_1, a_2, a_3,.....a_n$ is a sequence, then the expression $a_1 + a_2 + a_3 + ... + a_n$ is a series.

Example. (i) 1 + 2 + 3 + 4 + ... + n
(ii) 2 + 4 + 8 + 16 + ...

Progression : It is not necessary that the terms of a sequence always follow a certain pattern or they are described by some explicit formula for the nth term. Those sequences whose terms follow certain patterns are called progressions.

An arithmetic progression (A.P.):

A.P. is a sequence whose terms increase or decrease by a fixed number. This fixed number is called the common difference. If a is the first term & d the common difference, then A.P. can be written as a, a + d, a + 2 d,..... a + (n – 1) d,.....
Example – 4, – 1, 2, 5 .....

(i) n^{th} term of an A.P.

Let a be the first term and d be the common difference of an A.P., then
$t_n = a + (n – 1) d$ where $d = a_n – a_{n-1}$

Solved Example # 2 If t_{54} of an A.P. is – 61 and $t_4 = 64$, find t_{10}.

Solution. Let a be the first term and d be the common difference
so $t_{54} = a + 53d = – 61$ .....(i)
and $t_4 = a + 3d = 64$ .....(ii)
equation (i) – (ii)
$\Rightarrow$ $50d = – 125$

$d = -\frac{5}{2} \quad \Rightarrow \quad a = \frac{143}{2}$

so $t_{10} = \frac{143}{2} + 9\left(-\frac{5}{2}\right) = 49$

Solved Example # 3 Find the number of terms in the sequence 4, 12, 20, ...108.

Solution. a = 4, d = 8 so $108 = 4 + (n – 1)8 \Rightarrow n = 14$

(ii) The sum of first n terms of are A.P.

If a is first term and d is common difference then

$S_n = \frac{n}{2}[2a + (n – 1) d]$

$= \frac{n}{2}[a + \ell] = nt_{\left(\frac{n+1}{2}\right)}$,

where ℓ is the last term and $t_{\left(\frac{n+1}{2}\right)}$ is the middle term.

(iii) r^{th} term of an A.P. when sum of first r terms is given is $t_r = s_r – S_{r-1}$.

Solved Example # 4

Find the sum of all natural numbers divisible by 5, but less than 100.

Solution. All those numbers are 5, 10, 15, 20, ...95.

Here a = 5 n = 19 ℓ = 95 so $S = \frac{19}{2}(5 + 95) = 950$.

Solved Example # 5

Find the sum of all the three digit natural numbers which on division by 7 leaves remainder 3.

Solution. All these numbers are 101, 108, 115, ... 997, to find n.

$997 = 101 + (n-1)\,7 \quad \Rightarrow \quad n = 129$

so $S = \frac{129}{2}\,[101 + 997] = 70821.$

Solved Example # 6 The sum of n terms of two A.Ps. are in ratio $\frac{7n+1}{4n+27}$. Find the ratio of their 11th terms.

Sol. Let a_1 and a_2 be the first terms and d_1 and d_2 be the common differences of two A.P.s respectively then

$$\frac{\frac{n}{2}[2a_1+(n-1)d_1]}{\frac{n}{2}[2a_1+(n-1)d_2]} = \frac{7n+1}{4n+27} \quad \Rightarrow \quad \frac{a_1+\left(\frac{n-1}{2}\right)d_1}{a_2+\left(\frac{n-1}{2}\right)d_2} = \frac{7n+1}{4n+27}$$

For ratio of 11th terms

$$\frac{n-1}{2} = 10 \quad \Rightarrow \quad n = 21$$

so ratio of 11th terms is $\frac{7(21)+1}{4(21)+27}$

$$= \frac{148}{111}$$

Solved Example # 7 If sum of n terms of a sequence is given by $S_n = 2n^2 + 3n$, find its 50th term.

Solution. Let t_n is n^{th} term of the sequence so $t_n = s_n - s_{n-1}$.

$= 2n^2 + 3n - 2(n-1)^2 - 3(n-1)$

$= 4n + 1$

so $t_{50} = 201.$

Self Practice Problems :

1. Which term of the sequence 2005, 2000, 1995, 1990, 1985, ... contains the first negative term **Ans.** 403.

2. For an A.P. show that

$t_m + t_{2n+m} = 2\,t_{m+n}$

3. Find the maximum sum of the A.P. 40, 38, 36, 34, 32, ... **Ans.** 420

Properties of A.P.

(i) The common difference can be zero, positive or negative.

(ii) If a, b, c are in A.P. $\Rightarrow 2b = a + c$ & if a, b, c, d are in A.P. $\Rightarrow a + d = b + c$.

(iii) Three numbers in A.P. can be taken as $a-d, a, a+d$; four numbers in A.P. can be taken as $a-3d, a-d, a+d, a+3d$; five numbers in A.P. are $a-2d, a-d, a, a+d, a+2d$ & six terms in A.P. are $a-5d, a-3d, a-d, a+d, a+3d, a+5d$ etc.

(iv) The sum of the terms of an A.P. equidistant from the beginning & end is constant and equal to the sum of first & last terms.

(v) Any term of an A.P. (except the first) is equal to half the sum of terms which are equidistant from it. $a_n = 1/2\,(a_{n-k} + a_{n+k})$, $k < n$. For $k = 1$, $a_n = (1/2)\,(a_{n-1} + a_{n+1})$;
For $k = 2$, $a_n = (1/2)\,(a_{n-2} + a_{n+2})$ and so on.

(vi) If each term of an A.P. is increased, decreased, multiplied or divided by the sA.M.e non zero number, then the resulting sequence is also an A.P..

Solved Example # 8 The sum of three numbers in A.P. is 27 and the sum of their squares is 293, find them

Solution. Let the numbers be

$a-d,\ a,\ a+d$

so $3a = 27 \quad \Rightarrow \quad a = 9$

Also $(a-d)^2 + a^2 + (a+d)^2 = 293.$

$3a^2 + 2d^2 = 293$

$d^2 = 25 \quad \Rightarrow \quad d = \pm 5$

therefore numbers are 4, 9, 14.

Solved Example # 9 If a_1, a_2, a_3, a_4, a_5 are in A.P. with common difference $\neq 0$, then find the value of $\sum_{i=1}^{5} a_i$ when $a_3 = 2$.

Solution. As a_1, a_2, a_3, a_4, a_5, are in A.P., we have

$a_1 + a_5 = a_2 + a_4 = 2a_3.$

Hence $\sum_{i=1}^{5} a_i = 10.$

Solved Example # 10 If $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in A.P. prove that a^2, b^2, c^2 are also in A.P.

Solution. $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in A.P.

$$\Rightarrow \quad \frac{1}{c+a} - \frac{1}{b+c} = \frac{1}{a+b} - \frac{1}{c+a} \Rightarrow \quad \frac{b+c-c-a}{(c+a)(b+c)} = \frac{c+a-a-b}{(a+b)(c+a)}$$

$$\Rightarrow \quad \frac{b-a}{b+c} = \frac{c-b}{a+b} \quad \Rightarrow \quad b^2 - a^2 = c^2 - b^2 \quad \Rightarrow \quad a^2, b^2, c^2 \text{ are in A.P.}$$

Solved Example # 11 If $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P., then $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are also in A.P.

Solution. Given $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P.

Add 2 to each term

$\Rightarrow$ $\frac{b+c+a}{a}, \frac{c+a+b}{b}, \frac{a+b+c}{c}$ are in A.P.

divide each by a + b + c $\Rightarrow$ $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

Arithmetic Mean (Mean or Average) (A.M.):

If three terms are in A.P. then the middle term is called the A.M. between the other two, so if a, b, c are in A.P., b is A.M. of a & c.

(a) **n – Arithmetic Means Between Two Numbers:**

If a, b are any two given numbers & a, $A_1, A_2, \ldots, A_n$, b are in A.P. then $A_1, A_2, \ldots A_n$ are the n A.M.'s between a & b.

$$A_1 = a + \frac{b-a}{n+1}, A_2 = a + \frac{2(b-a)}{n+1}, \ldots\ldots, A_n = a + \frac{n(b-a)}{n+1}$$

NOTE : Sum of n A.M.'s inserted between a & b is equal to n times the single A.M. between a & b

i.e. $\sum_{r=1}^{n} A_r = nA$ where A is the single A.M. between a & b.

Solved Example # 12 Between two numbers whose sum is $\frac{13}{6}$, an even number of A.M.s is inserted, the sum of these means exceeds their number by unity. Find the number of means.

Solution. Let a and b be two numbers and 2n A.M.s are inserted between a and b then

$\frac{2n}{2}(a+b) = 2n + 1.$

$n\left(\frac{13}{6}\right) = 2n + 1.$ $\left[\text{given } a+b = \frac{13}{6}\right]$

$\Rightarrow$ n = 6. $\therefore$ Number of means = 12.

Solved Example # 13 Insert 20 A.M. between 2 and 86.

Solution. Here 2 is the first term and 86 is the 22nd term of A.P. so 86 = 2 + (21)d

$\Rightarrow$ d = 4

so the series is

2, 6, 10, 14,....., 82, 86 $\therefore$ required means are 6, 10, 14,...82.

Self Practice Problems :

4. If A.M. between p^{th} and q^{th} terms of an A.P. be equal to the A.M. between r^{th} and s^{th} term of the A.P. then prove that p + q = r + s.

5. If n A.M.s are inserted between 20 and 80 such that first means : last mean = 1 : 3, find n. **Ans. n = 11**

6. For what value of n, $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$, $a \neq b$ is the A.M. of a and b. **Ans.** n = 0

Geometric Progression (G.P.)

G.P. is a sequence of numbers whose first term is non zero & each of the succeeding terms is equal to the proceeding terms multiplied by a constant. Thus in a G.P. the ratio of successive terms is constant. This constant factor is called the **common ratio** of the series & is obtained by dividing any term by that which immediately proceeds it. Therefore a, ar, ar^2, ar^3, ar^4,..... is a G.P. with a as the first term & r as common ratio.

Example 2, 4, 8, 16 .....

Example $\frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \frac{1}{81}$ .....

(i) n^{th} term = ar^{n-1}

(ii) Sum of the first n terms i.e. $S_n = \begin{cases} \frac{a(r^n-1)}{r-1} & , r \neq 1 \\ na & , r = 1 \end{cases}$

(iii) Sum of an infinite G.P. when $|r| < 1$. When $n \to \infty$ $r^n \to 0$ if $|r| < 1$ therefore, $S_\infty = \frac{a}{1-r}\left(|r| < 1\right)$.

Solved Example # 14: If the first term of G.P. is 7, its nth term is 448 and sum of first n terms is 889, then find the fifth term of G.P.

Solution. Given a = 7 the first term

$t_n = ar^{n-1} = 7(r)^{n-1} = 448.$

$\Rightarrow$ $7r^n = 448\, r$

Also $S_n = \frac{a(r^n-1)}{r-1} = \frac{7(r^n-1)}{r-1}$ $\Rightarrow$ $889 = \frac{448r-7}{r-1}$

$\Rightarrow$ r = 2

Hence $T_5 = ar^4 = 7(2)^4 = 112.$

Solved Example # 15: The first term of an infinite G.P. is 1 and any term is equal to the sum of all the succeeding terms. Find the series.

Solution. Let the G.P. be 1, r, r^2, r^3, ...

given condition $\Rightarrow r = \frac{r^2}{1-r} \quad \Rightarrow \quad r = \frac{1}{2}$,

Hence series is $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots\ldots\ldots\ldots\infty$

Solved Example # 16: Let $S = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots\ldots\ldots$ find the sum of

(i) first 20 terms of the series (ii) infinite terms of the series.

Solution. (i) $S_{20} = \frac{\left(1-\left(\frac{1}{2}\right)^{20}\right)}{1-\frac{1}{2}} = \frac{2^{20}-1}{2^{19}}$. (ii) $S_\infty = \frac{1}{1-\frac{1}{2}} = 2$.

Self Practice Problems :

1. Find the G.P. if the common ratio of G.P. is 3, n^{th} term is 486 and sum of first n terms is 728.
Ans. 2, 6, 18, 54, 162, 486.

2. If the pth, q^{th}, r^{th} terms of a G.P. be a, b, c respectively, prove that $a^{q-r}\, b^{r-p}\, c^{p-q} = 1$.

3. A G.P. consist of 2n terms. If the sum of the terms occupying the odd places is S_1 and that of the terms occupying the even places is S_2 then find the common ratio of the progression. **Ans.** $\frac{S_2}{S_1}$.

4. The sum of infinite number of terms of a G.P. is 4, and the sum of their cubes is 192, find the series.
Ans. $6, -3, \frac{3}{2}, \ldots\ldots\ldots$

Properties of G.P.

(i) If a, b, c are in G.P. $\Rightarrow b^2 = ac$, in general if $a_1, a_2, a_3, a_4, \ldots\ldots\ldots a_{n-1}, a_n$ are in G.P., then $a_1 a_n = a_1 a_{n-1} = a_3 a_{n-2} = \ldots\ldots\ldots\ldots\ldots\ldots$

(ii) Any three consecutive terms of a G.P. can be taken as $\frac{a}{r}$, a , ar, in general we take

$\frac{a}{r^k}, \frac{a}{r^{k-1}}, \frac{a}{r^{k-2}}, \ldots\ldots\ldots a, ar, ar^2, \ldots\ldots\ldots ar^k$ in case we have to take 2k + 1 terms in a G.P.

(iii) Any four consecutive terms of a G.P. can be taken as $\frac{a}{r^3}, \frac{a}{r}$, ar, ar^3, in general we take

$\frac{a}{r^{2k-1}}, \frac{a}{r^{2k-3}}, \ldots\ldots\ldots \frac{a}{r}$, ar, $\ldots\ldots\ldots\ldots ar^{2k-1}$ in case we have to take 2k terms in a G.P.

(iv) If each term of a G.P. be multiplied or divided or raised to power by the some non–zero quantity, the resulting sequence is also a G.P..

(v) If $a_1, a_2, a_3, \ldots\ldots\ldots$ and $b_1, b_2, b_3, \ldots\ldots\ldots$ are two G.P's with common ratio r_1 and r_2 respectively then the sequence $a_1b_1, a_2b_2, a_3b_3, \ldots\ldots$ is also a G.P. with common ratio $r_1 r_2$.

(vi) If $a_1, a_2, a_3, \ldots\ldots\ldots$ are in G.P. where each $a_i > 0$, then $\log a_1, \log a_2, \log a_3, \ldots\ldots\ldots$ are in A.P. and its converse is also true.

Solved Example # 17: Find three numbers in G.P. having sum 19 and product 216.

Solution. Let the three numbers be $\frac{a}{r}$, a, ar so $a\left[\frac{1}{r}+1+r\right] = 19$ .....(i)

and $a^3 = 216 \quad \Rightarrow \quad a = 6$

so from (i) $6r^2 - 13r + 6 = 0$.

$\Rightarrow \quad r = \frac{3}{2}, \frac{2}{3}$ Hence the three numbers are 4, 6, 9.

Solved Example # 18: Find the product of 11 terms in G.P. whose 6th is 5.

Solution.: Using the property

$a_1 a_{11} = a_2 a_{10} = a_3 a_9 = \ldots\ldots\ldots\ldots = a_6^{\,2} = 25$

Hence product of terms $= 5^{11}$

Solved Example # 19: Using G.P. express $0.\overline{3}$ and $1.2\overline{3}$ as $\frac{p}{q}$ form.

Solution. Let $x = 0.\overline{3} = 0.3333\ldots\ldots\ldots\ldots$

$= 0.3 + 0.03 + 0.003 + 0.0003 + \ldots\ldots\ldots\ldots$

$= \frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + \frac{3}{10000} + \ldots\ldots\ldots\ldots$

$= \frac{\frac{3}{10}}{1-\frac{1}{10}} = \frac{3}{9} = \frac{1}{3}$.

Let $y = 1.2\overline{3}$

$= 1.233333$

$= 1.2 + 0.03 + 0.003 + 0.0003 + \ldots\ldots\ldots\ldots$

$= 1.2 + \frac{3}{10^2} + \frac{3}{10^3} + \frac{3}{10^4} + \ldots\ldots\ldots\ldots$

$$= 1.2 + \frac{\frac{3}{10^2}}{1-\frac{1}{10}} = 1.2 + \frac{1}{30} = \frac{37}{30}.$$

Solved Example # 20

Evaluate 7 + 77 + 777 + ... upto n terms.

Solution. Let S = 7 + 77 + 777 + ...upto n terms.

$$= \frac{7}{9}[9 + 99 + 999 + ...]$$

$$= \frac{7}{9}[(10-1) + (10^2-1) + (10^3-1) + ... + \text{upto n terms}]$$

$$= \frac{7}{9}[10 + 10^2 + 10^3 + ...+ 10^n - n]$$

$$= \frac{7}{9}\left[10\frac{(10^n)-1}{9} - n\right] \qquad = \frac{7}{81}[10^{n+1} - 9n - 10]$$

Geometric Means (Mean Proportional) (G.M.):

If a, b, c are in G.P., b is the G.M. between a & c.

$b^2 = ac$, therefore $b = \sqrt{ac}$; a > 0, c > 0.

(a) **n–Geometric Means Between a, b:**

If a, b are two given numbers & a, G_1, G_2,....., G_n, b are in G.P.. Then G_1, G_2, G_3,...., G_n are n G.M.s between a & b.

$G_1 = a(b/a)^{1/n+1}$, $G_2 = a(b/a)^{2/n+1}$,....., $G_n = a(b/a)^{n/n+1}$

NOTE : The product of n G.M.s between a & b is equal to the nth power of the single G.M. between a & b

i.e. $\prod_{r=1}^{n} G_r = (G)^n$ where G is the single G.M. between a & b.

Solved Example # 21 Insert 4 G.M.s between 2 and 486.

Solution. Common ratio of the series is given by $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}} = (243)^{1/5} = 3$

Hence four G.M.s are 6, 18, 54, 162.

Self Practice Problems :

1. The sum of three numbers in G.P. in 70, if the two extremes be multiplied each by 4 and the mean by 5, the products are in A.P. Find the numbers. **Ans.** 10, 20, 40

2. If $a = \underbrace{111.....1}_{55}$, $b = 1 + 10 + 10^2 + 10^3 + 10^4$ and $c = 1 + 10^5 + 10^{10} + ... + 10^{50}$, then prove that

(i) 'a' is a composite number (ii) a = bc.

Harmonic Progression (H.P.) : A sequence is said to H.P. if the reciprocals of its terms are in A.P.. If the sequence a_1, a_2, a_3,...., a_n is an H.P. then $1/a_1$, $1/a_2$,...., $1/a_n$ is an A.P. & converse. Here we do not have the formula for the sum of the n terms of a H.P.. For H.P. whose first term is a and second term is b, the nth term is $t_n = \frac{ab}{b + (n-1)(a-b)}$. If a, b, c are in H.P. $\Rightarrow b = \frac{2ac}{a+c}$ or $\frac{a}{c} = \frac{a-b}{b-c}$.

NOTE : (i) If a, b, c are in A.P. $\Rightarrow \frac{a-b}{b-c} = \frac{a}{a}$ (ii) If a, b, c are in G.P. $\Rightarrow \frac{a-b}{b-c} = \frac{a}{b}$

Harmonic Mean (H.M.):

If a, b, c are in H.P., b is the H.M. between a & c, then b = 2ac/[a + c].

If a_1, a_2 , ... a_n are 'n' non-zero numbers then H.M. H of these numbers is given by

$$\frac{1}{H} = \frac{1}{n}\left[\frac{1}{a_1} + \frac{1}{a_2} + ... + \frac{1}{a_n}\right]$$

Solved Example # 22: If m^{th} term of H.P. is n, while n^{th} term is m, find its $(m + n)^{th}$ term.

Solution.: Given $T_m = n$ or $\frac{1}{a+(m-1)d} = n$; where a is the first term and d is the common difference of the corresponding A.P.

so $a + (m-1)d = \frac{1}{n}$ and $a + (n-1)d = \frac{1}{m}$ $\Rightarrow (m-n)d = \frac{m-n}{mn}$ or $d = \frac{1}{mn}$

so $a = \frac{1}{n} - \frac{(m-1)}{mn} = \frac{1}{mn}$

Hence $T_{(m+n)} = \frac{1}{a+(m+n-d)d} = \frac{mn}{1+m+n-1} = \frac{mn}{m+n}$.

Solved Example # 23: Insert 4 H.M between 2/3 and 2/13.

Solution. Let d be the common difference of corresponding A.P. so $d = \frac{\frac{13}{2} - \frac{3}{2}}{5} = 1.$

$\therefore$ $\frac{1}{H_1} = \frac{3}{2} + 1 = \frac{5}{2}$ or $H_1 = \frac{2}{5}$

$\frac{1}{H_2} = \frac{3}{2} + 2 = \frac{7}{2}$ or $H_2 = \frac{2}{7}$

$\frac{1}{H_3} = \frac{3}{2} + 3 = \frac{9}{2}$ or $H_3 = \frac{2}{9}$

$\frac{1}{H_4} = \frac{3}{2} + 4 = \frac{11}{2}$ or $H_4 = \frac{2}{11}$.

Solved Example # 24: If p^{th}, q^{th}, r^{th} terms of a H.P. be a, b, c respectively, prove that
$(q - r)bc + (r - p)ac + (p - q)ab = 0$

Solution. Let x be the first term and d be the common difference of the corresponding A.P..

so $\frac{1}{a} = x + (p - 1)d$ .....(i)

$\frac{1}{b} = x + (q - 1)d$ .....(ii)

$\frac{1}{c} = x + (r - 1)d$ .....(iii)

(i) - (ii) $\Rightarrow$ $ab(p - q)d = b - a$ .....(iv)
(ii) - (iii) $\Rightarrow$ $bc(q - r)d = c - b$ .....(v)
(iii) - (i) $\Rightarrow$ $ac(r - p)d = a - c$ .....(vi)
(iv) + (v) + (vi) gives
$bc(q - r) + ac(r - p) + ab(p - q) = 0$.

Self Practice Problems : 1. If a, b, c be in H.P., show that $a : a - b = a + c : a - c$.

2. If the H.M. between two quantities is to their G.M.s as 12 to 13, prove that the quantities are in ratio 4 to 9.

3. If H be the harmonic mean of a and b then find the value of $\frac{H}{2a} + \frac{H}{2b} - 1$. **Ans.** 0

4. If a, b, c, d are in H.P., the show that $ab + bc + cd = 3ad$

Relation between means :

(i) If A, G, H are respectively A.M., G.M., H.M. between a & b both being unequal & positive then, $G^2 = AH$ i.e. A, G, H are in G.P.

Solved Example # 25: The A.M. of two numbers exceeds the G.M. by $\frac{3}{2}$ and the G.M. exceeds the H.M. by $\frac{6}{5}$; find the numbers.

Solution. Let the numbers be a and b, now using the relation
G^2 = A.H.

$= \left(G + \frac{3}{2}\right)\left(G - \frac{6}{5}\right)$

$= G^2 + \frac{3}{10}G - \frac{9}{5}$ $\Rightarrow$ $G = 6$

i.e. $ab = 36$
also $a + b = 15$ Hence the two numbers are 3 and 12.

(ii) **A.M. ≥ G.M. ≥ H.M.**
Let $a_1, a_2, a_3, ...a_n$ be n positive real numbers, then we define their

A.M. = $\frac{a_1 + a_2 + a_3 + ... + a_n}{n}$, their

G.M. = $(a_1 a_2 a_3 .....a_n)^{1/n}$ and their H.M. = $\frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + ... + \frac{1}{a_n}}$ It can be shown that

A.M. ≥ G.M. ≥ H.M. and equality holds at either places iff
$a_1 = a_2 = a_3 = ...= a_n$

Solved Example # 26 If a, b, c, > 0 prove that $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3$

Solution. Using the relation A.M. ≥ G.M. we have

$\frac{\frac{a}{b} + \frac{b}{c} + \frac{c}{a}}{3} \geq \left(\frac{a}{b} \cdot \frac{b}{c} \cdot \frac{c}{a}\right)^{\frac{1}{3}}$ $\Rightarrow$ $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3$

Solved Example # 27 For non-zero x, y, z prove that $(x + y + z)\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) \geq 9$

Solution. Using the relation A.M. ≥ H.M.

$\frac{x + y + z}{3} \geq \frac{3}{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}$

$\Rightarrow$ $(x + y + z)\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) \geq 9$

Sol. Ex. # 28: If $a_i > 0\ \forall\ i \in N$ such that $\prod_{i=1}^{n} a_i = 1$, then prove that $(1 + a_1)(1 + a_2)(1 + a_3) \ldots\ldots\ldots (1 + a_n) \geq 2^n$

Solution. Using A.M. $\geq$ G.M.

$1 + a_1 \geq 2\sqrt{a_1}$

$1 + a_2 \geq 2\sqrt{a_2}$

$1 + a_n \geq 2\sqrt{a_n}$ $\Rightarrow$ $(1 + a_1)(1 + a_2) \ldots\ldots\ldots (1 + a_n) \geq 2^n (a_1 a_2 a_3 \ldots\ldots a_n)^{1/n}$

As $a_1\ a_2\ a_3 \ldots\ldots a_n = 1$

Hence $(1 + a_1)(1 + a_2) \ldots\ldots\ldots (1 + a_n) \geq 2^n$.

Solved Example # 29 If $n > 0$ prove that $2^n > 1 + n\sqrt{2^{n-1}}$

Solution. Using the relation A.M. $\geq$ G.M. on the numbers $1, 2, 2^2, 2^3 \ldots\ldots\ldots 2^{n-1}$ we have

$$\frac{1 + 2 + 2^2 + \ldots\ldots + 2^{n-1}}{n} > (1.2\ 2^2\ 2^3 \ldots\ldots 2^{n-1})^{1/n}$$

Equality does not hold as all the numbers are not equal.

$$\Rightarrow \quad \frac{2^n - 1}{2 - 1} > n \left(2^{\frac{(n-1)n}{2}} \right)^{\frac{1}{n}} \qquad \Rightarrow \quad 2^n - 1 > n\ 2^{\frac{(n-1)}{2}}$$

$$\Rightarrow \quad 2^n > 1 + n\ 2^{\frac{(n-1)}{2}}$$

Sol. Ex. # 30 Find the greatest value of xyz for positive value of x, y, z subject to the condition $xy + yz + zx = 12$.

Solution. Using the relation A.M. $\geq$ G.M.

$$\frac{xy + yz + zx}{3} \geq (x^2 y^2 z^2)^{1/3} \qquad 4 \geq (x\ y\ z)^{2/3} \qquad \Rightarrow \qquad xyz \leq 8$$

Solved Example # 32 If a, b, c are in H.P. and they are distinct and positive then prove that $a^n + c^n > 2b^n$

Solution. Let a^n and c^n be two numbers

then $\frac{a^n + c^n}{2} > (a^n c^n)^{1/2}$

$a^n + c^n > 2\ (ac)^{n/2}$ .....(i)

Also G.M. > H.M.

i.e. $\sqrt{ac} > b$ $(ac)^{n/2} > b^n$ .....(ii)

hence from (i) and (ii) $a^n + c^n > 2b^n$

Self Practice Problems :

1. If a, b, c are real and distinct then show that $a^2 (1 + b^2) + b^2 (1 + c^2) + c^2 (1 + a^2) > 6abc$

2. Prove that $n^n > 1 \cdot 3 \cdot 5 \ldots\ldots\ldots (2n - 1)$

3. If a, b, c, d be four distinct positive quantities in G.P. then show that

(i) $a + d > b + c$ (ii) $\frac{1}{ab} + \frac{1}{cd} > 2\left(\frac{1}{bd} + \frac{1}{ac} - \frac{1}{ad} \right)$

4. Prove that ΔABC is an equilateral triangle iff $\tan A + \tan B + \tan C = 3\sqrt{3}$

5. If a, b, c > 0 prove that $[(1 + a)(1 + b)(1 + c)]^7 > 7^7\ a^4\ b^4\ c^4$

Arithmetico-Geometric Series:

A series each term of which is formed by multiplying the corresponding term of an A.P. & G.P. is called the AritH.M.etico–Geometric Series. e.g. $1 + 3x + 5x^2 + 7x^3 + \ldots\ldots$ Here 1, 3, 5,.... are in A.P. & 1, x, x^2, x^3.... are in G.P..

Sum of n terms of an Arithmetico–Geometric Series:

Let $S_n = a + (a + d)\ r + (a + 2\ d)\ r^2 + \ldots\ldots + [a + (n - 1)d]\ r^{n-1}$

then $S_n = \frac{a}{1 - r} + \frac{d\ r\left(1 - r^{n-1}\right)}{(1 - r)^2} - \frac{\left[a + (n-1)d\right] r^n}{1 - r}$, $r \neq 1$.

Sum To Infinity: If $|r| < 1$ & $n \to \infty$ then $\underset{n \to \infty}{\text{Limit}}\ r^n = 0$ $\Rightarrow$ $S_\infty = \frac{a}{1 - r} + \frac{d\ r}{(1 - r)^2}$.

Solved Example # 33 Find the sum of the series

$$1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \ldots\ldots \text{ to n terms.}$$

Solution. Let $S = 1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \ldots\ldots + \frac{3n - 2}{5^{n-1}}$ .....(i)

$$\left(\frac{1}{5}\right) S = \frac{1}{5} + \frac{4}{5^2} + \frac{7}{5^3} + \ldots\ldots + \frac{3n - 5}{5^{n-1}} + \frac{3n - 2}{5^n} \qquad \ldots\ldots\ldots (ii)$$

(i) – (ii) $\Rightarrow$

$$\frac{4}{5} S = 1 + \frac{3}{5} + \frac{3}{5^2} + \frac{3}{5^3} + \ldots\ldots + \frac{3}{5^{n-1}} - \frac{3n - 2}{5^n}.$$

$$\frac{4}{5} S = 1 + \frac{\frac{3}{5}\left(1 - \left(\frac{1}{5}\right)^{n-1}\right)}{1 - \frac{1}{5}} - \frac{3n - 2}{5^n}$$

$= 1 + \frac{3}{4} - \frac{3}{4} \times \frac{1}{5^{n-1}} - \frac{3n-2}{5^n}$

$= \frac{7}{4} - \frac{12n+7}{4.5\,n}$ $\quad\therefore\quad S = \frac{35}{16} - \frac{(12n+7)}{16\,.\,5^{n-1}}$.

Solved Example # 35: Evaluate $1 + 2x + 3x^2 + 4x^3 + ...$ upto infinity where $|x| < 1$.

Solution. Let $S = 1 + 2x + 3x^2 + 4x^3 + ...$ .....(i)

$xS = x + 2x^2 + 3x^3 + ...$ .....(ii)

(i) - (ii) $\Rightarrow$ $(1-x)\,S = 1 + x + x^2 + x^3 + ...$ or $S = \frac{1}{(1-x)^2}$

Solved Example # 36 Evaluate $1 + (1 + b)\,r + (1 + b + b^2)\,r^2 + ...$ to infinite terms for $|br| < 1$.

Solution. Let $S = 1 + (1 + b)r + (1 + b + b^2)\,r^2 + ...$ .....(i)

$rS = r + (1 + b)\,r^2 + ...$ .....(ii)

(i) - (ii) $\Rightarrow$ $(1-r)S = 1 + br + b^2r^2 + b^3r^3 + ...$

$\Rightarrow$ $S = \frac{1}{(1-br)(1-r)}$

Self Practice Problems :

1. Evaluate $1.2 + 2.2^2 + 3.2^3 + ... + 100.\,2^{100}$ **Ans.** $99.2^{101} + 2$.

2. Evaluate $1 + 3x + 6x^2 + 10x^3 + ...$ upto infinite term where $|x| < 1$. **Ans.** $\frac{1}{(1-x)^3}$

3. Sum to n terms of the series $1 + 2\left(1+\frac{1}{n}\right) + 3\left(1+\frac{1}{n}\right)^2 + ...$ **Ans.** n^2

Important Results

(i) $\sum_{r=1}^{n}(a_r \pm b_r) = \sum_{r=1}^{n} a_r \pm \sum_{r=1}^{n} b_r$. (ii) $\sum_{r=1}^{n} k\,a_r = k\sum_{r=1}^{n} a_r$.

(iii) $\sum_{r=1}^{n} k = k + k + k.....n \text{ times} = nk$; where k is a constant. (iv) $\sum_{r=1}^{n} r = 1 + 2 + 3 + ...+ n = \frac{n\,(n+1)}{2}$

(v) $\sum_{r=1}^{n} r^2 = 1^2 + 2^2 + 3^2 + ...+ n^2 = \frac{n\,(n+1)\,(2n+1)}{6}$ (vi) $\sum_{r=1}^{n} r^3 = 1^3 + 2^3 + 3^3 + ...+ n^3 = \frac{n^2\,(n+1)^2}{4}$

(vii) $2\sum_{i<j=1}^{n} a_i\,a_j = (a_1 + a_2 + ...+ a_n)^2 - (a_1^2 + a_2^2 + ... + a_n^2)$

Solved Example # 37: Find the sum of the series to n terms whose general term is $2n + 1$.

Solution. $S_n = \Sigma T_n = \Sigma(2n + 1)$

$= 2\Sigma n + \Sigma 1$

$= \frac{2(n+1)\,n}{2} + n$ $\quad = n^2 + 2n$ or $n(n + 2)$.

Solved Example # 38: $T_k = k^2 + 2^k$ then find $\sum_{k=1}^{n} T_k$.

Solution. $\sum_{k=1}^{n} T_k = \sum_{k=1}^{n} k^2 + \sum_{k=1}^{n} 2^k$

$= \frac{n\,(n+1)\,(2n+1)}{6} + \frac{2(2^n-1)}{2-1}$ $\quad = \frac{n\,(n+1)\,(2n+1)}{6} + 2^{n+1} - 2$.

Solved Example # 39: Find the value of the expression $\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j} 1$

Solution.: $\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j} 1 = \sum_{i=1}^{n}\sum_{j=1}^{i} j = \sum_{i=1}^{n}\frac{i\,(i+1)}{2}$

$= \frac{1}{2}\left[\sum_{i=1}^{n} i^2 + \sum_{i=1}^{n} i\right]$

$= \frac{1}{2}\left[\frac{n\,(n+1)\,(2n+1)}{6} + \frac{n\,(n+1)}{2}\right]$

$= \frac{n\,(n+1)}{12}\,[2n + 1 + 3] = \frac{n\,(n+1)\,(n+2)}{6}$.

METHOD OF DIFFERENCE

Type – 1 Let u_1, u_2, u_3 ..... be a sequence, such that $u_2 - u_1, u_3 - u_2$, ... is either an A.P. or a G.P. then nth term u_n of this sequence is obtained as follows

$S = u_1 + u_2 + u_3 + ... + u_n$ .....(i)

$S = u_1 + u_2 + ... + u_{n-1} + u_n$ .....(ii)

(i) – (ii) $\Rightarrow$ $u_n = u_1 + (u_2 - u_1) + (u_3 - u_2) + ... + (u_n - u_{n-1})$

Where the series $(u_2 - u_1) + (u_3 - u_2) + ... + (u_n - u_{n-1})$ is

either in A.P. or in G.P. then we can find u_n and hence sum of this series as $S = \sum_{r=1}^{k} u_r$

Solved Example # 40 Find the sum to n-terms 3 + 7 + 13 + 21 + ...

Solution. Let $S = 3 + 7 + 13 + 21 + ... + T_n$ .....(i)

$S = 3 + 7 + 13 + ... + T_{n-1} + T_n$ .....(ii)

(i) – (ii) $\Rightarrow T_n = 3 + 4 + 6 + 8 + ... + (T_n - T_{n-1})$

$= 3 + \frac{n-1}{2}[8 + (n-2)2]$

$= 3 + (n-1)(n+2)$

$= n^2 + n + 1$

Hence $S = \sum (n^2 + n + 1)$

$= \sum n^2 + \sum n + \sum 1$

$= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} + n = \frac{n}{3}(n^2 + 3n + 5)$

Solved Example # 41 Find the sum to n-terms 1 + 4 + 10 + 22 + ...

Solution. Let $S = 1 + 4 + 10 + 22 + ...+ T_n$ .....(i)

$S = 1 + 4 + 10 + ... + T_{n-1} + T_n$ .....(ii)

(i) – (ii) $\Rightarrow T_n = 1 + (3 + 6 + 12 + ... + T_n - T_{n-1})$

$T_n = 1 + 3\left(\frac{2^{n-1} - 1}{2-1}\right)$

$T_n = 3 \cdot 2^{n-1} - 2$

So $S^n = \sum T_n = 3\sum 2^{n-1} - \sum 2$

$= 3 \cdot \left(\frac{2^n - 1}{2-1}\right) - 2n = 3.2^n - 2n - 3$

Type – 2 If possible express r^{th} term as difference of two terms as $t_r = f(r) - f(r \pm 1)$. This can be explained with the help of examples given below.

Solved Example # 42 Find the sum to n-terms of the series 1.2 + 2.3 + 3.4 + ...

Solution. Let T_r be the general term of the series

So $T_r = r(r+1)$.

To express $t_r = f(r) - f(r-1)$ multiply and divide t_r by $[(r+2) - (r-1)]$

so $T_r = \frac{r}{3}(r+1)[(r+2) - (r-1)]$

$= \frac{1}{3}[r(r+1)(r+2) - (r-1)r(r+1)]$.

Let $f(r) = \frac{1}{3} r(r+1)(r+2)$

so $T_r = [f(r) - f(r-1)]$. Now $S = \sum_{r=1}^{n} T_r = T_1 + T_2 + T_3 + ...+ T_n$

$T_1 = \frac{1}{3}[1 \cdot 2 \cdot 3 - 0]$, $T_2 = \frac{1}{3}[2 \cdot 3 \cdot 4 - 1 \cdot 2 \cdot 3]$, $T_3 = \frac{1}{3}[3 \cdot 4 \cdot 5 - 2 \cdot 3 \cdot 4]$

$T_n = \frac{1}{3}[n(n+1)(n+2) - (n-1)n(n+1)]$ $\therefore$ $S = \frac{1}{3} n(n+1)(n+2)$

Hence sum of series is f(n) – f(0).

Solved Example # 43 Sum to n terms of the series $\frac{1}{(1+x)(1+2x)} + \frac{1}{(1+2x)(1+3x)} + \frac{1}{(1+3x)(1+4x)} + ...$

Solution. Let T_r be the general term of the series

$T_r = \frac{1}{(1+rx)(1+(r+1)x)}$ So $T_r = \frac{1}{x}\left[\frac{[1+(r+1)x] - (1+rx)}{(1+rx)(1+(r+1)x)}\right]$

$= \frac{1}{x}\left[\frac{1}{1+rx} - \frac{1}{1+(r+1)x}\right]$

$T_r = f(r) - f(r+1)$

$\therefore$ $S = \sum T_r = T_1 + T_2 + T_3 + ... + T_n$

$= \frac{1}{x}\left[\frac{1}{1+x} - \frac{1}{1+(n+1)x}\right] = \frac{n}{(1+x)[1+(n+1)x]}$

Solved Example # 44 Sun to n terms of the series $\frac{4}{1.2.3} + \frac{5}{2.3.4} + \frac{6}{3.4.5} + ...$

Solution. Let $T_r = \frac{r+3}{r(r+1)(r+2)}$

$= \frac{1}{(r+1)(r+2)} + \frac{3}{r(r+1)(r+2)} = \left[\frac{1}{r+1} - \frac{1}{r+2}\right] + \frac{3}{2}\left[\frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)}\right]$

$\therefore$ $S = \left[\frac{1}{2} - \frac{1}{n+2}\right] + \frac{3}{2}\left[\frac{1}{2} - \frac{1}{(n+1)(n+2)}\right]$

$= \frac{5}{4} - \frac{1}{n+2}\left[1+\frac{3}{2(n+1)}\right]$ $= \frac{5}{4} - \frac{1}{2(n+1)(n+2)}[2n+5]$

Note : It is not always necessary that the series of first order of differences i.e. $u_2 - u_1$, $u_3 - u_2$ ..... $u_n - u_{n-1}$, is always either in A.P. or in G.P. in such case let $u_1 = T_1$, $u_2 - u_1 = T_2$, $u_3 - u_2 = T_3$ ....., $u_n - u_{n-1} = T_n$.

So $u_n = T_1 + T_2 + ...+ T_n$ .....(i)

$u_n = T_1 + T_2 + ...+ T_{n-1} + T_n$ .....(ii)

(i) – (ii) $\Rightarrow$

$T_n = T_1 + (T_2 - T_1) + (T_3 - T_2) + ... + (T_n - T_{n-1})$

Now, the series $(T_2 - T_1) + (T_3 - T_2) + ... + (T_n - T_{n-1})$ is series of second order of differences and when it is either in A.P. or in G.P., then $u_n = u_1 + \sum T_r$

Otherwise in the similar way we find series of higher order of differences and the nth term of the series. With the help of following example this can be explained.

Solved Example # 45 Find the nth term and the sum of n term of the series

2, 12, 36, 80, 150, 252

Solution. Let $S = 2 + 12 + 36 + 80 + 150 + 252 + ...+T_n$ .....(i)

$S = 2 + 12 + 36 + 80 + 150 + 252 + ...+T_{n-1} + T_n$ .....(ii)

(i) – (ii) $\Rightarrow T_n = 2 + 10 + 24 + 44 + 70 + 102 + ... + (T_n - T_{n-1})$ .....(iii)

$T_n = 2 + 10 + 24 + 44 + 70 + 102 + ... + (T_{n-1} - T_{n-2}) + (T_n - T_{n-1})$ .....(iv)

(iii) – (iv) $\Rightarrow T_n - T_{n-1} = 2 + 8 + 14 + 20 + 26 + ...$

$= \frac{n}{2}[4 + (n-1)\,6] = n[3n-1] = T_n - T_{n-1} = 3n^2 - n$

$\therefore$ general term of given series is $\sum T_n - T_{n-1} = \sum 3n^2 - n = n^3 + n^2$.

Hence sum of this series is $S = \sum n^3 + \sum n^2$

$= \frac{n^2(n+1)^2}{4} + \frac{n(n+1)(2n+1)}{6}$ $= \frac{n(n+1)}{12}(3n^2 + 7n + 2)$

$\frac{1}{12} n (n+1)(n+2)(3n+1)$

Solved Example # 46: Find the general term and sum of n terms of the series 9, 16, 29, 54, 103

Sol. Let $S = 9 + 16 + 29 + 54 + 103 + ... + T_n$ .....(i)

$S = 9 + 16 + 29 + 54 + 103 + ... + T_{n-1} + T_n$ .....(ii)

(i) – (ii) $\Rightarrow T_n = 9 + 7 + 13 + 25 + 49 + ... + (T_n - T_{n-1})$ .....(iii)

$T_n = 9 + 7 + 13 + 25 + 49 + ... + (T_{n-1} - T_{n-2}) + (T_n - T_{n-1})$ .....(iv)

(iii) – (iv) $\Rightarrow T_n - T_{n-1} = 9 + (-2) + \underbrace{6 + 12 + 24 + ...}_{(n-2)\text{ terms}} = 7 + 6[2^{n-2} - 1] = 6(2)^{n-2} + 1.$

$\therefore$ General term is $T_n = 6(2)^{n-1} + n + 2$

Also sum $S = \sum T_n = 6\sum 2^{n-1} + \sum n + \sum 2$

$= 6 \cdot \frac{(2^n - 1)}{2-1} + \frac{n(n+1)}{2} + 2n$ $= 6(2^n - 1) + \frac{n(n+5)}{2}$

Self Practice Problems : **1.** Sum to n terms the following series

(i) $\frac{1}{1^3} + \frac{1+2}{1^3+2^3} + \frac{1+2+3}{1^3+2^3+3^3} + ...$ **Ans.** $\frac{2n}{n+1}$

(ii) $\frac{1}{1.3.5} + \frac{1}{3.5.7} + \frac{1}{5.7.9} + ...$ **Ans.** $\frac{1}{4}\left[\frac{1}{3} - \frac{1}{(2n+1)(2n+3)}\right]$

(iii) $1 . 5 . 9 + 2 . 6 . 10 + 3 . 7 . 11 + ...$ **Ans.** $\frac{n}{4}(n+1)(n+8)(n+9)$

(iv) $4 + 14 + 30 + 52 + 82 + 114 + ...$ **Ans.** $n(n+1)^2$

(v) $2 + 5 + 12 + 31 + 86 + ...$ **Ans.** $\frac{3^n + n^2 + n - 1}{2}$

SHORT REVESION
(SEQUENCES AND SERIES)

DEFINITION : A sequence is a set of terms in a definite order with a rule for obtaining the terms. e.g. 1 , 1/2 , 1/3 , ... , 1/n , ... is a sequence.

AN ARITHMETIC PROGRESSION (AP) : AP is a sequence whose terms increase or decrease by a fixed number. This fixed number is called the common difference. If a is the first term & d the common difference, then AP can be written as a, a + d, a + 2 d, ... a + (n – 1)d, ...

n^{th} term of this AP $t_n = a + (n-1)d$, where $d = a_n - a_{n-1}$.

The sum of the first n terms of the AP is given by ; $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}[a + l]$.

where l is the last term.

NOTES :(i) If each term of an A.P. is increased, decreased, multiplied or divided by the same non zero number, then the resulting sequence is also an AP.

(ii) Three numbers in AP can be taken as a – d , a , a + d ; four numbers in AP can be taken as a – 3d, a – d, a + d, a + 3d ; five numbers in AP are a – 2d , a – d , a, a + d, a + 2d & six terms in AP are a – 5d, a – 3d, a – d, a + d, a + 3d, a + 5d etc.

(iii) The common difference can be zero, positive or negative.

(iv) The sum of the two terms of an AP equidistant from the beginning & end is constant and equal to the sum of first & last terms.

(v) Any term of an AP (except the first) is equal to half the sum of terms which are equidistant from it.

(vi) $t_r = S_r - S_{r-1}$

(vii) If a , b , c are in AP $\Rightarrow$ 2 b = a + c.

GEOMETRIC PROGRESSION (GP) : GP is a sequence of numbers whose first term is non zero & each of the succeeding terms is equal to the proceeding terms multiplied by a constant . Thus in a GP the ratio of successive terms is constant. This constant factor is called the **COMMON RATIO** of the series & is obtained by dividing any term by that which immediately proceeds it. Therefore a, ar, ar^2, ar^3, ar^4, ... is a GP with a as the first term & r as common ratio.

(i) n^{th} term = $a r^{n-1}$

(ii) Sum of the I^{st} n terms i.e. $S_n = \frac{a(r^n - 1)}{r - 1}$, if $r \neq 1$.

(iii) Sum of an infinite GP when $|r| < 1$ when $n \to \infty$ $r^n \to 0$ if $|r| < 1$ therefore, $S_\infty = \frac{a}{1 - r}$ $(|r| < 1)$.

(iv) If each term of a GP be multiplied or divided by the same non-zero quantity, the resulting sequence is also a GP.

(v) Any 3 consecutive terms of a GP can be taken as a/r, a, ar ; any 4 consecutive terms of a GP can be taken as a/r^3, a/r, ar, ar^3 & so on.

(vi) If a, b, c are in GP $\Rightarrow$ $b^2 = ac$.

HARMONIC PROGRESSION (HP) : A sequence is said to HP if the reciprocals of its terms are in AP. If the sequence $a_1, a_2, a_3, ... , a_n$ is an HP then $1/a_1, 1/a_2, ... , 1/a_n$ is an AP & converse. Here we do not have the formula for the sum of the n terms of an HP. For HP whose first term is a & second term is b, the nth term is $t_n = \frac{ab}{b + (n-1)(a-b)}$.

If a, b, c are in HP $\Rightarrow$ $b = \frac{2ac}{a + c}$ or $\frac{a}{c} = \frac{a - b}{b - c}$.

MEANS

ARITHMETIC MEAN : If three terms are in AP then the middle term is called the AM between the other two, so if a, b, c are in AP, b is AM of a & c .

AM for any n positive number $a_1, a_2, ... , a_n$ is ; $A = \frac{a_1 + a_2 + a_3 + ... + a_n}{n}$.

n - ARITHMETIC MEANS BETWEEN TWO NUMBERS :

If a, b are any two given numbers & $a, A_1, A_2, ... , A_n, b$ are in AP then $A_1, A_2, ... A_n$ are the n AM's between a & b .

$A_1 = a + \frac{b - a}{n + 1}$, $A_2 = a + \frac{2(b - a)}{n + 1}$, ... , $A_n = a + \frac{n(b - a)}{n + 1}$

$= a + d$, $= a + 2d$, ... , $A_n = a + nd$, where $d = \frac{b - a}{n + 1}$

NOTE : Sum of n AM's inserted between a & b is equal to n times the single AM between a & b

i.e. $\sum_{r=1}^{n} A_r = nA$ where A is the single AM between a & b.

GEOMETRIC MEANS : If a, b, c are in GP, b is the GM between a & c.

$b^2 = ac$, therefore $b = \sqrt{ac}$; $a > 0$, $c > 0$.

n-GEOMETRIC MEANS BETWEEN a, b :

If a, b are two given numbers & a, $G_1, G_2, \ldots, G_n$, b are in GP. Then $G_1, G_2, G_3, \ldots, G_n$ are n GMs between a & b .

$G_1 = a(b/a)^{1/n+1}$, $G_2 = a(b/a)^{2/n+1}$, ... , $G_n = a(b/a)^{n/n+1}$

$= ar$, $= ar^2$, ... $= ar^n$, where $r = (b/a)^{1/n+1}$

NOTE : The product of n GMs between a & b is equal to the nth power of the single GM between a & b

i.e. $\prod_{r=1}^{n} G_r = (G)^n$ where G is the single GM between a & b.

HARMONIC MEAN : If a, b, c are in HP, b is the HM between a & c, then $b = 2ac/[a+c]$.

THEOREM : If A, G, H are respectively AM, GM, HM between a & b both being unequal & positive then,

(i) $G^2 = AH$ **(ii)** $A > G > H$ $(G > 0)$. Note that A, G, H constitute a GP.

ARITHMETICO-GEOMETRIC SERIES :

A series each term of which is formed by multiplying the corresponding term of an AP & GP is called the **Arithmetico-Geometric Series**. e.g. $1 + 3x + 5x^2 + 7x^3 + ...$

Here 1, 3, 5, ... are in AP & 1, x, x^2, x^3 ..... are in GP.

Standart appearance of an Arithmetico-Geometric Series is

Let $S_n = a + (a + d)r + (a + 2d)r^2 + ... + [a + (n-1)d]\, r^{n-1}$

SUM TO INFINITY : If $|r| < 1$ & $n \to \infty$ then $\lim_{n \to \infty} r^n = 0$. $S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$.

SIGMA NOTATIONS

THEOREMS :(i) $\sum_{r=1}^{n} (a_r \pm b_r) = \sum_{r=1}^{n} a_r \pm \sum_{r=1}^{n} b_r$. **(ii)** $\sum_{r=1}^{n} k\, a_r = k \sum_{r=1}^{n} a_r$.

(iii) $\sum_{r=1}^{n} k = nk$; where k is a constant.

RESULTS

(i) $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$ (sum of the first n natural nos.)

(ii) $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$ (sum of the squares of the first n natural numbers)

(iii) $\sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4} \left[\sum_{r=1}^{n} r\right]^2$ (sum of the cubes of the first n natural numbers)

(iv) $\sum_{r=1}^{n} r^4 = \frac{n}{30}(n+1)(2n+1)(3n^2+3n-1)$

METHOD OF DIFFERENCE : If $T_1, T_2, T_3, \ldots, T_n$ are the terms of a sequence then some times the terms $T_2 - T_1, T_3 - T_2, \ldots$ constitute an AP/GP. n^{th} term of the series is determined & the sum to n terms of the sequence can easily be obtained.

Remember that to find the sum of n terms of a series each term of which is composed of r factors in AP, the first factors of several terms being in the same AP, we "write down the nth term, affix the next factor at the end, divide by the number of factors thus increased and by the common difference and add a constant. Determine the value of the constant by applying the initial conditions".

EXERCISE–1

Q.1 If the 10th term of an HP is 21 & 21st term of the same HP is 10, then find the 210th term.

Q.2 Show that $\ln(4 \times 12 \times 36 \times 108 \times \ldots\ldots\ldots\ldots \text{ up to } n \text{ terms}) = 2n \ln 2 + \frac{n(n-1)}{2} \ln 3$

Q.3 There are n AM's between 1 & 31 such that 7th mean : $(n-1)^{th}$ mean = 5 : 9, then find the value of n.

Q.4 Find the sum of the series , 7 + 77 + 777 + ... to n terms.

Q.5 Express the recurring decimal $0.1\overline{576}$ as a rational number using concept of infinite geometric series.

Q.6 Find the sum of the n terms of the sequence $\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + ...$

Q.7 The first term of an arithmetic progression is 1 and the sum of the first nine terms equal to 369. The first and the ninth term of a geometric progression coincide with the first and the ninth term of the arithmetic progression. Find the seventh term of the geometric progression.

Q.8 If the pth, q^{th} & r^{th} terms of an AP are in GP . Show that the common ratio of the GP is $\frac{q-r}{p-q}$.

Q.9 If one AM 'a' & two GM's p & q be inserted between any two given numbers then show that $p^3 + q^3 = 2apq$.

Q.10 The sum of n terms of two arithmetic series are in the ratio of $(7n+1) : (4n+27)$. Find the ratio of their n^{th} term.

Q.11 If S be the sum, P the product & R the sum of the reciprocals of a GP, find the value of $P^2\left(\frac{R}{S}\right)^n$.

Q.12 The first and last terms of an A.P. are a and b. There are altogether (2n + 1) terms. A new series is formed by multiplying each of the first 2n terms by the next term. Show that the sum of the new series is $\frac{(4n^2-1)(a^2+b^2)+(4n^2+2)ab}{6n}$.

Q.13 In an AP of which 'a' is the Ist term, if the sum of the Ist p terms is equal to zero, show that the sum of the next q terms is $-a(p+q)\,q/(p-1)$.

Q.14(a) The interior angles of a polygon are in AP. The smallest angle is 120° & the common difference is 5°. Find the number of sides of the polygon.

(b) The interior angles of a convex polygon form an arithmetic progression with a common difference of 4°. Determine the number of sides of the polygon if its largest interior angle is 172°.

Q.15 An AP & an HP have the same first term, the same last term & the same number of terms; prove that the product of the rth term from the beginning in one series & the rth term from the end in the other is independent of r.

Q.16 Find three numbers a, b, c between 2 & 18 such that ;
(i) their sum is 25 (ii) the numbers 2, a, b are consecutive terms of an AP &
(iii) the numbers b, c, 18 are consecutive terms of a GP.

Q.17 Given that $a^x = b^y = c^z = d^u$ & a, b, c, d are in GP, show that x, y, z, u are in HP.

Q.18 In a set of four numbers, the first three are in GP & the last three are in AP, with common difference 6. If the first number is the same as the fourth, find the four numbers.

Q.19 Find the sum of the first n terms of the sequence : $1 + 2\left(1+\frac{1}{n}\right) + 3\left(1+\frac{1}{n}\right)^2 + 4\left(1+\frac{1}{n}\right)^3 + ...$

Q.20 Find the nth term and the sum to n terms of the sequence :
(i) $1 + 5 + 13 + 29 + 61 + ...$ (ii) $6 + 13 + 22 + 33 + ...$

Q.21 The AM of two numbers exceeds their GM by 15 & HM by 27. Find the numbers.

Q.22 The harmonic mean of two numbers is 4. The airthmetic mean A & the geometric mean G satisfy the relation $2A + G^2 = 27$. Find the two numbers.

Q.23 Sum the following series to n terms and to infinity :

(i) $\frac{1}{1.4.7} + \frac{1}{4.7.10} + \frac{1}{7.10.13} + ...$ (ii) $\sum_{r=1}^{n} r(r+1)(r+2)(r+3)$

(iii) $\sum_{r=1}^{n} \frac{1}{4r^2-1}$ (iv) $\frac{1}{4} + \frac{1.3}{4.6} + \frac{1.3.5}{4.6.8} + ...$

Q.24 Find the value of the sum (a) $\sum_{r=1}^{n}\sum_{s=1}^{n} \delta_{rs}\, 2^r\, 3^s$ where δ_{rs} is zero if $r \neq s$ & δ_{rs} is one if $r = s$.

(b) $\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j} 1$.

Q.25 For or $0 < \phi < \pi/2$, if :

$x = \sum_{n=0}^{\infty} \cos^{2n}\phi$, $y = \sum_{n=0}^{\infty} \sin^{2n}\phi$, $z = \sum_{n=0}^{\infty} \cos^{2n}\phi\, \sin^{2n}\phi$ then : Prove that

(i) $xyz = xy + z$ (ii) $xyz = x + y + z$

EXERCISE–2

Q.1 The series of natural numbers is divided into groups (1), (2, 3, 4), (5, 6, 7, 8, 9), ... & so on. Show that the sum of the numbers in the nth group is $(n-1)^3 + n^3$.

Q.2 The sum of the squares of three distinct real numbers, which are in GP is S^2. If their sum is αS, show that $\alpha^2 \in (1/3, 1) \cup (1, 3)$.

Q.3 If there be m AP's beginning with unity whose common difference is 1, 2, 3 .... m. Show that the sum of their n^{th} terms is $(m/2)(mn - m + n + 1)$.

Q.4 If S_n represents the sum to n terms of a GP whose first term & common ratio are a & r respectively, then prove that $S_1 + S_3 + S_5 + ... + S_{2n-1} = \frac{an}{1-r} - \frac{ar(1-r^{2n})}{(1-r)^2(1+r)}$.

Q.5 A geometrical & harmonic progression have the same p^{th}, q^{th} & r^{th} terms a, b, c respectively. Show that $a(b-c)\log a + b(c-a)\log b + c(a-b)\log c = 0$.

Q.6 A computer solved several problems in succession. The time it took the computer to solve each successive problem was the same number of times smaller than the time it took to solve the preceding problem. How many problems were suggested to the computer if it spent 63.5 min to solve all the problems except for the first, 127 min to solve all the problems except for the last one, and 31.5 min to solve all the problems except for the first two?

Q.7 If the sum of m terms of an AP is equal to the sum of either the next n terms or the next p terms of the same AP prove that $(m+n)[(1/m)-(1/p)] = (m+p)[(1/m)-(1/n)]$ $(n \neq p)$

Q.8 If the roots of $10x^3 - cx^2 - 54x - 27 = 0$ are in harmonic progression, then find c & all the roots.

Q.9(a) Let $a_1, a_2, a_3 \ldots\ldots a_n$ be an AP. Prove that :

$$\frac{1}{a_1\, a_n} + \frac{1}{a_2\, a_{n-1}} + \frac{1}{a_3\, a_{n-2}} + \ldots\ldots\ldots + \frac{1}{a_n\, a_1} = \frac{2}{a_1 + a_n}\left[\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \ldots\ldots\ldots + \frac{1}{a_n}\right]$$

(b) Show that in any arithmetic progression $a_1, a_2, a_3 \ldots\ldots$

$a_1^2 - a_2^2 + a_3^2 - a_4^2 + \ldots\ldots + a^2_{2K-1} - a^2_{2K} = [K/(2K-1)]\,(a_1^2 - a^2_{2K})$.

Q.10 Let $a_1, a_2, \ldots\ldots\ldots, a_n, a_{n+1}, \ldots\ldots$ be an A.P.

Let $S_1 = a_1 + a_2 + a_3 + \ldots\ldots\ldots\ldots + a_n$

$S_2 = a_{n+1} + a_{n+2} + \ldots\ldots\ldots\ldots + a_{2n}$

$S_3 = a_{2n+1} + a_{2n+2} + \ldots\ldots\ldots\ldots + a_{3n}$

.....

.....

Prove that the sequence $S_1, S_2, S_3, \ldots\ldots$ is an arithmetic progression whose common difference is n^2 times the common difference of the given progression.

Q.11 If a, b, c are in HP, b, c, d are in GP & c, d, e are in AP, Show that $e = ab^2/(2a-b)^2$.

Q.12 If a, b, c, d, e be 5 numbers such that a, b, c are in AP ; b, c, d are in GP & c, d, e are in HP then:

(i) Prove that a, c, e are in GP . (ii) Prove that $e = (2b-a)^2/a$.

(iii) If $a = 2$ & $e = 18$, find all possible values of b , c , d .

Q.13 The sequence $a_1, a_2, a_3, \ldots\ldots a_{98}$ satisfies the relation $a_{n+1} = a_n + 1$ for n = 1, 2, 3, ...97 and has the sum equal to 4949. Evaluate $\sum_{k=1}^{49} a_{2k}$.

Q.14 If n is a root of the equation $x^2(1-ac) - x(a^2+c^2) - (1+ac) = 0$ & if n HM's are inserted between a & c, show that the difference between the first & the last mean is equal to $ac(a-c)$.

Q.15 (a) The value of $x + y + z$ is 15 if a , x , y , z , b are in AP while the value of ; $(1/x)+(1/y)+(1/z)$ is 5/3 if a , x , y , z , b are in HP . Find a & b .

(b) The values of xyz is 15/2 or 18/5 according as the series a , x , y , z , b is an AP or HP . Find the values of a & b assuming them to be positive integer .

Q.16 An AP , a GP & a HP have 'a' & 'b' for their first two terms . Show that their $(n+2)^{th}$ terms will be in GP if $\frac{b^{2n+2} - a^{2n+2}}{ba\left(b^{2n} - a^{2n}\right)} = \frac{n+1}{n}$.

Q.17 Prove that the sum of the infinite series $\frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} + \ldots\ldots\ldots\infty = 23$.

Q.18 If there are n quantities in GP with common ratio r & S_m denotes the sum of the first m terms, show that the sum of the products of these m terms taken two & two together is $[r/(r+1)]\,[S_m]\,[S_{m-1}]$.

Q.19 Find the condition that the roots of the equation $x^3 - px^2 + qx - r = 0$ may be in A.P. and hence solve the equation $x^3 - 12x^2 + 39x - 28 = 0$.

Q.20 If $ax^2 + 2bx + c = 0$ & $a_1x^2 + 2b_1x + c_1 = 0$ have a common root & a/a_1 , b/b_1 , c/c_1 are in AP, show that a_1 , b_1 & c_1 are in GP.

Q.21 If a , b , c be in GP & $\log_c a, \log_b c, \log_a b$ be in AP , then show that the common difference of the AP must be 3/2.

Q.22 If $a_1 = 1$ & for $n > 1$, $a_n = a_{n-1} + \frac{1}{a_{n-1}}$, then show that $12 < a_{75} < 15$.

Q.23 Sum to n terms : (i) $\frac{1}{x+1} + \frac{2x}{(x+1)(x+2)} + \frac{3x^2}{(x+1)(x+2)(x+3)} + \ldots\ldots$

(ii) $\frac{a_1}{1+a_1} + \frac{a_2}{(1+a_1)(1+a_2)} + \frac{a_3}{(1+a_1)(1+a_2)(1+a_3)} + \ldots\ldots$

Q.24 In a GP the ratio of the sum of the first eleven terms to the sum of the last eleven terms is 1/8 and the ratio of the sum of all the terms without the first nine to the sum of all the terms without the last nine is 2. Find the number of terms in the GP.

Q.25 Given a three digit number whose digits are three successive terms of a G.P. If we subtract 792 from it, we get a number written by the same digits in the reverse order . Now if we subtract four from the hundred's digit of the initial number and leave the other digits unchanged, we get a number whose digits are successive terms of an A.P. Find the number.

EXERCISE–3

Q.1 For any odd integer $n \geq 1$, $n^3 - (n-1)^3 + \ldots\ldots + (-1)^{n-1}\,1^3 =$ _____ . [JEE '96, 1]

Q.2 $x = 1 + 3a + 6a^2 + 10a^3 + \ldots\ldots$ $|a| < 1$

$y = 1 + 4b + 10b^2 + 20b^3 + \ldots\ldots$ $|b| < 1$, find $S = 1 + 3ab + 5(ab)^2 + \ldots\ldots$ in terms of x & y.

Q.3 The real numbers x_1, x_2, x_3 satisfying the equation $x^3 - x^2 + \beta x + \gamma = 0$ are in A.P . Find the intervals in which β and γ lie . [JEE '96, 3]

Q.4 Let p & q be roots of the equation $x^2 - 2x + A = 0$, and let r & s be the roots of the equation $x^2 - 18x + B = 0$. If $p < q < r < s$ are in arithmatic progression, then A = _____, and B = _____.

Q.5 a, b, c are the first three terms of a geometric series. If the harmonic mean of a & b is 12 and that of b

& c is 36, find the first five terms of the series. [REE '98, 6]

Q.6 Select the correct alternative(s). [JEE '98, 2 + 2 + 8]

(a) Let T_r be the rth term of an AP, for r = 1, 2, 3, ... If for some positive integers m, n we have $T_m = \frac{1}{n}$ & $T_n = \frac{1}{m}$, then T_{mn} equals :

(A) $\frac{1}{mn}$ (B) $\frac{1}{m} + \frac{1}{n}$ (C) 1 (D) 0

(b) If x = 1, y > 1, z > 1 are in GP, then $\frac{1}{1+\ell n x}, \frac{1}{1+\ell n y}, \frac{1}{1+\ell n z}$ are in :

(A) AP (B) HP (C) GP (D) none of the above

(c) Prove that a triangle ABC is equilateral if & only if $\tan A + \tan B + \tan C = 3\sqrt{3}$.

Q.7(a) The harmonic mean of the roots of the equation $\left(5+\sqrt{2}\right) x^2 - \left(4+\sqrt{5}\right) x + 8 + 2\sqrt{5} = 0$ is

(A) 2 (B) 4 (C) 6 (D) 8

(b) Let $a_1, a_2, ..., a_{10}$, be in A.P. & $h_1, h_2, ..., h_{10}$ be in H.P. If $a_1 = h_1 = 2$ & $a_{10} = h_{10} = 3$ then $a_4 h_7$ is:

(A) 2 (B) 3 (C) 5 (D) 6

Q.8 The sum of an infinite geometric series is 162 and the sum of its first n terms is 160. If the inverse of its common ratio is an integer, find all possible values of the common ratio, n and the first terms of the series.

Q.9(a) Consider an infinite geometric series with first term 'a' and common ratio r . If the sum is 4 and the second term is 3/4, then :

(A) $a = \frac{7}{4}$, $r = \frac{3}{7}$ (B) $a = 2$, $r = \frac{3}{8}$

(C) $a = \frac{3}{2}$, $r = \frac{1}{2}$ (D) $a = 3$, $r = \frac{1}{4}$

(b) If a, b, c, d are positive real numbers such that $a + b + c + d = 2$, then $M = (a + b)(c + d)$ satisfies the relation :

(A) $0 \le M \le 1$ (B) $1 \le M \le 2$

(C) $2 \le M \le 3$ (D) $3 \le M \le 4$

(c) The fourth power of the common difference of an arithmetic progression with integer entries added to the product of any four consecutive terms of it . Prove that the resulting sum is the square of an integer.
[JEE 2000, Mains, 4 out of 100]

Q.10 Given that α, γ are roots of the equation, $A x^2 - 4x + 1 = 0$ and β, δ the roots of the equation, $B x^2 - 6x + 1 = 0$, find values of A and B, such that α, β, γ & δ are in H.P.
[REE 2000, 5 out of 100]

Q.11 The sum of roots of the equation $ax^2 + bx + c = 0$ is equal to the sum of squares of their reciprocals. Find whether bc^2, ca^2 and ab^2 in A.P., G.P. or H.P.? [REE 2001, 3 out of 100]

Q.12 Solve the following equations for x and y

$\log_2 x + \log_4 x + \log_{16} x + ... = y$

$\frac{5+9+13+.....+(4y+1)}{1+3+5+.....+(2y-1)} = 4\log_4 x$ [REE 2001, 5 out of 100]

Q.13(a) Let α, β be the roots of $x^2 - x + p = 0$ and γ, δ be the roots of $x^2 - 4x + q = 0$. If $\alpha, \beta, \gamma, \delta$ are in G.P., then the integral values of p and q respectively, are

(A) –2, –32 (B) –2, 3 (C) –6, 3 (D) –6, –32

(b) If the sum of the first 2n terms of the A.P. 2, 5, 8, ... is equal to the sum of the first n terms of the A.P. 57, 59, 61, ..., then n equals

(A) 10 (B) 12 (C) 11 (D) 13

(c) Let the positive numbers a, b, c, d be in A.P. Then abc, abd, acd, bcd are

(A) NOT in A.P./G.P./H.P. (B) in A.P.

(C) in G.P. (D) H.P. [JEE 2001, Screening, 1 + 1 + 1 out of 35]

(d) Let a_1, a_2 ..... be positive real numbers in G.P. For each n, let A_n, G_n, H_n, be respectively, the arithmetic mean, geometric mean and harmonic mean of $a_1, a_2, a_3, ...a_n$. Find an expression for the G.M. of $G_1, G_2, ...G_n$ in terms of $A_1, A_2 .....A_n, H_1, H_2, ...H_n$.

Q.14(a) Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. If $a < b < c$ and $a + b + c = \frac{3}{2}$, then the value of a is

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2\sqrt{3}}$ (C) $\frac{1}{2} - \frac{1}{\sqrt{3}}$ (D) $\frac{1}{2} - \frac{1}{\sqrt{2}}$

(b) Let a, b be positive real numbers. If a , A_1 , A_2 , b are in A.P. ; a , a_1 , a_2 , b are in G.P. and a , H_1 , H_2 , b are in H.P. , show that

$\frac{G_1 G_2}{H_1 H_2} = \frac{A_1 + A_2}{H_1 + H_2} = \frac{(2a+b)(a+2b)}{9ab}$ [JEE 2002 , Mains , 5 out of 60]

Q.15 If a, b, c are in A.P., a^2 , b^2 , c^2 are in H.P. , then prove that either $a = b = c$ or a, b, $-\frac{c}{2}$ form a G.P.

Q.16 The first term of an infinite geometric progression is x and its sum is 5. Then

(A) $0 \le x \le 10$ (B) $0 < x < 10$ (C) $-10 < x < 0$ (D) $x > 10$

Q.17 If a, b, c are positive real numbers, then prove that $[(1 + a)(1 + b)(1 + c)]^7 > 7^7 a^4 b^4 c^4$.

Q.18(a) In the quadratic equation $ax^2 + bx + c = 0$, if $\Delta = b^2 - 4ac$ and $\alpha + \beta$, $\alpha^2 + \beta^2$, $\alpha^3 + \beta^3$ are in G.P. where α, β are the roots of $ax^2 + bx + c = 0$, then

(A) $\Delta \neq 0$ (B) $b\Delta = 0$ (C) $c\Delta = 0$ (D) $\Delta = 0$

(b) If total number of runs scored in *n* matches is $\left(\frac{n+1}{4}\right)(2^{n+1} - n - 2)$ where $n > 1$, and the runs scored in the kth match are given by $k \cdot 2^{n+1-k}$, where $1 \le k \le n$. Find n. [JEE 2005 (Mains), 2]

Q.19 If $A_n = \left(\frac{3}{4}\right) - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + ... + (-1)^{n-1}\left(\frac{3}{4}\right)^n$ and $B_n = 1 - A_n$, then find the minimum natural number n_0 such that $B_n > A_n$. $\forall\, n > n_0$. [JEE 2006, 6]

EXERCISE–4

Part : (A) Only one correct option

1. If $x \in R$, the numbers $5^{1+x} + 5^{1-x}$, a/2, $25^x + 25^{-x}$ form an A.P. then 'a' must lie in the interval:
(A) [1, 5] (B) [2, 5] (C) [5, 12] (D) [12, ∞)

2. If $x > 1$ and $\left(\frac{1}{x}\right)^a$, $\left(\frac{1}{x}\right)^b$, $\left(\frac{1}{x}\right)^c$ are in G.P., then a, b, c are in
(A) A.P. (B) G.P. (C) H.P. (D) none of these

3. If A, G & H are respectively the A.M., G.M. & H.M. of three positive numbers a, b, & c, then the equation whose roots are a, b, & c is given by:
(A) $x^3 - 3Ax^2 + 3G^3x - G^3 = 0$ (B) $x^3 - 3Ax^2 + 3(G^3/H)x - G^3 = 0$
(C) $x^3 + 3Ax^2 + 3(G^3/H)x - G^3 = 0$ (D) $x^3 - 3Ax^2 - 3(G^3/H)x + G^3 = 0$

4. The sum $\sum_{r=2}^{\infty} \frac{1}{r^2 - 1}$ is equal to:
(A) 1 (B) 3/4 (C) 4/3 (D) none

5. If a, a_1, a_2, a_3,..., a_{2n} , b are in A.P. and a, g_1, g_2, g_3,.....g_{2n}, b are in G.P. and h is the harmonic mean of a and b, then $\frac{a_1 + a_{2n}}{g_1 g_{2n}} + \frac{a_2 + a_{2n-1}}{g_2 g_{2n-1}} + ... + \frac{a_n + a_{n+1}}{g_n g_{n+1}}$ is equal to
(A) $\frac{2n}{h}$ (B) 2nh (C) nh (D) $\frac{n}{h}$

6. One side of an equilateral triangle is 24 cm. The mid–points of its sides are joined to form another triangle whose mid – points are in turn joined to form still another triangle. This process continues indefinitely. Then the sum of the perimeters of all the triangles is
(A) 144 cm (B) 212 cm (C) 288 cm (D) none of these

7. If p is positive, then the sum to infinity of the series, $\frac{1}{1+p} - \frac{1-p}{(1+p)^2} + \frac{(1-p)^2}{(1+p)^3}$ -..... is:
(A) 1/2 (B) 3/4 (C) 1 (D) none of these

8. In a G.P. of positive terms, any term is equal to the sum of the next two terms. The common ratio of the G.P. is
(A) 2 cos 18° (B) sin 18° (C) cos 18° (D) 2 sin 18°

9. If $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + ...$ upto $\infty = \frac{\pi^2}{6}$, then $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + ... =$
(A) $\pi^2/12$ (B) $\pi^2/24$ (C) $\pi^2/8$ (D) none of these

10. The sum to 10 terms of the series $\sqrt{2} + \sqrt{6} + \sqrt{18} + \sqrt{54} + ...$ is
(A) 121 $(\sqrt{6} + \sqrt{2})$ (B) $\frac{121}{2}(\sqrt{3} + 1)$ (C) 243 $(\sqrt{3} + 1)$ (D) 243 $(\sqrt{3} - 1)$

11. If a_1, a_2,... a_n are in A.P. with common difference $d \neq 0$, then the sum of the series
(sin d) [cosec a_1 cosec a_2 + cosec a_2 cosec a_3 + ... + cosec a_{n-1} cosec a_n]
(A) sec a_1 – sec a_n (B) cosec a_1 – cosec a_n (C) cot a_1 – cot a_n (D) tan a_1 – tan a_n

12. Sum of the series
$S = 1^2 - 2^2 + 3^2 - 4^2 + ... - 2002^2 + 2003^2$ is
(A) 2007006 (B) 1005004 (C) 2000506 (D) none of these

13. If $H_n = 1 + \frac{1}{2} + \frac{1}{3} + ...+ \frac{1}{n}$, then value of $1 + \frac{3}{2} + \frac{5}{3} + ... + \frac{2n-1}{n}$ is
(A) $2n - H_n$ (B) $2n + H_n$ (C) $H_n - 2n$ (D) $H_n + n$

14. The sum of the series $\frac{1}{\log_2 4} + \frac{1}{\log_4 4} + \frac{1}{\log_8 4} + ... + \frac{1}{\log_{2^n} 4}$ is
(A) $\frac{1}{2}$ n (n + 1) (B) $\frac{1}{12}$ n (n + 1) (2n + 1) (C) $\frac{1}{n(n+1)}$ (D) $\frac{1}{4}$ n (n + 1)

15. If S_1 , S_2, S_3 are the sums of first n natural numbers, their squares, their cubes respectively, then $\frac{S_3(1+8S_1)}{S_2^2}$ is equal to

(A) 1 (B) 3 (C) 9 (D) 10.

16. If p and q are respectively the sum and the sum of the squares of n successive integers beginning with 'a', then $nq - p^2$ is

(A) independent of 'a' (B) independent of 'n' (C) dependent on 'a' (D) none of these

17. Sum of n terms of the series $1 + \frac{x}{a_1} + \frac{x(x+a_1)}{a_1a_2} + \frac{x(x+a_1)(x+a_2)}{a_1a_2a_3} + \ldots$ is

(A) $\frac{x(x+a_1)\ldots(x+a_{n-1})}{a_1a_2\ldots a_3}$ (B) $\frac{(x+a_1)(x+a_2)\ldots.(x+a_{n-1})}{a_1a_2\ldots a_{n-1}}$ (C) $\frac{x(x+a_1)\ldots.(x+a_n)}{a_1a_2\ldots a_n}$ (D) none of these

18. $\{a_n\}$ and $\{b_n\}$ are two sequences given by $a_n = (x)^{1/2^n} + (y)^{1/2^n}$ and $b_n = (x)^{1/2^n} - (y)^{1/2^n}$ for all $n \in N$. The value of $a_1a_2a_3\ldots\ldots a_n$ is equal to

(A) x – y (B) $\frac{x+y}{b_n}$ (C) $\frac{x-y}{b_n}$ (D) $\frac{xy}{b_n}$

19. If $a_1, a_2, a_3, \ldots\ldots, a_n$ are positive real numbers whose product is a fixed number c, then the minimum value of $a_1 + a_2 + a_3 + \ldots. + a_{n-1} + 2a_n$ is **[IIT - 2002, 3]**

(A) $n(2c)^{1/n}$ (B) $(n + 1)\, c^{1/n}$ (C) $2nc^{1/n}$ (D) $(n + 1)(2c)^{1/n}$

Part : (B) May have more than one options correct

20. If $\sum_{r=1}^{n} r(r+1)\,(2r + 3) = an^4 + bn^3 + cn^2 + dn + e$, then

(A) a + c = b + d (B) e = 0 (C) a, b – 2/3, c – 1 are in A.P. (D) c/a is an integer

21. The sides of a right triangle form a G.P. The tangent of the smallest angle is

(A) $\sqrt{\frac{\sqrt{5}+1}{2}}$ (B) $\sqrt{\frac{\sqrt{5}-1}{2}}$ (C) $\sqrt{\frac{2}{\sqrt{5}+1}}$ (D) $\sqrt{\frac{2}{\sqrt{5}-1}}$

22. Sum to n terms of the series $S = 1^2 + 2(2)^2 + 3^2 + 2(4^2) + 5^2 + 2(6^2) + \ldots$ is

(A) $\frac{1}{2}$ n $(n + 1)^2$ when n is even (B) $\frac{1}{2}n^2 (n + 1)$ when n is odd

(C) $\frac{1}{4}n^2 (n + 2)$ when n is odd (D) $\frac{1}{4}n(n + 2)^2$ when n is even.

23. If a, b, c are in H.P., then:

(A) $\frac{a}{b+c-a}, \frac{b}{c+a-b}, \frac{c}{a+b-c}$ are in H.P. (B) $\frac{2}{b} = \frac{1}{b-a} + \frac{1}{b-c}$

(C) $a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2}$ are in G.P. (D) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in H.P.

24. If b_1, b_2, b_3 $(b_i > 0)$ are three successive terms of a G.P. with common ratio r, the value of r for which the inequality $b_3 > 4b_2 - 3b_1$ holds is given by

(A) r > 3 (B) r < 1 (C) r = 3.5 (D) r = 5.2

EXERCISE–5

1. If a, b, c are in A.P., then show that:

(i) $a^2 (b + c)$, $b^2 (c + a)$, $c^2 (a + b)$ are also in A.P. (ii) b + c – a, c + a – b, a + b – c are in A.P.

2. If a, b, c, d are in G.P., prove that :

(i) $(a^2 - b^2)$, $(b^2 - c^2)$, $(c^2 - d^2)$ are in G.P. (ii) $\frac{1}{a^2+b^2}, \frac{1}{b^2+c^2}, \frac{1}{c^2+d^2}$ are in G.P.

3. Using the relation A.M. ≥ G.M. prove that

(i) $\tan\theta + \cot\theta \geq 2$; if $0 < \theta < \frac{\pi}{2}$ (ii) $(x^2y + y^2z + z^2x)\,(xy^2 + yz^2 + zx^2) > 9x^2y^2z^2$.

(iii) (a + b) . (b + c) . (c + a) ≥ abc ; if a, b, c are positive real numbers

4. Find the sum in the nth group of sequence,

(i) 1, (2, 3); (4, 5, 6, 7); (8, 9,....., 15); ..... (ii) (1), (2, 3, 4), (5, 6, 7, 8, 9),.....

5. If n is a root of the equation $x^2 (1 - ac) - x (a^2 + c^2) - (1 + ac) = 0$ & if n HM's are inserted between a & c, show that the difference between the first & the last mean is equal to ac(a – c).

6. The sum of the first ten terms of an AP is 155 & the sum of first two terms of a GP is 9. The first term of the AP is equal to the common ratio of the GP & the first term of the GP is equal to the common difference of the AP. Find the two progressions.

7. Find the sum of the series $\frac{5}{13} + \frac{55}{(13)^2} + \frac{555}{(13)^3} + \frac{5555}{(13)^4} + \ldots$ up to ∞

8. If $0 < x < \pi$ and the expression

$\exp\{(1 + |\cos x| + \cos^2 x + |\cos^3 x| + \cos^4 x + \ldots\ldots \text{ upto } \infty)\log_e 4\}$

satisfies the quadratic equation $y^2 - 20y + 64 = 0$ the find the value of x.

9. In a circle of radius R a square is inscribed, then a circle is inscribed in the square, a new square in the circle and so on for n times. Find the limit of the sum of areas of all the circles and the limit of the sum of areas of all the squares as $n \to \infty$.

10. The sum of the squares of three distinct real numbers, which are in GP is S^2. If their sum is αS, show that $\alpha^2 \in (1/3, 1) \cup (1, 3)$.

11. Let $S_1, S_2, \ldots S_p$ denote the sum of an infinite G.P. with the first terms 1, 2, ..., p and common ratios

1/2, 1/3, ..., 1/(p + 1) respectively. Show that $S_1 + S_2 + ... + S_p = \frac{1}{2}\, p(p+3)$

12. Circles are inscribed in the acute angle α so that every neighbouring circles touch each other. If the radius of the first circle is R then find the sum of the radii of the first n circles in terms of R and α.

13. Given that α, γ are roots of the equation, $A x^2 - 4x + 1 = 0$ and β, δ the roots of the equation, $B x^2 - 6x + 1 = 0$, find values of A and B, such that α, β, γ & δ are in H.P.

14. The airthmetic mean between m and n and the geometric mean between a and b are each equal to $\frac{ma+nb}{m+n}$: find the m and n in terms of a and b.

15. If a, b, c are positive real numbers then prove that (i) $b^2c^2 + c^2a^2 + a^2b^2 > abc\,(a + b + c)$.
(ii) $(a + b + c)^3 > 27abc$. (iii) $(a + b + c)^3 > 27\,(a + b - c)\,(c + a - b)\,(b + c - a)$

16. If 's' be the sum of 'n' positive unequal quantities a, b, c,....., then $\frac{s}{s-a}+\frac{s}{s-b}+\frac{s}{s-c}+...>\frac{n^2}{n-1}$.

17. Sum the following series to n terms and to infinity: (i) $\sum_{r=1}^{n} r\,(r+1)\,(r+2)\,(r+3)$

(ii) $\frac{1}{1+1^2+1^4}+\frac{2}{1+2^2+2^4}+\frac{3}{1+3^2+3^4}+.....$ (iii) $\frac{1}{3.5}+\frac{16}{3^2.5^2}+\frac{1}{5.7}+\frac{24}{5^2.7^2}+\frac{1}{7.9}+\frac{32}{7^2.9^2}+.....$

18. Let a, b, c d be real numbers in G.P. If u, v, w, satisfy the system of equations
$u + 2v + 3w = 6$; $4u + 5v + 6w = 12$
$6u + 9v = 4$ then show that the roots of the equation
$\left(\frac{1}{u}+\frac{1}{v}+\frac{1}{w}\right)x^2 + [(b-c)^2 + (c-a)^2 + (d-b)^2]\,x + u + v + w = 0$ and
$20x^2 + 10\,(a-d)^2\,x - 9 = 0$ are reciprocals of each other. **[IIT- 1999, 10]**

19. The fourth power of the common difference of an arithmetic progression with integer entries added to the product of any four consecutive terms of it. Prove that the resulting sum is the square of an integer. **[IIT - 2000, 4]**

20. If a, b & c are in arithmetic progression and a^2, b^2 & c^2 are in harmonic progression, then prove that either a = b = c or a, b & $-\frac{c}{2}$ are in geometric progression. **[IIT – 2003, 4]**

ANSWER KEY

EXERCISE–1

Q 1. 1 **Q 3.** $\mu = 14$
Q 4. $S = (7/81)\{10^{n+1} - 9n - 10\}$ **Q 5.** 35/222
Q 6. $n(n+1)/2\,(n^2+n+1)$ **Q 7.** 27
Q 10. $(14n-6)/(8n+23)$ **Q 11.** 1
Q 14. (a) 9 ; (b) 12 **Q 16.** a = 5 , b = 8 , c = 12
Q 18. (8 , –4 , 2 , 8) **Q 19.** n^2
Q 20. (i) $2^{n+1} - 3$; $2^{n+2} - 4 - 3n$ **(ii)** $n^2 + 4n + 1$; $(1/6)\,n\,(n+1)\,(2n+13) + n$
Q 21. 120 , 30 **Q 22.** 6 , 3
Q 23. (i) $s_n = (1/24) - [1/\{6(3n+1)(3n+4)\}]$; $s_\infty = 1/24$ **(ii)** $(1/5)\,n\,(n+1)\,(n+2)\,(n+3)\,(n+4)$
(iii) $n/(2n+1)$ **(iv)** $S_n = 2\left[\frac{1}{2} - \frac{1.3.5.....(2n-1)(2n+1)}{2.4.6.....(2n)(2n+2)}\right]$; $S_\infty = 1$
Q 24. (a) $(6/5)\,(6^n - 1)$ **(b)** $[n\,(n+1)\,(n+2)]/6$

EXERCISE–2

Q 6. 8 problems , 127.5 minutes **Q.8** C = 9 ; (3, –3/2 , –3/5)
Q 12. (iii) b = 4 , c = 6 , d = 9 **OR** b = – 2 , c = – 6 , d = – 18 **Q.13** 2499
Q 15. (a) a = 1 , b = 9 **OR** b = 1 , a = 9 **; (b)** a = 1 ; b = 3 or vice versa
Q.19 $2p^3 - 9pq + 27r = 0$; roots are 1, 4, 7

Q 23. (a) $1 - \frac{x^n}{(x+1)\,(x+2)\,.....\,(x+n)}$ **(b)** $1 - \frac{1}{(1+a_1)\,(1+a_2)\,.....\,(1+a_n)}$
Q 24. n = 38 **Q 25.** 931

EXERCISE–3

Q 1. $\frac{1}{4}\,(2n-1)\,(n+1)^2$
Q 2. $S = \frac{1+ab}{(1-ab)^2}$ Where $a = 1 - x^{-1/3}$ & $b = 1 - y^{-1/4}$ **Q3.** $\beta \le (1/3)$; $\gamma \ge -(1/27)$
Q 4. – 3, 77 **Q 5.** 8, 24, 72, 216, 648
Q 6. (a) C **(b)** B **Q 7. (a)** B **(b)** D

Q 8. $r = \pm 1/9$; $n = 2$; $a = 144/180$ **OR** $r = \pm 1/3$; $n = 4$; $a = 108$ **OR** $r = 1/81$; $n = 1$; $a = 160$

Q 9. (a) D **(b)** A **Q 10.** $A = 3$; $B = 8$

Q 11. A.P. **Q 12.** $x = 2\sqrt{2}$ and $y = 3$

Q 13. (a) A, **(b)** C, **(c)** D, **(d)** $\left[(A_1, A_2, \ldots\ldots\ldots A_n)(H_1, H_2, \ldots\ldots\ldots H_n)\right]^{\frac{1}{2n}}$

Q14. (a) D **Q.16** B **Q.18 (a)** C, **(b)** $n = 7$ **Q.19** $n_0 = 5$

EXERCISE–4

1. D **2.** A **3.** B **4.** B **5.** A **6.** A **7.** A **8.** D **9.** C **10.** A
11. C **12.** A **13.** A **14.** D **15.** C **16.** A **17.** B **18.** C **19.** A **20.** ABCD
21. BC **22.** AB **23.** ABCD **24.** ABCD

EXERCISE–5

4. (i) $2^{n-2}(2^n + 2^{n-1} - 1)$ (ii) $(n-1)^3 + n^3$

6. $(3 + 6 + 12 + \ldots\ldots)$; $(2/3 + 25/3 + 625/6 + \ldots\ldots)$ **7.** $\frac{65}{36}$ **8.** $\frac{\pi}{2}, \frac{2\pi}{3}, \frac{\pi}{3}$

9. $2\pi R^2$; $4R^2$ **12.** $\frac{R\left(1 - \sin\frac{\alpha}{2}\right)}{2\sin\frac{\alpha}{2}}\left[\left(\frac{1 + \sin\frac{\alpha}{2}}{1 - \sin\frac{\alpha}{2}}\right)^n - 1\right]$ **13.** $A = 3$; $B = 8$

14. $m = \frac{2b\sqrt{a}}{\sqrt{a} + \sqrt{b}}$, $n = \frac{2a\sqrt{b}}{\sqrt{a} + \sqrt{b}}$

17. (i) $(1/5)\,n(n+1)(n+2)(n+3)(n+4)$ (ii) $\frac{n(n+1)}{2(n^2 + n + 1)}$; $s_\infty = \frac{1}{2}$

(iii) $\frac{n}{3(2n+3)} + \frac{4}{9}\,\frac{n(n+3)}{(2n+3)^2}$

ASSERTION & REASON FOR SEQUANCE AND SERIES

Some questions (Assertion–Reason type) are given below. Each question contains **Statement – 1 (Assertion)** and **Statement – 2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. So select the correct choice:

(A) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is** a correct explanation for **Statement – 1**.
(B) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is NOT** a correct explanation for **Statement – 1**.
(C) **Statement – 1** is True, **Statement – 2** is False.
(D) **Statement – 1** is False, **Statement – 2** is True.

549. **Statement–1 :** In the expression $(x + 1)(x + 2) \ldots (x + 50)$, coefficient of x^{49} is equal to 1275.

Statement–2 : $\sum_{r=1}^{n} r = \frac{n(n+1)}{2},\ n \in N.$

550. Let a, b, c, d are four positive number

Statement–1 : $\left(\frac{a}{b}+\frac{b}{c}\right)\left(\frac{c}{d}+\frac{d}{e}\right) \geq 4\sqrt{\frac{a}{e}}$ **Statement–2** : $\frac{b}{a}+\frac{c}{b}+\frac{d}{c}+\frac{e}{d}+\frac{a}{e} \geq 5.$

551. Let a, b, c and d be distinct positive real numbers in H.P.

Statement–1 : $a + d > b + c$ **Statement–2** : $\frac{1}{a}+\frac{1}{d}=\frac{1}{b}+\frac{1}{c}$

552. Let $a, r \in R - \{0, 1, -1\}$ and n be an even number.
Statement–1 : $a.\ ar.\ ar^2 \ldots ar^{n-1} = (a^2 r^{n-1})^{n/2}$.
Statement–2 : Product of kth term from the beginning and from the end in a G.P. is independent of k.

553. **Statement–1** : Let $p, q, r \in R^+$ and $27pqr \geq (p + q + r)^3$ and $3p + 4q + 5r = 12$, then $p^3 + q^4 + r^5$ is equal to 4.
Statement–2 : If A,G, and H are A.M., G.M., and H.M. of positive numbers $a_1, a_2, a_3, \ldots, a_n$ then $H \leq G \leq A$.

554. **Statement–1** : The sum of series $n.n + (n - 1)(n + 1) + (n - 2)(n + 2) + \ldots 1.(2n - 1)$ is $\frac{1}{6}n(n+1)(4n+1)$.

Statement–2 : The sum of any series S_n can be given as, $S_n = \sum_{r=1}^{n} T_r$, where T_r is the general ten of the series.

555. **Statement–1** : P is a point (a, b, c). Let A, B, C be images of P in yz, zx and xy plane respectively, then equation of plane must be $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$.

Statement–2 : The direction ratio of the line joining origin and point (x, y, z) must be x, y, z.

556. **Statement–1** : If A, B, C, D be the vertices of a rectangle in order. The position vector of A, B, C, D be a, b, c, d respectively, then $\vec{a}.\vec{c} = \vec{b}.\vec{d}$.

Statement–2 : In a triangle ABC, let O, G and H be the circumcentre, centroid and orthocentre of the triangle ABC, then OA + OB + OC = OH.

557. **Statement-1:** 1 + 3 7 + 13 + ... upto n terms = $\frac{n(n+2)}{3}$ **Statement-2:** $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is HM of a & b if $n = -\frac{1}{2}$

558. **Statement-1:** 1111 .... 1 (up to 91 terms) is a prime number

Statement-2: If $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P., then $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are also in A.P.

559. **Statement-1:** For a infinite G.P. whose first term is 'a' and common ratio is r, then $S_\infty = \frac{a}{1-r}$ where $|r| \geq 1$

Statement-2: A, G, H are arithmetic mean, Geometric mean and harmonic mean of two positive real numbers a & b. Then A, G, H are in G.P.

560. **Statement-1:** 11 11 …… 1 (up to 91 terms) is a prime number.

Statement-2: If $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ Are in A.P., then $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are also in A.P.

561. **Statement-1:** The sum of all the products of the first n positive integers taken two at a time is $\frac{1}{24}$ (n – 1) (n + 1) n(3n + 2) **Statement-2:** $\sum_{i\le i<j\le n} a_i a_j = (a_1 + a_2 + ... + a_n)^2 - (a_1^2 + a_2^2 + a_n^2)$

562. **Statement-1:** Let the positive numbers a, b, c, d, e be in AP, then abcd, abce, abde, acde, bcde are in HP
Statement-2: If each term of an A.P. is divided by the same number k, the resulting sequence is also

563. **Statement-1:** If a, b, c are in G.P., $\frac{1}{\log a}, \frac{1}{\log b}, \frac{1}{\log c}$ are in H.P.
Statement-2: When we take logarithm of the terms in G.P., they occur in A.P.

564. **Statement-1:** If $3p + 4q + 5r = 12$ then $p^3q^4r^5 \ge 1$ here $p, q, r \in R^+$
S-2: If the quantities are positive then weighted arithmetic mean is greater than or equal to geometric mean.

565. **Statement-1:** $S = 1/4 - 1/2 + 1 - 2 + 2^2 - .... = \frac{1/4}{1+2} = \frac{1}{12}$
S-2: Sum of n terms of a G.P. with first term as 'a' and common ratio as r in given by $a\left(\frac{r^n - 1}{r - 1}\right)$, $|r| > 1$.

566. **Statement-1:** $-4 + 2 - 1 + 1/2 - 1/4 + ... \infty$ is a geometric sequence.
Statement-2: Terms of a sequence are positive numebrs.

567. **Statement-1:** The sum of the infinite A.P. $1 + 2 + 2^2 + 2^3 + ... + ...$ is given by $\frac{a}{1-r} = \frac{1}{1-2} = -1$
Statement-2: The sum of an infinite G.P. is given by $\frac{a}{1-r}$ where $|r| < 1$ a is first term and r is common ratio.

568. **Statement-1:** If $a_1, a_2, a_3, ... a_n$ are positive real numbers whose product is a fixed number C, then the minimum value of $a_1 + a_2 + ... + a_{n-1} + 2a_n$ is $n(2c)^{1/n}$.
Statement-2: If $a_1, a_2, a_3, ... a_n \in R^+$. then $\frac{a_1 + a_2 + a_3 + ... + a_n}{n} \ge (a_1 a_2 a_3 ..... a_n)^{1/n}$

569. **Statement-1:** If $a(b - c) x^2 + b (c - a) x + c(a - b) = 0$ has equal roots, then a, b, c are in H.P.
Statement-2: Sum of the roots and product of the root are equal

570. **Statement-1:** $\lim_{n\to\infty} \frac{x^n}{n!} = 0$ for every $n > 0$
Statement-2: Every sequence whose nth term contains n! in the denominator converges to zero.

571. **Statement-1:** Sum of an infinite geometric series with common ratio more than one is not possible to find out.
S-2: The geometric series (Infinite) with common ratio more than one becomes diverging and sum is not fixed.

572. **Statement-1:** If arithmetic mean of two numbers is 5/2, Geometric mean of the numbers is 2 then harmonic mean will be 8/5.
Statement-2: for a group of numbers $(GM)^2 = (AM) \times (HM)$.

573. **Statement-1:** If a, b, c, d be four distinct positive quantities in H.P. then $a + d > b + c$, $ad > bc$.
Statement-2: A.M. > G.M. > H.M.

574. **Statement-1:** The sum of n arithmetic means between two given numbers is n times the single arithmetic mean between them.
Statement-2: n^{th} term of the A.P. with first term a and common difference d is $a + (n + 1)d$.

575. **Statement-1:** If $a + b + c = 3$ $a > 0, b > 0, c > 0$, then greatest value of $a^2b^3c^4 = 3^{10}2^4 - 77$.
Statement-2: If $a_i > 0$ $i = 1, 2, 3, ... n$, then $\frac{a_1 + a_2 + a_3 + ... + a_n}{n} \ge (a_1 a_2 ..... a_n)^{1/n}$

ANSWER SHEET

549. A 550. B 551. B 552. B 553. D 554. D 555. B 556. B 557. C 558. D 559. D 560. D 561. A
562. A 563. A 564. D 565. D 566. D 567. D568. A 569. C 570. C 571. A 572. C 573. A 574. C 575. A

IMP QUESTION FROM COMPETITIVE EXAMS

1. If the angles of a quadrilateral are in A.P. whose common difference is 10°, then the angles of the quadrilateral are

(a) $65^o, 85^o, 95^o, 105^o$ (b) $75^o, 85^o, 95^o, 105^o$ (b) $65^o, 75^o, 85^o, 95^o$ (d) $65^o, 95^o, 105^o, 115^o$

2. If the sum of first n terms of an A.P. be equal to the sum of its first m terms, $(m \neq n)$, then the sum of its first $(m+n)$ terms will be **[MP PET 1984]**

(a) 0 (b) n (c) m (d) $m+n$

3. If p, q, r are in A.P. and are positive, the roots of the quadratic equation $px^2 + qx + r = 0$ are all real for **[IIT 1995]**

(a) $\left|\frac{r}{p} - 7\right| \geq 4\sqrt{3}$ (b) $\left|\frac{p}{r} - 7\right| < 4\sqrt{3}$ (c) All p and r (d) No p and r

4. The sums of n terms of three A.P.'s whose first term is 1 and common differences are 1, 2, 3 are S_1, S_2, S_3 respectively. The true relation is

(a) $S_1 + S_3 = S_2$ (b) $S_1 + S_3 = 2S_2$ (c) $S_1 + S_2 = 2S_3$ (d) $S_1 + S_2 = S_3$

5. The value of x satisfying

$\log_a x + \log_{\sqrt{a}} x + \log_{3\sqrt{a}} x + \ldots\ldots\ldots \log_{a\sqrt{a}} x = \frac{a+1}{2}$ will be

(a) $x = a$ (b) $x = a^a$ (c) $x = a^{-1/a}$ (d) $x = a^{1/a}$

6. Jairam purchased a house in Rs. 15000 and paid Rs. 5000 at once. Rest money he promised to pay in annual installment of Rs. 1000 with 10% per annum interest. How much money is to be paid by Jairam **[UPSEAT 1999]**

(a) Rs. 21555 (b) Rs. 20475 (c) Rs. 20500 (d) Rs. 20700

7. Let $S_1, S_2, \ldots\ldots$ be squares such that for each $n \geq 1$, the length of a side of S_n equals the length of a diagonal of S_{n+1}. If the length of a side of S_1 is $10\,cm$, then for which of the following values of n is the area of S_n less then $1\ sq\ cm$

(a) 7 (b) 8 (c) 9 (d) 10

8. If $S_1, S_2, S_3, \ldots\ldots\ldots\ldots S_m$ are the sums of n terms of m A.P.'s whose first terms are $1, 2, 3, \ldots\ldots\ldots\ldots, m$ and common differences are $1, 3, 5, \ldots\ldots\ldots\ldots 2m-1$ respectively, then $S_1 + S_2 + S_3 + \ldots\ldots S_m =$

(a) $\frac{1}{2}mn(mn+1)$ (b) $mn(m+1)$ (c) $\frac{1}{4}mn(mn-1)$ (d) None of the above

9. If $a_1, a_2, a_3, \ldots\ldots a_{24}$ are in arithmetic progression and $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$, then $a_1 + a_2 + a_3 + \ldots\ldots\ldots + a_{23} + a_{24} =$ **[MP PET 1999; AMU 1997]**

(a) 909 (b) 75 (c) 750 (d) 900

10. If the roots of the equation $x^3 - 12x^2 + 39x - 28 = 0$ are in A.P., then their common difference will be

(a) ± 1 (b) ± 2 (c) ± 3 (4) ± 4 **[UPSEAT 1994, 99, 2001; RPET 2001]**

11. If the first term of a G.P. $a_1, a_2, a_3, \ldots\ldots\ldots$ is unity such that $4a_2 + 5a_3$ is least, then the common ratio of G.P. is

(a) $-\frac{2}{5}$ (b) $-\frac{3}{5}$ (c) $\frac{2}{5}$ (d) None of these

12. If the sum of the n terms of G.P. is S product is P and sum of their inverse is R, than P^2 is equal to

(a) $\frac{R}{S}$ (b) $\frac{S}{R}$ (c) $\left(\frac{R}{S}\right)^n$ (d) $\left(\frac{S}{R}\right)^n$ **[IIT 1966; Roorkee 1981]**

13. Let $n(>1)$ be a positive integer, then the largest integer m such that $(n^m + 1)$ divides $(1 + n + n^2 + \ldots\ldots + n^{127})$, is

(a) 32 (b) 63 (c) 64 (d) 127 **[IIT 1995]**

14. A G.P. consists of an even number of terms. If the sum of all the terms is 5 times the sum of the terms occupying odd places, then the common ratio will be equal to

(a) 2 (b) 3 (c) 4 (d) 5

15. If $f(x)$ is a function satisfying $f(x+y) = f(x)f(y)$ for all $x, y \in N$ such that $f(1) = 3$ and $\sum_{x=1}^{n} f(x) = 120$. Then the value of n is **[IIT 1992]**

(a) 4 (b) 5 (c) 6 (d) None of these

16. If n geometric means between a and b be $G_1, G_2, \ldots\ldots G_n$ and a geometric mean be G, then the true relation is

(a) $G_1.G_2.....G_n = G$ (b) $G_1.G_2.....G_n = G^{1/n}$

(c) $G_1.G_2.....G_n = G^n$ (d) $G_1.G_2.....G_n = G^{2/n}$

17. α, β are the roots of the equation $x^2 - 3x + a = 0$ and γ, δ are the roots of the equation $x^2 - 12x + b = 0$. If $\alpha, \beta, \gamma, \delta$ form an increasing G.P., then $(a, b) =$ **[DCE 2000]**

(a) (3, 12) (b) (12, 3) (c) (2, 32) (d) (4, 16)

18. $2.\dot{3}\dot{5}\dot{7} =$ **[IIT 1983; RPET 1995]**

(a) $\frac{2355}{1001}$ (b) $\frac{2370}{997}$ (c) $\frac{2355}{999}$ (d) None of these

19. If $1 + \cos\alpha + \cos^2\alpha + ... \infty = 2 - \sqrt{2}$, then α, $(0 < \alpha < \pi)$ is **[Roorkee 2000; AMU 2005]**

(a) $\pi/8$ (b) $\pi/6$ (c) $\pi/4$ (d) $3\pi/4$

20. The first term of an infinite geometric progression is x and its sum is 5. Then **[IIT Screening 2004]**

(a) $0 \le x \le 10$ (b) $0 < x < 10$ (c) $-10 < x < 0$ (d) $x > 10$

21. If a, b, c are in H.P., then the value of $\left(\frac{1}{b} + \frac{1}{c} - \frac{1}{a}\right)\left(\frac{1}{c} + \frac{1}{a} - \frac{1}{b}\right)$, is **[MP PET 1998; Pb. CET 2000]**

(a) $\frac{2}{bc} + \frac{1}{b^2}$ (b) $\frac{3}{c^2} + \frac{2}{ca}$ (c) $\frac{3}{b^2} - \frac{2}{ab}$ (d) None of these

22. If m is a root of the given equation $(1 - ab)x^2 - (a^2 + b^2)x - (1 + ab) = 0$ and m harmonic means are inserted between a and b, then the difference between the last and the first of the means equals

(a) $b - a$ (b) $ab(b - a)$ (c) $a(b - a)$ (d) $ab(a - b)$

23. A boy goes to school from his home at a speed of *x km/hour* and comes back at a speed of *y km/hour*, then the average speed is given by **[DCE 2002]**

(a) A.M. (b) G.M. (c) H.M. (d) None of these

24. If a, b, c, d be in H.P., then

(a) $a^2 + c^2 > b^2 + d^2$ (b) $a^2 + d^2 > b^2 + c^2$ (c) $ac + bd > b^2 + c^2$ (d) $ac + bd > b^2 + d^2$

25. If a, b, c are the positive integers, then $(a + b)(b + c)(c + a)$ is **[DCE 2000]**

(a) $< 8abc$ (b) $> 8abc$ (c) $= 8abc$ (d) None of these

26. In a G.P. the sum of three numbers is 14, if 1 is added to first two numbers and subtracted from third number, the series becomes A.P., then the greatest number is **[Roorkee 1973]**

(a) 8 (b) 4 (c) 24 (d) 16

27. If a, b, c are in G.P. and $\log a - \log 2b, \log 2b - \log 3c$ and $\log 3c - \log a$ are in A.P., then a, b, c are the length of the sides of a triangle which is

(a) Acute angled (b) Obtuse angled (c) Right angled (d) Equilateral

28. If $A_1, A_2; G_1, G_2$ and H_1, H_2 be $AM's, GM's$ and $HM's$ between two quantities, then the value of $\frac{G_1G_2}{H_1H_2}$ is

(a) $\frac{A_1 + A_2}{H_1 + H_2}$ (b) $\frac{A_1 - A_2}{H_1 + H_2}$ (c) $\frac{A_1 + A_2}{H_1 - H_2}$ (d) $\frac{A_1 - A_2}{H_1 - H_2}$

29. The harmonic mean of two numbers is 4 and the arithmetic and geometric means satisfy the relation $2A + G^2 = 27$, the numbers are **[MNR 1987; UPSEAT 1999, 2000]**

(a) 6, 3 (b) 5, 4 (c) 5, – 2.5 (d) –3, 1

30. If the A.M. of two numbers is greater than G.M. of the numbers by 2 and the ratio of the numbers is $4 : 1$, then the numbers are **[RPET 1988]**

(a) 4, 1 (b) 12, 3 (c) 16, 4 (d) None of these

31. If the A.M. and G.M. of roots of a quadratic equations are 8 and 5 respectively, then the quadratic equation will be

[Pb. CET 1990]

(a) $x^2 - 16x - 25 = 0$ (b) $x^2 - 8x + 5 = 0$ (c) $x^2 - 16x + 25 = 0$ (d) $x^2 + 16x - 25 = 0$

32. The A.M., H.M. and G.M. between two numbers are $\frac{144}{15}$, 15 and 12, but not necessarily in this order. Then H.M., G.M. and A.M. respectively are

(a) $15, 12, \frac{144}{15}$ (b) $\frac{144}{15}, 12, 15$ (c) $12, 15, \frac{144}{15}$ (d) $\frac{144}{15}, 15, 12$

33. If a be the arithmetic mean of b and c and G_1, G_2 be the two geometric means between them, then $G_1^3 + G_2^3 =$

(a) $G_1 G_2 a$ (b) $2G_1 G_2 a$ (c) $3G_1 G_2 a$ (d) None of these

34. Three numbers form a G.P. If the 3rd term is decreased by 64, then the three numbers thus obtained will constitute an A.P. If the second term of this A.P. is decreased by 8, a G.P. will be formed again, then the numbers will be

(a) 4, 20, 36 (b) 4, 12, 36 (c) 4, 20, 100 (d) None of the above

35. If $x > 1, y > 1, z > 1$ are in G.P., then $\frac{1}{1+\text{In}\, x}, \frac{1}{1+\text{In}\, y}, \frac{1}{1+\text{In}\, z}$ are in **[IIT 1998; UPSEAT 2001]**

(a) A.P. (b) H.P. (c) G.P. (d) None of these

36. a, g, h are arithmetic mean, geometric mean and harmonic mean between two positive numbers x and y respectively. Then identify the correct statement among the following **[Karnataka CET 2001]**

(a) h is the harmonic mean between a and g (b) No such relation exists between a, g and h

(c) g is the geometric mean between a and h (d) A is the arithmetic mean between g and h

37. $2^{\sin\theta} + 2^{\cos\theta}$ is greater than **[AMU 2000]**

(a) $\frac{1}{2}$ (b) $\sqrt{2}$ (c) $2^{\frac{1}{\sqrt{2}}}$ (d) $2^{\left(1-\frac{1}{\sqrt{2}}\right)}$

38. If a, b, c, d are positive real numbers such that $a + b + c + d = 2$, then $M = (a+b)(c+d)$ satisfies the relation

[IIT Screening 2000]

(a) $0 < M \le 1$ (b) $1 \le M \le 2$

(c) $2 \le M \le 3$ (d) $3 \le M \le 4$

39. Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. If $a < b < c$ and $a + b + c = \frac{3}{2}$, then the value of a is

[IIT Screening 2002]

(a) $\frac{1}{2\sqrt{2}}$ (b) $\frac{1}{2\sqrt{3}}$ (c) $\frac{1}{2} - \frac{1}{\sqrt{3}}$ (d) $\frac{1}{2} - \frac{1}{\sqrt{2}}$

40. n^{th} term of the series $1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \ldots\ldots$ will be

(a) $\frac{3n+1}{5^{n-1}}$ (b) $\frac{3n-1}{5^n}$ (c) $\frac{3n-2}{5^{n-1}}$ (d) $\frac{3n+2}{5^{n-1}}$

41. The sum of the series $\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \ldots\ldots$ to n terms is

(a) $\frac{n(n^2+1)}{n^2+n+1}$ (b) $\frac{n(n+1)}{2(n^2+n+1)}$ (c) $\frac{n(n^2-1)}{2(n^2+n+1)}$ (d) None of these

42. For any odd integer $n \geq 1$,

$n^3 - (n-1)^3 + \ldots\ldots\ldots + (-1)^{n-1} 1^3 =$ **[IIT 1996]**

(a) $\frac{1}{2}(n-1)^2(2n-1)$ (b) $\frac{1}{4}(n-1)^2(2n-1)$ (c) $\frac{1}{2}(n+1)^2(2n-1)$ (d) $\frac{1}{4}(n+1)^2(2n-1)$

43. The sum of n terms of the series $\frac{1}{1+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{7}} + \ldots\ldots\ldots$ is **[UPSEAT 2002]**

(a) $\sqrt{2n+1}$ (b) $\frac{1}{2}\sqrt{2n+1}$ (c) $\sqrt{2n+1}-1$ (d) $\frac{1}{2}(\sqrt{2n+1}-1)$

44. n^{th} term of the series

$\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \ldots\ldots$ will be **[Pb. CET 2000]**

(a) n^2+2n+1 (b) $\frac{n^2+2n+1}{8}$ (c) $\frac{n^2+2n+1}{4}$ (d) $\frac{n^2-2n+1}{4}$

45. The sum of the series $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \ldots + \frac{1}{\sqrt{n^2-1}+\sqrt{n^2}}$

equals **[AMU 2002]**

(a) $\frac{(2n+1)}{\sqrt{n}}$ (b) $\frac{\sqrt{n}+1}{\sqrt{n}+\sqrt{n-1}}$ (c) $\frac{(n+\sqrt{n^2-1})}{2\sqrt{n}}$ (d) $n-1$

ANSWER

1	b	2	a	3	a	4	b	5	d
6	c	7	b,c,d	8	a	9	d	10	c
11	a	12	d	13	c	14	c	15	a
16	c	17	c	18	c	19	d	20	b
21	c	22	b	23	c	24	c	25	b
26	a	27	b	28	a	29	a	30	c
31	c	32	b	33	b	34	c	35	b
36	c	37	d	38	a	39	d	40	c
41	b	42	d	43	d	44	c	45	d

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 10 XI M 10. Probability

Index:

1. Key Concepts

2. Exercise I to V

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Probability

There are various phenomena in nature, leading to an outcome, which cannot be predicted apriori e.g. in tossing of a coin, a head or a tail may result. Probability theory aims at measuring the uncertainties of such outcomes.

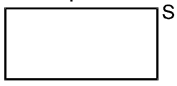
(I) Important terminology:

(i) *Random Experiment* :

It is a process which results in an outcome which is one of the various possible outcomes that are known to us before hand e.g. throwing of a die is a random experiment as it leads to fall of one of the outcome from {1, 2, 3, 4, 5, 6}. Similarly taking a card from a pack of 52 cards is also a random experiment.

(ii) *Sample Space :*

It is the set of all possible outcomes of a random experiment e.g. {H, T} is the sample space associated with tossing of a coin.
In set notation it can be interpreted as the universal set.



Solved Example # 1

Write the sample space of the experiment 'A coin is tossed and a die is thrown'.

Solution

The sample space S = {H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6}.

Solved Example # 2

Write the sample space of the experiment 'A coin is tossed, if it shows head a coin tossed again else a die is thrown.

Solution

The sample space S = {HH, HT, T1, T2, T3, T4, T5, T6}

Solved Example # 3

Find the sample space associated with the experiment of rolling a pair of dice (plural of die) once. Also find the number of elements of the sample space.

Sol. Let one die be blue and the other be grey. Suppose '1' appears on blue die and '2' appears on grey die. We denote this outcome by an ordered pair (1, 2). Similarly, if '3' appears on blue die and '5' appears on grey die, we denote this outcome by (3, 5) and so on. Thus, each outcome can be denoted by an ordered pair (x, y), where x is the number appeared on the first die (blue die) and y appeared on the second die (grey die). Thus, the sample space is given by

S = {(x, y) x is the number on blue die and y is the number on grey die}

We now list all the possible outcomes (figure)





	1	2	3	4	5	6
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

Figure

Number of elements (outcomes) of the above sample space is 6 × 6 i.e., 36

Self Practice Problems :

1. A coin is tossed twice, if the second throw results in head, a die is thrown.
Answer {HT, TT, HH1, HH2, HH3, HH4, HH5, HH6, TH1, TH2, TH3, TH4, TH5, TH6}.

2. An urn contains 3 red balls and 2 blue balls. Write sample space of the experiment 'Selection of a ball from the urn at random'.
Answer {R_1, R_2, R_3, B_1, B_2 }.

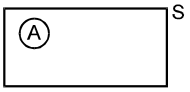
Note : -

Here the balls are distinguished from one and other by naming red balls as R_1, R_2 and R_3 and the blue

balls as B_1 and B_2.

(iii) ***Event :***

It is subset of sample space. e.g. getting a head in tossing a coin or getting a prime number is throwing a die. In general if a sample space consists 'n' elements, then a maximum of 2^n events can be associated with it.



(iv) ***Complement of event :***

The complement of an event 'A' with respect to a sample space S is the set of all elements of 'S' which are not in A. It is usually denoted by A′, $\overline{A}$ or A^C.

(v) ***Simple Event :***

If an event covers only one point of sample space, then it is called a simple event e.g. getting a head followed by a tail in throwing of a coin 2 times is a simple event.

(vi) ***Compound Event :***

When two or more than two events occur simultaneously, the event is said to be a compound event. Symbolically $A \cap B$ or AB represent the occurrence of both A & B simultaneously.

Note : "$A \cup B$" or A + B represent the occurrence of either A or B.

Solved Example # 4

Write down all the events of the experiment 'tossing of a coin'.

Solution

S = {H, T}
the events are ϕ , {H}, {T}, {H, T}

Solved Example # 5

A die is thrown. Let A be the event ' an odd number turns up' and B be the event 'a number divisible by 3 turns up'. Write the events (a) A or B (b) A and B

Solution

A = {1, 3, 5}, B = {3, 6}
$\therefore$ A or B = $A \cup B$ = {1, 3, 5, 6}
A and B = $A \cap B$ = {3}

Self Practice Problems :

3. A coin is tossed and a die is thrown. Let A be the event 'H turns up on the coin and odd number turns up on the die' and B be the event ' T turns up on the coin and an even number turns up on the die'. Write the events (a) A or B (b) A and B.
Answer (a) {H1, H3, H5, T2, T4, T6} (b) ϕ

4. In tossing of two coins, let A = {HH, HT} and B = {HT, TT}. Then write the events (a) A or B (b) A and B.
Answer (a) {HH, HT, TT} (b) {HT}

(vii) ***Equally likely Events :***

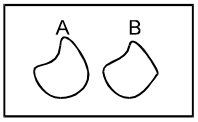
If events have same chance of occurrence, then they are said to be equally likely.

e.g

(i) In a single toss of a fair coin, the events {H} and {T} are equally likely.
(ii) In a single throw of an unbiased die the events {1}, {2}, {3} and {4}, are equally likely.
(iii) In tossing a biased coin the events {H} and {T} are not equally likely.

(viii) ***Mutually Exclusive / Disjoint / Incompatible Events :***

Two events are said to be mutually exclusive if occurrence of one of them rejects the possibility of occurrence of the other i.e. both cannot occur simultaneously.
In the vein diagram the events A and B are mutually exclusive. Mathematically, we write $A \cap B = \phi$



Solved Example # 6

In a single toss of a coin find whether the events {H}, {T} are mutually exclusive or not.

Solution

Since $\{H\} \cap \{T\} = \phi$,
$\therefore$ the events are mutually exclusive.

Solved Example # 7

In a single throw of a die, find whether the events {1, 2}, {2, 3} are mutually exclusive or not.

Solution

Since $\{1, 2\} \cap \{2, 3\} = \{2\} \neq \phi$
$\therefore$ the events are not mutually exclusive.

Self Practice Problems :

5. In throwing of a die write whether the events 'Coming up of an odd number' and 'Coming up of an even number' are mutually exclusive or not.
Answer Yes

6. An experiment involves rolling a pair of dice and recording the numbers that come up. Describe the following events :
A : the sum is greater than 8.
B : 2 occurs on either die.
C : the sum is at least 7 and a multiple of 3.
Also, find $A \cap B$, $B \cap C$ and $A \cap C$.
Are (i) A and B mutually exclusive ?
(ii) B and C mutually exclusive ?
(iii) A and C mutually exclusive ?
Ans. $A = \{(3, 6), (4, 5), (5, 4), (6, 3), (4, 6), (5, 5), (6, 4), (5, 6), (6, 5), (6, 6)\}$
$B = \{(1, 2), (2, 2), (3, 2), (4, 2), (5, 2), (6, 2), (2, 1), (2, 3), (2, 4). (2, 5), (2, 6)\}$
$C = \{(3, 6), (6, 3), (5, 4), (4, 5), (6, 6)\}$
$A \cap B = \phi$, $B \cap C = \phi$, $A \cap C = \{(3, 6), (6, 3), (5, 4), (4, 5), (6, 6)\}$
(i) Yes (ii) Yes (iii) No.

(ix) ***Exhaustive System of Events :***

If each outcome of an experiment is associated with at least one of the events E_1, E_2, E_3, ...E_n, then collectively the events are said to be exhaustive. Mathematically we write
$E_1 \cup E_2 \cup E_3$.....E_n = S. (Sample space)

Solved Example # 8
In throwing of a die, let A be the event 'even number turns up', B be the event 'an odd prime turns up' and C be the event 'a numbers less than 4 turns up'. Find whether the events A, B and C form an exhaustive system or not.
Solution
$A \equiv \{2, 4, 6\}$, $B \equiv \{3, 5\}$ and $C \equiv \{1, 2, 3\}$.
Clearly $A \cup B \cup C = \{1, 2, 3, 4, 5, 6\} = S$. Hence the system of events is exhaustive.

Solved Example # 9
Three coins are tossed. Describe
(i) two events A and B which are mutually exclusive
(ii) three events A, B and C which are mutually exclusive and exhaustive.
(iii) two events A and B which are not mutually exclusive.
(iv) two events A and B which are mutually exclusive but not exhaustive.
(v) three events A, B and C which are mutually exclusive but not exhaustive.
Ans. (i) A : "getting at least two heads" B : "getting at least two tails"
(ii) A : "getting at most one heads" B : "getting exactly two heads"
C : "getting exactly three heads"
(iii) A : "getting at most two tails" B : "getting exactly two heads"
(iv) A : "getting exactly one head" B : "getting exactly two heads"
(v) A : " getting exactly one tail" B : "getting exactly two tails"
C : "getting exactly three tails"
[Note : There may be other cases also]

Self Practice Problems :

7. In throwing of a die which of the following pair of events are mutually exclusive ?
(a) the events 'coming up of an odd number' and 'coming up of an even number'
(b) the events 'coming up of an odd number' and 'coming up of a number ≥ 4'
Answer (a)

8. In throwing of a die which of the following system of events are exhaustive ?
(a) the events 'an odd number turns up', 'a number ≤ 4 turns up' and 'the number 5 turns up'.
(b) the events 'a number ≤ 4 turns up', 'a number > 4 turns up'.
(c) the events 'an even number turns up', 'a number divisible by 3 turns up', 'number 1 or 2 turns up' and 'the number 6 turns up'.
Answer (b)

(II) Classical (A priori) Definition of Probability :

If an experiment results in a total of (m + n) outcomes which are equally likely and mutually exclusive with one another and if 'm' outcomes are favorable to an event 'A' while 'n' are unfavorable, then the probability of occurrence of the event 'A', denoted by P(A), is defined by $\frac{m}{m+n} = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$

i.e. $P(A) = \frac{m}{m+n}$.

We say that odds in favour of 'A' are m : n, while odds against 'A' are n : m.

Note that $P(\bar{A})$ or $P(A')$ or $P(A^c)$, i.e. probability of non-occurrence of $A = \frac{n}{m+n} = 1 - P(A)$

In the above we shall denote the number of out comes favourable to the event A by n(A) and the total number of out comes in the sample space S by n(S).

$\therefore \quad P(A) = \frac{n(A)}{n(S)}$.

Solved Example # 10

In throwing of a fair die find the probability of the event ' a number ≤ 4 turns up'.

Solution

Sample space S = {1, 2, 3, 4, 5, 6} ; event A = {1, 2, 3, 4}

$\therefore \quad n(A) = 4$ and $n(S) = 6$

$\therefore \quad P(A) = \frac{n(A)}{n(S)} = \frac{4}{6} = \frac{2}{3}$.

Solved Example # 11

In throwing of a fair die, find the probability of turning up of an odd number ≥ 4.

Solution

S = {1, 2, 3, 4, 5, 6}

Let E be the event 'turning up of an odd number ≥ 4'

then E = {5}

$\therefore \quad P(E) = \frac{n(E)}{n(S)} = \frac{1}{6}$.

Solved Example # 12

In throwing a pair of fair dice, find the probability of getting a total of 8.

Solution.

When a pair of dice is thrown the sample space consists

{(1, 1) (1, 2) ..... (1, 6)
(2, 1,) (2, 2,)..... (2, 6)
....
....
(6, 1), (6, 2) ..... (6, 6)}

Note that (1, 2) and (2, 1) are considered as separate points to make each outcome as equally likely.

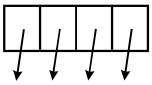
To get a total of '8', favourable outcomes are, (2, 6) (3, 5) (4, 4) (5, 3) and (6, 2).

Hence required probability = $\frac{5}{36}$

Solved Example # 13

A four digit number is formed using the digits 0, 1, 2, 3, 4 without repetition. Find the probability that it is divisible by 4

Solution

Total 4 digit numbers formed 

$4 \times 4 \times 3 \times 2 = 96$

Each of these 96 numbers are equally likely & mutually exclusive of each other.

Now, A number is divisible by 4, if last two digits of the number is divisible by 4

Hence we can have

		0	4	$\rightarrow$	first two places can be filled in $3 \times 2 = 6$ ways
		1	2	$\rightarrow$	first two places can be filled in $2 \times 2 = 4$ ways
		2	0	$\rightarrow$	6 ways
		2	4	$\rightarrow$	4 ways
		3	2	$\rightarrow$	4 ways
		4	0	$\rightarrow$	6 ways

Total number of ways 30 ways

probability = $\frac{\text{favorable outcomes}}{\text{Total outcomes}} = \frac{30}{96} = \frac{5}{16}$ Ans.

Self Practice Problems :

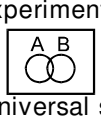
9. A bag contains 4 white, 3 red and 2 blue balls. A ball is drawn at random. Find the probability of the event (a) the ball drawn is white or red (b) the ball drawn is white as well as red.
Answer (a) 7/9 (b) 0

10. In throwing a pair of fair dice find the probability of the events ' a total of of less than or equal to 9"'.
Answer 5/36.

(III) Addition theorem of probability :

If 'A' and 'B' are any two events associated with an experiment, then

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$



De Morgan's Laws : If A & B are two subsets of a universal set U, then

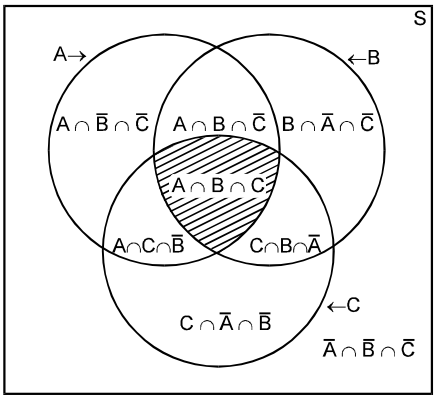
(a) $(A \cup B)^c = A^c \cap B^c$

(b) $(A \cap B)^c = A^c \cup B^c$

Distributive Laws : (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

For any three events A, B and C we have the figure



(i) $P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

(ii) $P(\text{at least two of A, B, C occur}) = P(B \cap C) + P(C \cap A) + P(A \cap B) - 2P(A \cap B \cap C)$

(iii) $P(\text{exactly two of A, B, C occur}) = P(B \cap C) + P(C \cap A) + P(A \cap B) - 3P(A \cap B \cap C)$

(iv) $P(\text{exactly one of A, B, C occur}) = P(A) + P(B) + P(C) - 2P(B \cap C) - 2P(C \cap A) - 2P(A \cap B) + 3P(A \cap B \cap C)$

Note : If three events A, B and C are pair wise mutually exclusive then they must be mutually exclusive, i.e. $P(A \cap B) = P(B \cap C) = P(C \cap A) = 0 \Rightarrow P(A \cap B \cap C) = 0$.
However the converse of this is not true.

Solved Example # 14

A bag contains 4 white, 3red and 4 green balls. A ball is drawn at random. Find the probability of the event 'the ball drawn is white or green'.

Solution

Let A be the event 'the ball drawn is white' and B be the event 'the ball drawn is green'.

$P(\text{The ball drawn is white or green}) = P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{8}{11}$

Solved Example # 15

In throwing of a die, let A be the event 'an odd number turns up', B be the event 'a number divisible by 3 turns up' and C be the event 'a number ≤ 4 turns up'. Then find the probability that exactly two of A, B and C occur.

Solution

Event $A = \{1, 3, 5\}$, event $B = \{3, 6\}$ and event $C = \{1, 2, 3, 4\}$

$\therefore$ $A \cap B = \{3\}$, $B \cap C = \{3\}$, $A \cap C = \{1, 3\}$ and $A \cap B \cap C = \{3\}$.

Thus $P(\text{exactly two of A, B and C occur}) = P(A \cap B) + P(B \cap C) + P(C \cap A) - 3P(A \cap B \cap C)$

$= \frac{1}{6} + \frac{1}{6} + \frac{2}{6} - 3 \times \frac{1}{6} = \frac{1}{6}$

Self Practice Problems :

11. In throwing of a die, let A be the event 'an odd number turns up', B be the event 'a number divisible by 3 turns up' and C be the event 'a number ≤ 4 turns up'. Then find the probability that atleast two of A, B and C occur. **Answer** $\frac{1}{3}$

12. In the problem number 11, find the probability that exactly one of A, B and C occurs. **Answer** $\frac{2}{3}$

(IV) Conditional Probability

If A and B are two events, then $P(A/B) = \frac{P(A \cap B)}{P(B)}$.

Note that for mutually exclusive events $P(A/B) = 0$.

Solved Example # 16

If P(A/B) = 0.2 and P(B) = 0.5 and P(A) = 0.2. Find $P(A \cap \overline{B})$.

Solution.

$P(A \cap \overline{B}) = P(A) - P(A \cap B)$

Also $P(A/B) = \frac{P(A \cap B)}{P(B)}$

$\Rightarrow$ $P(A \cap B) = 0.1$

From given data,

$P(A \cap \overline{B}) = 0.1$

Solved Example # 17

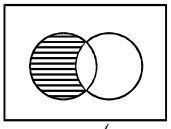
If P(A) = 0.25, P(B) = 0.5 and $P(A \cap B) = 0.14$, find probability that neither 'A' nor 'B' occurs. Also find $P\left(A \cap \overline{B}\right)$

Solution

We have to find $P\left(\overline{A} \cap \overline{B}\right) = 1 - P(A \cup B)$ (by De-Morgan's law)

Also, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

putting data we get, $P\left(\overline{A} \cap \overline{B}\right) = 0.39$



The shaded region denotes the simultaneous occurrence of A and $\overline{B}$

Hence $P\left(A \cap \overline{B}\right) = P(A) - P(A \cap B) = 0.11$

Self Practice Problem:-

13. If $P(\overline{A}/\overline{B}) = 0.2$, $P(A \cup B) = 0.9$, then find $P(A \cap \overline{B})$?
Ans. 0.4

5. Independent and dependent events

If two events are such that occurence or non-occurence of one does not affect the chances of occurence or non-occurence of the other event, then the events are said to be independent. Mathematically : if $P(A \cap B) = P(A)\ P(B)$, then A and B are independent.

Note: (i) If A and B are independent, then (a) A′ and B′ are independent, (b) A and B′ are independent and (c) A′ and B are independent.

(ii) If A and B are independent, then P(A / B) = P(A).

If events are not independent then they are said to be dependent.

Independency of three or more events

Three events A, B & C are independent if & only if all the following conditions hold :

$P(A \cap B) = P(A)\ .\ P(B)$; $P(B \cap C) = P(B)\ .\ P(C)$

$P(C \cap A) = P(C)\ .\ P(A)$; $P(A \cap B \cap C) = P(A)\ .\ P(B)\ .\ P(C)$

i.e. they must be independent in pairs as well as mutually independent.

Similarly for n events $A_1, A_2, A_3, \ldots\ldots A_n$ to be independent, the number of these conditions is equal to ${}^nC_2 + {}^nC_3 + \ldots\ldots + {}^cC_n = 2^{n^3} - n - 1$.

Solved Example # 18

In drawing two balls from a box containing 6 red and 4 white balls without replacement, which of the following pairs is independent ?

(a) Red on first draw and red on second draw

(b) Red on first draw and white on second draw

Solution

Let E be the event 'Red on first draw', F be the event 'Red on second draw' and G be the event 'white on second draw'.

$P(E) = \frac{6}{10}$, $P(F) = \frac{6}{10}$, $P(G) = \frac{4}{10}$

(a) $P(E \cap F) = \frac{{}^6P_2}{{}^{10}P_2} = \frac{1}{3}$

$P(E)\ .\ P(F) = \frac{3}{5} \times \frac{3}{5} = \frac{9}{25} \neq \frac{1}{3}$

$\therefore$ E and F are not independent

(b) $P(E)\ .\ P(G) = \frac{6}{10} \times \frac{4}{10} = \frac{6}{25}$

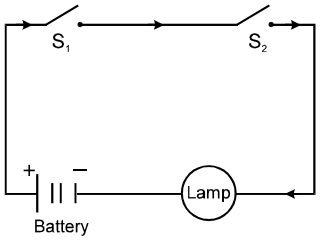
$$P(E \cap G) = \frac{{}^6P_1 \times {}^4P_1}{{}^{10}P_2} = \frac{4}{15}$$

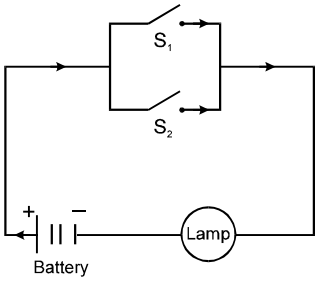
$\therefore$ $P(E) \cdot P(G) \neq P(E \cap G)$

$\therefore$ E and G are not independent

Solved Example # 19

If two switches S_1 and S_2 have respectively 90% and 80% chances of working. Find the probabilities that each of the following circuits will work.





Solution

Consider the following events :

A = Switch S_1 works,

B = Switch S_2 works,

We have,

$$P(A) = \frac{90}{100} = \frac{9}{10} \text{ and } P(B) = \frac{80}{100} = \frac{8}{10}$$

(i) The circuit will work if the current flows in the circuit. This is possible only when both the switches work together. Therefore,

Required probability

$= P(A \cap B) = P(A)\,P(B)$ [$\because$ A and B are independent events]

$= \frac{9}{10} \times \frac{8}{10} = \frac{72}{100} = \frac{18}{25}$

(ii) The circuit will work if the current flows in the circuit. This is possible only when at least one of the two switches S_1, S_2 works. Therefore,

Required Probability

$= P(A \cup B) = 1 - P(\overline{A})\,P(\overline{B})$ [$\because$ A, Bare independent events]

$= 1 - \left(1 - \frac{9}{10}\right)\left(1 - \frac{8}{10}\right)$

$= 1 - \frac{1}{10} \times \frac{2}{10} = \frac{49}{50}$

Solved Example # 20

A speaks truth in 60% of the cases and b in 90% of the cases. In what percentage of cases are they likely to contradict each other in stating the same fact?

Solution

Let E be the event that A speaks truth and F be the event that B speaks truth. Then E and F are independent events such that

$$P(E) = \frac{60}{100} = \frac{3}{5} \text{ and } P(F) = \frac{90}{100} = \frac{9}{10}$$

A and B will contradict each other in narrating the same fact in the following mutually exclusive ways :

(i) A speaks truth and B tells a lie i.e. $E \cap \overline{F}$

(ii) A tells a lie and B speaks truth lie i.e. $\overline{E} \cap F$

$\therefore$ P(A and B contradict each other)

$= P(\text{I or II}) = (\text{I} \cup \text{II})$

$= P[(E \cap \overline{F}) \cup (\overline{E} \cap F)]$

$= P(E \cap \overline{F}) + P(\overline{E} \cap F)$ [$\because$ $E \cap \overline{F}$ and $\overline{E} \cap F$ are mutually exclusive]

$= P(E)\,P(\overline{F}) + P(\overline{E})\,P(F)$ [$\because$ E and F are in dep.]

$= \frac{3}{5} \times \left(1 - \frac{9}{10}\right) + \left(1 - \frac{3}{5}\right) \times \frac{9}{10} = \frac{3}{5} \times \frac{1}{10} + \frac{2}{5} \times \frac{9}{10} = \frac{21}{50}$

Solved Example # 21

An urn contains 7 red and 4 blue balls. Two balls are drawn at random with replacement. Find the probability of getting

(i) 2 red balls (ii) 2 blue balls (iii) one red and one blue ball

Ans. (i) $\frac{49}{121}$ (ii) $\frac{16}{121}$ (iii) $\frac{56}{121}$

Solved Example # 22

Probabilities of solving a specific problem independently by A and B are $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If both try to solve the problem independently, find the probability that
(i) the problem is solved (ii) exactly one of them solves the problem.

Ans. (i) $\frac{2}{3}$ (ii) $\frac{1}{2}$

Solved Example # 23

A box contains 5 bulbs of which two are defective. Test is carried on bulbs one by one untill the two defective bulbs are found out. Find the probability that the process stops after
(i) Second test (ii) Third test

Solution

(i) Process will stop after second test. Only if the first and second bulb are both found to be defective

probability = $\frac{2}{5} \times \frac{1}{4} = \frac{1}{10}$ (Obviously the bulbs drawn are not kept back.)

(ii) Process will stop after third test when either

(a) DND $\rightarrow$ $\frac{2}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{1}{10}$ Here 'D' stands for defective

or (b) NDD $\rightarrow$ $\frac{3}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{1}{10}$ and 'N' is for not defective.

or (c) NNN $\rightarrow$ $\frac{3}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{1}{10}$

hence required probability = $\frac{3}{10}$

Solved Example # 24

If E_1 and E_2 are two events such that $P(E_1) = \frac{1}{4}$; $P(E_2) = \frac{1}{2}$; $P\left(\frac{E_1}{E_2}\right) = \frac{1}{4}$, then choose the correct options.
(i) E_1 and E_2 are independent (ii) E_1 and E_2 are exhaustive
(iii) E_1 and E_2 are mutually exclusive (iv) E_1 & E_2 are dependent

Also find $P\left(\frac{\overline{E}_1}{E_2}\right)$ and $\left(\frac{E_2}{\overline{E}_1}\right)$

Solution

Since $\left(\frac{E_2}{E_1}\right) = P(E_1)$ $\Rightarrow$ E_1 and E_2 are independent of each other

Also since $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1) \cdot P(E_2) \neq 1$
Hence events are not exhaustive. Independent events can't be mutually exclusive.
Hence only (i) is correct

Further since E_1 & E_2 are independent; E1 and $\overline{E}_2$ or $\overline{E}_1$, E_2 are $\overline{E}_1$, $\overline{E}_2$ are also independent.

Hence $P\left(\frac{\overline{E}_1}{E_2}\right) = P\left(\overline{E}_1\right) = \frac{3}{4}$ and $P\left(\frac{E_2}{\overline{E}_1}\right) = P(E_2) = \frac{1}{2}$

Solved Example # 25

If cards are drawn one by one from a well shuffled pack of 52 cards without replacement, until an ace appears, find the probability that the fourth card is the first ace to appear.

Solution

Probability of selecting 3 non-Ace and 1 Ace out of 52 cards is equal to $\frac{{}^{48}C_3 \times {}^4C_1}{{}^{52}C_4}$

Since we want 4th card to be first ace, we will also have to consider the arrangement, Now 4 cards in sample space can be arranged in 4! ways and, favorable they can be arranged in 3 ! ways as we want 4th position to be occupied by ace

Hence required probability = $\frac{{}^{48}C_3 \times {}^4C_1}{{}^{52}C_4} \times \frac{3!}{4!}$

Aliter : 'NNNA' is the arrangement than we desire in taking out cards, one by one

Hence required chance is $\frac{48}{52} \times \frac{47}{51} \times \frac{46}{50} \times \frac{4}{49}$

Self Practice Problems :

14. In throwing a pair of dies find the probability of getting an odd number on the first die and a total of 7 on both the dies.

Answer $\frac{1}{12}$

15. In throwing of a pair of dies, find the probability of getting a boublet or a total of 4.

Answer $\frac{2}{9}$

16. A bag contains 8 marbles of which 3 are blue and 5 are red. One marble is drawn at random, its colour is noted and the marble is replaced in the bag. A marble is again drawn from the bag and its colour is noted. Find the probability that the marbles will be
(i) blue followed by red (ii) blue and red in any order
(iii) of the same colour.

Ans. (i) $\frac{15}{64}$ (ii) $\frac{15}{32}$ (iii) $\frac{17}{32}$

17. A coin is tossed thrice. In which of the following cases are the events E and F independent ?
(i) E : "the first throw results in head".
F : "the last throw result in tail".
(ii) E : "the number of heads is two".
F : "the last throw result in head".
(iii) E : "the number of heads is odd ".
F : "the number of tails is odd".
Ans. (i)

6. Binomial Probability Theorem

If an experiment is such that the probability of success or failure does not change with trials, then the probability of getting exactly r success in n trials of an experiment is ${}^nC_r\, p^r\, q^{n-r}$, where 'p' is the probability of a success and q is the probability of a failure. Note that p + q = 1.

Solved Example 26

A pair of dice is thrown 5 times. Find the probability of getting a doublet twice.

Solution

In a single throw of a pair of dice probability of getting a doublet is $\frac{1}{6}$

con sidering it to be a success, $p = \frac{1}{6}$

$\therefore \quad q = 1 - \frac{1}{6} = \frac{5}{6}$

number of success r = 2

$\therefore \quad P(r = 2) = {}^5C_2\, p^2\, q^3 = 10 \cdot \left(\frac{1}{6}\right)^2 \cdot \left(\frac{5}{6}\right)^3 = \frac{625}{3888}$

Solved Example # 27

A pair of dice is thrown 4 times. If getting 'a total of 9' in a single throw is considered as a success then find the probability of getting 'a total of 9' thrice.

Solution

p = probability of getting 'a total of 9' $= \frac{4}{36} = \frac{1}{9}$

$\therefore \quad q = 1 - \frac{1}{9} = \frac{8}{9}$

r = 3, n = 4

$\therefore \quad P(r = 3) = {}^4C_3\, p^3\, q = 4 \times \left(\frac{1}{9}\right)^3 \cdot \frac{8}{9} = \frac{32}{6561}$

Solved Example # 28

In an examination of 10 multiple choice questions (1 or more can be correct out of 4 options). A student decides to mark the answers at random. Find the probability that he gets exactly two questions correct.

Solution

A student can mark 15 different answers to a MCQ with 4 option i.e. ${}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 15$

Hence if he marks the answer at random, chance that his answer is correct $= \frac{1}{15}$ and being incorrecting

$\frac{14}{15}$. Thus $p = \frac{1}{15}$, $q = \frac{14}{15}$.

$$P(2 \text{ success}) = {}^{10}C_2 \times \left(\frac{1}{15}\right)^2 \times \left(\frac{14}{15}\right)^8$$

Solved Example # 29

A family has three children. Event 'A' is that family has at most one boy, Event 'B' is that family has at least one boy and one girl, Event 'C' is that the family has at most one girl. Find whether events 'A' and 'B' are independent. Also find whether A, B, C are independent or not.

Solution

A family of three children can have
(i) All 3 boys (ii) 2 boys + 1 girl (iii) 1 boy + 2 girls (iv) 3 girls

(i) $P(3 \text{ boys}) = {}^3C_0 \left(\frac{1}{2}\right)^3 = \frac{1}{8}$ (Since each child is equally likely to be a boy or a girl)

(ii) $P(2 \text{ boys} + 1 \text{girl}) = {}^3C_1 \times \left(\frac{1}{2}\right)^2 \times \frac{1}{2} = \frac{3}{8}$ (Note that there are three cases BBG, BGB, GBB)

(iii) $P(1 \text{ boy} + 2 \text{ girls}) = {}^3C_2 \times \left(\frac{1}{2}\right)^1 \times \left(\frac{1}{2}\right)^2 = \frac{3}{8}$

(iv) $P(3 \text{ girls}) = \frac{1}{8}$

Event 'A' is associated with (iii) & (iv). Hence $P(A) = \frac{1}{2}$

Event 'B' is associated with (ii) & (iii). Hence $P(B) = \frac{3}{4}$

Event 'C' is associated with (i) & (ii). Hence $P(C) = \frac{1}{2}$

$P(A \cap B) = P(\text{iii}) = \frac{3}{8} = P(A) \cdot P(B)$. Hence A and B are independent of each other

$P(A \cap C) = 0 \neq P(A) \cdot P(C)$. Hence A, B, C are not independent

Self Practice Problems :

18. A box contains 2 red and 3 blue balls. Two balls are drawn successively without replacement. If getting 'a red ball on first draw and a blue ball on second draw' is considered a success, then find the probability of 2 successes in 3 performances.
Answer .189

19. Probability that a bulb produced by a factory will fuse after an year of use is 0.2. Find the probability that out of 5 such bulbs not more than 1 bulb will fuse after an year of use.
Answer $\frac{2304}{3125}$

7. Expectation :

If a value M_i is associated with a probability of p_i , then the expectation is given by $\sum p_i M_i$.

Solved Example # 30

There are 100 tickets in a raffle (Lottery). There is 1 prize each of Rs. 1000/-, Rs. 500/- and Rs. 200/-. Remaining tickets are blank. Find the expected price of one such ticket.

Solution

Expectation = $\sum p_i M_i$

outcome of a ticket can be	p_i	M_i	$p_i M_i$
(i) I prize	$\frac{1}{100}$	1000	10
(ii) II prize	$\frac{1}{100}$	500	5
(iii) III prize	$\frac{1}{100}$	200	2
(iv) Blank	$\frac{97}{100}$	0	0
			$\sum p_i M_i = 17$

Hence expected price of one such ticket Rs. 17

Solved Example # 31

A purse contains four coins each of which is either a rupee or two rupees coin. Find the expected value of a coin in that purse.

Solution

Various possibilities of coins in the purse can be

	p_i	M_i	p_iM_i
(i) 4 1 rupee coins	$\frac{1}{16}$	4	$\frac{4}{16}$
(ii) 3 one Rs. + 1 two Rs.	$\frac{4}{16}$	5	$\frac{20}{16}$
(iii) 2 one Rs. + 2 two Rs.	$\frac{6}{16}$	6	$\frac{36}{16}$
(iv) 1 one Rs. + 3 two Rs.	$\frac{4}{16}$	7	$\frac{28}{16}$
(iv) 4 two Rs.	$\frac{1}{16}$	8	$\frac{8}{16}$
			6 / -

Note that since each coin is equally likely to be one Rs. or two Rs. coin, the probability is determined using Binomial probability; unlike the case when the purse contained the coins with all possibility being equally likely, where we take $p_i = \frac{1}{5}$ for each.

Hence expected value is Rs. 6/-

Self Practice Problems :

20. From a bag containing 2 one rupee and 3 two rupee coins a person is allowed to draw 2 coins indiscriminately; find the value of his expectation.
Ans. Rs. 3.20

8. Total Probability Theorem

If an event A can occur with one of the n mutually exclusive and exhaustive events $B_1, B_2, ..., B_n$ and the probabilities $P(A/B_1), P(A/B_2) .... P(A/B_n)$ are known, then

$$P(A) = \sum_{i=1}^{n} P(B_i) \, . \, P(A/B_i)$$

Solved Example # 32

Box - I contains 5 red and 4 white balls whilst box - II contains 4 red and 2 white balls. A fair die is thrown. If it turns up a multiple of 3, a ball is drawn from box - I else a ball is drawn from box - II. Find the probability that the ball drawn is white.

Solution

Let A be the event 'a multiple of 3 turns up on the die' and R be the event 'the ball drawn is white' then P (ball drawn is white)

$$= P(A) \, . \, P(R / A) + P(\bar{A}) \; P(R / \bar{A})$$

$$= \frac{2}{6} \times \frac{4}{9} + \left(1 - \frac{2}{6}\right) \frac{2}{6} = \frac{10}{27}$$

Solved Example # 33

Cards of an ordinary deck of playing cards are placed into two heaps. Heap - I consists of all the red cards and heap - II consists of all the black cards. A heap is chosen at random and a card is drawn, find the probability that the card drawn is a king.

Solution

Let I and II be the events that heap - I and heap - II are choosen respectively. Then

$$P(I) = P(II) = \frac{1}{2}$$

Let K be the event 'the card drawn is a king'

$\therefore$ $P(K / I) = \frac{2}{26}$ and $P(K / II) = \frac{2}{26}$

$\therefore$ $P(K) = P(I) \, P(K / I) + P(II) \, P(K / II) = \frac{1}{2} \times \frac{2}{26} + \frac{1}{2} \times \frac{2}{26} = \frac{1}{13}$.

Self Practice Problems :

21. Box - I contains 3 red and 2 blue balls whilest box - II contains 2 red and 3 blue balls. A fair coin is tossed. If it turns up head, a ball is drawn from box - I, else a ball is drawn from box - II . Find the probability that the ball drawn is red.

Answer $\frac{1}{2}$

22. There are 5 brilliant students in class XI and 8 brilliant students in class XII. Each class has 50 students. The odds in favour of choosing the class XI are 2 : 3. If the class XI is not chosen then the class XII is chosen. Find the probability of selecting a brilliant student.

Answer $\frac{17}{125}$.

9. Bayes' Theorem :

If an event A can occur with one of the n mutually exclusive and exhaustive events $B_1, B_2, ..., B_n$ and the probabilities $P(A/B_1), P(A/B_2) .... P(A/B_n)$ are known, then

$$P(B_i / A) = \frac{P(B_i) . P(A/B_i)}{\sum_{i=1}^{n} P(B_i) . P(A/B_i)}$$

Proof :

The event A occurs with one of the n mutually exclusive and exhaustive events $B_1, B_2, B_3, ..., B_n$

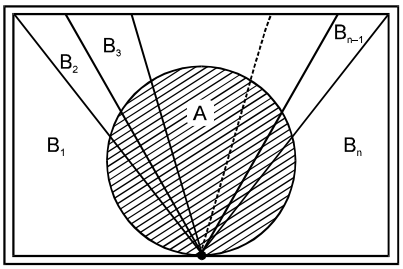
$A = (A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3) \cup ..... \cup (A \cap B_n)$

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + ... + P(A \cap B_n) = \sum_{i=1}^{n} P(A \cap B_i)$$

Note: $A \equiv$ event what we have ; B_i = event what we want ;

Now,

$P(A \cap B_i) = P(A) . P(B_i/A) = P(B_i) . P(A/B_i)$



$$P(B_i/A) = \frac{P(B_i) . P(A/B_i)}{P(A)} = \frac{P(B_i) . P(A/B_i)}{\sum_{i=1}^{n} P(A \cap B_i)}$$

$$P(B_i/A) = \frac{P(B_i) . P(A/B_i)}{\sum P(B_i) . P(A/B_i)}$$

Solved Example # 34

Pal's gardener is not dependable, the probability that he will forget to water the rose bush is $\frac{2}{3}$. The rose bush is in questionable condition any how, if watered the probability of its withering is $\frac{1}{2}$, if not watered, the probability of its withering is $\frac{3}{4}$. Pal went out of station and upon returning, he finds that the rose bush has withered, what is the probability that the gardener did not water the bush.

[Here result is known that the rose bush has withered, therefore. Bayes's theorem should be used]

Solution

Let A = the event that the rose bush has withered

Let A_1 = the event that the gardener did not water.

A_2 = the event that the gardener watered.

By Bayes's theorem required probability,

$$P(A_1/A) = \frac{P(A_1) . P(A/A_1)}{P(A_1) . P(A/A_1) + P(A_2) . P(A/A_2)} \quad .....(i)$$

Given, $P(A_1) = \frac{2}{3}$ $\quad \therefore \quad$ $P(A_2) = \frac{1}{3}$

$P(A/A_1) = \frac{3}{4}$, $P(A/A_2) = \frac{1}{2}$

From (1), $P(A_1/A) = \dfrac{\frac{2}{3}\cdot\frac{3}{4}}{\frac{2}{3}\cdot\frac{3}{4}+\frac{1}{3}\cdot\frac{1}{2}} = \dfrac{6}{6+2} = \dfrac{3}{4}$

Solved Example # 35

There are 5 brilliant students in class XI and 8 brilliant students in class XII. Each class has 50 students. The odds in favour of choosing the class XI are 2 : 3. If the class XI is not chosen then the class XII is chosen. A student is a chosen and is found to be brilliant, find the probability that the chosen student is from class XI.

Solution

Let E and F be the events 'Class XI is chosen' and 'Class XII is chosen' respectively.

Then $P(E) = \dfrac{2}{5}$, $P(F) = \dfrac{3}{5}$

Let A be the event 'Student chosen is brilliant'.

Then $P(A / E) = \dfrac{5}{50}$ and $P(A / F) = \dfrac{8}{50}$.

$\therefore \quad P(A) = P(E) \,.\, P(A / E) + P(F) \,.\, P(A / F) = \dfrac{2}{5}\cdot\dfrac{5}{50} + \dfrac{3}{5}\cdot\dfrac{8}{50} = \dfrac{34}{250}$.

$\therefore \quad P(E / A) = \dfrac{P(E).P(A/E)}{P(E).P(A/E)+P(F).P(A/F)} = \dfrac{5}{17}$.

Solved Example # 36

A pack of cards is counted with face downwards. It is found that one card is missing. One card is drawn and is found to be red. Find the probability that the missing card is red.

Solution

Let A be the event of drawing a red card when one card is drawn out of 51 cards (excluding missing card.) Let A_1 be the event that the missing card is red and A_2 be the event that the missing card is black.
Now by Bayes's theorem, required probability,

$$P(A_1/A) = \frac{P(A_1).(P(A/A_1)}{P(A_1).P(A/A_1)+P(A_2).P(A/A_2)} \qquad \text{.....(i)}$$

In a pack of 52 cards 26 are red and 26 are black.

Now $P(A_1)$ = probability that the missing card is red = $\dfrac{{}^{26}C_1}{{}^{52}C_1} = \dfrac{26}{52} = \dfrac{1}{2}$

$P(A_2)$ = probability that the missing card is black = $\dfrac{26}{52} = \dfrac{1}{2}$

$P(A/A_1)$ = probability of drawing a red card when the missing card is red.

$= \dfrac{25}{51}$

[$\because$ Total number of cards left is 51 out of which 25 are red and 26 are black as the missing card is red]

Again $P(A/A_2)$ = Probability of drawing a red card when the missing card is black = $\dfrac{26}{51}$

Now from (i), required probability,

$$P(A_1/A) = \frac{\frac{1}{2}\cdot\frac{25}{51}}{\frac{1}{2}\cdot\frac{25}{51}+\frac{1}{2}\cdot\frac{26}{51}} = \frac{25}{51}$$

Solved Example # 37

A bag contains 6 white and an unknown number of black balls (≤ 3). Balls are drawn one by one with replacement from this bag twice and is found to be white on both occassion. Find the probability that the bag had exactly '3' Black balls.

Solution

Apriori, we can think of the following possibilies

(i)	E_1	6W	,	0 B
(ii)	E_2	6W	,	1 B
(iii)	E_3	6W	,	2 B
(iv)	E_4	6W	,	3 B

Clearly $P(E_1) = P(E_2) = P(E_3) = P(E_4) = \dfrac{1}{4}$

Let 'A' be the event that two balls drawn one by one with replacement are both white therefore we have to find

$P\left(\dfrac{E_4}{A}\right)$

By Baye's theorem $P\left(\frac{E_4}{A}\right) = \frac{P\left(\frac{A}{E_4}\right)\times P(E_4)}{P\left(\frac{A}{E_1}\right)\times P(E_1)+P\left(\frac{A}{E_2}\right).P(E_2)+P\left(\frac{A}{E_3}\right).P(E_3)+P\left(\frac{A}{E_4}\right).P(E_4)}$

$P\left(\frac{A}{E_4}\right) = \frac{6}{9}\times\frac{6}{9}$; $P\left(\frac{A}{E_3}\right) = \frac{6}{8}\times\frac{6}{8}$; $P\left(\frac{A}{E_2}\right) = \frac{6}{7}\times\frac{6}{7}$;

$P\left(\frac{A}{E_1}\right) = \frac{6}{6}\times\frac{6}{6}$;

Putting values $P\left(\frac{E_4}{A}\right) = \frac{\frac{1}{81}}{\frac{1}{81}+\frac{1}{64}+\frac{1}{49}+\frac{1}{36}}$

Self Practice Problems :

23. Box-I contains 3 red and 2 blue balls whilest box-II contains 2 red and 3 blue balls. A fair coin is tossed. If it turns up head, a ball is drawn from box-I, else a ball is drawn from box-II. If the ball drawn is red, then find the probability that the ball is drawn from box-II.

Answer $\frac{3}{5}$

24. Cards of an ordinary deck of playing cards are placed into two heaps. Heap - I consists of all the red cards and heap - II consists of all the black cards. A heap is chosen at random and a card is drawn, if the card drawn is found to be a king, find the probability that the card drawn is from the heap - II.

Answer $\frac{1}{2}$

10. Value of Testimony

If p_1 and p_2 are the probabilities of speaking the truth of two independent witnesses A and B then P(their combined statement is true) = $\frac{p_1 p_2}{p_1 p_2+(1-p_1)(1-p_2)}$.

In this case it has been assumed that we have no knowledge of the event except the statement made by A and B.

However if p is the probability of the happening of the event before their statement, then

P(their combined statement is true) = $\frac{p\, p_1 p_2}{p\, p_1 p_2+(1-p)\,(1-p_1)(1-p_2)}$.

Here it has been assumed that the statement given by all the independent witnesses can be given in two ways only, so that if all the witnesses tell falsehoods they agree in telling the same falsehood. If this is not the case and c is the chance of their coincidence testimony then the

Probability that the statement is true = $P\, p_1\, p_2$

Probability that the statement is false = $(1-p).\, c\,(1-p_1)\,(1-p_2)$

However chance of coincidence testimony is taken only if the joint statement is not contradicted by any witness.

Solved Example # 38

A die is thrown, a man C gets a prize of Rs. 5 if the die turns up 1 and gets a prize of Rs. 3 if the die turns up 2, else he gets nothing. A man A whose probability of speaking the truth is $\frac{1}{2}$ tells C that the die has turned up 1 and another man B whose probability of speaking the truth is $\frac{2}{3}$ tells C that the die has turned up 2. Find the expectation value of C.

Solution

Let A and B be the events 'A speaks the truth' and 'B speaks the truth' respectively. Then $P(A) = \frac{1}{2}$ and $P(B) = \frac{2}{3}$.

The experiment consists of three hypothesis

(i) the die turns up 1

(ii) the die turns up 2

(iii) the die turns up 3, 4, 5 or 6

Let these be the events E_1, E_2 and E_3 respectively then $P(E_1) = P(E_2) = \frac{1}{6}$ and $P(E_3) = \frac{4}{6}$.

Let E be the event that the statements made by A and B agree to the same conclusion.

$\therefore \quad P(E / E_1) = P(A) \cdot P(\overline{B}) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$

$P(E / E_2) = P(\overline{A}) \cdot P(B) = \frac{1}{2} \times \frac{2}{3} = \frac{2}{6}$

$P(E / E_3) = P(\overline{A}) \cdot P(\overline{B}) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$

$\therefore \quad P(E) = P(E_1)\, P(E / E_1) + P(E_2)\, P(E / E_2) + P(E_3)\, P(E / E_3)$

$= \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{2}{6} + \frac{4}{6} \cdot \frac{1}{6} = \frac{7}{36}$

Thus $P(E_1 / E) = \frac{P(E_1)P(E/E_1)}{P(E)} = \frac{1}{7}$

$P(E_2 / E) = \frac{P(E_2)P(E/E_2)}{P(E)} = \frac{2}{7}$

$P(E_3 / E) = \frac{P(E_3)P(E/E_3)}{P(E)} = \frac{4}{7}$

$\therefore \quad$ expectation of C $= \frac{1}{7} \times 5 + \frac{2}{7} \times 3 + 0 = \text{Rs. } \frac{11}{7}$

Solved Example #39

A speaks the truth '3 times out of 4' and B speaks the truth '2 times out of 3'. A die is thrown. Both assert that the number turned up is 2. Find the probability of the truth of their assertion.

Solution

Let A and B be the events 'A speaks the truth' and 'B speaks the truth' repectively. Let C be the event 'the number turned up is not 2 but both agree to the same conclustion that the die has turned up 2'.

Then $P(A) = \frac{3}{4}$, $P(B) = \frac{2}{3}$ and $P(C) = \frac{1}{5} \times \frac{1}{5}$

There are two hypotheses

(i) the die turns up 2

(ii) the die does not turns up 2

Let these be the events E_1 and E_2 respectively, then

$P(E_1) = \frac{1}{6}$, $P(E_2) = \frac{5}{6}$ (a priori probabilities)

Now let E be the event 'the statement made by A and B agree to the same conclusion.

then $P(E / E_1) = P(A) \cdot P(B) = \frac{3}{4} \cdot \frac{2}{3} = \frac{1}{2}$

$P(E / E_2) = P(\overline{A}) \cdot P(\overline{B}) \cdot P(C) = \frac{1}{4} \cdot \frac{1}{3} \cdot \frac{1}{25} = \frac{1}{300}$

Thus $P(E) = P(E_1)\, P(E / E_1) + P(E_2)\, P(E / E_2)$

$= \frac{1}{6} \times \frac{1}{2} + \frac{5}{6} \times \frac{1}{300} = \frac{31}{360}$

$\therefore \quad P(E_1 / E) = \frac{P(E_1)\, P(E/E_1)}{P(E)} = \frac{30}{31}$

Self Practice Problems :

25. A ball is drawn from an urn containing 5 balls of different colours including white. Two men A and B whose probability of speaking the truth are $\frac{1}{3}$ and $\frac{2}{5}$ respectively assert that the ball drawn is white. Find the probability of the truth of their assertion.

Answer $\frac{4}{7}$

11. Binomial Probability Distribution :

(i) A probability distribution spells out how a total probability of 1 is distributed over several values of a random variable.

(ii) Mean of any probability distribution of a random variable is given by :

$\mu = \frac{\Sigma p_i x_i}{\Sigma p_i} = \Sigma\, p_i x_i$ (Since $\Sigma p_i = 1$)

(iii) *Variance of a random variable is given by,* $\sigma^2 = \Sigma (x_i - \mu)^2 \cdot p_i$

$\therefore \quad \sigma^2 = \Sigma p_i x_i^2 - \mu^2$ (Note that SD = $+\sqrt{\sigma^2}$)

(iv) The probability distribution for a binomial variate 'X' is given by :
$P(X = r) = {}^nC_r\, p^r\, q^{n-r}$ where P(X = r) is the probability of r successes.

The recurrence formula $\frac{P(r+1)}{P(r)} = \frac{n-r}{r+1} \cdot \frac{p}{q}$, is very helpful for quickly computing P(1) . P(2) . P(3) etc. if P(0) is known.

(v) Mean of BPD = np ; variance of BPD = npq.

(vi) If p represents a person's chance of success in any venture and 'M' the sum of money which he will receive in case of success, then his expectations or probable value = pM

Solved Example # 40

A random variable X has the following probability distribution :

X	0	1	2	3	4	5	6	7
P(X)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

Determine

(i) k (ii) P(X < 3) (iii) P(X > 6) (iv) P(0 < X < 3)

[**Hint :** Use $\sum P(X) = 1$ to determine k, P(X < 3) = P(0) + P(1) + P(2), P(X > 6) = P(7) etc.]

Solved Example # 41

A pair of dice is thrown 5 times. If getting a doublet is considered as a success, then find the mean and variance of successes.

Solution

In a single throw of a pair of dice, probability of getting a doublet = $\frac{1}{6}$

con sidering it to be a success, $p = \frac{1}{6}$

$\therefore \quad q = 1 - \frac{1}{6} = \frac{5}{6}$

mean = $5 \times \frac{1}{6} = \frac{5}{6}$

variance = $5 \times \frac{1}{6} \cdot \frac{5}{6} = \frac{25}{36}$

Solved Example # 42

A pair of dice is thrown 4 times. If getting a total of 9 in a single throw is considered as a success then find the mean and variance of successes.

Solution

p = probability of getting a total of 9 = $\frac{4}{36} = \frac{1}{9}$

$\therefore \quad q = 1 - \frac{1}{9} = \frac{8}{9}$

$\therefore \quad$ mean = np = $4 \times \frac{1}{9} = \frac{4}{9}$

variance = npq = $4 \times \frac{1}{9} \times \frac{8}{9} = \frac{32}{81}$

Solved Example # 43

Difference between mean and variance of a Binomial variate is '1' and difference between their squares is '11'. Find the probability of getting exactly three success

Solution

Mean = np & variance = npq

therefore, $np - npq = 1$ .....(i)

$n^2p^2 - n^2p^2q^2 = 11$ .....(ii)

Also, we know that p + q = 1 .....(iii)

Divide equation (ii) by square of (i) and solve, we get, $q = \frac{5}{6}$, $p = \frac{1}{6}$ & n = 36

Hence probability of '3' success = ${}^{36}C_3 \times \left(\frac{1}{6}\right)^3 \times \left(\frac{5}{6}\right)^{33}$ Ans.

Self Practice Problems :

26. A box contains 2 red and 3 blue balls. Two balls are drawn successively without replacement. If getting 'a red ball on first draw and a blue ball on second draw' is considered a success, then find the mean and variance of successes.

Answer mean = 2.1, σ^2 = .63

27. Probability that a bulb produced by a factory will fuse after an year of use is 0.2. If fusing of a bulb is considered an failure, find the mean and variance of successes for a sample of 10 bulbs.
Answer mean = 8 and variance = 1.6

28. A random variable X is specified by the following distribution law :

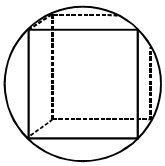
X	2	3	4
P(X = x)	0.3	0.4	0.3

Then the variance of this distribution is :
(A*) 0.6 (B) 0.7 (C) 0.77 (D) 1.55

12. Geometrical Applications:
The following statements are axiomatic :

(i) If a point is taken at random on a given straight line segment AB, the chance that it falls on a particular segment PQ of the line segment is PQ/AB.
(ii) If a point is taken at random on the area S which includes an area σ, the chance that the point falls on σ is σ/S.

Solved Example # 44
A sphere is circumscribed over a cube. Find the probability that a point lies inside the sphere, lies outside the cube.
Solution



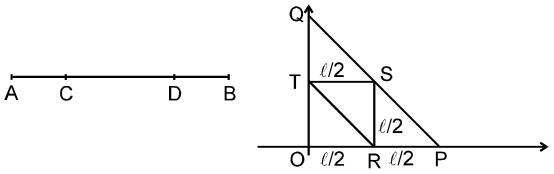
$$\text{Required probability} = \frac{\text{favorable volume}}{\text{total volume}}$$

Clearly if edge length of cube is a radius of sphere will be $\frac{a\sqrt{3}}{2}$

Thus, volume of sphere $= \frac{4}{3}\pi\left(\frac{a\sqrt{3}}{2}\right)^3 = \frac{\pi a^3\sqrt{3}}{2}$

Hence $P = 1 - \frac{1}{\pi\frac{\sqrt{3}}{2}} = 1 - \frac{2}{\pi\sqrt{3}}$

Solved Example # 45
A given line segment is divided at random into three parts. What is the probability that they form sides of a possible triangle ?
Solution
Let AB be the line segment of length ℓ.
Let C and D be the points which divide AB into three parts.
Let AC = x, CD = y. Then DB = $\ell - x - y$.
Clearly $x + y < \ell$
$\therefore$ the sample space is given by
the region enclosed by Δ OPQ, where OP = OQ = ℓ



Area of $\Delta OPQ = \frac{\ell^2}{2}$
Now if the parts AC, CD and DB form a triangle, then

$x + y > \ell - x - y$ i.e. $x + y > \frac{\ell}{2}$ .....(i)

$x + \ell - x - y > y$ i.e. $y < \frac{\ell}{2}$ .....(ii)

$y + \ell - x - y > x$ i.e. $x < \frac{\ell}{2}$ .....(iii)

from (i), (ii) and (iii), we get
the event is given by the region closed in ΔRST

$\therefore$ Probability of the event = $\dfrac{\text{ar }(\Delta RST)}{\text{ar }(\Delta OPQ)} = \dfrac{\frac{1}{2}\cdot\frac{\ell}{2}\cdot\frac{\ell}{2}}{\frac{\ell^2}{2}} = \dfrac{1}{4}$

Solved Example # 46

On a line segment of length L two points are taken at random, find the probability that the distance between them is ℓ , where $\ell < 1$

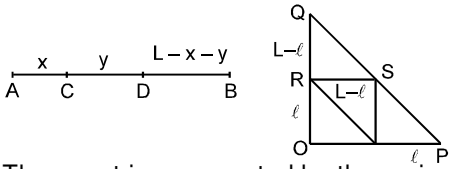
Solution

Let AB be the line segment

Let C and D be any two points on AB so that AC = x and CD = y. Then $x + y < L$, $y > \ell$

$\therefore$ sample space is represented by the region enclosed by ΔOPQ.

Area of $\Delta OPQ = \dfrac{1}{2}L^2$



The event is represented by the region, bounded by the ΔRSQ

Area of $\Delta RSQ = \dfrac{1}{2}(L - \ell)^2$

$\therefore$ probability of the event = $\left(\dfrac{L-\ell}{L}\right)^2$

Self Practice Problems :

29. A line segment of length a is divided in two parts at random by taking a point on it, find the probability that no part is greater than b, where $2b > a$

Answer $\dfrac{2b-a}{a}$

30. A cloth of length 10 meters is to be randomly distributed among three brothers, find the probability that no one gets more than 4 meters of cloth.

Answer $\dfrac{1}{25}$

KEY CONCEPTS

THINGS TO REMEMBER :

RESULT – 1

(i) ***SAMPLE–SPACE*** **:** The set of all possible outcomes of an experiment is called the **SAMPLE–SPACE(S)**.

(ii) ***EVENT*** **:** A sub set of sample–space is called an **EVENT**.

(iii) ***COMPLEMENT OF AN EVENT A*** **:** The set of all out comes which are in S but not in A is called the **COMPLEMENT OF THE EVENT A DENOTED BY** $\overline{A}$ **OR** A^c.

(iv) ***COMPOUND EVENT*** **:** If A & B are two given events then $A \cap B$ is called **COMPOUND EVENT** and is denoted by $A \cap B$ or AB or A & B .

(v) ***MUTUALLY EXCLUSIVE EVENTS*** **:** Two events are said to be **MUTUALLY EXCLUSIVE** (or disjoint or incompatible) if the occurence of one precludes (rules out) the simultaneous occurence of the other . If A & B are two mutually exclusive events then $P(A \& B) = 0$.

(vi) ***EQUALLY LIKELY EVENTS*** **:** Events are said to be **EQUALLY LIKELY** when each event is as likely to occur as any other event.

(vii) ***EXHAUSTIVE EVENTS*** **:** Events A,B,C ..... L are said to be **EXHAUSTIVE EVENTS** if no event outside this set can result as an outcome of an experiment . For example, if A & B are two events defined on a sample space S, then A & B are exhaustive $\Rightarrow A \cup B = S \Rightarrow P(A \cup B) = 1$.

(viii) ***CLASSICAL DEF. OF PROBABILITY*** **:** If n represents the total number of equally likely , mutually exclusive and exhaustive outcomes of an experiment and m of them are favourable to the happening of the event A, then the probability of happening of the event A is given by $P(A) = m/n$.

Note : **(1)** $0 \le P(A) \le 1$

(2) $P(A) + P(\overline{A}) = 1$, Where $\overline{A}$ = Not A .

(3) If x cases are favourable to A & y cases are favourable to $\overline{A}$ then $P(A) = \frac{x}{(x+y)}$ and $P(\overline{A}) = \frac{y}{(x+y)}$ We say that **ODDS IN FAVOUR OF A** are x: y & odds against A are y : x

Comparative study of Equally likely , Mutually Exclusive and Exhaustive events.

	Experiment	Events	E/L	M/E	Exhaustive
1.	Throwing of a die	A : throwing an odd face {1, 3, 5} B : throwing a composite face {4,. 6}	No	Yes	No
2.	A ball is drawn from an urn containing 2W, 3R and 4G balls	E_1 : getting a W ball E_2 : getting a R ball E_3 : getting a G ball	No	Yes	Yes
3.	Throwing a pair of dice	A : throwing a doublet {11, 22, 33, 44, 55, 66} B : throwing a total of 10 or more {46, 64, 55, 56, 65, 66}	Yes	No	No
4.	From a well shuffled pack of cards a card is drawn	E_1 : getting a heart E_2 : getting a spade E_3 : getting a diamond E_4 : getting a club	Yes	Yes	Yes
5.	From a well shuffled pack of cards a card is drawn	A = getting a heart B = getting a face card	No	No	No

RESULT – 2

$A \cup B = A + B$ = A or B denotes occurence of at least A or B. For 2 events A & B : **(See fig.1)**

(i) $P(A \cup B) = P(A) + P(B) - P(A \cap B) =$

$P(A.\overline{B}) + P(\overline{A}.B) + P(A.B) = 1 - P(\overline{A}.\overline{B})$

(ii) Opposite of " atleast A or B " is **NIETHER A NOR B**

i.e. $\overline{A+B} = 1\text{-}(A \text{ or } B) = \overline{A} \cap \overline{B}$

Note that $P(A+B) + P(\overline{A} \cap \overline{B}) = 1$.

(iii) If A & B are mutually exclusive then $P(A \cup B) = P(A) + P(B)$.

(iv) For any two events A & B, P(exactly one of A , B occurs)

$$= P(A \cap \overline{B}) + P(B \cap \overline{A}) = P(A) + P(B) - 2P(A \cap B)$$
$$= P(A \cup B) - P(A \cap B) = P(A^c \cup B^c) - P(A^c \cap B^c)$$

(v) If A & B are any two events $P(A \cap B) = P(A).P(B/A) = P(B).P(A/B)$, Where P(B/A) means conditional probability of B given A & P(A/B) means conditional probability of A given B. (This can be easily seen from the figure)

(vi) **DE MORGAN'S LAW** : – If A & B are two subsets of a universal set U , then

(a) $(A \cup B)^c = A^c \cap B^c$ & **(b)** $(A \cap B)^c = A^c \cup B^c$

(vii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ & $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

RESULT – 3

For any three events A,B and C we have **(See Fig. 2)**

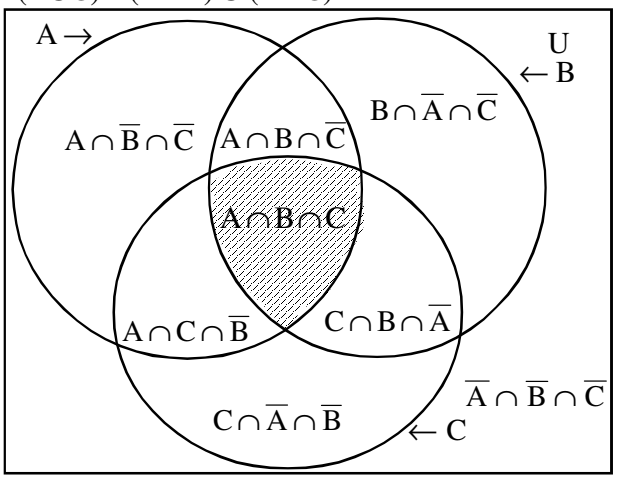


Fig. 2

(i) $P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

(ii) P (at least two of A,B,C occur) = $P(B \cap C) + P(C \cap A) + P(A \cap B) - 2P(A \cap B \cap C)$

(iii) P(exactly two of A,B,C occur) = $P(B \cap C) + P(C \cap A) + P(A \cap B) - 3P(A \cap B \cap C)$

(iv) P(exactly one of A,B,C occurs) = $P(A) + P(B) + P(C) - 2P(B \cap C) - 2P(C \cap A) - 2P(A \cap B) + 3P(A \cap B \cap C)$

NOTE :

If three events A, B and C are pair wise mutually exclusive then they must be mutually exclusive. i.e $P(A \cap B) = P(B \cap C) = P(C \cap A) = 0 \Rightarrow P(A \cap B \cap C) = 0$. However the converse of this is not true.

RESULT – 4

INDEPENDENT EVENTS **:** Two events A & B are said to be independent if occurence or non occurence of one does not effect the probability of the occurence or non occurence of other.

(i) If the occurence of one event affects the probability of the occurence of the other event then the events are said to be **DEPENDENT** or **CONTINGENT** . For two independent events A and B : $P(A \cap B) = P(A).\ P(B)$. Often this is taken as the definition of independent events.

(ii) Three events A , B & C are independent if & only if all the following conditions hold ;

$P(A \cap B) = P(A) . P(B)$; $P(B \cap C) = P(B) . P(C)$

$P(C \cap A) = P(C) . P(A)$ & $P(A \cap B \cap C) = P(A) . P(B) . P(C)$

i.e. they must be pairwise as well as mutually independent .

Similarly for n events $A_1, A_2, A_3, \ldots\ldots A_n$ to be independent , the number of these conditions is equal to ${}^nc_2 + {}^nc_3 + \ldots\ldots + {}^nc_n = 2^n - n - 1$.

(iii) The probability of getting exactly r success in n independent trials is given by

$P(r) = {}^nC_r\, p^r\, q^{n-r}$ where : p = probability of success in a single trial .

q = probability of failure in a single trial. note : $p + q = 1$.

Note : Independent events are not in general mutually exclusive & vice versa.

Mutually exclusiveness can be used when the events are taken from the same experiment & independence can be used when the events are taken from different experiments .

RESULT – 5 : BAYE'S THEOREM OR TOTAL PROBABILITY THEOREM :

If an event A can occur only with one of the n mutually exclusive and exhaustive events $B_1, B_2, \ldots. B_n$ & the probabilities $P(A/B_1)$, $P(A/B_2)$ ..... $P(A/B_n)$ are known then,

$$P(B_1/A) = \frac{P(B_i).P(A/B_i)}{\sum_{i=1}^{n} P(B_i).P(A/B_i)}$$

PROOF :

The events A occurs with one of the n mutually exclusive & exhaustive events $B_1, B_2, B_3, \ldots\ldots B_n$

$A = AB_1 + AB_2 + AB_3 + \ldots\ldots + AB_n$

$$P(A) = P(AB_1) + P(AB_2) + \ldots\ldots + P(AB_n) = \sum_{i=1}^{n} P(AB_i)$$

NOTE : $A \equiv$ event what we have ; $B_1 \equiv$ event what we want ;

$B_2, B_3, \ldots B_n$ are alternative event .

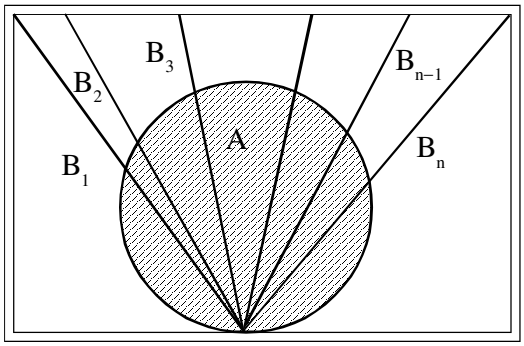


Fig. 3

Now,

$P(AB_i) = P(A) \cdot P(B_i/A) = P(B_i) \cdot P(A/B_i)$

$$P(B_i / A) = \frac{P(B_i) \cdot P(A/B_i)}{P(A)} = \frac{P(B_i) \cdot P(A/B_i)}{\sum_{i=1}^{n} P(AB_i)}$$

$$\boxed{P(B_i / A) = \frac{P(B_i) \cdot P(A/B_i)}{\sum P(B_i) \cdot P(A/B_i)}}$$

RESULT – 6

If p_1 and p_2 are the probabilities of speaking the truth of two indenpendent witnesses A and B then

P (their combined statement is true) = $\dfrac{p_1 p_2}{p_1 p_2 + (1-p_1)(1-p_2)}$.

In this case it has been assumed that we have no knowledge of the event except the statement made by A and B.

However if p is the probability of the happening of the event before their statement then

P (their combined statement is true) = $\dfrac{p\, p_1 p_2}{p\, p_1 p_2 + (1-p)(1-p_1)(1-p_2)}$.

Here it has been assumed that the statement given by all the independent witnesses can be given in two ways only, so that if all the witnesses tell falsehoods they agree in telling the same falsehood.

If this is not the case and c is the chance of their coincidence testimony then the

Pr. that the statement is true = $P\, p_1 p_2$

Pr. that the statement is false = $(1-p).c\,(1-p_1)(1-p_2)$

However chance of coincidence testimony is taken only if the joint statement is not contradicted by any witness.

RESULT – 7

(i) A **PROBABILITY DISTRIBUTION** spells out how a total probability of 1 is distributed over several values of a random variable .

(ii) Mean of any probability distribution of a random variable is given by :

$$\mu = \frac{\sum p_i x_i}{\sum p_i} = \sum p_i x_i \quad (\text{Since } \Sigma p_i = 1)$$

(iii) Variance of a random variable is given by, $\sigma^2 = \sum (x_i - \mu)^2 \cdot p_i$

$$\sigma^2 = \sum p_i x_i^2 - \mu^2 \qquad (\text{Note that SD} = +\sqrt{\sigma^2})$$

(iv) The probability distribution for a binomial variate 'X' is given by ;

$P(X = r) = {}^nC_r\, p^r q^{n-r}$ where all symbols have the same meaning as given in result 4.

The recurrence formula $\dfrac{P(r+1)}{P(r)} = \dfrac{n-r}{r+1} \cdot \dfrac{p}{q}$, is very helpful for quickly computing

P(1) , P(2). P(3) etc. if P(0) is known .

(v) Mean of BPD = np ; variance of BPD = npq .

(vi) If p represents a persons chance of success in any venture and 'M' the sum of money which he will receive in case of success, then his expectations or probable value = pM

expectations = pM

RESULT – 8 : GEOMETRICAL APPLICATIONS :

The following statements are axiomatic :

(i) If a point is taken at random on a given staright line AB, the chance that it falls on a particular

segment PQ of the line is PQ/AB .

(ii) If a point is taken at random on the area S which includes an area σ, the chance that the point falls on σ is σ/S .

EXERCISE-1

Q.1 Let a die be weighted so that the probability of a number appearing when the die is tossed is proportional to that number. Find the probability that,
(i) An even or a prime number appears (ii) An odd prime number appears
(iii) An even composite number appears (iv) An odd composite number appears.

Q.2 Numbers are selected at random , one at a time, from the two digit numbers 00, 01, 02, ... , 99 with replacement. An event E occurs if & only if the product of the two digits of a selected number is 18. If four numbers are selected, find the probability that the event E occurs at least 3 times.

Q.3 In a box , there are 8 alphabets cards with the letters : S, S, A, A,A, H, H, H . Find the probability that the word 'ASH' will form if :
(i) the three cards are drawn one by one & placed on the table in the same order that they are drawn.
(ii) the three cards are drawn simultaneously .

Q.4 There are 2 groups of subjects one of which consists of 5 science subjects & 3 engg. subjects & other consists of 3 science & 5 engg. subjects . An unbiased die is cast . If the number 3 or 5 turns up a subject is selected at random from first group, otherwise the subject is selected from 2nd group . Find the probability that an engg. subject is selected.

Q.5 A pair of fair dice is tossed. Find the probability that the maximum of the two numbers is greater than 4.

Q.6 In a building programme the event that all the materials will be delivered at the correct time is M, and the event that the building programme will be completed on time is F . Given that $P(M) = 0.8$ and $P(M \cap F) = 0.65$, find $P(F/M)$. If $P(F) = 0.7$, find the probability that the building programme will be completed on time if all the materials are not delivered at the correct time .

Q.7 In a given race, the odds in favour of four horses A, B, C & D are 1 : 3, 1 : 4, 1 : 5 and 1 : 6 respectively. Assuming that a dead heat is impossible, find the chance that one of them wins the race.

Q.8 A covered basket of flowers has some lilies and roses. In search of rose, Sweety and Shweta alternately pick up a flower from the basket but puts it back if it is not a rose. Sweety is 3 times more likely to be the first one to pick a rose. If sweety begin this 'rose hunt' and if there are 60 lilies in the basket, find the number of roses in the basket.

Q.9 Least number of times must a fair die be tossed in order to have a probability of at least 91/216, of getting atleast one six.

Q.10 Suppose the probability for A to win a game against B is 0.4. If A has an option of playing either a **"BEST OF THREE GAMES"** or a **"BEST OF 5 GAMES"** match against B , which option should A choose so that the probability of his winning the match is higher? (No game ends in a draw).

Q.11 A room has three electric lamps . From a collection of 10 electric bulbs of which 6 are good 3 are selected at random & put in the lamps. Find the probability that the room is lighted.

Q.12 A bomber wants to destroy a bridge . Two bombs are sufficient to destroy it.
If four bombs are dropped, what is the probability that it is destroyed, if the chance of a bomb hitting the target is 0.4.

Q.13 The chance of one event happening is the square of the chance of a 2nd event, but odds against the first are the cubes of the odds against the 2nd . Find the chances of each. (assume that both events are neither sure nor impossible).

Q.14 A box contains 5 radio tubes of which 2 are defective . The tubes are tested one after the other until the 2 defective tubes are discovered . Find the probability that the process stopped on the
(i) Second test; **(ii)** Third test. If the process stopped on the third test , find the probability that the first tube is non defective.

Q.15 Anand plays with Karpov 3 games of chess. The probability that he wins a game is 0.5, looses with probability 0.3 and ties with probability 0.2. If he plays 3 games then find the probability that he wins atleast two games.

Q.16 An aircraft gun can take a maximum of four shots at an enemy's plane moving away from it. The probability of hitting the plane at first, second, third & fourth shots are 0.4, 0.3, 0.2 & 0.1 respectively. What is the probability that the gun hits the plane .

Q.17 In a batch of 10 articles, 4 articles are defective. 6 articles are taken from the batch for inspection. If more than 2 articles in this batch are defective , the whole batch is rejected Find the probability that the batch will be rejected.

Q.18 Given $P(A \cup B) = 5/6$; $P(AB) = 1/3$; $P(\overline{B}) = 1/2$. Determine P(A) & P(B). Hence show that the events A & B are independent.

Q.19 One hundred management students who read at least one of the three business magazines are surveyed to study the readership pattern. It is found that 80 read Business India, 50 read Business world and 30 read Business Today. Five students read all the three magazines. A student was selected randomly. Find the probability that he reads exactly two magazines.

Q.20 An author writes a good book with a probability of 1/2. If it is good it is published with a probability of 2/3. If it is not, it is published with a probability of 1/4. Find the probability that he will get atleast one book published if he writes two.

Q.21 3 students {A, B, C} tackle a puzzle together and offers a solution upon which majority of the 3 agrees. Probability of A solving the puzzle correctly is p. Probability of B solving the puzzle correctly is also p. C is a dumb student who randomly supports the solution of either A or B. There is one more student D, whose probability of solving the puzzle correctly is once again, p. Out of the 3 member team {A, B, C} and one member team {D}, Which one is more likely to solve the puzzle correctly.

Q.22 A uniform unbised die is constructed in the shape of a regular tetrahedron with faces numbered 2, 2, 3 and 4 and the score is taken from the face on which the die lands. If two such dice are thrown together, find the probability of scoring.

(i) exactly 6 on each of 3 successive throws.

(ii) more than 4 on at least one of the three successive throws.

Q.23 Two cards are drawn from a well shuffled pack of 52 cards. Find the probability that one of them is a red card & the other is a queen.

Q.24 A cube with all six faces coloured is cut into 64 cubical blocks of the same size which are thoroughly mixed. Find the probability that the 2 randomly chosen blocks have 2 coloured faces each.

Q.25 Consider the following events for a family with children

A = {of both sexes} ; B = {at most one boy}

In which of the following (are/is) the events A and B are independent.

(a) if a family has 3 children (b) if a family has 2 children

Assume that the birth of a boy or a girl is equally likely mutually exclusive and exhaustive.

Q.26 A player tosses an unbiased coin and is to score two points for every head turned up and one point for every tail turned up. If P_n denotes the probability that his score is exactly n points, prove that

$$P_n - P_{n-1} = \frac{1}{2}(P_{n-2} - P_{n-1}) \qquad n \geq 3$$

Also compute P_1 and P_2 and hence deduce the pr that he scores exactly 4.

Q.27 Each of the 'n' passengers sitting in a bus may get down from it at the next stop with probability p . Moreover , at the next stop either no passenger or exactly one passenger boards the bus . The probability of no passenger boarding the bus at the next stop being p_o. Find the probability that when the bus continues on its way after the stop , there will again be 'n' passengers in the bus.

Q.28 The difference between the mean & variance of a Binomial Variate 'X' is unity & the difference of their square is 11. Find the probability distribution of 'X'.

Q.29 An examination consists of 8 questions in each of which the candidate must say which one of the 5 alternatives is correct one . Assuming that the student has not prepared earlier chooses for each of the question any one of 5 answers with equal probability.

(i) prove that the probability that he gets more than one correct answer is $(5^8 - 3 \times 4^8)/5^8$.

(ii) find the probability that he gets correct answers to six or more questions.

(iii) find the standard deviation of this distribution.

Q.30 Two bad eggs are accidently mixed with ten good ones. Three eggs are drawn at random without replacement, from this lot. Compute mean & S.D. for the number of bad eggs drawn.

EXERCISE-2

Q.1 The probabilities that three men hit a target are, respectively, 0.3, 0.5 and 0.4. Each fires once at the target. (As usual, assume that the three events that each hits the target are independent)

(a) Find the probability that they all : **(i)** hit the target ; **(ii)** miss the target

(b) Find the probability that the target is hit : **(i)** at least once, **(ii)** exactly once.

(c) If only one hits the target, what is the probability that it was the first man?

Q.2 Let A & B be two events defined on a sample space . Given P(A) = 0.4 ; P(B) = 0.80 and $P\left(\overline{A} / \overline{B}\right) = 0.10$. Then find ; (i) $P\left(\overline{A} \cup B\right)$ & $P\left[\left(\overline{A} \cap B\right) \cup \left(A \cap \overline{B}\right)\right]$.

Q.3 Three shots are fired independently at a target in succession. The probabilities that the target is hit in the first shot is 1/2 , in the second 2/3 and in the third shot is 3/4. In case of exactly one hit , the probability of destroying the target is 1/3 and in the case of exactly two hits, 7/11 and in the case of three hits is

1.0. Find the probability of destroying the target in three shots.

Q.4 In a game of chance each player throws two unbiased dice and scores the difference between the larger and smaller number which arise . Two players compete and one or the other wins if and only if he scores atleast 4 more than his opponent . Find the probability that neither player wins .

Q.5 A certain drug , manufactured by a Company is tested chemically for its toxic nature. Let the event **"THE DRUG IS TOXIC"** be denoted by H & the event **"THE CHEMICAL TEST REVEALS THAT THE DRUG IS TOXIC"** be denoted by S. Let $P(H) = a$, $P(S/H) = P\left(\overline{S}/\overline{H}\right) = 1 - a$. Then show that the probability that the drug is not toxic given that the chemical test reveals that it is toxic, is free from 'a'.

Q.6 A plane is landing. If the weather is favourable, the pilot landing the plane can see the runway. In this case the probability of a safe landing is p_1. If there is a low cloud ceiling, the pilot has to make a blind landing by instruments. The reliability (the probability of failure free functioning) of the instruments needed for a blind landing is P. If the blind landing instruments function normally, the plane makes a safe landing with the same probability p_1 as in the case of a visual landing. If the blind landing instruments fail, then the pilot may make a safe landing with probability $p_2 < p_1$. Compute the probability of a safe landing if it is known that in K percent of the cases there is a low cloud ceiling. Also find the probability that the pilot used the blind landing instrument, if the plane landed safely.

Q.7 A train consists of n carriages , each of which may have a defect with probability p. All the carriages are inspected , independently of one another , by two inspectors ; the first detects defects (if any) with probability p_1 , & the second with probability p_2 . If none of the carriages is found to have a defect, the train departs . Find the probability of the event ; **" THE TRAIN DEPARTS WITH ATLEAST ONE DEFECTIVE CARRIAGE ".**

Q.8 A is a set containing n distinct elements. A non-zero subset P of A is chosen at random. The set A is reconstructed by replacing the elements of P. A non-zero subset Q of A is again chosen at random. Find the probability that P & Q have no common elements.

Q.9 In a multiple choice question there are five alternative answers of which one or more than one is correct. A candidate will get marks on the question only if he ticks the correct answers. The candidate ticks the answers at random. If the probability of the candidate getting marks on the question is to be greater than or equal to 1/3 find the least number of chances he should be allowed.

Q.10 n people are asked a question successively in a random order & exactly 2 of the n people know the answer :

(a) If $n > 5$, find the probability that the first four of those asked do not know the answer.

(b) Show that the probability that the rth person asked is the first person to know the answer is : $\left[\frac{2(n-r)}{n(n-1)}\right]$, if $1 < r < n$.

Q.11 A box contains three coins two of them are fair and one two – headed. A coin is selected at random and tossed. If the head appears the coin is tossed again, if a tail appears, then another coin is selected from the remaining coins and tossed.

(i) Find the probability that head appears twice.

(ii) If the same coin is tossed twice, find the probability that it is two headed coin.

(iii) Find the probability that tail appears twice.

Q.12 The ratio of the number of trucks along a highway, on which a petrol pump is located, to the number of cars running along the same highway is 3 : 2. It is known that an average of one truck in thirty trucks and two cars in fifty cars stop at the petrol pump to be filled up with the fuel. If a vehicle stops at the petrol pump to be filled up with the fuel, find the probability that it is a car.

Q.13 A batch of fifty radio sets was purchased from three different companies A, B and C. Eighteen of them were manufactured by A, twenty of them by B and the rest were manufactured by C.
The companies A and C produce excellent quality radio sets with probability equal to 0.9 ; B produces the same with the probability equal to 0.6.
What is the probability of the event that the excellent quality radio set chosen at random is manufactured by the company B?

Q.14 The contents of three urns 1, 2 & 3 are as follows :

1 W,	2 R,	3B balls
2 W,	3 R,	1B balls
3 W,	1 R,	2B balls

An urn is chosen at random & from it two balls are drawn at random & are found to be **"1 RED & 1 WHITE "**. Find the probability that they came from the 2nd urn.

Q.15 Suppose that there are 5 red points and 4 blue points on a circle. Let $\frac{m}{n}$ be the probability that a convex polygon whose vertices are among the 9 points has at least one blue vertex where *m* and *n* are relatively

prime. Find $(m + n)$.

Q.16 There are 6 red balls & 8 green balls in a bag . 5 balls are drawn out at random & placed in a red box ; the remaining 9 balls are put in a green box . What is the probability that the number of red balls in the green box plus the number of green balls in the red box is not a prime number?

Q.17 Two cards are randomly drawn from a well shuffled pack of 52 playing cards, without replacement. Let x be the first number and y be the second number.
Suppose that Ace is denoted by the number 1; Jack is denoted by the number 11 ; Queen is denoted by the number 12 ; King is denoted by the number 13.
Find the probability that x and y satisfy $\log_3(x + y) - \log_3 x - \log_3 y + 1 = 0$.

Q.18(a) Two numbers x & y are chosen at random from the set {1,2,3,4,....3n}. Find the probability that $x^2 - y^2$ is divisible by 3 .

(b) If two whole numbers x and y are randomly selected, then find the probability that $x^3 + y^3$ is divisible by 8.

Q.19 A hunter's chance of shooting an animal at a distance r is $\frac{a^2}{r^2}$ $(r > a)$. He fires when $r = 2a$ & if he misses he reloads & fires when r = 3a, 4a, ... If he misses at a distance 'na', the animal escapes. Find the odds against the hunter.

Q.20 An unbiased normal coin is tossed 'n' times. Let :
E_1 : event that both **Heads and Tails** are present in 'n' tosses.
E_2 : event that the coin shows up **Heads** atmost once.
Find the value of 'n' for which E_1 & E_2 are independent.

Q.21 A coin is tossed $(m + n)$ times $(m>n)$. Show that the probability of at least m consecutive heads is $\frac{n+2}{2^{m+1}}$

Q.22 There are two lots of identical articles with different amount of standard and defective articles. There are N articles in the first lot, n of which are defective and M articles in the second lot, m of which are defective. K articles are selected from the first lot and L articles from the second and a new lot results. Find the probability that an article selected at random from the new lot is defective.

Q.23 *m* red socks and *n* blue socks $(m > n)$ in a cupboard are well mixed up, where $m + n \le 101$. If two socks are taken out at random, the chance that they have the same colour is 1/2. Find the largest value of *m*.

Q.24 With respect to a particular question on a multiple choice test (having 4 alternatives with only 1 correct) a student knows the answer and therefore can eliminate 3 of the 4 choices from consideration with probability 2/3, can eliminate 2 of the 4 choices from consideration with probability 1/6, can eliminate 1 choice from consideration with probability 1/9, and can eliminate none with probability 1/18. If the student knows the answer, he answers correctly, otherwise he guesses from among the choices not eliminated.
If the answer given by the student was found correct, then the probability that he knew the answer is $\frac{a}{b}$ where a and b are relatively prime. Find the value of $(a + b)$.

Q.25 In a knockout tournament 2^n equally skilled players; $S_1, S_2, \ldots\ldots\ldots S_{2^n}$ are participating. In each round players are divided in pairs at random & winner from each pair moves in the next round. If S_2 reaches the semifinal then find the probability that S_1 wins the tournament.

EXERCISE-3

Q.1 If p & q are chosen randomly from the set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} with replacement. Determine the probability that the roots of the equation $x^2 + px + q = 0$ are real. [JEE '97, 5]

Q.2 There is 30% chance that it rains on any particular day . What is the probability that there is at least one rainy day within a period of 7 – days ? Given that there is at least one rainy day, what is the probability that there are at least two rainy days ? [REE '97, 6]

Q.3 Select the correct alternative(s) . [JEE '98, $6 \times 2 = 12$]

(i) 7 white balls & 3 black balls are randomly placed in a row. The probability that no two black balls are placed adjacently equals :
(A) 1/2 (B) 7/15 (C) 2/15 (D) 1/3

(ii) If from each of the 3 boxes containing 3 white & 1 black, 2 white & 2 black, 1 white & 3 black balls, one ball is drawn at random, then the probability that 2 white & 1 black ball will be drawn is :
(A) 13/32 (B) 1/4 (C) 1/32 (D) 3/16

(iii) If $\overline{E}$ & $\overline{F}$ are the complementary events of events E & F respectively & if $0 < P(F) < 1$, then :

(A) $P(E|F) + P(\bar{E}|F) = 1$ (B) $P(E|F) + P(E|\bar{F}) = 1$

(C) $P(\bar{E}|F) + P(E|\bar{F}) = 1$ (D) $P(E|\bar{F}) + P(\bar{E}|\bar{F}) = 1$

(iv) There are 4 machines & it is known that exactly 2 of them are faulty . They are tested, one by one, in a random order till both the faulty machines are identified . Then the probability that only 2 tests are needed is :

(A) 1/3 (B) 1/6 (C) 1/2 (D) 1/4

(v) If E & F are events with $P(E) \le P(F)$ & $P(E \cap F) > 0$, then :

(A) occurrence of $E \Rightarrow$ occurrence of F

(B) occurrence of $F \Rightarrow$ occurrence of E

(C) non – occurrence of $E \Rightarrow$ non – occurrence of F

(D) none of the above implications holds.

(vi) A fair coin is tossed repeatedly. If tail appears on first four tosses, then the probability of head appearing on fifth toss equals :

(A) 1/2 (B) 1/32 (C) 31/32 (D) 1/5

Q.4 3 players A, B & C toss a coin cyclically in that order (that is A, B, C, A, B, C, A, B, ...) till a head shows . Let p be the probability that the coin shows a head. Let α, β & γ be respectively the probabilities that A, B and C gets the first head . Prove that $\beta = (1-p)\alpha$. Determine α, β & γ (in terms of p). [JEE '98, 8]

Q.5 Each co–efficient in the equation $ax^2 + bx + c = 0$ is determined by throwing an ordinary die . Find the probability that the equation will have equal roots. [REE '98, 6]

Q.6(a) If the integers m and n are chosen at random between 1 and 100, then the probability that a number of the form $7^m + 7^n$ is divisible by 5 equals

(A) $\frac{1}{4}$ (B) $\frac{1}{7}$ (C) $\frac{1}{8}$ (D) $\frac{1}{49}$

(b) The probability that a student passes in Mathematics, Physics and Chemistry are m, p and c respectively. Of these subjects, the student has a 75% chance of passing in at least one, a 50% chance of passing in at least two, and a 40% chance of passing in exactly two, which of the following relations are true?

(A) $p + m + c = \frac{19}{20}$ (B) $p + m + c = \frac{27}{20}$ (C) $pmc = \frac{1}{10}$ (D) $pmc = \frac{1}{4}$

(c) Eight players $P_1, P_2, P_3, \ldots\ldots\ldots P_8$ play a knock-out tournament. It is known that whenever the players P_i and P_j play, the player P_i will win if $i < j$. Assuming that the players are paired at random in each round, what is the probability that the player P_4 reaches the final. [JEE ' 99, 2 + 3 + 10 (out of 200)]

Q.7 Four cards are drawn from a pack of 52 playing cards. Find the probability (correct upto two places of decimals) of drawing exactly one pair. [REE'99, 6]

Q.8 A coin has probability 'p' of showing head when tossed. It is tossed 'n' times. Let p_n denote the probability that no two (or more) consecutive heads occur. Prove that,

$p_1 = 1$, $p_2 = 1 - p^2$ & $p_n = (1-p)\, p_{n-1} + p\,(1-p)\, p_{n-2}$, for all $n \ge 3$.

Q.9 A and B are two independent events. The probability that both occur simultaneously is 1/6 and the probability that neither occurs is 1/3. Find the probabilities of occurance of the events A and B separately.

Q.10 Two cards are drawn at random from a pack of playing cards. Find the probability that one card is a heart and the other is an ace. [REE ' 2001 (Mains), 3]

Q.11(a) An urn contains 'm' white and 'n' black balls. A ball is drawn at random and is put back into the urn along with K additional balls of the same colour as that of the ball drawn. A ball is again drawn at random. What is the probability that the ball drawn now is white.

(b) An unbiased die, with faces numbered 1, 2, 3, 4, 5, 6 is thrown n times and the list of n numbers showing up is noted. What is the probability that among the numbers 1, 2, 3, 4, 5, 6, only three numbers appear in the list. [JEE ' 2001 (Mains), 5 + 5]

Q.12 A box contains N coins, m of which are fair and the rest are biased. The probability of getting a head when a fair coin is tossed is 1/2, while it is 2/3 when a biased coin is tossed. A coin is drawn from the box at random and is tossed twice. The first time it shows head and the second time it shows tail. What is the probability that the coin drawn is fair? [JEE ' 2002 (mains)]

Q.13(a) A person takes three tests in succession. The probability of his passing the first test is p, that of his passing each successive test is p or p/2 according as he passes or fails in the preceding one. He gets selected provided he passes at least two tests. Determine the probability that the person is selected.

(b) In a combat, A targets B, and both B and C target A. The probabilities of A, B, C hitting their targets are 2/3 , 1/2 and 1/3 respectively. They shoot simultaneously and A is hit. Find the probability that B hits his target whereas C does not. [JEE' 2003, Mains-2 + 2 out of 60]

Q.14(a) Three distinct numbers are selected from first 100 natural numbers. The probability that all the three numbers are divisible by 2 and 3 is

(A) $\frac{4}{25}$ (B) $\frac{4}{35}$ (C) $\frac{4}{55}$ (D) $\frac{4}{1155}$

(b) If A and B are independent events, prove that $P(A \cup B) \cdot P(A' \cap B') \le P(C)$, where C is an event defined that exactly one of A or B occurs.

(c) A bag contains 12 red balls and 6 white balls. Six balls are drawn one by one without replacement of which atleast 4 balls are white. Find the probability that in the next two draws exactly one white ball is drawn (leave the answer in terms of ${}^{n}C_{r}$). [JEE 2004, 3 + 2 + 4]

Q.15(a) A six faced fair dice is thrown until 1 comes, then the probability that 1 comes in even number of trials is

(A) 5/11 (B) 5/6 (C) 6/11 (D) 1/6

(b) A person goes to office either by car, scooter, bus or train probability of which being $\frac{1}{7}, \frac{3}{7}, \frac{2}{7}$ and $\frac{1}{7}$ respectively. Probability that he reaches office late, if he takes car, scooter, bus or train is $\frac{2}{9}, \frac{1}{9}, \frac{4}{9}$ and $\frac{1}{9}$ respectively. Given that he reached office in time, then what is the probability that he travelled by a car. [JEE 2005 (Mains), 2]

Comprehension (3 questions)

There are n urns each containing $n + 1$ balls such that the ith urn contains i white balls and $(n + 1 - i)$ red balls. Let u_i be the event of selecting i^{th} urn, $i = 1, 2, 3, \ldots\ldots, n$ and w denotes the event of getting a white ball.

Q.16(a) If $P(u_i) \propto i$ where $i = 1, 2, 3, \ldots\ldots, n$ then $\lim_{n\to\infty} P(w)$ is equal to

(A) 1 (B) 2/3 (C) 3/4 (D) 1/4

(b) If $P(u_i) = c$, where c is a constant then $P(u_n/w)$ is equal to

(A) $\frac{2}{n+1}$ (B) $\frac{1}{n+1}$ (C) $\frac{n}{n+1}$ (D) $\frac{1}{2}$

(c) If n is even and E denotes the event of choosing even numbered urn ($P(u_i) = \frac{1}{n}$), then the value of $P(w/E)$, is

(A) $\frac{n+2}{2n+1}$ (B) $\frac{n+2}{2(n+1)}$ (C) $\frac{n}{n+1}$ (B) $\frac{1}{n+1}$

[JEE 2006, 5 marks each]

ANSWER KEY

EXERCISE-1

Q 1. (i) $\frac{20}{21}$ (ii) $\frac{8}{21}$ (iii) $\frac{10}{21}$ (iv) 0 **Q 2.** $97/(25)^4$ **Q 3. (i)** 3/56 **(ii)** 9/28

Q 4. 13/24 **Q 5.** 5/9 **Q 6.** $P(F/M) = \frac{13}{16}$; $P(F/\overline{M}) = \frac{1}{4}$

Q 7. 319/420 **Q 8.** 120 **Q 9.** 3

Q 10. best of 3 games **Q 11.** $\frac{29}{30}$ **Q 12.** $\frac{328}{625}$ **Q 13.** $\frac{1}{9}, \frac{1}{3}$

Q 14. (i) 1/10, **(ii)** 3/10, **(iii)** 2/3 **Q 15.** 1/2 **Q 16.** 0.6976

Q 17. 19/42 **Q 18.** P(A) = 2/3, P(B) = 1/2 **Q 19.** 1/2

Q 20. 407/576 **Q 21.** Both are equally likely **Q 22.** (i) $\frac{125}{16^3}$; (ii) $\frac{63}{64}$

Q 23. 101/1326 **Q 24.** $\frac{{}^{24}C_2}{{}^{64}C_2}$ or $\frac{23}{168}$

Q25. Independent in (a) and not independent in (b) **Q 26.** $P_1 = 1/2$, $P_2 = 3/4$

Q 27. $(1 - p)^{n-1} . [p_o (1 - p) + np(1 - p_0)]$ **Q 28.** $\left(\frac{5}{6} + \frac{1}{6}\right)^{36}$

Q 29. $\frac{481}{5^8}, \frac{4\sqrt{2}}{5}$ **Q 30.** mean = 0.5

EXERCISE-2

Q 1. (a) 6%, 21% ; (b) 79%, 44%, (c) 9/44 ≈ 20.45% **Q 2. (i)** 0.82, **(ii)** 0.76 **Q 3.** $\frac{5}{8}$

Q 4. 74/81 **Q 5.** $P\left(\overline{H}/S\right) = 1/2$

Q6. $P(E) = (1-\frac{K}{100})p_1 + \frac{K}{100}[P p_1 + (1-P) p_2]$; $P(H_2/A) = \frac{\frac{K}{100}[P p_1 + (1-P) p_2]}{\left(1-\frac{K}{100}\right)p_1 + \frac{K}{100}[P p_1 + (1-P) p_2]}$

Q 7. $1 - [1 - p(1-p_1)(1-p_2)]^n$ **Q 8.** $(3^n\text{-}2^{n+1}+1)/(4^n\text{-}2^{n+1}+1)$ **Q 9.** 11

Q 10. (a) $\frac{(n-4)(n-5)}{n((n-1)}$ **Q 11.** 1/2, 1/2, 1/12 **Q 12.** $\frac{4}{9}$

Q.13 $\frac{4}{13}$ **Q 14.** 6/11 **Q 15.** 458

Q 16. 213/1001 **Q.17** $\frac{11}{663}$ **Q 18. (a)** $\frac{(5n-3)}{(9n-3)}$ **(b)** $\frac{5}{16}$

Q 19. n+1 : n–1 **Q 20.** n = 3 **Q 22.** $\frac{K\,n\,M + L\,m\,N}{M\,N\,(K+L)}$

Q.23 55 **Q.24** 317 **Q.25** $\frac{3}{4\left(2^n-1\right)}$

EXERCISE-3

Q.1 31/50 **Q.2** $[1-(7/10)^7 - {}^7C_1\,(3/10)\,(7/10)^6] / 1-(7/10)^7$

Q.3 (i) B **(ii)** A **(iii)** A, D **(iv)** A **(v)** D **(vi)** A

Q.4 $\alpha = \frac{p}{1-(1-p)^3}$, $\beta = \frac{(1-p)p}{1-(1-p)^3}$, $\gamma = \frac{(1-p)^2 p}{1-(1-p)^3}$ **Q.5** 5/216

Q.6 (a) A **(b)** B, C **(c)** 4/35 **Q.7** 0.31 **Q.9** $\frac{1}{2}$ & $\frac{1}{3}$ or $\frac{1}{3}$ & $\frac{1}{2}$

Q.10 $\frac{1}{26}$ **Q.11** **(a)** $\frac{m}{m+n}$; **(b)** $\frac{{}^6C_3\left(3^n - 3.2^n + 3\right)}{6^n}$ **Q.12** $\frac{9m}{m+8N}$

Q.13 (a) $p^2(2-p)$; (b) 1/2 **Q.14** **(a)** D , **(c)** $\frac{{}^{12}C_2\,{}^6C_4\,{}^{10}C_1\,{}^2C_1 + {}^{12}C_1\,{}^6C_5\,{}^{11}C_1\,{}^1C_1}{{}^{12}C_2\left({}^{12}C_2\,{}^6C_4 + {}^{12}C_1\,{}^6C_5 + {}^{12}C_0\,{}^6C_6\right)}$

Q.15 **(a)** A, **(b)** $\frac{1}{7}$ **Q.16** (a) B, (b) A, (c) B

EXERCISE-4

Part : (A) Only one correct option

1. If A, B, C are 3 events, then the probability that exactly 2 of them occur is given by:
(A) $P(A \cap B) + P(B \cap C) + P(C \cap A) - 2P(A \cap B \cap C)$
(B) $P(A \cap B) + P(B \cap C) + P(C \cap A) - 3P(A \cap B \cap C)$
(C) $P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$
(D) none of these

2. In a series of 3 independent trials the probability of exactly 2 success is 12 times as large as the probability of 3 successes. The probability of a success in each trial is:
(A) 1/5 (B) 2/5 (C) 3/5 (D) 4/5

3. There are two urns. There are m white & n black balls in the first urn and p white & q black balls in the second urn. One ball is taken from the first urn & placed into the second. Now, the probability of drawing a white ball from the second urn is:

(A) $\frac{pm+(p+1)n}{(m+n)(p+q+1)}$ (B) $\frac{(p+1)m+pn}{(m+n)(p+q+1)}$ (C) $\frac{qm+(q+1)n}{(m+n)(p+q+1)}$ (D) $\frac{(q+1)m+qn}{(m+n)(p+q+1)}$

4. Box–I contains 3 red and 2 blue balls whilst box-II contains 2 red and 6 blue balls. A fair coin is tossed. If it turns up head, a ball is drawn from Box-I, else a ball is drawn from Box–II. Find the probabiliy of event 'ball drawn is from Box-I, if it is red'.

(A) $\frac{12}{17}$ (B) $\frac{10}{17}$ (C) $\frac{17}{20}$ (D) $\frac{3}{5}$

5. A local post office is to send M telegrams which are distributed at random over N communication channels, (N > M). Each telegram is sent over any channel with equal probability. Chance that not more than one telegram will be sent over each channel is:

(A) $\frac{{}^{N}C_{M} \,.\, M\,!}{N^{M}}$ (B) $\frac{{}^{N}C_{M} \,.\, N\,!}{M^{N}}$ (C) $1 - \frac{{}^{N}C_{M} \,.\, M\,!}{M^{N}}$ (D) $1 - \frac{{}^{N}C_{M} \,.\, N\,!}{N^{M}}$

6. A mapping is selected at random from all the mappings defined on the set A consisting of three distinct elements. The probability that the mapping selected is one to one is:
(A) 1/9 (B) 1/3 (C) 1/4 (D) 2/9

7. A bag contains 7 tickets marked with the numbers 0, 1, 2, 3, 4, 5, 6 respectively. A ticket is drawn & replaced. Then the chance that after 4 drawings the sum of the numbers drawn is 8 is:
(A) 165/2401 (B) 149/2401 (C) 3/49 (D) none

8. A biased coin with probability p, $0 < p < 1$ of heads is tossed until a head appears for the first time. If the probability that the number of tosses required is even is 2/5, then p equals
(A) 1/3 (B) 2/3 (C) 2/5 (D) 3/5

9. If 4 whole numbers taken at random are multiplied together, then the chance that the last digit in the product is 1, 3, 7 or 9 is:
(A) 16/625 (B) 4/125 (C) 8/81 (D) none

10. A letter is known to have come either from "KRISHNAGIRI" or "DHARMAPURI". On the post mark only the two consecutive letters "RI" are visible. Then the chance that it came from Krishnagiri is:
(A) 3/5 (B) 2/3 (C) 9/14 (D) none

11. If $\frac{(1+3p)}{3}, \frac{(1-p)}{4}$ & $\frac{(1-2p)}{2}$ are the probabilities of three mutually exclusive events then the set of all values of p is.

(A) $\left[\frac{1}{2}, \frac{2}{3}\right]$ (B) $\left[\frac{1}{3}, \frac{1}{2}\right]$ (C) $\left[\frac{1}{4}, \frac{1}{2}\right]$ (D) $\left[\frac{1}{3}, \frac{2}{3}\right]$

12. Let p be the probability that a man aged x years will die in a year time. The probability that out of 'n' men $A_1, A_2, A_3, \ldots\ldots, A_n$ each aged 'x' years. A_1 will die & will be the first to die is:

(A) $\frac{1-p^{n}}{n}$ (B) $\frac{p}{n}$ (C) $\frac{p\,(1-p)^{n-1}}{n}$ (D) $\frac{1-(1-p)^{n}}{n}$

13. 5 girls and 10 boys sit at random in a row having 15 chairs numbered as 1 to 15, then the probability that end seats are occupied by the girls and between any two girls an odd number of boys sit is:

(A) $\frac{20\times 10!\times 5!}{15!}$ (B) $\frac{10\times 10!\times 5!}{15!}$ (C) $\frac{20\times 10\,!\times 30}{15\,!}$ (D) $\frac{10\times 10!\times 5!}{25!}$

14. Two dice are rolled simultaneously. The probability that the sum of the two numbers on the top faces will be atleast 10 is:
(A) 1/6 (B) 1/12 (C) 1/18 (D) none

15. There are 4 urns. The first urn contains 1 white & 1 black ball, the second urn contains 2 white & 3 black balls, the third urn contains 3 white & 5 black balls & the fourth urn contains 4 white & 7 black balls. The selection of each urn is not equally likely. The probability of selecting i^{th} urn is $\frac{i^2+1}{34}$ (i = 1, 2, 3, 4). If we randomly select one of the urns & draw a ball, then the probability of ball being

white is :

(A) $\frac{569}{1496}$ (B) $\frac{27}{56}$ (C) $\frac{8}{73}$ (D) none of these

16. 2/3rd of the students in a class are boys & the rest girls. It is known that probability of a girl getting a first class is 0.25 & that of a boy is 0.28. The probability that a student chosen at random will get a first class is:

(A) 0.26 (B) 0.265 (C) 0.27 (D) 0.275

17. The contents of urn I and II are as follows,

Urn I: 4 white and 5 black balls

Urn II: 3 white and 6 black balls

One urn is chosen at random and a ball is drawn and its colour is noted and replaced back to the urn. Again a ball is drawn from the same urn, colour is noted and replaced. The process is repeated 4 times and as a result one ball of white colour and 3 of black colour are noted. Find the probability the chosen urn was I.

(A) $\frac{125}{287}$ (B) $\frac{64}{127}$ (C) $\frac{25}{287}$ (D) $\frac{79}{192}$

18. The sides of a rectangle are chosen at random, each less than 10 cm, all such lengths being equally likely. The chance that the diagonal of the rectangle is less than 10 cm is

(A) 1/10 (B) 1/20 (C) $\pi/4$ (D) $\pi/8$

19. The sum of two positive quantities is equal to 2n. The probability that their product is not less than 3/4 times their greatest product is

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) none of these

20. The probability that 4th power of a positive integer ends in the digit 6 is:

(A) 10 % (B) 20 % (C) 25 % (D) 40 %

21. Posteriory probability of the occurrance of the event 'The coin turns head and the die turns up an odd number' is

(A) $\frac{1}{7}$ (B) $\frac{2}{7}$ (C) $\frac{3}{7}$ (D) $\frac{4}{7}$

22. Expectation of D is

(A) $\frac{225}{7}$ (B) $\frac{150}{7}$ (C) $\frac{200}{7}$ (D) $\frac{300}{7}$

23. For the three events A, B & C, P(exactly one of the events A or B occurs) = P(exactly one of the events B or C occurs) = P(exactly one of the events C or A occurs) = p & P (all the three events occur simultaneously) = p^2, where $0 < p < 1/2$. Then the probability of at least one of the three events A, B & C occurring is: **[IIT -1996]**

(A) $\frac{3p+2p^2}{2}$ (B) $\frac{p+3p^2}{4}$ (C) $\frac{p+3p^2}{2}$ (D) $\frac{3p+2p^2}{4}$

Part : (B) May have more than one options correct

24. In throwing a die let A be the event 'coming up of an odd number', B be the event 'coming up of an even number', C be the event 'coming up of a number ≥ 4' and D be the event 'coming up of a number < 3', then

(A) A and B are mutually exclusive and exhaustive (B) A and C are mutually exclusive and exhaustive

(C) A, C and D form an exhaustive system (D) B, C and D form an exhaustive system

25. Let $0 < P(A) < 1$, $0 < P(B) < 1$ & $P(A \cup B) = P(A) + P(B) - P(A).P(B)$, then:

(A) $P(B/A) = P(B) - P(A)$ (B) $P(A^C \cup B^C) = P(A^C) + P(B^C)$

(C) $P((A \cup B)^C) = P(A^C).P(B^C)$ (D) $P(A/B) = P(A)$

26. For any two events A & B defined on a sample space,

(A) $P(A/B) \geq \frac{P(A)+P(B)-1}{P(B)}$, $P(B) \neq 0$ is always true

(B) $P(A \cup \overline{B}) = P(A) - P(A \cap B)$

(C) $P(A \cup B) = 1 - P(A^c).P(B^c)$, if A & B are independent

(D) $P(A \cup B) = 1 - P(A^c).P(B^c)$, if A & B are disjoint

27. If A, B & C are three events, then the probability that none of them occurs is given by:

(A) $P(\overline{A}) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

(B) $P(\overline{A}) + P(\overline{B}) + P(\overline{C})$

(C) $P(\overline{A}) - P(B) - P(C) + P(A \cap B) + P(B \cap C) + P(C \cap A) - P(A \cap B \cap C)$

(D) $P(\overline{A} \cup \overline{B} \cup \overline{C}) - P(A) - P(B) - P(C) + P(A \cap B) + P(B \cap C) + P(C \cap A)$

28. A student appears for tests I, II & III. The student is successful if he passes either in tests I & II or tests I & III. The probabilities of the student passing in the tests I, II & III are p, q &

1/2 respectively. If the probability that the student is successful is 1/2, then:
(A) p = 1, q = 0 (B) p = 2/3, q = 1/2
(C) p = 3/5, q = 2/3 (D) there are infinitely many values of p & q.

29. If the integers m and n are chosen at random between 1 and 100, then the probability that a number of the form $7^m + 7^n$ is divisible by 5 equals **[IIT - 1999]**
(A) 1/4 (B) 1/7 (C) 1/8 (D) 1/49

EXERCISE-5

1. A letter is known to have come either from London or Clifton; on the postmark only the two consecutive letters ON are legible; what is the chance that it came from London?

2. A speaks the truth 3 out of 4 times, and B 5 out of 6 times; what is the probability that they will contradict each other in starting the same fact?

3. A pair of dice is thrown 5 times. Find the mean and variance of the probability distribution of appearance of doublets on the throws.

4. If on a straight line 10 cm. two length of 6 cm and 4 cm are measured at random, find the probability that their common part does not exceed 3 cms.

5. Let p be the probability that a man aged x years will die in a year time. Find the probability that out of 'n' men A_1, A_2, A_3,....., A_n each aged 'x' years. A_1 will die & will be the second to die.

6. A car is parked by an owner amongst 25 cars in a row, not at either end. On his return he finds that exactly 15 placed are still occupied. Find the probability that both the neighbouring places are empty.

7. A gambler has one rupee in his pocket. He tosses an unbiased normal coin unless either he is ruined or unless the coin has been tossed for a maximum of five times. If for each head he wins a rupee and for each tail he looses a rupee, then find the probability that the gambler is ruined.

8. Mr. Dupont is a professional wine taster. When given a French wine, he will identify it with probability 0.9 correctly as French, and will mistake it for a Californian wine with probability 0.1. When given a Californian wine, he will identify it with probability 0.8 correctly as Californian, and will mistake it for a French wine with probability 0.2. Suppose that Mr. Dupont is given ten unlabelled glasses of wine, three with French and seven with Californian wines. He randomly picks a glass, tries the wine and solemnly says. “French”. Find the probability that the wine he tasted was Californian.

9. In ten trials of an experiment, if the probability of getting '4 successes' is maximum, then show that probability of failure in each trial can be equal to $\frac{3}{5}$.

10. Mean and variance of a Binomial variate are in the ratio of 3 : 2. Find the most probable number of happening of the variable in 10 trials of the experiment.

11. In a Nigerian hotel, among the english speaking people 40% are English & 60% Americans. The English & American spellings are **"RIGOUR" & "RIGOR"** respectively. An English speaking person in the hotel writes this word. A letter from this word is chosen at random & found to be a vowel. Find the probability that the writer is an Englishman.

12. There is a group of k targets, each of which independently of the other targets, can be detected by a radar unit with probability p. Each of 'm' radar units detects the targets independently of other units. Find the probability that not all the targets in the group will be detected.

13. 2 positive real numbers x and y satisfy $x \le 1$ and $y \le 1$ are chosen at random. Find the probability that $x + y \le 1$, given that $x^2 + y^2 \ge \frac{1}{4}$.

14. There are two lots of identical articles with different amounts of standard & defective articles. There are N articles in the first lot, n of which are defective & M articles in the second lot, m of which are defective. K articles are selected from the first lot & L articles from the second & a new lot results. Find the probability that an article selected at random from the new lot is defective.

15. The odds that a book will be favorably reviewed by three independent crities are 5 to 2, 4 to 3, and 3 to 4 respectively : what is the probability that of the three reviews a majority will be favourable?

16. Find the chance of throwing 10 exactly in one throw with 3 dice.

17. If 12 tickets numbered 0, 1, 2, ...11 are placed in a bag, and three are drawn out, show that the chance that the sum of the numbers on them is equal to 12 is $\frac{3n}{(6n-1)(6n-2)} = \frac{3}{55}$

18. A man has 10 coins and one of them is known to have two heads. He takes one at random and tosses it 5 times and it always falls head : what is the chance that it is the coins with two heads?

19. A purse contains five coins, each of which may be a rupees coin or a 50 ps coin ; two are drawn and found to be shillings : find the probable value of the remaining coins.

20. One of a pack of 52 cards has been lost; from the remainder of the pack two cards are drawn and are found to be spades; find the chance that the missing card is a spade.

21. A, B are two inaccurate arithmeticians whose chance of solving a given question correctly are $\frac{1}{8}$ and $\frac{1}{12}$ respectively; if they obtain the same result, and if it is 1000 to 1 against their making the same mistake, find the chance that the result is correct.

22. If n integers taken at random are multiplied together, show that the chance that the last digit of the product

is 1, 3, 7, or 9 is $\frac{2^n}{5^n}$; the chance of its being 2, 4, 6 or 8 is $\frac{4^n - 2^n}{5^n}$; of its being 5 is $\frac{5^n - 4^n}{10^n}$; and of its being 0 is $\frac{10^n - 8^n - 5^n + 4^n}{10^n}$.

23. A player tosses a coin and is to score one point for every head and 2 points if every tail turned up. He is to play until he reaches 'n'. If p_n is the chance of obtaining exactly 'n' crores, find p_n for n = 1, 2, 3, 4. Also show that $p_n = \frac{1}{2}(p_{n-1} + p_{n-2})$.

24. A lot contains 50 defective & 50 non defective bulbs. Two bulbs are drawn at random, one at a time, with replacement. The events A, B, C are defined as: **[IIT - 1992]**
A = { the first bulb is defective} ; B = { the second bulb is non defective}
C = { the two bulbs are both defective or both non defective}
Determine whether (i) A,B,C are pair wise independent (ii) A,B,C are independent

25. Eight players $P_1, P_2, P_3, \ldots\ldots\ldots P_8$ play a knock-out tournament. It is known that whenever the players P_i and P_j play, the player P_i will win if i < j. Assuming that the players are paired at random in each round, what is the probability that the players P_4 reaches the final. **[IIT - 1999]**

26. A box contains N coins, m of which are fair and the rest are biased. The probability of getting a head when a fair coin is tossed is 1/2, while it is 2/3 when a biased coin is tossed. A coin is drawn from the box at random and is tossed twice. The first time it shows head and the second time it shows tail. What is the probability that the coin drawn is fair? **[IIT - 2002]**

27. A person has to go through three successive tests. Probability of his passing first exam is P. Probability of passing successive test is P or P/2 according as he passed the last test or not. He is selected if he passes atleast two tests. Find the probability of his selection. **[IIT - 2003]**

28. Prove that $P(A \cup B)\ P\left(\overline{A} \cap \overline{B}\right) \le P(C)$ where A and B are independent events and P(C) is the probability of exactly one of A or B occurs. **[IIT - 2004]**

29. A person goes to office either by car, scooter, bus or train, the probability of which being $\frac{1}{7}, \frac{3}{7}, \frac{2}{7}$ and $\frac{1}{7}$ respectively. Probability that he reaches office late, if he takes car, scooter, bus or train is $\frac{2}{9}, \frac{1}{9}, \frac{4}{9}$ and $\frac{1}{9}$ respectively. Given that he reached office in time, then what is the probability that he travelled by a car. **[IIT - 2005]**

EXERCISE-4

1. B **2.** A **3.** B **4.** A **5.** A **6.** D **7.** B
8. A **9.** A **10.** C **11.** B **12.** D **13.** A **14.** A
15. A **16.** C **17.** A **18.** C **19.** B **20.** D **21.** B
22. A **23.** A **24.** AC **25.** CD **26.** AC **27.** CD
28. ABCD **29.** AD

EXERCISE-5

1. $\frac{12}{77}$ **2.** $\frac{1}{3}$ **3.** mean = $\frac{5}{6}$, variance = $\frac{25}{36}$

4. $\frac{9}{24}$ **5.** $\frac{1}{n}[1 - (1-p)^n - np(1-p)^{n-1}]$

6. $\frac{15}{92}$ **7.** $\frac{11}{16}$ **8.** $\frac{14}{41}$ **10.** 3 **11.** $\frac{5}{11}$

12. $1 - \{1 - (1-p)^m\}^k$ **13.** $\frac{8 - \pi}{16 - \pi}$ **14.** $\frac{KnM + LmN}{MN(K + L)}$

15. 209/343 **16.** 1/5 **18.** (32/41) **19.** 2.25 Rs

20. 11/50 **21.** (13/14)

23. $p_1 = \frac{1}{2}, p_2 = \frac{3}{4}, p_3 = \frac{5}{8}, p_4 = \frac{11}{16}$.

24. (i) A, B, C are pairwise independent
(ii) A, B, C are not independent

25. 4/35 **26.** $\frac{9m}{8N + m}$ **27.** $2P^2 - P^3$ **29.** 1/7

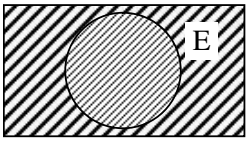
Some questions (Assertion–Reason type) are given below. Each question contains **Statement – 1 (Assertion)** and **Statement – 2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. So select the correct choice :

Choices are :

(A) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is** a correct explanation for **Statement – 1**.
(B) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is NOT** a correct explanation for **Statement – 1**.
(C) **Statement – 1** is True, **Statement – 2** is False.
(D) Statement – 1 is False, Statement – 2 is True.

PROBABILITY

429. $P(E) = \frac{n(E)}{n(S)} = \frac{m}{n}$ or $\frac{\text{Area of (E)}}{\text{Area of (S)}}$ **[Good]**



Statement-1: Always the probability of an event is a rational number and less than or equal to one
Statement-2: The equation $P(E) = |\sin\theta|$ is always consistent.

430. Let A and B be two event such that $P(A \cup B) \geq 3/4$ and $1/8 \leq P(A \cap B) \leq 3/8$
Statement–1 : $P(A) + P(B) \geq 7/8$
Statement–2 : $P(A) + P(B) \leq 11/8$

431. **Statement–1 :** The probability of drawing either a ace or a king from a pack of card in a single draw is $\frac{2}{13}$.
Statement–2 : For two events E_1 and E_2 which are not mutually exclusive, probability is given by $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$.

432. Let A and B be two independent events.
Statement–1 : If $P(A) = 0.3$ and $P(A \cup \bar{B}) = 0.8$ then P(B) is $\frac{2}{7}$.
Statement–2 : $P(\bar{E}) = 1 - P(E)$ where E is any event.

433. Let A and B be two independent events of a random experiment.
Statement–1 : $P(A \cap B) = P(A).\ P(B)$
Statement–2 : Probability of occurrence of A is independent of occurrence or non–occurrence of B.

434. A fair die is rolled once.
Statement–1 : The probability of getting a composite number is $\frac{1}{3}$
Statement–2 : There are three possibilities for the obtained number (i) the number is a prime number (ii) the number is a composite number (iii) the number is 1, and hence probability of getting a prime number = $\frac{1}{3}$

435. Let A and B are two events such that $P(A) = \frac{3}{5}$ and $P(B) = \frac{2}{3}$, then
Statement–1 : $\frac{4}{15} \leq P(A \cap B) \leq \frac{3}{5}$.
Statement–2 : $\frac{2}{5} \leq P\left(\frac{A}{B}\right) \leq \frac{9}{10}$.

436. **Statement–1 :** Three of the six vertices of a regular hexagon are chosen at random. The probability that the triangle with three vertices equilateral equals to $\frac{3}{10}$.

Statement–2 : A die is rolled three times. The probability of getting a large number than the previous number is $\frac{5}{64}$.

437. **Statement-1:** A coin is tossed thrice. The probability that exactly two heads appear, is 3/8
Statement-2: Probability of success r times out of total n trials = P(r) = ${}^{n}C_{r}$ = ${}^{n}C_{r}\, p^{r}\, q^{n-r}$ where p be the probability of success and q be the probability of failure.

438. **Statement-1 :** For any two events A and B in a sample space $P(A/B) \geq \frac{P(A)+P(B)}{P(B)}$, $P(B) \neq 0$ is always true

Statement-2 : For any two events A and B $0 < P(A \cup B) \leq 1$.

439. **Statement-1:** The letters of the English alphabet are written in random order. The probability that letters x & y are adjacent is $\frac{1}{13}$.

Statement-2: The probability that four lands deals at random from 94 ordinary deck of 52 cends will contain from an ordinary deck of 52 cends will contain from each suit is 1/4.

440. **Statement-1:** The probability of being at least one white ball selected from two balls drawn simultaneously from the bag containing 7 black & 4 white balls is 34/35.
Statement-2: Sample space = ${}^{11}C_{2}$ = 55, Number of fav. Cases = ${}^{4}C_{1} \times {}^{7}C_{1} + {}^{4}C_{2} \times {}^{7}C_{0}$

441. **Statement-1:** If A, B, C be three mutually independent events then A and B$\cup$C are also independent events.
Statement-2: Two events A and B are independent if and only if $P(A \cap B) = P(A)\, P(B)$.

442. **Statement-1:** If A and B be two events such that P(A) = 0.3 and $P(A \cup B) = 0.8$ also A and B are independent events P(B) is 0.5.
Statement-2: IF A & B are two independent events then $P(A \cap B) = P(A).P(B)$.

443. **Statement-1:** The probability of occurence of a doublet when two identical dies are thrown is 2/7.
Statement-2: When two identical dies are thrown then the total number of cases are 21 in place of 36 (when two distinct dies are thrown) because the cases like (5, 6). (6, 5) are considered to be same.

444. **Statement-1:** A = {2, 4, 6} , B = {1, 2, 3, } where A & B are the number occuring on a dice, then P(A) + P(B) = 1
Statement-2: If $A_1, A_2, A_3 \ldots A_n$ are all mutually exclusive events, then $P(A_1) + P(A_2) + \ldots + P(A_n) = 1$.

445. **Statement-1:** If $P(A/B) \geq P(A)$ then $P(B/A) \geq P(B)$
Statement-2:: $P(A/B) = \frac{P(A \cap B)}{P(B)}$

446. **Statement-1:** Balls are drawn one by one without replacement from a bag containing a white and b black balls, then probability that white balls will be first to exhaust is a/a+b.
Statement-2: Balls are drawn one by one without replacement from a bag containing a white and b black balls then probability that third drawn ball is white is a/a+b.

447. **Statement-1:** Out of 5 tickets consecutively numbers, three are drawn at random, the chance that the numbers on them are in A.P. is 2/15.
Statement-2: Out of (2n + 1) tickets consectively numbered, three are drawn at random, the chance that the numbers on them are in A.P. is $\frac{3n}{4n^2-1}$.

448. **Statement-1:** If the odds against an event is 2/3 then the probability of occurring of an event is 3/5.
Statement-2: For two events A and B $P(A' \cap B') = -1\, P(A \cup B)$

449. **Statement-1:** A, B, C are events such that P(A) = 0.3, P(B) = 0.4 P(C) = 0.8, $P(A \cap B) = 0.08$, $P(A \cap C) = 0.28$, $P(A \cap B \cap C) = 0.09$ then $P(B \cap C) \in (0.23, 0.48)$.
Statement-2: $0.75 \leq P(A \cup B \cup C) \leq 1$.

450. **Statement-1:** If P(A) = 0.25, P(B) = 0.50 and $P(A \cap B) = 0.14$ then the probability that neither A nor B occurs is 0.39.
Statement-2: $(\overline{A \cup B}) = \overline{A} \cup \overline{B}$

451. **Statement-1:** For events A and B of sample space if $P\left(\frac{A}{B}\right) \geq P(A)$ then $P\left(\frac{B}{A}\right) \geq P(B)$.
Statement-2: $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} \left(P(B) \neq 0\right)$

ANSWER

429. D	**430. D**	**431. B**	**432. A**	**433. A**	**434. C**	
435. A	**436. D**	**437. A**	**438. D**	**439. C**	**440. A**	**441. A**
442. D	**443. D**	**444. C**	**445. A**	**446. D**	**447. D**	**448. B**
449. A	**450. C**	**451. A**				

Details Solution

430. $P(A\cup B) = P(A) + P(B) - P(A\cap B)$
$\therefore\ 1 \geq P(A) + P(B) - P(A\cap B) \geq 3/4$
$\Rightarrow P(A) + P(B) - 1/8 \geq 1/8 \geq 3/4$ (since min. value of $P(A\cap B) = 1/8$)
$\Rightarrow P(A) + P(B) \leq 1/8 + 3/4 = 7/8$
As the max. value of $P(A\cap B) = 3/8$, we get
$1 \geq P(A) + P(B) - 3/8$
$\Rightarrow P(A) = P(B) \leq 1 + 3/8 = 11/8.$ Ans. D

431. (b) Clearly both are correct but statement – II is not the correct explanation for statement – I.

432. (A) $P\left(A\cup\overline{B}\right) = 1-\left(\overline{A\cup\overline{B}}\right) = 1-\left(\overline{A}\cap B\right) = 1-P\left(\overline{A}\right)P(B)$

$0.8 = 1 - 0.7\times P(B) \quad \Rightarrow\ P(B) = \frac{2}{7}.$

433. Statement –II is true as this is the definition of the independent events.

Statement – I is also true, as if events are independent, then $P\left(\frac{A}{B}\right) = P(A)$

$\Rightarrow \frac{P(A\cap B)}{P(B)} = P(A) \ \Rightarrow\ P(A\cap B) = P(A).\ P(B).$

Obviously statement – II is a correct reasoning of statement – I
Hence (a) is the correct answer.

434. Statement – I is true as there are six equally likely possibilities of which only two are favourable (4 and 6). Hence
P(obtained number is composite) = $\frac{2}{6} = \frac{1}{3}.$
Statement – II is not true, as the three possibilities are not equally likely.
Hence (c) is the correct answer.

435. $\because\ P(A\cap B) = P(A) + P(B) - P(A\cup B) \geq P(A) + P(B) - 1$

$\therefore\ P(A\cap B) \geq \frac{3}{5} + \frac{2}{3} - 1 \ \Rightarrow\ P(A\cap B) \geq \frac{4}{15} \quad \ldots$ (i)

$\because\ P(A\cap B) \leq P(A) \ \Rightarrow P(A\cap B) \leq \frac{3}{5} \quad \ldots$ (ii)

from (i) and (ii), $\frac{4}{15} \leq P(A\cap B) \leq \frac{3}{5} \quad \ldots$(iii)

from (iii), $\frac{4}{15P(B)} \leq \frac{P(A\cap B)}{P(B)} \leq \frac{3}{5P(B)} \Rightarrow \frac{2}{5} \leq P\left(\frac{A}{B}\right) \leq \frac{9}{10}$

Hence (a) is the correct answer.

436. For statement I, $n(S) = {}^{6}c_{3} = 20$

only two triangle formed are equilateral, they are $\Delta A_1A_3A_5$ and $\Delta A_2A_4A_6$. $\therefore\ n(E) = 2$

$\Rightarrow P(E) = \frac{n(E)}{n(S)} = \frac{2}{20} = \frac{1}{10}.$ For statement – II $n(S) = 216$

No. of favorable ways $= \sum_{i=1}^{6}(i-1)(6-i) = 20$

$\therefore$ Required probability $= \frac{20}{216} = \frac{5}{64}$.

Hence (d) is the correct answer.

440. $\therefore$ Reqd. probability = 35/55.
Option (A) is correct.

441. $P\{A\cap(B\cap C)\} = P(A\cap B\cap C) = P(A)\,P(B)\,P(C)$
$\therefore P[A\cup(B\cup C)] = P[(A\cap B)\cup(A\cap C)]$
$= P[(A\cap B) + (A\cap C) - P[(A\cap B)\cap(A\cap C)]$
$= p(A\cap B) + P(A\cap C) - P(A\cap B\cap C)$
$= P(A)\,P(B) + P(A)\,P(C) - P(A)\,P(B)\,P(C)$
$= P(A)\,[P(B) + P(C) - P(B)\,P(C)]$
$= P(A).P(B\cup C)$
$\therefore$ A & B $\cup$C are independent events Ans. (A)

442. $P(A\cup B) = P(A) + P(B) - P(A\cap B)$
$P(A\cup B) = P(A) + P(B) - P(A)\times P(B)$
$0.8 = 0.3 + P(B) - 0.3\times P(B)$
$P(B) = 5/7$ 'd' is correct.

445. (A) The statement-1 A is true and follows from statement-2
indeed $P(A/B) = \frac{P(A\cap B)}{P(B)} \le P(A)$

$\Rightarrow \frac{P(A\cap B)}{P(A)} \le P(B)$

$\Rightarrow P(B/A) \le P(B)$

446. Statement-1 is false. Since if the colour white is first to exhaust then last ball must be black.
$\Rightarrow$ favourable sample points
((a + b – 1)!)b
req. probability $= \frac{b(a+b-1)!}{a+b!} = \frac{b}{a+b}$

447. (D) 2n+1 = 5, n = 2
$P(E) = \frac{3n}{4n^2-1} = \frac{6}{15} = \frac{2}{5}$
For a, b, c are in A. P. a + c = 2b $\Rightarrow$ a + c is even
$\therefore$ a and c are both even or both odd.
So, number of ways of choosing a and c is ${}^{n}C_2 + {}^{n+1}C_2 = n^2$ ways.
$P(E) = \frac{n^2}{{}^{2n+1}C_3} = \frac{3n}{4n^2-1}$

448. (B) Both A and R are correct but R is not the correct explanation of A.

449. (A) $\because$ $P(A\cup B\cup C) = P(A) + P(B) + P(C) - P(A\cap B) - P(B\cap C) - P(A\cap C) + P(A\cap B\cap C)$
using all the given values we get that $P(B\cap C) \in (0.23, 0.48)$.

450. (C) Required probability is $P(\bar{A}\cap\bar{B})$
$= 1 - P(A\cup B) = 1 - [P(A) + P(B) - P(A\cap B)]$
$= 0.39$

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 11 XI M 11. Straight Lines

Index:

1. Key Concepts

2. Exercise I to V

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

The Point & Straight Line

1. Distance Formula:

The distance between the points $A(x_1,y_1)$ and $B(x_2,y_2)$ is $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$.

Solved Example # 1 **Find the value of x, if the distance between the points (x, –1) and (3, 2) is 5**

Solution.

Let P(x ,–1) and Q(3, 2) be the given points. Then PQ = 5 (given)

$\sqrt{(x-3)^2+(-1-2)^2} = 5 \quad \Rightarrow \quad (x-3)^2+9=25 \quad \Rightarrow$ **x = 7 or x = – 1 Ans.**

Self practice problems :

1. Show that four points (0, –1), (6, 7) (–2, 3) and (8, 3) are the vertices of a rectangle.

2. Find the coordinates of the circumcenter of the triangle whose vertices are (8, 6), (8, –2) and (2, –2). Also find its circumradius. **Ans.** (5, 2), 5

2. Section Formula :

If P(x, y) divides the line joining $A(x_1, y_1)$ & $B(x_2, y_2)$ in the ratio m : n, then;

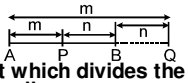
$$x = \frac{mx_2+nx_1}{m+n} \; ; \; y = \frac{my_2+ny_1}{m+n} .$$

NOTE : (i) If $\frac{m}{n}$ is positive, the division is internal, but if $\frac{m}{n}$ is negative, the division is external.

(ii) If P divides AB internally in the ratio m : n & Q divides AB externally in the ratio m : n then P & Q are said to be **harmonic conjugate** of each other w.r.t. AB.

Mathematically,

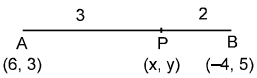
$\frac{2}{AB}=\frac{1}{AP}+\frac{1}{AQ}$ i.e. AP, AB & AQ are in H.P.



Solved Example# 2 **Find the coordinates of the point which divides the line segment joining the points (6, 3) and (–4, 5) in the ratio 3 : 2 (i) internally and (ii) externally.**

Solution. Let P (x, y) be the required point.

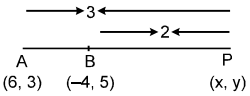
(i) For internal division :



$x = \frac{3\times-4+2\times6}{3+2}$ and $y = \frac{3\times5+2\times3}{3+2}$ or x = 0 and $y = \frac{21}{5}$

So the coordinates of P are $\left(0, \frac{21}{5}\right)$ **Ans.**

(ii) For external division



$x = \frac{3\times-4-2\times6}{3-2}$ and $y = \frac{3\times5-2\times3}{3-2}$

or x = –24 and y = 9

So the coordinates of P are **(–24, 9)** **Ans.**

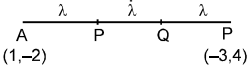
Solved Example # 3

Find the coordinates of points which trisect the line segment joining (1, – 2) and (– 3, 4).

Solution.

Let A (1, –2) and B(–3, 4) be the given points. Let the points of trisection be P and Q. Then

AP = PQ = QB = λ (say)



∴ PB = PQ + QB = 2λ and AQ = AP + PQ = 2λ

⇒ AP : PB = λ : 2λ = 1 : 2 and AQ : QB = 2λ : λ = 2 : 1

So P divides AB internally in the ratio 1 : 2 while Q divides internally in the ratio 2 : 1

∴ the coordinates of P are $\left(\frac{1\times-3+2\times1}{1+2}, \frac{1\times4+2\times-2}{1+2}\right)$ or $\left(-\frac{1}{3}, 0\right)$

and the coordinates of Q are $\left(\frac{2\times-3+1\times1}{2+1}, \frac{2\times4+1\times(-2)}{2+1}\right)$ or $\left(-\frac{5}{3}, 2\right)$

Hence, the points of trisection are $\left(-\frac{1}{3}, 0\right)$ and $\left(-\frac{5}{3}, 2\right)$ **Ans.**

Self practice problems :

3. In what ratio does the point (–1, –1) divide the line segment joining the points (4, 4) and (7, 7)? **Ans.** 5 : 8 externally

4. The three vertices of a parallelogram taken in order are (–1, 0), (3, 1) and (2, 2) respectively. Find the coordinates of the fourth vertex. **Ans.** (–2, 1)

3. Centroid, Incentre & Excentre:

If A (x_1, y_1), B(x_2, y_2), C(x_3, y_3) are the vertices of triangle ABC, whose sides BC, CA, AB are of lengths a, b, c respectively, then the co-ordinates of the special points of triangle ABC are as follows :

Centroid $G \equiv \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$

Incentre $I \equiv \left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)$, and Excentre (to A) $I_1 \equiv \left(\frac{-ax_1+bx_2+cx_3}{-a+b+c}, \frac{-ay_1+by_2+cy_3}{-a+b+c}\right)$ and so on.

NOTE :

(i) Incentre divides the angle bisectors in the ratio, (b + c) : a; (c + a) : b & (a + b) : c.

(ii) Incentre and excentre are harmonic conjugate of each other w.r.t. the angle bisector on which they lie.

(iii) Orthocenter, Centroid & Circumcenter are always collinear & centroid divides the line joining orthocentre & circumcenter in the ratio 2 : 1.

(iv) In an isosceles triangle G, O, I & C lie on the same line and in an equilateral triangle, all these four points coincide.

Sol. Ex. 4 Find the coordinates of (i) centroid (ii) in-centre of the triangle whose vertices are (0, 6), (8, 12) and (8, 0).

Solution (i) We know that the coordinates of the centroid of a triangle whose angular points are (x_1, y_1), (x_2, y_2) (x_3, y_3) are $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$

So the coordinates of the centroid of a triangle whose vertices are (0, 6), (8, 12) and (8, 0) are $\left(\frac{0+8+8}{3}, \frac{6+12+0}{3}\right)$ or $\left(\frac{16}{3}, 6\right)$ **Ans.**

(ii) Let A (0, 6), B (8, 12) and C(8,) be the vertices of triangle ABC.

Then c = AB = $\sqrt{(0-8)^2+(6-12)^2}$ = 10, b = CA = $\sqrt{(0-8)^2+(6-0)^2}$ = 10

and a = BC = $\sqrt{(8-8)^2+(12-0)^2}$ = 12.

The coordinates of the in-centre are $\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)$

or $\left(\frac{12\times0+10\times8+10\times8}{12+10+10}, \frac{12\times6+10\times12+10\times0}{12+10+10}\right)$

or $\left(\frac{160}{32}, \frac{192}{32}\right)$ or **(5, 6)** **Ans.**

Self practice problems :

5. Two vertices of a triangle are (3, –5) and (–7, 4). If the centroid is (2, –1), find the third vertex. **Ans.**(10, – 2)

6. Find the coordinates of the centre of the circle inscribed in a triangle whose vertices are (– 36, 7), (20, 7) and (0, – 8) **Ans.** (–1, 0)

4. Area of a Triangle:

If $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are the vertices of triangle ABC, then its area is equal to

$\Delta ABC = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$, provided the vertices are considered in the counter clockwise sense. The above formula will give a (–) ve area if the vertices (x_i, y_i), i = 1, 2, 3 are placed in the clockwise sense.

NOTE : Area of n-sided polygon formed by points (x_1, y_1) ; (x_2, y_2); .....(x_n, y_n) is given by

$$\frac{1}{2}\left(\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix} + \ldots\ldots\ldots\ldots \begin{vmatrix} x_{n-1} & x_n \\ y_{n-1} & y_n \end{vmatrix} + \begin{vmatrix} x_n & x_1 \\ y_n & y_1 \end{vmatrix}\right)$$

Solved Example # 5: If the coordinates of two points A and B are (3, 4) and (5, –2) respectively. Find the coordinates of any point P if PA = PB and Area of ΔPAB = 10.

Solution

Let the coordinates of P be (x, y). Then

PA = PB $\Rightarrow$ $PA^2 = PB^2$ $\Rightarrow$ $(x-3)^2 + (y-4)^2 = (x-5)^2 + (y+2)^2$

$\Rightarrow$ x – 3y – 1 = 0

Now, Area of ΔPAB = 10 $\Rightarrow$ $\frac{1}{2}\begin{vmatrix} x & y & 1 \\ 3 & 4 & 1 \\ 5 & -2 & 1 \end{vmatrix} = \pm 10 \Rightarrow 6x + 2y - 26 = \pm 20$

$\Rightarrow$ 6x + 2y – 46 = 0 or 6x + 2y – 6 = 0

$\Rightarrow$ 3x + y – 23 = 0 or 3x + y – 3 = 0

Solving x – 3 y – 1 = 0 and 3x + y – 23 = 0 we get x = 7, y = 2. Solving x – 3y – 1 = 0 and 3x + y – 3 = 0, we get x = 1, y = 0. Thus, the coordinates of P are **(7, 2) or (1, 0)** **Ans.**

Self practice problems :

7. The area of a triangle is 5. Two of its vertices are (2, 1) and (3, –2). The third vertex lies on y = x + 3. Find the third vertex. **Ans.** $\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(-\frac{3}{2}, \frac{3}{2}\right)$

8. The vertices of a quadrilateral are (6, 3), (–3, 5), (4, –2) and (x, 3x) and are denoted by A, B, C and D, respectively. Find the values of x so that the area of triangle ABC is double the area of triangle DBC.

Ans. $x = \frac{11}{8}$ or $-\frac{3}{8}$

5. Slope Formula:

If θ is the angle at which a straight line is inclined to the positive direction of x–axis, & $0° \le \theta < 180°$, $\theta \ne 90°$, then the slope of the line, denoted by m, is defined by $m = \tan\theta$. If θ is 90°, m does not exist, but the line is parallel to the y–axis. If θ = 0, then m = 0 & the line is parallel to the x-axis.

If A (x_1, y_1) & B (x_2, y_2), $x_1 \ne x_2$, are points on a straight line, then the slope m of the line is given by :

$m = \left(\frac{y_1 - y_2}{x_1 - x_2}\right)$.

Solved Example # 6: What is the slope of a line whose inclination is :
(i) 0º (ii) 90º (iii) 120º (iv) 150º

Solution

(i) Here θ = 0º

Slope = tan θ = tan 0º = **0** **Ans.**

(ii) Here θ = 90º

∴ The slope of line is **not defined** **Ans.**

(iii) Here θ = 120º

∴ Slope = tan θ = tan 120º = tan (180º – 60º) = – tan 30º = **– $\sqrt{3}$** **Ans.**

(iv) Here θ = 150º

∴ Slope = tan θ = tan 150º = tan (180º – 30º) = – tan 30º = $-\frac{1}{\sqrt{3}}$ **Ans.**

Solved Example # 7 : Find the slope of the line passing through the points :
(i) (1, 6) and (– 4, 2) (ii) (5, 9) and (2, 9)

Solution

(i) Let A = (1, 6) and B = (– 4, – 2)

∴ Slope of AB = $\frac{2-6}{-4-1} = \frac{-4}{-5} = \frac{4}{5}$ **Ans.** $\left(\text{Using slope} = \frac{y_2 - y_1}{x_2 - x_1}\right)$

(ii) Let A = (5, 9), B = (2, 9)

∴ Slope of AB = $\frac{9-9}{2-5} = \frac{0}{-3}$ = **0** **Ans.**

Self practice problems :

9. Find the value of x, if the slope of the line joining (1, 5) and (x, –7) is 4. **Ans.** – 2

10. What is the inclination of a line whose slope is

(i) 0 (ii) 1 (iii) –1 (iv) $-1/\sqrt{3}$

Ans. (i) 0º, (ii) 45º, (iii) 135º, (iv) 150º

6. Condition of collinearity of three points:

Points A (x_1, y_1), B (x_2, y_2), C(x_3, y_3) are collinear if

(i) $m_{AB} = m_{BC} = m_{CA}$ i.e. $\left(\frac{y_1 - y_2}{x_1 - x_2}\right) = \left(\frac{y_2 - y_3}{x_2 - x_3}\right)$ (ii) $\Delta ABC = 0$ i.e. $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

(iii) AC = AB + BC or AB ~ BC (iv) A divides the line segment BC in some ratio.

Solved Example # 8 Show that the points (1, 1), (2, 3) and (3, 5) are collinear.

Solution.

Let (1, 1) (2, 3) and (3, 5) be the coordinates of the points A, B and C respectively.

Slope of AB = $\frac{3-1}{2-1} = 2$ and Slope of BC = $\frac{5-3}{3-2} = 2$

$\therefore$ Slope of AB = slope of AC

$\therefore$ AB & BC are parallel $\therefore$ A, B, C are collinear because B is on both lines AB and BC.

Self practice problem :

11. Prove that the points (a, 0), (0, b) and (1, 1) are collinear if $\frac{1}{a} + \frac{1}{b} = 1$

7. Equation of a Straight Line in various forms:

(i) **Point - Slope form :** $y - y_1 = m(x - x_1)$ is the equation of a straight line whose slope is m & which passes through the point (x_1, y_1).

Solved Example # 9 : Find the equation of a line passing through (2, –3) and inclined at an angle of 135º with the positive direction of x-axis.

Solution.

Here, m = slope of the line = tan 135º = tan (90º + 45º) = – cot 45º = –1, $x_1 = 2$, $y_1 = -3$

So, the equation of the line is $y - y_1 = m(x - x_1)$

i.e. $y - (-3) = -1(x - 2)$ or $y + 3 = -x + 2$ or **x + y + 1 = 0** **Ans.**

Self practice problem :

12. Find the equation of the perpendicular bisector of the line segment joining the points A(2, 3) and B (6, –5). **Ans.** x – 2 y – 6 = 0

(ii) **Slope – intercept form :** $y = mx + c$ is the equation of a straight line whose slope is m & which makes an intercept c on the y–axis.

Solved Example # 10: Find the equation of a line with slope –1 and cutting off an intercept of 4 units on negative direction of y-axis.

Solution. Here m = –1 and c = – 4. So, the equation of the line is $y = mx + c$ i.e. $y = -x - 4$ or **x + y + 4 = 0** **Ans.**

Self practice problem :

13. Find the equation of a straight line which cuts off an intercept of length 3 on y-axis and is parallel to the line joining the points (3, –2) and (1, 4). **Ans.** 3x + y – 3 = 0

(iii) **Two point form :** $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$ is the equation of a straight line which passes through the points (x_1, y_1) & (x_2, y_2).

Solved Example # 11 Find the equation of the line joining the points (– 1, 3) and (4, – 2)

Solution.

Here the two points are $(x_1, y_1) = (-1, 3)$ and $(x_2, y_2) = (4, -2)$.

So, the equation of the line in two-point form is

$y - 3 = \frac{3-(-2)}{-1-4}(x + 1) \Rightarrow y - 3 = -x - 1 \Rightarrow$ **x + y – 2 = 0** **Ans.**

Self practice problem :

14. Find the equations of the sides of the triangle whose vertices are (–1, 8), (4, –2) and (–5, –3). Also find the equation of the median through (–1, 8)

Ans. 2x + y – 6 = 0, x – 9y – 22 = 0, 11x – 4y + 43 = 0, 21x + y + 13 = 0

(iv) **Determinant form :** Equation of line passing through (x_1, y_1) and (x_2, y_2) is $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

Solved Example # 12

Find the equation of line passing through (2, 4) & (– 1, 3).

Solution.

$\begin{vmatrix} x & y & 1 \\ 2 & 4 & 1 \\ -1 & 3 & 1 \end{vmatrix} = 0 \Rightarrow$ **x – 3y + 10 = 0** **Ans.**

Self practice problem :

15. Find the equation of the passing through (– 2, 3) & (– 1, – 1).

Ans. 4x + y + 5 = 0

(v) **Intercept form :** $\frac{x}{a} + \frac{y}{b} = 1$ is the equation of a straight line which makes intercepts a & b on OX & OY respectively.

Solved Example # 13: Find the equation of the line which passes through the point (3, 4) and the sum of its intercepts on the axes is 14.

Sol. Let the equation of the line be $\frac{x}{a} + \frac{y}{b} = 1$ ....(i)

This passes through (3, 4), therefore $\frac{3}{a} + \frac{4}{b} = 1$ ....(ii)

It is given that $a + b = 14 \Rightarrow b = 14 - a$. Putting $b = 14 - a$ in (ii), we get $\frac{3}{a} + \frac{4}{14-a} = 1$

$\Rightarrow a^2 - 13a + 42 = 0$

$\Rightarrow (a - 7)(a - 6) = 0 \Rightarrow a = 7, 6$

For a = 7, b = 14 – 7 = 7 and for a = 6, b = 14 – 6 = 8.

Putting the values of a and b in (i), we get the equations of the lines

$\frac{x}{7} + \frac{y}{7} = 1$ and $\frac{y}{6} + \frac{y}{8} = 1$ or **x + y = 7 and 4x + 3y = 24** **Ans.**

Self practice problem :

16. Find the equation of the line through (2, 3) so that the segment of the line intercepted between the axes is bisected at this point. **Ans.** $3x + 2y = 12$.

(vi) **Perpendicular/Normal form :** $x\cos\alpha + y\sin\alpha = p$ (where $p > 0$, $0 \le \alpha < 2\pi$) is the equation of the straight line where the length of the perpendicular from the origin O on the line is p and this perpendicular makes an angle α with positive x–axis.

Solved Example # 14: Find the equation of the line which is at a distance 3 from the origin and the perpendicular from the origin to the line makes an angle of 30º with the positive direction of the x-axis.

Solution. Here $p = 3$, $\alpha = 30º$

$\therefore$ Equation of the line in the normal form is

$$x\cos 30º + y\sin 30º = 3 \text{ or } x\,\frac{\sqrt{3}}{2} + \frac{y}{2} = 3 \text{ or } \mathbf{\sqrt{3}\,x + y = 6} \qquad \textbf{Ans.}$$

Self practice problem :

17. The length of the perpendicular from the origin to a line is 7 and the line makes an angle of 150º with the positive direction of y-axis. Find the equation of the line. **Ans.** $\sqrt{3}\,x + y - 14 = 0$

(vii) **Parametric form :** $P(r) = (x, y) = (x_1 + r\cos\theta, y_1 + r\sin\theta)$ or $\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$ is the equation of the line in parametric form, where 'r' is the parameter whose absolute value is the distance of any point (x, y) on the line from the fixed point (x_1, y_1) on the line.

Solved Example # 15: Find the equation of the line through the point A(2, 3) and making an angle of 45º with the x-axis. Also determine the length of intercept on it between A and the line x + y + 1 = 0

Solution. The equation of a line through A and making an angle of 45º with the x-axis is

$$\frac{x-2}{\cos 45º} = \frac{y-3}{\sin 45º} \text{ or } \frac{x-2}{\frac{1}{\sqrt{2}}} = \frac{y-3}{\frac{1}{\sqrt{2}}} \qquad \text{or} \qquad x - y + 1 = 0$$

Suppose this line meets the line $x + y + 1 = 0$ at P such that AP = r. Then the coordinates of P are given by

$$\frac{x-2}{\cos 45º} = \frac{y-3}{\sin 45º} = r \Rightarrow \qquad x = 2 + r\cos 45º,\ y = 3 + r\sin 45º$$

$$\Rightarrow \qquad x = 2 + \frac{r}{\sqrt{2}},\ y = 3 + \frac{r}{\sqrt{2}}$$

Thus, the coordinates of P are $\left(2 + \frac{r}{\sqrt{2}}, 3 + \frac{r}{\sqrt{2}}\right)$

Since P lies on $x + y + 1 = 0$, so $2 + \frac{r}{\sqrt{2}} + 3 + \frac{r}{\sqrt{2}} + 1 = 0$

$$\Rightarrow \qquad \sqrt{2}\,r = -6 \Rightarrow r = -3\sqrt{2} \quad \Rightarrow \text{ length AP} = |r| = 3\sqrt{2}$$

Thus, the length of the intercept = $\mathbf{3\sqrt{2}}$ **Ans.**

Self practice problem :

18. A straight line is drawn through the point $A\left(\sqrt{3}, 2\right)$ making an angle of $\pi/6$ with positive direction of the x-axis. If it meets the straight line $\sqrt{3x} - 4y + 8 = 0$ in B, find the distance between A and B. **Ans.** 6 units

(viii) **General Form :** $ax + by + c = 0$ is the equation of a straight line in the general form

In this case, slope of line $= -\frac{a}{b}$

x - intercept $= -\frac{c}{a}$

y - intercept $= -\frac{c}{b}$

Solved Example # 16 Find slope, x-intercept & y-intercept of the line $2x - 3y + 5 = 0$.

Solution. Here, $a = 2$, $b = -3$, $c = 5$

$\therefore$ slope $= -\frac{a}{b} = \mathbf{\frac{2}{3}}$ **Ans.**

x-intercept $= -\frac{c}{a} = \mathbf{-\frac{5}{2}}$ **Ans.**

y-intercept $= \mathbf{\frac{5}{3}}$ **Ans.**

Self practice problem :

19. Find the slope, x-intercept & y-intercept of the line $3x - 5y - 8 = 0$. **Ans** $\frac{3}{5}, \frac{8}{3}, -\frac{8}{5}$

8. Angle between two straight lines in terms of their slopes:

If m_1 & m_2 are the slopes of two intersecting straight lines ($m_1 m_2 \ne -1$) & θ is the acute angle between them, then $\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$.

NOTE :

(i) Let m_1, m_2, m_3 are the slopes of three lines $L_1 = 0; L_2 = 0; L_3 = 0$ where $m_1 > m_2 > m_3$ then the interior angles of the Δ ABC found by these lines are given by,

$$\tan A = \frac{m_1 - m_2}{1 + m_1 m_2};\ \tan B = \frac{m_2 - m_3}{1 + m_2 m_3} \ \&\ \tan C = \frac{m_3 - m_1}{1 + m_3 m_1}$$

(ii) The equation of lines passing through point (x_1, y_1) and making angle α with the line $y = mx + c$ are given by :

$(y - y_1) = \tan(\theta - \alpha)(x - x_1)$ &

$(y - y_1) = \tan(\theta + \alpha)(x - x1)$, where $\tan\theta = m$.

Solved Example # 17: The acute angle between two lines is $\pi/4$ and slope of one of them is 1/2. Find the slope of the other line.

Solution.

If θ be the acute angle between the lines with slopes m_1 and m_2, then $\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$

Let $\theta = \frac{\pi}{4}$ and $m_1 = \frac{1}{2}$

$\therefore \quad \tan\frac{\pi}{4} = \left|\frac{\frac{1}{2}-m_2}{1+\frac{1}{2}m_2}\right| \quad \Rightarrow \quad 1 = \left|\frac{1-2m_2}{2+m_2}\right| \quad \Rightarrow \quad \frac{1-2m_2}{2+m_2} = +1 \text{ or } -1$

Now $\frac{1-2m_2}{2+m_2} = 1 \quad \Rightarrow m_2 = -\frac{1}{3}$ and $\frac{1-2m_2}{2+m_2} = -1 \quad \Rightarrow \quad m_2 = 3.$

$\therefore$ The slope of the other line is either **– 1/3 or 3 Ans.**

Solved Example # 18: Find the equation of the straight line which passes through the origin and making angle 60º with the line $x + \sqrt{3}y + 3\sqrt{3} = 0$.

Solution. Given line is $x + \sqrt{3}y + 3\sqrt{3} = 0$.

$\Rightarrow \quad y = \left(-\frac{1}{\sqrt{3}}\right)x - 3 \quad \therefore$ Slope of (1) $= -\frac{1}{\sqrt{3}}$.

Let slope of the required line be m. Also between these lines is given to be 60º.

$\Rightarrow \quad \tan 60^\circ = \left|\frac{m-\left(-1/\sqrt{3}\right)}{1+m\left(-1/\sqrt{3}\right)}\right| \quad \Rightarrow \quad \sqrt{3} = \left|\frac{\sqrt{3}m+1}{\sqrt{3}-m}\right| \quad \Rightarrow \quad \frac{\sqrt{3}m+1}{\sqrt{3}-m} = \pm\sqrt{3}$

$\frac{\sqrt{3}m+1}{\sqrt{3}-m} = -\sqrt{3} \Rightarrow \sqrt{3}m + 1 = 3 - \sqrt{3}m \Rightarrow \quad m = \frac{1}{\sqrt{3}}$

Using $y = mx + c$, the equation of the required line is

$y = \frac{1}{\sqrt{3}}x + 0$ i.e. $x - \sqrt{3}y = 0$. ($\because$ This passes through origin, so c = 0)

$\frac{\sqrt{3}m+1}{\sqrt{3}-m} = -\sqrt{3} \quad \Rightarrow \quad \sqrt{3}m + 1 = -3 + \sqrt{3}m$

$\Rightarrow$ m is not defined

$\therefore$ The slope of the required line is not defined. Thus, the required line is a vertical line. This line is to pass through the origin.

$\therefore$ The equation of the required line is **x = 0 Ans.**

Self practice problem :

20. A vertex of an equilateral triangle is (2, 3) and the equation of the opposite side is x + y = 2. Find the equation of the other sides of the triangle. **Ans.** $(2+\sqrt{3})x - y + 2\sqrt{3} - 1 = 0$ and $(2+\sqrt{3})x - y - 2\sqrt{3} - 1 = 0$.

9. Parallel Lines:

(i) When two straight lines are parallel their slopes are equal. Thus any line parallel to $y = mx + c$ is of the type $y = mx + d$, where k is a parameter.

(ii) Two lines $ax + by + c = 0$ and $a'x + b'y + c' = 0$ are parallel if $\frac{a}{a'} = \frac{b}{b'} \neq \frac{c}{c'}$.

Thus any line parallel to $ax + by + c = 0$ is of the type $ax + by + k = 0$, where k is a parameter.

(iii) The distance between two parallel lines with equations $ax + by + c_1 = 0$ & $ax + by + c_2 = 0$ is $\left|\frac{c_1-c_2}{\sqrt{a^2+b^2}}\right|$.

NOTE: Coefficients of x & y in both the equations must be same.

(iv) The area of the parallelogram $= \frac{p_1p_2}{\sin\theta}$, where p_1 & p_2 are distances between two pairs of opposite sides & θ is the angle between any two adjacent sides. Note that area of the parallelogram bounded by the lines $y = m_1x + c_1$, $y = m_1x + c_2$ and $y = m_2x + d_1$, $y = m_2x + d_2$ is given by $\left|\frac{(c_1-c_2)(d_1-d_2)}{m_1-m_2}\right|$.

Sol. Ex. 19: Find the equation of the straight line that has y-intercept 4 and is parallel to the straight line $2x - 3y = 7$.

Solution. Given line is $2x - 3y = 7$

(1) $\Rightarrow \quad 3y = 2x - 7 \Rightarrow y = \frac{2}{3}x - \frac{7}{3} \quad \therefore$ Slope of (1) is 2/3

The required line is parallel to (1), so its slope is also 2/3, y-intercept of required line = 4

$\therefore$ By using $y = mx + c$ form, the equation of the required line is

$y = \frac{2}{3}x + 4$ or **$2x - 3y + 12 = 0$ Ans.**

Solved Example # 20: Two sides of a square lie on the lines x + y = 1 and x + y + 2 = 0. What is its area?

Solution. Clearly the length of the side of the square is equal to the distance between the parallel lines

$x + y - 1 = 0$ .....(i) and $x + y + 2 = 0$ .....(ii)

Putting x = 0 in (i), we get y = 1. So (0, 1) is a point on line (i).

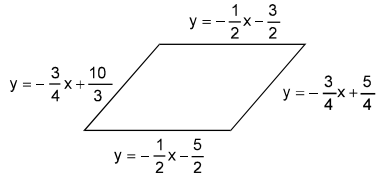
Now, Distance between the parallel lines

= length of the $\perp$ from (0, 1) to $x + y + 2 = 0 = \frac{|0+1+2|}{\sqrt{1^2+1^2}} = \frac{3}{\sqrt{2}}$

Thus, the length of the side of the square is $\frac{3}{\sqrt{2}}$ and hence its area $= \left(\frac{3}{\sqrt{2}}\right)^2 = \mathbf{\frac{9}{2}}$

Solved Example # 21: Find the area of the parallelogram whose sides are x + 2y + 3 = 0, 3x + 4y – 5 = 0, 2x + 4y + 5 = 0 and 3x + 4y – 10 = 0

Solution.



Here, $c_1 = -\frac{3}{2}$, $c_2 = \frac{5}{2}$, $d_1 = \frac{10}{3}$, $d_2 = -\frac{5}{2}$, $m_1 = -\frac{1}{2}$, $m_2 = -\frac{3}{4}$

$\therefore$ Area $= \left|\frac{\left(-\frac{3}{2}+\frac{5}{2}\right)\left(\frac{10}{3}+\frac{5}{2}\right)}{\left(-\frac{1}{2}+\frac{3}{4}\right)}\right| = \frac{70}{3}$ **sq. units** **Ans.**

Self practice problem :

21. Find the area of parallelogram whose sides are given by $4x - 5y + 1 = 0$, $x - 3y - 6 = 0$, $4x - 5y - 2 = 0$ and $2x - 6y + 5 = 0$ **Ans.** $\frac{51}{14}$ sq. units

10. Perpendicular Lines:

(i) If two lines of slopes m_1 & m_2 are at right angles, the product of their slopes is -1, i.e. $m_1 m_2 = -1$. Thus any line perpendicular to $y = mx + c$ is of the form $y = -\frac{1}{m}x + d$, where d is any parameter.

(ii) Two lines $ax + by + c = 0$ and $a'x + b'y + c' = 0$ are perpendicular if $aa' + bb' = 0$. Thus any line perpendicular to $ax + by + c = 0$ is of the form $bx - ay + k = 0$, where k is any parameter.

Solved Example # 22

Find the equation of the straight line that passes through the point (3, 4) and perpendicular to the line $3x + 2y + 5 = 0$

Solution.

The equation of a line perpendicular to $3x + 2y + 5 = 0$ is

$2x - 3y + \lambda = 0$ .....(i)

This passes through the point (3, 4)

$\therefore$ $3 \times 2 - 3 \times 4 + \lambda = 0 \Rightarrow \lambda = 6$

Putting $\lambda = 6$ in (i), we get **$2x - 3y + 6 = 0$**, which is the required equation. **Ans.**

Aliter The slope of the given line is –3/2. Since the required line is perpendicular to the given line. So, the slope of the required line is 2/3. As it passes through (3, 4). So, its equation is $y - 4 = \frac{2}{3}(x - 3)$ or **$2x - 3y + 6 = 0$ Ans.**

Self practice problem :

22. The vertices of a triangle are A(10, 4), B (–4, 9) and C(–2, –1). Find the equation of its altitudes. Also find its orthocentre.

Ans. $x - 5y + 10 = 0$, $12x + 5y + 3 = 0$, $14x - 5y + 23 = 0$, $\left(-1, \frac{9}{5}\right)$

11. Position of the point (x_1, y_1) relative of the line $ax + by + c = 0$:

If $ax_1 + by_1 + c$ is of the same sign as c, then the point (x_1, y_1) lie on the origin side of $ax + by + c = 0$. But if the sign of $ax_1 + by_1 + c$ is opposite to that of c, the point (x_1, y_1) will lie on the non–origin side of $ax + by + c = 0$.

In general two points (x_1, y_1) and (x_2, y_2) will lie on same side or opposite side of $ax + by + c = 0$ according as $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ are of same or opposite sign respectively.

Solved Example # 23

Show that (1, 4) and (0, –3) lie on the opposite sides of the line $x + 3y + 7 = 0$.

Solution.

At (1, 4), the value of $x + 3y + 7 = 1 + 3(4) + 7 = 20 > 0$.

At (0, – 3), the value of $x + 3y + 7 = 0 + 3(-3) + 7 = -2 < 0$

$\therefore$ **The points (1, 4) and (0, – 3) are on the opposite sides of the given line. Ans.**

Self practice problems :

23. Are the points (3, – 4) and (2, 6) on the same or opposite side of the line $3x - 4y = 8$?

Ans. Opposite sides

24. Which one of the points (1, 1), (–1, 2) and (2, 3) lies on the side of the line $4x + 3y - 5 = 0$ on which the origin lies?

Ans. (–1, 2)

12. The ratio in which a given line divides the line segment joining two points:

Let the given line $ax + by + c = 0$ divide the line segment joining $A(x_1, y_1)$ & $B(x_2, y_2)$ in the ratio m : n, then $\frac{m}{n} = -\frac{ax_1+by_1+c}{ax_2+by_2+c}$.

If A & B are on the same side of the given line then m/n is negative but if A & B are on opposite sides of the given line, then m/n is positive

Solved Example # 24

Find the ratio in which the line joining the points A (1, 2) and B(– 3, 4) is divided by the line $x + y - 5 = 0$.

Solution. Let the line $x + y = 5$ divides AB in the ratio k : 1 at P

$\therefore$ coordinate of P are $\left(\frac{-3k+1}{k+1}, \frac{4k+2}{k+1}\right)$

Since P lies on $x + y - 5 = 0$

$\therefore$ $\frac{-3k+1}{k+1} + \frac{4k+2}{k+1} - 5 = 0$. $\Rightarrow$ $k = -\frac{1}{2}$

$\therefore$ **Required ratio is 1 : 2 extrenally** **Ans.**

Aliter Let the ratio is m : n

$\therefore$ $\frac{m}{n} = -\frac{(1\times1+1\times2-5)}{1\times(-3)+1\times4-5} = -\frac{1}{2}$ $\therefore$ **ratio is 1 : 2 externally Ans.**

Self practice problem :

25. If the line $2x - 3y + \lambda = 0$ divides the line joining the points A (– 1, 2) & B(– 3, – 3) internally in the ratio 2 : 3, find λ.

Ans. $\frac{18}{5}$

13. Length of perpendicular from a point on a line:

The length of perpendicular from $P(x_1, y_1)$ on $ax + by + c = 0$ is $\left|\frac{ax_1 + by_1 + c}{\sqrt{a^2+b^2}}\right|$.

Solved Example # 25 Find the distance between the line $12x - 5y + 9 = 0$ and the point (2, 1)

Solution. The required distance $= \left|\frac{12\times2-5\times1+9}{\sqrt{12^2+(-5)^2}}\right| = \frac{|24-5+9|}{13} = \frac{28}{13}$ **Ans.**

Solved Example # 26
Find all points on x + y = 4 that lie at a unit distance from the line 4x + 3y – 10 = 0.
Solution. Note that the coordinates of an arbitrary point on x + y = 4 can be obtained by putting x = t (or y = t) and then obtaining y (or x) from the equation of the line, where t is a parameter.
Putting x = t in the equation x + y = 4 of the given line, we obtain y = 4 – t. So, coordinates of an arbitrary point on the given line are P(t, 4 – t). Let P(t, 4 – t) be the required point. Then, distance of P from the line 4x + 3y – 10 = 0 is unity i.e.

$\Rightarrow \quad \left|\frac{4t+3(4-t)-10}{\sqrt{4^2+3^2}}\right| = 1 \Rightarrow |t+2| = 5 \Rightarrow t+2 = \pm 5$

$\Rightarrow$ t = –7 or t = 3 Hence, required points are **(–7, 11) and (3, 1)** **Ans.**

Self practice problem :
26. Find the length of the altitudes from the vertices of the triangle with vertices :(–1, 1), (5, 2) and (3, –1).
Ans. $\frac{16}{\sqrt{13}}, \frac{8}{\sqrt{5}}, \frac{16}{\sqrt{37}}$

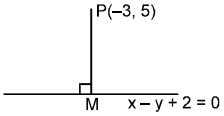
14. Reflection of a point about a line:

(i) Foot of the perpendicular from a point on the line is $\frac{x-x_1}{a} = \frac{y-y_1}{b} = -\frac{ax_1+by_1+c}{a^2+b^2}$

(ii) The image of a point (x_1, y_1) about the line ax + by + c = 0 is

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = -2\frac{ax_1+by_1+c}{a^2+b^2}.$$

Solved Example # 27 **Find the foot of perpendicular of the line drawn from P (– 3, 5) on the line x – y + 2 = 0.**
Solution.
Slope of PM = – 1



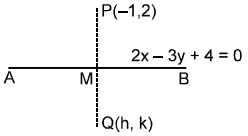
$\therefore$ Equation of PM is
x + y – 2 = 0 .....(i)
solving equation (i) with x – y + 2 = 0, we get coordinates of **M (0, 2)** **Ans.**

Aliter Here, $\frac{x+3}{1} = \frac{y-5}{-1} = -\frac{(1\times(-3)+(-1)\times 5+2)}{(1)^2+(-1)^2}$

$\Rightarrow \quad \frac{x+3}{1} = \frac{y-5}{-1} = 3 \quad \Rightarrow \quad x+3=3 \quad \Rightarrow \quad x=0$
and $y-5=-3 \quad \Rightarrow \quad y=2$

$\therefore$ **M is (0, 2)** **Ans.**

Solved Example # 28
Find the image of the point P(–1, 2) in the line mirror 2x – 3y + 4 = 0.
Solution. Let image of P is Q.



$\therefore$ PM = MQ & PQ $\perp$ AB
Let Q is (h, k)
$\therefore$ M is $\left(\frac{h-1}{2}, \frac{k+2}{2}\right)$
It lies on 2x – 3y + 4 = 0.
$\therefore \quad 2\left(\frac{h-1}{2}\right) - 3\left(\frac{k+2}{2}\right) + 4 = 0.$
or 2h – 3k = 0 .....(i)
slope of PQ = $\frac{k-2}{h+1}$
PQ $\perp$ AB
$\therefore \quad \frac{k-2}{h+1} \times \frac{2}{3} = -1.$
$\Rightarrow$ 3h + 2k – 1 = 0.....(ii)
soving (i) & (ii), we get
$h = \frac{3}{13}, k = \frac{2}{13}$
$\therefore$ Image of P(– 1, 2) is $Q\left(\frac{3}{13}, \frac{2}{13}\right)$ **Ans.**

Aliter The image of P (– 1, 2) about the line 2x – 3y + 4 = 0 is

$$\frac{x+1}{2} = \frac{y-2}{-3} = -2\frac{[2(-1)-3(2)+4]}{2^2+(-3)^2}$$

$$\frac{x+1}{2} = \frac{y-2}{-3} = \frac{8}{13}$$

$\Rightarrow \quad 13x + 13 = 16 \quad \Rightarrow \quad x = \frac{3}{13}$

& $13y - 26 = -24 \quad \Rightarrow \quad y = \frac{2}{13} \quad \therefore$ image is $\left(\frac{3}{13}, \frac{2}{13}\right)$ **Ans.**

Self practice problems :

27. Find the foot of perpendicular of the line drawn from (– 2, – 3) on the line 3x – 2y – 1 = 0. **Ans.** $\left(\frac{-23}{13}, \frac{-41}{13}\right)$

28. Find the image of the point (1, 2) in y-axis. **Ans.** (– 1, 2)

15. Bisectors of the angles between two lines:

Equations of the bisectors of angles between the lines $ax + by + c = 0$ &

$a'x + b'y + c' = 0$ $(ab' \neq a'b)$ are : $\frac{ax+by+c}{\sqrt{a^2+b^2}} = \pm \frac{a'x+b'y+c'}{\sqrt{a'^2+b'^2}}$

NOTE :

Equation of straight lines passing through $P(x_1, y_1)$ & equally inclined with the lines $a_1x + b_1y + c_1 = 0$ & $a_2x + b_2y + c_2 = 0$ are those which are parallel to the bisectors between these two lines & passing through the point P.

Solved Example # 29

Find the equations of the bisectors of the angle between the straight lines $3x - 4y + 7 = 0$ and $12x - 5y - 8 = 0$.

Solution.

The equations of the bisectors of the angles between $3x - 4y + 7 = 0$ and $12x - 5y - 8 = 0$ are $\frac{3x-4y+7}{\sqrt{3^2+(-4)^2}} = \pm \frac{12x-5y-8}{\sqrt{12^2+(-5)^2}}$

or $\frac{3x-4y+7}{5} = \pm \frac{12x-5y-8}{13}$

or $39x - 52y + 91 = \pm (60x - 25y - 8)$

Taking the positive sign, we get **$21x + 27y - 131 = 0$** as one bisector **Ans.**

Taking the negative sign, we get **$99x - 77y + 51 = 0$** as the other bisector. **Ans.**

Self practice problem :

29. Find the equations of the bisectors of the angles between the following pairs of straight lines $3x + 4y + 13 = 0$ and $12x - 5y + 32 = 0$

Ans. $21x - 77y - 9 = 0$ and $99x + 27y + 329 = 0$

16. Methods to discriminate between the acute angle bisector & the obtuse angle bisector:

(i) If θ be the angle between one of the lines & one of the bisectors, find $\tan\theta$.
If $|\tan\theta| < 1$, then $2\theta < 90°$ so that this bisector is the acute angle bisector.
If $|\tan\theta| > 1$, then we get the bisector to be the obtuse angle bisector.

(ii) Let $L_1 = 0$ & $L_2 = 0$ are the given lines & $u_1 = 0$ and $u_2 = 0$ are the bisectors between $L_1 = 0$ & $L_2 = 0$. Take a point P on any one of the lines $L_1 = 0$ or $L_2 = 0$ and drop perpendicular on $u_1 = 0$ & $u_2 = 0$ as shown.
If,
$|p| < |q| \Rightarrow u_1$ is the acute angle bisector.
$|p| > |q| \Rightarrow u_1$ is the obtuse angle bisector.
$|p| = |q| \Rightarrow$ the lines L_1 & L_2 are perpendicular.

(iii) If $aa' + bb' < 0$, then the equation of the bisector of this acute angle is

$$\frac{ax+by+c}{\sqrt{a^2+b^2}} = +\frac{a'x+b'y+c'}{\sqrt{a'^2+b'^2}}$$

If, however, $aa' + bb' > 0$, the equation of the bisector of the obtuse angle is :

$$\frac{ax+by+c}{\sqrt{a^2+b^2}} = +\frac{a'x+b'y+c'}{\sqrt{a'^2+b'^2}}$$

Solved Example # 30

For the straight lines $4x + 3y - 6 = 0$ and $5x + 12y + 9 = 0$, find the equation of the
(i) bisector of the obtuse angle between them;
(ii) bisector of the acute angle between them;

Solution.

(i) The equations of the given straight lines are
$4x + 3y - 6 = 0$ .....(1)
$5x + 12y + 9 = 0$ .....(2)
The equation of the bisectors of the angles between lines (1) and (2) are

$\frac{4x+3y-6}{\sqrt{4^2+3^2}} = \pm \frac{5x+12y+9}{\sqrt{5^2+12^2}}$ or $\frac{4x+3y-6}{5} = \pm \frac{5x+12y+9}{13}$

Taking the positive sign, we have

$$\frac{4x+3y-6}{5} = \frac{5x+12y+9}{13}$$

or $52x + 39y - 78 = 25x + 60y + 45$ or $27x - 21y - 123 = 0$
or $9x - 7y - 41 = 0$
Taking the negative sign, we have

$$\frac{4x+3y-6}{5} = -\frac{5x+12y+9}{13}$$

or $52x + 39y - 78 = -25x - 60y - 45$ or $77x + 99y - 33 = 0$
or $7x + 9y - 3 = 0$
Hence the equation of the bisectors are
$9x - 7y - 41 = 0$ .....(3)
and $7x + 9y - 3 = 0$ .....(4)

Now slope of line (1) $= -\frac{4}{3}$ and slope of the bisector (3) $= \frac{9}{7}$.

If θ be the acute angle between the line (1) and the bisector (3), then

$$\tan\theta = \left|\frac{\frac{9}{7}+\frac{4}{3}}{1+\frac{9}{7}\left(-\frac{4}{3}\right)}\right| = \left|\frac{27+28}{21-36}\right| = \left|\frac{55}{-15}\right| = \frac{11}{3} > 1$$

$\therefore$ $\theta > 45^0$

Hence **$9x - 7y - 41 = 0$** is the bisector of the obtuse angle between the given lines (1) and (2)
Ans.

(ii) Since $9x - 7y - 41$ is the bisector of the obtuse angle between the given lines, therefore the other bisector **$7x + 9y - 3 = 0$** will be the bisector of the acute angle between the given lines.

2nd Method *:*

Writing the equation of the lines so that constants become positive we have
$-4x - 3y + 6 = 0$ .....(1)

and $5x + 12y + 9 = 0$ .....(2)
Here $a_1 = -4$, $a_2 = 5$, $b_1 = -3$, $b_2 = 12$
Now $a_1a_2 + b_1b_2 = -20 - 36 = -56 < 0$
$\therefore$ origin does not lie in the obtuse angle between lines (1) and (2) and hence equation of the bisector of the obtuse angle between lines (1) and (2) will be

$$\frac{-4x-3y+6}{\sqrt{(-4)^2+(-3)^2}} = -\frac{5x+12y+9}{\sqrt{5^2+12^2}}$$

or $13(-4x - 3y + 6) = -5(5x + 12y + 9)$
or $27x - 21y - 123 = 0$ or **$9x - 7y - 41 = 0$ Ans.**
and the equation of the bisector of the acute angle will be (origin lies in the acute angle)

$$\frac{-4x-3y+6}{\sqrt{(-4)^2+(-3)^2}} = \frac{5x+12y+9}{\sqrt{5^2+12^2}}$$

or $77x + 99y - 33 = 0$ or **$7x + 9y - 3 = 0$ Ans.**

Self practice problem :

30. Find the equations of the bisectors of the angles between the lines $x + y - 3 = 0$ and $7x - y + 5 = 0$ and state which of them bisects the acute angle between the lines.
Ans. $x - 3y + 10 = 0$ (bisector of the obtuse angle); $4x + 1 = 0$ (bisector of the acute angle)

17. To discriminate between the bisector of the angle containing a point:

To discriminate between the bisector of the angle containing the origin & that of the angle not containing the origin. Rewrite the equations, $ax + by + c = 0$ & $a'x + b'y + c' = 0$ such that the constant terms c, c′ are positive. Then ;

$\frac{ax+by+c}{\sqrt{a^2+b^2}} = +\frac{a'x+b'y+c'}{\sqrt{a'^2+b'^2}}$ gives the equation of the bisector of the angle containing the origin & $\frac{ax+by+c}{\sqrt{a^2+b^2}} = -\frac{a'x+b'y+c'}{\sqrt{a'^2+b'^2}}$ gives the equation of the bisector of the angle not containing the origin. In general equation of the bisector which contains the point (α, β) is

$\frac{ax+by+c}{\sqrt{a^2+b^2}} = \frac{a'x+b'y+c'}{\sqrt{a'^2+b'^2}}$ or $\frac{ax+by+c}{\sqrt{a^2+b^2}} = -\frac{a'x+b'y+c'}{\sqrt{a'^2+b'^2}}$ according as
$a\alpha + b\beta + c$ and $a'\alpha + b'\beta + c'$ having same sign or otherwise.

Solved Example # 31

For the straight lines $4x + 3y - 6 = 0$ and $5x + 12y + 9 = 0$, find the equation of the bisector of the angle which contains the origin.
Solution.
For point O(0, 0), $4x + 3y - 6 = -6 < 0$ and $5x + 12y + 9 = 9 > 0$
Hence for point O(0, 0) $4x + 3y - 6$ and $5x + 12y + 9$ are of opposite signs.
Hence equation of the bisector of the angle between the given lines containing the origin will be

$$\frac{4x+3y-6}{\sqrt{(4)^2+(3)^2}} = -\frac{5x+12y+9}{\sqrt{5^2+12^2}}$$

or $\frac{4x+3y-6}{5} = -\frac{5x+12y+9}{13}$
or $52x + 39y - 78 = -25x - 60y - 45$.
or $77x + 99y - 33 = 0$
or **$7x + 9y - 3 = 0$ Ans.**
Self practice problem :
31. Find the equation of the bisector of the angle between the lines $x + 2y - 11 = 0$ and $3x - 6y - 5 = 0$ which contains the point $(1, -3)$. **Ans.** $3x - 19 = 0$

18. Condition of Concurrency:

Three lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ & $a_3x + b_3y + c_3 = 0$ are concurrent if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

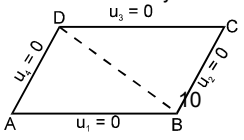
Alternatively : If three constants A, B & C (not all zero) can be found such that
$A(a_1x + b_1y + c_1) + B(a_2x + b_2y + c_2) + C(a_3x + b_3y + c_3) \equiv 0$, then the three straight lines are concurrent.
Solved Example # 32
Prove that the straight lines $4x + 7y = 9$, $5x - 8y + 15 = 0$ and $9x - y + 6 = 0$ are concurrent.
Solution.
Given lines are
$4x + 7y - 9 = 0$ .....(1)
$5x - 8y + 15 = 0$ .....(2)
and $9x - y + 6 = 0$ .....(3)

$$\Delta = \begin{vmatrix} 4 & 7 & -9 \\ 5 & -8 & 15 \\ 9 & -1 & 6 \end{vmatrix} = 4(-48 + 15) - 7(30 - 135) - 9(-5 + 72) = -132 + 735 - 603 = 0$$

Hence lines (1), (2) and (3) are concurrent. **Proved**
Self practice problem :
32. Find the value of m so that the lines $3x + y + 2 = 0$, $2x - y + 3 = 0$ and $x + my - 3 = 0$ may be concurrent.
Ans. 4

19. Family Of Straight Lines:

The equation of a family of straight lines passing through the point of intersection of the lines, $L_1 \equiv a_1x + b_1y + c_1 = 0$ & $L_2 \equiv a_2x + b_2y + c_2 = 0$ is given by $L_1 + k L_2 = 0$ i.e.
$(a_1x + b_1y + c_1) + k(a_2x + b_2y + c_2) = 0$, where k is an arbitrary real number.



NOTE :

(i) If $u_1 = ax + by + c$, $u_2 = a'x + b'y + d$, $u_3 = ax + by + c'$, $u_4 = a'x + b'y + d'$ then $u_1 = 0; u_2 = 0; u_3 = 0; u_4 = 0$ form a parallelogram. The diagonal BD can be given by $u_2u_3 - u_1u_4 = 0$.

(ii) The diagonal AC is also given by $u_1 + \lambda u_4 = 0$ and $u_2 + \mu u_3 = 0$, if the two equations are identical for some real λ and μ.

[For getting the values of λ & μ compare the coefficients of x, y & the constant terms].

Solved Example # 33

Find the equation of the straight line which passes through the point (2, –3) and the point of intersection of the lines x + y + 4 = 0 and 3x – y – 8 = 0.

Solution. Any line through the intersection of the lines $x + y + 4 = 0$ and $3x – y – 8 = 0$ has the equation

$(x + y + 4) + \lambda (3x – y – 8) = 0$ .....(i)

This will pass through (2, –3) if

$(2 – 3 + 4) + \lambda (6 + 3 – 8) = 0$ or $3 + \lambda = 0 \Rightarrow \lambda = – 3$.

Putting the value of λ in (i), the required line is

$(x + y + 4) + (–3) (3x – y – 8) = 0$

or $– 8x + 4y + 28 = 0$ or **2x – y – 7 = 0** **Ans.**

Aliter Solving the equations $x + y + 4 = 0$ and $3x – y – 8 = 0$ by cross-multiplication, we get $x = 1$, $y = –5$

So the two lines intersect at the point (1, –5). Hence the required line passes through (2, –3) and (1, –5) and so its equation is

$$y + 3 = -\frac{5+3}{1-2}(x - 2) \text{ or } \mathbf{2x – y – 7 = 0 \text{ Ans.}}$$

Solved Example # 34

Obtain the equations of the lines passing through the intersection of lines 4x – 3y – 1 = 0 and 2x – 5y + 3 = 0 and equally inclined to the axes.

Solution. The equation of any line through the intersection of the given lines is

$(4x – 3y – 1) + \lambda (2x – 5y + 3) = 0$

or $x (2\lambda + 4) – y (5\lambda + 3) + 3\lambda – 1 = 0$ .....(i)

Let m be the slope of this line. Then $m = \frac{2\lambda+4}{5\lambda+3}$

As the line is equally inclined with the axes, therefore

$m = \tan 45^\circ$ of $m = \tan 135^\circ \Rightarrow m = \pm 1$, $\frac{2\lambda+4}{5\lambda+3} = \pm 1 \Rightarrow \lambda = –1$ or $\frac{1}{3}$, putting the values of λ in (i), we get $2x + 2y – 4 = 0$ and $14x – 14y = 0$

i.e. x + y – 2 = 0 and x = y as the equations of the required lines. **Ans.**

Self practice problem :

33. Find the equation of the lines through the point of intersection of the lines $x – 3y + 1 = 0$ and $2x + 5y – 9 = 0$ and whose distance from the origin is $\sqrt{5}$ **Ans.** $2x + y – 5 = 0$

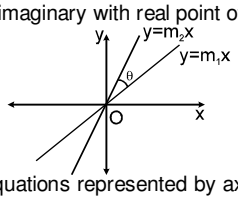
20. A Pair of straight lines through origin:

(i) A homogeneous equation of degree two, "$ax^2 + 2hxy + by^2 = 0$" always represents a pair of straight lines passing through the origin if :

(a) $h^2 > ab \Rightarrow$ lines are real & distinct .

(b) $h^2 = ab \Rightarrow$ lines are coincident .

(c) $h^2 < ab \Rightarrow$ lines are imaginary with real point of intersection i.e. (0, 0)



(ii) If $y = m_1x$ & $y = m_2x$ be the two equations represented by $ax^2 + 2hxy + by^2 = 0$, then;

$$m_1 + m_2 = -\frac{2h}{b} \quad \& \quad m_1 m_2 = \frac{a}{b}.$$

(iii) If θ is the acute angle between the pair of straight lines represented by,

$ax^2 + 2hxy + by^2 = 0$, then ; $\tan\theta = \left|\frac{2\sqrt{h^2-ab}}{a+b}\right|$.

(iv) The condition that these lines are :

(a) At right angles to each other is $a + b = 0$. i.e. co–efficient of x^2 + co–efficient of $y^2 = 0$.

(b) Coincident is $h^2 = ab$.

(c) Equally inclined to the axis of x is $h = 0$.i.e. coeff. of $xy = 0$.

NOTE : A homogeneous equation of degree n represents n straight lines passing through origin.

(v) The equation to the pair of straight lines bisecting the angle between the straight lines, $ax^2 + 2hxy + by^2 = 0$ is $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$.

Solved Example # 35

Show that the equation $6x^2 – 5xy + y^2 = 0$ represents a pair of distinct straight lines, each passing through the origin. Find the separate equations of these lines.

Solution.

The given equation is a homogeneous equation of second degree. So, it represents a pair of straight lines passing through the origin. Comparing the given equation with $ax^2 + 2hxy + by^2 = 0$, we obtain $a = 6$, $b = 1$ and $2h = – 5$.

$\therefore \quad h^2 – ab = \frac{25}{4} – 6 = \frac{1}{4} > 0 \Rightarrow \quad h^2 > ab$

Hence, the given equation represents a pair of distinct lines passing through the origin.

Now, $6x^2 – 5xy + y^2 = 0 \quad \Rightarrow \quad \left(\frac{y}{x}\right)^2 – 5\left(\frac{y}{x}\right) + 6 = 0$

$\Rightarrow \quad \left(\frac{y}{x}\right)^2 – 3\left(\frac{y}{x}\right) – 2\left(\frac{y}{x}\right) + 6 = 0 \Rightarrow \quad \left(\frac{y}{x} – 3\right)\left(\frac{y}{x} – 2\right) = 0$

$\Rightarrow \quad \frac{y}{x} – 3 = 0$ or $\frac{y}{x} – 2 = 0 \Rightarrow y – 3x = 0$ or $y – 2x = 0$

So the given equation represents the straight lines **$y – 3x = 0$ and $y – 2x = 0$** **Ans.**

Solved Example # 36 Find the equations to the pair of lines through the origin which are perpendicular to the lines represented by $2x^2 - 7xy + 2y^2 = 0$.

Solution.

We have $2x^2 - 7xy + 2y^2 = 0$.

$\Rightarrow \quad 2x^2 - 6xy - xy + 3y^2 = 0 \Rightarrow \quad 2x(x - 3y) - y(x - 3y) = 0$

$\Rightarrow \quad (x - 3y)(2x - y) = 0 \quad \Rightarrow \quad x - 3y = 0$ or $2x - y = 0$

Thus the given equation represents the lines $x - 3y = 0$ and $2x - y = 0$. The equations of the lines passing through the origin and perpendicular to the given lines are $y - 0 = -3(x - 0)$

and $y - 0 = -\frac{1}{2}(x - 0)$ [$\because$ (Slope of $x - 3y = 0$) is 1/3 and (Slope of $2x - y = 0$) is 2]

$\Rightarrow$ **$y + 3x = 0$ and $2y + x = 0$** **Ans.**

Solved Example # 37

Find the angle between the pair of straight lines $4x^2 + 24xy + 11y^2 = 0$

Solution. Given equation is $4x^2 + 24xy + 11y^2 = 0$

Here a = coeff. of $x^2 = 4$, b = coeff. of $y^2 = 11$

and 2h = coeff. of xy = 24 $\therefore$ h = 12

Now $\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right| = \left|\frac{2\sqrt{144 - 44}}{4 + 11}\right| = \frac{4}{3}$

Where θ is the acute angle between the lines.

$\therefore$ acute angle between the lines is $\tan^{-1}\left(\frac{4}{3}\right)$ and obtuse angle between them is

$\pi - \tan^{-1}\left(\frac{4}{3}\right)$ **Ans.**

Solved Example # 38 Find the equation of the bisectors of the angle between the lines represented by $3x^2 - 5xy + y^2 = 0$

Solution.

Given equation is $3x^2 - 5xy + y^2 = 0$ .....(1)

comparing it with the equation $ax^2 + 2hxy + by^2 = 0$ .....(2)

we have a = 3, 2h = –5; and b = 4

Now the equation of the bisectors of the angle between the pair of lines (1) is $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$

or $\frac{x^2 - y^2}{3 - 4} = \frac{xy}{-\frac{5}{2}}$; or $\frac{x^2 - y^2}{-1} = \frac{2xy}{-5}$

or **$5x^2 - 2xy - 5y^2 = 0$** **Ans.**

Self practice problems :

34. Find the area of the triangle formed by the lines $y^2 - 9xy + 18x^2 = 0$ and $y = 9$. **Ans.** $\frac{27}{4}$ sq. units

35. If the pairs of straight lines $x^2 - 2pxy - y^2 = 0$ and $x^2 - 2qxy - y^2 = 0$ be such that each pair bisects the angle between the other pair, prove that $pq = -1$.

21. General equation of second degree representing a pair of Straight lines:

(i) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if :

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$, i.e. if $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$.

(ii) The angle θ between the two lines representing by a general equation is the same as that between the two lines represented by its homogeneous part only.

Solved Example # 39 Prove that the equation $2x^2 + 5xy + 3y^2 + 6x + 7y + 4 = 0$ represents a pair of straight lines. Find the co-ordinates of their point of intersection and also the angle between them.

Solution. Given equation is

$2x^2 + 5xy + 2y^2 + 6x + 7y + 4 = 0$

Writing the equation (1) as a quadratic equation in x we have

$2x^2 + (5y + 6)x + 3y^2 + 7y + 4 = 0$

$\therefore \quad x = \frac{-(5y + 6) \pm \sqrt{(5y + 6)^2 - 4.2(3y^2 + 7y + 4)}}{4}$

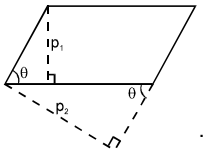
$= \frac{-(5y + 6) \pm \sqrt{25y^2 + 60y + 36 - 24y^2 - 56y - 32}}{4}$

$= \frac{-(5y + 6) \pm \sqrt{y^2 + 4y + 4}}{4} = \frac{-(5y + 6) \pm (y + 2)}{4}$

$\therefore \quad x = \frac{-5y - 6 + y + 2}{4}, \frac{-5y - 6 - y - 2}{4}$

or $4x + 4y + 4 = 0$ and $4x + 6y + 8 = 0$

or $x + y + 1 = 0$ and $2x + 3y + 4 = 0$



Hence equation (1) represents a pair of straight lines whose equation are **$x + y + 1 = 0$** .....(1)

and **$2x + 3y + 4 = 0$** .....(2) **Ans.**

Solving these two equations, the required point of intersection is **(1, – 2)** **Ans.**

Self practice problem :

36. Find the combined equation of the straight lines passing through the point (1, 1) and parallel to the lines represented by the equation $x^2 - 5xy + 4y^2 + x + 2y - 2 = 0$ and find the angle between them.

Ans. $x^2 - 5xy + 4y^2 + 3x - 3y = 0$, $\tan^{-1}\left(\frac{3}{5}\right)$

22. Homogenization :

The equation of a pair of straight lines joining origin to the points of intersection of the line $L \equiv \ell x + my + n = 0$ and a second degree curve,

$S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

is $ax^2 + 2hxy + by^2 + 2gx\left(\frac{\ell x+my}{-n}\right)+2fy\left(\frac{\ell x+my}{-n}\right)+c\left(\frac{\ell x+my}{-n}\right)^2 = 0.$

The equation is obtained by homogenizing the equation of curve with the help of equation of line.

NOTE : Equation of any curve passing through the points of intersection of two curves $C_1 = 0$ and $C_2 = 0$ is given by $\lambda C_1 + \mu C_2 = 0$ where λ & μ are parameters.

Solved Example # 40

Prove that the angle between the lines joining the origin to the points of intersection of the straight line $y = 3x + 2$ with the curve $x^2 + 2xy + 3y^2 + 4x + 8y - 11 = 0$ is $\tan^{-1}\frac{2\sqrt{2}}{3}$.

Solution.

Equation of the given curve is $x^2 + 2xy + 3y^2 + 4x + 8y - 11 = 0$

and equation of the given straight line is $y - 3x = 2$; $\therefore \quad \frac{y-3x}{2} = 1$

Making equation (1) homogeneous equation of the second degree in x any y with the help of (1), we have

$$x^2 + 2xy + 3y^2 + 4x\left(\frac{y-3x}{2}\right) + 8y\left(\frac{y-3x}{2}\right) - 11\left(\frac{y-3x}{2}\right)^2 = 0$$

or $\quad x^2 + 2xy + 3y^2 + \frac{1}{2}(4xy + 8y^2 - 12x^2 - 24xy) - \frac{11}{4}(y^2 - 6xy + 9x^2) = 0$

or $\quad 4x^2 + 8xy + 12y^2 + 2(8y^2 - 12x^2 - 20xy) - 11(y^2 - 6xy + 9x^2) = 0$

or $\quad -119x^2 + 34xy + 17y^2 = 0$ or $119x^2 - 34xy - 17y^2 = 0$

or $\quad 7x^2 - 2xy - y^2 = 0$

This is the equation of the lines joining the origin to the points of intersection of (1) and (2).

Comparing equation (3) with the equation $ax^2 + 2hxy + by^2 = 0$

we have $a = 7$, $b = -1$ and $2h = -2$ i.e. $h = -1$

If θ be the acute angle between pair of lines (3), then

$$\tan\theta = \left|\frac{2\sqrt{h^2-ab}}{a+b}\right| = \left|\frac{2\sqrt{1+7}}{7-1}\right| = \frac{2\sqrt{8}}{6} = \frac{2\sqrt{2}}{3} \qquad \therefore \quad \theta = \tan^{-1}\frac{2\sqrt{2}}{3}$$ **Proved**

Self practice problems :

37. Find the equation of the straight lines joining the origin to the points of intersection of the line $3x + 4y - 5 = 0$ and the curve $2x^2 + 3y^2 = 5$.
Ans. $x^2 - y^2 - 24xy = 0$

38. Find the equation of the straight lines joining the origin to the points of intersection of the line $lx + my + n = 0$ and the curve $y^2 = 4ax$. Also, find the condition of their perpendicularity.
Ans. $4alx^2 + 4amxy + ny^2 = 0$; $4al + n = 0$

SHORT REVISION

1. DISTANCE FORMULA : The distance between the points $A(x_1,y_1)$ and $B(x_2,y_2)$ is $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$.

2. SECTION FORMULA : If $P(x, y)$ divides the line joining $A(x_1, y_1)$ & $B(x_2, y_2)$ in the ratio $m : n$, then ;

$x = \frac{mx_2+nx_1}{m+n}$; $y = \frac{my_2+ny_1}{m+n}$.

If $\frac{m}{n}$ is positive, the division is internal, but if $\frac{m}{n}$ is negative, the division is external .

Note : If P divides AB internally in the ratio $m : n$ & Q divides AB externally in the ratio $m : n$ then P & Q are said to be harmonic conjugate of each other w.r.t. AB.

Mathematically ; $\frac{2}{AB}=\frac{1}{AP}+\frac{1}{AQ}$ i.e. AP, AB & AQ are in H.P.

3. CENTROID AND INCENTRE : If $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are the vertices of triangle ABC, whose sides BC, CA, AB are of lengths a, b, c respectively, then the coordinates of the centroid are : $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$

& the coordinates of the incentre are : $\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)$

Note that incentre divides the angle bisectors in the ratio
$(b+c) : a$; $(c+a) : b$ & $(a+b) : c$.

REMEMBER :

(i) Orthocentre , Centroid & circumcentre are always collinear & centroid divides the line joining orthocentre & cercumcentre in the ratio 2 : 1 .

(ii) In an isosceles triangle G, O, I & C lie on the same line .

4. SLOPE FORMULA :

If θ is the angle at which a straight line is inclined to the positive direction of x–axis, & $0° \le \theta < 180°$, $\theta \ne 90°$, then the slope of the line, denoted by m, is defined by $m = \tan\theta$. If θ is 90°, m does not exist, but the line is parallel to the y–axis.

If $\theta = 0$, then $m = 0$ & the line is parallel to the x–axis.

If $A(x_1, y_1)$ & $B(x_2, y_2)$, $x_1 \ne x_2$, are points on a straight line, then the slope m of the line is given by:

$m = \left(\frac{y_1-y_2}{x_1-x_2}\right)$.

5. CONDITION OF COLLINEARITY OF THREE POINTS – (SLOPE FORM) :

Points $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are collinear if $\left(\frac{y_1-y_2}{x_1-x_2}\right)=\left(\frac{y_2-y_3}{x_2-x_3}\right)$.

6. EQUATION OF A STRAIGHT LINE IN VARIOUS FORMS :

(i) Slope – intercept form: $y = mx + c$ is the equation of a straight line whose slope is m & which makes an intercept c on the y–axis .

(ii) Slope one point form: $y - y_1 = m(x - x_1)$ is the equation of a straight line whose slope is m & which passes through the point (x_1, y_1).

(iii) Parametric form : The equation of the line in parametric form is given by $\frac{x-x_1}{\cos\theta} = \frac{y-y_1}{\sin\theta} = r$ (say). Where 'r' is the distance of any point (x , y) on the line from the fixed point (x_1, y_1) on the line. r is positive if the point (x, y) is on the right of (x_1, y_1) and negative if (x, y) lies on the left of (x_1, y_1) .

(iv) Two point form : $y - y_1 = \frac{y_2-y_1}{x_2-x_1}(x - x_1)$ is the equation of a straight line which passes through the points (x_1, y_1) & (x_2, y_2) .

(v) Intercept form : $\frac{x}{a}+\frac{y}{b} = 1$ is the equation of a straight line which makes intercepts a & b on OX & OY respectively .

(vi) Perpendicular form : $x\cos\alpha + y\sin\alpha = p$ is the equation of the straight line where the length of the perpendicular from the origin O on the line is p and this perpendicular makes angle α with positive side of x–axis .

(vii) General Form : $ax + by + c = 0$ is the equation of a straight line in the general form

7. POSITION OF THE POINT (x_1, y_1) RELATIVE TO THE LINE $ax + by + c = 0$:

If $ax_1 + by_1 + c$ is of the same sign as c, then the point (x_1, y_1) lie on the origin side of $ax + by + c = 0$. But if the sign of $ax_1 + by_1 + c$ is opposite to that of c, the point (x_1, y_1) will lie on the non-origin side of $ax + by + c = 0$.

8. THE RATIO IN WHICH A GIVEN LINE DIVIDES THE LINE SEGMENT JOINING TWO POINTS :

Let the given line $ax + by + c = 0$ divide the line segment joining $A(x_1, y_1)$ & $B(x_2, y_2)$ in the ratio $m : n$, then $\frac{m}{n} = -\frac{ax_1+by_1+c}{ax_2+by_2+c}$. If A & B are on the same side of the given line then $\frac{m}{n}$ is negative but if A & B are on opposite sides of the given line , then $\frac{m}{n}$ is positive

9. LENGTH OF PERPENDICULAR FROM A POINT ON A LINE :

The length of perpendicular from $P(x_1, y_1)$ on $ax + by + c = 0$ is $\left|\frac{ax_1+by_1+c}{\sqrt{a^2+b^2}}\right|$.

10. ANGLE BETWEEN TWO STRAIGHT LINES IN TERMS OF THEIR SLOPES :

If m_1 & m_2 are the slopes of two intersecting straight lines ($m_1 m_2 \ne -1$) & θ is the acute angle between them, then

$\tan\theta = \left|\frac{m_1 - m_2}{1+m_1 m_2}\right|$.

Note : Let m_1, m_2, m_3 are the slopes of three lines $L_1 = 0$; $L_2 = 0$; $L_3 = 0$ where $m_1 > m_2 > m_3$ then the interior angles of the ΔABC found by these lines are given by,

$\tan A = \frac{m_1 - m_2}{1 + m_1 m_2}$; $\tan B = \frac{m_2 - m_3}{1 + m_2 m_3}$ & $\tan C = \frac{m_3 - m_1}{1 + m_3 m_1}$

11. PARALLEL LINES :

(i) When two straight lines are parallel their slopes are equal. Thus any line parallel to $ax + by + c = 0$ is of the type $ax + by + k = 0$. Where k is a parameter.

(ii) The distance between two parallel lines with equations $ax + by + c_1 = 0$ & $ax + by + c_2 = 0$ is $\left|\frac{c_1 - c_2}{\sqrt{a^2+b^2}}\right|$.

Note that the coefficients of x & y in both the equations must be same.

(iii) The area of the parallelogram $= \frac{p_1 p_2}{\sin\theta}$, where p_1 & p_2 are distances between two pairs of opposite sides & θ is the angle between any two adjacent sides . Note that area of the parallelogram bounded by the lines $y = m_1x + c_1$, $y = m_1x + c_2$ and $y = m_2x + d_1$, $y = m_2x + d_2$ is given by $\left|\frac{(c_1 - c_2)(d_1 - d_2)}{m_1 - m_2}\right|$.

12. PERPENDICULAR LINES :

(i) When two lines of slopes m_1 & m_2 are at right angles, the product of their slopes is -1, i.e. $m_1 m_2 = -1$. Thus any line perpendicular to $ax + by + c = 0$ is of the form $bx - ay + k = 0$, where k is any parameter.

(ii) Straight lines $ax + by + c = 0$ & $a'x + b'y + c' = 0$ are at right angles if & only if $aa' + bb' = 0$.

13. Equations of straight lines through (x_1, y_1) making angle α with $y = mx + c$ are:
$(y - y_1) = \tan(\theta - \alpha)(x - x_1)$ & $(y - y_1) = \tan(\theta + \alpha)(x - x_1)$, where $\tan\theta = m$.

14. CONDITION OF CONCURRENCY :

Three lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ & $a_3x + b_3y + c_3 = 0$ are concurrent if $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$. **Alternatively :** If three constants A, B & C can be found such that

$A(a_1x + b_1y + c_1) + B(a_2x + b_2y + c_2) + C(a_3x + b_3y + c_3) \equiv 0$, then the three straight lines are concurrent.

15. AREA OF A TRIANGLE :

If (x_i, y_i), i = 1, 2, 3 are the vertices of a triangle, then its area is equal to $\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$, provided the vertices are considered in the counter clockwise sense. The above formula will give a (–) ve area if the vertices (x_i, y_i), i = 1, 2, 3 are placed in the clockwise sense.

16. CONDITION OF COLLINEARITY OF THREE POINTS–(AREA FORM):

The points (x_i, y_i), i = 1 , 2 , 3 are collinear if $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$.

17. THE EQUATION OF A FAMILY OF STRAIGHT LINES PASSING THROUGH THE POINTS OF INTERSECTION OF TWO GIVEN LINES:

The equation of a family of lines passing through the point of intersection of $a_1x + b_1y + c_1 = 0$ & $a_2x + b_2y + c_2 = 0$ is given by $(a_1x + b_1y + c_1) + k(a_2x + b_2y + c_2) = 0$, where k is an arbitrary real number.

Note: If $u_1 = ax + by + c$, $u_2 = a'x + b'y + d$, $u_3 = ax + by + c'$, $u_4 = a'x + b'y + d'$ then, $u_1 = 0$; $u_2 = 0$; $u_3 = 0$; $u_4 = 0$ form a parallelogram.
$u_2u_3 - u_1u_4 = 0$ represents the diagonal BD.

Proof : Since it is the first degree equation in x & y it is a straight line. Secondly point B satisfies the equation because the co–ordinates of B satisfy $u_2 = 0$ and $u_1 = 0$.
Similarly for the point D. Hence the result.
On the similar lines $u_1u_2 - u_3u_4 = 0$ represents the diagonal AC.

Note: The diagonal AC is also given by $u_1 + \lambda u_4 = 0$ and $u_2 + \mu u_3 = 0$, if the two equations are identical for some λ and μ.
[For getting the values of λ & μ compare the coefficients of x, y & the constant terms] .

18. BISECTORS OF THE ANGLES BETWEEN TWO LINES :

(i) Equations of the bisectors of angles between the lines $ax + by + c = 0$ & $a'x + b'y + c' = 0$ $(ab' \neq a'b)$ are : $\frac{ax + by + c}{\sqrt{a^2+b^2}} = \pm \frac{a'x + b'y + c'}{\sqrt{a'^2+b'^2}}$

(ii) To discriminate between the acute angle bisector & the obtuse angle bisector

If θ be the angle between one of the lines & one of the bisectors, find $\tan\theta$.
If $|\tan\theta| < 1$, then $2\theta < 90°$ so that this bisector is the acute angle bisector .
If $|\tan\theta| > 1$, then we get the bisector to be the obtuse angle bisector .

(iii) To discriminate between the bisector of the angle containing the origin & that of the angle not containing the origin.
Rewrite the equations , $ax + by + c = 0$ &
$a'x + b'y + c' = 0$ such that the constant terms c , c′ are positive. Then;
$\frac{ax + by + c}{\sqrt{a^2+b^2}} = + \frac{a'x + b'y + c'}{\sqrt{a'^2+b'^2}}$ gives the equation of the bisector of the angle containing the origin & $\frac{ax + by + c}{\sqrt{a^2+b^2}} = - \frac{a'x + b'y + c'}{\sqrt{a'^2+b'^2}}$ gives the equation of the bisector of the angle not containing the origin.

(iv) To discriminate between acute angle bisector & obtuse angle bisector proceed as follows Write $ax + by + c = 0$ & $a'x + b'y + c' = 0$ such that constant terms are positive .

If $aa' + bb' < 0$, then the angle between the lines that contains the origin is acute and the equation of the bisector of this acute angle is $\frac{a x + b y + c}{\sqrt{a^2 + b^2}} = + \frac{a' x + b' y + c'}{\sqrt{a'^2 + b'^2}}$

therefore $\frac{a x + b y + c}{\sqrt{a^2 + b^2}} = - \frac{a' x + b' y + c'}{\sqrt{a'^2 + b'^2}}$ is the equation of other bisector.

If, however, $aa' + bb' > 0$, then the angle between the lines that contains the origin is obtuse & the equation of the bisector of this obtuse angle is:

$\frac{a x + b y + c}{\sqrt{a^2 + b^2}} = + \frac{a' x + b' y + c'}{\sqrt{a'^2 + b'^2}}$; therefore $\frac{a x + b y + c}{\sqrt{a^2 + b^2}} = - \frac{a' x + b' y + c'}{\sqrt{a'^2 + b'^2}}$

is the equation of other bisector.

(v) Another way of identifying an acute and obtuse angle bisector is as follows :
Let $L_1 = 0$ & $L_2 = 0$ are the given lines & $u_1 = 0$ and $u_2 = 0$ are the bisectors between $L_1 = 0$ & $L_2 = 0$. Take a point P on any one of the lines $L_1 = 0$ or $L_2 = 0$ and drop perpendicular on $u_1 = 0$ & $u_2 = 0$ as shown. If,
$|p| < |q| \Rightarrow u_1$ is the acute angle bisector .
$|p| > |q| \Rightarrow u_1$ is the obtuse angle bisector .
$|p| = |q| \Rightarrow$ the lines L_1 & L_2 are perpendicular .

Note : Equation of straight lines passing through $P(x_1, y_1)$ & equally inclined with the lines $a_1x + b_1y + c_1 = 0$ & $a_2x + b_2y + c_2 = 0$ are those which are parallel to the bisectors between these two lines & passing through the point P.

19. A PAIR OF STRAIGHT LINES THROUGH ORIGIN :

(i) A homogeneous equation of degree two of the type $ax^2 + 2hxy + by^2 = 0$ always represents a pair of straight lines passing through the origin & if :
(a) $h^2 > ab \Rightarrow$ lines are real & distinct .
(b) $h^2 = ab \Rightarrow$ lines are coincident .
(c) $h^2 < ab \Rightarrow$ lines are imaginary with real point of intersection i.e. (0, 0)

(ii) If $y = m_1x$ & $y = m_2x$ be the two equations represented by $ax^2 + 2hxy + by^2 = 0$, then;

$m_1 + m_2 = -\frac{2h}{b}$ & $m_1 m_2 = \frac{a}{b}$.

(iii) If θ is the acute angle between the pair of straight lines represented by,

$ax^2 + 2hxy + by^2 = 0$, then; $\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$.

The condition that these lines are:
(a) At right angles to each other is $a + b = 0$. i.e. co–efficient of x^2 + coefficient of $y^2 = 0$.
(b) Coincident is $h^2 = ab$. **(c)** Equally inclined to the axis of x is $h = 0$. i.e. coeff. of $xy = 0$.

Note: A homogeneous equation of degree n represents n straight lines passing through origin.

20. GENERAL EQUATION OF SECOND DEGREE REPRESENTING A PAIR OF STRAIGHT LINES:

(i) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if:

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$, i.e. if $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$.

(ii) The angle θ between the two lines representing by a general equation is the same as that between the two lines represented by its homogeneous part only .

21. The joint equation of a pair of straight lines joining origin to the points of intersection of the line given by $lx + my + n = 0$ ..... **(i)** &
the 2nd degree curve : $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ ..... **(ii)**

is $ax^2 + 2hxy + by^2 + 2gx\left(\frac{lx + my}{-n}\right) + 2fy\left(\frac{lx + my}{-n}\right) + c\left(\frac{lx + my}{-n}\right)^2 = 0$ ..... **(iii)**

(iii) is obtained by homogenizing (ii) with the help of **(i)**, by writing **(i)** in the form: $\left(\frac{lx + my}{-n}\right) = 1$.

22. The equation to the straight lines bisecting the angle between the straight lines,

$ax^2 + 2hxy + by^2 = 0$ is $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$.

23. The product of the perpendiculars, dropped from (x_1, y_1) to the pair of lines represented by the equation, $ax^2 + 2hxy + by^2 = 0$ is $\frac{a x_1^2 + 2h x_1 y_1 + b y_1^2}{\sqrt{(a-b)^2 + 4h^2}}$.

24. Any second degree curve through the four point of intersection of $f(xy) = 0$ & $xy = 0$ is given by $f(xy) + \lambda xy = 0$ where $f(xy) = 0$ is also a second degree curve.

EXERCISE–1

Q.1 The sides AB, BC, CD, DA of a quadrilateral have the equations $x + 2y = 3$, $x = 1$, $x - 3y = 4$, $5x + y + 12 = 0$ respectively. Find the angle between the diagonals AC & BD.

Q.2 Find the co-ordinates of the orthocentre of the triangle, the equations of whose sides are $x + y = 1$, $2x + 3y = 6$, $4x - y + 4 = 0$, without finding the co–ordinates of its vertices.

Q.3 Two vertices of a triangle are (4, –3) & (–2, 5). If the orthocentre of the triangle is at (1, 2), find the coordinates of the third vertex.

Q.4 The point A divides the join of P(–5, 1) & Q(3, 5) in the ratio K : 1. Find the two values of K for which the area of triangle ABC, where B is (1, 5) & C is (7, –2), is equal to 2 units in magnitude.

Q.5 Determine the ratio in which the point P(3, 5) divides the join of A(1, 3) & B(7, 9). Find the harmonic conjugate of P w.r.t. A & B.

Q.6 A line is such that its segment between the straight lines $5x - y - 4 = 0$ and $3x + 4y - 4 = 0$ is bisected at the point $(1, 5)$. Obtain the equation.

Q.7 A line through the point P(2, –3) meets the lines $x - 2y + 7 = 0$ and $x + 3y - 3 = 0$ at the points A and B respectively. If P divides AB externally in the ratio 3 : 2 then find the equation of the line AB.

Q.8 The area of a triangle is 5. Two of its vertices are (2, 1) & (3, –2). The third vertex lies on $y = x + 3$. Find the third vertex.

Q.9 A variable line, drawn through the point of intersection of the straight lines $\frac{x}{a}+\frac{y}{b} = 1$ & $\frac{x}{b}+\frac{y}{a} = 1$, meets the coordinate axes in A & B . Show that the locus of the mid point of AB is the curve $2xy(a + b) = ab(x + y)$.

Q.10 Two consecutive sides of a parallelogram are $4x + 5y = 0$ & $7x + 2y = 0$. If the equation to one diagonal is $11x + 7y = 9$, find the equation to the other diagonal.

Q.11 The line $3x + 2y = 24$ meets the y–axis at A & the x–axis at B. The perpendicular bisector of AB meets the line through (0, –1) parallel to x–axis at C. Find the area of the triangle ABC.

Q.12 If the straight line drawn through the point $P(\sqrt{3}, 2)$ & making an angle $\frac{\pi}{6}$ with the x–axis, meets the line $\sqrt{3}x - 4y + 8 = 0$ at Q. Find the length PQ.

Q.13 Find the condition that the diagonals of the parallelogram formed by the lines $ax + by + c = 0$; $ax + by + c' = 0$; $a'x + b'y + c = 0$ & $a'x + b'y + c' = 0$ are at right angles. Also find the equation to the diagonals of the parallelogram.

Q.14 If lines be drawn parallel to the axes of co-ordinates from the points where $x\cos\alpha + y\sin\alpha = p$ meets them so as to meet the perpendicular on this line from the origin in the points P and Q then prove that $|PQ| = 4p\,|\cos 2\alpha|\,\text{cosec}^2 2\alpha$.

Q.15 The points (1, 3) & (5, 1) are two opposite vertices of a rectangle. The other two vertices lie on the line $y = 2x + c$. Find c & the remaining vertices.

Q.16 A straight line L is perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line L & the coordinate axes is 5. Find the equation of the line.

Q.17 Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$ & its third side passes through the point (1, –10). Determine the equation of the third side.

Q.18 The vertices of a triangle OBC are O(0, 0), B(–3, –1), C(–1, –3). Find the equation of the line parallel to BC & intersecting the sides OB & OC, whose perpendicular distance from the point (0, 0) is half.

Q.19 Find the direction in which a straight line may be drawn through the point (2, 1) so that its point of intersection with the line $4y - 4x + 4 + 3\sqrt{2} + 3\sqrt{10} = 0$ is at a distance of 3 units from (2, 1).

Q.20 Consider the family of lines, $5x + 3y - 2 + K_1(3x - y - 4) = 0$ and $x - y + 1 + K_2(2x - y - 2) = 0$. Find the equation of the line belonging to both the families without determining their vertices.

Q.21 Given vertices A(1, 1), B(4, –2) & C(5, 5) of a triangle, find the equation of the perpendicular dropped from C to the interior bisector of the angle A.

Q.22 If through the angular points of a triangle straight lines be drawn parallel to the opposite sides, and if the intersections of these lines be joined to the opposite angular points of the traingle then using co-ordinate geometry, show that the lines so obtained are concurrent.

Q.23 Determine all values of α for which the point (α, α^2) lies inside the triangle formed by the lines $2x + 3y - 1 = 0$; $x + 2y - 3 = 0$; $5x - 6y - 1 = 0$.

Q.24 If the equation, $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represent a pair of straight lines, prove that the equation to the third pair of straight lines passing through the points where these meet the axes is, $ax^2 - 2hxy + by^2 + 2gx + 2fy + c + \frac{4fg}{c}xy = 0$.

Q.25 A straight line is drawn from the point (1, 0) to the curve $x^2 + y^2 + 6x - 10y + 1 = 0$, such that the intercept made on it by the curve subtends a right angle at the origin. Find the equations of the line.

Q.26 Determine the range of values of $\theta \in [0, 2\pi]$ for which the point $(\cos\theta, \sin\theta)$ lies inside the triangle formed by the lines $x + y = 2$; $x - y = 1$ & $6x + 2y - \sqrt{10} = 0$.

Q.27 Find the co-ordinates of the incentre of the triangle formed by the line $x + y + 1 = 0$; $x - y + 3 = 0$ & $7x - y + 3 = 0$. Also find the centre of the circle escribed to $7x - y + 3 = 0$.

Q.28 In a triangle ABC, D is a point on BC such that $\frac{BD}{DC} = \frac{AB}{AC}$. The equation of the line AD is $2x + 3y + 4 = 0$ & the equation of the line AB is $3x + 2y + 1 = 0$. Find the equation of the line AC.

Q.29 Show that all the chords of the curve $3x^2 - y^2 - 2x + 4y = 0$ which subtend a right angle at the origin are concurrent. Does this result also hold for the curve, $3x^2 + 3y^2 - 2x + 4y = 0$? If yes, what is the point of concurrency & if not, give reasons.

Q.30 Without finding the vertices or angles of the triangle, show that the three straight lines $au + bv = 0$; $au - bv = 2ab$ and $u + b = 0$ from an isosceles triangle where $u \equiv x + y - b$ & $v \equiv x - y - a$ & $a, b \neq 0$.

EXERCISE–2

Q.1 The equations of perpendiculars of the sides AB & AC of triangle ABC are $x - y - 4 = 0$ and $2x - y - 5 = 0$ respectively. If the vertex A is (–2, 3) and point of intersection of perpendiculars bisectors is $\left(\frac{3}{2}, \frac{5}{2}\right)$, find the equation of medians to the sides AB & AC respectively.

Q.2 A line $4x + y = 1$ through the point A(2, –7) meets the line BC whose equation is $3x - 4y + 1 = 0$ at a point B. Find the equation of the line AC, so that AB = AC.

Q.3 If $x\cos\alpha + y\sin\alpha = p$, where $p = -\frac{\sin^2\alpha}{\cos\alpha}$ be a straight line, prove that the perpendiculars on this straight line from the points $(m^2, 2m)$, $(mm', m + m')$, $(m'^2, 2m')$ form a G.P.

Q.4 A(3, 0) and B(6, 0) are two fixed points and $P(x_1, y_1)$ is a variable point. AP and BP meet the y-axis at C & D respectively and AD meets OP at Q where 'O' is the origin. Prove that CQ passes through a fixed point and find its co–ordinates.

Q.5 Find the equation of the straight lines passing through (–2, –7) & having an intercept of length 3 between the straight lines $4x + 3y = 12$, $4x + 3y = 3$.

Q.6 Let ABC be a triangle with AB = AC. If D is the mid point of BC, E the foot of the perpendicular from D to AC and F the midpoint of DE, prove analytically that AF is perpendicular to BE.

Q.7 Two sides of a rhombous ABCD are parallel to the lines $y = x + 2$ & $y = 7x + 3$. If the diagonals of the rhombous intersect at the point (1, 2) & the vertex A is on the y-axis, find the possible coordinates of A.

Q.8 The equations of the perpendicular bisectors of the sides AB & AC of a triangle ABC are $x - y + 5 = 0$ & $x + 2y = 0$, respectively. If the point A is $(1, -2)$, find the equation of the line BC.

Q.9 A pair of straight lines are drawn through the origin form with the line $2x + 3y = 6$ an isosceles triangle right angled at the origin. Find the equation of the pair of straight lines & the area of the triangle correct to two places of decimals.

Q.10 A triangle is formed by the lines whose equations are AB : $x + y - 5 = 0$, BC : $x + 7y - 7 = 0$ and CA : $7x + y + 14 = 0$. Find the bisector of the interior angle at B and the exterior angle at C. Determine the nature of the interior angle at A and find the equaion of the bisector.

Q.11 A point P is such that its perpendicular distance from the line $y - 2x + 1 = 0$ is equal to its distance from the origin. Find the equation of the locus of the point P. Prove that the line $y = 2x$ meets the locus in two points Q & R, such that the origin is the mid point of QR.

Q.12 A triangle has two sides $y = m_1x$ and $y = m_2x$ where m_1 and m_2 are the roots of the equation $b\alpha^2 + 2h\alpha + a = 0$. If (a, b) be the orthocentre of the triangle, then find the equation of the third side in terms of a, b and h.

Q.13 Find the area of the triangle formed by the straight lines whose equations are $x + 2y - 5 = 0$; $2x + y - 7 = 0$ and $x - y + 1 = 0$ without determining the coordinates of the vertices of the triangle. Also compute the tangent of the interior angles of the triangle and hence comment upon the nature of triangle.

Q.14 Find the equation of the two straight lines which together with those given by the equation $6x^2 - xy - y^2 + x + 12y - 35 = 0$ will make a parallelogram whose diagonals intersect in the origin.

Q.15 Find the equations of the sides of a triangle having $(4, -1)$ as a vertex, if the lines $x - 1 = 0$ and $x - y - 1 = 0$ are the equations of two internal bisectors of its angles.

Q.16 Equation of a line is given by $y + 2at = t(x - at^2)$, t being the parameter. Find the locus of the point of intersection of the lines which are at right angles.

Q.17 The ends A, B of a straight line line segment of a constant length 'c' slide upon the fixed rectangular axes OX & OY respectively. If the rectangle OAPB be completed then show that the locus of the foot of the perpendicular drawn from P to AB is $x^{2/3} + y^{2/3} = c^{2/3}$.

Q.18 A point moves so that the distance between the feet of the perpendiculars from it on the lines $bx^2 + 2hxy + ay^2 = 0$ is a constant 2d. Show that the equation to its locus is,
$(x^2 + y^2)(h^2 - ab) = d^2\{(a - b)^2 + 4h^2\}$

Q.19 The sides of a triangle are $U_r \equiv x\cos\alpha_r + y\sin\alpha_r - p_r = 0$, $(r = 1, 2, 3)$. Show that the orthocentre is given by $U_1\cos(\alpha_2 - \alpha_3) = U_2\cos(\alpha_3 - \alpha_1) = U_3\cos(\alpha_1 - \alpha_2)$.

Q.20 P is the point $(-1, 2)$, a variable line through P cuts the x & y axes at A & B respectively Q is the point on AB such that PA, PQ, PB are H.P. Show that the locus of Q is the line $y = 2x$.

Q.21 The equations of the altitudes AD, BE, CF of a triangle ABC are $x + y = 0$, $x - 4y = 0$ and $2x - y = 0$ respectively. The coordinates of A are $(t, -t)$. Find coordinates of B & C. Prove that if t varies the locus of the centroid of the triangle ABC is $x + 5y = 0$.

Q.22 A variable line is drawn through O to cut two fixed straight lines L_1 & L_2 in R & S. A point P is chosen on the variable line such that; $\frac{m+n}{OP} = \frac{m}{OR} + \frac{n}{OS}$. Show that the locus of P is a straight line passing the point of intersection of L_1 & L_2.

Q.23 If the lines $ax^2 + 2hxy + by^2 = 0$ from two sides of a parallelogram and the line $lx + my = 1$ is one diagonal, prove that the equation of the other diagonal is, $y(bl - hm) = x(am - hl)$

Q.24 The distance of a point (x_1, y_1) from each of two straight lines which passes through the origin of co-ordinates is δ; find the combined equation of these straight lines.

Q.25 The base of a triangle passes through a fixed point (f, g) & its sides are respectively bisected at right angles by the lines $y^2 - 8xy - 9x^2 = 0$. Determine the locus of its vertex.

EXERCISE–3

Q.1 The graph of the function, $\cos x \cos(x + 2) - \cos^2(x + 1)$ is:
(A) a straight line passing through $(0, -\sin^2 1)$ with slope 2 (B) a straight line passing through $(0, 0)$
(C) a parabola with vertex $(1, -\sin^2 1)$
(D) a straight line passing through the point $\left(\frac{\pi}{2}, -\sin^2 1\right)$ & parallel to the x–axis. [JEE '97, 2]

Q.2 One diagonal of a square is the portion of the line $7x + 5y = 35$ intercepted by the axes, obtain the extremities of the other diagonal. [REE '97, 6]

Q.3 A variable line L passing through the point B $(2, 5)$ intersects the line $2x^2 - 5xy + 2y^2 = 0$ at P & Q. Find the locus of the point R on L such that distances BP, BR & BQ are in harmonic progression. [REE '98, 6]

Q.4**(i)** Select the correct alternative(s): [JEE '98, 2 x 3 = 6]
(a) If P(1, 2), Q(4, 6), R(5, 7) & S(a, b) are the vertices of a parallelogram PQRS, then :
(A) $a = 2, b = 4$ (B) $a = 3, b = 4$ (C) $a = 2, b = 3$ (D) $a = 3, b = 5$

(b) The diagonals of a parallelogram PQRS are along the lines $x + 3y = 4$ and $6x - 2y = 7$. Then PQRS must be a :
(A) rectangle (B) square (C) cyclic quadrilateral (D) rhombus

(c) If the vertices P, Q, R of a triangle PQR are rational points, which of the following points of the triangle PQR is/are always rational point(s)?
(A) centriod (B) incentre (C) circumcentre (D) orthocentre

(ii) Using coordinate geometry, prove that the three altitudes of any triangle are concurrent. [JEE '98, 8]

Q.5 The equation of two equal sides AB and AC of an isosceles triangle ABC are $x + y = 5$ & $7x - y = 3$ respectively. Find the equations of the side BC if the area of the triangle of ABC is 5 units. [REE '99, 6]

Q.6 Let PQR be a right angled isosceles triangle, right angled at P (2, 1). If the equation of the line QR is $2x + y = 3$, then the equation representing the pair of lines PQ and PR is
(A) $3x^2 - 3y^2 + 8xy + 20x + 10y + 25 = 0$ (B) $3x^2 - 3y^2 + 8xy - 20x - 10y + 25 = 0$
(C) $3x^2 - 3y^2 + 8xy + 10x + 15y + 20 = 0$ (D) $3x^2 - 3y^2 - 8xy - 10x - 15y - 20 = 0$ [JEE'99, (2 out of 200)]

Q.7 (a) The incentre of the triangle with vertices $\left(1, \sqrt{3}\right)$, $(0, 0)$ and $(2, 0)$ is :

(A) $\left(1, \frac{\sqrt{3}}{2}\right)$ (B) $\left(\frac{2}{3}, \frac{1}{\sqrt{3}}\right)$ (C) $\left(\frac{2}{3}, \frac{\sqrt{3}}{2}\right)$ (D) $\left(1, \frac{1}{\sqrt{3}}\right)$

(b) Let PS be the median of the triangle with vertices, P(2, 2), Q(6, -1) and R(7, 3). The equation of the line passing through $(1, -1)$ and parallel to PS is :

(A) $2x - 9y - 7 = 0$ (B) $2x - 9y - 11 = 0$
(C) $2x + 9y - 11 = 0$ (D) $2x + 9y + 7 = 0$
[JEE 2000 (Screening) 1 + 1 out of 35]

(c) For points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ of the co-ordinate plane, a new distance d(P, Q) is defined by $d(P, Q) = |x_1 - x_2| + |y_1 - y_2|$. Let O = (0, 0) and A = (3, 2). Prove that the set of points in the first quadrant which are equidistant (with respect to the new distance) from O and A consists of the union of a line segment of finite length and an infinite ray. Sketch this set in a labelled diagram. [JEE 2000 (Mains) 10 out of 100]

Q.8 Find the position of point (4, 1) after it undergoes the following transformations successively.
(i) Reflection about the line, $y = x - 1$ (ii) Translation by one unit along x–axis in the positive direction .
(iii) Rotation through an angle $\frac{\pi}{4}$ about the origin in the anti–clockwise direction.
[REE 2000 (Mains) 3 out of 100]

Q.9 Area of the parallelogram formed by the lines $y = mx$, $y = mx + 1$, $y = nx$ and $y = nx + 1$ equals
(A) $\frac{|m+n|}{(m-n)^2}$ (B) $\frac{2}{|m+n|}$ (C) $\frac{1}{|m+n|}$ (D) $\frac{1}{|m-n|}$ [JEE 2001 (Screening)]

Q.10 (a) Let P = (–1, 0), Q = (0, 0) and $R = (3, 3\sqrt{3})$ be three points. Then the equation of the bisector of the angle PQR is
(A) $\frac{\sqrt{3}}{2}x + y = 0$ (B) $x + \sqrt{3}y = 0$ (C) $\sqrt{3}x + y = 0$ (D) $x + \frac{\sqrt{3}}{2}y = 0$

(b) A straight line through the origin O meets the parallel lines $4x + 2y = 9$ and $2x + y + 6 = 0$ at points P and Q respectively. Then the point O divides the segment PQ in the ratio
(A) 1 : 2 (B) 3 : 4 (C) 2 : 1 (D) 4 : 3

(c) The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is
(A) 1 (B) 2 (C) $2\sqrt{2}$ (D) 4
[JEE 2002 (Screening)]

(d) A straight line L through the origin meets the line $x + y = 1$ and $x + y = 3$ at P and Q respectively. Through P and Q two straight lines L_1 and L_2 are drawn, parallel to $2x - y = 5$ and $3x + y = 5$ respectively. Lines L_1 and L_2 intersect at R. Show that the locus of R, as L varies, is a straight line.
[JEE 2002 (Mains)]

Q.11 The area bounded by the angle bisectors of the lines $x^2 - y^2 + 2y = 1$ and the line $x + y = 3$, is
(A) 2 (B) 3 (C) 4 (D) 6 [JEE 2004 (Screening)]

Q.12 The area of the triangle formed by the intersection of a line parallel to x-axis and passing through P (h, k) with the lines $y = x$ and $x + y = 2$ is $4h^2$. Find the locus of the point P. [JEE 2005, Mains, 2]

EXERCISE–4

Part : (A) Only one correct option

1. The equation of the internal bisector of $\angle BAC$ of ΔABC with vertices A(5, 2), B(2, 3) and C(6, 5) is
(A) $2x + y + 12 = 0$ (B) $x + 2y - 12 = 0$ (C) $2x + y - 12 = 0$ (D) none of these

2. The lines $ax + by + c = 0$, where $3a + 2b + 4c = 0$ are concurrent at the point :
(A) $\left(\frac{1}{2}, \frac{3}{4}\right)$ (B) (1, 3) (C) (3, 1) (D) $\left(\frac{3}{4}, \frac{1}{2}\right)$

3. The equation of second degree $x^2 + 2\sqrt{2}xy + 2y^2 + 4x + 4\sqrt{2}y + 1 = 0$ represents a pair of straight lines. The distance between them is
(A) 4 (B) $\frac{4}{\sqrt{3}}$ (C) 2 (D) $2\sqrt{3}$

4. The straight lines joining the origin to the points of intersection of the line $2x + y = 1$ and curve $3x^2 + 4xy - 4x + 1 = 0$ include an angle :
(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

5. Given the points A (0, 4) and B (0, – 4), the equation of the locus of the point P (x, y) such that $|AP - BP| = 6$ is :
(A) $9x^2 - 7y^2 + 63 = 0$ (B) $9x^2 - 7y^2 - 63 = 0$
(C) $7x^2 - 9y^2 + 63 = 0$ (D) $7x^2 - 9y^2 - 63 = 0$

6. A triangle ABC with vertices A (– 1, 0), B (– 2, 3/4) & C (– 3, – 7/6) has its orthocentre H. Then the orthocentre of triangle BCH will be :
(A) (– 3, – 2) (B) (1, 3) (C) (– 1, 2) (D) none of these

7. Equation of a straight line passing through the origin and making with x – axis an angle twice the size of the angle made by the line y = 0.2 x with the x – axis, is :
(A) y = 0.4 x (B) y = (5/12) x (C) 6y – 5x = 0 (D) none of these

8. A variable straight line passes through a fixed point (a, b) intersecting the co–ordinates axes at A & B. If 'O' is the origin then the locus of the centroid of the triangle OAB is :
(A) $bx + ay - 3xy = 0$ (B) $bx + ay - 2xy = 0$
(C) $ax + by - 3xy = 0$ (D) $ax + by - 2xy = 0$

9. Area of the quadrilateral formed by the lines $|x| + |y| = 2$ is :
(A) 8 (B) 6 (C) 4 (D) none

10. The distance of the point (2, 3) from the line $2x - 3y + 9 = 0$ measured along a line $x - y + 1 = 0$ is :
(A) $5\sqrt{3}$ (B) $4\sqrt{2}$ (C) $3\sqrt{2}$ (D) $2\sqrt{2}$

11. The set of values of 'b' for which the origin and the point (1, 1) lie on the same side of the straight line, $a^2x + aby + 1 = 0 \ \forall \ a \in R, b > 0$ are :
(A) $b \in (2, 4)$ (B) $b \in (0, 2)$ (C) $b \in [0, 2]$ (D) $(2, \infty)$

12. Drawn from the origin are two mutually perpendicular straight lines forming an isosceles triangle together with the straight line, $2x + y = a$. Then the area of the triangle is :

(A) $\frac{a^2}{2}$ (B) $\frac{a^2}{3}$ (C) $\frac{a^2}{5}$ (D) none

13. The line joining two points A (2, 0);B (3, 1) is rotated about A in the anticlock wise direction through an angle of 15º. The equation of the line in the new position is :

(A) $x - \sqrt{3}y - 2 = 0$ (B) $x - 2y - 2 = 0$

(C) $\sqrt{3}x - y - 2\sqrt{3} = 0$ (D) none

14. The line $x + 3y - 2 = 0$ bisects the angle between a pair of straight lines of which one has equation $x - 7y + 5 = 0$. The equation of the other line is :
(A) $3x + 3y - 1 = 0$ (B) $x - 3y + 2 = 0$ (C) $5x + 5y - 3 = 0$ (D) none

15. On the portion of the straight line, $x + 2y = 4$ intercepted between the axes, a square is constructed on the side of the line away from the origin. Then the point of intersection of its diagonals has co–ordinates :
(A) (2, 3) (B) (3, 2) (C) (3, 3) (D) none

16. A light beam emanating from the point A(3, 10) reflects from the straight line $2x + y - 6 = 0$ and then passes through the point B(4, 3). The equation of the reflected beam is :
(A) $3x - y + 1 = 0$ (B) $x + 3y - 13 = 0$ (C) $3x + y - 15 = 0$ (D) $x - 3y + 5 = 0$

17. The equation of the bisector of the angle between two lines $3x - 4y + 12 = 0$ and $12x - 5y + 7 = 0$ which contains the points $(-1, 4)$ is :
(A) $21x + 27y - 121 = 0$ (B) $21x - 27y + 121 = 0$

(C) $21x + 27y + 191 = 0$ (D) $\frac{-3x+4y-12}{5} = \frac{12x-5y+7}{13}$

18. The equation of bisectors of two lines L_1 & L_2 are $2x - 16y - 5 = 0$ and $64x + 8y + 35 = 0$. If the line L_1 passes through $(-11, 4)$, the equation of acute angle bisector of L_1 & L_2 is :
(A) $2x - 16y - 5 = 0$ (B) $64x + 8y + 35 = 0$ (C) data insufficient (D) none of these

19. The equation of the pair of bisectors of the angles between two straight lines is, $12x^2 - 7xy - 12y^2 = 0$. If the equation of one line is $2y - x = 0$ then the equation of the other line is :
(A) $41x - 38y = 0$ (B) $38x - 41y = 0$ (C) $38x + 41y = 0$ (D) $41x + 38y = 0$

20. If the straight lines joining the origin and the points of intersection of the curve $5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ and $x + ky - 1 = 0$ are equally inclined to the x-axis then the value of k is equal to :
(A) 1 (B) -1 (C) 2 (D) 3

21. If the points of intersection of curves $C_1 = \lambda x^2 + 4y^2 - 2xy - 9x + 3$ & $C_2 = 2x^2 + 3y^2 - 4xy + 3x - 1$ subtends a right angle at origin, then the value of λ is :
(A) 19 (B) 9 (C) -19 (D) -9

Part : (B) May have more than one options correct

22. The equation of the bisectors of the angle between the two intersecting lines :
$\frac{x-3}{\cos\theta} = \frac{y+5}{\sin\theta}$ and $\frac{x-3}{\cos\phi} = \frac{y+5}{\sin\phi}$ are $\frac{x-3}{\cos\alpha} = \frac{y+5}{\sin\alpha}$ and $\frac{x-3}{\beta} = \frac{y+5}{\gamma}$ then

(A) $\alpha = \frac{\theta+\phi}{2}$ (B) $\beta = -\sin\alpha$ (C) $\gamma = \cos\alpha$ (D) $\beta = \sin\alpha$

23. Equation of a straight line passing through the point of intersection of $x - y + 1 = 0$ and $3x + y - 5 = 0$ are perpendicular to one of them is
(A) $x + y + 3 = 0$ (B) $x + y - 3 = 0$ (C) $x - 3y - 5 = 0$ (D) $x - 3y + 5 = 0$

24. Three lines $px + qy + r = 0$, $qx + ry + p = 0$ and $rx + py + q = 0$ are concurrent if
(A) $p + q + r = 0$ (B) $p^2 + q^2 + r^2 = pq + qr + rp$
(C) $p^3 + p^3 + r^3 = 3pqr$ (D) none of these

25. Equation of a straight line passing through the point (4, 5) and equally inclined to the lines, $3x = 4y + 7$ and $5y = 12x + 6$ is
(A) $9x - 7y = 1$ (B) $9x + 7y = 71$ (C) $7x + 9y = 73$ (D) $7x - 9y + 17 = 0$

26. If the equation, $2x^2 + kxy - 3y^2 - x - 4y - 1 = 0$ represents a pair of lines then the value of k can be:
(A) 1 (B) 5 (C) -1 (D) -5

27. If $a^2 + 9b^2 - 4c^2 = 6ab$ then the family of lines $ax + by + c = 0$ are concurrent at :
(A) (1/2, 3/2) (B) $(-1/2, -3/2)$ (C) $(-1/2, 3/2)$ (D) $(1/2, -3/2)$

EXERCISE–5

1. If the points (x_1, y_1), (x_2, y_2) and (x_3, y_3) be collinear, show that $\frac{y_2 - y_3}{x_2x_3} + \frac{y_3 - y_1}{x_3x_1} + \frac{y_1 - y_2}{x_1x_2} = 0$.

2. Find the length of the perpendicular from the origin upon the straight line joining the two points whose coordinates are $(a\cos\alpha, a\sin\alpha)$ and $(a\cos\beta, a\sin\beta)$.

3. Show that the product of the perpendiculars drawn from the two points $(\pm\sqrt{a^2 - b^2}, 0)$ upon the straight line $\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$ is b^2.

4. Find the equation of the bisector of the acute angle between the lines $3x - 4y + 7 = 0$ and $12x + 5y - 2 = 0$.

5. Find the equation to the pair of straight lines joining the origin to the intersections of the straight line $y = mx + c$ and the curve $x^2 + y^2 = a^2$. Prove that they are at right angles if $2c^2 = a^2(1 + m^2)$.

6. The variable line $x\cos\theta + y\sin\theta = 2$ cuts the x and y axes at A and B respectively. Find the locus of the vertex P of the rectangle OAPB, O being the origin.

7. If $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are the vertices of the triangle then show that :

(i) The median through A can be written in the form $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$

(ii) the line through A & parallel to BC can be written in the form ; $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} - \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$

(iii) equation to the angle bisector through A is $b\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + c\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$

where b = AC & c = AB.

8. Is there a real value of λ for which the image of the point $(\lambda, \lambda - 1)$ by the line mirror $3x + y = 6\lambda$ is the point $(\lambda^2 + 1, \lambda)$? If so, find λ.

9. If the straight lines, $ax + by + p = 0$ & $x\cos\alpha + y\sin\alpha - p = 0$ enclose an angle $\pi/4$ between them, and meet the straight line $x\sin\alpha - y\cos\alpha = 0$ in the same point, then find the value of $a^2 + b^2$.

10. Drive the conditions to be imposed on β so that $(0, \beta)$ should lie on or inside the triangle having sides $y + 3x + 2 = 0$, $3y - 2x - 5 = 0$ & $4y + x - 14 = 0$.

11. A straight line L is perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line L & the coordinate axes is 5. Find the equation of the line.

12. Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$ and its third side passes through the point $(1, -10)$. Determine the equation of the third side.

13. Find the equations of the straight lines passing through the point (1, 1) and parallel to the lines represented by the equation, $x^2 - 5xy + 4y^2 + x + 2y - 2 = 0$.

14. Find the coordinates of the vertices of a square inscribed in the triangle with vertices A (0, 0), B (2, 1), C (3, 0); given that two of its vertices are on the side AC.

15. The equations of perpendiculars of the sides AB & AC of Δ ABC are $x - y - 4 = 0$ and $2x - y - 5 = 0$ respectively. If the vertex A is $(-2, 3)$ and point of intersection of perpendiculars bisectors $\left(\frac{3}{2}, \frac{5}{2}\right)$ is, find the equation of medians to the sides AB and AC respectively.

16. The sides of a triangle are $4x + 3y + 7 = 0$, $5x + 12y = 27$ and $3x + 4y + 8 = 0$. Find the equations of the internal bisectors of the angles and show that they are concurrent.

17. A ray of light is sent along the line $x - 2y - 3 = 0$. Upon reaching the line $3x - 2y - 5 = 0$, the ray is reflected from it. Find the equation of the line containing the reflected ray.

18. A triangle is formed by the lines whose equations are AB : $x + y - 5 = 0$, BC : $x + 7y - 7 = 0$ and CA : $7x + y + 14 = 0$. Find the bisector of the interior angle at B and the exterior angle at C. Determine the nature of the interior angle at A and find the equation of the bisector.

19. Find the equations of the sides of a triangle having $(4, -1)$ as a vertex, if the lines $x - 1 = 0$ and $x - y - 1 = 0$ are the equations of two internal bisectors of its angles.

20. The equations of the altitudes AD, BE, CF of a triangle ABC are $x + y = 0$, $x - 4y = 0$ and $2x - y = 0$ respectively. Thr coordinates of A are $(t, -t)$. Find coordinates of B and C. Prove that it t varies the locus of the centroid of the triangle ABC is $x + 5y = 0$.

21. For points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ of the co-ordinate plane, a new distance d(P, Q) is defined by $d(P, Q) = |x_1 - x_2| + |y_1 - y_2|$. Let O = (0, 0) and A = (3, 2). Prove that the set of points in the first quadrant which are equidistant (with respect to the new distance) from O and A consists of the union of a line segment of finite length and an infinite ray. Sketch this set in a labelled diagram. **[IIT -2000, 10]**

22. Let ABC and PQR be any two triangles in the same plane. Assume that the prependiculars from the points A, B, C to the sides QR, RP, PQ respectively are concurrent. Using vector methods or otherwise, prove that the prependiculars from P, Q, R to BC, CA, AB respectively are also concurrent.
[IIT - 2000, 10]

23. A straight line L through the origin meets the lines $x + y = 1$ and $x + y = 3$ at P and Q respectively. Through P and Q two straight lines L_1 and L_2 are drawn parallel to $2x - y = 5$ and $3x + y = 5$ respectively. Lines L_1 and L_2 intersect at R. Show that the locus of R, as L varies, is a straight line.
[IIT - 2002, 5]

24. A straight line L with negative slope passes through the point (8, 2) and cuts the positive coordinate axes at points P and Q. Find the absolute minimum value of OP + OQ, as L varies, where O is the origin.
[IIT - 2002, 5]

25. The area of the triangle formed by the intersection of a line parallel to x-axis and passing through P(h, k) with the lines $y = x$ and $x + y = 2$ is $4h^2$. Find the locus of the point P. **[IIT - 2005, 2]**

ANSWER
EXERCISE–1

Q 1. $90°$ **Q 2.** $\left(\frac{3}{7}, \frac{22}{7}\right)$ **Q 3.** $(33, 26)$ **Q 4.** $K = 7$ or $\frac{31}{9}$

Q 5. $1:2$; $Q(-5, -3)$ **Q 6.** $83x - 35y + 92 = 0$ **Q 7.** $2x + y - 1 = 0$

Q 8. $\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(-\frac{3}{2}, \frac{3}{2}\right)$ **Q 10.** $x - y = 0$ **Q 11.** 91 sq.units

Q 12. 6 units **Q 13.** $a^2 + b^2 = a'^2 + b'^2$; $(a + a')x + (b + b')y + (c + c') = 0$; $(a - a')x + (b - b')y = 0$

Q 15. $c = -4$; $B(2, 0)$; $D(4, 4)$ **Q 16.** $x + 5y + 5\sqrt{2} = 0$ or $x + 5y - 5\sqrt{2} = 0$

Q 17. $x - 3y - 31 = 0$ or $3x + y + 7 = 0$ **Q 18.** $2x + 2y + \sqrt{2} = 0$ **Q 19.** $-9°, -81°$

Q 20. $5x - 2y - 7 = 0$ **Q.21** $x - 5 = 0$

Q23. $-\frac{3}{2} < \alpha < -1 \cup \frac{1}{2} < \alpha < 1$ **Q 25.** $x + y = 1$; $x + 9y = 21$ **Q 26.** $0 < \theta < \frac{5\pi}{6} - \tan^{-1} 3$

Q 27. (–1, 1) ; (4, 1) **Q 28.** $9x + 46y + 83 = 0$ **Q29.** (1, –2) , yes $\left(\frac{1}{3}, -\frac{2}{3}\right)$

EXERCISE–2

Q.1 $x + 4y = 4$; $5x + 2y = 8$ **Q 2.** $52x + 89y + 519 = 0$ **Q.4** (2, 0)

Q.5 $7x + 24y + 182 = 0$ or $x = -2$ **Q.7** (0, 0) or $\left(0, \frac{5}{2}\right)$ **Q.8** $14x + 23y = 40$

Q.9 $x - 5y = 0$ or $5x + y = 0$, Area = 2.77 sq.units
Q.10 $3x + 6y - 16 = 0$; $8x + 8y + 7 = 0$; $12x + 6y - 11 = 0$
Q.11 $x^2 + 4y^2 + 4xy + 4x - 2y - 1 = 0$ **Q.12** $(a + b)(ax + by) = ab(a + b - 2h)$

Q.13 $\frac{3}{2}$ sq. units, $\left(3, 3, \frac{3}{4}\right)$, isosceles **Q.14** $6x^2 - xy - y^2 - x - 12y - 35 = 0$

Q.15 $2x - y + 3 = 0, 2x + y - 7 = 0, x - 2y - 6 = 0$ **Q.16** $y^2 = a(x - 3a)$

Q.21 $B\left(-\frac{2t}{3}, -\frac{t}{6}\right), C\left(\frac{t}{2}, t\right)$

Q.24 $(y_1^2 - \delta^2)x^2 - 2x_1y_1xy + (x_1^2 - \delta^2)y^2 = 0$ **Q.25** $4(x^2 + y^2) + (4g + 5f)x + (4f - 5g)y = 0$

EXERCISE–3

Q.1 D **Q.2** (–1, 1) & (6, 6)
Q.3 $17x - 10y = 0$ **Q.4 (i) (a)** C **(b)** D **(c)** A, C, D
Q.5 $x - 3y + 21 = 0$, $x - 3y + 1 = 0$, $3x + y = 12$, $3x + y = 2$ **Q.6** B

Q.7 (a) D **(b)** D **Q.8** $(4, 1) \to (2, 3) \to (3, 3) \to \left(0, 3\sqrt{2}\right)$

Q.9 D **Q.10 (a)** C ; **(b)** B ; **(c)** B ; **(d)** $x - 3y + 5 = 0$
Q.11 A **Q.12** $y = 2x + 1, y = -2x + 1$

EXERCISE–4

1. C **2.** D **3.** C **4.** A **5.** A **6.** D **7.** B

8. A **9.** A **10.** B **11.** B **12.** C **13.** C **14.** C

15. C **16.** B **17.** A **18.** A **19.** A **20.** B **21.** C

22. ABC **23.** BD **24.** ABC **25.** AC **26.** AD

27. CD

EXERCISE–5

2. $a\cos\left(\frac{\alpha - \beta}{2}\right)$ **4.** $11x - 3y + 9 = 0$

6. $\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{4}$ **8.** 2 **9.** 2

10. $5/3 \le \beta \le 7/2$

11. $x + 5y + 5\sqrt{2} = 0$ or $x + 5y - 5\sqrt{2} = 0$

12. $x - 3y - 31 = 0$ or $3x + y + 7 = 0$

13. $(x - 4y + 3)(x - y) = 0$
or $x^2 - 5xy + 4y^2 + 3x - 3y = 0$

14. $\left(\frac{3}{2}, 0\right)\left(\frac{9}{4}, 0\right), \left(\frac{3}{2}, \frac{3}{4}\right), \left(\frac{9}{4}, \frac{3}{4}\right)$

15. $x + 4y = 4; 5x + 2y = 8$

17. $29x - 2y = 31$

18. $3x + 6y - 16 = 0; 8x + 8y + 7 = 0; 12x + 6y - 11 = 0$

19. $2x - y + 3 = 0, 2x + y - 7 = 0; x - 2y - 6 = 0$

20. $B\left(-\frac{2t}{3}, -\frac{t}{6}\right), C\left(\frac{t}{2}, t\right)$

24. 18 **25.** $y = 2x + 1$ or $y = -2x + 1$

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 12 XI M 12. Circles

Index:

1. Key Concepts

2. Exercise I to V

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Circle Theory

A circle is a locus of a point whose distance from a fixed point (called centre) is always constant (called radius).

1. Equation of a Circle in Various Form:

(a) The circle with centre as origin & radius 'r' has the equation; $x^2 + y^2 = r^2$.

(b) The circle with centre (h, k) & radius 'r' has the equation; $(x - h)^2 + (y - k)^2 = r^2$.

(c) The general equation of a circle is

$x^2 + y^2 + 2gx + 2fy + c = 0$

with centre as $(-g, -f)$ & radius = $\sqrt{g^2+f^2-c}$. If:

$g^2 + f^2 - c > 0 \Rightarrow$ real circle.

$g^2 + f^2 - c = 0 \Rightarrow$ point circle.

$g^2 + f^2 - c < 0 \Rightarrow$ imaginary circle, with real centre, that is $(-g, -f)$

Note : that every second degree equation in x & y, in which coefficient of x^2 is equal to coefficient of y^2 & the coefficient of xy is zero, always represents a circle.

(d) The equation of circle with (x_1, y_1) & (x_2, y_2) as extremeties of its diameter is:

$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.

Note that this will be the circle of least radius passing through (x_1, y_1) & (x_2, y_2).

Example : Find the equation of the circle whose centre is (1, –2) and radius is 4.

Solution : The equation of the circle is $(x - 1)^2 + (y - (-2))^2 = 4^2$

$\Rightarrow \quad (x - 1)^2 + (y + 2)^2 = 16$

$\Rightarrow \quad x^2 + y^2 - 2x + 4y - 11 = 0$ **Ans.**

Example : Find the equation of the circle which passes through the point of intersection of the lines $3x - 2y - 1 = 0$ and $4x + y - 27 = 0$ and whose centre is (2, – 3).

Solution : Let P be the point of intersection of the lines AB and LM whose equations are respectively

$3x - 2y - 1 = 0$ .....(i)

and $4x + y - 27 = 0$ .....(ii)

Solving (i) and (ii), we get x = 5, y = 7. So, coordinates of P are (5, 7). Let C(2, –3) be the centre of the circle. Since the circle passes through P, therefore

CP = radius $\Rightarrow \sqrt{(5-2)^2+(7+3)^2}$ = radius $\Rightarrow$ radius = $\sqrt{109}$.

Hence the equation of the required circle is

$(x - 2)^2 + (y + 3)^2 = \left(\sqrt{109}\right)^2$

Example : Find the centre & radius of the circle whose equation is $x^2 + y^2 - 4x + 6y + 12 = 0$

Solution : Comparing it with the general equation $x^2 + y^2 + 2gx + 2fy + c = 0$, we have

$2g = -4 \Rightarrow g = -2$

$2f = 6 \Rightarrow f = 3$

& $c = 12$

$\therefore$ centre is $(-g, -f)$ i.e. (2, –3)

and radius = $\sqrt{g^2+f^2-c} = \sqrt{(-2)^2+(3)^2-12} = 1$

Example : Find the equation of the circle, the coordinates of the end points of whose diameter are (–1, 2) and (4, –3)

Solution : We know that the equation of the circle described on the line segment joining (x_1, y_1) and (x_2, y_2) as a diameter is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.

Here, $x_1 = -1$, $x_2 = 4$, $y_1 = 2$ and $y_2 = -3$.

So, the equation of the required circle is

$(x + 1)(x - 4) + (y - 2)(y + 3) = 0 \Rightarrow x^2 + y^2 - 3x + y - 10 = 0$.

Self Practice Problems :

1. Find the equation of the circle passing through the point of intersection of the lines $x + 3y = 0$ and $2x - 7y = 0$ and whose centre is the point of intersection of the lines $x + y + 1 = 0$ and $x - 2y + 4 = 0$.
Ans. $x^2 + y^2 + 4x - 2y = 0$

2. Find the equation of the circle whose centre is (1, 2) and which passes through the point (4, 6)
Ans. $x^2 + y^2 - 2x - 4y - 20 = 0$

3. Find the equation of a circle whose radius is 6 and the centre is at the origin.
Ans. $x^2 + y^2 = 36$.

2. Intercepts made by a Circle on the Axes:

The intercepts made by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on the co–ordinate axes are $2\sqrt{g^2-c}$ & $2\sqrt{f^2-c}$ respectively. If

$g^2 - c > 0 \Rightarrow$ circle cuts the x axis at two distinct points.

$g^2 = c \Rightarrow$ circle touches the x–axis.

$g^2 < c \Rightarrow$ circle lies completely above or below the x–axis.

Example : Find the equation to the circle touching the y-axis at a distance – 3 from the origin and intercepting a length 8 on the x-axis.

Solution : Let the equation of the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$. Since it touches y-axis at (0, –3) and (0, – 3) lies on the circle.

$\therefore$ $c = f^2$ ...(i) $9 - 6f + c = 0$ .....(ii)

From (i) and (ii), we get $9 - 6f + f^2 = 0 \Rightarrow (f - 3)^2 = 0 \Rightarrow f = 3$.

Putting $f = 3$ in (i) we obtain $c = 9$.

It is given that the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ intercepts length 8 on x-axis

$\therefore$ $2\sqrt{g^2 - c} = 8 \Rightarrow 2\sqrt{g^2 - 9} = 8 \Rightarrow g^2 - 9 = 16 \Rightarrow g = \pm 5$

Hence, the required circle is $x^2 + y^2 \pm 10x + 6y + 9 = 0$.

Self Practice Problems :

1. Find the equation of a circle which touches the axis of y at a distance 3 from the origin and intercepts a distance 6 on the axis of x.

Ans. $x^2 + y^2 \pm 6\sqrt{2}\, x - 6y + 9 = 0$

2. Find the equation of a circle which touches y-axis at a distance of 2 units from the origin and cuts an intercept of 3 units with the positive direction of x-axis.

Ans. $x^2 + y^2 \pm 5x - 4y + 4 = 0$

3. Parametric Equations of a Circle:

The parametric equations of $(x - h)^2 + (y - k)^2 = r^2$ are: $x = h + r\cos\theta$; $y = k + r\sin\theta$; $-\pi < \theta \le \pi$ where (h, k) is the centre, r is the radius & θ is a parameter.

Example : Find the parametric equations of the circle $x^2 + y^2 - 4x - 2y + 1 = 0$

Solution : We have : $x^2 + y^2 - 4x - 2y + 1 = 0 \Rightarrow (x^2 - 4x) + (y^2 - 2y) = -1$

$\Rightarrow (x - 2)^2 + (y - 1)^2 = 2^2$

So, the parametric equations of this circle are

$x = 2 + 2\cos\theta$, $y = 1 + 2\sin\theta$.

Example : Find the equations of the following curves in cartesian form. Also, find the centre and radius of the circle $x = a + c\cos\theta$, $y = b + c\sin\theta$

Solution : We have : $x = a + c\cos\theta$, $y = b + c\sin\theta \Rightarrow \cos\theta = \dfrac{x-a}{c}$, $\sin\theta = \dfrac{y-b}{c}$

$\Rightarrow \left(\dfrac{x-a}{c}\right)^2 + \left(\dfrac{y-b}{c}\right)^2 = \cos^2\theta + \sin^2\theta \Rightarrow (x - a)^2 + (y - b)^2 = c^2$

Clearly, it is a circle with centre at (a, b) and radius c.

Self Practice Problems :

1. Find the parametric equations of circle $x^2 + y^2 - 6x + 4y - 12 = 0$

Ans. $x = 3 + 5\cos\theta$, $y = -2 + 5\sin\theta$

2. Find the cartesian equations of the curve $x = -2 + 3\cos\theta$, $y = 3 + 3\sin\theta$

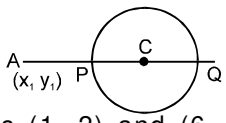
Ans. $(x + 2)^2 + (y - 3)^2 = 9$

4. Position of a point with respect to a circle:

The point (x_1, y_1) is inside, on or outside the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$.

according as $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c <$, = or > 0.

NOTE : The greatest & the least distance of a point A from a circle with centre C & radius r is AC+r& AC – r respectively.



Example : Discuss the position of the points (1, 2) and (6, 0) with respect to the circle $x^2 + y^2 - 4x + 2y - 11 = 0$

Solution : We have $x^2 + y^2 - 4x + 2y - 11 = 0$ or $S = 0$, where $S = x^2 + y^2 - 4x + 2y - 11$.

For the point (1, 2), we have $S_1 = 1^2 + 2^2 - 4 \times 1 + 2 \times 2 - 11 < 0$

For the point (6, 0), we have $S_2 = 6^2 + 0^2 - 4 \times 6 + 2 \times 0 - 11 > 0$

Hence, the point (1, 2) lies inside the circle and the point (6, 0) lies outside the circle.

Self Practice Problem :

1. How are the points (0, 1) (3, 1) and (1, 3) situated with respect to the circle $x^2 + y^2 - 2x - 4y + 3 = 0$?

Ans. (0, 1) lies on the circle ; (3, 1) lies outside the circle ; (1, 3) lies inside the circle.

5. Line and a Circle:

Let $L = 0$ be a line & $S = 0$ be a circle. If r is the radius of the circle & p is the length of the perpendicular from the centre on the line, then:

(i) $p > r \Leftrightarrow$ the line does not meet the circle i. e. passes out side the circle.

(ii) $p = r \Leftrightarrow$ the line touches the circle. (It is tangent to the circle)

(iii) $p < r \Leftrightarrow$ the line is a secant of the circle.

(iv) $p = 0 \Rightarrow$ the line is a diameter of the circle.

Also, if $y = mx + c$ is line and $x^2 + y^2 = a^2$ is circle then

(i) $c^2 > a^2(1 + m^2) \Leftrightarrow$ the line is a secant of the circle.

(ii) $c^2 = a^2(1 + m^2) \Leftrightarrow$ the line touches the circle. (It is tangent to the circle)

(iii) $c^2 < a^2(1 + m^2) \Leftrightarrow$ the line does not meet the circle i. e. passes out side the circle.

Example : For what value of c will the line $y = 2x + c$ be a tangent to the circle $x^2 + y^2 = 5$?

Solution : We have : $y = 2x + c$ or $2x - y + c = 0$ .....(i) and $x^2 + y^2 = 5$ .....(ii)

If the line (i) touches the circle (ii), then

length of the $\perp$ from the centre (0, 0) = radius of circle (ii)

$\Rightarrow \left|\dfrac{2\times 0 - 0 + c}{\sqrt{2^2 + (-1)^2}}\right| = \sqrt{5} \Rightarrow \left|\dfrac{c}{\sqrt{5}}\right| = \sqrt{5}$

$\Rightarrow \quad \frac{c}{\sqrt{5}} = \pm \sqrt{5} \quad \Rightarrow \quad c = \pm 5$

Hence, the line (i) touches the circle (ii) for $c = \pm 5$

Self Practice Problem :

1. For what value of λ, does the line $3x + 4y = \lambda$ touch the circle $x^2 + y^2 = 10x$. **Ans.** 40, –10

6. Tangent :

(a) Slope form :

$y = mx + c$ is always a tangent to the circle $x^2 + y^2 = a^2$ if $c^2 = a^2(1 + m^2)$. Hence, equation of tangent is $y = mx \pm a\sqrt{1+m^2}$ and the point of contact is $\left(-\frac{a^2 m}{c}, \frac{a^2}{c}\right)$.

(b) Point form :

(i) The equation of the tangent to the circle $x^2 + y^2 = a^2$ at its point (x_1, y_1) is, $x x_1 + y y_1 = a^2$.

(ii) The equation of the tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at its point (x_1, y_1) is: $xx_1 + yy_1 + g(x+x_1) + f(y+y_1) + c = 0$.

NOTE : In general the equation of tangent to any second degree curve at point (x_1, y_1) on it can be obtained by replacing x^2 by $x\,x_1$, y^2 by yy_1, x by $\frac{x+x_1}{2}$, y by $\frac{y+y_1}{2}$, xy by $\frac{x_1 y + x y_1}{2}$ and c remains as c.

(c) Parametric form :

The equation of a tangent to circle $x^2 + y^2 = a^2$ at $(a\cos\alpha, a\sin\alpha)$ is $x\cos\alpha + y\sin\alpha = a$.

NOTE : The point of intersection of the tangents at the points $P(\alpha)$ & $Q(\beta)$ is $\left(\frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{a\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$

Example : Find the equation of the tangent to the circle $x^2 + y^2 - 30x + 6y + 109 = 0$ at (4, –1).

Solution : Equation of tangent is

$$4x + (-y) - 30\left(\frac{x+4}{2}\right) + 6\left(\frac{y+(-1)}{2}\right) + 109 = 0$$

or $4x - y - 15x - 60 + 3y - 3 + 109 = 0$ or $-11x + 2y + 46 = 0$

or $11x - 2y - 46 = 0$

Hence, the required equation of the tangent is $11x - 2y - 46 = 0$

Example : Find the equation of tangents to the circle $x^2 + y^2 - 6x + 4y - 12 = 0$ which are parallel to the line $4x + 3y + 5 = 0$

Solution : Given circle is $x^2 + y^2 - 6x + 4y - 12 = 0$ .....(i)

and given line is $4x + 3y + 5 = 0$ .....(ii)

Centre of circle (i) is (3, –2) and its radius is 5. Equation of any line $4x + 3y + k = 0$ parallel to the line (ii) .....(iii)

If line (iii) is tangent to circle, (i) then

$$\frac{|4.3 + 3(-2) + k|}{\sqrt{4^2 + 3^2}} = 5 \text{ or } |6 + k| = 25$$

or $6 + k = \pm 25 \quad \therefore \quad k = 19, -31$

Hence equation of required tangents are $4x + 3y + 19 = 0$ and $4x + 3y - 31 = 0$

Self Practice Problem :

1. Find the equation of the tangents to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ which are (i) parallel, (ii) perpendicular to the line $3x - 4y - 1 = 0$

Ans. (i) $3x - 4y + 20 = 0$ and $3x - 4y - 10 = 0$ (ii) $4x + 3y + 5 = 0$ and $4x + 3y - 25 = 0$

7. Normal :

If a line is normal / orthogonal to a circle then it must pass through the centre of the circle. Using this fact normal to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at (x_1, y_1) is; $y - y_1 = \frac{y_1 + f}{x_1 + g}(x - x_1)$.

Exercise : Find the equation of the normal to the circle $x^2 + y^2 - 5x + 2y - 48 = 0$ at the point (5, 6).

Solution : The equation of the tangent to the circle $x^2 + y^2 - 5x + 2y - 48 = 0$ at (5, 6) is

$$5x + 6y - 5\left(\frac{x+5}{2}\right) + 2\left(\frac{x+6}{2}\right) - 48 = 0 \Rightarrow 10x + 12y - 5x - 25 + 2y + 12 - 96 = 0$$

$$\Rightarrow 5x + 14y - 109 = 0$$

$\therefore$ Slope of the tangent $= -\frac{5}{14} \Rightarrow$ Slope of the normal $= \frac{14}{5}$

Hence, the equation of the normal at (5, 6) is

$y - 6 = (14/5)(x - 5) \Rightarrow 14x - 5y - 40 = 0$

Self Practice Problem :

1. Find the equation of the normal to the circle $x^2 + y^2 - 2x - 4y + 3 = 0$ at the point (2, 3).

Ans. $x - y + 1 = 0$

8. Pair of Tangents from a Point:

The equation of a pair of tangents drawn from the point A (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is : $SS_1 = T^2$.

Where $S \equiv x^2 + y^2 + 2gx + 2fy + c$; $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

$T \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$.

Ex. : Find the equation of the pair of tangents drawn to the circle $x^2 + y^2 - 2x + 4y = 0$ from the point (0, 1)

Solution : Given circle is $S = x^2 + y^2 - 2x + 4y = 0$ .....(i)

Let $P \equiv (0, 1)$

For point P, $S_1 = 0^2 + 1^2 - 2.0 + 4.1 = 5$

Clearly P lies outside the circle

and $T \equiv x \cdot 0 + y \cdot 1 - (x + 0) + 2(y + 1)$

i.e. $T \equiv -x + 3y + 2$.

Now equation of pair of tangents from P(0, 1) to circle (1) is $SS_1 = T^2$

or $5(x^2 + y^2 - 2x + 4y) = (-x + 3y + 2)^2$

or $5x^2 + 5y^2 - 10x + 20y = x^2 + 9y^2 + 4 - 6xy - 4x + 12y$

or $4x^2 - 4y^2 - 6x + 8y + 6xy - 4 = 0$

or $2x^2 - 2y^2 + 3xy - 3x + 4y - 2 = 0$ .....(ii)

Note : Separate equation of pair of tangents : From (ii), $2x^2 + 3(y - 1)x - 2(2y^2 - 4y + 2) = 0$

$\therefore \quad x = \dfrac{3(y-1) \pm \sqrt{9(y-1)^2 + 8(2y^2 - 4y + 2)}}{4}$

or $4x - 3y + 3 = \pm \sqrt{25y^2 - 50y + 25} = \pm 5(y - 1)$

$\therefore$ Separate equations of tangents are $x - 2y + 2 = 0$ and $2x + y - 1 = 0$

Self Practice Problems :

1. Find the equation of the tangents through (7, 1) to the circle $x^2 + y^2 = 25$.
Ans. $12x^2 - 12y^2 + 7xy - 175x - 25y + 625 = 0$

9. Length of a Tangent and Power of a Point:

The length of a tangent from an external point (x_1, y_1) to the circle

$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ is given by $L = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2f_1y + c} = \sqrt{S_1}$.

Square of length of the tangent from the point P is also called the power of point w.r.t. a circle.

Power of a point w.r.t. a circle remains constant.

Power of a point P is positive, negative or zero according as the point 'P' is outside, inside or on the circle respectively.

Exercise : Find the length of the tangent drawn from the point (5, 1) to the circle $x^2 + y^2 + 6x - 4y - 3 = 0$

Solution : Given circle is $x^2 + y^2 + 6x - 4y - 3 = 0$ .....(i)

Given point is (5, 1). Let P = (5, 1)

Now length of the tangent from P(5, 1) to circle (i) = $\sqrt{5^2 + 1^2 + 6.5 - 4.1 - 3} = 7$

Self Practice Problems :

1. Find the area of the quadrilateral formed by a pair of tangents from the point (4, 5) to the circle $x^2 + y^2 - 4x - 2y - 11 = 0$ and a pair of its radii. **Ans.** 8 sq. units

2. If the length of the tangent from a point (f, g) to the circle $x^2 + y^2 = 4$ be four times the length of the tangent from it to the circle $x^2 + y^2 = 4x$, show that $15f^2 + 15g^2 - 64f + 4 = 0$

10. Director Circle:

The locus of the point of intersection of two perpendicular tangents is called the director circle of the given circle. The director circle of a circle is the concentric circle having radius equal to $\sqrt{2}$ times the original circle.

Example : Find the equation of director circle of the circle $(x - 2)^2 + (y + 1)^2 = 2$.

Solution : Centre & radius of given circle are (2, –1) & $\sqrt{2}$ respectively.

Centre and radius of the director circle will be (2, –1) & $\sqrt{2} \times \sqrt{2} = 2$ respectively.

$\therefore$ equation of director circle is $(x - 2)^2 + (y + 1)^2 = 4$

$\Rightarrow \quad x^2 + y^2 - 4x + 2y + 1 = 0$ **Ans.**

Self Practice Problems :

1. Find the equation of director circle of the circle whose diameters are $2x - 3y + 12 = 0$ and $x + 4y - 5 = 0$ and area is 154 square units. **Ans.** $(x + 3)^2 + (y + 2)^2 = 98$

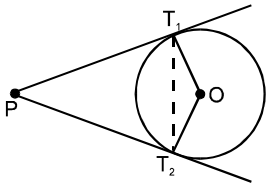
11. Chord of Contact:

If two tangents PT_1 & PT_2 are drawn from the point $P(x_1, y_1)$ to the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, then the equation of the chord of contact T_1T_2 is: $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$.

NOTE : Here R = radius; L = length of tangent.

(a) Chord of contact exists only if the point 'P' is not inside.

(b) Length of chord of contact $T_1T_2 = \dfrac{2LR}{\sqrt{R^2 + L^2}}$.



(c) Area of the triangle formed by the pair of the tangents & its chord of contact = $\dfrac{RL^3}{R^2 + L^2}$

(d) Tangent of the angle between the pair of tangents from $(x_1, y_1) = \left(\dfrac{2RL}{L^2 - R^2}\right)$

(e) Equation of the circle circumscribing the triangle PT_1T_2 is:
$(x - x_1)(x + g) + (y - y_1)(y + f) = 0$.

Example : Find the equation of the chord of contact of the tangents drawn from (1, 2) to the circle $x^2 + y^2 - 2x + 4y + 7 = 0$

Solution : Given circle is $x^2 + y^2 - 2x + 4y + 7 = 0$ .....(i)

Let P = (1, 2)

For point P (1, 2), $x^2 + y^2 - 2x + 4y + 7 = 1 + 4 - 2 + 8 + 7 = 18 > 0$

Hence point P lies outside the circle

For point P (1, 2), $T = x \cdot 1 + y \cdot 2 - (x + 1) + 2(y + 2) + 7$

i.e. $T = 4y + 10$

Now equation of the chord of contact of point P(1, 2) w.r.t. circle (i) will be

$4y + 10 = 0$ or $2y + 5 = 0$

Example : Tangents are drawn to the circle $x^2 + y^2 = 12$ at the points where it is met by the circle $x^2 + y^2 - 5x + 3y - 2 = 0$; find the point of intersection of these tangents.

Solution : Given circles are $S_1 \equiv x^2 + y^2 - 12 = 0$ ..... (i)

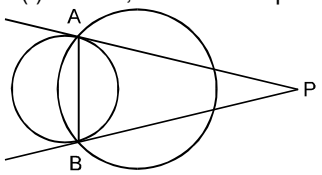
and $S_2 = x^2 + y^2 - 5x + 3y - 2 = 0$ ..... (ii)

Now equation of common chord of circle (i) and (ii) is

$S_1 - S_2 = 0$ i.e. $5x - 3y - 10 = 0$ ..... (iii)

Let this line meet circle (i) [or (ii)] at A and B

Let the tangents to circle (i) at A and B meet at P(α, β), then AB will be the chord of contact of the tangents to the circle (i) from P, therefore equation of AB will be



$x\alpha + y\beta - 12 = 0$ ..... (iv)

Now lines (iii) and (iv) are same, therefore, equations (iii) and (iv) are identical

$\therefore \quad \frac{\alpha}{5} = \frac{\beta}{-3} = \frac{-12}{-10} \quad \therefore \quad \alpha = 6, \beta = -\frac{18}{5}$

Hence P = $\left(6, -\frac{18}{5}\right)$

Self Practice Problems :

1. Find the co-ordinates of the point of intersection of tangents at the points where the line $2x + y + 12 = 0$ meets the circle $x^2 + y^2 - 4x + 3y - 1 = 0$ **Ans.** $(1, -2)$

2. Find the area of the triangle formed by the tangents drawn from the point (4, 6) to the circle $x^2 + y^2 = 25$ and their chord of contact. **Ans.** $\frac{405\sqrt{3}}{52}$; $4x + 6y - 25 = 0$

12. Pole and Polar:

(i) If through a point P in the plane of the circle there be drawn any straight line to meet the circle in Q and R, the locus of the point of intersection of the tangents at Q & R is called the Polar of the point P; also P is called the Pole of the Polar.

(ii) The equation to the polar of a point P (x_1, y_1) w.r.t. the circle $x^2 + y^2 = a^2$ is given by $xx_1 + yy_1 = a^2$, & if the circle is general then the equation of the polar becomes $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$ i.e. $T = 0$. Note that if the point (x_1, y_1) be on the circle then the tangent & polar will be represented by the same equation. Similarly if the point (x_1, y_1) be outside the circle then the chord of contact & polar will be represented by the same equation.

(iii) Pole of a given line $Ax + By + C = 0$ w.r.t. circle $x^2 + y^2 = a^2$ is $\left(-\frac{Aa^2}{C}, -\frac{Ba^2}{C}\right)$.

(iv) If the polar of a point P pass through a point Q, then the polar of Q passes through P.

(v) Two lines L_1 & L_2 are conjugate of each other if Pole of L_1 lies on L_2 & vice versa. Similarly two points P & Q are said to be conjugate of each other if the polar of P passes through Q & vice-versa.

Example : Find the equation of the polar of the point (2, –1) with respect to the circle $x^2 + y^2 - 3x + 4y - 8 = 0$

Solution : Given circle is $x^2 + y^2 - 3x + 4y - 8 = 0$ .....(i)

Given point is (2, –1) let P = (2, –1). Now equation of the polar of point P with respect to circle (i)

$$x.2 + y(-1) - 3\left(\frac{x+2}{2}\right) + 4\left(\frac{y-1}{2}\right) - 8 = 0$$

or $4x - 2y - 3x - 6 + 4y - 4 - 16 = 0$ or $x + 2y - 26 = 0$

Example : Find the pole of the line $3x + 5y + 17 = 0$ with respect to the circle $x^2 + y^2 + 4x + 6y + 9 = 0$

Solution : Given circle is $x^2 + y^2 + 4x + 6y + 9 = 0$ .....(i)

and given line is $3x + 5y + 17 = 0$ .....(ii)

Let P(α, β) be the pole of line (ii) with respect to circle (i)

Now equation of polar of point P(α, β) with respect to circle (i) is

$x\alpha + y\beta + 2(x + \alpha) + 3(y + \beta) + 9 = 0$

or $(\alpha + 2)x + (\beta + 3)y + 2\alpha + 3\beta + 9 = 0$ .....(iii)

Now lines (ii) and (iii) are same, therefore,

$$\frac{\alpha+2}{3} = \frac{\beta+3}{5} = \frac{2\alpha+3\beta+9}{17}$$
(i) (ii) (iii)

From (i) and (ii), we get

$5\alpha + 10 = 3\beta + 9$ or $5\alpha - 3\beta = -1$ .....(iv)

From (i) and (iii), we get

$17\alpha + 34 = 6\alpha + 9\beta + 27$ or $11\alpha - 9\beta = -7$ .....(v)

Solving (iv) & (v), we get $\alpha = 1$, $\beta = 2$ Hence required pole is (1, 2).

Self Practice Problems :

1. Find the co-ordinates of the point of intersection of tangents at the points where the line $2x + y + 12 = 0$ meets the circle $x^2 + y^2 - 4x + 3y - 1 = 0$. **Ans.** $(1, -2)$

2. Find the pole of the straight line $2x - y + 10 = 0$ with respect to the circle $x^2 + y^2 - 7x + 5y - 1 = 0$

Ans. $\left(\frac{3}{2}, \frac{3}{2}\right)$

13. Equation of the Chord with a given Middle Point:

The equation of the chord of the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ in terms of its mid point $M(x_1, y_1)$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$ which is designated by $T = S_1$.

NOTE : (i) The shortest chord of a circle passing through a point 'M' inside the circle is one chord whose middle point is M.

(ii) The chord passing through a point 'M' inside the circle and which is at a maximum distance from the centre is a chord with middle point M.

Ex. : Find the equation of the chord of the circle $x^2 + y^2 + 6x + 8y - 11 = 0$, whose middle point is $(1, -1)$

Solution : Equation of given circle is $S \equiv x^2 + y^2 + 6x + 8y - 11 = 0$

Let $L \equiv (1, -1)$

For point $L(1, -1)$, $S_1 = 1^2 + (-1)^2 + 6.1 + 8(-1) - 11 = -11$ and

$T \equiv x.1 + y(-1) + 3(x + 1) + 4(y - 1) - 11$ i.e. $T \equiv 4x + 3y - 12$

Now equation of the chord of circle (i) whose middle point is $L(1, -1)$ is

$T = S_1$ or $4x + 3y - 12 = -11$ or $4x + 3y - 1 = 0$

Second Method : Let C be the centre of the given circle, then$C \equiv (-3, -4)$. $L \equiv (1, -1)$ slope of $CL = \frac{-4+1}{-3-1} = \frac{3}{4}$

$\therefore$ Equation of chord of circle whose middle point is L, is

$\therefore$ $y + 1 = -\frac{4}{3}(x - 1)$ [$\because$ chord is perpendicular to CL)

or $4x + 3y - 1 = 0$

Self Practice Problems :

1. Find the equation of that chord of the circle $x^2 + y^2 = 15$, which is bisected at (3, 2) **Ans.**$3x + 2y - 13 = 0$

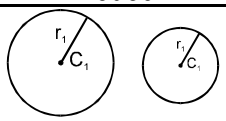
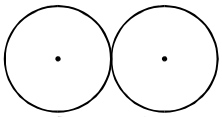
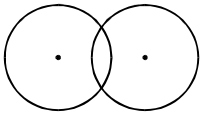
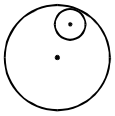
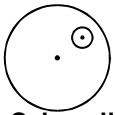
2. Find the co-ordinates of the middle point of the chord which the circle $x^2 + y^2 + 4x - 2y - 3 = 0$ cuts off on the line $y = x + 2$. **Ans.** $\left(-\frac{3}{2}, \frac{1}{2}\right)$

14. Equation of the chord joining two points of circle :

The equation of chord PQ to the circle $x^2 + y^2 = a^2$ joining two points $P(\alpha)$ and (β) on it is given by. The equation of a straight line joining two point α & β on the circle $x^2 + y^2 = a^2$ is

$$x \cos\frac{\alpha+\beta}{2} + y \sin\frac{\alpha+\beta}{2} = a \cos\frac{\alpha-\beta}{2}.$$

15. Common Tangents to two Circles:

Case		Number of Tangents	Condition
(i)		4 common tangents (2 direct and 2 transverse)	$r_1 + r_2 < c_1 c_2$.
(ii)		3 common tangents.	$r_1 + r_2 = c_1 c_2$.
(iii)		2 common tangents.	$\lvert r_1 - r_2 \rvert < c_1 c_2 < r_1 + r_2$
(iv)		1 common tangent.	$\lvert r_1 - r_2 \rvert = c_1 c_2$.
(v)		No common tangent.	$c_1 c_2 < \lvert r_1 - r_2 \rvert$.

(Here C_1C_2 is distance between centres of two circles.)

IMPORTANT NOTE :

(i) The direct common tangents meet at a point which divides the line joining centre of circles externally in the ratio of their radii.
Transverse common tangents meet at a point which divides the line joining centre of circles internally in the ratio of their radii.

(ii) Length of an external (or direct) common tangent & internal (or transverse) common tangent to the two circles are given by: $L_{ext} = \sqrt{d^2-(r_1-r_2)^2}$ & $L_{int} = \sqrt{d^2-(r_1+r_2)^2}$,
where d = distance between the centres of the two circles and r_1, r_2 are the radii of the two circles. Note that length of internal common tangent is always less than the length of the external or direct common tangent.

Example: Examine if the two circles $x^2 + y^2 - 2x - 4y = 0$ and $x^2 + y^2 - 8y - 4 = 0$ touch each other externally or internally.

Solution : Given circles are $x^2 + y^2 - 2x - 4y = 0$ .....(i)
and $x^2 + y^2 - 8y - 4 = 0$ .....(ii)
Let A and B be the centres and r_1 and r_2 the radii of circles (i) and (ii) respectively, then
$A \equiv (1, 2)$, $B \equiv (0, 4)$, $r_1 = \sqrt{5}$, $r_2 = 2\sqrt{5}$

Now $AB = \sqrt{(1-0)^2+(2-4)^2} = \sqrt{5}$ and $r_1 + r_2 = 3\sqrt{5}$, $|r_1 - r_2| = \sqrt{5}$
Thus $AB = |r_1 - r_2|$, hence the two circles touch each other internally.

Self Practice Problems :

1. Find the position of the circles $x^2 + y^2 - 2x - 6y + 9 = 0$ and $x^2 + y^2 + 6x - 2y + 1 = 0$ with respect to each other.
Ans. One circle lies completely outside the other circle.

16. Orthogonality Of Two Circles:

Two circles $S_1 = 0$ & $S_2 = 0$ are said to be orthogonal or said to intersect orthogonally if the tangents at their point of intersection include a right angle. The condition for two circles to be orthogonal is: $2 g_1 g_2 + 2 f_1 f_2 = c_1 + c_2$.

NOTE : (a) The centre of a variable circle orthogonal to two fixed circles lies on the radical axis of two circles.
(b) If two circles are orthogonal, then the polar of a point 'P' on first circle w.r.t. the second circle passes through the point Q which is the other end of the diameter through P. Hence locus of a point which moves such that its polars w.r.t. the circles $S_1 = 0$, $S_2 = 0$ & $S_3 = 0$ are concurrent in a circle which is orthogonal to all the three circles.
(c) The centre of a circle which is orthogonal to three given circles is the radical centre provided the radical centre lies outside all the three circles.

Example : Obtain the equation of the circle orthogonal to both the circles $x^2 + y^2 + 3x - 5y + 6 = 0$ and $4x^2 + 4y^2 - 28x + 29 = 0$ and whose centre lies on the line $3x + 4y + 1 = 0$.

Solution. Given circles are $x^2 + y^2 + 3x - 5y + 6 = 0$ .....(i)
and $4x^2 + 4y^2 - 28x + 29 = 0$

or $x^2 + y^2 - 7x + \frac{29}{4} = 0$. .....(ii)

Let the required circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ .....(iii)
Since circle (iii) cuts circles (i) and (ii) orthogonally

$\therefore$ $2g\left(\frac{3}{2}\right) + 2f\left(-\frac{5}{2}\right) = c + 6$ or $3g - 5f = c + 6$ .....(iv)

and $2g\left(-\frac{7}{2}\right) + 2f.0 = c + \frac{29}{4}$ or $-7g = c + \frac{29}{4}$ .....(v)

From (iv) & (v), we get $10g - 5f = -\frac{5}{4}$

or $40g - 20f = -5$. .....(vi)
Given line is $3x + 4y = -1$ .....(vii)
Since centre $(-g, -f)$ of circle (iii) lies on line (vii),
$\therefore$ $-3g - 4g = -1$ .....(viii)

Solving (vi) & (viii), we get $g = 0$, $f = \frac{1}{4}$

$\therefore$ from (5), $c = -\frac{29}{4}$
$\therefore$ from (iii), required circle is

$x^2 + y^2 + \frac{1}{2}y - \frac{29}{4} = 0$ or $4(x^2 + y^2) + 2y - 29 = 0$

Self Practice Problems :

1. For what value of k the circles $x^2 + y^2 + 5x + 3y + 7 = 0$ and $x^2 + y^2 - 8x + 6y + k = 0$ cut orthogonally.
Ans. -18

2. Find the equation to the circle which passes through the origin and has its centre on the line $x + y + 4 = 0$ and cuts the circle $x^2 + y^2 - 4x + 2y + 4 = 0$ orthogonally.
Ans. $3x^2 + 3y^2 + 4x + 20y = 0$

17. Radical Axis and Radical Centre:

The radical axis of two circles is the locus of points whose powers w.r.t. the two circles are equal. The equation of radical axis of the two circles $S_1 = 0$ & $S_2 = 0$ is given by

$S_1 - S_2 = 0$ i.e. $2(g_1 - g_2)x + 2(f_1 - f_2)y + (c_1 - c_2) = 0$.
The common point of intersection of the radical axes of three circles taken two at a time is called the radical centre of three circles. Note that the length of tangents from radical centre to the three circles are equal.

NOTE:

(a) If two circles intersect, then the radical axis is the common chord of the two circles.

(b) If two circles touch each other then the radical axis is the common tangent of the two circles at the common point of contact.

(c) Radical axis is always perpendicular to the line joining the centres of the two circles.

(d) Radical axis will pass through the mid point of the line joining the centres of the two circles only if the two circles have equal radii.

(e) Radical axis bisects a common tangent between the two circles.

(f) A system of circles, every two which have the same radical axis, is called a coaxal system.

(g) Pairs of circles which do not have radical axis are concentric.

Example : Find the co-ordinates of the point from which the lengths of the tangents to the following three circles be equal.

$3x^2 + 3y^2 + 4x - 6y - 1 = 0$
$2x^2 + 2y^2 - 3x - 2y - 4 = 0$
$2x^2 + 2y^2 - x + y - 1 = 0$

Solution : Here we have to find the radical centre of the three circles. First reduce them to standard form in which coefficients of x^2 and y^2 be each unity. Subtracting in pairs the three radical axes are

$$\frac{17}{6}x - y + \frac{5}{3} = 0 \quad ; \quad -x - \frac{3}{2}y - \frac{3}{2} = 0$$

$$-\frac{11}{6}x + \frac{5}{2}y - \frac{1}{6} = 0.$$

solving any two, we get the point $\left(-\frac{16}{21}, \frac{31}{63}\right)$ which satisfies the third also. This point is called the radical centre and by definition the length of the tangents from it to the three circles are equal.

Self Practice Problem :

1. Find the point from which the tangents to the three circles $x^2 + y^2 - 4x + 7 = 0$, $2x^2 + 2y^2 - 3x + 5y + 9 = 0$ and $x^2 + y^2 + y = 0$ are equal in length. Find also this length.
Ans. $(2, -1)$; 2.

18. Family of Circles:

(a) The equation of the family of circles passing through the points of intersection of two circles $S_1 = 0$ & $S_2 = 0$ is : $S_1 + K S_2 = 0$
($K \neq -1$ provided the co–efficient of x^2 & y^2 in S_1 & S_2 are same)

(b) The equation of the family of circles passing through the point of intersection of a circle $S = 0$ & a line $L = 0$ is given by $S + KL = 0$.

(c) The equation of a family of circles passing through two given points (x_1, y_1) & (x_2, y_2) can be written in the form:

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + K\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$ where K is a parameter.

(d) The equation of a family of circles touching a fixed line $y - y_1 = m(x - x_1)$ at the fixed point (x_1, y_1) is $(x - x_1)^2 + (y - y_1)^2 + K[y - y_1 - m(x - x_1)] = 0$, where K is a parameter.

(e) Family of circles circumscribing a triangle whose sides are given by $L_1 = 0$; $L_2 = 0$ and $L_3 = 0$ is given by; $L_1L_2 + \lambda L_2L_3 + \mu L_3L_1 = 0$ provided co–efficient of $xy = 0$ and co–efficient of x^2 = co–efficient of y^2.

(f) Equation of circle circumscribing a quadrilateral whose side in order are represented by the lines $L_1 = 0$, $L_2 = 0$, $L_3 = 0$ & $L_4 = 0$ are $u L_1L_3 + \lambda L_2L_4 = 0$ where values of u & λ can be found out by using condition that co–efficient of x^2 = co–efficient of y^2 and co–efficient of $xy = 0$.

Example : Find the equations of the circles passing through the points of intersection of the circles $x^2 + y^2 - 2x - 4y - 4 = 0$ and $x^2 + y^2 - 10x - 12y + 40 = 0$ and whose radius is 4.

Solution : Any circle through the intersection of given circles is $S_1 + \lambda S_2 = 0$

or $(x^2 + y^2 - 2x - 4y - 4) + l(x^2 + y^2 - 10x - 12y + 40) = 0$

or $(x^2 + y^2) - 2\frac{(1+5\lambda)}{1+\lambda}x - 2\frac{(2+6\lambda)}{1+\lambda}y + \frac{40\lambda - 4}{1+\lambda} = 0$ .....(i)

$r = \sqrt{g^2 + f^2 - c} = 4$, given

$\therefore \quad 16 = \frac{(1+5\lambda)^2}{(1+\lambda)^2} + \frac{(2+6\lambda)^2}{(1+\lambda)^2} - \frac{40\lambda - 4}{1+\lambda}$

$16(1 + 2\lambda + \lambda^2) = 1 + 10\lambda + 25\lambda^2 + 4 + 24\lambda + 36\lambda^2 - 40\lambda^2 - 40\lambda + 4 + 4\lambda$

or $16 + 32\lambda + 16\lambda^2 = 21\lambda^2 - 2\lambda + 9$ or $5\lambda^2 - 34\lambda - 7 = 0$

$\therefore \quad (\lambda - 7)(5\lambda + 1) = 0$ $\therefore \quad \lambda = 7, -1/5$

Putting the values of λ in (i) the required circles are

$2x^2 + 2y^2 - 18x - 22y + 69 = 0$ and $x^2 + y^2 - 2y - 15 = 0$

Example : Find the equations of circles which touche $2x - y + 3 = 0$ and pass through the points of intersection of the line $x + 2y - 1 = 0$ and the circle $x^2 + y^2 - 2x + 1 = 0$.

Solution : The required circle by $S + \lambda P = 0$ is

$x^2 + y^2 - 2x + 1 + \lambda (x + 2y - 1) = 0$

or $x^2 + y^2 - x (2 - \lambda) + 2\lambda y + (1 - \lambda) = 0$

centre $(-g, -f)$ is $[\{2 - \lambda)/2, -\lambda]$

$r = \sqrt{g^2 + f^2 - c}$

$= \sqrt{(2-\lambda)^2/4 + \lambda^2 - (1-\lambda)}$ $= \frac{1}{2}\sqrt{5\lambda^2} = (\lambda/2)\sqrt{5}$.

Since the circle touches the line $2x - y + 3 = 0$ therefore perpendicular from centre is equal to radius $\frac{2.[(2-\lambda)/2] - (-\lambda) + 3}{\pm\sqrt{5}} = \frac{\lambda}{2}\sqrt{5}$. or $5 = \pm \frac{\lambda}{2}.5$ $\therefore$ $\lambda = \pm 2$

Puttin the values of λ in (i) the required circles are

$x^2 + y^2 + 4y - 1 = 0$ and $x^2 + y^2 - 4x - 4y + 3 = 0$.

Example : Find the equation of circle pasing through the points A(1, 1) & B(2, 2) and whose radiu is 1.

Solution : Equation of AB is $x - y = 0$ $\therefore$ equation of circle is

$(x - 1)(x - 2) + (y - 1)(y - 2) + \lambda(x - y) = 0$ or $x^2 + y^2 + (\lambda - 3)x - (\lambda + 3)y + 4 = 0$

radius $= \sqrt{\frac{(\lambda-3)^2}{4} + \frac{(\lambda+3)^2}{4} - 4}$

But radius = 1 (given) $\therefore$ $\sqrt{\frac{(\lambda-3)^2}{4} + \frac{(\lambda+3)^2}{4} - 4} = 1$

or $(\lambda - 3)^2 + (\lambda + 3)^2 - 16 = 4$. or $2\lambda^2 = 2$ or $\lambda = \pm 1$

$\therefore$ equation of circle is

$x^2 + y^2 - 2x - 4y + 4 = 0$ & $x^2 + y^2 - 4x - 2y + 4 = 0$ **Ans.**

Example : Find the equation of the circle passing through the point (2, 1) and touching the line $x + 2y - 1 = 0$ at the point $(3, -1)$.

Solution : Equation of circle is

$(x - 3)^2 + (y + 1)^2 + \lambda(x + 2y - 1) = 0$

Since it passes through the point (2, 1)

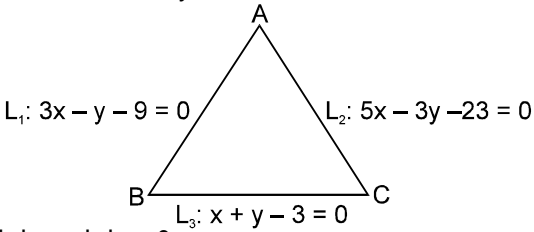
$1 + 4 + \lambda (2 + 2 - 1) = 0$ $\Rightarrow$ $\lambda = -5/3$

$\therefore$ circle is

$(x - 3)^2 + (y + 1)^2 - \frac{5}{3}(x + 2y - 1) = 0 \Rightarrow$ $3x^2 + 3y^2 - 23x - 4y + 35 = 0$ **Ans.**

Example : Find the equation of circle circumcscribing the triangle whose sides are $3x - y - 9 = 0$, $5x - 3y - 23 = 0$ & $x + y - 3 = 0$.

Solution :



$L_1L_2 + \lambda L_2L_3 + \mu L_1L_3 = 0$

$(3x - y - 9)(5x - 3y - 23) + \lambda(5x - 3y - 23)(x + y - 3) + \mu(3x - y - 9)(x + y - 3) = 0$

$(15x^2 + 3y^2 - 14xy - 114x + 50y + 207) + \lambda(5x^2 - 3y^2 + 2xy - 38x - 14y + 69)$

$+ \mu(3x^2 - y^2 + 2xy - 18x - 6y + 27) = 0$

$(5\lambda + 3\mu + 15)x^2 + (3 - 3\lambda - \mu)y^2 + xy(2\lambda + 2\mu - 14) - x(114 + 38\lambda + 18\mu) + y(50 - 14\lambda - 6\mu)$

$+ (207 + 69\lambda + 27\mu) = 0$ .....(i)

coefficient of x^2 = coefficient of y^2

$\Rightarrow$ $5\lambda + 3\mu + 15 = 3 - 3\lambda - \mu$

$8\lambda + 4\mu + 12 = 0$

$2\lambda + \mu + 3 = 0$ .....(ii)

coefficient of $xy = 0$ $\Rightarrow$ $2\lambda + 2\mu - 14 = 0$

$\Rightarrow$ $\lambda + \mu - 7 = 0$ .....(iii)

Solving (ii) and (iii), we have

$\lambda = -10$, $\mu = 17$

Puting these values of λ & μ in equation (i), we get

$2x^2 + 2y^2 - 5x + 11y - 3 = 0$

Self Practice Problems :

1. Find the equation of the circle passing through the points of intersection of the circles $x^2 + y^2 - 6x + 2y + 4 = 0$ and $x^2 + y^2 + 2x - 4y - 6 = 0$ and with its centre on the line $y = x$.

Ans. $7x^2 + 7y^2 - 10x - 10y - 12 = 0$

2. Find the equation of circle circumcribing the quadrilateral whose sides are $5x + 3y = 9$, $x = 3y$, $2x = y$ and $x + 4y + 2 = 0$.

Ans. $9x^2 + 9y^2 - 20x + 15y = 0$.

SHORT REVISION

STANDARD RESULTS :

1. EQUATION OF A CIRCLE IN VARIOUS FORM :

(a) The circle with centre (h, k) & radius 'r' has the equation ;
$(x-h)^2 + (y-k)^2 = r^2$.

(b) The general equation of a circle is $x^2 + y^2 + 2gx + 2fy + c = 0$ with centre as :
$(-g, -f)$ & radius $= \sqrt{g^2+f^2-c}$.

Remember that every second degree equation in x & y in which coefficient of x^2 = coefficient of y^2 & there is no xy term always represents a circle.

If $g^2 + f^2 - c > 0 \Rightarrow$ real circle.
$g^2 + f^2 - c = 0 \Rightarrow$ point circle.
$g^2 + f^2 - c < 0 \Rightarrow$ imaginary circle.

Note that the general equation of a circle contains three arbitrary constants, g, f & c which corresponds to the fact that a unique circle passes through three non collinear points.

(c) The equation of circle with (x_1, y_1) & (x_2, y_2) as its diameter is :
$(x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0$.

Note that this will be the circle of least radius passing through (x_1, y_1) & (x_2, y_2).

2. INTERCEPTS MADE BY A CIRCLE ON THE AXES :

The intercepts made by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on the co-ordinate axes are $2\sqrt{g^2-c}$ & $2\sqrt{f^2-c}$ respectively.

NOTE :

If $g^2 - c > 0 \Rightarrow$ circle cuts the x axis at two distinct points.

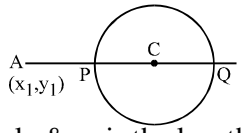
If $g^2 = c \Rightarrow$ circle touches the x-axis.

If $g^2 < c \Rightarrow$ circle lies completely above or below the x-axis.

3. POSITION OF A POINT w.r.t. A CIRCLE :

The point (x_1, y_1) is inside, on or outside the circle $x^2 + y^2 + 2gx + 2fy + c = 0$.
according as $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c \Leftrightarrow 0$.

Note : The greatest & the least distance of a point A from a circle with centre C & radius r is AC + r & AC − r respectively.



4. LINE & A CIRCLE :

Let L = 0 be a line & S = 0 be a circle. If r is the radius of the circle & p is the length of the perpendicular from the centre on the line, then :

(i) $p > r \Leftrightarrow$ the line does not meet the circle i. e. passes out side the circle.

(ii) $p = r \Leftrightarrow$ the line touches the circle.

(iii) $p < r \Leftrightarrow$ the line is a secant of the circle.

(iv) $p = 0 \Rightarrow$ the line is a diameter of the circle.

5. PARAMETRIC EQUATIONS OF A CIRCLE :

The parametric equations of $(x-h)^2 + (y-k)^2 = r^2$ are :
$x = h + r\cos\theta$; $y = k + r\sin\theta$; $-\pi < \theta \le \pi$ where (h, k) is the centre, r is the radius & θ is a parameter.

Note that equation of a straight line joining two point α & β on the circle $x^2 + y^2 = a^2$ is
$x\cos\frac{\alpha+\beta}{2} + y\sin\frac{\alpha+\beta}{2} = a\cos\frac{\alpha-\beta}{2}$.

6. TANGENT & NORMAL :

(a) The equation of the tangent to the circle $x^2 + y^2 = a^2$ at its point (x_1, y_1) is,
$xx_1 + yy_1 = a^2$. Hence equation of a tangent at $(a\cos\alpha, a\sin\alpha)$ is ;
$x\cos\alpha + y\sin\alpha = a$. The point of intersection of the tangents at the points $P(\alpha)$ and $Q(\beta)$ is
$\frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{a\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}$.

(b) The equation of the tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at its point (x_1, y_1) is
$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$.

(c) $y = mx + c$ is always a tangent to the circle $x^2 + y^2 = a^2$ if $c^2 = a^2(1 + m^2)$ and the point of contact is $\left(-\frac{a^2m}{c}, \frac{a^2}{c}\right)$.

(d) If a line is normal/orthogonal to a circle then it must pass through the centre of the circle. Using this fact normal to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at (x_1, y_1) is

$y - y_1 = \frac{y_1 + f}{x_1 + g}(x - x_1)$.

7. A FAMILY OF CIRCLES :

(a) The equation of the family of circles passing through the points of intersection of two circles $S_1 = 0$ & $S_2 = 0$ is : $S_1 + K S_2 = 0 \quad (K \neq -1)$.

(b) The equation of the family of circles passing through the point of intersection of a circle $S = 0$ & a line $L = 0$ is given by $S + KL = 0$.

(c) The equation of a family of circles passing through two given points (x_1, y_1) & (x_2, y_2) can be written in the form :

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + K \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$ where K is a parameter.

(d) The equation of a family of circles touching a fixed line $y - y_1 = m(x - x_1)$ at the fixed point (x_1, y_1) is $(x - x_1)^2 + (y - y_1)^2 + K[y - y_1 - m(x - x_1)] = 0$, where K is a parameter.
In case the line through (x_1, y_1) is parallel to y - axis the equation of the family of circles touching it at (x_1, y_1) becomes $(x - x_1)^2 + (y - y_1)^2 + K(x - x_1) = 0$.
Also if line is parallel to x - axis the equation of the family of circles touching it at (x_1, y_1) becomes $(x - x_1)^2 + (y - y_1)^2 + K(y - y_1) = 0$.

(e) Equation of circle circumscribing a triangle whose sides are given by $L_1 = 0$; $L_2 = 0$ & $L_3 = 0$ is given by ; $L_1L_2 + \lambda L_2L_3 + \mu L_3L_1 = 0$ provided co-efficient of $xy = 0$ & co-efficient of x^2 = co-efficient of y^2.

(f) Equation of circle circumscribing a quadrilateral whose side in order are represented by the lines $L_1 = 0$, $L_2 = 0$, $L_3 = 0$ & $L_4 = 0$ is $L_1L_3 + \lambda L_2L_4 = 0$ provided co-efficient of x^2 = co-efficient of y^2 and co-efficient of $xy = 0$.

8. LENGTH OF A TANGENT AND POWER OF A POINT :

The length of a tangent from an external point (x_1, y_1) to the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ is given by $L = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2f_1y + c} = \sqrt{S_1}$.
Square of length of the tangent from the point P is also called **THE POWER OF POINT** w.r.t. a circle. Power of a point remains constant w.r.t. a circle.
Note that : power of a point P is positive, negative or zero according as the point 'P' is outside, inside or on the circle respectively.

9. DIRECTOR CIRCLE :

The locus of the point of intersection of two perpendicular tangents is called the **DIRECTOR CIRCLE** of the given circle. The director circle of a circle is the concentric circle having radius equal to $\sqrt{2}$ times the original circle.

10. EQUATION OF THE CHORD WITH A GIVEN MIDDLE POINT :

The equation of the chord of the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ in terms of its mid point $M(x_1, y_1)$ is $y - y_1 = -\frac{x_1 + g}{y_1 + f}(x - x_1)$. This on simplication can be put in the form $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$
which is designated by $T = S_1$.
Note that : the shortest chord of a circle passing through a point 'M' inside the circle, is one chord whose middle point is M.

11. CHORD OF CONTACT :

If two tangents PT_1 & PT_2 are drawn from the point $P(x_1, y_1)$ to the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, then the equation of the chord of contact T_1T_2 is : $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$.

REMEMBER :

(a) Chord of contact exists only if the point 'P' is not inside .

(b) Length of chord of contact $T_1T_2 = \frac{2LR}{\sqrt{R^2 + L^2}}$.

(c) Area of the triangle formed by the pair of the tangents & its chord of contact $= \frac{RL^3}{R^2 + L^2}$
Where R is the radius of the circle & L is the length of the tangent from (x_1, y_1) on $S = 0$.

(d) Angle between the pair of tangents from $(x_1, y_1) = \tan^{-1}\left(\frac{2RL}{L^2 - R^2}\right)$

where R = radius ; L = length of tangent.

(e) Equation of the circle circumscribing the triangle PT_1T_2 is :
$(x - x_1)(x + g) + (y - y_1)(y + f) = 0$.

(f) The joint equation of a pair of tangents drawn from the point $A(x_1, y_1)$ to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is : $SS_1 = T^2$.
Where $S \equiv x^2 + y^2 + 2gx + 2fy + c$; $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$
$T \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$.

12. POLE & POLAR :

(i) If through a point P in the plane of the circle, there be drawn any straight line to meet the circle in Q and R, the locus of the point of intersection of the tangents at Q & R is called the **POLAR OF THE POINT P** ; also P is called the **POLE OF THE POLAR**.

(ii) The equation to the polar of a point $P(x_1, y_1)$ w.r.t. the circle $x^2 + y^2 = a^2$ is given by $\mathbf{xx_1 + yy_1 = a^2}$, & if the circle is general then the equation of the polar becomes $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$. Note that if the point (x_1, y_1) be on the circle then the chord of contact, tangent & polar will be represented by the same equation.

(iii) Pole of a given line $Ax + By + C = 0$ w.r.t. any circle $x^2 + y^2 = a^2$ is $\left(-\frac{Aa^2}{C}, -\frac{Ba^2}{C}\right)$.

(iv) If the polar of a point P pass through a point Q, then the polar of Q passes through P.

(v) Two lines L_1 & L_2 are conjugate of each other if Pole of L_1 lies on L_2 & vice versa Similarly two points P & Q are said to be conjugate of each other if the polar of P passes through Q & vice-versa.

13. COMMON TANGENTS TO TWO CIRCLES :

(i) Where the two circles neither intersect nor touch each other, there are FOUR common tangents, two of them are transverse & the others are direct common tangents.

(ii) When they intersect there are two common tangents, both of them being direct.

(iii) When they touch each other :

(a) **EXTERNALLY :** there are three common tangents, two direct and one is the tangent at the point of contact .

(b) **INTERNALLY :** only one common tangent possible at their point of contact.

(iv) Length of an external common tangent & internal common tangent to the two circles is given by:

$L_{ext} = \sqrt{d^2 - (r_1 - r_2)^2}$ & $L_{int} = \sqrt{d^2 - (r_1 + r_2)^2}$.

Where d = distance between the centres of the two circles . r_1 & r_2 are the radii of the two circles.

(v) The direct common tangents meet at a point which divides the line joining centre of circles externally in the ratio of their radii.
Transverse common tangents meet at a point which divides the line joining centre of circles internally in the ratio of their radii.

14. RADICAL AXIS & RADICAL CENTRE :

The radical axis of two circles is the locus of points whose powers w.r.t. the two circles are equal. The equation of radical axis of the two circles $S_1 = 0$ & $S_2 = 0$ is given ;
$S_1 - S_2 = 0$ i.e. $2(g_1 - g_2)x + 2(f_1 - f_2)y + (c_1 - c_2) = 0$.

NOTE THAT :

(a) If two circles intersect, then the radical axis is the common chord of the two circles.

(b) If two circles touch each other then the radical axis is the common tangent of the two circles at the common point of contact.

(c) Radical axis is always perpendicular to the line joining the centres of the two circles.

(d) Radical axis need not always pass through the mid point of the line joining the centres of the two circles.

(e) Radical axis bisects a common tangent between the two circles.

(f) The common point of intersection of the radical axes of three circles taken two at a time is called the radical centre of three circles.

(g) A system of circles, every two which have the same radical axis, is called a coaxal system.

(h) Pairs of circles which do not have radical axis are concentric.

15. ORTHOGONALITY OF TWO CIRCLES :

Two circles $S_1 = 0$ & $S_2 = 0$ are said to be orthogonal or said to intersect orthogonally if the tangents at their point of intersection include a right angle. The condition for two circles to be orthogonal is : $2g_1g_2 + 2f_1f_2 = c_1 + c_2$.

Note : (a) Locus of the centre of a variable circle orthogonal to two fixed circles is the radical axis between the

two fixed circles .

(b) If two circles are orthogonal, then the polar of a point 'P' on first circle w.r.t. the second circle passes through the point Q which is the other end of the diameter through P . Hence locus of a point which moves such that its polars w.r.t. the circles $S_1 = 0$, $S_2 = 0$ & $S_3 = 0$ are concurrent in a circle which is orthogonal to all the three circles.

EXERCISE–I

Q.1 Determine the nature of the quadrilateral formed by four lines $3x + 4y - 5 = 0$; $4x - 3y - 5 = 0$; $3x + 4y + 5 = 0$ and $4x - 3y + 5 = 0$. Find the equation of the circle inscribed and circumscribing this quadrilateral.

Q.2 Suppose the equation of the circle which touches both the coordinate axes and passes through the point with abscissa -2 and ordinate 1 has the equation $x^2 + y^2 + Ax + By + C = 0$, find all the possible ordered triplet (A, B, C).

Q.3 A circle $S = 0$ is drawn with its centre at $(-1, 1)$ so as to touch the circle $x^2 + y^2 - 4x + 6y - 3 = 0$ externally. Find the intercept made by the circle $S = 0$ on the coordinate axes.

Q.4 The line $lx + my + n = 0$ intersects the curve $ax^2 + 2hxy + by^2 = 1$ at the point P and Q. The circle on PQ as diameter passes through the origin. Prove that $n^2(a^2 + b^2) = l^2 + m^2$.

Q.5 One of the diameters of the circle circumscribing the rectangle ABCD is $4y = x + 7$. If A & B are the points $(-3, 4)$ & $(5,4)$ respectively, then find the area of the rectangle.

Q.6 Find the equation to the circle which is such that the length of the tangents to it from the points $(1, 0)$, $(2, 0)$ and $(3, 2)$ are 1, $\sqrt{7}$, $\sqrt{2}$ respectively.

Q.7 A circle passes through the points $(-1, 1)$, $(0, 6)$ and $(5, 5)$. Find the points on the circle the tangents at which are parallel to the straight line joining origin to the centre.

Q.8 Find the equations of straight lines which pass through the intersection of the lines $x - 2y - 5 = 0$, $7x + y = 50$ & divide the circumference of the circle $x^2 + y^2 = 100$ into two arcs whose lengths are in the ratio 2 : 1.

Q.9 A $(-a, 0)$; B $(a, 0)$ are fixed points. C is a point which divides AB in a constant ratio $\tan\alpha$. If AC & CB subtend equal angles at P, prove that the equation of the locus of P is $x^2 + y^2 + 2ax \sec 2\alpha + a^2 = 0$.

Q.10 A circle is drawn with its centre on the line $x + y = 2$ to touch the line $4x - 3y + 4 = 0$ and pass through the point $(0, 1)$. Find its equation.

Q.11(a) Find the area of an equilateral triangle inscribed in the circle $x^2 + y^2 + 2gx + 2fy + c = 0$.

(b) If the line $x \sin\alpha - y + a \sec\alpha = 0$ touches the circle with radius 'a' and centre at the origin then find the most general values of 'α' and sum of the values of 'α' lying in $[0, 100\pi]$.

Q.12 A point moving around circle $(x + 4)^2 + (y + 2)^2 = 25$ with centre C broke away from it either at the point A or point B on the circle and moved along a tangent to the circle passing through the point D $(3, -3)$. Find the following.

(i) Equation of the tangents at A and B. (ii) Coordinates of the points A and B.

(iii) Angle ADB and the maximum and minimum distances of the point D from the circle.

(iv) Area of quadrilateral ADBC and the ΔDAB.

(v) Equation of the circle circumscribing the ΔDAB and also the intercepts made by this circle on the coordinate axes.

Q.13 Find the locus of the mid point of the chord of a circle $x^2 + y^2 = 4$ such that the segment intercepted by the chord on the curve $x^2 - 2x - 2y = 0$ subtends a right angle at the origin.

Q.14 Find the equation of a line with gradient 1 such that the two circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 10x - 14y + 65 = 0$ intercept equal length on it.

Q.15 Find the locus of the middle points of portions of the tangents to the circle $x^2 + y^2 = a^2$ terminated by the coordinate axes.

Q.16 Tangents are drawn to the concentric circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ at right angle to one another. Show that the locus of their point of intersection is a 3rd concentric circle. Find its radius.

Q.17 Find the equation of the circle passing through the three points $(4, 7)$, $(5, 6)$ and $(1, 8)$. Also find the coordinates of the point of intersection of the tangents to the circle at the points where it is cut by the straight line $5x + y + 17 = 0$.

Q.18 Consider a circle S with centre at the origin and radius 4. Four circles A, B, C and D each with radius unity and centres $(-3, 0)$, $(-1, 0)$, $(1, 0)$ and $(3, 0)$ respectively are drawn. A chord PQ of the circle S touches the circle B and passes through the centre of the circle C. If the length of this chord can be expressed as $\sqrt{x}$, find x.

Q.19 Obtain the equations of the straight lines passing through the point A$(2, 0)$ & making $45°$ angle with the tangent at A to the circle $(x + 2)^2 + (y - 3)^2 = 25$. Find the equations of the circles each of radius 3 whose centres are on these straight lines at a distance of $5\sqrt{2}$ from A.

Q.20 Consider a curve $ax^2 + 2hxy + by^2 = 1$ and a point P not on the curve. A line is drawn from the point P intersects the curve at points Q & R. If the product PQ. PR is independent of the slope of the line, then

show that the curve is a circle.

Q.21 The line $2x - 3y + 1 = 0$ is tangent to a circle $S = 0$ at (1, 1). If the radius of the circle is $\sqrt{13}$. Find the equation of the circle S.

Q.22 Find the equation of the circle which passes through the point (1, 1) & which touches the circle $x^2 + y^2 + 4x - 6y - 3 = 0$ at the point (2, 3) on it.

Q.23 Let a circle be given by $2x(x - a) + y(2y - b) = 0$, $(a \neq 0,\ b \neq 0)$. Find the condition on a & b if two chords, each bisected by the x-axis, can be drawn to the circle from the point $\left(a, \frac{b}{2}\right)$.

Q.24 Show that the equation of a straight line meeting the circle $x^2 + y^2 = a^2$ in two points at equal distances 'd' from a point (x_1, y_1) on its circumference is $xx_1 + yy_1 - a^2 + \frac{d^2}{2} = 0$.

Q.25 The radical axis of the circles $x^2 + y^2 + 2gx + 2fy + c = 0$ and $2x^2 + 2y^2 + 3x + 8y + 2c = 0$ touches the circle $x^2 + y^2 + 2x - 2y + 1 = 0$. Show that either $g = 3/4$ or $f = 2$.

Q.26 Find the equation of the circle through the points of intersection of circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$ & cutting the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

Q.27 The centre of the circle $S = 0$ lie on the line $2x - 2y + 9 = 0$ & $S = 0$ cuts orthogonally the circle $x^2 + y^2 = 4$. Show that circle $S = 0$ passes through two fixed points & find their coordinates.

Q.28(a) Find the equation of a circle passing through the origin if the line pair, $xy - 3x + 2y - 6 = 0$ is orthogonal to it. If this circle is orthogonal to the circle $x^2 + y^2 - kx + 2ky - 8 = 0$ then find the value of k.

(b) Find the equation of the circle which cuts the circle $x^2 + y^2 - 14x - 8y + 64 = 0$ and the coordinate axes orthogonally.

Q.29 Find the equation of the circle whose radius is 3 and which touches the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ internally at the point $(-1, -1)$.

Q.30 Show that the locus of the centres of a circle which cuts two given circles orthogonally is a straight line & hence deduce the locus of the centers of the circles which cut the circles $x^2 + y^2 + 4x - 6y + 9 = 0$ & $x^2 + y^2 - 5x + 4y + 2 = 0$ orthogonally. Interpret the locus.

EXERCISE–II

Q.1 A variable circle passes through the point A (a, b) & touches the x-axis; show that the locus of the other end of the diameter through A is $(x - a)^2 = 4by$.

Q.2 Find the equation of the circle passing through the point $(-6, 0)$ if the power of the point (1, 1) w.r.t. the circle is 5 and it cuts the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ orthogonally.

Q.3 Consider a family of circles passing through two fixed points A(3, 7) & B(6, 5). Show that the chords in which the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ cuts the members of the family are concurrent at a point. Find the coordinates of this point.

Q.4 Find the equation of circle passing through (1, 1) belonging to the system of co–axal circles that are tangent at (2, 2) to the locus of the point of intersection of mutually perpendicular tangent to the circle $x^2 + y^2 = 4$.

Q.5 Find the locus of the mid point of all chords of the circle $x^2 + y^2 - 2x - 2y = 0$ such that the pair of lines joining (0, 0) & the point of intersection of the chords with the circles make equal angle with axis of x.

Q.6 The circle $C : x^2 + y^2 + kx + (1 + k)y - (k + 1) = 0$ passes through the same two points for every real number k. Find(i) the coordinates of these two points.(ii) the minimum value of the radius of a circle C.

Q.7 Find the equation of a circle which is co-axial with circles $2x^2 + 2y^2 - 2x + 6y - 3 = 0$ & $x^2 + y^2 + 4x + 2y + 1 = 0$. It is given that the centre of the circle to be determined lies on the radical axis of these two circles.

Q.8 Show that the locus of the point the tangents from which to the circle $x^2 + y^2 - a^2 = 0$ include a constant angle α is $(x^2 + y^2 - 2a^2)^2 \tan^2\alpha = 4a^2(x^2 + y^2 - a^2)$.

Q.9 A circle with center in the first quadrant is tangent to $y = x + 10$, $y = x - 6$, and the y-axis. Let (h, k) be the center of the circle. If the value of $(h + k) = a + b\sqrt{a}$ where $\sqrt{a}$ is a surd, find the value of $a + b$.

Q.10 A circle is described to pass through the origin and to touch the lines $x = 1$, $x + y = 2$. Prove that the radius of the circle is a root of the equation $\left(3 - 2\sqrt{2}\right)t^2 - 2\sqrt{2}\,t + 2 = 0$.

Q.11 Find the condition such that the four points in which the circle $x^2 + y^2 + ax + by + c = 0$ and $x^2 + y^2 + a'x + b'y + c' = 0$ are intercepted by the straight lines $Ax + By + C = 0$ & $A'x + B'y + C' = 0$ respectively, lie on another circle.

Q.12 A circle C is tangent to the x and y axis in the first quadrant at the points P and Q respectively. BC and AD are parallel tangents to the circle with slope – 1. If the points A and B are on the y-axis while C and D are on the x-axis and the area of the figure ABCD is $900\sqrt{2}$ sq. units then find the radius of the circle.

Q.13 The circle $x^2 + y^2 - 4x - 4y + 4 = 0$ is inscribed in a triangle which has two of its sides along the

coordinate axes. The locus of the circumcentre of the triangle is $x + y - xy + K\sqrt{x^2 + y^2} = 0$. Find K.

Q.14 Let A, B, C be real numbers such that
(i) (sin A, cos B) lies on a unit circle centred at origin. (ii) tan C and cot C are defined.

If the minimum value of $(\tan C - \sin A)^2 + (\cot C - \cos B)^2$ is $a + b\sqrt{2}$ where $a, b \in I$, find the value of $a^3 + b^3$.

Q.15 An isosceles right angled triangle whose sides are 1, 1, $\sqrt{2}$ lies entirely in the first quadrant with the ends of the hypotenuse on the coordinate axes. If it slides prove that the locus of its centroid is $(3x - y)^2 + (x - 3y)^2 = \frac{32}{9}$.

Q.16 Tangents are drawn to the circle $x^2 + y^2 = a^2$ from two points on the axis of x, equidistant from the point (k, 0). Show that the locus of their intersection is $ky^2 = a^2(k - x)$.

Q.17 Find the equation of a circle which touches the lines $7x^2 - 18xy + 7y^2 = 0$ and the circle $x^2 + y^2 - 8x - 8y = 0$ and is contained in the given circle.

Q.18 Let W_1 and W_2 denote the circles $x^2 + y^2 + 10x - 24y - 87 = 0$ and $x^2 + y^2 - 10x - 24y + 153 = 0$ respectively. Let *m* be the smallest possible value of 'a' for which the line $y = ax$ contains the centre of a circle that is externally tangent to W_2 and internally tangent to W_1. Given that $m^2 = \frac{p}{q}$ where *p* and *q* are relatively prime integers, find $(p + q)$.

Q.19 Find the equation of the circle which passes through the origin, meets the x-axis orthogonally & cuts the circle $x^2 + y^2 = a^2$ at an angle of 45°.

Q.20 The ends A, B of a fixed straight line of length 'a' & ends A′ & B′ of another fixed straight line of length 'b' slide upon the axis of x & the axis of y (one end on axis of x & the other on axis of y). Find the locus of the centre of the circle passing through A, B, A′ & B′.

EXERCISE–III

Q.1 (a) The intercept on the line $y = x$ by the circle $x^2 + y^2 - 2x = 0$ is AB. Equation of the circle with AB as a diameter is _____.

(b) The angle between a pair of tangents drawn from a point P to the circle $x^2 + y^2 + 4x - 6y + 9\sin^2\alpha + 13\cos^2\alpha = 0$ is 2α. The equation of the locus of the point P is
(A) $x^2 + y^2 + 4x - 6y + 4 = 0$ (B) $x^2 + y^2 + 4x - 6y - 9 = 0$
(C) $x^2 + y^2 + 4x - 6y - 4 = 0$ (D) $x^2 + y^2 + 4x - 6y + 9 = 0$

(c) Find the intervals of values of a for which the line $y + x = 0$ bisects two chords drawn from a point $\left(\frac{1+\sqrt{2}a}{2}, \frac{1-\sqrt{2}a}{2}\right)$ to the circle; $2x^2 + 2y^2 - (1+\sqrt{2}a)x - (1 - \sqrt{2}a)y = 0$. [JEE '96, 1+1+5]

Q.2 A tangent drawn from the point (4, 0) to the circle $x^2 + y^2 = 8$ touches it at a point A in the first quadrant. Find the coordinates of the another point B on the circle such that AB = 4. [REE '96, 6]

Q.3 (a) The chords of contact of the pair of tangents drawn from each point on the line $2x + y = 4$ to the circle $x^2 + y^2 = 1$ pass through the point _____.

(b) Let C be any circle with centre $\left(0, \sqrt{2}\right)$. Prove that at the most two rational point can be there on C. (A rational point is a point both of whose co-ordinate are rational numbers).[JEE'97, 2+5]

Q.4 (a) The number of common tangents to the circle $x^2 + y^2 = 4$ & $x^2 + y^2 - 6x - 8y = 24$ is :
(A) 0 (B) 1 (C) 3 (D) 4

(b) C_1 & C_2 are two concentric circles, the radius of C_2 being twice that of C_1. From a point P on C_2, tangents PA & PB are drawn to C_1. Prove that the centroid of the triangle PAB lies on C_1.[JEE '98, 2 + 8

Q.5 Find the equation of a circle which touches the line $x + y = 5$ at the point (–2, 7) and cuts the circle $x^2 + y^2 + 4x - 6y + 9 = 0$ orthogonally. [REE '98, 6]

Q.6 (a) If two distinct chords, drawn from the point (p, q) on the circle $x^2 + y^2 = px + qy$ (where $pq \neq q$) are bisected by the x–axis, then :
(A) $p^2 = q^2$ (B) $p^2 = 8q^2$ (C) $p^2 < 8q^2$ (D) $p^2 > 8q^2$

(b) Let L_1 be a straight line through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 & L_2 are equal, then which of the following equations can represent L_1 ?
(A) $x + y = 0$ (B) $x - y = 0$ (C) $x + 7y = 0$ (D) $x - 7y = 0$

(c) Let T_1, T_2 be two tangents drawn from (–2, 0) onto the circle $C : x^2 + y^2 = 1$. Determine the circles touching C and having T_1, T_2 as their pair of tangents. Further, find the equations of all possible common tangents to these circles, when taken two at a time.[JEE '99, 2 + 3 + 10]

Q.7 (a) The triangle PQR is inscribed in the circle, $x^2 + y^2 = 25$. If Q and R have co-ordinates (3, 4) &

$(-4, 3)$ respectively, then $\angle QPR$ is equal to :

(A) $\pi/2$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/6$

(b) If the circles, $x^2 + y^2 + 2x + 2ky + 6 = 0$ & $x^2 + y^2 + 2ky + k = 0$ intersect orthogonally, then 'k' is : [JEE '2000 (Screening) 1 + 1]

(A) 2 or $-3/2$ (B) -2 or $-3/2$ (C) 2 or $3/2$ (D) -2 or $3/2$

Q.8 (a) Extremities of a diagonal of a rectangle are (0, 0) & (4, 3). Find the equation of the tangents to the circumcircle of a rectangle which are parallel to this diagonal.

(b) Find the point on the straight line, $y = 2x + 11$ which is nearest to the circle, $16(x^2 + y^2) + 32x - 8y - 50 = 0$.

(c) A circle of radius 2 units rolls on the outerside of the circle, $x^2 + y^2 + 4x = 0$, touching it externally. Find the locus of the centre of this outer circle. Also find the equations of the common tangents of the two circles when the line joining the centres of the two circles makes on angle of 60° with x-axis. [REE '2000 (Mains) 3 + 3 + 5]

Q.9 (a) Let PQ and RS be tangents at the extremities of the diameter PR of a circle of radius r. If PS and RQ intersect at a point X on the circumference of the circle then 2r equals

(A) $\sqrt{PQ \cdot RS}$ (B) $\dfrac{PQ + RS}{2}$ (C) $\dfrac{2PQ \cdot RS}{PQ + RS}$ (D) $\sqrt{\dfrac{(PQ)^2 + (RS)^2}{2}}$

(b) Let $2x^2 + y^2 - 3xy = 0$ be the equation of a pair of tangents drawn from the origin 'O' to a circle of radius 3 with centre in the first quadrant. If A is one of the points of contact, find the length of OA.

Q.10 (a) Find the equation of the circle which passes through the points of intersection of circles $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 2x - 6y + 6 = 0$ and intersects the circle $x^2 + y^2 + 4x + 6y + 4 = 0$ orthogonally. [REE '2001 (Mains) 3 out of 100]

(b) Tangents TP and TQ are drawn from a point T to the circle $x^2 + y^2 = a^2$. If the point T lies on the line $px + qy = r$, find the locus of centre of the circumcircle of triangle TPQ.

Q.11 (a) If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the y-axis, then the length of PQ is

(A) 4 (B) $2\sqrt{5}$ (C) 5 (D) $3\sqrt{5}$

(b) If $a > 2b > 0$ then the positive value of m for which $y = mx - b\sqrt{1 + m^2}$ is a common tangent to $x^2 + y^2 = b^2$ and $(x - a)^2 + y^2 = b^2$ is [JEE '2002 (Scr)3 + 3 out of 270]

(A) $\dfrac{2b}{\sqrt{a^2 - 4b^2}}$ (B) $\dfrac{\sqrt{a^2 - 4b^2}}{2b}$ (C) $\dfrac{2b}{a - 2b}$ (D) $\dfrac{b}{a - 2b}$

Q.12 The radius of the circle, having centre at (2, 1), whose one of the chord is a diameter of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$

(A) 1 (B) 2 (C) 3 (D) $\sqrt{3}$ [JEE '2004 (Scr)]

Q.13 Line $2x + 3y + 1 = 0$ is a tangent to a circle at (1, -1). This circle is orthogonal to a circle which is drawn having diameter as a line segment with end points $(0, -1)$ and $(-2, 3)$. Find equation of circle.

Q.14 A circle is given by $x^2 + (y - 1)^2 = 1$, another circle C touches it externally and also the x-axis, then the locus of its centre is [JEE '2005 (Scr)]

(A) $\{(x, y) : x^2 = 4y\} \cup \{(x, y) : y \le 0\}$ (B) $\{(x, y) : x^2 + (y - 1)^2 = 4\} \cup \{x, y) : y \le 0\}$

(C) $\{(x, y) : x^2 = y\} \cup \{(0, y) : y \le 0\}$ (D) $\{(x, y) : x^2 = 4y\} \cup \{(0, y) : y \le 0\}$

ANSWER KEY EXERCISE–I

Q.1 square of side 2; $x^2 + y^2 = 1$; $x^2 + y^2 = 2$

Q.2 $x^2 + y^2 + 10x - 10y + 25 = 0$ **OR** $x^2 + y^2 + 2x - 2y + 1 = 0$, (10, – 10, 25) (2, – 2, 1)

Q.3 zero, zero Q.5 32 sq. unit Q.6 $2(x^2 + y^2) + 6x - 17y - 6 = 0$]

Q.7 (5, 1) & (–1, 5) Q.8 $4x - 3y - 25 = 0$ **OR** $3x + 4y - 25 = 0$

Q.10 $x^2 + y^2 - 2x - 2y + 1 = 0$ **OR** $x^2 + y^2 - 42x + 38y - 39 = 0$

Q.11 (a) $\dfrac{3\sqrt{3}}{4}(g^2 + f^2 - c)$; (b) $\alpha = n\pi$; 5050π

Q.12 (i) $3x - 4y = 45$; $4x + 3y = 3$; (ii) A(0, 1) and B (–1, – 6); (iii) 90°, $5\left(\sqrt{2} \pm 1\right)$ units
(iv) 12.5 sq. units; (v) $x^2 + y^2 + x + 5y - 6$, x intercept 5; y intercept 7]

Q.13 $x^2 + y^2 - 2x - 2y = 0$ Q.14 $2x - 2y - 3 = 0$ Q.15 $a^2(x^2 + y^2) = 4x^2y^2$

Q.16 $x^2 + y^2 = a^2 + b^2$; $r = \sqrt{a^2 + b^2}$ Q.17 $(-4, 2)$, $x^2 + y^2 - 2x - 6y - 15 = 0$ Q.18 63

Q.19 $x - 7y = 2$, $7x + y = 14$; $(x - 1)^2 + (y - 7)^2 = 3^2$; $(x - 3)^2 + (y + 7)^2 = 3^2$;
$(x - 9)^2 + (y - 1)^2 = 3^2$; $(x + 5)^2 + (y + 1)^2 = 3^2$

Q.21 $x^2 + y^2 - 6x + 4y = 0$ **OR** $x^2 + y^2 + 2x - 8y + 4 = 0$ Q.22 $x^2 + y^2 + x - 6y + 3 = 0$

Q.23 $a^2 > 2b^2$ Q.26 $x^2 + y^2 + 16x + 14y - 12 = 0$

Q.27 $(-4, 4)$; $(-1/2, 1/2)$ Q.28 (a) $x^2 + y^2 + 4x - 6y = 0$; k = 1; (b) $x^2 + y^2 = 64$

Q.29 $5x^2 + 5y^2 - 8x - 14y - 32 = 0$ Q.30 $9x - 10y + 7 = 0$; radical axis

EXERCISE–II

Q.2 $x^2 + y^2 + 6x - 3y = 0$ Q.3 $\left(2, \frac{23}{3}\right)$ Q.4 $x^2 + y^2 - 3x - 3y + 4 = 0$

Q.5 $x + y = 2$ Q.6 $(1, 0)$ & $(1/2, 1/2)$; $r = \frac{1}{2\sqrt{2}}$ Q.7 $4x^2 + 4y^2 + 6x + 10y - 1 = 0$

Q.9 10 Q.11 $\begin{vmatrix} a-a' & b-b' & c-c' \\ A & B & C \\ A' & B' & C' \end{vmatrix}$ Q.12 $r = 15$ Q.13 $K = 1$

Q.14 19 Q.17 $x^2 + y^2 - 12x - 12y + 64 = 0$ Q.18 169

Q.19 $x^2 + y^2 \pm a\sqrt{2}\, x = 0$ Q.20 $(2ax - 2by)^2 + (2bx - 2ay)^2 = (a^2 - b^2)^2$

EXERCISE–III

Q.1 (a) $\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{2}$, **(b)** D, **(c)** $(-\infty, -2) \cup (2, \infty)$ **Q.2** (2, -2) or (-2, 2) **Q.3 (a)** (1/2, 1/4)

Q.4 (a) B **Q.5** $x^2 + y^2 + 7x - 11y + 38 = 0$

Q.6 **(a)** D **(b)** B, C **(c)** $c_1 : (x-4)^2 + y^2 = 9$; $c_2 : \left(x + \frac{4}{3}\right)^2 + y^2 = \frac{1}{9}$

common tangent between c & c_1 : $T_1 = 0$; $T_2 = 0$ and $x - 1 = 0$;

common tangent between c & c_2 : $T_1 = 0$; $T_2 = 0$ and $x + 1 = 0$;

common tangent between c_1 & c_2 : $T_1 = 0$; $T_2 = 0$ and $y = \pm \frac{5}{\sqrt{39}}\left(x + \frac{4}{5}\right)$

where $T_1 : x - \sqrt{3}\, y + 2 = 0$ and $T_2 : x + \sqrt{3}\, y + 2 = 0$

Q.7 **(a)** C **(b)** A

Q.8 **(a)** $6x - 8y + 25 = 0$ & $6x - 8y - 25 = 0$; **(b)** $(-9/2, 2)$

(c) $x^2 + y^2 + 4x - 12 = 0$, T_1: $\sqrt{3}x - y + 2\sqrt{3} + 4 = 0$, T_2: $\sqrt{3}x - y + 2\sqrt{3} - 4 = 0$ (D.C.T.)

T_3: $x + \sqrt{3}\, y - 2 = 0$, T_4: $x + \sqrt{3}\, y + 6 = 0$ (T.C.T.)

Q.9 **(a)** A; **(b)** $OA = 3(3 + \sqrt{10})$ **Q.10** **(a)** $x^2 + y^2 + 14x - 6y + 6 = 0$; **(b)** $2px + 2qy = r$

Q.11 **(a)** C; **(b)** A **Q.12** C **Q.13** $2x^2 + 2y^2 - 10x - 5y + 1 = 0$ **Q.14** D

EXERCISE–IV

Part : (A) Only one correct option

1. If (–3, 2) lies on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, which is concentric with the circle $x^2 + y^2 + 6x + 8y - 5 = 0$, then c is

(A) 11 (B) –11 (C) 24 (D) none of these

2. The circle $x^2 + y^2 - 6x - 10y + c = 0$ does not intersect or touch either axis, & the point (1, 4) is inside the circle. Then the range of possible values of c is given by:

(A) $c > 9$ (B) $c > 25$ (C) $c > 29$ (D) $25 < c < 29$

3. The length of the tangent drawn from any point on the circle $x^2 + y^2 + 2gx + 2fy + p = 0$ to the circle $x^2 + y^2 + 2gx + 2fy + q = 0$ is:

(A) $\sqrt{q-p}$ (B) $\sqrt{p-q}$ (C) $\sqrt{q+p}$ (D) none

4. The angle between the two tangents from the origin to the circle $(x-7)^2 + (y+1)^2 = 25$ equals

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) none

5. The circumference of the circle $x^2 + y^2 - 2x + 8y - q = 0$ is bisected by the circle $x^2 + y^2 + 4x + 12y + p = 0$, then $p + q$ is equal to:

(A) 25 (B) 100 (C) 10 (D) 48

6. If $\left(a, \frac{1}{a}\right), \left(b, \frac{1}{b}\right), \left(c, \frac{1}{c}\right)$ & $\left(d, \frac{1}{d}\right)$ are four distinct points on a circle of radius 4 units then, abcd is equal to:

(A) 4 (B) 16 (C) 1 (D) none

7. The centre of a circle passing through the points (0, 0), (1, 0) & touching the circle $x^2 + y^2 = 9$ is :

(A) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, \frac{3}{2}\right)$ (C) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, -\sqrt{2}\right)$

8. Two thin rods AB & CD of lengths 2a & 2b move along OX & OY respectively, when 'O' is the origin. The equation of the locus of the centre of the circle passing through the extremities of the two rods is:

(A) $x^2 + y^2 = a^2 + b^2$ (B) $x^2 - y^2 = a^2 - b^2$ (C) $x^2 + y^2 = a^2 - b^2$ (D) $x^2 - y^2 = a^2 + b^2$

9. The value of 'c' for which the set, $\{(x, y) \mid x^2 + y^2 + 2x \le 1\} \cap \{(x, y) \mid x - y + c \ge 0\}$ contains only one point in common is:

(A) $(-\infty, -1] \cup [3, \infty)$ (B) $\{-1, 3\}$ (C) $\{-3\}$ (D) $\{-1\}$

10. Let x & y be the real numbers satisfying the equation $x^2 - 4x + y^2 + 3 = 0$. If the maximum and minimum values of $x^2 + y^2$ are M & m respectively, then the numerical value of M – m is:
(A) 2 (B) 8 (C) 15 (D) none of these

11. A line meets the co–ordinate axes in A & B. A circle is circumscribed about the triangle OAB. If d_1 & d_2 are the distances of the tangent to the circle at the origin O from the points A and B respectively, the diameter of the circle is:

(A) $\frac{2d_1 + d_2}{2}$ (B) $\frac{d_1 + 2d_2}{2}$ (C) $d_1 + d_2$ (D) $\frac{d_1 d_2}{d_1 + d_2}$

12. The distance between the chords of contact of tangents to the circle;
$x^2 + y^2 + 2gx + 2fy + c = 0$ from the origin & the point (g, f) is:

(A) $\sqrt{g^2 + f^2}$ (B) $\frac{\sqrt{g^2 + f^2 - c}}{2}$ (C) $\frac{g^2 + f^2 - c}{2\sqrt{g^2 + f^2}}$ (D) $\frac{\sqrt{g^2 + f^2 + c}}{2\sqrt{g^2 + f^2}}$

13. If tangent at (1, 2) to the circle c_1: $x^2 + y^2 = 5$ intersects the circle c_2: $x^2 + y^2 = 9$ at A & B and tangents at A & B to the second circle meet at point C, then the co–ordinates of C are:

(A) (4, 5) (B) $\left(\frac{9}{15}, \frac{18}{5}\right)$ (C) (4, –5) (D) $\left(\frac{9}{5}, \frac{18}{5}\right)$

14. The locus of the mid points of the chords of the circle $x^2 + y^2 + 4x - 6y - 12 = 0$ which subtend an angle of $\frac{\pi}{3}$ radians at its circumference is:
(A) $(x-2)^2 + (y+3)^2 = 6.25$ (B) $(x+2)^2 + (y-3)^2 = 6.25$
(C) $(x+2)^2 + (y-3)^2 = 18.75$ (D) $(x+2)^2 + (y+3)^2 = 18.75$

15. If the length of a common internal tangent to two circles is 7, and that of a common external tangent is 11, then the product of the radii of the two circles is:
(A) 36 (B) 9 (C) 18 (D) 4

16. Two circles whose radii are equal to 4 and 8 intersect at right angles. The length of their common chord is:

(A) $\frac{16}{\sqrt{5}}$ (B) 8 (C) $4\sqrt{6}$ (D) $\frac{8\sqrt{5}}{5}$

17. A circle touches a straight line $lx + my + n = 0$ & cuts the circle $x^2 + y^2 = 9$ orthogonally. The locus of centres of such circles is:
(A) $(lx + my + n)^2 = (l^2 + m^2)(x^2 + y^2 - 9)$ (B) $(lx + my - n)^2 = (l^2 + m^2)(x^2 + y^2 - 9)$
(C) $(lx + my + n)^2 = (l^2 + m^2)(x^2 + y^2 + 9)$ (D) none of these

18. If a circle passes through the point (a, b) & cuts the circle $x^2 + y^2 = K^2$ orthogonally, then the equation of the locus of its centre is:
(A) $2ax + 2by - (a^2 + b^2 + K^2) = 0$ (B) $2ax + 2by - (a^2 - b^2 + K^2) = 0$
(C) $x^2 + y^2 - 3ax - 4by + (a^2 + b^2 - K^2) = 0$ (D) $x^2 + y^2 - 2ax - 3by + (a^2 - b^2 - K^2) = 0$

19. The circle $x^2 + y^2 = 4$ cuts the circle $x^2 + y^2 + 2x + 3y - 5 = 0$ in A & B. Then the equation of the circle on AB as a diameter is:
(A) $13(x^2 + y^2) - 4x - 6y - 50 = 0$ (B) $9(x^2 + y^2) + 8x - 4y + 25 = 0$
(C) $x^2 + y^2 - 5x + 2y + 72 = 0$ (D) none of these

20. The length of the tangents from any point on the circle $15x^2 + 15y^2 - 48x + 64y = 0$ to the two circles $5x^2 + 5y^2 - 24x + 32y + 75 = 0$ and $5x^2 + 5y^2 - 48x + 64y + 300 = 0$ are in the ratio
(A) 1 : 2 (B) 2 : 3 (C) 3 : 4 (D) none of these

21. The normal at the point (3, 4) on a circle cuts the circle at the point (–1, –2). Then the equation of the circle is
(A) $x^2 + y^2 + 2x - 2y - 13 = 0$ (B) $x^2 + y^2 - 2x - 2y - 11 = 0$
(C) $x^2 + y^2 - 2x + 2y + 12 = 0$ (D) $x^2 + y^2 - 2x - 2y + 14 = 0$

22. The locus of poles whose polar with respect to $x^2 + y^2 = a^2$ always passes through (K, 0) is:
(A) $Kx - a^2 = 0$ (B) $Kx + a^2 = 0$ (C) $Ky + a^2 = 0$ (D) $Ky - a^2 = 0$

23. If two distinct chords, drawn from the point (p, q) on the circle $x^2 + y^2 = px + q$ (where $pq \neq 0$) are bisected by the x-axis, then **[IIT - 1999]**
(A) $p^2 = q^2$ (B) $p^2 = 8q^2$ (C) $p^2 < 8q^2$ (D) $p^2 > 8q^2$

24. The triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If Q and R have co-ordinates (3, 4) and (–4, 3) respectively, the $\angle$ QPR is equal to **[IIT - 2000]**

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

25. Let PQ and RS be tangents at the extremities of diameter PR of a circle of radius r. If PS and RQ intersect at a point X on the circumference of the circle, then 2r equals **[IIT- 2001]**

(A) $\sqrt{PQ \cdot RS}$ (B) $\frac{PQ + RS}{2}$ (C) $\frac{2PQ + RS}{PQ + RS}$ (D) $\frac{\sqrt{PQ^2 + RS^2}}{2}$

26. Let AB be a chord of the circle $x^2 + y^2 = r^2$ subtending a right angle at the centre. Then, locus of the centroid of the triangle PAB as P moves on the circles is **[IIT- 2001]**
(A) a parabola (B) a circle (C) an ellipse (D) a pair of straight line

27. If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the y-axis, then the length of PQ is **[IIT- 2002]**
(A) 4 (B) $2\sqrt{5}$ (C) 5 (D) $3\sqrt{5}$

28. Tangent to the curve $y = x^2 + 6$ at a point P(1, 7) touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at a point

Q. Then, the coordinates of Q are **[IIT- 2005]**
(A) (– 6, –11) (B) (– 9, – 13) (C) (– 10, – 15) (D) (– 6, – 7)

Part : (B) May have more than one options correct

29. A circle passes through the point $\left(3, \sqrt{\frac{7}{2}}\right)$ and touches the line pair $x^2 - y^2 - 2x + 1 = 0$. The co–ordinates of the centre of the circle are:
(A) (4, 0) (B) (5, 0) (C) (6, 0) (D) (0, 4)

30. The equation of the circle which touches both the axes and the line $\frac{x}{3} + \frac{y}{4} = 1$ and lies in the first quadrant is $(x - c)^2 + (y - c)^2 = c^2$ where c is
(A) 1 (B) 2 (C) 4 (D) 6

EXERCISE–V

1. If $y = 2x$ is a chord of the circle $x^2 + y^2 - 10x = 0$, find the equation of a circle with this chord as diameter.

2. Find the points of intersection of the line $x - y + 2 = 0$ and the circle $3x^2 + 3y^2 - 29x - 19y + 56 = 0$. Also determine the length of the chord intercepted.

3. Show that two tangents can be drawn from the point (9, 0) to the circle $x^2 + y^2 = 16$; also find the equation of the pair of tangents and the angle between them.

4. Given the three circles $x^2 + y^2 - 16x + 60 = 0$, $3x^2 + 3y^2 - 36x + 81 = 0$ and $x^2 + y^2 - 16x - 12y + 84 = 0$, find (1) the point from which the tangents to them are equal in length, and (2) this length.

5. On the line joining (1, 0) and (3, 0) an equilateral triangle is drawn having its vertex in the first quadrant. Find the equation to the circles described on its sides as diameter.

6. One of the diameters of the circle circumscribing the rectangle ABCD is $4y = x + 7$. If A & B are the points (–3, 4) & (5 4) respectively. Then find the area of the rectangle.

7. Let A be the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$. Suppose that the tangents at the points B (1, 7) & D (4, – 2) on the circle meet at the point C. Find the area of the quadrilateral ABCD.

8. Let a circle be given by $2x(x - a) + y(2y - b) = 0$, $(a \neq 0, b \neq 0)$. Find the condition on a & b if two chords, each bisected by the x–axis, can be drawn to the circle from (a, b/2).

9. Find the equation of the circle which cuts each of the circles, $x^2 + y^2 = 4$, $x^2 + y^2 - 6x - 8y + 10 = 0$ & $x^2 + y^2 + 2x - 4y - 2 = 0$ at the extremities of a diameter.

10. Find the equation and the length of the common chord of the two circles given by the equations, $x^2 + y^2 + 2x + 2y + 1 = 0$ & $x^2 + y^2 + 4x + 3y + 2 = 0$.

11. Find the values of a for which the point (2a, a + 1) is an interior point of the larger segment of the circle $x^2 + y^2 - 2x - 2y - 8 = 0$ made by the chord whose equation is $x - y + 1 = 0$.

12. If $4l^2 - 5m^2 + 6l + 1 = 0$. Prove that $lx + my + 1 = 0$ touches a definite circle. Find the centre & radius of the circle.

13. A circle touches the line $y = x$ at a point P such that $OP = 4\sqrt{2}$ where O is the origin. The circle contains the point (–10, 2) in its interior and the length of its chord on the line $x + y = 0$ is $6\sqrt{2}$. Find the equation of the circle.

14. Show that the equation of a straight line meeting the circle $x^2 + y^2 = a^2$ in two points at equal distances 'd' from a point (x_1, y_1) on its circumference is $xx_1 + yy_1 - a^2 + \frac{d^2}{2} = 0$.

15. For each natural number k, let C_k denote the circle with radius k centimetres and centre at the origin. On the circle C_k, α-particle moves k centrimetres in the counter - clockwise direction. After completing its motion on C_k, the particle moves to C_{k+1} in the radial direction. The motion of the particle continues in this manner. The particle starts at (1, 0). If the particle crosses the positive direction of the x-axis for the first time on the circle C_n then n = _____. **[IIT 1997]**

16. Let C be any circle with centre $(0, \sqrt{2})$. Prove that at the most two rational point can be there on C. (A rational point is a point both of whose co–ordinate are rational numbers). **[IIT - 1997]**

17. Let T_1, T_2 be two tangents drawn from (– 2, 0) onto the circle C: $x^2 + y^2 = 1$. Determine the circles touching C and having T_1, T_2 as their pair of tangents. Further, find the equations of all possible common tangents to these circles, when taken two at a time. **[IIT - 1999]**

18. Let C_1 and C_2 be two circles with C_2 lying inside C_1. A circle C lying inside C_1 touches C_1 internally C_1 internally and C_2 externally. Identify the locus of the centre of C. **[IIT 2001]**

19. Circles with radii 3, 4 and 5 touch each other externally. If P is the point of intersection of tangents to these circles at their points of contact, find the distance of P from the points of contact. **[IIT - 2005]**

EXERCISE–IV

1. B **2.** D **3.** A **4.** C **5.** C **6.** C **7.** D **8.** B **9.** D **10.** B **11.** C **12.** C **13.** D **14.** B
15. C **16.** A **17.** A **18.** A **19.** A **20.** A **21.** B **22.** A **23.** D **24.** C **25.** A **26.** B **27.** C **28.** D
29. AC **30.** AD

PTO

EXERCISE–V

1. $x^2 + y^2 - 2x - 4y = 0$ **2.** (1, 3), (5, 7), $4\sqrt{2}$

3. $16x^2 - 65y^2 - 288x + 1296 = 0$, $\tan^{-1}\left(\frac{8\sqrt{65}}{49}\right)$

4. $\left(\frac{33}{4}, 2\right)$; $\frac{1}{4}$

5. $x^2 + y^2 - 3x - \sqrt{3}y + 2 = 0$;

$x^2 + y^2 - 5x - \sqrt{3}y + 6 = 0$;

$x^2 + y^2 - 4x + 3 = 0$

6. 32 sq. unit **7.** 75 sq. units **8.** $(a^2 > 2b^2)$

9. $x^2 + y^2 - 4x - 6y - 4 = 0$

10. $2x + y + 1 = 0, \frac{2}{\sqrt{5}}$ **11.** $a \in (0, 9/5)$

12. Centre $\equiv (3, 0)$, (radius) $= \sqrt{5}$

13. $x^2 + y^2 + 18x - 2y + 32 = 0$

15. 7

17. $c_1: (x-4)^2 + y^2 = 9$; $c_2: \left(x + \frac{4}{3}\right)^2 + y^2 = \frac{1}{9}$

common tangent between c & c_1: $T_1 = 0$; $T_2 = 0$ and $x - 1 = 0$; common tangent between c & c_2: $T_1 = 0$; $T_2 = 0$ and $x + 1 = 0$; common tangent between c_1 & c_2: $T_1 = 0$; $T_2 = 0$ and

$y = \pm \frac{5}{\sqrt{39}}\left(x + \frac{4}{5}\right)$ where $T_1: x - \sqrt{3}y + 2 = 0$

and $T_2: x + \sqrt{3}y + 2 = 0$

18. ellipse **19.** $\sqrt{5}$

Some questions (Assertion–Reason type) are given below. Each question contains **Statement – 1 (Assertion)** and **Statement – 2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. So select the correct choice :***Choices are :***

(A)**Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is** a correct explanation for **Statement – 1**.

(B)**Statement – 1** is True, **Statmnt – 2** is True; **Statement – 2 is NOT** a correct explanation for **Statement – 1**.

(C) **Statement – 1** is True, **Statement – 2** is False.

(D) Statement – 1 is False, Statement – 2 is True.

253. Tangents are drawn from the origin to the circle $x^2 + y^2 - 2hx - 2hy + h^2 = 0$ $(h \geq 0)$

Statement 1: Angle between the tangents is $\pi/2$

Statement 2: The given circle is touching the co-ordinate axes.

254. Consider two circles $x^2 + y^2 - 4x - 6y - 8 = 0$ and $x^2 + y^2 - 2x - 3 = 0$

Statement 1: Both circles intersect each other at two distinct points

Statement 2: Sum of radii of two circles in greater than distance between the centres of two circles

255. C_1 is a circle of radius 2 touching x–axis and y–axis. C_2 is another circle of radius greater than 2 and touching the axes as well as the circle c_1.

Statement–1 : Radius of circle $c_2 = \sqrt{2}(\sqrt{2}+1)(\sqrt{2}+2)$

Statement–2 : Centres of both circles always lie on the line y = x.

256. From the point P($\sqrt{2}, \sqrt{6}$), tangents PA and PB are drawn to the circle $x^2 + y^2 = 4$.

Statement–1 : Area of the quadrilateral OAPB (obeying origin) is 4.

Statement–2 : Tangents PA and PB are perpendicular to each other and therefore quadrilateral OAPB is a square.

257. **Statement–1 :** Tangents drawn from ends points of the chord x + ay – 6 = 0 of the parabola $y^2 = 24x$ meet on the line x + 6 = 0

Statement–2 : Pair of tangents drawn at the end points of the parabola meets on the directrix of the parabola

258. **Statement–1 :** Number of focal chords of length 6 units that can be drawn on the parabola $y^2 - 2y - 8x + 17 = 0$ is zero **Statement–2 :** Lotus rectum is the shortest focal chord of the parabola

259. **Statement–1 :** Centre of the circle having x + y = 3 and x – y = 1 as its normal is (1, 2).

Statement–2 : Normals to the circle always passes through its centre.

260. **Statement–1 :** The number of common tangents to the circle $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y - 24 = 0$, is one

Statement–2 : If $C_1C_2 = |r_1 - r_2|$, then number of common tangents is three. Where C_1C_2 = Distance between the centres at both the circle and r_1, r_2 are the radius of the circle respectively

261. **Statement–1 :** The circle having equation $x^2 + y^2 - 2x + 6y + 5 = 0$ intersects both the coordinate axes.

Statement–2 : The lengths of x and y intercepts made by the circle having equation $x^2 + y^2 + 2gx + 2fy + c = 0$ are $2\sqrt{g^2 - c}$ and $2\sqrt{f^2 - c}$ respectively.

262. **Statement–1 :** The number of circles that pass through the points (1, – 7) and (– 5, 1) and of radius 4, is two.

Statement–2 : The centre of any circle that pass through the points A and B lies on the perpendicular bisector of AB.

263. The line OP and OQ are the tangents from (0, 0) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$.

Statement–1 : Equation of PQ is fx + gy + c = 0.

Statement–2 : Equation of circle OPQ is $x^2 + y^2 + gx + fy = 0$.

264. **Statement–1 :** $x^2 + y^2 + 2xy + x + y = 0$ represent circle passing through origin.

Statement–2 : Locus of point of intersection of perpendicular tangent is a circle

265. **Statement–1 :** Equation of circle touching x–axis at (1, 0) and passing through (1, 2) is $x^2 + y^2 - 2x - 2y + 1 = 0$

Statement–2 : If circle touches both the axis then its center lies on $x^2 - y^2 = 0$

266. **Statement-1:** Let C be any circle with centre (0, $\sqrt{2}$) has at the most two rational points on it

Statement-2: A straight line cuts a circle at atmost two points

267. Tangents are drawn from each point on the line $2x + y = 4$ to the circle $x^2 + y^2 = 1$
Statement-1: The chords of contact passes through a fixed point
Statement-2: Family of lines $(a_1x + b_1y + c_1) + k(a_2x + b_2y + c_2) = 0$ always pass through a fixed point.

268. **Statement-1:** The common tangents of the circles $x^2 + y^2 + 2x = 0$ and $x^2 + y^2 - 6 = 0$ form an equilateral triangle
Statement-2: The given circles touch each other externally.

269. **Statement-1:** The circle described on the segment joining the points (-2, -1), (0, -3) as diameter cuts the circle $x^2 + y^2 + 5x + y + 4 = 0$ orthogonally
Statement-2: Two circles $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ orthogonally if $2g_1g_2 + 2f_1f_2 = c_1 + c_2$

270. **Statement-1 :** The equation of chord of the circle $x^2 + y^2 - 6x + 10y - 9 = 0$, which is bisected at (-2, 4) must be $x + y - 2 = 0$.
Statement-2:In notations, the equation of the chord of the circle $S = 0$ bisected at (x_1, y_1) must be $T = S_1$.

271. **Statement-1 :** If two circles $x^2 + y^2 + 2gx + 2fy = 0$ and $x^2 + y^2 + 2g'x + 2f'y = 0$ touch each other, then $f'g = fg'$
Statement-2 : Two circles touch other, if line joining their centres is perpendicular to all possible common tangents.

272. **Statement-1 :** Number of circles passing through (1, 2), (4, 7) and (3, 0) is one.
Statement-2 : One and only circle can be made to pass through three non-collinear points.

273. **Statement-1 :** The chord of contact of tangent from three points A, B, C to the circle $x^2 + y^2 = a^2$ are concurrent, then A, B, C will be collinear.
Statement-2 : A, B, C always lies on the normal to the circle $x^2 + y^2 = a^2$

274. **Statement-1 :** Circles $x^2 + y^2 = 144$ and $x^2 + y^2 - 6x - 8y = 0$ do not have any common tangent.
Statement-2 : If one circle lies completely inside the other circle then both have no common tangent.

275. **Statement-1 :** The equation $x^2 + y^2 - 2x - 2ay - 8 = 0$ represents for different values of 'a' a system of circles passing through two fixed points lying on the x-axis.
Statement-2 : $S = 0$ is a circle & $L = 0$ is a straight line, then $S + \lambda L = 0$ represents the family of circles passing through the points of intersection of circle and straight line. (where λ is arbitrary parameter).

276. **Statement-1 :** Lengths of tangent drawn from any point on the line $x + 2y - 1 = 0$ to the circles $x^2 + y^2 - 16 = 0$ & $x^2 + y^2 - 4x - 8y - 12 = 0$ are equal
Statement-2 : Director circle is locus of point of intersection of perpendicular tangents.

277. **Statement-1 :** One & only one circle can be drawn through three given points
Statement-2 : Every triangle has a circumcircle.

278. **Statement-1 :** The circles $x^2 + y^2 + 2px + r = 0$, $x^2 + y^2 + 2qy + r = 0$ touch if $\frac{1}{p^2} + \frac{1}{q^2} = \frac{1}{r}$
Statement-2 : Two circles with centre C_1, C_2 and radii r_1, r_2 touch each other if $r_1 \pm r_2 = c_1c_2$

279. **Statement-1 :** The equation of chord of the circle $x^2 + y^2 - 6x + 10y - 9 = 0$ which is bisected at (-2, 4) must be $x + y - 2 = 0$
Statement-2 : In notations the equation of the chord of the circle $s = 0$ bisected at (x_1, y_1) must be $T=S_1$.

280. **Statement-1 :** The equation $x^2 + y^2 - 4x + 8y - 5 = 0$ represent a circle.
Statement-2 : The general equation of degree two $ax^2 + 2hxy + by^2 - 2gx + 2fy + c = 0$ represents a circle, if $a = b$ & $h = 0$. circle will be real if $g^2 + f^2 - c \geq 0$.

281. **Statement-1 :** The least and greatest distances of the point P(10, 7) from the circle $x^2 + y^2 - 4x - 2y - 20 = 0$ are 5 and 15 units respectively.
Statement-2 : A point (x_1, y_1) lies outside a circle $s = x^2 + y^2 + 2gx + 2fy + c = 0$ if $s_1 > 0$ where $s_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$.

282. **Statement-1 :** The point $(a, -a)$ lies inside the circle $x^2 + y^2 - 4x + 2y - 8 = 0$ when ever $a \in (-1, 4)$
Statement-2 : Point (x_1, y_1) lies inside the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, if $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c < 0$.

283. **Statement-1 :** If $n \geq 3$ then the value of n for which n circles have equal number of radical axes as well as radical centre is 5.
Statement-2 : If no two of n circles are concentric and no three of the centres are collinear then number of possible radical centre = ${}^{n}C_{3}$.

284. **Statement-1 :** Two circles $x^2 + y^2 + 2ax + c = 0$ and $x^2 + y^2 + 2by + c = 0$ touches if $\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c}$
Statement-2 : Two circles centres c_1, c_2 and radii r_1, r_2 touches each other if $r_1 \pm r_2 = c_1c_2$.

285. **Statement-1 :** Number of point $(a+1, \sqrt{3}a)$ $a \in I$, lying inside the region bounded by the circles $x^2 + y^2 - 2x - 3 = 0$ and $x^2 + y^2 - 2x - 15 = 0$ is 1.
Statement-2 : Sum of squares of the lengths of chords intercepted by the lines $x + y = n$, $n \in N$ on the circle $x^2 + y^2 = 4$ is 18.

253. A	**254. B**	**255. D**	**256. A**	**257. A**	**258. A**	**259. D**
260. C	**261. D**	**262. D**	**263. D**	**264. D**	**265. A**	**266. A**
267. A	**268. A**	**269. A**	**270. D**	**271. C**	**272. D**	**273. C**
274. A	**275. A**	**276. B**	**277. A**	**278. A**	**279. D**	**280. A**
281. B	**282. A**	**283. A**	**284. A**	**285. B**		

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 13 XI M 13. Parabola, Ellipse and Hyperbola

Index:

1. Key Concepts

2. Exercise I to XV

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Parabola

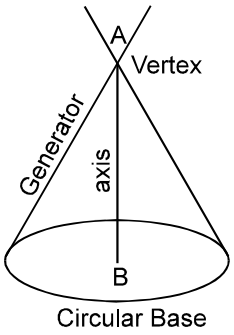
1. Conic Sections:

A conic section, or conic is the locus of a point which moves in a plane so that its distance from a fixed point is in a constant ratio to its perpendicular distance from a fixed straight line.

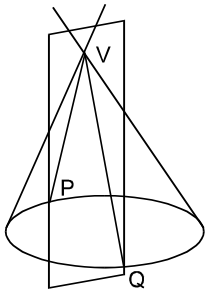
- The fixed point is called the **Focus.**
- The fixed straight line is called the **Directrix.**
- The constant ratio is called the **Eccentricity** denoted by e.
- The line passing through the focus & perpendicular to the directrix is called the **Axis**.
- A point of intersection of a conic with its axis is called a **Vertex**.

2. Section of right circular cone by different planes

A right circular cone is as shown in the

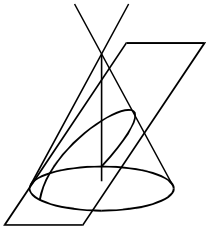


(i) Section of a right circular cone by a plane passing through its vertex is a pair of straight lines passing through the vertex as shown in the



(ii) Section of a right circular cone by a plane parallel to its base is a circle as shown in the **figure – 3**.

(iii) Section of a right circular cone by a plane parallel to a generator of the cone is a parabola as shown in the



Figure

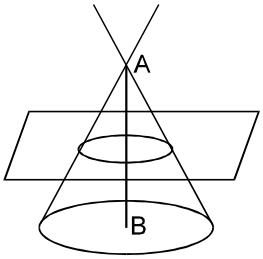


Figure- 3

(iv) Section of a right circular cone by a plane neither parallel to any generator of the cone nor perpendicular or parallel to the axis of the cone is an ellipse or hyperbola as shown in the **figure – 5 & 6**.

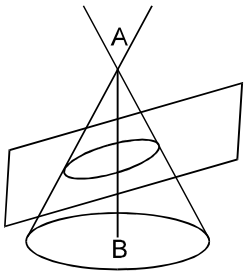


Figure -5

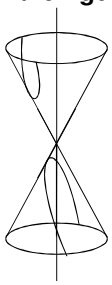
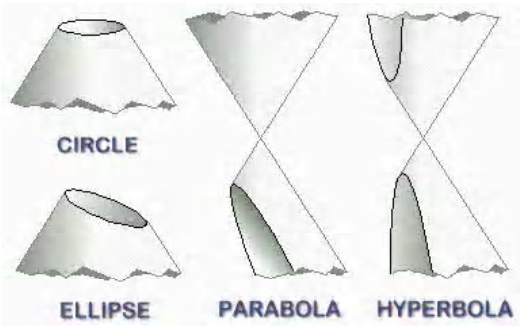


Figure -6

3D View :



3. General equation of a conic: Focal directrix property:

The general equation of a conic with focus (p, q) & directrix $lx + my + n = 0$ is:
$(l^2 + m^2)[(x - p)^2 + (y - q)^2] = e^2 (lx + my + n)^2 \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

4. Distinguishing various conics :

The nature of the conic section depends upon the position of the focus S w.r.t. the directrix & also upon the value of the eccentricity e. Two different cases arise.

Case (I) When The Focus Lies On The Directrix.

In this case $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ & the general equation of a conic represents a pair of straight lines if:

$e > 1 \equiv h^2 > ab$ the lines will be real & distinct intersecting at S.
$e = 1 \equiv h^2 \geq ab$ the lines will coincident.
$e < 1 \equiv h^2 < ab$ the lines will be imaginary.

Case (II) When The Focus Does Not Lie On Directrix.

a parabola	an ellipse	a hyperbola	rectangular hyperbola
$e = 1; \Delta \neq 0,$	$0 < e < 1; \Delta \neq 0;$	$e > 1; \Delta \neq 0;$	$e > 1; \Delta \neq 0$
$h^2 = ab$	$h^2 < ab$	$h^2 > ab$	$h^2 > ab; a + b = 0$

PARABOLA

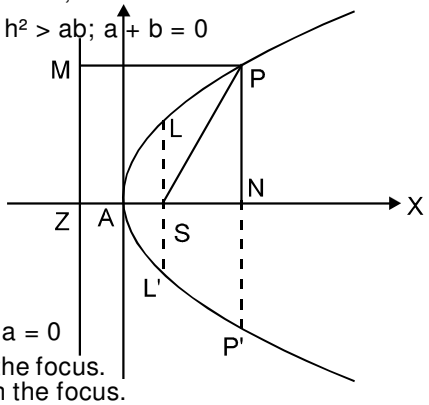
5. Definition and Terminology

A parabola is the locus of a point, whose distance from a fixed point (focus) is equal to perpendicular distance from a fixed straight line (directrix).

Four standard forms of the parabola are
$y^2 = 4ax$; $y^2 = -4ax$; $x^2 = 4ay$; $x^2 = -4ay$

For parabola $y^2 = 4ax$:

(i) Vertex is (0, 0) **(ii)** focus is (a, 0)
(iii) Axis is $y = 0$ **(iv)** Directrix is $x + a = 0$



Focal Distance: The distance of a point on the parabola from the focus.
Focal Chord : A chord of the parabola, which passes through the focus.
Double Ordinate: A chord of the parabola perpendicular to the axis of the symmetry.
Latus Rectum: A double ordinate passing through the focus or a focal chord perpendicular to the axis of parabola is called the Latus Rectum (L.R.).

For $y^2 = 4ax$. $\Rightarrow$ Length of the latus rectum = 4a.
$\Rightarrow$ ends of the latus rectum are L(a, 2a) & L' (a, – 2a).

NOTE :

(i) Perpendicular distance from focus on directrix = half the latus rectum.
(ii) Vertex is middle point of the focus & the point of intersection of directrix & axis.
(iii) Two parabolas are said to be equal if they have the same latus rectum.

Examples :

Find the equation of the parabola whose focus is at (– 1, – 2) and the directrix the line $x - 2y + 3 = 0$.

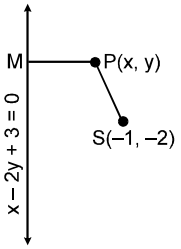
Solution.

Let P(x, y) be any point on the parabola whose focus is S(– 1, – 2) and the directrix $x - 2y + 3 = 0$. Draw PM perpendicular to directrix $x - 2y + 3 = 0$. Then by definition,

$SP = PM$
$\Rightarrow SP^2 = PM^2$

$$\Rightarrow (x + 1)^2 + (y + 2)^2 = \left(\frac{x - 2y + 3}{\sqrt{1+4}}\right)^2$$

$\Rightarrow 5[(x + 1)^2 + (y + 2)^2] = (x - 2y + 3)^2$
$\Rightarrow 5(x^2 + y^2 + 2x + 4y + 5) = (x^2 + 4y^2 + 9 - 4xy + 6x - 12y)$



$\Rightarrow \quad 4x^2 + y^2 + 4xy + 4x + 32y + 16 = 0$

This is the equation of the required parabola.

Example :

Find the vertex, axis, focus, directrix, latusrectum of the parabola, also draw their rough sketches.

$4y^2 + 12x - 20y + 67 = 0$

Solution.

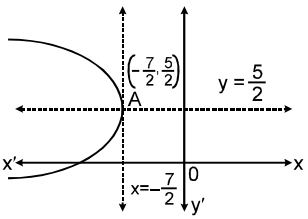
The given equation is

$4y^2 + 12x - 20y + 67 = 0 \quad \Rightarrow \quad y^2 + 3x - 5y + \frac{67}{4} = 0$

$\Rightarrow \quad y^2 - 5y = -3x - \frac{67}{4} \quad \Rightarrow \quad y^2 - 5y + \left(\frac{5}{2}\right)^2 = -3x - \frac{67}{4} + \left(\frac{5}{2}\right)^2$

$\Rightarrow \quad \left(y - \frac{5}{2}\right)^2 = -3x - \frac{42}{4} \quad \Rightarrow \quad \left(y - \frac{5}{2}\right)^2 = -3\left(x + \frac{7}{2}\right) \quad ....(i)$

Let $x = X - \frac{7}{2}, \; y = Y + \frac{5}{2}$ ....(ii)



Using these relations, equation (i) reduces to

$Y^2 = -3X$ ....(iii)

This is of the form $Y^2 = -4aX$. On comparing, we get $4a = 3 \Rightarrow a = 3/4$.

Vertex - The coordinates of the vertex are $(X = 0, Y = 0)$

So, the coordinates of the vertex are

$\left(-\frac{7}{2}, \frac{5}{2}\right)$ [Putting X = 0, Y = 0 in (ii)]

Axis: The equation of the axis of the parabola is $Y = 0$.

So, the equation of the axis is

$y = \frac{5}{2}$ [Putting Y = 0 in (ii)]

Focus- The coordinates of the focus are $(X = -a, Y = 0)$

i.e. $(X = -3/4, Y = 0)$.

So, the coordinates of the focus are

$(-17/4, 5/2)$ [Putting X = 3/4 in (ii)]

Directrix - The equation of the directrix is $X = a$ i.e. $X = \frac{3}{4}$.

So, the equation of the directrix is

$x = -\frac{11}{4}$ [Putting X = 3/4 in (ii)]

Latusrectum - The length of the latusrectum of the given parabola is $4a = 3$.

Self Practice Problems

1. Find the equation of the parabola whose focus is the point (0, 0)and whose directrix is the straight line $3x - 4y + 2 = 0$. **Ans.** $16x^2 + 9y^2 + 24xy - 12x + 16y - 4 = 0$

2. Find the extremities of latus rectum of the parabola $y = x^2 - 2x + 3$.

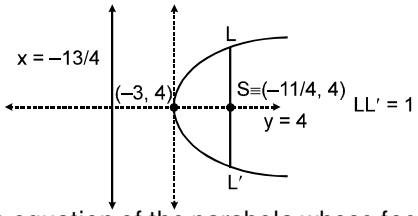
Ans. $\left(\frac{1}{2}, \frac{9}{4}\right) \left(\frac{3}{2}, \frac{9}{4}\right)$

3. Find the latus rectum & equation of parabola whose vertex is origin & directrix is $x + y = 2$.

Ans. $4\sqrt{2}$, $x^2 + y^2 - 2xy + 8x + 8y = 0$

4. Find the vertex, axis, focus, directrix, latusrectum of the parabola $y^2 - 8y - x + 19 = 0$. Also draw their roguht sketches.

Ans.



5. Find the equation of the parabola whose focus is (1, – 1) and whose vertex is (2, 1). Also find its axis and latusrectum.

Ans. $(2x - y - 3)^2 = -20\,(x + 2y - 4)$, Axis $2x - y - 3 = 0$. $LL' = 4\sqrt{5}$.

6. Parametric Representation:

The simplest & the best form of representing the co–ordinates of a point on the parabola is $(at^2, 2at)$ i.e. the equations $x = at^2$ & $y = 2at$ together represents the parabola $y^2 = 4ax$, t being the parameter.

Example : Find the parametric equation of the parabola $(x - 1)^2 = -12\,(y - 2)$

Solution. $\because \quad 4a = -12 \quad \Rightarrow \quad a = 3, \; y - 2 = at^2$

$x - 1 = 2\,at \quad \Rightarrow \quad x = 1 - 6t, \; y = 2 - 3t^2$

Self Practice Problems

1. Find the parametric equation of the parabola $x^2 = 4ay$ **Ans.** $x = 2at, y = at^2$.

7. Position of a point Relative to a Parabola:

The point $(x_1 y_1)$ lies outside, on or inside the parabola $y^2 = 4ax$ according as the expression $y_1^2 - 4ax_1$ is positive, zero or negative.

Example : Check weather the point (3, 4) lies inside or outside the paabola $y^2 = 4x$.

Solution. $y^2 - 4x = 0$

$\because$ $S_1 \equiv y_1^2 - 4x_1 = 16 - 12 = 4 > 0$

$\therefore$ (3, 4) lies outside the parabola.

Self Practice Problems

1. Find the set of value's of α for which $(\alpha, -2-\alpha)$ lies inside the parabola $y^2 + 4x = 0$.

Ans. $a \in (-4-2\sqrt{3}\ ,\ -4+2\sqrt{3}\ )$

8. Line & a Parabola:

The line $y = mx + c$ meets the parabola $y^2 = 4ax$ in two points real, coincident or imaginary according as $a \gtrless cm \Rightarrow$ condition of tangency is, $c = a/m$.

Length of the chord intercepted by the parabola on the line $y = mx + c$ is:

$$\left(\frac{4}{m^2}\right)\sqrt{a(1+m^2)\,(a-mc)}\ .$$

NOTE : 1. The equation of a chord joining t_1 & t_2 is $2x - (t_1 + t_2)\,y + 2at_1t_2 = 0$.

2. If t_1 & t_2 are the ends of a focal chord of the parabola $y^2 = 4ax$ then $t_1t_2 = -1$. Hence the co–ordinates at the extremities of a focal chord can be taken as $(at^2, 2at)$ & $\left(\frac{a}{t^2}\ ,\ -\frac{2a}{t}\right)$

3. Length of the focal chord making an angle α with the x– axis is $4a\,\text{cosec}^2\,\alpha$.

Example : Discuss the position of line $y = x + 1$ with respect to parabolas $y^2 = 4x$.

Solution. Solving we get $(x + 1)^2 = 4x \Rightarrow (x - 1)^2 = 0$

so $y = x + 1$ is tangent to the parabola.

Example :

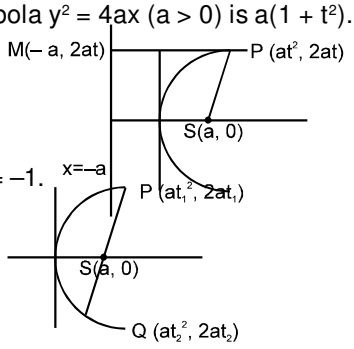
Prove that focal distance of a point $P(at^2, 2at)$ on parabola $y^2 = 4ax$ $(a > 0)$ is $a(1 + t^2)$.

Solution.

$\because$ PS = PM

$= a + at^2$

$PS = a(1 + t^2)$.



Example :

If t_1, t_2 are end points of a focal chord then show that $t_1 t_2 = -1$.

Solution.

Let parabola is $y^2 = 4ax$

since P, S & Q are collinear

$\therefore$ $m_{PQ} = m_{PS}$

$$\Rightarrow \quad \frac{2}{t_1 + t_2} = \frac{2t_1}{t_1^2 - 1}$$

$\Rightarrow$ $t_1^2 - 1 = t_1^2 + t_1t_2$

$\Rightarrow$ $t_1t_2 = -1$

Example :

If the endpoint t_1, t_2 of a chord satisfy the relation $t_1 t_2 = k$ (const.) then prove that the chord always passes through a fixed point. Find the point?

Solution.

Equation of chord joining $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ is

$$y - 2at_1 = \frac{2}{t_1 + t_2}\,(x - at_1^2)$$

$(t_1 + t_2)\,y - 2at_1^2 - 2at_1t_2 = 2x - 2at_1^2$

$$y = \frac{2}{t_1 + t_2}\,(x + ak) \qquad (\because \quad t_1t_2 = k)$$

$\therefore$ This line passes through a fixed point $(-ak, 0)$.

Self Practice Problems

1. If the line $y = 3x + \lambda$ intersect the parabola $y^2 = 4x$ at two distinct point's then set of value's of 'λ' is
Ans. $(-\infty, 1/3)$

2. Find the midpoint of the chord $x + y = 2$ of the parabola $y^2 = 4x$.
Ans. $(4, -2)$

3. If one end of focal chord of parabola $y^2 = 16x$ is (16, 16) then coordinate of other end is.
Ans. $(1, -4)$

4. If PSQ is focal chord of parabola $y^2 = 4ax$ $(a > 0)$, where S is focus then prove that

$$\frac{1}{PS} + \frac{1}{SQ} = \frac{1}{a}.$$

5. Find the length of focal chord whose one end point is 't'. **[Ans.** $a\left(t+\frac{1}{t}\right)^2$ **]**

9. Tangents to the Parabola $y^2 = 4ax$:

(i) $y y_1 = 2a(x + x_1)$ at the point (x_1, y_1) ; **(ii)** $y = mx + \frac{a}{m}$ $(m \neq 0)$ at $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$

(iii) $ty = x + at^2$ at $(at^2, 2at)$.

NOTE : Point of intersection of the tangents at the point t_1 & t_2 is $[at_1 t_2, a(t_1 + t_2)]$.

Example : Prove that the straight line $y = mx + c$ touches the parabola $y^2 = 4a(x + a)$ if $c = ma + \frac{a}{m}$

Solution. Equation of tangent of slope 'm' to the parabola $y^2 = 4a(x + a)$ is

$$y = m(x + a) + \frac{a}{m} \quad \Rightarrow \quad y = mx + a\left(m + \frac{1}{m}\right)$$

but the given tangent is $y = mx + c$ $\quad \therefore \quad c = am + \frac{a}{m}$

Example : A tangent to the parabola $y^2 = 8x$ makes an angel of 45° with the straight line $y = 3x + 5$. Find its equation and its point of contact.

Solution. Slope of required tangent's are

$$m = \frac{3 \pm 1}{1 \mp 3}$$

$$m_1 = -2, \qquad m_2 = \frac{1}{2}$$

$\because$ Equation of tangent of slope m to the parabola $y^2 = 4ax$ is

$$y = mx + \frac{a}{m}.$$

$\therefore$ tangent's $y = -2x - 1$ at $\left(\frac{1}{2}, -2\right)$

$$y = \frac{1}{2}x + 4 \text{ at } (8, 8)$$

Example :
Find the equation to the tangents to the paabola $y^2 = 9x$ which goes through the point (4, 10).

Solution.
Equation of tangent to parabola $y^2 = 9x$ is

$$y = mx + \frac{9}{4m}$$

Since it passes through (4, 10)

$\therefore \quad 10 = 4m + \frac{9}{4m} \quad \Rightarrow \quad 16m^2 - 40m + 9 = 0$

$$m = \frac{1}{4}, \frac{9}{4}$$

$\therefore$ equation of tangent's are $\quad y = \frac{x}{4} + 9 \quad$ & $\quad y = \frac{9}{4}x + 1.$

Example :
Find the equations to the common tangents of the parabolas $y^2 = 4ax$ and $x^2 = 4by$.

Solution.
Equation of tangent to $y^2 = 4ax$ is

$$y = mx + \frac{a}{m} \qquad \text{.....(i)}$$

Equation of tangent to $x^2 = 4by$ is

$$x = m_1 y + \frac{b}{m_1}$$

$$\Rightarrow \quad y = \frac{1}{m_1}x - \frac{b}{(m_1)^2} \qquad \text{.....(ii)}$$

for common tangent, (i) & (ii) must represent same line.

$\therefore \quad \frac{1}{m_1} = m \qquad \& \qquad \frac{a}{m} = -\frac{b}{m_1^2}$

$\Rightarrow \quad \frac{a}{m} = -bm^2 \quad \Rightarrow \quad m = \left(-\frac{a}{b}\right)^{1/3}$

$\therefore$ equation of common tangent is

$y = \left(-\frac{a}{b}\right)^{1/3} x + a\left(-\frac{b}{a}\right)^{1/3}$.

Self Practice Problems

1. Find equation tangent to parabola $y^2 = 4x$ whose intercept on y–axis is 2.

Ans. $y = \frac{x}{2} + 2$

2. Prove that perpendicular drawn from focus upon any tangent of a parabola lies on the tangent at the vertex.

3. Prove that image of focus in any tangent to parabola lies on its directrix.

4. Prove that the area of triangle formed by three tangents to the parabola $y^2 = 4ax$ is half the area of triangle formed by their points of contacts.

10. Normals to the parabola $y^2 = 4ax$:

(i) $y - y_1 = -\frac{y_1}{2a}(x - x_1)$ at (x_1, y_1) ;

(ii) $y = mx - 2am - am^3$ at $(am^2, -2am)$

(iii) $y + tx = 2at + at^3$ at $(at^2, 2at)$.

NOTE :

(i) Point of intersection of normals at t_1 & t_2 are, $a(t_1^2 + t_2^2 + t_1t_2 + 2); -at_1t_2(t_1 + t_2)$.

(ii) If the normals to the parabola $y^2 = 4ax$ at the point t_1, meets the parabola again at the point t_2, then $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$.

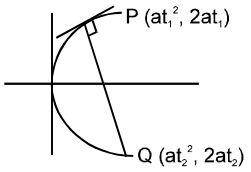
(iii) If the normals to the parabola $y^2 = 4ax$ at the points t_1 & t_2 intersect again on the parabola at the point 't_3' then $t_1t_2 = 2$; $t_3 = -(t_1 + t_2)$ and the line joining t_1 & t_2 passes through a fixed point $(-2a, 0)$.

Example :

If the normal at point 't_1' intersects the parabola again at 't_2' then show that $t_2 = -t_1 - \frac{2}{t_1}$

Solution.

Slope of normal at P $= -t_1$ and slope of chord PQ $= \frac{2}{t_1 + t_2}$



$\therefore \quad -t_1 = \frac{2}{t_1 + t_2}$

$t_1 + t_2 = -\frac{2}{t_1} \qquad \Rightarrow \qquad t_2 = -t_1 - \frac{2}{t_1}$.

Example :

If the normals at points t_1, t_2 meet at the point t_3 on the parabola then prove that

(i) $t_1t_2 = 2$ (ii) $t_1 + t_2 + t_3 = 0$

Solution.

Since normal at t_1 & t_2 meet the curve at t_3

$\therefore \quad t_3 = -t_1 - \frac{2}{t_1}$ .....(i)

$t_3 = -t_2 - \frac{2}{t_2}$ .....(ii)

$\Rightarrow \quad (t_1^2 + 2)t_2 = t_1(t_2^2 + 2)$

$t_1t_2(t_1 - t_2) + 2(t_2 - t_1) = 0$

$\because \quad t_1 \neq t_2$, $t_1t_2 = 2$ .....(iii)

Hence (i) $t_1t_2 = 2$

from equation (i) & (iii), we get

$t_3 = -t_1 - t_2$

Hence (ii) $t_1 + t_2 + t_3 = 0$

Example :

Find the locus of the point N from which 3 normals are drawn to the parabola $y^2 = 4ax$ are such that

(i) Two of them are equally inclined to x-axis

(ii) Two of them are perpendicular to each other

Solution.

Equation of normal to $y^2 = 4ax$ is

$y = mx - 2am - am^3$

Let the normal is passes through N(h, k)

$\therefore \quad k = mh - 2am - am^3 \quad \Rightarrow \quad am^3 + (2a - h)\, m + k = 0$

For given value's of (h, k) it is cubic in 'm'.

Let m_1, m_2 & m_3 are root's

$\therefore \quad m_1 + m_2 + m_3 = 0$ .....(i)

$$m_1m_2 + m_2m_3 + m_3m_1 = \frac{2a-h}{a} \quad \text{.....(ii)}$$

$$m_1m_2m_3 = -\frac{k}{a} \quad \text{.....(iii)}$$

(i) If two nromal are equally inclined to x-axis, then $m_1 + m_2 = 0$

$\therefore \quad m_3 = 0 \quad \Rightarrow \quad y = 0$

(ii) If two normal's are perpendicular

$\therefore \quad m_1 m_2 = -1$

from (3) $\quad m_3 = \frac{k}{a}$ .....(iv)

from (2) $\quad -1 + \frac{k}{a}(m_1 + m_2) = \frac{2a-h}{a}$ .....(v)

from (1) $\quad m_1 + m_2 = -\frac{k}{a}$ .....(vi)

from (5) & (6), we get

$$-1 - \frac{k^2}{a} = 2 - \frac{h}{a}$$

$y^2 = a(x - 3a)$

Self Practice Problems

1. Find the points of the parabola $y^2 = 4ax$ at which the normal is inclined at 30° to the axis.

Ans. $\left(\frac{a}{3}, -\frac{2a}{\sqrt{3}}\right), \left(\frac{a}{3}, \frac{2a}{\sqrt{3}}\right)$

2. If the normal at point P(1, 2) on the parabola $y^2 = 4x$ cuts it again at point Q then Q = ?

Ans. $(9, -6)$

3. Find the length of normal chord at point 't' to the parabola $y^2 = 4ax$.

Ans. $\ell = \frac{4a(t^2+1)^{\frac{3}{2}}}{t^2}$

4. If normal chord at a point 't' on the parabola $y^2 = 4ax$ subtends a right angle at the vertex then prove that $t^2 = 2$

5. Prove that the chord of the parabola $y^2 = 4ax$, whose equation is $y - x\sqrt{2} + 4a\sqrt{2} = 0$, is a normal to the curve and that its length is $6\sqrt{3}a$.

6. If the normals at 3 points P, Q & R are concurrent, then show that
(i) The sum of slopes of normals is zero, (ii) Sum of ordinates of points P, Q, R is zero
(iii) The centroid of ΔPQR lies on the axis of parabola.

11. Pair of Tangents:

The equation to the pair of tangents which can be drawn from any point (x_1, y_1) to the parabola $y^2 = 4ax$ is given by: $SS_1 = T^2$ where :

$S \equiv y^2 - 4ax$; $S_1 = y_1^2 - 4ax_1$; $T \equiv y\, y_1 - 2a(x + x_1)$.

Example :

Write the equation of pair of tangents to the parabola $y^2 = 4x$ drawn from a point P(–1, 2)

Solution.

We know the equation of pair of tangents are given by $SS_1 = T^2$

$\therefore \quad (y^2 - 4x)(4 + 4) = (2y + 2(x - 1))^2$

$\Rightarrow \quad 8y^2 - 32x = 4y^2 + 4x^2 + 4 + 8xy - 8y - 8x$

$\Rightarrow \quad y^2 - x^2 - 2xy - 6x + 2y = 1$

Example :

Find the focus of the point P from which tangents are drawn to parabola $y^2 = 4ax$ having slopes m_1, m_2 such that

(i) $m_1 + m_2 = m_0$ (const) (ii) $\theta_1 + \theta_2 = \theta_0$ (const)

Sol. Equation of tangent to $y^2 = 4ax$, is

$$y = mx + \frac{a}{m}$$

Let it passes through P(h, k)

$\therefore$ $m^2h - mk + a = 0$

(i) $m_1 + m_2 = m_0 = \dfrac{k}{h} \quad \Rightarrow \quad y = m_0x$

(ii) $\tan\theta_0 = \dfrac{m_1 + m_2}{1 - m_1 m_2} = \dfrac{k/h}{1 - a/h}$

$\Rightarrow \quad y = (x - a)\tan\theta_0$

Self Practice Problem

1. If two tangents to the parabola $y^2 = 4ax$ from a point P make angles θ_1 and θ_2 with the axis of the parabola, then find the locus of P in each of the following cases.

(i) $\tan^2\theta_1 + \tan^2\theta_2 = \lambda$ (a constant)

(ii) $\cos\theta_1 \cos\theta_2 = \lambda$ (a constant)

Ans. (i) $y^2 - 2ax = \lambda x^2$, (ii) $x^2 = \lambda^2\{(x - a)^2 + y^2\}$

12. Director Circle:

Locus of the point of intersection of the perpendicular tangents to a curve is called the Director Circle. For parabola $y^2 = 4ax$ it's equation is $x + a = 0$ which is parabola's own directrix.

13. Chord of Contact:

Equation to the chord of contact of tangents drawn from a point $P(x_1, y_1)$ is $yy_1 = 2a(x + x_1)$.

NOTE : The area of the triangle formed by the tangents from the point (x_1, y_1) & the chord of contact is $(y_1^2 - 4ax_1)^{3/2} \div 2a$.

Example :

Find the length of chord of contact of the tangents drawn from point (x_1, y_1) to the parabola $y^2 = 4ax$.

Solution.

Let tangent at $P(t_1)$ & $Q(t_2)$ meet at (x_1, y_1)

$\therefore$ $at_1t_2 = x_1 \quad$ & $\quad a(t_1 + t_2) = y_1$

$\because$ $PQ = \sqrt{(at_1^2 - at_2^2)^2 + (2a(t_1 - t_2))^2}$

$= a\sqrt{((t_1 + t_2)^2 - 4t_1t_2)((t_1 + t_2)^2 + 4)}$

$= \sqrt{\dfrac{(y_1^2 - 4ax_1)(y_1^2 + 4a^2)}{a^2}}$

Example :

If the line $x - y - 1 = 0$ intersect the parabola $y^2 = 8x$ at P & Q, then find the point of intersection of tangents at P & Q.

Solution.

Let (h, k) be point of intersection of tangents then chord of contact is

$yk = 4(x + h)$

$4x - yk + 4h = 0 \qquad$ .....(i)

But given is

$x - y - 1 = 0$

$\therefore \quad \dfrac{4}{1} = \dfrac{-k}{-1} = \dfrac{4h}{-1}$

$\Rightarrow \quad h = -1, k = 4$

$\therefore \quad$ point $\equiv (-1, 4)$

Example :

Find the locus of point whose chord of contact w.r.t to the parabola $y^2 = 4bx$ is the tangents of the parabola $y^2 = 4ax$.

Solution.

Equation of tangent to $y^2 = 4ax$ is $y = mx + \dfrac{a}{m} \qquad$ .....(i)

Let it is chord of contact for parabola $y^2 = 4bx$ w.r.t. the point P(h, k)

$\therefore$ Equation of chord of contact is

$yk = 2b(x + h)$

$y = \dfrac{2b}{k}x + \dfrac{2bh}{k} \qquad$ .....(ii)

From (i) & (ii)

$m = \dfrac{2b}{k}, \dfrac{a}{m} = \dfrac{2bh}{k} \quad \Rightarrow \quad a = \dfrac{4b^2h}{k^2}$

locus of P is

$y^2 = \frac{4b^2}{a}x$.

Self Practice Problems

1. Prove that locus of a point whose chord of contact w.r.t. parabola passes through focus is directrix

2. If from a variable point 'P' on the line $x - 2y + 1 = 0$ pair of tangent's are drawn to the parabola $y^2 = 8x$ then prove that chord of contact passes through a fixed point, also find that point.
Ans. (1, 8)

14. Chord with a given middle point:

Equation of the chord of the parabola $y^2 = 4ax$ whose middle point is

(x_1, y_1) is $y - y_1 = \frac{2a}{y_1}(x - x_1) \equiv T = S_1$

Example :
Find the locus of middle point of the chord of the parabola $y^2 = 4ax$ which pass through a given point (p, q).
Solution.
Let P(h, k) be the mid point of chord of parabola $y^2 = 4ax$,
so equation of chord is $yk - 2a(x + h) = k^2 - 4ah$.
Since it passes through (p, q)
$\therefore$ $qk - 2a(p + h) = k^2 - 4ah$
$\therefore$ Required locus is
$y^2 - 2ax - qy + 2ap = 0$.

Example :
Find the locus of middle point of the chord of the parabola $y^2 = 4ax$ whose slope is 'm'.
Solution.
Let P(h, k) be the mid point of chord of parabola $y^2 = 4ax$,
so equation of chord is $yk - 2a(x + h) = k^2 - 4ah$.

but slope $= \frac{2a}{k} = m$

$\therefore$ locus is $y = \frac{2a}{m}$

Self Practice Problems

1. Find the equation of chord of parabola $y^2 = 4x$ whose mid point is (4, 2).
Ans. $x - y - 2 = 0$

2. Find the locus of mid - point of chord of parabola $y^2 = 4ax$ which touches the parabola $x^2 = 4by$.
Ans. $y(2ax - y^2) = 4a^2b$

15. Important Highlights:

(i) If the tangent & normal at any point 'P' of the parabola intersect the axis at T & G then ST = SG = SP where 'S' is the focus. In other words the tangent and the normal at a point P on the parabola are the bisectors of the angle between the focal radius SP & the perpendicular from P on the directrix. From this we conclude that all rays emanating from S will become parallel to the axis of theparabola after reflection.

(ii) The portion of a tangent to a parabola cut off between the directrix & the curve subtends a right angle at the focus.

(iii) The tangents at the extremities of a focal chord intersect at right angles on the directrix, and hence a circle on any focal chord as diameter touches the directrix. Also a circle on any focal radii of a point P $(at^2, 2at)$ as diameter touches the tangent at the vertex and intercepts a chord of length $a\sqrt{1 + t^2}$ on a normal at the point P.

(iv) Any tangent to a parabola & the perpendicular on it from the focus meet on the tangent at the vertex.

(v) If the tangents at P and Q meet in T, then:
$\Rightarrow$ TP and TQ subtend equal angles at the focus S.
$\Rightarrow$ $ST^2 = SP.\ SQ$ & $\Rightarrow$ The triangles SPT and STQ are similar.

(vi) Semi latus rectum of the parabola $y^2 = 4ax$, is the harmonic mean between segments of any focal chord of the parabola.

(vii) The area of the triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents at these points.

(viii) If normal are drawn from a point P(h, k) to the parabola $y^2 = 4ax$ then
$k = mh - 2am - am^3$ i.e. $am^3 + m(2a - h) + k = 0$.

$m_1 + m_2 + m_3 = 0$; $m_1m_2 + m_2m_3 + m_3m_1 = \frac{2a-h}{a}$; $m_1m_2m_3 = -\frac{k}{a}$.

Where m_1, m_2, & m_3 are the slopes of the three concurrent normals. Note that

$\Rightarrow$ algebraic sum of the slopes of the three concurrent normals is zero.

$\Rightarrow$ algebraic sum of the ordinates of the three conormal points on the parabola is zero

$\Rightarrow$ Centroid of the Δ formed by three co–normal points lies on the x–axis.

$\Rightarrow$ Condition for three real and distinct normals to be drawn froma point P (h, k) is

$$h > 2a \text{ \& } k^2 < \frac{4}{27a}(h-2a)^3.$$

(ix) Length of subtangent at any point P(x, y) on the parabola $y^2 = 4ax$ equals twice the abscissa of the point P. Note that the subtangent is bisected at the vertex.

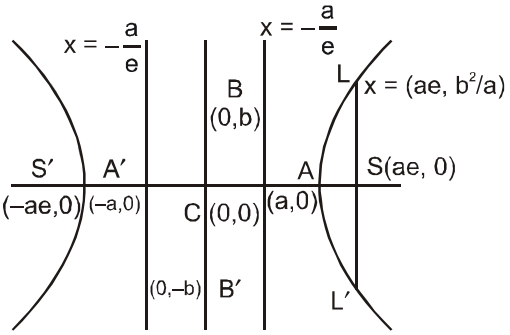
(x) Length of subnormal is constant for all points on the parabola & is equal to the semi latus rectum.

Note : Students must try to proof all the above properties.

Hyperbola

The **Hyperbola** is a conic whose eccentricity is greater than unity. (e > 1).

1. Standard Equation & Definition(s)



Standard equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$,

where $b^2 = a^2(e^2 - 1)$.

Eccentricity (e) : $e^2 = 1 + \frac{b^2}{a^2} = 1 + \left(\frac{C.A}{T.A}\right)^2$

Foci : $S \equiv (ae, 0)$ & $S' \equiv (-ae, 0)$.

Equations Of Directrices :

$x = \frac{a}{e}$ & $x = -\frac{a}{e}$.

Transverse Axis : The line segment A′A of length 2a in which the foci S′ & S both lie is called the transverse axis of the hyperbola.

Conjugate Axis : The line segment B′B between the two points $B' \equiv (0, -b)$ & $B \equiv (0, b)$ is called as the conjugate axis of the hyperbola.

Principal Axes : The transverse & conjugate axis together are called Principal Axes of the hyperbola.

Vertices : $A \equiv (a, 0)$ & $A' \equiv (-a, 0)$

Focal Chord : A chord which passes through a focus is called a focal chord.

Double Ordinate : A chord perpendicular to the transverse axis is called a double ordinate.

Latus Rectum (ℓ) :

The focal chord perpendicular to the transverse axis is called the latus rectum.

$$\ell = \frac{2b^2}{a} = \frac{(C.A.)^2}{T.A.} = 2a(e^2 - 1).$$

Note : ℓ (L.R.) = 2e (distance from focus to directrix)

Centre : The point which bisects every chord of the conic drawn through it is called the centre of the conic. $C \equiv (0, 0)$ the origin is the centre of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

General Note :

Since the fundamental equation to the hyperbola only differs from that to the ellipse in having $-b^2$ instead of b^2 it will be found that many propositions for the hyperbola are derived from those for the ellipse by simply changing the sign of b^2.

Example : Find the equation of the hyperbola whose directrix is 2x + y = 1, focus (1, 2) and eccentricity $\sqrt{3}$.

Solution. Let P 9x,y) be any point on the hyperbola.

Draw PM perpendicular from P on the directrix.

Then by definition SP = e PM

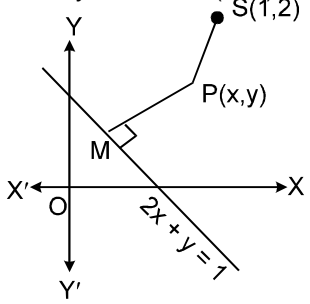
$\Rightarrow$ $(SP)^2 = e^2 (PM)^2$

$\Rightarrow$ $(x-1)^2 + (y-2)^2 = 3\left\{\frac{2x+y-1}{\sqrt{4+1}}\right\}^2$

$\Rightarrow$ $5(x^2 + y^2 - 2x - 4y + 5\}$
$= 3(4x^2 + y^2 + 1 + 4xy - 2y - 4x)$

$\Rightarrow$ $7x^2 - 2y^2 + 12xy - 2x + 14y - 22 = 0$

which is the required hyperbola.



Example : Find the eccentricity of the hyperbola whose latus rectum is half of its transverse axis.

Solution. Let the equation of hyperbola be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Then transverse axis = 2a and latus–rectum = $\frac{2b^2}{a}$

According to question $\frac{2b^2}{a} = \frac{1}{2}(2a)$

$\Rightarrow$ $2b^2 = a^2$ $(\because b^2 = a^2(e^2 - 1))$

$\Rightarrow$ $2a^2(e^2 - 1) = a^2$ $\Rightarrow$ $2e^2 - 2 = 1$

$\Rightarrow$ $e = \frac{3}{2}$ $\therefore$ $e = \sqrt{\frac{3}{2}}$

Hence the required eccentricity is $\sqrt{\frac{3}{2}}$.

2. Conjugate Hyperbola :

Two hyperbolas such that transverse & conjugate axes of one hyperbola are respectively the conjugate & the transverse axes of the other are called Conjugate Hyperbolas of each other.

eg. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ & $-\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are conjugate hyperbolas of each.

Note : **(a)** If e_1 & e_2 are the eccentrcities of the hyperbola & its conjugate then $e_1^{-2} + e_2^{-2} = 1$.

(b) The foci of a hyperbola and its conjugate are concyclic and form the vertices of a square.

(c) Two hyperbolas are said to be similiar if they have the same eccentricity.

(d) Two similar hyperbolas are said to be equal if they have same latus rectum.

(e) If a hyperbola is equilateral then the conjugate hyperbola is also equilateral.

Example : Find the lengths of transverse axis and conjugate axis, eccentricity, the co-ordinates of foci, vertices, lengths of the latus-rectum and equations of the directrices of the following hyperbolas $16x^2 - 9y^2 = -144$.

Solution. The equation $16x^2 - 9y^2 = -144$ can be written as

$$\frac{x^2}{9} - \frac{y^2}{16} = -1$$

This is of the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$

$\therefore$ $a^2 = 9, b^2 = 16$ $\Rightarrow$ $a = 3, b = 4$

Length of transverse axis : The length of transverse axis = 2b = 8

Length of conjugate axis : The length of conjugate axis = 2a = 6

Eccentricity : $e = \sqrt{\left(1 + \frac{a^2}{b^2}\right)} = \sqrt{\left(1 + \frac{9}{16}\right)} = \frac{5}{4}$

Foci : The co^{a}ordinates of the foci are $(0, \pm be)$ i.e., $(0, \pm 5)$

Vertices : The co–ordinates of the vertices are $(0, \pm b)$ i.e., $(0, \pm 4)$

Length of latus–rectum : The length of latus–rectum = $\frac{2a^2}{b}$

$$= \frac{2(3)^2}{4}\ \frac{9}{2}$$

Equation of directrices : The equation of directrices are

$$y = \pm \frac{b}{e}$$

$$y = \pm \frac{4}{(5/4)} \qquad y = \pm \frac{16}{5}$$

Self Practice Problems :

1. Find the equation of the hyperbola whose focis are (6, 4) and (– 4, 4) and eccentricity is 2.
Ans. $12x^2 - 4y^2 - 24x + 32y - 127 = 0$

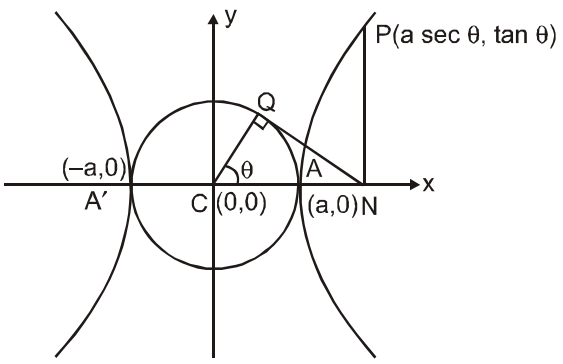
2. Obtain the equation of a hyperbola with coordinates axes as principal axes given that the distances of one of its vertices from the foci are 9 and 1 units. **Ans.** $\frac{x^2}{16} - \frac{y^2}{9} = 1,\ \frac{y^2}{16} - \frac{x^2}{9} = 1$

3. The foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$. Find the equation of the hyperbola if its eccentricity is 2. **Ans.** $3x^2 - y^2 - 12 = 0$.

3. Auxiliary Circle :

A circle drawn with centre C & T.A. as a diameter is called the **Auxiliary Circle** of the hyperbola. Equation of the auxiliary circle is $x^2 + y^2 = a^2$.

Note from the figure that P & Q are called the **"Corresponding Points"** on the hyperbola & the auxiliary circle.



4. Parametric Representation : The equations $x = a \sec\theta$ & $y = b \tan\theta$ together represents the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where θ is a parameter. The parametric equations; $x = a\cosh\phi$, $y = b\sinh\phi$ also represents the same hyperbola.

Note that if $P(\theta) \equiv (a \sec\theta, b \tan\theta)$ is on the hyperbola then;

$Q(\theta) \equiv (a\cos\theta, a\sin\theta)$ is on the auxiliary circle.

The equation to the chord of the hyperbola joining two points with eccentric angles α & β is given by $\frac{x}{a}\cos\frac{\alpha-\beta}{2} - \frac{y}{b}\sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha+\beta}{2}$.

5. Position Of A Point 'P' w.r.t. A Hyperbola :

The quantity $S_1 \equiv \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$ is positive, zero or negative according as the point (x_1, y_1) lies inside, on or outside the curve.

Example : Find the position of the point (5, – 4) relative to the hyperbola $9x^2 – y^2 = 1$.

Solution. Since $9(5)^2 – (–4)^2 = 1 = 225 – 16 – 1 = 208 > 0$,

So the point (5,–4) inside the hyperbola $9x^2 – y^2 = 1$.

6. Line And A Hyperbola :

The straight line $y = mx + c$ is a secant, a tangent or passes outside the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ according as : $c^2 >$ or $=$ or $< a^2m^2 – b^2$, respectively.

7. Tangents :

(i) Slope Form : $y = mx \pm \sqrt{a^2m^2 - b^2}$ can be taken as the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, having slope 'm'.

(ii) Point Form : Equation of tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point (x_1, y_1) is

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1.$$

(iii) Parametric Form : Equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point.

$(a \sec\theta, b \tan\theta)$ is $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$.

Note : **(i)** Point of intersection of the tangents at θ_1 & θ_2 is $x = a\frac{\cos\frac{\theta_1-\theta_2}{2}}{\cos\frac{\theta_1+\theta_2}{2}}$, $y = b\tan\left(\frac{\theta_1+\theta_2}{2}\right)$

(ii) If $|\theta_1 + \theta_2| = \pi$, then tangetns at these points (θ_1 & θ_2) are parallel.

(iii) There are two parallel tangents having the same slope m. These tangents touches the hyperbola at the extremities of a diameter.

Example :

Prove that the straight line $\ell x + my + n = 0$ touches the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if $a^2\ell^2 – b^2m^2 = n^2$.

Solution.

The given line is $\ell x + my + n = 0$

or $y = -\frac{\ell}{m}x - \frac{n}{m}$

Comparing this line with $y = Mx + c$

$\therefore$ $M = -\frac{\ell}{m}$ and $c = -\frac{n}{m}$ .....(1)

This line (1) will touch the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ if } c^2 = a^2M^2 - b^2$$

$\Rightarrow$ $\frac{n^2}{m^2} = \frac{a^2l^2}{m^2} - b^2$

or $a^2\ell^2 - b^2m^2 = n^2$

Example :

Find the equation of the tangent to the hyperbola $x^2 - 4y^2 = 36$ which is perpendicular to the line $x - y + 4 = 0$.

Solution.

Let m be the slope of the tangent. Since the tangent is perpendicular to the line $x - y = 0$

$\therefore$ $m \times 1 = -1 \Rightarrow m = -1$

Since $x^2 - 4y^2 = 36$ or $\frac{x^2}{36} - \frac{y^2}{9} = 1$

Comparing this with $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ $\therefore$ $a^2 = 36$ and $b^2 = 9$

So the equation of tangents are

$$y = (-1)\, x \pm \sqrt{36 \times (-1)^2 - 9}$$

$$y = -x \pm \sqrt{27}$$

$\Rightarrow$ $x + y \pm 3\sqrt{3} = 0$

Example : Find the equation and the length of the common tangents to hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$.

Solution. Tangent at $(a \sec \phi\ b \tan \phi)$ on the 1st hyperbola is

$$\frac{x}{a} \sec \phi - \frac{y}{b} \tan \phi = 1 \quad .....(1)$$

Similarly tangent at any point $(b \tan \theta, a \sec \theta)$ on 2nd hyperbolas is

$$\frac{y}{a} \sec \theta - \frac{x}{b} \tan \theta = 1 \quad .....(2)$$

If (1) and (2) are common tangents then they should be identical. Comparing the co–effecients of x and y

$\Rightarrow$ $\frac{\sec\theta}{a} = -\frac{\tan\phi}{b}$ .....(3)

and $-\frac{\tan\theta}{b} = \frac{\sec\phi}{a}$

or $\sec\theta = -\frac{a}{b}\tan\phi$ .....(4)

$\because$ $\sec^2\theta - \tan^2\theta = 1$

$\Rightarrow$ $\frac{a^2}{b^2}\tan^2\phi - \frac{b^2}{a^2}\sec^2\phi = 1$ {from (3) and (4)}

or $\frac{a^2}{b^2}\tan^2\phi - \frac{b^2}{a^2}(1 + \tan^2\phi) = 1$

or $\left(\frac{a^2}{b^2} - \frac{b^2}{a^2}\right)\tan^2\phi = 1 + \frac{b^2}{a^2}$

$$\tan^2\phi = \frac{b^2}{a^2 - b^2}$$

and $\sec^2\phi = 1 + \tan^2\phi = \frac{a^2}{a^2 - b^2}$

Hence the point of contanct are

$$\left\{\pm\frac{a^2}{\sqrt{(a^2-b^2)}}, \pm\frac{b^2}{\sqrt{(a^2-b^2)}}\right\} \text{ and } \left\{\pm\frac{b^2}{\sqrt{(a^2-b^2)}}, \pm\frac{a^2}{\sqrt{(a^2-b^2)}}\right\} \text{ {from (3) and (4)}}$$

Length of common tangent i.e., the distance between the above points is $\sqrt{2}\frac{(a^2+b^2)}{\sqrt{(a^2-b^2)}}$ and equation

of common tangent on putting the values of secϕ and tanϕ in (1) is

$$\pm \frac{x}{\sqrt{(a^2-b^2)}} \mp \frac{y}{\sqrt{(a^2-b^2)}} = 1 \qquad \text{or} \qquad x \mp y = \pm \sqrt{(a^2-b^2)}$$

Self Practice Problems :

1. Show that the line $x \cos\alpha + y \sin\alpha = p$ touches the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if $a^2\cos^2\alpha - b^2\sin^2\alpha = p^2$. **Ans.** $p^2 = a^2\cos^2\alpha - b^2\sin^2\alpha$

2. For what value of λ does the line $y = 2x + \lambda$ touches the hyperbola $16x^2 - 9y^2 = 144$?

Ans. $\lambda = \pm 2\sqrt{5}$

3. Find the equation of the tangent to the hyperbola $x^2 - y^2 = 1$ which is parallel to the line $4y = 5x + 7$.

Ans. $4y = 5x \pm 3$

8. NORMALS:

(a) The equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point P (x_1, y_1) on it is

$$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2 = a^2e^2.$$

(b) The equation of the normal at the point P (a sec θ, b tan θ) on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2 = a^2e^2$.

(c) Equation of normals in terms of its slope 'm' are $y = mx \pm \frac{\left(a^2+b^2\right)m}{\sqrt{a^2-b^2m^2}}$.

Example : A normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the axes in M and N and lines MP and NP are drawn perpendicular to the axes meeting at P. Prove that the locus of P is the hyperbola $a^2x^2 - b^2y^2 = (a^2 + b^2)^2$.

Solution. The equation of normal at the point Q (a sec ϕ, b tan ϕ) to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$ax \cos\phi + by \cot\phi = a^2 + b^2$ .....(1)

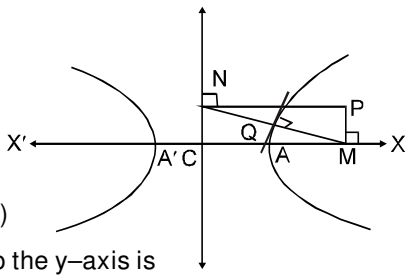
The normal (1) meets the x–axis in

$$M\left(\frac{a^2+b^2}{a}\sec\phi,\ 0\right)$$ and y–axis in

$$N\left(0, \frac{a^2+b^2}{b}\tan\phi\right)$$

$\therefore$ Equation of MP, the line through M and perpendicular to x–axis, is

$$x = \left(\frac{a^2+b^2}{a}\right)\sec\phi \text{ or } \sec\phi = \frac{ax}{(a^2+b^2)} \qquad .....(2)$$



and the equation of NP, the line through N and perpendicular to the y–axis is

$$y = \left(\frac{a^2+b^2}{b}\right)\tan\phi \text{ or } \tan\phi = \frac{by}{(a^2+b^2)} \qquad .....(3)$$

The locus of the point of intersection of MP and NP will be obtained by eliminating ϕ from (2) and (3), we have

$\sec^2\phi - \tan^2\phi = 1$

$$\Rightarrow \quad \frac{a^2x^2}{(a^2+b^2)^2} - \frac{b^2y^2}{(a^2+b^2)^2} = 1$$

or $a^2x^2 - b^2y^2 = (a^2 + b^2)^2$

is the required locus of P.

Self Practice Problems :

1. Prove that the line $lx + my - n = 0$ will be a normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if $\frac{a^2}{\ell^2} - \frac{b^2}{m^2} = \frac{(a^2+b^2)^2}{n^2}$.

Ans. $\frac{a^2}{\ell^2} - \frac{b^2}{m^2} = \frac{(a^2+b^2)^2}{n^2}$.

2. Find the locus of the foot of perpendicular from the centre upon any normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Ans. $(x^2 + y^2)^2 (a^2y^2 - b^2x^2) = x^2y^2 (a^2 + b^2)$

9. Pair of Tangents:

The equation to the pair of tangents which can be drawn from any point (x_1, y_1) to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is given by: $SS_1 = T^2$ where :

$S \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1$; $S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$; $T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$.

Example : How many real tangents can be drawn from the point (4, 3) to the ellipse $\frac{x^2}{16} - \frac{y^2}{9} = 1$. Find the equation these tangents & angle between them.

Solution.

Given point $P \equiv (4, 3)$

Hyperbola $S \equiv \frac{x^2}{16} - \frac{y^2}{9} - 1 = 0$

$\because$ $S_1 \equiv \frac{16}{16} - \frac{9}{9} - 1 = -1 < 0$

$\Rightarrow$ Point $P \equiv (4, 3)$ lies outside the hyperbola.

$\therefore$ Two tangents can be drawn from the point P(4, 3).

Equation of pair of tangents is

$SS_1 = T^2$

$\Rightarrow$ $\left(\frac{x^2}{16} - \frac{y^2}{9} - 1\right) . (-1) = \left(\frac{4x}{16} - \frac{3y}{9} - 1\right)^2$

$\Rightarrow$ $-\frac{x^2}{16} + \frac{y^2}{9} + 1 = \frac{x^2}{16} + \frac{y^2}{9} + 1 - \frac{xy}{6} - \frac{x}{2} + \frac{2y}{3}$

$\Rightarrow$ $3x^2 - 4xy - 12x + 16y = 0$

$\theta = \tan^{-1}\left(\frac{4}{3}\right)$

Example : Find the locus of point of intersection of perpendicular tangents to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Solution. Let P(h, k) be the point of intersection of two perpendicular tangents

equation of pair of tangents is $SS_1 = T^2$

$\Rightarrow$ $\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} - 1\right)\left(\frac{h^2}{a^2} - \frac{k^2}{b^2} - 1\right) = \left(\frac{hx}{a^2} - \frac{ky}{b^2} - 1\right)^2$

$\Rightarrow$ $\frac{x^2}{a^2}\left(-\frac{k^2}{b^2} - 1\right) - \frac{y^2}{b^2}\left(\frac{h^2}{a^2} - 1\right) + \ldots\ldots = 0$ .....(i)

Since equation (i) represents two perpendicular lines

$\therefore$ $\frac{1}{a^2}\left(-\frac{k^2}{b^2} - 1\right) - \frac{1}{b^2}\left(\frac{h^2}{a^2} - 1\right) = 0$

$\Rightarrow$ $-k^2 - b^2 - h^2 + a^2 = 0$ $\Rightarrow$ locus is $x^2 + y^2 = a^2 - b^2$ **Ans.**

10. Director Circle :

The locus of the intersection point of tangents which are at right angles is known as the Director Circle of the hyperbola. The equation to the director circle is : $x^2 + y^2 = a^2 - b^2$.

If $b^2 < a^2$ this circle is real.

If $b^2 = a^2$ (rectangular hyperbola) the radius of the circle is zero & it reduces to a point circle at the origin. In this case the centre is the only point from which the tangents at right angles can be drawn to the curve.

If $b^2 > a^2$, the radius of the circle is imaginary, so that there is no such circle & so no pair of tangents at right angle can be drawn to the curve.

11. Chord of Contact:

Equation to the chord of contact of tangents drawn from a point $P(x_1, y_1)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

T = 0, where $T = \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$

Example :

If tangents to the parabola $y^2 = 4ax$ intersect the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at A and B, then find the locus of point of intersection of tangents at A and B.

Solution.

Let P ≡ (h, k) be the point of intersection of tangents at A & B

$\therefore$ equation of chord of contact AB is $\frac{xh}{a^2} - \frac{yk}{b^2} = 1$ .....(i)

which touches the parabola

equation of tangent to parabola $y^2 = 4ax$

$y = mx - \frac{a}{m} \quad \Rightarrow \quad mx - y = -\frac{a}{m}$ .....(ii)

equation (i) & (ii) as must be same

$\therefore \quad \frac{m}{\left(\frac{h}{a^2}\right)} = \frac{-1}{\left(-\frac{k}{b^2}\right)} = \frac{-\frac{a}{m}}{1} \quad \Rightarrow \quad m = \frac{h}{k}\,\frac{b^2}{a^2}$ & $m = -\frac{ak}{b^2}$

$\therefore \quad \frac{hb^2}{ka^2} = -\frac{ak}{b^2} \quad \Rightarrow$ locus of P is $y^2 = -\frac{b^4}{a^3}\,.\,x$ **Ans.**

12. Chord with a given middle point:

Equation of the chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ whose middle point is (x_1, y_1) is $T = S_1$,

where $S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$; $T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$.

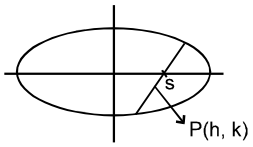
Example : Find the locus of the mid - point of focal chords of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Solution. Let P ≡ (h, k) be the mid-point

$\therefore$ equation of chord whose mid-point is given $\frac{xh}{a^2} - \frac{yk}{b^2} - 1 = \frac{h^2}{a^2} - \frac{k^2}{b^2} - 1$

since it is a focal chord,

$\therefore$ it passes through focus, either (ae, 0) or (–ae, 0)



If it passes trhrough (ae, 0)

$\therefore$ locus is $\frac{ex}{a} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$

If it passes through (–ae, 0)

$\therefore$ locus is $-\frac{ex}{a} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$ **Ans.**

Example : Find the condition on 'a' and 'b' for which two distinct chords of the hyperbola $\frac{x^2}{2a^2} - \frac{y^2}{2b^2} = 1$ passing through (a, b) are bisected by the line x + y = b.

Solution. Let the line x + y = b bisect the chord at $P(\alpha, b - \alpha)$

$\therefore$ equation of chord whose mid-point is $P(\alpha, b - \alpha)$

$$\frac{x\alpha}{2a^2} - \frac{y(b-\alpha)}{2b^2} = \frac{\alpha^2}{2a^2} - \frac{(b-\alpha)^2}{2b^2}$$

Since it passes through (a, b)

$\therefore$

$$\frac{\alpha}{2a} - \frac{(b-\alpha)}{2b} = \frac{\alpha^2}{2a^2} - \frac{(b-\alpha)^2}{2b^2}$$

$$\alpha^2\left(\frac{1}{a^2} - \frac{1}{b^2}\right) + \alpha\left(\frac{1}{b} - \frac{1}{a}\right) = 0$$

$\alpha = 0, \quad \alpha = \frac{1}{\frac{1}{a} + \frac{1}{b}} \qquad \therefore \quad a \neq \pm b$

Example : Find the locus of the mid point of the chords of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ which subtend a right angle at the origin.

Solution. let (h,k) be the mid–point of the chord of the hyperbola. Then its equation is

$$\frac{hx}{a^2} - \frac{ky}{b^2} - 1 = \frac{h^2}{b^2} - \frac{k^2}{b^2} - 1 \qquad \text{or} \qquad \frac{hx}{a^2} - \frac{ky}{b^2} = \frac{h^2}{a^2} - \frac{k^2}{b^2} \quad \text{.....(i)}$$

The equation of the lines joining the origin to the points of intersection of the hyperbola and the chored (1) is obtained by making homogeneous hyperbola with the help of (1)

$$\therefore \qquad \frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{\left(\frac{hx}{a^2} - \frac{ky}{b^2}\right)^2}{\left(\frac{h^2}{a^2} - \frac{k^2}{b^2}\right)^2}$$

$$\Rightarrow \qquad \frac{1}{a^2}\left(\frac{h^2}{a^2} - \frac{k^2}{b^2}\right)^2 x^2 - \frac{1}{b^2}\left(\frac{h^2}{a^2} - \frac{k^2}{b^2}\right)^2 y^2 = \frac{h^2}{a^4}x^2 + \frac{k^2}{b^4}y^2 - \frac{2hk}{a^2b^2}xy \quad \text{.....(2)}$$

The lines represented by (2) will be at right angle if coefficient of x^2 + coefficient of $y^2 = 0$

$$\Rightarrow \qquad \frac{1}{a^2}\left(\frac{h^2}{a^2} - \frac{k^2}{b^2}\right)^2 - \frac{h^2}{a^4} - \frac{1}{b^2}\left(\frac{h^2}{a^2} - \frac{k^2}{b^2}\right)^2 - \frac{k^2}{b^4} = 0$$

$$\Rightarrow \qquad \left(\frac{h^2}{a^2} - \frac{k^2}{b^2}\right)^2 \left(\frac{1}{a^2} - \frac{1}{b^2}\right) = \frac{h^2}{a^4} + \frac{k^2}{b^4}$$

hence, the locus of (h,k) is

$$\left(\frac{x^2}{a^2} - \frac{y^2}{b^2}\right)^2 \left(\frac{1}{a^2} - \frac{1}{b^2}\right) = \frac{x^2}{a^4} + \frac{y^2}{b^4}$$

Self Practice Problem

1. Find the equation of the chord $\frac{x^2}{36} - \frac{y^2}{9} = 1$ which is bisected at (2, 1). **Ans.** $x = 2y$

2. Find the point 'P' from which pair of tangents PA & PB are drawn to the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ in such a way that (5, 2) bisect AB **Ans.** $\left(\frac{375}{4}, 12\right)$

3. From the points on the circle $x^2 + y^2 = a^2$, tangent are drawn to the hyperbola $x^2 - y^2 = a^2$, prove that the locus of the middle points of the chords of contact is the curve $(x^2 - y^2)^2 = a^2(x^2 + y^2)$.
Ans. $(x^2 - y^2)^2 = a^2(x^2 + y^2)$.

13. Diameter :

The locus of the middle points of a system of parallel chords with slope 'm' of an hyperbola is called its diameter. It is a straight line passing through the centre of the hyperbola and has the equation $y = -\frac{b^2}{a^2 m}x$. **NOTE :** All diameters of the hyperbola passes through its centre.

14. Asymptotes : **Definition :** If the length of the perpendicular let fall from a point on a hyperbola to a straight line tends to zero as the point on the hyperbola moves to infinity along the hyperbola, then the straight line is called the Asymptote of the hyperbola.

Equations of Asymptote : $\frac{x}{a} + \frac{y}{b} = 0$ and $\frac{x}{a} - \frac{y}{b} = 0$.

Pair of asymptotes : $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$.

NOTE : (i) A hyperbola and its conjugate have the same asymptote.

(ii) The equation of the pair of asymptotes differs from the equation of hyperbola (or conjugate hyperbola) by the constant term only.

(iii) The asymptotes pass through the centre of the hyperbola & are equally inclined to the transverse axis of the hyperbola. Hence the bisectors of the angles between the asymptotes are the principle axes of the hyperbola.

(iv) The asymptotes of a hyperbola are the diagonals of the rectangle formed by the lines drawn through the extremities of each axis parallel to the other axis.

(v) Asymptotes are the tangent to the hyperbola from the centre.

(vi) A simple method to find the co-ordinates of the centre of the hyperbola expressed as a general equation of degree 2 should be remembered as:

Let $f(x, y) = 0$ represents a hyperbola.

Find $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y}$. Then the point of intersection of $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0$ gives the centre of the hyperbola.

Example : Find the asymptotes $xy - 3y - 2x = 0$.

Solution. Since equation of a hyperbola and its asymptotes differ in constant terms only,

$\therefore$ Pair of asymptotes is given by $xy - 3y - 2x + \lambda = 0$

where λ is any constant such that it represents two straight lines.

$\therefore$ $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$$\Rightarrow \quad 0 + 2 \times -\frac{3}{2} \times -1 \times \frac{1}{2} - 0 - 0 - \lambda \left(\frac{1}{2}\right)^2 = 0$$

$$\therefore \quad \lambda = 6$$

From (1), the asymptotes of given hyperbola are given by

$xy - 3y - 2x + 6 = 0$ or $(y - 2)(x - 3) = 0$

$\therefore$ Asymptotes are $x - 3 = 0$ and $y - 2 = 0$

Example : The asymptotes of a hyperbola having centre at the point (1, 2) are parallel to the lines $2x + 3y = 0$ and $3x + 2y = 0$. If the hyperbola passes through the point (5, 3), show that its equation is $(2x + 3y - 8)(3x + 2y + 7) = 154$

Solution. Let the asymptotes be $2x + 3y + \lambda = 0$ and $3x + 2y + \mu = 0$. Since asymptotes passes through (1,2), then $\lambda = -8$ and $\mu = -7$

Thus the equation of asympotes are

$$2x + 3y - 8 = 0 \text{ and } 3x + 2y - 7 = 0$$

Let the equation of hyperbola be

$$(2x + 3y - 8)(3x + 2y - 7) + v = 0 \qquad \text{.....(1)}$$

It passes through (5,3), then

$$(10 + 9 - 8)(15 + 6 - 7) + v = 0$$

$$\Rightarrow \quad 11 \times 14 + v = 0$$

$$\therefore \quad v = -154$$

putting the value of v in (1) we obtain

$$(2x + 3y - 8)(3x + 2y - 7) - 154 = 0$$

which is the equation of required hyperbola.

Self Practice Problems :

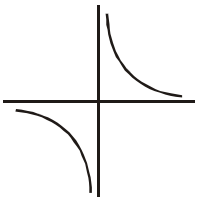
1. Show that the tangent at any point of a hyperbola cuts off a triangle of constant area from the asymptotes and that the portion of it intercepted between the asymptotes is bisected at the point of contact.

15. Rectangular Or Equilateral Hyperbola :

The particular kind of hyperbola in which the lengths of the transverse & conjugate axis are equal is called an Equilateral Hyperbola. Note that the eccentricity of the rectangular hyperbola is $\sqrt{2}$.

Rectangular Hyperbola ($xy = c^2$) :

It is referred to its asymptotes as axes of co–ordinates.



Vertices : (c, c) & $(-c, -c)$;

Foci : $\left(\sqrt{2}\,c, \sqrt{2}\,c\right)$ & $\left(-\sqrt{2}\,c, -\sqrt{2}\,c\right)$,

Directrices : $x + y = \pm\sqrt{2}\,c$

Latus Rectum (l) :

$\ell = 2\sqrt{2}\,c$ = T.A. = C.A.

Parametric equation $x = ct$, $y = c/t$, $t \in R - \{0\}$

Equation of a chord joining the points $P(t_1)$ & $Q(t_2)$ is $x + t_1 t_2 y = c(t_1 + t_2)$.

Equation of the tangent at $P(x_1, y_1)$ is $\frac{x}{x_1} + \frac{y}{y_1} = 2$ & at $P(t)$ is $\frac{x}{t} + ty = 2c$.

Equation of the normal at P (t) is $x t^3 - y t = c(t^4 - 1)$.

Chord with a given middle point as (h, k) is $kx + hy = 2hk$.

Example : A triangle has its vertices on a rectangle hyperbola. Prove that the orthocentre of the triangle also lies on the same hyperbola.

Solution. Let "t_1", "t_2" and "t_3" are the vertices of the triangle ABC, described on the rectangular hyperbola $xy = c^2$.

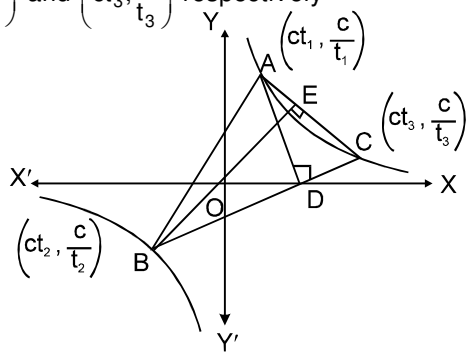
$\therefore$ Co–ordinates of A,B and C are $\left(ct_1, \frac{c}{t_1}\right)$, $\left(ct_2, \frac{c}{t_2}\right)$ and $\left(ct_3, \frac{c}{t_3}\right)$ respectively

Now lope of BC is $\frac{t_3 - t_2}{ct_3 - ct_2} = -\frac{1}{t_2 t_3}$

$\therefore$ Slope of AD is $t_2 t_3$

Equation of Altitude AD is

$$y - \frac{c}{t_1} = t_2 t_3 (x - ct_1)$$



or $t_1y - c = x\, t_1t_2t_3 - ct_1^2t_2t_3$ .....(1)

Similarly equation of altitude BE is

$t_2y - c = x\, t_1t_2t_3 - ct_1t_2^2t_3$ .....(2)

Solving (1) and (2), we get the orthocentre $\left(-\frac{c}{t_1t_2t_3}, -ct_1t_2t_3\right)$

Which lies on $xy = c^2$.

Example : A, B, C are three points on the rectangular hyperbola $xy = c^2$, find

(i) The area of the triangle ABC

(ii) The area of the triangle formed by the tangents A, B and C.

Sol. Let co–ordinates of A,B and C on the hyperbola $xy = c^2$ are $\left(ct_1, \frac{c}{t_1}\right)$, $\left(ct_2, \frac{c}{t_2}\right)$ and $\left(ct_3, \frac{c}{t_3}\right)$ respectively.

(i) $\therefore$ Area of triangle ABC $= \frac{1}{2}\begin{vmatrix} ct_1 & \frac{c}{t_1} \\ ct_2 & \frac{c}{t_2} \end{vmatrix} + \begin{vmatrix} ct_2 & \frac{c}{t_2} \\ ct_3 & \frac{c}{t_3} \end{vmatrix} + \begin{vmatrix} ct_3 & \frac{c}{t_3} \\ ct_1 & \frac{c}{t_1} \end{vmatrix}$

$= \frac{c^2}{2}\left|\frac{t_1}{t_2} - \frac{t_2}{t_1} + \frac{t_2}{t_3} - \frac{t_3}{t_2} + \frac{t_3}{t_1} - \frac{t_1}{t_3}\right|$

$= \frac{c^2}{2t_1t_2t_3}\left|t_3^2t_3 - t_2^2t_3 + t_1t_2^2 - t_3^2t_1 + t_2t_3^2 - t_1^2t_2\right|$

$= \frac{c^2}{2t_1t_2t_3}\ |\,(t_1 - t_2)\,(t_2 - t_3)\,(t_3 - t_1)\,|$

(ii) Equations of tangents at A,B,C are

$x + t_1^2 - 2ct_1 = 0$

$x + yt_2^2 - 2ct_2 = 0$

and $x + yt_3^2 - 2ct_3 = 0$

$\therefore$ Required Area $= \frac{1}{2\,|\,C_1C_2C_3\,|}\begin{vmatrix} 1 & t_1^2 & -2ct_1 \\ 1 & t_2^2 & -2ct_2 \\ 1 & t_3^2 & -2ct_3 \end{vmatrix}^2$ .....(1)

where $C_1 = \begin{vmatrix} 1 & t_1^2 \\ 1 & t_3^2 \end{vmatrix}$, $C_2 = -\begin{vmatrix} 1 & t_1^2 \\ 1 & t_3^2 \end{vmatrix}$ and $C_3 = \begin{vmatrix} 1 & t_1^2 \\ 1 & t_2^2 \end{vmatrix}$

$\therefore$ $C_1 = t_3^2 - t_2^2$, $C_2 = t_1^2 - t_3^2$ and $C_3 = t_2^2 - t_1^2$

From (1) $= \frac{1}{2\left|(t_3^2 - t_2^2)\,(t_1^2 - t_3^2)(t_2^2 - t_1^2)\right|}\ 4c^2.(t_1 - t_2)^2\,(t_2 - t_3)^2\,(t_3 - t_1)^2$

$= 2c^2\left|\frac{(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)}{(t_1 + t_2)\,(t_2 + t_3)\,(t_3 + t_1)}\right|$

$\therefore$ Required area is, $2c^2\left|\frac{(t_1 - t_2)\,(t_2 - t_3)(t_3 - t_1)}{(t_1 + t_2)\,(t_2 + t_3)\,(t_3 + t_1)}\right|$

Example : Prove that the perpendicular focal chords of a rectangular hyperbola are equal.

Solution. Let rectangular hyperbola is $x^2 - y^2 = a^2$

Let equations of PQ and DE are

$y = mx + c$ .....(1)

and $y = m_1x + c_1$ .....(2)

respectively.

Be any two focal chords of any rectangular hyperbola $x^2 - y^2 = a^2$ through its focus. We have to prove PQ = DE. Since PQ $\perp$ DE.

$\therefore$ $mm_1 = -1$ .....(3)

Also PQ passes through S $(a\sqrt{2}, 0)$ then from (1),

$0 = ma\sqrt{2} + c$

or $c^2 = 2a^2m^2$ .....(4)

Let (x_1, y_1) and (x_2, y_2) be the co–ordinates of P and Q then

$(PQ)^2 = (x_1 - x_2^2) + (y_1 - y_2)^2$ .....(5)

Since (x_1, y_1) and (x_2, y_2) lie on (1)

$\therefore$ $y_1 = mx_1 + c$ and $y_2 = mx_2 + c$

$\therefore$ $(y_1 - y_2) = m\,(x_1 - x_2)$ .....(6)

From (5) and (6)

$(PQ)^2 = (x_1 - x_2)^2\,(1 + m^2)$ .....(7)

Now solving $y = mx + c$ and $x^2 - y^2 = a^2$ then

$x^2 - (mx + c)^2 = a^2$

or $(m^2 - 1)x^2 + 2mcx + (a^2 + c^2) = 0$

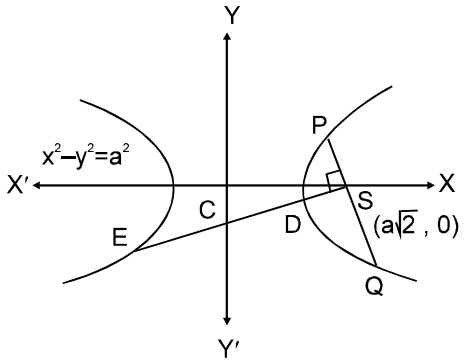
$\therefore \quad x_1 + x_2 = \dfrac{2mc}{m^2 - 1}$ and $x_1x_2 = \dfrac{a^2 + c^2}{m^2 - 1}$

$\Rightarrow \quad (x_1 - x_2)^2 = (x_1 + x_2)^2 - 4x_1x_2$

$$= \frac{4m^2c^2}{(m^2-1)^2} - \frac{4(a^2+c^2)}{(m^2-1)}$$

$$= \frac{4\{a^2 + c^2 - a^2m^2\}}{(m^2-1)^2}$$

$$= \frac{4a^2(m^2+1)}{(m^2-1)^2} \qquad \{\because c^2 = 2a^2m^2\}$$



From (7), $(PQ)^2 = 4a^2\left(\dfrac{m^2+1}{m^2-1}\right)$

Similarly, $(DE)^2 = 4a^2\left(\dfrac{m_1^2+1}{m_1^2-1}\right)^2$

$$= 4a^2\left(\frac{\left(1-\frac{1}{m}\right)^2+1}{\left(-\frac{1}{m}\right)^2-1}\right)^2 \qquad (\because mm_1 = -1)$$

$$= 4a^2\left(\frac{m^2+1}{m^2-1}\right)$$

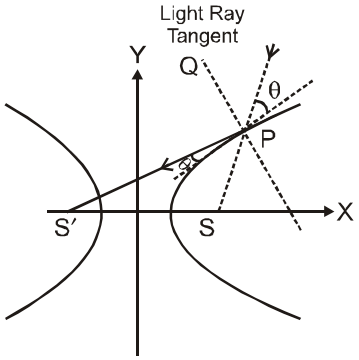
$= (PQ)^2$

Thus $(PQ)^2 = (DE)^2 \Rightarrow PQ = DE$.

Hence perpendicular focal chords of a rectangular hyperbola are equal.

15. Important Results :

- Locus of the feet of the perpendicular drawn from focus of the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ upon any tangent is its auxiliary circle i.e. $x^2 + y^2 = a^2$ & the product of these perpendiculars is b^2.
- The portion of the tangent between the point of contact & the directrix subtends a right angle at the corresponding focus.
- The tangent & normal at any point of a hyperbola bisect the angle between the focal radii. This spells the reflection property of the hyperbola as **"An incoming light ray"** aimed towards one focus is reflected from the outer surface of the hyperbola towards the other focus. It follows that if an ellipse and a hyperbola have the same foci, they cut at right angles at any of their common point.



Note that the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ & the hyperbola $\dfrac{x^2}{a^2 - k^2} - \dfrac{y^2}{k^2 - b^2} = 1$ $(a > k > b > 0)$ are confocal and therefore orthogonal.

- The foci of the hyperbola and the points P and Q in which any tangent meets the tangents at the vertices are concyclic with PQ as diameter of the circle.
- If from any point on the asymptote a straight line be drawn perpendicular to the transverse axis, the product of the segments of this line, intercepted between the point & the curve is always equal to the square of the semi conjugate axis.
- Perpendicular from the foci on either asymptote meet it in the same points as the corresponding directrix & the common points of intersection lie on the auxiliary circle.

- The tangent at any point P on a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with centre C, meets the asymptotes in Q and R and cuts off a Δ CQR of constant area equal to ab from the asymptotes & the portion of the tangent intercepted between the asymptote is bisected at the point of contact. This implies that locus of the centre of the circle circumscribing the Δ CQR in case of a rectangular hyperbola is the hyperbola itself & for a standard hyperbola the locus would be the curve, $4(a^2x^2 - b^2y^2) = (a^2 + b^2)^2$.
- If the angle between the asymptote of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is 2θ then the eccentricity of the hyperbola is $\sec\theta$.
- A rectangular hyperbola circumscribing a triangle also passes through the orthocentre of this triangle. If $\left(c t_i, \frac{c}{t_i}\right)$ i = 1, 2, 3 be the angular points P, Q, R then orthocentre is $\left(\frac{-c}{t_1 t_2 t_3}, -c t_1 t_2 t_3\right)$.
- If a circle and the rectangular hyperbola $xy = c^2$ meet in the four points t_1, t_2, t_3 & t_4, then
  (a) $t_1 t_2 t_3 t_4 = 1$
  (b) the centre of the mean position of the four points bisects the distance between the centres of the two curves.
  (c) the centre of the circle through the points t_1, t_2 & t_3 is :
  $$\left\{\frac{c}{2}\left(t_1 + t_2 + t_3 + \frac{1}{t_1 t_2 t_3}\right), \frac{c}{2}\left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + t_1 + t_2 + t_3\right)\right\}$$

Example :

A ray emanating from the point (5, 0) is incident on the hyperbola $9x^2 - 16y^2 = 144$ at the point P with abscissa 8. Find the equation of the reflected ray after first reflection and point P lies in first quadrant.

Solution.

Given hyperbola is
$9x^2 - 16y^2 = 144$. This equation can be

rewritten as $\frac{x^2}{16} - \frac{y^2}{9} = 1$ .....(1)

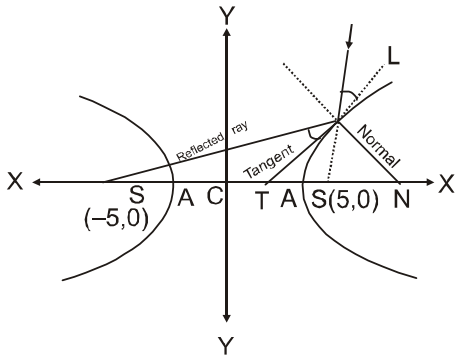
Since x co–ordinate of P is 8. Let y co–ordinate of P ia α.

$\because$ $(8,\alpha)$ lies on (1)

$\therefore$ $\frac{64}{16} - \frac{\alpha^2}{9} = 1$

$\Rightarrow$ $\alpha^2 = 27$

$\Rightarrow$ $a = 3\sqrt{3}$ ($\because$ P lies in first quadrant)



Hence coªordinate of point P is $(8, 3\sqrt{3})$.

$\because$ Equation of reflected ray passing through P $(8, 3\sqrt{3})$ and S′(–5,0)

$\therefore$ Its equation is $y - 3\sqrt{3} = \frac{0 - 3\sqrt{3}}{-5 - 8}(x - 8)$

or $13y - 39\sqrt{3} = 3\sqrt{3}x - 24\sqrt{3}$

or $3\sqrt{3}x - 13y + 15\sqrt{3} = 0$.

SHORT REVISION
PARABOLA

1. CONIC SECTIONS:

A conic section, or conic is the locus of a point which moves in a plane so that its distance from a fixed point is in a constant ratio to its perpendicular distance from a fixed straight line.

- ☞ The fixed point is called the **FOCUS.**
- ☞ The fixed straight line is called the **DIRECTRIX.**
- ☞ The constant ratio is called the **ECCENTRICITY** denoted by e.
- ☞ The line passing through the focus & perpendicular to the directrix is called the **AXIS.**
- ☞ A point of intersection of a conic with its axis is called a **VERTEX.**

2. GENERAL EQUATION OF A CONIC : FOCAL DIRECTRIX PROPERTY :

The general equation of a conic with focus (p, q) & directrix $lx + my + n = 0$ is :

$(l^2 + m^2)[(x - p)^2 + (y - q)^2] = e^2 (lx + my + n)^2 \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

3. DISTINGUISHING BETWEEN THE CONIC :

The nature of the conic section depends upon the position of the focus S w.r.t. the directrix & also upon the value of the eccentricity e. Two different cases arise.

CASE (I) : WHEN THE FOCUS LIES ON THE DIRECTRIX.

In this case $D \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ & the general equation of a conic represents a pair of straight lines if :

$e > 1$ the lines will be real & distinct intersecting at S.
$e = 1$ the lines will coincident.
$e < 1$ the lines will be imaginary.

CASE (II) : WHEN THE FOCUS DOES NOT LIE ON DIRECTRIX.

a parabola	an ellipse	a hyperbola	rectangular hyperbola
$e = 1 ; D \neq 0,$	$0 < e < 1 ; D \neq 0 ;$	$e > 1 ; D \neq 0 ;$	$e > 1 ; D \neq 0$
$h^2 = ab$	$h^2 < ab$	$h^2 > ab$	$h^2 > ab ; a + b = 0$

4. PARABOLA : DEFINITION :

A parabola is the locus of a point which moves in a plane, such that its distance from a fixed point (focus) is equal to its perpendicular distance from a fixed straight line (directrix).

Standard equation of a parabola is $y^2 = 4ax$. For this parabola :

(i) Vertex is (0, 0) (ii) focus is (a, 0) (iii) Axis is $y = 0$ (iv) Directrix is $x + a = 0$

FOCAL DISTANCE :

The distance of a point on the parabola from the focus is called the **FOCAL DISTANCE OF THE POINT.**

FOCAL CHORD :

A chord of the parabola, which passes through the focus is called a **FOCAL CHORD.**

DOUBLE ORDINATE :

A chord of the parabola perpendicular to the axis of the symmetry is called a **DOUBLE ORDINATE.**

LATUS RECTUM :

A double ordinate passing through the focus or a focal chord perpendicular to the axis of parabola is called the **LATUS RECTUM.** For $y^2 = 4ax$.

■ Length of the latus rectum = 4a. ■ ends of the latus rectum are L(a, 2a) & L' (a, – 2a).

Note that:
(i) Perpendicular distance from focus on directrix = half the latus rectum.
(ii) Vertex is middle point of the focus & the point of intersection of directrix & axis.
(iii) Two parabolas are laid to be equal if they have the same latus rectum.

Four standard forms of the parabola are $y^2 = 4ax$; $y^2 = -4ax$; $x^2 = 4ay$; $x^2 = -4ay$

5. POSITION OF A POINT RELATIVE TO A PARABOLA :

The point $(x_1\ y_1)$ lies outside, on or inside the parabola $y^2 = 4ax$ according as the expression $y_1^2 - 4ax_1$ is positive, zero or negative.

6. LINE & *A* PARABOLA :

The line $y = mx + c$ meets the parabola $y^2 = 4ax$ in two points real, coincident or imaginary according as $a \gtrless c\,m \Rightarrow$ condition of tangency is, $c = \frac{a}{m}$.

7. Length of the chord intercepted by the parabola on the line $y = mx + c$ is : $\left(\frac{4}{m^2}\right)\sqrt{a(1+m^2)(a-mc)}$.

Note: length of the focal chord making an angle α with the x– axis is $4a\text{Cosec}^2\alpha$.

8. PARAMETRIC REPRESENTATION :

The simplest & the best form of representing the co–ordinates of a point on the parabola is $(at^2, 2at)$. The equations $x = at^2$ & $y = 2at$ together represents the parabola $y^2 = 4ax$, t being the parameter. The equation of a chord joining t_1 & t_2 is $2x - (t_1 + t_2)y + 2at_1t_2 = 0$.

Note: If the chord joining t_1, t_2 & t_3, t_4 pass through a point $(c, 0)$ on the axis, then $t_1t_2 = t_3t_4 = -c/a$.

9. TANGENTS TO THE PARABOLA $y^2 = 4ax$:

(i) $yy_1 = 2a(x + x_1)$ at the point (x_1, y_1) ; **(ii)** $y = mx + \frac{a}{m}$ $(m \neq 0)$ at $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$

(iii) $ty = x + at^2$ at $(at^2, 2at)$.

Note : Point of intersection of the tangents at the point t_1 & t_2 is $[at_1t_2, a(t_1 + t_2)]$.

10. NORMALS TO THE PARABOLA $y^2 = 4ax$:

(i) $y - y_1 = -\frac{y_1}{2a}(x - x_1)$ at (x_1, y_1) ; **(ii)** $y = mx - 2am - am^3$ at $(am^2, -2am)$

(iii) $y + tx = 2at + at^3$ at $(at^2, 2at)$.

Note : Point of intersection of normals at t_1 & t_2 are, $a(t_1^2 + t_2^2 + t_1t_2 + 2)$; $-at_1t_2(t_1 + t_2)$.

11. THREE VERY IMPORTANT RESULTS :

(a) If t_1 & t_2 are the ends of a focal chord of the parabola $y^2 = 4ax$ then $t_1t_2 = -1$. Hence the co-ordinates at the extremities of a focal chord can be taken as $(at^2, 2at)$ & $\left(\frac{a}{t^2}, -\frac{2a}{t}\right)$.

(b) If the normals to the parabola $y^2 = 4ax$ at the point t_1 meets the parabola again at the point t_2, then $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$.

(c) If the normals to the parabola $y^2 = 4ax$ at the points t_1 & t_2 intersect again on the parabola at the point 't_3' then $t_1t_2 = 2$; $t_3 = -(t_1 + t_2)$ and the line joining t_1 & t_2 passes through a fixed point $(-2a, 0)$.

General Note :

(i) Length of subtangent at any point P(x, y) on the parabola $y^2 = 4ax$ equals twice the abscissa of the point P. Note that the subtangent is bisected at the vertex.

(ii) Length of subnormal is constant for all points on the parabola & is equal to the semi latus rectum.

(iii) If a family of straight lines can be represented by an equation $\lambda^2P + \lambda Q + R = 0$ where λ is a parameter and P, Q, R are linear functions of x and y then the family of lines will be tangent to the curve $Q^2 = 4PR$.

12. The equation to the pair of tangents which can be drawn from any point (x_1, y_1) to the parabola $y^2 = 4ax$ is given by : $SS_1 = T^2$ where :
$S \equiv y^2 - 4ax$; $S_1 = y_1^2 - 4ax_1$; $T \equiv yy_1 - 2a(x + x_1)$.

13. DIRECTOR CIRCLE :

Locus of the point of intersection of the perpendicular tangents to the parabola $y^2 = 4ax$ is called the **DIRECTOR CIRCLE.** It's equation is $x + a = 0$ which is parabola's own directrix.

14. CHORD OF CONTACT :

Equation to the chord of contact of tangents drawn from a point $P(x_1, y_1)$ is $yy_1 = 2a(x + x_1)$. Remember that the area of the triangle formed by the tangents from the point (x_1, y_1) & the chord of contact is $(y_1^2 - 4ax_1)^{3/2} \div 2a$. Also note that the chord of contact exists only if the point P is not inside.

15. POLAR & POLE :

(i) Equation of the Polar of the point $P(x_1, y_1)$ w.r.t. the parabola $y^2 = 4ax$ is,
$yy_1 = 2a(x + x_1)$

(ii) The pole of the line $lx + my + n = 0$ w.r.t. the parabola $y^2 = 4ax$ is $\left(\frac{n}{l}, -\frac{2am}{l}\right)$.

Note:

(i) The polar of the focus of the parabola is the directrix.

(ii) When the point (x_1, y_1) lies without the parabola the equation to its polar is the same as the equation to the chord of contact of tangents drawn from (x_1, y_1) when (x_1, y_1) is on the parabola the polar is the same as the tangent at the point.

(iii) If the polar of a point P passes through the point Q, then the polar of Q goes through P.

(iv) Two straight lines are said to be conjugated to each other w.r.t. a parabola when the pole of one lies on the other.

(v) Polar of a given point P w.r.t. any Conic is the locus of the harmonic conjugate of P w.r.t. the two points is which any line through P cuts the conic.

16. CHORD WITH A GIVEN MIDDLE POINT :

Equation of the chord of the parabola $y^2 = 4ax$ whose middle point is (x_1, y_1) is $y - y_1 = \frac{2a}{y_1}(x - x_1)$. This reduced to $T = S_1$

where $T \equiv y\,y_1 - 2a(x + x_1)$ & $S_1 \equiv y_1^2 - 4ax_1$.

17. DIAMETER :

The locus of the middle points of a system of parallel chords of a Parabola is called a **DIAMETER**. Equation to the diameter of a parabola is $y = 2a/m$, where m = slope of parallel chords.

Note:

(i) The tangent at the extremity of a diameter of a parabola is parallel to the system of chords it bisects.

(ii) The tangent at the ends of any chords of a parabola meet on the diameter which bisects the chord.

(iii) A line segment from a point P on the parabola and parallel to the system of parallel chords is called the ordinate to the diameter bisecting the system of parallel chords and the chords are called its double ordinate.

18. IMPORTANT HIGHLIGHTS :

(a) If the tangent & normal at any point 'P' of the parabola intersect the axis at T & G then ST = SG = SP where 'S' is the focus. In other words the tangent and the normal at a point P on the parabola are the bisectors of the angle between the focal radius SP & the perpendicular from P on the directrix. From this we conclude that all rays emanating from S will become parallel to the axis of the parabola after reflection.

(b) The portion of a tangent to a parabola cut off between the directrix & the curve subtends a right angle at the **focus.**

(c) The tangents at the extremities of a focal chord intersect at right angles on the **directrix**, and hence a circle on any focal chord as diameter touches the directrix. Also a circle on any focal radii of a point P $(at^2, 2at)$ as diameter touches the tangent at the vertex and intercepts a chord of length $a\sqrt{1 + t^2}$ on a normal at the point P.

(d) Any tangent to a parabola & the perpendicular on it from the focus meet on the tangtent at the vertex.

(e) If the tangents at P and Q meet in T, then :

- TP and TQ subtend equal angles at the focus S.
- $ST^2 = SP.\,SQ$ &
- The triangles SPT and STQ are similar.

(f) Tangents and Normals at the extremities of the latus rectum of a parabola $y^2 = 4ax$ constitute a square, their points of intersection being $(-a, 0)$ & $(3a, 0)$.

(g) Semi latus rectum of the parabola $y^2 = 4ax$, is the harmonic mean between segments of any focal chord of the parabola is ; $2a = \frac{2bc}{b+c}$ **i.e.** $\frac{1}{b} + \frac{1}{c} = \frac{1}{a}$.

(h) The circle circumscribing the triangle formed by any three tangents to a parabola passes through the focus.

(i) The orthocentre of any triangle formed by three tangents to a parabola $y^2 = 4ax$ lies on the directrix & has the co-ordinates $-a, a(t_1 + t_2 + t_3 + t_1t_2t_3)$.

(j) The area of the triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents at these points.

(k) If normal drawn to a parabola passes through a point P(h, k) then $k = mh - 2am - am^3$ i.e. $am^3 + m(2a - h) + k = 0$.

Then gives $m_1 + m_2 + m_3 = 0$; $m_1m_2 + m_2m_3 + m_3m_1 = \frac{2a-h}{a}$; $m_1\,m_2\,m_3 = -\frac{k}{a}$.

where m_1, m_2 & m_3 are the slopes of the three concurrent normals. Note that the algebraic sum of the:

- slopes of the three concurrent normals is zero.
- ordinates of the three conormal points on the parabola is zero.
- Centroid of the Δ formed by three co-normal points lies on the x-axis.

(*l*) A circle circumscribing the triangle formed by three co–normal points passes through the vertex of the parabola and its equation is, $2(x^2 + y^2) - 2(h + 2a)x - ky = 0$

***Suggested problems from Loney*: Exercise-25** (Q.5, 10, 13, 14, 18, 21), **Exercise-26 (Important)** (Q.4, 6, 7, 16, 17, 20, 22, 26, 27, 28, 34, 38), **Exercise-27** (Q.4, 7), **Exercise-28** (Q.2, 7, 11, 14, 17, 23), **Exercise-29** (Q.7, 8, 10, 19, 21, 24, 26, 27), **Exercise-30** (2, 3, 13, 18, 20, 21, 22, 25, 26, 30)

***Note*:** Refer to the figure on Pg.175 if necessary.

EXERCISE–1

Q.1 Show that the normals at the points (4a, 4a) & at the upper end of the latus ractum of the parabola $y^2 = 4ax$ intersect on the same parabola.

Q.2 Prove that the locus of the middle point of portion of a normal to $y^2 = 4ax$ intercepted between the curve & the axis is another parabola. Find the vertex & the latus rectum of the second parabola.

Q.3 Find the equations of the tangents to the parabola $y^2 = 16x$, which are parallel & perpendicular respectively to the line $2x - y + 5 = 0$. Find also the coordinates of their points of contact.

Q.4 A circle is described whose centre is the vertex and whose diameter is three-quarters of the latus rectum of a parabola $y^2 = 4ax$. Prove that the common chord of the circle and parabola bisects the distance between the vertex and the focus.

Q.5 Find the equations of the tangents of the parabola $y^2 = 12x$, which passes through the point (2,5).

Q.6 Through the vertex O of a parabola $y^2 = 4x$, chords OP & OQ are drawn at right angles to one another. Show that for all positions of P, PQ cuts the axis of the parabola at a fixed point. Also find the locus of the middle point of PQ.

Q.7 Let S is the focus of the parabola $y^2 = 4ax$ and X the foot of the directrix, PP' is a double ordinate of the curve and PX meets the curve again in Q. Prove that P'Q passes through focus.

Q.8 Three normals to $y^2 = 4x$ pass through the point (15, 12). Show that one of the normals is given by $y = x - 3$ & find the equations of the others.

Q.9 Find the equations of the chords of the parabola $y^2 = 4ax$ which pass through the point $(-6a, 0)$ and which subtends an angle of 45° at the vertex.

Q.10 Through the vertex O of the parabola $y^2 = 4ax$, a perpendicular is drawn to any tangent meeting it at P & the parabola at Q. Show that $OP \cdot OQ$ = constant.

Q.11 'O' is the vertex of the parabola $y^2 = 4ax$ & L is the upper end of the latus rectum. If LH is drawn perpendicular to OL meeting OX in H, prove that the length of the double ordinate through H is $4a\sqrt{5}$.

Q.12 The normal at a point P to the parabola $y^2 = 4ax$ meets its axis at G. Q is another point on the parabola such that QG is perpendicular to the axis of the parabola. Prove that $QG^2 - PG^2$ = constant.

Q.13 If the normal at P(18, 12) to the parabola $y^2 = 8x$ cuts it again at Q, show that $9PQ = 80\sqrt{10}$

Q.14 Prove that, the normal to $y^2 = 12x$ at $(3, 6)$ meets the parabola again in $(27, -18)$ & circle on this normal chord as diameter is $x^2 + y^2 - 30x + 12y - 27 = 0$.

Q.15 Find the equation of the circle which passes through the focus of the parabola $x^2 = 4y$ & touches it at the point (6, 9).

Q.16 P & Q are the points of contact of the tangents drawn from the point T to the parabola $y^2 = 4ax$. If PQ be the normal to the parabola at P, prove that TP is bisected by the directrix.

Q.17 Prove that the locus of the middle points of the normal chords of the parabola $y^2 = 4ax$ is $\frac{y^2}{2a} + \frac{4a^3}{y^2} = x - 2a$.

Q.18 From the point $(-1, 2)$ tangent lines are drawn to the parabola $y^2 = 4x$. Find the equation of the chord of contact. Also find the area of the triangle formed by the chord of contact & the tangents.

Q.19 Show that the locus of a point that divides a chord of slope 2 of the parabola $y^2 = 4x$ internally in the ratio 1 : 2 is a parabola. Find the vertex of this parabola.

Q.20 From a point A common tangents are drawn to the circle $x^2 + y^2 = a^2/2$ & parabola $y^2 = 4ax$. Find the area of the quadrilateral formed by the common tangents, the chord of contact of the circle and the chord of contact of the parabola.

Q.21 Prove that on the axis of any parabola $y^2 = 4ax$ there is a certain point K which has the property that, if a chord PQ of the parabola be drawn through it, then $\frac{1}{(PK)^2} + \frac{1}{(QK)^2}$ is same for all positions of the chord. Find also the coordinates of the point K.

Q.22 Prove that the two parabolas $y^2 = 4ax$ & $y^2 = 4c(x - b)$ cannot have a common normal, other than the axis, unless $\frac{b}{(a - c)} > 2$.

Q.23 Find the condition on 'a' & 'b' so that the two tangents drawn to the parabola $y^2 = 4ax$ from a point are normals to the parabola $x^2 = 4by$.

Q.24 Prove that the locus of the middle points of all tangents drawn from points on the directrix to the parabola $y^2 = 4ax$ is $y^2(2x + a) = a(3x + a)^2$.

Q.25 Show that the locus of a point, such that two of the three normals drawn from it to the parabola $y^2 = 4ax$ are perpendicular is $y^2 = a(x - 3a)$.

EXERCISE–2

Q.1 In the parabola $y^2 = 4ax$, the tangent at the point P, whose abscissa is equal to the latus rectum meets the axis in T & the normal at P cuts the parabola again in Q. Prove that PT : PQ = 4 : 5.

Q.2 Two tangents to the parabola $y^2 = 8x$ meet the tangent at its vertex in the points P & Q. If PQ = 4 units, prove that the locus of the point of the intersection of the two tangents is $y^2 = 8(x + 2)$.

Q.3 A variable chord $t_1 t_2$ of the parabola $y^2 = 4ax$ subtends a right angle at a fixed point t_0 of the curve. Show that it passes through a fixed point. Also find the co–ordinates of the fixed point.

Q.4 Two perpendicular straight lines through the focus of the parabola $y^2 = 4ax$ meet its directrix in T & T' respectively. Show that the tangents to the parabola parallel to the perpendicular lines intersect in the mid point of T T '.

Q.5 Two straight lines one being a tangent to $y^2 = 4ax$ and the other to $x^2 = 4by$ are right angles. Find the locus of their point of intersection.

Q.6 A variable chord PQ of the parabola $y^2 = 4x$ is drawn parallel to the line y = x. If the parameters of the points P & Q on the parabola are p & q respectively, show that p + q = 2. Also show that the locus of the point of intersection of the normals at P & Q is 2x – y = 12.

Q.7 Show that an infinite number of triangles can be inscribed in either of the parabolas $y^2 = 4ax$ & $x^2 = 4by$ whose sides touch the other.

Q.8 If (x_1, y_1), (x_2, y_2) and (x_3, y_3) be three points on the parabola $y^2 = 4ax$ and the normals at these points meet in a point then prove that $\frac{x_1 - x_2}{y_3} + \frac{x_2 - x_3}{y_1} + \frac{x_3 - x_1}{y_2} = 0$.

Q.9 Show that the normals at two suitable distinct real points on the parabola $y^2 = 4ax$ intersect at a point on the parabola whose abscissa > 8a.

Q.10 The equation $y = x^2 + 2ax + a$ represents a parabola for all real values of a.

(a) Prove that each of these parabolas pass through a common point and determine the coordinates of this point.

(b) The vertices of the parabolas lie on a curve. Prove that this curve is a parabola and find its equation.

Q.11 The normals at P and Q on the parabola $y^2 = 4ax$ intersect at the point R (x_1, y_1) on the parabola and the tangents at P and Q intersect at the point T. Show that,

$$l(TP) \cdot l(TQ) = \frac{1}{2}(x_1 - 8a)\sqrt{y_1^2 + 4a^2}$$

Also show that, if R moves on the parabola, the mid point of PQ lie on the parabola $y^2 = 2a(x + 2a)$.

Q.12 If Q (x_1, y_1) is an arbitrary point in the plane of a parabola $y^2 = 4ax$, show that there are three points on the parabola at which OQ subtends a right angle, where O is the origin. Show furhter that the normal at these three points are concurrent at a point R.,determine the coordinates of R in terms of those of Q.

Q.13 PC is the normal at P to the parabola $y^2 = 4ax$, C being on the axis. CP is produced outwards to Q so that PQ = CP; show that the locus of Q is a parabola, & that the locus of the intersection of the tangents at P & Q to the parabola on which they lie is $y^2(x + 4a) + 16a^3 = 0$.

Q.14 Show that the locus of the middle points of a variable chord of the parabola $y^2 = 4ax$ such that the focal distances of its extremities are in the ratio 2 : 1, is $9(y^2 - 2ax)^2 = 4a^2(2x - a)(4x + a)$.

Q.15 A quadrilateral is inscribed in a parabola $y^2 = 4ax$ and three of its sides pass through fixed points on the axis. Show that the fourth side also passes through fixed point on the axis of the parabola.

Q.16 Prove that the parabola $y^2 = 16x$ & the circle $x^2 + y^2 - 40x - 16y - 48 = 0$ meet at the point P(36, 24) & one other point Q. Prove that PQ is a diameter of the circle. Find Q.

Q.17 A variable tangent to the parabola $y^2 = 4ax$ meets the circle $x^2 + y^2 = r^2$ at P & Q. Prove that the locus of the mid point of PQ is $x(x^2 + y^2) + ay^2 = 0$.

Q.18 Find the locus of the foot of the perpendicular from the origin to chord of the parabola $y^2 = 4ax$ subtending an angle of 45^0 at the vertex.

Q.19 Show that the locus of the centroids of equilateral triangles inscribed in the parabola $y^2 = 4ax$ is the parabola $9y^2 - 4ax + 32a^2 = 0$.

Q.20 The normals at P, Q, R on the parabola $y^2 = 4ax$ meet in a point on the line y = k. Prove that the sides of the triangle PQR touch the parabola $x^2 - 2ky = 0$.

Q.21 A fixed parabola $y^2 = 4ax$ touches a variable parabola. Find the equation to the locus of the vertex of the variable parabola. Assume that the two parabolas are equal and the axis of the variable parabola remains parallel to the x-axis.

Q.22 Show that the circle through three points the normals at which to the parabola $y^2 = 4ax$ are concurrent at the point (h, k) is $2(x^2 + y^2) - 2(h + 2a)x - ky = 0$.

Q.23 Prove that the locus of the centre of the circle, which passes through the vertex of the parabola $y^2 = 4ax$ & through its intersection with a normal chord is $2y^2 = ax - a^2$.

Q.24 The sides of a triangle touch $y^2 = 4ax$ and two of its angular points lie on $y^2 = 4b(x + c)$. Show that the locus of the third angular point is $a^2y^2 = 4(2b - a)^2.(ax + 4bc)$

Q.25 Three normals are drawn to the parabola $y^2 = 4ax \cos\alpha$ from any point lying on the straight line $y = b \sin\alpha$. Prove that the locus of the orthocentre of the traingles formed by the corresponding tangents is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the angle α being variable.

EXERCISE–3

Q.1 Find the locus of the point of intersection of those normals to the parabola $x^2 = 8y$ which are at right angles to each other. [REE '97, 6]

Q.2 The angle between a pair of tangents drawn from a point P to the parabola $y^2 = 4ax$ is 45^0. Show that the locus of the point P is a hyperbola. [JEE '98, 8]

Q.3 The ordinates of points P and Q on the parabola $y^2 = 12x$ are in the ratio 1 : 2. Find the locus of the point of intersection of the normals to the parabola at P and Q. [REE '98, 6]

Q.4 Find the equations of the common tangents of the circle $x^2 + y^2 - 6y + 4 = 0$ and the parabola $y^2 = x$. [REE '99, 6]

Q.5(a) If the line $x - 1 = 0$ is the directrix of the parabola $y^2 - kx + 8 = 0$, then one of the values of 'k' is
(A) 1/8 (B) 8 (C) 4 (D) 1/4

(b) If $x + y = k$ is normal to $y^2 = 12x$, then 'k' is : [JEE'2000 (Scr), 1+1]
(A) 3 (B) 9 (C) – 9 (D) – 3

Q.6 Find the locus of the points of intersection of tangents drawn at the ends of all normal chords of the parabola $y^2 = 8(x - 1)$. [REE '2001, 3]

Q.7(a) The equation of the common tangent touching the circle $(x - 3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$ above the x – axis is
(A) $\sqrt{3}y = 3x + 1$ (B) $\sqrt{3}y = -(x + 3)$ (C) $\sqrt{3}y = x + 3$ (D) $\sqrt{3}y = -(3x + 1)$

(b) The equation of the directrix of the parabola, $y^2 + 4y + 4x + 2 = 0$ is
(A) $x = -1$ (B) $x = 1$ (C) $x = -\frac{3}{2}$ (D) $x = \frac{3}{2}$
[JEE'2001(Scr), 1+1]

Q.8 The locus of the mid-point of the line segment joining the focus to a moving point on the parabola $y^2 = 4ax$ is another parabola with directrix [JEE'2002 (Scr.), 3]
(A) $x = -a$ (B) $x = -\frac{a}{2}$ (C) $x = 0$ (D) $x = \frac{a}{2}$

Q.9 The equation of the common tangent to the curves $y^2 = 8x$ and $xy = -1$ is [JEE'2002 (Scr), 3]
(A) $3y = 9x + 2$ (B) $y = 2x + 1$
(C) $2y = x + 8$ (D) $y = x + 2$

Q.10(a) The slope of the focal chords of the parabola $y^2 = 16x$ which are tangents to the circle $(x - 6)^2 + y^2 = 2$ are
(A) ± 2 (B) – 1/2, 2 (C) ± 1 (D) – 2, 1/2
[JEE'2003, (Scr.)]

(b) Normals are drawn from the point 'P' with slopes m_1, m_2, m_3 to the parabola $y^2 = 4x$. If locus of P with $m_1 m_2 = \alpha$ is a part of the parabola itself then find α. [JEE 2003, 4 out of 60]

Q.11 The angle between the tangents drawn from the point (1, 4) to the parabola $y^2 = 4x$ is
(A) $\pi/2$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/6$
[JEE 2004, (Scr.)]

Q.12 Let P be a point on the parabola $y^2 - 2y - 4x + 5 = 0$, such that the tangent on the parabola at P intersects the directrix at point Q. Let R be the point that divides the line segment QP externally in the ratio $\frac{1}{2}:1$. Find the locus of R. [JEE 2004, 4 out of 60]

Q.13(a) The axis of parabola is along the line y = x and the distance of vertex from origin is $\sqrt{2}$ and that from its focus is $2\sqrt{2}$. If vertex and focus both lie in the 1st quadrant, then the equation of the parabola is
(A) $(x + y)^2 = (x - y - 2)$ (B) $(x - y)^2 = (x + y - 2)$
(C) $(x - y)^2 = 4(x + y - 2)$ (D) $(x - y)^2 = 8(x + y - 2)$
[JEE 2006, 3]

(b) The equations of common tangents to the parabola $y = x^2$ and $y = -(x - 2)^2$ is/are
(A) $y = 4(x - 1)$ (B) $y = 0$ (C) $y = -4(x - 1)$ (D) $y = -30x - 50$
[JEE 2006, 5]

(c) **Match the following**
Normals are drawn at points P, Q and R lying on the parabola $y^2 = 4x$ which intersect at (3, 0). Then

(i)	Area of ΔPQR	(A) 2
(ii)	Radius of circumcircle of ΔPQR	(B) 5/2
(iii)	Centroid of ΔPQR	(C) (5/2, 0)
(iv)	Circumcentre of ΔPQR	(D) (2/3, 0)

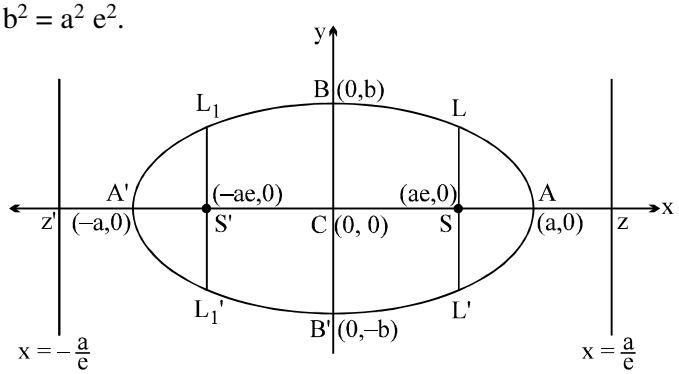
[JEE 2006, 6]

KEY CONCEPTS
ELLIPSE

1. STANDARD EQUATION & DEFINITIONS :

Standard equation of an ellipse referred to its principal axes along the co-ordinate axes is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Where $a > b$ & $b^2 = a^2(1 - e^2) \Rightarrow a^2 - b^2 = a^2 e^2$.
Where e = eccentricity $(0 < e < 1)$.
FOCI : $S \equiv (ae, 0)$ & $S' \equiv (-ae, 0)$.



EQUATIONS OF DIRECTRICES :

$x = \frac{a}{e}$ & $x = -\frac{a}{e}$.

VERTICES :

$A' \equiv (-a, 0)$ & $A \equiv (a, 0)$.

MAJOR AXIS :

The line segment A′A in which the foci S′ & S lie is of length 2a & is called the **major axis** (a > b) of the ellipse. Point of intersection of major axis with directrix is called **the foot of the directrix (z).**

MINOR AXIS :

The y–axis intersects the ellipse in the points $B' \equiv (0, -b)$ & $B \equiv (0, b)$. The line segment B′B of length 2b (b < a) is called the **Minor Axis** of the ellipse.

PRINCIPAL AXIS :

The major & minor axis together are called **Principal Axis** of the ellipse.

CENTRE :

The point which bisects every chord of the conic drawn through it is called the **centre** of the conic.

$C \equiv (0, 0)$ the origin is the centre of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

DIAMETER :

A chord of the conic which passes through the centre is called a **diameter** of the conic.

FOCAL CHORD : A chord which passes through a focus is called a **focal chord**.

DOUBLE ORDINATE :

A chord perpendicular to the major axis is called a **double ordinate.**

LATUS RECTUM :

The focal chord perpendicular to the major axis is called the **latus rectum**. Length of latus rectum

$(LL') = \frac{2b^2}{a} = \frac{(minor\ axis)^2}{major\ axis} = 2a(1-e^2) = 2e$ (distance from focus to the corresponding directrix)

NOTE :

(i) The sum of the focal distances of any point on the ellipse is equal to the major Axis. Hence distance of focus from the extremity of a minor axis is equal to semi major axis. **i.e. BS = CA.**

(ii) If the equation of the ellipse is given as $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ & nothing is mentioned, then the rule is to assume that $a > b$.

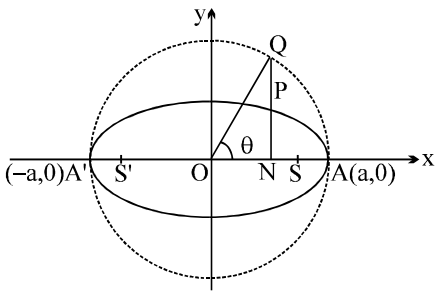
2. POSITION OF A POINT w.r.t. AN ELLIPSE :

The point $P(x_1, y_1)$ lies outside, inside or on the ellipse according as ; $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1 > < \text{or} = 0$.

3. AUXILIARY CIRCLE / ECCENTRIC ANGLE :

A circle described on major axis as diameter is called the **auxiliary circle**.

Let Q be a point on the auxiliary circle $x^2 + y^2 = a^2$ such that QP produced is perpendicular to the x-axis then P & Q are called as the **CORRESPONDING POINTS** on the ellipse & the auxiliary circle respectively 'θ' is called the **ECCENTRIC ANGLE** of the point P on the ellipse $(0 \le \theta < 2\pi)$.



Note that $\frac{\ell(PN)}{\ell(QN)} = \frac{b}{a} = \frac{Semi\ minor\ axis}{Semi\ major\ axis}$

Hence "If from each point of a circle perpendiculars are drawn upon a fixed diameter then the locus of the points dividing these perpendiculars in a given ratio is an ellipse of which the given circle is the auxiliary circle".

4. PARAMETRIC REPRESENTATION :

The equations $x = a\cos\theta$ & $y = b\sin\theta$ together represent the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Where θ is a parameter. Note that if $P(\theta) \equiv (a\cos\theta, b\sin\theta)$ is on the ellipse then ; $Q(\theta) \equiv (a\cos\theta, a\sin\theta)$ is on the auxiliary circle.

5. LINE AND AN ELLIPSE :

The line $y = mx + c$ meets the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in two points real, coincident or imaginary according as c^2 is $< = \text{or} > a^2m^2 + b^2$.

Hence $y = mx + c$ is tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ if $c^2 = a^2m^2 + b^2$.

The equation to the chord of the ellipse joining two points with eccentric angles α & β is given by $\frac{x}{a}\cos\frac{\alpha+\beta}{2}+\frac{y}{b}\sin\frac{\alpha+\beta}{2}=\cos\frac{\alpha-\beta}{2}$.

6. TANGENTS :

(i) $\frac{x\,x_1}{a^2}+\frac{y\,y_1}{b^2}=1$ is tangent to the ellipse at (x_1, y_1).

Note : The figure formed by the tangents at the extremities of latus rectum is rhoubus of area $\frac{2a^2}{e}$

(ii) $y = mx \pm \sqrt{a^2m^2 + b^2}$ is tangent to the ellipse for all values of m.

Note that there are two tangents to the ellipse having the same m, i.e. there are two tangents parallel to any given direction.

(iii) $\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$ is tangent to the ellipse at the point $(a\cos\theta, b\sin\theta)$.

(iv) The eccentric angles of point of contact of two parallel tangents differ by π. Conversely if the difference between the eccentric angles of two points is p then the tangents at these points are parallel.

(v) Point of intersection of the tangents at the point α & β is $a\frac{\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, b\frac{\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}$.

7. NORMALS :

(i) Equation of the normal at (x_1, y_1) is $\frac{a^2x}{x_1}-\frac{b^2y}{y_1}=a^2-b^2=a^2e^2$.

(ii) Equation of the normal at the point $(a\cos\theta, b\sin\theta)$ is ; $ax\cdot\sec\theta - by\cdot\operatorname{cosec}\theta = (a^2 - b^2)$.

(iii) Equation of a normal in terms of its slope 'm' is $y = mx - \frac{(a^2-b^2)m}{\sqrt{a^2+b^2m^2}}$.

8. DIRECTOR CIRCLE : Locus of the point of intersection of the tangents which meet at right angles is called the **Director Circle**. The equation to this locus is $x^2 + y^2 = a^2 + b^2$ i.e. a circle whose centre is the centre of the ellipse & whose radius is the length of the line joining the ends of the major & minor axis.

9. Chord of contact, pair of tangents, chord with a given middle point, pole & polar are to be interpreted as they are in parabola.

10. DIAMETER :

The locus of the middle points of a system of parallel chords with slope 'm' of an ellipse is a straight line passing through the centre of the ellipse, called its diameter and has the equation $y = -\frac{b^2}{a^2m}x$.

11. IMPORTANT HIGHLIGHTS : Refering to an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

H – 1 If P be any point on the ellipse with S & S′ as its foci then $\ell(SP) + \ell(S'P) = 2a$.

H – 2 The product of the length's of the perpendicular segments from the foci on any tangent to the ellipse is b^2 and the feet of these perpendiculars Y,Y′ lie on its auxiliary circle. The tangents at these feet to the auxiliary circle meet on the ordinate of P and that the locus of their point of intersection is a similiar ellipse as that of the original one. Also the lines joining centre to the feet of the perpendicular Y and focus to the point of contact of tangent are parallel.

H – 3 If the normal at any point P on the ellipse with centre C meet the major & minor axes in G & g respectively,

& if CF be perpendicular upon this normal then

(i) PF. PG = b^2 **(ii)** PF. Pg = a^2 **(iii)** PG. Pg = SP. S′P **(iv)** CG. CT = CS^2

(v) locus of the mid point of Gg is another ellipse having the same eccentricity as that of the original ellipse. [where S and S′ are the focii of the ellipse and T is the point where tangent at P meet the major axis]

H – 4 The tangent & normal at a point P on the ellipse bisect the external & internal angles between the focal distances of P. This refers to the well known reflection property of the ellipse which states that rays from one focus are reflected through other focus & vice–versa. Hence we can deduce that the straight lines joining each focus to the foot of the perpendicular from the other focus upon the tangent at any point P meet on the normal PG and bisects it where G is the point where normal at P meets the major axis.

H – 5 The portion of the tangent to an ellipse between the point of contact & the directrix subtends a right angle at the corresponding focus.

H – 6 The circle on any focal distance as diameter touches the auxiliary circle.

H – 7 Perpendiculars from the centre upon all chords which join the ends of any perpendicular diameters of the ellipse are of constant length.

H – 8 If the tangent at the point P of a standard ellipse meets the axis in T and t and CY is the perpendicular on it from the centre then,

(i) T t. PY = $a^2 - b^2$ and (ii) least value of Tt is a + b.

***Suggested problems from Loney*: Exercise-32** (Q.2 to 7, 11, 12, 16, 24), **Exercise-33 (Important)** (Q.3, 5, 6, 15, 16, 18, 19, 24, 25, 26, 34), **Exercise-35** (Q.2, 4, 6, 7, 8, 11, 12, 15)

EXERCISE–4

Q.1 Find the equation of the ellipse with its centre (1, 2), focus at (6, 2) and passing through the point (4, 6).

Q.2 The tangent at any point P of a circle $x^2 + y^2 = a^2$ meets the tangent at a fixed point A (a, 0) in T and T is joined to B, the other end of the diameter through A, prove that the locus of the intersection of AP and BT is an ellipse whose ettentricity is $\frac{1}{\sqrt{2}}$.

Q.3 The tangent at the point α on a standard ellipse meets the auxiliary circle in two points which subtends a right angle at the centre. Show that the eccentricity of the ellipse is $(1 + \sin^2\alpha)^{-1/2}$.

Q.4 An ellipse passes through the points (– 3, 1) & (2, –2) & its principal axis are along the coordinate axes in order. Find its equation.

Q.5 If any two chords be drawn through two points on the major axis of an ellipse equidistant from the centre, show that $\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}\cdot\tan\frac{\gamma}{2}\cdot\tan\frac{\delta}{2}=1$, where α, β, γ, δ are the eccentric angles of the extremities of the chords.

Q.6 If the normals at the points P, Q, R with eccentric angles α, β, γ on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are concurrent, then show that

$$\begin{vmatrix} \sin\alpha & \cos\alpha & \sin 2\alpha \\ \sin\beta & \cos\beta & \sin 2\beta \\ \sin\gamma & \cos\gamma & \sin 2\gamma \end{vmatrix} = 0$$

Q.7 Prove that the equation to the circle, having double contact with the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at the ends of a latus rectum, is $x^2 + y^2 - 2ae^3x = a^2(1 - e^2 - e^4)$.

Q.8 Find the equations of the lines with equal intercepts on the axes & which touch the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$.

Q.9 The tangent at $P\left(4\cos\theta, \frac{16}{\sqrt{11}}\sin\theta\right)$ to the ellipse $16x^2 + 11y^2 = 256$ is also a tangent to the circle $x^2 + y^2 - 2x - 15 = 0$. Find θ. Find also the equation to the common tangent.

Q.10 A tangent having slope $-\frac{4}{3}$ to the ellipse $\frac{x^2}{18}+\frac{y^2}{32}=1$, intersects the axis of x & y in points A & B respectively. If O is the origin, find the area of triangle OAB.

Q.11 'O' is the origin & also the centre of two concentric circles having radii of the inner & the outer circle as 'a' & 'b' respectively. A line OPQ is drawn to cut the inner circle in P & the outer circle in Q. PR is drawn parallel to the y-axis & QR is drawn parallel to the x-axis. Prove that the locus of R is an ellipse touching the two circles. If the focii of this ellipse lie on the inner circle, find the ratio of inner : outer radii & find also the eccentricity of the ellipse.

Q.12 ABC is an isosceles triangle with its base BC twice its altitude. A point P moves within the triangle such that the square of its distance from BC is half the rectangle contained by its distances from the two sides. Show that the locus of P is an ellipse with eccentricity $\sqrt{\frac{2}{3}}$ passing through B & C.

Q.13 Let d be the perpendicular distance from the centre of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ to the tangent drawn at a point P on the ellipse.If F_1 & F_2 are the two foci of the ellipse, then show that $(PF_1 - PF_2)^2 = 4a^2\left[1-\frac{b^2}{d^2}\right]$.

Q.14 Common tangents are drawn to the parabola $y^2 = 4x$ & the ellipse $3x^2 + 8y^2 = 48$ touching the parabola at A & B and the ellipse at C & D. Find the area of the quadrilateral.

Q.15 If the normal at a point P on the ellipse of semi axes a, b & centre C cuts the major & minor axes at G & g, show that $a^2.(CG)^2 + b^2.(Cg)^2 = (a^2 - b^2)^2$. Also prove that $CG = e^2CN$, where PN is the ordinate of P.

Q.16 Prove that the length of the focal chord of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ which is inclined to the major axis at angle θ is $\frac{2ab^2}{a^2\sin^2\theta + b^2\cos^2\theta}$.

Q.17 The tangent at a point P on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ intersects the major axis in T & N is the foot of the perpendicular from P to the same axis. Show that the circle on NT as diameter intersects the auxiliary circle orthogonally.

Q.18 The tangents from (x_1, y_1) to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ intersect at right angles. Show that the normals at the points of contact meet on the line $\frac{y}{y_1}=\frac{x}{x_1}$.

Q.19 Find the locus of the point the chord of contact of the tangent drawn from which to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

touches the circle $x^2 + y^2 = c^2$, where $c < b < a$.

Q.20 Prove that the three ellipse $\frac{x^2}{a_1^2} + \frac{y^2}{b_1^2} = 1, \frac{x^2}{a_2^2} + \frac{y^2}{b_2^2} = 1$ and $\frac{x^2}{a_3^2} + \frac{y^2}{b_3^2} = 1$ will have a common tangent if $\begin{vmatrix} a_1^2 & b_1^2 & 1 \\ a_2^2 & b_2^2 & 1 \\ a_3^2 & b_3^2 & 1 \end{vmatrix} = 0.$

EXERCISE–5

Q.1 PG is the normal to a standard ellipse at P, G being on the major axis. GP is produced outwards to Q so that PQ = GP. Show that the locus of Q is an ellipse whose eccentricity is $\frac{a^2 - b^2}{a^2 + b^2}$ & find the equation of the locus of the intersection of the tangents at P & Q.

Q.2 P & Q are the corresponding points on a standard ellipse & its auxiliary circle. The tangent at P to the ellipse meets the major axis in T. Prove that QT touches the auxiliary circle.

Q.3 The point P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is joined to the ends A, A′ of the major axis. If the lines through P perpendicular to PA, PA′ meet the major axis in Q and R then prove that l(QR) = length of latus rectum.

Q.4 Let S and S' are the foci, SL the semilatus rectum of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and LS' produced cuts the ellipse at P, show that the length of the ordinate of the ordinate of P is $\frac{(1-e^2)}{1+3e^2}a$, where 2a is the length of the major axis and e is the eccentricity of the ellipse.

Q.5 A tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ touches at the point P on it in the first quadrant & meets the coordinate axis in A & B respectively. If P divides AB in the ratio 3 : 1 find the equation of the tangent.

Q.6 PCP′ is a diameter of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (a > b) & QCQ′ is the corresponding diameter of the auxiliary circle, show that the area of the parallelogram formed by the tangent at P, P', Q & Q' is $\frac{8a^2b}{(a-b)\sin 2\phi}$ where ϕ is the eccentric angle of the point P.

Q.7 If the normal at the point P(θ) to the ellipse $\frac{x^2}{14} + \frac{y^2}{5} = 1$, intersects it again at the point Q(2θ), show that $\cos\theta = -(2/3)$.

Q.8 A normal chord to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ makes an angle of 45° with the axis. Prove that the square of its length is equal to $\frac{32a^4b^4}{(a^2+b^2)^3}$

Q.9 If (x_1, y_1) & (x_2, y_2) are two points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the tangents at which meet in (h, k) & the normals in (p, q), prove that $a^2p = e^2hx_1x_2$ and $b^4q = -e^2ky_1y_2a^2$ where 'e' is the eccentricity.

Q.10 A normal inclined at 45° to the axis of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is drawn. It meets the x-axis & the y-axis in P & Q respectively. If C is the centre of the ellipse, show that the area of triangle CPQ is $\frac{(a^2-b^2)^2}{2(a^2+b^2)}$ sq. units.

Q.11 Tangents are drawn to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ from the point $\left(\frac{a^2}{\sqrt{a^2-b^2}},\sqrt{a^2+b^2}\right)$. Prove that they intercept on the ordinate through the nearer focus a distance equal to the major axis.

Q.12 P and Q are the points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. If the chord P and Q touches the ellipse $\frac{4x^2}{a^2}+\frac{y^2}{b^2}-\frac{4x}{a}=0$, prove that $\sec\alpha+\sec\beta=2$ where α,β are the eccentric angles of the points P and Q.

Q.13 A straight line AB touches the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ & the circle $x^2 + y^2 = r^2$; where $a > r > b$. A focal chord of the ellipse, parallel to AB intersects the circle in P & Q, find the length of the perpendicular drawn from the centre of the ellipse to PQ. Hence show that PQ = 2b.

Q.14 Show that the area of a sector of the standard ellipse in the first quadrant between the major axis and a line drawn through the focus is equal to 1/2 ab $(\theta - e\sin\theta)$ sq. units, where θ is the eccentric angle of the point to which the line is drawn through the focus & e is the eccentricity of the ellipse.

Q.15 A ray emanating from the point $(-4, 0)$ is incident on the ellipse $9x^2 + 25y^2 = 225$ at the point P with abscissa 3. Find the equation of the reflected ray after first reflection.

Q.16 If p is the length of the perpendicular from the focus 'S' of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ on any tangent at 'P', then show that $\frac{b^2}{p^2}=\frac{2a}{\ell(SP)}-1$.

Q.17 In an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, n_1 and n_2 are the lengths of two perpendicular normals terminated at the major axis then prove that : $\frac{1}{n_1^2}+\frac{1}{n_2^2}=\frac{a^2+b^2}{b^4}$

Q.18 If the tangent at any point of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ makes an angle α with the major axis and an angle β with the focal radius of the point of contact then show that the eccentricity 'e' of the ellipse is given by the absolute value of $\frac{\cos\beta}{\cos\alpha}$.

Q.19 Using the fact that the product of the perpendiculars from either foci of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ upon a tangent is b^2, deduce the following loci. An ellipse with 'a' & 'b' as the lengths of its semi axes slides between two given straight lines at right angles to one another. Show that the locus of its centre is a circle & the locus of its foci is the curve, $(x^2 + y^2)(x^2 y^2 + b^4) = 4a^2 x^2 y^2$.

Q.20 If tangents are drawn to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ intercept on the x-axis a constant length c, prove that the locus of the point of intersection of tangents is the curve
$4y^2 (b^2x^2 + a^2y^2 - a^2b^2) = c^2 (y^2 - b^2)^2$.

EXERCISE–6

Q.1 If tangent drawn at a point $(t^2, 2t)$ on the parabola $y^2 = 4x$ is same as the normal drawn at a point $(\sqrt{5}\cos\phi, 2\sin\phi)$ on the ellipse $4x^2 + 5y^2 = 20$. Find the values of t & ϕ. [REE '96, 6]

Q.2 A tangent to the ellipse $x^2 + 4y^2 = 4$ meets the ellipse $x^2 + 2y^2 = 6$ at P & Q. Prove that the tangents at P & Q of the ellipse $x^2 + 2y^2 = 6$ are at right angles. [JEE '97, 5]

Q.3(i) The number of values of c such that the straight line $y = 4x + c$ touches the curve $(x^2/4) + y^2 = 1$ is
(A) 0 (B) 1 (C) 2 (D) infinite

(ii) If P = (x, y), $F_1 = (3, 0)$, $F_2 = (-3, 0)$ and $16x^2 + 25y^2 = 400$, then $PF_1 + PF_2$ equals
(A) 8 (B) 6 (C) 10 (D) 12

[JEE '98, 2 + 2]

Q.4(a) If x_1, x_2, x_3 as well as y_1, y_2, y_3 are in G.P. with the same common ratio, then the points (x_1, y_1), (x_2, y_2) & (x_3, y_3) :
(A) lie on a straight line (B) lie on on ellipse (C) lie on a circle (D) are vertices of a triangle.

(b) On the ellipse, $4x^2 + 9y^2 = 1$, the points at which the tangents are parallel to the line $8x = 9y$ are :

(A) $\left(\frac{2}{5}, \frac{1}{5}\right)$ (B) $\left(-\frac{2}{5}, \frac{1}{5}\right)$ (C) $\left(-\frac{2}{5}, -\frac{1}{5}\right)$ (D) $\left(\frac{2}{5}, -\frac{1}{5}\right)$

(c) Consider the family of circles, $x^2 + y^2 = r^2$, $2 < r < 5$. If in the first quadrant, the common tangent to a circle of the family and the ellipse $4x^2 + 25y^2 = 100$ meets the co–ordinate axes at A & B, then find the equation of the locus of the mid–point of AB. [JEE '99, 2 + 3 + 10 (out of 200)]

Q.5 Find the equation of the largest circle with centre (1, 0) that can be inscribed in the ellipse $x^2 + 4y^2 = 16$. [REE '99, 6]

Q.6 Let ABC be an equilateral triangle inscribed in the circle $x^2 + y^2 = a^2$. Suppose perpendiculars from A, B, C to the major axis of the ellipse, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $(a > b)$ meet the ellipse respectively at P, Q, R so that P, Q, R lie on the same side of the major axis as A, B, C respectively. Prove that the normals to the ellipse drawn at the points P, Q and R are concurrent. [JEE '2000, 7]

Q.7 Let C_1 and C_2 be two circles with C_2 lying inside C_1. A circle C lying inside C_1 touches C_1 internally and C_2externally. Identify the locus of the centre of C. [JEE '2001, 5]

Q.8 Find the condition so that the line $px + qy = r$ intersects the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in points whose eccentric angles differ by $\frac{\pi}{4}$. [REE '2001, 3]

Q.9 Prove that, in an ellipse, the perpendicular from a focus upon any tangent and the line joining the centre of the ellipse to the point of contact must on the corresponding directrix. [JEE ' 2002, 5]

Q.10(a) The area of the quadrilateral formed by the tangents at the ends of the latus rectum of the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ is

(A) $9\sqrt{3}$ sq. units (B) $27\sqrt{3}$ sq. units (C) 27 sq. units (D) none

(b) The value of θ for which the sum of intercept on the axis by the tangent at the point $\left(3\sqrt{3}\cos\theta, \sin\theta\right)$, $0 < \theta < \pi/2$ on the ellipse $\frac{x^2}{27} + y^2 = 1$ is least, is : [JEE ' 2003 (Screening)]

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{8}$

Q.11 The locus of the middle point of the intercept of the tangents drawn from an external point to the ellipse $x^2 + 2y^2 = 2$, between the coordinates axes, is

(A) $\frac{1}{x^2} + \frac{1}{2y^2} = 1$ (B) $\frac{1}{4x^2} + \frac{1}{2y^2} = 1$ (C) $\frac{1}{2x^2} + \frac{1}{4y^2} = 1$ (D) $\frac{1}{2x^2} + \frac{1}{y^2} = 1$

[JEE 2004 (Screening)]

Q.12(a) The minimum area of triangle formed by the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and coordinate axes is

(A) ab sq. units (B) $\frac{a^2 + b^2}{2}$ sq. units (C) $\frac{(a+b)^2}{2}$ sq. units (D) $\frac{a^2 + ab + b^2}{3}$ sq. units

[JEE 2005 (Screening)]

(b) Find the equation of the common tangent in 1st quadrant to the circle $x^2 + y^2 = 16$ and the ellipse $\frac{x^2}{25} + \frac{y^2}{4} = 1$. Also find the length of the intercept of the tangent between the coordinate axes.

[JEE 2005 (Mains), 4]

KEY CONCEPTS
HYPERBOLA

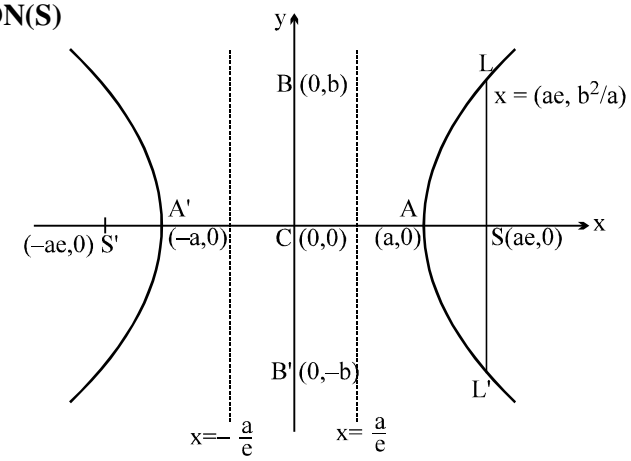
The **HYPERBOLA** is a conic whose eccentricity is greater than unity. (e > 1).

1. STANDARD EQUATION & DEFINITION(S)

Standard equation of the hyperbola is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. Where $b^2 = a^2(e^2-1)$

or $a^2e^2 = a^2 + b^2$ i.e. $e^2 = 1 + \frac{b^2}{a^2}$

$= 1 + \left(\frac{C.A}{T.A}\right)^2$



FOCI :

$S \equiv (ae, 0)$ & $S' \equiv (-ae, 0)$.

EQUATIONS OF DIRECTRICES :

$x = \frac{a}{e}$ & $x = -\frac{a}{e}$.

VERTICES : $A \equiv (a, 0)$ & $A' \equiv (-a, 0)$.

***l* (Latus rectum)** $= \frac{2b^2}{a} = \frac{(C.A.)^2}{T.A.} = 2a(e^2-1)$.

Note : *l* (L.R.) = 2e (distance from focus to the corresponding directrix)

TRANSVERSE AXIS : The line segment A′A of length 2a in which the foci S′ & S both lie is called the **T.A. OF THE HYPERBOLA**.

CONJUGATE AXIS : The line segment B′B between the two points $B' \equiv (0, -b)$ & $B \equiv (0, b)$ is called as the **C.A. OF THE HYPERBOLA**.

The T.A. & the C.A. of the hyperbola are together called the Principal axes of the hyperbola.

2. FOCAL PROPERTY :

The difference of the focal distances of any point on the hyperbola is constant and equal to transverse axis i.e. $\big||PS|-|PS'|\big| = 2a$. The distance SS' = focal length.

3. CONJUGATE HYPERBOLA :

Two hyperbolas such that transverse & conjugate axes of one hyperbola are respectively the conjugate & the transverse axes of the other are called **CONJUGATE HYPERBOLAS** of each other.

eg. $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ & $-\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are conjugate hyperbolas of each.

Note That :

(a) If e_1 & e_2 are the eccentrcities of the hyperbola & its conjugate then $e_1^{-2} + e_2^{-2} = 1$.

(b) The foci of a hyperbola and its conjugate are concyclic and form the vertices of a square.

(c) Two hyperbolas are said to be similiar if they have the same eccentricity.

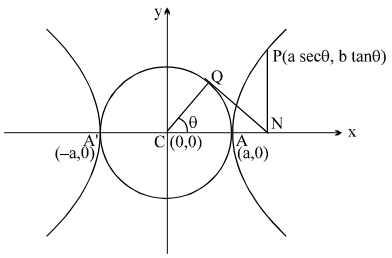
4. RECTANGULAR OR EQUILATERAL HYPERBOLA :

The particular kind of hyperbola in which the lengths of the transverse & conjugate axis are equal is called an **EQUILATERAL HYPERBOLA**. Note that the eccentricity of the rectangular hyperbola is $\sqrt{2}$ and the length of its latus rectum is equal to its transverse or conjugate axis.

5. AUXILIARY CIRCLE :

A circle drawn with centre C & T.A. as a diameter is called the **AUXILIARY CIRCLE** of the hyperbola. Equation of the auxiliary circle is $x^2 + y^2 = a^2$.

Note from the figure that P & Q are called the **"CORRESPONDING POINTS "** on the hyperbola & the auxiliary circle. 'θ' is called the eccentric angle of the point 'P' on the hyperbola. $(0 \le \theta < 2\pi)$.



Note : The equations $x = a \sec\theta$ & $y = b \tan\theta$ together represents the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where θ is a parameter. The parametric equations : $x = a \cos h\, \phi$, $y = b \sin h\, \phi$ also represents the same hyperbola.

General Note : Since the fundamental equation to the hyperbola only differs from that to the ellipse in having $-b^2$ instead of b^2 it will be found that many propositions for the hyperbola are derived from those for the ellipse by simply changing the sign of b^2.

6. POSITION OF A POINT 'P' w.r.t. A HYPERBOLA :

The quantity $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$ is positive, zero or negative according as the point (x_1, y_1) lies within, upon or without the curve.

7. LINE AND A HYPERBOLA :

The straight line $y = mx + c$ is a secant, a tangent or passes outside the hyperbola $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ according as: $c^2 >, =, < a^2 m^2 - b^2$.

8. TANGENTS AND NORMALS :

TANGENTS :

(a) Equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point (x_1, y_1) is $\frac{x\,x_1}{a^2} - \frac{y\,y_1}{b^2} = 1$.

Note: In general two tangents can be drawn from an external point $(x_1\, y_1)$ to the hyperbola and they are $y - y_1 = m_1(x - x_1)$ & $y - y_1 = m_2(x - x_2)$, where m_1 & m_2 are roots of the equation $(x_1^2 - a^2)m^2 - 2x_1y_1m + y_1^2 + b^2 = 0$. If $D < 0$, then no tangent can be drawn from $(x_1\, y_1)$ to the hyperbola.

(b) Equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $(a\sec\theta, b\tan\theta)$ is $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$.

Note : Point of intersection of the tangents at θ_1 & θ_2 is $x = a\frac{\cos\frac{\theta_1 - \theta_2}{2}}{\cos\frac{\theta_1 + \theta_2}{2}}$, $y = b\frac{\sin\frac{\theta_1 + \theta_2}{2}}{\cos\frac{\theta_1 + \theta_2}{2}}$

(c) $y = mx \pm \sqrt{a^2m^2 - b^2}$ can be taken as the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Note that there are two parallel tangents having the same slope m.

(d) Equation of a chord joining α & β is

$$\frac{x}{a}\cos\frac{\alpha - \beta}{2} - \frac{y}{b}\sin\frac{\alpha + \beta}{2} = \cos\frac{\alpha + \beta}{2}$$

NORMALS:

(a) The equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(x_1, y_1)$ on it is $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 - b^2 = a^2e^2$.

(b) The equation of the normal at the point P $(a\sec\theta, b\tan\theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{a\,x}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2 = a^2e^2$.

(c) Equation to the chord of contact, polar, chord with a given middle point, pair of tangents from an external point is to be interpreted as in ellipse.

9. DIRECTOR CIRCLE :

The locus of the intersection of tangents which are at right angles is known as the **DIRECTOR CIRCLE** of the hyperbola. The equation to the director circle is :

$x^2 + y^2 = a^2 - b^2$.

If $b^2 < a^2$ this circle is real; if $b^2 = a^2$ the radius of the circle is zero & it reduces to a point circle at the origin. In this case the centre is the only point from which the tangents at right angles can be drawn to the curve.

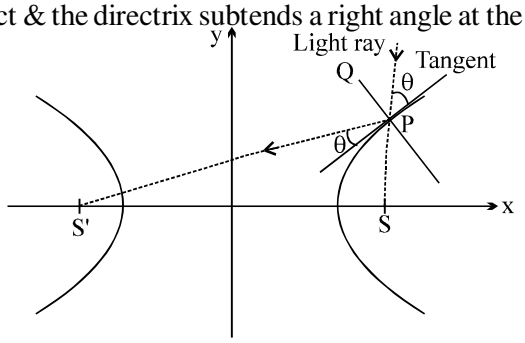
If $b^2 > a^2$, the radius of the circle is imaginary, so that there is no such circle & so no tangents at right angle can be drawn to the curve.

10. HIGHLIGHTS ON TANGENT AND NORMAL :

H–1 Locus of the feet of the perpendicular drawn from focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ upon any tangent is its auxiliary circle i.e. $x^2 + y^2 = a^2$ & the product of the feet of these perpendiculars is $b^2 \cdot (\text{semi C} \cdot \text{A})^2$

H–2 The portion of the tangent between the point of contact & the directrix subtends a right angle at the corresponding focus.

H–3 The tangent & normal at any point of a hyperbola bisect the angle between the focal radii. This spells the reflection property of the hyperbola as **"An incoming light ray"** aimed towards one focus is reflected from the outer surface of the hyperbola towards the other focus. It follows that if an ellipse and a hyperbola have the same foci, they cut at right angles at any of their common point.



Note that the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{a^2 - k^2} - \frac{y^2}{k^2 - b^2} = 1 (a > k > b > 0)$ Xare confocal and therefore orthogonal.

H–4 The foci of the hyperbola and the points P and Q in which any tangent meets the tangents at the vertices are concyclic with PQ as diameter of the circle.

11. ASYMPTOTES :

Definition : If the length of the perpendicular let fall from a point on a hyperbola to a straight line tends to zero as the point on the hyperbola moves to infinity along the hyperbola, then the straight line is called the Asymptote of the Hyperbola.

To find the asymptote of the hyperbola :

Let $y = mx + c$ is the asymptote of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Solving these two we get the quadratic as

$(b^2 - a^2m^2)\, x^2 - 2a^2 mcx - a^2 (b^2 + c^2) = 0$ ....(1)

In order that $y = mx + c$ be an asymptote, both roots of equation (1) must approach infinity, the conditions for which are :

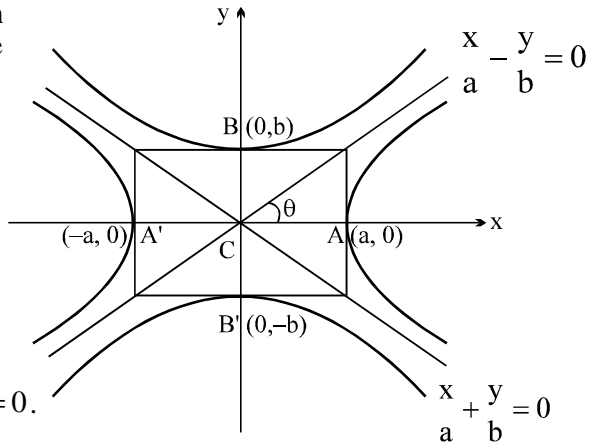
coeff of $x^2 = 0$ & coeff of $x = 0$.

$\Rightarrow$ $b^2 - a^2m^2 = 0$ or $m = \pm \frac{b}{a}$ &

$a^2 mc = 0 \Rightarrow c = 0$.

$\therefore$ equations of asymptote are $\frac{x}{a} + \frac{y}{b} = 0$

and $\frac{x}{a} - \frac{y}{b} = 0$.



combined equation to the asymptotes $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$.

PARTICULAR CASE :

When $b = a$ the asymptotes of the rectangular hyperbola.

$x^2 - y^2 = a^2$ are, $y = \pm x$ which are at right angles.

Note :

(i) Equilateral hyperbola $\Leftrightarrow$ rectangular hyperbola.

(ii) If a hyperbola is equilateral then the conjugate hyperbola is also equilateral.

(iii) A hyperbola and its conjugate have the same asymptote.

(iv) The equation of the pair of asymptotes differ the hyperbola & the conjugate hyperbola by the same constant only.

(v) The asymptotes pass through the centre of the hyperbola & the bisectors of the angles between the asymptotes are the axes of the hyperbola.

(vi) The asymptotes of a hyperbola are the diagonals of the rectangle formed by the lines drawn through the extremities of each axis parallel to the other axis.

(vii) Asymptotes are the tangent to the hyperbola from the centre.

(viii) A simple method to find the coordinates of the centre of the hyperbola expressed as a general equation of degree 2 should be remembered as:
Let $f(x, y) = 0$ represents a hyperbola.
Find $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y}$. Then the point of intersection of $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0$ gives the centre of the hyperbola.

12. HIGHLIGHTS ON ASYMPTOTES:

H–1 If from any point on the asymptote a straight line be drawn perpendicular to the transverse axis, the product of the segments of this line, intercepted between the point & the curve is always equal to the square of the semi conjugate axis.

H–2 Perpendicular from the foci on either asymptote meet it in the same points as the corresponding directrix & the common points of intersection lie on the auxiliary circle.

H–3 The tangent at any point P on a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with centre C, meets the asymptotes in Q and R and cuts off a Δ CQR of constant area equal to ab from the asymptotes & the portion of the tangent intercepted between the asymptote is bisected at the point of contact. This implies that locus of the centre of the circle circumscribing the Δ CQR in case of a rectangular hyperbola is the hyperbola itself & for a standard hyperbola the locus would be the curve, $4(a^2x^2 - b^2y^2) = (a^2 + b^2)^2$.

H–4 If the angle between the asymptote of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is 2θ then $e = \sec\theta$.

13. RECTANGULAR HYPERBOLA :

Rectangular hyperbola referred to its asymptotes as axis of coordinates.

(a) Equation is $xy = c^2$ with parametric representation $x = ct$, $y = c/t$, $t \in R - \{0\}$.

(b) Equation of a chord joining the points $P(t_1)$ & $Q(t_2)$ is $x + t_1t_2y = c(t_1 + t_2)$ with slope $m = -\frac{1}{t_1t_2}$.

(c) Equation of the tangent at $P(x_1, y_1)$ is $\frac{x}{x_1} + \frac{y}{y_1} = 2$ & at $P(t)$ is $\frac{x}{t} + ty = 2c$.

(d) Equation of normal : $y - \frac{c}{t} = t^2(x - ct)$

(e) Chord with a given middle point as (h, k) is $kx + hy = 2hk$.

***Suggested problems from Loney*: Exercise-36** (Q.1 to 6, 16, 22), **Exercise-37** (Q.1, 3, 5, 7, 12)

EXERCISE–7

Q.1 Find the equation to the hyperbola whose directrix is $2x + y = 1$, focus (1, 1) & eccentricity $\sqrt{3}$. Find also the length of its latus rectum.

Q.2 The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through the point of intersection of the lines, $7x + 13y - 87 = 0$ and $5x - 8y + 7 = 0$ & the latus rectum is $32\sqrt{2}/5$. Find 'a' & 'b'.

Q.3 For the hyperbola $\frac{x^2}{100} - \frac{y^2}{25} = 1$, prove that

(i) eccentricity $= \sqrt{5}/2$ (ii) SA. S′A = 25, where S & S′ are the foci & A is the vertex.

Q.4 Find the centre, the foci, the directrices, the length of the latus rectum, the length & the equations of the axes & the asymptotes of the hyperbola $16x^2 - 9y^2 + 32x + 36y - 164 = 0$.

Q.5 The normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ drawn at an extremity of its latus rectum is parallel to an asymptote. Show that the eccentricity is equal to the square root of $(1+\sqrt{5})/2$.

Q.6 If a rectangular hyperbola have the equation, $xy = c^2$, prove that the locus of the middle points of the chords of constant length $2d$ is $(x^2 + y^2)(xy - c^2) = d^2xy$.

Q.7 A triangle is inscribed in the rectangular hyperbola $xy = c^2$. Prove that the perpendiculars to the sides at the points where they meet the asymptotes are concurrent. If the point of concurrence is (x_1, y_1) for one asymptote and (x_2, y_2) for the other, then prove that $x_2y_1 = c^2$.

Q.8 The tangents & normal at a point on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ cut the $y-$axis at A & B. Prove that the circle on AB as diameter passes through the foci of the hyperbola.

Q.9 Find the equation of the tangent to the hyperbola $x^2 - 4y^2 = 36$ which is perpendicular to the line $x - y + 4 = 0$.

Q.10 Ascertain the co-ordinates of the two points Q & R, where the tangent to the hyperbola $\frac{x^2}{45} - \frac{y^2}{20} = 1$ at the point P(9, 4) intersects the two asymptotes. Finally prove that P is the middle point of QR. Also compute the area of the triangle CQR where C is the centre of the hyperbola.

Q.11 If θ_1 & θ_2 are the parameters of the extremities of a chord through (ae, 0) of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then show that $\tan\frac{\theta_1}{2} \cdot \tan\frac{\theta_2}{2} + \frac{e-1}{e+1} = 0$.

Q.12 If C is the centre of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, S, S′ its foci and P a point on it.
Prove that SP. S′P = $CP^2 - a^2 + b^2$.

Q.13 Tangents are drawn to the hyperbola $3x^2 - 2y^2 = 25$ from the point (0, 5/2). Find their equations.

Q.14 If the tangent at the point (h, k) to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ cuts the auxiliary circle in points whose ordinates are y_1 and y_2 then prove that $\frac{1}{y_1} + \frac{1}{y_2} = \frac{2}{k}$.

Q.15 Tangents are drawn from the point (α, β) to the hyperbola $3x^2 - 2y^2 = 6$ and are inclined at angles θ and ϕ to the x –axis. If $\tan\theta \cdot \tan\phi = 2$, prove that $\beta^2 = 2\alpha^2 - 7$.

Q.16 If two points P & Q on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ whose centre is C be such that CP is perpendicular to CQ & $a < b$, then prove that $\frac{1}{CP^2} + \frac{1}{CQ^2} = \frac{1}{a^2} - \frac{1}{b^2}$.

Q.17 The perpendicular from the centre upon the normal on any point of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets at R. Find the locus of R.

Q.18 If the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point P meets the transverse axis in G & the conjugate axis in g & CF be perpendicular to the normal from the centre C, then prove that $|PF.\ PG| = b^2$ & PF. Pg = a^2 where a & b are the semi transverse & semi-conjugate axes of the hyperbola.

Q.19 If the normal at a point P to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the $x-$axis at G, show that SG = e. SP, S being the focus of the hyperbola.

Q.20 An ellipse has eccentricity 1/2 and one focus at the point P (1/2, 1). Its one directrix is the common

tangent, nearer to the point P, to the circle $x^2 + y^2 = 1$ and the hyperbola $x^2 - y^2 = 1$. Find the equation of the ellipse in the standard form.

Q.21 Show that the locus of the middle points of normal chords of the rectangular hyperbola $x^2 - y^2 = a^2$ is $(y^2 - x^2)^3 = 4\,a^2x^2y^2$.

Q.22 Prove that infinite number of triangles can be inscribed in the rectangular hyperbola, $xy = c^2$ whose sides touch the parabola, $y^2 = 4ax$.

Q.23 A point P divides the focal length of the hyperbola $9x^2 - 16y^2 = 144$ in the ratio $S'P : PS = 2 : 3$ where S & S′ are the foci of the hyperbola. Through P a straight line is drawn at an angle of 135° to the axis OX. Find the points of intersection of this line with the asymptotes of the hyperbola.

Q.24 Find the length of the diameter of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ perpendicular to the asymptote of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ passing through the first & third quadrants.

Q.25 The tangent at P on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets one of the asymptote in Q. Show that the locus of the mid point of PQ is a similiar hyperbola.

EXERCISE–8

Q.1 The chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ whose equation is $x\cos\alpha + y\sin\alpha = p$ subtends a right angle at the centre. Prove that it always touches a circle.

Q.2 If a chord joining the points P $(a\sec\theta, a\tan\theta)$ & Q $(a\sec\phi, a\tan\phi)$ on the hyperbola $x^2 - y^2 = a^2$ is a normal to it at P, then show that $\tan\phi = \tan\theta\,(4\sec^2\theta - 1)$.

Q.3 Prove that the locus of the middle point of the chord of contact of tangents from any point of the circle $x^2 + y^2 = r^2$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is given by the equation $\left(\frac{x^2}{a^2} - \frac{y^2}{b^2}\right)^2 = \frac{(x^2 + y^2)}{r^2}$.

Q.4 A transversal cuts the same branch of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ in P, P' and the asymptotes in Q, Q'. Prove that (i) PQ = P'Q' & (ii) PQ' = P'Q

Q.5 Find the asymptotes of the hyperbola $2x^2 - 3xy - 2y^2 + 3x - y + 8 = 0$. Also find the equation to the conjugate hyperbola & the equation of the principal axes of the curve.

Q.6 An ellipse and a hyperbola have their principal axes along the coordinate axes and have a common foci separated by a distance $2\sqrt{13}$, the difference of their focal semi axes is equal to 4. If the ratio of their eccentricities is 3/7. Find the equation of these curves.

Q.7 The asymptotes of a hyperbola are parallel to $2x + 3y = 0$ & $3x + 2y = 0$. Its centre is (1, 2) & it passes through (5, 3). Find the equation of the hyperbola.

Q.8 Tangents are drawn from any point on the rectangular hyperbola $x^2 - y^2 = a^2 - b^2$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Prove that these tangents are equally inclined to the asymptotes of the hyperbola.

Q.9 The graphs of $x^2 + y^2 + 6x - 24y + 72 = 0$ & $x^2 - y^2 + 6x + 16y - 46 = 0$ intersect at four points. Compute the sum of the distances of these four points from the point (–3, 2).

Q.10 Find the equations of the tangents to the hyperbola $x^2 - 9y^2 = 9$ that are drawn from (3, 2). Find the area of the triangle that these tangents form with their chord of contact.

Q.11 A series of hyperbolas is drawn having a common transverse axis of length 2a. Prove that the locus of a point P on each hyperbola, such that its distance from the transverse axis is equal to its distance from an asymtote, is the curve $(x^2 - y^2)^2 = 4x^2(x^2 - a^2)$.

Q.12 A parallelogram is constructed with its sides parallel to the asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, and one of its diagonals is a chord of the hyperbola ; show that the other diagonal passes through the centre.

Q.13 The sides of a triangle ABC, inscribed in a hyperbola $xy = c^2$, makes angles α, β, γ with an asymptote. Prove that the nomals at A, B, C will meet in a point if $\cot 2\alpha + \cot 2\beta + \cot 2\gamma = 0$

Q.14 A line through the origin meets the circle $x^2 + y^2 = a^2$ at P & the hyperbola $x^2 - y^2 = a^2$ at Q. Prove that the locus of the point of intersection of the tangent at P to the circle and the tangent at Q to the hyperbola is curve $a^4(x^2 - a^2) + 4x^2y^4 = 0$.

Q.15 A straight line is drawn parallel to the conjugate axis of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ to meet it and the conjugate hyperbola in the points P & Q. Show that the tangents at P & Q meet on the curve $\frac{y^4}{b^4}\left(\frac{y^2}{b^2} - \frac{x^2}{a^2}\right) = \frac{4x^2}{a^2}$ and that the normals meet on the axis of x.

Q.16 A tangent to the parabola $x^2 = 4ay$ meets the hyperbola $xy = k^2$ in two points P & Q. Prove that the middle point of PQ lies on a parabola.

Q.17 Prove that the part of the tangent at any point of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ intercepted between the point of contact and the transverse axis is a harmonic mean between the lengths of the perpendiculars drawn from the foci on the normal at the same point.

Q.18 Let 'p' be the perpendicular distance from the centre C of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ to the tangent drawn at a point R on the hyperbola. If S & S′ are the two foci of the hyperbola, then show that $(RS + RS')^2 = 4a^2\left(1 + \frac{b^2}{p^2}\right)$.

Q.19 P & Q are two variable points on a rectangular hyperbola $xy = c^2$ such that the tangent at Q passes through the foot of the ordinate of P. Show that the locus of the point of intersection of tangent at P & Q is a hyperbola with the same asymptotes as the given hyperbola.

Q.20 Chords of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are tangents to the circle drawn on the line joining the foci as diameter. Find the locus of the point of intersection of tangents at the extremities of the chords.

Q.21 From any point of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, tangents are drawn to another hyperbola which has the same asymptotes. Show that the chord of contact cuts off a constant area from the asymptotes.

Q.22 The chord QQ′ of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is parallel to the tangent at P. PN, QM & Q′M′ are perpendiculars to an asymptote. Show that $QM \cdot Q'M' = PN^2$.

Q.23 If four points be taken on a rectangular hyperbola $xy = c^2$ such that the chord joining any two is perpendicular to the chord joining the other two and $\alpha, \beta, \gamma, \delta$ be the inclinations to either asymptotes of the straight lines joining these points to the centre. Then prove that ; $\tan\alpha \cdot \tan\beta \cdot \tan\gamma \cdot \tan\delta = 1$.

Q.24 The normals at three points P, Q, R on a rectangular hyperbola $xy = c^2$ intersect at a point on the curve. Prove that the centre of the hyperbola is the centroid of the triangle PQR.

Q.25 Through any point P of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ a line QPR is drawn with a fixed gradient m, meeting the asymptotes in Q & R. Show that the product, $(QP) \cdot (PR) = \frac{a^2b^2(1+m^2)}{b^2 - a^2m^2}$.

EXERCISE–9

Q.1 Find the locus of the mid points of the chords of the circle $x^2 + y^2 = 16$, which are tangent to the hyperbola $9x^2 - 16y^2 = 144$. [REE '97, 6]

Q.2 If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$, $S(x_4, y_4)$, then

(A) $x_1 + x_2 + x_3 + x_4 = 0$ (B) $y_1 + y_2 + y_3 + y_4 = 0$

(C) $x_1 x_2 x_3 x_4 = c^4$ (D) $y_1 y_2 y_3 y_4 = c^4$ [JEE '98, 2]

Q.3(a) The curve described parametrically by, $x = t^2 + t + 1$, $y = t^2 - t + 1$ represents:
(A) a parabola (B) an ellipse (C) a hyperbola (D) a pair of straight lines

(b) Let P (a sec θ, b tan θ) and Q (a sec ϕ, b tan ϕ), where $\theta + \phi = \frac{\pi}{2}$, be two points on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If (h, k) is the point of intersection of the normals at P & Q, then k is equal to:
(A) $\frac{a^2 + b^2}{a}$ (B) $-\left(\frac{a^2 + b^2}{a}\right)$ (C) $\frac{a^2 + b^2}{b}$ (D) $-\left(\frac{a^2 + b^2}{b}\right)$

(c) If $x = 9$ is the chord of contact of the hyperbola $x^2 - y^2 = 9$, then the equation of the corresponding pair of tangents, is :
(A) $9x^2 - 8y^2 + 18x - 9 = 0$ (B) $9x^2 - 8y^2 - 18x + 9 = 0$
(C) $9x^2 - 8y^2 - 18x - 9 = 0$ (D) $9x^2 - 8y^2 + 18x + 9 = 0$
[JEE '99, 2 + 2 + 2 (out of 200)]

Q.4 The equation of the common tangent to the curve $y^2 = 8x$ and $xy = -1$ is
(A) $3y = 9x + 2$ (B) $y = 2x + 1$ (C) $2y = x + 8$ (D) $y = x + 2$
[JEE 2002 Screening]

Q.5 Given the family of hyperbols $\frac{x^2}{\cos^2\alpha} - \frac{y^2}{\sin^2\alpha} = 1$ for $\alpha \in (0, \pi/2)$ which of the following does not change with varying α?
(A) abscissa of foci (B) eccentricity
(C) equations of directrices (D) abscissa of vertices [JEE 2003 (Scr.)]

Q.6 The line $2x + \sqrt{6}\,y = 2$ is a tangent to the curve $x^2 - 2y^2 = 4$. The point of contact is
(A) $(4, -\sqrt{6})$ (B) $(7, -2\sqrt{6})$ (C) (2, 3) (D) $(\sqrt{6}, 1)$ [JEE 2004 (Scr.)]

Q.7 Tangents are drawn from any point on the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ to the circle $x^2 + y^2 = 9$. Find the locus of midpoint of the chord of contact. [JEE 2005 (Mains), 4]

Q.8(a) If a hyperbola passes through the focus of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and its transverse and conjugate axis coincides with the major and minor axis of the ellipse, and product of their eccentricities is 1, then
(A) equation of hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ (B) equation of hyperbola $\frac{x^2}{9} - \frac{y^2}{25} = 1$
(C) focus of hyperbola (5, 0) (D) focus of hyperbola is $\left(5\sqrt{3}, 0\right)$ [JEE 2006, 5]

Comprehension: (3 questions)
Let ABCD be a square of side length 2 units. C_2 is the circle through vertices A, B, C, D and C_1 is the circle touching all the sides of the square ABCD. L is a line through A

(a) If P is a point on C_1 and Q in another point on C_2, then $\frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2}$ is equal to
(A) 0.75 (B) 1.25 (C) 1 (D) 0.5

(b) A circle touches the line L and the circle C_1 externally such that both the circles are on the same side of the line, then the locus of centre of the circle is
(A) ellipse (B) hyperbola (C) parabola (D) parts of straight line

(c) A line M through A is drawn parallel to BD. Point S moves such that its distances from the line BD and the vertex A are equal. If locus of S cuts M at T_2 and T_3 and AC at T_1, then area of $\Delta T_1T_2T_3$ is
(A) 1/2 sq. units (B) 2/3 sq. units (C) 1 sq. unit (D) 2 sq. units
[JEE 2006, 5 marks each]

ANSWER KEY

PARABOLA

EXERCISE–1

Q.2 (a, 0) ; a **Q.3** $2x - y + 2 = 0$, (1, 4) ; $x + 2y + 16 = 0$, (16, –16)

Q.5 $3x - 2y + 4 = 0$; $x - y + 3 = 0$ **Q.6** (4 , 0) ; $y^2 = 2a(x - 4a)$

Q.8 $y = -4x + 72$, $y = 3x - 33$ **Q.9** $7y \pm 2(x + 6a) = 0$

Q.15 $x^2 + y^2 + 18x - 28y + 27 = 0$ **Q.18** $x - y = 1$; $8\sqrt{2}$ sq. units

Q.19 $\left(y - \frac{8}{9}\right)^2 = \frac{4}{9}\left(x - \frac{2}{9}\right)$, vertex $\left(\frac{2}{9}, \frac{8}{9}\right)$ **Q.20** $15a^2/4$ **Q.21** (2a, 0) **Q.23** $a^2 > 8b^2$

EXERCISE–2

Q.3 $[a(t^2_o + 4), -2at_o]$ **Q.5** $(ax + by)(x^2 + y^2) + (bx - ay)^2 = 0$

Q.10 (a) $\left(-\frac{1}{2}, \frac{1}{4}\right)$; (b) $y = -(x^2 + x)$ **Q.12** $((x_1 - 2a), 2y_1)$ **Q.21** $y^2 = 8ax$

Q.16 Q(4, –8) **Q.18** $(x^2 + y^2 - 4ax)^2 = 16a(x^3 + xy^2 + ay^2)$

EXERCISE–3

Q.1 $x^2 - 2y + 12 = 0$ **Q.3** $x = 3\left[7\left(\frac{y}{18}\right)^{2/3} + 2\right]$ **Q.4** $x - 2y + 1 = 0$; $y = mx + \frac{1}{4m}$ where $m = \frac{-5 \pm \sqrt{30}}{10}$

Q.5 (a) C ; **(b)** B **Q.6** $(x + 3)y^2 + 32 = 0$ **Q.7** **(a)** C ; **(b)** D **Q.8** C

Q.9 D **Q.10** **(a)** C ; **(b)** $\alpha = 2$ **Q.11** B

Q.12 $xy^2 + y^2 - 2xy + x - 2y + 5 = 0$ **Q.13** (a) D, (b) A, B, (c) (i) A, (ii) B, (iii) D, (iv) C

ELLIPSE

EXERCISE–4

Q.1 $20x^2 + 45y^2 - 40x - 180y - 700 = 0$ **Q.4** $3x^2 + 5y^2 = 32$

Q.8 $x + y - 5 = 0$, $x + y + 5 = 0$ **Q.9** $\theta = \frac{\pi}{3}$ or $\frac{5\pi}{3}$; $4x \pm \sqrt{33}\, y - 32 = 0$

Q.10 24 sq.units **Q.11** $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ **Q.14** $55\sqrt{2}$ sq. units **Q.19** $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{c^2}$

EXERCISE–5

Q.1 $(a^2 - b^2)^2 x^2y^2 = a^2(a^2 + b^2)^2 y^2 + 4b^6x^2$ **Q.5** $bx + a\sqrt{3}\, y = 2ab$

Q.13 $\sqrt{r^2 - b^2}$ **Q.15** $12x + 5y = 48$; $12x - 5y = 48$

EXERCISE–6

Q.1 $\phi = \pi - \tan^{-1} 2$, $t = -\frac{1}{\sqrt{5}}$; $\phi = \pi + \tan^{-1} 2$, $t = \frac{1}{\sqrt{5}}$; $\phi = \pm\frac{\pi}{2}$, $t = 0$

Q.3 (i) C ; **(ii)** C **Q.4 (a)** A ; **(b)** B, D ; **(c)** $25y^2 + 4x^2 = 4x^2y^2$ **Q.5** $(x-1)^2 + y^2 = \frac{11}{3}$

Q.7 Locus is an ellipse with foci as the centres of the circles C_1 a nd C_2.

Q.8 $a^2p^2 + b^2q^2 = r^2\sec^2\frac{\pi}{8} = (4 - 2\sqrt{2})r^2$ **Q.10 (a)** C ; **(b)** A **Q.11** C **Q.12** (a) A, (b) $AB = \frac{14}{\sqrt{3}}$

HYPERBOLA

EXERCISE–7

Q.1 $7x^2 + 12xy - 2y^2 - 2x + 4y - 7 = 0$; $\sqrt{\frac{48}{5}}$ **Q.2** $a^2 = 25/2$; $b^2 = 16$

Q.4 $(-1, 2)$; $(4, 2)$ & $(-6, 2)$; $5x - 4 = 0$ & $5x + 14 = 0$; $\frac{32}{3}$; 6 ; 8 ; $y - 2 = 0$; $x + 1 = 0$; $4x - 3y + 10 = 0$; $4x + 3y - 2 = 0$.

Q.9 $x + y \pm 3\sqrt{3} = 0$ **Q.10** (15, 10) and (3, −2) and 30 sq. units

Q.13 $3x + 2y - 5 = 0$; $3x - 2y + 5 = 0$ **Q.17** $(x^2 + y^2)^2 (a^2y^2 - b^2x^2) = x^2y^2 (a^2 + b^2)^2$

Q.20 $\frac{\left(x - \frac{1}{3}\right)^2}{1/9} + \frac{(y-1)^2}{1/12} = 1$ **Q.23** $(-4, 3)$ & $\left(-\frac{4}{7}, -\frac{3}{7}\right)$ **Q.24** $\frac{150}{\sqrt{481}}$ **Q.25** $4\left(\frac{x^2}{a^2} - \frac{y^2}{b^2}\right) = 3$

EXERCISE–8

Q.5 $x - 2y + 1 = 0$; $2x + y + 1 = 0$; $2x^2 - 3xy - 2y^2 + 3x - y - 6 = 0$; $3x - y + 2 = 0$; $x + 3y = 0$

Q.6 $\frac{x^2}{49} + \frac{y^2}{36} = 1$; $\frac{x^2}{9} - \frac{y^2}{4} = 1$ **Q.7** $6x^2 + 13xy + 6y^2 - 38x - 37y - 98 = 0$

Q.9 40 **Q.10** $y = \frac{5}{12}x + \frac{3}{4}$ **;** $x - 3 = 0$; 8 sq. unit

Q.19 $xy = \frac{8}{9}c^2$ **Q.20** $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2 + b^2}$ **Q.21** ab

EXERCISE–9

Q.1 $(x^2 + y^2)^2 = 16x^2 - 9y^2$ **Q.2** A, B, C, D **Q.3 (a)** A ; **(b)** D ; (c) B **Q.4** D

Q.5 A **Q.6** A **Q.7** $\frac{x^2}{9} - \frac{y^2}{4} = \left(\frac{x^2 + y^2}{9}\right)^2$ **Q.8** (a) A, (b) C, (c) C

EXERCISE–10

Part : (A) Only one correct option

1. If $(2, 0)$ is the vertex & y – axis the directrix of a parabola, then its focus is:
(A) $(2, 0)$ (B) $(-2, 0)$ (C) $(4, 0)$ (D) $(-4, 0)$

2. A parabola is drawn with its focus at (3, 4) and vertex at the focus of the parabola $y^2 - 12x - 4y + 4 = 0$. The equation of the parabola is:
(A) $x^2 - 6x - 8y + 25 = 0$ (B) $y^2 - 8x - 6y + 25 = 0$
(C) $x^2 - 6x + 8y - 25 = 0$ (D) $x^2 + 6x - 8y - 25 = 0$

3. The length of the chord of the parabola, $y^2 = 12x$ passing through the vertex & making an angle of 60º with the axis of x is:
(A) 8 (B) 4 (C) 16/3 (D) none

4. The length of the side of an equilateral triangle inscribed in the parabola, $y^2 = 4x$ so that one of its angular point is at the vertex is:
(A) $8\sqrt{3}$ (B) $6\sqrt{3}$ (C) $4\sqrt{3}$ (D) $2\sqrt{3}$

5. The circles on focal radii of a parabola as diameter touch:
(A) the tangent at the vertex (B) the axis (C) the directrix (D) none of these

6. The equation of the tangent to the parabola $y = (x - 3)^2$ parallel to the chord joining the points (3, 0) and (4, 1) is:
(A) $2x - 2y + 6 = 0$ (B) $2y - 2x + 6 = 0$ (C) $4y - 4x + 11 = 0$ (D) $4x - 4y = 11$

7. The angle between the tangents drawn from a point $(-a, 2a)$ to $y^2 = 4ax$ is
(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

8. An equation of a tangent common to the parabolas $y^2 = 4x$ and $x^2 = 4y$ is
(A) $x - y + 1 = 0$ (B) $x + y - 1 = 0$ (C) $x + y + 1 = 0$ (D) $y = 0$

9. The line $4x - 7y + 10 = 0$ intersects the parabola, $y^2 = 4x$ at the points A & B. The co-ordinates of the point of intersection of the tangents drawn at the points A & B are:
(A) $\left(\frac{7}{2}, \frac{5}{2}\right)$ (B) $\left(-\frac{5}{2}, -\frac{7}{2}\right)$ (C) $\left(\frac{5}{2}, \frac{7}{2}\right)$ (D) $\left(-\frac{7}{2}, -\frac{5}{2}\right)$

10. AP & BP are tangents to the parabola, $y^2 = 4x$ at A & B. If the chord AB passes through a fixed point $(-1, 1)$ then the equation of locus of P is
(A) $y = 2(x - 1)$ (B) $y = 2(x + 1)$ (C) $y = 2x$ (D) $y^2 = 2(x - 1)$

11. Equation of the normal to the parabola, $y^2 = 4ax$ at its point $(am^2, 2am)$ is:
(A) $y = -mx + 2am + am^3$ (B) $y = mx - 2am - am^3$ (C) $y = mx + 2am + am^3$ (D) none

12. At what point on the parabola $y^2 = 4x$ the normal makes equal angles with the axes?
(A) (4, 4) (B) (9, 6) (C) (4, – 1) (D) (1, 2)

13. If on a given base, a triangle be described such that the sum of the tangents of the base angles is a constant, then the locus of the vertex is:
(A) a circle (B) a parabola (C) an ellipse (D) a hyperbola

14. A point moves such that the square of its distance from a straight line is equal to the difference between the square of its distance from the centre of a circle and the square of the radius of the circle. The locus of the point is:
(A) a straight line at right angles to the given line (B) a circle concentric with the given circle (C)a parabola with its axis parallel to the given line(D) a parabola with its axis perpendicular to the given line.

15. P is any point on the parabola, $y^2 = 4ax$ whose vertex is A. PA is produced to meet the directrix in D & M is the foot of the perpendicular from P on the directrix. The angle subtended by MD at the focus is:
(A) $\pi/4$ (B) $\pi/3$ (C) $5\pi/12$ (D) $\pi/2$

16. If the distances of two points P & Q from the focus of a parabola $y^2 = 4ax$ are 4 & 9, then the distance of the point of intersection of tangents at P & Q from the focus is:
(A) 8 (B) 6 (C) 5 (D) 13

17. Tangents are drawn from the point (– 1, 2) on the parabola $y^2 = 4x$. The length of intercept made by these tangents on the line x = 2 is:
(A) 6 (B) $6\sqrt{2}$ (C) $2\sqrt{6}$ (D) none of these

18. From the point $(4, 6)$ a pair of tangent lines are drawn to the parabola, $y^2 = 8x$. The area of the triangle formed by these pair of tangent lines & the chord of contact of the point $(4, 6)$ is:
(A) 8 (B) 4 (C) 2 (D) none of these

19. Locus of the intersection of the tangents at the ends of the normal chords of the parabola $y^2 = 4ax$ is
(A)$(2a + x)\ y^2 + 4a^3 = 0$ (B) $(2a + x) + y^2 = 0$
(C) $(2a + x)\ y^2 + 4a = 0$ (D) none of these

20. If the tangents & normals at the extremities of a focal chord of a parabola intersect at (x_1, y_1) and (x_2, y_2) respectively, then:
(A) $x_1 = x_2$ (B) $x_1 = y_2$ (C) $y_1 = y_2$ (D) $x_2 = y_1$

21. Tangents are drawn from the points on the line $x - y + 3 = 0$ to parabola $y^2 = 8x$. Then all the chords of contact passes through a fixed point whose coordinates are:
(A) (3, 2) (B) (2, 4) (C) (3, 4) (D) (4, 1)

22. The distance between a tangent to the parabola $y^2 = 4Ax$ $(A > 0)$ and the parallel normal with gradient 1 is:
(A) 4 A (B) $2\sqrt{2}$ A (C) 2 A (D) $\sqrt{2}$ A

23. A variable parabola of latus rectum ℓ, touches a fixed equal parabola, then axes of the two curves being

parallel. The locus of the vertex of the moving curve is a parabola, whole latus rectum is:
(A) ℓ (B) 2ℓ (C) 4ℓ (D) none

24. Length of the focal chord of the parabola $y^2 = 4ax$ at a distance p from the vertex is:

(A) $\frac{2a^2}{p}$ (B) $\frac{a^3}{p^2}$ (C) $\frac{4a^3}{p^2}$ (D) $\frac{p^2}{a}$

25. AB is a chord of the parabola $y^2 = 4ax$ with vertex at A. BC is drawn perpendicular to AB meeting the axis at C. The projection of BC on the axis of the parabola is
(A) a (B) 2a (C) 4a (D) 8a

26. The locus of the foot of the perpendiculars drawn from the vertex on a variable tangent to the parabola $y^2 = 4ax$ is:
(A) $x(x^2 + y^2) + ay^2 = 0$ (B) $y(x^2 + y^2) + ax^2 = 0$
(C) $x(x^2 - y^2) + ay^2 = 0$ (D) none of these

27. T is a point on the tangent to a parabola $y^2 = 4ax$ at its point P. TL and TN are the perpendiculars on the focal radius SP and the directrix of the parabola respectively. Then:
(A) SL = 2 (TN) (B) 3 (SL) = 2 (TN) (C) SL = TN (D) 2 (SL) = 3 (TN)

28. The point of contact of the tangent to the parabola $y^2 = 9x$ which passes through the point (4, 10) and makes an angle θ with the axis of the parabola such that $\tan\theta > 2$ is
(A) (4/9, 2) (B) (36, 18) (C) (4, 6) (D) (1/4, 3/2)

29. If the parabolas $y^2 = 4x$ and $x^2 = 32y$ intersect at (16, 8) at an angle θ, then θ is equal to

(A) $\tan^{-1}\left(\frac{3}{5}\right)$ (B) $\tan^{-1}\left(\frac{4}{5}\right)$ (C) π (D) $\frac{\pi}{2}$

30. From an external point P, pair of tangent lines are drawn to the parabola, $y^2 = 4x$. If θ_1 & θ_2 are the inclinations of these tangents with the axis of x such that, $\theta_1 + \theta_2 = \frac{\pi}{4}$, then the locus of P is:

(A) $x - y + 1 = 0$ (B) $x + y - 1 = 0$ (C) $x - y - 1 = 0$ (D) $x + y + 1 = 0$

31. Locus of the point of intersection of the normals at the ends of parallel chords of gradient m of the parabola $y^2 = 4ax$ is:
(A) $2xm^2 - ym^3 = 4a(2 + m^2)$ (B) $2xm^2 + ym^3 = 4a(2 + m^2)$
(C) $2xm + ym^2 = 4a(2 + m)$ (D) $2xm^2 - ym^3 = 4a(2 - m^2)$

32. The equation of the other normal to the parabola $y^2 = 4ax$ which passes through the intersection of those at $(4a, -4a)$ & $(9a, -6a)$ is:
(A) $5x - y + 115a = 0$ (B) $5x + y - 135a = 0$ (C) $5x - y - 115a = 0$ (D) $5x + y + 115 = 0$

33. The point(s) on the parabola $y^2 = 4x$ which are closest to the circle,
$x^2 + y^2 - 24y + 128 = 0$ is/are:

(A) (0, 0) (B) $\left(2, 2\sqrt{2}\right)$ (C) (4, 4) (D) none

34. If P_1Q_1 and P_2Q_2 are two focal chords of the parabola $y^2 = 4ax$, then the chords P_1P_2 and Q_1Q_2 intersect on the
(A) directrix (B) axis (C) tangent at the vertex (D) none of these

35. If $x + y = k$, is the normal to $y^2 = 12x$, then k is **[IIT - 2000]**
(A) 3 (B) 9 (C) –9 (D) –3

36. The equation of the common tangent touching the circle $(x-3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$ above the x-axis is **[IIT - 2001]**

(A) $\sqrt{3}y = 3x + 1$ (B) $\sqrt{3}y = -(x + 3)$ (C) $\sqrt{3}y = x + 3$ (D) $\sqrt{3}y = -(3x + 1)$

37. The focal chord to $y^2 = 16x$ is tangent to $(x-6)^2 + y^2 = 2$, then the possible values of the slope of this chord are: **[IIT - 2003]**
(A) {– 1, 1} (B) {– 2, 2} (C) {– 2, 1/2} (D) {2, – 1/2}

38. The normal drawn at a point $(at_1^2, -2at_1)$ of the parabola $y^2 = 4ax$ meets it again in the point $(at_2^2, 2at_2)$, then **[IIT - 2003]**

(A) $t_2 = t_1 + \frac{2}{t_1}$ (B) $t_2 = t_1 - \frac{2}{t_1}$ (C) $t_2 = -t_1 + \frac{2}{t_1}$ (D) $t_2 - t_1 - \frac{2}{t_1}$

39. The angle between the tangents drawn from the point (1, 4) to the parabola $y^2 = 4x$ is **[IIT - 2004]**

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

40. Let P be the point (1, 0) and Q a point of the locus $y^2 = 8x$. The locus of mid point of PQ is **[IIT - 2005]**
(A) $x^2 + 4y + 2 = 0$ (B) $x^2 - 4y + 2 = 0$ (C) $y^2 - 4x + 2 = 0$ (D) $y^2 + 4x + 2 = 0$

41. A parabola has its vertex and focus in the first quadrant and axis along the line y = x. If the distances of the vertex and focus from the origin are respectively $\sqrt{2}$ and $2\sqrt{2}$, then an equation of the parabola is **[IIT - 2006]**
(A) $(x + y)^2 = x - y + 2$ (B) $(x - y)^2 = x + y - 2$
(C) $(x - y)^2 = 8(x + y - 2)$ (D) $(x + y)^2 = 8(x - y + 2)$

Comprehension **[IIT - 2006]**

Let ABCD be a square of side length 2 units. C_2 is the circle through vertices A, B, C, D and C_1 is the circle touching all the sides of the square ABCD. L is a line through A.

42. If P is a point on C_1 and Q in another point on C_2, $\frac{PA^2+PB^2+PC^2+PD^2}{QA^2+QB^2+QC^2+QD^2}$ is equal to **[IIT - 2006]**

(A) 0.75 (B) 1.25 (C) 1 (D) 0.5

43. A circle touch the line L and the circle C_1 externally such that both the circles are on the same side of the line, then the locus of centre of the circle is **[IIT - 2006]**

(A) ellipse (B) hyperbola (C) parabola (D) parts of straight line

44. A line M through A is drawn parallel to BD. Point S moves such that its distances from the line BD and the vertex A are equal. If locus of S cuts M at T_2 and T_3 and AC at T_1, then area of $\Delta T_1T_2T_3$ is **[IIT - 2006)]**

(A) $\frac{1}{2}$ sq. units (B) $\frac{2}{3}$ sq. units (C) 1 sq. units (D) 2 sq. units

Part : (B) May have more than one options correct

45. If one end of a focal chord of the parabola $y^2 = 4x$ is (1, 2), the other end lies on

(A) $x^2 y + 2 = 0$ (B) $xy + 2 = 0$ (C) $xy - 2 = 0$ (D) $x^2 + xy - y - 1 = 0$

46. The tangents at the extremities of a focal chord of a parabola

(A) are perpendicular (B) are parallel

(C) intersect on the directrix (D) intersect at the vertex

47. If from a variable point 'P' pair of perpendicular tangents PA and PB are drawn to any parabola then

(A) P lies on directrix of parabola (B) chord of contact AB passes through focus

(C) chord of contact AB passes through of fixed point

(D) P lies on director circle

48. A normal chord of the parabola subtending a right angle at the vertex makes an acute angle θ with the x – axis, then θ =

(A) arc tan 2 (B) arc sec $\sqrt{3}$ (C) arc cot $\sqrt{2}$ (D) $\frac{\pi}{2}$ – arc cot $\sqrt{2}$

49. Variable chords of the parabola $y^2 = 4ax$ subtend a right angle at the vertex. Then:

(A) locus of the feet of the perpendiculars from the vertex on these chords is a circle

(B) locus of the middle points of the chords is a parabola

(C) variable chords passes through a fixed point on the axis of the parabola (D) none of these

50. Two parabolas have the same focus. If their directrices are the x – axis & the y – axis respectively, then the slope of their common chord is:

(A) 1 (B) – 1 (C) 4/3 (D) 3/4

51. P is a point on the parabola $y^2 = 4ax$ ($a > 0$) whose vertex is A. PA is produced to meet the directrix in D and M is the foot of the perpendicular from P on the directrix. If a circle is described on MD as a diameter then it intersects the x–axis at a point whose co–ordinates are:

(A) (– 3a, 0) (B) (– a, 0) (C) (– 2a, 0) (D) (a, 0)

EXERCISE–11

1. Find the vertex, axis, focus, directrix, latusrectum of the parabola $x^2 + 2y - 3x + 5 = 0$.

2. Find the set of values of α in the interval $[\pi/2, 3\pi/2]$, for which the point $(\sin\alpha, \cos\alpha)$ does not lie outside the parabola $2y^2 + x - 2 = 0$.

3. Two perpendicular chords are drawn from the origin 'O' to the parabola $y = x^2$, which meet the parabola at P and Q Rectangle POQR is completed. Find the locus of vertex R.

4. Find the equation of tangent & normal at the ends of the latus rectum of the parabola $y^2 = 4a(x - a)$.

5. Prove that the straight line $\ell x + my + n = 0$ touches the parabola $y^2 = 4ax$ if $\ell n = am^2$.

6. If tangent at P and Q to the parabola $y^2 = 4ax$ intersect at R then prove that mid point of R and M lies on the parabola, where M is the mid point of P and Q.

7. Find the equation of normal to the parabola $x^2 = 4y$ at (9, 6).

8. Find the equation of the chord of $y^2 = 8x$ which is bisected at (2, – 3)

9. Find the locus of the mid-points of the chords of the parabola $y^2 = 4ax$ which subtend a right angle at the vertex of the parabola.

10. Find the equation of the circle which passes through the focus of the parabola $x^2 = 4y$ & touches it at the point (6, 9).

11. Prove that the normals at the points, where the straight line $\ell x + my = 1$ meets the parabola $y^2 = 4ax$, meet on the normal at the point $\left(\frac{4am^2}{\ell^2}, \frac{4am}{\ell}\right)$ of the parabola.

12. If the normals at three points P, Q, and R on parabola $y^2 = 4ax$ meet in a point O and S be the focus, prove that $SP \cdot SQ \cdot SR = a \cdot SO^2$.

13. Show that the locus of the point of intersection of the tangents to $y^2 = 4ax$ which intercept a constant length d on the directrix is $(y^2 - 4ax)(x + a)^2 = d^2 x^2$.

14. Show that the distance between a tangent to the parabola $y^2 = 4ax$ and the parallel normal is $a \sec^2\theta \operatorname{cosec}\theta$, where θ is the inclination of the either with the axis of the parabola.

15. P and Q are the point of contact of the tangents drawn from a point R to the parabola $y^2 = 4ax$. If PQ be a normal to the parabola at P, prove that PR is bisected by the directrix.

16. A circle is described whose centre is the vertex and whose diameter is three-quarters of the latus rectum of the parabola $y^2 = 4ax$. If PQ is the common chord of the circle and the parabola and L_1L_2 is

the latus rectum, then prove that the area of the trapezium PL_1L_2Q is $\left(\frac{2+\sqrt{2}}{2}\right)a^2$.

17. If the normals from any point to the parabola $x^2 = 4y$ cuts the line $y = 2$ in points whose abscissa are in A.P., then prove that slopes of the tangents at the 3 conormal points are in GP.

18. Prove that the length of the intercept on the normal at the point $(at^2, 2at)$ made by the circle which is described on the focal distance of the given point as diameter is $a\sqrt{1+t^2}$.

19. A parabola is drawn to pass through A and B, the ends of a diameter of a given circle of radius a, and to have as directrix a tangent to a concentric circle of radius b; then axes being AB and a perpendicular diameter, prove that the locus of the focus of the parabola is $\frac{x^2}{b^2} + \frac{y^2}{b^2-a^2} = 1$

20. PNP′ is a double ordinate of the parabola then prove that the locus of the point of intersection of the normal at P and the straight line through P′ parallel to the axis is the equal parabola $y^2 = 4a(x - 4a)$.

21. Find the locus of the point of intersection of those normals to the parabola $x^2 = 8y$ which are at right angles to each other. **[IIT - 1997]**

22. Let C_1 and C_2 be respectively, the parabolas $x^2 = y - 1$ and $y^2 = x - 1$. Let P be any point on C_1 and Q be any point on C_2. Let P_1 and Q_1 be the reflections of P and Q, respectively, with respect to the line $y = x$. Prove that P_1 lies on C_2, Q_1 lies on C_1 and $PQ \geq \min\{PP_1, QQ_1\}$. Hence or otherwise determine points P_0 and Q_0 on the parabolas C_1 and C_2 respectively such that $P_0Q_0 \leq PQ$ for all pairs of points (P, Q) with P on C_1 and Q on C_2. **[IIT - 2000]**

23. Normals are drawn from the point P with slopes m_1, m_2, m_3 to the parabola $y^2 = 4x$. If locus of P with $m_2 m_2 = \alpha$ is a part of the parabola itself then find α. **[IIT - 2003]**

Answers

EXERCISE–10

1. C **2.** A **3.** A **4.** A **5.** A **6.** D **7.** B

8. C **9.** C **10.** A **11.** A **12.** D **13.** B **14.** D

15. D **16.** B **17.** B **18.** C **19.** A **20.** C **21.** C

22. B **23.** B **24.** C **25.** C **26.** A **27.** C **28.** A

29. A **30.** C **31.** A **32.** B **33.** C **34.** A **35.** B

36. C **37.** A **38.** A **39.** B **40.** C **41.** C **42.** A

43. C **44.** C **45.** ABD **46.** AC **47.** ABCD

48. BD **49.** ABC **50.** AB **51.** AD

EXERCISE–11

1. vertex $\equiv \left(\frac{3}{2}, -\frac{29}{8}\right)$, focus $\left(\frac{3}{2}, -\frac{33}{8}\right)$

axis $x = 3$, directrix $y = -\frac{29}{3}$. Latus rectum = 2.

2. $\alpha \in [\pi/2, 5\pi/6] \cup [\pi, 3\pi/2]$ **3.** $y^2 = x - 2$

4. Tangent $y = x$, $y = -x$,
Normal $x + y = 4a$, $x - y = 4a$

7. $2x + 9y = 72$ **8.** $4x + 3y + 1 = 0$

9. $y^2 - 2ax + 8a^2 = 0$

10. $x^2 + y^2 + 18x - 28y + 27 = 0$

21. $x^2 - 2y + 12 = 0$ **23.** $\alpha = 2$

EXERCISE–12

Part : (A) Only one correct option

1. The eccentricity of the ellipse $4x^2 + 9y^2 + 8x + 36y + 4 = 0$ is

(A) $\frac{5}{6}$ (B) $\frac{3}{5}$ (C) $\frac{\sqrt{2}}{3}$ (D) $\frac{\sqrt{5}}{3}$

2. The equation of the ellipse with its centre at (1, 2), focus at (6, 2) and passing through the point (4, 6) is

(A) $\frac{(x-1)^2}{45}+\frac{(y-2)^2}{20}=1$ (B) $\frac{(x-1)^2}{20}+\frac{(y-2)^2}{45}=1$

(C) $\frac{(x-1)^2}{25}+\frac{(y-2)^2}{16}=1$ (D) $\frac{(x-1)^2}{16}+\frac{(y-2)^2}{25}=1$

3. The eccentricity of the ellipse which meets the straight line $\frac{x}{7}+\frac{y}{2}=1$ on the axis of x and the straight line $\frac{x}{3}-\frac{y}{5}=1$ on the axis of y and whose axes lie along the axes of coordinates, is

(A) $\frac{3\sqrt{2}}{7}$ (B) $\frac{2\sqrt{6}}{7}$ (C) $\frac{\sqrt{3}}{7}$ (D) none of these

4. The curve represented by $x = 3(\cos t + \sin t)$, $y = 4(\cos t - \sin t)$, is
(A) ellipse (B) parabola (C) hyperbola (D) circle

5. Minimum area of the triangle by any tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with the coordinate axes is

(A) $\frac{a^2+b^2}{2}$ (B) $\frac{(a+b)^2}{2}$ (C) ab (D) $\frac{(a-b)^2}{2}$

6. A circle has the same centre as an ellipse & passes through the focii F_1 & F_2 of the ellipse, such that the two curves intersect in 4 points. Let 'P' be any one of their point of intersection. If the major axis of the ellipse is 17 & the area of the triangle PF_1F_2 is 30, then the distance between the focii is :

(A) 11 (B) 12 (C) 13 (D) 15

7. Q is a point on the auxiliary circle corresponding to the point P of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. If T is the foot of the perpendicular dropped from the focus S onto the tangent to the auxiliaryy circle at Q then the Δ SPT is :
(A) isosceles (B) equilateral (C) right angled (D) right isosceles

8. $x - 2y + 4 = 0$ is a common tangent to $y^2 = 4x$ & $\frac{x^2}{4}+\frac{y^2}{b^2}=1$. Then the value of 'b' and the other common tangent are given by :

(A) $b = \sqrt{3}$; $x + 2y + 4 = 0$ (B) $b = 3$; $x + 2y + 4 = 0$

(C) $b = \sqrt{3}$; $x + 2y - 4 = 0$ (D) $b = \sqrt{3}$; $x - 2y - 4 = 0$

9. The locus of point of intersection of tangents to an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at the points whose the sum of eccentric angles is constant, is :
(A) a hyperbola (B) an ellipse (C) a circle (D) a straight line

10. A tangent having slope of $-\frac{4}{3}$ to the ellipse $\frac{x^2}{18}+\frac{y^2}{32}=1$ intersects the major & minor axes in points A & B respectively. If C is the centre of the ellipse, then the area of the triangle ABC is :
(A) 12 sq. units (B) 24 sq. units (C) 36 sq. units (D) 48 sq. units

11. The normal at a variable point P on an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ of eccentricity 'e' meets the axes of the ellipse in Q and R then the locus of the mid-point of QR is a conic with an eccentricity e′ such that :

(A) e′ is independent of e (B) e′ = 1
(C) e′ = e (D) e′ = 1/e

12. $y = mx + c$ is a normal to the ellipse, $\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$, if c^2 is equal to :

(A) $\frac{(a^2-b^2)^2}{a^2m^2+b^2}$ (B) $\frac{(a^2-b^2)^2}{a^2m^2}$ (C) $\frac{(a^2-b^2)^2m^2}{a^2+b^2m^2}$ (D) $\frac{(a^2-b^2)^2m^2}{a^2m^2+b^2}$

13. An arc of a bridge is semi-elliptical with major axis horizontal. The length of the base is 9 meter and the highest part of the bridge is 3 meter from the horizontal. The best approximation of the Pillar 2 meter from the centre of the base is :
(A) 11/4 m (B) 8/3 m (C) 7/2 m (D) 2 m

14. Point 'O' is the centre of the ellipse with major axis AB & minor axis CD. Point F is one focus of the ellipse. If OF = 6 & the diameter of the inscribed circle of triangle OCF is 2, then the product (AB) (CD) is
(A) 64 (B) 12 (C) 65 (D) 3

15. An ellipse is such that the length of the latus rectum is equal to the sum of the lengths of its semi principal axes. Then:
(A) Ellipse bulges to a circle
(B) Ellipse becomes a line segment between the two foci
(C) Ellipse becomes a parabola
(D) none of these

16. A line of fixed length (a + b) moves so that its ends are always on two fixed perpendicular straight lines. The locus of the point which divided this line into portions of lengths a & b is:
(A) an ellipse (B) an hyperbola (C) a circle (D) none of these

17. The line $2x + y = 3$ cuts the ellipse $4x^2 + y^2 = 5$ at P and Q. If θ be the angle between the normals at these points, then $\tan\theta$ =
(A) 1/2 (B) 3/4 (C) 3/5 (D) 5

18. The focal chord of $y^2 = 16x$ is tangent to $(x-6)^2 + y^2 = 2$, then the possible values of the slope of this chord are **[IIT – 2003]**

(A) {– 1, 1} (B) {– 2, 2} (C) $\left\{-2, \frac{1}{2}\right\}$ (D) $\left\{2, -\frac{1}{2}\right\}$

19. A tangent is drawn to ellipse $x^2 + 2y^2 = 2$. Then the locus of mid point of portion of the tangent intercepted between coordinate axes. **[IIT - 2004]**

(A) $\frac{1}{2x^2}+\frac{1}{4y^2} = 1$ (B) $\frac{1}{4x^2}+\frac{1}{2y^2} = 1$ (C) $\frac{x^2}{2}+\frac{y^2}{4}=1$ (D) $\frac{x^2}{4}+\frac{y^2}{2}=1$

20. The locus of mid point of the intercept of the tangent drawn from an external point to the ellipse $x^2 + 2y^2 = 2$ between the coordinate axes, is **[IIT - 2004]**

(A) $\frac{1}{x^2} + \frac{1}{2y^2} = 1$ (B) $\frac{1}{4x^2} + \frac{1}{2y^2} = 1$ (C) $\frac{1}{2x^2} + \frac{1}{4y^2} = 1$ (D) $\frac{1}{2x^2}+\frac{1}{y^2} =1$

21. An ellipse has OB as semi-minor axis, F and F′ its foci and the angle FBF′ is a right angle. Then, the eccentricity of the ellipse is **[IIT - 2005]**

(A) $\frac{1}{4}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

Part : (B) May have more than one options correct

22. The tangent at any point 'P' on the standard ellipse with focii as S & S′ meets the tangents at the vertices A & A′ in the points V & V′, then :
(A) (AV) (A′V′) = b^2 (B) (AV) (A′V′) = a^2
(C) $\angle V'SV = 90^\circ$ (D) V′S′VS is a cyclic quadrilateral

23. Identify the statements which are True.
(A) the equation of the director circle of the ellipse, $5x^2 + 9y^2 = 45$ is $x^2 + y^2 = 14$.
(B) the sum of the focal distances of the point $(0, 6)$ on the ellipse $\frac{x^2}{25}+\frac{y^2}{36} = 1$ is 10.
(C) the point of intersection of any tangent to a parabola & the perpendicular to it from the focus lies on the tangent at the vertex.
(D) the line through focus and $(at^2_1, 2at_1)$ on $y^2 = 4ax$, meets it again in the point $(at^2_2, 2at_2)$ iff $t_1t_2 = -1$.

24. The Cartesian equation of the curve whose parametric equation is $x = 2t - 3$ and $y = 4t^2 - 1$ is given by
(A) $(x + 3)^2 - y - 1 = 0$ (B) $x^2 + 6x - y + 8 = 0$
(C) $(y + 1)^2 + x + 3 = 0$ (D) $y^2 + 6x - 2y + 4 = 0$

25. If P is a point of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, whose focii are S and S′. Let $\angle PSS' = \alpha$ and $\angle PS'S = \beta$, then

(A) PS + PS′ = 2a, if a > b

(B) PS + PS′ = 2b, if a < b

(C) $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{1-e}{1+e}$

(D) $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{\sqrt{a^2-b^2}}{b^2}$ $[a - \sqrt{a^2-b^2}]$ when a > b

26. If the distance between the focii of an ellipse is equal to the length of its latus rectum, the eccentricity of the ellipse is :

(A) $\frac{\sqrt{5}+1}{2}$ (B) $\frac{\sqrt{5}-1}{2}$ (C) $\frac{\sqrt{5}-2}{2}$ (D) $\frac{2}{\sqrt{5}+1}$

EXERCISE–13

1. Let use consider an ellipse whose major and minor axis are 3x + 4y – 7 = 0 and 4x – 3y – 1 = 0 respectively 'P' be a variable point on the ellipse at any instance, it is given that distance of 'P' from major and minor axis are 4 and 5 respectively. It is also given that maximum distance of 'P' from minor axis is $5\sqrt{2}$, then find its eccentricity.

2. Prove that the area of the triangle formed by the three points on an ellipse, whose eccentric angle are θ, ϕ, and ψ, is 2 ab $\sin\frac{\phi-\psi}{2}\sin\frac{\psi-\theta}{2}\sin\frac{\theta-\phi}{2}$.

3. Find the equation of tangents to the ellipse $\frac{x^2}{50} + \frac{y^2}{32} = 1$ which passes through a point (15, – 4).

4. If 'P' be a moving point on the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ in such a way that tangent at 'P' intersect $x = \frac{25}{3}$ at Q then circle on PQ as diameter passes through a fixed point. Find that fixed point.

5. Any tangent to an ellipse is cut by the tangents at the ends of major axis in the points T and T′. Prove that the circle, whose diameter is T T′ will pass through the foci of the ellipse.

6. If 3x + 4y = 12 intersect the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ at P and Q, then find the point of intersection of tangents at P and Q.

7. Find the equation of the largest circle with centre (1, 0) that can be inscribed in the ellipse $x^2 + 4y^2 = 16$.

8. If P is a variable point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ whose foci are S and S′, then prove that the locus of the incentre of $\Delta PSS'$ is an ellipse whose eccentricity is $\sqrt{\frac{2e}{1+e'}}$ where e is the eccentricity of the given ellipse.

9. The tangent at a point P (a cosθ, b sinθ) of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, meets its auxiliary circle in two points, the chord joining which subtends a right angle at the centre. Show that the eccentricity of the ellipse is $(1 + \sin^2\theta)^{-1/2}$.

10. A circle of radius r is concentric with the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Prove that the common tangent is inclined to the major axis at an angle $\tan^{-1}\sqrt{\left(\frac{r^2-b^2}{a^2-r^2}\right)}$.

11. 'O' is the origin & also the centre of two concentric circles having radii of the inner & the outer circle as 'a' & 'b' respectively. A line OPQ is drawn to cut the inner circle in P & the outer circle in Q. PR is drawn parallel to the y– axis & QR is drawn parallel to the x– axis. Prove that the locus of R is an ellipse touching the two circles. If the focii of this ellipse lie on the inner circle, find the ratio of inner : outer radii & find also the eccentricity of the ellipse.

12. If any two chords be drawn through two points on the major axis of an ellipse equidistant from the centre, show that $\tan\frac{\alpha}{2}.\tan\frac{\beta}{2}.\tan\frac{\gamma}{2}.\tan\frac{\delta}{2}=1$, where $\alpha, \beta, \gamma, \delta$ are the eccentric angles of the extremities of the chords.

13. The tangent at a point P on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ intersects the major axis in T & N is the foot of the perpendicular from P to the same axis. Show that the circle on NT as diameter intersects the auxiliary circle orthogonally.

14. Show that the equation of the tangents to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at the points of intersection with the line, $px + qy + 1 = 0$ is $\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}-1\right)(p^2a^2 + q^2b^2 - 1) = (px + qy + 1)^2$.

15. Common tangents are drawn to the parabola $y^2 = 4x$ & the ellipse $3x^2 + 8y^2 = 48$ touching the parabola at A & B and the ellipse at C & D. Find the area of the quadrilateral.

16. A tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meets the ellipse $\frac{x^2}{a}+\frac{y^2}{b}=a+b$ in the points P and Q; prove that the tangents at P and Q are at right angles.

17. Let P be a point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ for which the area of the ΔPON is the maximum where O is the origin and N is the foot of the perpendicular from O to the tangent at P. Find the maximum area and eccentric angle of point P.

18. Find the equation of the largest circle with centre (1, 0) that can be inscribed in the ellipse $x^2 + 4y^2 = 16$. **[IIT - 1999]**

19. Let P be point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $0 < b < a$. Let the line parallel to y–axis passing through P meet the circle $x^2 + y^2 = a^2$ at the point Q such that P and Q are on the same side of x–axis. For two positive real numbers r and s. Find the locus of the point R on PQ such that PR : RQ = r : s as P varies over the ellipse. **[IIT - 2001]**

20. Prove that in an ellipse, the perpendicular from a focus upon any tangent and the line joining the centre of the ellipse to the point of contact meet on the corresponding directrix. **[IIT - 2002]**

Answers

EXERCISE–12

1. D **2.** A **3.** B **4.** A **5.** C **6.** C **7.** A

8. A **9.** D **10.** B **11.** C **12.** C **13.** B **14.** C

15. A **16.** A **17.** B **18.** A **19.** A **20.** B **21.** C

22. ACD **23.** ACD **24.** AB **25.** ABD **26.** BD

EXERCISE–13

1. $\left(e=\frac{3}{5}\right)$ **3.** $4x + 5y = 40$, $4x - 35y = 200$.

4. (3, 0) **6.** $\left(\frac{25}{4},\frac{16}{3}\right)$ **11.** $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

18. $(x - 1)^2 + y^2 = \frac{11}{3}$ **19.** $\frac{x^2}{a^2}+\frac{y^2(r+s)^2}{(ra+sb)^2}=1$

EXERCISE–14

Part : (A) Only one correct option

1. An ellipse and a hyperbola have the same centre origin, the same foci and the minor-axis of the one is the same as the conjugate axis of the other. If e_1, e_2 be their eccentricities respectively, then $\frac{1}{e_1^2}+\frac{1}{e_2^2}=$

(A) 1 (B) 2 (C) 4 (D) none

2. The line $5x + 12y = 9$ touches the hyperbola $x^2 – 9y^2 = 9$ at the point

(A) $(–5, 4/3)$ (B) $(5, –4/3)$ (C) $(3, –1/2)$ (D) none of these

3. If the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{b^2}=1$ & the hyperbola $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ coincide then the value of b^2 is :

(A) 4 (B) 9 (C) 16 (D) none

4. The tangents from $(1, 2\sqrt{2})$ to the hyperbola $16x^2 – 25y^2 = 400$ include between them an angle equal to:

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

5. If $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$ and $S(x_4, y_4)$ are four concyclic points on the rectangular hyperbola $xy = c^2$, the coordinates of orthocentre of the ΔPQR are

(A) (x_4, y_4) (B) $(x_4, – y_4)$ (C) $(–x_4, – x_4)$ (D) $(– x_4, – y_4)$

6. The asymptotes of the hyperbola $xy = hx + ky$ are :

(A) $x – k = 0$ & $y – h = 0$ (B) $x + h = 0$ & $y + k = 0$
(C) $x – k = 0$ & $y + h = 0$ (D) $x + k = 0$ & $y – h = 0$

7. The combined equation of the asymptotes of the hyperbola $2x^2 + 5xy + 2y^2 + 4x + 5y = 0$ is

(A) $2x^2 + 5xy + 2y^2 + 4x + 5y + 2 = 0$ (B) $2x^2 + 5xy + 2y^2 + 4x + 5y – 2 = 0$
(C) $2x^2 + 5xy + 2y^2 = 0$ (D) none of these

8. If the hyperbolas, $x^2 + 3xy + 2y^2 + 2x + 3y + 2 = 0$ and $x^2 + 3xy + 2y^2 + 2x + 3y + c = 0$ are conjugate of each other, then the value of 'c' is equal to :

(A) – 2 (B) 4 (C) 0 (D) 1

9. P is a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, N is the foot of the perpendicular from P on the transverse axis. The tangent to the hyperbola at P meets the transverse axis at T. If O is the centre of the hyperbola, then OT. ON is equal to :

(A) e^2 (B) a^2 (C) b^2 (D) b^2/a^2

10. The locus of the foot of the perpendicular from the centre of the hyperbola $xy = c^2$ on a variable tangent is :

(A) $(x^2 – y^2)^2 = 4c^2xy$ (B) $(x^2 + y^2)^2 = 2c^2xy$ (C) $(x^2 + y^2) = 4x^2xy$ (D) $(x^2 + y^2)^2 = 4c^2xy$

11. If the chords of contact of tangents from two points (x_1, y_1) and (x_2, y_2) to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ are at right angles, then $\frac{x_1x_2}{y_1y_2}$ is equal to

(A) $-\frac{a^2}{b^2}$ (B) $-\frac{b^2}{a^2}$ (C) $-\frac{b^4}{a^4}$ (D) $-\frac{a^4}{b^4}$

12. The equations of the transverse and conjugate axes of a hyperbola are respectively $x + 2y – 3 = 0$, $2x – y + 4 = 0$, and their respective lengths are $\sqrt{2}$ and $2/\sqrt{3}$. The equation of the hyperbola is

(A) $\frac{2}{5}(x + 2y – 3)^2 – \frac{3}{5}(2x – y + 4)^2 = 1$ (B) $\frac{2}{5}(2x – y + 4)^2 – \frac{3}{5}(x + 2y – 3)^2 = 1$
(C) $2(2x – y + 4)^2 – 3(x + 2y – 3)^2 = 1$ (D) $2(x + 2y – 3)^2 – 3(2x – y + 4)^2 = 1$

13. The chord PQ of the rectangular hyperbola $xy = a^2$ meets the x-axis at A; C is the mid point of PQ & 'O' is the origin. Then the Δ ACO is :

(A) equilateral (B) isosceles (C) right angled (D) right isosceles.

14. The number those triangles that can be inscribed in the rectangular hyperbola $xy= c^2$ whose all sides touch the parabola $y^2 = 4ax$ is :

(A) 0 (B) 1 (C) 2 (D) Infinite

15. The number of points from where a pair of perpendicular tangents can be drawn to the hyperbola,

$x^2 \sec^2 \alpha - y^2 \operatorname{cosec}^2 \alpha = 1$, $\alpha \in (0, \pi/4)$, is :
(A) 0 (B) 1 (C) 2 (D) infinite

16. If hyperbola $\frac{x^2}{b^2} - \frac{y^2}{a^2} = 1$ passes through the focus of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then eccentricity of hyperbola is

(A) $\sqrt{2}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\sqrt{3}$ (D) None of these

17. The transverse axis of a hyperbola is of length 2a and a vertex divides the segment of the axis between the centre and the corresponding focus in the ratio 2 : 1, the equation of the hyperbola is :
(A) $4x^2 - 5y^2 = 4a^2$ (B) $4x^2 - 5y^2 = 5a^2$ (C) $5x^2 - 4y^2 = 4a^2$ (D) $5x^2 - 4y^2 = 5a^2$

18. If AB is a double ordinate of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that ΔOAB (O is the origin) is an equilateral triangle, then the eccentricity 'e' of the hyperbola satisfies

(A) $e > \sqrt{3}$ (B) $1 < e < \frac{2}{\sqrt{3}}$ (C) $e = \frac{2}{\sqrt{3}}$ (D) $e > \frac{2}{\sqrt{3}}$

19. If $x \cos \alpha + y \sin \alpha = p$, a variable chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{2a^2} = 1$ subtends a right angle at the centre of the hyperbola, then the chords touch a fixed circle whose radius is equal to

(A) $\sqrt{2}\,a$ (B) $\sqrt{3}\,a$ (C) $2\,a$ (D) $\sqrt{5}\,a$

20. Two conics $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $x^2 = -\frac{1}{b}y$ intersect if

(A) $0 < b \le \frac{1}{2}$ (B) $0 < a < \frac{1}{2}$ (C) $a^2 < b^2$ (D) $a^2 > b^2$

21. Number of points on hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ from where mutually perpendicular tangents can be drawn to circle $x^2 + y^2 = a^2$ $(a > b)$ is
(A) 2 (B) 3 (C) infinite (D) 4

22. The normal to the rectangular hyperbola $xy = c^2$ at the point 't_1' meets the curve again at the point 't_2'. The value of $t_1^3 t_2$ is
(A) -1 (B) $-|c|$ (C) $|c|$ (D) 1

23. If the tangent and the normal to a rectangular hyperbola cut off intercepts x_1 and x_2 on one axis and y_1 and y_2 on the other axis, then
(A) $x_1y_1 + x_2y_2 = 0$ (B) $x_1y_2 + x_2y_1 = 0$ (C) $x_1x_2 + y_1y_2 = 0$ (D) none of these

24. If $x = 9$ is the chord of contact of the hyperbola $x^2 - y^2 = 9$, then the equation of the corresponding pair of tangents is **[IIT - 1999]**
(A) $9x^2 - 8y^2 + 18x - 9 = 0$ (B) $9x^2 - 8y^2 + 18x + 9 = 0$
(C) $9x^2 - 8y^2 - 18x - 9 = 0$ (D) $9x^2 - 8y^2 + 18x + 9 = 0$

Part : (B) May have more than one options correct

25. The value of m for which $y = mx + 6$ is a tangent to the hyperbola $\frac{x^2}{100} - \frac{y^2}{49} = 1$ is

(A) $\sqrt{\left(\frac{17}{20}\right)}$ (B) $-\sqrt{\left(\frac{17}{20}\right)}$ (C) $\sqrt{\left(\frac{20}{17}\right)}$ (D) $-\sqrt{\left(\frac{20}{17}\right)}$

26. If $(a \sec \theta, b \tan \theta)$ and $(a \sec\phi, b \tan \phi)$ are the ends of a focal chord of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then $\tan\frac{\theta}{2} \tan\frac{\phi}{2}$ equals to

(A) $\frac{e-1}{e+1}$ (B) $\frac{1-e}{1+e}$ (C) $\frac{1+e}{1-e}$ (D) $\frac{e+1}{e-1}$

27. A common tangent to $9x^2 - 16y^2 = 144$ and $x^2 + y^2 = 9$ is

(A) $y = \frac{3}{\sqrt{7}}x + \frac{15}{\sqrt{7}}$ (B) $y = 3\sqrt{\frac{2}{7}}x + \frac{15}{\sqrt{7}}$

(C) $y = 2\sqrt{\frac{3}{7}}x + 15\sqrt{7}$ (D) $y = 3\sqrt{\frac{2}{7}}x - \frac{15}{\sqrt{7}}$

28. The equation of a hyperbola with co-ordinate axes as principal axes, if the distances of one of its vertices from the foci are 3 & 1 can be :
(A) $3x^2 - y^2 = 3$ (B) $x^2 - 3y^2 + 3 = 0$ (C) $x^2 - 3y^2 - 3 = 0$ (D) none

29. If (5, 12) and (24, 7) are the foci of a conic passing through the origin then the eccentricity of conic is
(A) $\sqrt{386}/12$ (B) $\sqrt{386}/13$ (C) $\sqrt{386}/25$ (D) $\sqrt{386}/38$

30. If the normal at P to the rectangular hyperbola $x^2 - y^2 = 4$ meets the axes in G and g and C is the centre of the hyperbola, then
(A) PG = PC (B) Pg = PC (C) PG = Pg (D) Gg = PC

31. The tangent to the hyperbola, $x^2 - 3y^2 = 3$ at the point $\left(\sqrt{3}, 0\right)$ when associated with two asymptotes constitutes :
(A) isosceles triangle (B) an equilateral triangle
(C) a triangles whose area is $\sqrt{3}$ sq. units (D) a right isosceles triangle.

32. Which of the following equations in parametric form can represent a hyperbolic profile, where 't' is a parameter.
(A) $x = \frac{a}{2}\left(t + \frac{1}{t}\right)$ & $y = \frac{b}{2}\left(t - \frac{1}{t}\right)$ (B) $\frac{tx}{a} - \frac{y}{b} + t = 0$ & $\frac{x}{a} + \frac{ty}{b} - 1 = 0$
(C) $x = e^t + e^{-t}$ & $y = e^t - e^{-t}$ (D) $x^2 - 6 = 2\cos t$ & $y^2 + 2 = 4\cos^2\frac{t}{2}$

33. If a hyperbola passes through the focii of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$. Its transverse and conjugate axes coincide respectively with the major and minor axes of the ellipse and if the product of eccentricities of hyperbola and ellipse is 1, then **[IIT - JEE]**
(A) the equation of hyperbola is $\frac{x^2}{9} - \frac{y^2}{16} = 1$ (B) the equation of hyperbola is $\frac{x^2}{9} - \frac{y^2}{25} = 1$
(C) focus of hyperbola is (5, 0) (D) focus of hyperbola is $(5\sqrt{3}, 0)$

EXERCISE–15

1. For the hyperbola $x^2/100 - y^2/25 = 1$, prove that
(i) eccentricity = $\sqrt{5}/2$
(ii) SA . S′A = 25, where S & S′ are the foci & A is the vertex .

2. Chords of the hyperbola, $x^2 - y^2 = a^2$ touch the parabola, $y^2 = 4ax$. Prove that the locus of their middle points is the curve, $y^2(x - a) = x^3$.

3. Find the asymptotes of the hyperbola $2x^2 - 3xy - 2y^2 + 3x - y + 8 = 0$. Also find the equation to the conjugate hyperbola & the equation of the principal axes of the curve .

4. Given the base of a triangle and the ratio of the tangent of half the base angles. Show that the vertex moves on a hyperbola whose foci are the extremities of the base.

5. If p_1 and p_2 are the perpendiculars from any point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ on its asymptotes, then prove that, $\frac{1}{p_1 p_2} = \frac{1}{a^2} + \frac{1}{b^2}$.

6. If two points P & Q on the hyperbola $x^2/a^2 - y^2/b^2 = 1$ whose centre is C be such that CP is perpendicular to CQ & a < b, then prove that $\frac{1}{CP^2} + \frac{1}{CQ^2} = \frac{1}{a^2} - \frac{1}{b^2}$.

7. If the normal at a point P to the hyperbola $x^2/a^2 - y^2/b^2 = 1$ meets the x–axis at G, show that SG = e . SP, S being the focus of the hyperbola .

8. A transversal cuts the same branch of a hyperbola $x^2/a^2 - y^2/b^2 = 1$ in P, P′ and the asymptotes in Q, Q′. Prove that (i) PQ = P′Q′ & (ii) PQ′ = P′Q

9. If PSP′ & QSQ′ are two perpendicular focal chords of the hyperbola $x^2/a^2 - y^2/b^2 = 1$ then prove that $\frac{1}{\ell(PS)\,.\,\ell(SP')} + \frac{1}{\ell(QS)\,.\,\ell(SQ')}$ is a constant .

10. A line through the origin meets the circle $x^2 + y^2 = a^2$ at P & the hyperbola $x^2 - y^2 = a^2$ at Q . Prove that the locus of the point of intersection of the tangent at P to the circle and the tangent at Q to the hyperbola is curve $a^4(x^2 - a^2) + 4x^2y^4 = 0$.

11. Prove that the part of the tangent at any point of the hyperbola $x^2/a^2 - y^2/b^2 = 1$ intercepted between the point of contact and the transverse axis is a harmonic mean between the lengths of the perpendiculars drawn from the foci on the normal at the same point .

12. Let 'p' be the perpendicular distance from the centre C of the hyperbola $x^2/a^2 - y^2/b^2 = 1$ to the tangent drawn at a point R on the hyperbola . If S & S′ are the two foci of the hyperbola, then show that $(RS + RS')^2 = 4a^2\left(1 + \frac{b^2}{p^2}\right)$.

13. Chords of the hyperbola $x^2/a^2 - y^2/b^2 = 1$ are tangents to the circle drawn on the line joining the foci as diameter . Find the locus of the point of intersection of tangents at the extremities of the chords .

14. A point P divides the focal length of the hyperbola $9x^2 - 16y^2 = 144$ in the ratio S′P : PS = 2 : 3 where S & S′ are the foci of the hyperbola. Through P a straight line is drawn at an angle of 135° to the axis OX. Find the points of intersection of this line with the asymptotes of the hyperbola.

15. The angle between a pair of tangents drawn from a point P to the parabola $y^2 = 4ax$ is 45º. Show that the locus of the point P is a hyperbola. **[IIT - 1998]**

16. Tangents are drawn from any point on the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ to the circle $x^2 + y^2 = 9$. Find the locus of mid-point of the chord of constant. **[IIT - 2005]**

Answers

EXERCISE–14

1.	A	**2.**	B	**3.**	C	**4.**	D	**5.**	D
6.	A	**7.**	A	**8.**	C	**9.**	B	**10.**	D
11.	D	**12.**	B	**13.**	B	**14.**	D	**15.**	D
16.	C	**17.**	D	**18.**	D	**19.**	A	**20.**	B
21.	D	**22.**	A	**23.**	C	**24.**	B	**25.**	AB
26.	BC	**27.**	BD	**28.**	AB	**29.**	AD	**30.**	ABC
31.	BC	**32.**	ACD			**33.**	AC		

EXERCISE–15

3. $x - 2y + 1 = 0$; $2x + y + 1 = 0$;
$2x^2 - 3xy - 2y^2 + 3x - y - 6 = 0$;
$3x - y + 2 = 0$; $x + 3y = 0$

13. $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2 + b^2}$ **14.** $(-4, 3)$ & $\left(-\frac{4}{7}, -\frac{3}{7}\right)$

16. $\frac{x^2}{9} - \frac{y^2}{4} = \left(\frac{x^2 + y^2}{9}\right)^2$

Some questions (Assertion–Reason type) are given below. Each question contains **Statement – 1 (Assertion)** and **Statement – 2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. So select the correct choice :

Choices are :

(A) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is** a correct explanation for **Statement – 1**.
(B) **Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is NOT** a correct explanation for **Statement – 1**.
(C) **Statement – 1** is True, **Statement – 2** is False.
(D) **Statement – 1** is False, **Statement – 2** is True.

PARABOLA

286. **Statement-1 :** Slope of tangents drawn from (4, 10) to parabola $y^2 = 9x$ are $\frac{1}{4}, \frac{9}{4}$.

Statement-2 : Every parabola is symmetric about its directrix.

287. **Statement-1 :** Though $(\lambda, \lambda + 1)$ there can't be more than one normal to the parabola $y^2 = 4x$, if $\lambda < 2$.

Statement-2 : The point $(\lambda, \lambda + 1)$ lies outside the parabola for all $\lambda \neq 1$.

288. **Statement-1 :** If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then k is 9.

Statement-2 : Equation of normal to the parabola $y^2 = 4ax$ is $y - mx + 2am + am^3 = 0$

289. **Statement-1 :** If b, k are the segments of a focal chord of the parabola $y^2 = 4ax$, then k is equal to ab/b-a.

Statement-2 : Latus rectum of the parabola $y^2 = 4ax$ is H.M. between the segments of any focal chord of the parabola

290. **Statement-1 :** Two parabolas $y^2 = 4ax$ and $x^2 = 4ay$ have common tangent $x + y + a = 0$

Statement-2 : $x + y + a = 0$ is common tangent to the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ and point of contacts lie on their respective end points of latus rectum.

291. **Statement-1 :** In parabola $y^2 = 4ax$, the circle drawn taking focal radii as diameter touches y-axis.

Statement-2 : The portion of the tangent intercepted between point of contact and directix subtends 90° angle at focus.

292. **Statement-1 :** The joining points (8, -8) & (1/2, –2), which are lying on parabola $y^2 = 4ax$, pass through focus of parabola.

Statement-2 : Tangents drawn at (8, -8) & (1/2, -2) on the parabola $y^2 = 4ax$ are perpendicular.

293. **Statement-1 :** There are no common tangents between circle $x^2 + y^2 - 4x + 3 = 0$ and parabola $y^2 = 2x$.

Statement-2 : Equation of tangents to the parabola $x^2 = 4ay$ is $x = my + a/m$ where m denotes slope of tangent.

294. **Statement-1 :** Three distinct normals of the parabola $y^2 = 12x$ can pass through a point (h ,0) where $h > 6$.

Statement-2 : If $h > 2a$ then three distinct nroamls can pass through the point (h, 0) to the parabola $y^2 = 4ax$.

295. **Statement-1 :** The normals at the point (4, 4) and $\left(\frac{1}{4}, -1\right)$ of the parabola $y^2 = 4x$ are perpendicular.

Statement-2 : The tangents to the parabola at the and of a focal chord are perpendicular.

296. **Statement-1 :** Through $(\lambda, \lambda + 1)$ there cannot be more than one-normal to the parabola $y^2 = 4x$ if $\lambda < 2$.

Statement-2 : The point $(\lambda, \lambda + 1)$ lines out side the parabola for all $\lambda \neq 1$.

297. **Statement-1 :** Slope of tangents drawn from (4, 10) to parabola $y^2 = 9x$ are 1/4, 9/4

Statement-2 : Every parabola is symmetric about its axis.

298. **Statement-1 :** If a parabola is defined by an equation of the form $y = ax^2 + bx + c$ where a, b, $c \in R$ and $a > 0$, then the parabola must possess a minimum.

Statement-2 : A function defined by an equation of the form $y = ax^2 + bx + c$ where a, b, $c \in R$ and $a \neq 0$, may not have an extremum.

299. **Statement-1 :** The point $(\sin\alpha, \cos\alpha)$ does not lie outside the parabola $2y^2 + x - 2 = 0$ when $\alpha \in \left[\frac{\pi}{2}, \frac{5\pi}{6}\right] \cup \left[\pi, \frac{3\pi}{2}\right]$

Statement-2 : The point (x_1, y_1) lies outside the parabola $y^2 = 4ax$ if $y_1^2 - 4ax_1 > 0$.

300. **Statement-1 :** The line $y = x + 2a$ touches the parabola $y^2 = 4a(x + a)$.

Statement-2 : The line $y = mx + c$ touches $y^2 = 4a(x + a)$ if $c = am + a/m$.

301. **Statement-1 :** If PQ is a focal chord of the parabola $y^2 = 32x$ then minimum length of PQ = 32.

Statement-2 : Latus rectum of a parabola is the shortest focal chord.

302. **Statement-1 :** Through $(\lambda, \lambda + 1)$, there can't be more than one normal to the parabola $y^2 = 4x$ if $\lambda < 2$.

Statement–2 : The point $(\lambda, \lambda + 1)$ lies outside the parabola for all $\lambda \in R \sim \{1\}$.

303. **Statement–1 :** Perpendicular tangents to parabola $y^2 = 8x$ meets on $x + 2 = 0$

Statement–2 : Perpendicular tangents of parabola meets on tangent at the vertex.

304. Let $y^2 = 4ax$ and $x^2 = 4ay$ be two parabolas
Statement-1: The equation of the common tangent to the parabolas is $x + y + a = 0$
Statement-2: Both the parabolas are reflected to each other about the line $y = x$.

305. Let $y^2 = 4a(x + a)$ and $y^2 = 4b(x + b)$ are two parabolas
Statement-1 : Tangents are drawn from the locus of the point are mutually perpendicular
State.-2: The locus of the point from which mutually perpendicular tangents can be drawn to the given comb is $x + y + b = 0$

ELLIPSE

306. Tangents are drawn from the point (-3, 4) to the curve $9x^2 + 16y^2 = 144$.
STATEMENT -1: The tangents are mutually perpendicular.
STATEMENT-2: The locus of the points from which mutually perpendicular tangents can be drawn to the given curve is $x^2 + y^2 = 25$.

307. **Statement–1 :** Circle $x^2 + y^2 = 9$, and the circle $(x - \sqrt{5})(\sqrt{2}x - 3) + y(\sqrt{2}y - 2) = 0$ touches each other internally.
Statement–2 : Circle described on the focal distance as diameter of the ellipse $4x^2 + 9y^2 = 36$ touch the auxiliary circle $x^2 + y^2 = 9$ internally

308. **Statement–1 :** If the tangents from the point $(\lambda, 3)$ to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ are at right angles then λ is equal to ± 2.
Statement–2 : The locus of the point of the intersection of two perpendicular tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, is $x^2 + y^2 = a^2 + b^2$.

309. **Statement–1 :** $x - y - 5 = 0$ is the equation of the tangent to the ellipse $9x^2 + 16y^2 = 144$.
Statement–2 : The equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is of the form $y = mx \pm \sqrt{a^2m^2 + b^2}$.

310. **Statement–1 :** At the most four normals can be drawn from a given point to a given ellipse.
Statement–2 : The standard equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ of an ellipse does not change on changing x by – x and y by – y.

311. **Statement–1 :** The focal distance of the point $\left(4\sqrt{3}, 5\right)$ on the ellipse $25x^2 + 16y^2 = 1600$ will be 7 and 13.
Statement–2 : The radius of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and having its centre at (0, 3) is 5.

312. **Statement-1 :** The least value of the length of the tangents to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ intercepted between the coordinate axes is $a + b$.
Statement-2 : If x_1 and x_2 be any two positive numbers then $\frac{x_1 + x_2}{2} \geq \sqrt{x_1 + x_2}$

313. **Statement-1 :** In an ellipse the sum of the distances between foci is always less than the sum of focal distances of any point on it.
Statement-2 : The eccentricity of any ellipse is less than 1.

314. **Statement-1 :** Any chord of the conic $x^2 + y^2 + xy = 1$, through (0, 0) is bisected at (0, 0)
Statement-2 : The centre of a conic is a point through which every chord is bisected.

315. **Statement-1 :** A tangent of the ellipse $x^2 + 4y^2 = 4$ meets the ellipse $x^2 + 2y^2 = 6$ at P & Q. The angle between the tangents at P and Q of the ellipse $x^2 + 2y^2 = 6$ is $\pi/2$
Statement-2 : If the two tangents from to the ellipse $x^2/a^2 + y^2/b^2 = 1$ are at right angle, then locus of P is the circle $x^2 + y^2 = a^2 + b^2$.

316. **Statement-1 :** The equation of the tangents drawn at the ends of the major axis of the ellipse $9x^2 + 5y^2 - 30y = 0$ is y = 0, y = 7.

Statement-1 : The equation of the tangent drawn at the ends of major axis of the ellipse $x^2/a^2 + y^2/b^2 = 1$ always parallel to y-axis

317. **Statement-1 :** Tangents drawn from the point (3, 4) on to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ will be mutually perpendicular

Statement-2 : The points (3, 4) lies on the circle $x^2 + y^2 = 25$ which is director circle to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$.

318. **Statement-1 :** For ellipse $\frac{x^2}{5}+\frac{y^2}{3}=1$, the product of the perpendicular drawn from focii on any tangent is 3.

Statement-2 : For ellipse $\frac{x^2}{5}+\frac{y^2}{3}=1$, the foot of the perpendiculars drawn from foci on any tangent lies on the circle $x^2 + y^2 = 5$ which is auxiliary circle of the ellipse.

319. **Statement-1 :** If line x + y = 3 is a tangent to an ellipse with foci (4, 3) & (6, y) at the point (1, 2), then y = 17.

Statement-2 : Tangent and normal to the ellipse at any point bisects the angle subtended by foci at that point.

320. **Statement-1 :** Tangents are drawn to the ellipse $\frac{x^2}{4}+\frac{y^2}{2}=1$ at the points, where it is intersected by the line 2x + 3y = 1. Point of intersection of these tangents is (8, 6).

Statement-2 : Equation of chord of contact to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ from an external point is given by $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1=0$

321. **Statement-1 :** In an ellipse the sum of the distances between foci is always less than the sum of focal distances of any point on it.

Statement-2 : The eccentricity of any ellipse is less than 1.

322. **Statement-1 :** The equation $x^2 + 2y^2 + \lambda xy + 2x + 3y + 1 = 0$ can never represent a hyperbola

Statement-2 : The general equation of second degree represent a hyperbola it $h^2 > ab$.

323. **Statement-1 :** The equation of the director circle to the ellipse $4x^2 + 9x^2 = 36$ is $x^2 + y^2 = 13$.

Statement-2 : The locus of the point of intersection of perpendicular tangents to an ellipse is called the director circle.

324. **Statement-1 :** The equation of tangent to the ellipse $4x^2 + 9y^2 = 36$ at the point (3, –2) is $\frac{x}{3}-\frac{y}{2}=1$.

Statement-2 : Tangent at (x_1, y_1) to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $\frac{xx_1}{a^2}-\frac{yy_1}{b^2}=1$

325. **Statement-1 :** The maximum area of ΔPS_1S_2 where S_1, S_2 are foci of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and P is any variable point on it, is abe, where e is eccentricity of the ellipse.

Statement-2 : The coordinates of pare (a sec θ, b tan θ).

326. **Statement-1 :** In an ellipse the sum of the distances between foci is always less than the sum of focal distance of any point on it.

Statement-2 : The eccentricity of ellipse is less than 1.

HYPERBOLA

327. Let $Y = \pm \frac{2}{3}\sqrt{x^2-9}$ $x \in [3, \infty)$ and $Y_1 = \pm \frac{2}{3}\sqrt{x^2-9}$ be $x \in (-\infty, -3]$ two curves.

Statement 1: The number of tangents that can be drawn from $\left(5,-\frac{10}{3}\right)$ to the curve $Y_1 = \pm \frac{2}{3}\sqrt{x^2-9}$ is zero

Statement 2: The point $\left(5, -\frac{10}{3}\right)$ **lies on the curve** $Y = \pm\frac{2}{3}\sqrt{x^2 - 9}$.

328. **Statement–1 :** If (3, 4) is a point of a hyperbola having focus (3, 0) and (λ, 0) and length of the transverse axis being 1 unit then λ can take the value 0 or 3.

Statement–2 : $|S'P - SP| = 2a$, where S and S′ are the two focus 2a = length of the transverse axis and P be any point on the hyperbola.

329. **Statement–1 :** The eccentricity of the hyperbola $9x^2 - 16y^2 - 72x + 96y - 144 = 0$ is $\frac{5}{4}$.

Statement–2 : The eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is equal to $\sqrt{1 + \frac{b^2}{a^2}}$.

330. Let a, b, $\alpha \in R - \{0\}$, where a, b are constants and α is a parameter.

Statement–1 : All the members of the family of hyperbolas $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{\alpha^2}$ have the same pair of asymptotes.

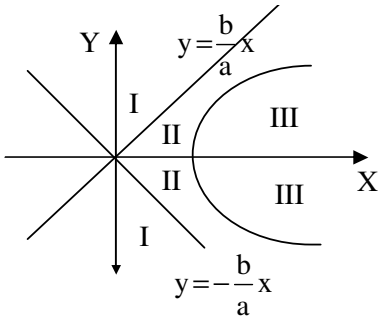
Statement–2 : Change in α, does not change the slopes of the asymptotes of a member of the family $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{\alpha^2}$.

331. **Statement–1 :** The slope of the common tangent between the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $-\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ may be 1 or – 1.

Statement–2 : The locus of the point of inteeersection of lines $\frac{x}{a} - \frac{y}{b} = m$ and $\frac{x}{a} + \frac{y}{b} = \frac{1}{m}$ is a hyperbola (where m is variable and $ab \neq 0$).

332. **Statement–1 :** The equation $x^2 + 2y^2 + \lambda xy + 2x + 3y + 1 = 0$ can never represent a hyperbola.
Statement–2 : The general equation of second degree represents a hyperbola if $h^2 > ab$.

333. **Statement–1** If a point (x_1, y_1) lies in the region II of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, shown in the figure, then $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} < 0$



Statement–2 If $(P(x_1, y_1)$ lies outside the a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} < 1$

334. **Statement–1** Equation of tangents to the hyperbola $2x^2 - 3y^2 = 6$ which is parallel to the line $y = 3x + 4$ is $y = 3x - 5$ and $y = 3x + 5$.
Statement–2 $y = mx + c$ is a tangent to $x^2/a^2 - y^2/b^2 = 1$ if $c^2 = a^2m^2 + b^2$.

335. **Statement–1 :** There can be infinite points from where we can draw two mutually perpendicular tangents on to the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$

Statement–2 : The director circle in case of hyperbola $\frac{x^2}{9}-\frac{y^2}{16}=1$ will not exist because $a^2 < b^2$ and director circle is $x^2 + y^2 = a^2 - b^2$.

336. **Statement–1 :** The average point of all the four intersection points of the rectangular hyperbola $xy = 1$ and circle $x^2 + y^2 = 4$ is origin (0, 0).
Statement–2 : If a rectangular hyperbola and a circle intersect at four points, the average point of all the points of intersection is the mid point of line-joining the two centres.

337. **Statement–1 :** No tangent can be drawn to the hyperbola $\frac{x^2}{2}-\frac{y^2}{1}=1,$ which have slopes greater than $\frac{1}{\sqrt{2}}$

Statement–2 : Line $y = mx + c$ is a tangent to hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If $c^2 = a^2m^2 - b^2$

338. **Statement–1 :** Eccentricity of hyperbola $xy - 3x - 3y = 0$ is 4/3

Statement–2 : Rectangular hyperbola has perpendicular asymptotes and eccentricity = $\sqrt{2}$

339. **Statement–1 :** The equation $x^2 + 2y^2 + \lambda xy + 2x + 3y + 1 = 0$ can never represent a hyperbola
Statement–2 : The general equation of second degree represent a hyperbola it $h^2 > ab$.

340. **Statement–1 :** The combined equation of both the axes of the hyperbola $xy = c^2$ is $x^2 - y^2 = 0$.
Statement–2 : Combined equation of axes of hyperbola is the combined equation of angle bisectors of the asymptotes of the hyperbola.

341. **Statement–1 :** The point (7, –3) lies inside the hyperbola $9x^2 - 4y^2 = 36$ where as the point (2, 7) lies outside this.

Statement–2 : The point (x_1, y_1) lies outside, on or inside the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ according as $\frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}-1$ < or = or > 0

342. **Statement–1 :** The equation of the chord of contact of tangents drawn from the point (2, –1) to the hyperbola $16x^2 - 9y^2 = 144$ is $32x + 9y = 144$.

Statement–2 : Pair of tangents drawn from (x_1, y_1) to $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $SS_1 = T^2$ $S=\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ $S_1=\frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}-1$

343. **Statement–1 :** If PQ and RS are two perpendicular chords of $xy = x^e$, and C be the centre of hyperbola $xy = c^2$. Then product of slopes of CP, CQ, CR and CS is equal to 1.
Statement–2 : Equation of largest circle with centre (1, 0) and lying inside the ellipse $x^2 + 4y^2$ 16 is $3x^2 + 3y^2 - 6x - 8 = 0$.

Answer

286. C **287. B** **288. A** **289. C** **290. B** **291. B** **292. B** **293. C** **294. A** **295. A** **296. B** **297. A** **298. C** **299. B** **300. A**
301. A **302. B** **303. C** **304. B** **305. A** **306. A** **307. A** **308. A** **309. A** **310. B** **311. C** **312. B** **313. A** **314. A** **315. A**
316. C **317. A** **318. B** **319. A** **320. D** **321. A** **322. A** **323. A** **324. C** **325. C** **326. A** **327. A** **328. D** **329. A** **330. A**
331. B **332. D** **333. D** **334. C** **335. D** **336. A** **337. A** **338. D** **339. A** **340. A** **341. A** **342. B** **343. B**

Solution

286. Option (C) is correct.

$y = mx + \frac{a}{m}$

$10 = 4m - 1.\ \frac{9/4}{m}$

$\Rightarrow 16m^2 - 40m + 9 = 0$ $\Rightarrow m_1 = \frac{1}{4}, m_2 = \frac{9}{4}$

Every $m_1 = \frac{1}{4}, m_2 = \frac{9}{4}$ Every parabola is symmetric about its axis.

287. Option (B) is correct
Any normal to $y^2 = 4x$ is

$Y + tx = 2t + t^3$
If this passes through $(\lambda, \lambda + 1)$, we get
$\lambda + 1 + \lambda = 2t + t^3$
$\Rightarrow t^3 + t(2 - \lambda) - \lambda - 1 = 0 = f(t)$ (say)
If $\lambda < 2$, then $f'(t) = 3t^2 + (2 - \lambda) > 0$
$\Rightarrow f(t) = 0$ will have only one real root. So A is true.
Statement 2 is also true b′ coz $(\lambda + 1)^2 > 4\lambda$ is true $\forall\ \lambda \neq 1$. The statement is true but does not follow true statement-2.

288. For the parabola $y^2 = 12x$, equation of a normal with slope -1 is $y = -x - 2.3(-1) - 3(-1)\ 3$
$\Rightarrow x + y = 9, \Rightarrow k = 9$ Ans. (A).

289. $SP = a + at_1^2 = a(1 + t_1^2)$

$SQ = a + a/t_1^2 = = \dfrac{a(1+t_1^2)}{t_1^2}$

$$\frac{1}{SP}+\frac{1}{SQ} = \frac{(1+t_1^2)}{a(1+t_1^2)}=\frac{1}{a}$$

$\dfrac{1}{SP}, \dfrac{1}{2a}, \dfrac{1}{SQ}$ are in A.P.

$\Rightarrow$ 2a is H.M. between SP & SQ

Hence $\dfrac{1}{b}+\dfrac{1}{k}=\dfrac{1}{a} \Rightarrow \dfrac{1}{k}=\dfrac{1}{a}-\dfrac{1}{b}$ $\Rightarrow k = ab/b\text{-}a = \dfrac{b-a}{ab}$ Ans. (C)

290. $y^2 = 4ax$
equation of tangent of slope 'm'

$y = mx + \dfrac{a}{m}$

If it touches $x^2 = 4ay$ then $x^2 = 4a\,(mx + a/m)$

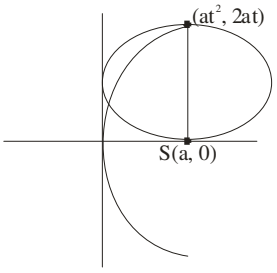
$x^2 - 4amx - \dfrac{4a^2}{m} = 0$ will have equal roots

$D = 0$

$16a^2 m^2 + \dfrac{16a^2}{m} = 0$

$m^3 = -1 \Rightarrow m = -1$
So $y = -x - a \Rightarrow x + y + a = 0$
(a, -2a) & (-2a, a) lies on it 'B' is correct.

291. $(x - a)(x - at^2) + y\ (y - 2at) = 0$
Solve with $x = 0$
$a^2t^2 + y\,(y - 2at) = 0$
$y^2 - 2aty + a^2t^2 = 0$
If it touches y-axis then above quadratic must have equal roots. SO, $D = 0$
$4a^2t^2 - 4a^2t^2 = 0$ which is correct.
'B' is correct.



296. (B) Any normal to the parabola $y^2 = 4x$ is $y + tx = 2t + t^3$
It this passes through $(\lambda, \lambda + 1)$
$\Rightarrow t^3 + t(2 - \lambda) - \lambda - 1 = 0 = f(t)$ say)
$\lambda < 2$ than $f'(t) = 3t^2 + (2 - \lambda) > 0$
$\Rightarrow f(t) = 0$ will have only one real root $\Rightarrow$ A is true
The statement-2 is also true since $(\lambda + 1)^2 > 4\lambda$ is true for all $\lambda \neq 1$. The statement-2 is true but does not follow true statement-2.

297. $y = mx + \dfrac{a}{m}$

$10 = 4m + \dfrac{9/4}{m} \Rightarrow 16m^2 - 40m + 9 = 0$

$m_1 = 1/4,\ m_2 = 9/4$

Every parabola is symmetric about its axis.

298. (C) Statement-1 is true but Statement-2 is false.

299. (B)

If the point $(\sin\alpha, \cos\alpha)$ lies inside or on the parabola $2y^2 + x - 2 = 0$ then $2\cos^2\alpha + \sin\alpha - 2 \le 0$

$\Rightarrow \sin\alpha(2\sin\alpha - 1) \ge 0 \qquad \Rightarrow \sin\alpha \le 0$, or $\sin\alpha \ge \dfrac{1}{2}$.

300. (A) $y = (x + a) + a$ is of the form

$y = m(x + a) + a/m$ where $m = 1$.

Hence the line touches the parabola.

302. Any normal to the parabola $y^2 = 4x$ is $y + xt = 2t + t^3$

If this passes through $(\lambda, \lambda + 1)$. We get $\lambda + 1 + \lambda t = 2t + t^3$.

$\Rightarrow t^3 + t(2 - \lambda) - (\lambda + 1) = 0 = f(t)$ (let)

if $\lambda < 2$, then, $f'(t) = 3t^2 + (2 - \lambda) > 0$

$\Rightarrow f(t) = 0$ will have only one real root.

$\Rightarrow$ statement–I is true. Statement–II is also true since $(\lambda + 1)^2 > 4\lambda$ is true for all $\lambda \in R \sim \{1\}$.

Statement – I is true but does ot follow true statement – II.

Hence (b) is the correct answer.

304. (B) Because the common tangent has to be perpendicular to $y = x$. Its slope is -1.

307. Ellipse is $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$

focus $\equiv (\sqrt{5}, 0)$, $e = \dfrac{\sqrt{5}}{3}$, Any point an ellipse $\equiv \left(\dfrac{3}{\sqrt{2}}, \dfrac{2}{\sqrt{2}}\right)$

equation of circle as the diameter, joining the points $\left(3/\sqrt{2},\ 2/\sqrt{2}\right)$ and focus $(\sqrt{5}, 0)$ is

$(x - \sqrt{5})(\sqrt{2}x - 3) + y(\sqrt{2}.y - 2) = 0$ (A) is the correct option.

308. (a) $(\lambda, 3)$ should satisfy the equation $x^2 + y^2 = 13$

$\therefore \quad \lambda = \pm 2.$

309. (A)

Here $a = 4$, $b = 3$ and $m = 1$

$\therefore$ equation of the tangent is $y = x \pm \sqrt{16 + 9}$

$y = x \pm 5.$

310. Statement – I is true as it is a known fact and statement – II is obviously true. However statement – II is not a true reasoning for statement – I, as coordinate system has nothing to do with statement – I.

311. Given ellipse is $\dfrac{x^2}{64} + \dfrac{y^2}{100} = 1$

$\Rightarrow a^2 = 64;\ b^2 = 100 \Rightarrow e = \dfrac{3}{5}\left(\because a < b\right)$

Now, focal distance of (x_1, y_1) on ellipse will be 7 and 13.

Now, for ellipse $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1 \Rightarrow a^2 = 16,\ b^2 = 9,\ e = \dfrac{\sqrt{7}}{4}.$ $\Rightarrow$ Focus is (ae, 0) or $\left(\sqrt{7}, 0\right)$.

Now radius of the circle = Distance between $\left(\sqrt{7}, 0\right)$ and (0, 3) = 4.

Hence (c) is the correct answer.

313. Option (A) is correct

Sum of the distance between foci = 2ae

Sum of the focal distances = $\frac{2a}{e}$

$ae < \frac{a}{e}$ b′coz e < 1.

Both are true and it is correct reason.

314. Option (A) is true.

Let y = mx be any chord through (0, 0). This will meet conic at points whose x-coordinates are given by $x^2 + m^2x^2 + mx^2 = 1$

$\Rightarrow (1 + m + m^2)\, x^2 - 1 = 0$

$\Rightarrow x_1 + x_2 = 0 \Rightarrow \frac{x_1 + x_2}{2} = 0$

Also $y_1 = mx_1$, $y_2 = mx_2$

$\Rightarrow y_1 + y_2 = m(x_1 + x_2) = 0$

$\Rightarrow \frac{y_1 + y_2}{2} = 0 \Rightarrow$ mid-point of chord is (0, 0) $\forall$m.

315. Equation of PQ (i.e., chord of contact) to the ellipse $x^2 + 2y^2 = 6$

$\frac{hx}{6} + \frac{ky}{3} = 1$ ... (1)

Any tangent to the ellipse $x^2 + 4y^2 = 4$ is

i.e., $x/2 \cos\theta + y\sin\theta = 1$ ... (2)

$\Rightarrow$ (1) & (2) represent the same line $h = 3\cos\theta$, $k = 3\sin\theta$

Locus of R (h, k) is $x^2 + y^2 = 9$ Ans. (A)

316. $x^2/5 + (y-3)^2/9 = 1$

Ends of the major axis are (0, 6) and (0, 0)

Equation of tangent at (0, 6) and (0, 0) is y = 6, and y = 0

Anc. (C)

317. $\frac{x^2}{16} + \frac{y^2}{9} = 1$ will have director circle $x^2 + y^2 = 16 + 9$

$\Rightarrow x^2 + y^2 = 25$

and we know that the locus of the point of intersection of two mutually perpendicular tangents drawn to any standard ellipse is its director circle.

'a' is correct.

318. By formula $p_1p_2 = b^2$

$= 3.$

also foot of perpendicular lies on auxiliary circle of the ellipse.

'B' is correct.

321. Sum of distances between foci = 2ae sum of the focal distances = 2a/e

ae < a/e since e < 1. (A)

322. The statement-1 is false. Since this will represent hyperbola if $h^2 > ab$

$\Rightarrow \frac{\lambda^2}{4} > 2 \Rightarrow |\lambda| > 2\sqrt{2}$

Thus reason R being a standard result is true. (A)

323. (a) Both Statement-1 and Statement-2 are True and Statement-2 is the correct explanation of Statement-1.

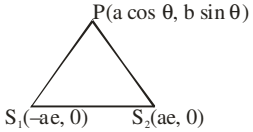
324. (C) Required tangent is

$\frac{3x}{9} - \frac{2y}{4} = 1$ or $\frac{x}{3} - \frac{y}{2} = 1$

325. (C)

area of $\Delta PS_1S_2 = abe \sin\theta$

clearly its maximum value is abe.



327. Tangents cannot be drawn from one branch of hyperbola to the other branch.
Ans. (A)

328. (d) $\sqrt{(\lambda-3)^2+16}-4=1 \Rightarrow \lambda = 0$ or 6.

329. (A)

Hyperbola is $\frac{(x-4)^2}{16}-\frac{(y-3)^2}{9}=1$

$\therefore e = \sqrt{1+\frac{9}{16}}=\frac{5}{4}$.

330. Both statements are true and statement – II is the correct reasoning for statement – I, as for any member, semi transverse and semi – conjugate axes are $\frac{a}{\alpha}$ and $\frac{b}{\alpha}$ respectively and hence asymptoters are always $y = \pm\frac{b}{a}x$.

Hence (a) is the correct answer

331. If y = mx + c be the common tangent,

then $c^2 = a^2m^2 - b^2$ . . . (i)

and $c^2 = -b^2m^2 + a^2$ . . . (ii)

on eliminating c^2, we get $m^2 = 1 \Rightarrow m = \pm 1$.

Now for statement – II,

On eliminating m, we get $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$,

Which is a hyperbola.

Hence (b) is the correct answer.

332. Option (D) is correct.

The statement-1 is false b′coz this will represent hyperbola if $h^2 > ab$

$\Rightarrow \frac{\lambda^2}{4}>2 \Rightarrow |\lambda| > 2\sqrt{2}$

The statmenet-2, being a standard result, is true.

333. The statement-1 is false b′coz points in region II lie below the line $y = b/a\,x \Rightarrow \frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}>0$

The region-2 is true (standard result). Indeed for points in region II

$0 < \frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}<1$.

334. $x^2/a^2 - y^2/b^2 = 1$

if $c^2 = a^2m^2 - b^2$

$\Rightarrow c^2 = 3.3^2 - 2 = 25$

$c = \pm 5$

real tangents are $y = 3x + 5$ Ans (C)

335. The locus of point of intersection of two mutually perpendicular tangents drawn on to hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is its director circle whose equation is $x^2 + y^2 = a^2 - b^2$. For $\frac{x^2}{9}-\frac{y^2}{16}=1$, $x^2 + y^2 = 9 - 16$

So director circle does not exist.
So 'd' is correct.

336. $\frac{x_1+x_2+x_3+x_4}{4}=0$

$\frac{y_1+y_2+y_3+y_4}{4}=0$

So (0, 0) is average point which is also the mid point of line joining the centres of circle & rectangular hyperbola
'a' is correct.

339. The statement-1 is false. Since this will represent hyperbola if $h^2 > ab$

$\Rightarrow \frac{\lambda^2}{4}>2 \Rightarrow |\lambda| > 2\sqrt{2}$

Thus reason R being a standard result is true.
(A)

340. (a)
Both Statement-1 and Statement-2 are True and Statement-2 is the correct explanation of Statement-2.

341. (A)

$\frac{7^2}{4}-\frac{(-3)^2}{9}-1>0$

and $\frac{2^2}{4}-\frac{7^2}{9}-1<0$

342. (B)

Required chord of contact is $32x + 9y = 144$ obtained from $\frac{xx_1}{a^2}-\frac{yy_1}{b^2}=1$.

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 14 XI M 14. Highlights on Conic Sections

Index:

1. Key Concepts

2. Exercise I to II

3. Answer Key

4. Assertion and Reasons

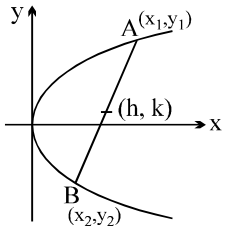
5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

HIGHLIGHTS ON PARABOLA, ELLIPSE & HYPERBOLA

1.HIGHLIGHTS OF PARABOLA

CHORD WITH A GIVEN MIDDLE POINT : General method for finding the equation of a chord of any conic with middle point (h, k).



Example: Let the parabola be $y^2 = 8x$ and (h, k) ≡ (2, – 3)
we have to find the equation of AB $y + 3 = m(x - 2)$ ....(1)

now $y_1^2 = 4ax_1$

and $y_2^2 = 4ax_2$

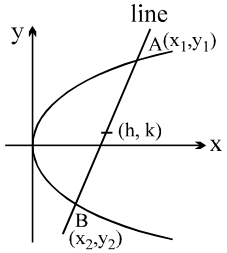
————————Subtraction

$y_1^2 - y_2^2 = 4a(x_1 - x_2)$ or $\dfrac{y_1 - y_2}{x_1 - x_2} = \dfrac{4a}{y_1 + y_2} = m$

but $y_1 + y_2 = 2k = -6$ and $4a = 8$ $\therefore$ $m = \dfrac{8}{-6} = -\dfrac{4}{3}$

Hence equation of AB = $y + 3 = -\dfrac{4}{3}(x - 2) \Rightarrow 4x + 3y + 1 = 0$]

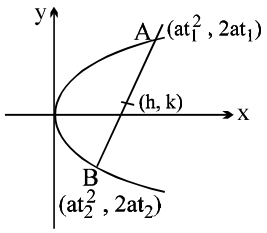
Conversely : To find the mid point of given chord :Let the equation of the line be 4x + 3y + 1= 0 given.To find the mid point (h, k) of AB



here $m = -\dfrac{4}{3} = \dfrac{4a}{y_1 + y_2} \Rightarrow -\dfrac{4}{3} = \dfrac{8}{2k} \Rightarrow k = -3$

since $4h + 3k + 1 = 0 \Rightarrow 4h - 9 + 1 = 0 \Rightarrow h = 2$ hence M is (2, – 3)

For a parabola in particular



Equation of AB $y - k = m(x - h)$ ....(1)

But $m_{AB} = \dfrac{2}{t_1 + t_2}$ also $2k = 2a(t_1 + t_2)$

$(t_1 + t_2) = \dfrac{k}{a}$ $\therefore$ $m_{AB} = \dfrac{2a}{k}$ Hence equation of a chord whose mid point is (h, k)

$y - k = \dfrac{2a}{k}(x - h)$ ....(2) or $4ah + ky - k^2 = 2a(x - h) + 4ah = 2a(x + h)$

$$\underbrace{ky - 2a(x + h)}_{T} = \underbrace{k^2 - 4ah}_{S_1}$$

EXAMPLES : Ex-1Find the locus of the middle point of chords of the parabola $y^2 = 4ax$ which

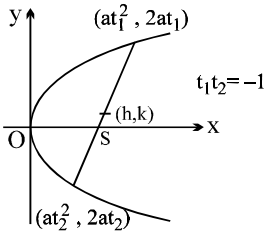
(i) passes through the focus [Ans. $y^2 = 2a(x - a)$]

(ii) are normal to the parabola [Ans. $y^2(y^2 - 2ax + 4a^2) + 8a^4 = 0$]

(iii)subtend a constant angle α at the vertex(Homogenise)Ans.$(8a^2+y^2-2ax)^2\tan^2\alpha = 16a^2(4ax - y^2)$]

(iv)are of given length (say 2*l*) (v) are such that the normals at their extremities meet on the parabola [Ans. $y^2 = 2a^2(x + 2a)$; Hint : use $t_1t_2 = 2$]

[Sol. (i)$2h = a(t_1^2 + t_2^2)$ ....(1)

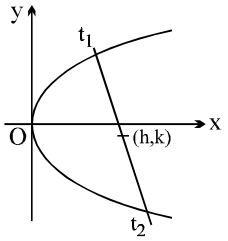


$2k = 2a(t_1 + t_2)$ .....(2) $k^2 = a^2[t_1^2 + t_2^2 + 2t_1t_2]$

$k^2 = a^2\left[\frac{2h}{a} - 2\right]$ $= 2ah - 2a^2$ $\therefore$ $y^2 = 2a(x - a)$

This could also be spelled as locus of the middle point of all focal chords of all the particles $y^2 = 4ax$. **or** Locus of the middle point of all the chord of contact of the pair of tangents drawn from any point on this directrix.

(ii) $2h = a(t_1^2 + t_2^2)$ ....(1) ; $k = a(t_1 + t_2)$ .....(2)



also $t_2 = -t_1 - \frac{2}{t_1} \Rightarrow t_1 + t_2 = -\frac{2}{t_1}$ using (2), $\frac{k}{a} = -\frac{2}{t_1} \Rightarrow t_1 = -\frac{2a}{k}$

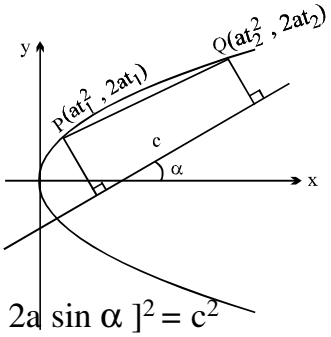
$\therefore t_2 = +\frac{2a}{k} + \frac{2k}{2a} \Rightarrow t_2 = \frac{2a}{k} + \frac{k}{a} \therefore t_1t_2 = -\frac{2a}{k}\left[\frac{2a}{k} + \frac{k}{a}\right] = -2 - \frac{4a^2}{k^2}$ ....(3)

from (1) $2h = a[(t_1 + t_2)^2 - 2t_1t_2] = a\left[\frac{k^2}{a^2} + 2\left(2 + \frac{4a^2}{k^2}\right)\right] = a\left[\frac{k^2}{a^2} + 4 + \frac{8a^2}{k^2}\right] = a\left[\frac{k^4 + 4a^2k^2 + 8a^4}{a^2k^2}\right]$

$2ahk^2 = k^4 + 4a^2k^2 + 8a^4$ $k^2(k^2 - 2ah + 4a^2) + 8a^4 = 0$ Ans.]

Ex-2 A series of chords is drawn so that their projections on the straight line which is inclined at an angle α to the axis are of constant length c. Prove that the locus of their middle point is the curve $(y^2 - 4ax)(y\cos\alpha + 2a\sin\alpha)^2 + a^2c^2 = 0$

[Sol. Let $\hat{n} = \cos\alpha\,\hat{i} + \sin\alpha\,\hat{j}$



$\overrightarrow{PQ} = \vec{v} = (at_2^2 - at_1^2)\hat{i} + 2a(t_2 - t_2)\hat{j}$;;;$2h = a(t_1^2 + t_2^2)$; $a(t_1 + t_2) = k$

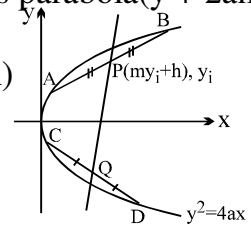
also projection of $\vec{v}$ on $\hat{n} = c$ $\left|\frac{\vec{v}\cdot\hat{n}}{|\hat{n}|}\right| = c$;;;

$\left|a(t_2^2 - t_1^2)\cos\alpha + 2a(t_2 - t_2)\sin\alpha\right| = c$;;;$a^2(t_2 - t_1)^2$ [$a(t_2 + t_1)\cos\alpha + 2a\sin\alpha$]$^2 = c^2$

$a^2[(t_2 + t_1)^2 - 4t_1t_2]$ [$a(t_2 + t_1)\cos\alpha + 2a\sin\alpha$]$^2 = c^2$]

Ex-3 Through each point of the straight line $x = my + h$ is drawn the chord of the parabola $y^2 = 4ax$ which is bisected at the point. Prove that it always touches parabola $(y + 2am)^2 = 8a(x - h)$.

[Sol. Equation of var. chord AB $y - y_i = \frac{2a}{y_i}(x - my_1 - h)$



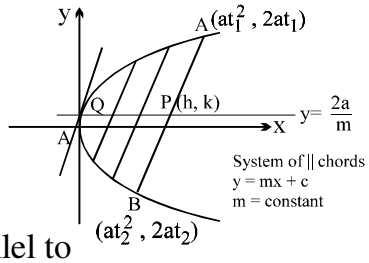
$yy_i - y_i^2 = 2ax - 2amy_1 - 2ah$

$y_i^2 - (y + 2am)y_i + 2a(x - h) = 0$

$(y + 2am)^2 = 8a(x - h)$]

DIAMETER : The locus of the middle points of a system of parallel chords of a Parabola is called a ***Diameter***. Equation to the diameter of a parabola is y = 2a/m, where m = slope of parallel chords.

Explanation : Slope of AB ism = $\frac{2}{t_1 + t_2}$ ....(1)

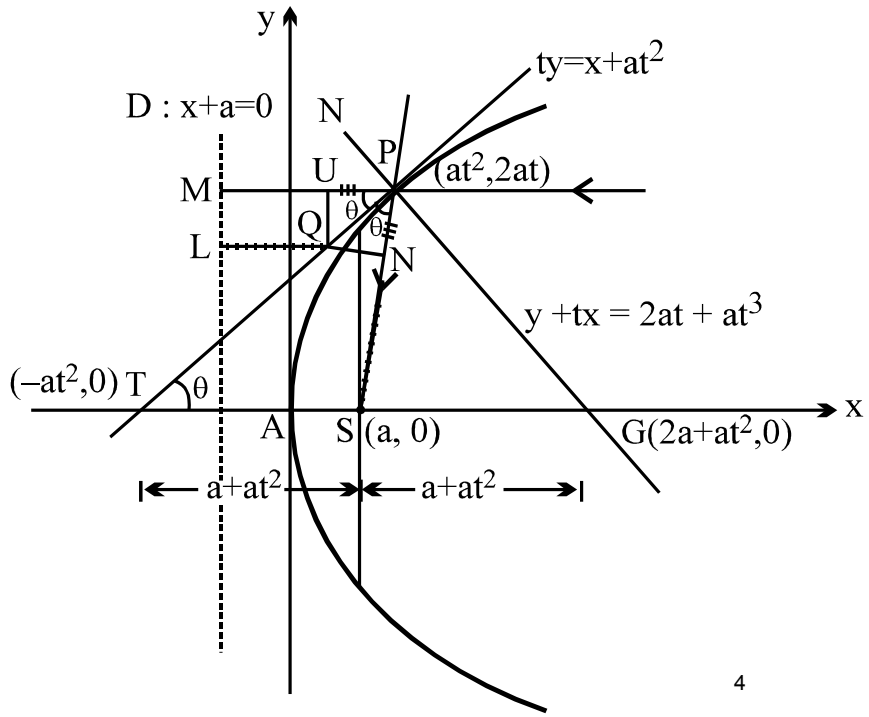


also $2k = 2a(t_1 + t_2) \Rightarrow t_1 + t_2 = \frac{k}{a} \therefore k = \frac{2a}{m}$ = constant

Hence equation of the diameter is $y = \frac{2a}{m}$ i.e. a line parallel to the axis of the parabola. Solving $y = \frac{2a}{m}$ with $y^2 = 4ax$, we have, $\frac{4a^2}{m^2} = 4ax$ or $x = \frac{a}{m^2}$ Hence coordinates of Q are $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$ Hence the tangent at the extremity of a diameter of a parabola is parallel to the system of chords it bisects.Since point of intersection of the two tangents are A and B is $at_1t_2, a(t_1 + t_2)$ or $\left(a\,t_1t_2, \frac{2a}{m}\right)$ Hence the tangent at the ends of any chords of a parabola meet on the diameter which bisects the chord. ☞ **Note:A line segment from a point P on the parabola and parallel to the system of parallel chords is called the ordinate to the diameter bisecting the system of parallel chords and the chords are called its double ordinate.**

IMPORTANT HIGHLIGHTS :

(a) If the tangent & normal at any point'P'of the parabola intersect the axis at T & G then



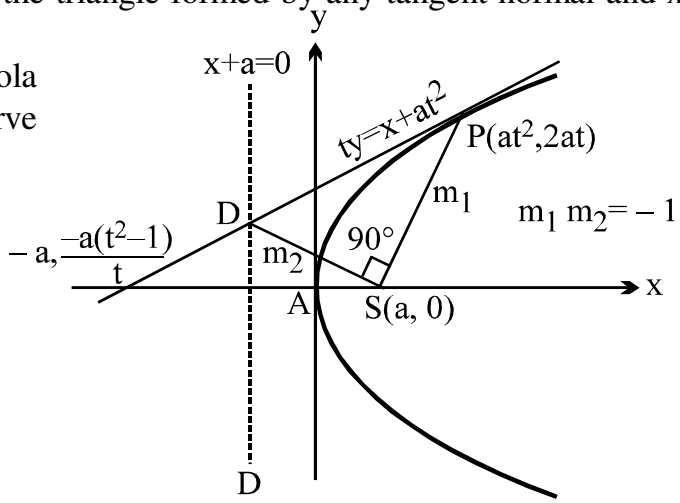
PM = PS
PU = PN
PM – PU = PS – PN
MU = SN
QL = SN

ST = SG = SP where 'S' is the focus. In other words the tangent and the normal at a point P on the parabola are the bisectors of the angle between the focal radius SP & theperpendicular from P on the directrix. From this we conclude that all rays emanating from S will become parallel to the axis of the parabola after reflection.

Deduce that, if Q is any point on the tangent and QN is the perpendicular from Q on focal radius and QL is the perpendicular on the directrix then QL = SN.

☞ **Note :** Circle circumscribing the triangle formed by any tangent normal and x-axis, has its centre at focus.

(b) The portion of a tangent to a parabola cut off between the directrix & the curve subtends a right angle at the **focus**.

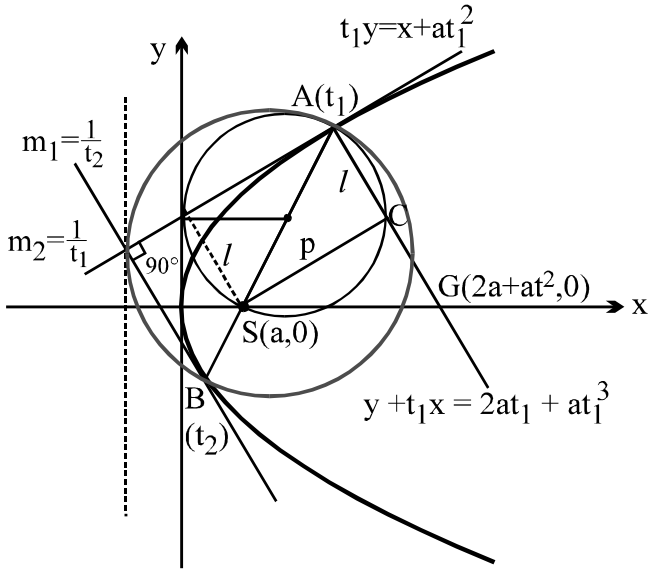


$$m_1 = \frac{2at}{at^2 - a} = \frac{2t}{t^2 - 1} \quad ;$$

$$m_2 = \frac{-a(t^2-1)}{t \cdot 2a} = -\frac{(t^2-1)}{2t}$$

$\therefore \quad m_1 m_2 = -1$]

(c) The tangents at the extremities of a focal chord intersect at right angles on the **directrix**, and hence a circle on any focal chord as diameter touches the directrix. Also a circle on any focal radii of a point P (at^2, 2at) as diameter touches the tangent at the vertex and intercepts a chord of length $a\sqrt{1+t^2}$ on a normal at the point P.



☞ **Note :** (1) For computing p draw a perpendicular from S (a, 0) on tangent at P.

(d) Any tangent to a parabola & the perpendicular on it from the focus meet on the tangent at the vertex.

i.e. locus of the feet of the perpendicular drawn from focus upon a variable tangent is the tangent drawn to the parabola at its vertex.

Explanation : Tangent 't' is

$$ty = x + at^2 \qquad .....(1)$$

Passing through (h, k) hence

$$tk = h + at^2 \qquad .....(A)$$

A line through (a, 0) with slope – t

$y = -t(x - a)$ ....(2)

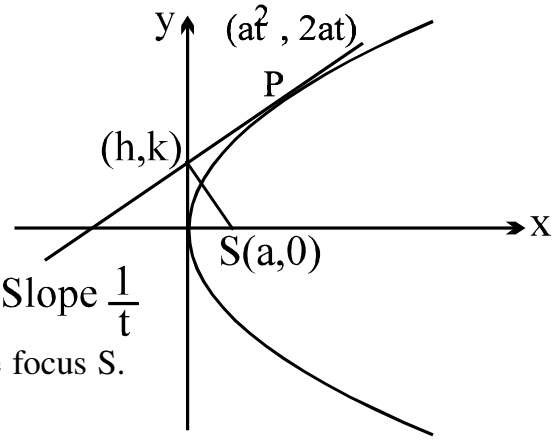
(2) also passes through (h, k)

$k = -th + at$

$tk = -t^2h + at^2$ ....(B)

(A) – (B) gives $0 = (1 + t^2)h$

$\Rightarrow$ $x = 0$ which is the tangent at the vertex.



(e) If the tangents at P and Q meet in T, then :

- TP and TQ subtend equal angles at the focus S.
- $ST^2 = SP \cdot SQ$ &
- The triangles SPT and STQ are similar.

(i) To prove that $\alpha = \beta$, it will be sufficient to prove that 'T' lies on the angle bisector of the angle $\angle PSQ$ i.e. perpendicular distance of 'T' from the line SP is equal to the perpendicular of T from SQ.

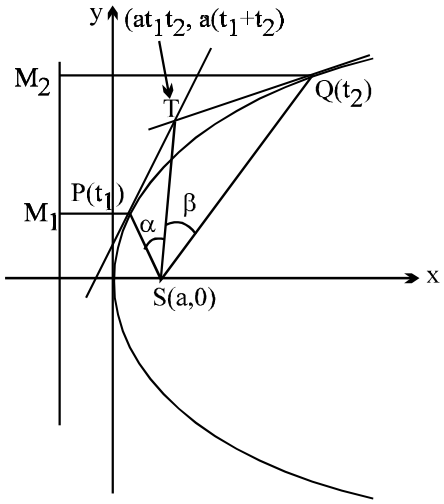
equation of SP

$$y = \frac{2at_1}{at_1^2 - a}(x - a)$$

$$2t_1x - (t_1^2 - 1)y - 2ab_1 = 0$$

$$p_1 = \left| \frac{2at_1^2t_2 - (t_1^2 - 1)a(t_1 + t_2) - 2at_1}{\sqrt{(t_1^2 - 1)^2 + 4t_1^2}} \right|$$

$= a\,|t_1 - t_2|$



Similarly $p_2 = a\,|t_2 - t_1| \Rightarrow \alpha = \beta$, Hence proved.

Now (ii) $SP \cdot SQ = (a + at_1^2)(a + at_2^2) = a^2(1 + t_1^2)(1 + t_2^2)$

also $(ST)^2 = a^2(t_1t_2 - 1)^2 + a^2(t_1 + t_2)^2 = a^2[t_1^2t_2^2 + 1 + t_2^2 + t_2^2]$

$= a^2(1 + t_1^2)(1 + t_2^2)$

Hence $(ST)^2 = SP \cdot SQ$

This is conclusive that product of the focal radii of two points P and Q is equal to the square of the distance of focus from the point of intersection of the tangents drawn at P and Q.

(iii) again, $\frac{ST}{SP} = \frac{SQ}{ST}$ and $\alpha = \beta$

hence the two triangles SPT and SQT are similar.

(f) Tangents and Normals at the extremities of the latus rectum of a parabola $y^2 = 4ax$ constitute a square, their points of intersection being (– a, 0) & (3a, 0).

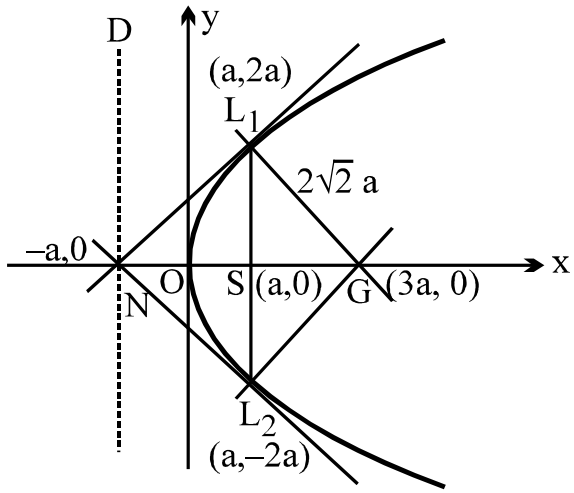
[Hint: 

figure is Self explanatory]

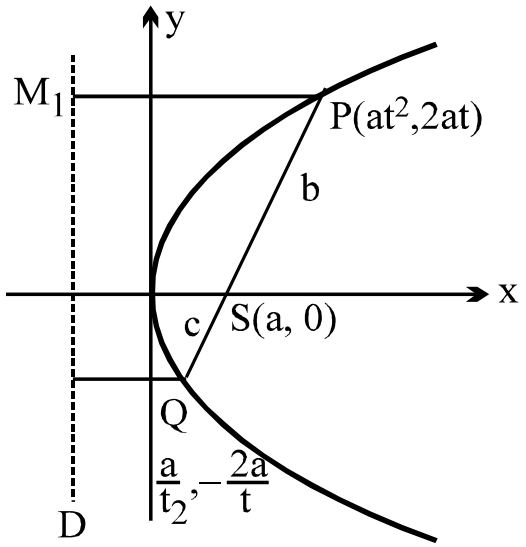
☞ **Note :**

(1) The two tangents at the extremities of focal chord meet on the foot of the directrix.

(2) Figure L_1NL_2G is square of side $2\sqrt{2}\,a$

(g) Semi latus rectum of the parabola $y^2 = 4ax$, is the harmonic mean between segments of any focal chord of the parabola is : $2a = \frac{2bc}{b+c}$ **i.e.** $\frac{1}{b}+\frac{1}{c}=\frac{1}{a}$.

[Sol. $2a = \frac{2bc}{b+c}$ i.e. $\frac{1}{a}=\frac{1}{b}+\frac{1}{c}$



$b = a + at^2$

$\Rightarrow \quad b = a(1 + t^2)$

$\Rightarrow \quad \frac{a}{b} = \frac{1}{1+t^2}$ ....(1)

$c = a + \frac{a}{t^2}$

$\Rightarrow \quad c = a\left(1+\frac{1}{t^2}\right) \Rightarrow \quad \frac{a}{c} = \frac{t^2}{t^2+1}$ ....(2)

from (1) and (2)

$\frac{a}{b} + \frac{a}{c} = 1 \qquad \Rightarrow \quad \frac{1}{a}=\frac{1}{b}+\frac{1}{c}$]

(h) The circle circumscribing the triangle formed by any three tangents to a parabola passes through the focus.

TPT $\alpha = \beta$

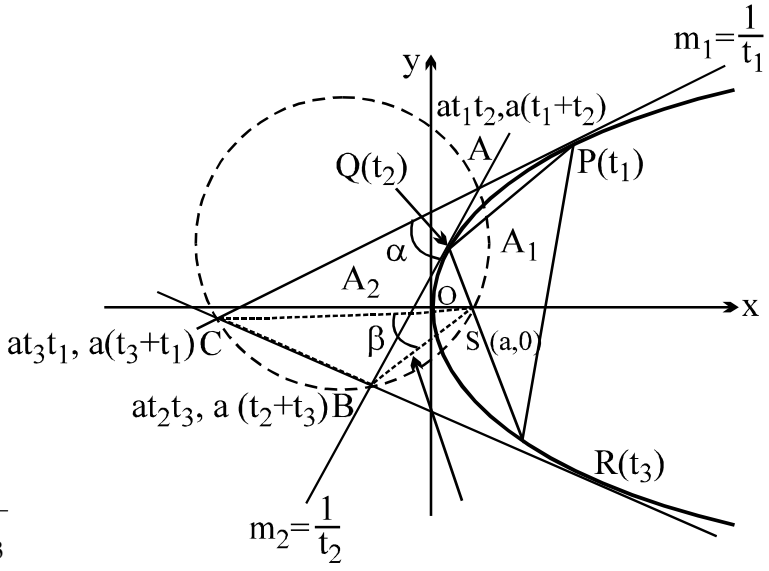
$\frac{1}{t_1} = \tan\theta_1$; $\frac{1}{t_2} = \tan\theta_2$; $\frac{1}{t_3} = \tan\theta_3$

$\therefore \quad \tan\alpha = \left| \frac{\tan\theta_1 - \tan\theta_2}{1 + \tan\theta_1 \cdot \tan\theta_2} \right|$

$\tan\alpha = |\tan(\theta_1 - \theta_2)| \quad ....(1)$

$m_1 = \frac{a(t_3 + t_1)}{at_3t_1 - a} = \frac{t_3 + t_1}{t_3t_1 - 1}$

$= \frac{\frac{1}{t_1} + \frac{1}{t_3}}{1 - \frac{1}{t_1t_3}} = \frac{\tan\theta_1 + \tan\theta_3}{1 - \tan\theta_1 \cdot \tan\theta_3}$



$m_1 = \tan(\theta_1 + \theta_3)$

Illly $m_2 = \tan(\theta_2 + \theta_3)$

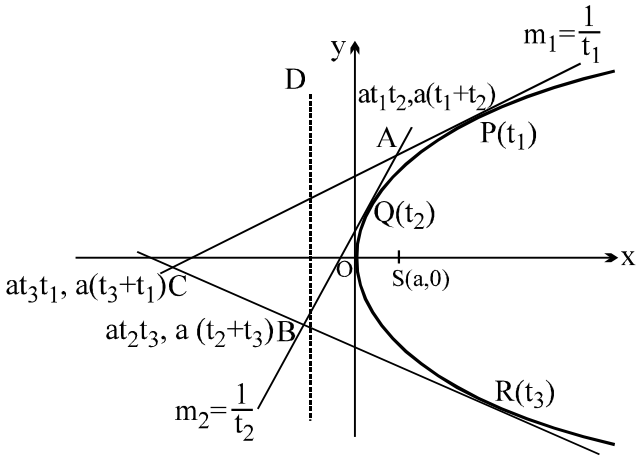
$\tan\beta = \left| \frac{\tan(\theta_1 + \theta_3) - \tan(\theta_2 + \theta_3)}{1 + \tan(\theta_1 + \theta_3) \cdot \tan(\theta_2 + \theta_3)} \right|$

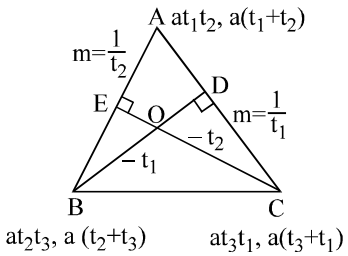
$\tan\beta = |\tan(\theta_1 - \theta_2)| \quad ....(2)$

from (1) and (2), we get

$\alpha = \beta$ hence proved]

(i) The orthocentre of any triangle formed by three tangents to a parabola $y^2 = 4ax$ lies on the directrix & has the co–ordinates $-a$, $a(t_1 + t_2 + t_3 + t_1t_2t_3)$.



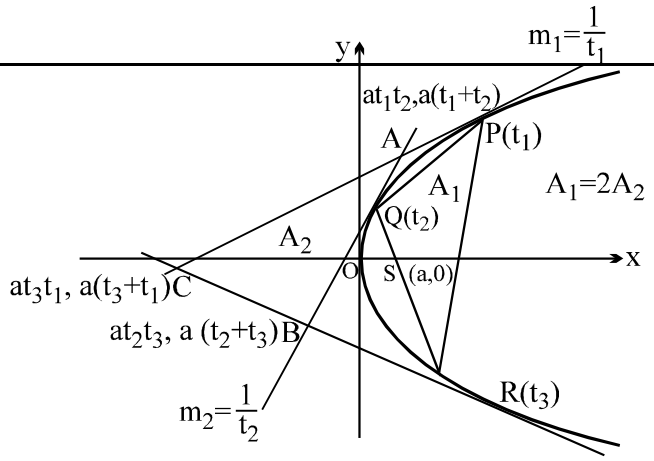


Find intersection of BD and CE to get 'O'.

(j) The area of the triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents at these points.

Refer figure of point (h)

$$\frac{A_1}{A_2} = \frac{\begin{vmatrix} at_1^2 & 2at_1 & 1 \\ at_2^2 & 2at_2 & 1 \\ at_3^2 & 2at_3 & 1 \end{vmatrix}}{\begin{vmatrix} at_1t_2 & a(t_1+t_2) & 1 \\ at_2t_3 & a(t_2+t_3) & 1 \\ at_3t_1 & a(t_3+t_1) & 1 \end{vmatrix}} = 2$$



MORE ABOUT NORMALS

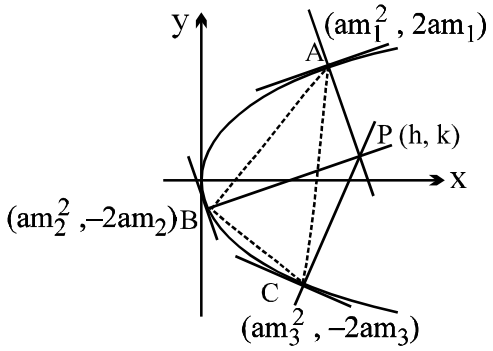
If normal drawn to a parabola passes through a point P(h , k) then $k = mh - 2am - am^3$ i.e. $am^3 + m(2a - h) + k = 0$.

Then gives $m_1 + m_2 + m_3 = 0$; $m_1m_2 + m_2m_3 + m_3m_1 = \frac{2a-h}{a}$; $m_1m_2m_3 = -\frac{k}{a}$.

where m_1, m_2, & m_3 are the slopes of the three concurrent normals. Note that the algebraic sum of the :

- slopes of the three concurrent normals is zero.
- ordinates of the three conormal points on the parabola is zero.
- Centroid of the triangle formed by three co–normal points lies on the x–axis.

Example : Consider of the triangle ABC is



$$x_1 = \frac{a(m_1^2 + m_2^2 + m_3^2)}{3}$$

$$y_1 = -\frac{2a(m_1 + m_2 + m_3)}{3} = 0$$

now $x_1 = \frac{a}{3}[(m_1 + m_2 + m_3)^2 - 2\sum m_1m_2]$

$$= -\frac{2a}{3}\frac{(2a-h)}{a} = \frac{2}{3}(h - 2a)$$

$\therefore$ centroid is $\frac{2}{3}(h - 2a), 0$ but $\frac{2}{3}(h - 2a) > 0$

$\therefore$ $h > 2a$

Hence abscissa of the point of concurrency of 3 concurrent normals > 2a.

EXAMPLES :

Ex.1 Find the locus of a point which is such that **(a)** two of the normals drawn from it to the parabola are at right angles, **(b)** the three normals through it cut the axis in points whose distances from the vertex are in arithmetical progression.

[Ans : (a) $y^2 = a(h - 3a)$; (b) $27ay^2 = 2(x - 2a)^3$] [Ex.237, Pg.212, Loney]

[Sol. (a) we have $m_1 m_2 = -1$

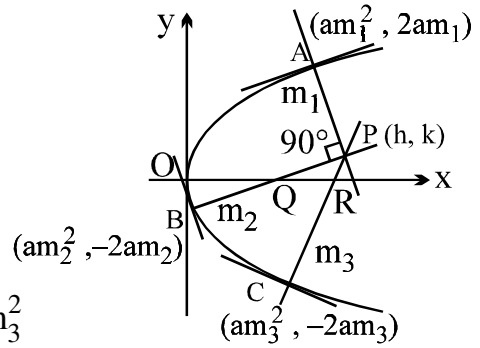
also $m_1 m_2 m_3 = -\frac{k}{a}$

$\therefore \quad m_3 = \dfrac{k}{a}$

put $m_3 = -\dfrac{k}{a}$ is a root of

$am^3 + (2a - h)m + k = 0$

(b) $y = mx - 2am - am^3$



hence $2a + am_1^2,\ 2a + am_2^2,\ 2a + am_3^2$

$\therefore \quad 2m_2^2 = m_1^2 + m_3^2$

$$3m_2^2 = m_1^2 + m_2^2 + m_3^2 = (m_1 + m_2 + m_3)^2 - 2\left(\sum m_1 m_2\right) = \frac{2(h-2a)}{a}$$

$$m_2^2 = \frac{2(h-2a)}{3a}$$ which is root of $am^3 + (2a - h)m + k = 0$]

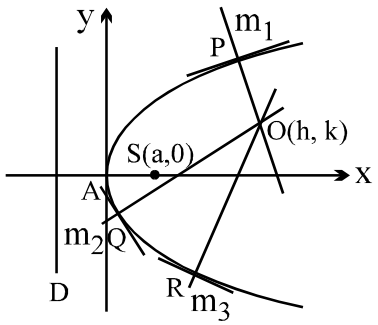
Ex.2 If the normals at three points P, Q and R meet in a point O and S be the focus, prove that SP · SQ · SR = a· SO^2. [Ex.238, Pg.213, Loney]

[Sol. $SP = a(1 + m_1^2)$;

$SQ = a(1 + m_2^2)$;

$SR = a(1 + m_3^2)$

$$\frac{SP \cdot SQ \cdot SR}{a^3} = (1 + m_1^2)(1 + m_2^2)(1 + m_3^2)$$



$$\left[1 + \left(\left(\sum m_1\right)^2 - 2\sum m_1 m_2\right) + \left(\left(\sum m_1 m_2\right)^2 - \underbrace{2m_1 m_2 m_3\left(\sum m_1\right)}_{\text{zero}}\right) + (m_1 m_2 m_3)^2\right]$$

Ex.3 A circle circumscribing the triangle formed by three co–normal points passes through the vertex of the parabola and its equation is, $2(x^2 + y^2) - 2(h + 2a)x - ky = 0$.

[Q.13, Ex-30, Loney]

[Sol. Equation of the normal at P

$y + tx = 2at + at^3$

passes through (h, k)

$at^3 + (2a - h)t - k = 0 \Leftarrow t_1, t_2, t_3$ ....(1)

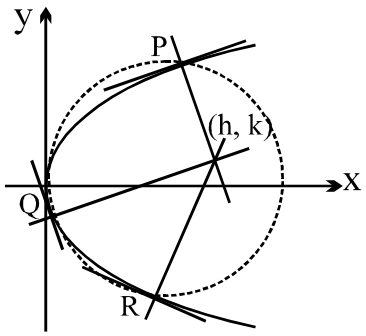
$t_1 + t_2 + t_3 = 0$

$$t_1t_2 + t_2t_3 + t_3t_1 = \frac{2a-h}{a}, \quad t_1t_2t_3 = \frac{k}{a}$$

Let the circle through PQR is

$x^2 + y^2 + 2gx + 2fy + c = 0$

solving circle $x = at^2$, $y = 2at$



$a^2t^4 + 4a^2t^2 + 2gat^2 + 2f \cdot 2at + c = 0$

$a^2t^4 + 2a(2a + g)t^2 + 4fat + c = 0$ ....(2)

$t_1 + t_2 + t_3 + t_4 = 0$

but $t_1 + t_2 + t_3 = 0 \quad \Rightarrow \quad t_4 = 0 \quad \Rightarrow$ circle passes through the origin

hence the equation of the circle

$x^2 + y^2 + 2gx + 2fy = 0$

now equation (2) becomes

$at^3 + 2(2a + g)t + 4f = 0 \begin{cases} t_1 \\ t_2 \\ t_3 \end{cases}$ ....(3)

(1) and (3) must have the same root

$2(2a + g) = 2a - h$

$2g = -(h + 2a)$

and $4f = -k \quad \Rightarrow \quad 2f = -\frac{k}{2}$

Hence the equation of the circle is

$x^2 + y^2 - (h + 2a)x - \frac{k}{2}y = 0 \quad \Rightarrow \quad 2(x^2 + y^2) - 2(h + 2a)x - ky = 0$]

Ex.4 Three normals are drawn to the parabola $y^2 = 4ax\cos\alpha$ from any point on the straight line $y = b\sin\alpha$. Prove that the locus of the orthocentre of the triangle formed by the corresponding tangent is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the angle α being variable.

[Q.15, Ex.30, Loney]

[Sol. $y^2 = 4Ax$ where $A = a\cos\alpha$

$y + tx = 2At + at^3$ passes through $\lambda, b\sin\alpha$

$b\sin\alpha + t\lambda = 2At + At^3$

$At^3 + (2A - \lambda)t - b\sin\alpha = 0$

$\therefore \quad t_1 + t_2 + t_3 = 0 \quad ; \quad t_1t_2t_3 = \frac{b\sin\alpha}{A}$

also $h = -A = -a\cos\alpha$ ....(1)

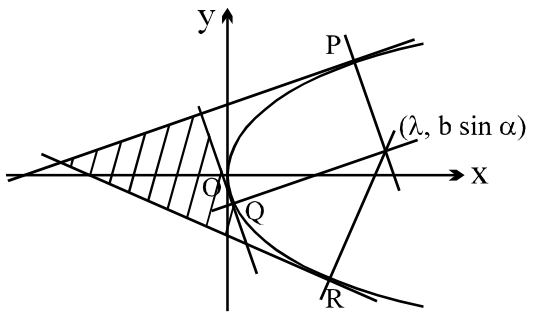
and $k = A(t_1 + t_2 + t_3 + t_1t_2t_3)$

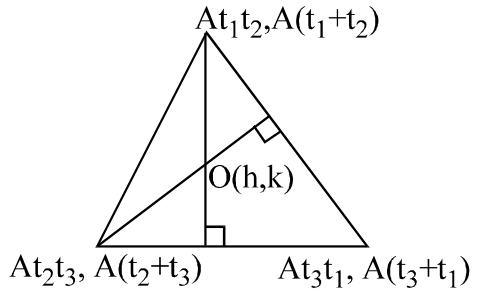
$= A\left(0 + \frac{b\sin\alpha}{A}\right)$

$k = b\sin\alpha$ ....(2)

from (1) and (2) locus is

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$]





LEVEL 3 PROBLEMS

Ex.1 Locus of a point P when the 3 normals drawn from it are such that area of the triangle

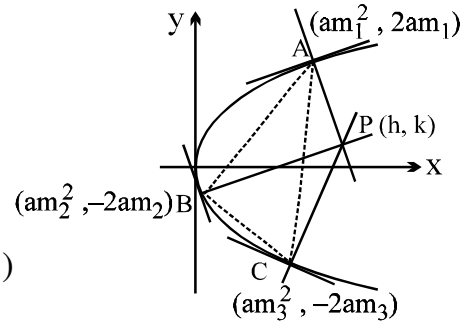
formed by their feet is constant. [Q.6, Ex-30, Loney]

[Hint: Area of ΔABC = constant

$$\begin{vmatrix} am_1^2 & -2am_1 & 1 \\ am_2^2 & -2am_2 & 1 \\ am_3^2 & -2am_3 & 1 \end{vmatrix} = C$$

$\Rightarrow \quad (m_1 - m_2)^2 (m_2 - m_3)^2 (m_3 - m_1)^2 = C \quad ....(1)$

consider



$$(m_1 - m_2)^2 = m_1^2 - 2m_1m_2 + m_2^2$$
$$= -[m_1(m_2 + m_3) + 2m_1m_2 + m_2(m_1 + m_3)$$
$$= -[(\sum m_1m_2) + 3m_1m_2] \quad \left(m_1m_2m_3 = -\frac{k}{a} \right)$$
$$= \frac{h - 2a}{a} + \frac{3k}{am_3}$$

$$\left[\underbrace{(h - 2a)}_{x} + \frac{3k}{\underbrace{m_3}_{l}} \right] \left[(h - 2a) + \frac{3k}{\underbrace{m_2}_{m}} \right] \left[(h - 2a) + \frac{3k}{\underbrace{m_1}_{n}} \right]$$

hence equation (1) becomes

$(x + l)(x + m)(x + n)$ = constant

$x^3 + (l + m + n)x^2 + (lm + mn + nl)x + lmn = 0$

$$(h - 2a)^3 + 3k\left(\frac{1}{m_1} + \frac{1}{m_2} + \frac{1}{m_3} \right)(h - 2a)^2 + 9k^2\left(\frac{1}{m_1m_2} + \frac{1}{m_2m_3} + \frac{1}{m_3m_4} \right)(h - 2a) + \frac{27k^3}{m_1m_2m_3}$$

= constant]

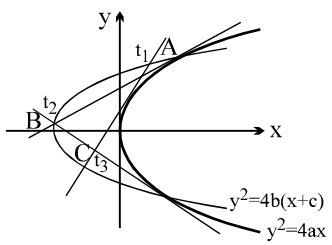
Ex.2 The sides of a triangle touch a parabola $y^2 = 4ax$ and two of its angular points lie on another parabola $y^2 = 4b(x + c)$ with its axis in the same direction, prove that the locus of the third angular point is another parabola. [Q.31, Ex-30, Loney]

[Sol. consider the equations abtained by putting the coordinates of A and B in $y^2 = 4b(x + c)$

$$a^2(t_1 + t_2)^2 = 4b(at_1t_2 + c) \quad ....(1)$$
$$a^2(t_2 + t_3)^2 = 4b(at_2t_3 + c) \quad ....(2)$$

this implies that t_1 and t_3 are the roots of the equation,

$a^2(t + t_2)^2 = 4b(a\,t\,t_2 + c)$ having t_1 and t_3 as its roots



i.e. $a^2(t^2 + 2tt_2 + t_2^2) = 4batt_2 + 4bc$

$\Rightarrow \quad a^2t^2 + 2t(a^2t_2 - 2abt_2) + a^2t_2^2 - 4bc = 0$

$\Rightarrow \quad t_1 + t_3 = -\dfrac{a^2t_2 - 2abt_2}{a^2} = \dfrac{t_2(2b-a)}{a}$

and $\quad t_1\, t_3 = \dfrac{a^2t_2^2 - 4bc}{a^2}$; $\quad$ hence $\dfrac{k}{a} = \dfrac{t_2(2b-a)}{a} \quad \Rightarrow \quad t_2 = \dfrac{k}{2b-a}$

$$\frac{h}{a} = \frac{at_2^2 - 4bc}{a^2} \quad \Rightarrow \quad \frac{ah+4bc}{a} = t_2^2 \quad \Rightarrow \quad \frac{k^2}{(2b-a)^2} = \frac{ah+4bc}{a}$$

hence locus is $ay^2 = (2b - a)^2 (ax + 4bc)$]

Ex.3 Circles are drawn through the vertex of the parabola to cut the parabola orthogonally at the other point of intersection. Prove that the locus of the centres of the circles is the curve,
$2y^2(2y^2 + x^2 - 12ax) = ax(3x - 4a)^2$ [Q.26, Ex.28 (Loney)]

Ex.4 If the normal at P and Q meet on the parabola, prove that the point of intersection of the tangents at P and Q lies either on a certain straight line, which is parallel to the tangent at the vertex, or on the curve whose equation is $y^2(x + 2a) + 4a^3 = 0$.
[Q.11, Ex.29 (Loney)]

Ex.5 (a) Prove that infinite number of triangles can be constructed in either of the parabolas $y^2 = 4ax$ and $x^2 = 4by$ whose sides touch the other parabola.

(b) Prove that the locus of the centre of the circle, which passes through the vertex of a parabola and through its intersections with a normal chord, is the parabola $2y^2 = ax - a^2$. [Q.25, Ex.30 (Loney)]

1.HIGHLIGHTS OF ELLIPSE :

Referring to an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

H-1 If P be any point on the ellipse with S & S′ as its foci then $\ell(SP) + \ell(S'P) = 2a$.

H-2

(a) Product of the length's of the perpendiculars from either focus on a variable tangent to an Ellipse / Hyperbola = (semi minor axis)^2 / (semi conjugate axis)^2 = b^2

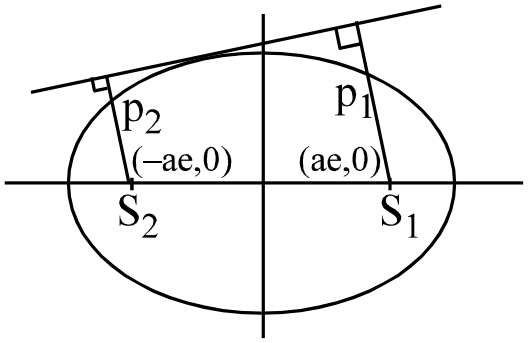
$mx - y + \sqrt{a^2m^2+b^2} = 0$

$$p_1p_2 = \left|\frac{mae+\sqrt{a^2m^2+b^2}}{\sqrt{1+m^2}}\cdot\frac{-mae+\sqrt{a^2m^2+b^2}}{\sqrt{1+m^2}}\right|$$

$$= \left|\frac{(a^2m^2+b^2)-m^2a^2e^2}{1+m^2}\right|$$

$$= \left|\frac{a^2m^2+b^2-m^2(a^2-b^2)}{1+m^2}\right|$$

$$= \left|\frac{b^2(1+m^2)}{1+m^2}\right| = b^2$$



(b) Feet of the perpendiculars from either foci on a variable tangent to an ellipse / hyperbola lies on its auxiliary circle. Hence deduce that the sum of the squares of the chords which the auxiliary circle intercept on two perpendicular tangents to an ellipse is constant and is equal to the square on the line joining the foci.

$y = mx + \sqrt{a^2m^2+b^2}$

$(k - mh)^2 = a^2m^2 + b^2$ **....(1)**

equation of line through F_1 &

slope = $-\frac{1}{m}$

$y - 0 = -\frac{1}{m}(x - ae)$

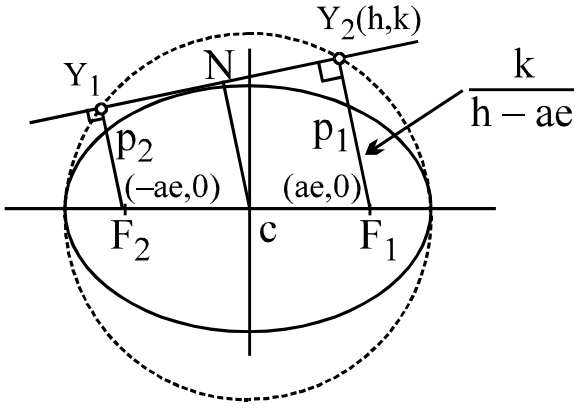
$k = -\frac{1}{m}(h - ae)$

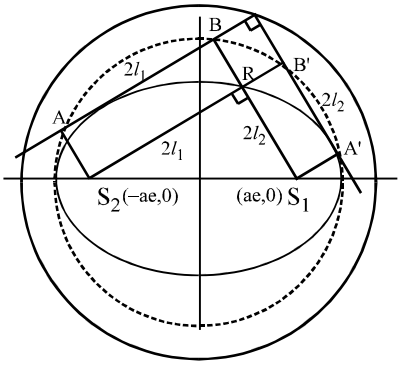
$(km + h)^2 = a^2e^2 = a^2 - b^2$ **....(2)**

adding (1) and (2), we get

$h^2 + k^2 + m^2(h^2 + k^2) = a^2m^2 + b^2 + a^2 - b^2$

$k^2(1 + m^2) + h^2(1 + m^2) = a^2(1 + m^2)$;14 $x^2 + y^2 = a^2$





now $A'B' = 2l_2$; $AB = 2l_1$

$(S_1R)^2 + (S_2R)^2 = (S_1S_2)^2$.]

(c) If Y_1 and Y_2 be the feet of the perpendiculars on the auxiliay circle from the foci upon any tangent, at P on the ellipse, then the point of intersection 'Q' of the tangents at Y_1 and Y_2 lies on the ordinate through P. If P varies i.e. θ varies then the locus of Q is an ellipse having the same eccentricity as the original ellipse.

Chord of Contact (C.O.C) w.r.t the circle $x^2 + y^2 = a^2$ is

$$hx + ky = a^2 \qquad ....(1)$$

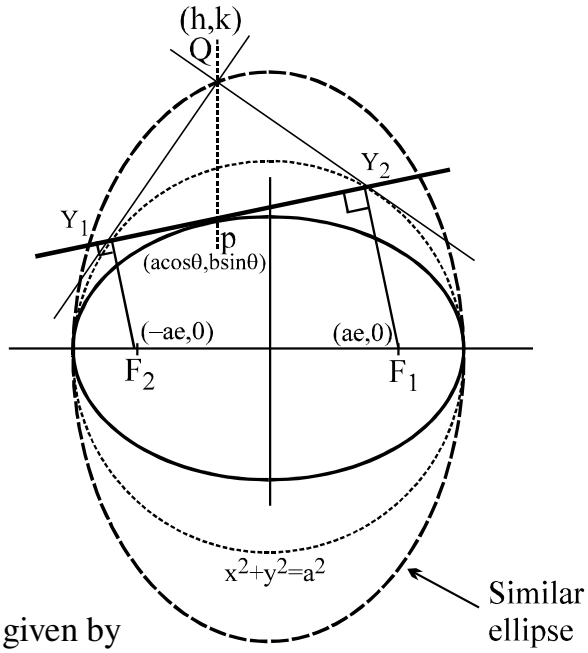
This must be the same tangent at P(θ)

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1 \qquad ....(2)$$

Comparing (1) and (2) $\dfrac{ha}{\cos\theta} + \dfrac{kb}{\sin\theta} = a^2$

$$\left.\begin{aligned} h &= a\cos\theta \\ k &= \frac{a^2\sin\theta}{b} \end{aligned}\right\}$$

$$\frac{x^2}{a^2} + \frac{y^2}{\left(\frac{a^2}{b}\right)^2} = 1$$



Which is an ellipse whose eccentricity e' is given by

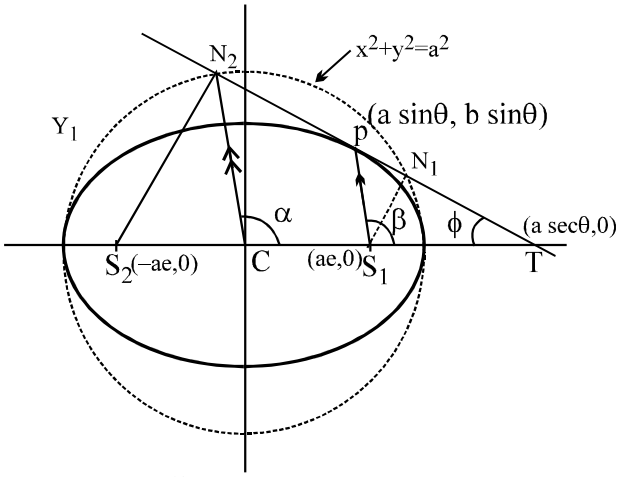
$$e'^2 = 1 - \frac{a^2b^2}{a^4} \quad \Rightarrow \quad e'^2 = 1 - \frac{b^2}{a^2} = e^2 \ ]$$

(d) Lines joining centre to the feet of perpendicular from a focus on any tangent at P and the line joining other focus to the point of contant 'P' are parallel.

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$$

$$\frac{CT}{S_1T} = \frac{a\sec\theta}{a\sec\theta - ae} = \frac{a\sec\theta}{a(\sec\theta - e)}$$

$$\frac{CT}{S_1T} = \frac{1}{1 - e\cos\theta}$$



Again $\dfrac{CN_2}{S_1P} = \dfrac{a}{e\left(\frac{a}{e} - a\cos\theta\right)}$

$$= \frac{a}{a - ae\cos\theta} = \frac{1}{1 - e\cos\theta} \Rightarrow \frac{CT}{ST} = \frac{CN_2}{S_1P}$$

Now use sine law in triangles CN_2T and S_1PT we can prove $\alpha + \phi = \beta + \phi$

i.e. $\dfrac{CT}{\sin(\alpha+\phi)} = \dfrac{a}{\sin\phi}$ and $\dfrac{S_1P}{\sin\phi} = \dfrac{S_1T}{\sin(\beta+\phi)}$

$\therefore$ $\dfrac{CT}{a} = \dfrac{\sin(\alpha+\phi)}{\sin\phi}$ and $\dfrac{S_1T}{S_1P} = \dfrac{\sin(\beta+\phi)}{\sin\phi}$

hence $\sin(\alpha + \phi) = \sin(\beta + \phi)$]

H-3 If the normal at any point P on the ellipse with centre C meet the major & minor axes in G & g respectively, & if CF be perpendicular upon this normal, then

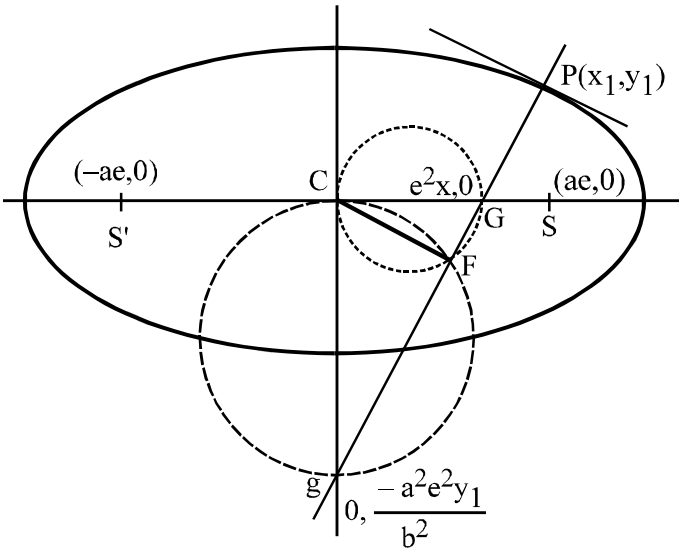
(i) $PF \cdot PG = b^2$ **(ii)** $PF \cdot Pg = a^2$

(iii) $PG \cdot Pg = SP \cdot S'P$ **(iv)** $CG \cdot CT = (CS)^2$

(v) locus of the mid point of Gg is another ellipse having the same eccentricity as that of the original ellipse.

[where S and S′ are the focii of the ellipse and T is the point where tangent at P meet the major axis]

(i) $PF \cdot PG = b^2 \dfrac{a^2x}{x_1} - \dfrac{b^2y}{y_1} = a^2e^2$



LHS=Power of the point P w.r.t. the circle on CG as diameter

$= x_1(x_1 - e^2x_1) + y_1^2$

$= x_1^2(1 - e^2) + y_1^2$

$= a^2\cos^2\theta \left(1 - 1 + \dfrac{b^2}{a^2}\right) + b^2\sin^2\theta$

$= b^2\cos^2\theta + b^2\sin^2\theta$

$= b^2$]

(ii) $PF \cdot Pg = a^2$

LHS = Power of the point P w.r.t. the circle on Cg as diameter

$= x_1^2 + y_1\left(y_1 + \dfrac{a^2e^2y_1}{b^2}\right) = a^2\cos^2\theta + b^2\sin^2\theta\left(1 + \dfrac{(a^2 - b^2)}{b^2}\right) + b^2\sin^2\theta$

$= a^2\cos^2\theta + b^2\sin^2\theta \cdot \dfrac{a^2}{h^2} = a^2$]

(iii) PG . Pg = SP . S′ P

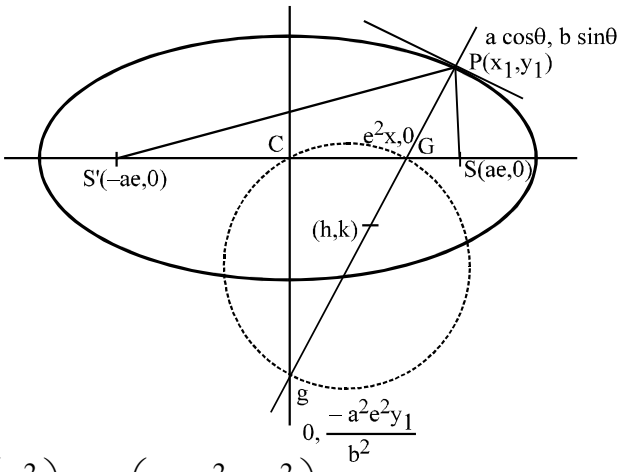
RHS = (a – ae cosθ)(a + ae cosθ)

$a^2 - a^2e^2\cos^2\theta$

$a^2 - (a^2 - b^2)\cos^2\theta$

$a^2\sin^2\theta + b^2\cos^2\theta$

LHS = Power of P w.r.t the circle on Gg as diameter



$$= x_1(x_1 - e^2x_1) + y_1\left(y_1 + \frac{a^2e^2y_1}{b^2}\right)$$

$$= x_1^2(1-e^2) + y_1^2\left(1 + \frac{a^2e^2}{b^2}\right) = x_1^2\left(\frac{b^2}{a^2}\right) + y_1^2\left(1 + \frac{a^2 - b^2}{b^2}\right)$$

$= b^2\cos^2\theta + a^2\sin^2\theta$ = RHS]

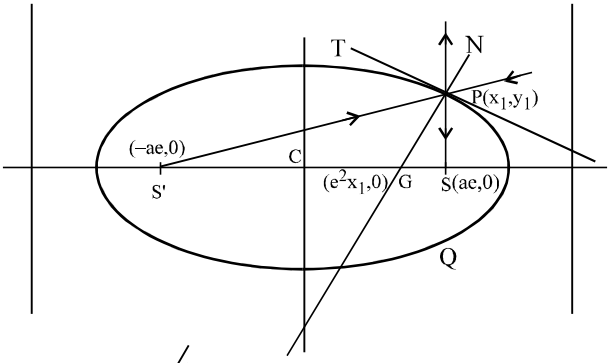
(iv) $CG \cdot CT = (CS)^2$

H-4 The tangent & normal at a point P on the ellipse bisect the external & internal angles between the focal distances of P. This refers to the well known reflection property of the ellipse which states that rays from one focus are reflected through other focus & vice–versa. Hence we can deduce that the straight lines joining each focus to the foot of the perpendicular from the other focus upon the tangent at any point P meet on the normal PG and bisects it where G is the point where normal at P meets the major axis.

Reflection property :

$$\frac{S_2G}{GS_1} = \frac{e^2x_1 + ae}{ae - e^2x_1} = \frac{a + ex_1}{a - ex_1}$$

or $$\frac{PS_2}{PS_1} = \frac{e\left(\frac{a}{e} + x_1\right)}{e\left(\frac{a}{e} - x_1\right)} = \frac{a + ex_1}{a - ex_1}$$



Also In ΔS_1PN_1 and ΔN_1PQ

$\angle N_1PQ = \angle N_2PS_1 = \theta$ (neglect property)

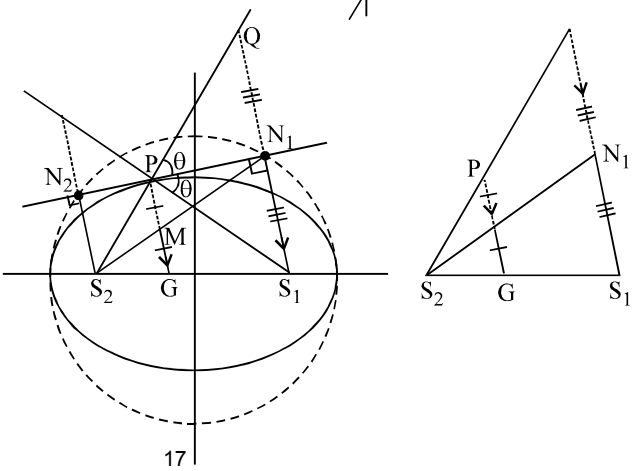
PN_1 is common

$\angle QN_1P = PN_1S_1 = 90°$

$\therefore$ $S_1PN_1 = N_1PQ_1$

$\therefore$ N_1 is the mid point of S_1Q

Now proceed



H-5 The portion of the tangent to an ellipse between the point of contact & the directrix subtends a right angle at the corresponding focus.

H-6 The circle on any focal distance as diameter touches the auxiliary circle.

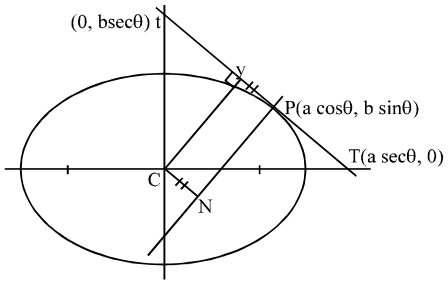
H-7 Perpendiculars from the centre upon all chords which join the ends of any perpendicular diameters of the ellipse are of constant length.

H-8 If the tangent at the point P of a standard ellipse meets the axis in T and t and CY is the perpendicular on it from the centre then,

(i) $Tt \cdot PY = a^2 - b^2$ and

(ii) least value of Tt is $a + b$.

$Tt \cdot py = a^2 - b^2$



$$\sqrt{a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta}$$

$$\sqrt{a^2 + b^2 + (a\sin\theta - b\cos\theta)^2 + 2ab}$$

Home Work : Tutorial Sheet, Ellipse.

TOUGH ELLIPSE

Ex.1 Chords at right angles are drawn through any point $P(\alpha)$ of the ellipse, and the line joining their extremities meets the normal in the point Q. Prove that Q is the same for all such chords, its coordinates being $\frac{a^3e^2\cos\alpha}{a^2+b^2}$ and $\frac{-a^2be^2\sin\alpha}{a^2+b^2}$.

Prove also that the major axis is the bisector of the angle PCQ, and that the locus of Q for different positions of P is the ellipse :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \left(\frac{a^2-b^2}{a^2+b^2}\right)^2 .$$ [Q.29, Ex-35, Loney]

Ex.2 Prove that the directrices of the two parabolas that can be drawn to have their foci at any given point P of the ellipse and to pass through its foci meet at an angle which is equal to twice the eccentric angle of P. [Q.28, Ex-35, Loney]

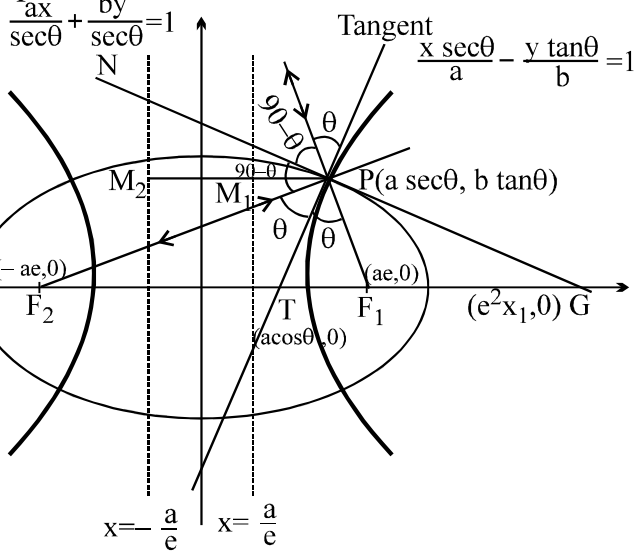
Ex.3 An ellipse is rotated through a right angle in its own plane about its centre, which is fixed, prove that the locus of the point of intersection of a tangent to the ellipse in its original position with the tangent at the same point of the curve in its new position is

$(x^2 + y^2)(x^2 + y^2 - a^2 - b^2) = 2(a^2 - b^2)\,xy$. [Q.26, Ex-35, Loney

3. HIGHLIGHTS ON HYPERBOLA (TANGENT AND NORMAL) :

H–1 Locus of the feet of the perpendicular drawn from focus of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ upon any tangent is its auxiliary circle i.e. $x^2 + y^2 = a^2$ & the product of the feet of these perpendiculars is $b^2 \cdot (\text{semi C} \cdot \text{A})^2$

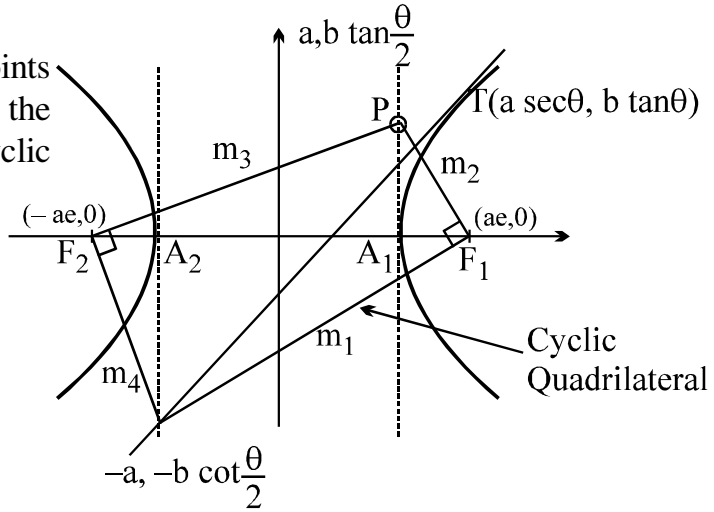
H–2 The portion of the tangent between the point of contact & the directrix subtends a right angle at the corresponding focus.



H–3 The tangent & normal at any point of a hyperbola bisect the angle between the focal radii. This spells the reflection property of the hyperbola as **"An incoming light ray "** aimed towards one focus is reflected from the outer surface of the hyperbola towards the other focus. It follows that if an ellipse and a hyperbola have the same foci, they cut at right angles at any of their common point.

Note that the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{a^2-k^2}-\frac{y^2}{k^2-b^2}=1 (a > k > b > 0)$ are confocal and therefore orthogonal.

H–4 The foci of the hyperbola and the points P and Q in which any tangent meets the tangents at the vertices are concyclic with PQ as diameter of the circle.



$$T: \frac{x\sec\theta}{a}-\frac{y\tan\theta}{b}=1$$

$$\frac{(\sec\theta-1)b}{\tan\theta}=y$$

$$y = b\left(\frac{1-\cos\theta}{\sin\theta}\right)$$

Family of circles through F_1F_2

$(x – ae)(x + ae) + y^2 + \lambda y = 0 \Rightarrow$ PQ as diameter]

13. ASYMPTOTES :

Definition : If the length of the perpendicular let fall from a point on a hyperbola to a straight line tends to zero as the point on the hyperbola moves to infinity along the hyperbola, then the straight line is called the Asymptote of the Hyperbola.

To find the asymptote of the hyperbola :

Let $y = mx + c$ is the asymptote of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Solving these two we get the quadratic as

$(b^2 - a^2m^2)\, x^2 - 2a^2 mcx - a^2 (b^2 + c^2) = 0$ ....(1)

In order that $y = mx + c$ be an asymptote, both roots of equation (1) must approach infinity, the conditions for which are :

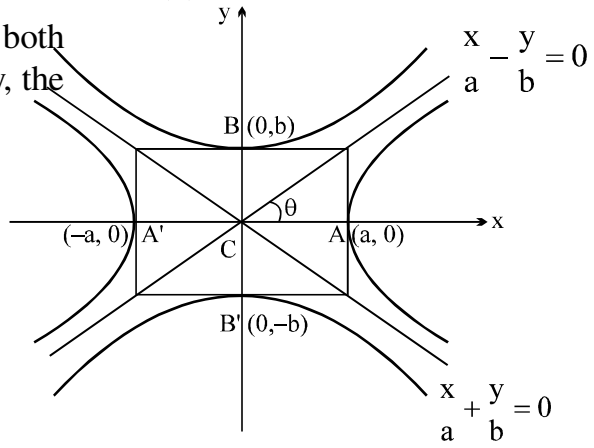
coeff of $x^2 = 0$ & coeff of $x = 0$.

$\Rightarrow$ $b^2 - a^2m^2 = 0$ or $m = \pm \frac{b}{a}$ &

$a^2 mc = 0 \Rightarrow c = 0$.

$\therefore$ equations of asymptote are $\frac{x}{a} + \frac{y}{b} = 0$

and $\frac{x}{a} - \frac{y}{b} = 0$.



combined equation to the asymptotes $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$.

PARTICULAR CASE :

When $b = a$ the asymptotes of the rectangular hyperbola.

$x^2 - y^2 = a^2$ are, $y = \pm x$ which are at right angles.

Note :

(i) Equilateral hyperbola $\Leftrightarrow$ rectangular hyperbola.

(ii) If a hyperbola is equilateral then the conjugate hyperbola is also equilateral.

(iii) A hyperbola and its conjugate have the same asymptote.

(iv) The equation of the pair of asymptotes differ the hyperbola & the conjugate hyperbola by the same constant only.

(v) The asymptotes pass through the centre of the hyperbola & the bisectors of the angles between the asymptotes are the axes of the hyperbola.

(vi) The asymptotes of a hyperbola are the diagonals of the rectangle formed by the lines drawn through the extremities of each axis parallel to the other axis.

(vii) Asymptotes are the tangent to the hyperbola from the centre.

(viii) A simple method to find the coordinates of the centre of the hyperbola expressed as a general equation of degree 2 should be remembered as:

Let $f(x, y) = 0$ represents a hyperbola.

Find $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y}$. Then the point of intersection of $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0$

gives the centre of the hyperbola.

EXAMPLES ON ASYMPTOTES

Ex.1 Find the asymptotes of the hyperbola, $3x^2 - 5xy - 2y^2 - 5x + 11y - 8 = 0$. Also find the equation of the conjugate hyperbola. [Solved Ex. 325 Pg.294]

Ex.2 Find the equation to the hyperbola whose asymptotes are the straight lines $2x + 3y + 3 = 0$ and $3x + 4y + 5 = 0$ and which passes through the point $(1, -1)$. Also write the equation to the conjugate hyperbola and the coordinates of its centre.

[Q.9, Ex-37, Loney]

Ex.3 A normal is drawn to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at P which meets the transverse axis (TA) at G. If perpendicular from G on the asymptote meets it at L, show that LP is parallel to conjugate axis. [Q.7, Ex-37, Loney]

[Sol. equation of GL with slope $-a/b$ and passing through $(e^2x_1, 0)$ is

$y - 0 = -a/b(x - e^2x_1)$

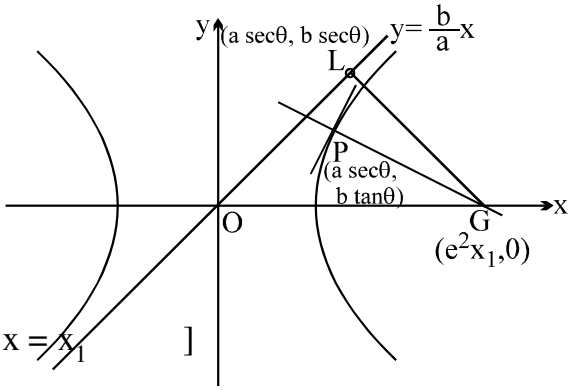
$ax + by = ae^2x_1$ ....(1)

put $y = \frac{b}{a}x$

$ax + b\cdot\frac{b}{a}x = ae^2x_1$

$x\left[\frac{a^2 + b^2}{a}\right] = a^2e^2x_1$

$x = x_1$]



Ex.4 A transversal cuts the same branch of a hyperbola $x^2/a^2 - y^2/b^2 = 1$ in P, P′ and the asymptotes in Q, Q′. Prove that : (i) PQ = P'Q' & (ii) PQ' = P'Q

[Sol. TPT PQ = P'Q' and PQ' = P'Q

$b^2x^2 - a^2y^2 = a^2b^2$; Let the transversal be $y = mx + c$

$b^2x^2 - a^2(mx + c)^2 = a^2b^2$

$(b^2 - a^2m^2)x^2 - 2a^2mcx - a^2(c^2 + b^2) = 0$

$\frac{x_1 + x_2}{2} = \frac{2a^2mc}{b^2 - a^2m^2}$ ....(1)

solving $y = mx + c$ with $b^2x^2 - a^2y^2 = 0$

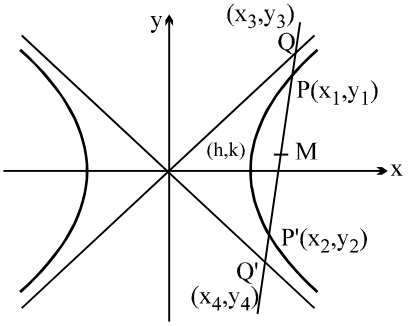
$(b^2 - a^2m^2)x^2 - 2a^2mcx - a^2c^2 = 0$ ....(2)

$\frac{x_3 + x_4}{2} = \frac{2a^2mc}{b^2 - a^2m^2}$ ....(3)

$MQ = MQ'$

also $MP = MP'$

$PQ = P'Q'$]



Q.5 The tangent at any point P of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ meets one of the asymptotes in Q and L, M are the feet of the perpendiculars from Q on the axes. Prove that LM passes through P. [Q.7, Ex.48, (Pillay)]

14. HIGHLIGHTS ON ASYMPTOTES :

H–1 If from any point on the asymptote a straight line be drawn perpendicular to the transverse axis, the product of the segments of this line, intercepted between the point & the curve is always equal to the square of the semi conjugate axis.

H–2 Perpendicular from the foci on either asymptote meet it in the same points as the corresponding directrix & the common points of intersection lie on the auxiliary circle.

$y = \frac{b}{a}x$ ....(1)

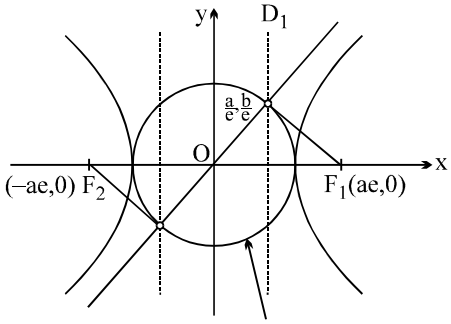
$y - 0 = -\frac{a}{b}(x - ae)$

$by + ax = a^2e$ ....(2)

$\left(\frac{b^2}{a}+a\right)x = a^2e$

$(b^2 + a^2)x = a \cdot a^2e$

$(a^2e^2)x = a^2ae \quad \Rightarrow \quad x = \frac{a}{e}$; hence $y = \frac{b}{a}\cdot\frac{a}{c} = \frac{b}{c}$]



$\frac{a^2}{e^2}+\frac{b^2}{e^2} = \frac{a^2+b^2}{a^2} = a^2$

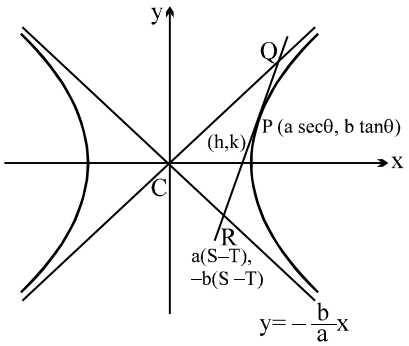
H–3 The tangent at any point P on a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ with centre C, meets the asymptotes in Q and R and cuts off a Δ CQR of constant area equal to ab from the asymptotes & the portion of the tangent intercepted between the asymptote is bisected at the point of contact. This implies that locus of the centre of the circle circumscribing the Δ CQR in case of a rectangular hyperbola is the hyperbola itself & for a standard hyperbola the locus would be the curve, $4(a^2x^2 - b^2y^2) = (a^2 + b^2)^2$.

Area of $\Delta QCR = \frac{1}{2}\begin{vmatrix} a(S+T) & b(S+T) & 1 \\ 0 & 0 & 1 \\ a(S-T) & -b(S-T) & 1 \end{vmatrix}$

$= -\frac{ab}{2}[-(1) - 1] = ab = \text{constant}$

Area of ΔQCR is constant = ab $\qquad \frac{x\sec\theta}{a}-\frac{y\tan\theta}{b}=1$;

solving it with $y = \frac{b}{a}x$, $\frac{x\sec\theta}{a} - \frac{(\tan\theta)x}{a} = 1$



$x = \frac{a}{S-T} = a(S + T) \quad \Rightarrow \quad x = a(S - T)$

TPT $4(a^2x^2 - b^2y^2) = (a^2 + b^2)^2$

$h^2 + k^2 = (h - a(S + T))^2 = 2(S + T)(ah + kb) = 0$ ....(1)

$(S - T)^2(a^2 + b^2) = 2(S - T)(ah - kb) = 0$ ....(2)

(1) × (2) $\Rightarrow$ $(a^2 + b^2)^2 = 4(a^2x^2 - b^2y^2)$]

H–4 If the angle between the asymptote of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is 2θ then $e = \sec\theta$.

. RECTANGULAR HYPERBOLA :

Rectangular hyperbola referred to its asymptotes as axis of coordinates.

(a) Equation is $xy = c^2$ with parametric representation $x = ct, y = c/t, t \in R - \{0\}$.

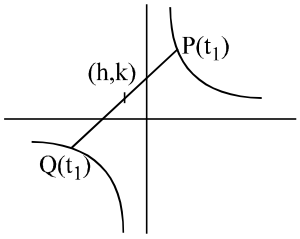
(b) Equation of a chord joining the points $P(t_1)$ & $Q(t_2)$ is $x + t_1 t_2 y = c(t_1 + t_2)$ with slope $m = -\frac{1}{t_1 t_2}$.

(c) Equation of the tangent at $P(x_1, y_1)$ is $\frac{x}{x_1} + \frac{y}{y_1} = 2$ & at $P(t)$ is $\frac{x}{t} + ty = 2c$.

(d) Equation of normal : $y - \frac{c}{t} = t^2(x - ct)$

(e) Chord with a given middle point as (h, k) is $kx + hy = 2hk$.

[**Explanation** : **(e)** Chord with a given middle point



$2h = c(t_1 + t_2)$

$2k = c\left(\frac{1}{t_1} + \frac{1}{t_2}\right) = c\left(\frac{t_1 + t_2}{t_1 t_2}\right) = \frac{2h}{t_1 t_2}$; $\quad \therefore \quad t_1 t_2 = \frac{h}{k}$

now equation of PQ is, $y - k = -\frac{1}{t_1 t_2}(x - h) = -\frac{k}{h}(x - h)$

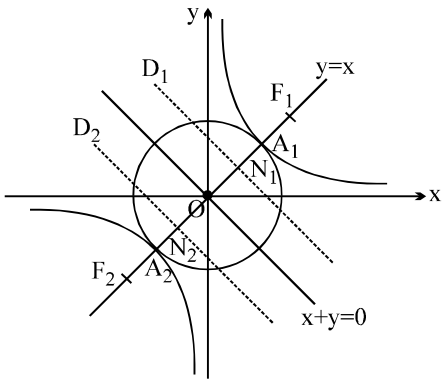
$hy - hk = -kx + hk \quad \Rightarrow \quad kx + hy = 2hk \quad \Rightarrow \quad \frac{x}{h} + \frac{y}{k} = 2$]

(f) Equation of the normal at $P(t)$ is $xt^3 - yt = c(t^4 - 1)$.

Ex.1 Find everything for the rectangular hyperbola $xy = c^2$.

[Sol.1. eccentricity $= \sqrt{2}$ (angle between the two asymptotes = 90°)

2. asymptotes $x = 0$; $y = 0$



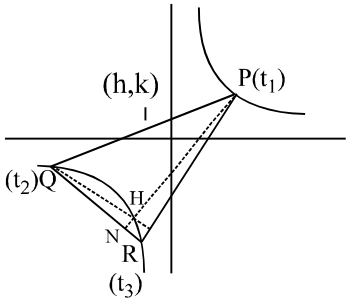
3. T.A. : y = x ; C.A. : y = – x
4. centre (0, 0)
5. vertex (c, c) & (– c, – c)
6. Foci $\left(c\sqrt{2}, c\sqrt{2}\right)$ and $(-c\sqrt{2}, -c\sqrt{2})$
7. Length of Latus rectum = Length of T.A. (in case of rectangular hyperbola) $= 2\sqrt{2}\,c$
8. Equation of auxilliary circle : $x^2 + y^2 = 2c^2$
9. Equation of director circle : $x^2 + y^2 = 0$
10. Equation of the directrices : $x + y = \pm\sqrt{2}\,c$
11. Co-ordinates of the extremities of latus rectum in the 1st quadrant A $\left((\sqrt{2}+1)c, (\sqrt{2}-1)c\right)$

Ex.2 A rectangular hyperbola $xy = c^2$ circumscribing a triangle also passes through the orthocentre of this triangle. If $\left(ct_i, \frac{c}{t_i}\right)$ i = 1, 2, 3 be the angular points P, Q, R then orthocentre is $\left(\frac{-c}{t_1t_2t_3}, -ct_1t_2t_3\right)$.

[Sol. slope of $QR = -\frac{1}{t_2t_3}$

$\therefore$ slope of $PN = t_2t_3$

$\therefore$ equation of altitude through P



$$y - \frac{c}{t_1} = t_2t_3(x - ct_1)$$

$$y + c\,t_1\,t_2\,t_3 = \frac{c}{t_1} + x\,t_2t_3$$

$$y + c\,t_1\,t_2\,t_3 = t_2t_3\left(x + \frac{c}{t_1t_2t_3}\right) \quad .....(1)$$

(1) is suggestive that orthocentre is $\left(\frac{-c}{t_1t_2t_3}, -ct_1t_2t_3\right)$

Ex.3 If a circle and the rectangular hyperbola $xy = c^2$ meet in the four points t_1, t_2, t_3 & t_4, then

(a) $t_1\,t_2\,t_3\,t_4 = 1$

(b) the centre of the mean position of the four points bisects the distance between the centres of

the two curves.

(c) the centre of the circle through the points t_1, t_2 & t_3 is :

$$\left\{\frac{c}{2}\left(t_1+t_2+t_3+\frac{1}{t_1t_2t_3}\right),\frac{c}{2}\left(\frac{1}{t_1}+\frac{1}{t_2}+\frac{1}{t_3}+t_1t_2t_3\right)\right\}$$

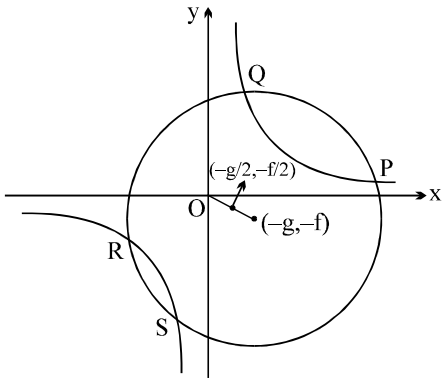
(d) If PQRS are the four points of intersection of the circle with rectangular hyperbola then $(OP)^2 + (OQ)^2 + (OR)^2 + (OS)^2 = 4r^2$ where r is the radius of circle.

[Sol.

(a) Let the equation of the circle be

$x^2 + y^2 + 2gx + 2fy + d = 0$ ....(1)

solving with $xy = c^2$



$$x^2 + \frac{c^4}{x^2} + 2gx + 2f\cdot\frac{c^2}{x} + d = 0$$

$x^4 + 2gx^3 + dx^2 + 2fc^2x + c^4 = 0$ ....(1)

from (1) $x_1x_2x_3x_4 = c^4$

$c^4[t_1t_2t_3t_4] = c^4 \quad \Rightarrow \quad t_1t_2t_3t_4 = 1 \quad \Rightarrow$ (a)

(b) again, centre of the mean position of 4 points of intersection = $\frac{\sum x_i}{4}, \frac{\sum y_i}{4}$

now from (1)

$x_1 + x_2 + x_3 + x_4 = -2g$ .....(2); hence $\frac{\sum x_i}{4} = -\frac{g}{2}$

using $xy = c^2$

$$y_1 + y_2 + y_3 + y_4 = c^2\left[\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}+\frac{1}{x_4}\right] = \frac{c^2}{x_1x_2x_3x_4}\sum x_1x_2x_3 = \frac{c^2}{c^4}(-2f\,c^2) = -2f$$

$$\therefore \quad \frac{\sum y_i}{4} = -\frac{f}{2} \quad ; \text{ Hence } \left(\frac{\sum x_i}{4}, \frac{\sum y_i}{4}\right) = \left(-\frac{g}{2}, -\frac{f}{2}\right)$$

(c) centre of the circle through PQR i.e. $(-g, -f)$ is given by

$$\frac{x_1+x_2+x_3+x_4}{2}, \quad \frac{y_1+y_2+y_3+y_4}{2} \quad (\text{using } t_1t_2t_3t_4 = 1)$$

$$\frac{c}{2}\left[(t_1+t_2+t_3)+\frac{1}{t_1t_2t_3}\right], \quad \frac{c}{2}\left[\frac{1}{t_1}+\frac{1}{t_2}+\frac{1}{t_3}+\frac{t_1t_2t_3}{1}\right]$$

(d) $(OP)^2 + (OQ)^2 + (OR)^2 + (OS)^2 = 4r^2$ where $r = g^2 + f^2 = d$

LHS $\left(x_1^2 + x_2^2 + x_3^2 + x_4^2\right) + \left(y_1^2 + y_2^2 + y_3^2 + y_4^2\right)$

$$\left[\left(\sum x_1\right)^2 - 4\sum x_1x_2\right] + c^4\left[\frac{1}{x_1^2}+\frac{1}{x_2^2}+\frac{1}{x_3^2}+\frac{1}{x_4^2}\right]$$

$$(4g^2 - 4d) + c^4\left[\left(\sum\frac{1}{x_1}\right)^2 - 2\sum\frac{1}{x_1x_2}\right]$$

$$(4g^2 - 4d) + c^4\left[\left\{\frac{1}{x_1x_2x_3x_4}\sum x_1x_2x_3\right\}^2 - \frac{4}{x_1x_2x_3x_4}\sum x_1x_2\right]$$

$$(4g^2 - 4d) + c^4\left[\left\{\frac{1}{c^4}(-2fc^2)\right\}^2 - \frac{4d}{c^4}\right]$$

$= (4g^2 - 4d) + (4f^2 - 4d) = 4[g^2 + f^2 - d] = 4r^2$]

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 15 XII M 8. Vector Algebra and 3-D Geometry

Index:

1. Key Concepts

2. Exercise I to VIII

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

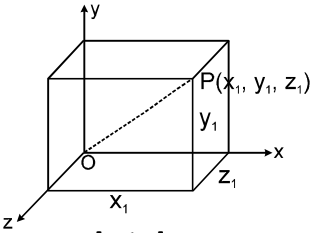
6. 10 Yrs. Que. from AIEEE

THREE DIMENSIONAL GEOMETRY

Coordinate of a point in space

There are infinite number of points in space. We want to identify each and every point of space with the help of three mutually perpendicular coordinate axes OX, OY and OZ.
Three mutually perpendicular lines OX, OY, OZ are considered as the three axes.

The plane formed with the help of x and y axes is called x-y plane, similarly y & z axes form y-z plane and z and x axes form z - x plane.
Consider any point P in the space, Drop a perpendicular from that point to x -y plane, then the algebraic length of this perpendicular is considered as z-coordinate and from foot of the perpendicular drop perpendiculars to x and y axes. These algebraic lengths of perpendiculars are considered as y and x coordinates respectively.



1. Vector representation of a point in space

If coordinate of a point P in space is (x, y, z) then the position vector of the point P with respect to the same origin is $x\hat{i} + y\hat{j} + z\hat{k}$.

2. Distance formula

Distance between any two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given as $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

Vector method We know that if position vector of two points A and B are given as $\overrightarrow{OA}$ and $\overrightarrow{OB}$ then

$$AB = |\overrightarrow{OB} - \overrightarrow{OA}|$$

$$\Rightarrow \quad AB = |(x_2 i + y_2 j + z_2 k) - (x_1 i + y_1 j + z_1 k)| \Rightarrow AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

3. Distance of a point P from coordinate axes

Let PA, PB and PC are distances of the point P(x, y, z) from the coordinate axes OX, OY and OZ respectively then

$$PA = \sqrt{y^2 + z^2}\ ,\ PB = \sqrt{z^2 + x^2}\ ,\ PC = \sqrt{x^2 + y^2}$$

Example : Show that the points (0, 7, 10), (– 1, 6, 6) and (– 4, 9, 6) form a right angled isosceles triangle.
Solution: Let A ≡ (0, 7, 10), B ≡ (–1, 6, 6), C ≡ (– 4, 9, 6)

$AB^2 = (0 + 1)^2 + (7 - 6)^2 + (10 - 6)^2 = 18 \quad \therefore \quad AB = 3\sqrt{2}$

Similarly $\therefore$ $BC = 3\sqrt{2}$,
& $AC = 6$

Clearly $AB^2 + BC^2 = AC^2 \quad \therefore \quad \angle ABC = 90°$
Also $AB = BC$
Hence ΔABC is right angled isosceles.

Example : Show by using distance formula that the points (4, 5, –5), (0, –11, 3) and (2, –3, –1) are collinear.
Solution Let A ≡ (4, 5, –5), B ≡ (0, –11, 3), C ≡ (2, –3, –1).

$$AB = \sqrt{(4-0)^2 + (5+11)^2 + (-5-3)^2} = \sqrt{336} = \sqrt{4 \times 84} = 2\sqrt{84}$$

$$BC = \sqrt{(0-2)^2 + (-11+3)^2 + (3+1)^2} = \sqrt{84} \qquad AC = \sqrt{(4-2)^2 + (5+3)^2 + (-5+1)^2} = \sqrt{84}$$

$BC + AC = AB$
Hence points A, B, C are collinear and C lies between A and B.

Example : Find the locus of a point which moves such that the sum of its distances from points A(0, 0, $-\alpha$) and B(0, 0, α) is constant.
Solution. Let the variable point whose locus is required be P(x, y, z)
Given PA + PB = constant = 2a (say)

$$\therefore \quad \sqrt{(x-0)^2 + (y-0)^2 + (z+\alpha)^2} + \sqrt{(x-0)^2 + (y-0)^2 + (z-\alpha)^2} = 2a$$

$$\Rightarrow \quad \frac{1}{2} = 2a - \sqrt{x^2 + y^2 + (z-\alpha)^2}$$

$$\Rightarrow \quad x^2 + y^2 + z^2 + \alpha^2 + 2z\alpha = 4a^2 + x^2 + y^2 + z^2 + \alpha^2 - 2z\alpha - 4a\sqrt{x^2 + y^2 + (z-\alpha)^2}$$

$$\Rightarrow \quad 4z\alpha - 4a^2 = -4a\sqrt{x^2 + y^2 + (z-\alpha)^2}$$

$$\Rightarrow \quad \frac{z^2\alpha^2}{a^2} + a^2 - 2z\alpha = x^2 + y^2 + z^2 + \alpha^2 - 2z\alpha$$

or, $x^2 + y^2 + z^2\left(1-\frac{\alpha^2}{a^2}\right) = a^2 - \alpha^2$ or, $\frac{x^2+y^2}{a^2-\alpha^2}+\frac{z^2}{a^2} = 1$ This is the required locus.

Self practice problems :

1. One of the vertices of a cuboid is (1, 2, 3) and the edges from this vertex are along the +ve x-axis, +ve y-axis and +z axis respectively and are of length 2, 3, 2 respectively find out the vertices.
Ans. (1, 2, 5), (3, 2, 5), (3, 2, 3), (1, – 1, 3), (1, – 1, 5), (3, – 1, 5), (3, – 1, 3).

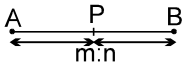
2. Show that the points (0, 4, 1), (2, 3, –1), (4, 5, 0) and (2, 6, 2) are the vertices of a square.

3. Find the locus of point P if $AP^2 - BP^2 = 18$, where A ≡ (1, 2, – 3) and B ≡ (3, – 2, 1)
Ans. $2x - 4y + 4z - 9 = 0$

4. Section Formula

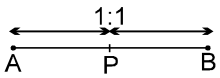
If point P divides the distance between the points A (x_1, y_1, z_1) and B (x_2, y_2, z_2) in the ratio of m : n, then coordinates of P are given as

$$\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}, \frac{mz_2+nz_1}{m+n}\right)$$



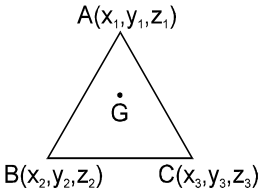
Note :- Mid point

$$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$



5. Centroid of a triangle

$$G \equiv \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}, \frac{z_1+z_2+z_3}{3}\right)$$



6. Incentre of triangle ABC: $\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}, \frac{az_1+bz_2+cz_3}{a+b+c}\right)$

Where AB = c, BC = a, CA = b

7. Centroid of a tetrahedron

A (x_1, y_1, z_1) B (x_2, y_2, z_2) C (x_3, y_3, z_3) and D (x_4, y_4, z_4) are the vertices of a tetrahedron then coordinate of its centroid (G) is given as

$$\left(\frac{\sum x_i}{4}, \frac{\sum y_i}{4}, \frac{\sum z_i}{4}\right)$$

Example : Show that the points A(2, 3, 4), B(–1, 2, –3) and C(–4, 1, –10) are collinear. Also find the ratio in which C divides AB.

Solution: Given A ≡ (2, 3, 4), B ≡ (–1, 2, –3), C ≡ (– 4, 1, –10).

A (2, 3, 4) B (–1, 2, –3)

Let C divide AB internally in the ratio k : 1, then

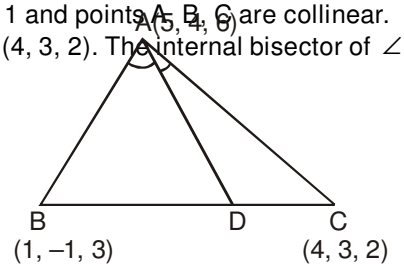
$$C \equiv \left(\frac{-k+2}{k+1}, \frac{2k+3}{k+1}, \frac{-3k+4}{k+1}\right)$$

$\therefore \quad \frac{-k+2}{k+1} = -4 \quad \Rightarrow \quad 3k = -6 \Rightarrow \quad k = -2$

For this value of k, $\frac{2k+3}{k+1} = 1$, and $\frac{-3k+4}{k+1} = -10$

Since k < 0, therefore k divides AB externally in the ratio 2 : 1 and points A, B, C are collinear.

Example : The vertices of a triangle are A(5, 4, 6), B(1, –1, 3) and C(4, 3, 2). The internal bisector of $\angle BAC$ meets BC in D. Find AD.



Solution $AB = \sqrt{4^2+5^2+3^2} = 5\sqrt{2}$

$AC = \sqrt{1^2+1^2+4^2} = 3\sqrt{2}$

Since AD is the internal bisector of $\angle BAC$

$\therefore \quad \frac{BD}{DC} = \frac{AB}{AC} = \frac{5}{3}$

$\therefore$ D divides BC internally in the ratio 5 : 3

$\therefore \quad D \equiv \left(\frac{5\times4+3\times1}{5+3}, \frac{5\times3+3(-1)}{5+3}, \frac{5\times2+3\times3}{5+3}\right)$ or, $D = \left(\frac{23}{8}, \frac{12}{8}, \frac{19}{8}\right)$

$\therefore \quad AD = \sqrt{\left(5-\frac{23}{8}\right)^2+\left(4-\frac{12}{8}\right)^2+\left(6-\frac{19}{8}\right)^2}$

$= \frac{\sqrt{1530}}{8}$ unit

Example :

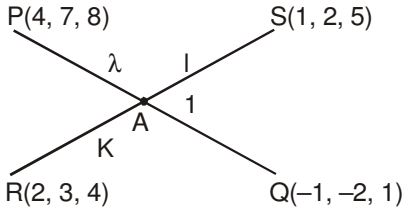
If the points P, Q, R, S are (4, 7, 8), (– 1, – 2, 1), (2, 3, 4) and (1, 2, 5) respectively, show that PQ and RS intersect. Also find the point of intersection.

Solution

Let the lines PQ and RS intersect at point A.

Let A divide PQ in the ratio λ : 1, then

$$A \equiv \left(\frac{-\lambda+4}{\lambda+1}, \frac{-2\lambda+7}{\lambda+1}, \frac{\lambda+8}{\lambda+1}\right). \quad \text{.... (1)}$$



Let A divide RS in the ratio k : 1, then

$$A \equiv \left(\frac{k+2}{k+1}, \frac{2k+3}{k+1}, \frac{5k+4}{k+1}\right) \quad \text{..... (2)}$$

From (1) and (2), we have,

$$\frac{-\lambda+4}{\lambda+1} = \frac{k+2}{k+1} \quad \text{..... (3)}$$

$$\frac{-2\lambda+7}{\lambda+1} = \frac{2k+3}{k+1} \quad \text{..... (4)}$$

$$\frac{\lambda+8}{\lambda+1} = \frac{5k+4}{k+1} \quad \text{..... (5)}$$

From (3), $-\lambda k - \lambda + 4k + 4 = \lambda k + 2\lambda + k + 2$

or $2\lambda k + 3\lambda - 3k - 2 = 0$ ..... (6)

From (4), $-2\lambda k - 2\lambda + 7k + 7 = 2\lambda k + 3\lambda + 2k + 3$

or $4\lambda k + 5\lambda - 5k - 4 = 0$ ..... (7)

Multiplying equation (6) by 2, and subtracting from equation (7), we get

$-\lambda + k = 0$ or, $\lambda = k$

Putting $\lambda = k$ in equation (6), we get

$2\lambda^2 + 3\lambda - 3\lambda - 2 = 0$

or, $\lambda = \pm 1$.

But $\lambda \neq -1$, as the co-ordinates of P would then be underfined and in this case PQ || RS, which is not true.

$\therefore$ $\lambda = 1 = k$.

Clearly $\lambda k = 1$ satisfies eqn. (5).

Hence our assumption is correct

$\therefore$ $A \equiv \left(\frac{-1+4}{2}, \frac{-2+7}{2}, \frac{1+8}{2}\right)$ or, $A \equiv \left(\frac{3}{2}, \frac{5}{2}, \frac{9}{2}\right)$.

Self practice problems:

1. Find the ratio in which xy plane divides the line joining the points A (1, 2, 3) and B (2, 3, 6).
Ans. – 1 : 2

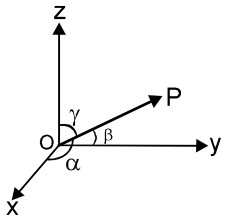
2. Find the co-ordinates of the foot of perpendicular drawn from the point A(1, 2, 1) to the line joining the point B(1, 4, 6) and C(5, 4, 4).
Ans. (3, 4, 5)

3. Two vertices of a triangle are (4, –6, 3) and (2, –2, 1) and its centroid is $\left(\frac{8}{3}, -1, 2\right)$. Find the third vertex.
Ans. (2, 5, 2)

4. If centroid of the tetrahedron OABC, where co-ordinates of A, B, C are (a, 2, 3), (1, b, 2) and (2, 1, c) respectively be (1, 2, 3), then find the distance of point (a, b, c) from the origin.
Ans. $\sqrt{107}$

5. Show that $\left(-\frac{1}{2}, 2, 0\right)$ is the circumcentre of the triangle whose vertices are A (1, 1, 0), B (1, 2, 1) and C (– 2, 2, –1) and hence find its orthocentre. **Ans.** (1, 11, 0)

8. Direction Cosines And Direction Ratios



(i) Direction cosines: Let α, β, γ be the angles which a directed line makes with the positive directions of the axes of x, y and z respectively, then $\cos\alpha, \cos\beta, \cos\gamma$ are called the direction cosines of the line. The direction cosines are usually denoted by (ℓ, m, n).
Thus $\ell = \cos\alpha$, $m = \cos\beta$, $n = \cos\gamma$.

(ii) If ℓ, m, n be the direction cosines of a line, then $\ell^2 + m^2 + n^2 = 1$

(iii) Direction ratios: Let a, b, c be proportional to the direction cosines ℓ, m, n then a, b, c are called the direction ratios.

If a, b, c, are the direction ratios of any line L then $a\hat{i} + b\hat{j} + c\hat{k}$ will be a vector parallel to the line L.

If ℓ, m, n are direction cosines of line L then $\ell\hat{i} + m\hat{j} + n\hat{k}$ is a unit vector parallel to the line L.

(iv) If ℓ, m, n be the direction cosines and a, b, c be the direction ratios of a vector, then

$$\left(\ell = \frac{a}{\sqrt{a^2+b^2+c^2}}, m = \frac{b}{\sqrt{a^2+b^2+c^2}}, n = \frac{c}{\sqrt{a^2+b^2+c^2}}\right)$$

or $\ell = \frac{-a}{\sqrt{a^2+b^2+c^2}}$, $m = \frac{-b}{\sqrt{a^2+b^2+c^2}}$, $n = \frac{-c}{\sqrt{a^2+b^2+c^2}}$

(v) If OP = r, when O is the origin and the direction cosines of OP are ℓ, m, n then the coordinates of P are (ℓr, mr, nr).

If direction cosines of the line AB are ℓ, m,n, |AB| = r, and the coordinates of A is (x_1, y_1, z_1) then the coordinates of B is given as $(x_1 + r\ell, y_1 + rm, z_1 + rn)$

(vi) If the coordinates P and Q are (x_1, y_1, z_1) and (x_2, y_2, z_2) then the direction ratios of line PQ are,

$a = x_2 - x_1$, $b = y_2 - y_1$ & $c = z_2 - z_1$ and the direction cosines of line PQ are $\ell = \frac{x_2 - x_1}{|PQ|}$,

$m = \frac{y_2 - y_1}{|PQ|}$ and $n = \frac{z_2 - z_1}{|PQ|}$

(vii) Direction cosines of axes**:** Since the positive x–axis makes angles 0º, 90º, 90º with axes of x, y and z respectively. Therefore

Direction cosines of x–axis are (1, 0, 0)

Direction cosines of y–axis are (0, 1, 0)

Direction cosines of z–axis are (0, 0, 1)

Example : If a line makes angles α, β, γ with the co-ordinate axes, prove that $\sin^2\alpha + \sin^2\beta + \sin\gamma^2 = 2$.

Solution Since a line makes angles α, β, γ with the co-ordinate axes,

hence $\cos\alpha$, $\cos\beta$, $\cos\gamma$ are its direction cosines

$\therefore$ $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow$ $(1 - \sin^2\alpha) + (1 - \sin^2\beta) + (1 - \sin^2\gamma) = 1$

$\Rightarrow$ $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$.

Example : Find the direction cosines ℓ, m, n of a line which are connected by the relations $\ell + m + n = 0$, $2mn + 2m\ell - n\ell = 0$

Solution Given, $\ell + m + n = 0$ ..... (1)

$2mn + 2m\ell - n\ell = 0$ ..... (2)

From (1), $n = -(\ell + m)$.

Putting $n = -(\ell + m)$ in equation (2), we get,

$-2m(\ell + m) + 2m\ell + (\ell + m)\ell = 0$

or, $-2m\ell - 2m^2 + 2m\ell + \ell^2 + m\ell = 0$

or, $\ell^2 + m\ell - 2m^2 = 0$

or, $\left(\frac{\ell}{m}\right)^2 + \left(\frac{\ell}{m}\right) - 2 = 0$ [dividing by m^2]

or $\frac{\ell}{m} = \frac{-1 \pm \sqrt{1+8}}{2} = \frac{-1 \pm 3}{2} = 1, -2$

Case I. when $\frac{\ell}{m} = 1$: In this case $m = \ell$

From (1), $2\ell + n = 0 \Rightarrow n = -2\ell$

$\therefore$ $\ell : m : n = 1 : 1 : -2$

$\therefore$ Direction ratios of the line are 1, 1, – 2

$\therefore$ Direction cosines are

$$\pm \frac{1}{\sqrt{1^2+1^2+(-2)^2}}, \pm \frac{1}{\sqrt{1^2+1^2+(-2)^2}}, \pm \frac{-2}{\sqrt{1^2+1^2+(-2)^2}}$$

or, $\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}$ or $-\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}$

Case II. When $\frac{\ell}{m} = -2$: In this case $\ell = -2m$

From (1), $-2m + m + n = 0 \Rightarrow n = m$

$\therefore$ $\ell : m : n = -2m : m : m$

$= -2 : 1 : 1$

$\therefore$ Direction ratios of the line are – 2, 1, 1.

$\therefore$ Direction cosines are

$$\frac{-2}{\sqrt{(-2)^2+1^2+1^2}}, \frac{-1}{\sqrt{(-2)^2+1^2+1^2}}, \frac{-1}{\sqrt{(-2)^2+1^2+1^2}} \quad \text{or,} \quad \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}.$$

Self practice problems:

1. Find the direction cosine of a line lying in the xy plane and making angle 30° with x-axis.

Ans. $m = \pm \frac{1}{2}$, $\ell = \frac{\sqrt{3}}{2}$, $n = 0$

2. A line makes an angle of 60° with each of x and y axes, find the angle which this line makes with z-axis. **Ans.** 45°

3. A plane intersects the co-ordinates axes at point A(a, 0, 0), B(0, b, 0), C(0, 0, c) O is origin. Find the direction ratio of the line joining the vertex B to the centroid of face AOC.

Ans. $\frac{a}{3}, -b, \frac{c}{3}$

4. A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube, prove that

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{4}{3}$.

9. Angle Between Two Line Segments:

If two lines have direction ratios a_1, b_1, c_1 and a_2, b_2, c_2 respectively then we can consider two vectors parallel to the lines as $a_1i + b_1j + c_1k$ and $a_2i + b_2j + c_2k$ and angle between them can be given as.

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}.$$

(i) The line will be perpendicular if $a_1a_2 + b_1b_2 + c_1c_2 = 0$ **(ii)** The lines will be parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(iii) Two parallel lines have same direction cosines i.e. $\ell_1 = \ell_2$, $m_1 = m_2$, $n_1 = n_2$

Example : What is the angle between the lines whose direction cosines are $-\frac{\sqrt{3}}{4}, \frac{1}{4}, -\frac{\sqrt{3}}{2}; -\frac{\sqrt{3}}{4}, \frac{1}{4}, \frac{\sqrt{3}}{2}$?

Solution Let θ be the required angle, then

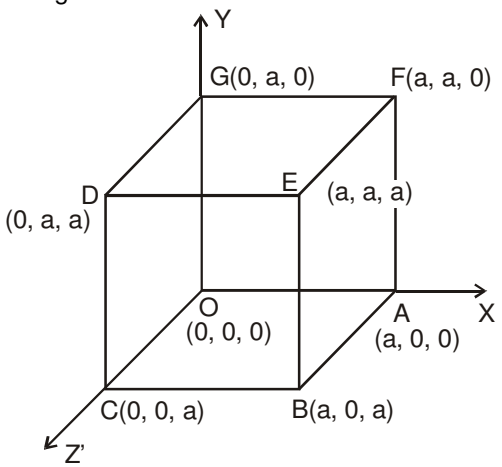
$\cos\theta = \ell_1\ell_2 + m_1m_2 + n_1n_2$

$$= \left(-\frac{\sqrt{3}}{4}\right)\left(-\frac{\sqrt{3}}{4}\right) + \left(\frac{1}{4}\right)\left(\frac{1}{4}\right) + \left(-\frac{\sqrt{3}}{2}\right)\cdot\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{3}{16} + \frac{1}{16} - \frac{3}{4} = -\frac{1}{2} \qquad \Rightarrow \qquad \theta = 120°,$$

Example : Find the angle between any two diagonals of a cube.

Solution The cube has four diagonals



OE, AD, CF and GB

The direction ratios of OE are

a, a, a or, 1, 1, 1

$\therefore$ its direction cosines are $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$.

Direction ratios of AD are – a, a, a. or, – 1, 1, 1.

$\therefore$ its direction cosines are $\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$.

Similarly, direction cosines of CF and GB respectively are

$\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}$ and $\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$.

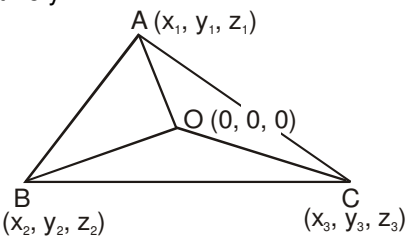
We take any two diagonals, say OE and AD

Let θ be the acute angle between them, then

$$\cos\theta = \left|\left(\frac{1}{\sqrt{3}}\right)\left(\frac{-1}{\sqrt{3}}\right) + \left(\frac{1}{\sqrt{3}}\right)\cdot\left(\frac{1}{\sqrt{3}}\right) + \left(\frac{1}{\sqrt{3}}\right)\cdot\left(\frac{1}{\sqrt{3}}\right)\right| = \frac{1}{3} \qquad \text{or,} \qquad \theta = \cos^{-1}\left(\frac{1}{3}\right).$$

Example : If two pairs of opposite edges of a tetrahedron are mutually perpendicular, show that the third pair will also be mutually perpendicular.

Solution: Let OABC be the tetrahedron where O is the origin and co-ordinates of A, B, C be (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, x_3) respectively.



Let OA ⊥ BC and OB ⊥ CA .

We have to prove that

OC ⊥ BA .

Now, direction ratios of OA are $x_1 - 0, y_1 - 0, z_1 - 0$ or, x_1, y_1, z_1

direction ratios of BC are $(x_3 - x_2), (y_3 - y_2), (z_3 - z_2)$.

$\because$ OA ⊥ BC .

$\therefore$ $x_1(x_3 - x_2) + y_1(y_3 - y_2) + z_1(z_3 - z_2) = 0$ ..... (1)

Similarly,

$\because$ OB ⊥ CA

$\therefore$ $x_2(x_1 - x_3) + y_2(y_1 - y_3) + z_2(z_1 - z_3) = 0$ ..... (2)

Adding equations (1) and (2), we get

$x_3(x_1 - x_2) + y_3(y_1 - y_2) + z_3(z_1 - z_2) = 0$

$\therefore$ OC ⊥ BA [$\because$ direction ratios of OC are x_3, y_3, z_3 and that of BA are $(x_1 - x_2), (y_1 - y_2), (z_1 - z_2)$]

Self practice problems:

1. Find the angle between the lines whose direction cosines are given by $\ell + m + n = 0$ and $\ell^2 + m^2 - n^2 = 0$ **Ans.** 60°

2. P (6, 3, 2)
Q (5, 1, 4)
R (3, 3, 5)
are vertices of a Δ find ∠Q. **Ans.** 90°

3. Show that the direction cosines of a line which is perpendicular to the lines having directions cosines $\ell_1\ m_1\ n_1$ and $\ell_2\ m_2\ n_2$ respectively are proportional to
$m_1n_2 - m_2n_1$, $n_1\ell_2 - n_2\ell_1$, $\ell_1m_2 - \ell_2m_1$

10. Projection of a line segment on a line

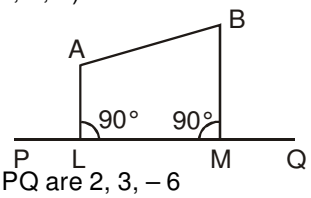
(i) If the coordinates P and Q are (x_1, y_1, z_1) and (x_2, y_2, z_2) then the projection of the line segments PQ on a line having direction cosines ℓ, m, n is $\left|\ell(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)\right|$

(ii) Vector form: projection of a vector $\vec{a}$ on another vector $\vec{b}$ is $\vec{a}\,.\,\hat{b} = \dfrac{\vec{a}\,.\,\vec{b}}{|\vec{b}|}$

In the above case we can consider $\overrightarrow{PQ}$ as $(x_2 - x_1)\,\hat{i} + (y_2 - y_1)\,\hat{j} + (z_2 - z_1)\,\hat{k}$ in place of $\vec{a}$ and $\ell\,\hat{i} + m\,\hat{j} + n\hat{k}$ in place of $\vec{b}$. **(iii)** $\ell\,|\vec{r}\,|$, $m\,|\vec{r}\,|$ & $n\,|\vec{r}\,|$ are the projection of $\vec{r}$ in OX, OY & OZ axes. **(iv)** $\vec{r} = |\vec{r}\,|\,(\ell\,\hat{i} + m\,\hat{j} + n\,\hat{k})$

Solved Example : Find the projection of the line joining (1, 2, 3) and (–1, 4, 2) on the line having direction ratios 2, 3, – 6.

Solution Let A ≡ (1, 2, 3), B ≡ (–1, 4, 2)



Direction ratios of the given line PQ are 2, 3, – 6

$\sqrt{2^2 + 3^2 + (-6)^2} = 7$ $\therefore$ direction cosines of PQ are

$\dfrac{2}{7}, \dfrac{3}{7}, -\dfrac{6}{7}$

Projection of AB on PQ

$= \ell\,(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$

$= \dfrac{2}{7}\,(-1 - 1) + \dfrac{3}{7}\,(4 - 2) - \dfrac{6}{7}\,(2 - 3)$ $= \dfrac{-4 + 6 + 6}{7} = \dfrac{8}{7}$

Self practice problems:

1. A (6, 3, 2), B (5, 1, 1,), C(3, –1, 3) D (0, 2, 5)
Find the projection of line segment AB on CD line. **Ans.** 5/7

2. The projections of a directed line segment on co-ordinate axes are – 2, 3, – 6. Find its length and direction cosines. **Ans.** 13 ; $\frac{12}{13}, \frac{4}{13}, \frac{3}{13}$

3. Find the projection of the line segment joining (2, – 1, 3) and (4, 2, 5) on a line which makes equal acute angles with co-ordinate axes. **Ans.** $\frac{7}{\sqrt{3}}$

A PLANE

If line joining any two points on a surface lies completely on it then the surface is a plane.

OR

If line joining any two points on a surface is perpendicular to some fixed straight line. Then this surface is called a plane. This fixed line is called the normal to the plane.

11. Equation Of A Plane

(i) Normal form of the equation of a plane is $\ell x + my + nz = p$, where, ℓ,m n are the direction cosines of the normal to the plane and p is the distance of the plane from the origin.

(ii) General form: $ax + by + cz + d = 0$ is the equation of a plane, where a, b, c are the direction ratios of the normal to the plane.

(iii) The equation of a plane passing through the point (x_1, y_1, z_1) is given by $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$ where a, b, c are the direction ratios of the normal to the plane.

(iv) Plane through three points: The equation of the plane through three non–collinear points $(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$ is $\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$

(v) Intercept Form: The equation of a plane cutting intercept a, b, c on the axes is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

(vi) Vector form: The equation of a plane passing through a point having position vector $\vec{a}$ & normal to vector $\vec{n}$ is $(\vec{r} - \vec{a}).\vec{n} = 0$ or $\vec{r}.\vec{n} = \vec{a}.\vec{n}$

Note: (a) Vector equation of a plane normal to unitvector $\hat{n}$ and at a distance d from the origin is $\vec{r} \cdot \vec{n} = d$

(b) **Coordinate planes (i)** Equation of yz–plane is $x = 0$ **(ii)** Equation of xz–plane is $y = 0$ **(iii)** Equation of xy–plane is $z = 0$

(c) **Planes parallel to the axes:**
If a = 0, the plane is parallel to x–axis i.e. equation of the plane parallel to the x–axis is $by + cz + d = 0$.
Similarly, equation of planes parallel to y–axis and parallel to z–axis are $ax + cz + d = 0$ and $ax + by + d = 0$ respectively.

(d) **Plane through origin:** Equation of plane passing through origin is $ax + by + cz = 0$.

(e) **Transformation of the equation of a plane to the normal form:** To reduce any equation $ax + by + cz - d = 0$ to the normal form, first write the constant term on the right hand side and make it positive, then divide each term by $\sqrt{a^2 + b^2 + c^2}$, where a, b, c are coefficients of x, y and z respectively e.g.

$$\frac{ax}{\pm\sqrt{a^2 + b^2 + c^2}} + \frac{by}{\pm\sqrt{a^2 + b^2 + c^2}} + \frac{cz}{\pm\sqrt{a^2 + b^2 + c^2}} = \frac{d}{\pm\sqrt{a^2 + b^2 + c^2}}$$

Where (+) sign is to be taken if $d > 0$ and (–) sign is to be taken if $d < 0$.

(f) Any plane parallel to the given plane $ax + by + cz + d = 0$ is $ax + by + cz + \lambda = 0$.
Distance between two parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is given as $\frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$

(g) **Equation of a plane passing through a given point & parallel to the given vectors:**
The equation of a plane passing through a point having position vector $\vec{a}$ and parallel to $\vec{b}$ & $\vec{c}$ is $\vec{r} = \vec{a} + \lambda\vec{b} + \mu\vec{c}$ (parametric form) where λ & μ are scalars.
or $\vec{r}.(\vec{b} \times \vec{c}) = \vec{a}.(\vec{b} \times \vec{c})$ (non parametric form)

(h) A plane $ax + by + cz + d = 0$ divides the line segment joining (x_1, y_1, z_1) and (x_2, y_2, z_2). in the ratio $\left(-\frac{ax_1 + by_1 + cz_1 + d}{ax_2 + by_2 + cz_2 + d}\right)$

(i) The xy–plane divides the line segment joining the points (x_1, y_1, z_1) and (x_2, y_2, z_2) in the ratio $-\frac{z_1}{z_2}$. Similarly yz–plane in $-\frac{x_1}{x_2}$ and zx–plane in $-\frac{y_1}{y_2}$

(j) Coplanarity of four points

The points $A(x_1\, y_1\, z_1)$, $B(x_2\, y_2 z_2)$ $C(x_3 y_3 z_3)$ and $D(x_4\, y_4\, z_4)$ are coplaner then

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \\ x_4 - x_1 & y_4 - y_1 & z_4 - z_1 \end{vmatrix} = 0$$

very similar in vector method the points A $(\vec{r}_1)$, B$(\vec{r}_2)$, C$(\vec{r}_3)$ and D$(\vec{r}_4)$ are coplanar if $[\vec{r}_4 - \vec{r}_1, \vec{r}_4 - \vec{r}_2, \vec{r}_4 - \vec{r}_3] = 0$

Example : Find the equation of the plane upon which the length of normal from origin is 10 and direction ratios of this normal are 3, 2, 6.

Solution If p be the length of perpendicular from origin to the plane and ℓ, m, n be the direction cosines of this normal, then its equation is

$\ell x + my + nz = p$ ..... (1)

Here p = 10; Direction ratios of normal to the plane are 3, 2, 6

$\sqrt{3^2 + 2^2 + 6^2} = 7$ $\therefore$ Direction cosines of normal to the required plane are

$\ell = \frac{3}{7}$, m = $\frac{2}{7}$, n = $\frac{6}{7}$

Putting the values of ℓ, m, n, p in (1), equation of required plane is

$\frac{3}{7}x + \frac{2}{7}y + \frac{6}{7}z = 10$ or, $3x + 2y + 6z = 70$

Example : Show that the points (0, – 1, 0), (2, 1, – 1), (1, 1, 1), (3, 3, 0) are coplanar.

Solution Let A ≡ (0, – 1, 0), B ≡ (2, 1, – 1), C ≡ (1, 1, 1) and D ≡ (3, 3, 0)

Equation of a plane through A (0, – 1, 0) is

$a(x - 0) + b(y + 1) + c(z - 0) = 0$

or, $ax + by + cz + b = 0$ ..... (1)

If plane (1) passes through B (2, 1, – 1) and C (1, 1, 1)

Then $2a + 2b - c = 0$ ..... (2)

and $a + 2b + c = 0$ ..... (3)

From (2) and (3), we have

$$\frac{a}{2+2} = \frac{b}{-1-2} = \frac{c}{4-2}$$

or, $\frac{a}{4} = \frac{b}{-3} = \frac{c}{2}$ = k (say)

Putting the value of a, b, c, in (1), equation of required plane is

$4kx - 3k(y + 1) + 2kz = 0$

or, $4x - 3y + 2z - 3 = 0$ ..... (2)

Clearly point D (3, 3, 0) lies on plane (2)

Thus points D lies on the plane passing through A, B, C and hence points A, B, C and D are coplanar.

Example : If P be any point on the plane $\ell x + my + nz = p$ and Q be a point on the line OP such that OP . OQ = p^2, show that the locus of the point Q is $p(\ell x + my + nz) = x^2 + y^2 + z^2$.

Solution Let P ≡ (α, β, γ), Q ≡ (x_1, y_1, z_1)

Direction ratios of OP are α, β, γ and direction ratios of OQ are x_1, y_1, z_1.

Since O, Q, P are collinear, we have

$$\frac{\alpha}{x_1} = \frac{\beta}{y_1} = \frac{\gamma}{z_1} = k \text{ (say)} \quad \text{..... (1)}$$

As P (α, β, γ) lies on the plane $\ell x + my + nz = p$,

$\ell\alpha + m\beta + n\gamma = p$ or $k(x_1 + my_1 + nz_1) = p$ ..... (2)

Given, OP . OQ = p^2

$\therefore$ $\sqrt{\alpha^2 + \beta^2 + \gamma^2}\ \sqrt{x_1^2 + y_1^2 + z_1^2} = p^2$

or, $\sqrt{k^2(x_1^2 + y_1^2 + z_1^2)}\ \sqrt{x_1^2 + y_1^2 + z_1^2} = p^2$

or, $k(x_1^2 + y_1^2 + z_1^2) = p^2$ ..... (3)

On dividing (2) by (3), we get,

$$\frac{\ell x_1 + my_1 + nz_1}{x_1^2 + y_1^2 + z_1^2} = \frac{1}{p}$$

or, $p(\ell x_1 + my_1 + nz_1) = x_1^2 + y_1^2 + z_1^2$

Hence the locus of point Q is

$p(\ell x + my + nz) = x^2 + y^2 + z^2$.

Example : A point P moves on a plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. A plane through P and perpendicular to OP meets the

co-ordinate axes in A, B and C. If the planes through A, B and C parallel to the planes x = 0, y = 0, z = 0 intersect in Q, find the locus of Q.

Solution Given plane is

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1 \qquad \text{..... (1)}$$

Let P ≡ (h, k, ℓ)

Then $\frac{h}{a}+\frac{k}{b}+\frac{\ell}{c}=1$ ..... (2)

$OP = \sqrt{h^2+k^2+\ell^2}$

Direction cosines of OP are $\frac{h}{\sqrt{h^2+k^2+\ell^2}}, \frac{k}{\sqrt{h^2+k^2+\ell^2}}, \frac{\ell}{\sqrt{h^2+k^2+\ell^2}}$

∴ Equation of the plane through P and normal to OP is

$$\frac{h}{\sqrt{h^2+k^2+\ell^2}}x+\frac{k}{\sqrt{h^2+k^2+\ell^2}}y+\frac{\ell}{\sqrt{h^2+k^2+\ell^2}}=\sqrt{h^2+k^2+\ell^2}$$

or, $hx + ky + \ell z = (h^2 + k^2 + \ell^2)$

∴ $A \equiv \left(\frac{h^2+k^2+\ell^2}{h}, 0, 0\right)$, $B \equiv \left(0, \frac{h^2+k^2+\ell^2}{k}, 0\right)$,

$C \equiv \left(0, 0, \frac{h^2+k^2+\ell^2}{\ell}\right)$

Let Q ≡ (α, β, γ), then

$$\alpha = \frac{h^2+k^2+\ell^2}{h}, \beta = \frac{h^2+k^2+\ell^2}{k}, \gamma = \frac{h^2+k^2+\ell^2}{\ell} \qquad \text{..... (3)}$$

Now $\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}=\frac{h^2+k^2+\ell^2}{(h^2+k^2+\ell^2)^2}=\frac{1}{(h^2+k^2+\ell^2)}$ ..... (4)

From (3), $h = \frac{h^2+k^2+\ell^2}{\alpha}$

∴ $\frac{h}{a}=\frac{h^2+k^2+\ell^2}{a\alpha}$

Similarly $\frac{k}{b}=\frac{h^2+k^2+\ell^2}{b\beta}$ and $\frac{\ell}{c}=\frac{h^2+k^2+\ell^2}{c\gamma}$

∴ $\frac{h^2+k^2+\ell^2}{a\alpha}+\frac{h^2+k^2+\ell^2}{b\beta}+\frac{h^2+k^2+\ell^2}{c\gamma}=\frac{h}{a}+\frac{k}{b}+\frac{\ell}{c}=1$ [from (2)]

or, $\frac{1}{a\alpha}+\frac{1}{b\beta}+\frac{1}{c\gamma}=\frac{1}{h^2+k^2+\ell^2}=\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}$ [from (4)]

∴ Required locus of Q (α, β, γ) is

$$\frac{1}{ax}+\frac{1}{by}+\frac{1}{cz}=\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}.$$

Self practice problems :

1. Check wether this point are coplanar if yes find the equation of plane containing them
A ≡ (1, 1, 1)
B ≡ (0, – 1, 0)
C ≡ (2, 1, –1)
D ≡ (3, 3, 0) **Ans.** yes, 4x – 3y + 2z = 3

2. Find the plane passing through point (– 3, – 3, 1) and perpendicular to the line joining the points (2, 6, 1) and (1, 3, 0). **Ans.** x + 3y + z + 11 = 0

3. Find the equation of plane parallel to x + 5y – 4z + 5 = 0 and cutting intercepts on the axes whose rum is 150. **Ans.** $x + 5y - 4z = \frac{3000}{19}$

4. Find the equation of plane passing through (2, 2, 1) and (9, 3, 6) and perpendicular to the plane x + 3y + 3z = 8. **Ans.** 3x + 4y – 5z = 9

5. Find the equation of the plane | | to $\hat{i}+\hat{j}+\hat{k}$ and $\hat{i}-\hat{j}$ and passing through (1, 1, 2).
Ans. x + y – 2z + 2 = 0

6. Find the equation of the plane passing through the point (1, 1, – 1) and perpendicular to the planes x + 2y + 3z – 7 = 0 and 2x – 3y + 4z = 0. **Ans.** 17x + 2y – 7z = 26

12. Sides of a plane:

A plane divides the three dimensional space in two equal parts. Two points A (x_1 y_1 z_1) and B (x_2 y_2 z_2) are on the same side of the plane ax + by + cz + d = 0 if $ax_1 + by_1 + cz_1 + d$ and

$ax_2 + by_2 + cz_2 + d$ are both positive or both negative and are opposite side of plane if both of these values are in opposite sign.

Example : Show that the points (1, 2, 3) and (2, – 1, 4) lie on opposite sides of the plane x + 4y + z – 3 = 0.

Solution Since the numbers 1+ 4 × 2 + 3 – 3 = 9 and 2 – 4 + 4 – 3 = – 1 are of opposite sign., the points are on opposite sides of the plane.

13. A Plane & A Point

(i) Distance of the point (x′, y′, z′) from the plane ax + by + cz+ d = 0 is given by $\frac{ax'+by'+cz'+d}{\sqrt{a^2+b^2+c^2}}$.

(ii) The length of the perpendicular from a point having position vector $\vec{a}$ to plane $\vec{r}\,.\,\vec{n} = d$ is given by $p = \frac{|\vec{a}\,.\,\vec{n} - d|}{|\vec{n}|}$.

(iii) The coordinates of the foot of perpendicular from the point (x, y, z) to the plane ax + by + cz + d = 0 aregain by $\frac{x'-x_1}{a} = \frac{y'-y_1}{b} = \frac{z'-z_1}{c} = -\frac{(ax_1+by_1+cz_1+d)}{a^2+b^2+c^2}$

(iv) To find image of a point w.r.t. a plane.

Let P (x_1, y_1, z_1) is a given point and ax + by + cz + d = 0 is given plane Let (x′, y′, z′) is the image point. then

(a) $x' - x_1 = \lambda a, \quad y' - y_1 = \lambda b, \quad z' - z_1 = \lambda c$

$\Rightarrow \quad x' = \lambda a + x_1, \quad y' = \lambda b + y_1, \quad z' = \lambda c + z_1$

(b) $a\left(\frac{x'+x_1}{2}\right) + b\left(\frac{y'+y_1}{2}\right) + c\left(\frac{z'+z_1}{2}\right) = 0$

from (i) put the values of x′, y′, z′ in (ii) and get the values of λ and resubtitute in (i) to get (x′ y′ z′).

The coordinate of the image of point (x_1, y_1, z_1) w.r.t the plane ax + by + cz + d = 0 are given by $\frac{x'-x_1}{a} = \frac{y'-y_1}{b} = \frac{z'-z_1}{c} = -2\,\frac{(ax_1+by_1+cz_1+d)}{a^2+b^2+c^2}$

(v) The distance between two parallel planes ax + by + cx + d = 0 and ax + by + cx + d' = 0 is given by $\frac{|d-d'|}{\sqrt{a^2+b^2+c^2}}$

Example : Find the image of the point P (3, 5, 7) in the plane 2x + y + z = 16.

Solution Given plane is 2x + y + z = 16 ..... (1)

P ≡ (3, 5, 7)

Direction ratios of normal to plane (1) are 2, 1, 1

Let Q be the image of point P in plane (1). Let PQ meet plane (1) in R

then PQ ⊥ plane (1)

Let R ≡ (2r + 3, r + 5, r + 7)

Since R lies on plane (1)

∴ 2(2r + 3) + r + 5 + r + 7 = 0 or, 6r + 18 = 0 ∴ r = – 3

∴ R ≡ (– 3, 2, 4)

Let Q ≡ (α, β, γ)

Since R is the middle point of PQ

∴ $-3 = \frac{\alpha+3}{2} \Rightarrow \alpha = -9$

$2 = \frac{\beta+5}{2} \Rightarrow \beta = -1$

$4 = \frac{\gamma+7}{2} \Rightarrow \gamma = 1$ ∴ Q = (– 9, – 1, 1).

Example : Find the distance between the planes 2x – y + 2z = 4 and 6x – 3y + 6z = 2.

Solution Given planes are

2x – y + 2z – 4 = 0 ..... (1)

and 6x – 3y + 6z – 2 = 0 ..... (2)

We find that $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ Hence planes (1) and (2) are parallel.

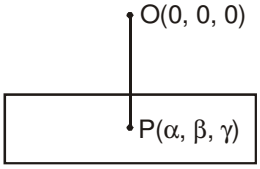
Plane (2) may be written as $2x - y + 2z - \frac{2}{3} = 0$ ..... (3)

∴ Required distance between the planes

$$= \frac{\left|4 - \frac{2}{3}\right|}{\sqrt{2^2+(-1)^2+2^2}} = \frac{10}{3.3} = \frac{10}{9}$$

Example : A plane passes through a fixed point (a, b, c). Show that the locus of the foot of perpendicular to it from the origin is the sphere $x^2 + y^2 + z^2 - ax - by - cz = 0$

Solution Let the equation of the variable plane be



$\ell x + my + nz + d = 0$ ..... (1)

Plane passes through the fixed point (a, b, c)

$\therefore$ $\ell a + mb + nc + d = 0$ ..... (2)

Let P (α, β, γ) be the foot of perpendicular from origin to plane (1).

Direction ratios of OP are

$\alpha - 0, \beta - 0, \gamma - 0$ i.e. α, β, γ

From equation (1), it is clear that the direction ratios of normal to the plane i.e. OP are ℓ, m, n ; α, β, γ and ℓ, m, n are the direction ratios of the same line OP

$\therefore$ $\dfrac{\alpha}{\ell} = \dfrac{\beta}{m} = \dfrac{\gamma}{n} = \dfrac{1}{k}$ (say)

$\therefore$ $\ell = k\alpha$, $m = k\beta$, $n = k\gamma$ ..... (3)

Putting the values of ℓ, m, n in equation (2), we get

$k\alpha + kb\beta + kc\gamma + d = 0$ ..... (4)

Since α, β, γ lies in plane (1)

$\therefore$ $\ell\alpha + m\beta + n\gamma + d = 0$ ..... (5)

Putting the values of ℓ, m, n from (3) in (5), we get

$k\alpha^2 + k\beta^2 + k\gamma^2 + d = 0$ ..... (6)

or $k\alpha^2 + k\beta^2 + k\gamma^2 - ka\alpha - kb\beta - kc\gamma = 0$

[putting the value of d from (4) in (6)]

or $\alpha^2 + \beta^2 + \gamma^2 - a\alpha - b\beta - c\gamma = 0$

Therefore, locus of foot of perpendicular P (α, β, γ) is

$x^2 + y^2 + z^2 - ax - by - cz = 0$ ..... (7)

Self practice problems:

1. Find the intercepts of the plane $3x + 4y - 7z = 84$ on the axes. Also find the length of perpendicular from origin to this line and direction cosines of this normal.

Ans. a = 28, b = 21, c = – 12, p = $\dfrac{1}{\sqrt{74}}; \dfrac{3}{\sqrt{74}}, \dfrac{4}{\sqrt{74}}, \dfrac{-7}{\sqrt{74}}$

2. Find : (i) perpendicular distance
(ii) foot of perpendicular
(iii) image of (1, 0, 2) in the plane $2x + y + z = 5$

Ans. (i) $\dfrac{1}{\sqrt{6}}$ (ii) $\left(\dfrac{4}{3}, \dfrac{1}{6}, \dfrac{13}{6}\right)$ (iii) $\left(\dfrac{5}{3}, \dfrac{1}{3}, \dfrac{7}{3}\right)$

14. Angle Between Two Planes:

(i) Consider two planes $ax + by + cz + d = 0$ and $a'x + b'y + c'z + d' = 0$. Angle between these planes is the angle between their normals. Since direction ratios of their normals are (a, b, c) and (a′, b′, c′) respectively, hence θ, the angle between them, is given by

$$\cos\theta = \frac{aa'+bb'+cc'}{\sqrt{a^2+b^2+c^2}\ \sqrt{a'^2+b'^2+c'^2}}$$

Planes are perpendicular if $aa' + bb' + cc' = 0$ and planes are parallel if $\dfrac{a}{a'} = \dfrac{b}{b'} = \dfrac{c}{c'}$

(ii) The angle θ between the planes $\vec{r}\,.\,\vec{n} = d_1$ and $\vec{r}\,.\,\vec{n}_2 = d_2$ is given by, $\cos\theta = \dfrac{\vec{n}_1\,.\,\vec{n}_2}{|\,\vec{n}_1\,|\,.\,|\,\vec{n}_2\,|}$

Planes are perpendicular if $\vec{n}_1\,.\,\vec{n}_2 = 0$ & planes are parallel if $\vec{n}_1 = \lambda\,\vec{n}_2$.

15. Angle Bisectors

(i) The equations of the planes bisecting the angle between two given planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ are

$$\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \pm\frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}}$$

(ii) Equation of bisector of the angle containing origin: First make both the constant terms positive.

Then the positive sign in $\dfrac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \pm\dfrac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}}$ gives the bisector of the angle which contains the origin.

(iii) Bisector of acute/obtuse angle: First make both the constant terms positive. Then

$a_1a_2 + b_1b_2 + c_1c_2 > 0 \Rightarrow$ origin lies on obtuse angle

$a_1a_2 + b_1b_2 + c_1c_2 < 0 \quad \Rightarrow$ origin lies in acute angle

16. Family of Planes

(i) Any plane passing through the line of intersection of non–parallel planes or equation of the plane through the given line in serval form.

$a_1x + b_1y + c_1z + d_1 = 0$ & $a_2x + b_2y + c_2z + d_2 = 0$ is

$a_1x + b_1y + c_1z + d_1 + \lambda (a_2x + b_2y + c_2z + d_2) = 0$

(ii) The equation of plane passing through the intersection of the planes $\vec{r} \cdot \vec{n}_1 = d_1$ & $\vec{r} \cdot \vec{n}_2 = d_2$ is $\vec{r} \cdot (n_1 + \lambda \vec{n}_2) = d_1 + \lambda d_2$ where λ is arbitrary scalar

Example : The plane $x – y – z = 4$ is rotated through 90° about its line of intersection with the plane $x + y + 2z = 4$. Find its equation in the new position.

Solution Given planes are

$x – y – z = 4$ ..... (1)

and $x + y + 2z = 4$ ..... (2)

Since the required plane passes through the line of intersection of planes (1) and (2)

$\therefore$ its equation may be taken as

$x + y + 2z – 4 + k (x – y – z – 4) = 0$

or $(1 + k)x + (1 – k)y + (2 – k)z – 4 – 4k = 0$ ..... (3)

Since planes (1) and (3) are mutually perpendicular,

$\therefore$ $(1 + k) – (1 – k) – (2 – k) = 0$

or, $1 + k – 1 + k – 2 + k = 0$ or, $k = \frac{2}{3}$

Putting $k = \frac{2}{3}$ in equation (3), we get,

$5x + y + 4z = 20$

This is the equation of the required plane.

Example : Find the equation of the plane through the point (1, 1, 1) which passes through the line of intersection of the planes $x + y + z = 6$ and $2x + 3y + 4z + 5 = 0$.

Solution Given planes are

$x + y + z – 6 = 0$ ..... (1)

and $2x + 3y + 4z + 5 = 0$ ..... (2)

Given point is P (1, 1, 1).

Equation of any plane through the line of intersection of planes (1) and (2) is

$x + y + z – 6 + k (2x + 3y + 4z + 5) = 0$ ..... (3)

If plane (3) passes through point P, then

$1 + 1 + 1 – 6 + k (2 + 3 + 4 + 5) = 0$ or, $k = \frac{3}{14}$

From (1) required plane is

$20x + 23y + 26z – 69 = 0$

Example : Find the planes bisecting the angles between planes

$2x + y + 2z = 9$ and $3x – 4y + 12z + 13 = 0$.

Which of these bisector planes bisects the acute angle between the given planes. Does origin lie in the acute angle or obtuse angle between the given planes ?

Solution Given planes are

$– 2x – y – 2z + 9 = 0$ ..... (1)

and $3x – 4y + 12z + 13 = 0$ ..... (2)

Equations of bisecting planes are

$$\frac{-2x - y - 2z + 9}{\sqrt{(-2)^2 + (-1)^2 + (-2)^2}} = \pm \frac{3x - 4y + 12z + 13}{\sqrt{3^2 + (-4)^2 + (12)^2}}$$

or, $13 [– 2x – y – 2z + 9] = \pm 3 (3x – 4y + 12z + 13)$

or, $35x + y + 62z = 78$, ... (3) [Taking +ve sign]

and $17x + 25y – 10z = 156$ ..... (4) [Taking – ve sign]

Now $a_1a_2 + b_1b_2 + c_1c_2 = (– 2) (3) + (– 1) (– 4) + (– 2) (12)$

$= – 6 + 4 – 24 = – 26 < 0$

$\therefore$ Bisector of acute angle is given by $35x + y + 62z = 78$

$\because$ $a_1a_2 + b_1b_2 + c_1c_2 < 0$, origin lies in the acute angle between the planes.

Example : If the planes $x – cy – bz = 0$, $cx – y + az = 0$ and $bx + ay – z = 0$ pass through a straight line, then find the value of $a^2 + b^2 + c^2 + 2abc$.

Solution Given planes are

$x – cy – bz = 0$ ..... (1)

$cx – y + az = 0$ ..... (2)

$bx + ay – z = 0$ ..... (3)

Equation of any plane passing through the line of intersection of planes (1) and (2) may be taken as

$x – cy – bz + \lambda (cx – y + az) = 0$

or, $x (1 + \lambda c) – y (c + \lambda) + z (– b + a\lambda) = 0$ ..... (4)

If planes (3) and (4) are the same, then equations (3) and (4) will be identical.

$$\therefore \quad \underset{(i)}{\frac{1 + c\lambda}{b}} = \underset{(ii)}{\frac{-(c + \lambda)}{a}} = \underset{(iii)}{\frac{-b + a\lambda}{-1}}$$

From (i) and (ii), $a + ac\lambda = – bc – b\lambda$

or, $\lambda = -\frac{(a+bc)}{(ac+b)}$ ..... (5)

From (ii) and (iii),

$c + \lambda = -ab + a^2\lambda$ or $\lambda = \frac{(ab+c)}{1-a^2}$ ..... (6)

From (5) and (6), we have,

$\frac{-(a+bc)}{ac+b} = \frac{-(ab+c)}{(1-a^2)}$.

or, $a - a^3 + bc - a^2bc = a^2bc + ac^2 + ab^2 + bc$

or, $a^2bc + ac^2 + ab^2 + a^3 + a^2bc - a = 0$

or, $a^2 + b^2 + c^2 + 2abc = 1$.

Self practice problems:

1. A tetrahedron has vertices at O(0, 0, 0), A(1, 2, 1), B(2, 1, 3) and C(–1, 1, 2). Prove that the angle between the faces OAB and ABC will be $\cos^{-1}\left(\frac{19}{35}\right)$.

2. Find the equation of plane passing through the line of intersection of the planes 4x – 5y – 4z = 1 and 2x + y + 2z = 8 and the point (2, 1, 3). **Ans.** 32x – 5y + 8z – 83 = 0, $\lambda = \frac{10}{3}$

3. Find the equations of the planes bisecting the angles between the planes x + 2y + 2z – 3 = 0, 3x + 4y + 12z + 1 = 0 and sepecify the plane which bisects the acute angle between them. **Ans.** 2x + 7y – 5z = 21, 11x + 19y + 31z = 18; 2x + 7y – 5z = 21

4. Show that the origin lies in the acute angle between the planes x + 2y + 2z – 9 = 0 and 4x – 3y + 12z + 13 = 0

5. Prove that the planes 12x – 15y + 16z – 28 = 0, 6x + 6y – 7z – 8 = 0 and 2x + 35y – 39z + 12 = 0 have a common line of intersection.

17. Area of a triangle:

Let A (x_1, y_1, z_1), B (x_2, y_2, z_2), C (x_3, y_3, z_3) be the vertices of a triangle, then $\Delta = \sqrt{(\Delta_x^2 + \Delta_y^2 + \Delta_z^2)}$

where $\Delta_x = \frac{1}{2}\begin{vmatrix} y_1 & z_1 & 1 \\ y_2 & z_2 & 1 \\ y_3 & z_3 & 1 \end{vmatrix}$, $\Delta_y = \frac{1}{2}\begin{vmatrix} z_1 & x_1 & 1 \\ z_2 & x_2 & 1 \\ z_3 & x_3 & 1 \end{vmatrix}$ and $\Delta_z = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

Vector Method – From two vector $\vec{AB}$ and $\vec{AC}$. Then area is given by

$$\frac{1}{2}\,|\vec{AB} \times \vec{AC}| = \frac{1}{2}\begin{vmatrix} i & j & k \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix}$$

Example : Through a point P (h, k, ℓ) a plane is drawn at right angles to OP to meet the co-ordinate axes in A, B and C. If OP = p, show that the area of ΔABC is $\frac{p^5}{2hk\ell}$.

Solution $OP = \sqrt{h^2 + k^2 + \ell^2} = p$

Direction cosines of OP are

$$\frac{h}{\sqrt{h^2+k^2+\ell^2}}, \frac{k}{\sqrt{h^2+k^2+\ell^2}}, \frac{\ell}{\sqrt{h^2+k^2+\ell^2}}$$

Since OP is normal to the plane, therefore, equation of the plane will be,

$$\frac{h}{\sqrt{h^2+k^2+\ell^2}}x + \frac{k}{\sqrt{h^2+k^2+\ell^2}}y + \frac{\ell}{\sqrt{h^2+k^2+\ell^2}}z = \sqrt{h^2+k^2+\ell^2}$$

or, $hx + ky + \ell z = h^2 + k^2 + \ell^2 = p^2$ ..... (1)

$\therefore$ $A \equiv \left(\frac{p^2}{h}, 0, 0\right)$, $B \equiv \left(0, \frac{p^2}{k}, 0\right)$, $C \equiv \left(0, 0, \frac{p^2}{\ell}\right)$

Now area of ΔABC, $\Delta = A^2_{xy} + A^2_{yz} + A^2_{zx}$

Now A_{xy} = area of projection of ΔABC on xy-plane = area of ΔAOB

$$= \text{Mod of } \frac{1}{2}\begin{vmatrix} \frac{p^2}{h} & 0 & 1 \\ 0 & \frac{p^2}{k} & 1 \\ 0 & 0 & 1 \end{vmatrix} = \frac{1}{2}\frac{p^4}{|hk|}$$

Similarly, $A_{yz} = \frac{1}{2}\frac{p^4}{|k\ell|}$ and $A_{zx} = \frac{1}{2}\frac{p^4}{|\ell h|}$

$\therefore \quad \Delta^2 = \frac{1}{4}\frac{p^8}{h^2k^2} + \frac{1}{4}\frac{p^8}{k^2\ell^2} + \frac{1}{4}\frac{p^8}{\ell^2h^2}$

$= \frac{p^8}{4h^2k^2\ell^2}(\ell^2 + k^2 + h^2) = \frac{p^{10}}{4h^2k^2\ell^2}$ or, $\Delta = \frac{p^5}{2hk\ell}$.

18. Volume Of A Tetrahedron:

Volume of a tetrahedron with vertices A (x_1, y_1, z_1), B(x_2, y_2, z_2), C (x_3, y_3, z_3) and D (x_4, y_4, z_4) is given by $V = \frac{1}{6}\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix}$

A LINE

19. Equation Of A Line

(i) A straight line in space is characterised by the intersection of two planes which are not parallel and therefore, the equation of a straight line is a solution of the system constituted by the equations of the two planes, $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + +c_2z + d_2 = 0$. This form is also known as non–symmetrical form.

(ii) The equation of a line passing through the point (x_1, y_1, z_1) and having direction ratios a, b, c is $\frac{x - x_1}{a}$ $= \frac{y - y_1}{b} = \frac{z - z_1}{c} = r$. This form is called symmetric form. A general point on the line is given by $(x_1 +$ *a*r, y_1 + br, z_1 + cr).

(iii) Vector equation: Vector equation of a straight line passing through a fixed point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is $\vec{r} = \vec{a} + \lambda\vec{b}$ where λ is a scalar.

(iv) The equation of the line passing through the points (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

(v) Vector equation of a straight line passing through two points with position vectors $\vec{a}$ & $\vec{b}$ is $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$.

(vi) Reduction of cartesion form of equation of a line to vector form & vice versa

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} \quad \Leftrightarrow \quad \vec{r} = (x_1\hat{i} + y_1\hat{j} + z_1\hat{k}) + \lambda(a\hat{i} + b\hat{j} + c\hat{k}).$$

Note: Straight lines parallel to co-ordinate axes:

Straight lines		Equation	Straight lines		Equation
(i)	Through origin	y = mx, z = nx	**(v)**	Parallel to x–axis	y = p, z = q
(ii)	x–axis	y = 0, z = 0	**(vi)**	Parallel to y–axis	x = h, z = q
(iii)	y–axis	x = 0, z = 0	**(vii)**	Parallel to z–axis	x = h, y = p
(iv)	z–axis	x = 0, y = 0			

Example : Find the equation of the line through the points (3, 4, –7) and (1, – 1, 6) in vector form as well as in cartesian form.

Solution Let A ≡ (3, 4, – 7), B ≡ (1, – 1, 6)

Now $\vec{a} = \vec{OA} = 3\vec{i} + 4\vec{j} - 7\vec{k}$,

$= \vec{b} = \vec{OB} = \vec{i} - \vec{j} + 6\vec{k}$

Equation of the line through A($\vec{a}$) and B($\vec{b}$) is $\vec{r} = \vec{a} + t(\vec{b} - \vec{a})$

or $\vec{r} = 3\vec{i} + 4\vec{j} - 7\vec{k} + t(-2\vec{i} - 5\vec{j} + 13\vec{k})$ ..... (1)

Equation in cartesian form :

Equation of AB is $\frac{x-3}{3-1} = \frac{y-4}{4+1} = \frac{z+7}{-7-6}$ or, $\frac{x-3}{2} = \frac{y-4}{5} = \frac{z+7}{-13}$

Example : Find the co-ordinates of those points on the line $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{6}$ which is at a distance of 3 units from point (1, –2, 3).

Solution Given line is $\frac{x-1}{2}=\frac{y+2}{3}=\frac{z-3}{6}$ ..... (1)

Let $P \equiv (1, -2, 3)$ Direction ratios of line (1) are 2, 3, 6

$\therefore$ Direction cosines of line (1) are $\frac{2}{7},\frac{3}{7},\frac{6}{7}$

Equation of line (1) may be written as

$$\frac{x-1}{\frac{2}{7}}=\frac{y+2}{\frac{3}{7}}=\frac{z-3}{\frac{6}{7}} \quad \text{..... (2)}$$

Co-ordinates of any point on line (2) may be taken as

$$\left(\frac{2}{7}r+1,\frac{3}{7}r-2,\frac{6}{7}r+3\right)$$

Let $Q \equiv \left(\frac{2}{7}r-1,\ \frac{3}{7}r-2,\ \frac{6}{7}r+3\right)$

Distance of Q from P = | r |

According to question | r | = 3 $\therefore$ $r = \pm 3$ Putting the value of r, we have

$$Q \equiv \left(-\frac{1}{7}, -\frac{5}{7}, \frac{39}{7}\right) \quad \text{or} \quad Q \equiv \left(-\frac{13}{7}, -\frac{23}{7}, \frac{3}{7}\right)$$

Example : Find the equation of the line drawn through point (1, 0, 2) to meet at right angles the line $\frac{x+1}{3}=\frac{y-2}{-2}=\frac{z+1}{-1}$

Solution Given line is

$$\frac{x+1}{3}=\frac{y-2}{-2}=\frac{z+1}{-1} \quad \text{..... (1)}$$

Let $P \equiv (1, 0, 2)$

Co-ordinates of any point on line (1) may be taken as

$Q \equiv (3r - 1, -2r + 2, -r - 1)$

Direction ratios of PQ are $3r - 2, -2r + 2, -r - 3$

Direction ratios of line AB are $3, -2, -1$

Since PQ $\perp$ AB

$\therefore$ $3(3r - 2) - 2(-2r + 2) - 1(-r - 3) = 0$

$\Rightarrow$ $9r - 6 + 4r - 4 + r + 3 = 0$ $\Rightarrow$ $14r = 7$ $\Rightarrow$ $r = \frac{1}{2}$

Therefore, direction ratios of PQ are

$$-\frac{1}{2}, 1, -\frac{7}{2} \quad \text{or,} \quad -1, 2, -7$$

Equation of line PQ is

$$\frac{x-1}{-1}=\frac{y=0}{2}=\frac{z-2}{-7} \quad \text{or,} \quad \frac{x-1}{1}=\frac{y}{-2}=\frac{z-2}{7}$$

Example : Show that the two lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-4}{5}=\frac{y-1}{2}$ = z intersect. Find also the point of intersection of these lines.

Solution Given lines are $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ ..... (1)

and $\frac{x-4}{5}=\frac{y-1}{2}=\frac{z-0}{1}$ ..... (2)

Any point on line (1) is P $(2r + 1, 3r + 2, 4r + 3)$

and any point on line (2) is Q $(5\lambda + 4, 2\lambda + 1, \lambda)$

Lines (1) and (2) will intersect if P and Q coincide for some value of λ and r.

$\therefore$ $2r + 1 = 5\lambda + 4 \Rightarrow 2r - 5\lambda = 3$ ..... (1)

$3r + 2 + 2\lambda + 1 \Rightarrow 3r - 2\lambda = -1$ ..... (2)

$4r + 3 = \lambda \Rightarrow 4r - \lambda = -3$ ..... (3)

Solving (1) and (2), we get $r = -1$, $\lambda = -1$

Clearly these values of r and λ satisfy eqn. (3)

Now $P \equiv (-1, -1, -1)$ Hence lines (1) and (2) intersect at $(-1, -1, -1)$.

Self practice problems:

1. Find the equation of the line passing through point (1, 0, 2) having direction ratio 3, – 1, 5. Prove that this line passes through (4, – 1, 7). **Ans.** $\frac{x-1}{3}-\frac{y}{-1}=\frac{z-2}{5}$

2. Find the equation of the line parallel to line $\frac{x-2}{3}=\frac{y+1}{1}=\frac{z-7}{9}$ and passing through the point (3, 0, 5).

Ans. $\frac{x-3}{3}=\frac{y}{1}=\frac{z-5}{3}$

3. Find the coordinates of the point when the line through (3, 4, 1) and (5, 1, 6) crosses the xy plane.

Ans. $\left(\frac{13}{5},\frac{23}{5},0\right)$

20. Reduction Of Non-Symmetrical Form To Symmetrical Form:

Let equation of the line in non–symmetrical form be $a_1x + b_1y + c_1z + d_1 = 0$, $a_2x + b_2y + c_2z + d_2 = 0$. To find the equation of the line in symmetrical form, we must know (i) its direction ratios (ii) coordinate of any point on it.

(i) Direction ratios: Let ℓ, m, n be the direction ratios of the line. Since the line lies in both the planes, it must be perpendicular to normals of both planes. So $a_1\ell + b_1m + c_1n = 0$, $a_2\ell + b_2m + c_2n = 0$. From these equations, proportional values of ℓ, m, n can be found by cross–multiplication as $\frac{\ell}{b_1c_2-b_2c_1}=\frac{m}{c_1a_2-c_2a_1}=\frac{n}{a_1b_2-a_2b_1}$

Alternative method The vector $\begin{vmatrix} i & j & k \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = i(b_1c_2-b_2c_1)+j(c_1a_2-c_2a_1)+k(a_1b_2-a_2b_1)$ will be parallel to the line of intersection of the two given planes. hence $\ell : m : n = (b_1c_2 - b_2c_1) : (c_1a_2 - c_2a_1) : (a_1b_2 - a_2b_1)$

(ii) Point on the line – Note that as ℓ, m, n cannot be zero simultaneously, so at least one must be non–zero. Let $a_1b_2 - a_2b_1 \neq 0$, then the line cannot be parallel to xy plane, so it intersect it. Let it intersect xy–plane in $(x_1, y_1, 0)$. Then $a_1x_1 + b_1y_1 + d_1 = 0$ and $a_2x_1 + b_2y_1 + d_2 = 0$.Solving these,we get a point on the line. Then its equation becomes.

$$\frac{x-x_1}{b_1c_2-b_2c_1}=\frac{y-y_1}{c_1a_2-c_2a_1}=\frac{z-0}{a_1b_2-a_2b_1} \text{ or } \frac{x-\frac{b_1d_2-b_2d_1}{a_1b_2-a_2b_1}}{b_1c_2-b_2c_1}=\frac{y-\frac{d_1a_2-d_2a_1}{a_1b_2-a_2b_1}}{c_1a_2-c_2a_1}=\frac{z-0}{a_1b_2-a_2b_1}$$

Note: If $\ell \neq 0$, take a point on yz–plane as $(0, y_1, z_1)$ and if $m \neq 0$, take a point on xz–plane as $(x_1, 0, z_1)$.

Alternative method

If $\frac{a_1}{a_2}\neq\frac{b_1}{b_2}$ Put z = 0 in both the equations and solve the equations $a_1x + b_1y + d_1 = 0$, $a_2x + b_2y + d_2 = 0$ otherwise Put y = 0 and solve the equations $a_1x + c_1z + d_1 = 0$ and $a_2x + c_2z + d_2 = 0$

Example : Find the equation of the line of intersection of planes $4x + 4y – 5z = 12$, $8x + 12y – 13z = 32$ in the symmetric form.

Solution Given planes are $4x + 4y – 5z – 12 = 0$ ..... (1)

and $8x + 12y – 13z – 32 = 0$ ..... (2)

Let ℓ, m, n be the direction ratios of the line of intersection :

then $4\ell + 4m – 5n = 0$ ..... (3)

and $8\ell + 12m – 13n = 0$

$\therefore \quad \frac{\ell}{-52+60}=\frac{m}{-40+52}=\frac{n}{48-32}$ or, $\frac{\ell}{8}=\frac{m}{12}=\frac{n}{16}$ or, $\frac{\ell}{2}=\frac{m}{3}=\frac{n}{4}$

Hence direction ratios of line of intersection are 2, 3, 4.

Here $4 \neq 0$, therefore line of intersection is not parallel to xy-plane.

Let the line of intersection meet the xy-plane at P $(\alpha, \beta, 0)$.

Then P lies on planes (1) and (2)

$\therefore \quad 4\alpha + 4\beta + 12 = 0$

or, $\alpha + \beta – 3 = 0$ ..... (5)

and $8\alpha + 12\beta – 32 = 0$

or, $2\alpha + 3\beta – 8 = 0$ ..... (6)

Solving (5) and (6), we get

$\frac{\alpha}{-8+9}=\frac{\beta}{-6+8}=\frac{1}{3-2}$ or, $\frac{\alpha}{1}=\frac{\beta}{2}=\frac{1}{1}$

$\therefore \quad \alpha = 1, \beta = 2$

Hence equation of line of intersection in symmetrical form is $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-0}{4}$.

Example : Find the angle between the lines $x – 3y – 4 = 0$, $4y – z + 5 = 0$ and $x + 3y – 11 = 0$, $2y – z + 6 = 0$.

Solution Given lines are

$$\left.\begin{aligned} x-3y-4=0 \\ 4y-z+5=0 \end{aligned}\right\} \quad \text{..... (1)}$$

and

$$\left.\begin{aligned} x+3y-11=0 \\ 2y-z+6=0 \end{aligned}\right\} \quad \text{..... (2)}$$

Let ℓ_1, m_1, n_1 and ℓ_1, m_2, n_2 be the direction cosines of lines (1) and (2) respectively

$\because$ line (1) is perpendicular to the normals of each of the planes
$x - 3y - 4 = 0$ and $4y - z + 5 = 0$
$\therefore$ $\ell_1 - 3m_1 + 0.n_1 = 0$ ..... (3)
and $0\ell_1 + 4m_1 - n_1 = 0$ ..... (4)
Solving equations (3) and (4), we get,

$$\frac{\ell_1}{3-0} = \frac{m_1}{0-(-1)} = \frac{n_1}{4-0} \quad \text{or,} \quad \frac{\ell_1}{3} = \frac{m_1}{1} = \frac{n_1}{4} = k \text{ (let).}$$

Since line (2) is perpendicular to the normals of each of the planes
$x + 3y - 11 = 0$ and $2y - z + 6 = 0$,
$\therefore$ $\ell_2 + 3m_2 = 0$ ..... (5)
and $2m_2 - n_2 = 0$ ..... (6)

$\therefore$ $\ell_2 = -3m_2$ or, $\frac{\ell_2}{-3} = m_2$

and $n_2 = 2m_2$ or, $\frac{n_2}{2} = m_2$.

$\therefore$ $\frac{\ell_2}{-3} = \frac{m_3}{1} = \frac{n_2}{2} = t$ (let).

If θ be the angle between lines (1) and (2), then
$\cos\theta = \ell_1\ell_2 + m_1m_2 + n_1n_2$
$= (3k)(-3t) + (k)(t) + (4k)(2t)$
$= -9kt + kt + 8kt = 0$ $\therefore$ $\theta = 90°$.

Self practice problems:

1. Find the equation of the line of intersection of the plane
$4x + 4y - 5z = 12$
$8x + 12y - 13z = 32$ **Ans.** $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-0}{4}$

2. Show that the angle between the two lines defined by the equations $x = y$ and $xy + yz + zx = 0$ is $\cos^{-1}\left(\frac{1}{3}\right)$

3. Prove that the three planes $2x + y - 4z - 17 = 0$, $3x + 2y - 2z - 25 = 0$, $2x - 4y + 3z + 25 = 0$ intersect at a point and find its co-ordinates. **Ans.** (3, 7, – 1)

21. Foot, Length And Equation Of Perpendicular From A Point To A Line:

(i) Cartesian form: Let equation of the line be $\frac{x-a}{\ell} = \frac{y-b}{m} = \frac{z-c}{n} = r$ (say) .....(i)
and A (α, β, γ) be the point.
Any point on line (i) is P $(\ell r + a, mr + b, nr + c)$ ..... (ii)
If it is the foot of the perpendicular from A on the line, then AP is perpendicular to the line. So $\ell(\ell r + a - \alpha) + m(mr + b - \beta) + n(nr + c - \gamma) = 0$ i.e. $r = (\alpha - a)\ell + (\beta - b)m + (\gamma - c)n$ since $\ell^2 + m^2 + n^2 = 1$. Putting this value of r in (ii), we get the foot of perpendicular from point A on the given line. Since foot of perpendicular P is known, then the length of perpendicular is given by AP $= \sqrt{(\ell r + a - \alpha)^2 + (mr + b - \beta)^2 + (nr + c - \gamma)^2}$ the equation of perpendicular is given by $\frac{x-\alpha}{\ell r + a - \alpha} = \frac{y-\beta}{mr + b - \beta} = \frac{z-\gamma}{nr + c - \gamma}$ **(ii)** Vector Form: Equation of a line passing through a point having position vector $\vec{\alpha}$ and perpendicular to the lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \lambda\vec{b}_2$ is parallel to $\vec{b}_1 \times \vec{b}_2$. So the vector equation of such a line is $\vec{r} = \vec{\alpha} + \lambda(\vec{b}_1 \times \vec{b}_2)$. Position vector $\vec{\beta}$ of the image of a point $\vec{\alpha}$ in a straight line $\vec{r} = \vec{a} + \lambda\vec{b}$ is given by $\vec{\beta} = 2\vec{a} - \left[\frac{2(\vec{a} - \vec{\alpha}).\vec{b}}{|\vec{b}|^2}\right]\vec{b} - \vec{\alpha}$. Position vector of the foot of the perpendicular on line is $\vec{f} = \vec{a} - \left[\frac{(\vec{a} - \vec{\alpha}).\vec{b}}{|\vec{b}|^2}\right]\vec{b}$. The equation of the perpendicular is $\vec{r} = \vec{\alpha} + \mu\left[(\vec{a} - \vec{\alpha}) - \left(\frac{(\vec{a} - \vec{\alpha}).\vec{b}}{|\vec{b}|^2}\right)\vec{b}\right]$.

22. To find image of a point w. r. t a line

Let $L \equiv \frac{x - x_2}{a} = \frac{y - y_2}{b} = \frac{z - z_2}{c}$ is a given line
Let (x', y', z') is the image of the point P (x_1, y_1, z_1) with respect to the line L. Then

(i) $a(x_1 - x') + b(y_1 - y') + c(z_1 - z') = 0$

(ii) $\dfrac{\frac{x_1+x'}{2}-x_2}{a}=\dfrac{\frac{y_1+y'}{2}-y_2}{b}=\dfrac{\frac{z_1+z'}{2}-z_2}{c}=\lambda$

from (ii) get the value of x′, y′, z′ in terms of λ as
$x' = 2a\lambda + 2x_2 - x_1$, $y' = 2b\alpha - 2y_2 - y_1$,
$z' = 2c\lambda + 2z_2 - z_1$
now put the values of x′, y′, z′ in (i) get λ and resubtitute the value of λ to to get (x′ y′ z′).

Example : Find the length of the perpendicular from P (2, – 3, 1) to the line $\dfrac{x+1}{2}=\dfrac{y-3}{3}=\dfrac{z+2}{-1}$.

Solution Given line is $\dfrac{x+1}{2}=\dfrac{y-3}{3}=\dfrac{z+2}{-1}$ ..... (1)

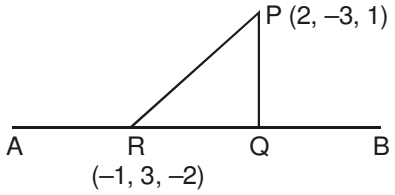
P ≡ (2, – 3, 1)
Co-ordinates of any point on line (1) may be taken as
Q ≡ (2r – 1, 3r + 3, – r – 2)
Direction ratios of PQ are 2r – 3, 3r + 6, – r – 3
Direction ratios of AB are 2, 3, – 1
Since PQ ⊥ AB
∴ 2 (2r – 3) + 3 (3r + 6) – 1 (– r – 3) = 0

or, 14r + 15 = 0 ∴ $r = \dfrac{-15}{14}$

∴ $Q \equiv \left(\dfrac{-22}{7}, \dfrac{-3}{14}, \dfrac{-13}{14}\right)$ ∴ $PQ = \sqrt{\dfrac{531}{14}}$ units.

Second method : Given line is

$\dfrac{x+1}{2}=\dfrac{y-3}{3}=\dfrac{z+2}{-1}$
P ≡ (2, – 3, 1)



Direction ratios of line (1) are $\dfrac{2}{\sqrt{14}}, \dfrac{3}{\sqrt{14}}, -\dfrac{1}{\sqrt{14}}$
RQ = length of projection of RP on AB

$$= \left|\frac{2}{\sqrt{14}}(2+1)+\frac{3}{\sqrt{14}}(-3-3)-\frac{1}{\sqrt{4}}(1+2)\right| = \frac{15}{\sqrt{14}}$$

$PR^2 = 3^2 + 6^2 + 3^2 = 54$

∴ $PQ = \sqrt{PR^2 - RQ^2} = \sqrt{54 - \dfrac{225}{14}} = \sqrt{\dfrac{531}{14}}$

Self practice problems:

1. Find the length and foot of perpendicular drawn from point (2, –1, 5) to the line $\dfrac{x-11}{10}=\dfrac{y+2}{-4}=\dfrac{z+8}{-11}$.
Also find the image of the point in the line. **Ans.** $\sqrt{14}$, N ≡ (1, 2, 3), I ≡ (0, 5, 1)

2. Find the image of the point (1, 6, 3) in the line $\dfrac{x}{1}=\dfrac{y-1}{2}=\dfrac{z-2}{3}$. **Ans.** (1, 0 , 7)

3. Find the foot and hence the length of perpendicular from (5, 7, 3) to the line $\dfrac{x-15}{3}=\dfrac{y-29}{8}=\dfrac{z-5}{-5}$.
Find also the equation of the perpendicular. **Ans.** (9, 13, 15) ; 14 ; $\dfrac{x-5}{2}=\dfrac{y-7}{3}=\dfrac{z-3}{6}$

23. Angle Between A Plane And A Line:

(i) If θ is the angle between line $\dfrac{x-x_1}{\ell}=\dfrac{y-y_1}{m}=\dfrac{z-z_1}{n}$ and the plane ax + by + cz + d = 0, then

$$\sin\theta = \left[\frac{a\ell + bm + cn}{\sqrt{(a^2+b^2+c^2)}\sqrt{\ell^2+m^2+n^2}}\right].$$

(ii) Vector form: If θ is the angle between a line $\vec{r} = (\vec{a} + \lambda\vec{b})$ and $\vec{r}\cdot\vec{n} = d$ then $\sin\theta = \left[\dfrac{\vec{b}\cdot\vec{n}}{|\vec{b}||\vec{n}|}\right]$.

(iii) Condition for perpendicularity $\dfrac{\ell}{a}=\dfrac{m}{b}=\dfrac{n}{c}$ $\vec{b}\times\vec{n} = 0$

(iv) Condition for parallel $a\ell + bm + cn = 0$ $\vec{b}\cdot\vec{n} = 0$

24. Condition For A Line To Lie In A Plane

(i) Cartesian form: Line $\frac{x-x_1}{\ell} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ would lie in a plane
$ax + by + cz + d = 0$, if $ax_1 + by_1 + cz_1 + d = 0$ & $a\ell + bm + cn = 0$.

(ii) Vector form: Line $\vec{r} = \vec{a} + \lambda\vec{b}$ would lie in the plane $\vec{r}.\vec{n} = d$ if $\vec{b}.\vec{n} = 0$ & $\vec{a}.\vec{n} = d$

25. Coplanar Lines:

(i) If the given lines are $\frac{x-\alpha}{\ell} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ and $\frac{x-\alpha'}{\ell'} = \frac{y-\beta'}{m'} = \frac{z-\gamma'}{n'}$, then condition for intersection/coplanarity is $\begin{vmatrix} \alpha-\alpha' & \beta-\beta' & \gamma-\gamma' \\ \ell & m & n \\ \ell' & m' & n' \end{vmatrix} = 0$ & plane containing the above two lines is $\begin{vmatrix} x-\alpha & y-\beta & z-\gamma \\ \ell & m & n \\ \ell' & m' & n' \end{vmatrix} = 0$

(ii) Condition of coplanarity if both the lines are in general form Let the lines be
$ax + by + cz + d = 0 = a'x + b'y + c'z + d'$ &
$\alpha x + \beta y + \gamma z + \delta = 0 = \alpha' x + \beta' y + \gamma' z + \delta'$

They are coplanar if $\begin{vmatrix} a & b & c & d \\ a' & b' & c' & d' \\ \alpha & \beta & \gamma & \delta \\ \alpha' & \beta' & \gamma' & \delta' \end{vmatrix} = 0$

Alternative method: get vector along the line of shortest distance as $\begin{vmatrix} i & j & k \\ \ell & m & n \\ \ell' & m' & n' \end{vmatrix}$

Now get unit vector along this vector

$\hat{u} = \ell i + mj + nk$

Let $\vec{v} = (\alpha - \alpha')\hat{i} + (B - B')\hat{j} + (y - y')$

S. D. $= u.v$

Example : Find the distance of the point (1, 0, – 3) from the plane $x - y - z = 9$ measured parallel to the line $\frac{x-2}{2} = \frac{y+2}{3} = \frac{z-6}{-6}$.

Solution Given plane is $x - y - z = 9$ ..... (1)

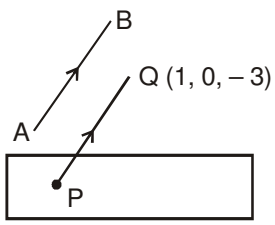
Given line AB is $\frac{x-2}{2} = \frac{y+2}{3} = \frac{z-6}{-6}$ ..... (2)

Equation of a line passing through the point Q(1, 0, – 3) and parallel to line (2) is

$\frac{x-1}{2} = \frac{y}{3} = \frac{z+3}{-6} = r.$ ..... (3)

Co-ordinates of point on line (3) may be taken as
P (2r + 1, 3r, – 6r – 3)
If P is the point of intersection of line (3) and plane (1), then P lies on plane (1),
$\therefore$ $(2r + 1) - (3r) - (-6r - 3) = 9$
$r = 1$
or, P ≡ (3, 3, – 9)
Distance between points Q (1, 0, – 3) and P (3, 3, – 9)

$PQ = \sqrt{(3-1)^2 + (3-0)^2 + (-9-(3))^2} = \sqrt{4+9+36} = 7.$



Example: Find the equation of the plane passing through (1, 2, 0) which contains the line $\frac{x+3}{3} = \frac{y-1}{4} = \frac{z-2}{-2}$.

Solution Equation of any plane passing through (1, 2, 0) may be taken as

$a(x-1)+b(y-2)+c(z-0)=0$ ..... (1)

where a, b, c are the direction ratios of the normal to the plane. Given line is

$$\frac{x+3}{3}=\frac{y-1}{4}=\frac{z-2}{-2} \quad \text{..... (2)}$$

If plane (1) contains the given line, then

$3a+4b-2c=0$ ..... (3)

Also point (– 3, 1, 2) on line (2) lies in plane (1)

$\therefore \quad a(-3-1)+b(1-2)+c(2-0)=0$

or, $-4a-b+2c=0$ ..... (4)

Solving equations (3) and (4), we get,

$$\frac{a}{8-2}=\frac{b}{8-6}=\frac{c}{-3+16}$$

or, $\frac{a}{6}=\frac{b}{2}=\frac{c}{13}=k$ (say). ..... (5)

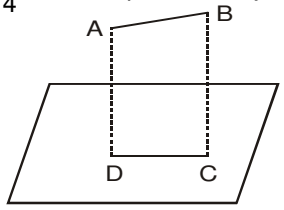
Substituting the values of a, b and c in equation (1), we get,

$6(x-1)+2(y-2)+13(z-0)=0.$

or, $6x+2y+13z-10=0$. This is the required equation.

Example : Find the equation of the projection of the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{4}$ on the plane $x+2y+z=9$.

Solution



Let the given line AB be

$$\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{4} \quad \text{..... (1)}$$

Given plane is

$x+2y+z=9$ ..... (2)

Let DC be the projection of AB on plane (2)

Clearly plane ABCD is perpendicular to plane (2).

Equation of any plane through AB may be taken as (this plane passes through the point (1, – 1, 3) on line AB)

$a(x-1)+b(y+1)+c(z-3)=0$ ..... (3)

where $2a-b+4c=0$ ..... (4)

[$\because$ normal to plane (3) is perpendicular to line (1)]

Since plane (3) is perpendicular to plane (2),

$\therefore \quad a+2b+c=0$ ..... (5)

Solving equations (4) & (5), we get,

$$\frac{a}{-9}=\frac{b}{2}=\frac{c}{5}.$$

Substituting these values of a, b and c in equation (3), we get

$9(x-1)-2(y+1)-5(z-3)=0$

or, $9x-2y-5z+4=0$ ..... (6)

Since projection DC of AB on plane (2) is the line of intersection of plane ABCD and plane (2), therefore equation of DC will be

and $\left.\begin{aligned} 9x-2y-5z+4=0 \quad &.....(i) \\ x+2y+z-9=0 \quad &.....(ii)\end{aligned}\right\}$ ..... (7)

Let ℓ, m, n be the direction ratios of the line of intersection of planes (i) and (ii)

$\therefore \quad 9\ell-2m-5n=0$ ..... (8)

and $\ell+2m+n=0$ ..... (9)

$$\therefore \quad \frac{\ell}{-2+10}=\frac{m}{-5-9}=\frac{n}{18+2}$$

Example : Show that the lines $\frac{x-3}{2}=\frac{y+1}{-3}=\frac{z+2}{1}$ and $\frac{x-7}{-3}=\frac{y}{1}=\frac{z+2}{2}$ are coplanar. Also find the equation of the plane containing them.

Solution Given lines are

$$\frac{x-3}{2}=\frac{y+1}{-3}=\frac{z+2}{1}=r \text{ (say)} \quad \text{..... (1)}$$

and $\frac{x-7}{-3}=\frac{y}{1}=\frac{z+7}{2}=R$ (say) ..... (2)

If possible, let lines (1) and (2) intersect at P.

Any point on line (1) may be taken as

$(2r+3, -3r-1, r-2)=P$ (let).

Any point on line (2) may be taken as

$(-3R+7, R, 2R-7)=P$ (let).

$\therefore \quad 2r+3=-3R+7$

or, $2r+3R=4$ ..... (3)

Also $-3r-1=R$

or, $-3r-R=1$ ..... (4)

and $r - 2 = 2R - 7$
or, $r - 2R = -5$. ..... (5)
Solving equations (3) and (4), we get,
$r = -1$, $R = 2$
Clearly $r = -1$, $R = 2$ satisfies equation (5).
Hence lines (1) and (2) intersect. $\therefore$ lines (1) and (2) are coplanar.
Equation of the plane containing lines (1) and (2) is

$$\begin{vmatrix} x-3 & y+1 & z+2 \\ 2 & -3 & 1 \\ -3 & 1 & 2 \end{vmatrix} = 0$$

or, $(x - 3)(-6 - 1) - (y + 1)(4 + 3) + (z + 2)(2 - 9) = 0$
or, $-7(x - 3) - 7(y + 1) - 7(z + 2) = 0$
or, $x - 3 + y + 1 + z + 2 = 0$ or, $x + y + z = 0$.

Self practice problems:

1. Find the values of a and b for which the line $\frac{x-2}{a} = \frac{y+3}{4} = \frac{z-6}{-2}$ is perpendicular to the plane $3x - 2y + bz + 10 = 0$. **Ans.** $a = 3$, $b = -2$

2. Prove that the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ are coplanar. Also find the equation of the plane in which they lie. **Ans.** $x - 2y + z = 0$

3. Find the plane containing the line $\frac{x-2}{2} = \frac{y-3}{3} = \frac{z-4}{5}$ and parallel to the line $\frac{x+1}{1} = \frac{y-1}{-2} = \frac{-z+1}{1}$
Ans. $13x + 3y - 72 - 7 = 0$

4. Show that the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ & $\frac{x-4}{5} = \frac{y-1}{2} = z$ are intersecting each other. Find thire intersection and the plane containing the line. **Ans.** $(-1, -1, -1)$ & $5x - 8y + 11z - 2 = 0$

5. Show that the lines $\vec{r} = (-\hat{i} - 3\hat{j} - 5\hat{k}) + \lambda(-3\hat{i} - 5\hat{j} - 7\hat{k})$ & $\vec{r}(2\hat{i} + 4\hat{j} + 6\hat{k}) + \mu(\hat{i} + 4\hat{j} + 7\hat{k})$ are coplanar and find the plane containing the line. **Ans.** $\vec{r} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 0$

26. Skew Lines:

(i) The straight lines which are not parallel and non–coplanar i.e. non–intersecting are called skew lines. If $\Delta = \begin{vmatrix} \alpha'-\alpha & \beta'-\beta & \gamma'-\gamma \\ \ell & m & n \\ \ell' & m' & n' \end{vmatrix} \neq 0$, then lines are skew.

(ii) Shortest distance: Suppose the equation of the lines are

$$\frac{x-\alpha}{\ell} = \frac{y-\beta}{m} = \frac{z-\gamma}{n} \text{ and } \frac{x-\alpha'}{\ell'} = \frac{y-\beta'}{m'} = \frac{z-\gamma'}{n'}$$

$$\text{S.D.} = \frac{(\alpha-\alpha')(mn'-m'n) + (\beta-\beta')(n\ell-n'\ell) + (\gamma-\gamma')(\ell m'-\ell' m)}{\sqrt{\sum (mn'-m'n)^2}}$$

$$= \begin{vmatrix} \alpha'-\alpha & \beta'-\beta & \gamma'-\gamma \\ \ell & m & n \\ \ell' & m' & n' \end{vmatrix} \div \sqrt{\sum (mn' - m'n)^2}$$

(iii) Vector Form: For lines $\vec{a}_1 + \lambda\vec{b}_1$ & $\vec{a}_2 + \lambda\vec{b}_2$ to be skew

$$(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1) \neq 0 \quad \text{or} \quad [\vec{b}_1\ \vec{b}_2\ (\vec{a}_2 - \vec{a}_1)] \neq 0.$$

(iv) Shortest distance between the two parallel lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}$ &

$$\vec{r} = \vec{a}_2 + \mu\vec{b} \text{ is } d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \times \vec{b}}{|\vec{b}|} \right|.$$

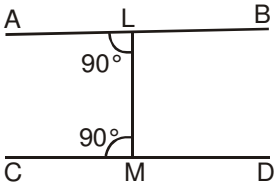
Example : Find the shortest distance and the vector equation of the line of shortest distance between the lines given by

$$\vec{r} = 3\vec{r} + 8\vec{j} + 3\vec{k} + \lambda(3\vec{r} - \vec{j} + \vec{k}) \quad \text{and} \quad \vec{r} = -3\vec{r} - 7\vec{j} + 6\vec{k} + \mu(-3\vec{i} + 2\vec{j} + 4\vec{k})$$

Solution Given lines are

$$\vec{r} = 3\vec{r} + 8\vec{j} + 3\vec{k} + \lambda(3\vec{i} - \vec{j} + \vec{k}) \quad \text{..... (1)}$$

and $\vec{r} = -3\vec{r} - 7\vec{j} + 6\vec{k} + \mu\left(-3\vec{i} + 2\vec{j} + 4\vec{k}\right)$ ..... (2)



Equation of lines (1) and (2) in cartesian form is

AB : $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1} = \lambda$

and CD : $\left(\frac{1}{3}\right) = \mu$

Let L ≡ (3λ + 3, – λ + 8, λ + 3)
and M ≡ (– 3μ – 3, 2μ – 7, 4μ + 6)
Direction ratios of LM are
3λ + 3μ + 6, – λ – 2μ + 15, λ – 4μ – 3.
Since LM ⊥ AB
∴ 3 (3λ + 3μ + 6) – 1 (– λ – 2μ + 15) + 1 (λ – 4μ – 3) = 0
or, 11λ + 7μ = 0 ..... (5)
Again LM ⊥ CD
∴ – 3 (3λ + 3μ + 6) + 2 (– λ – 2μ + 15) + 4 (λ – 4μ – 3) = 0
or, – 7λ – 29μ – 0 ..... (6)
Solving (5) and (6), we get λ = 0, μ = 0
∴ L ≡ (3, 8, 3), M ≡ (– 3, – 7, 6)
Hence shortest distance LM = $\sqrt{(3+3)^2 + (8+7)^2 + (3-6)^2}$

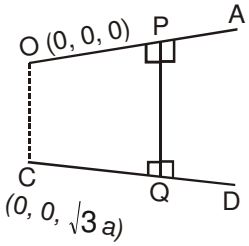
= $\sqrt{270} = 3\sqrt{30}$ units
Vector equation of LM is

$\vec{r} = 3\vec{i} + 8\vec{j} + 3\vec{k} + t\left(6\vec{i} + 15\vec{j} - 3\vec{k}\right)$

Note : Cartesian equation of LM is $\frac{x-3}{6} = \frac{y-8}{15} = \frac{z-3}{-3}$.

Example : Prove that the shortest distance between any two opposite edges of a tetrahedron formed by the planes y + z = 0, x + z = 0, x + y = 0, x + y + z = $\sqrt{3}$ a is $\sqrt{2}$ a.

Solution Given planes are y + z = 0 ..... (i) x + z = 0 ..... (ii)
x + y = 0 ..... (iii) x + y + z = $\sqrt{3}$ a ..... (iv)
Clearly planes (i), (ii) and (iii) meet at O(0, 0, 0)
Let the tetrahedron be OABC
Let the equation to one of the pair of opposite edges OA and BC be
y + z = 0, x + z = 0 ..... (1)



x + y = 0, x + y + z = $\sqrt{3}$ a ..... (2)
equation (1) and (2) can be expressed in symmetrical form as

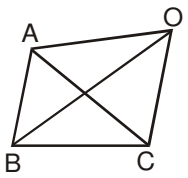
$\frac{x-0}{1} = \frac{y-0}{1} = \frac{z-0}{-1}$ ..... (3)

and, $\frac{x-0}{1} = \frac{y-0}{-1} = \frac{z-\sqrt{3}a}{0}$ ..... (4)

d. r. of OA and BC are (1, – 1) and (1, – 1, 0).
Let PQ be the shortest distance between OA and BC having direction cosine (ℓ, m, n)
∴ PQ is perpendicular to both OA and BC.
∴ ℓ + m – n = 0
and ℓ – m = 0
Solving (5) and (6), we get,

$\frac{\ell}{1} = \frac{m}{1} = \frac{n}{2}$ = k (say)

also, $\ell^2 + m^2 + n^2 = 1$

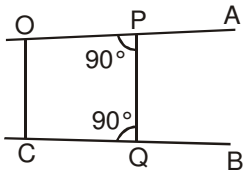


∴ $k^2 + k^2 + 4k^2 = 1 \Rightarrow k = \frac{1}{\sqrt{6}}$

∴ $\ell = \frac{1}{\sqrt{6}}, m = \frac{1}{\sqrt{6}}, n = \frac{2}{\sqrt{6}}$

Shortest distance between OA and BC
i.e. PQ = The length of projection of OC on PQ
= | $(x_2 - x_1)\ \ell + (y_2 - y_1)\ m + (z_2 - z_1)\ n$ |

= $\left|0 \cdot \frac{1}{\sqrt{6}} + 0 \cdot \frac{1}{\sqrt{6}} + \sqrt{3}a \cdot \frac{2}{\sqrt{6}}\right| = \sqrt{2}$ a.



Self practice problems:

1. Find the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$. Find also its equation. **Ans.** $\frac{1}{\sqrt{6}}$, $6x - y = 10 - 3y = 6z - 25$

2. Prove that the shortest distance between the diagonals of a rectangular parallelopiped whose sides are a, b, c and the edges not meeting it are $\frac{bc}{\sqrt{b^2+c^2}}, \frac{ca}{\sqrt{c^2+a^2}}, \frac{ab}{\sqrt{a^2+b^2}}$

27. Sphere

: General equation of a sphere is given by $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ $(-u, -v, -w)$ is the centre and $\sqrt{u^2+v^2+w^2-d}$ is the radius of the sphere.

Example : Find the equation of the sphere having centre at (1, 2, 3) and touching the plane $x + 2y + 3z = 0$.

Solution: Given plane is $x + 2y + 3z = 0$ ..... (1)

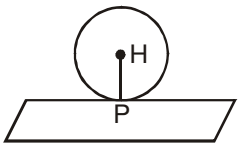
Let H be the centre of the required sphere.

Given $H \equiv (1, 2, 3)$

Radius of the sphere,

HP = length of perpendicular from H to plane (1)

$= \frac{|1+2\times2+3\times3|}{\sqrt{14}}$ $= \sqrt{14}$



Equation of the required sphere is $(x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 14$

or $x^2 + y^2 + z^2 - 2x - 4y - 6z = 0$

Example : Find the equation of the sphere if it touches the plane $\vec{r}.(2\vec{i}-2\vec{j}-\vec{k}) = 0$ and the position vector of its centre is $3\vec{i}+6\vec{j}-4\vec{k}$

Solution Given plane is $\vec{r}.(2\vec{i}-2\vec{j}-\vec{k}) = 0$ ..... (1)

Let H be the centre of the sphere, then

$\vec{OH} = 3\vec{i}+6\vec{j}-4\vec{k} = \vec{c}$ (say)

Radius of the sphere = length of perpendicular from H ot plane (1)

$= \frac{|\vec{c}.(2\vec{i}-2\vec{j}-\vec{k})|}{|2\vec{i}-2\vec{j}-\vec{k}|}$

$= \frac{|(3\vec{i}+6\vec{j}-4\vec{k}).(2\vec{i}-2\vec{j}-\vec{k})|}{|2\vec{i}-2\vec{j}-\vec{k}|}$

$= \frac{|6-12+4|}{3} = \frac{2}{3} = a$ (say)

Equation of the required sphere is

$|\vec{r}-\vec{c}| = a$

or $|x\vec{i}+y\vec{j}+z\vec{k}-(3\vec{i}+6\vec{j}-4\vec{k})| = \frac{2}{3}$ or $|(x-3)\vec{i}+(y-6)\vec{j}+(z+4)\vec{k}|^2 = \frac{4}{9}$

or $(x - 3)^2 + (y - 6)^2 + (z + 4)^2 = \frac{4}{9}$ or $9(x^2 + y^2 + z^2 - 6x - 12y + 8z + 61) = 4$

or $9x^2 + 9y^2 + 9z^2 - 54x - 108y + 72z + 545 = 0$

Example : Find the equation of the sphere passing through the points (3, 0, 0), (0, – 1, 0), (0, 0, – 2) and whose centre lies on the plane $3x + 2y + 4z = 1$

Solution Let the equation of the sphere be

$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ ..... (1)

Let $A \equiv (3, 0, 0)$, $B \equiv (0, -1, 0)$, $C \equiv (0, 0, -2)$

Since sphere (1) passes through A, B and C,

$\therefore$ $9 + 6u + d = 0$ ..... (2)

$1 - 2v + d = 0$ ..... (3)

$4 - 4w + d = 0$ ..... (4)

Since centre $(-u, -v, -w)$ of the sphere lies on plane

$3x + 2y + 4z = 1$

$\therefore$ $-3u - 2v - 4w = 1$ ..... (5)

(2) – (3) $\Rightarrow 6u + 2v = -8$ ..... (6)

(3) – (4) $\Rightarrow -2v + 4w = 3$ ..... (7)

From (6), $u = \frac{-2v-8}{6}$ ..... (8)

From (7), $4w = 3 + 2v$ ..... (9)

Putting the values of u, v and w in (5), we get

$$\frac{2v+8}{2} - 2v - 3 - 2v = 1$$

$\Rightarrow$ $2v + 8 - 4v - 6 - 4v = 2 \Rightarrow \quad v = 0$

From (8), $u = \frac{0-8}{6} = -\frac{4}{3}$

From (9), $4w = 3 \quad \therefore \quad w = \frac{3}{4}$

From (3), $d = 2v - 1 = 0 - 1 = -1$

From (1), equation of required sphere is

$$x^2 + y^2 + z^2 - \frac{0-8}{6} - \frac{8}{3}\ x + \frac{3}{2}z - 1 = 0$$

or $6x^2 + 6y^2 + 6z^2 - 16x + 9z - 6 = 0$

Example :

Find the equation of the sphere with the points (1, 2, 2) and (2, 3, 4) as the extremities of a diameter. Find the co-ordinates of its centre.

Solution

Let $A \equiv (1, 2, 2)$, $B \equiv (2, 3, 4)$

Equation of the sphere having (x_1, y_1, z_1) and (x_2, y_2, z_2) as the extremities of a diameter is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0$$

Here $x_1 = 1$, $x_2 = 2$, $y_1 = 2$, $y_2 = 3$, $z_1 = 2$, $z_2 = 4$

$\therefore$ required equation of the sphere is

$$(x - 1)(x - 2) + (y - 2)(y - 3) + (z - 2)(z - 4) = 0$$

or $x^2 + y^2 + z^2 - 3x - 5y - 6z + 16 = 0$

Centre of the sphere is middle point of AB

$\therefore$ Centre is $\left(\frac{3}{2}, \frac{5}{2}, 3\right)$

Self practice problems:

1. Find the value of k for which the plane $x + y + z = \sqrt{3}k$ touches the sphere $x^2 + y^2 + z^2 - 2x - 2y - 2z - 6 = 0$.

Ans. $\sqrt{3} \pm 3$

2. Find the equation to the sphere passing through (1, – 3, 4), (1, – 5, 2) and (1, – 3, 0) which has its centre in the plane $x + y + z = 0$

Ans. $x^2 + y^2 + z^2 - 2x + 6y - 4z + 10 = 0$

3. Find the equation of the sphere having centre on the line $2x - 3y = 0$, $5y + 2z = 0$ and passing through the points (0, – 2, – 4) and (2, – 1, – 1).

Ans. $x^2 + y^2 + z^2 - 6x - 4y + 10z + 12 = 0$

4. Find the centre and radius of the circle in which the plane $3x + 2y - z - 7\sqrt{14} = 0$ intersects the sphere $x^2 + y^2 + z^2 = 81$.

Ans. $4\sqrt{2}$ units

5. A plane passes through a fixed point (a, b, c) and cuts the axes in A, B, C. Show that the locus of the centre of the sphere OABC is

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2.$$

Vector

1. Vectors & Their Representation:

Vector quantities are specified by definite magnitude and definite directions. A vector is generally represented by a directed line segment, say $\overrightarrow{AB}$. A is called the **initial point** & B is called the **terminal point**. The magnitude of vector $\overrightarrow{AB}$ is expressed by $|\overrightarrow{AB}|$.

Zero Vector: A vector of zero magnitude is a zero vector. i.e. which has the same initial & terminal point, is called a **Zero Vector.** It is denoted by **O**. The direction of zero vector is indeterminate.

Unit Vector: A vector of unit magnitude in the direction of a vector $\vec{a}$ is called unit vector along $\vec{a}$ and is denoted by $\hat{a}$ symbolically, $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$.

Equal Vectors: Two vectors are said to be equal if they have the same magnitude, direction & represent the same physical quantity.

Collinear Vectors: Two vectors are said to be collinear if their directed line segments are parallel irrespective of their directions. Collinear vectors are also called **parallel vectors**. If they have the same direction they are named as **like vectors** otherwise **unlike vectors**.

Symbolically, two non zero vectors $\vec{a}$ and $\vec{b}$ are collinear if and only if, $\vec{a} = K\vec{b}$, where $K \in R$

Vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are collinear if $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$

Coplanar Vectors: A given number of vectors are called coplanar if their line segments are all parallel to the same plane. Note that **"TWO VECTORS ARE ALWAYS COPLANAR"**.

Solved Example Find unit vector of $\hat{i} - 2\hat{j} + 3\hat{k}$

Solution $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ if $\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$

then $|\vec{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$ $\therefore$ $|\vec{a}| = \sqrt{14}$

$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{1}{\sqrt{14}}\hat{i} - \frac{2}{14}\hat{j} + \frac{3}{\sqrt{14}}\hat{k}$

Solved Example Find values of x & y for which the vectors $\vec{a} = (x + 2)\hat{i} - (x - y)\hat{j} + \hat{k}$

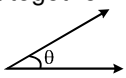
$\vec{b} = (x - 1)\hat{i} + (2x + y)\hat{j} + 2\hat{k}$ are parallel.

Solution $\vec{a}$ and $\vec{b}$ are parallel if $\frac{x+2}{x-1} = \frac{y-x}{2x+y} = \frac{1}{2}$

$x = -5, y = -20$

2. Angle Between two Vectors

It is the smaller angle formed when the initial points or the terminal points of the two vectors are brought together. It should be noted that $0^\circ \le \theta \le 180^\circ$.





3. Addition Of Vectors:

- If two vectors $\vec{a}$ & $\vec{b}$ are represented by $\overrightarrow{OA}$ & $\overrightarrow{OB}$, then their sum $\vec{a} + \vec{b}$ is a vector represented by $\overrightarrow{OC}$, where OC is the diagonal of the parallelogram OACB.
- $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (commutative)
- $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$ (associativity)
- $\vec{a} + \vec{0} = \vec{a} = \vec{0} + \vec{a}$
- $\vec{a} + (-\vec{a}) = \vec{0} = (-\vec{a}) + \vec{a}$
- $|\vec{a} + \vec{b}| \le |\vec{a}| + |\vec{b}|$
- $|\vec{a} - \vec{b}| \ge ||\vec{a}| - |\vec{b}||$
- $|\vec{a} \pm \vec{b}| = \sqrt{|\vec{a}|^2 + |\vec{b}|^2 \pm 2|\vec{a}||\vec{b}|\cos\theta}$ where θ is the angle between the vectors
- A vector in the direction of the bisector of the angle between the two vectors $\vec{a}$ & $\vec{b}$ is $\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}$. Hence bisector of the angle between the two vectors $\vec{a}$ and $\vec{b}$ is $\lambda\left(\hat{a} + \hat{b}\right)$, where $\lambda \in R^+$. Bisector of the exterior angle between $\vec{a}$ & $\vec{b}$ is $\lambda\left(\hat{a} - \hat{b}\right)$, $\lambda \in R^+$.

4. Multiplication Of A Vector By A Scalar:

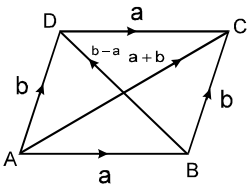
If $\vec{a}$ is a vector & m is a scalar, then m $\vec{a}$ is a vector parallel to $\vec{a}$ whose modulus is $|m|$ times that of $\vec{a}$. This multiplication is called **SCALAR MULTIPLICATION**. If $\vec{a}$ and $\vec{b}$ are vectors & m, n are scalars, then:

$m(\vec{a}) = (\vec{a})\, m = m\vec{a}$ $\qquad$ $m(n\vec{a}) = n(m\vec{a}) = (mn)\vec{a}$

$(m+n)\,\vec{a} = m\vec{a} + n\vec{a}$ $\qquad$ $m(\vec{a}+\vec{b}) = m\vec{a} + m\vec{b}$

Solved Example: If $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ represent two adjacent sides of a parallelogram, find unit vectors parallel to the diagonals of the parallelogram.

Solution. Let ABCD be a parallelogram such that $\overrightarrow{AB} = \vec{a}$ and $\overrightarrow{BC} = \vec{b}$.



Then, $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC} \Rightarrow \overrightarrow{AC} = \vec{a}+\vec{b} = 3\hat{i} + 6\hat{j} - 2\hat{k}$

and $\overrightarrow{AB} + \overrightarrow{BD} = \overrightarrow{AD} \Rightarrow \overrightarrow{AD} + \overrightarrow{AD} = \overrightarrow{AB}$

$\Rightarrow \overrightarrow{BD} = \overrightarrow{AD} - \overrightarrow{AB} = \vec{b} - \vec{a}$

Now, $\overrightarrow{AC} = 3\hat{i} + 6\hat{j} - 2\hat{k} \Rightarrow |\overrightarrow{AC}| = \sqrt{9+36+4} = 7$

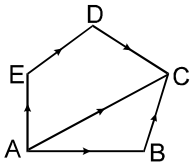
and, $\overrightarrow{BD} = \hat{i} + 2\hat{j} - 8\hat{k} \Rightarrow |\overrightarrow{BD}| = \sqrt{1+4+64} = \sqrt{69}$

$\therefore$ Unit vector along $\overrightarrow{AC} = \dfrac{\overrightarrow{AC}}{|\overrightarrow{AC}|} = \dfrac{1}{7}\left(3\hat{i} + 6\hat{j} - 2\hat{k}\right)$

and, Unit vector along $\overrightarrow{BD} = \dfrac{\overrightarrow{BD}}{|\overrightarrow{BD}|} = \dfrac{1}{\sqrt{69}}\left(\hat{i} + 2\hat{j} - 8\hat{k}\right)$

Solved Example ABCDE is a pentagon. Prove that the resultant of the forces $\overrightarrow{AB}$, $\overrightarrow{AE}$, $\overrightarrow{BC}$, $\overrightarrow{DC}$, $\overrightarrow{ED}$ and $\overrightarrow{AC}$ is $3\overrightarrow{AC}$.

Solution. Let R be the resultant force



$\therefore R = \overrightarrow{AB} + \overrightarrow{AE} + \overrightarrow{BC} + \overrightarrow{DC} + \overrightarrow{ED} + \overrightarrow{AC}$

$\therefore R = (\overrightarrow{AB} + \overrightarrow{BC}) + (\overrightarrow{AE} + \overrightarrow{ED} + \overrightarrow{DC}) + \overrightarrow{AC}$

$= \overrightarrow{AC} + \overrightarrow{AC} + \overrightarrow{AC}$

$= 3\overrightarrow{AC}$. Hence proved.

Self Practice Problems :

1. Express : (i) The vectors $\overrightarrow{BC}$ $\overrightarrow{CA}$ and $\overrightarrow{AB}$ in terms of the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$

(ii) The vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and in terms of the vectors $\overrightarrow{OC}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$.

Ans. (i) $\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB}$, $\overrightarrow{CA} = \overrightarrow{OA} - \overrightarrow{OC}$, $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

2. Given a regular hexagon ABCDEF with centre O, show that

(i) $\overrightarrow{OB} - \overrightarrow{OA} = \overrightarrow{OC} - \overrightarrow{OD}$ (ii) $\overrightarrow{OD} + \overrightarrow{OA} = 2\overrightarrow{OB} + \overrightarrow{OF}$ (iii) $\overrightarrow{AD} + \overrightarrow{EB} + \overrightarrow{PC} = 4\overrightarrow{AB}$

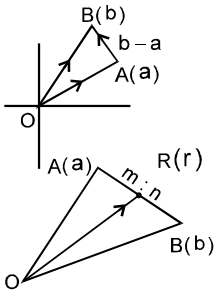
3. The vector $-\hat{i} + \hat{j} - \hat{k}$ bisects the angle between the vectors $\vec{c}$ and $3\hat{i} + 4\hat{j}$. Determine the unit vector along $\vec{c}$. **Ans.** $-\dfrac{1}{3}\hat{i} + \dfrac{2}{15}\hat{j} - \dfrac{14}{15}\hat{k}$

4. The sum of the two unit vectors is a unit vector. Show that the magnitude of the their difference is $\sqrt{3}$.

5. Position Vector Of A Point:

let O be a fixed origin, then the position vector of a point P is the vector $\overrightarrow{OP}$. If $\vec{a}$ and $\vec{b}$ are position vectors of two points A and B, then,

$\overrightarrow{AB} = \vec{b} - \vec{a}$ = pv of B – pv of **A.**



DISTANCE FORMULA

Distance between the two points A $(\vec{a})$ and B $(\vec{b})$ is AB = $|\vec{a} - \vec{b}|$

SECTION FORMULA

If $\vec{a}$ and $\vec{b}$ are the position vectors of two points A & B then the p.v. of a point which divides AB in the ratio m: n is given by: $\vec{r} = \dfrac{n\vec{a} + m\vec{b}}{m+n}$.

Note p.v. of mid point of AB = $\dfrac{\vec{a}+\vec{b}}{2}$.

Solved Example: ABCD is a parallelogram. If L, M be the middle point of BC and CD, express $\overrightarrow{AL}$ and $\overrightarrow{AM}$ in terms of $\overrightarrow{AB}$ and $\overrightarrow{AD}$, also show that $\overrightarrow{AL} + \overrightarrow{AM} = \dfrac{3}{2}\overrightarrow{AC}$.

Solution. Let the position vectors of points B and D be respectively $\vec{b}$ and $\vec{a}$ referred to A as origin of reference.

Then $\overrightarrow{AC} = \overrightarrow{AD} + \overrightarrow{DC} = \overrightarrow{AD} + \overrightarrow{AB}$ $\qquad$ [$\because$ $\overrightarrow{DC} = \overrightarrow{AB}$]

$= \vec{d} + \vec{b}$ $\quad\because\quad \overrightarrow{AB} = \vec{b}$, $\overrightarrow{AD} = \vec{d}$

i.e. position vector of C referred to A is $\vec{d} + \vec{b}$

$\therefore$ $\overrightarrow{AL}$ = p.v. of L, the mid point of $\overrightarrow{BC}$.

$= \frac{1}{2}$ [p.v. of D + p.v. of C] $= \frac{1}{2}\left(\vec{b}+\vec{d}+\vec{b}\right) = \overrightarrow{AB} + \frac{1}{2}\overrightarrow{AD}$

$\overrightarrow{AM} = \frac{1}{2}\left[\vec{d}+\vec{d}+\vec{b}\right] = \overrightarrow{AD} + \frac{1}{2}\overrightarrow{AB}$

$\therefore \overrightarrow{AL} + \overrightarrow{AM} = \vec{b} + \frac{1}{2}\vec{d} + \vec{d} + \frac{1}{2}\vec{b}$

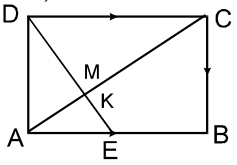
$= \frac{3}{2}\vec{b} + \frac{3}{2}\vec{d} \qquad = \frac{3}{2}(\vec{b} + \vec{d}) = \frac{3}{2}\overrightarrow{AC}$.

Solved Example If ABCD is a parallelogram and E is the mid point of AB, show by vector method that DE trisects and is trisected by AC.

Solution. Let $\overrightarrow{AB} = \vec{a}$ and $AD = \vec{b}$

Then $\overrightarrow{BC} = \overrightarrow{AD} = \vec{b}$ and $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{AD} = \vec{a} + \vec{b}$

Also let K be a point on AC, such that AK : AC = 1 : 3



or, $AK = \frac{1}{3}AC \quad \Rightarrow \quad \overrightarrow{AK} = \frac{1}{3}(\vec{a} + \vec{b})$ .....(i)

Again E being the mid point of AB, we have

$\overrightarrow{AE} = \frac{1}{2}\vec{a}$

Let M be the point on DE such that DM : ME = 2 : 1

$\therefore \quad \overrightarrow{AM} = \frac{\overrightarrow{AD}+2\overrightarrow{AE}}{1+2} = \frac{\vec{b}+\vec{a}}{3}$ .....(ii)

From (i) and (ii) we find that : $\overrightarrow{AK} = \frac{1}{3}(\vec{a} + \vec{b}) = \overrightarrow{AM}$, and so we conclude that K and M coincide. i.e. DE trisect AC and is trisected by AC. Hence proved.

Self Practice Problems

1. If $\vec{a}, \vec{b}$ are position vectors of the points (1, –1), (–2, m), find the value of m for which $\vec{a}$ and $\vec{b}$ are collinear. **Ans.** m = 2

2. The position vectors of the points A, B, C, D are $\hat{i}+\hat{j}+\hat{k}$, $2\hat{i}+5\hat{j}$, $3\hat{i}+2\hat{j}-3\hat{k}$, $\hat{i}-6\hat{j}-\hat{k}$ respectively. Show that the lines AB and CD are parallel and find the ratio of their lengths. **Ans.** 1 : 2

3. The vertices P, Q and S of a triangle PQS have position vectors $\vec{p}, \vec{q}$ and $\vec{s}$ respectively.

(i) Find $\vec{m}$, the position vector of M, the mid-point of PQ, in terms of $\vec{p}$ and $\vec{q}$.

(ii) Find $\vec{t}$, the position vector of T on SM such that ST : TM = 2 : 1, in terms of $\vec{p}, \vec{q}$ and $\vec{s}$.

(iii) If the parallelogram PQRS is now completed. Express $\vec{r}$, the position vector of the point R in terms of $\vec{p}, \vec{q}$ and $\vec{s}$

Prove that P, T and R are collinear.

Ans. $\vec{m} = \frac{1}{2}(\vec{p}+\vec{q})$, $\quad \vec{t} = \frac{1}{2}(\vec{p}+\vec{q}+\vec{s})$, $\quad \vec{r} = \frac{1}{2}\vec{q}+\vec{p}-\vec{s}$

4. D, E, F are the mid-points of the sides BC, CA, AB respectively of a triangle. Show $\overrightarrow{FE} = 1/2\ \overrightarrow{BC}$ and that the sum of the vectors $\overrightarrow{AD}$, $\overrightarrow{BE}$, $\overrightarrow{CF}$ is zero.

5. The median AD of a triangle ABC is bisected at E and BE is produced to meet the side AC in F; show that AF = 1/3 AC and EF = 1/4 BF.

6. Point L, M, N divide the sides BC, CA, AB of ΔABC in the ratios 1 : 4, 3 : 2, 3 : 7 respectively. Prove that $\overrightarrow{AL} + \overrightarrow{BM} + \overrightarrow{CN}$ is a vector parallel to $\overrightarrow{CK}$, when K divides AB in the ratio 1 : 3.

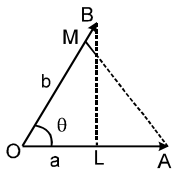
6. Scalar Product Of Two Vectors:

Geometrical interpretation of Scalar Product

Let $\vec{a}$ and $\vec{b}$ be vectors represented by $\overrightarrow{OA}$ and $\overrightarrow{OB}$ respectively. Let θ be the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$. Draw BL $\perp$ OA and AM $\perp$ OB.

From Δs OBL and OAM, we have OL = OB cos θ and OM = OA cos θ. Here OL and OM are known as

projections of $\vec{b}$ on $\vec{a}$ and $\vec{a}$ on $\vec{b}$ respectively.



Now, $\vec{a}\,.\,\vec{b} = |\vec{a}|\,|\vec{b}|\cos\theta$

$= |\vec{a}|\,(OB\cos\theta)$

$= |\vec{a}|\,(OL)$

$=$ (Magnitude of $\vec{a}$) (Projection of $\vec{b}$ on $\vec{a}$) .....(i)

Again $\vec{a}\,.\,\vec{b} = |\vec{a}|\,|\vec{b}|\cos\theta = |\vec{b}|\,(|\vec{a}|\cos\theta)$

$= |\vec{b}|\,(OA\cos\theta)$

$= |\vec{b}|\,(OM)$

$=$ (magnitude of $\vec{b}$) (Projection of $\vec{a}$ on $\vec{b}$) .....(ii)

Thus geometrically interpreted, the scalar product of two vectors is the product of modulus of either vector and the projection of the other in its direction.

1. i.i = j.j = k.k = 1; i.j = j.k = k.i = 0 ☞ projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{a}\vec{b}}{|\vec{b}|}$

2. if $\vec{a} = a_1i + a_2j + a_3k$ & $\vec{b} = b_1i + b_2j + b_3k$ then $\vec{a}\,.\,\vec{b} = a_1b_1 + a_2b_2 + a_3b_3$

$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$, $|\vec{b}| = \sqrt{b_1^2 + b_2^2 + b_3^2}$

3. the angle ϕ between $\vec{a}$ & $\vec{b}$ is given by $\cos\phi = \dfrac{\vec{a}\vec{b}}{|\vec{a}|\,|\vec{b}|}$ $0 \le \phi \le \pi$

4. $\vec{a}\,.\,\vec{b} = |\vec{a}|\,|\vec{b}|\cos\theta$ $(0 \le \theta \le \pi)$, note that if θ is acute then $\vec{a}\,.\,\vec{b} > 0$ & if θ is obtuse then $\vec{a}\,.\,\vec{b} < 0$

5. $\vec{a}\,.\,\vec{a} = |\vec{a}|^2 = \vec{a}^2$, $\vec{a}\vec{b} = \vec{b}\vec{a}$ (commutative) ☞ $\vec{a}\,.\,(\vec{b}+\vec{c}) = \vec{a}\,.\,\vec{b} + \vec{a}\,.\,\vec{c}$ (distributive)

6. $\vec{a}\,.\,\vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$ $(\vec{a} \ne 0\ \vec{b} \ne 0)$

7. $(m\,\vec{a})\,.\,\vec{b} = \vec{a}\,.\,(m\,\vec{b}) = m\,(\vec{a}\,.\,\vec{b})$ (associative) where m is scalar.

Note : (i) Maximum value of $\vec{a}\,.\,\vec{b}$ is $|\vec{a}|\,|\vec{b}|$ **(ii)** Minimum value of $\vec{a}\,.\,\vec{b}$ is $-|\vec{a}|\,|\vec{b}|$

(iii) Any vector $\vec{a}$ can be written as, $\vec{a} = (\vec{a}\,.\,\hat{i})\,\hat{i} + (\vec{a}\,.\,\hat{j})\,\hat{j} + (\vec{a}\,.\,\hat{k})\,\hat{k}$.

Solved Example Find the value of p for which the vectors $\vec{a} = 3\hat{i} + 2\hat{j} + 9\hat{k}$ and $\vec{b} = \hat{i} + p\hat{j} + 3\hat{k}$ are

(i) perpendicular (ii) parallel

Solution. (i) $\vec{a} \perp \vec{b} \Rightarrow \vec{a}\,.\,\vec{b} = 0 \Rightarrow (3\hat{i} + 2\hat{j} + 9\hat{k})\,.\,(\hat{i} + p\hat{j} + 3\hat{k}) = 0$

$\Rightarrow 3 + 2p + 27 = 0 \Rightarrow p = -15$

(ii) We know that the vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are parallel iff

$\vec{a} = \lambda\vec{b} \Leftrightarrow (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) = \lambda\,(b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) \Leftrightarrow a_1 = \lambda b_1,\ a_2 = \lambda b_2,\ a_3 = \lambda b_3$

$\Leftrightarrow \dfrac{a_1}{b_1} = \dfrac{a_2}{b_2} = \dfrac{a_3}{b_3}$ $(=\lambda)$

So, vectors $\vec{a} = 3\hat{i} + 2\hat{j} + 9\hat{k}$ and $\vec{b} = \hat{i} + p\hat{j} + 3\hat{k}$ are parallel iff

$\dfrac{3}{1} = \dfrac{2}{p} = \dfrac{9}{3} \Rightarrow 3 = \dfrac{2}{p} \Rightarrow p = 2/3$

Solved Example: If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 3$, $|\vec{b}| = 5$ and $|\vec{c}| = 7$, find the angle between $\vec{a}$ and $\vec{b}$.

Solution. We have, $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

$\Rightarrow \vec{a} + \vec{b} = -\vec{c} \Rightarrow (\vec{a} + \vec{b})\,.\,(\vec{a} + \vec{b}) = (-\vec{c})\,.\,(-\vec{c})$

$\Rightarrow |\vec{a} + \vec{b}|^2 = |\vec{c}|^2 \Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a}\,.\,\vec{b} = |\vec{c}|^2$

$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + 2\,|\vec{a}|\,|\vec{b}|\cos\theta = |\vec{c}|^2$

$\Rightarrow 9 + 25 + 2\,(3)\,(5)\cos\theta = 49 \Rightarrow \cos\theta = \dfrac{1}{2} \Rightarrow \theta = \dfrac{\pi}{3}$.

Solved Example Find the values of x for which the angle between the vectors $\vec{a} = 2x^2\,\hat{i} + 4x\,\hat{j} + \hat{k}$ and $\vec{b} = 7\hat{i} - 2\hat{j} + x\,\hat{k}$ is obtuse.

Solution. The angle q between vectors $\vec{a}$ and $\vec{b}$ is given by $\cos\theta = \dfrac{\vec{a}\,.\,\vec{b}}{|\vec{a}|\,|\vec{b}|}$

Now, θ is obtuse $\Rightarrow$ $\cos\theta < 0$ $\Rightarrow$ $\dfrac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|} < 0$

$\Rightarrow$ $\vec{a}.\vec{b} < 0$ $[\because, |\vec{a}|, |\vec{b}| > 0]$

$\Rightarrow$ $14x^2 - 8x + x < 0$

$\Rightarrow$ $17x(2x-1) < 0$ $\Rightarrow x(2x-1) < 0 \Rightarrow 0 < x < \dfrac{1}{2}$

Hence, the angle between the given vectors is obtuse if $x \in (0, 1/2)$

Solved Example: D is the mid point of the side BC of a triangle ABC, show that $AB^2 + AC^2 = 2(AD^2 + BD^2)$

Solution. We have

$\overrightarrow{AB} = \overrightarrow{AD} + \overrightarrow{DB}$

$\Rightarrow$ $AB^2 = (\overrightarrow{AD} + \overrightarrow{DB})^2$

$= AD^2 + DB + 2\overrightarrow{AD} \cdot \overrightarrow{DB}$ .....(i)

Also we have

$\overrightarrow{AC} = \overrightarrow{AD} + \overrightarrow{DC}$ $\Rightarrow$ $AC^2 = (\overrightarrow{AD} + \overrightarrow{DC})^2$

$= AD^2 + DC^2 + 2\overrightarrow{AD} \cdot \overrightarrow{DC}$ .....(ii)

Adding (i) and (ii), we get

$AB^2 + AC^2 = 2AD^2 + 2BD^2 + 2\overrightarrow{AD} \cdot (\overrightarrow{DB} + \overrightarrow{DC})$

$= 2(DA^2 + DB^2)$, for $\overrightarrow{DB} + \overrightarrow{DC} = 0$

Solved Example

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$, then find

(i) Component of $\vec{b}$ along $\vec{a}$. (ii) Component of $\vec{b}$ perpendicular to along $\vec{a}$.

Solution. (i) Component of $\vec{b}$ along $\vec{a}$ is

$\left(\dfrac{\vec{a}.\vec{b}}{|\vec{a}|^2}\right)\vec{a}$

Here $\vec{a} \cdot \vec{b} = 2 - 1 + 3 = 4$

$|\vec{a}|^2 = 3$

Hence $\left(\dfrac{\vec{a}.\vec{b}}{|\vec{a}|^2}\right)\vec{a} = \dfrac{4}{3}\vec{a} = \dfrac{4}{3}(\hat{i} + \hat{j} + \hat{k})$

(ii) Component of $\vec{b}$ perpendicular to along $\vec{a}$ is $\vec{b} - \left(\dfrac{\vec{a}.\vec{b}}{|\vec{a}|^2}\right)\vec{a}. = \dfrac{1}{3}\left(2\hat{i} - 7\hat{j} + 5\hat{k}\right)$

Self Practice Problems :1. If $\vec{a}$ and $\vec{b}$ are unit vectors and θ is angle between them, prove that $\tan\dfrac{\theta}{2} = \dfrac{|\vec{a} - \vec{b}|}{|\vec{a} + \vec{b}|}$. **2.** Find the values of x and y is the vectors $\vec{a} = 3\hat{i} + x\hat{j} - \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} + y\hat{k}$ are mutually perpendicular vectors of equal magnitude. **Ans.** $x = -\dfrac{31}{12}$, $y = \dfrac{41}{12}$

3. Let $\vec{a} = x^2\hat{i} + 2\hat{j} - 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $c = x^2\hat{i} + 5\hat{j} - 4\hat{k}$ be three vectors. Find the values of x for which the angle between $\vec{a}$ and $\vec{b}$ is acute and the angle between $\vec{b}$ and $\vec{c}$ is obtuse. **Ans.** $(-3, -2) \cup (2, 3)$

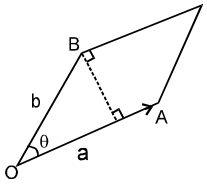
4. The points O, A, B, C, D, are such that $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, $\overrightarrow{OC} = 2\vec{a} + 3\vec{b}$, $\overrightarrow{OD} = \vec{a} + 2\vec{b}$. Give that the length of $\overrightarrow{OA}$ is three times the length of $\overrightarrow{OB}$ show that $\overrightarrow{BD}$ and $\overrightarrow{AC}$ are perpendicular.

5. ABCD is a tetrahedron and G is the centroid of the base BCD. Prove that
$AB^2 + AC^2 + AD^2 = GB^2 + GC^2 + GD^2 + 3GA^2$

7. Vector Product Of Two Vectors:

1. If $\vec{a}$ & $\vec{b}$ are two vectors & θ is the angle between them then $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\vec{n}$, where $\vec{n}$ is the unit vector perpendicular to both $\vec{a}$ & $\vec{b}$ such that $\vec{a}$, $\vec{b}$ & $\vec{n}$ forms a right handed screw system.

2. Geometrically $|\vec{a} \times \vec{b}|$ = area of the parallelogram whose two adjacent sides are represented by $\vec{a}$ & $\vec{b}$.



3. $\hat{i}\times\hat{i}=\hat{j}\times\hat{j}=\hat{k}\times\hat{k}=\vec{0}$; $\hat{i}\times\hat{j}=\hat{k},\ \hat{j}\times\hat{k}=\hat{i},\ \hat{k}\times\hat{i}=\hat{j}$

4. If $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ & $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ then $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\a_1&a_2&a_3\\b_1&b_2&b_3\end{vmatrix}$

5. $\vec{a}\ x\ \vec{b}\neq\vec{b}\ x\ \vec{a}$ (not commutative)

6. $(m\vec{a})\times\vec{b}=\vec{a}\times(m\vec{b})=m(\vec{a}\times\vec{b})$ (associative) where m is a scalar.

7. $\vec{a}\ x\ (\vec{b}+\vec{c})=(\vec{a}\ x\ \vec{b})+(\vec{a}\ x\ \vec{c})$ (distributive)

8. $\vec{a}\ x\ \vec{b}=0\Leftrightarrow\vec{a}$ & $\vec{b}$ are parallel (collinear) $(\vec{a}\neq0\ ,\ \vec{b}\neq0)$ i.e. $\vec{a}=K\vec{b}$, where K is a scalar.

9. Unit vector perpendicular to the plane of $\vec{a}$ & $\vec{b}$ is $\hat{n}=\pm\dfrac{\vec{a}\ x\ \vec{b}}{|\vec{a}\ x\ \vec{b}|}$

☞ A vector of magnitude 'r' & perpendicular to the palne of $\vec{a}$ & $\vec{b}$ is $\pm\dfrac{r(\vec{a}\ x\ \vec{b})}{|\vec{a}\ x\ \vec{b}|}$

☞ If θ is the angle between $\vec{a}$ & $\vec{b}$ then $\sin\theta=\dfrac{|\vec{a}\ x\ \vec{b}|}{|\vec{a}||\vec{b}|}$

☞ If $\vec{a},\vec{b}$ & $\vec{c}$ are the pv's of 3 points A, B & C then the vector area of triangle ABC = $\dfrac{1}{2}\left[\vec{a}\ x\ \vec{b}+\vec{b}\ x\ \vec{c}+\vec{c}\ x\ \vec{a}\right]$. The points A, B & C are collinear if $\vec{a}\ x\ \vec{b}+\vec{b}\ x\ \vec{c}+\vec{c}\ x\ \vec{a}=\vec{0}$

☞ Area of any quadrilateral whose diagonal vectors are $\vec{d}_1$ & $\vec{d}_2$ is given by $\dfrac{1}{2}|\vec{d}_1\ x\ \vec{d}_2|$

☞ Lagrange's Identity: for any two vectors $\vec{a}$ & $\vec{b}$; $(\vec{a}\ x\ \vec{b})^2=|\vec{a}|^2|\vec{b}|^2-(\vec{a}.\vec{b})^2=\begin{vmatrix}\vec{a}.\vec{a}&\vec{a}.\vec{b}\\\vec{a}.\vec{b}&\vec{b}.\vec{b}\end{vmatrix}$

Solved Example

Find a vector of magnitude 9, which is perpendicular to both the vectors $4\hat{i}+\hat{j}+3\hat{k}$ and $-2\hat{i}+\hat{j}-2\hat{k}$.

Solution. Let $\vec{a}=4\hat{i}-\hat{j}+3\hat{k}$ and $\vec{b}=-2\hat{i}+\hat{j}-2\hat{k}$. Then,

$$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\4&-1&3\\-2&1&-2\end{vmatrix}=(2-3)\ \hat{i}-(-8+6)\ \hat{j}+(4-2)\ \hat{k}=-\hat{i}+2\hat{j}+2\hat{k}$$

$$\Rightarrow\quad|\vec{a}\times\vec{b}|=\sqrt{(-1)^2+2^2+2^2}=3$$

$$\therefore\quad\text{Required vector}=9\left(\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}\right)=\frac{9}{3}\ (-\hat{i}+2\hat{j}+2\hat{k})=-3\hat{i}+6\hat{j}+6\hat{k}$$

Solved Example

For any three vectors $\vec{a},\vec{b},\vec{c}$. Show that $\vec{a}\times(\vec{b}+\vec{c})+\vec{b}\times(\vec{c}+\vec{a})+\vec{c}\times(\vec{a}+\vec{b})=\vec{0}$.

Solution. We have, $\vec{a}\times(\vec{b}+\vec{c})+\vec{b}\times(\vec{c}+\vec{a})+\vec{c}\times(\vec{a}+\vec{b})$

$=\vec{a}\times\vec{b}+\vec{a}\times\vec{c}+\vec{b}\times\vec{c}+\vec{b}\times\vec{a}+\vec{c}\times\vec{a}+\vec{c}\times\vec{b}$ [Using distributive law]

$=\vec{a}\times\vec{b}+\vec{a}\times\vec{c}+\vec{b}\times\vec{c}-\vec{a}\times\vec{b}-\vec{a}\times\vec{c}-\vec{b}\times\vec{c}$ [$\because\ \vec{a}\times\vec{b}=-\vec{b}\times\vec{a}$ etc]

Solved Example: For any vector $\vec{a}$, prove that $|\vec{a}\times\hat{i}|^2+|\vec{a}\times\hat{j}|^2+|\vec{a}\times\hat{k}|^2=2|\vec{a}|^2$

Solution. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$. Then

$\vec{a}\times\hat{i}=(a_1\hat{i}+a_2\hat{j}+a_3\hat{k})\times\hat{i}=a_1(\hat{i}\times\hat{i})+a_2(\hat{j}\times\hat{i})+a_3(\hat{k}\times\hat{i})=-a_2\hat{k}+a_3\hat{i}$

$\Rightarrow\quad|\vec{a}\times\hat{i}|^2=a_2^2+a_3^2$

$\vec{a}\times\hat{j}=(a_1\hat{i}+a_2\hat{j}+a_3\hat{k})\times\hat{j}=a_1\hat{k}-a_3\hat{i}$

$\Rightarrow\quad|\vec{a}\times\hat{j}|^2=a^2_1+a_3^2\qquad\Rightarrow\quad|\vec{a}\times\hat{k}|^2=a_1^2+a_2^2$

$\therefore\quad|\vec{a}\times\hat{i}|^2+|\vec{a}\times\hat{j}|^2+|\vec{a}\times\hat{k}|^2=a_2^2+a_3^3+a_1^2+a_3^2+a_1^2+a_2^2$

$2(d_1^2+a_2^2+a_3^2)=2|\vec{a}|^2$

Solved Example: Let $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = 10\vec{a} + 2\vec{b}$ and $\overrightarrow{OC} = \vec{b}$ where O is origin. Let p denote the area of the quadrilateral OABC and q denote the area of the parallelogram with OA and OC as adjacent sides. Prove that p = 6q.

Solution. We have,

p = Area of the quadrilateral OABC

$$= \frac{1}{2}\,|\overrightarrow{OB}\times\overrightarrow{AC}|$$

$$= \frac{1}{2}\,|\overrightarrow{OB}\times(\overrightarrow{OC}-\overrightarrow{OA})|$$

$$= \frac{1}{2}\,|(10\vec{a}+2\vec{b})\times(\vec{b}-\vec{a})|$$

$$= \frac{1}{2}\,|10(\vec{a}\times\vec{b}-10(\vec{a}\times\vec{a})+2(\vec{b}\times\vec{b})-2(\vec{b}\times\vec{a})|$$

$$= \frac{1}{2}\,|10(\vec{a}\times\vec{b})-0+0+2(\vec{a}\times\vec{b})|$$

and, q = Area of the parallelogram with OA and OC as adjacent sides

$$= |\overrightarrow{OA}\times\overrightarrow{OC}| = |\vec{a}\times\vec{b}| \qquad \text{.....(ii)}$$

From (i) and (ii), we get p = 6q

Self Practice Problems :

1. If $\vec{p}$ and $\vec{q}$ are unit vectors forming an angle of 30º; find the area of the parallelogram having $\vec{a} = \vec{p} + 2\vec{q}$ and $\vec{b} = 2\vec{p} + \vec{q}$ as its diagonals. **Ans.** 3/4 sq. units

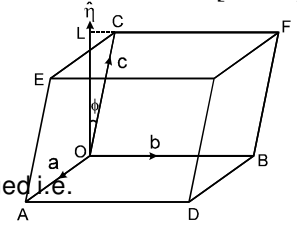
2. Show that $\{(\vec{a} + \vec{b} + \vec{c}) \times (\vec{c} - \vec{b})\} \cdot \vec{a} = 2[\vec{a}\ \vec{b}\ \vec{c}]$.

3. Prove that the normal to the plane containing the three points whose position vectors are $\vec{a}, \vec{b}, \vec{c}$ lies in the direction $\vec{b}\times\vec{c} + \vec{c}\times\vec{a} + \vec{a}\times\vec{b}$

4. ABC is a triangle and EF is any straight line parallel to BC meeting AC, AB in E, F respectively. If BR and CQ be drawn parallel to AC, AB respectively to meet EF in R and Q respectively, prove that Δ ARB = ΔACQ.

8. Scalar Triple Product:

- The scalar triple product of three vectors $\vec{a}$, $\vec{b}$ & $\vec{c}$ is defined as: $\vec{a}\times\vec{b}\cdot\vec{c} = |\vec{a}||\vec{b}||\vec{c}|\sin\theta\cos\phi$ where θ is the angle between $\vec{a}$ & $\vec{b}$ & ϕ is the angle between $\vec{a}\times\vec{b}$ & $\vec{c}$. It is also written as $[\vec{a}\ \vec{b}\ \vec{c}]$ and spelled as box product.



- Scalar triple product geometrically represents the volume of the parallelopiped whose three coterminous edges are represented by $\vec{a}, \vec{b} \& \vec{c}$ i.e. $V = [\vec{a}\ \vec{b}\ \vec{c}]$
- In a scalar triple product the position of dot & cross can be interchanged i.e.

  $\vec{a}.(\vec{b}\times\vec{c}) = (\vec{a}\times\vec{b}).\vec{c}$ OR $[\vec{a}\ \vec{b}\ \vec{c}] = [\vec{b}\ \vec{c}\ \vec{a}] = [\vec{c}\ \vec{a}\ \vec{b}]$
- $\vec{a}.(\vec{b}\times\vec{c}) = -\vec{a}.(\vec{c}\times\vec{b})$ i.e. $[\vec{a}\ \vec{b}\ \vec{c}] = -[\vec{a}\ \vec{c}\ \vec{b}]$
- If $\vec{a} = a_1i+a_2j+a_3k$; $\vec{b} = b_1i+b_2j+b_3k$ & $\vec{c} = c_1i+c_2j+c_3k$ then $[\vec{a}\,\vec{b}\,\vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$.

  In general, if $\vec{a} = a_1\vec{l} + a_2\vec{m} + a_3\vec{n}$; $\vec{b} = b_1\vec{l} + b_2\vec{m} + b_3\vec{n}$ & $\vec{c} = c_1\vec{l} + c_2\vec{m} + c_3\vec{n}$

  then $\left[\vec{a}\,\vec{b}\,\vec{c}\right] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \left[\vec{l}\,\vec{m}\,\vec{n}\right]$; where $\vec{\ell}$, $\vec{m}$ & $\vec{n}$ are non coplanar vectors.
- If $\vec{a}$, $\vec{b}$, $\vec{c}$ are coplanar $\Leftrightarrow [\vec{a}\ \vec{b}\ \vec{c}] = 0$.
- Scalar product of three vectors, two of which are equal or parallel is 0 i.e. $[\vec{a}\,\vec{b}\,\vec{c}] = 0$,
- If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non – coplanar then $[\vec{a}\ \vec{b}\ \vec{c}] > 0$ for right handed system & $[\vec{a}\ \vec{b}\ \vec{c}] < 0$ for left handed system.
- $[i\ j\ k] = 1$ ☞ $[K\vec{a}\ \vec{b}\ \vec{c}] = K[\vec{a}\ \vec{b}\ \vec{c}]$ ☞ $[(\vec{a}+\vec{b})\ \vec{c}\ \vec{d}] = [\vec{a}\ \vec{c}\ \vec{d}] + [\vec{b}\ \vec{c}\ \vec{d}]$
- The volume of the tetrahedron OABC with O as origin & the pv's of A, B and C being $\vec{a}$, $\vec{b}$ & $\vec{c}$ respectively is given by $V = \frac{1}{6}[\vec{a}\ \vec{b}\ \vec{c}]$

☞ The positon vector of the centroid of a tetrahedron if the pv's of its vertices are $\vec{a}$, $\vec{b}$, $\vec{c}$ & $\vec{d}$ are given by $\frac{1}{4}[\vec{a}+\vec{b}+\vec{c}+\vec{d}]$.

Note that this is also the point of concurrency of the lines joining the vertices to the centroids of the opposite faces and is also called the centre of the tetrahedron. In case the tetrahedron is regular it is equidistant from the vertices and the four faces of the tetrahedron.

Remember that: $\left[\vec{a}-\vec{b}\quad \vec{b}-\vec{c}\quad \vec{c}-\vec{a}\right] = 0$ & $\left[\vec{a}+\vec{b}\quad \vec{b}+\vec{c}\quad \vec{c}+\vec{a}\right] = 2\left[\vec{a}\ \vec{b}\ \vec{c}\right]$.

Solved Example

Find the volume of a parallelopiped whose sides are given by $-3\hat{i}+7\hat{j}+5\hat{k}$, $-5\hat{i}+7\hat{j}-3\hat{k}$ and $7\hat{i}-5\hat{j}-3\hat{k}$

Solution. Let $\vec{a}=-3\hat{i}+7\hat{j}+5\hat{k}$, $\vec{b}=-5\hat{i}+7\hat{j}+3\hat{k}$ and $\vec{c}=7\hat{i}-5\hat{j}-3\hat{k}$.

We know that the volume of a parallelopiped whose three adjacent edges are $\vec{a},\vec{b},\vec{c}$ is $\left|[\vec{a},\vec{b},\vec{c}]\right|$.

Now, $[\vec{a}\,\vec{b}\,\vec{c}] = \begin{vmatrix} -3 & 7 & 5 \\ -5 & 7 & -3 \\ 7 & -5 & -3 \end{vmatrix} = -3(-21-15) - 7(15+21) + 5(25-49)$

$= 108 - 252 - 120 = -264$

So, required volume of the parallelopiped = $\left|[\vec{a},\vec{b},\vec{c}]\right| = |-264| = 264$ cubic units

Solved Example: Simplify $[\vec{a}-\vec{b}\,\vec{b}-\vec{c}\,\vec{c}-\vec{a}]$

Solution. We have :

$[\vec{a}-\vec{b}\,\vec{b}-\vec{c}\,\vec{c}-\vec{a}] = \{(\vec{a}-\vec{b})\times(\vec{b}-\vec{c})\}\cdot(\vec{c}-\vec{a})$ [by def.]

$= (\vec{a}\times\vec{b}-\vec{a}\times\vec{c}-\vec{b}\times\vec{b}+\vec{b}\times\vec{c})\cdot(\vec{c}-\vec{a})$ [by dist. law]

$= (\vec{a}\times\vec{b}+\vec{c}\times\vec{a}+\vec{b}\times\vec{c})\cdot(\vec{c}-\vec{a})$ [$\because \vec{b}\times\vec{b}=0$]

$= (\vec{a}\times\vec{b})\cdot\vec{c} - (\vec{a}\times\vec{b})\cdot\vec{a} + (\vec{c}\times\vec{a})\cdot\vec{c} - (\vec{c}\times\vec{a})\cdot\vec{a} + (\vec{b}\times\vec{c})\cdot\vec{c} - (\vec{b}\times\vec{c})\cdot\vec{a}$ [by dist. law]

$= [\vec{a}\,\vec{b}\,\vec{c}] - [\vec{a}\,\vec{b}\,\vec{a}] + [\vec{c}\,\vec{a}\,\vec{c}] - [\vec{c}\,\vec{a}\,\vec{a}] + [\vec{b}\,\vec{c}\,\vec{c}] - [\vec{b}\,\vec{c}\,\vec{a}]$

$= [\vec{a}\,\vec{b}\,\vec{c}] - [\vec{b}\,\vec{c}\,\vec{a}]$ [$\because$ scalar triple product when any two vectors are equal is zero]

$= [\vec{a}\,\vec{b}\,\vec{c}] - [\vec{a}\,\vec{b}\,\vec{c}] = 0$ [$\because [\vec{b}\,\vec{c}\,\vec{a}] = [\vec{a}\,\vec{b}\,\vec{c}]$]

Solved Example: Find the volume of the tetrahedron whose four vertices have position vectors $\vec{a}$ $\vec{b}$ $\vec{c}$ and $\vec{d}$.

Solution. Let four vertices be A, B, C, D with p. v. $\vec{a}$ $\vec{b}$ $\vec{c}$ and $\vec{d}$. respectively.

$\therefore$ $\overrightarrow{DA} = (\vec{a}-\vec{d})$

$\overrightarrow{DB} = (\vec{b}-\vec{d})$

$\overrightarrow{DC} = (\vec{c}-\vec{d})$

Hence volume $= \frac{1}{6}[\vec{a}-\vec{d}\ \ \vec{b}-\vec{d}\ \ \vec{c}-\vec{d}]$

$= \frac{1}{6}(\vec{a}-\vec{d})\cdot[(\vec{b}-\vec{d})\times(\vec{c}-\vec{d})]$

$= \frac{1}{6}(\vec{a}-\vec{d})\cdot[\vec{b}\times\vec{c}-\vec{b}\times\vec{d}+\vec{c}\times\vec{d}]$

$= \frac{1}{6}\{[\vec{a}\ \ \vec{b}\ \ \vec{c}]-[\vec{a}\ \ \vec{b}\ \ \vec{d}]+[\vec{a}\ \ \vec{c}\ \ \vec{d}]-[\vec{d}\ \ \vec{b}\ \ \vec{c}]\}$

$= \frac{1}{6}\{[\vec{a}\ \ \vec{b}\ \ \vec{c}]-[\vec{a}\ \ \vec{b}\ \ \vec{d}]+[\vec{a}\ \ \vec{c}\ \ \vec{d}]-[\vec{b}\ \ \vec{c}\ \ \vec{d}]\}$

Solved Example: Show that the vectors $\vec{a}=-2\hat{i}+4\hat{j}-2\hat{k}$, $\vec{b}=4\hat{i}-2\hat{j}-2\hat{k}$ and $\vec{c}=-2\hat{i}-2\hat{j}+4\hat{k}$ are coplanar.

Solution: The vectors are coplanar since $[\vec{a}\,\vec{b}\,\vec{c}] = \begin{vmatrix} -2 & 4 & -2 \\ 4 & -2 & -2 \\ -2 & -2 & 4 \end{vmatrix} = 0$

Self Practice Problems : 1. Show that $\vec{a}\cdot(\vec{b}+\vec{c})\times(\vec{a}+\vec{b}+\vec{c}) = 0$

2. One vertex of a parallelopiped is at the point A (1, –1, –2) in the rectangular cartesian co-ordinate. If three adjacent vertices are at B(–1, 0, 2), C(2, –2, 3) and D(4, 2, 1), then find the volume of the parallelopiped.
Ans. 72

3. Find the value of m such that the vectors $2\hat{i}-\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}-3\hat{k}$ and $3\hat{i}+m\hat{j}+5\hat{k}$ are coplanar.
Ans. – 4

4. Show that the vector $\vec{a}, \vec{b}, \vec{c}$, are coplanar if and only if $\vec{b}+\vec{c}$, $\vec{c}+\vec{a}$, $\vec{a}+\vec{b}$ are coplanar.

9. Vector Triple Product:

Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be any three vectors, then the expression $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector & is called a vector triple product.

Geometrical Interpretation of $\vec{a} \times (\vec{b} \times \vec{c})$

Consider the expression $\vec{a} \times (\vec{b} \times \vec{c})$ which itself is a vector, since it is a cross product of two vectors $\vec{a}$ & $(\vec{b} \times \vec{c})$. Now $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector perpendicular to the plane containing $\vec{a}$ & $(\vec{b} \times \vec{c})$ but $\vec{b} \times \vec{c}$ is a vector perpendicular to the plane containing $\vec{b}$ & $\vec{c}$, therefore $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector which lies in the plane of $\vec{b}$ & $\vec{c}$ and perpendicular to $\vec{a}$. Hence we can express $\vec{a} \times (\vec{b} \times \vec{c})$ in terms of $\vec{b}$ & $\vec{c}$ i.e. $\vec{a} \times (\vec{b} \times \vec{c}) = x\vec{b} + y\vec{c}$ where x & y are scalars.

- ☞ $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$ ☞ $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$
- ☞ $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$, in general

Solved Example For any vector $\vec{a}$, prove that $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k}) = 2\vec{a}$

Solution. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$. Then, $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\hat{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k})$

$= \{(\hat{i} \cdot \hat{i})\vec{a} - (\hat{i} \cdot \vec{a})\hat{i}\} + \{(\hat{j} \cdot \hat{j})\vec{a} - (\hat{j} \cdot \vec{a})\hat{j}\} + \{(\hat{k} \cdot \hat{k})\vec{a} - (\hat{k} \cdot \vec{a})\hat{k}\}$

$= \{(\vec{a} - (\hat{i} \cdot \vec{a})\,\hat{i}\} + \{\vec{a} - (\hat{j} \cdot \vec{a})\hat{j}\} + \{\vec{a} - (\hat{k} \cdot \vec{a})\hat{k}\}$

$= 3\vec{a} - \{(\hat{i} \cdot \vec{a})\hat{i} + (\hat{j} \cdot \vec{a})\hat{j} + (\hat{k} \cdot \vec{a})\hat{k}$

$= 3\vec{a} - (a_1\hat{i} + a_2\hat{j} + a_3\hat{k})$

$= 3\vec{a} - \vec{a} = 2\vec{a}$

Solved Example Prove that $\vec{a} \times \{\vec{b} \times (\vec{c} \times \vec{a})\} = (\vec{b} \cdot \vec{d})(\vec{a} \times \vec{c}) - (\vec{b} \cdot \vec{c})\ (\vec{a} \times \vec{d})$

Solution. We have,

$\vec{a} \times \{\vec{b} \times (\vec{c} \times \vec{a})\} = \vec{a} \times \{(\vec{b} \cdot \vec{d})\,\vec{c} - (\vec{b} \cdot \vec{c})\,\vec{d}\}$

$= \vec{a} \times \{(\vec{b} \cdot \vec{d})\,\vec{c}\} - \vec{a} \times \{(\vec{b} \cdot \vec{c})\,\vec{d}\}$ [by dist. law]

$= (\vec{b} \cdot \vec{d})\,(\vec{a} \times \vec{c}) - (\vec{b} \cdot \vec{c})\,(\vec{a} \times \vec{d})$

Solved Example: Let $\vec{a} = \alpha\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{b} = \hat{i} + 2\alpha\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} - \alpha\hat{j} + \hat{k}$. Find the value(s) of α, if any, such that $\{(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c})\} \times (\vec{c} \times \vec{a}) = 0$. Find the vector product when $\alpha = 0$.

Solution.: $\{(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c})\} \times (\vec{c} \times \vec{a})$

$= [\vec{a}\ \vec{b}\ \vec{c}]\ \vec{b} \times (\vec{c} \times \vec{a}) = [\vec{a}\ \vec{b}\ \vec{c}]\ \{(\vec{a} \cdot \vec{b})\vec{c} - (\vec{b} \cdot \vec{c})\vec{a}\}$

which vanishes if (i) $(\vec{a} \cdot \vec{b})\ \vec{c} = (\vec{b} \cdot \vec{c})\ \vec{a}$ (ii) $[\vec{a}\ \vec{b}\ \vec{c}] = 0$

(i) $(\vec{a} \cdot \vec{b})\ \vec{c} = (\vec{b} \cdot \vec{c})\ \vec{a}$ leads to the equation $2\alpha^3 + 10\alpha + 12 = 0$, $\alpha^2 + 6\alpha = 0$ and $6\alpha - 6 = 0$, which do not have a common solution. (ii) $[\vec{a}\ \vec{b}\ \vec{c}] = 0$

$\Rightarrow \begin{vmatrix} \alpha & 2 & -3 \\ 1 & 2\alpha & -2 \\ 2 & -\alpha & 1 \end{vmatrix} = 0 \quad \Rightarrow \quad 3\alpha = 2 \quad \Rightarrow \quad \alpha = \frac{2}{3}$

when $\alpha = 0$, $[\vec{a}\ \vec{b}\ \vec{c}] = -10$, $\vec{a} \cdot \vec{b} = 6$, $\vec{b} \cdot \vec{c} = 0$ and the vector product is $-60\,(2\hat{i} + \hat{k})$.

Sol Exa If $\vec{A} + \vec{B} = \vec{a}$, $\vec{A} \cdot \vec{a} = 1$ and $\vec{A} \times \vec{B} = \vec{b}$, then prove that $\vec{A} = \frac{\vec{a} \times \vec{b} + \vec{a}}{|\vec{a}|^2}$ and $\vec{B} = \frac{\vec{b} \times \vec{a} + \vec{a}\,(|\vec{a}|^2 - 1)}{|\vec{a}|^2}$.

Solution.: Given $\vec{A} + \vec{B} = \vec{a}$ .....(i)

$\Rightarrow \vec{a} \cdot (\vec{A} + \vec{B}) = \vec{a} \cdot \vec{a} \quad \Rightarrow \quad \vec{a} \cdot \vec{A} + \vec{a} \cdot \vec{B} = \vec{a} \cdot \vec{a} \quad \Rightarrow \quad 1 + \vec{a} \cdot \vec{B} = |\vec{a}|^2$

$\Rightarrow \vec{a} \cdot \vec{B} = |\vec{a}|^2 - 1$ Given $\vec{A} \times \vec{B} = \vec{b} \Rightarrow \vec{a} \times (\vec{A} \times \vec{B}) = \vec{a} \times \vec{b}$

$\Rightarrow (\vec{a} \cdot \vec{B})\ \vec{A} - (\vec{a} \cdot \vec{A})\ \vec{B} = \vec{a} \times \vec{b}$

$\Rightarrow (|\vec{a}|^2 - 1)\ \vec{A} - \vec{B} = \vec{a} \times \vec{b}$ [using equation (2)]

solving equation (1) and (5), simultaneously, we get

$\vec{A} = \frac{\vec{a} \times \vec{b} + \vec{a}}{|\vec{a}|^2}$ and $\vec{B} = \frac{\vec{b} \times \vec{a} + a\,(|\vec{a}|^2 - 1)}{|\vec{a}|^2}$

Sol. Ex. Solve for $\vec{r}$, the simultaneous equations $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$, $\vec{r}\,.\,\vec{a}=0$ provided $\vec{a}$ is not perpnedicular to $\vec{b}$.

Solution $(\vec{r}-\vec{c})\times\vec{b}=0 \quad\Rightarrow\quad \vec{r}-\vec{c}$ and $\vec{b}$ are collinear

$\therefore \quad \vec{r}-\vec{c}=k\vec{b} \quad\Rightarrow\quad r=\vec{c}+k\vec{b}$ .....(i)

$\vec{r}\,.\,\vec{a}=0 \quad\Rightarrow\quad (\vec{c}+k\vec{b})\,.\,\vec{a}=0$

$\Rightarrow \quad k=-\dfrac{\vec{a}\,.\,\vec{c}}{\vec{a}\,.\,\vec{b}}$ putting in (i) we get $\vec{r}=\vec{c}-\dfrac{\vec{a}\,.\,\vec{c}}{\vec{a}\,.\,\vec{b}}\,\vec{b}$

Solved Example :If $\vec{x}\times\vec{a}+k\vec{x}=\vec{b}$, where k is a scalar and $\vec{a},\vec{b}$ are any two vectors, then determine $\vec{x}$ in terms of $\vec{a},\vec{b}$ and k.

Solution: $\vec{x}\times\vec{a}+k\vec{x}=\vec{b}$ .....(i)

Premultiple the given equation vectorially by $\vec{a}$

$\vec{a}\times(\vec{x}\times\vec{a})+k\,(\vec{a}\times\vec{x})=\vec{a}\times\vec{b}$

$\Rightarrow \quad (\vec{a}\,.\,\vec{a})\,\vec{x}-(\vec{a}\,.\,\vec{x})\,\vec{a}+k(\vec{a}\times\vec{x})=\vec{a}\times\vec{b}$ .....(ii)

Premultiply (i) scalarly by $\vec{a}$

$[\vec{a}\;\vec{x}\;\vec{a}]+k\,(\vec{a}\,.\,\vec{x})=\vec{a}\,.\,\vec{b}$

$k(\vec{a}\,.\,\vec{x})=\vec{a}\,.\,\vec{b}$ .....(iii)

Substituting $\vec{x}\times\vec{a}$ from (i) and $\vec{a}\,.\,\vec{x}$ from (iii) in (ii) we get

$$\vec{x}=\frac{1}{a^2+k^2}\left[k\vec{b}+(\vec{a}\times\vec{b})+\frac{(\vec{a}\,.\,\vec{b})}{k}\,\vec{a}\right]$$

Self Practice Problems : **1.** Prove that $\vec{a}\times(\vec{b}\times\vec{c})+\vec{b}\times(\vec{c}\times\vec{a})+\vec{c}\times(\vec{a}\times\vec{b})=0$.

2. Find the unit vector coplanar with $\hat{i}+\hat{j}+2\hat{k}$ and $\hat{i}+2\hat{j}+\hat{k}$ and perpendicular to $\hat{i}+\hat{j}+\hat{k}$.

Ans. $\dfrac{1}{\sqrt{2}}(-\hat{j}+\hat{k})$ or, $\dfrac{1}{\sqrt{2}}(\hat{j}-\hat{k})$

3. Prove that $\vec{a}\times\{\vec{a}\times(\vec{a}\times\vec{b})\}=(\vec{a}\,.\,\vec{a})\,(\vec{b}\times\vec{a})$.

4. Given that $\vec{x}+\dfrac{1}{\vec{p}^2}\,(\vec{p}\,.\,\vec{x})\,\vec{p}=\vec{q}$, show that $\vec{p}\,.\,\vec{x}=\dfrac{1}{2}\vec{p}\,.\,\vec{q}$ and find $\vec{x}$ in terms of $\vec{p}$ and $\vec{q}$.

5. If $\vec{x}\,.\,\vec{a}=0$, $\vec{x}\,.\,\vec{b}=0$ and $\vec{x}\,.\,\vec{c}=0$ for some non-zero vector $\vec{x}$, then show that $[\vec{a}\,\vec{b}\,\vec{c}]=0$

6. Prove that $\vec{r}=\dfrac{(\vec{r}\,.\,\vec{a})\,(\vec{b}\times\vec{c})}{[abc]}+\dfrac{(\vec{r}\,.\,\vec{b})\,(\vec{c}\times\vec{a})}{[abc]}+\dfrac{(\vec{r}\,.\,\vec{c})\,(\vec{a}\times\vec{b})}{[abc]}$ where $\vec{a},\vec{b},\vec{c}$ are three non-coplanar vectors

10. Reciprocal System Of Vectors:

If $\vec{a},\vec{b},\vec{c}$ & $\vec{a}',\vec{b}',\vec{c}'$ are two sets of non coplanar vectors such that $\vec{a}.\vec{a}'=\vec{b}.\vec{b}'=\vec{c}.\vec{c}'=1$ then the two systems are called Reciprocal System of vectors.

Note: $\vec{a}=\dfrac{\vec{b}\times\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}\quad\vec{b}=\dfrac{\vec{c}\times\vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}\quad\vec{c}=\dfrac{\vec{a}\times\vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$

Solved Example If $\vec{a}\,\vec{b}\,\vec{c}$ and $\vec{a}',\vec{b}',\vec{c}'$ be the reciprocal system of vectors, prove that

(i) $\vec{a}\,.\,\vec{a}'+\vec{b}\,.\,\vec{b}'+\vec{c}\,.\,\vec{c}'=3$ (ii) $\vec{a}\times\vec{a}'+\vec{b}\times\vec{b}'+\vec{c}\times\vec{c}'=\vec{0}$

Solution. (i) We have : $\vec{a}\,.\,\vec{a}'=\vec{b}\,.\,\vec{b}'=\vec{c}\,.\,\vec{c}'=1$

$\vec{a}\,.\,\vec{a}'+\vec{b}\,.\,\vec{b}'+\vec{c}\,.\,\vec{c}'=1+1+1=3$

(ii) We have : $\vec{a}'=\lambda\,(\vec{b}\times\vec{c})$, $\vec{b}'=\lambda\,(\vec{c}\times\vec{a})$ and $\vec{c}'=\lambda\,((\vec{a}\times\vec{b})$, where $\lambda=\dfrac{1}{[\vec{a}\,\vec{b}\,\vec{c}]}$

$\vec{a}\times\vec{a}'=\vec{a}\times\lambda(\vec{b}\times\vec{c})=\lambda\{\vec{a}\times(\vec{b}\times\vec{c})\}=\lambda\{(\vec{a}\,.\,\vec{c})\,\vec{b}-(\vec{a}\,.\,\vec{b})\,\vec{c}\}$

$\vec{b}\times\vec{b}'=\vec{b}\times\lambda(\vec{c}\times\vec{a})=\lambda\{\vec{b}\times(\vec{c}\times\vec{a})\}=\lambda\{(\vec{b}\,.\,\vec{a})\,\vec{c}-(\vec{b}\,.\,\vec{c})\,\vec{a}\}$

and $\vec{c}\times\vec{c}'=\vec{c}\times\lambda(\vec{a}\times\vec{b})=\lambda\{\vec{c}\times(\vec{a}\times\vec{b})\}=\lambda\{(\vec{c}\,.\,\vec{b})\,\vec{a}-(\vec{c}\,.\,\vec{a})\,\vec{b}\}$

$\therefore \quad \vec{a}\times\vec{a}'+\vec{b}\times\vec{b}'+\vec{c}\times\vec{c}'$

$=\lambda\,\{(\vec{a}\,.\,\vec{c})\,\vec{b}-(\vec{a}\,.\,\vec{b})\,\vec{c}\}+\lambda\{(\vec{b}\,.\,\vec{a})\,\vec{c}-(\vec{b}\,.\,\vec{c})\,\vec{a}\}+\lambda\{(\vec{c}\,.\,\vec{b})\,\vec{a}-(\vec{c}\,.\,\vec{a})\,\vec{b}\}$

$=\lambda\,[(\vec{a}\,.\,\vec{c})\,\vec{b}-(\vec{a}\,.\,\vec{b})\,\vec{c}+(\vec{b}\,.\,\vec{a})\,\vec{c}-(\vec{b}\,.\,\vec{c})\,\vec{a}+(\vec{c}\,.\,\vec{b})\,\vec{a}-(\vec{c}\,.\,\vec{a})\,\vec{b}]$

$=\lambda\,[(\vec{a}\,.\,\vec{c})\,\vec{b}-(\vec{a}\,.\,\vec{b})\,\vec{c}+(\vec{a}\,.\,\vec{b})\,\vec{c}-(\vec{b}\,.\,\vec{c})\,\vec{a}+(\vec{b}\,.\,\vec{c})\,\vec{a}-(\vec{a}\,.\,\vec{c})\,\vec{b}]$

$=\lambda\vec{0}=\vec{0}$

11. Linear Combinations:

Given a finite set of vectors $\vec{a},\vec{b},\vec{c},.....$ then the vector $\vec{r}=x\vec{a}+y\vec{b}+z\vec{c}+.....$ is called a linear combination of $\vec{a},\vec{b},\vec{c},.....$ for any x, y, z..... $\in$ R. We have the following results:

(a) If $\vec{a},\vec{b}$ are non zero, non–collinear vectors then $x\vec{a}+y\vec{b}=x'\vec{a}+y'\vec{b}\Rightarrow x=x';y=y'$

(b) **FundamentalTheorem:** Let $\vec{a},\vec{b}$ be non zero, non collinear vectors. Then any vector $\vec{r}$ coplanar with $\vec{a},\vec{b}$ can be expressed uniquely as a linear combination of $\vec{a},\vec{b}$

i.e. There exist some uniquly x, y $\in$ R such that $x\vec{a}+y\vec{b}=\vec{r}$.

(c) If $\vec{a},\vec{b},\vec{c}$ are non–zero, non–coplanar vectors then:

$$x\vec{a}+y\vec{b}+z\vec{c}=x'\vec{a}+y'\vec{b}+z'\vec{c}\Rightarrow x=x',\ y=y',\ z=z'$$

(d) **Fundamental Theorem In Space:** Let $\vec{a},\vec{b},\vec{c}$ be non–zero, non–coplanar vectors in space. Then any vector $\vec{r}$, can be uniquly expressed as a linear combination of $\vec{a},\vec{b},\vec{c}$ i.e. There exist some unique x,y $\in$ R such that $x\vec{a}+y\vec{b}+z\vec{c}=\vec{r}$.

(e) If $\vec{x}_1,\vec{x}_2,.....\vec{x}_n$ are n non zero vectors, & k_1, k_2,.....k_n are n scalars & if the linear combination $k_1\vec{x}_1+k_2\vec{x}_2+.....k_n\vec{x}_n=0\Rightarrow k_1=0,k_2=0.....k_n=0$ then we say that vectors $\vec{x}_1,\vec{x}_2,.....\vec{x}_n$ are **Linearly Independent Vectors**.

(f) If $\vec{x}_1,\vec{x}_2,.....\vec{x}_n$ are not **Linearly Independent** then they are said to be **Linearly Dependent** vectors. i.e. if $k_1\vec{x}_1+k_2\vec{x}_2+.....+k_n\vec{x}_n=0$ & if there exists at least one $k_r\neq 0$ then $\vec{x}_1,\vec{x}_2,.....\vec{x}_n$ are said to be

Linearly Dependent.

Note 1: If $k_r\neq 0$; $k_1\vec{x}_1+k_2\vec{x}_2+k_3\vec{x}_3+.....+k_r\vec{x}_r+.....+k_n\vec{x}_n=0$

$$-k_r\vec{x}_r=k_1\vec{x}_1+k_2\vec{x}_2+.....+k_{r-1}.\vec{x}_{r-1}+k_{r+1}.\vec{x}_{r+1}+.....+k_n\vec{x}_n$$

$$-k_r\frac{1}{k_r}\vec{x}_r=k_1\frac{1}{k_r}\vec{x}_1+k_2\frac{1}{k_r}\vec{x}_2+.....+k_{r-1}.\frac{1}{k_r}\vec{x}_{r-1}+.....+k_n\frac{1}{k_r}\vec{x}_n$$

$$\vec{x}_r=c_1\vec{x}_1+c_2\vec{x}_2+.....+c_{r-1}\vec{x}_{r-1}+c_r\vec{x}_{r-1}+.....+c_n\vec{x}_n$$

i.e. $\vec{x}_r$ is expressed as a linear combination of vectors.

$\vec{x}_1,\ \vec{x}_2,.....\vec{x}_{r-1},\vec{x}_{r+1},.....\vec{x}_n$

Hence $\vec{x}_r$ with $\vec{x}_1,\vec{x}_2,....\vec{x}_{r-1},\vec{x}_{r+1}....\vec{x}_n$ forms a linearly dependent set of vectors.

Note 2: ☞ If $\vec{a}=3\hat{i}+2\hat{j}+5\hat{k}$ then $\vec{a}$ is expressed as a **Linear Combination** of vectors $\hat{i},\ \hat{j},\ \hat{k}$ Also, $\vec{a},\ \hat{i},\ \hat{j},\ \hat{k}$ form a linearly dependent set of vectors. In general, every set of four vectors is a linearly dependent system.

☞ $\hat{i},\hat{j},\hat{k}$ are **Linearly Independent** set of vectors. For $K_1\hat{i}+K_2\hat{j}+K_3\hat{k}=0\Rightarrow K_1=K_2=K_3=0$

☞ Two vectors $\vec{a}$ & $\vec{b}$ are linearly dependent $\Rightarrow\vec{a}$ is parallel to $\vec{b}$ i.e. $\vec{a}\times\vec{b}=0\Rightarrow$ linear dependence of $\vec{a}$ & $\vec{b}$. Conversely if $\vec{a}\times\vec{b}\neq 0$ then $\vec{a}$ & $\vec{b}$ are linearly independent.

☞ If three vectors $\vec{a},\ \vec{b},\ \vec{c}$ are linearly dependent, then they are coplanar i.e. $[\vec{a},\ \vec{b},\ \vec{c}]=0$, conversely, if $[\vec{a},\ \vec{b},\ \vec{c}]\neq 0$, then the vectors are linearly independent.

Solved Example: Given A that the points $\vec{a}-2\vec{b}+3\vec{c}$, $2\vec{a}+3\vec{b}-4\vec{c}$, $-7\vec{b}+10\vec{c}$, A, B, C have position vector prove that vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are linearly dependent.

Solution. Let A, B, C be the given points and O be the point of reference then

$\overrightarrow{OA}=\vec{a}-2\vec{b}+3\vec{c}$, $\overrightarrow{OB}=2\vec{a}+3\vec{b}-4\vec{c}$ and $\overrightarrow{OC}=-7\vec{b}+10\vec{c}$

Now $\overrightarrow{AB}$ = p.v. of B – p.v. of A

$=\overrightarrow{OB}-\overrightarrow{OA}=(\vec{a}+5\vec{b}-7\vec{c})=-\overrightarrow{AB}$

$\therefore\ \overrightarrow{AC}=\lambda\overrightarrow{AB}$ where $\lambda=-1$. Hence $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are linearly dependent

Solved Example: Prove that the vectors $5\vec{a}+6\vec{b}+7\vec{c}$, $7\vec{a}-8\vec{b}+9\vec{c}$ and $3\vec{a}+20\vec{b}+5\vec{c}$ are linearly dependent $\vec{a},\vec{b},\vec{c}$ being linearly independent vectors.

Solution. We know that if these vectors are linearly dependent , then we can express one of them as a linear combination of the other two.

Now let us assume that the given vector are coplanar, then we can write

$5\vec{a} + 6\vec{b} + 7\vec{c} = \ell(7\vec{a} - 8\vec{b} + 9\vec{c}) + m(3\vec{a} + 20\vec{b} + 5\vec{c})$
where ℓ, m are scalars

Comparing the coefficients of $\vec{a}, \vec{b}$ and $\vec{c}$ on both sides of the equation
$5 = 7\ell + 3$ .....(i) $6 = -8\ell + 20\,m$ .....(ii)
$7 = 9\ell + 5m$ .....(iii)
From (i) and (iii) we get

$4 = 8\ell \quad \Rightarrow \quad \ell = \frac{1}{2}$ = m which evidently satisfies (ii) equation too.

Hence the given vectors are linearly dependent .

Self Practice Problems :

1. Does there exist scalars u, v, w such that $u\vec{e}_1 + v\vec{e}_2 + w\vec{e}_3 = \vec{i}$ where $\vec{e}_1 = \vec{k}$, $\vec{e}_2 = \vec{j} + \vec{k}$, $\vec{e}_3 = -\vec{j} + 2\vec{k}$?
Ans. No

2. Consider a base $\vec{a}, \vec{b}, \vec{c}$ and a vector $-2\vec{a} + 3\vec{b} - \vec{c}$. Compute the co-ordinates of this vector relatively to the base p, q, r where $\vec{p} = 2\vec{a} - 3\vec{b}$, $\vec{q} = \vec{a} - 2\vec{b} + \vec{c}$, $\vec{r} = -3\vec{a} + \vec{b} + 2\vec{c}$. **Ans.** (0, –7/5, 1/5)

3. If $\vec{a}$ and $\vec{b}$ are non-collinear vectors and $\vec{A} = (x + 4y)\vec{a} + (2x + y + 1)\vec{b}$ and $\vec{B} = (y - 2x + 2)\vec{a} + (2x - 3y - 1)\vec{b}$, find x and y such that $3\vec{A} = 2\vec{B}$. **Ans.** x = 2, y = –1

4. If vectors $\vec{a}, \vec{b}, \vec{c}$ be linearly independent, then show that :(i) $\vec{a} - 2\vec{b} + 3\vec{c}$, $-2\vec{a} + 3\vec{b} - 4\vec{c}$, $-\vec{b} + 2\vec{c}$ are linearly dependent (ii) $\vec{a} - 3\vec{b} + 2\vec{c}$, $-2\vec{a} - 4\vec{b} - \vec{c}$, $3a + 2\vec{b} - \vec{c}$ are linearly independent.

5. Given that $\hat{i} - \hat{j}$, $\hat{i} - 2\hat{j}$ are two vectors. Find a unit vector coplanar with these vectors and perpendicular to the first vector $\hat{i} - \hat{j}$. Find also the unit vector which is perpendicular to the plane of the two given vectors. Do you thus obtain an orthonormal triad? **Ans.** $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$; k; Yes

6. If with reference to a right handed system of mutually perpendicular unit vectors $\hat{i}, \hat{j}, \hat{k}$ $\vec{\alpha} = 3\vec{i} - \vec{j}$, $\vec{\beta} = 2\vec{i} + \vec{j} - 3\vec{k}$ express $\vec{\beta}$ in the form $\vec{\beta} = \vec{\beta}_1 + \vec{\beta}_2$ where $\vec{\beta}_1$ is parallel to $\vec{\alpha}$ & $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$.
Ans. $\vec{\beta}_1 = \frac{3}{2}\vec{i} - \frac{1}{2}\vec{j}$, $\vec{\beta}_2 = \frac{1}{2}\vec{i} + \frac{3}{2}\vec{j} - 3\vec{k}$

7. Prove that a vector $\vec{r}$ in space can be expressed linearly in terms of three non-coplanar, non-null vectors $\vec{a}, \vec{b}, \vec{c}$ in the form $\vec{r} = \frac{[\vec{r}\,\vec{b}\,\vec{c}]\vec{a} + [\vec{r}\,\vec{c}\,\vec{a}]\vec{b} + [\vec{r}\,\vec{a}\,\vec{b}]\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}$

Note: Test Of Collinearity: Three points A,B,C with position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ respectively are collinear, if & only if there exist scalars x, y, z not all zero simultaneously such that; $x\vec{a} + y\vec{b} + z\vec{c} = 0$, where x + y + z = 0.

Note: Test Of Coplanarity: Four points A, B, C, D with position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ respectively are coplanar if and only if there exist scalars x, y, z, w not all zero simultaneously such that $x\vec{a} + y\vec{b} + z\vec{c} + w\vec{d} = 0$ where, x + y + z + w = 0.

Solved Example Show that the vectors $2\vec{a} - \vec{b} + 3\vec{c}$, $\vec{a} + \vec{b} - 2\vec{c}$ and $\vec{a} + \vec{b} - 3\vec{c}$ are non-coplanar vectors.
Solution. Let, the given vectors be coplanar.
Then one of the given vectors is expressible in terms of the other two.

Let $2\vec{a} - \vec{b} + 3\vec{c} = x(\vec{a} + \vec{b} - 2\vec{c}) + y(\vec{a} + \vec{b} - 3\vec{c})$, for some scalars x and y.

$\Rightarrow \quad 2\vec{a} - \vec{b} + 3\vec{c} = (x + y)\vec{a} + (x + y)\vec{b} + (-2x - 3y)\vec{c}$
$\Rightarrow \quad 2 = x + y, -1 = x + y$ and $3 = 2x - 3y$.
Solving, first and third of these equations, we get x = 9 and y = –7.
Clearly, these values do not satisfy the thrid equation.
Hence, the given vectors are not coplanar.

Solved Example: Prove that four points $2\vec{a} + 3\vec{b} - \vec{c}$, $\vec{a} - 2\vec{b} + 3\vec{c}$, $3\vec{a} + 4\vec{b} - 2\vec{c}$ and $\vec{a} - 6\vec{b} + 6\vec{c}$ are coplanar.

Solution. Let the given four points be P, Q, R and S respectively. These points are coplanar if the vectors $\overrightarrow{PQ}$, $\overrightarrow{PR}$ and $\overrightarrow{PS}$ are coplanar. These vectors are coplanar iff one of them can be expressed as a linear combination of other two. So, let

$$\overrightarrow{PQ} = x\,\overrightarrow{PR} + y\,\overrightarrow{PS}$$

$\Rightarrow -\vec{a} - 5\vec{b} + 4\vec{c} = x(\vec{a} + \vec{b} - \vec{c}) + y(-\vec{a} - 9\vec{b} - 7\vec{c}) \Rightarrow -\vec{a} - 5\vec{b} + 4\vec{c} = (x - y)\vec{a} + (x - 9y)\vec{b} + (-x + 7y)\vec{c}$

$\Rightarrow x - y = -1,\ x - 9y = -5,\ -x + 7y = 4$ [Equating coeff. of $\vec{a}, \vec{b}, \vec{c}$ on both sides]

Solving the first of these three equations, we get $x = -\frac{1}{2}$, $y = \frac{1}{2}$.

These values also satisfy the third equation. Hence, the given four points are coplanar.

Self Practice Problems :

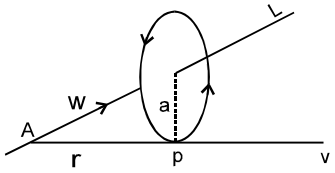
1. If, $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are any four vectors in 3-dimensional space with the same initial point and such that $3\vec{a} - 2\vec{b} + \vec{c} - 2\vec{d} = \vec{0}$, show that the terminal A, B, C, D of these vectors are coplanar. Find the point at which AC and BD meet. Find the ratio in which P divides AC and BD.

2. Show that the vector $\vec{a} - \vec{b} + \vec{c}$, $\vec{b} - \vec{c} - \vec{a}$ and $2\vec{a} - 3\vec{b} - 4\vec{c}$ are non-coplanar, where $\vec{a}, \vec{b}, \vec{c}$, are any non-coplanar vectors.

3. Find the value of λ for which the four points with position vectors $-\hat{j} - \hat{k}$, $4\hat{i} + 5\hat{j} + \lambda\hat{k}$, $3\hat{i} + 9\hat{j} + 4\hat{k}$ and $-4\hat{i} + 4\hat{j} + 4\hat{k}$ are coplanar. **Ans.** $\lambda = 1$

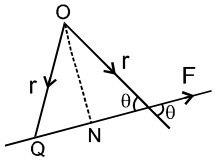
12. Application Of Vectors:

(a) Work done against a constant force $\vec{F}$ over a displacement $\vec{s}$ is defined as $\vec{W} = \vec{F} . \vec{s}$ **(b)** The tangential velocity $\vec{V}$ of a body moving in a circle is given by $\vec{V} = \vec{w} \times \vec{r}$ where $\vec{r}$ is the pv of the point P.



(c) The moment of $\vec{F}$ about 'O' is defined as $\vec{M} = \vec{r} \times \vec{F}$ where $\vec{r}$ is the pv of P wrt 'O'. The direction of $\vec{M}$ is along the normal to the plane OPN such that $\vec{r}, \vec{F}$ & $\vec{M}$ form a right handed system.

(d) Moment of the couple $= (\vec{r}_1 - \vec{r}_2) \times \vec{F}$ where $\vec{r}_1$ & $\vec{r}_2$ are pv's of the point of the application of the forces $\vec{F}$ & $-\vec{F}$.



Solved Example: Forces of magnitudes 5 and 3 units acting in the directions $6\hat{i} + 2\hat{j} + 3\hat{k}$ and $3\hat{i} + 2\hat{j} + 6\hat{k}$ respectively act on a particle which is displaced from the point (2, 2, –1) to (4, 3, 1). Find the work done by the forces.

Solution. Let $\vec{F}$ be the resultant force and $\vec{d}$ be the displacement vector. Then,

$$\vec{F} = 5\frac{(6\hat{i} + 2\hat{j} + 3\hat{k})}{\sqrt{36 + 4 + 9}} + 3\frac{(3\hat{i} + 2\hat{j} + 6\hat{k})}{\sqrt{9 + 4 + 36}} = \frac{1}{7}(39\hat{i} + 4\hat{j} + 33\hat{k})$$

and, $\vec{d} = (4\hat{i} + 3\hat{j} + \hat{k}) - (2\hat{i} + 2\hat{j} - \hat{k}) = 2\hat{i} + \hat{j} + 2\hat{k}$

$\therefore$ Total work done $= \vec{F} \cdot \vec{d} = \frac{1}{7}(39\hat{i} + 4\hat{j} + 33\hat{k}) \cdot (2\hat{i} + \hat{j} + 2\hat{k})$

$$= \frac{1}{7}(78 + 4 + 66) = \frac{148}{7} \text{ units.}$$

Self Practice Problems :1. A point describes a circle uniformly in the $\hat{i}$, $\hat{j}$ plane taking 12 seconds to complete one revolution. If its initial position vector relative to the centre is $\hat{i}$, and the rotation is from $\hat{i}$ to $\hat{j}$, find the position vector at the end of 7 seconds. Also find the velocity vector. **Ans.** 1 / 2 $\left(\hat{j} - \sqrt{3}\,\hat{i}\right)$, p/12 $\left(\hat{i} - \sqrt{3}\,\hat{j}\right)$

2. The force represented by $3\hat{i} + 2\hat{k}$ is acting through the point $5\hat{i} + 4\hat{j} - 3\hat{k}$. Find its moment about the point $\hat{i} + 3\hat{j} + \hat{k}$. **Ans.** $2\hat{i} - 20\hat{j} - 3\hat{k}$

3. Find the moment of the comple formed by the forces $5\hat{i} + \hat{k}$ and $-5\hat{i} - \hat{k}$ acting at the points (9, –1, 2) and (3, –2, 1) respectively **Ans.** $\hat{i} - \hat{j} - 5\hat{k}$

Miscellaneous Solved Examples

Solved Example: Show that the points A, B, C with position vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$ respectively, are the vertices of a right angled triangle. Also find the remaining angles of the triangle.

Solution. We have, $\overrightarrow{AB}$ = Position vector of B – Position vector of A

$= (\hat{i} - 3\hat{j} - 5\hat{k}) - (2\hat{i} - \hat{j} + \hat{k}) = -\hat{i} - 2\hat{j} - 6\hat{k}$

$\overrightarrow{BC}$ = Position vector of C – Position vector of B

$= (3\hat{i} - 4\hat{j} - 4\hat{k}) - (\hat{i} - 3\hat{j} - 5\hat{k}) = 2\hat{i} - \hat{j} + \hat{k}$

and, $\overrightarrow{CA}$ = Position vector of A – Position vector of C

$= (2\hat{i} - \hat{j} + \hat{k}) - (3\hat{i} - 4\hat{j} - 4\hat{k}) = -\hat{i} + 3\hat{j} + 5\hat{k}$

Since $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = (-\hat{i} - 2\hat{j} - 6\hat{k}) + (2\hat{i} - \hat{j} + \hat{k}) + (-\hat{i} + 3\hat{j} + 5\hat{k}) = \vec{0}$
So, A, B and C are the vertices of a triangle.

Now, $\overrightarrow{BC} \cdot \overrightarrow{CA} = (2\hat{i} - \hat{j} + \hat{k}) \cdot (-\hat{i} + 3\hat{j} + 5\hat{k}) = -2 - 3 + 5 = 0$

$\Rightarrow \quad \overrightarrow{BC} \perp \overrightarrow{CA} \quad \Rightarrow \quad \angle BCA = \frac{\pi}{2}$ Hence, ABC is a right angled triangle.

Since a is the angle between the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$. Therefore

$$\cos A = \frac{\overrightarrow{AB} \cdot \overrightarrow{AC}}{|\overrightarrow{AB}||\overrightarrow{AC}|} = \frac{(-\hat{i} - 2\hat{j} - 6\hat{k}) \cdot (\hat{i} - 3\hat{j} - 5\hat{k})}{\sqrt{(-1)^2 + (-2)^2 + (-6)^2}\sqrt{1^2 + (-3)^2 + (-5)^2}}$$

$$= \frac{-1 + 6 + 30}{\sqrt{1 + 4 + 36}\sqrt{1 + 9 + 25}} = \frac{35}{\sqrt{41}\sqrt{35}} = \sqrt{\frac{35}{41}} \qquad A = \cos^{-1}\sqrt{\frac{35}{41}}$$

$$\cos B = \frac{\overrightarrow{BA}}{|\overrightarrow{BA}||\overrightarrow{BC}|} = \frac{(\hat{i} + 2\hat{j} + 6\hat{k})(2\hat{i} - \hat{j} + \hat{k})}{\sqrt{1^2 + 2^2 + 6^2}\sqrt{2^2 + (-1)^2 + (1)^2}} \Rightarrow \cos B = \frac{2 - 2 + 6}{\sqrt{41}\sqrt{6}} = \sqrt{\frac{6}{41}} \Rightarrow B = \cos^{-1}\sqrt{\frac{6}{41}}$$

Solved Example: If $\vec{a}, \vec{b}, \vec{c}$ are three mutually perpendicular vectors of equal magnitude, prove that $\vec{a} + \vec{b} + \vec{c}$ is equally inclined with vectors $\vec{a}, \vec{b}$ and $\vec{c}$.

Solution.: Let $|\vec{a}| = |\vec{b}| = |\vec{c}| = \lambda$ (say). Since $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors, therefore $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$ .....(i)

Now, $\left|\vec{a} + \vec{b} + \vec{c}\right|^2$

$= \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} + \vec{c} \cdot \vec{c} + 2\vec{a} \cdot \vec{b} + 2\vec{b} \cdot \vec{c} + 2\vec{c} \cdot \vec{a}$

$= |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2$ [Using (i)]

$= 3\lambda^2$ [$\because |\vec{a}| = |\vec{b}| = |\vec{c}| = \lambda$]

$\therefore \quad |\vec{a} + \vec{b} + \vec{c}| = \sqrt{3}\lambda$ .....(ii)

Suppose $\vec{a} + \vec{b} + \vec{c}$ makes angles $\theta_1, \theta_2, \theta_3$ with $\vec{a}, \vec{b}$ and $\vec{c}$ respectively. Then,

$$\cos\theta_1 = \frac{\vec{a} \cdot (\vec{a} + \vec{b} + \vec{c})}{|\vec{a}||\vec{a} + \vec{b} + \vec{c}|} = \frac{\vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}}{|\vec{a}||\vec{a} + \vec{b} + \vec{c}|}$$

$$= \frac{|\vec{a}|^2}{|\vec{a}||\vec{a} + \vec{b} + \vec{c}|} = \frac{|\vec{a}|}{|\vec{a} + \vec{b} + \vec{c}|} = \frac{\lambda}{\sqrt{3}\lambda} = \frac{1}{\sqrt{3}} \qquad \text{[Using (ii)]}$$

$\therefore \quad \theta_1 \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Similarly, $\theta_2 = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ and $\theta_3 = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ $\therefore \quad \theta_1 = \theta_2 = \theta_3$.

Hence, $\vec{a} + \vec{b} + \vec{c}$ is equally inclineded with $\vec{a}, \vec{b}$ and $\vec{c}$

Solved Example: Prove using vectors : If two medians of a triangle are equal, then it is isosceles.

Solution. : Let ABC be a triangle and let BE and CF be two equal medians. Taking A as the origin, let the position vectors of B and C be $\vec{b}$ and $\vec{c}$ respectively. Then,

P.V. of E = $\frac{1}{2}\vec{c}$ and, P.V. of F = $\frac{1}{2}\vec{b}$ $\therefore \quad \overrightarrow{BE} = \frac{1}{2}(\vec{c} - 2\vec{b})$

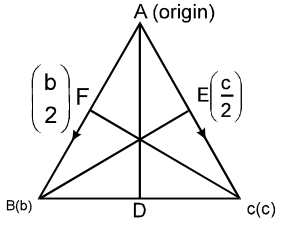
$\overrightarrow{CF} = \frac{1}{2}(\vec{b} - 2\vec{c})$

Now, BE = CF $\Rightarrow \quad |\overrightarrow{BE}| = |\overrightarrow{CF}|$

$$\Rightarrow \quad |\overrightarrow{BE}|^2 = |\overrightarrow{CF}|^2 \Rightarrow \left|\frac{1}{2}(\vec{c} - 2\vec{b})\right|^2 = \left|\frac{1}{2}(\vec{b} - 2\vec{c})\right|^2$$

$$\Rightarrow \quad \frac{1}{4}|\vec{c} - 2\vec{b}|^2 = \frac{1}{4}|\vec{b} - 2\vec{c}|^2 \Rightarrow |\vec{c} - 2\vec{b}|^2 = |\vec{b} - 2\vec{c}|^2$$

$$\Rightarrow \quad (\vec{c} - 2\vec{b}) \cdot (\vec{c} - 2\vec{b}) = (\vec{b} - 2\vec{c}) \cdot (\vec{b} - 2\vec{c})$$



$\Rightarrow \quad \vec{c}\,.\,\vec{c} - 4\vec{b}\,.\,\vec{c} + 4\vec{b}\,.\,\vec{b} = \vec{b}\,.\,\vec{b} - 4\vec{b}\,.\,\vec{c} + 4\vec{c}\,.\,\vec{c}$

$\Rightarrow \quad |\vec{c}|^2 - 4\vec{b}\,.\,\vec{c} + 4\,|\vec{b}|^2 = |\vec{b}|^2 - 4\vec{b}\,.\,\vec{c} + 4\,|\vec{c}|^2$

$\Rightarrow \quad 3\,|b|^2 = 3\,|\vec{c}|^2 \qquad \Rightarrow \quad |b|^2 = |\vec{c}|^2$

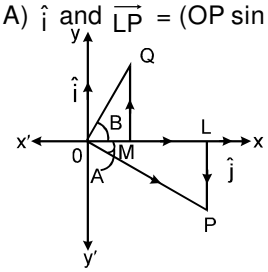
$\Rightarrow$ AB = AC Hence, triangle ABC is an isosceles triangle.

Solved Example: Using vectors : Prove that cos (A + B) = cos A cos B – sin A sin B

Solution. Let OX and OY be the coordinate axes and let $\hat{i}$ and $\hat{j}$ be unit vectors along OX and OY respectively. Let $\angle XOP = A$ and $\angle XOQ = B$. Drawn PL $\perp$ OX and QM $\perp$ OX.

Clearly angle between $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ is A + B

In ΔOLP, OL = OP cos A and LP = OP sin A. Therefore $\overrightarrow{OL} = (OP \cos A)\,\hat{i}$ and $\overrightarrow{LP} = (OP \sin A)\,(-\hat{j})$



Now. $\overrightarrow{OL} + \overrightarrow{LP} = \overrightarrow{OP}$

$\Rightarrow \quad \overrightarrow{OP} = OP\,[(\cos A)\,\hat{i} - (\sin A)\,\hat{j}]$

In ΔOMQ, OM = OQ cos B and MQ = OQ sin B.
Therefore,

$\overrightarrow{OM} = (OQ \cos B)\,\hat{i}$, $\overrightarrow{MQ} = (OQ \sin B)\,\hat{j}$

Now, $\overrightarrow{OM} + \overrightarrow{MQ} = \overrightarrow{OQ}$

From (i) and (ii), we get

$\overrightarrow{OP}\,.\,\overrightarrow{OQ} = OP\,[(\cos A)\,\hat{i} - (\sin A)\,\hat{j}]\,.\,OQ\,[(\cos B)\,\hat{i} + (\sin B)\,\hat{j}]$

$= OP\,.\,OQ\,[\cos A \cos B - \sin A \sin B]$

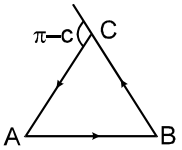
But, $\overrightarrow{OP}\,.\,\overrightarrow{OQ} = |\overrightarrow{OP}|\,|\overrightarrow{OQ}| \cos (A + B) = OP\,.\,OQ \cos (A + B)$

$\therefore \quad OP\,.\,OQ \cos (A + B) = OP\,.\,OQ\,[\cos A \cos B - \sin A \sin B]$

$\Rightarrow \quad \cos (A + B) = \cos A \cos B - \sin A \sin B$

Solved Example: Prove that in any triangle ABC

(i) $c^2 = a^2 + b^2 - 2ab \cos C$ (ii) $c = b\cos A + a\cos B$.



Solution. (i) In ΔABC, $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = 0$

or, $\overrightarrow{BC} + \overrightarrow{CA} = -\overrightarrow{AB}$ .....(i)

Squaring both sides

$(\overrightarrow{BC})^2 + (\overrightarrow{CA})^2 + (\overrightarrow{BC})\,.\,\overrightarrow{CA} + (\overrightarrow{AB})^2$

$\Rightarrow \quad a^2 + b^2 + 2\,(\overrightarrow{BC}\,.\,\overrightarrow{CA}) = c^2 \qquad \Rightarrow \quad c^2 = a^2 + b^2 = 2\,ab \cos (\pi - C)$

$\Rightarrow \quad c^2 = a^2 + b^2 - 2ab \cos C$

(ii) $(\overrightarrow{BC} + \overrightarrow{CA})\,.\,\overrightarrow{AB} = -\overrightarrow{AB}\,.\,\overrightarrow{AB}$

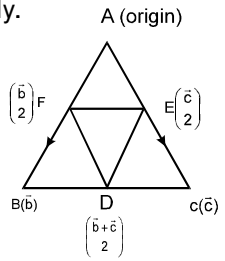
$\overrightarrow{BC}\,.\,\overrightarrow{AB} + \overrightarrow{CA}\,.\,\overrightarrow{AB} = -c^2$

$-ac \cos B - bc \cos A = -c^2$

$a\cos B + b\cos A = c.$

Solved Ex.: If D, E, F are the mid-points of the sides of a triangle ABC, prove by vector method that area of $\Delta DEF = \frac{1}{4}$ (area of ΔABC)

Solution. Taking A as the origin, let the position vectors of B and C be $\vec{b}$ and $\vec{c}$ respectively. Then, the position vectors of D, E and F are $\frac{1}{2}(\vec{b}+\vec{c})$, $\frac{1}{2}\vec{c}$ and $\frac{1}{2}\vec{b}$ respectively.



Now, $\overrightarrow{DE} = \frac{1}{2}\,\vec{c} - \frac{1}{2}\,(\vec{b}+\vec{c}) = \frac{-\vec{b}}{2}$

and $\overrightarrow{DF} = \frac{1}{2}\,\vec{b} - \frac{1}{2}\,(\vec{b}+\vec{c}) = \frac{-\vec{c}}{2}$

$\therefore$ Vector area of $\Delta DEF\ \frac{1}{2}\,(\overrightarrow{DE}\times\overrightarrow{DF}) = \frac{1}{2}\left(\frac{-\vec{b}}{2}\times\frac{-\vec{c}}{2}\right)$

$= \frac{1}{8}\,(\vec{b}\times\vec{c}) = \frac{1}{4}\left\{\frac{1}{2}(\overrightarrow{AB}\times\overrightarrow{AC})\right\}$

$= \frac{1}{4}$(vector area of ΔABC) Hence, area of $\Delta DEF = \frac{1}{4}$ area of ΔABC.

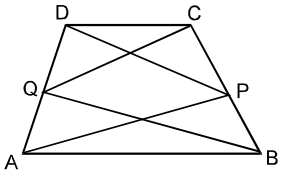
Solved Example: P, Q are the mid-points of the non-parallel sides BC and AD of a trapezium ABCD. Show that $\Delta APD = \Delta CQB$.

Solution. Let $\overrightarrow{AB} = \vec{b}$ and $\overrightarrow{AD} = \vec{d}$

Now DC is parallel to AB $\Rightarrow$ there exists a acalar t sush that $\overrightarrow{DC} = t\,\overrightarrow{DB} = t\,\vec{b}$

$\therefore \quad \overrightarrow{AC} = \overrightarrow{AD} + \overrightarrow{DC} = \vec{d} + t\,\vec{b}$

The position vectors of P and Q are $\frac{1}{2}(\vec{b}+\vec{d}+t\vec{b})$ and $\frac{1}{2}\vec{d}$ respectively.



Now $2\Delta\ \overrightarrow{APD} = \overrightarrow{AP} \times \overrightarrow{AD}$

$= \frac{1}{2}(\vec{b}+\vec{d}+t\vec{b}) \times \vec{d} = \frac{1}{2}(1+t)(\vec{b}\times\vec{d})$

Also $2\Delta\ \overrightarrow{CQB} = \overrightarrow{CQ} \times \overrightarrow{CB} = \left[\frac{1}{2}\vec{d}-(\vec{d}+t\vec{b}\right] \times [\vec{b}-(\vec{d}+t\vec{b})]$

$= \left[-\frac{1}{2}d - t\vec{b}\right]\times[-d+(1-t)b] = -\frac{1}{2}(1-t)\vec{d}\times\vec{b} + t\vec{b}\times\vec{d}$

$= \frac{1}{2}(1-t+2t)\vec{b}\times\vec{d} = \frac{1}{2}(1+t)\vec{b}\times\vec{d} = 2\Delta\overrightarrow{APD}$ Hence the result.

Solved Example: Let $\vec{u}$ and $\vec{v}$ are unit vectors and $\vec{w}$ is a vector such that $\vec{u}\times\vec{v}+\vec{u} = \vec{w}$ and $\vec{w}\times\vec{u} = \vec{v}$ then find the value of $[\vec{u}\ \vec{v}\ \vec{w}]$.

Solution. Given $\vec{u}\times\vec{v}+\vec{u} = \vec{w}$ and $\vec{w}\times\vec{u} = \vec{v}$

$\Rightarrow$ $(\vec{u}\times\vec{v}+\vec{u}) \times \vec{u} = \vec{w}\times\vec{u}$ $\Rightarrow$ $(\vec{u}\times\vec{v}) \times \vec{u}+\vec{u}\times\vec{u} = \vec{v}$ (as, $\vec{w}\times\vec{u} = \vec{v}$)

$\Rightarrow$ $(\vec{u}.\vec{u})\ \vec{v}-(v.\vec{u})\ \vec{u}+\vec{u}\times\vec{u} = \vec{v}$ (using $\vec{u}\ .\ \vec{u} = 1$ and $\vec{u}\times\vec{u} = 0$, since unit vector)

$\Rightarrow$ $\vec{v}-(\vec{v}.\vec{u})\vec{u} = \vec{v}$ $\Rightarrow$ $(\vec{u}.\vec{v})\vec{u} = \vec{0}$

$\Rightarrow$ $\vec{u}.\vec{v} = 0$ (as; $\vec{u} \neq 0$) .....(i)

$\Rightarrow$ $\vec{u}.(\vec{v}\times\vec{w})$

$= \vec{u}.(\vec{v}\times(\vec{u}\times\vec{v}+\vec{u}))$ (given $\vec{w} = \vec{u}\times\vec{v}+u$)

$= \vec{u}.(\vec{v}\times(\vec{u}\times\vec{v})+\vec{v}\times\vec{u})$ $= \vec{u}.((\vec{v}.\vec{v})\vec{u}-(\vec{v}.\vec{u})\vec{v}+\vec{v}\times\vec{u})$

$= \vec{u}.(|\vec{v}|^2\ u-0+\vec{v}\times\vec{u})$ (as; $\vec{u}.\vec{v} = 0$ from (i))

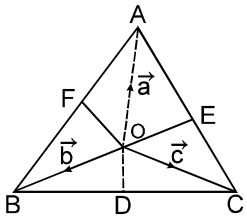
$= |\vec{v}|^2\ (\vec{u}.\vec{u}) - \vec{u}\ .\ (\vec{v}\times\vec{u})$ $= |\vec{v}|^2\ |\vec{u}|^2 - 0$ (as, $[\vec{u}\ \vec{v}\ \vec{u}] = 0$)

$= 1$ (as; $|\vec{u}| = |\vec{v}| = 1$) $\therefore$ $[\vec{u}\ \vec{v}\ \vec{w}] = 1$

Sol. Ex.: In any triangle, show that the perpendicular bisectors of the sides are concurrent.

Solution. Let ABC be the triangle and D, E and F are respectively middle points of sides BC, CA and AB. Let the perpendicular of D and E meet at O join OF. We are required to prove that OF is $\perp$ to AB. Let the position vectors of A, B, C with O as origin of reference be $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively.

$\therefore$ $\overrightarrow{OD} = \frac{1}{2}(\vec{b}+\vec{c})$, $\overrightarrow{OE} = \frac{1}{2}(\vec{c}+\vec{a})$ and $\overrightarrow{OF} = \frac{1}{2}(\vec{a}+\vec{b})$



Also $\overrightarrow{BC} = \vec{c}-\vec{b}$, $\overrightarrow{CA} = \vec{a}-\vec{c}$ and $\overrightarrow{AB} = \vec{b}-\vec{a}$

Since OD $\perp$ BC, $\frac{1}{2}(\vec{b}+\vec{c}).(\vec{c}-\vec{b}) = 0$

$\Rightarrow b^2 = c^2$ .....(i)

Similarly $\frac{1}{2}(\vec{c}+\vec{a}).(\vec{a}+\vec{c}) = 0$

$\Rightarrow a^2 = c^2$ .....(ii)

from (i) and (ii) we have $a^2-b^2 = 0$

$\Rightarrow (\vec{a}+\vec{b}).(\vec{b}+\vec{a}) = 0$ $\Rightarrow \frac{1}{2}(\vec{b}+\vec{a}).(\vec{b}-\vec{a}) = 0$

Solved Example: A, B, C, D are four points in space. using vector methods, prove that $AC^2 + BD^2 + AD^2 + BC^2 \geq AB^2 + CD^2$ what is the implication of the sign of equaility.

Solution.: Let the position vector of A, B, C, D be $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ respectively then

$AC^2 + BD^2 + AD^2 + BC^2 = (\vec{c}-\vec{a}).(\vec{c}-\vec{a}) + (\vec{d}-\vec{b}).(\vec{d}-\vec{b}) + (\vec{d}-\vec{a}).(\vec{d}-\vec{a}) + (\vec{c}-\vec{b}).(\vec{c}-\vec{b})$

$= |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{a}.\vec{c} + |\vec{d}|^2 + |\vec{b}|^2 - 2\vec{d}.\vec{b} + |\vec{d}|^2 + |\vec{a}|^2 - 2\vec{a}.\vec{d} + |\vec{c}|^2 + |\vec{b}|^2 - 2\vec{b}.\vec{c}$

$= |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a}.\vec{b} + |\vec{c}|^2 + |\vec{d}|^2 - 2\vec{c}.\vec{d} + |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + |\vec{d}|^2$

$+ 2\vec{a}.\vec{b} + 2\vec{c}.\vec{d} - 2\vec{a}.\vec{c} - 2\vec{b}.\vec{d} - 2\vec{a}.\vec{d} - 2\vec{b}.\vec{c}$

$= (\vec{a}-\vec{b}).(\vec{a}-\vec{b}) + (\vec{c}-\vec{d}).(\vec{c}-\vec{d}) + (\vec{a}-\vec{b}-\vec{c}-\vec{d}) \geq AB^2 + CD^2$

$= AB^2 + CD^2 + (\vec{a}+\vec{b}-\vec{c}-\vec{d}).(\vec{a}+\vec{b}-\vec{c}-\vec{d}) \leq AB^2 + CD^2$

$\therefore$ $AC^2 + BD^2 + AD^2 + BC^2 \geq AB^2 + CD^2$

for the sign of equality to hold, $\vec{a}+\vec{b}-\vec{c}-\vec{d} = 0$

$\vec{a}-\vec{c} = \vec{d}-\vec{b}$

$\Rightarrow$ $\overrightarrow{AC}$ and $\overrightarrow{BD}$ are collinear the four points A, B, C, D are collinear

SHORT REVISION

1. DEFINITIONS: A VECTOR may be described as a quantity having both magnitude & direction. A vector is generally represented by a directed line segment, say $\overrightarrow{AB}$. A is called the **initial point** & B is called the **terminal point**. The magnitude of vector $\overrightarrow{AB}$ is expressed by $|\overrightarrow{AB}|$.

ZERO VECTOR a vector of zero magnitude i.e.which has the same initial & terminal point, is called a **ZERO VECTOR**. It is denoted by O.

UNIT VECTOR a vector of unit magnitude in direction of a vector $\vec{a}$ is called unit vector along $\vec{a}$ and is denoted by $\hat{a}$ symbolically $\hat{a}=\frac{\vec{a}}{|\vec{a}|}$.

EQUAL VECTORS two vectors are said to be equal if they have the same magnitude, direction & represent the same physical quantity.

COLLINEAR VECTORS two vectors are said to be collinear if their directed line segments are parallel disregards to their direction. Collinear vectors are also called **PARALLEL VECTORS**. If they have the same direction they are named as like vectors otherwise unlike vectors.

Simbolically, two non zero vectors $\vec{a}$ and $\vec{b}$ are collinear if and only if, $\vec{a}=K\vec{b}$, where $K \in R$

COPLANAR VECTORS a given number of vectors are called coplanar if their line segments are all parallel to the same plane. Note that **"TWO VECTORS ARE ALWAYS COPLANAR"**.

POSITION VECTOR let O be a fixed origin, then the position vector of a point P is the vector $\overrightarrow{OP}$. If $\vec{a}$ & $\vec{b}$ & position vectors of two point A and B, then, $\overrightarrow{AB} = \vec{b}-\vec{a}$ = pv of B − pv of A.

2. VECTOR ADDITION : ☞ If two vectors $\vec{a}$ & $\vec{b}$ are represented by $\overrightarrow{OA}$ & $\overrightarrow{OB}$, then their sum $\vec{a}+\vec{b}$ is a vector represented by $\overrightarrow{OC}$, where OC is the diagonal of the parallelogram OACB.

☞ $\vec{a}+\vec{b}=\vec{b}+\vec{a}$ (commutative) ☞ $(\vec{a}+\vec{b})+\vec{c}=\vec{a}+(\vec{b}+\vec{c})$ (associativity)

☞ $\vec{a}+\vec{0}=\vec{a}=\vec{0}+\vec{a}$ ☞ $\vec{a}+(-\vec{a})=\vec{0}=(-\vec{a})+\vec{a}$

3. MULTIPLICATION OF VECTOR BY SCALARS :

If $\vec{a}$ is a vector & m is a scalar, then $m\vec{a}$ is a vector parallel to $\vec{a}$ whose modulus is $|m|$ times that of $\vec{a}$. This multiplication is called **SCALAR MULTIPLICATION**. If $\vec{a}$ & $\vec{b}$ are vectors & m, n are scalars, then:

$m(\vec{a})=(\vec{a})m=m\vec{a}$ $\quad$ $m(n\vec{a})=n(m\vec{a})=(mn)\vec{a}$

$(m+n)\vec{a}=m\vec{a}+n\vec{a}$ $\quad$ $m(\vec{a}+\vec{b})=m\vec{a}+m\vec{b}$

4. SECTION FORMULA :

If $\vec{a}$ & $\vec{b}$ are the position vectors of two points A & B then the p.v. of a point which divides AB in the ratio m : n is given by : $\vec{r}=\frac{n\vec{a}+m\vec{b}}{m+n}$. Note p.v. of mid point of AB = $\frac{\vec{a}+\vec{b}}{2}$.

5. DIRECTION COSINES :

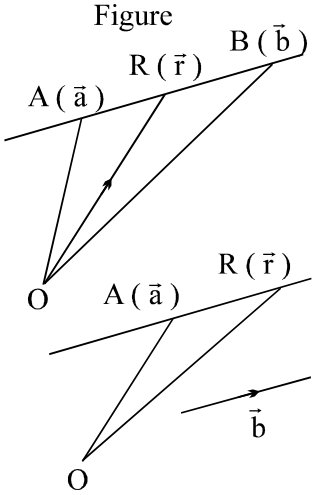
Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ the angles which this vector makes with the +ve directions OX,OY & OZ are called **DIRECTION ANGLES** & their cosines are called the **DIRECTION COSINES**.

$\cos\alpha=\frac{a_1}{|\vec{a}|}$, $\cos\beta=\frac{a_2}{|\vec{a}|}$, $\cos\Gamma=\frac{a_3}{|\vec{a}|}$. **Note that, $\cos^2\alpha+\cos^2\beta+\cos^2\Gamma=1$**

6. VECTOR EQUATION OF A LINE :

Parametric vector equation of a line passing through two point $A(\vec{a})$ & $B(\vec{b})$ is given by, $\vec{r}=\vec{a}+t(\vec{b}-\vec{a})$ where t is a parameter. If the line passes through the point $A(\vec{a})$ & is parallel to the vector $\vec{b}$ then its equation is, $\vec{r}=\vec{a}+t\vec{b}$

Figure



Note that the equations of the bisectors of the angles between the lines $\vec{r} = \vec{a} + \lambda \vec{b}$ & $\vec{r} = \vec{a} + \mu \vec{c}$ is : $\vec{r} = \vec{a} + t\left(\hat{b}+\hat{c}\right)$ & $\vec{r} = \vec{a} + p\left(\hat{c}-\hat{b}\right)$.

7. TEST OF COLLINEARITY : Three points A,B,C with position vectors $\vec{a}, \vec{b}, \vec{c}$ respectively are collinear, if & only if there exist scalars x, y, z not all zero simultaneously such that ; $x\vec{a} + y\vec{b} + z\vec{c} = 0$, where $x + y + z = 0$.

8. SCALAR PRODUCT OF TWO VECTORS :

☞ $\vec{a}.\vec{b} = |\vec{a}||\vec{b}| \cos\theta (0 \le \theta \le \pi)$,

note that if θ is acute then $\vec{a}.\vec{b} > 0$ & if θ is obtuse then $\vec{a}.\vec{b} < 0$

☞ $\vec{a}.\vec{a} = |\vec{a}|^2 = \vec{a}^2, \vec{a}.\vec{b} = \vec{b}.\vec{a}$ (commutative) ☞ $\vec{a} . (\vec{b} + \vec{c}) = \vec{a}.\vec{b} + \vec{a}.\vec{c}$ (distributive)

☞ $\vec{a}.\vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$ $(\vec{a} \ne 0 \quad \vec{b} \ne 0)$

☞ $\hat{i}.\hat{i} = \hat{j}.\hat{j} = \hat{k}.\hat{k} = 1$; $\hat{i}.\hat{j} = \hat{j}.\hat{k} = \hat{k}.\hat{i} = 0$ ☞ projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{a}.\vec{b}}{|\vec{b}|}$.

Note: That vector component of $\vec{a}$ along $\vec{b} = \left(\dfrac{\vec{a} \cdot \vec{b}}{\vec{b}^2}\right)\vec{b}$ and perpendicular to $\vec{b} = \vec{a} - \left(\dfrac{\vec{a} \cdot \vec{b}}{\vec{b}^2}\right)\vec{b}$.

☞ the angle ϕ between $\vec{a}$ & $\vec{b}$ is given by $\cos\phi = \dfrac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|}$ $0 \le \phi \le \pi$

☞ if $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then $\vec{a}.\vec{b} = a_1b_1 + a_2b_2 + a_3b_3$

$|\vec{a}| = \sqrt{a_1^{\,2} + a_2^{\,2} + a_3^{\,2}}$, $|\vec{b}| = \sqrt{b_1^{\,2} + b_2^{\,2} + b_3^{\,2}}$

Note : (i) Maximum value of $\vec{a} . \vec{b} = |\vec{a}|\,|\vec{b}|$

(ii) Minimum values of $\vec{a} . \vec{b} = \vec{a} . \vec{b} = -|\vec{a}|\,|\vec{b}|$

(iii) Any vector $\vec{a}$ can be written as , $\vec{a} = (\vec{a} . \hat{i})\,\hat{i} + (\vec{a} . \hat{j})\,\hat{j} + (\vec{a} . \hat{k})\,\hat{k}$.

(iv) A vector in the direction of the bisector of the angle between the two vectors $\vec{a}$ & $\vec{b}$ is $\dfrac{\vec{a}}{|\vec{a}|} + \dfrac{\vec{b}}{|\vec{b}|}$. Hence bisector of the angle between the two vectors $\vec{a}$ & $\vec{b}$ is $\lambda\left(\hat{a} + \hat{b}\right)$, where $\lambda \in R^+$. Bisector of the exterior angle between $\vec{a}$ & $\vec{b}$ is $\lambda\left(\hat{a} - \hat{b}\right)$, $\lambda \in R^+$.

9. VECTOR PRODUCT OF TWO VECTORS :

(i) ☞ If $\vec{a}$ & $\vec{b}$ are two vectors & θ is the angle between them then $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin\theta\,\vec{n}$, where $\vec{n}$ is the unit vector perpendicular to both $\vec{a}$ & $\vec{b}$ such that $\vec{a}, \vec{b}$ & $\vec{n}$ forms a right handed screw system .

(ii) ☞ Lagranges Identity : for any two vectors $\vec{a}$ & $\vec{b}$; $(\vec{a}\times\vec{b})^2 = |\vec{a}|^2|\vec{b}|^2 - (\vec{a}.\vec{b})^2 = \begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} \\ \vec{a}.\vec{b} & \vec{b}.\vec{b} \end{vmatrix}$

(iii) ☞ Formulation of vector product in terms of scalar product:

The vector product $\vec{a} \times \vec{b}$ is the vector $\vec{c}$, such that

(i) $|\vec{c}| = \sqrt{\vec{a}^2\vec{b}^2 - (\vec{a}\cdot\vec{b})^2}$ (ii) $\vec{c}\cdot\vec{a} = 0$; $\vec{c}\cdot\vec{b} = 0$ and

(iii) $\vec{a}$, $\vec{b}$, $\vec{c}$ form a right handed system

(iv) ☞ $\vec{a}\times\vec{b} = 0 \Leftrightarrow \vec{a}$ & $\vec{b}$ are parallel (collinear) $(\vec{a} \neq 0, \vec{b} \neq 0)$ i.e. $\vec{a} = K\vec{b}$, where K is a scalar.

☞ $\vec{a}\times\vec{b} \neq \vec{b}\times\vec{a}$ (not commutative)

☞ $(m\vec{a})\times\vec{b} = \vec{a}\times(m\vec{b}) = m(\vec{a}\times\vec{b})$ where m is a scalar .

☞ $\vec{a}\times(\vec{b}+\vec{c}) = (\vec{a}\times\vec{b}) + (\vec{a}\times\vec{c})$ (distributive)

☞ $\hat{i}\times\hat{i} = \hat{j}\times\hat{j} = \hat{k}\times\hat{k} = 0$ ☞ $\hat{i}\times\hat{j} = \hat{k},\ \hat{j}\times\hat{k} = \hat{i},\ \hat{k}\times\hat{i} = \hat{j}$

(v) ☞ If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then $\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$

(vi) ☞ Geometrically $|\vec{a}\times\vec{b}|$ = area of the parallelogram whose two adjacent sides are represented by $\vec{a}$ & $\vec{b}$.

(vii) ☞ Unit vector perpendicular to the plane of $\vec{a}$ & $\vec{b}$ is $\hat{n} = \pm\dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$

☞ A vector of magnitude 'r' & perpendicular to the palne of $\vec{a}$ & $\vec{b}$ is $\pm\dfrac{r(\vec{a}\times\vec{b})}{|\vec{a}\times\vec{b}|}$

☞ If θ is the angle between $\vec{a}$ & $\vec{b}$ then $\sin\theta = \dfrac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}$

(viii) Vector area

☞ If $\vec{a}, \vec{b}$ & $\vec{c}$ are the pv's of 3 points A, B & C then the vector area of triangle ABC = $\frac{1}{2}\left[\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\right]$. The points A, B & C are collinear if $\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a} = 0$

☞ Area of any quadrilateral whose diagonal vectors are $\vec{d}_1$ & $\vec{d}_2$ is given by $\frac{1}{2}|\vec{d}_1\times\vec{d}_2|$

10. SHORTEST DISTANCE BETWEEN TWO LINES :

If two lines in space intersect at a point, then obviously the shortest distance between them is zero. Lines which do not intersect & are also not parallel are called **SKEW LINES**. For Skew lines the direction of the shortest distance would be perpendicular to both the lines. The magnitude of the shortest distance vector would be equal to that of the projection of $\vec{AB}$ along the direction of the line of shortest distance, $\vec{LM}$ is parallel to $\vec{p}\times\vec{q}$ i.e. $\vec{LM} = \left|\text{Projection of } \vec{AB} \text{ on } \vec{LM}\right| = \left|\text{Projection of } \vec{AB} \text{ on } \vec{p}\times\vec{q}\right|$

$$= \left|\frac{\vec{AB}\,.\,(\vec{p}\times\vec{q})}{\vec{p}\times\vec{q}}\right| = \left|\frac{(\vec{b}-\vec{a})\,.\,(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}\right|$$

1. The two lines directed along $\vec{p}$ & $\vec{q}$ will intersect only if shortest distance = 0 i.e. $(\vec{b}-\vec{a}).(\vec{p}\times\vec{q}) = 0$ i.e. $(\vec{b}-\vec{a})$ lies in the plane containing $\vec{p}$ & $\vec{q}$. $\Rightarrow \left[(\vec{b}-\vec{a})\ \vec{p}\ \vec{q}\right] = 0$.

2. If two lines are given by $\vec{r}_1 = \vec{a}_1 + K\vec{b}$ & $\vec{r}_2 = \vec{a}_2 + K\vec{b}$ i.e. they are parallel then , $d = \left|\dfrac{\vec{b}\times(\vec{a}_2 - \vec{a}_1)}{|\vec{b}|}\right|$

11. SCALAR TRIPLE PRODUCT / BOX PRODUCT / MIXED PRODUCT :

☞ The scalar triple product of three vectors $\vec{a}, \vec{b}$ & $\vec{c}$ is defined as :
$\vec{a} \times \vec{b} . \vec{c} = |\vec{a}||\vec{b}||\vec{c}| \sin\theta \cos\phi$ where θ is the angle between $\vec{a}$ & $\vec{b}$ & ϕ is the angle between $\vec{a} \times \vec{b}$ & $\vec{c}$.
It is also defined as $[\vec{a}\ \vec{b}\ \vec{c}]$, spelled as box product .

☞ Scalar triple product geometrically represents the volume of the parallelopiped whose three couterminous edges are represented by $\vec{a}, \vec{b}$ & $\vec{c}$ i.e. $V = [\vec{a}\ \vec{b}\ \vec{c}]$

☞ In a scalar triple product the position of dot & cross can be interchanged i.e.
$\vec{a} . (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) . \vec{c}$ OR $[\vec{a}\ \vec{b}\ \vec{c}] = [\vec{b}\ \vec{c}\ \vec{a}] = [\vec{c}\ \vec{a}\ \vec{b}]$

☞ $\vec{a} . (\vec{b} \times \vec{c}) = -\vec{a} . (\vec{c} \times \vec{b})$ i.e. $[\vec{a}\ \vec{b}\ \vec{c}] = -[\vec{a}\ \vec{c}\ \vec{b}]$

☞ If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ & $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ then $[\vec{a}\,\vec{b}\,\vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$.

In general , if $\vec{a} = a_1\vec{l} + a_2\vec{m} + a_3\vec{n}$; $\vec{b} = b_1\vec{l} + b_2\vec{m} + b_3\vec{n}$ & $\vec{c} = c_1\vec{l} + c_2\vec{m} + c_3\vec{n}$

then $\left[\vec{a}\,\vec{b}\,\vec{c}\right] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \left[\vec{l}\,\vec{m}\,\vec{n}\right]$; where $\vec{\ell}, \vec{m}$ & $\vec{n}$ are non coplanar vectors .

☞ If $\vec{a}, \vec{b}, \vec{c}$ are coplanar $\Leftrightarrow [\vec{a}\ \vec{b}\ \vec{c}] = 0$.

☞ Scalar product of three vectors, two of which are equal or parallel is 0 i.e. $[\vec{a}\,\vec{b}\,\vec{c}] = 0$,

Note : If $\vec{a}, \vec{b}, \vec{c}$ are non – coplanar then $[\vec{a}\ \vec{b}\ \vec{c}] > 0$ for right handed system & $[\vec{a}\ \vec{b}\ \vec{c}] < 0$ for left handed system .

☞ $[i\ j\ k] = 1$ ☞ $[K\vec{a}\ \vec{b}\ \vec{c}] = K[\vec{a}\ \vec{b}\ \vec{c}]$ ☞ $[(\vec{a} + \vec{b})\ \vec{c}\ \vec{d}] = [\vec{a}\ \vec{c}\ \vec{d}] + [\vec{b}\ \vec{c}\ \vec{d}]$

☞ The volume of the tetrahedron OABC with O as origin & the pv's of A, B and C being $\vec{a}, \vec{b}$ & $\vec{c}$ respectively is given by $V = \frac{1}{6}[\vec{a}\ \vec{b}\ \vec{c}]$

☞ The positon vector of the centroid of a tetrahedron if the pv's of its angular vertices are $\vec{a}, \vec{b}, \vec{c}$ & $\vec{d}$ are given by $\frac{1}{4}[\vec{a} + \vec{b} + \vec{c} + \vec{d}]$.

Note that this is also the point of concurrency of the lines joining the vertices to the centroids of the opposite faces and is also called the centre of the tetrahedron. In case the tetrahedron is regular it is equidistant from the vertices and the four faces of the tetrahedron .

Remember that : $\left[\vec{a}-\vec{b}\quad \vec{b}-\vec{c}\quad \vec{c}-\vec{a}\right] = 0$ & $\left[\vec{a}+\vec{b}\quad \vec{b}+\vec{c}\quad \vec{c}+\vec{a}\right] = 2\left[\vec{a}\ \vec{b}\ \vec{c}\right]$.

***12. VECTOR TRIPLE PRODUCT :** Let $\vec{a}, \vec{b}, \vec{c}$ be any three vectors, then the expression $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector & is called a vector triple product .

GEOMETRICAL INTERPRETATION OF $\vec{a} \times (\vec{b} \times \vec{c})$

Consider the expression $\vec{a} \times (\vec{b} \times \vec{c})$ which itself is a vector, since it is a cross product of two vectors $\vec{a}$ & $(\vec{b} \times \vec{c})$. Now $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector perpendicular to the plane containing $\vec{a}$ & $(\vec{b} \times \vec{c})$ but $\vec{b} \times \vec{c}$ is a vector perpendicular to the plane $\vec{b}$ & $\vec{c}$, therefore $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector lies in the plane of $\vec{b}$ & $\vec{c}$ and perpendicular to $\vec{a}$. Hence we can express $\vec{a} \times (\vec{b} \times \vec{c})$ in terms of $\vec{b}$ & $\vec{c}$ i.e. $\vec{a} \times (\vec{b} \times \vec{c}) = x\vec{b} + y\vec{c}$ where x & y are scalars .

☞ $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} . \vec{c})\vec{b} - (\vec{a} . \vec{b})\vec{c}$ ☞ $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} . \vec{c})\vec{b} - (\vec{b} . \vec{c})\vec{a}$

☞ $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$

13. LINEAR COMBINATIONS / Linearly Independence and Dependence of Vectors :

Given a finite set of vectors $\vec{a}, \vec{b}, \vec{c}, \ldots\ldots$ then the vector $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c} + \ldots\ldots$ is called a linear

combination of $\vec{a},\vec{b},\vec{c},.....$ for any x, y, z ..... $\in$ R. We have the following results :

(a) **FUNDAMENTAL THEOREM IN PLANE :** Let $\vec{a},\vec{b}$ be non zero , non collinear vectors . Then any vector $\vec{r}$ coplanar with $\vec{a},\vec{b}$ can be expressed uniquely as a linear combination of $\vec{a},\vec{b}$ i.e. There exist some unique x,y $\in$ R such that $x\vec{a} + y\vec{b}=\vec{r}$.

(b) **FUNDAMENTAL THEOREM IN SPACE :** Let $\vec{a},\vec{b},\vec{c}$ be non–zero, non–coplanar vectors in space. Then any vector $\vec{r}$, can be uniquily expressed as a linear combination of $\vec{a},\vec{b},\vec{c}$ i.e. There exist some unique x,y $\in$ R such that $x\vec{a} + y\vec{b} + z\vec{c} = \vec{r}$.

(c) If $\vec{x}_1,\vec{x}_2,..... \vec{x}_n$ are n non zero vectors, & $k_1, k_2, ...k_n$ are n scalars & if the linear combination $k_1\vec{x}_1 + k_2\vec{x}_2 + ... k_n\vec{x}_n = 0 \Rightarrow k_1 = 0, k_2 = 0 ..... k_n = 0$ then we say that vectors $\vec{x}_1,\vec{x}_2,..... \vec{x}_n$ are **LINEARLY INDEPENDENT VECTORS** .

(d) If $\vec{x}_1,\vec{x}_2,..... \vec{x}_n$ are not **LINEARLY INDEPENDENT** then they are said to be **LINEARLY DEPENDENT** vectors . i.e. if $k_1\vec{x}_1 + k_2\vec{x}_2 + ... + k_n\vec{x}_n = 0$ & if there exists at least one $k_r \neq 0$ then $\vec{x}_1,\vec{x}_2,..... \vec{x}_n$ are said to be **LINEARLY DEPENDENT** .

Note : ☞ If $\vec{a} = 3i + 2j + 5k$ then $\vec{a}$ is expressed as a **LINEAR COMBINATION** of vectors $\hat{i}, \hat{j}, \hat{k}$. Also, $\vec{a}$, $\hat{i}, \hat{j}, \hat{k}$ form a linearly dependent set of vectors. In general , every set of four vectors is a linearly dependent system.

☞ $\hat{i}, \hat{j}, \hat{k}$ are **LINEARLY INDEPENDENT** set of vectors. For $K_1\hat{i} + K_2\hat{j} + K_3\hat{k} = 0 \Rightarrow K_1 = 0 = K_2 = K_3$.

☞ Two vectors $\vec{a}$ & $\vec{b}$ are linearly dependent $\Rightarrow \vec{a}$ is parallel to $\vec{b}$ i.e. $\vec{a} \times \vec{b} = 0 \Rightarrow$ linear dependence of $\vec{a}$ & $\vec{b}$. Conversely if $\vec{a} \times \vec{b} \neq 0$ then $\vec{a}$ & $\vec{b}$ are linearly independent .

☞ If three vectors $\vec{a},\vec{b},\vec{c}$ are linearly dependent, then they are coplanar i.e. $[\vec{a}, \vec{b}, \vec{c}] = 0$, conversely, if $[\vec{a}, \vec{b}, \vec{c}] \neq 0$, then the vectors are linearly independent.

14. COPLANARITY OF VECTORS :

Four points A, B, C, D with position vectors $\vec{a},\vec{b},\vec{c},\vec{d}$ respectively are coplanar if and only if there exist scalars x, y, z, w not all zero simultaneously such that $x\vec{a} + y\vec{b} + z\vec{c} + w\vec{d} = 0$ where, $x + y + z + w = 0$.

15. RECIPROCAL SYSTEM OF VECTORS :

If $\vec{a},\vec{b},\vec{c}$ & $\vec{a}',\vec{b}',\vec{c}'$ are two sets of non coplanar vectors such that $\vec{a}.\vec{a}' = \vec{b}.\vec{b}' = \vec{c}.\vec{c}' = 1$ then the two systems are called Reciprocal System of vectors.

Note : $a' = \dfrac{\vec{b} \times \vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}$; $b' = \dfrac{\vec{c} \times \vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}$; $c' = \dfrac{\vec{a} \times \vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$

16. EQUATION OF A PLANE :

(a) The equation $(\vec{r} - \vec{r}_0).\vec{n} = 0$ represents a plane containing the point with p.v. $\vec{r}_0$ where $\vec{n}$ is a vector normal to the plane . $\vec{r}.\vec{n} = d$ is the general equation of a plane.

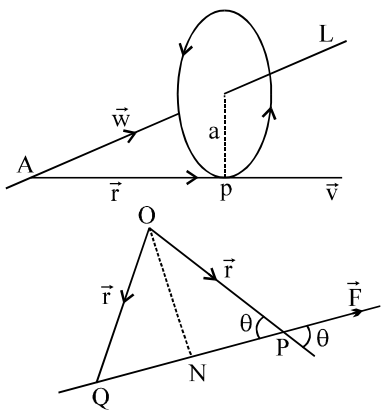
(b) Angle between the 2 planes is the angle between 2 normals drawn to the planes and the angle between a line and a plane is the compliment of the angle between the line and the normal to the plane.

17. APPLICATION OF VECTORS :

(a) Work done against a constant force $\vec{F}$ over a displacement $\vec{s}$ is defined as $\vec{W} = \vec{F}.\vec{s}$

(b) The tangential velocity $\vec{V}$ of a body moving in a circle is given by $\vec{V} = \vec{w} \times \vec{r}$ where $\vec{r}$ is the pv of the point P.

(c) The moment of $\vec{F}$ about 'O' is defined as $\vec{M} = \vec{r} \times \vec{F}$ where $\vec{r}$ is the pv of P wrt 'O'. The direction of $\vec{M}$ is along the normal to the plane OPN such that $\vec{r},\vec{F}$ & $\vec{M}$ form a right handed system.



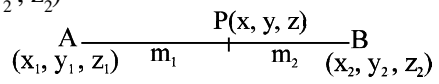
(d) Moment of the couple $= (\vec{r}_1 - \vec{r}_2) \times \vec{F}$ where $\vec{r}_1$ & $\vec{r}_2$ are pv's of the point of the application of the forces $\vec{F}$ & $-\vec{F}$.

3 -D COORDINATE GEOMETRY USEFUL RESULTS

A General :

(1) Distance (d) between two points (x_1, y_1, z_1) and (x_2, y_2, z_2)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



(2) **Section Fomula**

$$x = \frac{m_2 x_1 + m_1 x_2}{m_1 + m_2} \quad ; \quad y = \frac{m_2 y_1 + m_1 y_2}{m_1 + m_2} \quad ; \quad z = \frac{m_2 z_1 + m_1 z_2}{m_1 + m_2}$$

(For external division take –ve sign)

Direction Cosine and direction ratio's of a line

(3) Direction cosine of a line has the same meaning as d.c's of a vector.

(a) Any three numbers a, b, c proportional to the direction cosines are called the direction ratios i.e.

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \pm \frac{1}{\sqrt{a^2 + b^2 + c^2}}$$

same sign either +ve or –ve should be taken through out.

note that d.r's of a line joining x_1, y_1, z_1 and x_2, y_2, z_2 are proportional to $x_2 - x_1$, $y_2 - y_1$ and $z_2 - z_1$

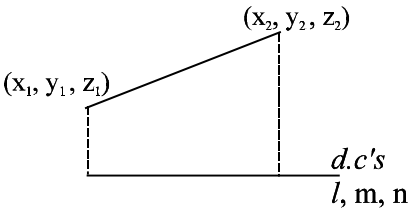
(b) If θ is the angle between the two lines whose d.c's are l_1, m_1, n_1 and l_2, m_2, n_2

$\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$

hence if lines are perpendicular then $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$

if lines are parallel then $\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$

note that if three lines are coplanar then $\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} = 0$



(4) Projection of the join of two points on a line with d.c's l, m, n are

$l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$

B PLANE: (i) General equation of degree one in x, y, z i.e. $ax + by + cz + d = 0$ represents a plane.

(ii) Equation of a plane passing through (x_1, y_1, z_1) is

$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$

where a, b, c are the direction ratios of the normal to the plane.

(iii) Equation of a plane if its intercepts on the co-ordinate axes are x_1, y_1, z_1 is

$$\frac{x}{x_1} + \frac{y}{y_1} + \frac{z}{z_1} = 1.$$

(iv) Equation of a plane if the length of the perpendicular from the origin on the plane is p and d.c's of the perpendicular as l, m, , n is $lx + my + nz = p$

(v) **Parallel and perpendicular planes** – Two planes

$a_1 x + b_1 y + c_1 z + d_1 = 0$ and $a_2 x + b_2 y + c_2 z + d_2 = 0$ are

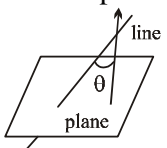
perpendicular if $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ and

coincident if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{d_1}{d_2}$

(vi) Angle between a plane and a line is the compliment of the angle between the normal to the plane and the line . If $\left.\begin{matrix} \text{Line} : \vec{r} = \vec{a} + \lambda \vec{b} \\ \text{Plane} : \vec{r} \cdot \vec{n} = d \end{matrix}\right]$ then $\cos(90 - \theta) = \sin\theta = \frac{\vec{b} \cdot \vec{n}}{|\vec{b}| . |\vec{n}|}$.

where θ is the angle between the line and normal to the plane.



(vii) Length of the perpendicular from a point (x_1, y_1, z_1) to a plane $ax + by + cz + d = 0$ is

$$p = \left| \frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}} \right|$$

(viii) Distance between two parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is

$$\left|\frac{d_1 - d_2}{\sqrt{a^2+b^2+c^2}}\right|$$

(ix) Planes bisecting the angle between two planes
$a_1x + b_1y + c_1z + d_1 = 0$ and $a_2 + b_2y + c_2z + d_2 = 0$ is given by

$$\left|\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}}\right| = \pm \left|\frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}}\right|$$

Of these two bisecting planes , one bisects the acute and the other obtuse angle between the given planes.

(x) Equation of a plane through the intersection of two planes P_1 and P_2 is given by $P_1 + \lambda P_2 = 0$

C STRAIGHT LINE IN SPACE

(i) Equation of a line through $A(x_1, y_1, z_1)$ and having direction cosines l ,m , n are

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$$

and the lines through (x_1, y_1, z_1) and (x_2, y_2, z_2)

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

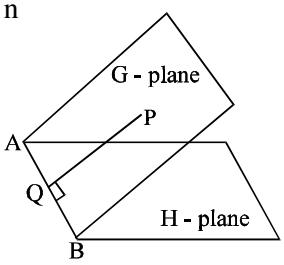
(ii) Intersection of two planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ together represent the unsymmetrical form of the straight line.

(iii) General equation of the plane containing the line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ is

$A(x-x_1) + B(y-y_1) + c(z-z_1) = 0$ where $Al + bm + cn = 0$.

LINE OF GREATEST SLOPE

AB is the line of intersection of G-plane and H is the horizontal plane. Line of greatest slope on a given plane, drawn through a given point on the plane, is the line through the point 'P' perpendicular to the line of intersetion of the given plane with any horizontal plane.



EXERCISE–1

Q.1 If $\vec{a}$ & $\vec{b}$ are non collinear vectors such that, $\vec{p} = (x+4y)\vec{a} + (2x+y+1)\vec{b}$ & $\vec{q} = (y-2x+2)\vec{a} + (2x-3y-1)\vec{b}$, find x & y such that $3\vec{p} = 2\vec{q}$.

Q.2 (a) Show that the points $\vec{a} - 2\vec{b} + 3\vec{c}; 2\vec{a} + 3\vec{b} - 4\vec{c}$ & $-7\vec{b} + 10\vec{c}$ are collinear .

(b) Prove that the points A= (1,2,3), B (3,4,7), C (–3,–2,–5) are collinear & find the ratio in which B divides AC.

Q.3 Points X & Y are taken on the sides QR & RS , respectively of a parallelogram PQRS, so that $\vec{QX} = 4\vec{XR}$ & $\vec{RY} = 4\vec{YS}$. The line XY cuts the line PR at Z . Prove that $\vec{PZ} = \left(\frac{21}{25}\right)\vec{PR}$.

Q.4 Find out whether the following pairs of lines are parallel, non-parallel & intersecting, or non-parallel & non-intersecting.

(i) $\vec{r}_1 = \hat{i} + \hat{j} + 2\hat{k} + \lambda\left(3\hat{i} - 2\hat{j} + 4\hat{k}\right)$
$\vec{r}_2 = 2\hat{i} + \hat{j} + 3\hat{k} + \mu\left(-6\hat{i} + 4\hat{j} - 8\hat{k}\right)$

(ii) $\vec{r}_1 = \hat{i} - \hat{j} + 3\hat{k} + \lambda\left(\hat{i} - \hat{j} + \hat{k}\right)$
$\vec{r}_2 = 2\hat{i} + 4\hat{j} + 6\hat{k} + \mu\left(2\hat{i} + \hat{j} + 3\hat{k}\right)$

(iii) $\vec{r}_1 = \hat{i} + \hat{k} + \lambda\left(\hat{i} + 3\hat{j} + 4\hat{k}\right)$
$\vec{r}_2 = 2\hat{i} + 3\hat{j} + \mu\left(4\hat{i} - \hat{j} + \hat{k}\right)$

Q.5 Let OACB be paralelogram with O at the origin & OC a diagonal. Let D be the mid point of OA. Using vector method prove that BD & CO intersect in the same ratio. Determine this ratio.

Q.6 A line EF drawn parallel to the base BC of a Δ ABC meets AB & AC in F & E respectively. BE & CF meet in L. Use vectors to show that AL bisects BC.

Q.7 'O' is the origin of vectors and A is a fixed point on the circle of radius 'a' with centre O. The vector $\vec{OA}$

is denoted by $\vec{a}$. A variable point 'P' lies on the tangent at A & $\vec{OP} = \vec{r}$. Show that $\vec{a}.\vec{r} = |a|^2$. Hence if P ≡ (x,y) & A ≡ (x_1,y_1) deduce the equation of tangent at A to this circle.

Q.8 (a) By vector method prove that the quadrilateral whose diagonals bisect each other at right angles is a rhombous.

(b) By vector method prove that the right bisectors of the sides of a triangle are concurrent.

Q.9 The resultant of two vectors $\vec{a}$ & $\vec{b}$ is perpendicular to $\vec{a}$. If $|\vec{b}| = \sqrt{2}|\vec{a}|$ show that the resultant of $2\vec{a}$ & $\vec{b}$ is perpendicular to $\vec{b}$.

Q.10 $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are the position vectors of the points A ≡ (x, y, z) ; B ≡ (y, – 2z, 3x) ; C ≡ (2z, 3x, – y) and D≡(1,–1, 2) respectively. If $|\vec{a}| = 2\sqrt{3}$; $(\hat{\vec{a}}\,\hat{\vec{b}}) = (\hat{\vec{a}}\,\hat{\vec{c}})$; $(\hat{\vec{a}}\,\hat{\vec{d}}) = \frac{\pi}{2}$ and $(\hat{\vec{a}}\,\hat{j})$ is obtuse, then find x, y, z.

Q.11 If $\vec{r}$ and $\vec{s}$ are non zero constant vectors and the scalar b is chosen such that $|\vec{r} + b\vec{s}|$ is minimum, then show that the value of $|b\vec{s}|^2 + |\vec{r} + b\vec{s}|^2$ is equal to $|\vec{r}|^2$.

Q.12 Use vectors to prove that the diagonals of a trapezium having equal non parallel sides are equal & conversely.

Q.13(a) Find a unit vector â which makes an angle (π/4) with axis of z & is such that â + i + j is a unit vector.

(b) Prove that $\left(\frac{\vec{a}}{\vec{a}^2} - \frac{\vec{b}}{\vec{b}^2}\right)^2 = \left(\frac{\vec{a} - \vec{b}}{|\vec{a}|\,|\vec{b}|}\right)^2$

Q.14 Given four non zero vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$. The vectors $\vec{a}, \vec{b}$ & $\vec{c}$ are coplanar but not collinear pair by pair and vector $\vec{d}$ is not coplanar with vectors $\vec{a}, \vec{b}$ & $\vec{c}$ and $(\hat{\vec{a}}\,\vec{b}) = (\hat{\vec{b}}\,\vec{c}) = \frac{\pi}{3}, (\hat{\vec{d}}\,\vec{a}) = \alpha, (\hat{\vec{d}}\,\vec{b}) = \beta$ then prove that $(\hat{\vec{d}}\,\vec{c}) = \cos^{-1}(\cos\beta - \cos\alpha)$.

Q.15 (a) Use vectors to find the acute angle between the diagonals of a cube .

(b) Prove cosine & projection rule in a triangle by using dot product .

Q.16 In the plane of a triangle ABC, squares ACXY, BCWZ are described , in the order given, externally to the triangle on AC & BC respectively. Given that $\vec{CX} = \vec{b}$, $\vec{CA} = \vec{a}$, $\vec{CW} = \vec{x}$, $\vec{CB} = \vec{y}$. Prove that $\vec{a}.\vec{y} + \vec{x}.\vec{b} = 0$. Deduce that $\vec{AW} . \vec{BX} = 0$.

Q.17 A Δ OAB is right angled at O ; squares OALM & OBPQ are constructed on the sides OA and OB externally. Show that the lines AP & BL intersect on the altitude through 'O' .

Q.18 Given that $\vec{u} = \hat{i} - 2\hat{j} + 3\hat{k}$; $\vec{v} = 2\hat{i} + \hat{j} + 4\hat{k}$; $\vec{w} = \hat{i} + 3\hat{j} + 3\hat{k}$ and $(\vec{u}\cdot\vec{R} - 10)\hat{i} + (\vec{v}\cdot\vec{R} - 20)\hat{j} + (\vec{w}\cdot\vec{R} - 20)\hat{k} = 0$. Find the unknown vector $\vec{R}$.

Q.19 If O is origin of reference, point A($\vec{a}$) ; B($\vec{b}$) ; C($\vec{c}$) ; D($\vec{a} + \vec{b}$) ; E($\vec{b} + \vec{c}$) ; F($\vec{c} + \vec{a}$) ; G($\vec{a} + \vec{b} + \vec{c}$) where $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ then prove that these points are vertices of a cube having length of its edge equal to unity provided the matrix.

$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$ is orghogonal. Also find the length XY such that X is the point of intersection of CM and GP ; Y is the point of intersection of OQ and DN where P, Q, M, N are respectively the midpoint of sides CF, BD, GF and OB.

Q.20 (a) If $\vec{a} + \vec{b} + \vec{c} = 0$, show that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$. Deduce the Sine rule for a Δ ABC.

(b) Find the minimum area of the triangle whose vertices are A(–1, 1, 2); B(1, 2, 3) and C(t, 1, 1) where t is a real number.

Q.21 (a) Determine vector of magnitude 9 which is perpendicular to both the vectors :

$$4\hat{i} - \hat{j} + 3\hat{k} \;\&\; -2\hat{i} + \hat{j} - 2\hat{k}$$

(b) A triangle has vertices (1, 1, 1) ; (2, 2, 2), (1, 1, y) and has the area equal to $\csc\left(\frac{\pi}{4}\right)$ sq. units. Find the value of y.

Q.22 The internal bisectors of the angles of a triangle ABC meet the opposite sides in D, E, F ; use vectors to prove that the area of the triangle DEF is given by

$\frac{(2abc)\,\Delta}{(a+b)(b+c)(c+a)}$ where Δ is the area of the triangle.

Q.23 If $\vec{a},\vec{b},\vec{c},\vec{d}$ are position vectors of the vertices of a cyclic quadrilateral ABCD prove that :
$\frac{|\vec{a}\times\vec{b}+\vec{b}\times\vec{d}+\vec{d}\times\vec{a}|}{(\vec{b}-\vec{a})\,.\,(\vec{d}-\vec{a})}+\frac{|\vec{b}\times\vec{c}+\vec{c}\times\vec{d}+\vec{d}\times\vec{b}|}{(\vec{b}-\vec{c})\,.\,(\vec{d}-\vec{c})}=0$

Q.24 The length of the edge of the regular tetrahedron P – ABC is 'a' . Point E and F are taken on the edges AD and BD respectively such that E divides $\vec{DA}$ and F divides $\vec{BD}$ in the ratio 2:1 each . Then find the area of triangle CEF.

Q.25 Let $\vec{a}=\sqrt{3}\,\hat{i}-\hat{j}$ and $\vec{b}=\frac{1}{2}\hat{i}+\frac{\sqrt{3}}{2}\hat{j}$ and $\vec{x}=\vec{a}+(q^2-3)\vec{b}$, $\vec{y}=-p\vec{a}+q\vec{b}$. If $\vec{x}\perp\vec{y}$, then express p as a function of q, say $p=f(q)$, $(p\neq 0\ \&\ q\neq 0)$ and find the intervals of monotonicity of $f(q)$.

EXERCISE–2

Q.1 $A(\vec{a})$; $B(\vec{b})$; $C(\vec{c})$ are the vertices of the triangle ABC such that $\vec{a}=\frac{1}{2}(2\hat{i}-\vec{r}-7\hat{k})$; $\vec{b}=3\vec{r}+\hat{j}-4\hat{k}$; $\vec{c}=22\hat{i}-11\hat{j}-9\vec{r}$. A vector $\vec{p}=2\hat{j}-\hat{k}$ is such that $(\vec{r}+\vec{p})$ is parallel to $\hat{i}$ and $(\vec{r}-2\hat{i})$ is parallel to $\vec{p}$. Show that there exists a point $D(\vec{d})$ on the line AB with $\vec{d}=2t\,\hat{i}+(1-2t)\hat{j}+(t-4)\hat{k}$. Also find the shortest distance C from AB.

Q.2 The position vectors of the points A, B, C are respectively (1, 1, 1) ; (1, –1, 2) ; (0, 2, –1). Find a unit vector parallel to the plane determined by ABC & perpendicular to the vector (1, 0, 1) .

Q.3 Let $\begin{vmatrix}(a_1-a)^2 & (a_1-b)^2 & (a_1-c)^2\\ (b_1-a)^2 & (b_1-b)^2 & (b_1-c)^2\\ (c_1-a)^2 & (c_1-b)^2 & (c_1-c)^2\end{vmatrix}=0$ and if the vectors $\vec{\alpha}=\hat{i}+a\hat{j}+a^2\hat{k}$; $\vec{\beta}=\hat{i}+b\hat{j}+b^2\hat{k}$; $\vec{\gamma}=\hat{i}+c\hat{j}+c^2\hat{k}$ are non coplanar, show that the vectors $\vec{\alpha}_1=\hat{i}+a_1\hat{j}+a_1^2\hat{k}$; $\vec{\beta}_1=\hat{i}+b_1\hat{j}+b_1^2\hat{k}$ and $\vec{\gamma}_1=\hat{i}+c_1\hat{j}+c_1^2\hat{k}$ are coplaner.

Q.4 Given non zero number x_1, x_2, x_3 ; y_1, y_2, y_3 and z_1, z_2 and z_3 such that $x_i>0$ and $y_i<0$ for all i = 1, 2, 3.

(i) Can the given numbers satisfy
$\begin{vmatrix}x_1 & x_2 & x_3\\ y_1 & y_2 & y_3\\ z_1 & z_2 & z_3\end{vmatrix}=0$ and $\begin{cases}x_1x_2+y_1y_2+z_1z_2=0\\ x_2x_3+y_2y_3+z_2z_3=0\\ x_3x_1+y_3y_1+z_3z_1=0\end{cases}$

(ii) If P = (x_1, x_2, x_3) ; Q (y_1, y_2, y_3) and O (0, 0, 0) can the triangle POQ be a right angled triangle?

Q.5 The pv's of the four angular points of a tetrahedron are : A $\left(\hat{j}+2\hat{k}\right)$; B $\left(3\hat{i}+\hat{k}\right)$; C $\left(4\hat{i}+3\hat{j}+6\hat{k}\right)$ & D $\left(2\hat{i}+3\hat{j}+2\hat{k}\right)$. Find :

(i) the perpendicular distance from A to the line BC. (ii) the volume of the tetrahedron ABCD.
(iii) the perpendicular distance from D to the plane ABC.
(iv) the shortest distance between the lines AB & CD.

Q.6 The length of an edge of a cube $ABCDA_1B_1C_1D_1$ is equal to unity. A point E taken on the edge $\vec{AA_1}$ is such that $\left|\vec{AE}\right|=\frac{1}{3}$. A point F is taken on the edge $\vec{BC}$ such that $\left|\vec{BF}\right|=\frac{1}{4}$. If O_1 is the centre of the cube, find the shortest distance of the vertex B_1 from the plane of the $\Delta\, O_1EF$.

Q.7 The vector $\vec{OP}=\hat{i}+2\hat{j}+2\hat{k}$ turns through a right angle, passing through the positive x-axis on the way. Find the vector in its new position.

Q.8 Find the point R in which the line AB cuts the plane CDE where
$\vec{a}=\hat{i}+2\hat{j}+\hat{k}$, $\vec{b}=2\hat{i}+\hat{j}+2\hat{k}$, $\vec{c}=-4\hat{j}+4\hat{k}$, $\vec{d}=2\hat{i}-2\hat{j}+2\hat{k}$ & $\vec{e}=4\hat{i}+\hat{j}+2\hat{k}$.

Q.9 If $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$; $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ and $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ then show that the value of the scalar triple product $[n\vec{a}+\vec{b}\quad n\vec{b}+\vec{c}\quad n\vec{c}+\vec{a}]$ is $(n^3+1)\begin{vmatrix}\vec{a}\cdot\hat{i} & \vec{a}\cdot\hat{j} & \vec{a}\cdot\hat{k}\\ \vec{b}\cdot\hat{i} & \vec{b}\cdot\hat{j} & \vec{b}\cdot\hat{k}\\ \vec{c}\cdot\hat{i} & \vec{c}\cdot\hat{j} & \vec{c}\cdot\hat{k}\end{vmatrix}$

Q.10 Find the scalars α & β if $\vec{a}\times(\vec{b}\times\vec{c})+(\vec{a}.\vec{b})\vec{b}=(4-2\beta-\sin\alpha)\vec{b}+(\beta^2-1)\vec{c}$ & $(\vec{c}.\vec{c})\vec{a}=\vec{c}$ while $\vec{b}$ & $\vec{c}$ are non zero non collinear vectors.

Q.11 If the vectors $\vec{b},\vec{c},\vec{d}$ are not coplanar, then prove that the vector

$(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})+(\vec{a}\times\vec{c})\times(\vec{d}\times\vec{b})+(\vec{a}\times\vec{d})\times(\vec{b}\times\vec{c})$ is parallel to $\vec{a}$.

Q.12 $\hat{a}$, $\hat{b}$, $\hat{c}$ are non–coplanar unit vectors. The angle between $\hat{b}$ & $\hat{c}$ is α, between $\hat{c}$ & $\hat{a}$ is β and between $\hat{a}$ & $\hat{b}$ is γ. If $A(\hat{a}\cos\alpha)$, $B(\hat{b}\cos\beta)$, $C(\hat{c}\cos\gamma)$, then show that in ΔABC,

$$\frac{\left|\hat{a}\times(\hat{b}\times\hat{c})\right|}{\sin A}=\frac{\left|\hat{b}\times(\hat{c}\times\hat{a})\right|}{\sin B}=\frac{\left|\hat{c}\times(\hat{a}\times\hat{b})\right|}{\sin C}=\frac{\prod\left|\hat{a}\times(\hat{b}\times\hat{c})\right|}{\left|\sum\sin\alpha\cos\beta\cos\gamma\,\hat{n}_1\right|}$$ where

$\hat{n}_1=\dfrac{\hat{b}\times\hat{c}}{\left|\hat{b}\times\hat{c}\right|}$, $\hat{n}_2=\dfrac{\hat{c}\times\hat{a}}{\left|\hat{c}\times\hat{a}\right|}$ & $\hat{n}_3=\dfrac{\hat{a}\times\hat{b}}{\left|\hat{a}\times\hat{b}\right|}$.

Q.13 Given that $\vec{a},\vec{b},\vec{p},\vec{q}$ are four vectors such that $\vec{a}+\vec{b}=\mu\vec{p}$, $\vec{b}.\vec{q}=0$ & $(\vec{b})^2=1$, where μ is a scalar then prove that $\left|(\vec{a}.\vec{q})\vec{p}-(\vec{p}.\vec{q})\vec{a}\right|=\left|\vec{p}.\vec{q}\right|$.

Q.14 Show that $\vec{a}=\vec{p}\times(\vec{q}\times\vec{r})$; $\vec{b}=\vec{q}\times(\vec{r}\times\vec{p})$ & $\vec{c}=\vec{r}\times(\vec{p}\times\vec{q})$ represents the sides of a triangle. Further prove that a unit vector perpendicular to the plane of this triangle is

$$\pm\frac{\hat{n}_1\tan(\vec{p}\wedge\vec{q})+\hat{n}_2\tan(\vec{q}\wedge\vec{r})+\hat{n}_3\tan(\vec{r}\wedge\vec{p})}{\left|\hat{n}_1\tan(\vec{p}\wedge\vec{q})+\hat{n}_2\tan(\vec{q}\wedge\vec{r})+\hat{n}_3\tan(\vec{r}\wedge\vec{p})\right|}$$ where $\vec{a},\vec{b},\vec{c},\vec{p},\vec{q}$ are non zero vectors and no two of $\vec{p},\vec{q},\vec{r}$ are mutually perpendicular & $\hat{n}_1=\dfrac{\vec{p}\times\vec{q}}{\left|\vec{p}\times\vec{q}\right|}$; $\hat{n}_2=\dfrac{\vec{q}\times\vec{r}}{\left|\vec{q}\times\vec{r}\right|}$ & $\hat{n}_3=\dfrac{\vec{r}\times\vec{p}}{\left|\vec{r}\times\vec{p}\right|}$

Q.15 Given four points P_1, P_2, P_3 and P_4 on the coordinate plane with origin O which satisfy the condition

$$\overrightarrow{OP}_{n-1}+\overrightarrow{OP}_{n+1}=\frac{3}{2}\overrightarrow{OP}_n,\ n=2,3$$

(i) If P_1, P_2 lie on the curve $xy=1$, then prove that P_3 does not lie on the curve.

(ii) If P_1, P_2, P_3 lie on the circle $x^2+y^2=1$, then prove that P_4 lies on this circle.

Q.16 Let $\vec{a}=\alpha\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}+2\alpha\hat{j}-2\hat{k}$ and $\vec{c}=2\hat{i}-\alpha\hat{j}+\hat{k}$. Find the value(s) of α, if any, such that $\left\{(\vec{a}\times\vec{b})\times(\vec{b}\times\vec{c})\right\}\times(\vec{c}\times\vec{a})=0$. Find the vector product when $\alpha=0$.

Q.17 Prove the result (Lagrange's identity) $(\vec{p}\times\vec{q})\cdot(\vec{r}\times\vec{s})=\begin{vmatrix}\vec{p}.\vec{r} & \vec{p}.\vec{s}\\ \vec{q}.\vec{r} & \vec{q}.\vec{s}\end{vmatrix}$ & use it to prove the following. Let (ab)denote the plane formed by the lines a,b. If (ab) is perpendicular to (cd) and (ac) is perpendicular to (bd) prove that (ad) is perpendicular to (bc).

Q.18 (a) If $p\vec{x}+(\vec{x}\times\vec{a})=\vec{b};(p\neq0)$ prove that $\vec{x}=\dfrac{p^2\vec{b}+(\vec{b}.\vec{a})\vec{a}-p(\vec{b}\times\vec{a})}{p(p^2+\vec{a}^2)}$.

(b) Solve the following equation for the vector $\vec{p}$; $\vec{p}\times\vec{a}+(\vec{p}.\vec{b})\vec{c}=\vec{b}\times\vec{c}$ where $\vec{a},\vec{b},\vec{c}$ are non zero non coplanar vectors and $\vec{a}$ is neither perpendicular to $\vec{b}$ nor to $\vec{c}$, hence show that $\left(\vec{p}\times\vec{a}+\dfrac{[\vec{a}\,\vec{b}\,\vec{c}]}{\vec{a}\cdot\vec{c}}\vec{c}\right)$ is perpendicular to $\vec{b}-\vec{c}$.

Q.19 Find a vector $\vec{v}$ which is coplanar with the vectors $\hat{i}+\hat{j}-2\hat{k}$ & $\hat{i}-2\hat{j}+\hat{k}$ and is orthogonal to the vector $-2\hat{i}+\hat{j}+\hat{k}$. It is given that the projection of $\vec{v}$ along the vector $\hat{i}-\hat{j}+\hat{k}$ is equal to $6\sqrt{3}$.

Q.20 Consider the non zero vectors $\vec{a},\vec{b},\vec{c}$ & $\vec{d}$ such that no three of which are coplanar then prove that $\vec{a}[\vec{b}\,\vec{c}\,\vec{d}]+\vec{c}[\vec{a}\,\vec{b}\,\vec{d}]=\vec{b}[\vec{a}\,\vec{c}\,\vec{d}]+\vec{d}[\vec{a}\,\vec{b}\,\vec{c}]$. Hence prove that $\vec{a},\vec{b},\vec{c}$ & $\vec{d}$ represent the position vectors of the vertices of a plane quadrilateral if $\dfrac{[\vec{b}\,\vec{c}\,\vec{d}]+[\vec{a}\,\vec{b}\,\vec{d}]}{[\vec{a}\,\vec{c}\,\vec{d}]+[\vec{a}\,\vec{b}\,\vec{c}]}=1$.

Q.21 The base vectors $\vec{a}_1,\vec{a}_2,\vec{a}_3$ are given in terms of base vectors $\vec{b}_1,\vec{b}_2,\vec{b}_3$ as, $\vec{a}_1=2\vec{b}_1+3\vec{b}_2-\vec{b}_3$; $\vec{a}_2=\vec{b}_1-2\vec{b}_2+2\vec{b}_3$ & $\vec{a}_3=-2\vec{b}_1+\vec{b}_2-2\vec{b}_3$. If $\vec{F}=3\vec{b}_1-\vec{b}_2+2\vec{b}_3$, then express $\vec{F}$ in terms of

$\vec{a}_1, \vec{a}_2$ & $\vec{a}_3$.

Q.22 If $A(\vec{a})$; $B(\vec{b})$ & $C(\vec{c})$ are three non collinear points , then for any point $P(\vec{p})$ in the plane of the ΔABC , prove that ; (i) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \vec{p}\,.\left(\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\right)$

(ii) The vector $\vec{v}$ perpendicular to the plane of the triangle ABC drawn from the origin 'O' is given by $\vec{v} = \pm \dfrac{\left[\vec{a}\,\vec{b}\,\vec{c}\right]\left(\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right)}{4\Delta^2}$ where $\vec{\Delta}$ is the vector area of the triangle ABC.

Q.23 Given the points P (1, 1, –1), Q (1, 2, 0) and R (–2, 2, 2). Find

(a) $\overrightarrow{PQ}\times\overrightarrow{PR}$

(b) Equation of the plane in

(i) scalar dot product form (ii) parametric form (iii) cartesian form

(iv) if the plane through PQR cuts the coordinate axes at A, B, C then the area of the ΔABC

Q.24 Let $\vec{a}, \vec{b}$ & $\vec{c}$ be non coplanar unit vectors, equally inclined to one another at an angle θ. If $\vec{a}\times\vec{b} + \vec{b}\times\vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$. Find scalars p , q & r in terms of θ.

Q.25 Solve the simultaneous vector equations for the vectors $\vec{x}$ and $\vec{y}$.

$\vec{x} + \vec{c}\times\vec{y} = \vec{a}$ and $\vec{y} + \vec{c}\times\vec{x} = \vec{b}$ where $\vec{c}$ is a non zero vector.

EXERCISE–3

Q.1 Find the angle between the two straight lines whose direction cosines l, m, n are given by $2l + 2m - n = 0$ and $mn + nl + lm = 0$.

Q.2 If two straight line having direction cosines l, m, n satisfy $al + bm + cn = 0$ and $f\,mn + g\,nl + h\,lm = 0$ are perpendicular, then show that $\dfrac{f}{a} + \dfrac{g}{b} + \dfrac{h}{c} = 0$.

Q.3 P is any point on the plane $lx + my + nz = p$. A point Q taken on the line OP (where O is the origin) such that $OP.\,OQ = p^2$. Show that the locus of Q is $p(lx + my + nz) = x^2 + y^2 + z^2$.

Q.4 Find the equation of the plane through the points (2, 2, 1), (1, –2, 3) and parallel to the x-axis.

Q.5 Through a point P (f, g, h), a plane is drawn at right angles to OP where 'O' is the origin, to meet the coordinate axes in A, B, C. Prove that the area of the triangle ABC is $\dfrac{r^5}{2fgh}$ where OP = r.

Q.6 The plane $lx + my = 0$ is rotated about its line of intersection with the plane $z = 0$ through an angle θ. Prove that the equation to the plane in new position is $lx + my \pm z\sqrt{l^2 + m^2}\,\tan\theta = 0$

Q.7 Find the equations of the straight line passing through the point (1, 2, 3) to intersect the straight line $x + 1 = 2(y - 2) = z + 4$ and parallel to the plane $x + 5y + 4z = 0$.

Q.8 Find the equations of the two lines through the origin which intersect the line $\dfrac{x-3}{2} = \dfrac{y-3}{1} = \dfrac{z}{1}$ at an angle of $\dfrac{\pi}{3}$.

Q.9 A variable plane is at a constant distance p from the origin and meets the coordinate axes in points A, B and C respectively. Through these points, planes are drawn parallel to the coordinates planes. Find the locus of their point of intersection.

Q.10 Find the distance of the point P (– 2, 3, – 4) from the line $\dfrac{x+2}{3} = \dfrac{2y+3}{4} = \dfrac{3z+4}{5}$ measured parallel to the plane $4x + 12y - 3z + 1 = 0$.

Q.11 Find the equation to the line passing through the point (1, –2, –3) and parallel to the line $2x + 3y - 3z + 2 = 0 = 3x - 4y + 2z - 4$.

Q.12 Find the equation of the line passing through the point (4, –14, 4) and intersecting the line of intersection of the planes : $3x + 2y - z = 5$ and $x - 2y - 2z = -1$ at right angles.

Q.13 Let P = (1, 0, – 1) ; Q = (1, 1, 1) and R = (2, 1, 3) are three points.

(a) Find the area of the triangle having P, Q and R as its vertices.

(b) Give the equation of the plane through P, Q and R in the form $ax + by + cz = 1$.

(c) Where does the plane in part (b) intersect the y-axis.

(d) Give parametric equations for the line through R that is perpendicular to the plane in part (b).

Q.14 Find the point where the line of intersection of the planes $x - 2y + z = 1$ and $x + 2y - 2z = 5$, intersects the plane $2x + 2y + z + 6 = 0$.

Q.15 Feet of the perpendicular drawn from the point P (2, 3, –5) on the axes of coordinates are A, B and C. Find the equation of the plane passing through their feet and the area of ΔABC.

Q.16 Find the equations to the line which can be drawn from the point (2, –1, 3) perpendicular to the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-4}{4}=\frac{y}{5}=\frac{z+3}{3}$ at right angles.

Q.17 Find the equation of the plane containing the straight line $\frac{x-1}{2}=\frac{y+2}{-3}=\frac{z}{5}$ and perpendicular to the plane $x-y+z+2=0$.

Q.18 Find the value of p so that the lines $\frac{x+1}{-3}=\frac{y-p}{2}=\frac{z+2}{1}$ and $\frac{x}{1}=\frac{y-7}{-3}=\frac{z+7}{2}$ are in the same plane. For this value of p, find the coordinates of their point of intersection and the equation of the plane containing them.

Q.19 Find the equations to the line of greatest slope through the point (7, 2 , –1) in the plane $x-2y+3z=0$ assuming that the axes are so placed that the plane $2x+3y-4z=0$ is horizontal.

Q.20 Let ABCD be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the area of triangles ABC, ACD and ADB be denoted by x, y and z sq. units respectively. Find the area of the triangle BCD.

Q.21 The position vectors of the four angular points of a tetrahedron OABC are (0, 0, 0); (0, 0, 2); (0, 4, 0) and (6, 0, 0) respectively. A point P inside the tetrahedron is at the same distance 'r' from the four plane faces of the tetrahedron. Find the value of 'r'.

Q.22 The line $\frac{x+6}{5}=\frac{y+10}{3}=\frac{z+14}{8}$ is the hypotenuse of an isosceles right angled triangle whose opposite vertex is (7, 2, 4). Find the equation of the remaining sides.

Q.23 Find the foot and hence the length of the perpendicular from the point (5, 7, 3) to the line $\frac{x-15}{3}=\frac{y-29}{8}=\frac{5-z}{5}$. Also find the equation of the plane in which the perpendicular and the given straight line lie.

Q.24 Find the equation of the line which is reflection of the line $\frac{x-1}{9}=\frac{y-2}{-1}=\frac{z+3}{-3}$ in the plane $3x-3y+10z=26$.

Q.25 Find the equation of the plane containing the line $\frac{x-1}{2}=\frac{y}{3}=\frac{z}{2}$ and parallel to the line $\frac{x-3}{2}=\frac{y}{5}=\frac{z-2}{4}$. Find also the S.D. between the two lines.

EXERCISE–4

Q.1(a) Let $\vec{OA}=\vec{a}$, $\vec{OB}=10\vec{a}+2\vec{b}$ and $\vec{OC}=\vec{b}$ where O, A & C are non–collinear points . Let p denote the area of the quadrilateral OABC, and let q denote the area of the parallelogram with OA and OC as adjacent sides. If p = kq, then k = _____ .

(b) If $\vec{A},\vec{B}$ & $\vec{C}$ are vectors such that $|\vec{B}|=|\vec{C}|$, Prove that ;

$$\left[(\vec{A}+\vec{B})\times(\vec{A}+\vec{C})\right]\times(\vec{B}\times\vec{C}).(\vec{B}+\vec{C})=0$$ [JEE ' 97, 2 + 5]

Q.2(a) Vectors $\vec{x},\vec{y}$ & $\vec{z}$ each of magnitude $\sqrt{2}$, make angles of 60^0 with each other. If $\vec{x}\times(\vec{y}\times\vec{z})=\vec{a},\vec{y}\times(\vec{z}\times\vec{x})=\vec{b}$ and $\vec{x}\times\vec{y}=\vec{c}$ then find $\vec{x}$, $\vec{y}$ and $\vec{z}$ in terms of $\vec{a}$, $\vec{b}$ and $\vec{c}$.

(b) The position vectors of the points P & Q are $5\hat{i}+7\hat{j}-2\hat{k}$ and $-3\hat{i}+3\hat{j}+6\hat{k}$ respectively. The vector $\vec{A}=3\hat{i}-\hat{j}+\hat{k}$ passes through the point P & the vector $\vec{B}=-3\hat{i}+2\hat{j}+4\hat{k}$ passes through the point Q. A third vector $2\hat{i}+7\hat{j}-5\hat{k}$ intersects vectors $\vec{A}$ & $\vec{B}$. Find the position vectors of the points of intersection. [REE ' 97, 6 + 6]

Q.3**(a)** Select the correct alternative(s)

(i) If $\vec{a}=\hat{i}+\hat{j}+\hat{k},\vec{b}=4\hat{i}+3\hat{j}+4\hat{k}$ and $\vec{c}=\hat{i}+\alpha\hat{j}+\beta\hat{k}$ are linearly dependent vectors & $|\vec{c}|=\sqrt{3}$, then:
(A) $\alpha=1,\beta=-1$ (B) $\alpha=1,\beta=\pm1$ (C) $\alpha=-1,\beta=\pm1$ (D) $\alpha=\pm1,\beta=1$

(ii) For three vectors $\vec{u}$, $\vec{v}$, $\vec{w}$ which of the following expressions is not equal to any of the remaining three?
(A) $\vec{u}.(\vec{v}\times\vec{w})$ (B) $(\vec{v}\times\vec{w}).\vec{u}$ (C) $\vec{v}.(\vec{u}\times\vec{w})$ (D) $(\vec{u}\times\vec{v}).\vec{w}$

(iii) Which of the following expressions are meaningful ?

(A) $\vec{u}\,.\,(\vec{v}\times\vec{w})$ (B) $(\vec{u}\,.\,\vec{v})\,.\,\vec{w}$ (C) $(\vec{u}\,.\,\vec{v})\,\vec{w}$ (D) $\vec{u}\times(\vec{v}\,.\,\vec{w})$

(b) Prove, by vector methods or otherwise, that the point of intersection of the diagonals of a trapezeum lies on the line passing through the mid–points of the parallel sides.
(You may assume that the trapezeum is not a parallelogram.)

(c) For any two vectors $\vec{u}$ & $\vec{v}$, prove that [JEE '98 , 2 + 2 + 2 + 8 + 8]

(i) $(\vec{u}.\vec{v})^2+|\vec{u}\times\vec{v}|^2=|\vec{u}|^2|\vec{v}|^2$ & (ii) $(1+|\vec{u}|^2)(1+|\vec{v}|^2)=(1-\vec{u}.\vec{v})^2+|\vec{u}+\vec{v}+(\vec{u}\times\vec{v})|^2$

Q.4(a) If $\vec{x}\times\vec{y}=\vec{a}$, $\vec{y}\times\vec{z}=\vec{b}$, $\vec{x}.\vec{b}=\gamma$, $\vec{x}.\vec{y}=1$ and $\vec{y}.\vec{z}=1$ then find $\vec{x}$, $\vec{y}$ & $\vec{z}$ in terms of $\vec{a}$, $\vec{b}$ and γ.

(b) Vectors $\vec{AB}=3\hat{i}-\hat{j}+\hat{k}$ & $\vec{CD}=-3\hat{i}+2\hat{j}+4\hat{k}$ are not coplanar. The position vectors of points A and C are $6\hat{i}+7\hat{j}+4\hat{k}$ and $-9\hat{j}+2\hat{k}$ respectively. Find the position vectors of a point P on the line AB & a point Q on the line CD such that $\vec{PQ}$ is perpendicular to $\vec{AB}$ and $\vec{CD}$ both.

Q.5(a) Let $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ & $\vec{b}=\hat{i}+\hat{j}$. If $\vec{c}$ is a vector such that $\vec{a}\cdot\vec{c}=|\vec{c}|$, $|\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $(\vec{a}\times\vec{b})$ and $\vec{c}$ is 30º, then $\left|(\vec{a}\times\vec{b})\times\vec{c}\right|=$

(A) 2/3 (B) 3/2 (C) 2 (D) 3

(b) Let $\vec{a}=2\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$ and a unit vector $\vec{c}$ be coplanar. If $\vec{c}$ is perpendicular to $\vec{a}$, then $\vec{c}=$

(A) $\frac{1}{\sqrt{2}}(-\hat{j}+\hat{k})$ (B) $\frac{1}{\sqrt{3}}(-\hat{i}-\hat{j}-\hat{k})$ (C) $\frac{1}{\sqrt{5}}(\hat{i}-2\hat{j})$ (D) $\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k})$

(c) Let $\vec{a}$ & $\vec{b}$ be two non-collinear unit vectors . If $\vec{u}=\vec{a}-(\vec{a}\,.\,\vec{b})\,\vec{b}$ & $\vec{v}=\vec{a}\times\vec{b}$, then $|\vec{v}|$ is :

(A) $|\vec{u}|$ (B) $|\vec{u}|+|\vec{u}\,.\,\vec{a}|$ (C) $|\vec{u}|+|\vec{u}\,.\,\vec{b}|$ (D) $\vec{u}+\vec{u}\,.\,(\vec{a}+\vec{b})$

(d) Let $\vec{u}$ & $\vec{v}$ be unit vectors . If $\vec{w}$ is a vector such that $\vec{w}+(\vec{w}\times\vec{u})=\vec{v}$, then prove that $|(\vec{u}\times\vec{v})\,.\,\vec{w}|\le\frac{1}{2}$ and the equality holds if and only if $\vec{u}$ is perpendicular to $\vec{v}$.

Q.6(a) An arc AC of a circle subtends a right angle at the centre O. The point B divides the arc in the ratio 1 : 2. If $\vec{OA}=\vec{a}$ & $\vec{OB}=\vec{b}$, then calculate $\vec{OC}$ in terms of $\vec{a}$ & $\vec{b}$.

(b) If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-coplanar vectors and $\vec{d}$ is a unit vector, then find the value of,
$\left|(\vec{a}\,.\,\vec{d})(\vec{b}\times\vec{c})+(\vec{b}\,.\,\vec{d})(\vec{c}\times\vec{a})+(\vec{c}\,.\,\vec{d})(\vec{a}\times\vec{b})\right|$ independent of $\vec{d}$. [REE '99, 6 + 6]

Q.7**(a)** Select the correct alternative :

(i) If the vectors $\vec{a}$, $\vec{b}$ & $\vec{c}$ form the sides BC, CA & AB respectively of a triangle ABC, then

(A) $\vec{a}\,.\,\vec{b}+\vec{b}\,.\,\vec{c}+\vec{c}\,.\,\vec{a}=0$ (B) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$

(C) $\vec{a}\,.\,\vec{b}=\vec{b}\,.\,\vec{c}=\vec{c}\,.\,\vec{a}$ (D) $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}=0$

(ii) Let the vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ & $\vec{d}$ be such that $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})=\vec{0}$. Let P_1 & P_2 be planes determined by the pairs of vectors $\vec{a}$, $\vec{b}$ & $\vec{c}$, $\vec{d}$ respectively . Then the angle between P_1 and P_2 is :

(A) 0 (B) $\pi/4$ (C) $\pi/3$ (4) $\pi/2$

(iii) If $\vec{a}$, $\vec{b}$ & $\vec{c}$ are unit coplanar vectors, then the scalar triple product

$\left[2\vec{a}-\vec{b}\quad 2\vec{b}-\vec{c}\quad 2\vec{c}-\vec{a}\right]=$ [JEE ,2000 (Screening) 1 + 1 + 1 out of 35]

(A) 0 (B) 1 (C) $-\sqrt{3}$ (D) $\sqrt{3}$

(b) Let ABC and PQR be any two triangles in the same plane . Assume that the perpendiculars from the points A, B, C to the sides QR, RP, PQ respectively are concurrent . Using vector methods or otherwise, prove that the perpendiculars from P, Q, R to BC, CA, AB respectively are also concurrent .
[JEE '2000 (Mains) 10 out of 100]

Q.8. (i) If $\vec{a}=\hat{i}+\hat{j}-\hat{k}$, $\vec{b}=-\hat{i}+2\hat{j}+2\hat{k}$ & $\vec{c}=-\hat{i}+2\hat{j}-\hat{k}$, find a unit vector normal to the vectors $\vec{a}+\vec{b}$ and $\vec{b}-\vec{c}$.

(ii) Given that vectors $\vec{a}$ & $\vec{b}$ are perpendicular to each other, find vector $\vec{v}$ in terms of $\vec{a}$ & $\vec{b}$ satisfying the equations, $\vec{v}\,.\,\vec{a}=0$, $\vec{v}\,.\,\vec{b}=1$ and $\left[\vec{v},\vec{a},\vec{b}\right]=1$

(iii) $\vec{a}$, $\vec{b}$ & $\vec{c}$ are three unit vectors such that $\vec{a} \times \left(\vec{b} \times \vec{c}\right) = \frac{1}{2}\left(\vec{b} + \vec{c}\right)$. Find angle between vectors $\vec{a}$ & $\vec{b}$ given that vectors $\vec{b}$ & $\vec{c}$ are non-parallel.

(iv) A particle is placed at a corner P of a cube of side 1 meter . Forces of magnitudes 2, 3 and 5 kg weight act on the particle along the diagonals of the faces passing through the point P . Find the moment of these forces about the corner opposite to P . [REE '2000 (Mains) 3 + 3 + 3 + 3 out of 100]

Q.9(a) The diagonals of a parallelogram are given by vectors $2\hat{i} + 3\hat{j} - 6\hat{k}$ and $3\hat{i} - 4\hat{j} - \hat{k}$. Determine its sides and also the area.

(b) Find the value of λ such that a, b, c are all non-zero and
$\left(-4\hat{i} + 5\hat{j}\right)a + (3\hat{i} - 3\hat{j} + \hat{k})b + (\hat{i} + \hat{j} + 3\hat{k})c = \lambda(a\hat{i} + b\hat{j} + c\hat{k})$ [REE '2001 (Mains) 3 + 3]

Q.10(a) Find the vector $\vec{r}$ which is perpendicular to $\vec{a} = \hat{i} - 2\hat{j} + 5\hat{k}$ and $\vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{r} \cdot \left(2\hat{i} + \hat{j} + \hat{k}\right) + 8 = 0$.

(b) Two vertices of a triangle are at $-\hat{i} + 3\hat{j}$ and $2\hat{i} + 5\hat{j}$ and its orthocentre is at $\hat{i} + 2\hat{j}$. Find the position vector of third vertex. [REE '2001 (Mains) 3 + 3]

Q.11 (a) If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors, then $\left|\vec{a} - \vec{b}\right|^2 + \left|\vec{b} - \vec{c}\right|^2 + \left|\vec{c} - \vec{a}\right|^2$ does NOT exceed
(A) 4 (B) 9 (C) 8 (D) 6

(b) Let $\vec{a} = \hat{i} - \hat{k}, \vec{b} = x\hat{i} + \hat{j} + (1-x)\hat{k}$ and $\vec{c} = y\hat{i} + x\hat{j} + (1+x-y)\hat{k}$. Then $[\vec{a}, \vec{b}, \vec{c}]$ depends on
(A) only x (B) only y (C) NEITHER x NOR y (D) both x and y
[JEE '2001 (Screening) 1 + 1 out of 35]

Q.12(a) Show by vector methods, that the angular bisectors of a triangle are concurrent and find an expression for the position vector of the point of concurrency in terms of the position vectors of the vertices.

(b) Find 3–dimensional vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3$ satisfying
$\vec{v}_1 \cdot \vec{v}_1 = 4$, $\vec{v}_1 \cdot \vec{v}_2 = -2$, $\vec{v}_1 \cdot \vec{v}_3 = 6$, $\vec{v}_2 \cdot \vec{v}_2 = 2$, $\vec{v}_2 \cdot \vec{v}_3 = -5$, $\vec{v}_3 \cdot \vec{v}_3 = 29$.

(c) Let $\vec{A}(t) = f_1(t)\hat{i} + f_2(t)\hat{j}$ and $\vec{B}(t) = g_1(t)\hat{i} + g_2(t)\hat{j}$, $t \in [0, 1]$, where f_1, f_2, g_1, g_2 are continuous functions. If $\vec{A}(t)$ and $\vec{B}(t)$ are nonzero vectors for all t and $\vec{A}(0) = 2\hat{i} + 3\hat{j}$, $\vec{A}(1) = 6\hat{i} + 2\hat{j}$, $\vec{B}(0) = 3\hat{i} + 2\hat{j}$ and $\vec{B}(1) = 2\hat{i} + 6\hat{j}$, then show that $\vec{A}(t)$ and $\vec{B}(t)$ are parallel for some t. [JEE '2001 (Mains) 5 + 5 + 5 out of 100]

Q.13(a) If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a} + 2\vec{b}$ and $5\vec{a} - 4\vec{b}$ are perpendicular to each other then the angle between $\vec{a}$ and $\vec{b}$ is

(A) 45^0 (B) 60^0 (C) $\cos^{-1}\left(\frac{1}{3}\right)$ (D) $\cos^{-1}\left(\frac{2}{7}\right)$

(b) Let $\vec{V} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{W} = \hat{i} + 3\hat{k}$. If $\vec{U}$ is a unit vector, then the maximum value of the scalar triple product $\left[\vec{U}\ \vec{V}\ \vec{W}\right]$ is [JEE 2002(Screening), 3 + 3]

(A) -1 (B) $\sqrt{10} + \sqrt{6}$ (C) $\sqrt{59}$ (D) $\sqrt{60}$

Q.14 Let V be the volume of the parallelopiped formed by the vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$. If a_r, b_r, c_r, where r = 1, 2, 3, are non-negative real numbers and $\sum_{r=1}^{3}(a_r + b_r + c_r) = 3L$, show that $V < L^3$. [JEE 2002(Mains), 5]

Q.15 If $\vec{a} = \hat{i} + a\hat{j} + \hat{k}$, $\vec{b} = \hat{j} + a\hat{k}$, $\vec{c} = a\hat{i} + \hat{k}$, then find the value of 'a' for which volume of parallelopiped formed by three vectors as coterminous edges, is minimum, is

(A) $\frac{1}{\sqrt{3}}$ (B) $-\frac{1}{\sqrt{3}}$ (C) $\pm\frac{1}{\sqrt{3}}$ (D) none [JEE 2003(Scr.), 3]

Q.16(i) Find the equation of the plane passing through the points (2, 1, 0) , (5, 0, 1) and (4, 1, 1).

(ii) If P is the point (2, 1, 6) then find the point Q such that PQ is perpendicular to the plane in (i) and the mid

point of PQ lies on it. [JEE 2003, 4 out of 60]

Q.17 If $\vec{u}$, $\vec{v}$, $\vec{w}$ are three non-coplanar unit vectors and α, β, γ are the angles between $\vec{u}$ and $\vec{v}$, $\vec{v}$ and $\vec{w}$, $\vec{w}$ and $\vec{u}$ respectively and $\vec{x}$, $\vec{y}$, $\vec{z}$ are unit vectors along the bisectors of the angles α, β, γ respectively. Prove that $[\vec{x}\times\vec{y}\ \ \vec{y}\times\vec{z}\ \ \vec{z}\times\vec{x}] = \frac{1}{16}[\vec{u}\ \vec{v}\ \vec{w}]^2 \sec^2\frac{\alpha}{2}\ \sec^2\frac{\beta}{2}\ \sec^2\frac{\gamma}{2}$.
[JEE 2003, 4 out of 60]

Q.18(a) If the lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-1}{4}$ and $\frac{x-3}{1}=\frac{y-k}{2}=\frac{z}{1}$ intersect, then k =

(A) $\frac{2}{9}$ (B) $\frac{9}{2}$ (C) 0 (D) – 1

(b) A unit vector in the plane of the vectors $2\hat{i}+\hat{j}+\hat{k}$, $\hat{i}-\hat{j}+\hat{k}$ and orthogonal to $5\hat{i}+2\hat{j}+6\hat{k}$

(A) $\frac{6\hat{i}-5\hat{k}}{\sqrt{61}}$ (B) $\frac{3\hat{j}-\hat{k}}{\sqrt{10}}$ (C) $\frac{2\hat{i}-5\hat{k}}{\sqrt{29}}$ (D) $\frac{2\hat{i}+\hat{j}-2\hat{k}}{3}$

(c) If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{a}\cdot\vec{b}=1$ and $\vec{a}\times\vec{b}=\hat{j}-\hat{k}$, then $\vec{b}$ = [JEE 2004 (screening)]

(A) $\hat{i}$ (B) $\hat{i}-\hat{j}+\hat{k}$ (C) $2\hat{j}-\hat{k}$ (D) $2\hat{i}$

Q.19(a) Let $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ are four distinct vectors satisfying $\vec{a}\times\vec{b}$ = $\vec{c}\times\vec{d}$ and $\vec{a}\times\vec{c}=\vec{b}\times\vec{d}$. Show that $\vec{a}\cdot\vec{b}+\vec{c}\cdot\vec{d}\neq\vec{a}\cdot\vec{c}+\vec{b}\cdot\vec{d}$.

(b) T is a parallelopiped in which A, B, C and D are vertices of one face. And the face just above it has corresponding vertices A', B', C', D'. T is now compressed to S with face ABCD remaining same and A', B', C', D' shifted to A., B., C., D. in S. The volume of parallelopiped S is reduced to 90% of T. Prove that locus of A. is a plane.

(c) Let P be the plane passing through (1, 1, 1) and parallel to the lines L_1 and L_2 having direction ratios 1, 0, –1 and –1, 1, 0 respectively. If A, B and C are the points at which P intersects the coordinate axes, find the volume of the tetrahedron whose vertices are A, B, C and the origin.

Q.20(a) If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero, non-coplanar vectors and $\vec{b}_1=\vec{b}-\frac{\vec{b}\cdot\vec{a}}{|\vec{a}|^2}\vec{a}$, $\vec{b}_2=\vec{b}+\frac{\vec{b}\cdot\vec{a}}{|\vec{a}|^2}\vec{a}$,

$\vec{c}_1=\vec{c}-\frac{\vec{c}\cdot\vec{a}}{|\vec{a}|^2}\vec{a}+\frac{\vec{b}\cdot\vec{c}}{|\vec{c}|^2}\vec{b}_1, \vec{c}_2=\vec{c}-\frac{\vec{c}\cdot\vec{a}}{|\vec{a}|^2}\vec{a}-\frac{\vec{b}_1\cdot\vec{c}}{|\vec{b}_1|^2}\vec{b}_1, \vec{c}_3=\vec{c}-\frac{\vec{c}\cdot\vec{a}}{|\vec{c}|^2}\vec{a}+\frac{\vec{b}\cdot\vec{c}}{|\vec{c}|^2}\vec{b}_1, \vec{c}_4=\vec{c}-\frac{\vec{c}\cdot\vec{a}}{|\vec{c}|^2}\vec{a}-\frac{\vec{b}\cdot\vec{c}}{|\vec{b}|^2}\vec{b}_1$

then the set of orthogonal vectors is

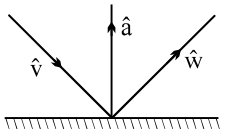
(A) $(\vec{a}, \vec{b}_1, \vec{c}_3)$ (B) $(\vec{a}, \vec{b}_1, \vec{c}_2)$ (C) $(\vec{a}, \vec{b}_1, \vec{c}_1)$ (D) $(\vec{a}, \vec{b}_2, \vec{c}_2)$

(b) A variable plane at a distance of 1 unit from the origin cuts the co-ordinate axes at A, B and C. If the centroid D (x, y, z) of triangle ABC satisfies the relation $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=k$, then the value of k is

(A) 3 (B) 1 (C) 1/3 (D) 9 [JEE 2005 (Screening), 3]

(c) Find the equation of the plane containing the line $2x - y + z - 3 = 0, 3x + y + z = 5$ and at a distance of $\frac{1}{\sqrt{6}}$ from the point (2, 1, – 1).

(d) Incident ray is along the unit vector $\hat{v}$ and the reflected ray is along the unit vector $\hat{w}$. The normal is along unit vector $\hat{a}$ outwards. Express $\hat{w}$ in terms of $\hat{a}$ and $\hat{v}$.



[JEE 2005 (Mains), 2 + 4 out of 60]

Q.21(a) A plane passes through (1, –2, 1) and is perpendicular to two planes 2x – 2y + z = 0 and x – y + 2z = 4. The distance of the plane from the point (1, 2, 2) is

(A) 0 (B) 1 (C) $\sqrt{2}$ (D) $2\sqrt{2}$

(b) Let $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}+\hat{j}-\hat{k}$. A vector in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{1}{\sqrt{3}}$, is [JEE 2006,3 marks each]

(A) $4\hat{i}-\hat{j}+4\hat{k}$ (B) $3\hat{i}+\hat{j}-3\hat{k}$ (C) $2\hat{i}+\hat{j}-2\hat{k}$ (D) $4\hat{i}+\hat{j}-4\hat{k}$

(c) Let $\vec{A}$ be vector parallel to line of intersection of planes P_1 and P_2 through origin. P_1 is parallel to the vectors $2\hat{j}+3\hat{k}$ and $4\hat{j}-3\hat{k}$ and P_2 is parallel to $\hat{j}-\hat{k}$ and $3\hat{i}+3\hat{j}$, then the angle between vector $\vec{A}$ and $2\hat{i}+\hat{j}-2\hat{k}$ is

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{3\pi}{4}$ [JEE 2006, 5]

(d) **Match the following**

(i) Two rays in the first quadrant $x + y = |a|$ and $ax - y = 1$ intersects each other in the interval $a \in (a_0, \infty)$, the value of a_0 is (A) 2

(ii) Point (α, β, γ) lies on the plane $x + y + z = 2$.
Let $\vec{a} = \alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$, $\hat{k}\times(\hat{k}\times\vec{a}) = 0$, then $\gamma =$ (B) 4/3

(iii) $\left|\int_0^1 (1-y^2)\,dy\right| + \left|\int_1^0 (y^2-1)\,dy\right|$ (C) $\left|\int_0^1 \sqrt{1-x}\,dx\right| + \left|\int_{-1}^0 \sqrt{1+x}\,dx\right|$

(iv) If $\sin A \sin B \sin C + \cos A \cos B = 1$, then the value of $\sin C =$ (D) 1 [JEE 2006, 6]

(e) **Match the following**

(i) $\sum_{i=1}^{\infty} \tan^{-1}\left(\frac{1}{2i^2}\right) = t$, then $\tan t =$ (A) 0

(ii) Sides a, b, c of a triangle ABC are in A.P.
and $\cos\theta_1 = \frac{a}{b+c}$, $\cos\theta_2 = \frac{b}{a+c}$, $\cos\theta_3 = \frac{c}{a+b}$,
then $\tan^2\frac{\theta_1}{2} + \tan^2\frac{\theta_3}{2} =$ (B) 1

(iii) A line is perpendicular to $x + 2y + 2z = 0$ and passes through (0, 1, 0). The perpendicular distance of this line from the origin is (C) $\frac{\sqrt{5}}{3}$

(D) 2/3 [JEE 2006, 6]

ANSWER KEY

EXERCISE–1

Q.1 $x = 2$, $y = -1$ Q.2 **(b)** externally in the ratio 1 : 3

Q.4 **(i)** parallel **(ii)** the lines intersect at the point p.v. $-2\hat{i} + 2\hat{j}$ **(iii)** lines are skew

Q.5 2 : 1 Q.7 $xx_1 + yy_1 = a^2$ Q.10 $x = 2, y = -2, z = -2$

Q.13 **(a)** $\frac{-1}{2}i - \frac{1}{2}j + \frac{1}{\sqrt{2}}k$ Q.15 (a) $\arccos\frac{1}{3}$ Q.18 $-\hat{i} + 2\hat{j} + 5\hat{k}$

Q.19 $\frac{\sqrt{11}}{3}$ Q.20 **(b)** $\frac{\sqrt{3}}{2}$ Q.21 (a) $\pm 3(\hat{i} - 2\hat{j} - 2\hat{k})$, (b) $y = 3$ or $y = -1$ Q.24 $\frac{5a^2}{12\sqrt{3}}$ sq. units

Q.25 $p = \frac{q(q^3 - 3)}{4}$; decreasing in $q \in (-1, 1)$, $q \neq 0$

EXERCISE–2

Q.1 $2\sqrt{17}$ Q.2 $\pm\frac{1}{3\sqrt{3}}(\hat{i} + 5\hat{j} - \hat{k})$ Q.4 NO, NO

Q.5 **(i)** $\frac{6}{7}\sqrt{14}$ **(ii)** 6 **(iii)** $\frac{3}{5}\sqrt{10}$ **(iv)** $\sqrt{6}$ Q.6 $\frac{11}{\sqrt{170}}$ Q.7 $\frac{4}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$

Q.8 p.v. of $\vec{R} = r = 3i + 3k$ Q.10 $\alpha = n\pi + \frac{(-1)^n\pi}{2}, n \in I \& \beta = 1$

Q.16 $\alpha = 2/3$; if $\alpha = 0$ then vector product is $-60\left(2\hat{i} + \hat{k}\right)$

Q.18 **(b)** $\left\{\vec{p} = \frac{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}{(\vec{a}.\vec{c})(\vec{a}.\vec{b})}\left(\vec{a} + \vec{c}\times\vec{b}\right) + \frac{(\vec{b}.\vec{c})\vec{b}}{(\vec{a}.\vec{b})} - \frac{(\vec{b}.\vec{b})\vec{c}}{(\vec{a}.\vec{b})}\right\}$ Q.19 $9\left(-\hat{j} + \hat{k}\right)$

Q.21 $F = 2\vec{a}_1 + 5\vec{a}_2 + 3\vec{a}_3$

Q.23 (a) $2\hat{i} - 3\hat{j} + 3\hat{k}$, (b) (i) -4, (ii) $\hat{r} = \hat{i} + \hat{j} - \hat{k} + \lambda(\hat{j} + \hat{k}) + \mu(-3\hat{i} + \hat{j} + 3\hat{k})$, (iii) -4, (iv) $\frac{4\sqrt{22}}{9}$

Q.24 $p = -\frac{1}{\sqrt{1 + 2\cos\theta}}$; $q = \frac{2\cos\theta}{\sqrt{1 + 2\cos\theta}}$; $r = -\frac{1}{\sqrt{1 + 2\cos\theta}}$

or $p = \frac{1}{\sqrt{1 + 2\cos\theta}}$; $q = -\frac{2\cos\theta}{\sqrt{1 + 2\cos\theta}}$; $r = \frac{1}{\sqrt{1 + 2\cos\theta}}$

Q.25 $\vec{x} = \frac{\vec{a} + (\vec{c}.\vec{a})\vec{c} + \vec{b}\times\vec{c}}{1 + \vec{c}^2}$, $y = \frac{\vec{b} + (\vec{c}.\vec{b})\vec{c} + \vec{a}\times\vec{c}}{1 + \vec{c}^2}$

EXERCISE–3

Q.1 $\theta = 90^0$ Q.4 $y + 2z = 4$ Q.7 $\frac{x - 1}{2} = \frac{y - 2}{2} = \frac{z - 3}{-3}$

Q.8 $\frac{x}{1}=\frac{y}{2}=\frac{z}{-1}$ or $\frac{x}{-1}=\frac{y}{1}=\frac{z}{-2}$ Q.9 $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{p^2}$ Q.10 $\frac{17}{2}$

Q.11 $\frac{x-1}{6}=\frac{y+2}{13}=\frac{z+3}{17}$ Q.12 $\frac{x-4}{3}=\frac{y+14}{10}=\frac{z-4}{4}$

Q.13 (a) $\frac{3}{2}$; (b) $\frac{2x}{3}+\frac{2y}{3}-\frac{z}{3}=1$; (c) $\left(0,\frac{3}{2},0\right)$; (d) x = 2t + 2 ; y = 2t + 1 and z = – t + 3

Q.14 (1, –2, – 4) Q.15 $\frac{x}{2}+\frac{y}{3}+\frac{z}{-5}=1$, Area = $\frac{19}{2}$ sq. units Q.16 $\frac{x-2}{11}=\frac{y+1}{-10}=\frac{z-3}{2}$

Q.17 2x + 3y + z + 4 = 0 Q.18 p = 3, (2, 1, –3) ; x + y + z = 0

Q.19 $\frac{x-7}{22}=\frac{y-2}{5}=\frac{z+1}{-4}$ Q.20 $\sqrt{(x^2+y^2+z^2)}$ Q.21 $\frac{2}{3}$

Q.22 $\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}$; $\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6}$

Q.23 (9, 13, 15) ; 14 ; 9x – 4y – z = 14 Q.24 $\frac{x-4}{9}=\frac{y+1}{-1}=\frac{z-7}{-3}$

Q.25 x – 2y + 2z – 1 = 0; 2 units

EXERCISE–4

Q.1 (a) 6

Q.2 (a) $\vec{x}=\vec{a}\times\vec{c}$; $\vec{y}=\vec{b}\times\vec{c}$; $\vec{z}=\vec{b}+\vec{a}\times\vec{c}$ or $\vec{b}\times\vec{c}-\vec{a}$ (b) (2, 8, –3) ; (0, 1, 2)

Q.3 (a) (i) D (ii) C (iii) A, C

Q.4 (a) $x=\frac{\frac{\vec{a}\times\vec{b}}{\gamma}-\vec{a}\times\frac{\vec{a}\times\vec{b}}{\gamma}}{\left(\frac{\vec{a}\times\vec{b}}{\gamma}\right)^2}$; $y=\frac{\vec{a}\times\vec{b}}{\gamma}$; $z=\frac{\frac{\vec{a}\times\vec{b}}{\gamma}+\vec{b}\times\frac{\vec{a}\times\vec{b}}{\gamma}}{\left(\frac{\vec{a}\times\vec{b}}{\gamma}\right)^2}$ (b) P ≡ (3, 8, 3) & Q ≡ (–3, –7, 6)

Q.5 (a) B (b) A (c) A, C Q.6 (a) $\vec{c}=-\sqrt{3}\,\vec{a}+2\vec{b}$ (b) $\left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right|$

Q.7 (a) (i) B (ii) A (iii) A

Q.8 (i) $\pm\hat{i}$; (ii) $\frac{\vec{b}}{\vec{b}^2}+\frac{\vec{a}x\vec{b}}{(\vec{a}\times\vec{b})^2}$; (iii) $\frac{2\pi}{3}$; (iv) $|\vec{M}|=\sqrt{7}$

Q.9 (a) $\frac{1}{2}\left(5\hat{i}-\hat{j}-7\hat{k}\right), \frac{1}{2}(-\hat{i}+7\hat{j}-5\hat{k})$; $\frac{1}{2}\sqrt{1274}$ sq. units (b) $\lambda=0,\ \lambda=-2\pm\sqrt{29}$

Q.10 (a) $\vec{r}=-13\hat{i}+11\hat{j}+7\hat{k}$; (b) $\frac{5}{7}\hat{i}+\frac{17}{7}\hat{j}$

Q.11 (a) B (b) C Q.12 (b) $\vec{v}_1=2\hat{i},\ \vec{v}_2=-\hat{i}\pm\hat{j},\ \vec{v}_3=3\hat{i}\pm2\hat{j}\pm4\hat{k}$

Q.13 (a) B ; (b) C Q.15 D Q.16 (i) x + y – 2z = 3 ; (ii) (6, 5, –2)

Q.18 (a) B, (b) B, (c) A Q.19 (c) 9/2 cubic units

Q.20 (a) B, (b) D ; (c) 2x – y + z – 3 = 0 and 62x + 29y + 19z – 105 = 0, (d) $\hat{w}=\hat{v}-2(\hat{a}\cdot\hat{v})\hat{a}$

Q.21 **(a)** D; **(b)** A; **(c)** B, D; **(d)** (i) D, (ii) A, (iii) B, C, (iv) D; **(e)** (i) B, (ii) D, (iii) C

EXERCISE–5

Part : (A) Only one correct option

1. The locus of a point P which moves such that $PA^2 - PB^2 = 2k^2$ where A and B are (3, 4, 5) and (– 1, 3 – 7) respectively is
(A) $8x + 2y + 24z - 9 + 2k^2 = 0$ (B) $8x + 2y + 24z - 2k^2 = 0$
(C) $8x + 2y + 24z + 9 + 2k^2 = 0$ (D) none of these

2. The position vectors of three points A, B, C are $\hat{i} + 2\hat{j} + 3\hat{k}, 2\hat{i} + 3\hat{j} + \hat{k}$ & $3\hat{i} + \hat{j} + 2\hat{k}$. A unit vector perpendicular to the plane of the triangle ABC is:

(A) $\left(-\frac{1}{\sqrt{3}}\right)\left(\hat{i} + \hat{j} + \hat{k}\right)$ (B) $\left(\frac{1}{\sqrt{3}}\right)\left(\hat{i} - \hat{j} + \hat{k}\right)$ (C) $\left(\frac{1}{\sqrt{3}}\right)\left(\hat{i} + \hat{j} - \hat{k}\right)$ (D) none

3. The square of the perpendicular distance of a point P (p, q, r) from a line through A(a, b, c) and whose direction cosine are ℓ, m, n is
(A) $\Sigma\{(q - b)n - (r - c)m\}^2$ (B) $\Sigma\{(q + b)n - (r + c)m\}^2$
(C) $\Sigma\{(q - b)n + (r - c)m\}^2$ (D) none of these

4. A variable plane passes through a fixed point (1, 2, 3). The locus of the foot of the perpendicular drawn from origin to this plane is:
(A) $x^2 + y^2 + z^2 - x - 2y - 3z = 0$ (B) $x^2 + 2y^2 + 3z^2 - x - 2y - 3z = 0$
(C) $x^2 + 4y^2 + 9z^2 + x + 2y + 3 = 0$ (D) $x^2 + y^2 + z^2 + x + 2y + 3z = 0$

5. The equation of the plane which bisects the angle between the planes $3x - 6y + 2z + 5 = 0$ and $4x - 12y + 3z - 3 = 0$ which contains the origin is
(A) $33x - 13y + 32z + 45 = 0$ (B) $x - 3y + z - 5 = 0$ (C) $33x + 13y + 32z + 45 = 0$ (D) None

6. The distance of the point of intersection of the line $x - 3 = (1/2)(y-4) = (1/2)(z-5)$ and the plane $x + y + z = 17$ from the point (3, 4, 5)
(A) 2 (B) 3 (C) 1/3 (D) 1/2

7. The lines $x = ay + b$, $z = cy + d$ and $x = a'y + b'$, $z = c'y + d'$ will be mutually perpendicular provided
(A) $(a + a')(b + b')(c + c')$ (B) $aa' + cc' + 1 = 0$
(C) $aa' + bb' + cc' + 1 = 0$ (D) $(a + a')(b + b')(c + c') + 1 = 0$

8. A straight line $\vec{r} = \vec{a} + \lambda\vec{b}$ meets the plane $\vec{r} \cdot \hat{n} = p$ in the point P whose position vector is

(A) $\vec{a} + \left(\frac{\vec{a}.\hat{n}}{\vec{b}.\hat{n}}\right)\vec{b}$ (B) $\vec{a} + \left(\frac{p - \vec{a}.\hat{n}}{\vec{b}.\hat{n}}\right)\vec{b}$ (C) $\vec{a} - \left(\frac{\vec{a}.\hat{n}}{\vec{b}.\hat{n}}\right)\vec{b}$ (D) $\vec{a} - \left(\frac{p - \vec{a}.\hat{n}}{\vec{b}.\hat{n}}\right)\vec{b}$

9. Equation of the angle bisector of the angle between the lines $\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{1}$ & $\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{-1}$ is

(A) $\frac{x-1}{2} = \frac{y-2}{2}; z - 3 = 0$ (B) $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$

(C) $x - 1 = 0; \frac{y-2}{1} = \frac{z-3}{1}$ (D) None of these

10. The distance of the point, (– 1, – 5, – 10) from the point of intersection of the line, $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane, $x - y + z = 5$, is:
(A) 10 (B) 11 (C) 12 (D) 13

11. If a plane cuts off intercepts OA = a, OB = b, OC = c from the coordinate axes, then the area of the triangle ABC =

(A) $\frac{1}{2}\sqrt{b^2c^2 + c^2a^2 + a^2b^2}$ (B) $\frac{1}{2}$ (bc + ca + ab) (C) $\frac{1}{2}$ abc (D) $\frac{1}{2}\sqrt{(b-c)^2 + (c-a)^2 + (a-b)^2}$

12. The angle between the lines whose direction cosines satisfy the equations $\ell + m + n = 0$ and $\ell^2 = m^2 + n^2$ is

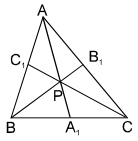
(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

13. If a_1, b_1, c_1 and a_2, b_2, c_2 are the direction ratios of two lines and θ is the angle between the lines then $\tan\theta$ is equal to

(A) $\frac{\sqrt{\Sigma(b_1c_2 - b_2c_1)^2}}{a_1b_1 + a_2b_2 + c_1c_2}$ (B) $\frac{\sqrt{\Sigma(b_1c_2 - b_2c_1)^2}}{a_1a_2 + b_1b_2 + c_1c_2}$ (C) $\frac{\sqrt{\Sigma(b_1c_2 + b_2c_1)^2}}{a_1a_2 + b_1b_2 + c_1c_2}$ (D) none of these

14. A point moves so that the sum of the squares of its distances from the six faces of a cube given by $x = \pm 1$, $y = \pm 1$, $z = \pm 1$ is 10 units. The locus of the point is
(A) $x^2 + y^2 + z^2 = 1$ (B) $x^2 + y^2 + z^2 = 2$ (C) $x + y + z = 1$ (D) $x + y + z = 2$

15. In the adjacent figure 'P' is any arbitrary interior point of the triangle ABC such that the lines AA_1, BB_1 and CC_1 are concurrent at P. Value of $\frac{PA_1}{AA_1} + \frac{PB_1}{BB_1} + \frac{PC_1}{CC_1}$ is always equal to .



(A) 1 (B) 2 (C) 3 (D) None of these

16. The plane $ax + by + cz = d$, meets the coordinate axes at the points A, B and C respectively. Area of triangle ABC is equal to

(A) $\frac{d^2\sqrt{a^2 + b^2 + c^2}}{|abc|}$ (B) $\frac{d^2\sqrt{a^2 + b^2 + c^2}}{2|abc|}$ (C) $\frac{d^2\sqrt{a^2 + b^2 + c^2}}{4|abc|}$ (D) None of these

17. The length of projection, of the line segment joining the points (1, –1, 0) and (–1, 0, 1), to the plane $2x + y + 6z = 1$, is equal to

(A) $\sqrt{\frac{255}{61}}$ (B) $\sqrt{\frac{237}{61}}$ (C) $\sqrt{\frac{137}{61}}$ (D) $\sqrt{\frac{155}{61}}$

18. Two systems of rectangular axes have the same origin. If a plane cuts them at distances a, b, c and a_1, b_1, c_1 from the origin, then

(A) $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2} = \frac{1}{a_1^2}+\frac{1}{b_1^2}+\frac{1}{c_1^2}$ (B) $\frac{1}{a^2}-\frac{1}{b^2}+\frac{1}{c^2} = \frac{1}{a_1^2}-\frac{1}{b_1^2}+\frac{1}{c_1^2}$

(C) $a^2 + b^2 + c^2 = a_1^2 + b_1^2 + c_1^2$ (D) $a^2 - b^2 + c^2 = a_1^2 - b_1^2 + c_1^2$

19. The angle between the plane $2x – y + z = 6$ and a plane perpendicular to the planes $x + y + 2z = 7$ and $x – y = 3$ is :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

20. The non zero value of 'a' for which the lines $2x – y + 3z + 4 = 0 = ax + y – z + 2$ and $x – 3y + z = 0 = x + 2y + z + 1$ are co-planar is :

(A) – 2 (B) 4 (C) 6 (D) 0

21. The equation of the plane through the point (–1, 2 , 0) and parallel to the lines

$\frac{x}{3} = \frac{y+1}{0} = \frac{z-2}{-1}$ and $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$ is -

(A) $x + 2y + 3z - 1 = 0$ (B) $x – 2y + 3z + 5 = 0$
(C) $x + y – 3z + 1 = 0$ (D) $x + y + 3z – 1 = 0$

22. The equation of the plane bisecting the acute angle between the planes $2x + y + 2z = 9$ and $3x – 4y + 12z + 13 = 0$ is :

(A) $11x + 33y – 34z – 172 = 0$ (B) $11x + 33y – 34z – 182 = 0$
(C) $41x – 7y + 86z – 52 = 0$ (D) $41x – 7y + 86z – 62 = 0$

23. The base of the pyramid AOBC is an equilateral triangle OBA with each side equal to $4\sqrt{2}$, 'O' is the origin of reference, AO is perpendicular to the plane of Δ OBC and $|\vec{AO}| = 2$. Then the cosine of the angle between the skew straight lines one passing through A and the mid point of OB and the other passing through O and the mid point of BC is :

(A) $-\frac{1}{\sqrt{2}}$ (B) 0 (C) $\frac{1}{\sqrt{6}}$ (D) $\frac{1}{\sqrt{2}}$

24. The coplanar points A , B , C , D are (2 – x , 2 , 2) , (2 , 2 – y , 2) , (2 , 2 , 2 – z) and (1 , 1 , 1) respectively . Then :

(A) $\frac{1}{x}+\frac{1}{y}+\frac{1}{z} = 1$ (B) $x + y + z = 1$ (C) $\frac{1}{1-x} + \frac{1}{1-y} + \frac{1}{1-z} = 1$ (D) none of these

25. Let the centre of the parallelopiped formed by $\vec{PA} = \hat{i} + 2\hat{j} + 2\hat{k}$; $\vec{PB} = 4\hat{i} - 3\hat{j} + \hat{k}$; $\vec{PC} = 3\hat{i} + 5\hat{j} - \hat{k}$ is given by the position vector (7, 6, 2). Then the position vector of the point P is:

(A) (3, 4, 1) (B) (6, 8, 2) (C) (1, 3, 4) (D) (2, 6, 8)

26. Taken on side $\vec{AC}$ of a triangle ABC, a point M such that $\vec{AM} = \frac{1}{3}\vec{AC}$. A point N is taken on the side $\vec{CB}$ such that $\vec{BN} = \vec{CB}$ then, for the point of intersection X of $\vec{AB}$ & $\vec{MN}$ which of the following holds good?

(A) $\vec{XB} = \frac{1}{3}\vec{AB}$ (B) $\vec{AX} = \frac{1}{3}\vec{AB}$ (C) $\vec{XN} = \frac{3}{4}\vec{MN}$ (D) $\vec{XM} = 3\vec{XN}$

27. If the acute angle that the vector, $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ makes with the plane of the two vectors $2\hat{i} + 3\hat{j} - \hat{k}$ & $\hat{i} - \hat{j} + 2\hat{k}$ is $\cot^{-1}\sqrt{2}$ then:

(A) $\alpha(\beta + \gamma) = \beta\gamma$ (B) $\beta(\gamma + \alpha) = \gamma\alpha$ (C) $\gamma(\alpha + \beta) = \alpha\beta$ (D) $\alpha\beta + \beta\gamma + \gamma\alpha = 0$

28. Locus of the point P for which $\vec{OP}$ represents a vector with direction cosine $\cos\alpha = \frac{1}{2}$ ('O' is the origin) is:

(A) A circle parallel to y z plane with centre on the x – axis
(B) a cone concentric with positive x – axis having vertex at the origin and the slant height equal to the magnitude of the vector
(C) a ray emanating from the origin and making an angle of 60º with x – axis
(D) a disc parallel to y z plane with centre on x – axis & radius equal to $|\vec{OP}|$ sin 60º

29. Equation of the plane passing through $A(x_1, y_1, z_1)$ and containing the line $\frac{x-x_2}{d_1} = \frac{y-y_2}{d_2} = \frac{z-z_2}{d_3}$ is

(A) $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$ (B) $\begin{vmatrix} x-x_2 & y-y_2 & z-z_2 \\ x_1-x_2 & y_1-y_2 & z_1-z_2 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$

(C) $\begin{vmatrix} x-d_1 & y-d_2 & z-d_3 \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} = 0$ (D) $\begin{vmatrix} x & y & z \\ x_1-x_2 & y_1-y_2 & z_1-z_2 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$

30. The equations of the line of shortest distance between the lines $\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z-2}{2}$ are

(A) 3(x – 21) = 3y + 92 = 3z – 32 (B) $\frac{x-(62/3)}{1/3} = \frac{y+31}{1/3} = \frac{z-(31/3)}{1/3}$

(C) $\frac{x-21}{1/3} = \frac{y+(92/3)}{1/3} = \frac{z-(32/3)}{1/3}$ (D) $\frac{x-2}{1/3} = \frac{y+3}{1/3} = \frac{z-1}{1/3}$

31. A line passes through a point A with p.v. $3\hat{i} + \hat{j} - \hat{k}$ & is parallel to the vector $2\hat{i} - \hat{j} + 2\hat{k}$. If P is a point on this line such that AP = 15 units, then the p.v. of the point P is:

(A) $13\hat{i} + 4\hat{j} - 9\hat{k}$ (B) $13\hat{i} - 4\hat{j} + 9\hat{k}$ (C) $7\hat{i} - 6\hat{j} + 11\hat{k}$ (D) $-7\hat{i} + 6\hat{j} - 11\hat{k}$

32. The equations of the planes through the origin which are parallel to the line $\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z+1}{-2}$ and distant $\frac{5}{3}$ from it are

(A) 2x + 2y + z = 0 (B) x + 2y + 2z = 0 (C) 2x – 2y + z = 0 (D) x – 2y + 2z = 0

33. The value(s) of k for which the equation $x^2 + 2y^2 - 5z^2 + 2kyz + 2zx + 4xy = 0$ represents a pair of planes passing through origin is/are

(A) 2 (B) – 2 (C) 6 (D) – 6

34. The equation of lines AB is $\frac{x}{2} = \frac{y}{-3} = \frac{2}{6}$. Through a point P(1, 2, 5), line PN is drawn perpendicular to AB and line PQ is drawn parallel to the plane 3x + 4y + 5z = 0 to meet AB is Q. Then

(A) coordinate of N is $\left(\frac{52}{49}, -\frac{78}{49}, \frac{156}{49}\right)$ (B) the coordinates of Q is $\left(3, -\frac{9}{2}, 9\right)$

(C) the equation of PN is $\frac{x-1}{3} = \frac{y-2}{-176} = \frac{z-5}{-89}$ (D) the equation of PQ is $\frac{x-1}{4} = \frac{y-2}{-13} = \frac{z-5}{8}$

35. Let a perpendicular PQ be drawn from P (5, 7, 3) to the line $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$ when Q is the foot. Then

(A) Q is (9, 13, – 15) (B) PQ = 14

(C) the equation of plane containing PQ and the given line is 9x – 4y – z – 14 = 0 (D) none

EXERCISE–6

1. Find the equation of the plane which contains the origin and the line of intersection of the planes $\vec{r} \cdot \vec{a} = p$ and $\vec{r} \cdot \vec{b} = q$

2. If the lines $\frac{x-a}{a'} = \frac{y-b}{b'} = \frac{z-c}{c'}$ and $\frac{x-a'}{a} = \frac{y-b'}{b} = \frac{z-c'}{c}$ intersect at a point then the coordinate of the point of intersection.

3. The locus of a point which is a equidistant from the two given points with position vectors $\vec{a}$ and $\vec{b}$ is the plane $\left[\vec{r} - \frac{1}{2}(\vec{a}+\vec{b})\right] \cdot (\vec{a} - \vec{b}) = 0$ bisecting the line joining the points normally.

4. The foot of the perpendicular from (a, b, c) on the line x = y = z is the point (r, r, r) where 3r = a + b + c.

5. **Match the following :**

	Column A	**Column B**
(a)	Sum of the square of the direction cosines of line is	(P) 0
(b)	All the points on the z-axis have their x and y coordinate equal to	(Q) 1
(c)	Distance between the points (1, 3, 2) and (2, 3, 1) is	(R) 9
(d)	Shortest distance between the lines $\frac{x-6}{1} = \frac{y-2}{-2} = \frac{z-2}{2}$ and $\frac{x+4}{3} = \frac{y}{-2} = \frac{z+1}{-2}$ is	(S) $\sqrt{2}$

6. Show that the angle between the straight lines whose direction cosines are given by the equations $\ell + m + n = 0$ and $amn + bn\ell + c\ell m = 0$ is $\frac{\pi}{3}$ if $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$.

7. Prove that the two lines whose direction cosines are given by the relations $p\ell + qm + rn = 0$ & $a\ell^2 + bm^2 + cn^2 = 0$ are perpendicular if, $p^2(b + c) + q^2(c + a) + r^2(a + b) = 0$ and parallel if $\frac{p^2}{a}+\frac{q^2}{b}+\frac{r^2}{c}=0$.

8. Find the plane π passing through the points of intersection of the planes $2x + 3y - z + 1 = 0$ and $x + y - 2z + 3 = 0$ and is perpendicular to the plane $3x - y - 2z = 4$. Find the image of point (1, 1, 1) in plane π.

9. Given parallel planes $\vec{r}\cdot(2\hat{i} - \lambda\hat{j} + \hat{k}) = 3$ and $\vec{r}\cdot(4\hat{i} + \hat{j} - \mu\hat{k}) = 5$ for what values of α, planes $\vec{r}\cdot(\mu\hat{i} - \alpha\hat{j} + 3\hat{k}) = 0$ & $\vec{r}\cdot(\alpha\hat{i} - 3\hat{j} + 2\lambda\hat{k}) = 0$ would be perpendicular.

10. The edges of a rectangular parallelepiped are a, b, c; show that the angles between the four diagonals are given by $\cos^{-1}\frac{a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2}$.

11. Prove that the line of intersection of the planes $\vec{r}\cdot(\hat{i} + 2\hat{j} + 3\hat{k}) = 0$ and $\vec{r}\cdot(3\hat{i} + 2\hat{j} + \hat{k}) = 0$ is $\vec{r} = t(\hat{i} - 2\hat{j} + \hat{k})$. Show that the line is equally inclined to $\hat{i}$ and $\hat{k}$ and makes an angle $(1/2)\sec^{-1} 3$ with. $\hat{j}$.

12. Find the shortest distance between the lines $\frac{x-1}{2} = \frac{y+1}{3} = z$ & $\frac{x+1}{3} = (y - 2); z = 2$

13. Show that the line L whose equation is, $\vec{r} = (2\hat{i} - 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + 4\hat{k})$ is parallel to the plane π whose vector $\vec{r}\cdot(\hat{i} + 5\hat{j} + \hat{k}) = 5$. Find the distance between them.

14. A sphere has an equation $|\vec{r} - \vec{a}|^2 + |\vec{r} - \vec{b}|^2 = 72$ where $\vec{a} = \hat{i} + 3\hat{j} - 6\hat{k}$ and $\vec{b} = 2\hat{i} + 4\hat{j} + 2\hat{k}$. Find:

(i) the centre of the sphere **(ii)** the radius of the sphere

(iii) perpendicular distance from the centre of the sphere to the plane $\vec{r}\cdot(2\hat{i} + 2\hat{j} - \hat{k}) = -3$.

15. Find the equation of the sphere which is tangential to the plane $x - 2y - 2z = 7$ at $(3, -1, -1)$ and passes through the point $(1, 1, -3)$.

16. P_1 and P_2 are planes passing through origin. L_1 and L_2 also passes through origin. L_1 lies on P_1 not on P_2 and L_2 lies on P_2 but not on P_1. Show that there exists points A, B, C and whose permutation A′.B′.C′ can be chosen such that **[IIT - 2004]**

(i)A is on L_1, B on P_1 but not on L_1 and C not on P_1.(ii)A′ in on L_2, B′ on P_2 but not on L_2 and C′ not on P_2.

17. A parallelopiped 'S' has base points A, B, C and D and upper face points A′, B′, C′ and D′. This parallelopiped is compressed by upper face A′B′C′D′ to form a new parallelopiped 'T' having upper face points A″, B″, C″ and D″. Volume of parallelpiped T is 90 percent of the volume of parallelopiped S. Prove that the locus of 'A″' is a plane. **[IIT - 2004]**

EXERCISE–5

1.	C	**2.**	A	**3.**	A
4.	A	**5.**	D	**6.**	B
7.	B	**8.**	B	**9.**	A
10.	D	**11.**	A	**12.**	C
13.	B	**14.**	B	**15.**	A
16.	B	**17.**	B	**18.**	A
19.	D	**20.**	A	**21.**	A
22.	C	**23.**	D	**24.**	A
25.	A	**26.**	C	**27.**	A
28.	B	**29.**	AB	**30.**	ABC
31.	AB	**32.**	AD	**33.**	BC
34.	ABCD	**35.**	BC		

EXERCISE–6

1. $\vec{r}\cdot(\vec{a}q - p\vec{b}) = 0$

2. $(a + a', b + b', c + c')$

3. True **4.** True

5. (a) → (Q), (b) → (P), (c) → (S), (d) → (R)

8. $7x + 13y + 4z - 9 = 0$; $\left(\frac{12}{117}, \frac{-78}{117}, \frac{57}{117}\right)$

9. $\alpha = +3$ **12.** $\frac{3}{\sqrt{59}}$ **13.** $\frac{10}{3\sqrt{3}}$

14. (i) (0, 5, 5) (ii) 9 (iii) $\frac{8}{3}$

15. $(x - 2)^2 + (y - 1)^2 + (z - 1)^2 = 5$

EXERCISE–7

Part : (A) Only one correct option

1. The lengths of the diagonals of a parallelogram constructed on the vectors $\vec{p} = 2\vec{a} + \vec{b}$ & $\vec{q} = \vec{a} - 2\vec{b}$, where $\vec{a}$ & $\vec{b}$ are unit vectors forming an angle of 60º are:

(A) 3 & 4 (B) $\sqrt{7}$ & $\sqrt{13}$ (C) $\sqrt{5}$ & $\sqrt{11}$ (D) none

2. $\left[\frac{\vec{a}}{|\vec{a}|^2} - \frac{\vec{b}}{|\vec{b}|^2}\right]^2 =$

(A) $|\vec{a}|^2 - |\vec{b}|^2$ (B) $\left[\frac{\vec{a} - \vec{b}}{|\vec{a}|\,|\vec{b}|}\right]^2$ (C) $\left[\frac{\vec{a}\,|\vec{a}| - \vec{b}\,|\vec{b}|}{|\vec{a}|\,|\vec{b}|}\right]^2$ (D) none

3. A, B, C & D are four points in a plane with pv's $\vec{a}$, $\vec{b}$, $\vec{c}$ & $\vec{d}$ respectively such that $(\vec{a} - \vec{d}).(\vec{b} - \vec{c}) = (\vec{b} - \vec{d}).(\vec{c} - \vec{a}) = 0$. Then for the triangle ABC, D is its:

(A) incentre (B) circumcentre (C) orthocentre (D) centroid

4. Vectors $\vec{a}$ & $\vec{b}$ make an angle $\theta = \frac{2\pi}{3}$. If $|\vec{a}| = 1, |\vec{b}| = 2$ then $\left\{(\vec{a} + 3\vec{b}) \times (3\vec{a} - \vec{b})\right\}^2 =$

(A) 225 (B) 250 (C) 275 (D) 300

5. Consider a tetrahedron with faces f_1, f_2, f_3, f_4. Let $\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4$ be the vectors whose magnitudes are respectively equal to the areas of f_1, f_2, f_3, f_4 & whose directions are perpendicular to these faces in the outward direction. Then,

(A) $\vec{a}_1 + \vec{a}_2 + \vec{a}_3 + \vec{a}_4 = 0$ (B) $\vec{a}_1 + \vec{a}_3 = \vec{a}_2 + \vec{a}_4$ (C) $\vec{a}_1 + \vec{a}_2 = \vec{a}_3 + \vec{a}_4$ (D) none

6. For non–zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, $|\vec{a} \times \vec{b}.\vec{c}| = |\vec{a}|\,|\vec{b}|\,|\vec{c}|$ holds if and only if;

(A) $\vec{a}.\vec{b} = 0, \vec{b}.\vec{c} = 0$ (B) $\vec{c}.\vec{a} = 0, \vec{a}.\vec{b} = 0$ (C) $\vec{a}.\vec{c} = 0, \vec{b}.\vec{c} = 0$ (D) $\vec{a}.\vec{b} = \vec{b}.\vec{c} = \vec{c}.\vec{a} = 0$

7. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} - \hat{j} + \hat{k}, \vec{c} = \hat{i} + 2\hat{j} - \hat{k}$, then the value of $\begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix} =$

(A) 2 (B) 4 (C) 16 (D) 64

8. If $\vec{a}$, $\vec{b}$ & $\vec{c}$ are any three vectors, then $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$ is true if:

(A) $\vec{b}$ & $\vec{c}$ are collinear (B) $\vec{a}$ & $\vec{c}$ are collinear (C) $\vec{a}$ & $\vec{b}$ are collinear (D) none

9. $(\vec{r}.i)(i \times r) + (\vec{r}.j)(j \times r) + (\vec{r}.k)(k \times r) =$

(A) 0 (B) $\vec{r}$ (C) $2\vec{r}$ (D) $3\vec{r}$

10. A point taken on each median of a triangle divides the median in the ratio 1 : 3, reckoning from the vertex. Then the ratio of the area of the triangle with vertices at these points to that of the original triangle is:

(A) 5 : 13 (B) 25 : 64 (C) 13 : 32 (D) none

11. Given a parallelogram ABCD. If $|\vec{AB}| = a, |\vec{AD}| = b$ & $|\vec{AC}| = c$, then $\vec{DB}.\vec{AB}$ has the value:

(A) $\frac{3a^2 + b^2 - c^2}{2}$ (B) $\frac{a^2 + 3b^2 - c^2}{2}$ (C) $\frac{a^2 - b^2 + 3c^2}{2}$ (D) none

12. The points whose position vectors are $p\hat{i} + q\hat{j} + r\hat{k}$; $q\hat{i} + r\hat{j} + p\hat{k}$ & $r\hat{i} + p\hat{j} + q\hat{k}$ are collinear if:

(A) $p + q + r = 0$ (B) $p^2 + q^2 + r^2 - pq - qr - rp = 0$
(C) $p^3 + q^3 + r^3 - 3pqr = 0$ (D) none of these

13. If $\vec{p}$ & $\vec{s}$ are not perpendicular to each other and $\vec{r} \times \vec{p} = \vec{q} \times \vec{p}$ & $\vec{r}.\vec{s} = 0$, then $\vec{r} =$

(A) $\vec{p}.\vec{s}$ (B) $\vec{q} - \left(\frac{\vec{q}.\vec{s}}{\vec{p}.\vec{s}}\right)\vec{p}$ (C) $\vec{q} + \left(\frac{\vec{q}.\vec{p}}{\vec{p}.\vec{s}}\right)\vec{p}$ (D) $\vec{q} + \mu\vec{p}$ for all scalars μ

14. If a, b, c are pth, qth, rth terms of an H.P. and

$\vec{u} = (q - r)\vec{i} + (r - p)\vec{j} + (p - q)\vec{k}$, $\vec{v} = \frac{\vec{i}}{a} + \frac{\vec{j}}{b} + \frac{\vec{k}}{c}$, then:

(A) $\vec{u}, \vec{v}$ are parallel vectors (B) $\vec{u}, \vec{v}$ are orthogonal vectors
(C) $\vec{u}.\vec{v} = 1$ (D) $\vec{u} \times \vec{v} = \vec{i} + \vec{j} + \vec{k}$

15. If $\vec{p}, \vec{q}$ are two noncollinear and nonzero vectors such that $(b-c)\vec{p}\times\vec{q}+(c-a)\vec{p}+(a-b)\vec{q}=0$, where a, b, c are the length of the sides of a triangle, then the triangle is
(A) right angled (B) obtuse angled (C) equilateral (D) isoceles

16. If $\cos\alpha\,\hat{i}+\hat{j}+\hat{k}, \hat{i}+\cos\beta\,\hat{j}+\hat{k}$ & $\hat{i}+\hat{j}+\cos\gamma\,\hat{k}$ $(\alpha\neq\beta\neq\gamma\neq 2n\pi)$ are coplanar then the value of $\left[\operatorname{cosec}^2\frac{\alpha}{2}+\operatorname{cosec}^2\frac{\beta}{2}+\operatorname{cosec}^2\frac{\gamma}{2}\right]=$
(A) 1 (B) 2 (C) 3 (D) none of these

17. If $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ & $\vec{r}\cdot\vec{a}=0$ where $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}-\hat{j}+\hat{k}$ & $\vec{c}=\hat{i}+\hat{j}+3\hat{k}$, then $\vec{r}$ is equal to:
(A) $2(\hat{i}-\hat{j}+\hat{k})$ (B) $2(\hat{i}+\hat{j}-\hat{k})$ (C) $2(-\hat{i}+\hat{j}+\hat{k})$ (D) $2(\hat{i}+\hat{j}+\hat{k})$

18. The value of $[\vec{d}\ \vec{b}\ \vec{c}]\,\vec{a}+[\vec{d}\ \vec{c}\ \vec{a}]\,\vec{b}+[\vec{d}\ \vec{a}\ \vec{b}]\,\vec{c}-\vec{d}\,[\vec{a}\ \vec{b}\ \vec{c}]$ is equal to:
(A) 0 (B) $2[\vec{a}\ \vec{b}\ \vec{c}]\,\vec{d}$ (C) $-2[\vec{a}\ \vec{b}\ \vec{c}]\,\vec{d}$ (D) none of these

19. The three vectors $\hat{i}+\hat{j}, \hat{j}+\hat{k}, \hat{k}+\hat{i}$ taken two at a time form three planes. The three unit vectors drawn perpendicular to these three planes form a parallelopiped of volume:
(A) $\frac{1}{3}$ (B) 4 (C) $\frac{3\sqrt{3}}{4}$ (D) $\frac{4}{3\sqrt{3}}$

20. For any four points P, Q, R, S, $\left|\vec{PQ}\times\vec{RS}-\vec{QR}\times\vec{PS}+\vec{RP}\times\vec{QS}\right|$ is equal to 4 times the area of the triangle:
(A) PQR (B) QRS (C) PRS (D) PQS

21. If $\vec{a}, \vec{b}, \vec{c}$ are three non–coplanar & $\vec{p}, \vec{q}, \vec{r}$ are reciprocal vectors, then:
$\left(\ell\vec{a}+m\vec{b}+n\vec{c}\right).\left(\ell\vec{p}+m\vec{q}+n\vec{r}\right)$ is equal to:
(A) $\ell^2+m^2+n^2$ (B) $\ell m+m n+n\ell$ (C) 0 (D) none of these

22. In a quadrilateral ABCD, $\vec{AC}$ is the bisector of the $\left(\vec{AB}\wedge\vec{AD}\right)$ which is $\frac{2\pi}{3}$, $15|\vec{AC}| = 3|\vec{AB}| = 5|\vec{AD}|$ then $\cos\left(\vec{BA}\wedge\vec{CD}\right)$ is:
(A) $-\frac{\sqrt{14}}{7\sqrt{2}}$ (B) $-\frac{\sqrt{21}}{7\sqrt{3}}$ (C) $\frac{2}{\sqrt{7}}$ (D) $\frac{2\sqrt{7}}{14}$

23. In the isosceles triangle ABC $|\vec{AB}| = |\vec{BC}| = 8$, a point E divides AB internally in the ratio 1: 3, then the cosine of the angle between $\vec{CE}$ & $\vec{CA}$ is (where $|\vec{CA}| = 12$)
(A) $-\frac{3\sqrt{7}}{8}$ (B) $\frac{3\sqrt{8}}{17}$ (C) $\frac{3\sqrt{7}}{8}$ (D) $\frac{-3\sqrt{8}}{17}$

24. If $\vec{p}=3\vec{a}-5\vec{b}$; $\vec{q}=3\vec{a}+\vec{b}$; $\vec{r}=\vec{a}+4\vec{b}$; $\vec{s}=-\vec{a}+\vec{b}$ are four vectors such that $\sin\left(\vec{p}\wedge\vec{q}\right)=1$ and $\left(\vec{r}\wedge\vec{s}\right)=1$ then $\cos\left(\vec{a}\wedge\vec{b}\right)$ is:
(A) $-\frac{19}{5\sqrt{43}}$ (B) 0 (C) $\frac{19}{5\sqrt{43}}$ (D) 1

25. If $\vec{p},\vec{q},\vec{r}$ be three mutually perpendicular vectors of the same magnitude. If a vector $\vec{x}$ satisfies the equation $\vec{p}\times((\vec{x}-\vec{q})\times\vec{p})+\vec{q}\times((\vec{x}-\vec{r})\times\vec{q})+\vec{r}\times((\vec{x}-\vec{p})\times\vec{r})=\vec{0}$, then $\vec{x}$ is given by **[IIT - 1997]**
(A) $\frac{1}{2}(\vec{p}+\vec{q}-2\vec{r})$ (B) $\frac{1}{2}(\vec{p}+\vec{q}+\vec{r})$ (C) $\frac{1}{3}(\vec{p}+\vec{q}+\vec{r})$ (D) $\frac{1}{3}(2\vec{p}+\vec{q}-\vec{r})$

26. Let $\vec{a}$ & $\vec{b}$ be two non–collinear unit vectors. If $\vec{u}=\vec{a}-(\vec{a}\cdot\vec{b})\,\vec{b}$ & $\vec{v}=\vec{a}\times\vec{b}$, then $|\vec{v}|$ is **[IIT - 1999]**
(A) $|\vec{u}|$ (B) $|\vec{u}|+|\vec{u}\cdot\vec{a}|$ (C) $|\vec{u}|+|\vec{u}\cdot\vec{b}|$ (D) $\vec{u}+\vec{u}\cdot(\vec{a}+\vec{b})$

27. If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero, non coplanar vectors and $\vec{b}_1=\vec{b}-\frac{\vec{b}\cdot\vec{a}}{|\vec{a}|^2}\vec{a}$, $\vec{b}_2=\vec{b}+\frac{\vec{b}\cdot\vec{a}}{|\vec{a}|^2}\vec{a}$,

$\vec{c}_1 = \vec{c} - \frac{\vec{c}\cdot\vec{a}}{|\vec{a}|^2}\vec{a} + \frac{\vec{b}\cdot\vec{c}}{|\vec{c}|^2}\vec{b}_1$, $\vec{c}_2 = \vec{c} - \frac{\vec{c}\cdot\vec{a}}{|\vec{a}|^2}\vec{a} - \frac{\vec{b}_1\cdot\vec{c}}{|\vec{b}_1|^2}\vec{b}_1$, $\vec{c}_3 = \vec{c} - \frac{\vec{c}\cdot\vec{a}}{|\vec{c}|^2}\vec{a} + \frac{\vec{b}\cdot\vec{c}}{|\vec{c}|^2}\vec{b}_1$,

$\vec{c}_4 = \vec{c} - \frac{\vec{c}\cdot\vec{a}}{|\vec{c}|^2}\vec{a} - \frac{\vec{b}\cdot\vec{c}}{|\vec{b}|^2}\vec{b}_1$, then the set of orthogonal vectors is **[IIT - 2005]**

(A) $(\vec{a}, \vec{b}_1, \vec{c}_3)$ (B) $(\vec{a}, \vec{b}_1, \vec{c}_2)$ (C) $(\vec{a}, \vec{b}_1, \vec{c}_1)$ (D) $(\vec{a}, \vec{b}_2, \vec{c}_2)$

28. Let $\vec{A}$ be vector parallel to line of intersection of planes P_1 and P_2 through origin, P_1 is parallel to the vectors $2\hat{i} + 3\hat{k}$ and $4\hat{j} - 3\hat{k}$ and P_2 is parallel to $\hat{j} - \hat{k}$ and $3\hat{i} + 3\hat{j}$, then the angle between vector $\vec{A}$ and $2\hat{i} + \hat{j} - 2\hat{k}$ is **[IIT - 2006]**

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{3\pi}{4}$

Part : (B) May have more than one options correct

29. If $\vec{a}, \vec{b}, \vec{c}$ & $\vec{d}$ are linearly independent set of vectors & $K_1\vec{a} + K_2\vec{b} + K_3\vec{c} + K_4\vec{d} = 0$ then:
(A) $K_1 + K_2 + K_3 + K_4 = 0$ (B) $K_1 + K_3 = K_2 + K_4 = 0$ (C) $K_1 + K_4 = K_2 + K_3 = 0$ (D) none of these

30. Given three vectors $\vec{a}, \vec{b}, \vec{c}$ such that they are non – zero, non – coplanar vectors, then which of the following are coplanar.

(A) $\vec{a}+\vec{b}, \vec{b}+\vec{c}, \vec{c}+\vec{a}$ (B) $\vec{a}-\vec{b}, \vec{b}+\vec{c}, \vec{c}+\vec{a}$ (C) $\vec{a}+\vec{b}, \vec{b}-\vec{c}, \vec{c}+\vec{a}$ (D) $\vec{a}+\vec{b}, \vec{b}-\vec{c}, \vec{c}-\vec{a}$

31. Let $\vec{p} = 2\hat{i} + 3\hat{j} + a\hat{k}, \vec{q} = b\hat{i} + 5\hat{j} - \hat{k}$ & $\vec{r} = \hat{i} + \hat{j} + 3\hat{k}$. If $\vec{p}, \vec{q}, \vec{r}$ are coplanar and $\vec{p}\cdot\vec{q} = 20$, a & b have the values:
(A) 1, 3 (B) 9, 7 (C) 5, 5 (D) 13, 9

32. If $\vec{z}_1 = a\hat{i} + b\hat{j}$ & $\vec{z}_2 = c\hat{i} + d\hat{j}$ are two vectors in $\hat{i}$ & $\hat{j}$ system where $|\vec{z}_1| = |\vec{z}_2| = r$ & $\vec{z}_1\cdot\vec{z}_2 = 0$ then $\vec{w}_1 = a\hat{i} + c\hat{j}$ & $\vec{w}_2 = b\hat{i} + d\hat{j}$ satisfy:
(A) $|\vec{w}_1| = r$ (B) $|\vec{w}_2| = r$ (C) $\vec{w}_1\cdot\vec{w}_2 = 0$ (D) none of these

33. If $\vec{a}$ & $\vec{b}$ are two non colinear unit vectors & $\vec{a}, \vec{b}, x\vec{a} - y\vec{b}$ form a triangle, then:

(A) $x = -1; y = 1$ & $|\vec{a} + \vec{b}| = 2\cos\left(\frac{\vec{a}\,\hat{}\,\vec{b}}{2}\right)$

(B) $x = -1; y = 1$ & $\cos\left(\vec{a}\,\hat{}\,\vec{b}\right) + |\vec{a} + \vec{b}|\cos\left[\vec{a}, \hat{-}\left(\vec{a}+\vec{b}\right)\right] = -1$

(C) $|\vec{a} + \vec{b}| = -2\cot\left(\frac{\vec{a}\,\hat{}\,\vec{b}}{2}\right)\cos\left(\frac{\vec{a}\,\hat{}\,\vec{b}}{2}\right)$ & $x = -1, y = 1$ (D) none

34. The value(s) of $\alpha \in [0, 2\pi]$ for which vector $\vec{a} = \hat{i} + 3\hat{j} + (\sin 2\alpha)\hat{k}$ makes an obtuse angle with the Z-axis and the vectors $\vec{b} = (\tan\alpha)\hat{i} - \hat{j} + 2\sqrt{\sin\frac{\alpha}{2}}\,\hat{k}$ and $\vec{c} = (\tan\alpha)\hat{i} + (\tan\alpha)\hat{j} - 3\sqrt{\operatorname{cosec}\frac{\alpha}{2}}\,\hat{k}$ are orthogonal, is/are:
(A) $\tan^{-1}3$ (B) $\pi - \tan^{-1}2$ (C) $\pi + \tan^{-1}3$ (D) $2\pi - \tan^{-1}2$

35. A parallelogram is constructed on the vectors $\vec{p}$ & $\vec{q}$. A vector which coincides with the altitude of the parallelogram & perpendicular to the side $\vec{p}$ expressed in terms of the vectors $\vec{p}$ & $\vec{q}$ is:

(A) $\vec{q} - \frac{\vec{q}\cdot\vec{p}}{(\vec{p})^2}\vec{p}$ (B) $\frac{(\vec{p}\times\vec{q})\times\vec{p}}{\vec{p}^2}$ (C) $\frac{\vec{q}\cdot\vec{p}}{\vec{p}^2}\vec{p} - \vec{q}$ (D) $\frac{\vec{p}\times(\vec{p}\times\vec{q})}{\vec{p}^2}$

36. Identify the statement(s) which is/are incorrect?

(A) $\vec{a}\times\left[\vec{a}\times(\vec{a}\times\vec{b})\right] = (\vec{a}\times\vec{b})(\vec{a}^2)$

(B) If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors and $\vec{v}\cdot\vec{a} = \vec{v}\cdot\vec{b} = \vec{v}\cdot\vec{c} = 0$ then $\vec{v}$ must be a null vector

(C) If $\vec{a}$ and $\vec{b}$ lie in a plane normal to the plane containing the vectors $\vec{c}$ and $\vec{d}$ then $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = 0$

(D) If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{a}', \vec{b}', \vec{c}'$ are reciprocal system of vectors then $\vec{a}\cdot\vec{b}' + \vec{b}\cdot\vec{c}' + \vec{c}\cdot\vec{a}' = 3$

37. If $a = \hat{i} + 2\hat{j} + 4\hat{k}, \vec{b} = 2\hat{i} - 3\hat{j} + \hat{k}, \vec{c} = \hat{i} + 4\hat{j} - 4\hat{k}$, then the vector $\vec{a}\times(\vec{b}\times\vec{c})$ is orthogonal to:

(A) $\vec{a}$ (B) $\vec{b}$ (C) $\vec{c}$ (D) $\vec{a}+\vec{b}+\vec{c}$

38. If $\vec{a}, \vec{b}, \vec{c}$ are non-zero, non-collinear vectors such that a vector $\vec{p}$ = a b $\cos\left(2\pi-\left(\vec{a}\wedge\vec{b}\right)\right)\vec{c}$ and a vector $\vec{q}$ = a c $\cos\left(\pi-\left(\vec{a}\wedge\vec{c}\right)\right)\vec{b}$ then p + $\vec{q}$ is

(A) parallel to $\vec{a}$ (B) perpendicular to $\vec{a}$ (C) coplanar with $\vec{b}$ & $\vec{c}$ (D) none of these

39. Which of the following statement(s) is/are true?

(A) If $\vec{n}\cdot\vec{a}=0, \vec{n}\cdot\vec{b}=0$ & $\vec{n}\cdot\vec{c}=0$ for some non zero vector $\vec{n}$, then $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=0$

(B) there exist a vector having direction angles $\alpha = 30^\circ$ & $\beta = 45^\circ$

(C) locus of point for which x = 3 & y = 4 is a line parallel to the z - axis whose distance from the z axis is 5

(D) the vertices of a regular tetrahedron are OABC where 'O' is the origin. The vector $\vec{OA}+\vec{OB}+\vec{OC}$ is perpendicular to the plane ABC.

40. In a Δ ABC, let M be the mid point of segment AB and let D be the foot of the bisector of $\angle$ C. Then the ratio $\frac{\text{Area }\Delta\text{ CDM}}{\text{Area }\Delta\text{ ABC}}$ is:

(A) $\frac{1}{4}\ \frac{a-b}{a+b}$ (B) $\frac{1}{2}\ \frac{a-b}{a+b}$ (C) $\frac{1}{2}\tan\frac{A-B}{2}\cot\frac{A+B}{2}$ (D) $\frac{1}{4}\cot\frac{A-B}{2}\tan\frac{A+B}{2}$

41. The vectors $\vec{a}, \vec{b}, \vec{c}$ are of the same length & pairwise form equal angles. If $\vec{a}=\hat{i}+\hat{j}$ & $\vec{b}=\hat{j}+\hat{k}$, the pv's of $\vec{c}$ can be:

(A) (1, 0, 1) (B) $\left(-\frac{4}{3},\frac{1}{3},-\frac{4}{3}\right)$ (C) $\left(\frac{1}{3},-\frac{4}{3},\frac{1}{3}\right)$ (D) $\left(-\frac{1}{3},\frac{4}{3},-\frac{1}{3}\right)$

EXERCISE–8

1. Through the middle point M of the side AD of a parallelogram ABCD the straight line BM is drawn cutting AC at R and CD produced at Q prove that QR = 2RB

2. Show that the perpendicular distance of the point $\vec{c}$ from the line joining $\vec{a}$ & $\vec{b}$ is,

$$\frac{\left|\vec{b}\times\vec{c}+\vec{c}\times\vec{a}+\vec{a}\times\vec{b}\right|}{\left|\vec{b}-\vec{a}\right|}.$$

3. If $\vec{\alpha}=\hat{i}+2\hat{j}+3\hat{k}; \vec{\beta}=2\hat{i}-\hat{j}+\hat{k}; \vec{\gamma}=3\hat{i}+2\hat{j}+\hat{k}$ and $\vec{\alpha}\times\left(\vec{\beta}\times\vec{\gamma}\right)=p\vec{\alpha}+q\vec{\beta}+r\vec{\gamma}$ then find the values of p, q, r

4. If $\vec{a}=2\hat{i}-3\hat{j}+\hat{k}, \vec{b}=\hat{i}-\hat{j}+2\hat{k}, \vec{c}=2\hat{i}+\hat{j}-\hat{k}$ & $\vec{d}=3\hat{i}-\hat{j}-2\hat{k}$, then find the value of $(\vec{a}\times\vec{b})\times(\vec{a}\times\vec{c}).\vec{d}$

5. Show that $\vec{a}\times\left((\vec{q}\times\vec{c})\times(\vec{p}\times\vec{b})\right)=\vec{b}\times\left((\vec{p}\times\vec{c})\times(\vec{q}\times\vec{a})\right)+\vec{c}\times\left((\vec{p}\times\vec{a})\times(\vec{q}\times\vec{b})\right)$

6. It is given that $\vec{x}=\frac{\vec{b}\times\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}$; $\vec{y}=\frac{\vec{c}\times\vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}$; $\vec{z}=\frac{\vec{a}\times\vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$ where $\vec{a},\vec{b},\vec{c}$ are non – coplanar vectors. Show that $\vec{x},\vec{y},\vec{z}$ also forms a non – coplanar system. Find the value of

$\vec{x}.(\vec{a}+\vec{b})+\vec{y}.(\vec{b}+\vec{c})+\vec{z}.(\vec{c}+\vec{a})$.

7. The median AD of a triangle ABC is bisected at E and BE is produced to meet the side AC in F. Prove that AF = (1/3) AC and EF = (1/4) BF.

8. Points X and Y are taken on the sides QR and RS, respectively of a parallelogram PQRS, so that QX = 4XR and RY = 4YS. The line XY cuts the line PR at Z. Find the ratio PZ: ZR.

9. Forces $\vec{P},\vec{Q}$ act at O & have a resultant $\vec{R}$. If any transversal cuts their line of action at A,B,C respectively, then show that $\frac{P}{OA}+\frac{Q}{OB}=\frac{R}{OC}$.

10. In a tetrahedron, if two pairs of opposite edges are perpendicular, then show that the third pair of opposite edges is also perpendicular & in this case the sum of the squares of two opposite edges is the same for each pair. Also show that the segment joining the mid points of opposite edges bisect one another.

11. Use vectors to prove that the diagonals of a trapezium having equal non parallel sides are equal & conversely.

12. Given four non zero vectors $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$. The vectors $\vec{a},\vec{b}$ & $\vec{c}$ are coplanar but not collinear pair by

pair and vector $\vec{d}$ is not coplanar with vectors $\vec{a}, \vec{b}$ & $\vec{c}$ and $(\vec{a}\,\hat{}\,\vec{b}) = (\vec{b}\,\hat{}\,\vec{c}) = \frac{\pi}{3}$, $(\vec{d}\,\hat{}\,\vec{a}) = \alpha$, $(\vec{d}\,\hat{}\,\vec{b}) = \beta$, prove that $(\vec{d}\,\hat{}\,\vec{c}) = \cos^{-1}(\cos\beta - \cos\alpha)$.

13. If $\vec{p}, \vec{q}$ & $\vec{r}$ are three non-coplanar vectors, prove that,

$$\vec{a} \times \vec{b} = \frac{1}{\sqrt{[\vec{q}\times\vec{r},\ \vec{r}\times\vec{p},\ \vec{p}\times\vec{q}]}} \begin{vmatrix} \vec{p} & \vec{q} & \vec{r} \\ \vec{p}\,.\,\vec{a} & \vec{q}\,.\,\vec{a} & \vec{r}\,.\,\vec{a} \\ \vec{p}\,.\,\vec{b} & \vec{q}\,.\,\vec{b} & \vec{r}\,.\,\vec{b} \end{vmatrix}$$

14. Consider the non zero vectors $\vec{a}, \vec{b}, \vec{c}$ & $\vec{d}$ such that no three of which are coplanar then prove that $\vec{a}[\vec{b}\,\vec{c}\,\vec{d}] + \vec{c}[\vec{a}\,\vec{b}\,\vec{d}] = \vec{b}[\vec{a}\,\vec{c}\,\vec{d}] + \vec{d}[\vec{a}\,\vec{b}\,\vec{c}]$. Hence prove that $\vec{a}, \vec{b}, \vec{c}$ & $\vec{d}$ represent the position vectors of the vertices of a plane quadrilateral if and only if $\frac{[\vec{b}\,\vec{c}\,\vec{d}] + [\vec{a}\,\vec{b}\,\vec{d}]}{[\vec{a}\,\vec{c}\,\vec{d}] + [\vec{a}\,\vec{b}\,\vec{c}]} = 1$.

15. Solve the following equation for the vector $\vec{p}$; $\vec{p} \times \vec{a} + (\vec{p}\,.\,\vec{b})\vec{c} = \vec{b} \times \vec{c}$ where $\vec{a}, \vec{b}, \vec{c}$ are non zero non coplanar vectors and $\vec{a}$ is neither perpendicular to $\vec{b}$ nor to $\vec{c}$, hence show that $\left[\vec{p} \times \vec{a} + \frac{[\vec{a}\,\vec{b}\,\vec{c}]}{\vec{a}\,.\,\vec{c}}\,\vec{c}\right]$ is perpendicular to $\vec{b} - \vec{c}$.

16. If $\vec{a}, \vec{b}, \vec{c}$ & $\vec{a}', \vec{b}', \vec{c}'$ are reciprocal system of vectors then prove that:

(i) $[\vec{a}\ \vec{b}\ \vec{c}]\,[\vec{a}'\ \vec{b}'\ \vec{c}'] = 1$ **(ii)** $(\vec{a}' \times \vec{b}') + (\vec{b}' \times \vec{c}') + (\vec{c}' \times \vec{a}') = \frac{\vec{a} + \vec{b} + \vec{c}}{[\vec{a}\vec{b}\vec{c}]}$.

17. Let $\vec{A} = 2i + k$; $\vec{B} = i + j + k$ & $\vec{C} = 4i - 3j + 7k$. Determine a vector $\vec{R}$ satisfying $\vec{R} \times \vec{B} = \vec{C} \times \vec{B}$ & $\vec{R}\,.\,\vec{A} = 0$

18. For any two vectors $\vec{u}$ & $\vec{v}$, prove that **[IIT - 1998]**

(a) $(\vec{u}.\vec{v})^2 + |\vec{u} \times \vec{v}|^2 = |\vec{u}|^2 |\vec{v}|^2$ & **(b)** $(1+|\vec{u}|^2)(1+|\vec{v}|^2) = (1-\vec{u}.\vec{v})^2 + |\vec{u}+\vec{v}+(\vec{u}\times\vec{v})|^2$

19. Let ABC and PQR be any two triangles in the same plane. Assume that the perpendiculars from the points A, B, C to the sides QR, RP, PQ respectively are concurrent. Using vector methods or otherwise, prove that the perpendiculars from P, Q, R to BC, CA, AB respectively are also concurrent. **[IIT - 2000]**

20. Find 3 – dimensional vectors $\vec{v}_1, \vec{v}_2, v_3$ satisfying

$\vec{v}_1\,.\,\vec{v}_1 = 4$, $\vec{v}_1\,.\,\vec{v}_2 = -2$, $\vec{v}_1\,.\,\vec{v}_3 = 6$, $\vec{v}_2\,.\,\vec{v}_2 = 2$, $\vec{v}_2\,.\,\vec{v}_3 = -5$, $\vec{v}_3\,.\,\vec{v}_3 = 29$. **[IIT - 2001]**

21. If $\hat{u}, \hat{v}, \hat{w}$ be three non-coplanar unit vectors with angles between $\hat{u}$ & $\hat{v}$ is α, between $\hat{v}$ & $\hat{w}$ is β and between $\hat{w}$ & $\hat{u}$ is γ. If $\vec{a}, \vec{b}, \vec{c}$ are the unit vectors along angle bisectors of α, β, γ respectively, then prove that, $[\vec{a} \times \vec{b},\ \vec{b} \times \vec{c},\ \vec{c} \times \vec{a}] = \frac{1}{16}[\hat{u}\ \hat{v}\ \hat{w}]^2 \sec^2\left(\frac{\alpha}{2}\right) \sec^2\left(\frac{\beta}{2}\right) \sec^2\left(\frac{\gamma}{2}\right)$. **[IIT - 2003]**

EXERCISE–7

1. B **2.** B **3.** C **4.** D **5.** A **6.** D
7. C **8.** B **9.** A **10.** B **11.** A **12.** B
13. B **14.** B **15.** C **16.** B **17.** C **18.** A
19. D **20.** B **21.** A **22.** C **23.** C **24.** C
25. B **26.** A **27.** B **28.** D **29.** ABC
30. BCD **31.** AD **32.** ABC **33.** AB **34.** BD
35. BD **36.** ACD **37.** AD **38.** BC **39.** ACD
40. BC **41.** AD

EXERCISE–8

3. p = 0; q = 10; r = – 3

4. – 98 **6.** 3

20. $\vec{v}_1 = 2\hat{i}$, $\vec{v}_2 = -\hat{i} \pm \hat{j}$, $\vec{v}_3 = 3\hat{i} \pm 2\hat{j} \pm 4\hat{k}$ are some possible values

Assertion- Reason

Some questions (Assertion–Reason type) are given below. Each question contains **Statement – 1 (Assertion)** and **Statement – 2 (Reason)**. Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. So select the correct choice :***Choices are :***

(A)**Statement – 1** is True, **Statement – 2** is True; **Statement – 2 is** a correct explanation for **Statement – 1**.

(B)**Statement – 1** is True, **Statmnt – 2** is True; **Statement – 2 is NOT** a correct explanation for **Statement – 1**.

(C) **Statement – 1** is True, **Statement – 2** is False.

(D) Statement – 1 is False, Statement – 2 is True.

452. Let $\bar{a}, \bar{b}, \bar{c}$ be three non-coplanar vectors then $(\bar{b}-\bar{c}).[(\bar{c}-\bar{a})\times(\bar{a}-\bar{b})] = 0$

Statement 1: $\bar{b}-\bar{c}$ can be expressed as linear combination of $\bar{c}-\bar{a}$ and $\bar{a}-\bar{b}$.

Statement 2: Given non-coplanar vectors one vector can be expressed as a linear combination of other two.

453. A vector has components p and 1 with respect to a rectangular cartesian system. If the axes are rotated through an angle α about the origin in the anticlockwise sense.

Statement–1 : If the vector has component $p+2$ and 1 with respect to the new system then $p = -1$

Statement–2 : Magnitude of vector original and new system remains same

454. Let $|\vec{a}|=4$, $|\vec{b}|=2$ and angle between $\vec{a}$ and $\vec{b}$ is $\pi/6$

Statement–1 : $(\vec{a}\times\vec{b})^2 = 4$ **Statement–2 :** $(\vec{a}\times\vec{b})^2 = |\vec{a}|^2$

455. **Statement–1 :** $\left[\vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}\ \ \vec{a}\times\vec{b}\right]=0$

Statement–2 : If $\vec{a}$, $\vec{b}$, $\vec{r}$ are linearly dependent vectors then they are coplanar.

456. **Statement–1 :** If $\left|\vec{a}+\vec{b}\right|=\left|\vec{a}-\vec{b}\right|$ then $\vec{a}$ is parallel to $\vec{b}$.

Statement–2 : If $\left|\vec{a}+\vec{b}\right| = \left|\vec{a}-\vec{b}\right|$ then $\vec{a}.\vec{b} = 0$.

457. Let $\vec{r}$ be a non-zero vector satisfying $\vec{r}.\vec{a} = \vec{r}.\vec{b} = \vec{r}.\vec{c} = 0$ for given non–zero vectors $\vec{a}, \vec{b}$ and $\vec{c}$.

Statement–1 : $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar vectors.

Statement–2 : $\vec{r}$ is perpendicular to the vectors $\vec{a}, \vec{b}$ and $\vec{c}$.

458. Let $\vec{a}$ and $\vec{r}$ be two non–collinear vectors.

Statement–1 : vector $\vec{a} \times (\vec{a}\times\vec{r})$ is a vector in the plane of $\vec{a}$ and $\vec{r}$, perpendicular to $\vec{a}$.

Statement–2 : $\vec{a}\times(\vec{a}\times\vec{b}) = \vec{0}$, for any vector $\vec{b}$.

459. **Statement–1 :** If three points P, Q, R have position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ respectively and $2\vec{a}+3\vec{b}-5\vec{c}=0$, then the points P, Q, R must be collinear. **Statement–2 :** If for three points A, B, C; $\overrightarrow{AB}=\lambda\overrightarrow{AC}$, then the points A, B, C must be collinear.

460. **Statement–1 :** Let $\vec{a}$ and $\vec{b}$ be two non collinear unit vectors. If $\vec{u}=\vec{a}-(\vec{a}.\vec{b})\vec{b}$ and $\vec{v}=\vec{a}\times\vec{b}$ then $|\vec{v}|=|\vec{u}|$.

Statement–2 : The vector $\frac{1}{3}(2\hat{i}-2\hat{j}+\hat{k})$ is makes an angle of $\frac{\pi}{3}$ with the vector $(5\hat{i}-4\hat{j}+3\hat{k})$.

461. **Statement-1:** If $\vec{u}$ & $\vec{v}$ are unit vectors inclined at an angle α and $\vec{x}$ is a unit vector bisecting the angle between them, then $\vec{x}=\dfrac{\vec{u}+\vec{v}}{2\cos\dfrac{\alpha}{2}}$

Statement-2: If ΔABC is an isosceles triangle with AB = AC = 1, then vector representing bisector of angle A is given by $\overrightarrow{AD} = \dfrac{\vec{AB}+\vec{AC}}{2}$

462. **Statement-1:** The direction ratios of line joining origin and point (x, y, z) must be x, y, z.

Statement-2: If P is a point (x, y, z) in space and OP = r, then direction cosines of OP are $\frac{x}{r}, \frac{y}{r}, \frac{z}{r}$.

463. **Statement-1:** If the vectors $2\hat{i}-\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}-3\hat{k}$ and $3\hat{i}-\lambda\hat{j}+5\hat{k}$ are coplanar, then $|\lambda|^2$ is equal to 16.

Statement-2: The vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar iff $\vec{a}, (\vec{b}\times\vec{c}) = 0$

464. **Statement-1:** A line L is perpendicular to the plane $3x - 4y + 5z = 10$

Statement-2: Direction co-sines of L be $< \frac{3}{5\sqrt{2}}, -\frac{4}{5\sqrt{2}}, \frac{1}{\sqrt{2}} >$

465. **Statement-1 :** The points with position vectors $\vec{a}-2\vec{b}+3\vec{c}, -2\vec{a}+3\vec{b}-\vec{c}$, $4\vec{a}-7\vec{b}+7\vec{c}$ are collinear.

Statement-2: The position vectors $\vec{a}-2\vec{b}+3\vec{c}$, $-2\vec{a}+3\vec{b}-\vec{c}$, $4\vec{a}-7\vec{b}+7\vec{c}$ are linearly dependent vectors.

466. **Statement-1:** If $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors such that $\vec{a}\times(\vec{b}\times\vec{c})=\frac{1}{2}\vec{b}$ then the angle between $\vec{a}$ & $\vec{b}$ is $\pi/2$

Statement-2: If $\vec{a}\times(\vec{b}\times\vec{c}) = \frac{1}{2}\vec{b}$, then $\vec{a}.\vec{b} = 0$.

467. **Statement-1:** If $\cos\alpha, \cos\beta, \cos\gamma$ are the direction cosine of any line segment, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

Statement-2: If $\cos\alpha, \cos\beta, \cos\gamma$ are the direction cosine of line segment,
$\cos2\alpha + \cos2\beta + \cos2\gamma = -1$.

468. **Statement-1:** The direction cosines of one of the angular bisector of two intersecting lines having direction cosines as l_1, m_1, n_1, & l_2, m_2, n_2 is proportional to $l_1 + l_2$, $m_1 + m_2$, $n_1 + n_2$.

Statement-2: The angle between the two intersecting lines having direction cosines as l_1, m_1, n_1 & l_2, m_2, n_2 is given by $\cos\theta = l_1 l_2 + m_1m_2 + n_1n_2$.

469. **Statement-1:** If $\vec{a}.\vec{b} = 0 \Rightarrow \vec{a} \perp \vec{b}$ **Statement-2:** $\vec{a}.\vec{b} = 0 \Rightarrow$ either $\vec{a}=0$ or $\vec{b} = 0$ or $\vec{a} \perp \vec{b}$

470. **Statement-1:** $\vec{A}\times\vec{B} = \vec{B}\times\vec{A}$

Statement-2: $\vec{A}\times\vec{B} = |\vec{A}||\vec{B}| \sin\theta\ \hat{n}$, when θ is angle, when your fingers curls from A to B

471. **Statement-1 :** A vector $\perp^r$ the plane of (1, -1, 0), (2, 1, -1) & (-1, 1, 2) is $6\hat{i}+6\hat{k}$

Statement-2 : $\vec{A}\times\vec{B}$ always gives a vector perpendicular to plane of $\vec{A}$ & $\vec{B}$

472. **Statement-1 :** Angle between planes $\vec{r}.\vec{n}_1 = \vec{q}_1$ & $\vec{r}.\vec{n}_2 = \vec{q}_2$.

(acute angle) is given by $\cos\theta = \vec{n}_1.\vec{n}_2$

Statement-2 : Angle between the planes in same as acute angle formed by their normals.

473. **Statement-1:** In ΔABC, $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA} = 0$

Statement-2 : If $\overrightarrow{OA}=\vec{a}$, $\overrightarrow{OB}=\vec{b}$ then $\overrightarrow{AB}=\vec{a}+\vec{b}$

474. **Statement-1:** $\vec{a}=3\vec{i}+p\vec{j}+3\vec{k}$ and $\vec{b}=2\vec{i}+3\vec{j}+q\vec{k}$ are parallel vectors it p = 9/2 and q = 2.

Statement-2 : If $\vec{a}=a_1\vec{i}+a_2\vec{j}+a_3\vec{k}$ and $\vec{b}=b_1\vec{i}+b_2\vec{j}+b_3\vec{k}$ are parallel $\frac{a_1}{b_1}=\frac{a_2}{b_2}=\frac{a_3}{b_3}$

475. **Statement-1:** The direction ratios of line joining origin and point (x, y, z) must be x, y, z

Statement-2: If P is a point (x,y, z) in space and OP = r then directions cosines of OP are $\frac{x}{r}, \frac{y}{r}, \frac{z}{r}$

476. **Statement-1:** The shortest distance between the skew lines $\vec{r}=\vec{a}+\alpha\vec{b}$ and $\vec{r}=\vec{c}+\beta\vec{d}$ is $\frac{|[\vec{a}-\vec{c}\ \vec{b}\vec{d}]|}{|\vec{b}\times\vec{d}|}$

Statement-2: Two lines are skew lines if three axist no plane passing through them.

477. **Statement-1:** $\vec{a}=\hat{i}+p\hat{j}+2\hat{k}$, $\vec{b}=2\hat{i}+3\hat{j}+q\hat{k}$ are parallel vectors of p = 3/2 and q = 4.

Statement-2: $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ are parallel if $\frac{a_1}{b_1}=\frac{a_2}{b_2}=\frac{a_3}{b_3}$.

478. **Statement-1:** If $\vec{a}=2\hat{i}+\hat{k},\vec{b}=3\hat{j}+4\hat{k}$ and $\vec{c}=8\hat{i}-3\hat{j}$ are coplanar then $\vec{c}=4\vec{a}-\vec{b}$.

Statement-2: A set of vectors $\vec{a}_1,\vec{a}_2...\vec{a}_n$ is said to be linearly independent if every relation of the form $l_1\ \vec{a}_1+l_2\ \vec{a}_2+\ldots+l_n\ \vec{a}_n=0$ implies that $l_1=l_2=\ldots.=l_n=0$ (scalars).

479. **Statement-1:** The shortest distance between the skew lines $\vec{r}=\vec{a}+\alpha\vec{b}$ and $\vec{r}=\vec{c}+\beta\vec{d}$ is $\left|\frac{(\vec{a}-\vec{c}).(\vec{b}\times\vec{d})}{|\vec{b}\times\vec{d}|}\right|$

Statement-2: Two lines are skew lines if there exists no plane passing through them.

480. **Statement-1:** The curve which is tangent to a sphere at a given point is the equation of a plane.
Statement-2: Infinite number of lines touch the sphere at a given point.

481. **Statement-1:** In $\Delta ABC\ \overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA}=\overrightarrow{O}$

Statement-2: If $\overrightarrow{OA}=\vec{a},\ \overrightarrow{OB}=\vec{b},$ then $\overrightarrow{AB}=\vec{a}+\vec{b}$ (Δ law of addition).

482. **Statement-1:** $\vec{a}=\hat{i}+p\hat{j}+2\hat{k}$ and $\vec{b}=2\hat{i}+3\hat{j}+q\hat{k}$ are parallel vectors if $P=\frac{3}{2},\ q=4$

Statement-2: If $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ are parallel then $\frac{a_1}{b_1}=\frac{a_2}{b_2}=\frac{a_3}{b_3}$.

483. **Statement-1:** If $\vec{a}=2\hat{i}+\hat{k}\ \vec{a}_1,\vec{a}_3,\vec{a}_3.....\vec{a}_n,\vec{b}=3\hat{j}+4\hat{k}$ and $\vec{c}=8\hat{i}-3\hat{j}$ are coplanar then $\vec{c}=4\vec{a}-\vec{b}$
Statement-2: A set of vectors is said to be linearly independent if every relation of the form
$l_1\vec{a}+l_2\vec{a}_2+.....+l_n\vec{a}_n=0\Rightarrow\ l_1=l_2=.....=l_n=0.$

484. **Statement-1:** The shortest distance between the skew lines $\vec{r}=\vec{a}_1+\alpha\vec{b}_1$ and $\vec{r}=\vec{a}_2+\beta\vec{b}_2$ is $\left|\frac{[\vec{b}_1\ \vec{b}_2(\vec{a}_2-\vec{a}_1)]}{(\vec{b}_1\times\vec{b}_2)}\right|$ **Statement-2:** Two lines are skew lines if there exists no plane passing through them.

485. **Statement-1 :** The value of expression $\hat{i}(\hat{j}\times\hat{k})+\hat{j}(\hat{k}\times\hat{i})+\hat{k}(\hat{i}\times\hat{j})=3$

Statement-2 : $\hat{i}(\hat{j}\times\hat{k})=[\hat{i}.\hat{j}.\hat{k}]=1$

486. **Statement-1:** A relation between the vectors $\vec{r},\vec{a}$ and $\vec{b}$ is $\vec{r}\times\vec{a}=\vec{b}\Rightarrow\vec{r}=\frac{\vec{a}\times\vec{b}}{\vec{a}.\vec{a}}$ **Statement-2 :** $\vec{r}.\vec{a}=0$

3-Dimension

487. The equation of two straight line are $\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{-3}$ and $\frac{x-2}{1}=\frac{y-1}{-3}=\frac{z+3}{2}$

Statement–1 : The given lines are coplanar
Statement–2 : The equation $2x_1-y_1=1,\ x_1+3y_1=4,\ 3x_1+2y_1=5$ are consistent.

488. **Statement–1 :** The distance between the planes $4x-5y+3z=5$ and $4x-5y+3z+2=0$ is $\frac{3}{5\sqrt{2}}$. **Statement–2** The distance between $ax+by+cz+d_1=0$ and $ax+by+cz+d_2=0$ is $\left|\frac{d_1-d_2}{\sqrt{a^2+b^2+c^2}}\right|$.

489. Given the line $\frac{x-1}{3}=\frac{y+1}{2}=\frac{z-3}{-1}$ and the plane $\pi: x-2y-z=0$

Statement-1: L lies in π **Statement-2:** L is parallel to π

490. The image of the point (1, b, 3) in the **Statement-1:** Line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ will be (1, 0, 7)

Statement-2: Length of the perpendicular from the point A($\bar{\alpha}$) on the line $\vec{r}=\vec{a}+t\vec{b}$, is given by d = $\frac{|(\bar{a}-\bar{\alpha})\times\bar{b}|}{|\bar{b}|}$

Answer

452. C	453. A	454. D	455. D	456. D	457. A	458. C
459. A	460. C	461. A	462. A	463. A	464. A	465. A
466. A	467. B	468. B	469. D	470. D	471. A	472. A
473. C	474. A	475. A	476. B	477. A	478. B	479. B
480. A	481. C	482. A	483. B	484. B	485. A	486. A
487. A	488. D	489. C	490. B			

Que from Compt. Exams

Co-ordinate Geometry of Three Dimensions

1. The direction cosines of a line segment AB are $-2/\sqrt{17}$, $3/\sqrt{17}$, $-2/\sqrt{17}$. If $AB=\sqrt{17}$ and the co-ordinates of A are (3, –6, 10), then the co-ordinates of B are

(a) (1, –2, 4) (b) (2, 5, 8) (c) (–1, 3, –8) (d) (1, – 3, 8)

2. The projection of any line on co-ordinate axes be respectively 3, 4, 5 then its length is **[MP PET 1995; RPET 2001]**

(a) 12 (b) 50 (c) $5\sqrt{2}$ (d) None of these

3. If centroid of the tetrahedron $OABC$, where A, B, C are given by $(a, 2, 3),(1, b, 2)$ and $(2, 1, c)$ respectively be $(1, 2, -1)$, then distance of $P(a,b,c)$ from origin is equal to

(a) $\sqrt{107}$ (b) $\sqrt{14}$ (c) $\sqrt{107/14}$ (d) None of these

4. If $P\equiv(0,1,0), Q\equiv(0,0,1)$, then projection of PQ on the plane $x+y+z=3$ is **[EAMCET 2002]**

(a) $\sqrt{3}$ (b) 3 (c) $\sqrt{2}$ (d) 2

5. The points $A(4,5,1), B(0,-1,-1), C(3,9,4)$ and $D(-4,4,4)$ are **[Kurukshetra CEE 2002]**

(a) Collinear (b) Coplanar (c) Non- coplanar
(d) Non- Collinear and non-coplanar

6. The angle between two diagonals of a cube will be **[MP PET 1996, 2000; RPET 2000, 02; UPSEAT 2004]**

(a) $\sin^{-1}1/3$ (b) $\cos^{-1}1/3$ (c) Variable (d) None of these

7. The equations of the line passing through the point (1,2,–4) and perpendicular to the two lines $\frac{x-8}{3}=\frac{y+19}{-16}=\frac{z-10}{7}$ and $\frac{x-15}{3}=\frac{y-29}{8}=\frac{z-5}{-5}$, will be **[AI CBSE 1983]**

(a) $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z+4}{6}$ (b) $\frac{x-1}{-2}=\frac{y-2}{3}=\frac{z+4}{8}$ (c) $\frac{x-1}{3}=\frac{y-2}{2}=\frac{z+4}{8}$ (d) None of these

8. If three mutually perpendicular lines have direction cosines $(l_1,m_1,n_1),(l_2,m_2,n_2)$ and (l_3,m_3,n_3), then the line having direction cosines $l_1+l_2+l_3$, $m_1+m_2+m_3$ and $n_1+n_2+n_3$ make an angle of ..... with each other

(a) 0° (b) 30° (c) 60° (d) 90°

9. The straight lines whose direction cosines are given by $al+bm+cn=0, fmn+gnl+hlm=0$ are perpendicular, if

(a) $\frac{f}{a}+\frac{g}{b}+\frac{h}{c}=0$ (b) $\sqrt{\frac{a}{f}}+\sqrt{\frac{b}{g}}+\sqrt{\frac{c}{h}}=0$ (c) $\sqrt{af}=\sqrt{bg}=\sqrt{ch}$ (d) $\sqrt{\frac{a}{f}}=\sqrt{\frac{b}{g}}=\sqrt{\frac{c}{h}}$

10. If the straight lines $x=1+s$, $y=-3-\lambda s$, $z=1+\lambda s$ and $x=t/2, y=1+t, z=2-t$, with parameters s and t respectively, are co-planar, then λ equals **[AIEEE 2004]**

(a) 0 (b) –1 (c) –1/2 (d) – 2

11. The co-ordinates of the foot of perpendicular drawn from point $P(1,0,3)$ to the join of points $A(4,7,1)$ and $B(3,5,3)$ is **[RPET 01]**

(a) (5, 7, 1) (b) $\left(\frac{5}{3},\frac{7}{3},\frac{17}{3}\right)$ (c) $\left(\frac{2}{3},\frac{5}{3},\frac{7}{3}\right)$ (d) $\left(\frac{5}{3},\frac{2}{3},\frac{7}{3}\right)$

12. If the lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-1}{4}$ and $\frac{x-3}{1}=\frac{y-k}{1}=\frac{z}{1}$ intersect, then $k=$ **[IIT Screening 2004]**

(a) $\frac{2}{9}$ (b) $\frac{9}{2}$ (c) 0 (d) None of these

13. A square $ABCD$ of diagonal $2a$ is folded along the diagonal AC so that the planes DAC and BAC are at right angle. The shortest distance between DC and AB is **[Kurukshetra CEE 1998]**

(a) $\sqrt{2}a$ (b) $2a/\sqrt{3}$ (c) $2a/\sqrt{5}$ (d) $(\sqrt{3}/2)a$

14. A line with direction cosines proportional to 2,1, 2 meets each of the lines $x = y + a = z$ and $x + a = 2y = 2z$. The co-ordinates of each of the points of intersection are given by **[AIEEE 2004]**

(a) $(2a, a, 3a), (2a, a, a)$ (b) $(3a, 2a, 3a), (a, a, a)$ (c) $(3a, 2a, 3a), (a, a, 2a)$ (d) $(3a, 3a, 3a), (a, a, a)$

15. The equation of the planes passing through the line of intersection of the planes $3x - y - 4z = 0$ and $x + 3y + 6 = 0$ whose distance from the origin is 1, are

(a) $x - 2y - 2z - 3 = 0,\ 2x + y - 2z + 3 = 0$ (b) $x - 2y + 2z - 3 = 0,\ 2x + y + 2z + 3 = 0$

(c) $x + 2y - 2z - 3 = 0,\ 2x - y - 2z + 3 = 0$ (d) None of these

16. The co-ordinates of the points A and B are (2, 3, 4) (–2, 5,– 4) respectively. If a point P moves so that $PA^2 - PB^2 = k$ where k is a constant, then the locus of P is

(a) A line (b) A plane (c) A sphere (d) None of these

17. The equation of the plane passing through the points (1,–3,–2) and perpendicular to planes $x + 2y + 2z = 5$ and $3x + 3y + 2z = 8$, is **[AISSE 1987]**

(a) $2x - 4y + 3z - 8 = 0$ (b) $2x - 4y - 3z + 8 = 0$ (c) $2x + 4y + 3z + 8 = 0$ (d) None of these

18. A variable plane at a constant distance p from origin meets the co-ordinates axes in A, B, C. Through these points planes are drawn parallel to co-ordinate planes. Then locus of the point of intersection is

(a) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{p^2}$ (b) $x^2 + y^2 + z^2 = p^2$ (c) $x + y + z = p$ (d) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = p$

19. P is a fixed point (a, a, a) on a line through the origin equally inclined to the axes, then any plane through P perpendicular to OP, makes intercepts on the axes, the sum of whose reciprocals is equal to

(a) a (b) $\frac{3}{2a}$ (c) $\frac{3a}{2}$ (d) None of these

20. The equation of the plane through the intersection of the planes $x + 2y + 3z - 4 = 0,\ 4x + 3y + 2z + 1 = 0$ and passing through the origin will be **[MP PET 1998]**

(a) $x + y + z = 0$ (b) $17x + 14y + 11z = 0$ (c) $7x + 4y + z = 0$ (d) $17x + 14y + z = 0$

21. The d.r's of normal to the plane through $(1, 0, 0), (0, 1, 0)$ which makes an angle $\frac{\pi}{4}$ with plane $x + y = 3$, are **[AIEEE 2002]**

(a) $1, \sqrt{2}, 1$ (b) $1, 1, \sqrt{2}$ (c) $1, 1, 2$ (d) $\sqrt{2}, 1, 1$

22. Two systems of rectangular axes have the same origin. If a plane cuts them at distance *a, b, c* and *a', b', c'* from the origin, then

(a) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} + \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2} = 0$ (b) $\frac{1}{a^2} + \frac{1}{b^2} - \frac{1}{c^2} + \frac{1}{a'^2} + \frac{1}{b'^2} - \frac{1}{c'^2} = 0$

(c) $\frac{1}{a^2} - \frac{1}{b^2} - \frac{1}{c^2} + \frac{1}{a'^2} - \frac{1}{b'^2} - \frac{1}{c'^2} = 0$ (d) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} - \frac{1}{a'^2} - \frac{1}{b'^2} - \frac{1}{c'^2} = 0$ **[AIEEE 2003]**

23. If $4x + 4y - kz = 0$ is the equation of the plane through the origin that contains the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{4}$, then $k =$

(a) 1 (b) 3 (c) 5 (d) 7 **[MP PET 1992]**

24. The distance of the point (1, –2, 3) from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$, is

(a) 1 (b) 6/7 (c) 7/6 (d) None of these **[AI CBSE 1984]**

25. The distance of the point of intersection of the line $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ and the plane $x + y + z = 17$ from the point (3, 4, 5) is given by

(a) 3 (b) 3/2 (c) $\sqrt{3}$ (d) None of these

26. The lines $\frac{x-a+d}{\alpha-\delta} = \frac{y-a}{\alpha} = \frac{z-a-d}{\alpha+\delta}$ and $\frac{x-b+c}{\beta-\gamma} = \frac{y-b}{\beta} = \frac{z-b-c}{\beta+\gamma}$ are coplanar and then equation to the plane in which they lie, is

(a) $x + y + z = 0$ (b) $x - y + z = 0$ (c) $x - 2y + z = 0$ (d) $x + y - 2z = 0$

27. The line $\frac{x-3}{2} = \frac{y-4}{3} = \frac{z-5}{4}$ lies in the plane $4x + 4y - kz - d = 0$. The values of k and d are

(a) 4, 8 (b) –5, – 3 (c) 5, 3 (d) – 4, – 8

28. The value of k such that $\frac{x-4}{1} = \frac{y-2}{1} = \frac{z-k}{2}$ lies in the plane $2x - 4y + z = 7$, is **[IIT Screening 2003]**

(a) 7 (b) -7 (c) No real value (d) 4

The shortest distance from the plane $12x + 4y + 3z = 327$ to the sphere $x^2 + y^2 + z^2 + 4x - 2y - 6z = 155$ is **[AIEEE 2003]**

(a) 26 (b) $11\frac{4}{13}$ (c) 13 (d) 39

29. The radius of the circle in which the sphere $x^2 + y^2 + z^2 + 2x - 2y - 4z - 19 = 0$ is cut by the plane $x + 2y + 2z + 7 = 0$ is

(a) 1 (b) 2 (c) 3 (d) 4 **[AIEEE 2003]**

30. The equation of motion of a rocket are: $x = 2t, y = -4t, \quad z = 4t$ where the time 't' is given in seconds, and the co-ordinates of a moving point in kilometers. What is the path of the rocket? At what distance will be the rocket be from the starting point 0(0, 0, 0) in 10 seconds

(a) Straight line, 60 *km* (b) Straight line, 30 *km* (c) Parabola, 60 *km* (d) Ellipse, 60 *km*

31. The plane $lx + my = 0$ is rotated an angle α about its line of intersection with the plane $z = 0$, then the equation to the plane in its new position is

(a) $lx + my \pm z\sqrt{(l^2 + m^2)}\tan\alpha = 0$

(b) $lx - my \pm z\sqrt{(l^2 + m^2)}\tan\alpha = 0$

(c) $lx + my \pm z\sqrt{(l^2 + m^2)}\cos\alpha = 0$

(d) $lx - my \pm z\sqrt{(l^2 + m^2)}\cos\alpha = 0$

32. The distance between two points P and Q is d and the length of their projections of PQ on the co-ordinate planes are d_1, d_2, d_3. Then $d_1^2 + d_2^2 + d_3^2 = kd^2$ where 'k' is

(a) 1 (b) 5

(c) 3 (d) 2

33. If P_1 and P_2 are the lengths of the perpendiculars from the points (2,3,4) and (1,1,4) respectively from the plane $3x - 6y + 2z + 11 = 0$, then P_1 and P_2 are the roots of the equation

(a) $P^2 - 23P + 7 = 0$ (b) $7P^2 - 23P + 16 = 0$

(c) $P^2 - 17P + 16 = 0$ (d) $P^2 - 16P + 7 = 0$

34. The edge of a cube is of length 'a' then the shortest distance between the diagonal of a cube and an edge skew to it is

(a) $a\sqrt{2}$ (b) a

(c) $\sqrt{2}/a$ (d) $a/\sqrt{2}$

Que from Compt. Exams

Co-ordinate Geometry of Three Dimensions

1	d	2	c	3	a	4	c	5	b
6	b	7	a	8	a	9	a	10	d
11	b	12	b	13	b	14	b	15	a
16	b	17	a	18	a	19	d	20	b
21	b	22	d	23	c	24	a	25	a
26	c	27	c	28	a	29	c	30	c
31	a	32	a	33	d	34	b	35	d

Que from Compt. Exams

Vector Algebra

1. Three forces of magnitudes 1, 2, 3 dynes meet in a point and act along diagonals of three adjacent faces of a cube. The resultant force is **[MNR 1987]**

(a) 114 dyne (b) 6 dyne (c) 5 dyne (d) None of these

2. The vectors **b** and **c** are in the direction of north-east and north-west respectively and $|\mathbf{b}|=|\mathbf{c}|= 4$. The magnitude and direction of the vector $\mathbf{d} = \mathbf{c} - \mathbf{b}$, are **[Roorkee 2000]**

(a) $4\sqrt{2}$, towards north (b) $4\sqrt{2}$, towards west (c) 4, towards east (d) 4, towards south

3. If **a, b** and **c** are unit vectors, then $|\mathbf{a} - \mathbf{b}|^2 + |\mathbf{b} - \mathbf{c}|^2 + |\mathbf{c} - \mathbf{a}|^2$ does not exceed **[IIT Screening 2001]**

(a) 4 (b) 9 (c) 8 (d) 6

4. The vectors $\overrightarrow{AB} = 3\mathbf{i} + 5\mathbf{j} + 4\mathbf{k}$ and $\overrightarrow{AC} = 5\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}$ are the sides of a triangle *ABC*. The length of the median through *A* is **[UPSEAT 2004]**

(a) $\sqrt{13}$ *unit* (b) $2\sqrt{5}$ *unit* (c) 5 *unit* (d) 10 *unit*

5. Let the value of $\mathbf{p} = (x + 4y)\mathbf{a} + (2x + y + 1)\mathbf{b}$ and $\mathbf{q} = (y - 2x + 2)\mathbf{a} + (2x - 3y - 1)\mathbf{b}$, where **a** and **b** are non-collinear vectors. If $3\mathbf{p} = 2\mathbf{q}$, then the value of x and y will be **[RPET 1984; MNR 1984]**

(a) $-1, 2$ (b) $2, -1$ (c) $1, 2$ (d) $2, 1\backslash$

6. The points D, E, F divide BC, CA and AB of the triangle ABC in the ratio $1:4, 3:2$ and $3:7$ respectively and the point K divides AB in the ratio $1:3$, then $(\overrightarrow{AD}+\overrightarrow{BE}+\overrightarrow{CF}):\overrightarrow{CK}$ is equal to **[MNR 1987]**

(a) $1:1$ (b) $2:5$ (c) $5:2$ (d) None of these

7. If two vertices of a triangle are $\mathbf{i}-\mathbf{j}$ and $\mathbf{j}+\mathbf{k}$, then the third vertex can be **[Roorkee 1995]**

(a) $\mathbf{i}+\mathbf{k}$ (b) $\mathbf{i}-2\mathbf{j}-\mathbf{k}$ (c) $\mathbf{i}-\mathbf{k}$ (d) $2\mathbf{i}-\mathbf{j}$

(e) All the above

8. If $\mathbf{a}$ of magnitude 50 is collinear with the vector $\mathbf{b}=6\mathbf{i}-8\mathbf{j}-\frac{15\mathbf{k}}{2}$, and makes an acute angle with the positive direction of z-axis, then the vector $\mathbf{a}$ is equal to **[Pb. CET 2004]**

(a) $24\mathbf{i}-32\mathbf{j}+30\mathbf{k}$ (b) $-24\mathbf{i}+32\mathbf{j}+30\mathbf{k}$ (c) $16\mathbf{i}-16\mathbf{j}-15\mathbf{k}$ (d) $-12\mathbf{i}+16\mathbf{j}-30\mathbf{k}$

9. If three non-zero vectors are $\mathbf{a}=a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}$, $\mathbf{b}=b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k}$ and $\mathbf{c}=c_1\mathbf{i}+c_2\mathbf{j}+c_3\mathbf{k}$. If $\mathbf{c}$ is the unit vector perpendicular to the vectors $\mathbf{a}$ and $\mathbf{b}$ and the angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{6}$, then $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2$ is equal to

(a) 0 (b) $\frac{3(\Sigma a_1^2)(\Sigma b_1^2)(\Sigma c_1^2)}{4}$ (c) 1 (d) $\frac{(\Sigma a_1^2)(\Sigma b_1^2)}{4}$

10. Let the unit vectors $\mathbf{a}$ and $\mathbf{b}$ be perpendicular and the unit vector $\mathbf{c}$ be inclined at an angle θ to both $\mathbf{a}$ and $\mathbf{b}$. If $\mathbf{c}=\alpha\mathbf{a}+\beta\mathbf{b}+\gamma(\mathbf{a}\times\mathbf{b})$, then **[Orissa JEE 2003]**

(a) $\alpha=\beta=\cos\theta,\ \gamma^2=\cos 2\theta$ (b) $\alpha=\beta=\cos\theta,\ \gamma^2=-\cos 2\theta$

(c) $\alpha=\cos\theta,\ \beta=\sin\theta,\ \gamma^2=\cos 2\theta$ (d) None of these

11. The vector $\mathbf{a}+\mathbf{b}$ bisects the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$, if

(a) $|\mathbf{a}|=|\mathbf{b}|$ (b) $|\mathbf{a}|=|\mathbf{b}|$ or angle between $\mathbf{a}$ and $\mathbf{b}$ is zero

(c) $|\mathbf{a}|=m|\mathbf{b}|$ (d) None of these

12. The points O, A, B, C, D are such that $\overrightarrow{OA}=\mathbf{a}$, $\overrightarrow{OB}=\mathbf{b}$, $\overrightarrow{OC}=2\mathbf{a}+3\mathbf{b}$ and $\overrightarrow{OD}=\mathbf{a}-2\mathbf{b}$. If $|\mathbf{a}|=3|\mathbf{b}|$, then the angle between $\overrightarrow{BD}$ and $\overrightarrow{AC}$ is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{6}$ (d) None of these

13. If $\vec{A}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$, $\vec{B}=-\mathbf{i}+2\mathbf{j}+\mathbf{k}$ and $\vec{C}=3\mathbf{i}+\mathbf{j}$, then the value of t such that $\vec{A}+t\vec{B}$ is at right angle to vector $\vec{C}$, is **[RPET 2002]**

(a) 2 (b) 4 (c) 5 (d) 6

14. Let $\mathbf{b}=4\mathbf{i}+3\mathbf{j}$ and $\mathbf{c}$ be two vectors perpendicular to each other in the xy-plane. All vectors in the same plane having projections 1 and 2 along $\mathbf{b}$ and $\mathbf{c}$ respectively, are given by **[IIT 1987]**

(a) $2\mathbf{i}-\mathbf{j}, \frac{2}{5}\mathbf{i}+\frac{11}{5}\mathbf{j}$ (b) $2\mathbf{i}+\mathbf{j}, -\frac{2}{5}\mathbf{i}+\frac{11}{5}\mathbf{j}$ (c) $2\mathbf{i}+\mathbf{j}, -\frac{2}{5}\mathbf{i}-\frac{11}{5}\mathbf{j}$ (d) $2\mathbf{i}-\mathbf{j}, -\frac{2}{5}\mathbf{i}+\frac{11}{5}\mathbf{j}$

15. Let $\mathbf{a}=2\mathbf{i}-\mathbf{j}+\mathbf{k}$, $\mathbf{b}=\mathbf{i}+2\mathbf{j}-\mathbf{k}$ and $\mathbf{c}=\mathbf{i}+\mathbf{j}-2\mathbf{k}$ be three vectors. A vector in the plane of $\mathbf{b}$ and $\mathbf{c}$ whose projection on $\mathbf{a}$ is of magnitude $\sqrt{2/3}$ is **[IIT 1993; Pb. CET 2004]**

(a) $2\mathbf{i}+3\mathbf{j}-3\mathbf{k}$ (b) $2\mathbf{i}+3\mathbf{j}+3\mathbf{k}$ (c) $-2\mathbf{i}-\mathbf{j}+5\mathbf{k}$ (d) $2\mathbf{i}+\mathbf{j}+5\mathbf{k}$

16. A vector $\mathbf{a}$ has components $2p$ and 1 with respect to a rectangular cartesian system. The system is rotated through a certain angle about the origin in the anti-clockwise sense. If $\mathbf{a}$ has components $p+1$ and 1 with respect to the new system, then **[IIT 1984]**

(a) $p=0$ (b) $p=1$ or $-\frac{1}{3}$ (c) $p=-1$ or $\frac{1}{3}$ (d) $p=1$ or -1

17. If $\mathbf{u}=2\mathbf{i}+2\mathbf{j}-\mathbf{k}$ and $\mathbf{v}=6\mathbf{i}-3\mathbf{j}+2\mathbf{k}$, then a unit vector perpendicular to both $\mathbf{u}$ and $\mathbf{v}$ is **[MP PET 1987]**

(a) $\mathbf{i}-10\mathbf{j}-18\mathbf{k}$ (b) $\frac{1}{\sqrt{17}}\left(\frac{1}{5}\mathbf{i}-2\mathbf{j}-\frac{18}{5}\mathbf{k}\right)$ (c) $\frac{1}{\sqrt{473}}(7\mathbf{i}-10\mathbf{j}-18\mathbf{k})$ (d) None of these

18. If $\mathbf{a}=2\mathbf{i}+\mathbf{k}$, $\mathbf{b}=\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\mathbf{c}=4\mathbf{i}-3\mathbf{j}+7\mathbf{k}$. If $\mathbf{d}\times\mathbf{b}=\mathbf{c}\times\mathbf{b}$ and $\mathbf{d}.\mathbf{a}=0$, then $\mathbf{d}$ will be **[IIT 1990]**

(a) $\mathbf{i}+8\mathbf{j}+2\mathbf{k}$ (b) $\mathbf{i}-8\mathbf{j}+2\mathbf{k}$ (c) $-\mathbf{i}+8\mathbf{j}-\mathbf{k}$ (d) $-\mathbf{i}-8\mathbf{j}+2\mathbf{k}$

19. If $\mathbf{a}\times\mathbf{r}=\mathbf{b}+\lambda\mathbf{a}$ and $\mathbf{a}.\mathbf{r}=3$, where $\mathbf{a}=2\mathbf{i}+\mathbf{j}-\mathbf{k}$ and $\mathbf{b}=-\mathbf{i}-2\mathbf{j}+\mathbf{k}$, then $\mathbf{r}$ and λ are equal to

(a) $\mathbf{r}=\frac{7}{6}\mathbf{i}+\frac{2}{3}\mathbf{j},\ \lambda=\frac{6}{5}$ (b) $\mathbf{r}=\frac{7}{6}\mathbf{i}+\frac{2}{3}\mathbf{j},\ \lambda=\frac{5}{6}$ (c) $\mathbf{r}=\frac{6}{7}\mathbf{i}+\frac{2}{3}\mathbf{j},\ \lambda=\frac{6}{5}$ (d) None of these

20. Let the vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ be such that $(\mathbf{a}\times\mathbf{b})\times(\mathbf{c}\times\mathbf{d})=0$. Let P_1 and P_2 be planes determined by pair of vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, $\mathbf{d}$ respectively. Then the angle between P_1 and P_2 is **[IIT Screening 2000; MP PET 2004]**

(a) 0° (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

21. If $\mathbf{a}=\mathbf{i}+\mathbf{j}+\mathbf{k}$, $\mathbf{a}.\mathbf{b}=1$ and $\mathbf{a}\times\mathbf{b}=\mathbf{j}-\mathbf{k}$, then $\mathbf{b}=$ **[IIT Screening 2004]**

(a) $\mathbf{i}$ (b) $\mathbf{i}-\mathbf{j}+\mathbf{k}$ (c) $2\mathbf{j}-\mathbf{k}$ (d) $2\mathbf{i}$

22. The position vectors of the vertices of a quadrilateral $ABCD$ are $\mathbf{a},\mathbf{b},\mathbf{c}$ and $\mathbf{d}$ respectively. Area of the quadrilateral formed by joining the middle points of its sides is **[Roorkee 2000]**

(a) $\frac{1}{4}|\mathbf{a}\times\mathbf{b}+\mathbf{b}\times\mathbf{d}+\mathbf{d}\times\mathbf{a}|$ (b) $\frac{1}{4}|\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{d}+\mathbf{a}\times\mathbf{d}+\mathbf{b}\times\mathbf{a}|$

(c) $\frac{1}{4}|\mathbf{a}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{d}+\mathbf{d}\times\mathbf{a}|$ (d) $\frac{1}{4}|\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{d}+\mathbf{d}\times\mathbf{b}|$

23. The moment about the point $M(-2,4,-6)$ of the force represented in magnitude and position by $\overrightarrow{AB}$ where the points A and B have the co-ordinates $(1,2,-3)$ and $(3,-4,2)$ respectively, is **[MP PET 2000]**

(a) $8\mathbf{i}-9\mathbf{j}-14\mathbf{k}$ (b) $2\mathbf{i}-6\mathbf{j}+5\mathbf{k}$ (c) $-3\mathbf{i}+2\mathbf{j}-3\mathbf{k}$ (d) $-5\mathbf{i}+8\mathbf{j}-8\mathbf{k}$

24. If the vectors $a\mathbf{i}+\mathbf{j}+\mathbf{k}$, $\mathbf{i}+b\mathbf{j}+\mathbf{k}$ and $\mathbf{i}+\mathbf{j}+c\mathbf{k}$ $(a\neq b\neq c\neq 1)$ are coplanar, then the value of $\frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}=$ **[BIT Ranchi 1988; RPET 1987;*IIT 1987; DCE 2001; MP PET 2004; ORISSA JEE 2005*]**

(a) -1 (b) $-\frac{1}{2}$ (c) $\frac{1}{2}$ (d) 1

25. If $\alpha(\mathbf{a}\times\mathbf{b})+\beta(\mathbf{b}\times\mathbf{c})+\gamma(\mathbf{c}\times\mathbf{a})=\mathbf{0}$ and at least one of the numbers α, β and γ is non-zero, then the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are

(a) Perpendicular (b) Parallel (c) Coplanar (d) None of these

26. The volume of the tetrahedron, whose vertices are given by the vectors $-\mathbf{i}+\mathbf{j}+\mathbf{k}$, $\mathbf{i}-\mathbf{j}+\mathbf{k}$ and $\mathbf{i}+\mathbf{j}-\mathbf{k}$ with reference to the fourth vertex as origin, is

(a) $\frac{5}{3}$ *cubic unit* (b) $\frac{2}{3}$ *cubic unit* (c) $\frac{3}{5}$ *cubic unit* (d) None of these

27. Let $\mathbf{a}=\mathbf{i}-\mathbf{j}$, $\mathbf{b}=\mathbf{j}-\mathbf{k}$, $\mathbf{c}=\mathbf{k}-\mathbf{i}$. If $\hat{\mathbf{d}}$ is a unit vector such that $\mathbf{a}.\hat{\mathbf{d}}=0=[\mathbf{b}\ \mathbf{c}\ \hat{\mathbf{d}}]$, then $\hat{\mathbf{d}}$ is equal to **[IIT 1995]**

(a) $\pm\frac{\mathbf{i}+\mathbf{j}-\mathbf{k}}{\sqrt{3}}$ (b) $\pm\frac{\mathbf{i}+\mathbf{j}+\mathbf{k}}{\sqrt{3}}$ (c) $\pm\frac{\mathbf{i}+\mathbf{j}-2\mathbf{k}}{\sqrt{6}}$ (d) $\pm\mathbf{k}$

28. The value of 'a' so that the volume of parallelopiped formed by $\mathbf{i}+a\mathbf{j}+\mathbf{k}, \mathbf{j}+a\mathbf{k}$ and $a\mathbf{i}+\mathbf{k}$ becomes minimum is **[IIT Screening 2003]**

(a) -3 (b) 3 (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$

29. If $\mathbf{b}$ and $\mathbf{c}$ are any two non-collinear unit vectors and $\mathbf{a}$ is any vector, then $(\mathbf{a}.\mathbf{b})\mathbf{b}+(\mathbf{a}.\mathbf{c})\mathbf{c}+\frac{\mathbf{a}.(\mathbf{b}\times\mathbf{c})}{|\mathbf{b}\times\mathbf{c}|}(\mathbf{b}\times\mathbf{c})=$ **[IIT 1996]**

(a) $\mathbf{a}$ (b) $\mathbf{b}$ (c) $\mathbf{c}$ (d) $\mathbf{o}$

30. If $\mathbf{a},\mathbf{b},\mathbf{c}$ are non-coplanar unit vectors such that $\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=\frac{\mathbf{b}+\mathbf{c}}{\sqrt{2}}$, then the angle between $\mathbf{a}$ and $\mathbf{b}$ is **[IIT 1995]**

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $\frac{3\pi}{4}$ (d) π

31. $[(\mathbf{a}\times\mathbf{b})\times(\mathbf{b}\times\mathbf{c})(\mathbf{b}\times\mathbf{c})\times(\mathbf{c}\times\mathbf{a})(\mathbf{c}\times\mathbf{a})\times(\mathbf{a}\times\mathbf{b})]=$

(a) $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]^2$ (b) $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]^3$ (c) $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]^4$ (d) None of these

32. Unit vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are coplanar. A unit vector $\mathbf{d}$ is perpendicular to them. If $(\mathbf{a}\times\mathbf{b})\times(\mathbf{c}\times\mathbf{d})=\frac{1}{6}\mathbf{i}-\frac{1}{3}\mathbf{j}+\frac{1}{3}\mathbf{k}$ and the angle between $\mathbf{a}$ and $\mathbf{b}$ is 30°, then $\mathbf{c}$ is **[Roorkee Qualifying 1998]**

(a) $\frac{(\mathbf{i}-2\mathbf{j}+2\mathbf{k})}{3}$ (b) $\frac{(2\mathbf{i}+\mathbf{j}-\mathbf{k})}{3}$ (c) $\frac{(-\mathbf{i}+2\mathbf{j}-2\mathbf{k})}{3}$ (d) $\frac{(-\mathbf{i}+2\mathbf{j}+\mathbf{k})}{3}$

33. The radius of the circular section of the sphere $|\mathbf{r}|=5$ by the plane $\mathbf{r}.(\mathbf{i}+\mathbf{j}+\mathbf{k})=3\sqrt{3}$ is **[DCE 1999]**

(a) 1 (b) 2 (c) 3 (d) 4

34. If $\mathbf{x}$ is parallel to $\mathbf{y}$ and $\mathbf{z}$ where $\mathbf{x}=2\mathbf{i}+\mathbf{j}+\alpha\mathbf{k}$, $\mathbf{y}=\alpha\mathbf{i}+\mathbf{k}$ and $\mathbf{z}=5\mathbf{i}-\mathbf{j}$, then α is equal to **[J & K 2005]**

(a) $\pm\sqrt{5}$ (b) $\pm\sqrt{6}$ (c) $\pm\sqrt{7}$ (d) None of these

35. The vector $\mathbf{c}$ directed along the internal bisector of the angle between the vectors $\mathbf{a} = 7\mathbf{i} - 4\mathbf{j} - 4\mathbf{k}$ and $\mathbf{b} = -2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ with $|\mathbf{c}| = 5\sqrt{6}$, is

(a) $\frac{5}{3}(\mathbf{i} - 7\mathbf{j} + 2\mathbf{k})$ (b) $\frac{5}{3}(5\mathbf{i} + 5\mathbf{j} + 2\mathbf{k})$ (c) $\frac{5}{3}(\mathbf{i} + 7\mathbf{j} + 2\mathbf{k})$ (d) $\frac{5}{3}(-5\mathbf{i} + 5\mathbf{j} + 2\mathbf{k})$

36. The distance of the point $B(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$ from the line which is passing through $A(4\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})$ and which is parallel to the vector $\vec{C} = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$ is **[Roorkee 1993]**

(a) 10 (b) $\sqrt{10}$ (c) 100 (d) None of these

37. Let **a, b, c** are three non-coplanar vectors such that
$\mathbf{r}_1 = \mathbf{a} - \mathbf{b} + \mathbf{c}$, $\mathbf{r}_2 = \mathbf{b} + \mathbf{c} - \mathbf{a}$, $\mathbf{r}_3 = \mathbf{c} + \mathbf{a} + \mathbf{b}$,
$\mathbf{r} = 2\mathbf{a} - 3\mathbf{b} + 4\mathbf{c}$. If $\mathbf{r} = \lambda_1\mathbf{r}_1 + \lambda_2\mathbf{r}_2 + \lambda_3\mathbf{r}_3$, then

(a) $\lambda_1 = 7$ (b) $\lambda_1 + \lambda_3 = 3$ (c) $\lambda_1 + \lambda_2 + \lambda_3 = 4$ (d) $\lambda_3 + \lambda_2 = 2$

38. Let $\mathbf{a} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{b} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and a unit vector $\mathbf{c}$ be coplanar. If $\mathbf{c}$ is perpendicular to $\mathbf{a}$, then $\mathbf{c}$ = **[IIT 1999; Pb. CET 2003; DCE 2005]**

(a) $\frac{1}{\sqrt{2}}(-\mathbf{j} + \mathbf{k})$ (b) $\frac{1}{\sqrt{3}}(-\mathbf{i} - \mathbf{j} - \mathbf{k})$ (c) $\frac{1}{\sqrt{5}}(\mathbf{i} - 2\mathbf{j})$ (d) $\frac{1}{\sqrt{3}}(\mathbf{i} - \mathbf{j} - \mathbf{k})$

39. Let **p, q, r** be three mutually perpendicular vectors of the same magnitude. If a vector **x** satisfies equation $\mathbf{p}\times\{(\mathbf{x}-\mathbf{q})\times\mathbf{p}\} + \mathbf{q}\times\{(\mathbf{x}-\mathbf{r})\times\mathbf{q}\} + \mathbf{r}\times\{(\mathbf{x}-\mathbf{p})\times\mathbf{r}\} = 0$, then **x** is given by **[IIT 1997 Cancelled]**

(a) $\frac{1}{2}(\mathbf{p} + \mathbf{q} - 2\mathbf{r})$ (b) $\frac{1}{2}(\mathbf{p} + \mathbf{q} + \mathbf{r})$ (c) $\frac{1}{3}(\mathbf{p} + \mathbf{q} + \mathbf{r})$ (d) $\frac{1}{3}(2\mathbf{p} + \mathbf{q} - \mathbf{r})$

40. The point of intersection of $\mathbf{r}\times\mathbf{a} = \mathbf{b}\times\mathbf{a}$ and $\mathbf{r}\times\mathbf{b} = \mathbf{a}\times\mathbf{b}$, where $\mathbf{a} = \mathbf{i} + \mathbf{j}$ and $\mathbf{b} = 2\mathbf{i} - \mathbf{k}$ is **[Orissa JEE 2004]**

(a) $3\mathbf{i} + \mathbf{j} - \mathbf{k}$ (b) $3\mathbf{i} - \mathbf{k}$ (c) $3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ (d) None of these

Que from Compt. Exams

Vector Algebra

1	c	2	b	3	b	4	c	5	b
6	b	7	e	8	b	9	d	10	b
11	b	12	d	13	c	14	d	15	a,c
16	b	17	b	18	d	19	b	20	a
21	a	22	c	23	a	24	d	25	c
26	b	27	c	28	c	29	a	30	c
31	c	32	a,c	33	d	34	c	35	a
36	b	37	b,c	38	a	39	b	40	a

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 16 XI M 16. Limits

Index:

1. Key Concepts

2. Exercise I to V

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Limit

1. Limit of a function f(x) is said to exist as, $x \to a$ when,

$\lim_{h\to 0^+} f(a-h) = \lim_{h\to 0^+} f(a+h) =$ some finite value M.

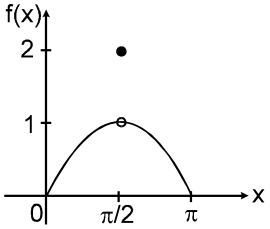
(Left hand limit) (Right hand limit)

Note that we are not interested in knowing about what happens at x = a. Also note that if L.H.L. & R.H.L. are both tending towards '∞' or '$-\infty$' then it is said to be infinite limit.

Remember, $\lim_{x\to a} \Rightarrow x \neq a$

Solved Example # 1

Find $\lim_{x\to \pi/2} f(x)$

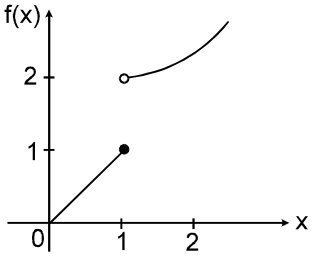


Solution.

Here $\lim_{x\to \pi/2} f(x) = 1$

Solved Example # 2

Find $\lim_{x\to 1} f(x)$



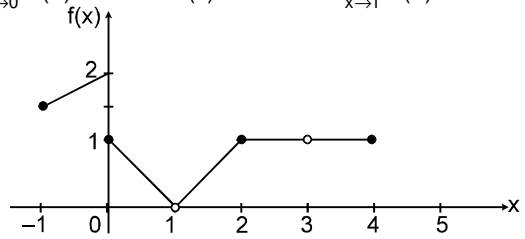
Solution.

Left handed limit = 1 Right handed limit = 2

Hence $\lim_{x\to 1} f(x)$ = does not exist.

Solved Example # 3

(i) Find $\lim_{x\to 0} f(x)$ (ii) Find $\lim_{x\to 1} f(x)$ (iii) Find $\lim_{x\to 3} f(x)$



Solution.

(i) $\lim_{x\to 0} f(x)$ = does not exists

because left handed limit ≠ right handed limit

(ii) $\lim_{x\to 1} f(x) = 0$ (iii) $\lim_{x\to 3} f(x) = 1$

2. Indeterminant Forms:

$\frac{0}{0}, \frac{\infty}{\infty}$, $0 \times \infty$, $\infty - \infty$, ∞^0, 0^0, and 1^∞.

Solved Example # 4

Which of the following limits are forming indeterminant from also indicate the form

(i) $\lim_{x\to 0} \frac{1}{x}$ (ii) $\lim_{x\to 0} \frac{1-x}{1-x^2}$

(iii) $\lim_{x\to 0} x \ell n\, x$ (iv) $\lim_{x\to 0} \left(\frac{1}{x} - \frac{1}{x^2}\right)$

(v) $\lim_{x\to 0} (\sin x)^x$ (vi) $\lim_{x\to 0} (\ell n\, x)^x$

(vii) $\lim_{x\to 0} (1+\sin x)^{\frac{1}{x}}$ (viii) $\lim_{x\to 0} (1)^{1/x}$

Solution

(i) No (ii) Yes $\frac{0}{0}$ from

(iii) Yes $0 \times \infty$ form (iv) Yes $(\infty - \infty)$ form

(v) Yes, $(0)^0$ form (vi) Yes $(\infty)^0$ form

(vii) Yes $(1)^\infty$ form (viii)

NOTE :

(i) '0' doesn't means exact zero but represent a value approaching towards zero similary to '1' and infinity.

(ii) $\infty + \infty = \infty$

(iii) $\infty \times \infty = \infty$

(iv) $(a/\infty) = 0$ if a is finite

(v) $\frac{a}{0}$ is not defined for any $a \in R$.

(vi) $ab = 0$, if & only if $a = 0$ or $b = 0$ and a & b are finite.

3. Method of Removing Indeterminancy

To evaluate a limit, we must always put the value where 'x' is approaching to in the function. If we get a determinate form, then that value becomes the limit otherwise if an indeterminant form comes. Then apply one of the following methods:

(i) Factorisation (ii) Rationalisation or double rationalisation

(iii) Substitution (iv) Using standard limits

(v) Expansions of functions.

1. Factorization method :-

We can cancel out the factors which are leading to indeterminancy and find the limit of the remaing expression.

Solved Example # 5

$\lim_{x\to 2} \frac{x^6 - 24x - 16}{x^3 + 2x - 12}$

Solution.

$$\lim_{x\to 2} \frac{x^6 - 24x - 16}{x^3 + 2x - 12} = \lim_{x\to 2} \frac{(x-2)(x^5 + 2x^4 + 4x^3 + 8x^2 + 16x + 8)}{(x^2 + 2x + 6)(x-2)}$$

$$= \frac{168}{14} = \mathbf{12}$$

2. Rationalization /Double Rationalization.

We can rationalize the irrational expression by multiplying with their conjugates to remove the indeterminancy.

Solved Example # 6 $\lim_{x\to 1} \frac{4 - \sqrt{5x+1}}{2 - \sqrt{3x+1}}$.

Solution. $\lim_{x\to 1} \frac{4 - \sqrt{5x+1}}{2 - \sqrt{3x+1}}$

$$= \lim_{x\to 1} \frac{(4 - \sqrt{5x+1})(2 + \sqrt{3x+1})(4 + \sqrt{5x+1})}{(2 - \sqrt{3x+1})(4 + \sqrt{5x+1})(2 + \sqrt{3x+1})}$$

$$= \lim_{x\to 1} \frac{(15 - 5x)}{(3 - 3x)} \times \frac{2 + \sqrt{3x+1}}{4 + \sqrt{5x+1}} = \frac{5}{6}$$

Solved Example # 7

Evaluate : (i) $\lim_{x\to 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x^3 - 3x^2 + 2x}\right]$ (ii) $\lim_{x\to 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$

(iii) $\lim_{x\to 1} \frac{(2x-3)\left(\sqrt{x} - 1\right)}{2x^2 + x - 3}$

Solution (i) We have

$$\lim_{x\to 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x^3 - 3x^2 + 2x}\right] = \lim_{x\to 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x(x-1)(x-2)}\right]$$

$$= \lim_{x\to 2} \left[\frac{x(x-1) - 2(2x-3)}{x(x-1)(x-2)}\right]$$

$$= \lim_{x\to 2}\left[\frac{x^2-5x+6}{x(x-1)(x-2)}\right]$$

$$= \lim_{x\to 2}\left[\frac{(x-2)(x-3)}{x(x-1)(x-2)}\right] \qquad = \lim_{x\to 2}\left[\frac{x-3}{x(x-1)}\right] = -\frac{1}{2}$$

(ii) The given limit taken the form $\frac{0}{0}$ when $x \to 0$. Rationalising the numerator, we get

$$\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{x} = \lim_{x\to 0}\left[\frac{\sqrt{1+x}-\sqrt{1-x}}{x}\times\frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right]$$

$$= \lim_{x\to 0}\left[\frac{(1+x)-(1-x)}{x\left(\sqrt{1+x}+\sqrt{1-x}\right)}\right]$$

$$= \lim_{x\to 0}\left[\frac{2x}{x\left(\sqrt{1+x}+\sqrt{1-x}\right)}\right] \qquad = \lim_{x\to 0}\left[\frac{2}{\sqrt{1+x}+\sqrt{1-x}}\right] = \frac{2}{2} = 1$$

(iii) We have

$$\lim_{x\to 1}\left[\frac{(2x-3)\left(\sqrt{x}-1\right)}{2x^2+x-3}\right] = \lim_{x\to 1}\left[\frac{(2x-3)\left(\sqrt{x}-1\right)}{(2x+3)(x-1)}\right]$$

$$= \lim_{x\to 1}\left[\frac{(2x-3)\left(\sqrt{x}-1\right)}{(2x+3)\left(\sqrt{x}-1\right)\left(\sqrt{x}+1\right)}\right]$$

$$= \lim_{x\to 1}\left[\frac{2x-3}{(2x+3)\left(\sqrt{x}+1\right)}\right] \qquad = \frac{-1}{(5)(2)} = \frac{-1}{10}$$

4. Fundamental Theorems on Limits:

Let $\underset{x\to a}{\text{Limit}}\ f(x) = \ell$ & $\underset{x\to a}{\text{Limit}}\ g(x) = m$. If ℓ & m exists then:

(i) $\underset{x\to a}{\text{Limit}}\ \{f(x) \pm g(x)\} = \ell \pm m$ (ii) $\underset{x\to a}{\text{Limit}}\ \{f(x).\ g(x)\} = \ell . m$

(iii) $\underset{x\to a}{\text{Limit}}\ \frac{f(x)}{g(x)} = \frac{\ell}{m}$, provided $m \neq 0$

(iv) $\underset{x\to a}{\text{Limit}}\ k f(x) = k \underset{x\to a}{\text{Limit}}\ f(x)$; where k is a constant.

(v) $\underset{x\to a}{\text{Limit}}\ f[g(x)] = f\left(\underset{x\to a}{\text{Limit}}\, g(x)\right) = f(m)$; provided f is continuous at $g(x) = m$.

Solved Example # 8 Evaluate

(i) $\lim_{x\to 2}(x+2)$ (ii) $\lim_{x\to 2} x(x-1)$ (iii) $\lim_{x\to 2}\frac{x^2+4}{x+2}$ (iv) $\lim_{x\to 0}\cos(\sin x)$

(v) $\lim_{x\to 1}\frac{x^2-3x+2}{x^2-1}$ (vi) $\lim_{x\to 1}\frac{x^2+3x+2}{x^2-1}$

Solution (i) $x + 2$ being a polynomial in x, its limit as $x \to 2$ is given by $\lim_{x\to 2}(x+2) = 2 + 2 = 4$

(ii) Again $x(x-1)$ being a polynomial in x, its limit as $x \to 2$ is given by

$$\lim_{x\to 2} x(x-1) = 2\,(2-1) = 2$$

(iii) By (II) above, we have $\lim_{x\to 2}\frac{x^2+4}{x+2} = \frac{(2)^2+4}{2+2} = 2$

(iv) $\lim_{x\to 0}\cos(\sin x) = \cos\left(\lim_{x\to 0}\sin x\right) = \cos 0 = 1$

(v) Note that for $x = 1$ both the numerator and the denominator of the given fraction vanish. Therefore by (III) above, we have $\lim_{x\to 1}\frac{x^2-3x+2}{x^2-1} = \lim_{x\to 1}\frac{(x-1)(x-2)}{(x-1)(x+1)} = \lim_{x\to 1}\frac{x-2}{x+1} = -\frac{1}{2}$

(vi) Note that for $x = 1$, the numerator of the given expression is a non-zero constant 6 and the denominator is zero. Therefore, the given limit is of the form $\frac{6}{0}$. Hence, by (IV) above, we conclude that $\lim_{x\to 1}\frac{x^2+3x+2}{x^2-1}$ does not exist

5. Standard Limits:

(a) $\underset{x\to 0}{\text{Limit}}\ \frac{\sin x}{x} = 1 = \underset{x\to 0}{\text{Limit}}\ \frac{\tan x}{x} = \underset{x\to 0}{\text{Limit}}\ \frac{\tan^{-1}x}{x} = \underset{x\to 0}{\text{Limit}}\ \frac{\sin^{-1}x}{x}$

[Where x is measured in radians]

(b) $\underset{x\to 0}{\text{Limit}}\ (1+x)^{1/x} = e$; $\underset{x\to\infty}{\text{Limit}}\left(1+\frac{1}{x}\right)^x = e$

(c) $\lim\limits_{x\to 0} \dfrac{e^x-1}{x} = 1$; $\lim\limits_{x\to 0} \dfrac{a^x-1}{x} = \log_e a,\ a > 0$

(d) $\lim\limits_{x\to 0} \dfrac{\ell n(1+x)}{x} = 1$ (e) $\lim\limits_{x\to a} \dfrac{x^n-a^n}{x-a} = na^{n-1}$.

Solved Example # 9: Find $\lim\limits_{x\to 0} \dfrac{\sin 2x}{x}$

Solution. $\lim\limits_{x\to 0} \dfrac{\sin 2x}{x} \Rightarrow \lim\limits_{x\to 0} \dfrac{\sin 2x}{2x} \cdot 2 = 2$

Solved Example # 10: $\lim\limits_{x\to 0} \dfrac{e^{3x}-1}{x/2}$

Solution. $\lim\limits_{x\to 0} \dfrac{e^{3x}-1}{x/2} \quad \lim\limits_{x\to 0} 2\times 3\ \dfrac{e^{3x}-1}{3x} = -6.$

Solved Example # 11 $\lim\limits_{x\to 0} \dfrac{\tan x-\sin x}{x^3}$

Solution. $\lim\limits_{x\to 0} \dfrac{\tan x-\sin x}{x^3}$

$= \lim\limits_{x\to 0} \dfrac{\tan x(1-\cos x)}{x^3}$

$= \lim\limits_{x\to 0} \dfrac{\tan x \cdot 2\sin^2 \frac{x}{2}}{x^3} = \lim\limits_{x\to 0} \dfrac{\tan x}{x} \cdot \left(\dfrac{\sin \frac{x}{2}}{\frac{x}{2}}\right)^2 = 1.$

Solved Example # 12 Compute $\lim\limits_{x\to 0} \dfrac{\sin 2x}{\sin 3x}$

Solution We have $\lim\limits_{x\to 0} \dfrac{\sin 2x}{\sin 3x} = \lim\limits_{x\to 0} \left[\dfrac{\sin 2x}{2x} \cdot \dfrac{2x}{3x} \cdot \dfrac{3x}{\sin 3x}\right]$

$= \left[\lim\limits_{2x\to 0} \dfrac{\sin 2x}{2x}\right] \cdot \dfrac{2}{3} \cdot \left[\lim\limits_{3x\to 0} \dfrac{3x}{\sin 2x}\right],\ x \neq 0$

$= 1 \cdot \dfrac{2}{3} + \left[\lim\limits_{3x\to 0} \dfrac{\sin 3x}{3x}\right] = \dfrac{2}{3} \times 1 = \dfrac{2}{3}$

Solved Example # 13 Evaluate $\lim\limits_{x\to\infty} \left(1+\dfrac{2}{x}\right)^x$

Solution $\lim\limits_{x\to\infty} \left(1+\dfrac{2}{x}\right)^x = e^{\lim\limits_{x\to\infty} \frac{2}{x}.x} = e^2.$

Solved Example # 14 Compute (i) $\lim\limits_{x\to 3} \dfrac{e^x-e^3}{x-3}$ (ii) $\lim\limits_{x\to 0} \dfrac{x(e^x-1)}{1-\cos x}$

Solution (i) Put $y = x - 3$. So, as $x \to 3$, $y \to 0$. Thus

$\lim\limits_{x\to 3} \dfrac{e^x-e^3}{x-3} = \lim\limits_{y\to 0} \dfrac{e^{3+y}-e^3}{y}$

$= \lim\limits_{y\to 0} \dfrac{e^3 \cdot e^y - e^3}{y}$

$= e^3 \lim\limits_{y\to 0} \dfrac{e^y-1}{y} = e^3 \cdot 1 = e^3$

(ii) We have

(ii) $\lim\limits_{x\to 0} \dfrac{x(e^x-1)}{1-\cos x} = \lim\limits_{x\to 0} \dfrac{x(e^x-1)}{2\sin^2 \frac{x}{2}}$

$= \dfrac{1}{2} \cdot \lim\limits_{x\to 0} \left[\dfrac{e^x-1}{x} \cdot \dfrac{x^2}{\sin^2 \frac{x}{2}}\right] = 2.$

Solved Example # 15 Evaluate $\lim\limits_{x\to 2} \dfrac{x^3-8}{x^2-4}$

Solution **(First Method)**

The given expression is of the form

$\dfrac{x^3-(2)^3}{x^2-(2)^2} = \dfrac{x^3-(2)^3}{x-2} \div \dfrac{x^2-(2)^2}{x-2}$

$\Rightarrow \quad \lim\limits_{x\to 2} \dfrac{x^3-8}{x^2-4} = \lim\limits_{x\to 2} \dfrac{x^3-(2)^3}{x-2} \div \lim\limits_{x\to 2} \dfrac{x^2-(2)^2}{x-2}$

$= 3(2^2) \div 2(2^1)$ (using $\lim\limits_{x\to a} \dfrac{x^n-a^n}{x-a} = na^{n-1}$)

$= 12 \div 4 = 3$

(Second Method)

The numerator and denominator have a common factor $(x-2)$. Cancelling this factor, we obtain

$$\frac{x^3-8}{x^2-4} = \frac{x^2+2x+4}{x+2} \quad \Rightarrow \quad \lim_{x\to 2} \frac{x^3-8}{x^2-4} = \lim_{x\to 2} \frac{x^2+2x+4}{x+2}$$

$$= \frac{(2)^2+2(2)+4}{2+2} = \frac{12}{4} = 3$$

Note : Since $x \to 2$, $x-2$ is not zero, so the cancellation of the factor $x-2$ in the above example is carried out.

6. Use of Subsitution in Solving Limit Problems

Sometimes in solving limit problem we convert $\lim\limits_{x\to a} f(x)$ by subtituting $x = a + h$ or $x = a - h$ as $\lim\limits_{h\to 0} f(a+h)$ or $\lim\limits_{h\to 0} f(a-h)$ according as need of the problem.

Solved Example # 16 $\lim\limits_{x\to \pi/4} \dfrac{1-\tan x}{1-\sqrt{2}\sin x}$

Solution. Put $x = \dfrac{\pi}{4} + h$ $\because$ $x \to \dfrac{\pi}{4} \Rightarrow h \to 0$

$$\lim_{h\to 0} \frac{1-\tan\left(\frac{\pi}{4}+h\right)}{1-\sqrt{2}\sin\left(\frac{\pi}{4}+h\right)}$$

$$= \lim_{h\to 0} = \frac{1-\dfrac{1-\tan h}{1-\tan h}}{1-\sin h-\cos h}$$

$$= \lim_{h\to 0} \frac{\dfrac{-2\tan h}{1-\tan h}}{2\sin^2\dfrac{h}{2}-2\sin\dfrac{h}{2}\cos\dfrac{h}{2}}$$

$$= \lim_{h\to 0} \frac{-2\tan h}{2\sin^2\dfrac{h}{2}\left[2\sin\dfrac{h}{2}-\cos\dfrac{h}{2}\right]} \frac{1}{(1-\tanh)}$$

$$= \lim_{h\to 0} \frac{-2\dfrac{\tanh}{h}}{\dfrac{\sin\frac{h}{2}}{\frac{h}{2}}\left[\sin\dfrac{h}{2}-\cos\dfrac{h}{2}\right]} \frac{1}{(1-\tanh)} = \frac{-2}{-1} = 2.$$

7. Limit When $x \to \infty$

Since $x \to \infty \quad \Rightarrow \quad \dfrac{1}{x} \to 0$ hence in this type of problem we express most of the part of expression in terms of $\dfrac{1}{x}$ and apply $\dfrac{1}{x} \to 0$. We can see this approch in the given solve examples.

Solved Example # 17 $\lim\limits_{x\to\infty} x \sin \dfrac{1}{x}$

Solution. $\lim\limits_{x\to\infty} x \sin \dfrac{1}{x}$

$= \lim\limits_{x\to\infty} \dfrac{\sin 1/x}{1/x} = 1$

Solved Example # 18 $\lim\limits_{x\to\infty} \dfrac{x-2}{2x-3}$

Solution. $\lim\limits_{x\to\infty} \dfrac{x-2}{2x-3}$

$\lim\limits_{x\to\infty} \dfrac{1-2/x}{2-3/x} = \dfrac{1}{2}.$

Solved Example # 19 $\lim\limits_{x\to\infty} \dfrac{x^2-4x+5}{3x^2-x^3+2}$

Solution. $\lim\limits_{x\to\infty} \dfrac{x^2-4x+5}{3x^2-x^3+2}$

$= \lim\limits_{x\to\infty} \dfrac{\frac{1}{x}-\frac{4}{x^2}+\frac{5}{x^3}}{\frac{3}{x}-1+\frac{2}{x^3}} = 0$

Solved Example # 20 $\lim\limits_{x\to-\infty} \dfrac{\sqrt{3x^2+2}}{x-2}$

Solution. $\lim\limits_{x\to-\infty} \dfrac{\sqrt{3x^2+2}}{x-2}$

Put $x = \dfrac{-1}{t}$ $\quad x\to-\infty$ $\quad t\to 0^+$

$= \lim\limits_{x\to 0^+} \dfrac{\sqrt{3+2t^2}.\frac{1}{\sqrt{t^2}}}{\frac{1-2t}{t}}$

$= \lim\limits_{x\to 0^+} \dfrac{\sqrt{3+2t^2}}{-(1+2t)} \dfrac{t}{|t|} = \dfrac{\sqrt{3}}{-1} = -\sqrt{3}$.

8. Limits Using Expansion

(i) $a^x = 1+\dfrac{x\ln a}{1!}+\dfrac{x^2\ln^2 a}{2!}+\dfrac{x^3\ln^3 a}{3!}+\ldots\ldots\ldots a>0$ (ii) $e^x = 1+\dfrac{x}{1!}+\dfrac{x^2}{2!}+\dfrac{x^3}{3!}+\ldots\ldots$

(iii) $\ln(1+x) = x-\dfrac{x^2}{2}+\dfrac{x^3}{3}-\dfrac{x^4}{4}+\ldots\ldots\ldots \text{for } -1<x\le 1$ (iv) $\sin x = x-\dfrac{x^3}{3!}+\dfrac{x^5}{5!}-\dfrac{x^7}{7!}+\ldots\ldots$

(v) $\cos x = 1-\dfrac{x^2}{2!}+\dfrac{x^4}{4!}-\dfrac{x^6}{6!}+\ldots\ldots$ (vi) $\tan x = x+\dfrac{x^3}{3}+\dfrac{2x^5}{15}+\ldots\ldots$

(vii) $\tan^{-1}x = x-\dfrac{x^3}{3}+\dfrac{x^5}{5}-\dfrac{x^7}{7}+\ldots\ldots$ (viii) $\sin^{-1}x = x+\dfrac{1^2}{3!}x^3+\dfrac{1^2.3^2}{5!}x^5+\dfrac{1^2.3^2.5^2}{7!}x^7+\ldots\ldots$

(ix) $\sec^{-1}x = 1+\dfrac{x^2}{2!}+\dfrac{5x^4}{4!}+\dfrac{61x^6}{6!}+\ldots\ldots$

(x) for $|x|<1$, $n\in R$ $(1+x)^n = 1+nx+\dfrac{n(n-1)}{1.2}x^2+\dfrac{n(n-1)(n-2)}{1.2.3}x^3+\ldots\ldots\ldots\infty$

Solved Example # 21 $\lim\limits_{x\to 0} \dfrac{e^x-1-x}{x^2}$

Solution. $\lim\limits_{x\to 0} \dfrac{e^x-1-x}{x^2}$

$= \lim\limits_{x\to 0} \dfrac{\left(1+x+\frac{x^2}{2!}\ldots\ldots\right)-1-x}{x^2} = \dfrac{1}{2}$

Solved Example # 22 $\lim\limits_{x\to 0} \dfrac{\tan x-\sin x}{x^3}$

Solution. $\lim\limits_{x\to 0} \dfrac{\tan x-\sin x}{x^3}$

$= \lim\limits_{x\to 0} \dfrac{\left(x+\frac{x^3}{3}\ldots\ldots\right)-\left(x-\frac{x^3}{3!}\ldots\ldots\right)}{x^3} = \dfrac{1}{3}+\dfrac{1}{6} = \dfrac{1}{2}$.

Solved Example # 23 $\lim\limits_{x\to 0} \dfrac{(7+x)^{1/3}-2}{x-1}$

Solution. Put $x\to 1+h$ $\lim\limits_{h\to 0} \dfrac{(8+h)^{1/3}-2}{h}$

$\lim\limits_{h\to 0} \dfrac{2.\left(1+\frac{h}{8}\right)^{1/3}-2}{h}$

$= \lim\limits_{h\to 0} \dfrac{2\left\{1+\frac{1}{3}.\frac{h}{8}+\frac{\frac{1}{3}\left(\frac{1}{3}-1\right)\left(\frac{h}{8}\right)^2}{1.2}+\ldots\ldots-1\right\}}{h}$

$= \lim_{h \to 0} 2 \times \frac{1}{24} = \frac{1}{12}$

Solved Example # 24 $\lim_{x \to 0} \frac{\ell n(1+x) - \sin x + \frac{x^2}{2}}{x \tan x \sin x}$

$$\lim_{x \to 0} \frac{\ell n(1+x) - \sin x + \frac{x^2}{2}}{x \tan x \sin x}$$

$$= \lim_{x \to 0} \frac{\left(x - \frac{x^2}{2} + \frac{x^3}{3} .....\right) - \left(x + \frac{x^3}{3!} + \frac{x^5}{5!} + ...\right) + \frac{x^2}{2}}{x^3 \cdot \frac{\tan x}{x} \cdot \frac{\sin x}{x}} \qquad = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

9. Limits of form 1^∞, 0^0, ∞^0

All these forms can be convered into $\frac{0}{0}$ form in the following ways

(i) If $x \to 1$, $y \to \infty$, then $z = (x)^y$

$\Rightarrow \quad \ell n\, z = y\, \ell n\, x \quad \Rightarrow \quad \ell n\, z = \frac{\ell n x}{(1/y)}$

Since $y \to \infty$ hence $\frac{1}{y} \to 0$ and $x \to 1$ hence $\ell n x \to 0$

(ii) If $x \to 0$, $y \to 0$, then $z = x^y \quad \Rightarrow \quad \ell n\, z = y\, \ell n\, x$

$\Rightarrow \quad \ell n\, z = \frac{y}{1/\ell n y} = \frac{0}{0}$ form

(iii) If $x \to \infty$, $y \to 0$, then $z = x^y \quad \Rightarrow \quad \ell n\, z = y\, \ell n\, x$

$\Rightarrow \quad \ell n\, z = \frac{y}{1/\ell n x} = \frac{0}{0}$ form

also for $(1)^\infty$ type of problems we can use following rules.

(i) $\lim_{x \to 0} (1 + x)^{1/x} = e$

(ii) $\lim_{x \to a} [f(x)]^{g(x)}$ where $f(x) \to 1$; $g(x) \to \infty$ as $x \to a$

$= \lim_{x \to a} [1 + f(x) - 1]^{\frac{1}{f(x)-1}\{f(x)-1\} \cdot g(x)} \qquad = e^{\lim_{x \to a} [f(x)-1]\, g(x)}$

Solved Example # 25 $\lim_{x \to \infty} \left(\frac{2x^2 - 1}{2x^2 + 3}\right)^{4x^2+2}$

Solution. Since it is in the form of 1^∞

$\lim_{x \to \infty} \left(\frac{2x^2 - 1}{2x^2 + 3}\right)^{4x^2+2} = e^{\lim_{x \to \infty} \left(\frac{2x^2 - 1 - 2x^2 - 3}{2x^2 + 3}\right)(4x^2 + 2)} = e^{-8}$

Solved Example # 26: $\lim_{x \to \frac{\pi}{4}} (\tan x)^{\tan 2x}$

Solution $= e^{\lim_{x \to \frac{\pi}{4}} (\tan x - 1) \tan 2x}$

$= e^{\lim_{x \to \frac{\pi}{4}} (\tan x - 1) \frac{2 \tan x}{1 - \tan^2 x}}$

$= e^{2 \times \frac{\tan \pi/4}{-1(1 + \tan \pi/4)}} \qquad = e^{-1} = \frac{1}{e}$

Solved Example # 27 Evaluate $\lim_{x \to a} \left(2 - \frac{a}{x}\right)^{\tan \frac{\pi x}{2a}}$.

Solution. $\lim_{x \to a} \left(2 - \frac{a}{x}\right)^{\tan \frac{\pi x}{2a}}$

put $x = a + h \quad \Rightarrow \quad \lim_{h \to 0} \left(1 + \frac{h}{(a+h)}\right)^{\tan\left(\frac{\pi}{2} + \frac{\pi h}{2a}\right)}$

$\Rightarrow \quad \lim_{h \to 0} \left(1 + \frac{h}{a+h}\right)^{-\cot\left(\frac{\pi h}{2a}\right)} \quad \Rightarrow \quad e^{\lim_{h \to 0} -\cot \frac{\pi h}{2a} \cdot \left(1 + \frac{h}{a+h} - 1\right)}$

$\Rightarrow \quad e^{\lim_{h \to 0} -\left(\frac{\frac{\pi h}{2a}}{\tan \frac{\pi h}{2a}}\right) \cdot \frac{2a}{\pi} \cdot \frac{1}{a+h}} = e^{-2/\pi}$

Solved Example # 28: $\lim_{x\to 0^+} x^x$

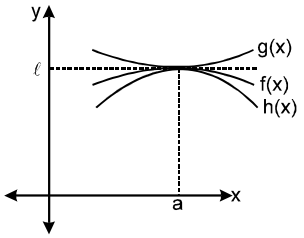
Solution. $y = \lim_{x\to 0} x^x$

$\ell n\, y = \lim_{x\to 0} x\, \ell n\, x$

$= \lim_{x\to 0} -\dfrac{\ell n \frac{1}{x}}{\frac{1}{x}} = 0 \because \quad \dfrac{1}{x} \to \infty \qquad y = 1$

10. Sandwich Theorem or Squeeze Play Theorem:

If $f(x) \le g(x) \le h(x)\ \forall\ x$ & $\underset{x\to a}{\text{Limit}}\ f(x) = \ell = \underset{x\to a}{\text{Limit}}\ h(x)$ then $\underset{x\to a}{\text{Limit}}\ g(x) = \ell$.



Solved Example # 29: Evaluate $\lim_{n\to\infty} \dfrac{[x]+[2x]+[3x]+....+[nx]}{n^2}$

Where [ ] denotes the greatest integer function.

Solution.

We know that, $x - 1 < |x| \le x$

$\Rightarrow \quad 2x - 1 < [2x] \le 2x$

$\Rightarrow \quad 3x - 1 < [3x] \le 3x$

.....

.....

$\Rightarrow \quad nx - 1 < [nx] \le nx$

$\therefore \quad (x + 2x + 3x + ... + nx) - n < [x] + [2x] + ... + [nx] \le (x + 2x + ... + nx)$

$\Rightarrow \quad \dfrac{xn(n+1)}{2} - n < \sum_{r=1}^{n} [r\ x] \le \dfrac{x.n(n+1)}{2}$

Thus, $\lim_{n\to\infty} \dfrac{[x]+[2x]+....+[nx]}{n^2}$

$\Rightarrow \quad \lim_{n\to\infty} \dfrac{x}{2}\left(1+\dfrac{1}{n}\right) - \dfrac{1}{n} < \lim_{n\to\infty} \dfrac{[x]+[2x]+....+[nx]}{n^2} \le \lim_{n\to\infty} \dfrac{x}{2}\left(1+\dfrac{1}{n}\right)$

$\Rightarrow \quad \dfrac{x}{2} < \lim_{n\to\infty} \dfrac{[x]+[2x]+....+[nx]}{n^2} \le \dfrac{x}{2} \qquad \Rightarrow \quad \lim_{n\to\infty} \dfrac{[x]+[2x]+....+[nx]}{n^2} = \dfrac{x}{2}$

Aliter We know that $[x] = x - \{x\}$

$$\sum_{r=1}^{n} r\ x = [x] + [2x] + ... + nx - [nx]$$
$$= (x + 2x + 3x + ... + nx) - (\{x\} + \{2x\} + ... + \{nx\})$$
$$= \frac{xn(n+1)}{2} - (\{x\} + \{2x\} + .. + \{nx\})$$

$\therefore \quad \dfrac{1}{n^2}\sum_{r=1}^{n}[r\ x] = \dfrac{x}{2}\left(1+\dfrac{1}{n}\right) - \dfrac{\{x\}+\{2x\}+....+\{nx\}}{n^2}$

Since, $0 \le \{rx\} < 1, \quad \therefore \quad 0 \le \sum_{r=1}^{n}[r\ x] < n$

$\Rightarrow \quad \lim_{n\to\infty} \dfrac{\sum_{r=1}^{n}[rx]}{n^2} = 0 \qquad \therefore \quad \lim_{n\to\infty} \dfrac{\sum_{r=1}^{n}[rx]}{n^2} = \lim_{n\to\infty} \dfrac{x}{2}\left(1+\dfrac{1}{n}\right) - \lim_{n\to\infty} \dfrac{\sum_{r=1}^{n}\{rx\}}{n^2}$

$$\lim_{n\to\infty} \frac{\sum_{r=1}^{n}[rx]}{n^2} = \frac{x}{2}$$

Solved Example # 30 $\underset{x\to 0}{\text{limit}}\ x \sin\dfrac{1}{x}$

Solution. $\underset{x\to 0}{\text{limit}}\ x \sin\dfrac{1}{x}$

$= 0 \times$ (some value in $[-1, 1]$) $= 0$

11. Some Important Notes :

(i) $\lim\limits_{x\to\infty} \frac{\ell nx}{x} = 0$ (ii) $\lim\limits_{x\to\infty} \frac{x}{e^x} = 0$

As $x \to \infty$, $\ell n\, x$ increnes much slower than any (+ve) power of x where e^x increases much faster than (+ve) power of x

(iii) $\underset{n\to\infty}{\text{Limit}}\ (1-h)^n = 0$ & $\underset{n\to\infty}{\text{Limit}}\ (1+h)^n \to \infty$, where $h > 0$.

(iv) If $\underset{x\to a}{\text{Limit}}\ f(x) = A > 0$ & $\underset{x\to a}{\text{Limit}}\ \phi(x) = B$ (a finite quantity) then;

$\underset{x\to a}{\text{Limit}}\ [f(x)]^{\phi(x)} = e^z$ where $z = \underset{x\to a}{\text{Limit}}\ \phi(x).\ \ln[f(x)] = e^{B\ln A} = A^B$

Solved Example # 31 $\lim\limits_{x\to\infty} \frac{x^{1000}}{e^x}$

Solution. $\lim\limits_{x\to\infty} \frac{x^{1000}}{e^x} = 0$

Short Revesion (LIMIT)

THINGS TO REMEMBER :

1. Limit of a function f(x) is said to exist as, $x \to a$ when

$\underset{x\to a^-}{\text{Limit}}\ f(x) = \underset{x\to a^+}{\text{Limit}}\ f(x) =$ finite quantity.

2. **FUNDAMENTAL THEOREMS ON LIMITS :**

Let $\underset{x\to a}{\text{Limit}}\ f(x) = l$ & $\underset{x\to a}{\text{Limit}}\ g(x) = m$. If l & m exists then :

(i) $\underset{x\to a}{\text{Limit}}\ f(x) \pm g(x) = l \pm m$ **(ii)** $\underset{x\to a}{\text{Limit}}\ f(x)\cdot g(x) = l.m$

(iii) $\underset{x\to a}{\text{Limit}}\ \frac{f(x)}{g(g)} = \frac{\ell}{m}$, provided $m \neq 0$

(iv) $\underset{x\to a}{\text{Limit}}\ k\,f(x) = k\ \underset{x\to a}{\text{Limit}}\ f(x)$; where k is a constant.

(v) $\underset{x\to a}{\text{Limit}}\ f[g(x)] = f\left(\underset{x\to a}{\text{Limit}}\ g(x)\right) = f(m)$; provided f is continuous at $g(x) = m$.

For example $\underset{x\to a}{\text{Limit}}\ l\,n\,(f(x) = l\,n\left[\underset{x\to a}{\text{Limit}}\ f(x)\right]\ l\,n\,l\ (l>0)$. **REMEMBER**

$$\underset{x\to a}{\text{Limit}} \Rightarrow x \neq a$$

3. **STANDARD LIMITS :**

(a) $\underset{x\to 0}{\text{Limit}}\ \frac{\sin x}{x} = 1 = \underset{x\to 0}{\text{Limit}}\ \frac{\tan x}{x} = \underset{x\to 0}{\text{Limit}}\ \frac{\tan^{-1}x}{x} = \underset{x\to 0}{\text{Limit}}\ \frac{\sin^{-1}x}{x}$

[Where x is measured in radians]

(b) $\underset{x\to 0}{\text{Limit}}\ (1+x)^{1/x} = e = \underset{x\to\infty}{\text{Limit}}\ \left(1+\frac{1}{x}\right)^x$ note however there $\underset{\substack{h\to 0\\ n\to\infty}}{\text{Limit}}\ (1-h)^n = 0$

and $\underset{\substack{h\to 0\\ n\to\infty}}{\text{Limit}}\ (1+h)^n \to \infty$

(c) If $\underset{x\to a}{\text{Limit}}\ f(x) = 1$ and $\underset{x\to a}{\text{Limit}}\ \phi(x) = \infty$, then ;

$\underset{x\to a}{\text{Limit}}\ [f(x)]^{\phi(x)} = e^{\underset{x\to a}{\text{Limit}}\ \phi(x)[f(x)-1]}$

(d) If $\underset{x\to a}{\text{Limit}}\ f(x) = A > 0$ & $\underset{x\to a}{\text{Limit}}\ \phi(x) = B$ (a finite quantity) then ;

$\underset{x\to a}{\text{Limit}}\ [f(x)]^{\phi(x)} = e^z$ where $z = \underset{x\to a}{\text{Limit}}\ \phi(x).\ \ln[f(x)] = e^{B\ln A} = A^B$

(e) $\underset{x\to 0}{\text{Limit}}\ \frac{a^x-1}{x} = 1n\,a\ (a>0)$. In particular $\underset{x\to 0}{\text{Limit}}\ \frac{e^x-1}{x} = 1$

(f) $\underset{x\to a}{\text{Limit}}\ \frac{x^n-a^n}{x-a} = n\,a^{n-1}$

4. **SQUEEZE PLAY THEOREM :** If $f(x) \le g(x) \le h(x)\ \forall\ x$ & $\underset{x\to a}{\text{Limit}}\ f(x) = l = \underset{x\to a}{\text{Limit}}\ h(x)$ then $\underset{x\to a}{\text{Limit}}\ g(x) = l$.

5. **INDETERMINANT FORMS :** $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0\times\infty$, 0° , ∞° , $\infty-\infty$ and 1^∞

Note : (i) We cannot plot ∞ on the paper. Infinity (∞) is a symbol & not a number. It does not obey the laws of elementry algebra. **(ii)** $\infty + \infty = \infty$ **(iii)** $\infty \times \infty = \infty$

(iv) $(a/\infty) = 0$ if a is finite **(v)** $\frac{a}{0}$ is not defined , if $a \neq 0$.

(vi) $a\,b = 0$, if & only if $a = 0$ or $b = 0$ and a & b are finite.

6. The following strategies should be born in mind for evaluating the limits:

(a) Factorisation **(b)** Rationalisation or double rationalisation

(c) Use of trigonometric transformation ;

appropriate substitution and using standard limits

(d) Expansion of function like Binomial expansion, exponential & logarithmic expansion, expansion of sinx , cosx , tanx should be remembered by heart & are given below :

(i) $a^x = 1 + \frac{x \ln a}{1!} + \frac{x^2 \ln^2 a}{2!} + \frac{x^3 \ln^3 a}{3!} + \ldots\ldots\ldots a > 0$ **(ii)** $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots\ldots\ldots$

(iii) $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots\ldots\ldots \text{for} -1 < x \le 1$ **(iv)** $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots\ldots\ldots$

(v) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots\ldots$ **(vi)** $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \ldots\ldots\ldots$

(vii) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots\ldots\ldots$

(viii) $\sin^{-1} x = x + \frac{1^2}{3!}x^3 + \frac{1^2 . 3^2}{5!}x^5 + \frac{1^2 . 3^2 . 5^2}{7!}x^7 + \ldots\ldots\ldots$ **(ix)** $\sec^{-1} x = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \frac{61x^6}{6!} + \ldots\ldots$

EXERCISE–1

Q 1. $\lim\limits_{x \to 1} \frac{x^2 - \sqrt{x}}{\sqrt{x} - 1}$ **Q 2.** $\lim\limits_{x \to 1} \frac{\sqrt[13]{x} - \sqrt[7]{x}}{\sqrt[5]{x} - \sqrt[3]{x}}$ **Q3.** $\lim\limits_{x \to 1} \frac{x^2 - x.\ln x + \ln x - 1}{x - 1}$

Q 4. $\lim\limits_{x \to 1} \left(\frac{p}{1 - x^p} - \frac{q}{1 - x^q} \right) p, q \in N$ **Q 5.** $\lim\limits_{x \to \infty} \frac{2\sqrt{x} + 3x^{1/3} + 5x^{1/5}}{\sqrt{3x - 2} + (2x - 3)^{1/3}}$ **Q 6.** $\lim\limits_{x \to \frac{3\pi}{4}} \frac{1 + \sqrt[3]{\tan x}}{1 - 2\cos^2 x}$

Q 7. (a) $\lim\limits_{x \to 0} \tan^{-1} \frac{a}{x^2}$ where $a \in R$

(b) Plot the graph of the function $f(x) = \lim\limits_{t \to 0} \left(\frac{2x}{\pi} \tan^{-1} \frac{x}{t^2} \right)$

Q 8. $\lim\limits_{x \to 1} \frac{\left[\sum\limits_{K=1}^{100} x^k \right] - 100}{x - 1}$

Q 9. Find the sum of an infinite geometric series whose first term is the limit of the function

$f(x) = \frac{\tan x - \sin x}{\sin^3 x}$ as $x \to 0$ and whose common ratio is the limit of the function

$g(x) = \frac{1 - \sqrt{x}}{(\cos^{-1} x)^2}$ as $x \to 1$. (Use of series expansion or L' Hospital's rule is not allowed.)

Q 10. $\lim\limits_{x \to \infty} (x - l \, n \cosh x)$ where $\cosh t = \frac{e^t + e^{-t}}{2}$.

Q 11. $\lim\limits_{x \to \frac{\pi}{2}} \cos^{-1} [\cot x]$ where [ ] denotes greatest integer function **Q 12.** $\lim\limits_{x \to \frac{\pi}{4}} \frac{1 - \tan x}{1 - \sqrt{2} \sin x}$

Q 13. $\lim\limits_{x \to 0} [l n (1 + \sin^2 x) . \cot(l n^2 (1 + x))]$ **Q 14.** $\lim\limits_{x \to 0} \frac{\sec 4x - \sec 2x}{\sec 3x - \sec x}$

Q 15. $\lim\limits_{\theta \to \frac{\pi}{4}} \frac{\sqrt{2} - \cos\theta - \sin\theta}{(4\theta - \pi)^2}$ **Q 16.** $\lim\limits_{x \to \frac{\pi}{2}} \frac{2^{-\cos x} - 1}{x(x - \frac{\pi}{2})}$

Q 17. If $\lim\limits_{x \to 0} \frac{a \sin x - \sin 2x}{\tan^3 x}$ is finite then find the value of 'a' & the limit.

Q 18. $\lim\limits_{x \to 0} \frac{8}{x^8} \left[1 - \cos\frac{x^2}{2} - \cos\frac{x^2}{4} + \cos\frac{x^2}{2} \cos\frac{x^2}{4} \right]$ **Q 19.** $\lim\limits_{x \to 1} \frac{(\ln(1+x) - \ln 2)(3.4^{x-1} - 3x)}{[(7+x)^{\frac{1}{3}} - (1+3x)^{\frac{1}{2}}] . \sin(x-1)}$

Q 20. Using Sandwich theorem to evaluate $\lim\limits_{n \to \infty} \left(\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \ldots\ldots\ldots + \frac{1}{\sqrt{n^2+2n}} \right)$

Q21. Given $f(x) = \lim\limits_{n \to \infty} \tan^{-1}(nx)$; $g(x) = \lim\limits_{n \to \infty} \sin^{2n} x$ and $\sin(h(x)) = \frac{1}{2} [\cos \pi(g(x)) + \cos(2 f(x))]$

Find the domain and range of h (x).

Q 22. $\lim\limits_{x \to \frac{\pi}{4}} \frac{\sqrt{1 - \sqrt{\sin 2x}}}{\pi - 4x}$ **Q 23.** $\lim\limits_{x \to 3} \frac{(x^3 + 27) \ln(x - 2)}{x^2 - 9}$

Q 24. $\lim\limits_{x \to 2} \frac{(\cos\alpha)^x + (\sin\alpha)^x - 1}{x - 2}$ **Q 25.** $\lim\limits_{x \to 0} \frac{27^x - 9^x - 3^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}}$

Q 26. Let $f(x)=\dfrac{x}{\sin x},\ x>0$ and $g(x)=x+3,\ x<1$

$=2-x,\ x\le 0$ $=x^2-2x-2,\ 1\le x<2$

$=x-5,\ x\ge 2$

find LHL and RHL of $g(f(x))$ at $x=0$ and hence find $\lim\limits_{x\to 0} g(f(x))$.

Q 27. Let $P_n=a^{P_{n-1}}-1,\ \forall\ n=2,3,.....$and Let $P_1=a^x-1$ where $a\in R^+$ then evaluate $\lim\limits_{x\to 0}\dfrac{P_n}{x}$.

Q 28. $\lim\limits_{x\to-\infty}\dfrac{(3x^4+2x^2)\sin\frac{1}{x}+|x|^3+5}{|x|^3+|x|^2+|x|+1}$

Q.29 If $f(x)=\ell n\operatorname{cosec}(x\pi)\quad 0<x<1$ and $g(x)=\dfrac{2^{f(x)}+1}{3^{f(x)}+1}$ then

$=\ell n\sin(2x\pi)\quad 1<x<3/2$

find $\tan^{-1}(g(1^-))$ and $\sec^{-1}(g(1^+))$.

Q.30 At the end-points and the midpoint of a circular arc AB tangent lines are drawn, and the points A and B are joined with a chord. Prove that the ratio of the areas of the two triangles thus formed tends to 4 as the arc AB decreases indefinitely.

EXERCISE–2

Q 1. $\lim\limits_{x\to\infty}\left[\dfrac{2x^2+3}{2x^2+5}\right]^{8x^2+3}$ **Q 2.** $\lim\limits_{x\to\infty}\left(\dfrac{x+c}{x-c}\right)^x=4$ then find c **Q 3.** $\lim\limits_{x\to 0}\left[\dfrac{(1+x)^{1/x}}{e}\right]^{1/x}$

Q 4. $\lim\limits_{x\to 0}\left[\sin^2\left(\dfrac{\pi}{2-ax}\right)\right]^{\sec^2\left(\frac{\pi}{2-bx}\right)}$ **Q.5** $\lim\limits_{x\to\infty} x^2\sin\left(\ell n\sqrt{\cos\dfrac{\pi}{x}}\right)$

Q 6. $\lim\limits_{x\to\infty}\left[\cos\left(2\pi\left(\dfrac{x}{1+x}\right)^a\right)\right]^{x^2}$ $a\in R$ **Q 7.** $\lim\limits_{x\to 1}\left(\tan\dfrac{\pi x}{4}\right)^{\tan\frac{\pi x}{2}}$

Q 8. $\lim\limits_{x\to 0}\left(\dfrac{x-1+\cos x}{x}\right)^{\frac{1}{x}}$ **Q 9.** $\lim\limits_{x\to\infty}\left(\dfrac{a_1^{\frac{1}{x}}+a_2^{\frac{1}{x}}+a_3^{\frac{1}{x}}+.....+a_n^{\frac{1}{x}}}{n}\right)^{nx}$ where $a_1,a_2,a_3,.....a_n>0$

Q 10. Let $f(x)=\dfrac{\sin^{-1}(1-\{x\}).\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}\ .(1-\{x\})}$ then find $\lim\limits_{x\to 0^+} f(x)$ and $\lim\limits_{x\to 0^-} f(x)$, where $\{x\}$ denotes the fractional part function.

Q 11. Find the values of a, b & c so that $\lim\limits_{x\to 0}\dfrac{ae^x-b\cos x+ce^{-x}}{x.\sin x}=2$

Q 12. $\lim\limits_{x\to a}\dfrac{1}{(a^2-x^2)^2}\left(\dfrac{a^2+x^2}{ax}-2\sin\left(\dfrac{a\pi}{2}\right)\sin\left(\dfrac{\pi x}{2}\right)\right)$ where a is an odd integer

Q 13. $\lim\limits_{x\to 0}\dfrac{\tan^2x-x^2}{x^2\tan^2x}$ **Q 14.** $\lim\limits_{n\to\infty}\dfrac{x^n f(x)+g(x)}{x^n+1}$ $x\ \varepsilon\ R$

Q 15. $\lim\limits_{n\to\infty}\dfrac{[1.x]+[2.x]+[3.x]+.....+[n.x]}{n^2}$, Where [.] denotes the greatest integer function.

Q 16. Without using series expansion or L'Hospital's rule evaluate, $\lim\limits_{x\to 1}\dfrac{1-x+\ell n\,x}{1+\cos\pi x}$

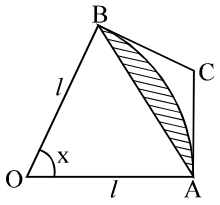
Q 17. $\lim\limits_{y\to 0}\left[\operatorname*{Limit}_{x\to\infty}\dfrac{\exp\left(x\ln(1+\frac{ay}{x})\right)-\exp\left(x\ln(1+\frac{by}{x})\right)}{y}\right]$

Q 18. If s_n be the sum of n terms of the series, $\sin x+\sin 2x+\sin 3x+.....+\sin nx$ then show that $\operatorname*{Limit}_{n\to\infty}\dfrac{s_1+s_2+.....+s_n}{n}=\dfrac{1}{2}\cot\dfrac{x}{2}$ $(x\ne 2k\pi,\ k\in I)$

Q 19. $\lim\limits_{x\to 0}\left[\dfrac{\ell n\,(1+x)^{1+x}}{x^2}-\dfrac{1}{x}\right]$

Q 20. Let $P_n=\dfrac{2^3-1}{2^3+1}.\dfrac{3^3-1}{3^3+1}.\dfrac{4^3-1}{4^3+1}.....\dfrac{n^3-1}{n^3+1}$. Evaluate $\lim\limits_{n\to\infty} P_n$

Q 21. A circular arc of radius 1 subtends an angle of x radians, $0<x<\dfrac{\pi}{2}$ as shown in the figure. The point C is the intersection of the two tangent lines at A & B. Let T(x) be the area of triangle ABC & let S(x) be the area of the shaded region. Compute :



(a) T(x) (b) S(x) & (c) the limit of $\frac{T(x)}{S(x)}$ as $x \to 0$.

Q 22. (a) $\lim\limits_{x\to\infty} \dfrac{\sqrt{x}}{\sqrt{x+\sqrt{x+\sqrt{x}}}}$ **(b)** $\lim\limits_{x\to\infty}\left[\sqrt{x+\sqrt{x+\sqrt{x}}}-\sqrt{x}\right]$

Q 23. If $f(n,\theta)=\prod_{r=1}^{n}\left(1-\tan^2\frac{\theta}{2^r}\right)$, then compute $\lim\limits_{n\to\infty} f(n,\theta)$

Q 24. Let $l=\lim\limits_{x\to a}\dfrac{x^x-a^x}{x-a}$ & $m=\lim\limits_{x\to a}\dfrac{a^x-x^a}{x-a}$ where $a>0$. If $l = m$ then find the value of 'a'.

Q 25. $\lim\limits_{x\to\infty}\left(\dfrac{\cosh\frac{\pi}{x}}{\cos\frac{\pi}{x}}\right)^{x^2}$ where $\cosh t = \dfrac{e^t+e^{-t}}{2}$ **Q 26.** $\lim\limits_{x\to 0}\dfrac{2(\tan x-\sin x)-x^3}{x^5}$

Q 27. Through a point A on a circle, a chord AP is drawn & on the tangent at A a point T is taken such that AT = AP. If TP produced meet the diameter through A at Q, prove that the limiting value of AQ when P moves upto A is double the diameter of the circle.

Q 28. Using Sandwich theorem, evaluate

(a) $\lim\limits_{n\to\infty}\dfrac{1}{1+n^2}+\dfrac{2}{2+n^2}+\ldots\ldots+\dfrac{n}{n+n^2}$ **(b)** $\lim\limits_{n\to\infty}(a^n+b^n)^{\frac{1}{n}}$, $0<a<b$

Q 29. Find a & b if : **(i)** $\lim\limits_{x\to\infty}\left[\dfrac{x^2+1}{x+1}-ax-b\right]=0$ **(ii)** $\lim\limits_{x\to-\infty}\left[\sqrt{x^2-x+1}-ax-b\right]=0$

Q 30. Show that $\lim\limits_{h\to 0}\dfrac{(\sin(x+h))^{x+h}-(\sin x)^x}{h}=(\sin x)^x[\,x\cot x+l\,n\sin x\,]$

EXERCISE–3

Q.1 $\lim\limits_{x\to 0}\left[\dfrac{1+5x^2}{1+3x^2}\right]^{\frac{1}{x^2}}=$ _____. [IIT'96, 1]

Q.2 $\lim\limits_{x\to 1}\dfrac{\sqrt{1-\cos 2(x-1)}}{x-1}$ [IIT '98, 2]

(A) exits and it equals $\sqrt{2}$ (B) exits and it equals $-\sqrt{2}$
(C) does not exist because $x-1\to 0$
(D) does not exist because left hand limit is not equal to right hand limit.

Q.3 $\lim\limits_{x\to 0}\dfrac{x\tan 2x-2x\tan x}{(1-\cos 2x)^2}$ is : [JEE '99, 2 (out of 200)]

(A) 2 (B) -2 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

Q.4 For $x\in R$, $\lim\limits_{x\to\infty}\left(\dfrac{x-3}{x+2}\right)^x=$ [JEE 2000, Screening]

(A) e (B) e^{-1} (C) e^{-5} (D) e^5

Q.5 $\lim\limits_{x\to 0}\dfrac{\sin(\pi\cos^2 x)}{x^2}$ equals [JEE 2001, Screening]

(A) $-\pi$ (B) π (C) $\frac{\pi}{2}$ (D) 1

Q.6 Evaluate $\lim\limits_{x\to 0}\dfrac{a^{\tan x}-a^{\sin x}}{\tan x-\sin x}$, $a>0$. [REE 2001, 3 out of 100]

Q.7 The integer n for which $\lim\limits_{x\to 0}\dfrac{(\cos x-1)(\cos x-e^x)}{x^n}$ is a finite non-zero number is

(A) 1 (B) 2 (C) 3 (D) 4
[JEE 2002 (screening), 3]

Q.8 If $\lim\limits_{x\to 0}\dfrac{\sin(nx)[(a-n)nx-\tan x]}{x^2}=0$ $(n>0)$ then the value of 'a' is equal to

(A) $\frac{1}{n}$ (B) n^2+1 (C) $\frac{n^2+1}{n}$ (D) None

[JEE 2003 (screening)]

Q.9 Find the value of $\lim_{n\to\infty}\left[\frac{2}{\pi}(n+1)\cos^{-1}\left(\frac{1}{n}\right)-n\right]$. [JEE ' 2004, 2 out of 60]

EXERCISE–4

1. $\lim_{n\to\infty}\frac{5^{n+1}+3^n-2^{2n}}{5^n+2^n+3^{2n+3}}$ = (A) 5 (B) 3 (C) 1 (D) zero

2. $\lim_{x\to -1}\frac{\cos 2-\cos 2x}{x^2-|x|}$ =

(A) $2\cos 2$ (B) $-2\cos 2$ (C) $2\sin 2$ (D) $-2\sin 2$

3. The value of $\lim_{x\to 0}\frac{1}{x}\sqrt{\frac{1-\cos 2x}{2}}$ is:

(A) 1 (B) -1 (C) 0 (D) none

4. $\lim_{x\to 0}\sin^{-1}(\sec x)$.

(A) is equal to $\pi/2$ (B) is equal to 1 (C) is equal to zero (D) none of these

5. $\lim_{x\to 5}\frac{x^2-9x+20}{x-[x]}$ where [x] is the greatest integer not greater than x:

(A) is equal to 1 (B) 0 (C) 4 (D) none

6. $\lim_{x\to -\pi}\frac{|x+\pi|}{\sin x}$:

(A) is equal to -1 (B) is equal to 1 (C) is equal to π (D) does not exist

7. $\lim_{x\to 3}\frac{(x^3+27)\ln(x-2)}{(x^2-9)}$ =

(A) -8 (B) 8 (C) 9 (D) -9

8. $\lim_{x\to 1}\frac{\sum_{k=1}^{100}x^k-100}{x-1}$ =

(A) 0 (B) 5050 (C) 4550 (D) -5050

9. $\lim_{x\to\infty}\left(\sqrt{(x+a)(x+b)}-x\right)$ =

(A) $\sqrt{ab}$ (B) $\frac{a+b}{2}$ (C) ab (D) none

10. $\lim_{x\to\infty}\frac{x^3.\sin\frac{1}{x}+x+1}{x^2+x+1}$ =

(A) 0 (B) 1/2 (C) 1 (D) none

11. $\lim_{n\to\infty}\frac{(n+2)!+(n+1)!}{(n+3)!}$, $n\in N$ =

(A) 0 (B) 1 (C) 2 (D) -1

12. $\lim_{x\to 0}|x|^{\sin x}$ =

(A) 0 (B) 1 (C) -1 (D) none of these

13. $\lim_{x\to\infty}\left(\frac{x^2-2x+1}{x^2-4x+2}\right)^x$ =

(A) 1 (B) 2 (C) e^2 (D) e

14. The values of a and b such that $\lim_{x\to 0}\frac{x(1+a\cos x)-b\sin x}{x^3}=1$ are

(A) $\frac{5}{2},\frac{3}{2}$ (B) $\frac{5}{2},-\frac{3}{2}$ (C) $-\frac{5}{2},-\frac{3}{2}$ (D) $-\frac{5}{2},\frac{3}{2}$

15. $\lim_{x\to 0}\frac{2\left(\sqrt{3}\sin\left(\frac{\pi}{6}+x\right)-\cos\left(\frac{\pi}{6}+x\right)\right)}{x\sqrt{3}\left(\sqrt{3}\cos x-\sin x\right)}$ =

(A) $-1/3$ (B) 2/3 (C) 4/3 (D) $-4/3$

16. If $f(x)=\begin{cases}x-1, & x\ge 1\\ 2x^2-2, & x<1\end{cases}$, $g(x)=\begin{cases}x+1, & x>0\\ -x^2+1, & x\le 0\end{cases}$ and $h(x)=|x|$

then find $\lim_{x\to 0}f(g(h(x)))$

(A) 1 (B) 0 (C) –1 (D) does not exists

17. $\underset{x\to1}{\text{Limit}}\ (1 - x + [x - 1] + [1 - x])$ = where [x] denotes greatest integer function.

(A) 0 (B) 1 (C) -1 (D) does not exist

18. $\underset{x\to0}{\text{Lt}}\left[\frac{\sin[x-3]}{[x-3]}\right]$, where [.] denotes greatest integer function is :

(A) 0 (B) 1 (C) does not exist (D) sin 1

19. Let f (x) = $\begin{cases} x\sin\left(\frac{1}{x}\right)+\sin\left(\frac{1}{x^2}\right) & x\neq0 \\ 0 & x=0 \end{cases}$, then $\lim_{x\to\infty}$ f(x) equals

(A) 0 (B) $-1/2$ (C) 1 (D) none of these.

20. $\lim_{x\to a^-}\left(\frac{|x|^3}{a}-\left[\frac{x}{a}\right]^3\right)$ $(a > 0)$, where [x] denotes the greatest integer less than or equal to x is

(A) a^2+1 (B) a^2-1 (C) a^2 (D) $-a^2$

21. Let α, β be the roots of $ax^2 + bx + c = 0$, where $1 < \alpha < \beta$. Then $\underset{x\to x_0}{\text{Lim}}\ \frac{|ax^2+bx+c|}{ax^2+bx+c} = 1$ then which of the following statements is incorrect

(A) $a > 0$ and $x_0 < 1$ (B) $a > 0$ and $x_0 > \beta$

(C) $a < 0$ and $\alpha < x_0 < \beta$ (D) $a < 0$ and $x_0 < 1$

22. $\underset{n\to\infty}{\text{Limit}}\ \frac{1.n+2(n-1)+3(n-2)+\ldots\ldots+n.1}{1^2+2^2+3^2+\ldots\ldots+n^2}$ has the value :

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) 1

23. $\underset{x\to0}{\text{Lim}}\left[\left(1-e^x\right)\frac{\sin x}{|x|}\right]$ is (where [·] represents greatest integral part function)

(A) -1 (B) 1 (C) 0 (D) does not exist

24. If $\ell = \lim_{x\to\infty}(\sin\sqrt{x+1}-\sin\sqrt{x})$ and m = $\lim_{x\to-\infty}[\sin\sqrt{x+1}-\sin\sqrt{x}]$ where [.] denotes the greatest integer function then :

(A) ℓ = m = 0 (B) ℓ = 0 ; m is undefined

(C) ℓ, m both do not exist (D) ℓ = 0, m $\neq$ 0 (although m exist)

25. If $f(x) = \sum_{\lambda=1}^{n}\left(x-\frac{1}{\lambda}\right)\left(x-\frac{1}{\lambda+1}\right)$ then $\lim_{n\to\infty}$ f(0) is.

(A) 1 (B) -1 (C) 2 (D) None

26. The limit $\lim_{\theta\to0}\left(\left[\frac{n\sin\theta}{\theta}\right]+\left[\frac{n\tan\theta}{\theta}\right]\right)$, where [x] is the greatest integer function and $n \in I$, is

(A) 2n (B) 2n + 1 (C) 2n – 1 (D) does not exist

27. The limit $\underset{x\to\infty}{\text{Limit}}\ x - x^2\,\ell n\left(1+\frac{1}{x}\right)$ is equal to :

(A) 1/2 (B) 3/2 (C) 1/3 (D) 1

28. $\lim_{x\to\pi/2}\left[\frac{x-\frac{\pi}{2}}{\cos x}\right]$ is : (where [.] represents greatest integer function.

(A) -1 (B) 0 (C) -2 (D) does not exist

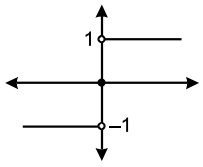
29. If f(x) = sin x , $x \neq n\pi$, n = 0 , ± 1 , ± 2 , ± 3, ,....
= 2 , otherwise and

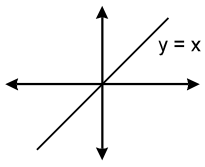
g(x) = x^2 + 1 , x ≠ 0 , 2
= 4 , x = 0
= 5 , x = 2

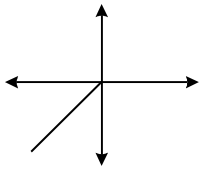
then $\underset{x\to0}{\text{Limit}}$ g [f(x)] is :

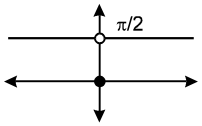
(A) 1 (B) 0 (C) 4 (D) does not exists

30. The graph of the function f(x) = $\underset{t\to0}{\text{Lim}}\left(\frac{2x}{\pi}\cot^{-1}\frac{x}{t^2}\right)$, is

(A) 

(B) 

(C) 

(D) 

31. The value of $\underset{x\to0}{\text{Limit}}\ \frac{\cos(\sin x)-\cos x}{x^4}$ is equal to:

(A) 1/5 (B) 1/6 (C) 1/4 (D) 1/2

32. $\lim\limits_{x\to\infty} \dfrac{e^x\left(\left(2^{x^n}\right)^{\frac{1}{e^x}}-\left(3^{x^n}\right)^{\frac{1}{e^x}}\right)}{x^n}$, $n \in N$ is equal to :

(A) 0 (B) ln(2/3) (C) ln(3/2) (D) none

33. $\lim\limits_{y\to 0}\left(\lim\limits_{x\to\infty}\dfrac{\exp\left(x\ln\left(1+\frac{ay}{x}\right)\right)-\exp\left(x\ln\left(1+\frac{by}{x}\right)\right)}{y}\right)=$

(A) a + b (B) a − b (C) b − a (D) − (a + b)

EXERCISE–5

1. Evaluate the following limits, where [.] represents greatest integer function and { . } represents fractional part function

(i) $\lim\limits_{x\to\frac{\pi}{2}}$ [sin x] (ii) $\lim\limits_{x\to 2}\left\{\dfrac{x}{2}\right\}$ (iii) $\lim\limits_{x\to\pi}$ sgn [tan x]

2. If $f(x)=\begin{cases}x^2+2 & , \ x\ge 2\\ 1-x & , \ x<2\end{cases}$ and $g(x)=\begin{cases}2x & , \ x>1\\ 3-x & , \ x\le 1\end{cases}$, evaluate $\lim\limits_{x\to 1}$ f (g(x)).

3. Evaluate each of the following limits, if exists

(i) $\lim\limits_{x\to 2}\dfrac{x^2-4}{\sqrt{x+2}-\sqrt{3x-2}}$ (ii) $\lim\limits_{x\to a}\dfrac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}$, $a\neq 0$

4. Evaluate the following limits, if exists

(i) $\lim\limits_{x\to 0}\dfrac{\tan 3x-2x}{3x-\sin^2 x}$ (ii) $\lim\limits_{x\to a}\dfrac{(x+2)^{\frac{5}{2}}-(a+2)^{\frac{5}{2}}}{x-a}$ (iii) $\lim\limits_{x\to 0}\dfrac{x\left(e^{2+x}-e^2\right)}{1-\cos x}$

5. Evaluate the following limits, if exist :

(i) $\lim\limits_{x\to\infty}\sqrt{x^2+x-1}-x$ (ii) $\lim\limits_{x\to\infty}\left(\dfrac{1}{x^2}+\dfrac{2}{x^2}+....+\dfrac{x}{x^2}\right)$

(iii) $\lim\limits_{x\to\infty}\left\{\cos\left(\sqrt{x+1}\right)-\cos\left(\sqrt{x}\right)\right\}$ (iv) $\lim\limits_{x\to\infty}\sqrt{x^2-8x}+x$

6. Evaluate the following limits using expansions : (i) $\lim\limits_{x\to 0}\dfrac{e^x-1-\sin x-\frac{\tan^2 x}{2}}{x^3}$

(ii) If $\lim\limits_{x\to 0}\dfrac{a+b\sin x-\cos x+ce^x}{x^3}$ exists, then find values of a, b, c. Also find the limit

7. Evaluate $\lim\limits_{x\to\infty}\dfrac{[1\,.\,2x]+[2\,.\,3x]+.....+[n\,.\,(n+1)\,x]}{n^3}$ where [.] denotes greatest integer function

8. If $f(x)=\lim\limits_{n\to\infty}\dfrac{x^{2n}-1}{x^{2n}+1}$, find range of f(x).

9. Evaluate the following limits

(i) $\lim\limits_{x\to 1}\dfrac{(\ln(1+x)-\ln 2)(3.4^{x-1}-3x)}{[(7+x)^{\frac{1}{3}}-(1+3x)^{\frac{1}{2}}].\sin(x-1)}$ (ii) $\lim\limits_{x\to 4}\dfrac{(\cos\alpha)^x-(\sin\alpha)^x-\cos 2\alpha}{x-4}$, $\alpha\in\left(0\,,\,\dfrac{\pi}{2}\right)$

10. Evaluate the following limits

(i) $\lim\limits_{x\to\infty} x^3\left\{\sqrt{x^2+\sqrt{1+x^4}}-x\sqrt{2}\right\}$ (ii) $\lim\limits_{x\to-\infty}\dfrac{x^5\tan\left(\frac{1}{\pi x^2}\right)+3|x|^2+7}{|x|^3+7|x|+8}$

11. Evaluate the following limits (i) $\lim\limits_{x\to 0}\left[\sin^2\left(\dfrac{\pi}{2-ax}\right)\right]^{\sec^2\left(\frac{\pi}{2-bx}\right)}$

(ii) $\lim\limits_{x\to\infty}\left(\dfrac{a_1^{1/x}+a_2^{1/x}+a_3^{1/x}+.....+a_n^{1/x}}{n}\right)^{nx}$, where $a_1, a_2, a_3, ..., a_n > 0$.

12. Find the values of a & b so that: (i) $\lim\limits_{x\to 0}\dfrac{(1+ax\sin x)-(b\cos x)}{x^4}$ may find to a definite limit.

(ii) $\lim\limits_{x\to\infty}\left(\sqrt{x^4+ax^3+3x^2+bx+2}-\sqrt{x^4+2x^3-cx^2+3x-d}\right)=4$

13. Find the limits using expansion : $\lim\limits_{x\to 0}\left[\dfrac{\ell n(1+x)^{(1+x)}}{x^2}-\dfrac{1}{x}\right]$

14. Let $f(x) = \dfrac{\sin^{-1}(1-\{x\}).\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}.(1-\{x\})}$ then find $\underset{x\to 0^+}{\text{Limit}}\ f(x)$ and $\underset{x\to 0^-}{\text{Limit}}\ f(x)$, where {.} denotes the fractional part function.

15. Let $f(x) = \underset{m\to\infty}{\text{Limit}}\left\{\underset{n\to\infty}{\text{Limit}}\left(\cos^{2m}(n!\pi x)\right)\right\}$ where $x \in R$. Prove that

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}.$$

16. Evaluate $\underset{x\to 0^+}{\text{Limit}}\left\{\underset{n\to\infty}{\text{Limit}}\left(\dfrac{[1^2(\sin x)^x]+[2^2(\sin x)^x]+.....+[n^2(\sin x)^x]}{n^3}\right)\right\}$,

where [.] denotes the greatest integer function.

17. Evaluate the following limits

(i) $\underset{n\to\infty}{\text{Limit}}\ \cos\dfrac{x}{2}\cos\dfrac{x}{4}\cos\dfrac{x}{8}.....\cos\dfrac{x}{2^n}$

(ii) $\underset{n\to\infty}{\text{Limit}}\ \dfrac{1}{2}\tan\dfrac{x}{2}+\dfrac{1}{2^2}\tan\dfrac{x}{2^2}+\dfrac{1}{2^3}\tan\dfrac{x}{2^3}+.....+\dfrac{1}{2^n}\tan\dfrac{x}{2^n}$.

(iii) $\underset{x\to\infty}{\text{Limit}}\ \log_{x-1}(x).\log_x(x+1).\log_{x+1}(x+2).\log_{x+2}(x+3).....\log_k(x^5)$; where $k = x^5-1$.

(iv) Let $P_n = \dfrac{2^3-1}{2^3+1}.\dfrac{3^3-1}{3^3+1}.\dfrac{4^3-1}{4^3+1}.....\dfrac{n^3-1}{n^3+1}$. Prove that $\underset{n\to\infty}{\text{Limit}}\ P_n = \dfrac{2}{3}$.

ANSWER

EXERCISE–1

Q 1. 3 **Q 2.** $\dfrac{45}{91}$ **Q 3.** 2 **Q 4.** $\dfrac{p-q}{2}$ **Q 5.** $\dfrac{2}{\sqrt{3}}$ **Q.6** $-\dfrac{1}{3}$

Q 7. (a) $\pi/2$ if $a > 0$; 0 if $a = 0$ and $-\pi/2$ if $a < 0$ (b) $f(x) = |x|$

Q 8. 5050 **Q 9.** $a = \dfrac{1}{2}$; $r = \dfrac{1}{4}$; $S = \dfrac{2}{3}$ **Q 10.** ln 2 **Q 11.** does not exist **Q 12.** 2

Q 13. 1 **Q 14.** $\dfrac{3}{2}$ **Q 15.** $\dfrac{1}{16\sqrt{2}}$ **Q 16.** $\dfrac{2\ln 2}{\pi}$

Q 17. $a = 2$; limit = 1 **Q 18.** $\dfrac{1}{32}$ **Q 19.** $-\dfrac{9}{4}\ln\dfrac{4}{e}$ **Q 20.** 2

Q 21. Domain, $x \in R$, Range, $x = \dfrac{n\pi}{2}$, $n \in I$ **Q 22.** does not exist **Q 23.** 9

Q 24. $\cos^2\alpha\ l\text{n}\cos\alpha + \sin^2\alpha\ l\text{n}\sin\alpha$ **Q 25.** $8\sqrt{2}(\ln 3)^2$ **Q 26.** – 3, –3, – 3

Q 27. $(l\text{n}\ a)^n$ **Q 28.** –2 **Q 29.** 0, 0 **Q. 30** 4

EXERCISE–2

Q 1. e^{-8} **Q 2.** $c = \ln 2$ **Q 3.** $e^{-\frac{1}{2}}$ **Q 4.** e^{-a^2/b^2} **Q.5** $-\dfrac{\pi^2}{4}$ **Q 6.** $e^{-2\pi^2a^2}$ **Q 7.** e^{-1}

Q 8. $e^{-1/2}$ **Q 9.** $(a_1.a_2.a_3....a_n)$ **Q 10.** $\dfrac{\pi}{2}, \dfrac{\pi}{2\sqrt{2}}$ **Q 11.** $a = c = 1, b = 2$ **Q 12.** $\dfrac{\pi^2a^2+4}{16a^4}$

Q 13. $\dfrac{2}{3}$ **Q 14.** f(x) when $|x| > 1$; g(x) when $|x| < 1$; $\dfrac{g(x)+f(x)}{2}$ when $x = 1$ & not defined when $x = -1$

Q 15. $\dfrac{x}{2}$ **Q 16.** $-\dfrac{1}{\pi^2}$ **Q 17.** a - b **Q 19.** 1/2 **Q 20.** $\dfrac{2}{3}$

Q 21. $T(x) = \dfrac{1}{2}\tan^2\dfrac{x}{2}.\sin x$ or $\tan\dfrac{x}{2} - \dfrac{\sin x}{2}$, $S(x) = \dfrac{1}{2}x - \dfrac{1}{2}\sin x$, limit $= \dfrac{3}{2}$

Q 22. (a) 1 **(b)** $\dfrac{1}{2}$ **Q 23.** $\dfrac{\theta}{\tan\theta}$ **Q 24.** $a = e^2$ **Q 25.** e^{π^2}

Q 26. $\dfrac{1}{4}$ **Q 28.** (a) 1/2, (b) b **Q 29. (i)** $a = 1, b = -1$ **(ii)** $a = -1, b = \dfrac{1}{2}$

EXERCISE–3

Q 1. e^2 **Q 2.** D **Q 3.** C **Q 4.** C **Q5.** B

Q6. $l$na **Q7.** C **Q8.** C **Q.9** $1-\dfrac{2}{\pi}$

EXERCISE–4

1. D **2.** C **3.** D **4.** D **5.** D **6.** D **7.** C **8.** B **9.** B **10.** C
11. A **12.** B **13.** C **14.** C **15.** C **16.** B **17.** C **18.** C **19.** C **20.** C
21. D **22.** A **23.** A **24.** B **25.** A **26.** C **27.** A **28.** C **29.** A **30.** C
31. B **32.** B **33.** B

EXERCISE–5

1. (i) 0 (ii) Limit does not exists (iii) Limit does not exists

2. 6 **3.** (i) (– 8) (ii) $\frac{2}{3\sqrt{3}}$ **4.** (i) 1/3 (ii) $\frac{5}{2}(a+2)^{3/2}$ (iii) $2e^2$

5. (i) 1/2 (ii) 1/2 (iii) zero (iv) ∞ **6.** (i) $\frac{1}{3}$ (ii) $a = 2, b = 1, c = -1$ and value $= -\frac{1}{3}$

7. $\frac{x}{3}$ **8.** {–1, 0, 1} ***9.*** *(i)* $-\frac{9}{4}\ln\frac{4}{e}$ *(ii)* $\cos^4 a \ln(\cos a) - \sin^4 a \ln(\sin a)$

10. (i) $\frac{1}{4\sqrt{2}}$ (ii) $-\frac{1}{\pi}$ **11.** (i) $e^{-\frac{a^2}{b^2}}$ (ii) $(a_1\, a_2\, a_3 \ldots\ldots a_n)$

12. (i) $a = -\frac{1}{2}, b = 1$ (ii) $a = 2, b \in R, c = 5, d \in R$ **13.** $\frac{1}{2}$

14. $\frac{\pi}{2}, \frac{\pi}{2\sqrt{2}}$ **16.** $\frac{1}{3}$ **17.** (i) $\frac{\sin x}{x}$ (ii) $\frac{1}{x} - \cot x$ (iii) 5

STUDY PACKAGE

Target: IIT-JEE (Advanced)

SUBJECT: MATHEMATICS

TOPIC: 17 XI M 17. Differentiation

Index:

1. Key Concepts

2. Exercise I to II

3. Answer Key

4. Assertion and Reasons

5. 34 Yrs. Que. from IIT-JEE

6. 10 Yrs. Que. from AIEEE

Differentiation

A. First Principle Of Differentiation

1. **The derivative of a given function f at a point x = a on its domain is defined as:**

$\underset{h\to 0}{\text{Limit}}\ \frac{f(a+h)-f(a)}{h}$, provided the limit exists & is denoted by f′(a).

i.e. $f'(a) = \underset{x\to a}{\text{Limit}}\ \frac{f(x)-f(a)}{x-a}$, provided the limit exists.

2 If x and x + h belong to the domain of a function f defined by y = f(x), then

$\underset{h\to 0}{\text{Limit}}\ \frac{f(x+h)-f(x)}{h}$ if it exists, is called the Derivative of f at x & is denoted by f′(x) or $\frac{dy}{dx}$. i.e., $f'(x) = \underset{h\to 0}{\text{Limit}}\ \frac{f(x+h)-f(x)}{h}$ This method of differentiation is also called ab-initio method or first principle.

Solved Example # 1 Find derivative of following functions by first principle

(i) $f(x) = x^2$ (ii) $f(x) = \tan x$ (iii) $f(x) = e^{\sin x}$

Solution (i) $f'(x) = \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h\to 0} \frac{2xh+h^2}{h} = 2x.$

(ii) $f'(x) = \lim_{h\to 0} \frac{\tan(x+h)-\tan x}{h}$

$= \lim_{h\to 0} \frac{\tan(x+h-x)[1+\tan x\tan(x+h)]}{h} = \lim_{h\to 0} \frac{\tan h}{h}\,.\,(1+\tan^2 x) = \sec^2 x.$

(iii) $f'(x) = \lim_{h\to 0} \frac{e^{\sin(x+h)} - e^{\sin x}}{h}$

$= \lim_{h\to 0} e^{\sin x}\ \frac{\left[e^{\sin(x+h)-\sin x}-1\right]}{\sin(x+h)-\sin x}\left(\frac{\sin(x+h)-\sin x}{h}\right)$

$= e^{\sin x} \lim_{h\to 0} \frac{\sin(x+h)-\sin x}{h} = e^{\sin x}\cos x$

3. **Differentiation of some elementary functions**

	f(x)	f′(x)	
1.	x^n	nx^{n-1}	$(x \in R, n \in R)$
2.	a^x	$a^x \ell n\, a$	
3.	$\ell n\, \lvert x\rvert$	$\frac{1}{x}$	
4.	$\log_a x$	$\frac{1}{x\,\ell n\, a}$	
5.	$\sin x$	$\cos x$	
6.	$\cos x$	$-\sin x$	
7.	$\sec x$	$\sec x \tan x$	
8.	$\text{cosec}\, x$	$-\text{cosec}\, x \cot x$	
9.	$\tan x$	$\sec^2 x$	
10.	$\cot x$	$-\text{cosec}\, x$	

4. **Basic Theorems**

1. $\frac{d}{dx}(f \pm g) = f'(x) \pm g'(x)$

2. $\frac{d}{dx}(k\, f(x)) = k\, \frac{d}{dx} f(x)$

3. $\frac{d}{dx}(f(x)\,.\,g(x)) = f(x)\, g'(x) + g(x)\, f'(x)$

4. $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)\, f'(x) - f(x) g'(x)}{g^2(x)}$

5. $\frac{d}{dx}(f(g(x))) = f'(g(x))\, g'(x)$

This rule is also called the chain rule of differentiation and can be written as $\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx}$

Note that an important inference obtained from the chain rule is that $\frac{dy}{dy} = 1 = \frac{dy}{dx} \cdot \frac{dx}{dy}$

$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{dx/dy}$

another way of expressing the same concept is by considering y = f(x) and x = g(y) as inverse functions of each other.

$\frac{dy}{dx} = f'(x)$ and $\frac{dx}{dy} = g'(y) \Rightarrow g'(y) = \frac{1}{f'(x)}$

Solved Example # 2

Find the differential of the following functions with respect to x.

(i) $f(x) = e^{\sin x}$ (ii) $f(x) = \sqrt{\sin(2x+3)}$ (iii) $f(x) = \frac{x}{1+x^2}$ (iv) $f(x) = x \cdot \sin x$

Solution. (i) $f(x) = e^{\sin x}$

$f'(x) = e^{\sin x} \frac{d}{dx}(\sin x) = e^{\sin x} \cos x$

(ii) $f(x) = \sqrt{\sin(2x+3)}$

$= \frac{1}{2\sqrt{\sin(2x+3)}} \cdot \frac{d}{dx}(\sin(2x+3)) = \frac{\cos(2x+3)}{\sqrt{\sin(2x+3)}}$

(iii) $f(x) = \frac{x}{1+x^2}$

$f'(x) = \frac{(1+x^2)-x(2x)}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$

(iv) $f(x) = x \sin x$

$f'(x) = x.\cos x + \sin x$

Solved Example # 3 If $f(x) = \sin(x + \tan x)$ then find value of $f'(0)$.

Solution. $f'(x) = \cos(x + \tan x)(1 + \sec^2 x)$ $f'(0) = 2$

Self Practice Problems :

1. Find the derviative of following functions using first principle.

(i) $f(x) = x \sin x$ (ii) $f(x) = \sin^2 x$

Ans. (i) $x \cos x + \sin x$ (ii) $2 \sin x \cos x$

2. Evaluate if $f'(5) = 7$, then $\lim_{t \to 0} \frac{f(5+t)-f(5-t)}{2t}$ **Ans.** 7.

3. Differentiate the following functions

(i) $(1 + 3x^2)(2x^3 - 1)$ (ii) $\frac{(x-1)}{(x-2)(x-3)}$ (iii) $\sqrt{1+x^2}$

(iv) $\sqrt{\frac{1+x}{1-x}}$ (v) $\cos^3 x \sin x$ (vi) $x e^x \sin x$

(vii) $\frac{\sin x}{1+\cos x}$ (viii) $\ell n(\sin x - \cos x)$

Ans. (i) $6x(5x^3 + x - 1)$ (ii) $\frac{-x^2+2x+1}{(x-2)^2(x-3)^2}$ (iii) $\frac{x}{\sqrt{1+x^2}}$ (iv) $\frac{1}{(1+x)^{1/2}(1-x)^{3/2}}$

(v) $\cos^4 x - 3\cos^2 x \sin^2 x$ (vi) $e^x((\sin x + \cos x)x + \sin x)$ (vii) $\frac{1}{2}\sec^2\frac{x}{2}$ (viii) $\frac{\cos x + \sin x}{\sin x - \cos x}$

B. Derivative Of Inverse Trigonometric Functions.

$y = \sin^{-1} x \Rightarrow -\frac{\pi}{2} \le y \le \frac{\pi}{2} \Rightarrow x = \sin y$

$\frac{dx}{dy} = \cos y$

$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}}$ $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$ $-1 < x < 1.$

Note here that $\cos y \neq \sqrt{1-\sin^2 y}$, rather $\cos y = \pm\sqrt{1-\sin^2 y}$ but for values of $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, cos y is always positive and hence the result. similarly let us find derivative of other inverse trigonometric functions.

Let $y = \tan^{-1} x$

$x = \tan y$

$\frac{dx}{dy} = \sec^2 y = 1 + \tan^2 y \Rightarrow \frac{dx}{dy} = 1 + x^2$

$\frac{dy}{dx} = \frac{1}{1+x^2}$ $(x \in R)$

Also if $y = \sec^{-1} x$ $y \in [0, \pi] - \left\{\frac{\pi}{2}\right\} \Rightarrow x = \sec y$

$\frac{dx}{dy} = \sec y \tan y \Rightarrow \frac{dy}{dx} = \frac{1}{x.\tan y} \Rightarrow \frac{dy}{dx} = \frac{1}{\pm x\sqrt{\sec^2 y - 1}}$

$$\frac{dy}{dx} = \begin{cases} \frac{1}{x\sqrt{x^2-1}} & \sec y > 1 \\ -\frac{1}{x\sqrt{x^2-1}} & \sec y < -1 \end{cases} \Rightarrow \frac{dy}{dx} = \frac{1}{|x|\sqrt{x^2-1}} \qquad x \in (-\infty, -1) \cup (1, \infty)$$

results for the derivative of inverse trigonometric functions can be summarized as :

f(x)	f′(x)	
$\sin^{-1}x$	$\frac{1}{\sqrt{1-x^2}}$	; $\lvert x\rvert < 1$
$\cos^{-1}x$	$\frac{-1}{\sqrt{1-x^2}}$	; $\lvert x\rvert < 1$
$\tan^{-1}x$	$\frac{1}{1+x^2}$	; $x \in R$
$\cot^{-1}x$	$\frac{-1}{1+x^2}$	; $x \in R$
$\sec^{-1}x$	$\frac{1}{\lvert x\rvert\sqrt{x^2-1}}$	; $\lvert x\rvert > 1$
$\text{cosec}^{-1}x$	$\frac{-1}{\lvert x\rvert\sqrt{x^2-1}}$	; $\lvert x\rvert > 1$

Solved Example # 4 If $f(x) = \ell n\,(\sin^{-1}x^2)$ find $f'(x)$

Solution. $f'(x) = \frac{1}{(\sin^{-1}x^2)} \cdot \frac{1}{\sqrt{1-(x^2)^2}} \cdot 2x = \frac{2x}{(\sin^{-1}x^2)\sqrt{1-x^4}}$

Solved Example # 5 If $f(x) = 2x\sec^{-1}x - \text{cosec}^{-1}(x)$ then find $f'(-2)$

Solution. $f'(x) = 2\sec^{-1}(x) - \frac{2x}{|x|\sqrt{x^2-1}} + \frac{1}{|x|\sqrt{x^2-1}}$

$f'(-2) = 2.\sec^{-1}(-2) + \frac{2}{\sqrt{3}} + \frac{1}{2\sqrt{3}}$ $\qquad f'(-2) = \frac{4\pi}{3} + \frac{5}{2\sqrt{3}}$.

C. Methods Of Differentiation

1. Logrithmic Differentiation

The process of taking logarithm of the function first and then differentiate is called **Logarithmic Differentiation**.
It is useful if
(i) a function is the product or quotient of a number of functions OR
(ii) a function is of the form $[f(x)]^{g(x)}$ where f & g are both derivable,

Solved Example # 6 If $y = x^x$ find $\frac{dy}{dx}$

Solution. $\ell n\, y = x\,\ell n\, x \Rightarrow \frac{1}{y}\cdot\frac{dy}{dx} = x\cdot\frac{1}{x} + \ell n\, x \Rightarrow \frac{dy}{dx} = x^x(1 + \ell n\, x)$

Solved Example # 7 If $y = (\sin x)^{\ell n\, x}$, find $\frac{dy}{dx}$

Solution. $\ell n\, y = \ell n\, x \,.\, \ell n\,(\sin x)$

$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x}\ell n\,(\sin x) + \ell n\, x.\frac{\cos x}{\sin x} \Rightarrow \frac{dy}{dx} = (\sin x)^{\ell n\, x}\left[\frac{\ell n(\sin x)}{x} + \cot x\ \ell n\, x\right]$$

Solved Example # 8 If $y = \frac{x^{1/2}(1-2x)^{2/3}}{(2-3x)^{3/4}(3-4x)^{4/5}}$ find $\frac{dy}{dx}$

Solution. $\ell n\, y = \frac{1}{2}\ell n\, x + \frac{2}{3}\ell n\,(1-2x) - \frac{3}{4}\ell n\,(2-3x) - \frac{4}{5}\ell n\,(3-4x)$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{1}{2x} - \frac{4}{3(1-2x)} + \frac{9}{4(2-3x)} + \frac{16}{5(3-4x)}$$

$$\frac{dy}{dx} = y\left(\frac{1}{2x} - \frac{4}{3(1-2x)} + \frac{9}{4(2-3x)} + \frac{16}{5(3-4x)}\right)$$

2. Implicit differentiation

If $f(x, y) = 0$, is an implicit function then in order to find dy/dx, we differentiate each term w.r.t. x regarding y as a functions of x & then collect terms in dy/dx.

Solved Example # 9 If $x^3 + y^3 = 3xy$ find $\frac{dy}{dx}$

Solution. Differentiation both sides w.r.t.x, we get

$$3x^2 + 3y^2\frac{dy}{dx} = 3x\frac{dy}{dx} + 3y \qquad \frac{dy}{dx} = \frac{y-x^2}{y^2-x}$$

Note that above result holds only for points where $y^2 - x \neq 0$

Solved Example # 10 If $x^y = e^{x-y}$, then find $\frac{dy}{dx}$

Solution.

Taking log on both sides

$y\,\ell n\,x = (x - y)$ .....(i)

differentiating w.r.t x, we get

$$\frac{y}{x} + \ln x\,\frac{dy}{dx} = 1 - \frac{dy}{dx} \quad \Rightarrow \quad \frac{dy}{dx} = \frac{1-\frac{y}{x}}{1+\ell n\,x} \quad \Rightarrow \quad \frac{dy}{dx} = \frac{x-y}{x(1+\ell n\,x)}$$

Solved Example # 11 If $x^y + y^x = 2$ then find $\frac{dy}{dx}$

Solution. $u + v = 2 \quad \Rightarrow \quad \frac{du}{dx} + \frac{dv}{dx} = 0$

where $u = x^y$ & $v = y^x$

$\Rightarrow \quad \ell n\,u = y\,\ell n\,x$ & $\ell n\,v = x\,\ell n\,y$

$$\Rightarrow \quad \frac{1}{u}\frac{du}{dx} = \frac{y}{x} + \ell n\,x\,\frac{dy}{dx} \;\&\; \frac{1}{v}\frac{dv}{dx} = \ell n\,y + \frac{x}{y}\frac{dy}{dx}$$

$$\Rightarrow \quad \frac{du}{dx} = x^y\left(\frac{y}{x} + \ell n\,x\frac{dy}{dx}\right) \;\&\; \frac{dv}{dx} = y^x\left(\ell n\,y + \frac{x}{y}\frac{dy}{dx}\right)$$

$$\Rightarrow \quad x^y\left(\frac{y}{x} + \ell n\,x\frac{dy}{dx}\right) + y^x\left(\ell n\,y + \frac{x}{y}\frac{dy}{dx}\right) = 0. \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{\left(y^x\ell n\,y + x^y.\frac{y}{x}\right)}{\left(x^y\ell n\,x + y^x.\frac{x}{y}\right)}$$

Self Practice Problems

1. Differentiate the following functions : (i) $y = \sec^{-1}(x^2)$ (ii) $y = \tan^{-1}\left(\frac{1+x}{1-x}\right)$

(iii) $y = \left(1+\frac{1}{x}\right)^x$ (iv) $y = e^{x^x}$ (v) $y = (\ln x)^x + (x)^{\sin x}$

2. Find $\frac{dy}{dx}$ if

(i) $y = \cos(x + y)$ (ii) $x^{2/3} + y^{2/3} = a^{2/3}$ (iii) $x = y\,\ell n\,(x - y)$

3. If $x^y = e^{x-y}$, then prove that $\frac{dy}{dx} = \frac{\ell n\,x}{(1+\ell n\,x)^2}$.

4. If $\frac{x}{x-y} = \log\frac{a}{x-y}$, prove that $\frac{dy}{dx} = 2 - \frac{x}{y}$.

Ans. **1.** (i) $\frac{2}{x\sqrt{x^4-1}}$ (ii) $\frac{1}{1+x^2}$ (iii) $\left(1+\frac{1}{x}\right)^x\left[\ell n\left(1+\frac{1}{x}\right)+\frac{1}{1+x}\right]$

(iv) $x^x.e^{x^x}(\ell nx + 1)$ (v) $\left(\ell n\,(\ell nx) + \left(\frac{1}{\ell nx}\right)\right)(\ell n\,x)^x + x^{\sin x}\left(\frac{\sin x}{x} + \cos x\,\ell nx\right)$

2. (i) $\frac{-\sin(x+y)}{1+\sin(x+y)}$ (ii) $-\left(\frac{y}{x}\right)^{1/3}$ (iii) $\frac{y(x-y)}{x(x+y)}$

3. Differentiation using substitution

Following substitutions are normally used to sumplify these expression.

(i) $\sqrt{x^2+a^2}$ $\Rightarrow$ $x = a\tan\theta$ or $a\cot\theta$

(ii) $\sqrt{a^2-x^2}$ $\Rightarrow$ $x = a\sin\theta$ or $a\cos\theta$

(iii) $\sqrt{x^2-a^2}$ $\Rightarrow$ $x = a\sec\theta$ or $a\,\text{cosec}\,\theta$

(iv) $\sqrt{\frac{x+a}{a-x}}$ $\Rightarrow$ $x = a\cos\theta$

Solved Example # 12 : Differentiate $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$.

Solution. Let $x = \tan\theta \quad \Rightarrow \quad \theta = \tan^{-1}x \quad ; \quad \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$y = \tan^{-1}\left(\frac{|\sec\theta| - 1}{\tan\theta}\right)$ $\left[|\sec\theta| = \sec\theta\ \forall\ \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\right]$

$\Rightarrow \quad y = \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right) \qquad \Rightarrow \quad y = \tan^{-1}\left(\tan\frac{\theta}{2}\right)$

$\Rightarrow \quad y = \frac{\theta}{2} \qquad [\tan^{-1}(\tan x) = x \text{ for } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)]$

$\Rightarrow \quad y = \frac{1}{2}\tan^{-1} x \qquad \Rightarrow \quad \frac{dy}{dx} = \frac{1}{2(1+x^2)}$

Solved Example # 13 : Find $\frac{dy}{dx}$ where $y = \tan^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$

Solution. $x = \cos\theta$
$\theta = \cos^{-1}(x) \quad ; \quad \theta \in [0, \pi]$

$\Rightarrow \quad y = \tan^{-1}\left(\frac{\sqrt{1+\cos\theta}-\sqrt{1-\cos\theta}}{\sqrt{1+\cos\theta}+\sqrt{1-\cos\theta}}\right) \qquad \Rightarrow \quad y = \tan^{-1}\left(\frac{\sqrt{2}\cos\frac{\theta}{2}-\sqrt{2}\sin\frac{\theta}{2}}{\sqrt{2}\cos\frac{\theta}{2}+\sqrt{2}\sin\frac{\theta}{2}}\right)$

$\Rightarrow \quad y = \tan^{-1}\left(\frac{1-\tan\frac{\theta}{2}}{1+\tan\frac{\theta}{2}}\right) \qquad \Rightarrow \quad y = \frac{\pi}{4} - \frac{\theta}{2}$

$\Rightarrow \quad y = \frac{\pi}{4} - \frac{1}{2}\cos^{-1}x \qquad \Rightarrow \quad \frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}}$

Note that $\sqrt{1+\cos\theta} = \left|\sqrt{2}\cos\frac{\theta}{2}\right|$ but for $\frac{\theta}{2} \in \left(0, \frac{\pi}{2}\right)$, $\left|\sqrt{2}\cos\frac{\theta}{2}\right| = \sqrt{2}\cos\frac{\theta}{2}$

Also $\tan^{-1}(\tan x) = x$ for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Solved Example # 14 If $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ then find

(i) $f'(2)$ (ii) $f'\left(\frac{1}{2}\right)$ (iii) $f'(1)$

Solution. $x = \tan\theta$

$\Rightarrow \quad \theta = \tan^{-1}(x) \quad ; \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2} \qquad \Rightarrow \quad y = \sin^{-1}(\sin 2\theta)$

$$y = \begin{cases} \pi - 2\theta & \frac{\pi}{2} < 2\theta < \pi \\ 2\theta & -\frac{\pi}{2} \le 2\theta \le \frac{\pi}{2} \\ -(\pi + 2\theta) & -\pi < 2\theta < -\frac{\pi}{2} \end{cases} \Rightarrow f(x) = \begin{cases} \pi - 2\tan^{-1}x & x > 1 \\ 2\tan^{-1}x & -1 \le x \le 1 \\ -(\pi + 2\tan^{-1}x) & x < -1 \end{cases} \Rightarrow f'(x) = \begin{cases} -\frac{2}{1+x^2} & x > 1 \\ \frac{2}{1+x^2} & -1 < x < 1 \\ \frac{-2}{1+x^2} & x < -1 \end{cases}$$

(i) $f'(2) = -\frac{2}{5}$ (ii) $f'\left(\frac{1}{2}\right) = \frac{8}{5}$ (iii) $f'(1^+) = -1$ & $f'(1^-) = +1$

$\Rightarrow \quad f'(1)$ does not exist.

Aliter Above problem can also be solved without any substution also, but in a little tedious way.

$f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$

$f'(x) = \frac{1}{\sqrt{1-\frac{4x^2}{(1+x^2)^2}}} \cdot \frac{2\{(1+x^2)-2x^2\}}{(1+x^2)^2}$

$= \frac{(1+x^2)}{\sqrt{(1-x^2)^2}} \cdot \frac{2(1-x^2)}{(1+x^2)^2}$

$f'(x) = \frac{2}{(1+x^2)} \cdot \frac{(1-x^2)}{|1-x^2|}$ thus $f'(x) = \begin{cases} \frac{2}{1+x^2} & |x| < 1 \\ \frac{-2}{1+x^2} & |x| > 1 \end{cases}$

Solved Example # 15 If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, then prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.

Solution. Put $x = \sin\alpha \Rightarrow \alpha = \sin^{-1}(x)$
$y = \sin\beta \Rightarrow \beta = \sin^{-1}(y)$

$\Rightarrow \quad \cos\alpha + \cos\beta = a(\sin\alpha - \sin\beta)$

$\Rightarrow \quad 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = 2a\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right) \quad \Rightarrow \quad \cot\left(\frac{\alpha-\beta}{2}\right) = a$

$\Rightarrow \quad \alpha - \beta = 2\cot^{-1}(a) \quad \Rightarrow \quad \sin^{-1}x - \sin^{-1}y = 2\cot^{-1}(a)$

differentiating w.r.t to x.

$$\frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-y^2}}\frac{dy}{dx} = 0 \quad \Rightarrow \quad \frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$$

Aliter Using implicit differention.

$$\frac{-x}{\sqrt{1-x^2}} - \frac{y}{\sqrt{1-y^2}}\frac{dy}{dx} = a\left(1-\frac{dy}{dx}\right)$$

$$\Rightarrow \quad \left(a - \frac{y}{\sqrt{1-y^2}}\right)\frac{dy}{dx} = a + \frac{x}{\sqrt{1-x^2}} \quad \Rightarrow \quad \frac{dy}{dx} = \frac{a+\frac{x}{\sqrt{1-x^2}}}{a-\frac{y}{\sqrt{1-y^2}}}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{\frac{\sqrt{1-x^2}+\sqrt{1-y^2}}{x-y} + \frac{x}{\sqrt{1-x^2}}}{\frac{\sqrt{1-x^2}+\sqrt{1-y^2}}{x-y} - \frac{y}{\sqrt{1-y^2}}}$$

$$\frac{dy}{dx} = \frac{(1-x^2)+\sqrt{(1-x^2)(1-y^2)}+x^2-xy}{\sqrt{(1-x^2)(1-y^2)}+(1-y^2)-xy+y^2} \cdot \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}} = \frac{1+\sqrt{(1-x^2)(1-y^2)}-xy}{1+\sqrt{(1-x^2)(1-y^2)}-xy} \cdot \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

$$\Rightarrow \quad \frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}} \qquad \textbf{Hence proved}$$

4. Parametric Differentiation If $y = f(\theta)$ & $x = g(\theta)$ where θ is a parameter, then $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$.

Solved Example # 16 If $x = a\cos^3 t$ and $y = a\sin^3 t$. Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-3a\sin^2 t\cos t}{3a\cos^2 t\sin t} = -\tan t$$

Solved Example # 17 If $y = a\cos t$ and $x = a(t - \sin t)$ find the value of $\frac{dy}{dx}$ at $t = \frac{\pi}{2}$.

$$\frac{dy}{dx} = \frac{-a\sin t)}{a(1-\cos t)}$$

$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{2}} = -1.$$

5. Derivative of one function with respect to another

Let $y = f(x)$; $z = g(x)$ then $\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{f'(x)}{g'(x)}$.

Solved Example # 18

Find derivative of $y = \ell n\, x$ with respect to $z = e^x$.

$$\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{1}{xe^x}$$

Self Practice Problems :

1. Find $\frac{dy}{dx}$ when

(i) $x = a(\cos t + t\sin t)$ & $y = a(\sin t - t\cos t)$

(ii) $x = a\left(\frac{1-t^2}{1+t^2}\right)$ & $y = b\cdot\left(\frac{2t}{1+t^2}\right)$

Ans. (i) $\tan t$ (ii) $\frac{(t^2-1)b}{2at}$

2. If $y = \sin^{-1}\left(\frac{x^2}{\sqrt{x^4+a^4}}\right)$ then prove that $\frac{dy}{dx} = \frac{2xa^2}{x^4+a^4}$.

3. If $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ then prove that $\frac{dy}{dx} = \frac{2}{1+x^2}$ $(|x| \neq 1)$

4. If $u = \sin(m\cos^{-1}x)$ and $v = \cos(m\sin^{-1}x)$ then prove that $\frac{du}{dv} = \sqrt{\frac{1-u^2}{1-v^2}}$.

D. Derivatives of Higher Order

Let a function $y = f(x)$ be defined on an open interval (a, b). It's derivative, if it exists on (a, b) is a certain function $f'(x)$ [or (dy/dx) or y'] & is called the first derivative of y w. r. t. x.
If it happens that the first derivative has a derivative on (a, b) then this derivative is called the second derivative of y w. r. t. x & is denoted by $f''(x)$ or (d^2y/dx^2) or y''.

Similarly, the 3rd order derivative of y w. r. t. x, if it exists, is defined by $\frac{d^3y}{dx^3} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right)$ It is also denoted by $f'''(x)$ or y'''.

Solved Example # 19
If $y = x^3\,\ell n\,x$ then y'' and y'''

Solution. $y' = 3x^2\,\ell n\,x + x^3\,\frac{1}{x}$

$y' = 3x^2\,\ell n\,x + x^2$

$y'' = 6x\,\ell n\,x + 3x^2\,.\,\frac{1}{x} + 2x$

$y'' = 6x\,\ell n\,x + 5x$

$y''' = 6\,\ell n\,x + 11$

Solved Example # 20 If $y = \left(\frac{1}{x}\right)^x$ then find $y''(1)$

Solution.

$\ell n\,y = -x\,\ell n\,x$ when $x = 1 \Rightarrow y = 1$

$\Rightarrow \frac{y'}{y} = -(1 + \ell n\,x) \Rightarrow y' = -y\,(1 + \ell n\,x)$ .....(i)

again diff. w.r.t. to x,

$y'' = -y'(1 + \ell n\,x) - y\,.\,\frac{1}{x} \Rightarrow y'' = y\,(1 + \ln x)^2 - \frac{y}{x}$ (using (i)) $\Rightarrow y''(1) = 0$

It must be carefully noted that in case of parametric functions

although $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ **but** $\frac{d^2y}{dx^2} \neq \frac{d^2y/dt^2}{dx^2/dt^2}$ **rather** $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy/dt}{dx/dt}\right)$

which on applying chain rule can be resolved as

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left\{\frac{dy/dt}{dx/dt}\right\}.\frac{dt}{dx} \Rightarrow \frac{d^2y}{dx^2} = \frac{\left(\frac{dx}{dt}.\frac{d^2y}{dt^2} - \frac{dy}{dt}.\frac{d^2x}{dt^2}\right)}{\left(\frac{dx}{dt}\right)^2}.\frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{\left[\frac{dx}{dt}.\frac{d^2y}{dt^2} - \frac{dy}{dt}.\frac{d^2x}{dt^2}\right]}{\left(\frac{dx}{dt}\right)^3}$$

Solved Example # 21 If $x = t + 1$ and $y = t^2 + t^3$ then find $\frac{d^2y}{dx^2}$.

Solution. $\frac{dy}{dt} = 2t + 3t^2$; $\frac{dx}{dt} = 1$

$\Rightarrow \frac{dy}{dx} = 2t + 3t^2 \Rightarrow \frac{d^2y}{dx^2} = \frac{d}{dt}(2t + 3t^2)\,.\,\frac{dt}{dx}$

$\frac{d^2y}{dx^2} = 2 + 6t.$

Solved Example # 22 If $x = 2\cos t - \cos 2t$ and $y = 2\sin t - \sin 2t$ then find value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{2}$.

Solution. $\frac{dy}{dt} = 2\cos t - 2\cos 2t$ $\frac{dx}{dt} = 2\sin 2t - 2\sin t$

$$\frac{dy}{dx} = \frac{\cos t - \cos 2t}{\sin 2t - \sin t} = \frac{2\sin\frac{3t}{2}.\sin\frac{t}{2}}{2\cos\frac{3t}{2}.\sin\frac{t}{2}}.$$

$\Rightarrow \frac{dy}{dx} = \tan\frac{3t}{2} \Rightarrow \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\tan\frac{3t}{2}\right) \Rightarrow \frac{d^2y}{dx^2} = \frac{d}{dt}\left(\tan\frac{3t}{2}\right).\frac{dt}{dx}$

$$\frac{d^2y}{dx^2} = \frac{\frac{3}{2}\cdot\sec^2\frac{3t}{2}}{2(\sin 2t - \sin t)} \qquad \Rightarrow \qquad \left.\frac{d^2y}{dx^2}\right|_{t=\frac{\pi}{2}} = -\frac{3}{2}$$

Solved Example # 23 Find second order derivative of $y = \sin x$ with respect to $z = e^x$.

Solution. $\Rightarrow \quad \frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{\cos x}{e^x}$

$$\Rightarrow \quad \frac{d^2y}{dz^2} = \frac{d}{dz}\left(\frac{\cos x}{e^x}\right) \qquad \Rightarrow \qquad \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{\cos x}{e^x}\right)\cdot\frac{dx}{dz}$$

$$= \frac{-e^x\sin x - \cos x e^x}{(e^x)^2}\cdot\frac{1}{e^x}$$

$$\frac{d^2y}{dz^2} = -\frac{(\sin x + \cos x)}{e^{2x}}$$

Solved Example # 24: $y = f(x)$ and $x = g(y)$ are inverse functions of each other than express $g'(y)$ and $g''(y)$ in terms of derivative of $f(x)$.

Solution. $\frac{dy}{dx} = f'(x)$ and $\frac{dx}{dy} = g'(y)$

$$\Rightarrow \quad g'(y) = \frac{1}{f'(x)} \qquad \text{.....(i)} \qquad \text{again differentiating w.r.t. to y}$$

$$g''(y) = \frac{d}{dy}\left(\frac{1}{f'(x)}\right)$$

$$= \frac{d}{dx}\left(\frac{1}{f'(x)}\right)\cdot\frac{dx}{dy}$$

$$= -\frac{f''(x)}{f'(x)^2}\cdot g'(y)$$

$$\Rightarrow \quad g''(y) = -\frac{f''(x)}{f'(x)^3} \qquad \text{.....(ii) which can also be remembered as}$$

$$\frac{d^2x}{dy^2} = -\frac{\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx}\right)^3}\cdot$$

Solved Example # 25 $y = \sin(\sin x)$ then prove that $y'' + (\tan x)\,y' + y\cos^2 x = 0$

Solution. Such expression can be easily proved using implict differention.

$\Rightarrow \quad y' = \cos(\sin x)\cos x \quad \Rightarrow \quad \sec x\, y' = \cos(\sin x)$

again differentiating w.r.t x, we can get

$\sec x\, y'' + y'\sec x\tan x = -\sin(\sin x)\cos x$

$\Rightarrow \quad \tan x\, y' = -y\cdot\cos^2 x \quad \Rightarrow \quad y'' + (\tan x)\,y' + y\cos^2 x = 0$

Self Practice Problems :

1. If $y = \frac{\ell n x}{x}$ then find $\frac{d^2y}{dx^2}$ **Ans.** $\frac{2\ell n x - 3}{x^3}$

2. Prove that $y = x + \tan x$ satisfies the differentiation equation

$$\cos^2 x\,\frac{d^2y}{dx^2} - 2y + 2x = 0.$$

3. If $x = a(\cos\theta + \theta\sin\theta)$ and $y = a(\sin\theta - \theta\cos\theta)$ then find $\frac{d^2y}{dx^2}$. **Ans.** $\frac{\sec^3\theta}{a\theta}$

4. Find second derivative of ℓnx with respect to $\sin x$. **Ans.** $\frac{x\sin x - \cos x}{x^2\cos^3 x}$

5. if $y = e^{-x}(A\cos x + B\sin x)$, prove that

$$\frac{d^2y}{dx^2} + 2\cdot\frac{dy}{dx} + 2y = 0.$$

Solved Example # 26 If $y = (\tan^{-1}x)^2$ then prove that $(1 + x^2)^2\frac{d^2y}{dx^2} + 2x(1 + x^2)\frac{dy}{dx} = 2$.

Solution.

$$\frac{dy}{dx} = \frac{2\tan^{-1}x}{1+x^2}$$

$$\Rightarrow \quad (1 + x^2)\frac{dy}{dx} = 2\tan^{-1}(x) \qquad \Rightarrow \qquad (1 + x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = \frac{2}{(1+x^2)}$$

$$\Rightarrow \quad (1 + x^2)\frac{d^2y}{dx^2} + 2x(1 + x^2)\frac{dy}{dx} = 2$$

11. If $F(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ l(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix}$, where f, g, h, l, m, n, u, v, w are differentiable functions of x then $F'(x)$

$= \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ l(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ l'(x) & m'(x) & n'(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ l(x) & m(x) & n(x) \\ u'(x) & v'(x) & w'(x) \end{vmatrix}$

12. L' Hospital's Rule:

If f(x) & g(x) are functions of x such that:

(i) $\lim\limits_{x\to a} f(x) = 0 = \lim\limits_{x\to a} g(x)$ OR $\lim\limits_{x\to a} f(x) = \infty = \lim\limits_{x\to a} g(x)$ &

(ii) Both f(x) & g(x) are continuous at x = a &

(iii) Both f(x) & g(x) are differentiable at x = a &

(iv) Both $f'(x)$ & $g'(x)$ are continuous at x = a, Then

$\lim\limits_{x\to a} \frac{f(x)}{g(x)} = \lim\limits_{x\to a} \frac{f'(x)}{g'(x)} = \lim\limits_{x\to a} \frac{f''(x)}{g''(x)}$ & so on till indeterminant form vanishes

QUESTION BANK ON METHOD OF DIFFERENTIATION

***Select the correct alternative* : (Only one is correct)**

Q.1 If g is the inverse of f & $f'(x) = \frac{1}{1+x^5}$ then $g'(x) =$

(A) $1 + [g(x)]^5$ (B) $\frac{1}{1+[g(x)]^5}$ (C) $-\frac{1}{1+[g(x)]^5}$ (D) none

Q.2 If $y = \tan^{-1}\left(\frac{\ell n \frac{e}{x^2}}{\ell n e x^2}\right) + \tan^{-1}\frac{3+2\ell n x}{1-6\ell n x}$ then $\frac{d^2y}{dx^2} =$

(A) 2 (B) 1 (C) 0 (D) -1

Q.3 If $y = f\left(\frac{3x+4}{5x+6}\right)$ & $f'(x) = \tan x^2$ then $\frac{dy}{dx} =$

(A) $\tan x^3$ (B) $-2\tan\left[\frac{3x+4}{5x+6}\right]^2 . \frac{1}{(5x+6)^2}$ (C) $f\left(\frac{3\tan x^2+4}{5\tan x^2+6}\right)\tan x^2$ (D) none

Q.4 If $y = \sin^{-1}\left(x\sqrt{1-x} + \sqrt{x}\sqrt{1-x^2}\right)$ & $\frac{dy}{dx} = \frac{1}{2\sqrt{x(1-x)}} + p$, then $p =$

(A) 0 (B) $\sin^{-1} x$ (C) $\sin^{-1}\sqrt{x}$ (D) none of these

Q.5 If $y = f\left(\frac{2x-1}{x^2+1}\right)$ & $f'(x) = \sin x$ then $\frac{dy}{dx} =$

(A) $\frac{1+x-x^2}{\left(1+x^2\right)^2}\sin\left(\frac{2x-1}{x^2+1}\right)$ (B) $\frac{2\left(1+x-x^2\right)}{\left(1+x^2\right)^2}\sin\left(\frac{2x-1}{x^2+1}\right)$ (C) $\frac{1-x+x^2}{\left(1+x^2\right)^2}\sin\left(\frac{2x-1}{x^2+1}\right)$ (D) none

Q.6 Let g is the inverse function of f & $f'(x) = \frac{x^{10}}{\left(1+x^2\right)}$. If $g(2) = a$ then $g'(2)$ is equal to

(A) $\frac{5}{2^{10}}$ (B) $\frac{1+a^2}{a^{10}}$ (C) $\frac{a^{10}}{1+a^2}$ (D) $\frac{1+a^{10}}{a^2}$

Q.7 If $\sin(xy) + \cos(xy) = 0$ then $\frac{dy}{dx} =$

(A) $\frac{y}{x}$ (B) $-\frac{y}{x}$ (C) $-\frac{x}{y}$ (D) $\frac{x}{y}$

Q.8 If $y = \sin^{-1}\frac{2x}{1+x^2}$ then $\left.\frac{dy}{dx}\right]_{x=-2}$ is :

(A) $\frac{2}{5}$ (B) $\frac{2}{\sqrt{5}}$ (C) $-\frac{2}{5}$ (D) none

Q.9 The derivative of $\sec^{-1}\left(\frac{1}{2x^2-1}\right)$ w.r.t. $\sqrt{1-x^2}$ at $x=\frac{1}{2}$ is :

(A) 4 (B) 1/4 (C) 1 (D) none

Q.10 If $y^2 = P(x)$, is a polynomial of degree 3, then $2\left(\frac{d}{dx}\right)\left(y^3 \cdot \frac{d^2y}{dx^2}\right)$ equals :

(A) $P'''(x) + P'(x)$ (B) $P''(x) \cdot P'''(x)$ (C) $P(x) \cdot P'''(x)$ (D) a constant

Q.11 Let f(x) be a quadratic expression which is positive for all real x . If $g(x) = f(x) + f'(x) + f''(x)$, then for any real x, which one is correct .

(A) $g(x) < 0$ (B) $g(x) > 0$ (C) $g(x) = 0$ (D) $g(x) \geq 0$

Q.12 If $x^p \cdot y^q = (x+y)^{p+q}$ then $\frac{dy}{dx}$ is :

(A) independent of p but dependent on q (B) dependent on p but independent of q
(C) dependent on both p & q (D) independent of p & q both .

Q.13 Let $f(x) = \begin{cases} g(x) \cdot \cos\frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ where g(x) is an even function differentiable at $x = 0$, passing through the origin . Then $f'(0)$:(A) is equal to 1 (B) is equal to 0 (C) is equal to 2 (D) does not exist

Q.14 If $y = \frac{1}{1+x^{n-m}+x^{p-m}} + \frac{1}{1+x^{m-n}+x^{p-n}} + \frac{1}{1+x^{m-p}+x^{n-p}}$ then $\frac{dy}{dx}$ at $e^{m^{np}}$ is equal to:

(A) e^{mnp} (B) $e^{mn/p}$ (C) $e^{np/m}$ (D) none

Q.15 $\lim\limits_{x\to 0} \frac{\log_{\sin^2 x} \cos x}{\log_{\sin^2 \frac{x}{2}} \cos\frac{x}{2}}$ has the value equal to

(A) 1 (B) 2 (C) 4 (D) none of these

Q.16 If f is differentiable in (0, 6) & $f'(4) = 5$ then $\lim\limits_{x\to 2} \frac{f(4) - f\left(x^2\right)}{2-x} =$

(A) 5 (B) 5/4 (C) 10 (D) 20

Q.17 Let $l = \lim\limits_{x\to 0^+} x^m (\ln x)^n$ where $m, n \in N$ then :

(A) l is independent of m and n (B) l is independent of m and depends on m
(C) l is independent of n and dependent on m (D) l is dependent on both m and n

Q.18 Let $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2\sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$. Then $\lim\limits_{x\to 0} \frac{f'(x)}{x} =$

(A) 2 (B) -2 (C) -1 (D) 1

Q.19 Let $f(x) = \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2\cos 2x \\ \cos 3x & \sin 3x & 3\cos 3x \end{vmatrix}$ then $f'\left(\frac{\pi}{2}\right) =$

(A) 0 (B) -12 (C) 4 (D) 12

Q.20 People living at Mars, instead of the usual definition of derivative D f(x), define a new kind of derivative, D*f(x) by the formula

$D^*f(x) = \lim\limits_{h\to 0} \frac{f^2(x+h) - f^2(x)}{h}$ where $f^2(x)$ means $[f(x)]^2$. If $f(x) = x\,\ln x$ then

$D^*f(x)\big|_{x=e}$ has the value

(A) e (B) 2e (C) 4e (D) none

Q.21 If $f(4) = g(4) = 2$; $f'(4) = 9$; $g'(4) = 6$ then $\lim\limits_{x\to 4} \frac{\sqrt{f(x)} - \sqrt{g(x)}}{\sqrt{x} - 2}$ is equal to :

(A) $3\sqrt{2}$ (B) $\frac{3}{\sqrt{2}}$ (C) 0 (D) none

Q.22 If f(x) is a differentiable function of x then $\lim\limits_{h\to 0} \frac{f(x+3h) - f(x-2h)}{h} =$

(A) $f'(x)$ (B) $5f'(x)$ (C) 0 (D) none

Q.23 If $y = x + e^x$ then $\frac{d^2x}{dy^2}$ is :

(A) e^x (B) $-\frac{e^x}{\left(1+e^x\right)^3}$ (C) $-\frac{e^x}{\left(1+e^x\right)^2}$ (D) $\frac{-1}{\left(1+e^x\right)^3}$

Q.24 If $x^2y + y^3 = 2$ then the value of $\frac{d^2y}{dx^2}$ at the point (1, 1) is :

(A) $-\frac{3}{4}$ (B) $-\frac{3}{8}$ (C) $-\frac{5}{12}$ (D) none

Q.25 If $f(a) = 2,\ f'(a) = 1,\ g(a) = -1,\ g'(a) = 2$ then the value of $\underset{x \to a}{\text{Limit}} \frac{g(x) \cdot f(a) - g(a) \cdot f(x)}{x - a}$ is:

(A) -5 (B) 1/5 (C) 5 (D) none

Q.26 If f is twice differentiable such that $f''(x) = -f(x),\ f'(x) = g(x)$

$h'(x) = \left[f(x)\right]^2 + \left[g(x)\right]^2$ and

$h(0) = 2,\ h(1) = 4$

then the equation $y = h(x)$ represents :

(A) a curve of degree 2 (B) a curve passing through the origin

(C) a straight line with slope 2 (D) a straight line with y intercept equal to -2.

Q.27 The derivative of the function, $f(x)=\cos^{-1}\left\{\frac{1}{\sqrt{13}}(2\cos x - 3\sin x)\right\} + \sin^{-1}\left\{\frac{1}{\sqrt{13}}(2\cos x + 3\sin x)\right\}$ w.r.t. $\sqrt{1+x^2}$ at $x = \frac{3}{4}$ is :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{10}{3}$ (D) 0

Q.28 Let f(x) be a polynomial in x. Then the second derivative of $f(e^x)$, is :

(A) $f''(e^x) \cdot e^x + f'(e^x)$ (B) $f''(e^x) \cdot e^{2x} + f'(e^x) \cdot e^{2x}$

(C) $f''(e^x)\, e^{2x}$ (D) $f''(e^x) \cdot e^{2x} + f'(e^x) \cdot e^x$

Q.29 The solution set of $f'(x) > g'(x)$, where $f(x) = \frac{1}{2}(5^{2x+1})$ & $g(x) = 5^x + 4x\,(\ln 5)$ is :

(A) $x > 1$ (B) $0 < x < 1$ (C) $x \le 0$ (D) $x > 0$

Q.30 If $y = \sin^{-1}\frac{x^2-1}{x^2+1} + \sec^{-1}\frac{x^2+1}{x^2-1},\ |x| > 1$ then $\frac{dy}{dx}$ is equal to :

(A) $\frac{x}{x^4-1}$ (B) $\frac{x^2}{x^4-1}$ (C) 0 (D) 1

Q.31 If $y = \frac{x}{a+}\ \frac{x}{b+}\ \frac{x}{a+}\ \frac{x}{b+}\ \frac{x}{a+}\ \frac{x}{b+}\ \ldots\ldots \infty$ then $\frac{dy}{dx}$ =

(A) $\frac{a}{ab+2ay}$ (B) $\frac{b}{ab+2by}$ (C) $\frac{a}{ab+2by}$ (D) $\frac{b}{ab+2ay}$

Q.32 Let f (x) be a polynomial function of second degree. If f (1) = f (–1) and a, b, c are in A.P., then f '(a), f '(b) and f '(c) are in

(A) G.P. (B) H.P. (C) A.G.P. (D) A.P.

Q.33 If $y = \sin mx$ then the value of $\begin{vmatrix} y & y_1 & y_2 \\ y_3 & y_4 & y_5 \\ y_6 & y_7 & y_8 \end{vmatrix}$ (where subscripts of y shows the order of derivatiive) is:

(A) independent of x but dependent on m (B) dependent of x but independent of m

(C) dependent on both m & x (D) independent of m & x .

Q.34 If $x^2 + y^2 = R^2\ (R > 0)$ then $k = \frac{y''}{\sqrt{\left(1+y'^2\right)^3}}$ where k in terms of R alone is equal to

(A) $-\frac{1}{R^2}$ (B) $-\frac{1}{R}$ (C) $\frac{2}{R}$ (D) $-\frac{2}{R^2}$

Q.35 If f & g are differentiable functions such that $g'(a) = 2$ & $g(a) = b$ and if fog is an identity function then $f'(b)$ has the value equal to :

(A) 2/3 (B) 1 (C) 0 (D) 1/2

Q.36 Given $f(x) = -\frac{x^3}{3} + x^2 \sin 1.5\,a - x \sin a \cdot \sin 2a - 5 \arcsin\,(a^2 - 8a + 17)$ then :

(A) f(x) is not defined at $x = \sin 8$ (B) $f'(\sin 8) > 0$

(C) $f'(x)$ is not defined at $x = \sin 8$ (D) $f'(\sin 8) < 0$

Q.37 A function f, defined for all positive real numbers, satisfies the equation $f(x^2) = x^3$ for every $x > 0$. Then the value of $f'(4) =$
(A) 12 (B) 3 (C) 3/2 (D) cannot be determined

Q.38 Given : $f(x) = 4x^3 - 6x^2 \cos 2a + 3x \sin 2a \cdot \sin 6a + \sqrt{\ell n\left(2a - a^2\right)}$ then :
(A) f(x) is not defined at $x = 1/2$ (B) $f'(1/2) < 0$
(C) $f'(x)$ is not defined at $x = 1/2$ (D) $f'(1/2) > 0$

Q.39 If $y = (A + Bx)\, e^{mx} + (m-1)^{-2} e^x$ then $\frac{d^2y}{dx^2} - 2m \frac{dy}{dx} + m^2 y$ is equal to :
(A) e^x (B) e^{mx} (C) e^{-mx} (D) $e^{(1-m)x}$

Q.40 Suppose $f(x) = e^{ax} + e^{bx}$, where $a \neq b$, and that $f''(x) - 2f'(x) - 15f(x) = 0$ for all x. Then the product ab is equal to
(A) 25 (B) 9 (C) – 15 (D) – 9

Q.41 Let h (x) be differentiable for all x and let $f(x) = (kx + e^x)\, h(x)$ where k is some constant. If $h(0) = 5$, $h'(0) = -2$ and $f'(0) = 18$ then the value of k is equal to
(A) 5 (B) 4 (C) 3 (D) 2.2

Q.42 Let $e^{f(x)} = \ell n\, x$. If g(x) is the inverse function of f(x) then $g'(x)$ equals to :
(A) e^x (B) $e^x + x$ (C) $e^{(x + e^x)}$ (D) $e^{(x + \ell n\, x)}$

Q.43 The equation $y^2 e^{xy} = 9e^{-3} \cdot x^2$ defines y as a differentiable function of x. The value of $\frac{dy}{dx}$ for $x = -1$ and $y = 3$ is
(A) $-\frac{15}{2}$ (B) $-\frac{9}{5}$ (C) 3 (D) 15

Q.44 Let $f(x) = \left(x^x\right)^x$ and $g(x) = x^{\left(x^x\right)}$ then :
(A) $f'(1) = 1$ and $g'(1) = 2$ (B) $f'(1) = 2$ and $g'(1) = 1$
(C) $f'(1) = 1$ and $g'(1) = 0$ (D) $f'(1) = 1$ and $g'(1) = 1$

Q.45 The function $f(x) = e^x + x$, being differentiable and one to one, has a differentiable inverse $f^{-1}(x)$. The value of $\frac{d}{dx}(f^{-1})$ at the point $f(\ell n 2)$ is
(A) $\frac{1}{\ell n 2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) none

Q.46 If $f(x) = \dfrac{\log_{\sin|x|} \cos^3 x}{\log_{\sin|3x|} \cos^3\left(\frac{x}{2}\right)}$ for $|x| < \frac{\pi}{3}$ $x \neq 0$

$= 4$ for $x = 0$

then, the number of points of discontinuity of f in $\left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$ is
(A) 0 (B) 3 (C) 2 (D) 4

Q.47 If $y = \dfrac{(a-x)\sqrt{a-x} - (b-x)\sqrt{x-b}}{\sqrt{a-x} + \sqrt{x-b}}$ then $\frac{dy}{dx}$ wherever it is defined is equal to :
(A) $\dfrac{x + (a+b)}{\sqrt{(a-x)(x-b)}}$ (B) $\dfrac{2x - (a+b)}{2\sqrt{(a-x)(x-b)}}$ (C) $-\dfrac{(a+b)}{2\sqrt{(a-x)(x-b)}}$ (D) $\dfrac{2x + (a+b)}{2\sqrt{(a-x)(x-b)}}$

Q.48 If y is a function of x then $\frac{d^2y}{dx^2} + y\frac{dy}{dx} = 0$. If x is a function of y then the equation becomes :
(A) $\frac{d^2x}{dy^2} + x\frac{dx}{dy} = 0$ (B) $\frac{d^2x}{dy^2} + y\left(\frac{dx}{dy}\right)^3 = 0$ (C) $\frac{d^2x}{dy^2} - y\left(\frac{dx}{dy}\right)^2 = 0$ (D) $\frac{d^2x}{dy^2} - x\left(\frac{dx}{dy}\right)^2 = 0$

Q.49 A function $f(x)$ satisfies the condition, $f(x) = f'(x) + f''(x) + f'''(x) + \ldots\ldots \infty$ where $f(x)$ is a differentiable function indefinitely and dash denotes the order of derivative . If $f(0) = 1$, then $f(x)$ is :
(A) $e^{x/2}$ (B) e^x (C) e^{2x} (D) e^{4x}

Q.50 If $y = \dfrac{\cos 6x + 6\cos 4x + 15\cos 2x + 10}{\cos 5x + 5\cos 3x + 10\cos x}$, then $\frac{dy}{dx} =$
(A) 2 sinx + cosx (B) –2sinx (C) cos2x (D) sin2x

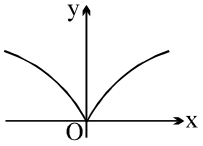
Q.51 If $\frac{d^2x}{dy^2}\left(\frac{dy}{dx}\right)^3 + \frac{d^2y}{dx^2} = K$ then the value of K is equal to
(A) 1 (B) –1 (C) 2 (D) 0

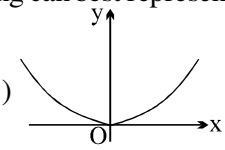
Q.52 If $f(x) = 2\sin^{-1}\sqrt{1-x} + \sin^{-1}\left(2\sqrt{x(1-x)}\right)$ where $x \in \left(0, \frac{1}{2}\right)$ then $f'(x)$ has the value equal to

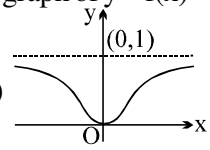
(A) $\frac{2}{\sqrt{x(1-x)}}$ (B) zero (C) $-\frac{2}{\sqrt{x(1-x)}}$ (D) π

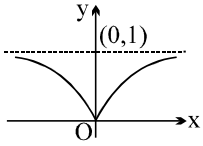
Q.53 Let $y = f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Then which of the following can best represent the graph of $y = f(x)$?

(A) 

(B) 

(C) 

(D) 

Q.54 Diffrential coefficient of $\left(x^{\frac{\ell+m}{m-n}}\right)^{\frac{1}{n-\ell}} \cdot \left(x^{\frac{m+n}{n-\ell}}\right)^{\frac{1}{\ell-m}} \cdot \left(x^{\frac{n+\ell}{\ell-m}}\right)^{\frac{1}{m-n}}$ w.r.t. x is

(A) 1 (B) 0 (C) – 1 (D) $x^{\ell mn}$

Q.55 Let $f(x)$ be diffrentiable at $x = h$ then $\lim_{x\to h} \frac{(x+h)\, f(x) - 2h\, f(h)}{x - h}$ is equal to

(A) $f(h) + 2hf'(h)$ (B) $2f(h) + hf'(h)$ (C) $hf(h) + 2f'(h)$ (D) $hf(h) - 2f'(h)$

Q.56 If $y = at^2 + 2bt + c$ and $t = ax^2 + 2bx + c$, then $\frac{d^3y}{dx^3}$ equals

(A) $24\, a^2 (at + b)$ (B) $24\, a (ax + b)^2$ (C) $24\, a (at + b)^2$ (D) $24\, a^2 (ax + b)$

Q.57 $\lim_{x\to 0^+} \frac{1}{x\sqrt{x}} \left(a \arctan\frac{\sqrt{x}}{a} - b \arctan\frac{\sqrt{x}}{b} \right)$ has the value equal to

(A) $\frac{a-b}{3}$ (B) 0 (C) $\frac{(a^2-b^2)}{6a^2b^2}$ (D) $\frac{a^2-b^2}{3a^2b^2}$

Q.58 Let $f(x)$ be defined for all $x > 0$ & be continuous. Let $f(x)$ satisfy $f\left(\frac{x}{y}\right) = f(x) - f(y)$ for all x, y & $f(e) = 1$.

Then :

(A) $f(x)$ is bounded (B) $f\left(\frac{1}{x}\right) \to 0$ as $x \to 0$ (C) $x.f(x) \to 1$ as $x \to 0$ (D) $f(x) = \ln x$

Q.59 Suppose the function $f(x) - f(2x)$ has the derivative 5 at $x = 1$ and derivative 7 at $x = 2$. The derivative of the function $f(x) - f(4x)$ at $x = 1$, has the value equal to

(A) 19 (B) 9 (C) 17 (D) 14

Q.60 If $y = \frac{x^4 - x^2 + 1}{x^2 + \sqrt{3}x + 1}$ and $\frac{dy}{dx} = ax + b$ then the value of $a + b$ is equal to

(A) $\cot\frac{5\pi}{8}$ (B) $\cot\frac{5\pi}{12}$ (C) $\tan\frac{5\pi}{12}$ (D) $\tan\frac{5\pi}{8}$

Q.61 Suppose that $h(x) = f(x) \cdot g(x)$ and $F(x) = f\left(g(x)\right)$, where $f(2) = 3$; $g(2) = 5$; $g'(2) = 4$; $f'(2) = -2$ and $f'(5) = 11$, then

(A) $F'(2) = 11\, h'(2)$ (B) $F'(2) = 22h'(2)$ (C) $F'(2) = 44\, h'(2)$ (D) none

Q.62 Let $f(x) = x^3 + 8x + 3$

which one of the properties of the derivative enables you to conclude that $f(x)$ has an inverse?

(A) $f'(x)$ is a polynomial of even degree. (B) $f'(x)$ is self inverse.

(C) domain of $f'(x)$ is the range of $f'(x)$. (D) $f'(x)$ is always positive.

Q.63 Which one of the following statements is NOT CORRECT ?

(A) The derivative of a diffrentiable periodic function is a periodic function with the same period.

(B) If $f(x)$ and $g(x)$ both are defined on the entire number line and are aperiodic then the function $F(x) = f(x) \cdot g(x)$ can not be periodic.

(C) Derivative of an even differentiable function is an odd function and derivative of an odd differentiable function is an even function.

(D) Every function $f(x)$ can be represented as the sum of an even and an odd function

Select the correct alternatives : (More than one are correct)

Q.64 If $y = \tan x \tan 2x \tan 3x$ then $\frac{dy}{dx}$ has the value equal to :

(A) $3\sec^2 3x\tan x\tan 2x+\sec^2 x\tan 2x\tan 3x+2\sec^2 2x\tan 3x\tan x$
(B) $2y(\operatorname{cosec} 2x+2\operatorname{cosec} 4x+3\operatorname{cosec} 6x)$
(C) $3\sec^2 3x-2\sec^2 2x-\sec^2 x$ (D) $\sec^2 x+2\sec^2 2x+3\sec^2 3x$

Q.65 If $y=e^{\sqrt{x}}+e^{-\sqrt{x}}$ then $\frac{dy}{dx}$ equals

(A) $\frac{e^{\sqrt{x}}-e^{-\sqrt{x}}}{2\sqrt{x}}$ (B) $\frac{e^{\sqrt{x}}-e^{-\sqrt{x}}}{2x}$ (C) $\frac{1}{2\sqrt{x}}\sqrt{y^2-4}$ (D) $\frac{1}{2\sqrt{x}}\sqrt{y^2+4}$

Q.66 If $y=x^{x^2}$ then $\frac{dy}{dx}=$ (A) $2\,ln\,x\,.\,x^{x^2}$ (B) $(2\,ln\,x+1).x^{x^2}$ (C) $(2\,ln\,x+1).x^{x^2+1}$ (D) $x^{x^2+1}\,.\,ln\,ex^2$

Q.67 Let $y=\sqrt{x+\sqrt{x+\sqrt{x+.....\infty}}}$ then $\frac{dy}{dx}=$

(A) $\frac{1}{2y-1}$ (B) $\frac{x}{x+2y}$ (C) $\frac{1}{\sqrt{1+4x}}$ (D) $\frac{y}{2x+y}$

Q.68 If $2^x+2^y=2^{x+y}$ then $\frac{dy}{dx}$ has the value equal to :

(A) $-\frac{2^y}{2^x}$ (B) $\frac{1}{1-2^x}$ (C) $1-2^y$ (D) $\frac{2^x\left(1-2^y\right)}{2^y\left(2^x-1\right)}$

Q.69 The functions $u=e^x\sin x$; $v=e^x\cos x$ satisfy the equation :

(A) $v\frac{du}{dx}-u\frac{dv}{dx}=u^2+v^2$ (B) $\frac{d^2u}{dx^2}=2v$ (C) $\frac{d^2v}{dx^2}=-2u$ (D) none of these

Q.70 Let $f(x)=\frac{\sqrt{x-2\sqrt{x-1}}}{\sqrt{x-1}-1}.x$ then :

(A) $f'(10)=1$ (B) $f'(3/2)=-1$ (C) domain of $f(x)$ is $x\ge 1$ (D) none

Q.71 Two functions f & g have first & second derivatives at $x=0$ & satisfy the relations,

$f(0)=\frac{2}{g(0)}$, $f'(0)=2\,g'(0)=4g(0)$, $g''(0)=5\,f''(0)=6\,f(0)=3$ then :

(A) if $h(x)=\frac{f(x)}{g(x)}$ then $h'(0)=\frac{15}{4}$ (B) if $k(x)=f(x)\,.\,g(x)\sin x$ then $k'(0)=2$

(C) $\lim\limits_{x\to 0}\frac{g'(x)}{f'(x)}=\frac{1}{2}$ (D) none

Q.72 If $y=x^{(\ell nx)^{\ell n(\ell nx)}}$, then $\frac{dy}{dx}$ is equal to :

(A) $\frac{y}{x}\left(\ell nx^{\ell nx-1}+2\,\ell nx\;\ell n\left(\ell nx\right)\right)$ (B) $\frac{y}{x}(ln\,x)^{ln\,(ln\,x)}(2\,ln\,(ln\,x)+1)$

(C) $\frac{y}{x\,\ell nx}((ln\,x)^2+2\,ln\,(ln\,x))$ (D) $\frac{y\,\ell ny}{x\,\ell nx}(2\,ln\,(ln\,x)+1)$

ANSWER KEY

Q.1	A	Q.2	C	Q.3	B	Q.4	D	Q.5	B
Q.6	B	Q.7	B	Q.8	C	Q.9	A	Q.10	C
Q.11	B	Q.12	D	Q.13	B	Q.14	D	Q.15	C
Q.16	D	Q.17	A	Q.18	B	Q.19	C	Q.20	C
Q.21	A	Q.22	B	Q.23	B	Q.24	B	Q.25	C
Q.26	C	Q.27	C	Q.28	D	Q.29	D	Q.30	C
Q.31	D	Q.32	D	Q.33	D	Q.34	B	Q.35	D
Q.36	D	Q.37	B	Q.38	D	Q.39	A	Q.40	C
Q.41	C	Q.42	C	Q.43	D	Q.44	D	Q.45	B
Q.46	C	Q.47	B	Q.48	C	Q.49	A	Q.50	B
Q.51	D	Q.52	B	Q.53	C	Q.54	B	Q.55	A
Q.56	D	Q.57	D	Q.58	D	Q.59	A	Q.60	B
Q.61	B	Q.62	D	Q.63	B				
Q.64	A,B,C	Q.65	A,C	Q.66	C,D	Q.67	A,C,D		
Q.68	A,B,C,D	Q.69	A,B,C	Q.70	A,B	Q.71	A,B,C	Q.72	B,D

EXERCISE -1

Part : (A) Only one correct option

1. If $f'(x) = \sqrt{2x^2 - 1}$ and $y = f(x^2)$ then $\frac{dy}{dx}$ at $x = 1$ is
(A) 2 (B) 1 (C) – 2 (D) –1

2. If $y = x^{x^2}$ then $\frac{dy}{dx}$ =
(A) $2\,\ell n\, x.\ x^{x^2}$ (B) $(2\,\ell n\, x + 1).\ x^{x^2}$ (C) $(2\,\ell n\, x + 1).\ x^{x^2+1}$ (D) $x^{x^2+1}.\ \ell n\, ex^2$

3. If $f(x) = e^{\tan^{-1}\left(\sin\frac{x}{2}\right)}$, then $f'(0)$.
(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 1 (D) –1

4. If $y = \cfrac{x}{a+\cfrac{x}{b+\cfrac{x}{a+.....}}}$ then $\frac{dy}{dx}$ =

(A) $\frac{a}{ab+2ay}$ (B) $\frac{b}{ab+2by}$ (C) $\frac{a}{ab+2by}$ (D) $\frac{b}{ab+2ay}$

5. Let $f(x) = \sin x$; $g(x) = x^2$ & $h(x) = \log_e x$ & $F(x) = h[g(f(x))]$ then $\frac{d^2F}{dx^2}$ is equal to:
(A) $2\,\text{cosec}^3 x$ (B) $2\cot(x^2) - 4x^2\,\text{cosec}^2(x^2)$ (C) $2x\cot x^2$ (D) $-2\,\text{cosec}^2 x$

6. If $y = (1 + x)(1 + x^2)(1 + x^4)\ldots..(1 + x^{2^n})$, then $\frac{dy}{dx}$ at $x = 0$ is
(A) –1 (B) 1 (C) 0 (D) 2^n

7. If $y = \sin^{-1}\left(x\sqrt{1-x} + \sqrt{x}\sqrt{1-x^2}\right)$ and $\frac{dy}{dx} = \frac{1}{2\sqrt{x(1-x)}} + p$, then p =

(A) 0 (B) $\frac{1}{\sqrt{1-x}}$ (C) $\sin^{-1}\sqrt{x}$ (D) $\frac{1}{\sqrt{1-x^2}}$

8. If $\sqrt{x^2+y^2} = e^t$ where $t = \sin^{-1}\left(\frac{y}{\sqrt{x^2+y^2}}\right)$ then $\frac{dy}{dx}$:

(A) $\frac{x-y}{x+y}$ (B) $\frac{x+y}{x-y}$ (C) $\frac{2x+y}{x-y}$ (D) $\frac{x-y}{2x+y}$

9. If $y = \sin^{-1}\frac{x^2-1}{x^2+1} + \sec^{-1}\frac{x^2+1}{x^2-1}$, $|x| > 1$ then $\frac{dy}{dx}$ is equal to:

(A) $\frac{x}{x^4-1}$ (B) $\frac{x^2}{x^4-1}$ (C) 0 (D) 1

10. The differential coefficient of $\sin^{-1}\frac{t}{\sqrt{1+t^2}}$ w.r.t. $\cos^{-1}\frac{1}{\sqrt{1+t^2}}$ is:

(A) 1 (B) t (C) $\frac{1}{\sqrt{1+t^2}}$ (D) none

11. Differentiation of $\left(\frac{\tan^{-1}x}{1+\tan^{-1}x}\right)$ w.r.t. $\tan^{-1}x$ is:

(A) $\left(\frac{1}{1+\tan^{-1}x}\right)$ (B) – 1 (C) $\frac{1}{\left(1+\tan^{-1}x\right)^2}$ (D) $\frac{-1}{(1+\tan^{-1}x)^2}$

12. Let f(x) be a polynomial in x. Then the second derivative of $f(e^x)$, is:
(A) $f''(e^x).e^x + f'(e^x)$ (B) $f''(e^x).e^{2x} + f'(e^x).e^{2x}$ (C) $f''(e^x)\,e^{2x}$ (D) $f''(e^x).e^{2x} + f'(e^x).e^x$

13. If f(x), g(x), h(x) are polynomials in x of degree 2 and $F(x) = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix}$, then $F'(x)$ is equal to
(A) 1 (B) 0 (C) –1 (D) f(x) . g(x) . h(x)

14. If $y = \sin mx$ then the value of $\begin{vmatrix} y & y_1 & y_2 \\ y_3 & y_4 & y_5 \\ y_6 & y_7 & y_8 \end{vmatrix}$ (where settings of y shows the order of derivative) is:

(A) independent of x but dependent on m (B) dependent of x but independent of m
(C) dependent on both m & x (D) independent of m & x.

15. If $f'(5) = 7$ then $\lim_{t\to 0} \frac{f(5+t) - f(5-t)}{2t}$ =

(A) 0 (B) 3.5 (C) 7 (D) 14

16. Let $e^{f(x)} = \ln x$. If g(x) is the inverse function of f(x) then g′(x) equals to:

(A) e^x (B) $e^x + x$ (C) e^{x+e^x} (D) $e^{x + \ln x}$

17. If $u = ax + b$ then $\frac{d^n}{dx^n}[f(ax+b)]$ is equal to:

(A) $\frac{d^n}{du^n}[f(u)]$ (B) $a\frac{d^n}{du^n}[f(u)]$ (C) $a^n\frac{d^n}{du^n}[f(u)]$ (D) $a^{-n}\frac{d^n}{dx^n}[f(u)]$

18. If $y = f\left(\frac{2x-1}{x^2+1}\right)$ & $f'(x) = \sin x$ then $\frac{dy}{dx}$ =

(A) $\frac{1+x-x^2}{\left(1+x^2\right)^2}\sin\left(\frac{2x-1}{x^2+1}\right)$ (B) $\frac{2\left(1+x-x^2\right)}{\left(1+x^2\right)^2}\sin\left(\frac{2x-1}{x^2+1}\right)$

(C) $\frac{1-x+x^2}{\left(1+x^2\right)^2}\sin\left(\frac{2x-1}{x^2+1}\right)$ (D) none

19. If $y^2 = P(x)$, is a polynomial of degree 3, then $2\left(\frac{d}{dx}\right)\left(y^3 \cdot \frac{d^2y}{dx^2}\right)$ equals:

(A) $P'''(x) + P'(x)$ (B) $P''(x).P'''(x)$ (C) $P(x).P'''(x)$ (D) a constant

Part : (B) May have more than one options correct

20. Two functions f & g have first & second derivatives at x = 0 & satisfy the relations,

$f(0) = \frac{2}{g(0)}$, $f'(0) = 2\,g'(0) = 4g(0)$, $g''(0) = 5\,f''(0) = 6\,f(0) = 3$ then:

(A) if $h(x) = \frac{f(x)}{g(x)}$ then $h'(0) = \frac{15}{4}$ (B) if $k(x) = f(x).g(x)\sin x$ then $k'(0) = 2$

(C) $\lim_{x\to 0}\frac{g'(x)}{f'(x)} = \frac{1}{2}$ (D) none

21. If $f_n(x) = e^{f_{n-1}(x)}$ for all $n \in N$ and $f_o(x) = x$, then $\frac{d}{dx}\{f_n(x)\}$ is equal to:

(A) $f_n(x).\frac{d}{dx}\{f_{n-1}(x)\}$ (B) $f_n(x).f_{n-1}(x)$

(C) $f_n(x).f_{n-1}(x)\ldots\ldots f_2(x).f_1(x)$ (D) none of these

22. If f is twice differentiable such that $f''(x) = -f(x)$ and $f'(x) = g(x)$. If h(x) is a twice differentiable function such that $h'(x) = [f(x)]^2 + [g(x)]^2$. If $h(0) = 2$, $h(1) = 4$, then the equation $y = h(x)$ represents:
(A) a curve of degree 2 (B) a curve passing through the origin
(C) a straight line with slope 2 (D) a straight line with y intercept equal to 2.

23. Given $f(x) = -\frac{x^3}{3} + x^2\sin 1.5\,a - x\sin a.\sin 2a - 5\sin^{-1}(a^2 - 8a + 17)$ then:

(A) $f'(x) = -x^2 + 2x\sin 6 - \sin 4\sin 8$ (B) $f'(\sin 8) > 0$
(C) $f'(x)$ is not defined at $x = \sin 8$ (D) $f'(\sin 8) < 0$

24. If $f(x) = x^3 + x^2 f'(1) + xf''(3)$ for all $x \in R$ then

(A) $f(0) + f(2) = f(1)$ (B) $f(0) + f(3) = 0$ (C) $f(1) + f(3) = f(2)$ (D) none of these

25. If $f(x) = (ax + b)\sin x + (cx + d)\cos x$, then the values of a, b, c and d such that $f'(x) = x\cos x$ for all x are
(A) $a = d = 1$ (B) $b = 0$ (C) $c = 0$ (D) $b = c$

EXERCISE -2

1. If $y = A\,e^{-kt}\cos(p\,t + c)$ then prove that $\frac{d^2y}{dt^2} + 2\,k\frac{dy}{dt} + n^2 y = 0$, where $n^2 = p^2 + k^2$.

Evaluate the following limits using L′ hospitale rule as otherwise

2. $\lim_{x\to 0}\log_{\tan^2 x}(\tan^2 2x)$

3. If $f(x) = \begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$ then $f'(x) = \lambda.\begin{vmatrix} (x-a)^4 & (x-a)^2 & 1 \\ (x-b)^4 & (x-b)^2 & 1 \\ (x-c)^4 & (x-c)^2 & 1 \end{vmatrix}$. Find the value of λ.

4. If $x = a t^3$ & $y = b t^2$, where t is a parameter, then prove that $\frac{d^3y}{dx^3} = \frac{8.b}{27a^3.t^7}$

5. If $\sin y = x \sin(a + y)$, show that $\frac{dy}{dx} = \frac{\sin a}{1 - 2x\cos a + x^2}$.

6. If $F(x) = f(x).\, g(x)$ & $f'(x).\, g'(x) = c$, prove that $\frac{F''}{F} = \frac{f''}{f} + \frac{g''}{g} + \frac{2c}{fg}$ & $\frac{F'''}{F} = \frac{f'''}{f} + \frac{g'''}{g}$.

7. If α be a repeated root of a quadratic equation $f(x) = 0$ & A(x), B(x), C(x) be the polynomials of degree 3, 4 & 5 respectively, then show that $\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ is divisible by f(x), where dash denotes the derivative.

8. Show that $R = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$ can be reduced to the form $R^{2/3} = \frac{1}{\left(\frac{d^2y}{dx^2}\right)^{\frac{2}{3}}} + \frac{1}{\left(\frac{d^2x}{dy^2}\right)^{\frac{2}{3}}}$.

Also show that, if $x = a \sin 2\theta\,(1 + \cos 2\theta)$ & $y = a \cos 2\theta\,(1 - \cos 2\theta)$ then the value of R equals to $4a \cos 3\theta$.

9. Differentiate the following functions with respect to x.

(i) $x^2.\, \ell n\, x.\, e^x$ (i) $\frac{\sin x - x\cos x}{x\sin x + \cos x}$ (iii) $\tan\left(\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}}\right)$

Exercise # 1

1. A **2.** C **3.** A **4.** D **5.** D **6.** B **7.** D **8.** B **9.** C **10.** A **11.** C
12. D **13.** B **14.** D **15.** C **16.** C **17.** C **18.** B **19.** C **20.** ABC
21. AC **22.** CD **23.** AD **24.** ABC **25.** ABC

Exercise # 2

2. 1 **3.** 3

9. (i) $e^x x\,(2\, \ell n\, x + 1 + x\, \ell n\, x)$ (ii) $\frac{x^2}{(x\sin x + \cos x)^2}$ (iii) $\frac{1}{2}\sec^2\frac{x}{2}$